TCS Corporate Website — Project Handover

Adobe Experience Manager Edge Delivery Services with Universal Editor: architecture, authoring, components, operations and support

Date: October 13, 2026

Adobe

Table of Contents

Document Control 6
Document Information 6
Revision History 6
Review and Approval 6
About This Document 7
Audience Guide 7
Conventions Used 7
Executive Summary 8
Part 1 — Project Overview 9
1.1 Project Name and Summary 9
1.2 Business Objectives 9
1.3 Program Goals 10
1.4 Key Stakeholders 11
1.5 Business Benefits 12
1.6 Scope 12
1.7 Non-Functional Requirements 15
1.8 Assumptions 18
1.9 Dependencies 18
1.10 Success Criteria 19
1.11 Contact and Ownership Matrix 20
Part 2 — Solution Architecture 22
2.1 High-Level Architecture 22
2.2 Solution Context Diagram 23
2.3 Component Architecture 24
2.4 Application Landscape 27
2.5 Deployment Architecture 28
2.6 Environment Topology 30
2.7 Security Architecture 30
2.8 Integration Architecture 32
2.9 System Catalogue 33
Part 3 — Site Structure 42
3.1 Site Hierarchy 42
3.2 Locale Structure 43
3.3 Page Model 45
3.4 Folder Structure 47
3.5 Navigation Model 48
3.6 Content Organization Strategy 49
3.7 Error Page Strategy 50
3.8 Redirect Strategy 50
Part 4 — Content Architecture 52
4.1 Templates 52
4.2 Content Types 57
4.3 Metadata Model 59
4.4 Taxonomy Model 62
4.5 Placeholders 62
4.6 References 68
4.7 Data Sheets 70
4.8 Shared Content 70
4.9 Localization Strategy 71
Part 5 — Authoring Fundamentals 75

5.1 Getting access 75
5.2 Universal Editor tour 77
5.3 Creating a page 80
5.4 Working with sections and blocks 82
5.5 Images and assets 84
5.6 The Additional tab 85
5.7 Preview, publish, validation and rollback 89
5.8 Authoring do's and don'ts 92
Part 6 — Component Library 93
6.1 Header Block 93
6.2 Footer Block 113
6.3 Fragment Block 130
6.4 Page Header Block 143
6.5 Section Block 166
6.6 Text Component 181
6.7 Title Component 193
6.8 Image Component 203
6.9 Button Component 213
6.10 Hero Block 226
6.11 Teaser Block 263
6.12 Quote Block 322
6.13 Testimonial Block 343
6.14 Columns Block 366
6.15 Cards Block 382
6.16 List Block 436
6.17 Filter List Block 500
6.18 Table List Block 541
6.19 Anchor List Block 568
6.20 Accordion Block 582
6.21 Tabs Block 597
6.22 Tab Block 625
6.23 Modal Block 638
6.24 Gen Carousel Block 653
6.25 Table Block 671
6.26 Separator Block 691
6.27 Embed Block 728
6.28 Video Block 743
6.29 Form Variations Block 775
6.30 Conversation Canvas Block 805
6.31 Interactive Dashboard Block 835
Part 7 — Taxonomy Documentation 845
7.1 Taxonomy Structure 845
7.2 Tag Hierarchy 850
7.3 Business Taxonomy 851
7.4 Tag Governance 852
7.5 Tag Ownership 853
7.6 Tag Lifecycle 854
7.7 Parent-Child Relationships 856
7.8 Usage Examples 857
7.9 Tags in the Query Index and Analytics 860
Part 8 — Metadata Documentation 862
8.1 How Metadata Works 862

8.2 Metadata Field Reference 867
8.3 Metadata Cheat Sheet 881
Part 9 — Data Sheets Documentation 884
9.1 Placeholders 885
9.2 Metadata 887
9.3 Taxonomy 893
9.4 Redirects 895
9.5 Authors 897
9.6 Locations 899
9.7 Partners 901
9.8 Social Links 903
9.9 Media Contacts 905
Part 10 — Error Pages 908
10.1 Overview 908
10.2 404 Not Found 908
10.3 403 Forbidden 913
10.4 500 Server Error 913
10.5 Maintenance Page 914
10.6 Custom Error Handling 914
10.7 Redirection Rules 916
10.8 Recovery Behaviour 916
Part 11 — Analytics & Data Layer 918
11.1 Overview 918
11.2 Page state 920
11.3 Component state 921
11.4 Events 922
11.5 data-cmp-* attributes and data-analytics 926
11.6 Block-specific tracking 928
11.7 Integrating Adobe Tags and the Web SDK (recommended next step) 931
11.8 Validation 934
11.9 Privacy 936
Part 12 — Publishing, Deployment & Release 937
12.1 Content Publishing Lifecycle 937
12.2 Code Deployment 942
12.3 Branching and Versioning 945
12.4 Release Checklist 945
12.5 Configuration Changes 947
12.6 Rollback 948
Part 13 — Global Troubleshooting Guide 950
13.1 Triage Approach and Diagnostic Toolkit 950
13.2 Page Not Updating After Publish 951
13.3 Block Not Rendering ("failed to load module") 951
13.4 Pages Published at the Wrong Path or a Locale Missing 952
13.5 Component Dialog Fields Missing or Outdated in Universal Editor 952
13.6 Wrong Header (Navigation) or Footer 952
13.7 404 Page in the Wrong Locale or Built-in English 404 953
13.8 Images Not Optimised or Too Large 953
13.9 Page Missing from the Query Index 954
13.10 List or Filter List Empty in Universal Editor 954
13.11 List or Filter List Empty or Incomplete on the Published Site 955
13.12 Redirects Not Working 955
13.13 Placeholders Showing English Text 955

13.14 Tags Showing IDs or Technical Labels 956
13.15 Lint or PageSpeed (aem-psi-check) Failures in Pull Requests 956
13.16 Layout Shift (CLS) or LCP Regression 956
13.17 Bulk Metadata Sheet Not Applied 957
13.18 Hide Block Confusion 957
13.19 Anchor ID Not Applied or Anchor Links Not Jumping 958
13.20 Data Layer Events Missing 958
13.21 Form Submit Does Nothing (No Data Received) 958
13.22 Code Changes Not Visible After Merge (Code Sync) 959
13.23 Content Security Policy Blocking Scripts 959
13.24 Embedded Content Not Loading (YouTube, Vimeo, X) 959
13.25 Sitemap Uses the Wrong Host 960
13.26 Template (Page Type) Not Applied 960
13.27 Page Header or Article Author Details Missing 960
13.28 Site Root Returns 404 961
13.29 Universal Editor Reloads the Page on Every Edit or Shows Stale Rendering 961
13.30 Video or Animation Not Playing Automatically 961
13.31 Fragment, Modal or Tab Content Not Showing 962
13.32 Issue Index 962
Part 14 — Operational Runbook & AMS Handover 963
14.1 Support Model and Ownership 963
14.2 Monitoring Requirements 964
14.3 Logs 965
14.4 Common Errors 967
14.5 Escalation Matrix 967
14.6 Recovery Procedures 968
14.7 Health Checks 969
14.8 Operational Procedures 970
14.9 Known Limitations 973
14.10 AMS Handover Checklist 975
Part 15 — Appendix 978
15.1 Glossary 978
15.2 Acronyms 979
15.3 Reference Documents 981
15.4 Content Modeling Standards 982
15.5 Authoring Cheat Sheets 984
15.6 Metadata Cheat Sheet 986
15.7 Taxonomy Cheat Sheet 988
15.8 Component Inventory 989
15.9 Screenshot Register 992
Part 16 — Handover Findings and Open Items Register 994
16.1 How to Use This Register 994
16.2 Findings 994
16.3 Open Items (To Be Confirmed) 998
16.4 Not Implemented (as of this handover) 998

Document Control

Document Information

Attribute	Value
Document title	TCS Corporate Website — Project Handover
Document type	Official production handover artifact
Version	1.0
Date	13 October 2026
Status	Draft for review [To be confirmed: approval — owner: Product Owner]
Solution	Adobe Experience Manager (AEM) as a Cloud Service, Universal Editor authoring, Edge Delivery Services delivery (xwalk)
Code repository	GitHub `sradha-tcs/tcs-corp`, branch `main`
AEM program / environment	p220207 / e2271469 (author `author-p220207-e2271469.adobeaemcloud.com`)
Edge Delivery site	org `sradha-tcs`, site `tcs-corp`
Source of truth	The code on `main` as of 13 October 2026 (after pull request #62) and the published preview on that date
Classification	Internal — project and support teams [To be confirmed: classification — owner: Information Security]

Revision History

Version	Date	Author	Change
0.1	11 October 2026	Delivery team	First handover guides (author, developer and admin guides)
1.0	13 October 2026	Delivery team	Complete enterprise handover: architecture, every component and variation, field specifications, metadata, taxonomy, sheets, operations, AMS runbook, findings register

Review and Approval

Role	Name	Responsibility	Signature / Date
Product Owner	[To be confirmed]	Business scope, content model, open items	[Pending]
Solution Architect	[To be confirmed]	Architecture, integrations	[Pending]
Technical Lead	[To be confirmed]	Code, components, deployment	[Pending]
Content Lead	[To be confirmed]	Authoring guide, governance	[Pending]
QA Lead	[To be confirmed]	Validation processes	[Pending]
AMS / Support Lead	[To be confirmed]	Runbook, escalation, monitoring	[Pending]

About This Document

This document is the official handover of the TCS Corporate Website. It describes the solution as it is implemented — its architecture, site and content structure, every authorable component and each of its variations, the metadata, taxonomy and data sheets, the analytics data layer, the release process and the operational and support procedures — so that a team member with no prior project knowledge can author, maintain, troubleshoot and enhance the site.

All statements were verified against the code on `main` and the published preview on 13 October 2026. Where a business fact (a name, an SLA, a production domain) wasn't available, the text says `[To be confirmed: … — owner: …]`; Part 16 collects these open items.

Audience Guide

Audience	Start with	Then read
Business teams and Product Owners	Part 1 Project Overview	Part 2.1–2.2 architecture overview, Part 6 component purposes (section 1 of each chapter), Part 16 open items
Content authors (beginners)	Part 5 Authoring Fundamentals	The Authoring Guide (section 5) and Field Specification (section 6) of each component you use; Part 15.5 cheat sheets
Content managers	Part 4 Content Architecture	Parts 7 Taxonomy, 8 Metadata, 9 Data Sheets, 10 Error Pages
Site administrators	Part 3 Site Structure	Parts 9, 10, 12 Publishing & Release, 14 Runbook
Developers	Part 2 Solution Architecture	Part 6 sections 8–11 of each component, Parts 11 Analytics, 12, 15.4 Content Modeling Standards
Architects	Part 2	Parts 3, 4, 11, 16
QA teams	Part 5.7 validation process	Section 5 Validation Process and section 11 Business Rules of each component, Part 13
Release teams	Part 12	Part 14.7 health checks
AMS, operations and support	Part 14 Operational Runbook & AMS Handover	Part 13 Global Troubleshooting, section 13–14 of each component, Part 16

Conventions Used

Convention	Meaning
Enforced	A rule the dialog (model validation) or the code applies; content that breaks it is rejected, corrected or not shown
Recommended	Editorial guidance (lengths, image sizes, wording); not checked by the system
`[To be confirmed: … — owner: …]`	A fact not available at handover; listed in Part 16
Not implemented (as of this handover)	A capability that doesn't exist in the code today
`> **Note:**`, `> **Important:**`, `> **Tip:**`, `> **Warning:**`	Call-outs

Convention	Meaning
[Screenshot Placeholder]	A screenshot to capture and store under `docs/handover/screenshots/` with the given file name (see 15.9 Screenshot Register)
Breakpoints	Mobile < 600px, tablet 600–1199px, desktop ≥ 1200px (some layouts and the header switch at 900px; each chapter states where)
Paths	AEM paths (`/content/tcs-corp/us/en/...`) and published paths (`/en/...`); Part 3.2 maps them

Executive Summary

The TCS Corporate Website runs on Adobe Experience Manager as a Cloud Service with Universal Editor for authoring and Edge Delivery Services for delivery. Content is authored in AEM and published to Edge Delivery; the code (vanilla JavaScript and CSS, no build step) lives in GitHub and is deployed by AEM Code Sync on every merge to `main`.

Area	State at handover
Sites	13 locale sites (`/en`, `/de-de`, `/fr-fr`, `/fr-ca`, `/es-es`, `/pt-pt`, `/nl-nl`, `/sv-se`, `/da-dk`, `/no-no`, `/zh-cn`, `/ja-jp`, `/ko-kr`), each with references, error pages, block library, placeholders, metadata, taxonomy, redirects and data sheets
Templates	Home, Article, Blog, Press Release (Page Type), plus pages without a template
Components	31 documented components: 4 page-level (Header, Footer, Fragment, Page Header), 5 page structure and default content (Section, Text, Title, Image, Button) and 22 content blocks (Part 6, inventory in 15.8)
Authoring	Universal Editor with component models for every block; an Additional tab (Anchor ID, Hide block, padding, margin) on every block but header and footer; UI text from placeholders
Dynamic content	List and Filter List read the query index; Page Header, Teaser and cards read page metadata; Footer reads the Social Links sheet; tags are labelled from the taxonomy
Accessibility	Built to WCAG 2.1 AA (keyboard, focus, names, headings, reduced motion, pause controls); open contrast points in Part 16
Performance	Three-phase loading (eager, lazy, delayed); PageSpeed check on every pull request
Analytics	Adobe Client Data Layer for every page and block (state and events) — ready for Adobe Tags and the Web SDK, which aren't installed yet
Integrations	Not implemented yet: tag manager (Analytics, CJA, Target, Journey Optimizer), form back end (Salesforce / Marketo / API), search service, production domain and CDN
Operations	Runbooks, troubleshooting and AMS handover in Parts 13–14; open items and findings in Part 16

> **Important:** Part 16 lists findings from the code review done for this handover (for example, the contact form confirms a submission although no back end receives it). Review them before go-live.

Part 1 — Project Overview

1.1 Project Name and Summary

Attribute	Value
Project name	TCS Corporate Website
Code name	`tcs-corp`
Document version	1.0
Document date	13 October 2026
Platform	Adobe Experience Manager (AEM) as a Cloud Service with Edge Delivery Services
Authoring	Universal Editor (AEM authoring for Edge Delivery Services, also called "xwalk")
Content source	AEM as a Cloud Service author `https://author-p220207-e2271469.adobeaemcloud.com` (program p220207, environment e2271469)
Delivery bridge	`bin/franklin.delivery/sradha-tcs/tcs-corp/main` (configured in `fstab.yaml`)
Code repository	GitHub `sradha-tcs/tcs-corp`, default branch `main`
Boilerplate	`adobe-rnd/aem-boilerplate-xwalk`
Edge Delivery org / site	`sradha-tcs` / `tcs-corp`
Technology	Vanilla JavaScript (ES modules), CSS3, semantic HTML5; no framework, no transpiling, no build step
Sites	13 locale sites under `/content/tcs-corp/{country}/{language}`, published at language-only paths (`/en`, `/de-de`, …)
Production domain	[To be confirmed: production domain and CDN — owner: Site Administrator]

Summary. The TCS Corporate Website is the multi-locale corporate web presence of TCS, delivered on Adobe Edge Delivery Services. Content is authored in AEM as a Cloud Service with the Universal Editor, a visual, in-context editor, and published to Edge Delivery Services, which renders pages from the authored HTML and the code in the GitHub repository. Pages are decorated in the browser by a small set of vanilla JavaScript and CSS modules: 26 block folders (two of them, Gen Carousel and Tabs, also provide section types), four default-content components (Text, Title, Image, Button), four page templates (Home, Article, Blog, Press Release) and a shared library of decorators (code that builds UI) and utils (helpers that build no UI).

The site is structured as 13 locale sites (United States English plus 12 European and Asia-Pacific locales). Each locale site owns its pages and its own navigation, footer, error pages, UI text (placeholders), bulk metadata, taxonomy, redirects and data sheets (authors, locations, partners, social links, media contacts). The code resolves the locale of every page from its path and loads the matching locale resources, so one code base serves all locales.

Every page and block pushes its state and user interactions to the Adobe Client Data Layer (`window.adobeDataLayer`) in the AEM Core Components schema. The data layer is ready for Adobe Tags with the Adobe Experience Platform Web SDK; no tag manager is loaded yet.

1.2 Business Objectives

The objectives below are derived from the delivered implementation. Business KPIs and target values are not recorded in the repository.

#	Objective	How the implementation supports it
BO-1	One corporate web platform for all markets	13 locale sites served by one code base; locale-aware navigation, footer, error pages, UI text, taxonomy and data (`utils/site.js`)
BO-2	Fast, search-friendly pages	Edge Delivery Services rendering with three-phase loading (eager / lazy / delayed), image optimisation, self-hosted variable font, Lighthouse / PageSpeed target of 100
BO-3	Autonomous, efficient content authoring	Universal Editor with component models for every block, an **Additional** tab (anchor, hide, spacing) on every block except header and footer, Site Global Templates authors copy, block libraries per locale
BO-4	Consistent brand experience	TCS.com Design System (Figma) variations implemented as blocks; design tokens in `styles/styles.css` (spacing, colour, radii, border widths, type scale)
BO-5	Content discovery	List, Filter List, Cards and Conversation Canvas blocks over the query index (`/query-index.json`) and taxonomy
BO-6	Measurable digital experience	Adobe Client Data Layer for every page and block (page, component state; `cmp:click`, `cmp:show`, `cmp:hide`, `cmp:media`, `cmp:search`, `cmp:submit`, `cmp:change`)
BO-7	Inclusive experience	WCAG 2.1 AA target: semantic markup, ARIA on interactive controls, keyboard support, reduced-motion handling, screen-reader announcements
BO-8	Low operational overhead	Serverless delivery (no application servers for the public site), Git-based code deployment through AEM Code Sync, no build pipeline
BO-9	Business KPIs	[To be confirmed: business KPIs (traffic, engagement, lead targets) — owner: Business Owner]

1.3 Program Goals

#	Goal	Status (as of this handover)
PG-1	Launch the corporate site on Edge Delivery Services with Universal Editor authoring	Implemented on preview / live hosts (`*.aem.page` / `*.aem.live`); production domain [To be confirmed: production domain and CDN — owner: Site Administrator]
PG-2	Provide a block library that covers the TCS.com Design System page patterns	Implemented (see 1.6.1); Interactive Dashboard is a placeholder only
PG-3	Set up 13 locale sites with their own references, error pages, sheets and data	Implemented in code and in `paths.json`; content population per locale [To be confirmed: content status per locale — owner: Content Lead]
PG-4	Page templates for recurring page types	Implemented: Home, Article, Blog, Press Release (`templates/`)
PG-5	Analytics readiness	Data layer implemented; Adobe Tags / Web SDK / Analytics / CJA / Target / Journey Optimizer not implemented
PG-6	Accessibility (WCAG 2.1 AA)	Implemented in the blocks and decorators; formal third-party audit [To be confirmed: accessibility audit — owner: QA Lead]
PG-7	Performance (Lighthouse / PageSpeed 100)	Architecture follows "Keeping it 100"; PageSpeed results per release [To be confirmed: PageSpeed baseline — owner: Development Lead]
PG-8	Handover to the run organisation (AMS)	This document

1.4 Key Stakeholders

Role	Responsibility	Name
Executive Sponsor	Funding, strategic direction, final sign-off	[To be confirmed: name — owner: Program Management]
Business Owner / Product Owner	Business requirements, prioritisation, acceptance	[To be confirmed: name — owner: Program Management]
Program / Project Manager	Plan, scope, risks, reporting	[To be confirmed: name — owner: Program Management]
Digital Marketing Lead	Campaigns, analytics requirements, personalisation roadmap	[To be confirmed: name — owner: Business Owner]
Content Lead / Managing Editor	Content strategy, authoring standards, publishing approvals	[To be confirmed: name — owner: Business Owner]
Locale Content Owners (13)	Content, translations, placeholders and data sheets of each locale	[To be confirmed: name per locale — owner: Content Lead]
Taxonomy Owner	Tag namespaces, taxonomy page per locale	[To be confirmed: name — owner: Content Lead]
SEO Lead	Metadata, sitemap, redirects, canonical strategy	[To be confirmed: name — owner: Digital Marketing Lead]
Solution Architect	Architecture, integrations, technical decisions	[To be confirmed: name — owner: Program Management]
Development Lead	Code base, reviews, releases	[To be confirmed: name — owner: Solution Architect]
Front-end Developers	Blocks, templates, decorators, utils	[To be confirmed: names — owner: Development Lead]
QA / Accessibility Lead	Functional, accessibility and performance testing	[To be confirmed: name — owner: Program Management]
AEM Platform Administrator	AEM author environment, users and groups (Adobe Admin Console / IMS), Cloud Manager	[To be confirmed: name — owner: IT Operations]
Edge Delivery Site Administrator	Site configuration (Configuration Service), paths, metadata source, admin API, cache purges	[To be confirmed: name — owner: IT Operations]
CDN / Network Owner	Production domain, DNS, CDN rules, redirects, error responses	[To be confirmed: name — owner: IT Operations]
Security Officer	Security review, CSP, data protection	[To be confirmed: name — owner: IT Security]
Privacy / Legal	Consent, cookie policy, form data processing	[To be confirmed: name — owner: Legal]
AMS (Application Management Services) Lead	Run support, incident management	[To be confirmed: name — owner: IT Operations]
Adobe Customer Success / Support contact	Adobe product support, escalations	[To be confirmed: name — owner: AEM Platform Administrator]

1.5 Business Benefits

Benefit	Description	Source in the implementation
Performance and SEO	Pages are pre-rendered by Edge Delivery Services and served from its CDN; only the code needed for the visible blocks is loaded; the first section is rendered before anything else (LCP). Fast pages improve search ranking and conversion	`scripts/scripts.js` (`loadEager`, `loadLazy`, `loadDelayed`), `scripts/aem.js`
Faster time to market	Authors build pages from ready blocks in the Universal Editor and publish without developer involvement; new code is deployed by merging to `main` (AEM Code Sync), without a build or deployment window	Component models (`blocks/*/_*.json`, `models/*.json`), GitHub workflow
Lower cost of ownership	No application server for delivery; no front-end framework or build chain to maintain; one code base for 13 locales	Repository structure, `utils/site.js`
Global consistency with local autonomy	Shared design and blocks; each locale owns its navigation, footer, 404 page, UI text, taxonomy, metadata, redirects and data sheets	`utils/site.js`, `utils/placeholders.js`, `utils/inherited-metadata.js`
Data-driven optimisation (once tags are deployed)	Every block and interaction is described in the Adobe Client Data Layer, so Adobe Tags rules and the Web SDK can be configured without block changes	`decorators/data-layer-decorator.js`, `utils/analytics.js`
Inclusive reach	WCAG 2.1 AA practices built into shared decorators reduce legal risk and widen the audience	`decorators/a11y-decorator.js`, `decorators/link-decorator.js`, block code
Reuse	Templates add structure (Page Header, About the author, Media contacts) from metadata and sheets, so authors don't repeat it	`templates/`, `decorators/template-decorator.js`

1.6 Scope

1.6.1 In Scope (implemented)

Locale sites. 13 locale sites under `/content/tcs-corp`, each published at a language-only path: `us/en` → `/en`, `de/de-de` → `/de-de`, `fr/fr-fr` → `/fr-fr`, `ca/fr-ca` → `/fr-ca`, `es/es-es` → `/es-es`, `pt/pt-pt` → `/pt-pt`, `nl/nl-nl` → `/nl-nl`, `se/sv-se` → `/sv-se`, `dk/da-dk` → `/da-dk`, `no/no-no` → `/no-no`, `cn/zh-cn` → `/zh-cn`, `jp/ja-jp` → `/ja-jp`, `kr/ko-kr` → `/ko-kr` (see Part 3).

Blocks inventory. The component palette of the Universal Editor (`component-definition.json`) and the block folders (`blocks/`):

Component	Kind	Purpose	Status
Text, Title, Image, Button	Default content	Rich text, headings with Title Type, images with alt text, links styled as buttons (Views, Type, Icon, Alignment)	Implemented
Section	Section	Container with Style (Highlight), Width Type (Full / Fixed), Rounded edge, Background Image, Image alt text, Anchor name	Implemented
Gen Carousel	Section type	A section whose content (Hero, Teaser) becomes carousel slides; Navigation and Autoplay interval	Implemented
Tabs Container	Section type	A section whose Tab items and following components become tabs	Implemented
Accordion (+ Accordion Item)	Block	Collapsible items built on `<details>`	Implemented

Component	Kind	Purpose	Status
Anchor List	Block	Chapter navigation to sections by Anchor name, optional sticky	Implemented
Cards (+ Card)	Block	Manually authored cards in the List presentation	Implemented
Columns	Block	Multi-column layout of default content	Implemented
Conversation Canvas (+ Button)	Block	Search / conversation entry point with menu, speech input and buttons; dispatches `conversation-canvas:search`	Implemented (no search service connected)
Embed	Block	YouTube, Vimeo and X (Twitter) embeds	Implemented
Filter List	Block	Content discovery over the query index with filter pills, advanced filter, featured cards	Implemented
Footer	Block	Page footer (footer page fragment and Social Links) and the authored footer bar	Implemented
Form Variations (+ field items)	Block	Forms with text, select, textarea, consent checkbox, contact card output and submit; client-side validation	Implemented (no submit back end)
Fragment	Block	Includes another page's content	Implemented
Header	Block (auto)	Navigation from the Navigation page with dropdowns and mobile menu	Implemented
Hero	Block	Hero with image or Lottie animation, eyebrow, title, description, CTA list	Implemented
Interactive Dashboard	Block	Placeholder: renders nothing on the site; labelled placeholder in the editor	Not implemented (placeholder)
List (+ List Item)	Block	AEM Core List equivalent: child pages, fixed list or tagged pages from the query index	Implemented
Modal	Block	Opens a fragment in a dialog; links to `/modals/` paths open as modals	Implemented
Page Header	Block	Page title area from page metadata (layouts, author, dates, tags, listen / share / download)	Implemented
Quote	Block	Quote card overlapping a portrait image	Implemented
Separator	Block	AEM Core Separator equivalent (`<hr>` with variants)	Implemented
Tab	Item block	Marks the start of a tab in a Tabs Container section	Implemented
Table (+ rows)	Block	Data tables (2–5 columns) with header options and scroll region	Implemented
Table List (+ Header, Item)	Block	Industries / solutions / products lists	Implemented
Tabs (+ Tab)	Block	Tabs whose panels show fragments	Implemented
Teaser	Block	Image and text layouts (several Figma patterns)	Implemented
Testimonial	Block	Testimonial with author and stats	Implemented
Video	Block	AEM Assets video, YouTube or Vimeo with playback options	Implemented

Templates. Home, Article, Blog, Press Release (`templates/{name}/`), selected by the Page Type page property or the metadata sheet's `page-type` column; template pages live in `/content/tcs-corp/site-global-templates` (author only).

Data layer. Adobe Client Data Layer for page, components and interactions (`decorators/data-layer-decorator.js`, `utils/analytics.js`).

Accessibility. WCAG 2.1 AA practices in all blocks and shared decorators.

Universal Editor authoring. Component definitions, models and filters (`component-definition.json`, `component-models.json`, `component-filters.json`, generated from `blocks/*/_*.json` and `models/_*.json` by `npm run build:json`); editor live updates (`scripts/editor-support.js`); editor-only hints and placeholders in blocks.

Site data. Placeholders, metadata, taxonomy, redirects and data sheets per locale; the query index (`helix-query.yaml`) and sitemap (`helix-sitemap.yaml`); path mapping (`paths.json`).

SEO. Bulk metadata SEO / social columns mapped to standard head tags (`decorators/seo-decorator.js`); `<html lang>` per locale or `language` metadata; sitemap from the query index.

1.6.2 Out of Scope / Not Implemented (as of this handover)

Item	Status	Evidence	Note
Tag manager (Adobe Tags / Launch)	Not implemented	`scripts/delayed.js` contains only a comment; no Tags embed code anywhere	Integration-ready: the embed code goes into `scripts/delayed.js` (delayed phase)
Adobe Experience Platform Web SDK	Not implemented	No `alloy` / Web SDK code	Deployed through Adobe Tags
Adobe Analytics / Customer Journey Analytics	Not implemented	No report suite or data stream configuration	Data layer is ready
Adobe Target	Not implemented	No Target library; the Teaser has a Personalization flag that only adds a `personalization` class and `data-personalization="true"`	Integration-ready markers only
Adobe Journey Optimizer	Not implemented	—	—
Form back end (Salesforce, Marketo, API)	Not implemented	Form Variations dispatches a `form-variations:submit` DOM event and shows the success message; no data is sent	[To be confirmed: form back end — owner: Solution Architect]
Search service	Not implemented	Conversation Canvas dispatches `conversation-canvas:search`; no listener in the code base	[To be confirmed: search service — owner: Solution Architect]
Production CDN and domain	Not configured	`helix-sitemap.yaml` origin is `https://main--tcs-corp--sradha-tcs.aem.live`	[To be confirmed: production domain and CDN — owner: Site Administrator]
Per-locale redirects applied at the edge	Not implemented	Edge Delivery applies only a root `/redirects.json`; locale sheets are read only by the List block	Needs CDN rules or another mechanism
Root `/` → `/en/` redirect	Not implemented in code	The site root is outside all locales and isn't published	Needs a root `/redirects.json` entry or CDN rule
403 / 500 / maintenance pages served by the platform	Not implemented	Edge Delivery serves only `404.html`	Authored 403 / 500 pages exist per locale for a CDN to use
hreflang alternates	Not implemented	No `hreflang` in code or sitemap configuration	Recommended for multi-locale SEO
Interactive Dashboard	Placeholder	`blocks/interactive-dashboard/interactive-dashboard.js` renders nothing	Dialog and logic to be added
Authentication / gated content	Not implemented	Public site, no login	—

Item	Status	Evidence	Note
Consent management (cookie banner)	Not implemented	No consent code	Required before tags / marketing cookies [To be confirmed: consent platform — owner: Privacy / Legal]
Language switcher	Not implemented as a dedicated component	Locales are reached by path; a navigation link list can be authored in the nav page	[To be confirmed: language selector requirement — owner: Business Owner]

1.7 Non-Functional Requirements

1.7.1 Performance

Requirement	Implementation	Type
Lighthouse / PageSpeed Insights score of 100 (mobile and desktop) on feature preview URLs before merge	Project publishing process (developer guidelines in the repository), step "PageSpeed Insights check"	Recommended (process); not enforced by CI (CI runs lint only)
Largest Contentful Paint from the first section	`loadEager` decorates `main` and loads only the first section (`loadSection(main.querySelector('.section'), waitForFirstImage)`); the first image is set to `loading="eager"`	Enforced by code
Three-phase loading	Eager (head decoration, template, first section), Lazy (header, remaining sections, footer, `lazy-styles.css`, fonts), Delayed (after 3 seconds: late block tracking and `scripts/delayed.js`)	Enforced by code
Fonts don't block LCP	`fonts.css` (self-hosted Inter variable font, `font-display: swap`) loads in the eager phase only on viewports ≥ 900px or when already cached in the session (`sessionStorage` flag `fonts-loaded`), otherwise in the lazy phase; an `inter-fallback` font face reduces layout shift	Enforced by code
Per-block code splitting	Each block's JS and CSS are loaded only when the block is on the page (`loadBlock` in `scripts/aem.js`)	Enforced by code
Image optimisation	Authored images are delivered as optimised `/media_…` resources with responsive `<picture>` sources (`createOptimizedPicture`, `decorators/responsive-image-decorator.js`); section backgrounds use 2000px (≥ 600px) and 750px widths	Enforced by code
No third-party scripts on load	No external script in `head.html`; X (Twitter) widgets script loads only when an X embed is rendered; Lottie player is vendored (`scripts/lib/lottie`) and loaded when an animation is near the viewport	Enforced by code
Shared data loaded once	Query index, placeholders, taxonomy, redirects, data sheets are fetched once per page view (in-memory promise cache); inherited metadata cached per browser session	Enforced by code
Real-user monitoring	`sampleRUM` (Edge Delivery Operational Telemetry) in `scripts/aem.js`; `404.html` sends a `404` checkpoint with the referrer	Enforced by code

1.7.2 Accessibility

Requirement	Implementation
WCAG 2.1 AA conformance (target)	Semantic HTML (`nav`, `main`, `header`, `footer`, `details`, `dialog`, `hr`, tables with headers); buttons with `aria-expanded` / `aria-controls`; accessible names from placeholders
Keyboard operability	Header dropdowns and mobile menu close on Escape with focus returned; focus leaving the nav closes it; carousels, tabs and scrollers support keyboard; scrollable tables are focusable labelled regions
New-window links announced	`announceNewTab` adds visually hidden text or extends the `aria-label` (WCAG 3.2.5); `rel="noopener noreferrer"` set
Motion	`prefers-reduced-motion` respected (animations, autoplay, smooth scrolling); Lottie animations have a pause / play control (WCAG 2.2.2)
Images	Alt text fields on images; empty alt marks decorative images; section background alt text field
Language	`<html lang>` from the locale (or `language` metadata); content from another locale (e.g. `/en` 404 fallback) is marked with its `lang`
Forced colours / print	`forced-colors: active` and `print` media queries present in the CSS
Heading hierarchy	Title Type fields on blocks; authors are responsible for a logical order (Recommended)

1.7.3 SEO

Requirement	Implementation
Server-rendered content	Edge Delivery Services serves the full authored HTML; blocks decorate it in the browser
Title, description, keywords	Page properties Title, Description, Keywords; bulk metadata `meta-title`, `meta-description`, `meta-keywords` mapped by `decorators/seo-decorator.js` (in the browser)
Open Graph / Twitter	`og-*`, `twitter-*` columns mapped to standard tags, with fallbacks (og-* → meta-, *twitter-* → og-*)
Canonical	`canonical-url` → `<link rel="canonical">` and `og:url`
Robots	`robots` metadata (standard, works server side)
Sitemap	`helix-sitemap.yaml`: `/sitemap.xml` from `/query-index.json`, `lastmod` YYYY-MM-DD, origin currently the `.aem.live` host
Index hygiene	`helix-query.yaml` excludes references, error pages, data, nav / footer, block libraries, templates, fragments and modals
Multi-locale	Language-only paths; `<html lang>` per locale; hreflang not implemented

> **Note:** The SEO decorator runs in the browser. Search engines that execute JavaScript see the mapped tags; the HTML delivered by Edge Delivery Services carries the raw columns as `<meta name="meta-title">` etc. The page's own Title and Description page properties are rendered server side.

1.7.4 Security

Requirement	Implementation
Content Security Policy	`head.html` and `404.html`: `script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:; base-uri 'self'; object-src 'none';` with `move-to-http-header="true"` (Edge Delivery Services sends it as an HTTP header and replaces the `aem` nonce per response)
No secrets in client code	No API keys, tokens or credentials in the repository; everything is public client-side code

Requirement	Implementation
Safe links	`isSafeUrl` (`utils/card-list-config.js`) rejects `javascript:`, `data:` and other non-http(s) schemes in List and Cards links
Fragments same-origin only	`loadFragment` accepts only paths starting with `/` (not `//`)
Editor updates sanitised	Universal Editor patches are sanitised with DOMPurify 3.2.4 before re-rendering (`scripts/editor-support.js`)
Form data never in analytics	`cmp:submit` events carry field names only, never values
Author access	Adobe IMS through the Adobe Admin Console; AEM permissions
Served files	`.hlxignore` excludes dot files, `*.md`, `package*.json`, `test/*`, `docs/*`, `_*` (model partials), `LICENSE`, `karma.config.js` from the code bus

1.7.5 Browser Support

Item	Value
Supported browsers	Current evergreen browsers (Chrome, Edge, Firefox, Safari, including mobile Safari and Chrome on Android)
Rationale	The CSS uses media query range syntax (`width >= 600px`), `:has()`, custom properties, grid; the JS uses ES modules and modern DOM APIs. Internet Explorer is not supported
Formal support matrix	[To be confirmed: browser and device support matrix — owner: QA Lead]

1.7.6 Responsiveness

Breakpoint	Range	Notes
Mobile	< 600px	Mobile-first base styles
Tablet	600px – 1199px	`@media (width >= 600px)`; some type tokens step up at 900px
Desktop	≥ 1200px	`@media (width >= 1200px)`
Header	switches to the desktop navigation at 900px	`blocks/header/header.js` `matchMedia('(min-width: 900px)')`
Units	rem (1rem = 16px); only breakpoints in px	Code style rule

1.7.7 Localisation

Requirement	Implementation
13 locales	`LOCALES` in `utils/site.js`; `paths.json` generated from it (`npm run build:paths`)
UI text	`/{locale}/placeholders.json` with English fallbacks in code (`utils/placeholders.js`); keys match regardless of case and separators
Tag labels	`/{locale}/taxonomy.json`, the locale's language sheet when translated, else labels derived from tag IDs
Locale resources	Navigation, footer, 404 page, metadata, redirects, data sheets per locale
Typography	Inter variable font covers Latin (the `unicode-range` of `fonts.css`); Chinese, Japanese and Korean text falls back to the system sans-serif font
Dates	`utils/dates.js` formats (DD-MM-YYYY, MM-DD-YYYY, MMM d, yyyy, DD/MM/YY, DD MMM YY)

1.7.8 Maintainability

Requirement	Implementation
Code style	Airbnb ESLint, Stylelint standard, xwalk model lint rules (`eslint-plugin-xwalk`); `npm run lint`
CI	GitHub Actions `Build` workflow runs `npm ci` and `npm run lint` on every push
Pre-commit	Husky pre-commit hook rebuilds `component-*.json` when a `_*.json` model partial is staged
Unit tests	`npm test` (Node test runner): `test/list`, `test/cards`, `test/separator`
Shared code	One purpose per file in `decorators/` and `utils/`, each with a README; utils never import decorators
Dependencies	No runtime npm dependencies; dev dependencies kept up to date by Renovate (`.renovaterc.json`, dev dependencies auto-merged, ESLint pinned to v8)
Core library	`scripts/aem.js` must never be modified (boilerplate rule)

1.8 Assumptions

1. AEM as a Cloud Service (author `p220207-e2271469`) remains the single content source; no other mount points are added to `fstab.yaml`.
2. The Edge Delivery site configuration (Configuration Service) for `sradha-tcs/tcs-corp` holds the `public.json` paths mapping matching the repository `paths.json`; AEM uses the configuration, not the repository file. [To be confirmed: configuration in sync — owner: Site Administrator]
3. The bulk metadata of the 13 locales is applied only after all 13 sheets are listed in the site configuration's `metadata.source`. [To be confirmed: metadata.source configured — owner: Site Administrator]
4. Each locale has, at least, `references/nav`, `references/footer` and `error-pages/404` published; otherwise the header stays empty, the footer shows only Social Links, and the 404 page falls back to `/en`.
5. Authors use the Universal Editor; documents (Google Drive / SharePoint) are not a content source.
6. The site is public; no personal data is stored by the site itself.
7. A production domain and CDN (Adobe-managed or customer-managed) will be configured before go-live. [To be confirmed: production domain and CDN — owner: Site Administrator]
8. Analytics, personalisation and form processing will be added after this handover using the data layer and DOM events provided.
9. Translations are managed in AEM (e.g. language copies or a translation integration). [To be confirmed: translation process — owner: Content Lead]
10. Users run current evergreen browsers.

1.9 Dependencies

Dependency	Type	Used for	Impact if unavailable
AEM as a Cloud Service author (p220207 / e2271469)	Adobe service	Authoring, preview / publish of content to Edge Delivery	No content changes can be made or published; the live site keeps serving the last published content
Universal Editor	Adobe service	Visual authoring	Authors can't edit pages
Edge Delivery Services (code, content and media bus; admin API)	Adobe service	Rendering and delivery	Site unavailable on `aem.page` / `aem.live`
AEM Code Sync (GitHub App)	Adobe service	Synchronising code from GitHub	Code changes not deployed

Dependency	Type	Used for	Impact if unavailable
GitHub `sradha-tcs/tcs-corp`	SaaS	Code repository, pull requests, CI	No code changes
Adobe IMS / Admin Console	Adobe service	Author identity and access	Authors can't sign in
AEM Assets (`/content/dam/tcs-corp/`)	Adobe service	Images, videos, PDFs	Assets can't be picked; published assets keep working
YouTube, Vimeo, X (Twitter)	Third party	Embeds and videos, only where authored	Those embeds don't render
Browser Web Speech API	Browser	Conversation Canvas speech input, only where supported	The microphone button is not functional; typing works
Production CDN / DNS	Customer or Adobe	Production domain	[To be confirmed: production domain and CDN — owner: Site Administrator]
Adobe Tags, Web SDK, Analytics, CJA, Target, AJO	Adobe services	Future analytics and personalisation	Not implemented; no current impact
Node.js (20 or newer; CI uses 24) and npm	Tooling	Lint, tests, JSON build, paths generation	Developer tooling only
AEM CLI (`@adobe/aem-cli`)	Tooling	Local development server	Developer tooling only

1.10 Success Criteria

#	Criterion	Measure	Target
SC-1	All 13 locale sites published	Home, nav, footer, 404 published per locale	13 / 13 [To be confirmed: launch scope per locale — owner: Content Lead]
SC-2	Performance	PageSpeed Insights (mobile) on key templates	100 (project standard); [To be confirmed: accepted minimum — owner: Development Lead]
SC-3	Core Web Vitals (field)	LCP, INP, CLS from Operational Telemetry / CrUX	"Good" thresholds [To be confirmed: targets — owner: Development Lead]
SC-4	Accessibility	Automated and manual WCAG 2.1 AA checks	No AA violations [To be confirmed: audit — owner: QA Lead]
SC-5	Lint and tests	`npm run lint`, `npm test`	Pass on `main`
SC-6	Authoring autonomy	Authors create and publish pages from templates without developer help	[To be confirmed: authoring acceptance — owner: Content Lead]
SC-7	Analytics readiness	Data layer entries present for page, every block and interactions	Verified in the browser console (`window.adobeDataLayer`)
SC-8	SEO	Sitemap generated; no indexed references / error pages; metadata applied	Verified on preview
SC-9	Business KPIs	Traffic, engagement, leads	[To be confirmed: KPIs and targets — owner: Business Owner]
SC-10	Operational readiness	AMS runbooks accepted; escalation contacts named	[To be confirmed: AMS acceptance — owner: AMS Lead]

1.11 Contact and Ownership Matrix

R = Responsible, A = Accountable, C = Consulted, I = Informed.

Area	Content Lead	Locale Content Owners	Taxonomy Owner	SEO Lead	Development Lead	Site Administrator	AEM Platform Admin	CDN Owner	Digital Marketing / Analytics	AMS
Page content and publishing	A	R	I	C	I	I	I	I	I	C
Navigation and footer pages (references)	A	R	I	C	C	I	I	I	I	C
Error pages (404 / 403 / 500)	A	R	I	C	C	C	I	C	I	C
Taxonomy (tags, translations)	C	C	A/R	C	C	I	C	I	C	I
Bulk metadata sheets	C	R	I	A	C	C	I	I	C	I
Placeholders sheets (UI text)	A	R	I	I	C	I	I	I	I	I
Data sheets (authors, locations, partners, social links, media contacts)	A	R	I	I	C	I	I	I	I	I
Redirects sheets	C	R	I	A	C	C	I	R	I	C
Code (blocks, templates, decorators, utils)	C	I	I	C	A/R	I	I	I	C	C
Component models (Universal Editor dialogs)	C	I	I	I	A/R	I	I	I	I	I
Site configuration (paths, metadata source, admin API)	I	I	I	C	C	A/R	C	C	I	C
AEM environment (users, groups, Cloud Manager)	I	I	I	I	C	C	A/R	I	I	C
CDN, DNS, production domain, edge redirects	I	I	I	C	C	C	C	A/R	I	C
Analytics and tags (future)	I	I	I	C	C	C	I	I	A/R	I
Run support (AMS)	I	I	I	I	C	C	C	C	I	A/R

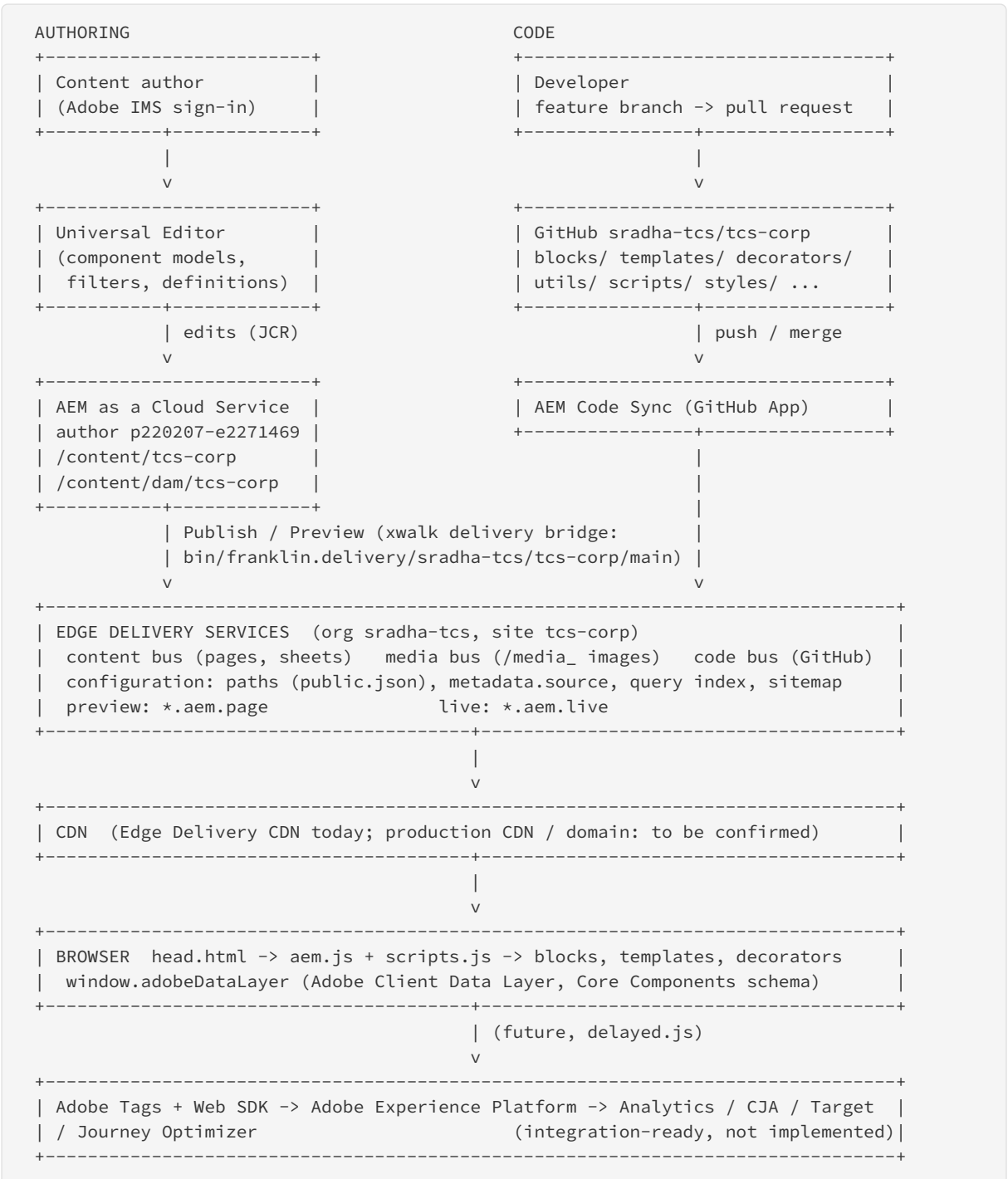
Area	Named owner	Contact
Content	[To be confirmed: name — owner: Business Owner]	[To be confirmed: contact — owner: Business Owner]
Taxonomy	[To be confirmed: name — owner: Content Lead]	[To be confirmed: contact — owner: Content Lead]
Metadata and SEO	[To be confirmed: name — owner: Digital Marketing Lead]	[To be confirmed: contact — owner: Digital Marketing Lead]
Sheets (placeholders, data, redirects)	[To be confirmed: name — owner: Content Lead]	[To be confirmed: contact — owner: Content Lead]
Code	[To be confirmed: name — owner: Solution Architect]	[To be confirmed: contact — owner: Solution Architect]
Configuration (Edge Delivery site)	[To be confirmed: name — owner: IT Operations]	[To be confirmed: contact — owner: IT Operations]

Area	Named owner	Contact
AEM environment	[To be confirmed: name — owner: IT Operations]	[To be confirmed: contact — owner: IT Operations]
CDN	[To be confirmed: name — owner: IT Operations]	[To be confirmed: contact — owner: IT Operations]
Analytics	[To be confirmed: name — owner: Digital Marketing Lead]	[To be confirmed: contact — owner: Digital Marketing Lead]
AMS	[To be confirmed: name — owner: IT Operations]	[To be confirmed: contact — owner: IT Operations]

Part 2 — Solution Architecture

2.1 High-Level Architecture

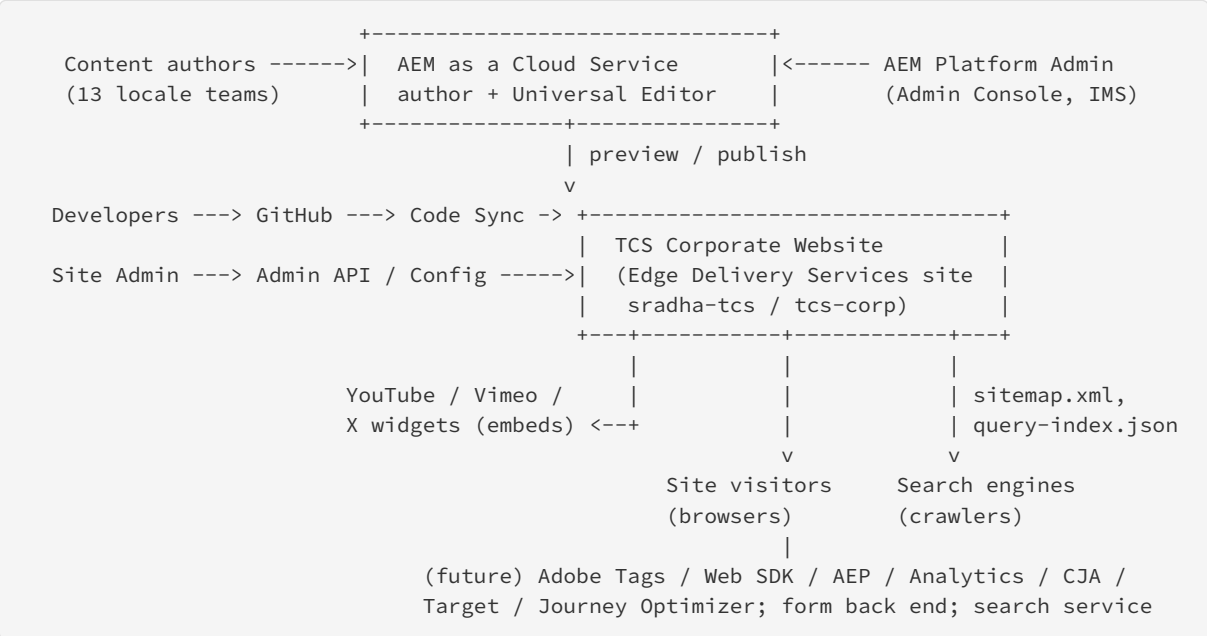
The solution separates three concerns: **authoring** in AEM as a Cloud Service with the Universal Editor, **code** in GitHub synchronised by AEM Code Sync, and **delivery** by Edge Delivery Services, which combines the published content with the code and serves it through its CDN. The browser decorates the delivered HTML with the project's JavaScript and CSS.



Key characteristics

- Content is stored in AEM (JCR) and authored with the Universal Editor. On preview / publish, AEM renders the page as Edge Delivery markup through the xwalk delivery bridge configured in `fstab.yaml` (`https://author-p220207-e2271469.adobeaemcloud.com/bin/franklin.delivery/sradha-tcs/tcs-corp/main`, type `markup`, suffix `.html`).
- AEM paths are mapped to published paths by `paths.json` (applied through the site configuration's `public.json` → `paths`): `/content/tcs-corp/de/de-de/...` is published at `/de-de/...`.
- Edge Delivery Services stores pages and sheets on the **content bus**, images on the **media bus** (served as `/media_<hash>` URLs), and code from GitHub on the **code bus**.
- The browser receives server-rendered HTML with `head.html` injected; `scripts/scripts.js` decorates sections and blocks and loads each block's JS / CSS on demand.
- The data layer is populated in the browser; a tag manager (Adobe Tags) is not loaded yet.

2.2 Solution Context Diagram



Actor / System	Interaction with the site
Content authors	Create, edit, preview and publish pages and sheets in AEM with the Universal Editor; use AEM Sidekick on preview / live pages
Developers	Change code in GitHub through pull requests; preview changes on feature-branch hosts
Site Administrator	Configures the Edge Delivery site (paths, metadata source, access), uses the admin API for preview / publish / cache operations
AEM Platform Administrator	Manages the AEM environment, users and groups, Cloud Manager
Site visitors	Browse the 13 locale sites
Search engines	Crawl pages and `/sitemap.xml`
Third-party media	YouTube (including `youtube-nocookie.com`), Vimeo players and the X widgets script, loaded only for authored embeds / videos

Actor / System	Interaction with the site
Future systems	Adobe Tags, Web SDK, AEP, Analytics, CJA, Target, Journey Optimizer, form back end, search service (not implemented)

2.3 Component Architecture

2.3.1 Code Layers

```
+-------------------------------------------------------------------------+
| head.html          CSP, viewport, aem.js, scripts.js, styles.css        |
+-------------------------------------------------------------------------+
| scripts/aem.js     Edge Delivery core library (NEVER modified): sections, |
|                    blocks, loadBlock, loadCSS, loadScript, RUM, metadata  |
+-------------------------------------------------------------------------+
| scripts/scripts.js page lifecycle: loadEager / loadLazy / loadDelayed,    |
|                    decorateMain, section options, templates, 404 loading  |
+-------------------------------------------------------------------------+
| templates/{name}/  home, article, blog, press-release (structure + CSS)   |
+-------------------------------------------------------------------------+
| blocks/{name}/     26 blocks: {name}.js, {name}.css, _{name}.json (models) |
+-------------------------------------------------------------------------+
| decorators/        shared UI builders (buttons, links, images, data layer, |
|                    SEO, block options, card list, media, ...)             |
+-------------------------------------------------------------------------+
| utils/             no-UI helpers (site/locale, metadata, placeholders,     |
|                    taxonomy, data sheets, query index, redirects, ...)    |
+-------------------------------------------------------------------------+
| styles/            styles.css (tokens, global, LCP), lazy-styles.css,      |
|                    fonts.css, card-list.css                               |
+-------------------------------------------------------------------------+
| scripts/editor-support.js (+ editor-support-rte.js, dompurify.min.js)     |
|                    Universal Editor live updates (editor context only)    |
+-------------------------------------------------------------------------+
```

Layering rules (from `decorators/README.md` and `utils/README.md`): blocks and templates import decorators and utils; **decorators may import utils; utils never import decorators**; each file has one purpose. `scripts/aem.js` is the boilerplate core library and must not be changed.

2.3.2 Page Lifecycle (`scripts/scripts.js`)

`loadPage()` runs `loadEager(document)`, then `loadLazy(document)`, then `loadDelayed()`.

```
loadEager                          loadLazy                         loadDelayed (+3 s)
---------                          --------                         ------------------
decorateHead (SEO, <html lang>)    autolinkModals (/modals/ links)  trackBlocks(document)
trackInteractions (data layer)     loadHeader (nav fragment)        import('./delayed.js')
decorateTemplateAndTheme           loadSections (all sections)        (empty today: place
decoratePageType (body class)      trackBlocks(main)                   for Adobe Tags)
404: loadErrorPage(main)           scroll to #hash element
  else: loadTemplate(main)         loadFooter (footer fragment)
decorateMain(main)                 lazy-styles.css
body.appear                        loadFonts()
loadSection(first, waitForFirstImage)
fonts.css if >= 900px or cached
```

Eager phase (`loadEager`)

1. `decorateHead()` (`decorators/seo-decorator.js`): sets `<html lang>` (`language` metadata if it is a valid language tag, else the locale's `lang`), maps `meta-title`, `meta-description`, `meta-keywords`, `og-*`, `twitter-*`, `canonical-url` to standard head tags.
2. `trackInteractions()` (`decorators/data-layer-decorator.js`): pushes the page entry to `window.adobeDataLayer` and adds capture-phase listeners for click, toggle, close, play / pause / ended, submit and change.
3. `decorateTemplateAndTheme()` (aem.js) adds the `template` / `theme` metadata as body classes; `decoratePageType()` adds the Page Type (`page-type` metadata) as a body class.
4. On the 404 page (`window.isErrorPage`), `loadErrorPage(main)` replaces the built-in content with the authored 404 page (Part 10). Otherwise `loadTemplate(main)` loads `templates/{name}/{name}.js` and `.css` when the Page Type is one of `home`, `article`, `blog`, `press-release` (`TEMPLATES`), and calls its default export with `main` before decoration. A template failure is logged (`failed to load template {name}`) and the page continues.
5. `decorateMain(main)`: `decorateIcons` → `buildAutoBlocks` (empty placeholder) → `decorateSectionsWithOptions` (sections plus the section options Style, Width Type, Rounded edge, Background Image, Image alt text, Anchor name; then Gen Carousel and Tabs Container sections are wrapped in synthetic `gen-carousel` / `tabs` blocks) → `decorateBlocks` → `applyBlockOptions` (Additional tab: `anchor-…` class becomes a unique id) → `decorateButtons` (authored links become buttons).
6. `body.appear` is added; the first section is loaded and the first image is awaited (`waitForFirstImage` sets it to `loading="eager"`) — this is the LCP.
7. Fonts: `fonts.css` is loaded now only if the viewport is ≥ 900px or the session already loaded it.

Lazy phase (`loadLazy`)

1. `autolinkModals(doc)`: a document click listener opens links whose path contains `/modals/` with the Modal block (`openModal`), unless another handler already prevented the default.
2. `loadHeader()`: builds and loads the `header` block (Navigation page as a fragment, see 3.5).
3. `loadSections(main)`: loads all remaining sections in order (each block's JS and CSS); after the first section, RUM enhancement runs.
4. `trackBlocks(main)`: pushes the component state of every loaded block to the data layer.
5. Scrolls to the element of the URL hash, if any.
6. `loadFooter()`: builds and loads the `footer` block.
7. Loads `styles/lazy-styles.css` and `fonts.css`.

Delayed phase (`loadDelayed`)

After 3 seconds: `trackBlocks(document)` (header, footer and late blocks such as modals) and `import('./delayed.js')`. `scripts/delayed.js` is empty (`// add delayed functionality here`); this is where the Adobe Tags embed code belongs.

2.3.3 Block Loading (`scripts/aem.js`)

- `decorateBlock(block)` marks the element as a block (`.block`, `data-block-name`, `data-block-status="initialized"`) and wraps cell content.
- `loadBlock(block)` sets `data-block-status="loading"`, loads `/blocks/{name}/{name}.css` and imports `/blocks/{name}/{name}.js` in parallel, calls its default export, then sets `data-block-status="loaded"`. A failure in the module is caught and logged (`failed to load module for {name}`); the block is still marked loaded so the page continues.
- `loadSection()` loads the blocks of a section one after another and then shows the section (sections are hidden until loaded).

2.3.4 Templates

Page Type	Template code	Adds
home	templates/home/	Layout styles only
article	templates/article/	A Page Header (unless authored, ensurePageHeader); "About the author" from the Authors sheet (author-id), loaded without blocking (appendLoadedSection)
blog	templates/blog/	The article structure and styles
press-release	templates/press-release/	A Page Header (unless authored); "Media contacts" from the Media Contacts sheet filtered by the page's region (else all)

Template pages authors copy live in /content/tcs-corp/site-global-templates/ (excluded from publishing in paths.json).

2.3.5 Decorators and Utils

Layer	Modules (selection)	Responsibility
Decorators	button-decorator.js, icon-button-decorator.js, content-buttons-decorator.js, link-decorator.js, heading-decorator.js, content-decorator.js, image-decorator.js, responsive-image-decorator.js, scroller-decorator.js, card-list-decorator.js, masonry-decorator.js, date-decorator.js, tag-decorator.js, media-decorator.js, animation-decorator.js, article-meta-decorator.js, stats-decorator.js, seo-decorator.js, social-links-decorator.js, template-decorator.js, author-bio-decorator.js, contact-list-decorator.js, a11y-decorator.js, block-options-decorator.js, data-layer-decorator.js	Create or decorate elements consistently in every block
Utils	site.js, inherited-metadata.js, metadata.js, placeholders.js, taxonomy.js, data.js, properties.js, multifield.js, links.js, buttons.js, dates.js, query-index.js, redirects.js, analytics.js, cells.js, card-list-config.js, tags.js, urls.js, ids.js, environment.js, viewport.js, media.js, lottie.js	Read authored values, site data and context without building UI

2.3.6 Universal Editor Support (scripts/editor-support.js)

- Loaded only in the Universal Editor context (it isn't referenced by head.html).
- Listens on main for aue:content-patch, aue:content-update, aue:content-add, aue:content-move, aue:content-remove, aue:content-copy.
- For each change it loads scripts/dompurify.min.js, sanitises the returned HTML (DOMPurify.sanitize(content, { USE_PROFILES: { html: true } })), and re-decorates only the changed part: the whole main (decorateMain, loadSections), a block (decorateButtons, decorateIcons, decorateBlock, applyBlockOptions, decorateRichtext, loadBlock), a section (decorateSectionsWithOptions, decorateBlocks, applyBlockOptions, loadSections) or default content.
- If the change can't be applied in place, the page is reloaded.
- editor-support-rte.js groups rich-text instrumentation (data-richtext-*) and is re-run through a MutationObserver.
- Blocks detect the editor with isEditing() (html.adobe-ue-edit) to show editor-only hints (empty List / Cards / Filter List messages, Tabs hints, Separator warnings, Interactive Dashboard placeholder) and to stop animations, autoplay and sticky behaviour while editing; Hide block keeps hidden blocks visible in the editor.

2.3.7 Component Model Files

File	Content	Generated from
`component-definition.json`	Component palette: groups Default Content, Sections, Blocks	`models/_component-definition.json` + `blocks/*/_*.json` (`npm run build:json`)
`component-models.json`	Dialog fields per model	`models/_component-models.json` + partials
`component-filters.json`	Allowed children: `main` (section, gen-carousel, tabs), `section` (27 components), item filters (accordion, cards, columns, form-variations, list, table, tabs, …)	`models/_component-filters.json` + partials
`models/_page.json`	Page properties (Title, Description, Keywords, Page Type, Navigation, Footer, 404 Page)	—
`models/_section.json`	Section properties	—
`xwalk.json`	`public.xwalk.multi-field.enabled: true` (composite multifields)	—

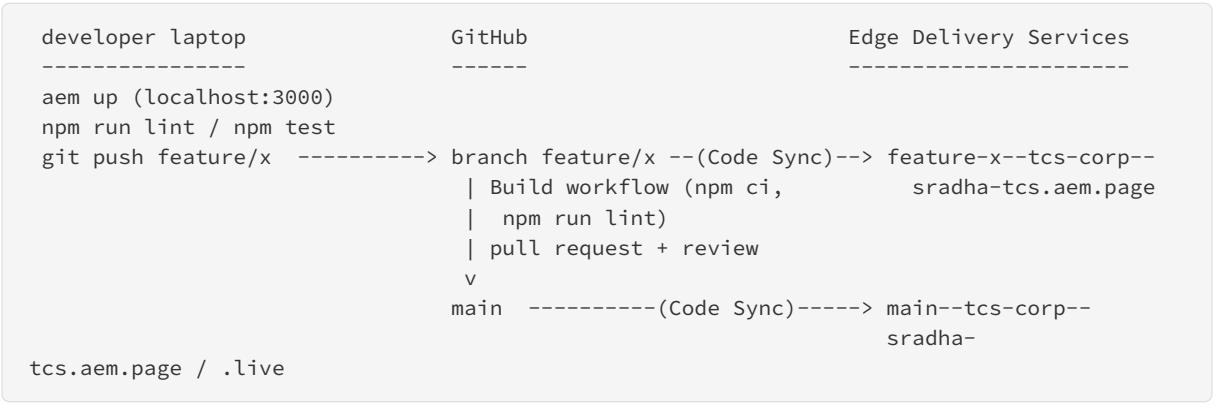
2.4 Application Landscape

System	Category	Role in this solution	Status
AEM as a Cloud Service (author p220207-e2271469)	Content management	Content repository, authoring, publishing to Edge Delivery	Implemented
Universal Editor	Authoring UI	Visual in-context editing with the project's component models	Implemented
Edge Delivery Services	Delivery	Rendering, content / media / code bus, preview and live hosts, query index, sitemap	Implemented
xwalk delivery bridge (`bin/franklin.delivery`)	Integration	Renders AEM pages as Edge Delivery markup	Implemented
AEM Assets (`/content/dam/tcs-corp/`)	DAM	Images, videos, PDFs picked in the Universal Editor	Implemented
GitHub (`sradha-tcs/tcs-corp`)	Source control / CI	Code, pull requests, lint workflow	Implemented
AEM Code Sync	Deployment	Pushes code changes to the code bus per branch	Implemented
AEM Sidekick	Authoring tool	Preview / publish / edit actions on site pages (`tools/sidekick/config.json`)	Implemented
Adobe IMS / Admin Console	Identity	Author authentication and product access	Implemented (platform)
Edge Delivery CDN	CDN	Serves `*.aem.page` / `*.aem.live`	Implemented (platform)
Production CDN / domain	CDN	Production hostname, edge redirects, error responses	[To be confirmed: production domain and CDN — owner: Site Administrator]

System	Category	Role in this solution	Status
Adobe Client Data Layer	Analytics data	`window.adobeDataLayer` page, component and event entries	Implemented
Adobe Tags + Web SDK	Tag management / collection	Would read the data layer and send XDM to AEP	Integration-ready, not implemented
Adobe Analytics / CJA	Analytics	Reporting	Not implemented
Adobe Target	Personalisation	A/B tests and personalisation	Not implemented (Teaser personalisation marker only)
Adobe Journey Optimizer	Journeys	Messaging, journeys, offers	Not implemented
Salesforce / Marketo / API	Lead management	Form submissions	Not implemented (`form-variations:submit` event)
Search service	Search	Site search / conversational search	Not implemented (`conversation-canvas:search` event)
YouTube / Vimeo / X	Third-party media	Embeds and videos	Implemented (per authored component)
Operational Telemetry (RUM)	Monitoring	Sampled real-user data from `sampleRUM`	Implemented (platform)

2.5 Deployment Architecture

2.5.1 Code Deployment



1. Developers work on a feature branch; the AEM CLI dev server (`aem up`) serves local code with previewed content.
2. A push triggers AEM Code Sync, which updates the code bus for that branch: `https://{branch}--tcs-corp--sradha-tcs.aem.page/`.
3. The GitHub `Build` workflow (`.github/workflows/main.yaml`) runs `npm ci` and `npm run lint` on every push.
4. A pull request (template `.github/pull_request_template.md`) includes before / after test URLs; a reviewer merges to `main`.
5. Code Sync updates `main`; preview and live hosts serve the new code immediately (code isn't versioned per content publish).
6. Rollback: revert the commit on `main` (a new commit); Code Sync deploys the revert.

Note: There is no build step: the files in the repository are the files served. `.hlxignore` keeps dot files, Markdown, `package*.json`, `test/*`, `docs/*`, `_*` model partials, `LICENSE` and `karma.config.js` off the code bus.

2.5.2 Content Deployment

1. Authors edit pages in the Universal Editor on the AEM author.
2. **Preview** (Universal Editor or AEM Sites console / AEM Sidekick) sends the page through the delivery bridge to the content bus preview partition → `*.aem.page`.
3. **Publish** sends it to the live partition → `*.aem.live` (and the production CDN once configured).
4. Images are moved to the media bus and served as `/media_<hash>.<ext>` URLs; AEM Assets that are linked (videos, PDFs) are published at their DAM path (`/content/dam/tcs-corp/` is in `paths.json` includes).
5. Sheets (placeholders, metadata, redirects, data) are published as JSON (`.json` appended to the mapped path); the taxonomy page is mapped explicitly to `/{locale}/taxonomy.json`.

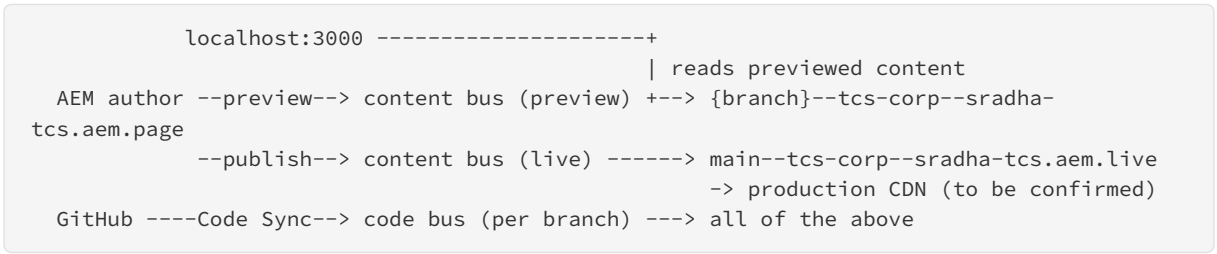
2.5.3 Configuration Files

File	Purpose	Key content
`fstab.yaml`	Content source mount point	`/` → AEM author delivery bridge, type `markup`, suffix `.html`
`paths.json`	AEM path → published path mappings, includes, excludes (generated by `npm run build:paths` from `LOCALES`)	143 mappings (13 locale folders, 13 locale Home pages, 13 × 8 sheets, 13 taxonomy pages); 14 includes (13 locale pages + `/content/dam/tcs-corp/`); 28 excludes (26 block-library entries + `site-global-templates` and its children)
`helix-query.yaml`	Query index definition	Index `pages`, target `/query-index.json`, includes `/**`, excludes `/**.json`, `/*/references/**`, `/*/error-pages/**`, `/*/data/**`, `/*/nav`, `/*/footer`, `/*/block-library(/**)`, `/site-global-templates/**`, `/**/fragments/**`, `/**/modals/**`; properties for all metadata columns
`helix-sitemap.yaml`	Sitemap	`default`: origin `https://main--tcs-corp--sradha-tcs.aem.live`, source `/query-index.json`, destination `/sitemap.xml`, `lastmod: YYYY-MM-DD`
`component-definition.json`, `component-models.json`, `component-filters.json`	Universal Editor palette, dialogs and filters	Generated by `npm run build:json` (also by the pre-commit hook)
`head.html`	Global head	CSP, viewport, `aem.js`, `scripts.js`, `styles.css`
`404.html`	Not-found page	Built-in content and the authored 404 loader (Part 10)
`xwalk.json`	xwalk feature flags	Multi-field enabled
`tools/sidekick/config.json`	AEM Sidekick project config	`editUrlLabel: "AEM Editor"`, `editUrlPattern: "{{contentSourceUrl}}{{pathname}}?cmd=open"`; project label still "AEM XWalk Boilerplate"
`.hlxignore`	Files not served	See note above
`.well-known/adobe/cloud-manager-challenge`	Cloud Manager domain / ownership challenge file	Present in the repository

> **Important:** AEM reads the path mappings from the site configuration (`public.json` → `paths`), not from the repository `paths.json`. After `npm run build:paths`, the Site Administrator must apply the same mappings to the configuration.

2.6 Environment Topology

Environment	URL	Content	Code	Purpose
Local development	`http://localhost:3000` (`aem up`)	Previewed content (same as preview) or `drafts/` with `--html-folder drafts`	Local working copy (uncommitted)	Development and testing
Feature preview	`https://{branch}--tcs-corp--sradha-tcs.aem.page/`	Previewed content	Feature branch	Pull-request validation, PageSpeed checks
Preview (main)	`https://main--tcs-corp--sradha-tcs.aem.page/`	Previewed content	`main`	Author review before publishing
Live (main)	`https://main--tcs-corp--sradha-tcs.aem.live/`	Published content	`main`	Published site until the production domain exists
Production	[To be confirmed: production domain and CDN — owner: Site Administrator]	Published content	`main`	Public website
AEM author	`https://author-p220207-e2271469.adobeaemcloud.com`	Source content (JCR)	—	Authoring in Universal Editor
AEM publish	[To be confirmed: AEM publish tier usage — owner: AEM Platform Admin]	—	—	Not used for page delivery (Edge Delivery serves pages)
Other AEM environments (dev / stage)	[To be confirmed: other Cloud Manager environments — owner: AEM Platform Admin]	—	—	—



2.7 Security Architecture

2.7.1 Content Security Policy

`head.html` (all pages) and `404.html` declare the policy below (the `content` value is a single line in the files; it's wrapped here for readability):

```
<meta
  http-equiv="Content-Security-Policy"
  content="script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:;
```

```
    base-uri 'self'; object-src 'none';"
  move-to-http-header="true"
>
```

Directive	Effect
`script-src 'nonce-aem' 'strict-dynamic'`	Only scripts carrying the nonce (the `<script nonce="aem">` tags of `head.html` / `404.html`) and the scripts they load (modules, `loadScript`) run; `'unsafe-inline'` `http: https:` are fallbacks for old browsers and ignored where `strict-dynamic` is supported
`base-uri 'self'`	Prevents `<base>` injection
`object-src 'none'`	Blocks plugins (`<object>`, `<embed>`)
`move-to-http-header="true"`	Edge Delivery Services sends the policy as an HTTP response header
Not restricted	`style-src`, `img-src`, `frame-src`, `connect-src` aren't set (embeds from YouTube, Vimeo and X work without changes)

> **Note:** When Adobe Tags is added, its embed must be loaded by the project code (`scripts/delayed.js`) so `strict-dynamic` trusts it; inline scripts injected by tag rules need a review against the policy.

2.7.2 Client-Side Security Measures

Measure	Where	Detail
No secrets in code	Whole repository	Everything is public client-side code; no API keys or tokens
DOMPurify (3.2.4)	`scripts/dompurify.min.js`, used only by `scripts/editor-support.js`	Sanitises Universal Editor update HTML before it's parsed and re-rendered; not loaded on the public site
Same-origin fragments	`blocks/fragment/fragment.js` `loadFragment`	Only paths that start with `/` and not `//` are fetched (`{path}.plain.html`); used by header, footer, modal, tabs, footer canvas
Error page content	`scripts/scripts.js` `loadPageContent`	Fetches same-origin `.plain.html` paths built from locale paths or the resolved 404 Page property
Safe URL checks	`utils/card-list-config.js` `isSafeUrl` (List, Cards)	Allows `http:`, `https:`, `mailto:`, `tel:` and relative paths; rejects `javascript:`, `data:` and other schemes
Embeds built from the DOM	`decorators/media-decorator.js`, `utils/media.js`	iframes created with DOM APIs; YouTube privacy-enhanced host (`youtube-nocookie.com`) available; Vimeo URLs encoded
New window links	`decorators/link-decorator.js` `setLinkTarget`	`target="_blank"` with `rel="noopener noreferrer"`
Form data	`decorators/data-layer-decorator.js`	`cmp:submit` events contain field names only, never values; no submission endpoint exists
Analytics never breaks pages	`utils/analytics.js`	`pushToDataLayer` is wrapped in `try / catch`
innerHTML	`fragment.js`, `scripts.js`	Only for same-origin Edge Delivery `.plain.html` responses (trusted content bus)

2.7.3 Platform Security

Area	Implementation
Author authentication	Adobe IMS (Adobe Admin Console product profiles); AEM permissions on `/content/tcs-corp` and `/content/dam/tcs-corp`
Public site	No login, no gated content, no cookies set by project code (the `fonts-loaded` and `tcs-inherited-metadata` values are `sessionStorage`)
Edge Delivery admin	Admin API operations (preview, publish, cache, configuration) require authorised users [To be confirmed: admin roles in the site configuration — owner: Site Administrator]
Preview protection	[To be confirmed: whether preview / live hosts are access-restricted — owner: Site Administrator]
Served files	`.hlxignore` keeps non-runtime files off the code bus
Dependency hygiene	No runtime npm dependencies; Renovate updates dev dependencies
Unpublished areas	Block libraries and Site Global Templates are excluded in `paths.json`

2.8 Integration Architecture

#	Integration	Direction	Mechanism	Status
1	AEM author → Edge Delivery Services	Outbound from AEM	xwalk delivery bridge (`fstab.yaml`), preview / publish	Implemented
2	GitHub → Edge Delivery Services	Outbound from GitHub	AEM Code Sync GitHub App	Implemented
3	AEM Assets → pages	Internal	Asset picker in Universal Editor; media bus `/media_` URLs; DAM path includes	Implemented
4	Query index	Internal	`helix-query.yaml` → `/query-index.json`; read by `utils/query-index.js` (pages of 500, cached per page)	Implemented
5	Sitemap	Outbound to crawlers	`helix-sitemap.yaml` → `/sitemap.xml`	Implemented (origin to be replaced by the production domain)
6	Locale sheets (placeholders, metadata, taxonomy, redirects, data)	Internal	Published sheets read by utils	Implemented
7	Adobe Client Data Layer	Browser	`window.adobeDataLayer` array (Core Components schema)	Implemented (the ACDL library itself is provided by the future Tags extension)
8	Adobe Tags (Launch)	Browser → Adobe	Embed in `scripts/delayed.js`	Integration-ready, not implemented
9	Adobe Experience Platform Web SDK	Browser → AEP Edge Network	Through Adobe Tags	Integration-ready, not implemented
10	Adobe Analytics	AEP → Analytics	Data stream	Not implemented
11	Customer Journey Analytics	AEP → CJA	Connection / data view	Not implemented
12	Adobe Target	Browser ↔ Target	Web SDK personalisation; Teaser personalisation marker	Not implemented

#	Integration	Direction	Mechanism	Status
13	Adobe Journey Optimizer	AEP ↔ AJO	Web SDK / AEP profile	Not implemented
14	Form back end (Salesforce / Marketo / API)	Browser → back end	Listener on `form-variations:submit`	Not implemented
15	Search service	Browser → search API	Listener on `conversation-canvas:search`	Not implemented
16	YouTube / Vimeo / X	Browser → third party	iframes, X widgets script	Implemented
17	Web Speech API	Browser	`SpeechRecognition` in Conversation Canvas	Implemented (browser support dependent)
18	Operational Telemetry	Browser → Adobe	`sampleRUM`	Implemented (platform)
19	Production CDN	CDN ↔ Edge Delivery	Origin `*.aem.live`, push invalidation	[To be confirmed: production domain and CDN — owner: Site Administrator]
20	Authentication for visitors	—	—	Not implemented (public site)

Integration hooks provided by the code

```
// Form Variations: a valid submit (no network request is made)
document.addEventListener('form-variations:submit', (e) => {
  const { data } = e.detail; // field values of the form
});

// Conversation Canvas: the search query
document.addEventListener('conversation-canvas:search', (e) => {
  const { query } = e.detail;
});

// Data layer: page, components and events (Core Components schema)
window.adobeDataLayer = window.adobeDataLayer || [];
```

> **Warning:** The Form Variations block shows its success message (`formSuccess`, default "Thank you, your details were sent.") after a valid submit even though no data is sent anywhere. Connect a back end before forms are used in production.

2.9 System Catalogue

2.9.1 AEM Author

Purpose: The AEM as a Cloud Service author environment (`https://author-p220207-e2271469.adobeaemcloud.com`, program p220207, environment e2271469) is the system of record for all content of the TCS Corporate Website: pages under `/content/tcs-corp`, assets under `/content/dam/tcs-corp`, sheets, taxonomy and data.

Responsibilities:

- Store pages, sections and blocks as JCR content using the xwalk (`core/franklin/components/...`) resource types.

- Host the 13 locale sites (`/content/tcs-corp/{country}/{language}`) with their `references/`, `error-pages/`, `block-library/`, `data/` folders and the placeholders, metadata, taxonomy and redirects sheets; host `site-global-templates/`.
- Provide page properties (Title, Description, Keywords, Page Type, Navigation, Footer, 404 Page — `models/_page.json`).
- Trigger preview and publish to Edge Delivery Services; manage workflows, versions and permissions.

Dependencies: Adobe IMS (sign-in), Universal Editor, the component models served from the code repository (`component-*.json`), Edge Delivery Services (target of preview / publish), AEM Assets.

Data Flow: Author edits in Universal Editor → JCR → on preview / publish AEM renders the page through `bin/franklin.delivery/sradha-tcs/tcs-corp/main` as Edge Delivery markup → content bus (preview or live). The AEM path is mapped to the published path by the site configuration's `paths` (same as `paths.json`).

Failure Handling: If the author is unavailable, the live site is unaffected (Edge Delivery serves the last published content); authors can't edit or publish until it's restored. Failed publications show in the AEM publish status / Universal Editor; re-publish after the cause is fixed. Escalate platform outages to Adobe Support through the AEM Platform Administrator.

Security Considerations: Access through Adobe IMS and Admin Console product profiles; AEM groups control who may edit and publish per locale [To be confirmed: group and permission model per locale — owner: AEM Platform Admin]. Block libraries and site-global-templates must stay excluded from publishing (`paths.json` excludes).

Performance Considerations: Authoring performance depends on AEM; it doesn't affect site performance. Large pages with many blocks re-render more slowly in the Universal Editor.

Implementation Status: Implemented (content source in `fstab.yaml`).

2.9.2 Universal Editor

Purpose: The visual, in-context editor in which authors build pages from the project's components.

Responsibilities:

- Show the component palette (`component-definition.json`: Default Content, Sections, Blocks), dialogs (`component-models.json`) and allowed children (`component-filters.json`).
- Render the page with the project code; apply edits live through `scripts/editor-support.js`.
- Provide asset picking (AEM Assets) and content picking (`aem-content` fields such as Navigation, Footer, 404 Page, links).

Dependencies: AEM author, Adobe IMS, the project code (served from the code bus), `component-*.json`.

Data Flow: Author changes a field → Universal Editor writes to AEM → dispatches `aue:content-*` events → `editor-support.js` fetches no extra data; it sanitises the returned HTML with DOMPurify and re-decorates the changed main, section, block or default content; otherwise reloads the page.

Failure Handling: If a change can't be applied in place (`applyChanges` returns false), the page reloads. Blocks show editor hints instead of empty output (List, Cards, Filter List, Tabs, Separator, Interactive Dashboard). Hidden blocks (Additional tab → Hide block) stay visible in the editor. Query-index-based blocks show "The pages show on the published page (from the query index)" because the index isn't available in the editor.

Security Considerations: Editor update HTML is sanitised (DOMPurify `USE_PROFILES: { html: true }`). Access requires IMS sign-in and AEM permissions.

Performance Considerations: Animations, autoplay and sticky behaviour are disabled while editing (`isEditing()`), which keeps the editor responsive.

Implementation Status: Implemented.

2.9.3 Edge Delivery Services

Purpose: Adobe's delivery platform that renders and serves the website from published content and GitHub code.

Responsibilities:

- Content bus (pages and sheets), media bus (optimised images at `/media_…`), code bus (repository files).
- Hosts: preview `*.aem.page`, live `*.aem.live`, per branch.
- Inject `head.html`, apply metadata (page properties, bulk metadata sheets listed in `metadata.source`), serve `404.html` for missing pages, build the query index and sitemap.
- Admin API for preview, publish, cache purge, index, configuration.

Dependencies: AEM author (content), GitHub and AEM Code Sync (code), the site configuration (Configuration Service).

Data Flow: Preview / publish → content bus; Code Sync → code bus; request → CDN → HTML with `head.html` → browser loads `/scripts`, `/styles`, `/blocks` from the code bus and `.json` sheets, `.plain.html` fragments and `/query-index.json` from the content bus.

Failure Handling: Missing pages return HTTP 404 with `404.html` (authored 404 loading, Part 10). Missing sheets or fragments return 404 to the code, which falls back gracefully (English placeholders, empty lists, empty header). Platform incidents: check Adobe status and escalate to Adobe Support.

Security Considerations: CSP moved to the HTTP header; `.hlxignore` hides non-runtime files; admin operations require authorised users; no server-side code runs for the project.

Performance Considerations: Pages served from CDN cache; push invalidation on publish; per-block code loading; image optimisation; Operational Telemetry (RUM) sampling.

Implementation Status: Implemented (org `sradha-tcs`, site `tcs-corp`).

2.9.4 xwalk (Franklin) Content Delivery Bridge

Purpose: The AEM-side renderer (`/bin/franklin.delivery/sradha-tcs/tcs-corp/main`) that converts AEM pages built with the `core/franklin/components/*` resource types into Edge Delivery markup (sections, blocks as `div` tables, metadata in the head).

Responsibilities:

- Render sections (`core/franklin/components/section/v1/section`), blocks (`.../block/v1/block`), block items (`.../block/v1/block/item`) and default content.
- Render block fields as rows (or key-value rows), `classes_*` fields as block classes (Additional tab), section properties as `data-*` attributes or section metadata, page properties as `<meta>` tags.
- Apply the path mapping so links and references use published paths.

Dependencies: AEM author, component models (field names drive the markup), `paths.json` / site configuration `paths`.

Data Flow: AEM page (JCR) → delivery bridge → HTML (`.html` suffix per `fstab.yaml`) → Edge Delivery content bus.

Failure Handling: If a field is missing, the cell is left out; blocks handle missing cells (e.g. `utils/cells.js` `matchCells` maps cells to model fields when AEM leaves out unset fields). Markup errors appear as undecorated blocks; re-publish after fixing the content or model.

Security Considerations: Runs inside AEM with AEM permissions; only published content leaves AEM.

Performance Considerations: Rendering happens at publish time, not at request time.

Implementation Status: Implemented.

2.9.5 Adobe Data Collection (Tags) and Web SDK

Purpose: Adobe Tags (formerly Launch) would load tracking and marketing tools and map the Adobe Client Data Layer to XDM for the Adobe Experience Platform Web SDK.

Responsibilities (target state):

- Load the Adobe Client Data Layer extension and Web SDK extension.
- Rules on data layer events: `cmp:click` (with `xdm:linkURL`, `xdm:linkType` other / exit / download, `position`, `breakpoint`), `cmp:show` / `cmp:hide`, `cmp:media`, `cmp:search` (`xdm:searchTerm`), `cmp:submit` (field names only), `cmp:change`; page and component state.
- Send events to the AEP Edge Network data stream.

Dependencies: `window.adobeDataLayer` (implemented), a Tags property and data stream [To be confirmed: Tags property and data stream — owner: Digital Marketing Lead], consent management [To be confirmed: consent platform — owner: Privacy / Legal].

Data Flow: Page code pushes entries to `window.adobeDataLayer` (`trackPage` once per page, `trackBlocks` after sections load and again after 3 seconds, interaction events on capture listeners) → (future) Tags reads the data layer → Web SDK → AEP Edge Network.

Failure Handling: `pushToDataLayer` never throws; analytics can't break rendering or links. A failing Tags library would not affect the page because it would load in the delayed phase.

Security Considerations: The Tags embed must be loaded by project code in `scripts/delayed.js` so that the CSP (`strict-dynamic`) trusts it; review custom code in Tags rules against the CSP; no personal data is in the data layer (form values are never pushed).

Performance Considerations: Load in the delayed phase (3 seconds after lazy loading) to protect LCP and the PageSpeed score.

Implementation Status: Integration-ready, not implemented (`scripts/delayed.js` is empty; no Tags embed).

2.9.6 Adobe Analytics

Purpose: Web analytics reporting (traffic, content performance, link and media tracking).

Responsibilities (target state): Receive Web SDK events (through a data stream with Adobe Analytics enabled), map XDM to report suite variables.

Dependencies: Adobe Tags and Web SDK, data stream, report suite [To be confirmed: report suite — owner: Digital Marketing Lead].

Data Flow: Data layer → Tags → Web SDK → AEP Edge Network → Adobe Analytics.

Failure Handling: Not applicable until implemented; the site works without it.

Security Considerations: Consent before collection; no form values or personal data in the data layer.

Performance Considerations: Collection must run in the delayed phase.

Implementation Status: Not implemented.

2.9.7 Customer Journey Analytics

Purpose: Cross-channel analysis on Adobe Experience Platform datasets.

Responsibilities (target state): Report on web events collected through the Web SDK into AEP datasets (connection, data view).

Dependencies: Web SDK, AEP schema / dataset, CJA connection [To be confirmed: CJA licence and connection — owner: Digital Marketing Lead].

Data Flow: Data layer → Tags → Web SDK → AEP dataset → CJA.

Failure Handling: Not applicable until implemented.

Security Considerations: Data governance labels in AEP; consent.

Performance Considerations: No site impact beyond the Web SDK.

Implementation Status: Not implemented.

2.9.8 Adobe Target

Purpose: A/B testing and personalisation of page content.

Responsibilities (target state): Deliver offers through the Web SDK; swap or modify block content.

Dependencies: Web SDK with Target enabled in the data stream; markers in content. The Teaser block's personalisation option (`isOn(props.personalization)`) adds the `personalization` class and `data-personalization="true"` with no visible change, as a hook for a future personalisation script. The Conversation Canvas exports `setConversationCanvasFilters(block, tags)` so a script (e.g. a personalisation service) can supply its filter tags.

Data Flow: (future) Web SDK request → Target decisions → DOM changes on marked blocks.

Failure Handling: Not applicable until implemented; any future implementation must avoid flicker and fall back to the authored content.

Security Considerations: Consent; offer content must respect the CSP.

Performance Considerations: Personalising above-the-fold content conflicts with the eager LCP phase; a pre-hiding strategy would need a performance review.

Implementation Status: Not implemented (markers only).

2.9.9 Adobe Journey Optimizer

Purpose: Journeys, messaging and web / in-app offers based on AEP profiles.

Responsibilities (target state): Web channel experiences and code-based experiences delivered through the Web SDK.

Dependencies: AEP, Web SDK, profile and consent.

Data Flow: (future) Web SDK → AEP Edge → AJO decisions.

Failure Handling: Not applicable until implemented.

Security Considerations: Profile data governance, consent.

Performance Considerations: Same as Target.

Implementation Status: Not implemented.

2.9.10 DAM / AEM Assets

Purpose: Digital asset management for images, videos and documents (`/content/dam/tcs-corp/`).

Responsibilities:

- Asset picking in the Universal Editor (`reference` fields: images, section Background Image, Poster Image, author images).
- Images published to the Edge Delivery media bus and delivered as optimised `/media_<hash>` URLs with responsive `<picture>` (`createOptimizedPicture`, `decorators/responsive-image-decorator.js`).
- Videos and PDFs linked from pages are published at their DAM path (`/content/dam/tcs-corp/` is in `paths.json` includes).
- Lottie animations (Hero / Teaser "Lotto" image type) as `.json` links.

Dependencies: AEM author, Edge Delivery media bus.

Data Flow: Asset in DAM → referenced by a page → on publish the image is copied to the media bus (content-addressed) → page HTML references `/media_…`; documents / videos are served from their published DAM path.

Failure Handling: An unpublished or missing asset shows as a broken image; `waitForFirstImage` resolves on image errors so LCP isn't blocked. Re-publish the page after fixing the asset.

Security Considerations: Only assets referenced by published pages (or below the included DAM folder) are delivered; access to DAM controlled by AEM permissions.

Performance Considerations: Upload images at sufficient resolution; Edge Delivery generates widths (e.g. section backgrounds 2000px ≥ 600px, 750px below). Assets committed to Git (icons, fonts) must be optimised.

Implementation Status: Implemented.

2.9.11 Salesforce

Purpose: CRM for leads and contacts (potential form destination).

Responsibilities (target state): Receive Form Variations submissions (Web-to-Lead or an API).

Dependencies: A listener on `form-variations:submit`, a server-side proxy or service, consent text.

Data Flow: (future) Form → `form-variations:submit` (`detail.data`) → proxy / API → Salesforce.

Failure Handling: Not implemented; today the form shows its success message without sending data.

Security Considerations: Never put Salesforce credentials in client code; use a server-side endpoint; spam protection; data protection review.

Performance Considerations: Load any form integration script only on pages with forms.

Implementation Status: Not implemented. [To be confirmed: form back end — owner: Solution Architect]

2.9.12 Marketo

Purpose: Marketing automation (potential form and campaign destination).

Responsibilities (target state): Receive form submissions, track campaign membership.

Dependencies: A listener on `form-variations:submit`; Marketo Forms API or server-side integration; the `campaign-id` metadata column exists in the query index and metadata for campaign attribution.

Data Flow: (future) Form → event → integration → Marketo.

Failure Handling: Not implemented.

Security Considerations: No client-side secrets; consent.

Performance Considerations: Marketo Munchkin or Forms 2.0 scripts must load in the delayed phase or on demand.

Implementation Status: Not implemented. [To be confirmed: marketing automation platform — owner: Digital Marketing Lead]

2.9.13 APIs

Purpose: Data interfaces used by the site.

Responsibilities:

API	Provider	Used by	Notes
`/query-index.json?offset=&limit=500`	Edge Delivery	List, Filter List, Conversation Canvas (`utils/query-index.js`)	Paged, cached per page view, filtered to the current locale root
`/{locale}/placeholders.json`	Edge Delivery sheet	All blocks with UI text	English fallbacks
`/{locale}/taxonomy.json`	Edge Delivery (taxonomy page)	Teaser, Page Header, Filter List, Conversation Canvas	Language sheet, else default
`/{locale}/metadata.json`	Edge Delivery sheet	Applied by the platform (when in `metadata.source`)	Not fetched by code
`/{locale}/redirects.json`	Edge Delivery sheet	List (`utils/redirects.js`)	Not applied by the platform
`/{locale}/data/{authors,locations,partners,social-links,media-contacts}.json`	Edge Delivery sheets	Page Header, Filter List, footer, templates (`utils/data.js`)	Missing sheets give empty lists
`{path}.plain.html`	Edge Delivery	Fragments (header, footer, modal, tabs, footer canvas), authored 404	Same-origin only
Edge Delivery Admin API	Adobe	Site Administrator (preview, publish, cache, config)	Authenticated

Dependencies: Edge Delivery Services.

Data Flow: Browser `fetch` to same-origin JSON / HTML resources; no third-party APIs are called by project code (except the X widgets script for X embeds).

Failure Handling: Every util catches failures: placeholders → English fallbacks; taxonomy → labels from tag IDs; data sheets, redirects, query index → empty lists; fragments → `null` (the caller keeps or empties its content).

Security Considerations: Public JSON; don't put confidential data in sheets.

Performance Considerations: Each resource is loaded once per page view; the query index is paged (500 rows per request).

Implementation Status: Implemented (site-internal); no external business APIs.

2.9.14 CDN

Purpose: Cache and serve the site close to users; host the production domain.

Responsibilities:

- Today: the Edge Delivery CDN serves `*.aem.page` and `*.aem.live` with push invalidation on preview / publish.
- Target: a production CDN (Adobe-managed or customer-managed BYO CDN) for the production domain, with edge redirects (including `/` → `/en/` and the per-locale redirects sheets), 403 / 500 / maintenance error responses using the authored error pages, and push invalidation.

Dependencies: Edge Delivery origin, DNS, TLS certificates, site configuration `cdn.prod.host` [To be confirmed: production domain and CDN — owner: Site Administrator].

Data Flow: Browser → CDN → (cache miss) Edge Delivery origin `main--tcs-corp--sradha-tcs.aem.live`.

Failure Handling: CDN errors (5xx) can serve the locale's authored `error-pages/500` page; maintenance via a CDN rule (Part 10). [To be confirmed: CDN error configuration — owner: CDN Owner]

Security Considerations: TLS, WAF / DDoS protection [To be confirmed — owner: CDN Owner]; preserve the CSP header from the origin.

Performance Considerations: Respect origin cache headers; configure push invalidation so publishes appear immediately.

Implementation Status: Edge Delivery CDN implemented; production CDN not configured.

2.9.15 Search Services

Purpose: Full-text or conversational site search.

Responsibilities (current): Content discovery uses the query index: List (child pages, fixed list, tagged pages), Filter List (filters over metadata), Conversation Canvas menu (pages below its Menu root path).

Responsibilities (target state): A search service listening on `conversation-canvas:search` (`detail.query`) and rendering results; the data layer already emits `cmp:search` with `xdm:searchTerm`.

Dependencies: Query index (implemented); a search service [To be confirmed: search service — owner: Solution Architect].

Data Flow: (current) `/query-index.json` → blocks. (future) Conversation Canvas → event → search service → results.

Failure Handling: Without a listener the search form does nothing visible after submit; an empty query is ignored.

Security Considerations: Search queries may contain personal data; don't send them to analytics beyond `xdm:searchTerm` without review.

Performance Considerations: The query index is downloaded fully per page view where used; very large indexes increase transfer size.

Implementation Status: Query-index discovery implemented; search service not implemented.

2.9.16 Authentication Services

Purpose: Identity for authors and administrators.

Responsibilities: Adobe IMS authenticates AEM author, Universal Editor, AEM Sidekick and admin API users; Adobe Admin Console assigns product profiles. Visitors aren't authenticated.

Dependencies: Adobe IMS, the organisation's identity provider (federated) [To be confirmed: SSO / federated ID — owner: AEM Platform Admin].

Data Flow: User → IMS → token for AEM / Universal Editor / Sidekick / admin API.

Failure Handling: IMS outage prevents authoring; the live site isn't affected.

Security Considerations: Least-privilege product profiles and AEM groups; offboarding process; no tokens in the repository.

Performance Considerations: None for the public site.

Implementation Status: Implemented (platform) for authors; not applicable for visitors (public site).

2.9.17 GitHub and AEM Code Sync

Purpose: Source control, review and deployment of the code.

Responsibilities:

- Repository `sradha-tcs/tcs-corp` (default branch `main`), pull requests with test URLs (`.github/pull_request_template.md`).
- GitHub Actions `Build` (`npm ci`, `npm run lint`) on every push.
- AEM Code Sync GitHub App: syncs every branch to the code bus (`{branch}--tcs-corp--sradha-tcs`).
- Renovate for dev dependency updates; Husky pre-commit rebuilds `component-*.json`.

Dependencies: GitHub, AEM Code Sync app installed on the repository.

Data Flow: Commit → push → Code Sync → code bus → served immediately on that branch's hosts.

Failure Handling: If Code Sync fails, the branch host serves the previous code; re-push or check the app installation. Bad deployments are rolled back by reverting on `main`.

Security Considerations: Branch protection on `main` and required reviews [To be confirmed: branch protection settings — owner: Development Lead]; never commit secrets; everything in the repository not in `.hlxignore` is public.

Performance Considerations: No build step; changes are live within seconds.

Implementation Status: Implemented.

2.9.18 AEM Sidekick

Purpose: Browser extension for authors to preview, publish and open the editor from site pages.

Responsibilities: Configured by `tools/sidekick/config.json`: edit button "AEM Editor" opening `{{contentSourceUrl}}{{pathname}}?cmd=open` (the page in AEM); preview / publish / unpublish actions according to the user's permissions.

Dependencies: Edge Delivery admin API, IMS sign-in, the site configuration.

Data Flow: Sidekick on `*.aem.page` / `*.aem.live` → admin API → content bus / AEM editor.

Failure Handling: If actions fail, authors use the Universal Editor or AEM Sites console to publish.

Security Considerations: Actions require the corresponding admin role.

Performance Considerations: None for visitors.

Implementation Status: Implemented. The configuration still uses the boilerplate project label ("AEM XWalk Boilerplate") — a cosmetic change recommended.

Part 3 — Site Structure

3.1 Site Hierarchy

The AEM content tree `/content/tcs-corp` holds 13 locale sites and the Site Global Templates. Every locale site has the same layout; `us/en` is shown in full.

```
/content/tcs-corp/                    site root (not published; "/" needs a redirect
to /en/)
├── us/                               country folder
│   └── en/                           locale page = Home of /en (published at /en/)
│       ├── …                         content pages (published below /en/)
│       ├── references/               fragments and reference pages
│       │   ├── nav                   header (Navigation page)
│       │   ├── footer                footer page (Footer block)
│       │   └── …                     other fragments, e.g. alternative nav / footer
pages
│       ├── error-pages/
│       │   ├── 404                   used by 404.html
│       │   ├── 403                   for the CDN (not used by code)
│       │   └── 500                   for the CDN (not used by code)
│       ├── block-library/            a page per block with every variation (not
published)
│       ├── placeholders              sheet: Key, Text, Description, Category
│       ├── metadata                  sheet: bulk metadata (URL patterns /en/**)
│       ├── taxonomy                  taxonomy page (tag, title; a sheet per language)
│       ├── redirects                 sheet: source, destination
│       └── data/
│           ├── authors               sheet
│           ├── locations             sheet
│           ├── partners              sheet
│           ├── social-links          sheet
│           └── media-contacts        sheet
├── de/de-de/  fr/fr-fr/  ca/fr-ca/  es/es-es/  pt/pt-pt/  nl/nl-nl/
├── se/sv-se/  dk/da-dk/  no/no-no/  cn/zh-cn/  jp/ja-jp/  kr/ko-kr/
│                                     12 more locale sites, same layout
└── site-global-templates/            template pages authors copy (not published):
                                      home, press release, blog, article, ...

/content/dam/tcs-corp/                AEM Assets (images, videos, PDFs); published with
pages
```

Published structure (example for German):

```
/de-de/                         Home (AEM /content/tcs-corp/de/de-de, published as /de-de/
index)
/de-de/{page path}              content pages
/de-de/references/nav           header fragment     (.plain.html loaded by the header)
/de-de/references/footer        footer fragment     (.plain.html loaded by the footer)
/de-de/error-pages/404          authored 404 content (.plain.html loaded by 404.html)
/de-de/placeholders.json        UI text
/de-de/metadata.json            bulk metadata (applied when listed in metadata.source)
/de-de/taxonomy.json            tag labels
/de-de/redirects.json           redirects (read by the List block only)
/de-de/data/authors.json        and locations, partners, social-links, media-contacts
```

```
/query-index.json              one index for all locales (blocks filter by /de-de/)
/sitemap.xml                   one sitemap for all locales
```

3.2 Locale Structure

3.2.1 Locale Table

The locales are defined once in `utils/site.js` (`LOCALES`); the first entry (`en`) is the default locale (`DEFAULT_LOCALE`).

#	AEM path	Published path	lang (`<html lang>`)	Country	Region (descriptive)
1	`/content/tcs-corp/us/en`	`/en`	en-US	United States	North America
2	`/content/tcs-corp/de/de-de`	`/de-de`	de-DE	Germany	Europe
3	`/content/tcs-corp/fr/fr-fr`	`/fr-fr`	fr-FR	France	Europe
4	`/content/tcs-corp/ca/fr-ca`	`/fr-ca`	fr-CA	Canada (French)	North America
5	`/content/tcs-corp/es/es-es`	`/es-es`	es-ES	Spain	Europe
6	`/content/tcs-corp/pt/pt-pt`	`/pt-pt`	pt-PT	Portugal	Europe
7	`/content/tcs-corp/nl/nl-nl`	`/nl-nl`	nl-NL	Netherlands	Europe
8	`/content/tcs-corp/se/sv-se`	`/sv-se`	sv-SE	Sweden	Europe
9	`/content/tcs-corp/dk/da-dk`	`/da-dk`	da-DK	Denmark	Europe
10	`/content/tcs-corp/no/no-no`	`/no-no`	no-NO	Norway	Europe
11	`/content/tcs-corp/cn/zh-cn`	`/zh-cn`	zh-CN	China	Asia Pacific
12	`/content/tcs-corp/jp/ja-jp`	`/ja-jp`	ja-JP	Japan	Asia Pacific
13	`/content/tcs-corp/kr/ko-kr`	`/ko-kr`	ko-KR	South Korea	Asia Pacific

> **Note:** The "Region" column is descriptive only; the code doesn't group locales by region. The `region` metadata column (pages) and the `region` column of the Authors and Media Contacts sheets are separate, authored values (e.g. the Press Release template shows the media contacts of the page's `region`).

3.2.2 How the Locale Is Resolved (`getLocale`)

`getLocale(pathname = window.location.pathname)` in `utils/site.js`:

1. Detects an AEM content path: the pattern `^/content/(?!dam/)[^/]+` (e.g. `/content/tcs-corp`), present when the page is opened in the Universal Editor on the AEM author.
2. **Editing (AEM path):** strips `/content/tcs-corp` and matches the remainder against each locale's AEM folders `/{country}/{code}` (e.g. `/de/de-de`).
3. **Site (published path):** matches the path against each locale's published root `/{code}` (e.g. `/de-de`). Matching is case-insensitive and requires the full folder (`/de-de` or `/de-de/...`, not `/de-dex`).
4. **No match** (e.g. `/`, `/site-global-templates/...`, drafts): the default locale `en`.
5. Returns `{ country, code, lang, language, root, aemRoot, contentRoot }`, e.g. for `/de-de/about`: `code: 'de-de'`, `lang: 'de-DE'`, `language: 'de'`, `root: '/de-de'`, `aemRoot: '/de/de-de'`, `contentRoot: ''`.

Related helpers:

Function	Purpose	Example
`localePagePath(path)`	A page of the current locale	`references/nav` → `/de-de/references/nav` (editing: `/content/tcs-corp/de/de-de/references/nav`)
`resolvePagePath(value, fallback)`	A configured page (AEM path, published path or path relative to the locale root), else the fallback in the current locale	`/content/tcs-corp/de/de-de/references/nav-careers` → `/de-de/references/nav-careers` on the site
`localeSheetPath(name)`	A sheet URL of the current locale	`placeholders` → `/de-de/placeholders.json`
`toPublishedPath(path)`	AEM path → published path	`/content/tcs-corp/de/de-de/about` → `/de-de/about`
`toAemPath(path, root)`	Published path → AEM path	`/de-de/about` → `/content/tcs-corp/de/de-de/about`
`getParentPages(path)`	Parent pages up to the locale Home, nearest first	`/de-de/insights/blogs/x` → `/de-de/insights/blogs`, `/de-de/insights`, `/de-de/`
`sheetRows(json, sheet)`	Rows of a (multi-)sheet JSON	Taxonomy language sheet

The language used for `<html lang>` is the page's `language` metadata when it is a valid language tag (`getPageLanguage` in `utils/metadata.js`), otherwise the locale's `lang`.

3.2.3 Adding or Changing a Locale

1. Add the locale to `LOCALES` in `utils/site.js` (`{ country, code, lang }`), e.g. `{ country: 'it', code: 'it-it', lang: 'it-IT' }`.
2. Run `npm run build:paths` (`tools/paths/generate-paths.mjs`). It regenerates `paths.json`: the locale folder mapping, the Home page mapping, the 8 sheet mappings (`placeholders`, `metadata`, `redirects`, `data/authors`, `data/locations`, `data/partners`, `data/social-links`, `data/media-contacts`), the taxonomy mapping (`.../taxonomy:/{code}/taxonomy.json`), the include and the block-library excludes.
3. Commit `utils/site.js` and `paths.json` together; run `npm run lint`.
4. Site Administrator: apply the new mappings to the site configuration (`public.json` → `paths`) and add `/{code}/metadata.json` to `metadata.source`.
5. AEM: create `/content/tcs-corp/{country}/{code}` with `references/nav`, `references/footer`, `error-pages/404` (and 403 / 500), `placeholders`, `metadata`, `taxonomy`, `redirects`, `data/*`, `block-library/`; publish the references, error pages and sheets.
6. If the new language needs a non-Latin script, check typography (the Inter font covers Latin only).
7. Verify on preview: header, footer, 404, placeholders and `<html lang>`.

Important: `paths.json` is generated — never edit it by hand. A locale missing from `LOCALES` is treated as the default `en` locale (its sheets, nav and footer won't be found).

3.3 Page Model

3.3.1 Page Properties (`models/_page.json`, model `page-metadata`)

Field	Name	Type	Inherited by child pages	Description
Title	`jcr:title`	text	No	Page title (`<title>`, `og:title`, used by Page Header and cards)
Description	`jcr:description`	text	No	Meta description, card description fallback
Keywords	`keywords`	text, multi	No	Meta keywords
Page Type	`page-type`	select: Default (empty), Home (`home`), Article (`article`), Blog (`blog`), Press Release (`press-release`)	No	Selects the Site Global Template code and body class; overrides the metadata sheet's `page-type` column
Navigation	`nav`	aem-content	Yes	The header page; empty → nearest parent's Navigation (or the metadata sheet's `nav` column), else `references/nav`
Footer	`footer`	aem-content	Yes	The footer page; empty → nearest parent's Footer (or `footer` column), else `references/footer`
404 Page	`error-page`	aem-content	Yes	The page shown when a page below this one isn't found; empty → nearest parent's 404 Page (or `error-page` column), else the locale's `error-pages/404`

All other page-level values (SEO, social, card, author, dates, region, featured, …) are columns of the locale's bulk metadata sheet (columns: `url`, `page-type`, `meta-title`, `meta-description`, `meta-keywords`, `author-id`, `publish-date`, `modified-date`, `reading-time`, `og-title`, `og-description`, `og-image`, `og-type`, `twitter-card`, `twitter-title`, `twitter-description`, `twitter-image`, `canonical-url`, `robots`, `region`, `country`, `language`, `persona`, `thumbnail-image`, `featured-image`, `campaign-id`, `content-type`, `featured`, `priority`, `listen-link`, `author`, `card-title`, `card-description`, `card-image`, `card-image-alt`, `image-alt`, `eyebrow`, `category`, `hide-in-nav` (the inherited `nav`, `footer` and `error-page` values can also be set as columns)). Each column is published as `<meta name="{column}">`.

3.3.2 Sections (`models/_section.json`)

Field	Name	Type	Effect
Section Name	`name`	text	Label in the Content Tree only
Style	`style`	multiselect: Highlight	Section class `highlight`
Width Type	`sectionWidth`	select: Full Width (default), Fixed Width	`fixed-width` class for Fixed Width

Field	Name	Type	Effect
Rounded edge	`roundedEdge`	boolean	`rounded` class
Background Image	`backgroundImage`	reference	`.section-background` wrapper with an optimised image (2000px ≥ 600px, 750px below), `has-background` class
Image alt text	`backgroundImageAlt`	text	Alt of the background image; empty = decorative
Anchor name	`anchorName`	text	Section `id` (normalised with `toClassName`, leading # removed; not set if the id is already used) — target of Anchor List links

Section types available at the top level of `main` (filter `main`): **Section**, **Gen Carousel** (style `gen-carousel-section`; Navigation, Autoplay interval; allowed children Hero, Teaser) and **Tabs Container** (style `tabs-section`; ID, Width Type, Label, Style, Alignment, Orientation; children Tab, Columns, Title, Text, Image, Button). `scripts.js` wraps the content of Gen Carousel and Tabs Container sections in synthetic `gen-carousel` / `tabs` blocks.

3.3.3 Blocks

A section (filter `section`) accepts 27 components: Text, Image, Button, Title, Page Header, Hero, Teaser, Quote, Testimonial, Anchor List, Accordion, Tabs, Table, Table List, List, Separator, Embed, Video, Modal, Form Variations, Filter List, Interactive Dashboard, Conversation Canvas, Cards, Columns, Fragment, Footer. Every block except header and footer has an **Additional** tab: Anchor ID (`anchor-…`), Hide block, Padding left / right / top / bottom and Margin left / right / top / bottom from the spacing tokens (applied by `decorators/block-options-decorator.js` and `styles/styles.css`).

3.3.4 Default Content

Component	Model fields
Text	Rich text
Title	Title, Title Type
Image	Image (reference), Alt Text
Button	Link, Text, Title, Views, Type, Icon, Alignment

Authored links in default content become buttons (`decorators/content-buttons-decorator.js`).

3.3.5 Markup the Code Receives

```
<main>
  <div>                                  <!-- section -->
    <h1>Page title</h1>                  <!-- default content -->
    <p>Intro text</p>
    <div class="teaser anchor-overview pt-200">   <!-- block with Additional classes -->
      <div><div>title</div><div><h2>Our services</h2></div></div>
    </div>
    <div class="section-metadata">       <!-- section options (document style) -->
      <div><div>style</div><div>highlight</div></div>
    </div>
  </div>
</main>
```

3.4 Folder Structure

3.4.1 Repository Folders

Path	Purpose	Notes
`blocks/{name}/`	Block code: `{name}.js`, `{name}.css`, `_{name}.json` (definitions, models, filters); some have `README.md`, data modules (`list-data.js`, `cards-data.js`, `separator-config.js`)	26 blocks
`templates/{name}/`	Page Type code (`home`, `article`, `blog`, `press-release`)	See `templates/README.md`
`decorators/`	Shared UI builders	See `decorators/README.md`
`utils/`	Shared helpers without UI	See `utils/README.md`
`scripts/`	`aem.js` (core, never modified), `scripts.js`, `delayed.js`, `editor-support.js`, `editor-support-rte.js`, `dompurify.min.js`, `lib/lottie/` (vendored Lottie light player)	—
`styles/`	`styles.css` (tokens, global, LCP), `lazy-styles.css`, `fonts.css`, `card-list.css` (shared by List and Cards)	—
`models/`	`_page.json`, `_section.json`, `_text.json`, `_title.json`, `_image.json`, `_button.json`, `_card-list.json` and the aggregation entry points `_component-*.json`	—
`fonts/`	`inter-variable.woff2`	Self-hosted
`icons/`	SVG icons (arrows, chevrons, close, download, social, filter, headphones, menu, mic, pause, play, search, share)	Used with `:icon:` syntax / code
`tools/paths/`	`generate-paths.mjs`	`npm run build:paths`
`tools/sidekick/`	`config.json`	AEM Sidekick
`test/`	Node unit tests (`list`, `cards`, `separator`)	`npm test`; not served
`docs/`	Documentation	Not served (`.hlxignore`)
`.github/`	`workflows/main.yaml` (lint), `cleanup-on-create.yaml` (boilerplate), PR template	—
`.husky/`	Pre-commit hook (rebuild component JSON)	—
Root files	`head.html`, `404.html`, `fstab.yaml`, `paths.json`, `helix-query.yaml`, `helix-sitemap.yaml`, `component-*.json`, `xwalk.json`, `.hlxignore`, `package.json`, `README.md`, `favicon.ico`	—

3.4.2 AEM Folders

AEM path	Published	Purpose	Indexed
`/content/tcs-corp/{country}/{code}`	`/{code}/`	Locale Home and content pages	Yes
`…/references/`	`/{code}/references/`	Nav, footer and other fragments	No (excluded)
`…/error-pages/`	`/{code}/error-pages/`	404 / 403 / 500	No (excluded)

AEM path	Published	Purpose	Indexed
…/block-library/	Not published	Block test pages	No
…/data/	/{code}/data/*.json	Data sheets	No
…/placeholders, metadata, redirects	/{code}/*.json	Sheets	No (/**.json excluded)
…/taxonomy	/{code}/taxonomy.json	Taxonomy	No
Fragment folders …/fragments/, modal folders …/modals/ (if used)	Published	Content included in pages / opened as modals	No (excluded)
/content/tcs-corp/site-global-templates	Not published	Template pages	No
/content/dam/tcs-corp/	At their DAM path	Assets	No

3.5 Navigation Model

3.5.1 Header

The header is not authored on each page; scripts/aem.js loadHeader creates a header block in <header> during the lazy phase.

Which Navigation page is used (blocks/header/header.js):

```
page's own Navigation (page property `nav`, or the metadata sheet's `nav` column)
  └─ else the nearest parent page's Navigation, up to the locale Home page
       └─ else the locale's references/nav  (e.g. /de-de/references/nav)
```

- getInheritedMetadata('nav') (utils/inherited-metadata.js) loads the published HTML of all parent pages in parallel (getParentPages), reads <meta name="nav">, and takes the nearest non-empty value. Values are cached per URL in sessionStorage (tcs-inherited-metadata) for the browser session — a changed parent property shows in a new session.
- resolvePagePath(value, 'references/nav') accepts AEM paths, published paths or paths relative to the locale root (e.g. references/nav-careers).
- The page is loaded as a fragment (loadFragment → {path}.plain.html, decorated like a page).

Navigation page structure (authored in references/nav): three sections, in order:

Section	Class	Content
1	nav-brand	Logo / brand link (its button decoration is removed, a plain link)
2	nav-sections	A bulleted list of links; a list item with a nested list becomes a dropdown
3	nav-tools	Tools (e.g. links, icons)

Dropdowns. Each top-level item with a sub-list gets a toggle <button class="nav-drop-toggle"> with aria-controls (sub-list id nav-submenu-…) and aria-expanded. A plain label becomes the button; a linked label keeps its link and gets a separate toggle button labelled "Show {label} menu" (placeholder showSubmenu). On desktop, opening one dropdown closes the others; the sub-list opens as a 12.5rem panel below the item.

Mobile menu (< 900px). A hamburger button (aria-controls="nav-sections", label "Open navigation" / "Close navigation" from placeholders openNavigation / closeNavigation) opens the sections full height; the sub-lists

are expanded in the menu; page scrolling is locked (`body overflow-y: hidden`) while open. The header wrapper is fixed on mobile and relative on desktop; the nav height is `--nav-height` (4rem).

Keyboard and focus. Escape closes the open dropdown (desktop) or the menu (mobile) and returns focus to its button; focus leaving the nav closes the dropdowns / the menu. Crossing the 900px breakpoint resets the state.

Accessibility name. `<nav id="nav" aria-label="Main navigation">` (placeholder `mainNavigation`).

Failure. No Navigation page published (e.g. a new locale): the header stays empty.

3.5.2 Footer

`loadFooter` creates a `footer` block in `<footer>`:

```
page's own Footer (page property `footer`, or the metadata sheet's `footer` column)
  └─ else the nearest parent page's Footer, up to the locale Home page
       └─ else the locale's references/footer
```

- The footer page is loaded as a fragment together with the locale's Social Links sheet (`/{locale}/data/social-links.json`, in `sort-order`), rendered as an icon list (`footer-social`, labels "Follow us on {platform}", placeholder `followUsOn`).
- On the footer page, the authored **Footer** block (model `_footer.json`) renders: the Conversation Canvas Page (a page with the Conversation Canvas block, loaded as a fragment), the Copyright Text (rich text; `{year}` becomes the current year) and the Link List (Link Title, Link Path, Link Type Default / Open in new window) as `<nav class="footer-links" aria-label="Footer links">`.
- Missing footer page: only the Social Links (if any) are shown.

3.5.3 In-Page Navigation

- **Anchor List** links jump to sections by Anchor name (section property) or any element id; optional sticky.
- **Additional tab Anchor ID** (`anchor-…`) gives blocks unique ids for `#` links; on load, the URL hash scrolls to its element after the sections are loaded.
- **Modal links**: links to a path containing `/modals/` open as a modal dialog.

3.5.4 Alternative Navigation per Area

Because Navigation and Footer are inherited, a site area (e.g. Careers) can use its own header and footer by setting the property once on the area's top page, for example Navigation = `/content/tcs-corp/us/en/references/nav-careers`. All pages below inherit it unless they set their own.

3.6 Content Organization Strategy

Principle	Implementation	Type
One locale tree per market	`/content/tcs-corp/{country}/{code}`; every locale owns its pages, references, error pages, sheets and data	Enforced (code resolves resources per locale)
Locale resources in fixed locations	`references/nav`, `references/footer`, `error-pages/404`, `placeholders`, `metadata`, `taxonomy`, `redirects`, `data/*`	Enforced (default paths in code and `paths.json`)
Hierarchy drives inheritance	Navigation, Footer and 404 Page are inherited from parent pages up to Home	Enforced
Hierarchy drives discovery	List "child pages" and Filter List Root path use the published path hierarchy from the query index	Enforced

Principle	Implementation	Type
Page types for recurring layouts	Page Type property or `page-type` metadata column (e.g. `/en/newsroom/**` → `press-release`)	Enforced
Metadata at scale	Bulk metadata sheet per locale with URL patterns starting with the locale (`/de-de/**`)	Recommended (applies only when listed in `metadata.source`)
Tags from taxonomy	Tags resolved to labels from the locale's taxonomy	Enforced (fallback to ID-derived labels)
Hidden from listings	`hide-in-nav = yes` excludes a page from List results	Enforced
Templates as starting points	Authors copy pages from `site-global-templates`	Recommended
Fragments and modals in their own folders	`references/`, `fragments/`, `modals/` are excluded from the query index	Enforced (by path)
Block libraries for validation	`block-library/` per locale shows every block variation in the locale's language	Recommended; excluded from publishing
Short, lowercase, hyphenated URLs	Page names become URLs	Recommended

3.7 Error Page Strategy

Summary (details in Part 10):

Error	Served by	Content
404 Not Found	Edge Delivery Services (`404.html`)	The inherited **404 Page** property of the nearest parent, else `/{locale}/error-pages/404`, else `/en/error-pages/404`, else the built-in "404 / Page Not Found / Go home"
403 Forbidden	Not served by Edge Delivery	Authored `error-pages/403` for a CDN to serve [To be confirmed: CDN — owner: CDN Owner]
500 Server Error	Not served by Edge Delivery	Authored `error-pages/500` for a CDN to serve
Maintenance	Not implemented	Recommended: CDN rule (Part 10.5)

3.8 Redirect Strategy

3.8.1 Current State

Mechanism	Status	Detail
Per-locale redirects sheets `/{locale}/redirects.json` (Source, Destination)	Authored and published per locale; **not applied by the platform**	Edge Delivery Services applies only a root `/redirects.json`; there's no setting for more sheets
Root `/redirects.json`	Not present	The site root `/content/tcs-corp` is outside the locales and isn't published
`/` → `/en/`	Not implemented	Needs a root redirect or a CDN rule
CDN redirect rules	Not configured	[To be confirmed: production domain and CDN — owner: Site Administrator]

Mechanism	Status	Detail
List block	Implemented	Links to pages that redirect point to the final target

3.8.2 List Block Redirect Resolution (`utils/redirects.js`)

- `loadRedirects()` loads the current locale's sheet once per page (`localeSheetPath('redirects')`), accepting `Source` / `source` and `Destination` / `destination` columns. Rows are kept only when the source starts with `/` and a destination exists.
- `normalizeRedirectPath()` compares paths without query, hash, `.html` and trailing slash (except `/`).
- `resolveRedirect(map, href)` follows internal redirects up to **5 hops**, stops on loops and at external destinations; non-internal hrefs are returned unchanged.
- Used only by the List block (`blocks/list`); other links (navigation, default content) aren't rewritten.

3.8.3 Recommended Target State

1. Configure the production CDN with redirect rules generated from the 13 `/{locale}/redirects.json` sheets (sources must start with the locale, e.g. `/de-de/old` → `/de-de/new`), returning HTTP 301.
2. Add `/` → `/en/` (or a language-negotiating rule) at the CDN; alternatively, publish a root `/redirects.json` with the root redirect and global redirects.
3. Keep the sheets as the single source authors maintain; define a sync process (manual export or automation) [To be confirmed: redirect sync mechanism — owner: CDN Owner].
4. Avoid chains: point sources to final destinations (the List block follows at most 5 hops).
5. Test redirects on preview with the List block and at the CDN after each change.

Part 4 — Content Architecture

This part describes how the content of the TCS Corporate Website is organised: the Site Global Templates and the code behind them, the page and content types, the metadata and taxonomy models, the placeholders (user interface text), the reference pages (navigation, footer, fragments), the data sheets, the shared content and the localization strategy across the 13 locale sites.

Every statement in this part was verified against the repository (`sradha-tcs/tcs-corp`, branch `main`) and, where noted, against the published preview (`https://main--tcs-corp--sradha-tcs.aem.page/`) on 13 October 2026.

```
/content/tcs-corp/                        AEM as a Cloud Service (author)
|
+-- us/en/  (Home page = the locale page itself, published at /en/)
|   +-- ...content pages...                -> /en/...
|   +-- references/  nav, footer, fragments  -> /en/references/...
|   +-- error-pages/ 404, 403, 500          -> /en/error-pages/...
|   +-- block-library/                      (not published)
|   +-- placeholders  (sheet)               -> /en/placeholders.json
|   +-- metadata      (sheet)               -> /en/metadata.json
|   +-- taxonomy      (taxonomy page)       -> /en/taxonomy.json
|   +-- redirects     (sheet)               -> /en/redirects.json
|   +-- data/ authors, locations, partners, social-links, media-contacts
|                                            -> /en/data/{name}.json
+-- de/de-de/ ... kr/ko-kr/  (12 more locale sites, same layout)
+-- site-global-templates/  home, article, blog, press-release  (not published)
```

4.1 Templates

A **template** in this project has two halves:

1. **A template page** under `/content/tcs-corp/site-global-templates/` (Site Global Templates) that authors copy to create a new page of that kind. The page name is the Page Type (`home`, `article`, `blog`, `press-release`). The template pages are excluded from publishing in `paths.json` (`/content/tcs-corp/site-global-templates` and `/content/tcs-corp/site-global-templates/**`) and from the query index (`/site-global-templates/**` in `helix-query.yaml`, as a safeguard).
2. **Template code** in `templates/{name}/{name}.js` and `templates/{name}/{name}.css`, which runs on every page whose Page Type is that name and can add structure to the page before it is decorated.

How the Page Type selects the template (enforced by code, `scripts/scripts.js`):

```
loadEager()
  decorateHead()            SEO / social metadata, <html lang>
  decorateTemplateAndTheme() body classes from "template" and "theme" metadata (aem.js)
  decoratePageType()        body class from the "page-type" metadata (e.g. body.article)
  loadTemplate(main)        name = page-type, else "template" metadata (class-name form)
                            if name is in TEMPLATES = [home, article, blog, press-
release]:
                              import templates/{name}/{name}.js
                              + loadCSS templates/{name}/{name}.css   (in parallel)
                              await template.default(main)      (before main is decorated)
                            on error: console.error("failed to load template {name}",
error)
  decorateMain(main)        sections, blocks, buttons, block options
  loadSection(first)        LCP
```

- The **Page Type** value comes from the **Page Type** page property (`page-type`, a select in `models/_page.json` with the options Default (empty), Home, Article, Blog, Press Release) or, when the page property is empty, from the `page-type` column of the locale's bulk metadata sheet (for example a row `/en/newsroom/**` with `page-type` = `press-release`). Page-level values win over the sheet.
- The value is converted to a class name (lower case, hyphenated). `decoratePageType()` adds it as a body class (only from `page-type`); `loadTemplate()` also accepts the standard `template` metadata as a fallback.
- Only values listed in `TEMPLATES` load template code. Any other value (for example `landing`, `listing`, `report`, set through the metadata sheet) only becomes a body class and is still available to blocks (the Page Header uses `landing` and `listing`; the Filter List filters on page types).
- The template JavaScript and CSS are loaded **eagerly** (before the first section is shown), in parallel. Templates keep the LCP content authored and load data sheet content without blocking (see `appendLoadedSection` below).
- Templates run in the Universal Editor too, because the editor renders the same page code. Content that a template adds is generated, not authored: it has no editor instrumentation and cannot be selected or edited in the Universal Editor.

Shared template building blocks (`decorators/template-decorator.js`):

Function	What it does	Used by
`ensurePageHeader(main)`	If the page has no authored Page Header block (`.page-header`), it inserts an empty Page Header block as the first section. The Page Header block then builds its content from the page metadata (title, description, tags, content type, publish date, reading time, author, featured image, listen link).	Article, Blog, Press Release
`appendLoadedSection(main, className, build)`	Appends a new last section containing an `<aside class="{className}">`, then fills it when `build()` resolves (for example after a data sheet is loaded). If `build()` returns nothing or fails, the section is removed. The page is never delayed.	Article, Blog (author bio); Press Release (media contacts)

> **Note:** A page without a Page Type (Default) loads no template code. This is the "Default (no template)" case described in 4.1.5.

4.1.1 Home Template

Purpose: The layout of a locale's Home page (the locale page itself, for example `/content/tcs-corp/us/en`, published at `/en/`). The Home page is composed entirely of authored blocks (for example Hero, Gen Carousel sections, Teasers); the template adds no generated content.

Ownership: Template page `site-global-templates/home`: [To be confirmed: content owner of the Home template page — owner: Digital marketing / web content lead]. Template code `templates/home/`: development team.

Usage:

- Page Type value: `home` (Page Type page property option "Home").
- `templates/home/home.js` exports an empty `decorate(main)` (no generated structure).
- `templates/home/home.css` removes the top margin of the first section and the top padding of its content (`body.home main > .section:first-of-type { margin-block-start: 0; }` and `... > div { padding-block-start: 0; }`), so the first section (usually a Hero or a Gen Carousel) meets the header without a gap.
- To create a locale Home page: the locale page itself is the Home page; set its Page Type to **Home** and build it like `site-global-templates/home` (copy the blocks from the template page).

Dependencies: `scripts/scripts.js` (`TEMPLATES`, `loadTemplate`, `decoratePageType`); the blocks the authors place on the page. No data sheets, no placeholders.

Governance:

- Enforced: the template CSS only applies when the body has the `home` class (Page Type = `home`).
- Recommended: only the locale Home pages use the Home Page Type; other landing pages use Default (or the `landing` page type via the metadata sheet, for the Page Header Landing layout).
- Changes to `templates/home/` follow the development process (feature branch, pull request, review).

Best Practices:

- Make the first section the LCP element (a Hero image or the first Gen Carousel slide) and keep it authored.
- Do not add a Page Header block to the Home page unless the design calls for it; the Home template does not add one.
- Keep the Home page's **Navigation**, **Footer** and **404 Page** properties set to the locale defaults (the published preview of `/en/` sets `nav = /en/references/nav`, `footer = /en/references/footer`, `error-page = /en/error-pages/404`); every page of the locale inherits them.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/template-home-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** The /en/ Home page at 1440 px wide showing the first section (Hero or Gen Carousel) directly under the header with no top gap, followed by the authored blocks.

4.1.2 Article Template

Purpose: Thought-leadership and insight articles. The template guarantees a consistent article header and an "About the author" box without the author having to build them.

Ownership: Template page `site-global-templates/article`: [To be confirmed: content owner — owner: Editorial / content strategy lead]. Template code `templates/article/`: development team. Author data: the locale's Authors data sheet (see 9.5).

Usage:

- Page Type value: `article` (Page Type option "Article").
- `templates/article/article.js`:
  1. `ensurePageHeader(main)`: adds a Page Header block as the first section unless one is authored. With Page Type `article` and no Layout chosen, the Page Header uses the **Article** layout (meta row with tags and content type, published date, reading time, listen / download / share actions; title and author; description and CTAs).
  2. Reads `author-id` from the page metadata (`getPageMetadata().authorId`). If there is none, it stops.
  3. Otherwise appends a section `<aside class="template-author">` and fills it, without blocking the page, with `buildAuthorBio()` (`decorators/author-bio-decorator.js`) from the Authors sheet entry (`getAuthor(authorId)`, case-insensitive ID match).
- The author bio card shows: the author image (round, 6rem), the heading "About the author" (placeholder `aboutTheAuthor`), the name, the designation and region ("designation, region"), the bio text, and a "View LinkedIn profile" link (placeholder `viewLinkedinProfile`) when the sheet has a LinkedIn URL. If the author is not found, or the entry has no name, the section is removed.
- `templates/article/article.css` styles the bio card (`main .template-author .author-bio`): a bordered, rounded card; from 600 px the image and the text sit side by side (`grid-template-columns: auto 1fr`).

Dependencies: `decorators/template-decorator.js`, `decorators/author-bio-decorator.js`, `utils/data.js` (`getAuthor`), `utils/metadata.js` (`getPageMetadata`), `utils/placeholders.js`; the Page Header block; the

locale's `data/authors` sheet; page metadata `author-id`, `publish-date`, `reading-time`, `content-type`, tags, `listen-link`.

Governance:

- Enforced: the author box only appears when `author-id` is set and matches a row (with a `name`) of the locale's Authors sheet.
- Enforced: an authored Page Header block always wins over the generated one.
- Recommended: set `author-id`, `publish-date` and `reading-time` for every article (bulk metadata sheet row covering the article path, or page-level metadata).
- Recommended: copy new articles from `site-global-templates/article` so the Page Type is already set.

Best Practices:

- Keep author IDs stable and lower case with hyphens (for example `jane-doe`); the template matches them case-insensitively, but the List and Filter List blocks match them exactly.
- Maintain each author in every locale's Authors sheet where they publish (sheets are per locale).
- Use a square author image (it is cropped to a circle with `object-fit: cover`).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/template-article-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** An article page at 1440 px showing the generated Page Header (Article layout: tags and content type row, published date, reading time, listen / download / share icons, title, author lockup, description) and, at the end of the page, the "About the author" card with image, name, designation and LinkedIn link.

4.1.3 Blog Template

Purpose: Blog posts. They share the article structure (Page Header and "About the author") and styles, but keep a separate Page Type so they can be filtered, indexed and styled separately.

Ownership: Template page `site-global-templates/blog`: [To be confirmed: content owner — owner: Editorial / blog lead]. Template code `templates/blog/`: development team.

Usage:

- Page Type value: `blog` (Page Type option "Blog").
- `templates/blog/blog.js` calls the Article template's `decorate(main)`: the same generated Page Header and "About the author" section.
- `templates/blog/blog.css` imports `../article/article.css` (identical styles today). Blog-specific styles can be added to `blog.css`, scoped to `body.blog` or to the content the template adds.
- With no Layout chosen, the generated Page Header uses the Article layout (the page type `blog` is not a landing page type).

Dependencies: As the Article template (4.1.2), plus `templates/article/article.js` and `templates/article/article.css`.

Governance:

- Enforced: the body class `blog` and the query index `pageType` value `blog` distinguish blog posts from articles (List and Filter List can select them by page type).
- Recommended: use the Blog Page Type only for blog posts; use `content-type` for the editorial label shown on cards and in the Page Header.

Best Practices: As for articles. Set `author-id`, `publish-date`, `reading-time` and tags for every post.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/template-blog-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A blog post at 1440 px with the same generated Page Header and "About the author" card as an article; the body class `blog` visible in the browser developer tools.

4.1.4 Press Release Template

Purpose: Newsroom press releases with a consistent header and the media contacts at the end of every release.

Ownership: Template page `site-global-templates/press-release`: [To be confirmed: content owner — owner: Corporate communications / newsroom lead]. Template code `templates/press-release/`: development team. Media contacts: the locale's Media Contacts data sheet (see 9.9).

Usage:

- Page Type value: `press-release` (Page Type option "Press Release"). A metadata sheet row such as `/en/newsroom/**` with `page-type` = `press-release` gives every page in the newsroom folder this template.
- `templates/press-release/press-release.js`:
  1. `ensurePageHeader(main)`: a Page Header from the page metadata (title, publish date, tags, …) unless one is authored. With no Layout chosen, the Article layout is used.
  2. `appendLoadedSection(main, 'template-media-contacts', ...)`: loads the Media Contacts sheet (`getMediaContacts()`) and the placeholders in parallel; keeps the contacts whose `region` matches the page's `region` metadata (case-insensitive, `inRegion`); if no contact matches (or the page has no region), **all** contacts are shown; builds the list with `buildContactList()` (`decorators/contact-list-decorator.js`) under the heading "Media contacts" (placeholder `mediaContacts`).
  3. If the sheet has no contacts, the section is removed.
- Each contact shows the name, "designation, region", the email as a `mailto:` link and the phone as a `tel:` link (non-digit characters except `+` are removed from the `tel:` URL; the visible text stays as authored).
- `templates/press-release/press-release.css`: a top border above the list; contacts in one column on mobile, two columns from 600 px, three columns from 900 px.

Dependencies: `decorators/template-decorator.js`, `decorators/contact-list-decorator.js`, `utils/data.js` (`getMediaContacts`, `inRegion`), `utils/metadata.js`, `utils/placeholders.js`; the Page Header block; the locale's `data/media-contacts` sheet; page metadata `region`, `publish-date`, tags.

Governance:

- Enforced: contact selection by region (exact, case-insensitive match of the `region` values) with the fall back to all contacts.
- Recommended: use the same region vocabulary in the metadata sheet (`region` column) and in the Media Contacts sheet (for example `Americas`, `Europe`, `Asia Pacific`, `Middle East Africa`, matching the `tcs-corp:region/*` tag titles).
- Recommended: corporate communications reviews the Media Contacts sheet at least quarterly ([To be confirmed: review cadence — owner: Corporate communications]).

Best Practices:

- Set `publish-date` (YYYY-MM-DD) on every release; the Page Header shows it as "Published DD/MM/YY".
- Do not type the media contacts into the release body; maintain them in the sheet so every release stays current.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/template-press-release-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A press release at 1440 px with the generated Page Header and, at the end, the "Media contacts" section: a top border, the heading and three columns of contacts (name, designation and region, email and phone links).

4.1.5 Default (no template)

Purpose: Every page whose Page Type is empty ("Default") or a value without template code: corporate pages, service and industry pages, landing pages, listing pages, legal pages.

Ownership: The page owner of each section of the site ([To be confirmed: section owners — owner: Web content lead]).

Usage:

- No template JavaScript or CSS is loaded; the page consists only of what is authored.
- If the Page Type (page property or `page-type` sheet column) holds another value, it still becomes a body class and an index column (`pageType`), and blocks can use it: the Page Header defaults to the **Landing** layout for `landing` and `listing`, and to the **Article** layout otherwise; the Filter List can restrict its results to page types (`article`, `blog`, `report`, `case-study`, `news`, `event`).
- A Page Header is not added automatically; add the Page Header block where the design needs it.

Dependencies: `scripts/scripts.js` only.

Governance:

- Enforced: values outside `TEMPLATES` never load code (no error).
- Recommended: values used in the `page-type` column follow a controlled list (see 4.2) so listings and filters stay consistent.

Best Practices: Create Default pages with **Create** > **Page** in the locale folder; add a Page Header block as the first block when the page needs a title area.

> **Tip:** To add a new template (for example `case-study`), follow `templates/README.md`: create the template page under Site Global Templates (page name = Page Type), add `templates/case-study/case-study.js` and `.css`, add the name to `TEMPLATES` in `scripts/scripts.js` and as an option of Page Type in `models/_page.json`, then run `npm run build:json`. This is a development task.

4.2 Content Types

The implementation distinguishes two classification fields. Both are page metadata (page property or bulk metadata sheet column) and both are indexed in `/query-index.json`.

Field	Metadata name	Index column	Purpose	Where it is set
Page Type	`page-type`	`pageType`	The template (layout and template code) and a structural class of page	Page Type page property (Default, Home, Article, Blog, Press Release); or the `page-type` sheet column (any value)

Field	Metadata name	Index column	Purpose	Where it is set
Content Type	`content-type`	`contentType`	The editorial label of a page (Report, Article, Case Study, …), shown on cards and in the Page Header	`content-type` column of the bulk metadata sheet (there is no page property for it)

4.2.1 Page Types

Value	Source	Template code	Effect in code
(empty)	Page Type "Default"	None	No template; Page Header (if authored) uses the Article layout by default
`home`	Page Type "Home"	`templates/home`	First section meets the header
`article`	Page Type "Article"	`templates/article`	Generated Page Header, "About the author"
`blog`	Page Type "Blog"	`templates/blog`	As article
`press-release`	Page Type "Press Release"	`templates/press-release`	Generated Page Header, "Media contacts"
`landing`	Metadata sheet only	None	Page Header defaults to the Landing layout (with the featured image)
`listing`	Metadata sheet only	None	Page Header defaults to the Landing layout
`report`, `case-study`, `news`, `event`	Metadata sheet only	None	Selectable in the Filter List **Page types** filter

> **Note:** The Filter List's Page types option list contains `article`, `blog`, `report`, `case-study`, `news` and `event`; `press-release` is not an option there (a Filter List can select press releases by Content type or by Root path instead).

4.2.2 Content Types

Values offered by the code (Filter List "Content types" option list, `blocks/filter-list/_filter-list.json`): Article (`article`), Blog (`blog`), Report (`report`), Case Study (`case-study`), News (`news`), Event (`event`), Whitepaper (`whitepaper`), Podcast (`podcast`), Video (`video`).

Content-type tags in the taxonomy (published `/en/taxonomy.json`, verified 13 October 2026): `tcs-corp:content-type/article` (Article), `.../blog` (Blog), `.../case-study` (Case Study), `.../press-release` (Press Release), `.../whitepaper` (Whitepaper), `.../research` (Research), `.../webinar` (Webinar), `.../event` (Event).

How the code uses `content-type`:

Component	Behaviour	Rule
Page Header	Landing and Secondary layouts: the content type is the eyebrow above the title (else the tags). Article layout: the content type is shown next to the tags in the meta row (`page-header-collection`). Display layout: tags only.	Displayed as authored (text)
List (query index source) and Cards	The card's content type label: next to the date on image cards; the kicker on report cards. With Card style **Mixed**, a card whose content type contains "report" (case-insensitive, `/report/i`) becomes a **report card**; cards without an image become panels; others image cards.	`kindOf()` in `utils/card-list-config.js`

Component	Behaviour	Rule
Filter List	**Content types** filter: a page is kept when `toClassName(content-type)` is one of the selected values (for example `Case Study` matches `case-study`). The card eyebrow comes from the **Eyebrow source** column (default `content-type`). With Card style **Mixed**, a page whose eyebrow text contains "report" is shown as a report card with a "Download" action.	`candidatesOf()`, `isReport()` in `blocks/filter-list/filter-list.js`
Query index	Indexed as `contentType`	`helix-query.yaml`

Governance:

- Enforced: the Filter List matches content types by class name; any spelling that converts to the same class name matches (`Case Study`, `case-study`, `CASE STUDY`).
- Enforced: the word "Report" anywhere in the content type (List / Cards) or eyebrow (Filter List) triggers report cards in Mixed style.
- Recommended: use the display form (Title Case, as in the option names: `Article`, `Report`, `Case Study`, `Whitepaper`, …) in the `content-type` column, because the value is shown to visitors as written.
- Recommended: keep the content-type values, the Filter List options and the `tcs-corp:content-type/*` tags aligned. Today they differ (the taxonomy has Press Release, Research and Webinar; the Filter List has Report, News, Podcast and Video). [To be confirmed: the master list of content types — owner: Content strategy lead].

> **Important:** Content type is a **metadata value**, not a tag. Tagging a page with `tcs-corp:content-type/article` does not set its `content-type` metadata; the Filter List "Content types" filter and the Mixed card style read the metadata.

4.3 Metadata Model

Page metadata has two sources, and both end up as `<meta>` tags in the published page's `<head>`, which the code then reads with `getMetadata()` (`scripts/aem.js`) and `getPageMetadata()` (`utils/metadata.js`).

4.3.1 Page properties

The page model (`models/_page.json`, model `page-metadata`) has exactly these fields:

Field	Name	Type	Published as	Inherited by child pages
Title	`jcr:title`	text	`<title>`, `og:title`, `twitter:title`	No
Description	`jcr:description`	text	`description`, `og:description`	No
Keywords	`keywords`	text, multi-value	`keywords`	No
Page Type	`page-type`	select: Default, Home, Article, Blog, Press Release	`<meta name="page-type">`	No
Navigation	`nav`	AEM content reference	`<meta name="nav">`	Yes (by code, `utils/inherited-metadata.js`)
Footer	`footer`	AEM content reference	`<meta name="footer">`	Yes (by code)
404 Page	`error-page`	AEM content reference	`<meta name="error-page">`	Yes (by code)

Page tags are not part of the page model; the code reads tags from `<meta property="article:tag">` (one per tag, as AEM publishes the page Tags) and, as a fallback, from a comma-separated `tags` metadata value (for example a `tags` column of the bulk metadata sheet). [To be confirmed: verify on a published, tagged page that the AEM page Tags are published as `article:tag` — owner: development team].

Everything else that is page-level (author, dates, SEO and social overrides, card overrides, featured flags, region) is a **column of the bulk metadata sheet**. Page-specific values that are not metadata (CTAs, author image overrides, share / download switches) are block fields.

4.3.2 Bulk metadata sheet

- One sheet per locale, at `/content/tcs-corp/{country}/{language}/metadata`, published at `/{locale}/metadata.json` (mapped in `paths.json`).
- Each row has a `url` pattern (glob, for example `/en/newsroom/**`) and values for any of the columns. Every column is published on the matching pages as `<meta name="{column}">`.
- The platform applies the sheets **only when all 13 are listed in the site configuration's `metadata.source`** (Configuration Service, `POST https://admin.hlx.page/config/sradha-tcs/sites/tcs-corp/metadata.json`). Until then no bulk metadata applies, English included. [To be confirmed: whether `metadata.source` has been configured — owner: site administrator]
- Each locale's URL patterns must start with that locale's published path (`/de-de/**` in the German sheet); rows for other locales have no effect.
- Page-level values (page properties) take precedence over sheet values.

4.3.3 How metadata reaches the head (SEO decorator)

`decorators/seo-decorator.js` (`decorateHead()`, first step of the eager phase) maps the SEO and social columns to the standard tags in the browser:

Sheet column	Head element	Fallback
`meta-title`	`<title>`	the page Title
`meta-description`	`<meta name="description">`	the page Description
`meta-keywords`	`<meta name="keywords">`	the page Keywords
`og-title`	`og:title`	`meta-title`

Sheet column	Head element	Fallback
`og-description`	`og:description`	`meta-description`
`og-image`	`og:image` (absolute URL)	none (platform default image)
`og-type`	`og:type`	none
`twitter-card`	`twitter:card`	none (platform default `summary_large_image`)
`twitter-title`	`twitter:title`	`og-title`, then `meta-title`
`twitter-description`	`twitter:description`	`og-description`, then `meta-description`
`twitter-image`	`twitter:image` (absolute URL)	`og-image`
`canonical-url`	`<link rel="canonical">` and `og:url`	the platform canonical
`language`	`<html lang>`	the locale's language tag (see 4.9)
`robots`	`<meta name="robots">` (standard name, applied server side)	none

> **Important:** The SEO decorator runs in the browser. Search engines that render JavaScript see the overrides; crawlers and link-preview bots that read the raw HTML see the platform values (page Title and Description). `robots` is a standard name and works server side.

4.3.4 Inheritance

Edge Delivery publishes each page with its own metadata only. The code adds AEM-style inheritance for three properties (`utils/inherited-metadata.js`, `INHERITED_METADATA = ['nav', 'footer', 'error-page']`):

1. The page's own value (page property, or the `nav` / `footer` / `error-page` column of the metadata sheet).
2. Else the value of the nearest parent page, up to the locale's Home page (`getParentPages()` in `utils/site.js`; the parent pages' published HTML heads are loaded in parallel and read).
3. Else the locale default: `references/nav`, `references/footer`, `error-pages/404`.

Parent values are cached in `sessionStorage` (key `tcs-inherited-metadata`) for the browser session: a changed parent property shows in a new session. On a 404 page the lookup starts at the missing page's nearest existing parent.

All other metadata is **not inherited**; use URL patterns in the bulk metadata sheet (`/en/insights/**`) to apply values to a whole folder.

4.3.5 Typed access in code

`getPageMetadata()` returns the values blocks use: `title` and `description` (as authored, read before the SEO decorator changes them), `image` (`featured-image`, else `og:image`), `pageType` (`page-type`, else `template`), `contentType`, `category`, `tags`, `authorId`, `author`, `publishDate` (`publish-date`, else `publication-date`), `modifiedDate`, `readingTime`, `listenLink`, `thumbnailImage`, `featuredImage`, `region`, `country` (else the locale's country), `language` (see 4.9), `persona`, `campaignId`, `featured` (true for yes / true / 1 / on / y) and `priority` (a number or null).

> **Note:** The complete field-by-field specification of every page property and every metadata column (type, allowed values, consumers, SEO impact) is in **Part 8**. The sheet itself (columns, examples, authoring) is documented in **9.2**.

4.4 Taxonomy Model

- **Tags** are AEM tags in the `tcs-corp:` namespace (tag IDs `namespace:path`, for example `tcs-corp:region/europe`). The published English taxonomy (verified 13 October 2026) holds the roots `tcs-corp:content-type`, `tcs-corp:industry` and `tcs-corp:region` with 12 child tags.
- **Taxonomy page per locale**: `/content/tcs-corp/{country}/{language}/taxonomy`, published as `/{locale}/taxonomy.json` (each mapped explicitly with `.json` in `paths.json`). Its rows have the columns `tag` (the tag ID) and `title` (the label).
- **Translated titles**: when the tags have translated titles, the taxonomy JSON is a multi-sheet; the code uses the sheet named after the locale's ISO 639-1 language (`de`, `fr`, `ja`, …), else the `default` sheet, else the first sheet (`sheetRows(json, getLocale().language)`).
- **Resolution** (`utils/taxonomy.js`): `loadTaxonomy()` loads the locale's taxonomy once per page into a map of tag ID to title; `resolveTags(values)` turns each value into `{ id, label }`: the taxonomy title when the ID is listed; else, for an ID-like value (contains `:` or `/`), a label derived from the last segment (`parseTag`: `tcs-corp:industry/banking-financial` becomes "Banking financial"); else the value as authored. A missing taxonomy gives an empty map, so labels fall back to derived names.
- **Consumers**: Page Header (page tags), Teaser (its Tags field), List and Cards (card tags), Filter List (filter pills, advanced filter categories and card tags; labels never show raw IDs), Conversation Canvas (suggestion pills from the page tags).
- **Page tags** come from `<meta property="article:tag">` (or a `tags` metadata value) through `getPageTags()`, and are indexed in the query index column `tags`.

> **Note:** The tag structure, the governance of tag creation and translation, and the per-component use of tags are documented in **Part 7**. The taxonomy sheet is documented in **9.3**.

4.5 Placeholders

Placeholders are the user-interface texts that the code shows or announces: button labels, accessible names, counters, empty-state and validation messages. They are authored per locale in a spreadsheet, so every text can be translated without a code change. The code never hard-codes user-facing strings without a placeholder key; every key has an English fallback in the code.

4.5.1 The placeholders sheet

Item	Value
AEM path	`/content/tcs-corp/{country}/{language}/placeholders` (for example `/content/tcs-corp/de/de-de/placeholders`)
Published path	`/{locale}/placeholders.json` (for example `/en/placeholders.json`, `/de-de/placeholders.json`)
Columns	`Key`, `Text`, `Description`, `Category`
Read by code	`Key` and `Text` (lower-case `key` / `text` headers are accepted too)

Item	Value
For authors only	`Description` (where the text appears), `Category` (grouping, for example CTA, Navigation, Forms)
Current state	`/en/placeholders.json` is published with one row (`read-more` / "Read More" / "CTA label" / "CTA"); the other 12 locales return 404 (verified 13 October 2026)

4.5.2 How `utils/placeholders.js` works

- **Loading:** `loadPlaceholders()` fetches `localeSheetPath('placeholders')`, which is the current locale's `/{locale}/placeholders.json` (from the AEM path in the Universal Editor, from the published path on the site; pages outside a locale use `/en/`). The request is made **once per page view**; the promise is kept in a module variable and shared by every block. There is no `sessionStorage` cache; the browser and the Edge Delivery CDN cache the JSON like any published resource.
- **Key matching:** keys are compared without case and without separators (`normalizeKey`: lower case, every character other than `a-z` and `0-9` removed). The code uses camelCase keys (`pauseSlides`), and the sheet may write them in any form: `pauseSlides`, `pause-slides`, `Pause Slides` and `PAUSE_SLIDES` all match. The published English row `read-more` therefore already overrides the code key `readMore`.
- **Lookup:** `getPlaceholders()` resolves to a function `t(key, fallback, values)`. It returns the sheet's Text for the key; if the key is missing, or its Text is empty, it returns the **English fallback** passed by the code.
- **Variables:** `fillPlaceholder(text, values)` replaces `{name}` tokens with the values passed by the code (for example `t('slideOf', '{n} of {total}', { n: 2, total: 5 })` gives "2 of 5"). A token the code does not pass stays visible as written. Translations must keep the tokens (the order may change: "{total} ⊠ {n}").
- **Failure handling:** if the sheet is not published (404), not valid JSON, or the request fails, the lookup map is empty and every text uses its English fallback. No error is shown.
- **Early rendering:** some components (Header, Conversation Canvas, Form Variations, Table List, the Lottie animation toggle) start with a fallback-only lookup and switch to the sheet once it has loaded; the animation toggle updates its label when the placeholders arrive.

```
import { getPlaceholders } from '../../utils/placeholders.js';

const t = await getPlaceholders();
button.setAttribute('aria-label', t('nextSlide', 'Next slide'));
label.textContent = t('slideOf', '{n} of {total}', { n: 1, total: 4 });
```

4.5.3 Complete list of placeholder keys

The table lists **every** placeholder key used in `blocks/`, `decorators/`, `scripts/`, `templates/` and `utils/` (91 keys, extracted from the code on 13 October 2026). "Shown as" tells translators where the text appears: **Visible** (on the page), **Accessible name** (`aria-label`, read by screen readers, not visible), **Status** (announced live region text), **Editor only** (shown to authors in the Universal Editor, never to visitors).

Key	English fallback	Variables	Component	Shown as
`aboutTheAuthor`	About the author	none	Article and Blog templates (author bio)	Visible heading
`advancedFilters`	Advanced filters	none	Filter List	Visible panel heading and accessible name of the filter button
`all`	All	none	Filter List	Visible "All" filter pill
`back`	Back	none	Conversation Canvas	Accessible name of the menu Back button

Key	English fallback	Variables	Component	Shown as
`carousel`	Carousel	none	Gen Carousel	Accessible name of the carousel region (when not authored)
`carouselRole`	carousel	none	Gen Carousel	Screen reader role description (`aria-roledescription`)
`cards`	Cards	none	Cards	Accessible name of the scrolling card region (when the block has no heading)
`cardsEditorHint`	Add a Card to the block and give it a title, a description or an image.	none	Cards	Editor only (empty block)
`clearAll`	Clear all	none	Filter List	Visible button in the advanced filter panel
`close`	Close	none	Modal	Accessible name of the close button
`closeNavigation`	Close navigation	none	Header	Accessible name of the mobile menu button when open
`contentFrom`	Content from {source}	{source} (host name)	Embed	`title` of the embed iframe
`dialog`	Dialog	none	Modal	Accessible name of a dialog without a heading
`download`	Download	none	Filter List; List and Cards (card list)	Visible call to action of report cards
`downloadThisPage`	Download this page	none	Page Header; Teaser (article meta)	Accessible name of the download (print) icon button
`explore`	Explore	none	Filter List; Table List; List and Cards (card list)	Visible call to action of panel cards and Table List item buttons
`exploreItem`	Explore {title}	{title}	Table List	Accessible name of an item's arrow link
`fieldEmail`	Enter a valid email address.	none	Form Variations	Visible validation message
`fieldFormat`	Enter a valid value.	none	Form Variations	Visible validation message
`fieldRequired`	This field is required.	none	Form Variations	Visible validation message
`fieldTooLong`	This text is too long.	none	Form Variations	Visible validation message
`filterListEditor`	The results show on the published page, from the pages' metadata.	none	Filter List	Editor only (empty state)
`filters`	Filters	none	Filter List	Visible toolbar label (when no Filters label is authored) and accessible name of the pill group
`followUsOn`	Follow us on {platform}	{platform} (Social Links platform)	Footer	Accessible name of each social icon link
`footerLinks`	Footer links	none	Footer	Accessible name of the footer link navigation
`formSuccess`	Thank you, your details were sent.	none	Form Variations	Status message after a valid submit
`itemLinkLabel`	{text}: {title}	{text}, {title}	Table List	Accessible name of an item button ("Explore: Item title")

Key	English fallback	Variables	Component	Shown as
`linkCopied`	Link copied to clipboard	none	Page Header; Teaser (article meta)	Status message after Share copies the link
`list`	List	none	List and Cards (card list)	Accessible name of the scrolling card region (no heading, no list label)
`listEditorHint`	Set up the list in its properties: {problems}.	{problems} (configuration warnings)	List	Editor only (misconfigured list)
`listEditorPages`	The pages show on the published page (from the query index).	none	List	Editor only (query index not available in the editor)
`listEmpty`	There are no items to show.	none	List; Cards	Visible empty message (when no Empty message is authored)
`listenToThisPage`	Listen to this page	none	Page Header; Teaser (article meta)	Visible text / accessible name of the Listen link
`listNext`	Next items	none	List and Cards (card list)	Accessible name of the next arrow (horizontal scroll)
`listPrevious`	Previous items	none	List and Cards (card list)	Accessible name of the previous arrow (horizontal scroll)
`loadMore`	Load more	none	Filter List	Visible button
`mainNavigation`	Main navigation	none	Header	Accessible name of the header `<nav>`
`mediaContacts`	Media contacts	none	Press Release template	Visible heading
`menu`	Menu	none	Conversation Canvas	Accessible name of the menu toggle and menu
`minRead`	{n} min read	{n} (minutes)	Page Header; Teaser (article meta)	Visible reading time (numeric reading-time only)
`minutesShort`	{n}m	{n} (minutes)	Filter List; List and Cards (card list)	Visible reading time on cards (numeric reading-time only)
`nextSlide`	Next slide	none	Gen Carousel	Accessible name of the next button
`noPagesFound`	No pages found	none	Conversation Canvas	Visible empty menu message
`noResources`	No resources found.	none	Filter List	Visible empty-state title
`noResourcesHint`	Try selecting another filter.	none	Filter List	Visible empty-state hint
`onThisPage`	On this page	none	Anchor List	Accessible name of the anchor navigation
`open`	Open	none	Modal	Visible button text of a Modal link without its own text
`openNavigation`	Open navigation	none	Header	Accessible name of the mobile menu button when closed
`opensInNewTab`	(opens in a new tab)	none	Page Header (author LinkedIn link); List and Cards (card links)	Hidden text / accessible name suffix of links opening a new tab
`pauseAnimation`	Pause animation	none	Hero; Teaser (Lottie animation toggle)	Accessible name and title of the toggle

Key	English fallback	Variables	Component	Shown as
pauseSlides	Pause slides	none	Gen Carousel	Accessible name of the auto-play toggle while playing
pauseVideo	Pause video	none	Video	Accessible name of the play / pause button (video file playing)
play	Play	none	Embed	Accessible name of the play button of a click-to-load embed
playAnimation	Play animation	none	Hero; Teaser (Lottie animation toggle)	Accessible name and title of the toggle while paused
playSlides	Play slides	none	Gen Carousel	Accessible name of the auto-play toggle while paused
playVideo	Play video	none	Video	Accessible name of the play button (untitled video)
playVideoTitle	Play video: {title}	{title} (video title)	Video	Accessible name of the play button (titled video)
previousSlide	Previous slide	none	Gen Carousel	Accessible name of the previous button
published	Published	none	Page Header	Visible label before the publish date
readMore	Read more	none	Filter List	Visible call to action of featured-style cards
readTheTranscript	Read the transcript	none	Video	Visible transcript link text (when no text is authored)
requiredNote	* Required fields	none	Form Variations	Visible note above the form
resetFilters	Reset filters	none	Filter List	Visible button
resultCount	{n} results	{n}	Filter List	Visible / status result count (more than one)
resultCountOne	1 result	none	Filter List	Visible / status result count (exactly one)
searchByVoice	Search by voice	none	Conversation Canvas	Accessible name of the microphone button (when no alt text is authored)
searchPlaceholder	Ask us a question	none	Conversation Canvas	Visible search field placeholder (when none is authored)
selectAnOption	Select an option	none	Form Variations	Visible first option of dropdowns
selectedCount	{n} selected	{n}	Filter List	Accessible name suffix of the advanced filter button
send	Send	none	Form Variations	Visible submit button text (when no label is authored)
separatorHidden	Separator hidden on all devices	none	Separator	Editor only
separatorInvisible	Semantic separator without a line: screen readers announce a break nobody sees	none	Separator	Editor only
shareThisPage	Share this page	none	Page Header; Teaser (article meta)	Accessible name of the share icon button
showingCount	Showing {n} of {total}	{n}, {total}	Filter List	Visible status under the results
showSlide	Show slide {n} of {total}	{n}, {total}	Gen Carousel	Accessible name of each dot

Key	English fallback	Variables	Component	Shown as
`showSubmenu`	Show {label} menu	{label} (nav item text)	Header	Accessible name of a submenu toggle
`shownWhenValue`	Shown when a drop-down value is "{value}"	{value}	Form Variations	Editor only (conditional field note)
`siteNavigation`	Site	none	Conversation Canvas	Accessible name of the site menu navigation
`slideOf`	{n} of {total}	{n}, {total}	Gen Carousel	Accessible name of each slide
`slideRole`	slide	none	Gen Carousel	Screen reader role description of each slide
`stopVideo`	Stop video	none	Video	Accessible name of the play / stop button (embedded video playing)
`submit`	Submit	none	Conversation Canvas	Accessible name of the search submit button (when no label is authored)
`suggestions`	Suggestions	none	Conversation Canvas	Accessible name of the suggestion pill list
`table`	Table	none	Table	Accessible name of a scrollable table region without a heading
`tabNumber`	Tab {n}	{n}	Tabs	Visible tab label of a tab without a title
`tabs`	Tabs	none	Tabs	Accessible name of the tab list (when no Accessibility label is authored)
`tabsBlockEmpty`	Add a Tab, then choose the Fragment it shows in the Tab properties.	none	Tabs (block)	Editor only
`tabsEmpty`	Add a Tab, then the components of the tab after it.	none	Tabs (section)	Editor only
`video`	Video	none	Video	Default video title (when none is authored)
`viewAll`	View all	none	Table List	Visible View all button text (when no text is authored)
`viewLinkedinProfile`	View LinkedIn profile	none	Article and Blog templates (author bio)	Visible link text

Note: Texts that are **not** placeholders (as of this handover): the built-in content of `404.html` ("Page not found", "Go back"), shown only when no authored 404 page can be loaded; the English month abbreviations of the date formats with "mmm" (`utils/dates.js`, see 4.9.7); and the Filter List card action accessible names, which combine a placeholder with the page title ("Download: {title}"). Keys in the JSDoc example of `utils/placeholders.js` (`nextSlide`, `slideOf`) are the Gen Carousel keys listed above.

4.5.4 Governance and best practices

- **Ownership:** [To be confirmed: owner of the placeholders sheets — owner: Localization / content operations lead]. Each locale's translator or content manager owns that locale's sheet.
- **Enforced:** the code falls back to English for every missing key, so an incomplete sheet never breaks a page; it only shows English text.

- **Recommended:** create the full key list (the table above) in the English sheet first, with a Description and a Category per key, then copy the sheet to every locale and translate the Text column.
- **Recommended:** keep `{variables}` exactly as written (same name, braces included).
- **Recommended:** write keys in camelCase as in the code; any casing works, but one convention avoids duplicates (`read-more` and `readMore` are the same key; if both exist, the later row wins).
- **Recommended:** publish the sheet after every change (preview, then live) and check `/{locale}/placeholders.json`.

4.6 References

The **references** folder of each locale (`/content/tcs-corp/{country}/{language}/references/`, published at `/{locale}/references/`) holds the pages that are not content pages but are included in other pages: the header navigation, the footer, alternative headers and footers, the Conversation Canvas page of the footer, modal pages and fragments. The query index excludes `/*/references/**`, so these pages never appear in listings or the sitemap.

Page	AEM path (us/en)	Published	Required	Used by
`nav`	`/content/tcs-corp/us/en/references/nav`	`/en/references/nav`	Yes	Header block (default navigation of the locale)
`footer`	`/content/tcs-corp/us/en/references/footer`	`/en/references/footer`	Yes	Footer block (default footer of the locale)
`nav-{name}`	for example `references/nav-careers`	`/en/references/nav-careers`	Optional	Header, when selected in a page's Navigation property (or the `nav` sheet column)
`footer-{name}`	for example `references/footer-campaign`	`/en/references/footer-campaign`	Optional	Footer, when selected in a page's Footer property (or the `footer` sheet column)
`footer-conversation-canvas`	`references/footer-conversation-canvas`	`/en/references/footer-conversation-canvas`	Optional	The authored Footer block's Conversation Canvas Page (a page with a Conversation Canvas block)
`modals/{name}`	for example `references/modals/contact-us`	`/en/references/modals/contact-us`	Optional	Any link to a path containing `/modals/` opens as a modal (`autolinkModals` in `scripts.js`); the Modal block
Fragments	for example `references/cta-banner`	`/en/references/cta-banner`	Optional	Fragment block; Tabs (block mode) Fragment field

4.6.1 How the header and footer page is resolved

```
Header (blocks/header)
value = getInheritedMetadata('nav')
  1. this page's Navigation / nav column
  2. nearest parent page's value up to Home
  3. ''
```

```
Footer (blocks/footer, page footer role)
value = getInheritedMetadata('footer')
  1. this page's Footer / footer column
  2. nearest parent page's value up to Home
  3. ''
```

```
path = resolvePagePath(value,
         'references/nav')
loadFragment(path)  ->  {path}.plain.html
no page: header left empty
```

```
path = resolvePagePath(value,
         'references/footer')
loadFragment(path) + Social Links sheet
no page: footer shows only the social
links
```

`resolvePagePath(value, fallback)` (`utils/site.js`) accepts three forms of value and returns the path to fetch in the current context (the AEM path in the Universal Editor, the published path on the site):

Value form	Example	Result on the site
Empty	(none)	`/{locale}/references/nav` (the fallback in the current locale)
AEM path	`/content/tcs-corp/de/de-de/references/nav-jobs`	`/de-de/references/nav-jobs`
Published path	`/de-de/references/nav-jobs`	`/de-de/references/nav-jobs`
Relative to the locale root	`references/nav-jobs`	`/{current locale}/references/nav-jobs`

The 404 page uses the same mechanism with `error-page` and the fallback `error-pages/404`, then `/en/error-pages/404`, then the built-in content (`loadErrorPage` in `scripts.js`); content from another locale is marked with its `lang`.

4.6.2 Authoring references

1. In the AEM Sites console open the locale's `references` folder (create it as an empty folder page if it does not exist).
2. Create the page (**Create** > **Page**) with the exact page name (`nav`, `footer`, `nav-careers`, ...). The page name is the URL; the title is free.
3. Open it in the Universal Editor and author it:
   - **nav:** three sections in this order: brand (logo link), sections (navigation lists; nested lists become submenus), tools. The Header adds the classes `nav-brand`, `nav-sections`, `nav-tools` by position.
   - **footer:** the content shown in the footer, typically a Footer block (Conversation Canvas Page, Copyright Text with `{year}`, Link List). The social icons are added from the locale's Social Links sheet.
   - **fragments:** any sections and blocks.
4. Publish the reference page (preview, then live). An unpublished `nav` leaves the header empty.
5. To use an alternative nav or footer for a section of the site, set the **Navigation** or **Footer** page property on the top page of that section. All pages below inherit it unless they set their own.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/references-folder-sites-console.png`
> **Purpose:** References folder in AEM
> **Description:** The AEM Sites console column view of `/content/tcs-corp/us/en/references` listing `nav`, `footer`, `footer-conversation-canvas` and a `modals` folder.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-properties-navigation-footer.png`
> **Purpose:** Inherited Navigation, Footer and 404 Page properties

Description: The Universal Editor page properties panel of a section top page (for example /en/careers) with the Navigation field set to `references/nav-careers` and the Footer and 404 Page fields empty.

Warning: Parent values are cached for the browser session. After changing a Navigation or Footer property, test in a new private window.

4.7 Data Sheets

Each locale has a `data/` folder (`/content/tcs-corp/{country}/{language}/data/`, published at `/{locale}/data/`) with five spreadsheets: `authors`, `locations`, `partners`, `social-links` and `media-contacts`. They are read by `utils/data.js` (`loadDataSheet(name)`, once per page and sheet), which trims every value, converts the rows to objects with camelCase properties and returns an empty list for a sheet that is missing or fails to load.

Sheet	Read by	Status
`authors`	Page Header (author lockup), Article / Blog templates (About the author), List (feature card author), Filter List (hero / featured card author)	In use
`social-links`	Footer (social icons)	In use
`media-contacts`	Press Release template (Media contacts)	In use
`locations`	`getLocations()` only	No component reads it yet (as of this handover)
`partners`	`getPartners()` only	No component reads it yet (as of this handover)

The query index excludes `/*/data/**`. All five sheets are published for `/en/` with their column headers and no rows (verified 13 October 2026); the other locales return 404.

Note: Columns, types, examples, business rules and authoring steps for each data sheet are in **Part 9** (9.5 to 9.9).

4.8 Shared Content

Mechanism	Where	What is shared	How it is included	Notes
Fragment block	Any page	A reference page (usually under `references/`)	Fragment block, **Reference** field (AEM content reference)	`blocks/fragment/fragment.js` loads `{path}.plain.html`, decorates its sections and blocks and replaces the block with them. The path must start with `/`. Media paths are rebased to the fragment.
Header / Footer	Every page	`references/nav`, `references/footer` (or the inherited alternatives)	Automatic (4.6)	One nav and one footer per locale by default
Footer Conversation Canvas	Footer	A page with a Conversation Canvas block	Footer block, Conversation Canvas Page field	Loaded as a fragment above the footer bar

Mechanism	Where	What is shared	How it is included	Notes
Tabs (block mode)	Tabs block	A fragment per tab	Tab item Fragment field	Each tab panel shows its fragment
Modals	Any link	Pages under a `/modals/` path	A link to the page (or the Modal block)	Opens in a dialog
Data sheets	Templates and blocks	Authors, social links, media contacts	Automatic	Per locale (4.7)
Placeholders	All components	UI text	Automatic	Per locale (4.5)
Block library	`/{locale}/block-library/` (author only)	One page per block with every variation, in the locale's language	Not included in pages	Excluded from publishing (`paths.json` excludes each locale's `block-library` and `block-library/**`) and from the index. Used to validate blocks and as an authoring reference.
Site Global Templates	`/content/tcs-corp/site-global-templates/`	Starter pages (home, article, blog, press-release)	Copied by authors (copy / paste in the Sites console)	Not published. Changing a template page does not change pages already copied from it.

Governance:

- Enforced: a Fragment whose page is not published (or whose path does not start with `/`) renders nothing; the block stays empty.
- Recommended: keep all shared fragments under the locale's `references/` so they are excluded from the index and the sitemap. Fragments elsewhere are indexed unless they sit in a `fragments/` or `modals/` folder (`/**/fragments/**` and `/**/modals/**` are excluded).
- Recommended: name fragments by purpose (`cta-contact-sales`, `legal-disclaimer-financials`), and list them with their owners. [To be confirmed: fragment inventory and owners — owner: Web content lead]
- Recommended: before deleting or renaming a fragment, use the AEM **References** panel to find the pages that include it.
- Changes to a fragment are visible on every including page after the fragment is published; the including pages need no republish.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-block-reference-field.png`
> **Purpose:** Selecting a fragment
> **Description:** The Universal Editor properties panel of a Fragment block with the Reference field open on the content picker at `/content/tcs-corp/us/en/references/`.

4.9 Localization Strategy

4.9.1 Locales

The 13 locale sites are defined once in code, in `LOCALES` (`utils/site.js`), and `paths.json` is generated from that list (`npm run build:paths`, `tools/paths/generate-paths.mjs`).

AEM folder	Published root	Language tag (`<html lang>`)	ISO 639-1 language (taxonomy sheet)	Country
us/en	/en	en-US	en	us
de/de-de	/de-de	de-DE	de	de
fr/fr-fr	/fr-fr	fr-FR	fr	fr
ca/fr-ca	/fr-ca	fr-CA	fr	ca
es/es-es	/es-es	es-ES	es	es
pt/pt-pt	/pt-pt	pt-PT	pt	pt
nl/nl-nl	/nl-nl	nl-NL	nl	nl
se/sv-se	/sv-se	sv-SE	sv	se
dk/da-dk	/da-dk	da-DK	da	dk
no/no-no	/no-no	no-NO	no	no
cn/zh-cn	/zh-cn	zh-CN	zh	cn
jp/ja-jp	/ja-jp	ja-JP	ja	jp
kr/ko-kr	/ko-kr	ko-KR	ko	kr

- **Locale detection** (`getLocale()`): in the Universal Editor from the AEM path (`/content/tcs-corp/de/de-de/...`), on the site from the published path (`/de-de/...`). Pages outside a locale (Site Global Templates, drafts) use the default locale `en` (en-US).
- **Default locale:** the first entry, `us/en` (`DEFAULT_LOCALE`). It is the final fallback for the 404 page.
- **Root path:** nothing is published at `/`; a redirect `/` to `/en/` is needed and is not applied automatically (see 9.4). [To be confirmed: CDN redirect for `/` and language selection — owner: site administrator]

4.9.2 Page language (`<html lang>`)

`decorateHead()` (`decorators/seo-decorator.js`) sets `document.documentElement.lang = getPageLanguage()`:

1. The page's `language` metadata, when it is a valid language code (pattern `^[a-z]{2,3}(-[a-z0-9]{2,8})*$`, case-insensitive; for example `en`, `en-GB`).
2. Otherwise the language tag of the locale site (`de-DE` for `/de-de/...`).

The same value is sent to the data layer (`xdm:language`). On the 404 page, content loaded from another locale (the `/en/` fallback) gets that locale's `lang` on `<main>`. Sorting in the List (alphabetical) and the Filter List uses `Intl.Collator` with the page language.

4.9.3 Placeholders per locale

Each locale has its own `placeholders` sheet (4.5). A locale without a published sheet, or a key missing from it, shows the English fallback from the code. There is no fallback from a locale's sheet to the English sheet: a missing key in `/de-de/placeholders.json` shows the **code's** English text, not the text of `/en/placeholders.json`.

4.9.4 Taxonomy per language

Each locale publishes its own taxonomy page (`/{locale}/taxonomy.json`). When tag titles are translated, the code reads the sheet named after the locale's ISO 639-1 language (for example `de`), else `default`, else the first sheet. `fr-fr` and `fr-ca` both read the `fr` sheet, and `zh-cn` reads `zh`. Tags missing from the taxonomy show a label derived from the tag ID (English words from the ID).

4.9.5 Metadata per locale

Each locale has its own bulk metadata sheet (`/{locale}/metadata.json`) with URL patterns starting with its root. All 13 must be listed in the site configuration's `metadata.source`. Values are not shared between locales: SEO titles, descriptions, social images, author IDs and dates are set per locale.

4.9.6 Translating content

Topic	Implemented in code	Process (outside the code)
Locale structure	13 locale sites, locale detection, per-locale nav / footer / 404 / sheets / taxonomy / data	Each locale is a separate AEM page tree; locales share no content
Creating a new locale's content	Not implemented (no automation)	Copy `us/en` (or a closer language) in the Sites console and translate it; or use AEM's translation features (language copies, translation projects with a translation connector) ([To be confirmed: whether AEM translation projects / a translation vendor connector are configured — owner: AEM administrator])
Keeping locales in sync (live copy / MSM)	Not implemented	[To be confirmed: whether MSM live copies or translation projects are used — owner: Localization lead]
Language navigation / language selector	Not implemented (no language switcher block or code)	Can be authored as links in each locale's nav
`hreflang` alternates	Not implemented (no code adds `<link rel="alternate" hreflang>`; `helix-sitemap.yaml` defines one default sitemap without language alternates)	[To be confirmed: hreflang strategy — owner: SEO lead]
Adding a locale	`LOCALES` + `npm run build:paths` (developer); see `utils/site.js`	Create the AEM folders, sheets and references; configure `metadata.source`; publish
UI text	Placeholders per locale	Translate the Text column of each sheet
Tag labels	Taxonomy per locale and language sheet	Translate tag titles in AEM Tagging, republish the taxonomy
Data sheets	Per locale	Maintain each locale's authors, social links and media contacts

4.9.7 Date formats

Dates are authored as `YYYY-MM-DD` (optionally with a time) and formatted from their calendar date as written, without time-zone conversion (`utils/dates.js` `formatDate`, `decorators/date-decorator.js` `buildTime`, which outputs `<time datetime="YYYY-MM-DD">`). Values that are not dates are shown as written.

Format value	Output for 2026-09-04	Used by
`dd-mm-yyyy`	04-09-2026	Teaser default; option in List, Cards
`mm-dd-yyyy`	09-04-2026	Option in Teaser, List, Cards
`mmm-d-yyyy`	Sep 4, 2026	Option in Teaser, List, Cards
`dd/mm/yy`	04/09/26	Page Header (fixed, after "Published"); option in Teaser, List, Cards
`dd-mmm-yy`	04 Sep 26	List and Cards default; Filter List (fixed)

> **Important:** The month abbreviations (Jan to Dec) are English in every locale; they are not placeholders and are not localised with `Intl.DateTimeFormat` (as of this handover). For non-English locales, prefer the numeric formats (`dd-mm-yyyy`, `dd/mm/yy`) where the block offers a choice. The Page Header and Filter List formats are fixed in code.

Part 5 — Authoring Fundamentals

This part explains how content authors work on the TCS Corporate Website day to day: how to get access, how the Universal Editor is laid out, how pages, sections and blocks are created, how images are handled, the **Additional** tab that every block shares, and how content is previewed, published, validated and rolled back. It applies to all 13 locale sites (`/content/tcs-corp/{country}/{language}`, published at `/en`, `/de-de`, `/fr-fr`, `/fr-ca`, `/es-es`, `/pt-pt`, `/nl-nl`, `/sv-se`, `/da-dk`, `/no-no`, `/zh-cn`, `/ja-jp`, `/ko-kr`). Block-by-block instructions are in Part 6 (Component Library).

> **Note:** In the Universal Editor, blocks are called **components**. This document uses "block" for a component from the **Blocks** group (Hero, Teaser, List, …) and "default content" for the four simple components Text, Title, Image and Button.

5.1 Getting access

5.1.1 Systems an author uses

System	Purpose	Address
Adobe IMS (Adobe ID / enterprise ID)	Single sign-on for every Adobe system below	https://experience.adobe.com (sign-in page)
Adobe Admin Console	Where an administrator grants access (product profiles)	https://adminconsole.adobe.com (administrators only)
AEM as a Cloud Service author (Sites console)	Browse the content tree, create, copy, move, publish and unpublish pages, page versions	https://author-p220207-e2271469.adobeaemcloud.com/ui#/aem/sites.html/content/tcs-corp
Universal Editor	Visual editing of a page (canvas, content tree, properties panel)	Opened from the Sites console with **Edit**; URL pattern in 5.1.3
AEM Assets	Images and documents used on pages	`/content/dam/tcs-corp/` on the author environment
Preview site (Edge Delivery Services)	Check content before it goes live	https://main–tcs-corp–sradha-tcs.aem.page/en/ (and `/de-de/`, …)
Live site (Edge Delivery Services)	The public site	https://main–tcs-corp–sradha-tcs.aem.live/en/ (and `/de-de/`, …)
Production domain	The final public domain	[To be confirmed: production domain and CDN — owner: site administrator]
AEM Sidekick (browser extension)	Preview / publish status and switching between preview and live from the site	Chrome Web Store, "AEM Sidekick"

The author environment is program **p220207**, environment **e2271469**. Content is delivered to Edge Delivery Services through `bin/franklin.delivery/sradha-tcs/tcs-corp/main` (the mount point in `fstab.yaml`); authors never call this URL directly.

5.1.2 Requesting access

Step 1 — Request an Adobe account

Action: Ask your AEM administrator for access to the AEM as a Cloud Service author environment `author-p220207-e2271469`. The administrator adds you in the Adobe Admin Console to the product profile of that environment (typically the "AEM Users" profile of the author instance) and to the AEM user group for your locale.

Expected Result: You receive an Adobe invitation e-mail and can sign in at https://experience.adobe.com.

Step 2 — Get the right permissions in AEM

Action: Ask the administrator to add you to the group that matches your role: edit rights for your locale folder (for example `/content/tcs-corp/us/en` or `/content/tcs-corp/de/de-de`), publish (replicate) rights or an approval workflow, and read/write access to `/content/dam/tcs-corp/`. Content managers also need edit rights on the locale's sheets (`placeholders`, `metadata`, `taxonomy`, `redirects`, `data/*`) and the `references/` pages.

Expected Result: The locale folder shows in the Sites console and the **Edit**, **Create** and **Publish** actions are available.

> **Note:** The group names and the approval workflow are not part of the code repository. [To be confirmed: AEM user groups, permissions per locale and publish approval workflow — owner: AEM administrator]

Step 3 — Install AEM Sidekick (optional, recommended)

Action: Install the AEM Sidekick extension in Chrome (or Edge) and add the project `sradha-tcs/tcs-corp`. The repository's Sidekick configuration (`tools/sidekick/config.json`) labels the edit link **AEM Editor**; it opens the AEM source of the page you are viewing.

Expected Result: On a `*.aem.page` or `*.aem.live` page the Sidekick toolbar shows the page status and lets you switch between Preview and Live.

5.1.3 Universal Editor URL pattern

The Universal Editor is normally opened from the Sites console (select the page, then **Edit**). The editor address has this form:

```
https://experience.adobe.com/#/@{ims-org}/aem/editor/canvas/
  author-p220207-e2271469.adobeaemcloud.com/content/tcs-corp/{country}/{language}/
{page}.html
```

Example for the English home page: `.../canvas/author-p220207-e2271469.adobeaemcloud.com/content/tcs-corp/us/en.html`.

> **Note:** `{ims-org}` is the technical name of the Adobe organisation. [To be confirmed: IMS organisation name used in the Universal Editor URL — owner: AEM administrator]

> **Tip:** Bookmark the Universal Editor URL of the pages you edit most. The editor keeps you signed in through Adobe IMS.

5.1.4 Access troubleshooting

Symptom	Likely cause	Resolution
"You don't have access" after sign-in	Not in the AEM product profile	Ask the administrator to add you in the Admin Console
The locale folder is missing in the Sites console	No read permission on `/content/tcs-corp/{country}/{language}`	Ask for the locale group
Publish is greyed out	No replicate permission	Ask for publish rights or use the approval workflow
The canvas stays blank in the Universal Editor	Browser blocks third-party cookies or the session expired	Allow cookies for `adobe.com` and `adobeaemcloud.com`, sign in again
Images cannot be picked	No read access to `/content/dam/tcs-corp/`	Ask for Assets access

5.2 Universal Editor tour

The Universal Editor shows the real page (rendered by the project code) and lets you select and edit each part of it. Everything you see on the canvas is the same code that runs on the preview and live site, so what you see is very close to what visitors see.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/ue-overview.png`
> **Purpose:** Universal Editor overview
> **Description:** The Universal Editor with a TCS page open: the top toolbar (location, device size, Preview toggle, Publish), the page canvas in the middle, and the right rail with the Properties panel and the Content tree. Label each area A to E.

5.2.1 The main areas

Area	What it is	What you do there
Header bar	Adobe Experience Cloud header and the editor toolbar	Open another page (location field), switch device size (desktop / tablet / mobile emulation), switch **Preview** mode on and off, open the **Publish** dialog
Canvas	The page as rendered by the site code	Click a component to select it; double-click text to edit it inline (Text and Title)
Properties panel (right rail)	The fields of the selected component, grouped in tabs	Fill in the fields; changes are saved automatically
Content tree (right rail)	The page structure: sections, the components in each section and their items	Select, add, reorder (drag), duplicate and delete components
Component actions	Add (+), Duplicate, Delete buttons shown for the selected item	Insert a component from the picker; copy or remove the selected one

> **[Screenshot Placeholder]**
> **File name:** `screenshots/ue-content-tree.png`
> **Purpose:** Content tree

> **Description:** The Content tree of a page expanded to show Main, two Sections, default content (Title, Text, Button) and a Cards block with three Card items; the Add (+), Duplicate and Delete actions are visible for the selected section.

5.2.2 Page structure in the content tree

```
Main (page)                          allowed: Section, Gen Carousel, Tabs Container
 +-- Section                         allowed: Text, Image, Button, Title and the blocks
 |    +-- Title / Text / Image / Button        (default content)
 |    +-- Hero, Teaser, Cards, List, ...       (blocks)
 |         +-- Card, List Item, ...            (block items, only inside their block)
 +-- Gen Carousel (a carousel section)      allowed: Hero, Teaser
 +-- Tabs Container (a tabs section)        allowed: Tab, Columns, Title, Text, Image,
Button
```

What may be added where is defined by the **filters** (see 5.4.3). The **Add (+)** picker only lists the components the selected container allows.

5.2.3 Adding, duplicating, deleting and moving

Task	How
Add a component	Select the section (or container block) in the content tree, select **Add (+)**, choose the component
Add a section	Select **Main** (the page) in the content tree, select **Add (+)**, choose **Section** (or **Gen Carousel** / **Tabs Container**)
Duplicate	Select the component, select **Duplicate**; the copy appears after the original with the same field values
Delete	Select the component, select **Delete**, confirm
Move	Drag the component in the content tree to its new position (inside the same or another allowed container)
Edit text	Double-click a Text or Title on the canvas and type; formatting toolbar for rich text
Edit fields	Select the component; use the Properties panel

> **Note:** After each change the editor sends the update to AEM and the project code re-renders only the changed part (`scripts/editor-support.js`: it re-decorates the changed block, section or default content). If a change cannot be applied in place, the whole canvas reloads; this is expected.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/ue-add-component-picker.png`
> **Purpose:** Component picker
> **Description:** The Add (+) picker opened on a Section, showing the groups Default Content (Text, Title, Image, Button) and Blocks (Accordion, Anchor List, Cards, ...).

5.2.4 Edit mode and Preview mode

The Universal Editor has two modes. The code detects the mode from a class on the page's `html` element: `adobe-ue-edit` in edit mode, `adobe-ue-preview` in preview mode (`utils/environment.js` `isEditing()`; `isInEditor()` is true in both modes because the editor instrumentation `data-aue-resource` is present).

Behaviour	Edit mode (`html.adobe-ue-edit`)	Preview mode, preview and live site
Clicking a link	Selects the component	Follows the link
Blocks with **Hide block** on	Shown faded (opacity 0.45) with a dashed outline, so you can still select them	Not displayed (`display: none`, its wrapper too)
Gen Carousel	Autoplay does not run; slides are not made inert	Autoplay runs (Auto modes); off-screen slides are inert
Hero and Table List entrance motion	Not applied (content always visible)	Applied when the option is set
Separator hidden on all breakpoints	Shown as a placeholder in edit mode	Not displayed
Interactive Dashboard (empty)	An editor placeholder is shown	Nothing
Tabs Container	Tab item markers visible in panels	Normal tabs
List, Cards, Filter List	Editor hints when the block is not set up (for example "Set up the list in its properties: …")	Empty-state message or nothing
Form Variations	"Shown when a dropdown value is …" notes on conditional fields	Fields shown or hidden by the condition

[Screenshot Placeholder]
File name: `screenshots/ue-edit-mode-hidden-block.png`
Purpose: Hidden block in edit mode
Description: A Teaser with Hide block switched on, shown faded with a dashed grey outline in edit mode, next to the same page in Preview mode where the teaser is absent.

[Screenshot Placeholder]
File name: `screenshots/ue-preview-mode.png`
Purpose: Preview mode toggle
Description: The Preview toggle in the editor toolbar switched on; the canvas behaves like the website (links are clickable, hidden blocks are gone).

5.2.5 Device sizes

Use the device switch in the toolbar to check the page at the project breakpoints: mobile (below 600 px), tablet (600 to 1199 px) and desktop (1200 px and above). Some text sizes step at 900 px, and the header switches to the desktop navigation at 900 px.

5.3 Creating a page

5.3.1 Where pages live

Location	Contents	Published
`/content/tcs-corp/{country}/{language}` (for example `us/en`)	The locale's Home page (the locale page itself) and all content pages below it	Yes, at `/{language}/` (for example `/en/`)
`{locale}/references/`	Navigation (`nav`), `footer` and other fragments	Yes (used by other pages)
`{locale}/error-pages/`	`404` (used by the site), `403` and `500` (for the CDN)	Yes
`{locale}/block-library/`	Pages with every block variation, in the locale's language	No (author only)
`/content/tcs-corp/site-global-templates/`	Template pages to copy: home, article, blog, press release	No (author only)

5.3.2 Creating a page from a Site Global Template

Step 1 — Copy the template page

Action: In the Sites console open **TCS Corp > site-global-templates**, select the template page you need (home, article, blog or press-release) and select **Copy**.

Expected Result: The page is on the clipboard.

Step 2 — Paste it into your locale

Action: Open the target folder in your locale (for example `TCS Corp > us > en > newsroom`) and select **Paste**. Then select the copy and **Rename** / **Move** it: set the page title and the page name. The name becomes the URL, for example `newsroom/q3-results` in `us/en` is published as `/en/newsroom/q3-results`.

Expected Result: The new page appears in the folder with the template's Page Type already set.

Step 3 — Open it in the Universal Editor

Action: Select the page and select **Edit**.

Expected Result: The page opens in the Universal Editor with the template's sections and components.

Step 4 — Set the page properties

Action: Open the page properties (Sites console: select the page, **Properties**; in the Universal Editor the same page-level fields are shown when the page is selected [To be confirmed: page-properties entry point in the installed Universal Editor version]). Fill in the fields listed in 5.3.4.

Expected Result: Title and Description are filled in; Page Type matches the template.

Step 5 — Add the page's metadata sheet values

Action: If the page needs values that are not page properties (for example `author-id`, `publish-date`, `reading-time`, `region`, `content-type`, `featured-image`, card values or SEO / social columns), make sure a row of the locale's bulk metadata sheet (`{locale}/metadata`) covers the page URL, and publish the sheet.

Expected Result: The values show on the page once the sheet is published (and, for the bulk metadata sheets, once the administrator has listed them in the site configuration; see the note in 5.3.4).

Step 6 — Edit the content, then preview and publish

Action: Edit the sections and components (5.4), then follow 5.7.

Expected Result: The page is available on `.aem.page` and, after publishing to live, on `.aem.live`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sites-console-copy-template.png`
> **Purpose:** Copying a Site Global Template
> **Description:** The Sites console showing `site-global-templates` with the press-release page selected and the Copy action highlighted.

For other pages (no matching template) use **Create** > **Page** in the locale folder. [To be confirmed: AEM page template offered by Create > Page — owner: AEM administrator]

5.3.3 Page Types and templates

The **Page Type** selects the site global template code (`scripts/scripts.js` `TEMPLATES` = `home`, `article`, `blog`, `press-release`; code in `templates/{name}/`). The value is also added to `<body>` as a class.

Page Type option	Value	What the template code adds
Default	(empty)	Nothing; the page shows exactly what is authored
Home	`home`	Layout styles only (the first section meets the header)
Article	`article`	A Page Header built from the page metadata (unless one is authored) and "About the author" from the Authors data sheet (`author-id`)
Blog	`blog`	The article structure and styles
Press Release	`press-release`	A Page Header (unless authored) and "Media contacts" from the Media Contacts data sheet (matching the page's `region`, else all)

Order of precedence: the **Page Type** page property, else the `page-type` column of the locale's metadata sheet, else the standard `template` metadata. A value that is not one of the four names loads no template code but is still added as a body class.

5.3.4 Page properties (models/_page.json)

Field Name	Field Type	Mandatory	Default	Purpose	Notes
Title (`jcr:title`)	Text	Recommended (no model validation)	(empty)	Browser title, SEO title, social title, Page Header heading, card title fallback	Keep it unique per page
Description (`jcr:description`)	Text	Recommended (no model validation)	(empty)	Meta description and Page Header description	Recommended 120–160 characters
Keywords (`keywords`)	Text, multi-value	No	(empty)	Meta keywords	One keyword per value
Page Type (`page-type`)	Select	No	Default (empty)	Site Global Template: Default, Home, Article, Blog, Press Release	Overrides the sheet's `page-type`

Field Name	Field Type	Mandatory	Default	Purpose	Notes
Navigation (`nav`)	AEM content picker	No	(empty)	Header page of this page and, unless they set their own, its child pages	Empty: nearest parent page's value (or the sheet's `nav`), else `{locale}/references/nav`
Footer (`footer`)	AEM content picker	No	(empty)	Footer page of this page and its child pages	Empty: nearest parent page's value (or the sheet's `footer`), else `{locale}/references/footer`
404 Page (`error-page`)	AEM content picker	No	(empty)	The "page not found" page for missing pages below this one	Empty: nearest parent's 404 Page (or the sheet's `error-page`), else `{locale}/error-pages/404`, else `/en/error-pages/404`, else the built-in text

Important: Navigation, Footer and 404 Page are **inherited**: leave them empty on ordinary pages. Set them on the top page of a sub-site (for example `/en/careers`) to change them for every page below it. Inherited values are cached for the browser session, so check changes in a new private window.

Note: All other page-level values (author, dates, reading time, region, images, card values, SEO and social values) come from the locale's **bulk metadata sheet** (`{locale}/metadata`, published as `/en/metadata.json`, `/de-de/metadata.json`, …). The platform applies these sheets only after the administrator lists all 13 in the site configuration (`metadata.source`). [To be confirmed: bulk metadata sheets listed in the site configuration — owner: site administrator]

5.4 Working with sections and blocks

5.4.1 Sections

A **section** is a horizontal band of the page. It contains default content (Text, Title, Image, Button) and blocks. Each section is rendered as `main > div.section`, and its content as one wrapper per block or run of default content (`div.default-content-wrapper`, `div.{block}-wrapper`). Spacing between sections, the content width and the side padding come from the layout tokens in `styles/styles.css` (see 6.5).

To add a section: select **Main** in the content tree, select **Add (+)** and choose **Section**. Then select the section and fill in its properties.

5.4.2 Section properties (models/_section.json)

Field	Type	Options / default	Effect
Section Name	Text	(empty)	Label of the section in the content tree only; not shown on the site
Style	Multi-select	Highlight	Adds the class `highlight`: light grey background (`--color-bg-secondary`), no outer margin, vertical padding of one section spacing
Width Type	Select	Full Width (default), Fixed Width	Fixed Width limits the section box (and its background) to the content width (90rem) and centres it; Full Width caps the content at the page width (120rem)

Field	Type	Options / default	Effect
Rounded edge	Boolean	off	Adds `rounded`: corners with radius 1rem (`--radius-l`), content clipped
Background Image	AEM Assets reference	(empty)	An image behind the section content, scaled to cover the section
Image alt text	Text	(empty)	Alt text of the background image; empty marks it as decorative (`alt=""`)
Anchor name	Text	(empty)	The section's `id` (for Anchor List links and `#links`), normalised to lower case with hyphens

Full details, including the rules for anchors and backgrounds, are in chapter 6.5.

5.4.3 What each container allows (filters)

The filters are defined in `models/_component-filters.json` (page level), `models/_section.json` (sections) and each block's `_{block}.json`; the aggregated result is `component-filters.json`.

Container	Allowed components
Main (the page)	Section; Gen Carousel; Tabs Container
Section	Text; Image; Button; Title; Page Header; Hero; Teaser; Quote; Testimonial; Anchor List; Accordion; Tabs; Table; Table List; List; Separator; Embed; Video; Modal; Form Variations; Filter List; Interactive Dashboard; Conversation Canvas; Cards; Columns; Fragment; Footer
Gen Carousel (section)	Hero; Teaser
Tabs Container (section)	Tab; Columns; Title; Text; Image; Button
Accordion	Accordion Item
Cards	Card
Columns	Column (each Column allows Text, Image, Button, Title)
Conversation Canvas	Conversation Canvas Button
Form Variations	Text Input Field; Dropdown / Select Field; Textarea Field; Consent Checkbox Field; Contact Card Output; Submit Button
List	List Item
Table List	Table List Item
Table	Row (the 2-, 3-, 4- and 5-column variants have their own Row items)
Tabs (block)	Tab
Video	(nothing)

> **Note: Tab** (the Tabs Container item) is not available inside an ordinary section, and Gen Carousel and Tabs Container can only be added at page level. The **Footer** block is allowed in sections so the footer reference page (`{locale}/references/footer`) can be authored.

5.4.4 Default content

Component	Model	Fields	Chapter
Text	`models/_text.json` (no model fields; rich text)	The rich text itself	6.6
Title	`models/_title.json`	Title; Title Type (h1–h6)	6.7
Image	`models/_image.json`	Image; Alt Text	6.8
Button	`models/_button.json`	Link; Text; Title; Views; Type; Icon; Alignment	6.9

5.4.5 Working with blocks

1. Select the section in the content tree and select **Add (+)**.
2. Choose the block from the **Blocks** group.
3. Fill in its tabs in the Properties panel (most blocks have several tabs and always an **Additional** tab last; see 5.6).
4. For blocks with items (Cards, List, Accordion, Tabs, Table, Form Variations, …) select the block and use **Add (+)** to add items.
5. Check the block at the three device sizes and in Preview mode.

> **Tip:** Each locale has a `block-library/` folder with pages showing every block variation in that language. Open them on author to see what each option looks like before using it.

5.5 Images and assets

5.5.1 Where images come from

Images are stored in **AEM Assets** under `/content/dam/tcs-corp/`. Image fields in the Universal Editor (component type `reference`: the Image component, the section Background Image, the image fields of blocks) open the **Assets picker**. You can also drag an asset from the Assets panel onto an image field.

Step 1 — Upload

Action: In AEM Assets open `/content/dam/tcs-corp/` (use the folder agreed for your locale or campaign), select **Create** > **Files** and upload the original image.

Expected Result: The asset is listed with a thumbnail.

Step 2 — Pick the image

Action: In the Universal Editor select the component, select the image field and choose the asset in the picker.

Expected Result: The image appears on the canvas.

Step 3 — Write the alt text

Action: Fill in the component's alt text field (Alt Text, Image alt text, …) describing what the image shows and why it matters on the page. Leave it empty only for purely decorative images.

Expected Result: The rendered `<img>` carries the alt text (or `alt=""` for decorative images).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/ue-assets-picker.png`
> **Purpose:** Assets picker
> **Description:** The Assets picker opened from the Image field of an Image component, browsing `/content/dam/tcs-corp/` with one image selected.

5.5.2 Image optimisation

When a page is published, Edge Delivery Services copies each referenced image into the site's media store (URLs of the form `./media_{hash}.{ext}`) and serves it in several widths and formats. Images placed in content arrive as responsive `<picture>` elements built by Edge Delivery Services. Where the code builds an image from a URL itself, it uses `createOptimizedPicture` (`scripts/aem.js`, through `decorators/responsive-image-decorator.js`) for site media (`/media_` URLs), for example section backgrounds (2000 px wide from 600 px viewport width, 750 px below); other URLs are used as they are, lazily loaded. The first image of the first section is loaded eagerly (it is usually the Largest Contentful Paint element); all other images are loaded lazily.

Do	Don't
Upload high-quality originals (JPEG for photos, PNG for graphics with transparency, SVG for icons where the field allows it)	Upload pre-shrunk thumbnails; they look blurred on large screens
Use descriptive file names (`tcs-cloud-team-mumbai.jpg`)	Use camera file names (`IMG_0042.jpg`)
Use images of at least 2000 px width for full-width backgrounds and heroes	Use images with text baked in (not translatable, not accessible)
Check the crop at all three device sizes (backgrounds use `object-fit: cover`)	Rely on a specific crop of a cover image; edges are cut off on narrow screens
Publish (or let **Manage Publication** include) the referenced assets	Reference assets you have no right to use [To be confirmed: image licensing rules — owner: brand team]

> **Note:** For social sharing images (`og-image` in the metadata sheet) use at least 1200 × 630 px.

5.5.3 Alt text rules

Image type	Alt text
Informative (shows something the text does not say)	Describe it briefly, for example "TCS engineers reviewing a cloud dashboard"
Functional (inside a link or button)	Describe the destination, not the picture
Decorative (adds nothing; backgrounds, mood images)	Leave empty; the code then sets `alt=""`
Contains text	Repeat the text in the alt text

Recommended length: up to about 125 characters (not enforced by the models).

5.6 The Additional tab

Every block dialog has an **Additional** tab as its last tab, with the same ten fields. The tab is present in the models of: Accordion, Anchor List, Cards, Columns, Conversation Canvas, Embed, Filter List, Form Variations, Fragment,

Gen Carousel, Hero, Interactive Dashboard, List, Modal, Page Header, Quote, Separator, Tab, Table, Table List, Tabs (block), Tabs Container, Teaser, Testimonial and Video (25 models).

It is **not** present on: the header and footer, block items (Accordion Item, Card, List Item, Tab items, Table rows, Table List items, form fields), the section, and default content (Text, Title, Image, Button).

The fields are "classes" fields (`classes_*`): AEM delivers their values as CSS classes of the block, never as content rows. `decorators/block-options-decorator.js` turns the Anchor ID class into the block's `id`; the other options are pure CSS in `styles/styles.css`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/ue-additional-tab.png`
> **Purpose:** The Additional tab
> **Description:** The Properties panel of a Teaser with the Additional tab open, showing Anchor ID, Hide block and the eight padding and margin selects.

5.6.1 Field table

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	Text	No	No	(empty)	Gives the block an `id` so links can jump to it (`#anchor-...`)	Start with `anchor-`; lower-case letters, digits and hyphens; unique on the page	Enforced: regular expression `^anchor-[a-z0-9][a-z0-9-]*$`, message "Start with anchor-, then lower-case letters, digits and hyphens."	`anchor-our-services`
Hide block (`classes_hideblock`)	Boolean	No	No	false	Temporarily hides the whole block on the site without deleting it	Use for content that is prepared but not yet released, or temporarily withdrawn	None	true
Padding left (`classes_paddingleft`)	Select	No	No	Default (empty)	Extra inner space on the left of the block	Use only when the design asks for it	Enforced: one of the options	`pl-600` (1.5rem)
Padding right (`classes_paddingright`)	Select	No	No	Default (empty)	Extra inner space on the right	As above	Enforced: one of the options	`pr-600`
Padding top (`classes_paddingtop`)	Select	No	No	Default (empty)	Extra inner space above the content	As above	Enforced: one of the options	`pt-1200` (3rem)
Padding bottom (`classes_paddingbottom`)	Select	No	No	Default (empty)	Extra inner space below the content	As above	Enforced: one of the options	`pb-1200`
Margin left (`classes_marginleft`)	Select	No	No	Default (empty)	Outer space on the left of the block	As above	Enforced: one of the options	`ml-0`
Margin right (`classes_marginright`)	Select	No	No	Default (empty)	Outer space on the right	As above	Enforced: one of the options	`mr-0`
Margin top (`classes_margintop`)	Select	No	No	Default (empty)	Outer space above the block	Use to separate a block from the previous one	Enforced: one of the options	`mt-800` (2rem)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Margin bottom (`classes_margin-bottom`)	Select	No	No	Default (empty)	Outer space below the block	As above	Enforced: one of the options	`mb-0`

5.6.2 Spacing options and their values

Each padding and margin select offers the same 15 options. The option label shows the size; the stored value is a class `{prefix}-{token}` with the prefixes `pl`, `pr`, `pt`, `pb` (padding left, right, top, bottom) and `ml`, `mr`, `mt`, `mb` (margin).

Option label	Class suffix	Token	Size (rem)	Size (px at 16 px root)
Default	(none)	the block's own CSS	—	—
0 (none)	`-0`	`--space-0`	0	0
0.25rem (space-100)	`-100`	`--space-100`	0.25	4
0.5rem (space-200)	`-200`	`--space-200`	0.5	8
0.75rem (space-300)	`-300`	`--space-300`	0.75	12
1rem (space-400)	`-400`	`--space-400`	1	16
1.25rem (space-500)	`-500`	`--space-500`	1.25	20
1.5rem (space-600)	`-600`	`--space-600`	1.5	24
2rem (space-800)	`-800`	`--space-800`	2	32
2.5rem (space-1000)	`-1000`	`--space-1000`	2.5	40
3rem (space-1200)	`-1200`	`--space-1200`	3	48
4rem (space-1600)	`-1600`	`--space-1600`	4	64
5rem (space-2000)	`-2000`	`--space-2000`	5	80
6rem (space-2400)	`-2400`	`--space-2400`	6	96
7.5rem (space-3000)	`-3000`	`--space-3000`	7.5	120

The rules are written as `main .section .block.pl-600 { padding-left: var(--space-600); }` (and so on for every prefix and token). The `main .section` part makes them stronger than a block's own spacing rules, so the authored value wins. The values are the same at every breakpoint; there are no per-device spacing options. The larger tokens `--space-4000` to `--space-6000` exist in `styles/styles.css` but are not offered.

5.6.3 Anchor ID rules

1. **Format (Enforced by the model):** the value must match `^anchor-[a-z0-9][a-z0-9-]*$`: it starts with `anchor-`, the next character is a lower-case letter or digit, then lower-case letters, digits and hyphens only. `anchor-our-services` and `anchor-2026-results` are valid; `Anchor-Services`, `our-services`, `anchor-` and `anchor-our services` are rejected with "Start with anchor-, then lower-case letters, digits and hyphens.".
2. **Becomes the id:** `applyBlockOptions()` finds the class matching the same pattern and sets it as the block element's `id`, once per block. Link to it with `#anchor-our-services` (same page) or `/en/page#anchor-our-services`.
3. **Uniqueness (Enforced by code):** if another element on the page already has that id (another block, a section's Anchor name, a heading id generated from the heading text, a tab id), the block gets the first free

suffix: `anchor-our-services-2`, then `-3`, and so on (`utils/ids.js` `availableId`). Links to the plain id then go to the first element. **Recommended:** use each Anchor ID once per page.

4. **Own ids win:** a block that already has an id is not changed. List, Cards and Separator have their own **Component ID** field, which they use as their id; set the Anchor ID only when no Component ID is set.
5. **Scrolling:** after the page has loaded, the site scrolls to the element named in the URL hash (`scripts/scripts.js` `loadLazy`).
6. **Analytics:** the id is also the block's component id in the data layer (Part 11), so changing an Anchor ID changes the analytics id of the block.
7. **Section anchors are different:** a section's **Anchor name** (section properties) has no `anchor-` prefix rule; it is normalised to lower case with hyphens and is skipped (not suffixed) if the id is already taken.

5.6.4 Hide block behaviour

Where	What happens
Live and preview site	The block and its wrapper are not displayed (`html:not(.adobe-ue-edit) main .block.hideblock` and `main .section > div:has(> .block.hideblock)` get `display: none`)
Universal Editor, Preview mode	Not displayed (same rule: the page has no `adobe-ue-edit` class)
Universal Editor, Edit mode	Shown faded (opacity 0.45) with a 0.125rem dashed outline (`--color-border-secondary`) 0.25rem outside the block, so it can be selected and switched back on
Page source	The block's content is still in the delivered HTML and its JavaScript still runs; search engines can read the text
Data layer	The block is still listed in the data layer component state (Part 11)
Section	The section stays; if every block in it is hidden, the empty section's spacing remains

> **Warning:** Hide block is not a security or embargo mechanism. Confidential content (unreleased results, names) must not be published at all, hidden or not.

> **Note:** For the two section-based containers (Gen Carousel and Tabs Container) the Additional values are stored on the section. The code that applies them reads block classes only, so their effect on these two containers could not be verified from the code. [To be confirmed: Additional tab on Gen Carousel and Tabs Container — owner: development team]

5.6.5 Padding and margin guidance

- **Recommended:** leave all eight selects on **Default**. Every block already follows the design system spacing, and sections already add spacing between each other (2.5rem on mobile, 5rem from 900 px, 7.5rem from 1200 px).
- Use **Margin top / bottom** to tighten or loosen the gap between two blocks in the same section.
- Use **Padding** when a block with a coloured background needs more inner space.
- Avoid large left / right values on mobile: the same value applies at every breakpoint and narrows the content.
- Margins of neighbouring blocks are added to the default layout; check the page at all three device sizes.

5.7 Preview, publish, validation and rollback

5.7.1 Environments

Environment	Address	Who sees it	How content gets there
Author	`author-p220207-e2271469.adobeaemcloud.com`	Signed-in authors	Saved automatically while editing
Preview	`https://main--tcs-corp--sradha-tcs.aem.page/{language}/...`	Anyone with the link (not indexed by search engines)	Publish to **Preview**
Live	`https://main--tcs-corp--sradha-tcs.aem.live/{language}/...`	Public	Publish to **Live** (Publish)
Production domain	[To be confirmed: production domain and CDN — owner: site administrator]	Public	Served from Live through the CDN
Feature preview (developers)	`https://{branch}--tcs-corp--sradha-tcs.aem.page/`	Developers and testers	Code from a Git branch, content from Preview

```
 Author (Universal Editor) --Publish to Preview--> Preview  (*.aem.page)
        |                                            |
        +-----------------Publish (Live)-----------> Live     (*.aem.live) --> CDN /
domain
```

5.7.2 Publishing from the Universal Editor

Step 1 — Save and check

Action: Finish editing; check the page in Preview mode and at the three device sizes.

Expected Result: No editor hints (“Set up the list …”) are visible; hidden blocks are absent in Preview mode.

Step 2 — Open the Publish dialog

Action: Select **Publish** in the editor toolbar.

Expected Result: The dialog lists the page and the references it uses (assets, fragments) and offers the destination(s) configured for this project (Preview and / or Live). [To be confirmed: publish destinations enabled in the Universal Editor for this program — owner: AEM administrator]

> **[Screenshot Placeholder]**
> **File name:** `screenshots/ue-publish-dialog.png`
> **Purpose:** Universal Editor publish dialog
> **Description:** The Publish dialog with the page and its referenced assets listed, and the destination choice Preview / Live.

Step 3 — Publish to Preview and verify

Action: Choose **Preview**, publish, then open `https://main--tcs-corp--sradha-tcs.aem.page/{language}/{path}`.

Expected Result: The page shows the new content on `.aem.page`.

Step 4 — Publish to Live and verify

Action: When approved, publish again with destination **Live** (or use **Quick Publish** / **Manage Publication** in the Sites console). Open `https://main--tcs-corp--sradha-tcs.aem.live/{language}/{path}`.

Expected Result: The page is updated on `.aem.live`, normally within one to two minutes.

5.7.3 Publishing from the Sites console

Action	Use
Quick Publish	Publish the selected page(s) and their references now
Manage Publication	Publish or unpublish several pages, include child pages and referenced assets, or schedule a later date
Unpublish (Manage Publication > Unpublish)	Remove the page from the live site

5.7.4 What must be published separately

Changes to these items only take effect when the item itself is published: navigation and footer pages (`references/`), fragments, error pages, the locale's sheets (`placeholders`, `metadata`, `taxonomy`, `redirects`) and data sheets (`data/authors`, `locations`, `partners`, `social-links`, `media-contacts`), and images in AEM Assets. Listings (List, Filter List) read the query index (`/query-index.json`), which is updated when a page is published; republish a page if its card values in the metadata sheet changed.

5.7.5 Using AEM Sidekick

On a `.aem.page` or `.aem.live` page, open the Sidekick to see whether the page is previewed and published and when, to switch between **Preview** and **Live**, and to open the page in AEM (**AEM Editor**). Content is authored in AEM, so publish from AEM (Universal Editor or Sites console) rather than from the Sidekick. [To be confirmed: whether authors may use Sidekick preview / publish buttons in this project — owner: site administrator]

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sidekick-status.png`
> **Purpose:** AEM Sidekick on the preview site
> **Description:** The Sidekick toolbar on a `.aem.page` page showing the Preview / Live switch and the last preview and publish times.

5.7.6 Caching

Edge Delivery Services purges its CDN cache for a page when it is published, so the new version is normally visible within a minute or two. If you still see the old version:

1. Hard-refresh the browser (Ctrl+Shift+R / Cmd+Shift+R) or open a private window.
2. Check that you published to **Live**, not only to **Preview**.
3. Check that the referenced item (fragment, sheet, image) was published too.
4. For inherited Navigation / Footer / 404 Page, use a new browser session (the values are cached per session).
5. If a production CDN is in front of `.aem.live`, its cache may add delay. [To be confirmed: production CDN cache and purge configuration — owner: site administrator]

5.7.7 Validation checklist (every page)

- ☐ Title and Description set in page properties; Page Type correct.
- ☐ One `h1` on the page (from the Page Header, a Hero with H1 or a Title with h1), headings in order.

- ☐ Every informative image has alt text; decorative images have none.
- ☐ Links work and point to published pages (`/en/...`, not `/content/tcs-corp/...`).
- ☐ No block is unintentionally hidden (check edit mode for faded blocks).
- ☐ Anchor links jump to the right block or section.
- ☐ The page looks right on mobile, tablet and desktop.
- ☐ Translated UI labels appear (locale placeholders sheet published).
- ☐ The preview URL (`.aem.page`) shows the change before you publish to Live.

5.7.8 Unpublishing a page

1. If the URL has been public, add a redirect from the old URL to a replacement in the locale's `redirects` sheet and publish it. **Note:** the per-locale redirects sheets are not applied by Edge Delivery on their own; another mechanism (for example CDN rules) must read them. [To be confirmed: redirect mechanism — owner: site administrator]
2. In the Sites console select the page, **Manage Publication** > **Unpublish** (or Quick Unpublish).
3. Check that the live URL now shows the locale's 404 page.
4. Delete the page on author only after it is unpublished.

5.7.9 Rolling back to an earlier version

AEM keeps versions of a page (created on publish and when you create one manually).

Step 1 — Open the timeline

Action: In the Sites console select the page, open the **Timeline** rail and filter on **Versions**.

Expected Result: The list of versions with date and author.

Step 2 — Restore

Action: Select the version and choose **Revert to this Version** (or use **Create** > **Restore** > **Restore Version** on the parent to restore a deleted child page).

Expected Result: The author page content returns to that version (the current state is kept as a new version).

Step 3 — Republish

Action: Open the page in the Universal Editor, check it, then publish to Preview and Live.

Expected Result: The live site shows the restored content.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sites-console-revert-version.png`
> **Purpose:** Restoring a page version
> **Description:** The Sites console Timeline rail of a page with the Versions filter, one version selected and the Revert to this Version action visible.

> **Note:** Sheets, fragments and navigation pages are rolled back the same way, then republished. Code changes are rolled back by the development team (Git revert), not by authors.

5.8 Authoring do's and don'ts

Do	Don't
Start a templated page by copying it from Site Global Templates	Build article, blog or press-release pages from scratch
Use one `h1` per page and keep heading levels in order (h2, then h3)	Pick a heading level for its size
Fill in alt text for every informative image	Put text inside images
Keep standalone links in their own paragraph to make them buttons; use the Button component for explicit control	Put several links in one paragraph and expect buttons
Use the locale's own references, sheets and data	Link to another locale's fragments or sheets
Use Anchor IDs with the `anchor-` prefix, once per page	Rename an Anchor ID that campaigns or analytics already use without telling the analytics team
Leave padding and margin on Default unless the design asks otherwise	Use spacing options to fix layout problems at one breakpoint only
Use Hide block for short-term withdrawals	Use Hide block to hide confidential content
Preview first, then publish to Live	Publish to Live without checking the preview
Publish sheets, fragments and assets after changing them	Expect a changed sheet to apply without publishing it
Add a redirect before unpublishing or moving a public page	Delete public pages without a redirect
Use the locale's block library pages to try options	Experiment on live pages
Keep UI labels in the locale's placeholders sheet	Type button labels or UI text into code-driven fields that come from placeholders
Keep personal data (names, e-mail addresses, phone numbers of individuals) out of free-text fields used for analytics labels	Put personal data in Anchor IDs, Component IDs or analytics labels

Part 6 — Component Library

This part documents every component authors can use on the TCS Corporate Website. Each chapter covers one component and follows the same fourteen-section template, so readers can find the same information in the same place for every component: 1 Component Purpose, 2 Visual Overview, 3 Component Variations, 4 Authoring Experience, 5 Authoring Guide, 6 Authorable Field Specification, 7 Field Annotation Mapping, 8 Component Specification, 9 Input Model, 10 Output Model, 11 Business Rules, 12 Author Best Practices, 13 Troubleshooting Guide and 14 Support Runbook. Field names, options, defaults and validation rules are taken from the component models (`blocks/{name}/_{name}.json`, `models/*.json`, aggregated in `component-models.json`); behaviour is taken from the component code. Rules are marked **Enforced** when the model or the code validates them and **Recommended** when they are editorial guidance. Chapters 6.5 to 6.9 cover the section container and the four default-content components (Text, Title, Image and Button), which have no Additional tab; the other blocks are documented in their own chapters. The complete inventory of components, items, models and filters is in the appendix.

6.1 Header Block

The Header block is the site header and main navigation of every page. It isn't added to pages by authors: the page loader (`scripts/aem.js` `loadHeader`, called from `scripts/scripts.js` `loadLazy`) creates it inside the page's `<header>` element. The block loads a **navigation page** (the "nav" reference page) as a fragment and turns its three sections into the brand (logo), the navigation sections (with dropdowns) and the tools area. Authors change the header by editing the nav page, not by editing a block.

Source files: `blocks/header/header.js`, `blocks/header/header.css`. The Header has **no** `_header.json` model: it has no dialog, no properties panel and no Additional tab.

6.1.1 Component Purpose

Business Objective. Give every page of every locale site a consistent, accessible, responsive main navigation with the TCS brand, the top-level sections of the site and a tools area, maintained centrally by authors without code changes.

Business Use Cases.

- Global navigation of a locale site (`/en`, `/de-de`, …) from the locale's `references/nav` page.
- A different navigation for one area of the site (for example a Careers area using `references/nav-careers`), set once on the area's top page and inherited by all pages below it.
- Campaign or microsite pages with a reduced navigation (a dedicated nav page chosen in the page's Navigation property or the metadata sheet's `nav` column).
- Access to second-level pages through dropdowns (desktop) or an expanded menu (mobile and tablet).

When To Use. Always: the header is loaded on every page automatically. Authors decide only which nav page a page uses and what the nav page contains.

When Not To Use.

- Don't add the Header as a block on a page; it isn't in the component palette and isn't meant to be authored inside content.
- Don't use the nav page for page-specific content (promotions, banners). Use a Fragment, Teaser or Hero on the page instead.
- Don't build mega menus with more than two levels: only the first level gets dropdown behaviour (see Business Rules).

Dependencies.

Type	Dependency	Use
Code	`blocks/fragment/fragment.js` `loadFragment`	Loads `{navPath}.plain.html` and decorates it like a page
Code	`utils/inherited-metadata.js` `getInheritedMetadata('nav')`	The page's Navigation, else the nearest parent page's, up to the locale Home page
Code	`utils/site.js` `resolvePagePath`	Turns the configured value (AEM path, published path or locale-relative path) into the path to fetch; default `references/nav` of the locale
Code	`decorators/button-decorator.js` `removeButtonDecoration`	Keeps the brand (logo) link a plain link
Code	`utils/ids.js` `createIdFactory`	Unique ids for the dropdown sub-lists (`nav-submenu-…`)
Code	`utils/placeholders.js` `getPlaceholders`, `fillPlaceholder`	Translated accessible names
Content	Nav page per locale: `/content/tcs-corp/{country}/{language}/references/nav` (published `/{locale}/references/nav`)	The header content
Content	Optional alternative nav pages, e.g. `references/nav-careers`	Area-specific navigation
Page property	**Navigation** (`nav`) in Page Properties (`models/_page.json`)	Chooses the nav page; inherited by child pages
Sheet	Bulk metadata sheet column `nav` (`/{locale}/metadata.json`)	Alternative way to choose the nav page by URL pattern
Placeholders	`mainNavigation` ("Main navigation"); `openNavigation` ("Open navigation"); `closeNavigation` ("Close navigation"); showSubmenu ("Show {label} menu")	Accessible names of the nav and its controls
Styles	`styles/styles.css`: `--nav-height` (4rem), `--background-color`, `--light-color`, `--body-font-size-s`, `--heading-font-size-s`; `header { height: var(--nav-height) }`; the header is hidden until its `data-block-status` is `loaded`	Layout and loading state

Related Components. Footer (loaded the same way from `references/footer`); Fragment (the loading mechanism); Conversation Canvas (its menu can also list site pages); Page Header (the first block below the header on content pages); 404 handling in `scripts.js` (uses the same inheritance for the 404 Page property).

Applicable Templates. All pages and all templates (home, article, blog, press-release and pages without a Page Type), including the 404 page.

Supported Content Types. All content types; the header doesn't read page content type metadata.

6.1.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** The header at 1440px wide: logo on the left, the top-level navigation sections in a row, the tools area on the right, one dropdown open below its section label with the light grey panel and the small pointer triangle.

Desktop (>= 1200px, and the same from 900px). The navigation is a single horizontal row (flexbox, `justify-content: space-between`, 2rem gap) centred in the page with a maximum width of 79rem and 2rem padding left and right; the row height is `--nav-height` (4rem). The brand is on the left (an image is 8rem wide; text is shown at `--heading-font-size-s`, bold). The sections are a horizontal list (1.5rem gap, no wrapping: `white-space: nowrap`), first level medium weight (500), at `--body-font-size-s` (1rem from 900px). A section with a sub-list shows a small chevron drawn with borders to the right of its label; the chevron points down when closed and up when open. An open dropdown is an absolutely positioned panel 12.5rem wide, starting 150% of the item height below the label and 1.5rem to the left of it, with 1rem padding, the `--light-color` background (`--color-bg-secondary`) and a 0.5rem triangle pointing at the label; its items have 0.5rem vertical padding and regular weight (400). The hamburger button is hidden. The header scrolls away with the page (`.nav-wrapper` is `position: relative` from 900px).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The header at 768px wide with the menu closed: hamburger icon on the left, logo next to it, tools on the right; the navigation sections are hidden.

Tablet (600–899px). Same as mobile: the header uses the mobile menu below 900px (the desktop navigation starts at 900px, not at the 1200px desktop breakpoint of the rest of the site). Between 900px and 1199px the desktop navigation is shown.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The header at 390px wide in two states: closed (hamburger, logo, tools in one 4rem row) and open (full-height menu with the close icon, the sections and their sub-lists stacked, the tools at the bottom).

Mobile (< 600px) and tablet (< 900px). The navigation is a CSS grid with the areas `hamburger brand tools` in one 4rem row and 1.5rem padding (maximum width 78rem). The `.nav-wrapper` is `position: fixed` at the top (z-index 2), so the header stays visible while the page scrolls; the `<header>` element reserves its 4rem height. The hamburger icon (three 1.25rem bars, 0.125rem thick) is drawn in CSS. When the menu is open, the nav switches to the areas `hamburger brand` / `sections` / `tools`, takes at least the full viewport height (`100dvh`), scrolls internally (`overflow-y: auto`), the page behind it doesn't scroll (`body overflow-y: hidden`), the icon turns into an "X" (two rotated 1.5rem bars), the sections are listed vertically with all sub-lists visible, and the tools move to the bottom row.

6.1.3 Component Variations

The header has no style options. Its variations are the nav page it shows, the breakpoint mode and the way each navigation section is authored.

Variation 1 — Default Locale Navigation

Variation Name: Default locale navigation (`references/nav`).

Business Purpose: One standard navigation per locale site, used by every page that doesn't choose another one.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-default-locale-navigation.png`
> **Purpose:** Default navigation of the /en locale
> **Description:** An /en content page showing the header built from /en/references/nav: TCS logo, top-level sections, tools.

Use Cases: All locale pages; the Home page of each locale; error pages.

Authoring Guidelines: Leave the **Navigation** page property empty on the pages (and their parents) and don't set a `nav` value in the metadata sheet. Maintain the content in `/content/tcs-corp/{country}/{language}/references/nav`.

Business Rules: Used when the page and none of its parent pages (up to the locale Home page) has a Navigation value. Each locale has its own nav page: `/de-de/...` pages use `/de-de/references/nav`. Pages outside a locale (drafts, templates) use the `/en` locale.

Rendering Rules: `resolvePagePath('', 'references/nav')` returns `/{locale}/references/nav` on the published site and `/content/tcs-corp/{country}/{language}/references/nav` when editing in Universal Editor. If the nav page doesn't exist or isn't published, the header stays empty (4rem high, no content).

Responsive Behaviour: Desktop and from 900px: horizontal navigation with dropdowns. Tablet below 900px and mobile: hamburger menu.

SEO Impact: Navigation links are rendered in the browser (the header is built by JavaScript after the main content), so search engines that render JavaScript see them; the nav page itself is excluded from the query index (`/*/references/**`) and sitemap.

Accessibility Compliance: `<nav aria-label="Main navigation">` landmark; buttons with `aria-expanded` / `aria-controls`; Escape and focus handling (see 6.1.8).

Variation 2 — Alternate Navigation Page

Variation Name: Alternate navigation (for example `references/nav-careers`).

Business Purpose: A different header for an area of the site (Careers, Investors, a campaign) without changing the global navigation.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-alternate-navigation.png`
> **Purpose:** Area-specific navigation
> **Description:** A page below /en/careers showing the navigation from /en/references/nav-careers, with Careers-specific sections, compared side by side with the default navigation.

Use Cases: Section-specific navigation; microsites; landing pages with a reduced navigation.

Authoring Guidelines:

1. Create the nav page in the locale's `references/` folder (for example `references/nav-careers`), with the same three-section structure as `references/nav`.
2. Open the top page of the area (for example `/content/tcs-corp/us/en/careers`), open **Page Properties** and select the nav page in **Navigation**.
3. Publish the nav page and the top page.

Alternatively, add a row to the locale's metadata sheet with the URL pattern (for example `/en/careers/**`) and the `nav` column set to the nav page.

Business Rules: The value is inherited: the page's own Navigation (page property or metadata sheet `nav` column), else the nearest parent page's, up to the locale's Home page, else `references/nav`. Accepted values: an AEM path (`/content/tcs-corp/us/en/references/nav-careers`), a published path (`/en/references/nav-careers`) or a path relative to the locale root (`references/nav-careers`). A value with `.html` or `.plain.html` is accepted (the extension is removed).

Rendering Rules: Parent pages are looked up by loading their published HTML head in parallel (`getInheritedMetadata`), and the found values are cached in `sessionStorage` (key `tcs-inherited-metadata`) for the browser session.

Responsive Behaviour: As Variation 1.

SEO Impact: As Variation 1. Keep the brand link and main sections consistent so users and crawlers find the global pages.

Accessibility Compliance: As Variation 1. Keep the same order (brand, sections, tools) so users of assistive technology find the same structure in every area.

Variation 3 — Desktop Dropdown Navigation

Variation Name: Desktop dropdown navigation (viewport 900px and wider).

Business Purpose: Compact horizontal navigation for large screens, revealing second-level pages on demand.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-desktop-dropdown.png`
> **Purpose:** An open dropdown
> **Description:** Desktop header with one section's dropdown open: the chevron rotated up, the grey 12.5rem panel with the pointer triangle and the second-level links.

Use Cases: All desktop and large-tablet visits.

Authoring Guidelines: On the nav page, in the second section, write the top-level items as a bulleted list; indent the second-level items below an item to create its dropdown.

Business Rules: Only one dropdown is open at a time: opening a dropdown closes the others. A dropdown closes when its toggle is selected again, when Escape is pressed (focus returns to its toggle) or when focus leaves the navigation. There is no hover behaviour: dropdowns open on click / Enter / Space only.

Rendering Rules: Each first-level `li` with a nested `ul` gets the class `nav-drop`, `data-expanded="false|true"` and a `<button class="nav-drop-toggle" aria-expanded aria-controls="nav-submenu-…">`. The sub-list is `display: none` until the item's `data-expanded` is `true`.

Responsive Behaviour: Desktop and 900–1199px: dropdown behaviour. Below 900px the same markup is shown as the mobile menu (Variation 4). Resizing across 900px resets the state (`matchMedia` change listener calls `toggleMenu`).

SEO Impact: Second-level links are in the DOM (hidden with `display: none` until opened), so they are discoverable after rendering.

Accessibility Compliance: Toggle buttons expose `aria-expanded` and `aria-controls`; the chevron is decorative CSS; Escape returns focus to the toggle (WCAG 2.1.1, 2.4.3, 4.1.2).

Variation 4 — Mobile and Tablet Menu

Variation Name: Mobile / tablet hamburger menu (viewport below 900px).

Business Purpose: A touch-friendly full-screen menu on small screens.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-mobile-menu.png`
> **Purpose:** Open mobile menu
> **Description:** 390px viewport, the menu open: "X" icon, logo, vertical list of sections with their second-level items visible, tools at the bottom, page behind not scrollable.

Use Cases: Phones and tablets in portrait and landscape below 900px.

Authoring Guidelines: Same nav page as desktop; no separate authoring. Keep labels short so they fit on narrow screens.

Business Rules: The menu button (`.nav-hamburger button`) opens and closes the menu. When the menu opens, every dropdown is marked expanded. Escape closes the menu and focuses the menu button; focus leaving the navigation closes the menu.

Rendering Rules: `nav[data-expanded="true"]` shows `.nav-sections`; `body.style.overflowY = 'hidden'` while open. The button's `aria-expanded` and `aria-label` switch between "Open navigation" and "Close navigation".

Responsive Behaviour: Mobile and tablet only (the hamburger is hidden from 900px). Sub-lists are always visible inside the open menu; on mobile the dropdown toggles change `aria-expanded` but the CSS doesn't hide collapsed sub-lists (see Known Limitations).

SEO Impact: None beyond Variation 1.

Accessibility Compliance: Menu button with `aria-controls="nav-sections"`, `aria-expanded` and a translated `aria-label`; the icon is `aria-hidden`.

Variation 5 — Unlinked Section Label

Variation Name: Section label without a link (label is the dropdown button).

Business Purpose: A top-level grouping (for example "Services") that has no landing page of its own and only opens its sub-pages.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-unlinked-section-label.png`
> **Purpose:** Section label rendered as a button
> **Description:** Desktop header: the "Services" label with its chevron, focused with the keyboard focus ring, dropdown open.

Use Cases: Grouping labels such as "What we do", "Industries", "Insights".

Authoring Guidelines: In the nav page's list, type the first-level item as plain text (no link) and indent its sub-items below it.

Business Rules: The label text becomes the button text; the button has no separate `aria-label` (its text is its accessible name).

Rendering Rules: The label node (the first element or non-empty text of the `li`) is replaced by `<button type="button" class="nav-drop-toggle">Services</button>`. If the item has no label at all, an empty button is inserted at the start of the item.

Responsive Behaviour: Desktop: the button opens the dropdown. Mobile: shown as a heading-like label above its always-visible sub-list.

SEO Impact: The label isn't a link; give important groups a landing page (Variation 6) if they need to rank.

Accessibility Compliance: Native `<button>` (keyboard operable with Enter and Space), `aria-expanded`, `aria-controls`.

Variation 6 — Linked Section Label

Variation Name: Section label with a link (label keeps its link, a separate toggle button is added).

Business Purpose: A top-level section that has its own landing page and sub-pages.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-linked-section-label.png`
> **Purpose:** Linked label with dropdown toggle
> **Description:** Desktop header: "Industries" as a link, with its chevron; keyboard focus on the toggle button after the link.

Use Cases: Sections such as "Industries" with an overview page.

Authoring Guidelines: In the nav page's list, make the first-level label a link to the section page and indent its sub-items below it. The link must be inside an element (for example a paragraph or bold formatting) of the list item to be kept (see Business Rules).

Business Rules: The link is kept and a toggle button is added right after the label element, with the accessible name "Show {label} menu" (placeholder `showSubmenu`). The code checks for a link **inside** the label element (`label.querySelector('a[href]')`): when the list item's first child is the link itself (`<li><a href>…</a><ul>…`), the link isn't detected, so it is replaced by a text toggle button and the section page link is lost. Verify the rendered header after editing a linked label.

Rendering Rules: `<p><a href="/en/industries">Industries</a></p><button class="nav-drop-toggle" aria-label="Show Industries menu" aria-expanded="false" aria-controls="nav-submenu-industries"></button><ul id="nav-submenu-industries">…`. The toggle button has no visible content and no size of its own in the CSS (measured 0 × 0 px in a browser check); the chevron is drawn on the list item (`.nav-drop::after`), not on the button.

Responsive Behaviour: Desktop: the link navigates; the dropdown can be opened by keyboard (Tab to the toggle, Enter). Because the toggle has no visible area, mouse and touch users can't open this dropdown on desktop (see Known Limitations). Mobile: the link and the always-visible sub-list are shown.

SEO Impact: The section landing page is linked from every page (good for internal linking).

Accessibility Compliance: Separate link and button, each with an accessible name. The zero-size toggle has no visible focus indicator area; this is a gap for WCAG 2.4.7 (Focus Visible) on desktop and must be fixed in code before relying on linked labels with dropdowns.

Variation 7 — Simple Section Link (No Dropdown)

Variation Name: First-level link without a sub-list.

Business Purpose: Direct links to key pages ("Contact", "Careers").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-simple-section-link.png`
> **Purpose:** Plain first-level link
> **Description:** Desktop header with a first-level link without a chevron next to two dropdown sections.

Use Cases: Single destination pages.

Authoring Guidelines: Add a first-level list item with a link and no indented sub-items.

Business Rules: Items without a nested list aren't decorated (no button, no `data-expanded`).

Rendering Rules: Rendered as the authored `<li><a>`; links inherit the text colour (`color: currentcolor`).

Responsive Behaviour: Desktop: inline in the row. Mobile: one line in the open menu.

SEO Impact: Standard internal link.

Accessibility Compliance: Standard link; the link text is its accessible name.

Variation 8 — Tools Area

Variation Name: Tools area (third section of the nav page).

Business Purpose: Utility links or buttons such as "Contact us", a language link or a search link.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-tools-area.png`
> **Purpose:** Tools on the right of the header
> **Description:** Desktop header with a "Contact us" primary button and a text link in the tools area at the right end; mobile closed state with the tools in the top row.

Use Cases: Primary call to action, secondary utility links.

Authoring Guidelines: Add a third section to the nav page with Text or Button components. A standalone link in a paragraph becomes a link button; bold or italic formatting (or the Button component's options) gives a primary / secondary button (`decorators/content-buttons-decorator.js`).

Business Rules: Only the first three sections of the nav page are used as brand, sections and tools; a fourth section is appended to the nav without a role class or specific styling. There is no built-in search, language switcher or login in the tools area (Not implemented as of this handover).

Rendering Rules: The section gets `nav-tools` and the grid area `tools`; there are no other tools-specific styles.

Responsive Behaviour: Desktop: at the right end of the row. Mobile closed: in the top row at the right. Mobile open: in the bottom row of the menu.

SEO Impact: Standard links.

Accessibility Compliance: Buttons and links keep their accessible names from the authored text; icon-only buttons get an `aria-label` from the button decorator.

6.1.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-authoring-view.png`
> **Purpose:** Universal Editor canvas of the nav page
> **Description:** The page /content/tcs-corp/us/en/references/nav open in Universal Editor, showing three sections with the letters A to E marked.

Annotations:

- A = Section 1 (brand): the logo image or the brand link (Image, Text or Button component).
- B = Section 2 (sections): the Text component with the bulleted list of first-level items.
- C = A first-level item with indented second-level items (becomes a dropdown).
- D = A second-level item (a link in the dropdown).
- E = Section 3 (tools): Text or Button components with utility links or buttons.

> **Note:** When the nav page is opened directly in Universal Editor, it is rendered as an ordinary page (sections and default content), not as a header; the header of that page is itself loaded from the nav page. Check the result on a content page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-properties-panel.png`
> **Purpose:** Properties used to choose the nav page
> **Description:** Page Properties of a content page with the Navigation field (and Footer, 404 Page) highlighted; plus the properties panel of the Text component holding the list on the nav page.

Properties panel description. The Header block itself has no properties panel. The fields involved are:

- **Page Properties > Navigation** (`nav`, AEM content picker): chooses the nav page for this page and, unless they set their own, its child pages.
- **Text component** (rich text) on the nav page: the list of sections; formatting, links and list indentation are edited in the rich text editor.
- **Image component** (Image, Image alt text) on the nav page: the logo.
- **Button component** (Link, Link Text, Link Title, Button View, Link Type, Button Icon, Button Alignment) on the nav page: buttons in the tools area; in the brand section any button styling is removed.
- **Section properties** of the nav page: not used by the header (section styles apply to the nav's sections but the header CSS doesn't plan for them).

6.1.5 Authoring Guide

Step 1 — Open the nav page in Universal Editor

Action: In AEM Sites, navigate to `/content/tcs-corp/{country}/{language}/references/`, select `nav` (or the alternate nav page) and choose **Edit**.

Expected Result: The nav page opens in Universal Editor with its sections.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-step-01-open-nav-page.png`
> **Purpose:** Opening the nav page
> **Description:** AEM Sites console with the references folder of /us/en and the nav page selected, the Edit action highlighted.

Step 2 — Check the three-section structure

Action: In the Content tree, confirm the page has three sections in this order: brand, sections, tools. Add a missing section with **Add** > **Section**.

Expected Result: Three sections are listed; their order matches brand, sections, tools.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-step-02-section-structure.png`
> **Purpose:** Section order
> **Description:** The Content tree of the nav page with Section 1 (brand), Section 2 (sections), Section 3 (tools).

Step 3 — Edit the brand

Action: In section 1, select the Image component and pick the logo from AEM Assets, and fill **Image alt text** (for example "TCS home"); or use a Text / Button component with a link to the locale Home page.

Expected Result: The logo or brand link shows in the first section.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-step-03-brand.png`
> **Purpose:** Brand section
> **Description:** Section 1 with the logo image selected and the Image alt text field filled.

Step 4 — Edit the navigation sections

Action: In section 2, select the Text component and edit the bulleted list: one first-level item per section; indent (increase indent) the second-level items under their section; add links with the link tool.

Expected Result: A nested bulleted list with first-level and second-level items.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-step-04-sections-list.png`
> **Purpose:** Navigation list
> **Description:** Rich text editor with a two-level bulleted list, the indent button highlighted.

Step 5 — Decide linked or unlinked labels

Action: For each first-level item with sub-items, either leave the label as plain text (it becomes the dropdown button) or make it a link wrapped in formatting, for example a bold link (it stays a link and a toggle is added).

Expected Result: Labels are authored as intended; see Variations 5 and 6.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-step-05-section-labels.png`
> **Purpose:** Label types
> **Description:** The list with one plain text label and one linked label.

Step 6 — Edit the tools area

Action: In section 3, add or edit Text or Button components for the utility links and the call to action.

Expected Result: The tools appear in section 3.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-step-06-tools.png`
> **Purpose:** Tools section
> **Description:** Section 3 with a Button component “Contact us” selected and its properties panel open.

Step 7 — (Optional) Assign an alternate nav page

Action: Open the top page of the area that needs the alternate navigation, open **Page Properties**, and pick the nav page in **Navigation**. Save.

Expected Result: The Navigation field holds the nav page path.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-step-07-navigation-property.png`
> **Purpose:** Navigation page property
> **Description:** Page Properties dialog of /en/careers with Navigation set to /content/tcs-corp/us/en/references/nav-careers.

Step 8 — Preview

Action: Publish the nav page to Preview (AEM Sidekick or **Publish** > **Preview**), then open a content page on `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/...`.

Expected Result: The preview page shows the updated header.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-step-08-preview.png`
> **Purpose:** Preview check
> **Description:** A content page on the .aem.page host with the changed header.

Step 9 — Validate

Action: Run the Validation Process checklist below on desktop and mobile widths.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/header-step-09-validate.png`

> **Purpose:** Validation at two widths
> **Description:** The same page at 1440px with a dropdown open and at 390px with the menu open.

Step 10 — Publish

Action: Publish the nav page (and any page whose Navigation property changed) to Live.

Expected Result: The live site (`https://main--tcs-corp--sradha-tcs.aem.live/`) shows the new header after the CDN cache refresh.

> [**Screenshot Placeholder**]
> **File name:** `screenshots/header-step-10-publish.png`
> **Purpose:** Publishing
> **Description:** The Publish dialog with the nav page selected and the Live destination.

Preview Process

1. Publish the nav page to Preview; the header on every preview page of that locale uses it immediately (no page re-publication is needed, because the header is loaded in the browser from the nav page).
2. Open several pages of different areas on `.aem.page` and check they use the expected nav page.
3. If you changed a Navigation page property, publish that page to Preview too: child pages read the value from the parent's published head.
4. Open a new private browser window when checking inheritance: inherited values are cached per browser session.

Publishing Process

1. Publish the nav page to Live.
2. Publish pages whose Navigation property changed.
3. If a new alternate nav page is used, publish it before (or together with) the pages that reference it; otherwise the header of those pages is empty.

Validation Process

- ☐ The logo shows and links to the locale Home page; it is a plain link, not a button.
- ☐ The first-level items appear in the authored order on desktop (900px and wider).
- ☐ Each section with sub-items shows a chevron; selecting its label opens the dropdown; opening another closes the first.
- ☐ Linked labels still link to their pages (and their dropdown opens with the keyboard).
- ☐ Escape closes an open dropdown and focus returns to its button.
- ☐ Tabbing out of the navigation closes the dropdown.
- ☐ Below 900px the hamburger opens a full-height menu; the page behind doesn't scroll; Escape closes it and focus returns to the menu button.
- ☐ The tools area shows its links / buttons at the right (desktop) and in the top row (mobile, closed).
- ☐ Screen reader: the landmark is announced as "Main navigation" (or the translated text); toggles announce expanded / collapsed.
- ☐ Pages in an area with an alternate nav page show that nav; other pages show the default.
- ☐ Translated labels (placeholders) are correct in non-English locales.

Rollback Process

1. In AEM, open the nav page's version history (**Timeline** or **Versions**), restore the previous version and publish it to Preview and Live.
2. If an alternate nav page caused the problem, clear the **Navigation** page property on the area's top page (or remove the `nav` value from the metadata sheet row) and publish; the pages fall back to the parent's or the locale's `references/nav`.
3. Tell testers to open a new browser session (inherited values are cached in `sessionStorage`).

6.1.6 Authorable Field Specification

The Header block has no block model and no Additional tab. The authorable inputs are the Navigation page property and the content of the nav page.

Page Properties (model `page-metadata`, `models/_page.json`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Navigation (`nav`)	aem-content (content picker)	No	No	Empty (inherit)	Chooses the nav page of the page and its children	Set only on the top page of an area that needs another navigation	None in the model; code accepts AEM, published or locale-relative paths	`/content/tcs-corp/us/en/references/nav-careers`

Nav page content (default content of the nav page)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Section 1 — Brand	Section with Image / Text / Button	Yes (Recommended)	No	None	Logo and Home link	One logo (image 8rem wide) or a brand link	None; button styling is removed in code	TCS logo linked to `/en/`
Image (brand)	reference (AEM Assets)	No	No	None	Logo image	Use an optimized SVG or PNG	None	`/content/dam/tcs/logo.svg`
Image alt text (brand)	text	Recommended	No	Empty	Accessible name of the logo	Describe the destination, e.g. "TCS home"	None	`TCS home`
Section 2 — Sections	Section with a Text component	Yes (Recommended)	No	None	First-level navigation	One bulleted list	Only lists directly in the section's default content are decorated	List of 4–7 items
First-level item	Rich text list item	Yes	Yes	None	A navigation section	Plain text label or a wrapped link	None	`Services`
Second-level item	Rich text nested list item	No	Yes	None	Dropdown entry	A link per item	None	`Consulting → /en/services/consulting`
Section 3 — Tools	Section with Text / Button	No	No	None	Utilities and call to action	Keep to 1–3 items	None	`Contact us` button

Field details

- **Navigation (nav).** Purpose: selects the header for a page tree. Allowed Values: any page path; typically `references/nav-*` pages of the same locale. Character Limits: none. Business Rules: inherited by child pages; the page's own value wins; the metadata sheet `nav` column is an equivalent source. Dependency Rules: the referenced page must be published. Visibility Rules: always visible (no `condition`). Sample Content: `/content/tcs-corp/us/en/references/nav-careers`.
- **Section 1 — Brand.** Purpose: the logo. Allowed Values: Image, Text or Button components. Character Limits: Recommended a single short brand text if not an image. Business Rules: the first `.button` / `.link-button` is turned back into a plain link. Dependency Rules: none. Visibility Rules: always shown. Sample Content: the TCS logo.
- **Section 2 — Sections.** Purpose: navigation tree. Allowed Values: one bulleted list with up to two levels. Character Limits: Recommended labels of 1–3 words (desktop row doesn't wrap). Business Rules: items with a nested list become dropdowns; others stay links. Dependency Rules: the list must be in the default content of the second section. Visibility Rules: hidden below 900px until the menu is opened. Sample Content: Services (Consulting, Cloud, AI), Industries, Insights, Careers.
- **Section 3 — Tools.** Purpose: utilities. Allowed Values: Text and Button components. Character Limits: Recommended up to 3 items of 1–3 words. Business Rules: standalone links become buttons per the content buttons decorator. Dependency Rules: none. Visibility Rules: always visible. Sample Content: "Contact us" (primary button).

6.1.7 Field Annotation Mapping

Marker	Field	Purpose
A	Section 1 — Brand (Image / Text / Button)	Logo and Home link
B	Section 2 — Sections (Text with list)	First-level navigation
C	First-level item with sub-items	Dropdown section (button label or linked label)
D	Second-level item	Dropdown link
E	Section 3 — Tools	Utility links and call to action

6.1.8 Component Specification

Attribute	Value
Component Name	Header (`blocks/header`)
Version	1.0
Owner	[To be confirmed: product owner — owner: Digital Marketing]; technical owner: development team
Dependencies	`blocks/fragment/fragment.js`; `utils/inherited-metadata.js`; `utils/site.js`; `utils/ids.js`; `utils/placeholders.js`; `decorators/button-decorator.js`; nav pages in `references/`; Navigation page property; metadata sheet `nav` column
Component Type	Block (page-level, loaded automatically into `<header>`; not authorable on pages)
Supported Variants	Default nav; alternate nav pages; desktop dropdowns (>= 900px); mobile / tablet menu (< 900px); unlinked label; linked label; simple link; tools area
Supported Templates	All (home, article, blog, press-release, default, 404)
Supported Metadata	`nav` (page property Navigation, metadata sheet column), inherited from parent pages
Supported Tags	None

Attribute	Value
Supported Personalization	None. Different navigation per area via nav pages only. Adobe Target personalization would be possible in future through the data layer and Adobe Tags (not implemented)
Caching Considerations	The nav page `.plain.html` is served from the Edge Delivery CDN and the browser HTTP cache; parent-page Navigation values are cached in `sessionStorage` (`tcs-inherited-metadata`) for the session; placeholders are loaded once per page
Performance Considerations	Loaded in the lazy phase after the first section (LCP) is shown; one fetch for the nav page plus up to one fetch per parent page (in parallel) when the page has no own Navigation value; header area reserves 4rem to avoid layout shift; the logo isn't optimized by code (author an optimized asset)
Accessibility Compliance	nav landmark with translated `aria-label`; native buttons with `aria-expanded` / `aria-controls`; Escape closes and returns focus; focus leaving the nav closes it; hamburger icon `aria-hidden`; known gaps: zero-size toggle for linked labels, mobile toggles don't collapse visually
SEO Compliance	Links are standard `<a href>`; navigation is client-rendered; nav pages excluded from query index and sitemap

6.1.9 Input Model

- **Content Inputs:** the nav page's sections (brand, sections list, tools).
- **Metadata Inputs:** `<meta name="nav">` of the page (Navigation page property or metadata sheet `nav` column), else the parent pages' `nav` meta values.
- **Tags Inputs:** none.
- **Page Context Inputs:** the current path (locale and parent pages, published or AEM author path); viewport width (`(min-width: 900px)`).
- **Query Inputs:** none (no query index).
- **Manual Inputs:** the nav page's rich text and components.

Authored HTML received (nav page `.plain.html`, simplified):

```
<div>
  <p><a href="/en/"><picture><img src="./media_1a2b.png" alt="TCS home"></picture></a></p>
</div>
<div>
  <ul>
    <li>Services
      <ul>
        <li><a href="/en/services/consulting">Consulting</a></li>
        <li><a href="/en/services/cloud">Cloud</a></li>
      </ul>
    </li>
    <li><p><strong><a href="/en/industries">Industries</a></strong></p>
      <ul><li><a href="/en/industries/banking">Banking</a></li></ul>
    </li>
    <li><a href="/en/careers">Careers</a></li>
  </ul>
</div>
<div>
  <p><strong><a href="/en/contact">Contact us</a></strong></p>
</div>
```

The header block itself, as built by `aem.js`, is `<div class="header block" data-block-name="header">` with one empty row; its authored content is ignored.

6.1.10 Output Model

Rendered Output (DOM structure).

```
header
└─ div.header.block
   └─ div.nav-wrapper
      └─ nav#nav[aria-label][data-expanded]
         ├─ div.nav-hamburger > button[aria-controls][aria-expanded][aria-label]
         │                        > span.nav-hamburger-icon[aria-hidden]
         ├─ div.section.nav-brand      (section 1)
         ├─ div.section.nav-sections#nav-sections   (section 2)
         │  └─ div.default-content-wrapper > ul > li.nav-drop[data-expanded]
         │       ├─ button.nav-drop-toggle[aria-expanded][aria-controls]
         │       └─ ul#nav-submenu-{label}
         └─ div.section.nav-tools      (section 3)
```

Generated HTML (desktop, simplified).

```
<nav id="nav" aria-label="Main navigation" data-expanded="true">
  <div class="nav-hamburger">
    <button type="button" aria-controls="nav-sections" aria-expanded="false"
      aria-label="Open navigation"><span class="nav-hamburger-icon" aria-hidden="true"></
span>
    </button>
  </div>
  <div class="section nav-brand">…</div>
  <div class="section nav-sections" id="nav-sections">
    <div class="default-content-wrapper"><ul>
      <li class="nav-drop" data-expanded="false">
        <button type="button" class="nav-drop-toggle" aria-controls="nav-submenu-services"
          aria-expanded="false">Services</button>
        <ul id="nav-submenu-services">…</ul>
      </li>
    </ul></div>
  </div>
  <div class="section nav-tools">…</div>
</nav>
```

On desktop the `nav` keeps `data-expanded="true"` (the code opens the "menu" on desktop so the sections are visible) while the hamburger reports `aria-expanded="false"`; on mobile `data-expanded` follows the menu state.

Generated JSON: none.

Generated Metadata: none.

Analytics Tracking. The header block is a `.block`, so the data layer decorator (`decorators/data-layer-decorator.js`) tracks it:

- State (pushed in the delayed phase, about 3 seconds after load, by `trackBlocks(document)`):

```
{ "component": { "header-1": {
  "@type": "tcs-corp/components/header",
  "parentId": "page-1x2y3z" } } }
```

- Clicks on header links and buttons push `cmp:click` with `interaction` `link` or `button`, `dc:title` (the `aria-label` or text, up to 100 characters), `xdm:linkURL`, `xdm:linkType` (`other` for site links, `exit` for other domains, `download` for files) and `breakpoint`:

```
{ "event": "cmp:click", "eventInfo": {
  "path": "component.header-1", "interaction": "link", "dc:title": "Consulting",
```

```
  "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/services/consulting",
  "xdm:linkType": "other", "breakpoint": "desktop" } }
```

Blocks placed on the nav page are tracked as their own components with `parentId` `header-1`. No tag manager is loaded yet; events wait in `window.adobeDataLayer`.

Personalization Output: none.

Search Index Output: none; the nav pages are excluded from `/query-index.json` (`/*/references/**`) and from the sitemap.

6.1.11 Business Rules

1. The nav page is the page's Navigation value (page property or metadata sheet `nav` column), else the nearest parent page's up to the locale Home page, else `{locale}/references/nav`.
2. The Navigation value may be an AEM path, a published path or a path relative to the locale root; `.html` / `.plain.html` endings are removed; an unparsable value falls back to `references/nav`.
3. When editing in Universal Editor the nav page is loaded from its AEM author path (`/content/tcs-corp/...`); on the published site from its published path.
4. Parent-page values are looked up by loading the parents' published HTML in parallel and cached for the browser session (`sessionStorage` key `tcs-inherited-metadata`); a changed parent value is seen in a new session.
5. If the nav page can't be loaded (missing, not published, network error, or a path not starting with `/`), the header is left empty; no error is shown.
6. The nav page's first section is the brand, the second the sections, the third the tools; further sections are appended without a role.
7. In the brand section, the first button or link button is turned back into a plain link.
8. Only first-level list items (`.default-content-wrapper > ul > li` in the sections section) that contain a nested list become dropdowns.
9. An unlinked label becomes the toggle button's text; a label element containing a link keeps the link and gets an additional toggle button with the accessible name "Show {label} menu"; a list item whose first child is the link itself loses the link (it's replaced by a text button).
10. Sub-list ids are `nav-submenu-{label}` (lower-case, hyphenated), unique on the page (suffix `-2`, `-3`, ...); an existing id is kept.
11. Desktop (>= 900px): selecting a toggle closes all other dropdowns and toggles its own; only one is open.
12. Mobile (< 900px): the menu button toggles the whole menu; opening it marks all dropdowns expanded and locks page scrolling; closing it closes all dropdowns.
13. Escape: on desktop closes the open dropdown and focuses its toggle (nothing happens if none is open); on mobile closes the open menu and focuses the menu button.
14. Focus moving outside the nav closes all dropdowns (desktop) or the menu (mobile).
15. Crossing the 900px breakpoint resets the state for the new mode.
16. Accessible names come from the placeholders sheet (`mainNavigation`, `openNavigation`, `closeNavigation`, `showSubmenu`) with English fallbacks.
17. The header is invisible until the block has loaded (`data-block-status="loaded"`), and its height is reserved (4rem) to avoid layout shift.
18. There is no hover-to-open behaviour and no outside-click handler; closing on outside clicks relies on focus leaving the nav.
19. The header has no Additional tab (no anchor, hide or spacing options).

6.1.12 Author Best Practices

- **Recommended Content Length:** 4–7 first-level items; 1–3 words per label (the desktop row doesn't wrap); up to about 8 second-level items per dropdown (the panel is 12.5rem wide and its text wraps); 1–3 tools.
- **Image Dimensions:** the logo is displayed 8rem (128px) wide, height automatic; provide an SVG or a PNG at least 256px wide (2x) with a transparent background. The image isn't processed by `createOptimizedPicture` in the header code, so keep the file small (Recommended under 20 KB).
- **CTA Writing Guidelines:** one primary action in the tools area ("Contact us"); use verbs; avoid "Click here".
- **Accessibility Guidelines:** give the logo an alt text naming the destination ("TCS home"); use unlinked labels for dropdown sections until the linked-label toggle is fixed; keep the same structure across locales; don't use only icons without text.
- **SEO Guidelines:** link the main hub pages from the first level or the dropdowns; use descriptive link text; avoid duplicate links with different texts.
- **Metadata Guidelines:** set Navigation only on area top pages, not on every page; keep the metadata sheet `nav` column and page properties consistent and avoid setting both for the same page (both are published as the page's `nav` meta; on Edge Delivery, page-level metadata normally takes precedence over bulk metadata).
- **Taxonomy Guidelines:** not applicable (the header doesn't use tags).
- **Localization Guidelines:** each locale maintains its own `references/nav` with translated labels and locale links (`/de-de/...`); translate the four header placeholders in each locale's placeholders sheet; check that long translations (German, Dutch) still fit in one desktop row.

6.1.13 Troubleshooting Guide

Issue 1 — Header is empty

Symptoms: A 4rem white band at the top, no logo or links.

Root Cause: The nav page (default or inherited) doesn't exist, isn't published, or the Navigation value points to a wrong path.

Diagnosis Steps: Open the browser developer tools Network tab and look for the `.plain.html` request of the nav page and its status (404). Check the page's `<meta name="nav">` and its parents' values.

Resolution Steps: Publish the nav page, or correct / clear the Navigation property on the page or its parent, then publish.

Validation Steps: Reload in a new private window; the header shows.

Preventive Measures: Publish nav pages before pages that reference them; include `references/nav` in every new locale's go-live checklist.

Issue 2 — Area shows the default nav instead of the alternate nav

Symptoms: Pages below `/en/careers` still show the global navigation.

Root Cause: The Navigation property was set but the top page wasn't published, or the browser session still has the old cached value.

Diagnosis Steps: View the source of the published top page and check `<meta name="nav">`; inspect `sessionStorage` key `tcs-inherited-metadata`.

Resolution Steps: Publish the top page; test in a new browser session.

Validation Steps: All pages of the area show the alternate nav.

Preventive Measures: Always publish the page whose properties changed; tell testers to use a new session.

Issue 3 — A section page link disappeared

Symptoms: A first-level label that was a link is now a button that only opens the dropdown.

Root Cause: The link is the list item's direct first child; the code only keeps links inside an element (Business Rule 9).

Diagnosis Steps: Inspect the rendered `li.nav-drop`: there is no `<a>` before the `button.nav-drop-toggle`.

Resolution Steps: Wrap the label link in formatting (for example bold) or a paragraph, or add the section page as the first sub-item ("Overview"); for a permanent fix, development changes `decorateDrop` to also detect a label that is itself a link.

Validation Steps: The label navigates; the dropdown still opens.

Preventive Measures: Validate linked labels after every nav change.

Issue 4 — Dropdown of a linked label can't be opened with the mouse

Symptoms: On desktop, clicking the chevron of a linked section does nothing; only keyboard users can open it.

Root Cause: The toggle button for a linked label has no visible content or size; the chevron is drawn on the list item, which has no click handler.

Diagnosis Steps: Inspect `button.nav-drop-toggle[aria-label]`: its box is 0 × 0.

Resolution Steps: Short term: use an unlinked label and add an "Overview" sub-item. Long term: development gives the toggle a visible size and moves the chevron into it.

Validation Steps: Mouse, touch and keyboard all open the dropdown; the focus ring is visible.

Preventive Measures: Include mouse and keyboard checks in the Validation Process.

Issue 5 — Mobile menu stays open or the page doesn't scroll

Symptoms: After closing the menu or rotating the device the page can't be scrolled.

Root Cause: `body overflow-y: hidden` is set while the menu is open; a script error or an interrupted state change can leave it set.

Diagnosis Steps: Check `document.body.style.overflowY` and `nav[data-expanded]`; look for console errors.

Resolution Steps: Press Escape or the menu button; reload. Report persistent cases to development with the device and browser.

Validation Steps: The page scrolls with the menu closed.

Preventive Measures: Test open / close / rotate on iOS Safari and Android Chrome after header code changes.

Issue 6 — Labels or accessible names are in English on a non-English locale

Symptoms: Screen readers announce "Open navigation" on `/de-de`.

Root Cause: The locale's placeholders sheet lacks the keys `openNavigation`, `closeNavigation`, `mainNavigation`, `showSubmenu`, or the sheet isn't published.

Diagnosis Steps: Open `/{locale}/placeholders.json` and search for the keys.

Resolution Steps: Add the keys with translated Text and publish the sheet.

Validation Steps: Check `aria-label` values in the rendered header.

Preventive Measures: Keep the placeholders sheet complete in every locale.

Issue 7 — Dropdown doesn't close when clicking elsewhere (Safari)

Symptoms: On desktop Safari, a dropdown stays open after clicking in the page.

Root Cause: Closing relies on focus leaving the nav; Safari doesn't focus buttons on click, so no focus-out happens. There is no outside-click handler.

Diagnosis Steps: Reproduce in Safari and Chrome; compare.

Resolution Steps: Close with the toggle or Escape; development can add an outside-click handler.

Validation Steps: Dropdowns close on outside clicks in all browsers after the fix.

Preventive Measures: Cross-browser testing of navigation changes.

Issue 8 — Navigation wraps or overflows on smaller desktops

Symptoms: At 900–1100px the sections overlap the tools or are cut off.

Root Cause: Labels don't wrap (`white-space: nowrap`) and too many or too long items are authored.

Diagnosis Steps: Resize the browser from 900px upward.

Resolution Steps: Shorten labels or reduce the number of first-level items.

Validation Steps: No overlap at 900px, 1024px and 1200px.

Preventive Measures: Respect the Recommended content length.

6.1.14 Support Runbook

Monitoring Requirements. No dedicated monitoring exists. Recommended: a synthetic check per locale that loads the Home page and asserts `nav#nav` has children; Real User Monitoring (Edge Delivery RUM) for CLS on page load; data layer `cmp:click` events on `component.header-1` once Adobe Tags is installed.

Logs.

- Browser console: the header code emits no messages of its own; a failed nav fetch shows as a network 404 for `{path}.plain.html`; an exception during decoration (for example a network error) is logged by `aem.js` as `failed to load module for header`.
- Admin API status: `https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{locale}/references/nav` shows preview / live status of the nav page.
- Author logs: AEM publication and replication logs for the nav page; Universal Editor shows errors when saving content.

Common Errors.

Error	Likely Cause	Fix
Empty header	Nav page missing / unpublished	Publish or correct the Navigation value
Wrong nav in an area	Parent not published; stale session cache	Publish parent; new session
Linked label lost its link	Link not wrapped	Wrap link or add Overview item
Dropdown not mouse-operable	Linked label toggle has no size	Use unlinked label; code fix
English accessible names	Placeholders missing	Add keys to the locale sheet
Logo shown as a button	Logo authored outside section 1	Move it into section 1

Escalation Matrix.

Level	Team	Checks
L1	Author support	Nav page content, section order, publication status, Navigation property, placeholders sheet
L2	AMS / site admin	Admin API status of the nav and parent pages, `paths.json` mapping, CDN cache purge, metadata sheet `nav` column and `metadata.source` configuration
L3	Development team	`blocks/header/header.js` / `.css`, `utils/inherited-metadata.js`, `utils/site.js`, console errors, browser-specific behaviour
L4	Adobe Support	Edge Delivery delivery errors, Universal Editor or AEM publishing defects

Recovery Procedures. Restore the previous nav page version and publish; clear a faulty Navigation property; if a code release broke the header, revert the commit on `main` (AEM Code Sync deploys it within minutes).

Health Checks. For each of the 13 locales: `/{locale}/references/nav.plain.html` returns 200; the Home page header shows brand, sections and tools; a dropdown opens on desktop; the menu opens on mobile.

Operational Procedures. Nav changes follow Preview → validation → Live. New locale: create and publish `references/nav` with the three sections and translated placeholders before go-live. New alternate nav: create the page, set Navigation on the area's top page, publish both.

Known Limitations.

- Linked labels: a link that is the list item's direct child is replaced by the toggle; the extra toggle for wrapped links has no visible area (not mouse-operable, no visible focus on desktop).
- On mobile, dropdown toggles change `aria-expanded` but sub-lists stay visible.
- No hover opening, no outside-click closing, no arrow-key navigation within dropdowns.
- Only two levels; deeper lists aren't styled as nested dropdowns.
- No built-in search, language switcher, sticky desktop header or mega menu (Not implemented as of this handover).
- Inherited values are cached per session; parent changes appear in a new session.
- The logo isn't optimized by code.

6.2 Footer Block

The Footer block has **two roles under one block name** (`blocks/footer/footer.js`):

1. **The page footer.** The page loader (`scripts/aem.js loadFooter`, called from `scripts/scripts.js loadLazy`) creates a `footer` block inside the page's `<footer>` element. It loads the **footer page** (the locale's `references/footer`, or the inherited Footer page property) as a fragment and appends the locale's **Social Links** data sheet as a list of social profile links.
2. **The authored Footer block** placed on the footer page. Authors add it in Universal Editor; it renders an optional **Conversation Canvas** (a page with the Conversation Canvas block, loaded as a fragment) above the **footer bar** with the copyright text and the footer link list.

The code decides the role by the block's parent: a block directly inside `<footer>` is the page footer, any other Footer block is the authored footer bar. Source files: `blocks/footer/footer.js`, `blocks/footer/footer.css`, `blocks/footer/_footer.json`. The Footer has **no** Additional tab.

6.2.1 Component Purpose

Business Objective. Close every page with a consistent, brand-compliant footer (legal copyright, policy links, social channels and, optionally, the conversational search entry point), maintained per locale by authors.

Business Use Cases.

- Legal footer of every page: copyright with the current year, Privacy / Terms / Cookie / Accessibility links.
- Social channel links (LinkedIn, X, Facebook, Instagram, YouTube) per locale from a data sheet, without editing the footer page.
- The Conversation Canvas ("Ask us a question" search and quick links) at the bottom of every page, as in the TCS design (Figma "Footer – Desktop / Tablet / Mobile").
- A different footer for an area or campaign (for example `references/footer-campaign`) set once on the area's top page.

When To Use. The page footer is always loaded. Use the authored Footer block only on footer pages in `references/` (for example `references/footer`, `references/footer-campaign`).

When Not To Use.

- Don't add the Footer block to content pages: it is offered in the section's component list (`models/_section.json` filter), but its styles are scoped to the `<footer>` element, so inside a page it renders unstyled.
- Don't put social links into the Link List; maintain them in the Social Links data sheet so every footer page shows the same channels.
- Don't use the footer for page-specific calls to action.

Dependencies.

Type	Dependency	Use
Code	`blocks/fragment/fragment.js` `loadFragment`	Loads the footer page and the conversation canvas page
Code	`utils/inherited-metadata.js` `getInheritedMetadata('footer')`	The page's Footer, else the nearest parent's up to the locale Home page
Code	`utils/site.js` `resolvePagePath`	Path to fetch (AEM path when editing, published path on the site)
Code	`utils/data.js` `getSocialLinks`	Rows of `/{locale}/data/social-links.json` in Sort Order
Code	`decorators/social-links-decorator.js` `buildSocialLinks`	Social list with icons and accessible names
Code	`utils/links.js` `readLinkItems`	Link List items and their Link Type
Code	`decorators/button-decorator.js` `removeButtonDecoration`; `decorators/link-decorator.js` `setLinkTarget`	Plain links; new-window target
Code	`decorators/content-decorator.js` `fromCell`; `scripts/scripts.js` `moveInstrumentation`	Copyright rich text and editor instrumentation
Code	`utils/placeholders.js`	`footerLinks` ("Footer links"), `followUsOn` ("Follow us on {platform}")
Content	Footer page per locale `references/footer`; optional alternatives `references/footer-*`	Footer content
Content	Conversation canvas page, e.g. `references/footer-conversation-canvas` (the `/en` sample)	Canvas above the bar
Page property	**Footer** (`footer`, `models/_page.json`)	Chooses the footer page; inherited by child pages
Sheet	Metadata sheet column `footer`; data sheet `data/social-links` (platform, url, icon, sort-order)	Footer choice; social links
Icons	`/icons/linkedin.svg`, `x.svg`, `facebook.svg`, `instagram.svg`, `youtube.svg`	Social icons
Styles	`styles/styles.css` tokens (`--space-*`, `--color-bg-default`, `--color-bg-brand`, `--color-bg-warning`, `--color-text-default`, `--font-weight-medium`, `--page-max-width`); footer hidden until loaded	Layout

Related Components. Header (same loading and inheritance); Fragment; Conversation Canvas (the canvas shown in the footer; its sticky dock hides when the footer comes into view); 404 page (uses the same footer).

Applicable Templates. All pages and templates, including the 404 page.

Supported Content Types. All; the footer doesn't read content type metadata.

6.2.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** The footer at 1440px: the conversation canvas title on two lines, the floating menu button and search field, the buttons in a row, and the bottom bar with the copyright on the left and the links on the right, over the white background with the soft blue and yellow glows at the bottom.

Desktop (>= 1200px). White background (`--color-bg-default`) with two radial-gradient glows at the bottom edge: a blue glow (`--color-bg-brand` at 30%) and a yellow glow (`--color-bg-warning` at 30%). Content is a vertical flex column, maximum width `--page-max-width` (120rem), padding 7.5rem top, 5rem left and right, 2.5rem bottom, 5rem gap between the footer page's sections. With a conversation canvas the footer is at least 50.125rem high and the bar sits at the bottom; the canvas title is 6rem (line height 7.1875rem, letter-spacing –3%) with a maximum width of 61.5rem (two lines). The bar is a wrapping flex row (`justify-content: space-between`): copyright left (at least 19.375rem wide), links right with 1.5rem column gap; text 1rem, line height 1.375rem, medium weight. Social links follow as a wrapping row with 1rem gap and 1.5rem icons.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The footer at 1024px: canvas title one line wide (no maximum width), menu button visible, buttons in a row, bar below.

Tablet (600–1199px). Padding 7.5rem top, 1.25rem left and right, 2.5rem bottom; with a canvas the minimum height is 59.75rem, the canvas gap 5rem, the title 6rem without a maximum width, the canvas buttons in a row, the menu button of the canvas visible (`display: grid`), the search field padding 1.5rem at the start; bar text 1rem / 1.375rem. The glows move to 25% (blue) and 88% (yellow) of the width.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The footer at 390px: canvas title 3.25rem over several lines, buttons stacked, no menu button, the copyright above the links.

Mobile (< 600px). Padding 3.5rem top, 1.25rem sides, 1.5rem bottom; with a canvas the minimum height is 50rem, the canvas gap 4rem, the title 3.25rem (line height 3.875rem) limited to 20rem wide and centred, the canvas buttons stacked in a column, the canvas menu button hidden (the site header has the navigation), the search field padding 1rem at the start. Bar text 0.75rem, line height 1.0625rem; the copyright takes at least 19.375rem (or the full width), so the links wrap below it. Glows: blue near the left edge (6%), yellow mostly beyond the right edge (130%).

6.2.3 Component Variations

Variation 1 — Default Locale Footer
Variation Name: Default locale footer (`references/footer`).

Business Purpose: One standard footer per locale for every page.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-default-locale-footer.png`
> **Purpose:** Default footer of /en
> **Description:** An /en content page with the footer from /en/references/footer: canvas, copyright, Privacy & Terms link, social icons.

Use Cases: All pages without their own Footer value.

Authoring Guidelines: Leave the **Footer** page property empty; maintain `/content/tcs-corp/{country}/{language}/references/footer`.

Business Rules: Used when neither the page nor its parents up to the locale Home page has a Footer value. Each locale has its own footer page.

Rendering Rules: The footer page's sections are moved into `footer > .footer > div`; the Social Links list is appended after them.

Responsive Behaviour: See 6.2.2.

SEO Impact: Footer links are client-rendered; the footer page is excluded from the query index and sitemap.

Accessibility Compliance: The page `<footer>` is the contentinfo landmark; the link list is a `nav` landmark named "Footer links".

Variation 2 — Alternate Footer Page

Variation Name: Alternate footer (for example `references/footer-campaign`).

Business Purpose: A different footer for an area, campaign or microsite.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-alternate-footer.png`
> **Purpose:** Area-specific footer
> **Description:** A campaign page with a footer bar only (no canvas) from references/footer-campaign.

Use Cases: Campaign pages; areas with different legal links.

Authoring Guidelines: Create the page in `references/`, add a Footer block, then set **Footer** in the Page Properties of the area's top page (or the `footer` column of the metadata sheet). Values: AEM path, published path or locale-relative path (`references/footer-campaign`).

Business Rules: Inherited like Navigation (see 6.1); parent values cached per browser session.

Rendering Rules: As Variation 1 with the alternate page; the Social Links list is always appended (it doesn't depend on the footer page).

Responsive Behaviour: As Variation 1.

SEO Impact: As Variation 1.

Accessibility Compliance: As Variation 1.

Variation 3 — Footer Bar (Copyright and Links)

Variation Name: Footer bar only (no Conversation Canvas Page).

Business Purpose: A compact legal footer.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-bar-only.png`
> **Purpose:** Bar without canvas
> **Description:** Footer with only the copyright text and the link list in one row (desktop) and stacked (mobile).

Use Cases: Campaign footers, simple microsites.

Authoring Guidelines: In the Footer block leave **Conversation Canvas Page** empty; fill **Copyright Text** and add **Link List** items.

Business Rules: `{year}` in the copyright is replaced by the current year (browser time). The bar is rendered even if both copyright and links are empty (an empty `div.footer-bar`).

Rendering Rules: `div.footer-bar` with `div.footer-copyright` (the rich text) and `nav.footer-links > ul > li > a`; links are plain links (button styling removed), underlined on hover and focus.

Responsive Behaviour: Desktop and tablet: copyright and links in one row if they fit, else wrapping. Mobile: links below the copyright.

SEO Impact: Policy links on every page.

Accessibility Compliance: `nav` with `aria-label` from placeholder `footerLinks`; links underlined on hover / focus.

Variation 4 — Footer with Conversation Canvas

Variation Name: Footer with Conversation Canvas.

Business Purpose: The TCS design footer: a large conversational prompt with search and quick links above the bar.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-with-conversation-canvas.png`
> **Purpose:** Canvas in the footer
> **Description:** Desktop footer with the canvas title, the floating menu button and "Ask us a question" search field, secondary buttons with dark 2px borders, then the bar.

Use Cases: Global footers of the corporate locales.

Authoring Guidelines: Create a page in `references/` (for example `references/footer-conversation-canvas`) containing a Conversation Canvas block; in the Footer block select it in **Conversation Canvas Page**.

Business Rules: The value may be a link or plain text path; it is resolved like Navigation values (AEM, published or locale-relative path). If the page can't be loaded or is empty, no canvas is shown and the bar remains.

Rendering Rules: `div.footer-canvas` holds the canvas page's sections; the canvas loses its own surface, radius, glow and padding (`background: none`) and uses footer sizes; the footer gets a minimum height and the bar is pushed to the bottom (`justify-content: space-between`). Canvas secondary buttons: 2px `#494c4e` border, white background, medium weight, 50px tall.

Responsive Behaviour: Mobile: title 3.25rem, buttons stacked, menu button hidden. Tablet: title 6rem, buttons in a row, menu button shown. Desktop: title max-width 61.5rem.

SEO Impact: The canvas title is part of every page's DOM (client-rendered); avoid heading levels that compete with the page H1 (see Conversation Canvas chapter).

Accessibility Compliance: Inherits the Conversation Canvas accessibility (search field label, buttons); the search dispatches `conversation-canvas:search` (no search service is connected).

Variation 5 — Social Links with Platform Icons

Variation Name: Social links with /icons icons.

Business Purpose: Recognizable channel icons from the Social Links sheet.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-social-links-icons.png`
> **Purpose:** Social icon row
> **Description:** A row of LinkedIn, X, Facebook, Instagram and YouTube icons (1.5rem) at the end of the footer.

Use Cases: Standard corporate channels.

Authoring Guidelines: In `/{locale}/data/social-links` add rows with `platform` (LinkedIn), `url`, optional `icon` (an icon name: `linkedin`, `x`, `facebook`, `instagram`, `youtube`) and `sort-order` (1, 2, 3…). Publish the sheet.

Business Rules: Rows without a `url` are skipped. Order: by `sort-order` ascending; rows without a number come last in sheet order. Without an `icon`, the platform name is used as the icon name when it is one of the five platforms (`twitter` maps to `x`).

Rendering Rules: `ul.footer-social > li > a > span.icon.icon-{name}` plus a visually hidden label "Follow us on {platform}"; links to another domain open in a new tab (`target="_blank" rel="noopener noreferrer"`).

Responsive Behaviour: Wrapping row with 1rem gap at all sizes.

SEO Impact: Outbound links to social profiles.

Accessibility Compliance: Each icon link has a text name (hidden label). The new tab isn't announced (no "(opens in a new tab)" text, a recommended improvement for WCAG 3.2.5, which is AAA).

Variation 6 — Social Links with Image Icons or Text Labels

Variation Name: Social links with an image icon, or as text.

Business Purpose: Channels without an icon in `/icons` (for example a regional network).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-social-links-text-image.png`
> **Purpose:** Image and text fallbacks
> **Description:** Social row with one custom image icon and one text-only link "Follow us on Xing".

Use Cases: Additional or regional channels.

Authoring Guidelines: Put an image URL in `icon` (any value containing `.` or `/`, for example a published `/media_…png` or a DAM URL) for an image icon; leave `icon` empty for a platform without an icon to show text.

Business Rules: An icon value with `.` or `/` is an image (site media is optimized, other URLs used as is with `loading="lazy"`); an icon name that has no file in `/icons` produces a broken icon (the `/icons/{name}.svg` request fails). Without an icon and an unknown platform, the label "Follow us on {platform}" is shown as text.

Rendering Rules: Image icon: `<picture>` or `<img alt="">` (decorative) plus hidden label; text: visible `<span>` label.

Responsive Behaviour: As Variation 5; images are 1.5rem × 1.5rem.

SEO Impact: As Variation 5.

Accessibility Compliance: Decorative images (`alt=""`) with a hidden text label; text links readable as is.

Variation 7 — Footer Link Opening in a New Window

Variation Name: Link List item with Link Type "Open in new window".

Business Purpose: External policy or group pages opened without leaving the site.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/footer-link-new-window.png`
> **Purpose:** New-window link
> **Description:** Footer bar with an "Investor relations" link to an external domain, the link's target highlighted in developer tools.

Use Cases: Links to external sites (group company pages, regulatory sites).

Authoring Guidelines: In a Link List item set **Link Type** to **Open in new window**.

Business Rules: `new-window` adds `target="_blank"` and `rel="noopener noreferrer"`; Default leaves the link in the same tab.

Rendering Rules: Same markup as other footer links.

Responsive Behaviour: As Variation 3.

SEO Impact: None beyond the link itself.

Accessibility Compliance: The new tab isn't announced to screen readers in the footer (no hidden hint); Recommended: add "(opens in a new tab)" to the link text or avoid new windows.

6.2.4 Authoring Experience

> [Screenshot Placeholder]
> **File name:** `screenshots/footer-authoring-view.png`
> **Purpose:** Universal Editor canvas of the footer page
> **Description:** /content/tcs-corp/us/en/references/footer open in Universal Editor with the Footer block selected and markers A to E.

Annotations:

- A = The Footer block on the footer page (select it to open its properties).
- B = The Conversation Canvas area (rendered from the Conversation Canvas Page).
- C = Copyright Text.
- D = A Link List item (Link Title, Link Path, Link Type).

- E = The Social Links row (not editable here: comes from the `data/social-links` sheet, shown only in the page footer of content pages).

> **Note:** When the footer page itself is opened in Universal Editor, its own page footer is also loaded from the footer page; edit the block in the main content area, and check the final result on a content page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-properties-panel.png`
> **Purpose:** Footer block properties
> **Description:** The properties panel of the Footer block with Conversation Canvas Page, Copyright Text and the Link List with two items expanded.

Properties panel description. The Footer model has no tabs (and no Additional tab). Fields, in order:

- **Conversation Canvas Page** (content picker): a page with the Conversation Canvas block, shown above the copyright and links.
- **Copyright Text** (rich text): for example "©TATA Consultancy Services {year}"; `{year}` becomes the current year.
- **Link List** (multifield of items), each with **Link Title** (text), **Link Path** (content picker) and **Link Type** (Default / Open in new window).

6.2.5 Authoring Guide

Step 1 — Open the footer page in Universal Editor

Action: In AEM Sites, open `/content/tcs-corp/{country}/{language}/references/footer` with **Edit**.

Expected Result: The footer page opens with its Footer block.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-01-open-footer-page.png`
> **Purpose:** Opening the footer page
> **Description:** Sites console, references folder, footer page selected, Edit highlighted.

Step 2 — Add the Footer block (new footer pages only)

Action: Select the section, choose **Add** (+) and pick **Footer**.

Expected Result: An empty Footer block is added.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-02-add-block.png`
> **Purpose:** Adding the block
> **Description:** The component picker with Footer highlighted.

Step 3 — Select the Conversation Canvas Page (optional)

Action: Select the Footer block; in **Conversation Canvas Page** choose the canvas page (for example `references/footer-conversation-canvas`), or leave it empty.

Expected Result: The canvas appears above the bar after the block re-renders.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-03-canvas-page.png`
> **Purpose:** Canvas page picker
> **Description:** The content picker dialog with the canvas page selected.

Step 4 — Enter the Copyright Text

Action: In **Copyright Text** type, for example, "©TATA Consultancy Services {year}".

Expected Result: The bar shows the copyright with the current year.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-04-copyright.png`
> **Purpose:** Copyright field
> **Description:** The rich text field with {year} and the canvas preview showing 2026.

Step 5 — Add Link List items

Action: In **Link List** choose **Add**; fill **Link Title** ("Privacy & Terms"), pick **Link Path**, set **Link Type**. Repeat; reorder by dragging.

Expected Result: The links appear in the bar in the item order.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-05-link-list.png`
> **Purpose:** Link items
> **Description:** Two expanded Link List items with title, path and type.

Step 6 — Maintain the Social Links sheet

Action: Open `/content/tcs-corp/{country}/{language}/data/social-links`; edit rows (platform, url, icon, sort-order); publish the sheet.

Expected Result: `/{locale}/data/social-links.json` lists the rows.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-06-social-links-sheet.png`
> **Purpose:** Social Links data
> **Description:** The social-links sheet with five rows and the four column headers.

Step 7 — (Optional) Assign an alternate footer

Action: On the area's top page, open **Page Properties** and pick the footer page in **Footer**.

Expected Result: The field shows the footer page path.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-07-footer-property.png`
> **Purpose:** Footer page property
> **Description:** Page Properties with Footer set to references/footer-campaign.

Step 8 — Preview

Action: Publish the footer page (and canvas page, sheet, changed pages) to Preview; open a content page on `.aem.page`.

Expected Result: The content page shows the new footer.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-08-preview.png`
> **Purpose:** Preview check
> **Description:** A preview content page scrolled to the footer.

Step 9 — Validate

Action: Run the Validation Process checklist at mobile, tablet and desktop widths.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-09-validate.png`
> **Purpose:** Validation
> **Description:** The footer at 390px, 1024px and 1440px.

Step 10 — Publish

Action: Publish the same pages and sheets to Live.

Expected Result: The live footer is updated after cache refresh.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/footer-step-10-publish.png`
> **Purpose:** Publishing
> **Description:** Publish dialog listing the footer page, the canvas page and the social-links sheet.

Preview Process

1. Preview the footer page, the conversation canvas page and the `data/social-links` sheet.
2. Check the footer on several content pages of the locale on `.aem.page`.
3. For inherited Footer values, preview the area's top page and test in a new browser session.

Publishing Process

1. Publish the canvas page and the social-links sheet first, then the footer page, then any page with a changed Footer property.
2. Never unpublish `references/footer` of a live locale: pages without a footer show only the social links.

Validation Process

- ☐ The copyright shows the current year (no literal `{year}`).
- ☐ All Link List links work and open in the right window.
- ☐ Links aren't styled as buttons.
- ☐ The canvas (if configured) shows above the bar; the bar sits at the bottom of the footer.
- ☐ Mobile: canvas buttons stacked, menu button hidden, links below the copyright.

- ☐ Tablet / desktop: canvas buttons in a row, menu button visible.
- ☐ Social icons appear in Sort Order; each has an accessible name (“Follow us on LinkedIn”); no broken icons.
- ☐ Screen reader: “Footer links” navigation landmark announced.
- ☐ Localized placeholders (`footerLinks`, `followUsOn`) in non-English locales.
- ☐ Areas with an alternate footer show it.

Rollback Process

1. Restore the previous version of the footer page (or the canvas page / sheet) in AEM and publish it.
2. Clear a faulty Footer page property (or metadata sheet `footer` value) and publish; pages fall back to the parent’s or the locale’s `references/footer`.
3. Retest in a new browser session.

6.2.6 Authorable Field Specification

Footer block (model `footer`, `blocks/footer/_footer.json`; no tabs)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Conversation Canvas Page (`conversationCanvas`)	aem-content	No	No	Empty	Canvas above the bar	Pick a `references/` page with a Conversation Canvas block	None in the model; ignored if the page doesn't load	`/content/tcs-corp/us/en/references/footer-conversation-canvas`
Copyright Text (`copyright`)	richtext	No (Recommended)	No	Empty	Legal copyright	Use `{year}` for the current year	None	`©TATA Consultancy Services {year}`
Link List (`links`)	container (composite multifield)	No	Yes	No items	Footer links	3–6 policy links	None	Privacy & Terms; Cookie Policy

Link List item (fields of the `links` container)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Link Title (`linkText`)	text	Yes (Recommended)	No	Empty	Visible link text	Short, descriptive	None	`Privacy & Terms`
Link Path (`link`)	aem-content	Yes (Enforced by code: items without a link are skipped)	No	Empty	Destination	AEM page or external URL	Items without a link are dropped	`/content/tcs-corp/us/en/privacy`
Link Type (`linkTarget`)	select	No	No	Default (`""`)	Same or new window	Use new window only for external sites	Options: Default (`""`); Open in new window (`new-window`)	`new-window`

Social Links data sheet (`/{locale}/data/social-links`, not a dialog)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
platform	sheet text	Recommended	No	Empty	Name used in the label and as the default icon	Use the platform name	None	`LinkedIn`
url	sheet text	Yes (Enforced: rows without url skipped)	No	Empty	Profile URL	Full https URL	Rows without url skipped	`https://www.linkedin.com/company/tata-consultancy-services`
icon	sheet text	No	No	Empty (platform name)	Icon name or image URL	`linkedin`, `x`, `facebook`, `instagram`, `youtube`, or an image URL	Value with `.` or `/` = image	`linkedin`
sort-order	sheet number	No	No	Empty (last)	Order	1, 2, 3…	Non-numbers treated as empty	`1`

There is no Additional tab (Anchor ID, Hide block, padding and margin fields don't exist on the Footer model).

Field details

- **Conversation Canvas Page.** Purpose: shows a conversation canvas above the bar. Allowed Values: a page path (link or text); AEM, published or locale-relative. Character Limits: none. Business Rules: loaded as a fragment; ignored when it can't be loaded or has no content. Dependency Rules: the canvas page must be published and contain a Conversation Canvas block. Visibility Rules: always visible (no `condition`). Sample Content: `references/footer-conversation-canvas`.
- **Copyright Text.** Purpose: legal line. Allowed Values: rich text (bold, italic, links). Character Limits: Recommended up to about 80 characters (the copyright column is at least 19.375rem wide). Business Rules: every `{year}` is replaced by the current year; empty text renders no copyright element. Dependency Rules: none. Visibility Rules: always. Sample Content: "©TATA Consultancy Services {year}".
- **Link List.** Purpose: legal / policy links. Allowed Values: items with a link. Character Limits: Recommended up to 6 items. Business Rules: items without a link are skipped; links are plain links. Dependency Rules: none. Visibility Rules: always. Sample Content: Privacy & Terms; Cookies; Accessibility.
- **Link Title.** Purpose: link text. Allowed Values: plain text. Character Limits: Recommended up to 30 characters. Business Rules: collapsed with Link Path into one anchor by AEM. Dependency Rules: requires Link Path. Visibility Rules: always. Sample Content: "Privacy & Terms".
- **Link Path.** Purpose: destination. Allowed Values: AEM page or URL. Character Limits: none. Business Rules: mandatory in practice. Dependency Rules: none. Visibility Rules: always. Sample Content: `/content/tcs-corp/us/en/privacy`.
- **Link Type.** Purpose: window. Allowed Values: Default, Open in new window. Character Limits: n/a. Business Rules: `new-window` opens a new tab with `noopener noreferrer`. Dependency Rules: requires Link Path. Visibility Rules: always. Sample Content: Default.
- **Social Links columns.** See the table; the sheet is loaded once per page; missing sheet = no social links.

6.2.7 Field Annotation Mapping

Marker	Field	Purpose
A	Footer block	Holds the footer bar fields
B	Conversation Canvas Page	Canvas above the bar

Marker	Field	Purpose
C	Copyright Text	Legal copyright with `{year}`
D	Link List item (Link Title, Link Path, Link Type)	Footer links
E	Social Links sheet rows (platform, url, icon, sort-order)	Social icon row in the page footer

6.2.8 Component Specification

Attribute	Value
Component Name	Footer (`blocks/footer`)
Version	1.0
Owner	[To be confirmed: product owner — owner: Digital Marketing]; technical owner: development team
Dependencies	Fragment; inherited-metadata, site, data, links, placeholders utils; social-links, button, link, content decorators; Conversation Canvas block; footer pages; Social Links sheet
Component Type	Block with two roles: page-level footer (automatic, in `<footer>`) and authored block on footer pages (with a Link List composite multifield)
Supported Variants	Default footer; alternate footer; bar only; with Conversation Canvas; social icons; social image / text; new-window links
Supported Templates	All
Supported Metadata	`footer` (page property Footer, metadata sheet column), inherited
Supported Tags	None
Supported Personalization	None; alternate footers per area only; future Target personalization through Adobe Tags (not implemented)
Caching Considerations	Footer and canvas pages and the social-links JSON are cached by the Edge Delivery CDN and the browser; inherited values cached per session; data sheets and placeholders loaded once per page
Performance Considerations	Lazy phase, after main sections; footer page, social links and placeholders load in parallel; the canvas page is a second fragment request; social image icons lazy-loaded; footer hidden until loaded (no flash)
Accessibility Compliance	contentinfo landmark (`<footer>`); `nav` "Footer links"; social icon links with hidden text names; decorative icon images `alt=""`; hover and focus underline; new-tab links not announced (recommended improvement)
SEO Compliance	Plain `<a href>` links; client-rendered; footer pages excluded from index and sitemap

6.2.9 Input Model

- **Content Inputs:** the footer page's sections; the Footer block rows (canvas page, copyright, links); the canvas page.
- **Metadata Inputs:** `<meta name="footer">` of the page or its parents.
- **Tags Inputs:** none.
- **Page Context Inputs:** current path (locale, parent pages; AEM path when editing); current year (browser clock).
- **Query Inputs:** `/{locale}/data/social-links.json`; `/{locale}/placeholders.json`.
- **Manual Inputs:** Footer block fields; Social Links sheet rows.

Authored HTML received (footer page `.plain.html`, based on the `/en` sample in `content/en/references/footer.plain.html`, which has the year typed as 2027 instead of `{year}`):

```
<div>
  <div class="footer">
    <div><div><a href="/en/references/footer-conversation-canvas">…</a></div></div>
    <div><div>© TATA Consultancy Services {year}</div></div>
    <div><div><hr><p><a href="/en/privacy">Privacy &amp;amp; Terms</a></p><hr></div></div>
  </div>
  <div class="section-metadata">
    <div><div>sectionWidth</div><div>full-width</div></div>
  </div>
</div>
```

Rows are positional: row 1 Conversation Canvas Page, row 2 Copyright Text, row 3 Link List (items as `<ul><li>` or runs separated by `<hr>`; a `new-window` value in an item marks it for a new window).

6.2.10 Output Model

Rendered Output (DOM structure).

```
footer
└─ div.footer.block                (page footer)
   └─ div
      ├─ div.section …             (footer page sections)
      │  └─ div.footer-wrapper > div.footer.block   (authored Footer block)
      │     ├─ div.footer-canvas > div.section > … div.conversation-canvas.block
      │     └─ div.footer-bar
      │        ├─ div.footer-copyright
      │        └─ nav.footer-links[aria-label] > ul > li > a
      └─ ul.footer-social > li > a > span.icon.icon-{name} + span.visually-hidden
```

Generated HTML (bar and social links, simplified).

```
<div class="footer-bar">
  <div class="footer-copyright">© TATA Consultancy Services 2026</div>
  <nav class="footer-links" aria-label="Footer links">
    <ul><li><a href="/en/privacy">Privacy &amp;amp; Terms</a></li></ul>
  </nav>
</div>
<ul class="footer-social">
  <li><a href="https://www.linkedin.com/company/…" target="_blank"
    rel="noopener noreferrer"><span class="icon icon-linkedin"><img …></span>
    <span class="visually-hidden">Follow us on LinkedIn</span></a></li>
</ul>
```

Generated JSON: none (reads the social-links JSON).

Generated Metadata: none.

Analytics Tracking. Both footer blocks are `.block` elements and tracked in the delayed phase (`trackBlocks(document)`), for example `footer-1` (page footer, `parentId` the page) and `footer-2` (authored block, `parentId footer-1`); a Conversation Canvas inside is its own component with its own events. Clicks push `cmp:click` to the nearest block:

```
{ "event": "cmp:click", "eventInfo": {
  "path": "component.footer-1", "interaction": "link",
  "dc:title": "Follow us on LinkedIn",
```

```
  "xdm:linkURL": "https://www.linkedin.com/company/tata-consultancy-services",
  "xdm:linkType": "exit", "breakpoint": "mobile" } }
```

Footer bar links report `component.footer-2`, `xdm:linkType` `other` (same site) or `exit` (other domain).

Personalization Output: none.

Search Index Output: none (footer and canvas pages excluded from `/query-index.json`).

6.2.11 Business Rules

1. A Footer block whose parent is `<footer>` is the page footer; every other Footer block is the authored footer bar.
2. The footer page is the page's Footer value (page property or metadata `footer` column), else the nearest parent's up to the locale Home page, else `{locale}/references/footer`; values may be AEM, published or locale-relative paths.
3. The footer page, Social Links and placeholders load in parallel; the footer page's sections are moved into the page footer.
4. If the footer page can't be loaded, the page footer contains only the Social Links (or nothing).
5. Social Links are always appended after the footer page content, regardless of which footer page is used.
6. Social rows without `url` are skipped; rows are ordered by numeric `sort-order`, rows without one last in sheet order; a missing sheet gives no list.
7. Social icon: a value with `.` or `/` is an image; otherwise the icon name, else the platform alias (`twitter` → `x`) or name; without an icon value and an unknown platform, no icon and the text label is visible.
8. Social links to another origin open in a new tab; links on the same origin stay in the tab.
9. The social link's accessible name is "Follow us on {platform}" (placeholder `followUsOn`); when the platform is empty the URL is used in its place ("Follow us on https://..."); the name is visually hidden when an icon is shown.
10. Authored block rows are read by position: canvas page, copyright, link list.
11. The canvas is loaded only when the first row has a link or text; it is skipped when the page doesn't load or has no sections.
12. `{year}` (all occurrences) in the copyright is replaced with the current four-digit year from the visitor's clock.
13. An empty copyright renders no copyright element; a link list without links renders no `nav`.
14. The bar element is always rendered (it may be empty).
15. Link List items without a link are skipped; links lose any button styling; `new-window` items open in a new tab.
16. Editor instrumentation is kept on the canvas and copyright so they stay selectable in Universal Editor.
17. With a canvas, the footer has a minimum height (50rem / 59.75rem / 50.125rem) and the bar sits at its bottom.
18. The canvas's own background, radius, glow and padding are removed inside the footer; its menu button is hidden below 600px.
19. The footer is invisible until loaded (`data-block-status="loaded"`).
20. There are no Additional tab options; the Footer block offered in sections isn't styled outside `<footer>`.

6.2.12 Author Best Practices

- **Recommended Content Length:** copyright up to about 80 characters; 3–6 links of up to 30 characters; 3–6 social channels; canvas title as in the Conversation Canvas chapter (it shows at 6rem on tablet and desktop, so keep it to a short sentence).
- **Image Dimensions:** social image icons display at 1.5rem × 1.5rem (24px); provide square SVG or PNG at 48px or more. Icons from `/icons` need no upload.

- **CTA Writing Guidelines:** footer links are navigational, not CTAs: use nouns ("Privacy & Terms", "Cookie Policy"). Canvas buttons follow the Conversation Canvas guidance.
- **Accessibility Guidelines:** give each social row a `platform` value (it builds the accessible name); avoid new-window links or say so in the link text; keep link text unique.
- **SEO Guidelines:** link the legal and key corporate pages; avoid duplicate links to the header's pages with different texts.
- **Metadata Guidelines:** set Footer only on area top pages; don't set it on every page.
- **Taxonomy Guidelines:** not applicable.
- **Localization Guidelines:** each locale maintains `references/footer`, its canvas page and `data/social-links` (regional channel URLs); translate `footerLinks` and `followUsOn` in the placeholders sheet; keep `{year}` in translated copyright text.

6.2.13 Troubleshooting Guide

Issue 1 — Footer shows only social icons

Symptoms: The footer has social icons but no copyright, links or canvas.

Root Cause: The footer page (default or inherited) is missing or unpublished, or the Footer value is wrong.

Diagnosis Steps: Network tab: the footer page's `.plain.html` request returns 404; check `<meta name="footer">` on the page and its parents.

Resolution Steps: Publish the footer page or correct the Footer value.

Validation Steps: The full footer shows in a new session.

Preventive Measures: Publish footer pages before referencing them.

Issue 2 — `{year}` is not replaced, or the year is wrong

Symptoms: The copyright shows "{year}" or a fixed year.

Root Cause: The placeholder is written differently (for example `{Year}` or `{ year }`), or the year was typed as a number (the `/en` sample has a typed year).

Diagnosis Steps: Open the footer page and check the exact text.

Resolution Steps: Use exactly `{year}`.

Validation Steps: The current year shows.

Preventive Measures: Use the sample "©TATA Consultancy Services {year}".

Issue 3 — Social links missing or in the wrong order

Symptoms: No social icons, or a channel is missing or misplaced.

Root Cause: The `data/social-links` sheet isn't published, a row has no `url`, `sort-order` isn't a number, or the column headers differ (`sort-order`, lower-case).

Diagnosis Steps: Open `/{locale}/data/social-links.json` and check rows and headers.

Resolution Steps: Fix headers and values; publish the sheet.

Validation Steps: Icons appear in Sort Order.

Preventive Measures: Keep the header row exactly `platform | url | icon | sort-order`.

Issue 4 — Broken social icon

Symptoms: An empty square or nothing where an icon should be.

Root Cause: The `icon` value is a name without a file in `/icons` (only linkedin, x, facebook, instagram, youtube exist), or an image URL that doesn't load.

Diagnosis Steps: Network tab: a failing `/icons/{name}.svg` or image request.

Resolution Steps: Use an existing icon name, an image URL, or leave `icon` empty for a text link; request new icons from development.

Validation Steps: All icons render.

Preventive Measures: Document allowed icon names in the sheet's guidance.

Issue 5 — Conversation canvas not shown in the footer

Symptoms: Only the bar shows, although a canvas page is selected.

Root Cause: The canvas page isn't published, is empty, or the path doesn't resolve in this locale.

Diagnosis Steps: Network tab for the canvas `.plain.html`; open it directly.

Resolution Steps: Publish the canvas page; check its Conversation Canvas block; reselect the page.

Validation Steps: The canvas appears above the bar.

Preventive Measures: Publish canvas pages together with footer pages.

Issue 6 — Footer links look like buttons, or the footer is unstyled

Symptoms: Footer bar shown in a content page without styles, or links appear as buttons.

Root Cause: The Footer block was added to a content page (styles are scoped to `<footer>`), or a link was authored outside the Link List (as content text).

Diagnosis Steps: Check where the block is placed; inspect the link classes.

Resolution Steps: Remove the Footer block from content pages; put footer links into the Link List.

Validation Steps: Footer links are plain text links.

Preventive Measures: Use the Footer block only in `references/` footer pages.

Issue 7 — Wrong footer in an area

Symptoms: An area shows the global footer instead of its own.

Root Cause: The area's top page with the Footer property isn't published, or the session cache holds old values.

Diagnosis Steps: Check the top page's published `<meta name="footer">`; `sessionStorage tcs-inherited-metadata`.

Resolution Steps: Publish the top page; new session.

Validation Steps: The area shows its footer.

Preventive Measures: Publish property changes.

6.2.14 Support Runbook

Monitoring Requirements. Recommended synthetic checks per locale: `/{locale}/references/footer.plain.html` and `/{locale}/data/social-links.json` return 200; the Home page footer contains `.footer-bar`. Once Adobe Tags is installed, monitor `cmp:click` on footer components for exit links.

Logs.

- Browser console: the footer code emits no messages of its own; failures appear as network errors (footer page, canvas page, social-links JSON, icon SVGs); an exception during decoration is logged by `aem.js` as `failed to load module for footer`.

- Admin API status: `https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{locale}/references/footer` (and the canvas page, `data/social-links.json`).
- Author logs: AEM publication logs for the footer page, canvas page and sheet.

Common Errors.

Error	Likely Cause	Fix
Only social icons	Footer page missing / unpublished	Publish / correct Footer value
No social icons	Sheet missing, no url, wrong headers	Fix and publish sheet
Broken icon	Unknown icon name	Use a supported name or image URL
`{year}` visible	Wrong placeholder spelling	Use `{year}`
No canvas	Canvas page unpublished / empty	Publish / fix canvas page
Unstyled footer bar in content	Block placed on a content page	Remove it

Escalation Matrix.

Level	Team	Checks
L1	Author support	Footer page content and publication, canvas page, Link List, Social Links sheet, Footer property
L2	AMS / site admin	Admin API status, `paths.json` mapping of `data/social-links`, CDN cache purge, metadata sheet `footer` column / `metadata.source`
L3	Development team	`blocks/footer/*`, `decorators/social-links-decorator.js`, `utils/data.js`, icons, console errors
L4	Adobe Support	Edge Delivery delivery or AEM publishing defects

Recovery Procedures. Restore and republish the previous footer / canvas page or sheet version; clear a faulty Footer property; revert faulty code on `main`.

Health Checks. Per locale: footer page and social-links JSON available; the footer shows copyright with the current year, links and social icons; canvas renders where configured; layout correct at 390px, 1024px and 1440px.

Operational Procedures. New locale: create `references/footer` with a Footer block, the canvas page (if used) and `data/social-links`; translate placeholders; publish all. Yearly: no action needed for the year when `{year}` is used.

Known Limitations.

- New-tab links (footer links, external social links) aren't announced to screen readers.
- Unknown icon names produce broken icons (no fallback to text).
- The bar is rendered even when empty.
- The Footer block is offered in every section's component list but only styled inside the page footer.
- No newsletter sign-up, language selector or back-to-top control (Not implemented as of this handover).
- Inherited values cached per browser session.

6.3 Fragment Block

The Fragment block includes the content of another page (a **fragment page**) in the current page. The module `blocks/fragment/fragment.js` has two parts:

- `loadFragment(path)`, an exported function used by other code to load and fully decorate a page's content: the Header (nav page), the Footer (footer page and conversation canvas page), the Tabs block (a tab's Fragment) and the Modal block (`openModal`, also used for links to `/modals/` pages).
- The authored **Fragment** block (`decorate`), which reads the Reference field and replaces its own content with the fragment page's sections.

Source files: `blocks/fragment/fragment.js`, `blocks/fragment/fragment.css` (empty: the fragment's content keeps the styles of its own blocks and sections), `blocks/fragment/_fragment.json`. Based on the AEM Block Collection Fragment block.

6.3.1 Component Purpose

Business Objective. Author shared content once and reuse it on many pages, so a change is made in one place (for example a legal disclaimer, a contact banner or a recurring call-to-action section).

Business Use Cases.

- A shared "Talk to us" or newsletter section on many pages.
- Regulatory disclaimers repeated on all investor or press pages.
- Shared promotional banners for a campaign period.
- Technical reuse: the header, footer, tab panels and modal dialogs are all loaded as fragments.

When To Use. When the same content must be identical on several pages of the same locale and maintained centrally.

When Not To Use.

- For content that differs per page (author it on the page).
- For the header and footer (they are loaded automatically; use the Navigation / Footer page properties).
- For content that must be indexed as part of the page or appear in the page's server HTML for SEO (fragments are loaded in the browser).
- For the first, LCP-critical section of a page (fragments add a network round trip).
- Never let a fragment page include itself, directly or through another fragment (there is no protection against loops).

Dependencies.

Type	Dependency	Use
Code	`scripts/scripts.js` `decorateMain`	Decorates the fragment's icons, sections (with section options), blocks, block options and buttons
Code	`scripts/aem.js` `loadSections`	Loads the fragment's blocks (JS and CSS)
Content	Fragment pages, by convention in the locale's `references/` folder (for example `references/cta-banner`)	Shared content
Styles	The styles of the blocks and sections inside the fragment; `styles/styles.css` Additional tab classes (`main .section .block.pl-*` etc., `hideblock`)	Layout
Placeholders	None	—

Related Components. Header, Footer, Tabs (tab Fragment field), Modal (fragment pages in dialogs), Conversation Canvas (shown in the footer through a fragment).

Applicable Templates. Any page; all templates.

Supported Content Types. Any content that can be authored on a page: default content (text, title, image, button) and any block allowed in a section.

6.3.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A content page at 1440px with a shared "Talk to our experts" section loaded from references/cta-banner between two authored sections, looking identical to authored content.

Desktop (>= 1200px). The Fragment block has no visual style of its own (`fragment.css` is empty). After loading, it contains the fragment page's sections, each rendered with its own section options (style, width, background) and blocks, inside the section of the current page that holds the Fragment block. Spacing comes from the inner sections and from the Additional tab padding / margin of the Fragment block.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same shared section at 768px, its blocks in their tablet layout.

Tablet (600–1199px). The inner blocks follow their own tablet rules.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The same shared section at 390px, stacked.

Mobile (< 600px). The inner blocks follow their own mobile rules. Before the fragment is loaded (or if it fails), the block shows its authored content: the Reference link.

6.3.3 Component Variations

The Fragment block has no style options. Its variations are the ways fragments are used.

Variation 1 — Authored Fragment Block

Variation Name: Fragment block on a page.

Business Purpose: Reuse a shared section on content pages.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-authored-block.png`
> **Purpose:** Fragment block in a page
> **Description:** Universal Editor with a Fragment block selected, showing the content of references/cta-banner inside it.

Use Cases: Shared banners, disclaimers, contact sections.

Authoring Guidelines: Add **Fragment** to a section; in **Reference** pick the fragment page; optionally set the Additional tab.

Business Rules: The reference must be a site path starting with `/` (the content picker gives one). Full URLs (`https://…`) and protocol-relative paths (`//…`) are ignored.

Rendering Rules: The block's children are replaced by the fragment page's sections (`div.section`), already decorated and loaded.

Responsive Behaviour: That of the inner content.

SEO Impact: The included content is added in the browser; it isn't in the page's delivered HTML and isn't part of the query index entry of the page (the index reads the page head).

Accessibility Compliance: Depends on the fragment's content; check its heading levels fit the host page.

Variation 2 — Header Navigation Fragment

Variation Name: Nav page loaded by the Header.

Business Purpose: Header content managed as a page.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-header-navigation.png`
> **Purpose:** Nav page as fragment
> **Description:** Developer tools Network tab with the request /en/references/nav.plain.html and the rendered header.

Use Cases: Every page.

Authoring Guidelines: Edit the nav page (see 6.1).

Business Rules: The path comes from the inherited Navigation value or `references/nav` (resolved for the locale and for author / published mode by `utils/site.js`).

Rendering Rules: The header moves the decorated sections into its `nav` (see 6.1.10).

Responsive Behaviour: See 6.1.

SEO Impact: See 6.1.

Accessibility Compliance: See 6.1.

Variation 3 — Footer and Conversation Canvas Fragments

Variation Name: Footer page and canvas page loaded by the Footer.

Business Purpose: Footer content managed as pages.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-footer.png`
> **Purpose:** Footer fragments
> **Description:** Network tab with /en/references/footer.plain.html and /en/references/footer-conversation-canvas.plain.html, and the rendered footer.

Use Cases: Every page.

Authoring Guidelines: See 6.2.

Business Rules: The footer page path comes from the inherited Footer value or `references/footer`; the canvas page from the Footer block's Conversation Canvas Page (a nested fragment: the footer page's Footer block loads the canvas page).

Rendering Rules: See 6.2.10.

Responsive Behaviour: See 6.2.

SEO Impact: See 6.2.

Accessibility Compliance: See 6.2.

Variation 4 — Tab Panel Fragment

Variation Name: Fragment of a Tabs item.

Business Purpose: Rich tab panels (columns, images, buttons) authored on their own page.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/fragment-tab-panel.png`
> **Purpose:** Fragment in a tab
> **Description:** A Tabs block whose second tab panel shows the content of a fragment page with Columns.

Use Cases: Tab content too complex for the tab's own fields.

Authoring Guidelines: In the Tabs block's tab item, set **Fragment** (content picker: "A page whose content … is shown in the tab").

Business Rules: A full URL is ignored like in Variation 1; if the fragment can't be loaded the panel stays without that content.

Rendering Rules: The fragment's sections are wrapped in `div.tabs-fragment` in the tab panel (see the Tabs chapter).

Responsive Behaviour: As the inner content within the tab panel.

SEO Impact: As Variation 1; hidden panels' content is in the DOM.

Accessibility Compliance: The panel content is inside the tab panel's `role="tabpanel"`.

Variation 5 — Modal Fragment

Variation Name: Fragment shown in a modal dialog.

Business Purpose: Show forms, videos or details in a dialog without leaving the page.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/fragment-modal.png`
> **Purpose:** Fragment in a modal
> **Description:** A dialog open over a page with the content of /en/modals/contact-form.

Use Cases: Contact forms, video players, terms.

Authoring Guidelines: Use the Modal block with a link to the fragment page, or link any text to a page whose path contains `/modals/` (opened as a modal automatically by `scripts.js` `autolinkModals`).

Business Rules: `openModal` converts a full URL (`http…`) to its path before loading, so absolute links work for modals (unlike the Fragment block).

Rendering Rules: The fragment's content is placed in the modal dialog (see the Modal chapter).

Responsive Behaviour: As the Modal block.

SEO Impact: Modal content isn't in the page HTML; `/modals/` pages are excluded from the query index.

Accessibility Compliance: As the Modal block (dialog semantics, focus handling).

6.3.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A content page in Universal Editor with a Fragment block selected (marker A) and the included content (marker B).

Annotations:

- A = The Fragment block (select it to set Reference and the Additional options).
- B = The included content of the fragment page (not editable here; edit it on the fragment page).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Fragment block's properties with the Content tab (Reference) and the Additional tab (Anchor ID, Hide block, padding and margin selects).

Properties panel description.

- **Content tab: Reference** (content picker): the fragment page.
- **Additional tab: Anchor ID** (text, must match `^anchor-[a-z0-9][a-z0-9-]*$`); **Hide block** (switch); **Padding left / right / top / bottom** and **Margin left / right / top / bottom** (selects: Default, 0, 0.25rem … 7.5rem from the spacing tokens).

6.3.5 Authoring Guide

Step 1 — Create or open the fragment page

Action: In AEM Sites, create a page in `/content/tcs-corp/{country}/{language}/references/` (for example `cta-banner`) and author its sections and blocks; publish it.

Expected Result: The fragment page exists and is published.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-step-01-fragment-page.png`
> **Purpose:** Fragment page
> **Description:** The references folder with the new cta-banner page.

Step 2 — Open the target page in Universal Editor

Action: Open the content page that should show the fragment with **Edit**.

Expected Result: The page opens in Universal Editor.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-step-02-open-page.png`
> **Purpose:** Target page
> **Description:** The content page open in Universal Editor.

Step 3 — Add the Fragment block

Action: Select the section, choose **Add** (+) and pick **Fragment**.

Expected Result: An empty Fragment block is added.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-step-03-add-block.png`
> **Purpose:** Adding the block
> **Description:** The component picker with Fragment highlighted.

Step 4 — Set the Reference

Action: In the Content tab, choose **Reference** and pick the fragment page.

Expected Result: After re-rendering, the block shows the fragment's content.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-step-04-reference.png`
> **Purpose:** Reference picker
> **Description:** The content picker with references/cta-banner selected.

Step 5 — Fill the Additional tab (optional)

Action: Set an **Anchor ID** (for example `anchor-contact`), spacing selects, or **Hide block**.

Expected Result: Spacing applies; hidden blocks show faded in the editor.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-step-05-additional.png`
> **Purpose:** Additional tab
> **Description:** The Additional tab with Anchor ID and Margin top set.

Step 6 — Preview

Action: Publish the page (and the fragment page) to Preview and open it on `.aem.page`.

Expected Result: The fragment content shows in place.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-step-06-preview.png`
> **Purpose:** Preview
> **Description:** The preview page with the included section.

Step 7 — Validate

Action: Run the Validation Process checklist.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-step-07-validate.png`
> **Purpose:** Validation
> **Description:** The page at desktop and mobile widths.

Step 8 — Publish

Action: Publish the fragment page and the page to Live.

Expected Result: The live page shows the fragment.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/fragment-step-08-publish.png`
> **Purpose:** Publishing
> **Description:** Publish dialog with both pages.

Preview Process

1. Preview the fragment page first; every page using it shows the change immediately on Preview (no page re-publication needed).
2. Check a sample of the pages that use the fragment.

Publishing Process

1. Publish the fragment page to Live; all pages using it update (after CDN cache refresh).
2. Publish new pages that use the fragment after the fragment page.

Validation Process

- ☐ The fragment content appears (not the bare Reference link).
- ☐ Its blocks are fully styled and interactive.
- ☐ Heading levels fit the host page (no second H1).
- ☐ Images load (relative media paths are rebased to the fragment page).
- ☐ Spacing from the Additional tab is as intended.
- ☐ The Anchor ID link (`#anchor-…`) jumps to the block.
- ☐ The fragment is in the same locale as the page.
- ☐ No fragment includes itself.

Rollback Process

1. Restore the previous version of the fragment page and publish it (all pages update).
2. Or remove / hide the Fragment block on the affected page and publish the page.

6.3.6 Authorable Field Specification

Fragment block (model `fragment`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Content tab — Reference (`reference`)	aem-content	Yes (Recommended; without it nothing is included)	No	Empty	The fragment page	Pick a page in the locale's `references/`	None in the model; code requires a path starting with `/`	`/content/tcs-corp/us/en/references/cta-banner`
Additional tab — Anchor ID (`classes_anchor`)	text	No	No	Empty	Block id for # links	Start with `anchor-`	Enforced: `^anchor-[a-z0-9][a-z0-9-]*$` ("Start with anchor-, then lower-case letters, digits and hyphens.")	`anchor-contact`
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide without deleting	Use for temporary removal	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default (`""`)	Inner spacing	Use tokens only	Options: Default; 0; 0.25rem; 0.5rem; 0.75rem; 1rem; 1.25rem; 1.5rem; 2rem; 2.5rem; 3rem; 4rem; 5rem; 6rem; 7.5rem (`pl-0` … `pl-3000`)	`pl-400`
Padding right (`classes_paddingright`)	select	No	No	Default	Inner spacing	As above	Same options (`pr-*`)	`pr-400`
Padding top (`classes_paddingtop`)	select	No	No	Default	Inner spacing	As above	Same options (`pt-*`)	`pt-800`
Padding bottom (`classes_paddingbottom`)	select	No	No	Default	Inner spacing	As above	Same options (`pb-*`)	`pb-800`
Margin left (`classes_marginleft`)	select	No	No	Default	Outer spacing	As above	Same options (`ml-*`)	`ml-0`
Margin right (`classes_marginright`)	select	No	No	Default	Outer spacing	As above	Same options (`mr-*`)	`mr-0`
Margin top (`classes_margintop`)	select	No	No	Default	Outer spacing	As above	Same options (`mt-*`)	`mt-1200`
Margin bottom (`classes_marginbottom`)	select	No	No	Default	Outer spacing	As above	Same options (`mb-*`)	`mb-1200`

Field details

- **Reference.** Purpose: the page to include. Allowed Values: a page path from the content picker (AEM path; published as the site path). Character Limits: none. Business Rules: a path not starting with `/`, or starting with `//`, is ignored; `.html` / `.plain.html` endings are removed; the page's `.plain.html` is fetched. Dependency Rules: the page must be published (Preview for `.aem.page`, Live for `.aem.live`). Visibility Rules: always visible. Sample Content: `references/cta-banner`.
- **Anchor ID.** Purpose: makes the block a jump target. Allowed Values: `anchor-` followed by lower-case letters, digits and hyphens. Character Limits: none enforced (Recommended up to 40 characters). Business Rules: becomes the block's `id`; a duplicate on the page gets `-2`, `-3`… (`decorators/block-options-decorator.js`). Dependency Rules: none. Visibility Rules: always. Sample Content: `anchor-contact`.

- **Hide block.** Purpose: hide the block on the site. Allowed Values: on / off. Business Rules: `hideblock` class: hidden on the site (`display: none` for the block and its wrapper), shown at 45% opacity with a dashed outline in Universal Editor. Dependency Rules: none. Visibility Rules: always. Sample Content: off.
- **Padding / Margin selects (8 fields).** Purpose: spacing around / inside the block. Allowed Values: Default or one of 14 tokens (0, 0.25rem (space-100), 0.5rem (space-200), 0.75rem (space-300), 1rem (space-400), 1.25rem (space-500), 1.5rem (space-600), 2rem (space-800), 2.5rem (space-1000), 3rem (space-1200), 4rem (space-1600), 5rem (space-2000), 6rem (space-2400), 7.5rem (space-3000)). Business Rules: applied by `styles/styles.css` (`main .section .block.pl-400` etc.); Default leaves the block's own spacing. Dependency Rules: none. Visibility Rules: always. Sample Content: Margin top 3rem.

6.3.7 Field Annotation Mapping

Marker	Field	Purpose
A	Fragment block (Reference; Additional tab fields)	Chooses the fragment page and block options
B	Included content (fragment page)	Shared content, edited on the fragment page

6.3.8 Component Specification

Attribute	Value
Component Name	Fragment (`blocks/fragment`)
Version	1.0
Owner	[To be confirmed: product owner — owner: Digital Marketing]; technical owner: development team
Dependencies	`scripts/scripts.js` `decorateMain`; `scripts/aem.js` `loadSections`; fragment pages; block options decorator and styles
Component Type	Block (container of included sections); also a shared loader function (`loadFragment`)
Supported Variants	Authored block; header nav; footer and canvas; tab panel; modal
Supported Templates	All
Supported Metadata	None (the fragment page's metadata isn't used)
Supported Tags	None
Supported Personalization	None; future: different fragment per audience through Adobe Target (not implemented)
Caching Considerations	`.plain.html` served by the Edge Delivery CDN and cached by the browser; no caching in code: each call fetches (the same fragment used twice on a page is requested twice, normally from the browser cache)
Performance Considerations	One extra request per fragment, made when the block's section loads; avoid in the first section (LCP); nested fragments add sequential requests
Accessibility Compliance	Inherits the fragment content's compliance; headings must fit the host page's hierarchy
SEO Compliance	Content added in the browser; not in the delivered page HTML; fragment pages in `references/`, `/fragments/` and `/modals/` are excluded from the query index

6.3.9 Input Model

- **Content Inputs:** the fragment page's `.plain.html` (its sections).
- **Metadata Inputs:** none.
- **Tags Inputs:** none.
- **Page Context Inputs:** `window.location` (to rebase `./media_` image URLs).

- **Query Inputs:** none.
- **Manual Inputs:** Reference; Additional tab.

Authored HTML received:

```
<div class="fragment mt-1200">
  <div>
    <div><a href="/en/references/cta-banner">
      /content/tcs-corp/us/en/references/cta-banner</a></div>
  </div>
</div>
```

6.3.10 Output Model

Rendered Output (DOM structure).

```
div.fragment.block[data-block-name=fragment]
├─ div.section …[data-section-status=loaded]   (fragment section 1)
│   ├─ div.default-content-wrapper
│   └─ div.{block}-wrapper > div.{block}.block
└─ div.section …                               (fragment section 2)
```

Generated HTML (simplified).

```
<div class="fragment block mt-1200" id="anchor-contact" data-block-name="fragment"
  data-block-status="loaded">
  <div class="section" data-section-status="loaded">
    <div class="default-content-wrapper"><h2>Talk to our experts</h2>…</div>
  </div>
</div>
```

Generated JSON: none. **Generated Metadata:** none.

Analytics Tracking. The Fragment block is tracked as a component (for example `fragment-1`, `@type tcs-corp/components/fragment`); blocks inside it are tracked with `parentId fragment-1`; clicks are reported for the nearest block (`cmp:click`, see 6.1.10 for the event shape). Clicks on links in the fragment's default content are reported for the fragment block.

```
{ "component": {
  "fragment-1": { "@type": "tcs-corp/components/fragment", "parentId": "page-1x2y3z" },
  "teaser-3": { "@type": "tcs-corp/components/teaser", "parentId": "fragment-1",
    "dc:title": "Talk to our experts" } } }
```

Personalization Output: none. **Search Index Output:** none for the host page (the index reads the page head); the fragment page itself is excluded when in `references/`, `/fragments/` or `/modals/`.

6.3.11 Business Rules

1. The path is the `href` of the first link in the block, else the block's text.
2. Only paths starting with `/` and not `//` are loaded; anything else (empty, full URL, relative) is ignored and the block keeps its authored content.
3. The first `.plain.html` or `.html` in the path is removed, then `{path}.plain.html` is fetched.
4. A non-OK response (404, 401…) returns nothing: the authored block stays as it is (showing the Reference link).

5. A network error throws; `aem.js` logs `failed to load module for fragment` and marks the block loaded with its authored content.
6. Image `src` and `srcset` values starting with `./media_` are rebased to the fragment page's URL so images load on the host page.
7. The fragment is decorated like a page (`decorateMain`): icons, sections with their options (style, width, rounded edge, background, anchor name), Gen Carousel and Tabs sections, blocks, block options (anchor ids) and content buttons; then all its sections and blocks are loaded.
8. The Fragment block's content is replaced by the fragment's sections.
9. The path isn't localized: the page referenced is used as is (pick a page of the same locale).
10. When editing in Universal Editor the reference is an AEM path and the AEM author `.plain.html` is loaded.
11. There is no loop protection: a fragment including itself never finishes loading.
12. No caching in code; each use fetches the page.
13. The Additional tab applies to the Fragment block (anchor id, hide, spacing).

6.3.12 Author Best Practices

- **Recommended Content Length:** one to three sections per fragment; a fragment should be a self-contained unit (for example one banner).
- **Image Dimensions:** as required by the blocks inside the fragment (see their chapters).
- **CTA Writing Guidelines:** CTAs in fragments appear on many pages: keep them generic and valid for all host pages.
- **Accessibility Guidelines:** start fragment content at H2 (pages have their H1 in the Page Header or Hero); don't rely on surrounding page text for context.
- **SEO Guidelines:** don't put unique page content in fragments; use fragments for boilerplate only.
- **Metadata Guidelines:** the fragment page's metadata isn't used; give it a clear title for authors.
- **Taxonomy Guidelines:** not applicable.
- **Localization Guidelines:** create one fragment page per locale (`/de-de/references/cta-banner`) and reference the locale's own page.

6.3.13 Troubleshooting Guide

Issue 1 — The block shows a bare link instead of the content

Symptoms: The page shows the fragment's path as a link.

Root Cause: The fragment page isn't published (404), or the Reference is a full URL.

Diagnosis Steps: Network tab: the `.plain.html` request status; check the Reference value.

Resolution Steps: Publish the fragment page; pick the page with the content picker (site path).

Validation Steps: The content renders.

Preventive Measures: Publish fragment pages before host pages.

Issue 2 — Images in the fragment are broken

Symptoms: Missing images in included content.

Root Cause: Image paths that aren't `./media_` relative paths and don't resolve from the host page, or unpublished assets.

Diagnosis Steps: Inspect the `img src`; check it loads directly.

Resolution Steps: Re-select the images in the fragment page and republish.

Validation Steps: Images load on the host page.

Preventive Measures: Use AEM Assets via the image picker.

Issue 3 — Page never finishes loading / browser hangs

Symptoms: The section with the fragment stays empty and the browser becomes slow.

Root Cause: The fragment includes itself (directly or through another fragment).

Diagnosis Steps: Network tab: repeated requests for the same `.plain.html`.

Resolution Steps: Remove the self-reference from the fragment page; publish.

Validation Steps: One request per fragment.

Preventive Measures: Never place Fragment blocks pointing to the same page in a fragment page.

Issue 4 — Wrong language content

Symptoms: English content on a German page.

Root Cause: The Reference points to another locale's fragment (copied page kept the old reference).

Diagnosis Steps: Check the Reference path's locale.

Resolution Steps: Re-point to the locale's fragment page.

Validation Steps: Content in the page language.

Preventive Measures: Review references after copying pages between locales.

Issue 5 — Changes to the fragment don't appear

Symptoms: The old content is shown.

Root Cause: The fragment page was saved but not published, or caches still hold the old version.

Diagnosis Steps: Admin API status of the fragment page; open its `.plain.html` directly.

Resolution Steps: Publish; wait for or request a CDN cache purge.

Validation Steps: The new content shows after reload.

Preventive Measures: Publish the fragment page, not only the host page.

Issue 6 — Two H1 headings on the page

Symptoms: Accessibility checker reports multiple H1.

Root Cause: The fragment contains an H1 (or a Page Header).

Diagnosis Steps: Inspect headings in the included content.

Resolution Steps: Change the fragment's headings to H2 or lower.

Validation Steps: One H1 per page.

Preventive Measures: Start fragment headings at H2.

Issue 7 — Anchor link doesn't jump to the fragment

Symptoms: `#anchor-contact` stays at the top.

Root Cause: Invalid Anchor ID (rejected by validation), or a duplicate id renamed to `anchor-contact-2`.

Diagnosis Steps: Inspect the block's `id`.

Resolution Steps: Use a unique, valid Anchor ID; update the link.

Validation Steps: The link jumps to the block.

Preventive Measures: Keep anchors unique per page.

6.3.14 Support Runbook

Monitoring Requirements. Recommended: periodic checks that frequently used fragment pages return 200 on `.aem.live`; RUM for layout shift on pages with fragments.

Logs.

- Browser console: `failed to load module for fragment` (network error during the fetch or an error in an inner block); 404s for `.plain.html` in the Network tab (no console message).
- Admin API status: `https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}` for the fragment page.
- Author logs: publication logs of the fragment page.

Common Errors.

Error	Likely Cause	Fix
Bare link shown	Fragment unpublished or full URL	Publish; pick with content picker
Broken images	Unresolvable image path	Reselect images; republish
Endless loading	Self-including fragment	Remove the loop
Wrong language	Reference to another locale	Re-point to the locale's fragment
Stale content	Fragment not published	Publish fragment page

Escalation Matrix.

Level	Team	Checks
L1	Author support	Reference value, fragment page publication, heading levels, locale
L2	AMS / site admin	Admin API status, CDN purge, `paths.json` coverage of the fragment path
L3	Development team	`blocks/fragment/fragment.js`, `decorateMain`, console errors, inner block errors
L4	Adobe Support	Edge Delivery delivery of `.plain.html`, AEM publishing

Recovery Procedures. Restore the previous fragment page version and publish; or hide (Additional tab > Hide block) or remove the Fragment block on affected pages.

Health Checks. Fragment pages used site-wide load with 200; host pages render the content; no console errors.

Operational Procedures. Keep an inventory of shared fragment pages per locale ([To be confirmed: fragment inventory owner — owner: Content Lead]); review fragments before campaigns end; unpublish fragments only after removing their references.

Known Limitations.

- Full URLs in Reference aren't supported (only site paths).
- No loop protection; no caching in code.
- Fragment content isn't in the delivered HTML or the page's index entry.
- Fragments aren't localized automatically.
- Personalized or scheduled fragments: Not implemented (as of this handover).

6.4 Page Header Block

The Page Header is the header area at the top of a content page: eyebrow, H1 title, description, calls to action and, depending on the layout, tags, publish date, reading time, the author lockup, the listen / download / share actions

or the featured image. **All its text and images come from the page's metadata** (page properties and the locale's bulk metadata sheet) and the locale's **Authors** data sheet; authors only choose the **Layout** and author the **CTA List** in the block. The Article, Blog and Press Release templates add a Page Header automatically when the page has none.

Source files: `blocks/page-header/page-header.js`, `blocks/page-header/page-header.css`, `blocks/page-header/_page-header.json` (a key-value block). Shared code: `decorators/article-meta-decorator.js` (date, reading time, actions, author lookup), `decorators/tag-decorator.js`, `decorators/button-decorator.js`, `decorators/link-decorator.js`, `decorators/responsive-image-decorator.js`, `utils/metadata.js`, `utils/data.js`, `utils/taxonomy.js`, `utils/links.js`, `utils/properties.js`, `utils/placeholders.js`. Design reference: TCS.com design system Figma "Secondary Page Header" / "L3 Header".

6.4.1 Component Purpose

Business Objective. Give every secondary page (landing, listing, article, blog, press release, insight) a consistent, design-system header whose content is always in sync with the page's metadata, so the title, description, tags, dates and author shown to visitors are the same values used for SEO, the query index, cards and listings.

Business Use Cases.

- Landing pages (services, industries) with a large title, description, CTAs and a featured image.
- Article, blog and insight pages with tags, content type, publish date, reading time, author with LinkedIn link, and listen / download / share actions.
- Press releases (automatic Page Header from the template).
- Secondary pages with a title on the left and the description and CTAs on the right.
- Campaign or statement pages with a very large display title.

When To Use. As the first block of every page that isn't the Home page and doesn't start with a Hero; on landing, listing, article, blog and press-release pages.

When Not To Use.

- On the Home page (use Hero or Gen Carousel).
- When the header needs content that isn't page metadata (a video, a custom image per breakpoint): use Hero or Teaser.
- More than once per page (it renders an H1).
- Don't type the title or description into the block: there are no such fields; change the page properties or the metadata sheet.

Dependencies.

Type	Dependency	Use
Page properties	Title (`jcr:title`); Description (`jcr:description`); Page Type (`page-type`); page Tags	Title (H1), description, default layout, tags
Bulk metadata sheet (`/{locale}/metadata.json`)	`page-type`, `content-type`, `publish-date`, `reading-time`, `author-id`, `author`, `featured-image`, `listen-link`; `tags` (comma-separated fallback)	Eyebrow / collection, date, reading time, author, image, listen link
Data sheet	`/{locale}/data/authors.json` (author-id, name, designation, bio, image, linkedin, region)	Author lockup
Taxonomy	`/{locale}/taxonomy.json` (tag, title; per-language sheets)	Tag labels
Placeholders	`published` ("Published"); `minRead` ("{n} min read"); `listenToThisPage`; `downloadThisPage`; `shareThisPage`; `linkCopied` ("Link copied to clipboard"); `opensInNewTab` ("(opens in a new tab)")	UI text

Type	Dependency	Use
Icons	`/icons/headphones.svg`, `download.svg`, `share.svg`	Action icons
Code	`decorators/article-meta-decorator.js` `buildPublished`, `buildReadingTime`, `buildPageActions`, `buildAuthorLockup`	Article meta parts
Code	`decorators/tag-decorator.js` `buildTagList`; `utils/taxonomy.js` `resolveTags`	Tags
Code	`decorators/button-decorator.js` `decorateButton`; `utils/links.js` `readLinkItems`	CTA List
Code	`decorators/link-decorator.js` `setLinkTarget`, `announceNewTab`	LinkedIn link
Code	`decorators/responsive-image-decorator.js` `buildImage`	Featured image and author image
Code	`utils/metadata.js` `getPageMetadata`; `utils/data.js` `getAuthor`; `utils/properties.js` `readProperties`, `optionOf`	Inputs
Templates	`templates/article`, `templates/blog`, `templates/press-release` via `decorators/template-decorator.js` `ensurePageHeader`	Automatic Page Header

Related Components. Hero (alternative first block); Teaser (shares the article meta decorator); Filter List and List (show the same metadata on cards); "About the author" (article template, from the same Authors sheet); SEO decorator (uses the same metadata for the document head).

Applicable Templates. Article and Blog (added automatically), Press Release (added automatically), pages with Page Type Default; landing / listing pages (Page Type set to `landing` / `listing` in the metadata sheet). Not recommended for Home.

Supported Content Types. Any `content-type` value (for example Industry, Service, Insight, Blog, Press Release, Case Study); it is shown as the eyebrow (Landing, Secondary) or the collection label (Article).

6.4.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** An article page at 1440px: the meta row (tags, content type on the left; "Published dd/mm/yy", reading time and the three icon actions on the right), then two columns with the large title and author lockup on the left and the description and CTAs on the right.

Desktop (>= 1200px; layout changes start at 900px). The block is a vertical flex column with 2.5rem gaps and 7.5rem top padding. Secondary, Article and Display place the intro (title side) and the summary (description and CTAs) in two equal columns (`repeat(2, minmax(0, 1fr))`, 2.5rem gap); Secondary and Article move the summary down by 0.75rem; Display centres both columns vertically. Landing with a featured image is a two-column grid (text | image, 5rem gap, vertically centred). Title sizes from 900px: Landing and Display 6rem (letter-spacing –0.18rem and –0.12rem), Article `--display-font-size-2` (4rem), Secondary `--heading-font-size-xxl` (3rem, semibold). Description `--body-font-size-l` (1.5rem, line height 1.417); Landing description `--body-font-size-m` (1.125rem, line height 1.61).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same article page at 1024px (two columns) and at 768px (one column).

Tablet (600–1199px). From 900px as desktop. Between 600px and 899px as mobile (single column); the type tokens step at 900px.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The article page at 390px: meta row wrapping onto two lines, title, author, description and CTAs stacked.

Mobile (< 600px, and up to 899px). Everything stacks in one column with 2.5rem gaps and 2.5rem top padding; the Landing image follows the text. Title sizes: Landing and Display `--display-font-size-1` (2.75rem), Article `--display-font-size-2` (2.5rem), Secondary `--heading-font-size-xxl` (2rem). Description 1rem. Eyebrow and tags 0.75rem uppercase; the meta row wraps (1rem row gap, 2.5rem column gap). Icon actions are 2.75rem square touch targets.

6.4.3 Component Variations

Variation 1 — Default Layout (from the Page Type)

Variation Name: Default (from the Page Type).

Business Purpose: The right layout without a decision by the author.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-default-layout.png`
> **Purpose:** Layout chosen by Page Type
> **Description:** Two pages with Layout = Default: a page with page-type landing rendering the Landing layout, and an article page rendering the Article layout.

Use Cases: Pages created from the Site Global Templates; the automatic Page Header of the templates.

Authoring Guidelines: Leave **Layout** on "Default (from the Page Type)".

Business Rules: Page Type (`page-type` metadata, else the `template` metadata) `landing` or `listing` (any case) → Landing; any other value, including empty, Home, Article, Blog, Press Release → Article. The Page Type page property only offers Default, Home, Article, Blog and Press Release; `landing` and `listing` are set in the metadata sheet's `page-type` column (they have no template code).

Rendering Rules: The resolved layout is added as a class (`landing` or `article`) after removing any other layout class.

Responsive Behaviour: That of the resolved layout.

SEO Impact: As the resolved layout.

Accessibility Compliance: As the resolved layout.

Variation 2 — Landing with Featured Image

Variation Name: Landing (with the featured image).

Business Purpose: Hub pages (services, industries) that need a strong visual.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-landing-with-image.png`
> **Purpose:** Landing with image
> **Description:** Desktop: eyebrow "INDUSTRY", 6rem semibold title, description, two CTAs on the left, the featured image on the right; mobile: image below the text.

Use Cases: Industry, service and capability landing pages; listing pages.

Authoring Guidelines: Set **Layout** to "Landing (with the featured image)" (or Default with `page-type` landing / listing); set `featured-image` in the metadata sheet; optionally `content-type`; add CTAs.

Business Rules: Eyebrow = `content-type`; if empty, the tags (as a list); title, description and CTAs all on the text side; the image has an empty alt (decorative).

Rendering Rules: `div.page-header-intro` (eyebrow, H1, description, CTAs) + `div.page-header-media` (picture); the block gets `landing page-header-has-image`. Site media (`/media_…` on the same origin) is a responsive picture (1200px wide from 900px, 750px below) loaded eagerly with `fetchpriority="high"` (LCP candidate). Other image URLs are used as a plain `<img>` with `loading="lazy"` (eager loading isn't applied to them).

Responsive Behaviour: From 900px: two-column grid, 5rem gap, centred vertically. Below 900px: text then image, stacked. Title 2.75rem → 6rem.

SEO Impact: The H1 is the page Title; the featured image is the likely LCP element: use a site-media image so it loads eagerly.

Accessibility Compliance: Single H1; decorative image (`alt=""`): don't put essential information in the image.

Variation 3 — Landing without Featured Image

Variation Name: Landing (text only).

Business Purpose: Landing or listing pages without a hero image.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-landing-text-only.png`
> **Purpose:** Landing without image
> **Description:** Eyebrow, very large title, description and CTAs stacked across the full width.

Use Cases: Listing pages ("All insights"), simple landing pages.

Authoring Guidelines: Landing layout and no `featured-image`.

Business Rules: Without an image the block isn't a grid; all text stacks.

Rendering Rules: Only `div.page-header-intro`; no `page-header-has-image` class.

Responsive Behaviour: Single column at all widths; title 2.75rem → 6rem.

SEO Impact: H1 is the LCP candidate (text).

Accessibility Compliance: As Variation 2.

Variation 4 — Secondary

Variation Name: Secondary (title left, text right).

Business Purpose: Secondary pages with a balanced header.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-secondary.png`
> **Purpose:** Secondary layout
> **Description:** Eyebrow above both columns; semibold title on the left; description and CTAs on the right, 0.75rem lower.

Use Cases: About, leadership, careers subpages.

Authoring Guidelines: Set **Layout** to "Secondary (title left, text right)"; optional `content-type`.

Business Rules: Eyebrow = `content-type`, else the tags; it sits above both columns; no image, author or meta row.

Rendering Rules: `p.page-header-eyebrow` (or `ul.page-header-tags`) then `div.page-header-columns` with `div.page-header-intro` (H1) and `div.page-header-summary` (description, CTAs); empty columns are left out.

Responsive Behaviour: From 900px: two equal columns. Below: stacked. Title 2rem → 3rem, semibold.

SEO Impact: H1 page Title; description visible.

Accessibility Compliance: Reading order eyebrow → title → description → CTAs.

Variation 5 — Article

Variation Name: Article (meta row, author).

Business Purpose: Editorial pages showing context, credibility and sharing tools.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-article.png`
> **Purpose:** Article layout
> **Description:** Meta row (tag list, content type | "Published 12/03/26", "6 min read", listen, download, share icons), then title with the author lockup (round photo, LinkedIn-linked name, designation and region) on the left and description and CTAs on the right.

Use Cases: Articles, blogs, insights, press releases (template default).

Authoring Guidelines: Set **Layout** to "Article (meta row, author)" or keep Default on article / blog / press-release pages. Fill `content-type`, `publish-date` (YYYY-MM-DD), `reading-time`, `author-id` (or `author`), `listen-link` and page tags.

Business Rules: The meta row has the labels (tags; `content-type` as the "collection") and the info (Published date, reading time, actions). Download and share are always offered, so the meta row always exists; listen only with `listen-link`. The author lockup follows the title.

Rendering Rules: `div.page-header-meta-row` (`div.page-header-labels`, `div.page-header-info` with `p.page-header-date`, `p.page-header-reading-time`, `div.page-header-actions`) then `div.page-header-columns` (intro: H1 and `div.page-header-author`; summary).

Responsive Behaviour: Meta row: labels left, info right (`space-between`), wrapping on small screens. Columns from 900px. Title 2.5rem → 4rem regular weight.

SEO Impact: Visible date and author support E-E-A-T signals; no structured data (JSON-LD) is generated (Not implemented).

Accessibility Compliance: `<time datetime>` for ISO dates; icon buttons with `aria-label` and `title`; share feedback in a `role="status"` live region; LinkedIn link announces "(opens in a new tab)".

Variation 6 — Display

Variation Name: Display (large title).

Business Purpose: Statement pages with a very large title and tags as eyebrow.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/page-header-display.png`
> **Purpose:** Display layout
> **Description:** Tags "CLOUD | AI | SUSTAINABILITY" pipe-separated above a 6rem medium-weight dark slate title on the left; description and CTAs vertically centred on the right.

Use Cases: Thought leadership, campaign pages, topic hubs.

Authoring Guidelines: Set **Layout** to "Display (large title)"; tag the page.

Business Rules: The eyebrow is always the tags (never `content-type`); tags separated by "|".

Rendering Rules: `div.page-header-columns` with intro (`ul.page-header-tags`, H1) and summary; separators are CSS (`li + li::before { content: '|' }`).

Responsive Behaviour: From 900px: two columns centred vertically, title 6rem (–0.12rem). Below: stacked, title 2.75rem.

SEO Impact: As Secondary.

Accessibility Compliance: Tags are a list (`ul`); the "|" separators are CSS generated content (some screen readers may read them).

Variation 7 — Automatic Page Header (Templates)

Variation Name: Page Header added by the Article, Blog and Press Release templates.

Business Purpose: Every article, blog and press release gets a header without authoring.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/page-header-template-automatic.png`
> **Purpose:** Template-generated header
> **Description:** A press release page without an authored Page Header, showing the Article layout built from its metadata.

Use Cases: Pages with Page Type Article, Blog or Press Release.

Authoring Guidelines: Nothing to author. To add CTAs or choose another layout, add a Page Header block yourself (an authored one always wins).

Business Rules: `ensurePageHeader` adds an empty `page-header` block in a new first section only if the page has no `.page-header`; with no Layout and these page types the layout is Article; there are no CTAs.

Rendering Rules: As Variation 5.

Responsive Behaviour: As Variation 5.

SEO Impact: As Variation 5.

Accessibility Compliance: As Variation 5. The generated block isn't editable in Universal Editor (it has no content resource).

Variation 8 — CTA Style: Primary Button

Variation Name: CTA Link Style "Primary Button" (default).

Business Purpose: The main action.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-cta-primary.png`
> **Purpose:** Primary CTA
> **Description:** A filled brand-colour primary button "Talk to an expert" in the summary column.

Use Cases: "Contact us", "Talk to an expert".

Authoring Guidelines: CTA List item with **CTA Link Style** = Primary Button.

Business Rules: Default value of the field; one primary per header is Recommended.

Rendering Rules: `a.button.primary`.

Responsive Behaviour: CTAs wrap (1rem row gap, 2.5rem column gap).

SEO Impact: Standard link.

Accessibility Compliance: Link with visible text; focus styles from `styles/styles.css`.

Variation 9 — CTA Style: Secondary Button

Variation Name: CTA Link Style "Secondary Button".

Business Purpose: A second, lower-emphasis action.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-cta-secondary.png`
> **Purpose:** Secondary CTA
> **Description:** An outlined secondary button next to the primary button.

Use Cases: "Download the report", "Explore services".

Authoring Guidelines: CTA Link Style = Secondary Button.

Business Rules: None beyond the style.

Rendering Rules: `a.button.secondary`.

Responsive Behaviour: As Variation 8.

SEO Impact: Standard link.

Accessibility Compliance: As Variation 8.

Variation 10 — CTA Style: Text Link

Variation Name: CTA Link Style "Text link".

Business Purpose: A subtle tertiary action.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-cta-text-link.png`
> **Purpose:** Text link CTA
> **Description:** A tertiary text-style link after the buttons.

Use Cases: "Learn more", "View all".

Authoring Guidelines: CTA Link Style = Text link.

Business Rules: Any value other than primary / secondary renders the tertiary style.

Rendering Rules: `a.button.tertiary`.

Responsive Behaviour: As Variation 8.

SEO Impact: Standard link.

Accessibility Compliance: Use descriptive text (not "Learn more" alone where several appear).

Variation 11 — Author from the Authors Sheet

Variation Name: Author lockup from `author-id`.

Business Purpose: Consistent author presentation with photo and LinkedIn profile.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-author-sheet.png`
> **Purpose:** Author lockup
> **Description:** Round 2.5rem photo, the name linked to LinkedIn, and "Chief Technology Officer, Europe" below.

Use Cases: Articles and blogs by registered authors.

Authoring Guidelines: Set `author-id` (metadata sheet or page metadata) to an ID of `/{locale}/data/authors`.

Business Rules: ID matched case-insensitively; name, image, designation and region from the sheet; designation and region joined with ","; name linked to `linkedin` (new tab) when present. Article layout only.

Rendering Rules: `div.page-header-author` > `div.page-header-author-image` (picture, `alt=""`) + `div.page-header-author-details` (`p.page-header-author-name` [with `a`], `p.page-header-author-designation`).

Responsive Behaviour: Same at all sizes; text 0.875rem → 1rem.

SEO Impact: Visible authorship.

Accessibility Compliance: Decorative photo; LinkedIn link with hidden "(opens in a new tab)".

Variation 12 — Author from Author Metadata

Variation Name: Author name only (`author` metadata).

Business Purpose: Guest or external authors not in the Authors sheet.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-author-metadata.png`
> **Purpose:** Name-only author
> **Description:** The author name in light weight below the title, without photo or designation.

Use Cases: Guest posts, external contributors.

Authoring Guidelines: Leave `author-id` empty (or use an ID not in the sheet) and set `author`.

Business Rules: The name comes from `author` when the Authors sheet has no match; no image, designation or link.

Rendering Rules: Lockup with the name only.

Responsive Behaviour: As Variation 11.

SEO Impact: As Variation 11.

Accessibility Compliance: Plain text.

Variation 13 — Article with Listen Action

Variation Name: Listen action (`listen-link`).

Business Purpose: Offer an audio version of the page.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-listen-action.png`
> **Purpose:** Listen icon
> **Description:** The headphones icon before the download and share icons in the meta row.

Use Cases: Articles with a recorded or generated audio file or podcast page.

Authoring Guidelines: Set `listen-link` to the audio file or audio page URL.

Business Rules: Only in the Article layout; the link opens in the same tab; no player is embedded (Not implemented).

Rendering Rules: `a.button.tertiary.icon-button` with `--icon-url` headphones, `aria-label` / `title` "Listen to this page", `data-cmp-interaction="listen"`.

Responsive Behaviour: 2.75rem icon button at all sizes.

SEO Impact: None.

Accessibility Compliance: Accessible name from placeholder `listenToThisPage`.

6.4.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** An article page in Universal Editor with the Page Header selected and markers A to H on the rendered parts.

Annotations:

- A = The Page Header block (select it to open Layout and CTA List).
- B = Eyebrow / tags (from `content-type` or the page tags; not editable in the block).
- C = Title H1 (from the page Title in Page Properties).
- D = Description (from the page Description in Page Properties).
- E = Meta row: Published date, reading time (metadata `publish-date`, `reading-time`) and actions (listen from `listen-link`; download and share from code).
- F = Author lockup (`author-id` and the Authors sheet, or `author`).
- G = CTA List items (authored in the block).
- H = Featured image (Landing; metadata `featured-image`).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Page Header properties with the Style tab (Layout select open), the CTAs tab with two CTA List items, and the Additional tab.

Properties panel description.

- **Style tab: Layout** (select): Default (from the Page Type); Landing (with the featured image); Secondary (title left, text right); Article (meta row, author); Display (large title). Its help text lists the metadata the block reads.
- **CTAs tab: CTA List** (multifield); each item: **CTA Title** (text), **CTA Link** (content picker), **CTA Link Type** (Default / Open in New Window), **CTA Link Style** (Text link / Primary Button / Secondary Button; default Primary Button).
- **Additional tab: Anchor ID**, **Hide block**, **Padding left / right / top / bottom**, **Margin left / right / top / bottom**.

6.4.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites select the page and choose **Edit**.

Expected Result: The page opens in Universal Editor.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** Sites console with the page selected and Edit highlighted.

Step 2 — Fill the page metadata

Action: Open **Page Properties**: enter **Title** (the H1), **Description**, page tags, and the **Page Type**. In the locale's metadata sheet, add or update the row for the page URL with `content-type`, `publish-date` (YYYY-MM-DD), `reading-time` (minutes), `author-id`, `featured-image` and `listen-link` as needed. Publish the sheet.

Expected Result: The page metadata contains the values.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-02-metadata.png`
> **Purpose:** Metadata inputs
> **Description:** Page Properties with Title and Description, and the metadata sheet row with content-type, publish-date, reading-time and author-id.

Step 3 — Add the Page Header block

Action: Select the first section, choose **Add** (+) and pick **Page Header**; move it to the top of the first section.

Expected Result: The block renders from the metadata (Article or Landing layout by default).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-03-add-block.png`
> **Purpose:** Adding the block
> **Description:** The component picker with Page Header highlighted.

Step 4 — Choose the Layout (Style tab)

Action: Select the block; in **Style** > **Layout** choose the layout or keep Default.

Expected Result: The block re-renders in the chosen layout.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-04-layout.png`
> **Purpose:** Layout select
> **Description:** The Layout select open with its five options.

Step 5 — Add CTAs (CTAs tab)

Action: In **CTAs** > **CTA List** choose **Add**; fill **CTA Title**, pick **CTA Link**, choose **CTA Link Type** and **CTA Link Style**. Repeat and reorder.

Expected Result: The CTAs appear below the description.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-05-ctas.png`
> **Purpose:** CTA List
> **Description:** Two CTA items: "Talk to an expert" (Primary) and "Explore services" (Secondary).

Step 6 — Set the Additional tab (optional)

Action: Set Anchor ID, spacing or Hide block.

Expected Result: Options apply to the block.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-06-additional.png`
> **Purpose:** Additional tab
> **Description:** The Additional tab with Padding bottom set.

Step 7 — Check the author data

Action: For the Article layout, verify the `author-id` exists in `/{locale}/data/authors` with name, designation, region, image and LinkedIn.

Expected Result: The author lockup shows photo, name and designation.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-07-author-data.png`
> **Purpose:** Authors sheet
> **Description:** The authors sheet row for the author-id used on the page.

Step 8 — Reload and preview

Action: Reload the editor (metadata changes are read when the page loads), then publish the page to Preview and open it on `.aem.page`.

Expected Result: The preview shows all metadata-driven parts.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-08-preview.png`
> **Purpose:** Preview
> **Description:** The preview page header at desktop width.

Step 9 — Validate

Action: Run the Validation Process checklist at 390px, 768px, 1024px and 1440px.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-09-validate.png`
> **Purpose:** Validation
> **Description:** The header at mobile and desktop widths side by side.

Step 10 — Publish

Action: Publish the page (and the metadata / authors sheets if changed) to Live.

Expected Result: The live page shows the header.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/page-header-step-10-publish.png`
> **Purpose:** Publishing
> **Description:** Publish dialog with the page and the metadata sheet.

Preview Process

1. Publish the metadata sheet, the authors sheet and the taxonomy (if changed) to Preview, then the page.
2. Check the header on `.aem.page`; bulk metadata applies only once all 13 locale sheets are listed in the site configuration's `metadata.source` (see Part on administration); until then use page properties.

Publishing Process

1. Publish sheets first, then the page, to Live.
2. Changes to metadata sheets affect every page matching the row's URL pattern.

Validation Process

- ☐ Exactly one H1 on the page, equal to the page Title.
- ☐ The layout is the intended one (check Default resolution by Page Type).
- ☐ Eyebrow / tags show the right labels (taxonomy titles in the locale's language).
- ☐ Article: date as dd/mm/yy after "Published"; reading time "{n} min read"; listen (if set), download and share icons work; "Link copied to clipboard" shows on browsers without a share dialog.
- ☐ Author: photo, name, designation and region correct; LinkedIn link opens a new tab.
- ☐ CTAs in the right styles and order; new-window CTAs open a new tab.
- ☐ Landing: featured image loads, sharp at 1200px; no layout shift.
- ☐ Display: tags separated by "|".
- ☐ Translated UI text in non-English locales.
- ☐ No empty spaces from missing metadata.

Rollback Process

1. Restore the previous page version (Layout and CTAs) and publish.
2. Restore the previous metadata / authors sheet version and publish.
3. To remove a broken authored header on a template page, delete it: the template adds the automatic header again.

6.4.6 Authorable Field Specification

Page Header block (model `page-header`, key-value)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Style tab — Layout (`classes`)	select	No	No	Default (`""`)	Layout	Keep Default unless the page needs another layout	Options: Default (from the Page Type) `""`; Landing `landing`; Secondary `secondary`; Article `article`; Display `display`	`secondary`
CTAs tab — CTA List (`ctas`)	container (composite multifield)	No	Yes	No items	Calls to action	1–2 CTAs	Items without a link are skipped	Talk to an expert; Explore services
Additional tab — Anchor ID (`classes_anchor`)	text	No	No	Empty	Block id	Start with `anchor-`	Enforced: `^anchor-[a-z0-9][a-z0-9-]*$`	`anchor-intro`

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide on site	Temporary use	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default	Spacing	Tokens only	Default; 0; 0.25rem … 7.5rem (`pl-*`)	`pl-0`
Padding right (`classes_paddingright`)	select	No	No	Default	Spacing	Tokens only	Same (`pr-*`)	`pr-0`
Padding top (`classes_paddingtop`)	select	No	No	Default	Spacing	Overrides the block's 2.5rem / 7.5rem top padding	Same (`pt-*`)	`pt-1200`
Padding bottom (`classes_paddingbottom`)	select	No	No	Default	Spacing	Tokens only	Same (`pb-*`)	`pb-800`
Margin left (`classes_marginleft`)	select	No	No	Default	Spacing	Tokens only	Same (`ml-*`)	`ml-0`
Margin right (`classes_marginright`)	select	No	No	Default	Spacing	Tokens only	Same (`mr-*`)	`mr-0`
Margin top (`classes_margintop`)	select	No	No	Default	Spacing	Tokens only	Same (`mt-*`)	`mt-0`
Margin bottom (`classes_marginbottom`)	select	No	No	Default	Spacing	Tokens only	Same (`mb-*`)	`mb-1600`

CTA List item (fields of `ctas`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
CTA Title (`ctaText`)	text	Yes (Recommended)	No	Empty	Visible CTA text	Verb first, 2–4 words	None	`Talk to an expert`
CTA Link (`cta`)	aem-content	Yes (Enforced by code: items without a link are skipped)	No	Empty	Destination	Page or URL	Items without link dropped	`/content/tcs-corp/us/en/contact`
CTA Link Type (`ctaTarget`)	select	No	No	Default (`""`)	Window	New window for external only	Default `""`; Open in New Window `new-window`	`""`
CTA Link Style (`ctaStyle`)	select	No	No	Primary Button (`primary`)	Emphasis	One primary	Text link `""`; Primary Button `primary`; Secondary Button `secondary`	`secondary`

Metadata read by the block (not block fields; page properties and bulk metadata sheet)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Title (page property)	text	Yes (Recommended)	No	Page name	H1	40–70 characters	None; read as authored (not `meta-title`)	`Cloud transformation for banks`
Description (page property)	text	Recommended	No	Empty	Description paragraph	120–200 characters	None; read as authored (not `meta-description`)	`How banks modernise …`
Page tags	tags	No	Yes	None	Tags eyebrow / labels	1–4 tags	Labels from taxonomy, else derived from the tag ID	`tcs:topics/cloud`
`page-type`	sheet / page property	No	No	Empty	Default layout	`landing` / `listing` for Landing	Case-insensitive	`landing`
`content-type`	sheet	No	No	Empty	Eyebrow / collection	One short term	None	`Insight`
`publish-date` (or `publication-date`)	sheet	No	No	Empty	Published date (Article)	YYYY-MM-DD	Non-ISO text shown as is	`2026-03-12`
`reading-time`	sheet	No	No	Empty	Reading time (Article)	Minutes as a number	Digits → "{n} min read"; other text as is	`6`
`author-id`	sheet	No	No	Empty	Authors sheet lookup	An ID of `data/authors`	Case-insensitive match	`jdoe`
`author`	sheet	No	No	Empty	Fallback author name	When no author-id	None	`Jane Doe`
`featured-image`	sheet	No	No	Empty	Landing image	Site media URL	`/media_` same origin = optimized and eager	`/en/media_1ab2.jpg`
`listen-link`	sheet	No	No	Empty	Listen action (Article)	Audio URL	None	`https://…/article.mp3`

Field details

- **Layout.** Purpose: chooses the presentation. Allowed Values: Default, Landing, Secondary, Article, Display. Character Limits: n/a. Business Rules: Default → Landing for `landing` / `listing` page types, else Article; a layout class on the block (from any source) is honoured if the select is empty. Dependency Rules: Landing uses `featured-image`; Article uses the article metadata; Display uses tags. Visibility Rules: no `condition`; always visible. Sample Content: Secondary.
- **CTA List.** Purpose: calls to action. Allowed Values: items with a link. Character Limits: Recommended up to 3 items. Business Rules: items without a link are skipped; order is kept. Dependency Rules: none. Visibility Rules: always. Sample Content: Talk to an expert (Primary); Explore services (Secondary).
- **CTA Title.** Purpose: link text. Allowed Values: text. Character Limits: Recommended up to 25 characters (buttons don't wrap well). Business Rules: collapsed with CTA Link into one anchor. Dependency Rules: CTA Link. Visibility Rules: always. Sample Content: "Talk to an expert".
- **CTA Link.** Purpose: destination. Allowed Values: page path or URL. Character Limits: none. Business Rules: required for the item to render. Dependency Rules: none. Visibility Rules: always. Sample Content: `/content/tcs-corp/us/en/contact`.

- **CTA Link Type.** Purpose: same / new window. Allowed Values: Default, Open in New Window. Business Rules: new window adds `target="_blank" rel="noopener noreferrer"`; the new tab isn't announced. Dependency Rules: CTA Link. Visibility Rules: always. Sample Content: Default.
- **CTA Link Style.** Purpose: emphasis. Allowed Values: Text link (tertiary), Primary Button, Secondary Button. Business Rules: default Primary; values other than primary / secondary render tertiary. Dependency Rules: CTA Link. Visibility Rules: always. Sample Content: Secondary Button.
- **Anchor ID, Hide block, Padding / Margin.** As in 6.3.6: Anchor ID Enforced pattern `^anchor-[a-z0-9][a-z0-9-]*$`, duplicates get `-2`…; Hide block hides on the site and fades in the editor; 14 spacing tokens plus Default.
- **Metadata inputs.** See the table: they are edited in Page Properties or the metadata sheet, never in the block. Title and Description are the page values as authored, captured before the SEO decorator replaces the document title and description with `meta-title` / `meta-description`.

6.4.7 Field Annotation Mapping

Marker	Field	Purpose
A	Page Header block (Layout; Additional tab)	Layout and block options
B	`content-type` / page tags	Eyebrow or tag labels
C	Page Title	H1
D	Page Description	Description paragraph
E	`publish-date`, `reading-time`, `listen-link`	Meta row (Article)
F	`author-id` (Authors sheet) or `author`	Author lockup (Article)
G	CTA List (CTA Title, CTA Link, CTA Link Type, CTA Link Style)	Calls to action
H	`featured-image`	Landing image

6.4.8 Component Specification

Attribute	Value
Component Name	Page Header (`blocks/page-header`)
Version	1.0
Owner	[To be confirmed: product owner — owner: Digital Marketing]; technical owner: development team
Dependencies	article-meta, tag, button, link, responsive-image decorators; metadata, data, taxonomy, links, properties, placeholders utils; metadata sheet; Authors sheet; taxonomy; placeholders; icons
Component Type	Block (key-value) with a CTA List composite multifield
Supported Variants	Default; Landing with image; Landing text only; Secondary; Article; Display; automatic (templates); CTA styles Primary / Secondary / Text link; author from sheet / from metadata; listen action
Supported Templates	Article, Blog, Press Release (automatic), Default, landing / listing page types; not Home
Supported Metadata	Title, Description, tags, page-type / template, content-type, publish-date / publication-date, reading-time, author-id, author, featured-image, listen-link
Supported Tags	Page tags (`article:tag` meta, else a comma-separated `tags` meta), labels from `/{locale}/taxonomy.json`
Supported Personalization	None; content follows metadata. Future: Target offers via Adobe Tags (not implemented)

Attribute	Value
Caching Considerations	Metadata is in the page HTML (CDN cached); Authors sheet, taxonomy and placeholders are fetched once per page and cached by CDN / browser
Performance Considerations	Usually in the first section (eager phase: LCP); waits for placeholders, taxonomy and Authors sheet in parallel before rendering; Landing image eager with high fetch priority when it is site media; author image lazy
Accessibility Compliance	One H1; `<time datetime>`; icon buttons 2.75rem with names and tooltips; `role="status"` share feedback; LinkedIn new-tab hint; decorative images `alt=""`; CTA new windows not announced
SEO Compliance	H1 = page Title; visible description and dates; no JSON-LD (Not implemented); client-rendered from the head metadata already in the page

6.4.9 Input Model

- **Content Inputs:** CTA List items.
- **Metadata Inputs:** `og:title` / document title (Title), `description`, `page-type` / `template`, `content-type`, `publish-date` / `publication-date`, `reading-time`, `author-id`, `author`, `featured-image`, `listen-link`.
- **Tags Inputs:** `meta[property="article:tag"]` values (or a `tags` meta), resolved through the locale taxonomy.
- **Page Context Inputs:** locale (for sheets and taxonomy language); `window.location.href` (share); viewport (`(min-width: 900px)` picture source).
- **Query Inputs:** `/{locale}/data/authors.json`, `/{locale}/taxonomy.json`, `/{locale}/placeholders.json`.
- **Manual Inputs:** Layout, CTA List, Additional tab.

Authored HTML received (illustrative; key-value rows):

```
<div class="page-header secondary">
  <div>
    <div>ctas</div>
    <div>
      <p><a href="/en/contact">Talk to an expert</a></p><p>primary</p>
      <hr>
      <p><a href="/en/services">Explore services</a></p><p>secondary</p>
    </div>
  </div>
</div>
```

The template-generated block is just `<div class="page-header"></div>`.

6.4.10 Output Model

Rendered Output (DOM structure, Article).

```
div.page-header.block.article
├─ div.page-header-meta-row
│   ├─ div.page-header-labels
│   │   ├─ ul.page-header-tags > li[data-tag]
│   │   └─ p.page-header-collection
│   └─ div.page-header-info
│       ├─ p.page-header-date > time[datetime]
│       ├─ p.page-header-reading-time
│       └─ div.page-header-actions
│           ├─ a|button.button.tertiary.icon-button (listen, download, share)
│           └─ p.page-header-status[role=status]
└─ div.page-header-columns
```

```
    ├─ div.page-header-intro > h1.page-header-title ; div.page-header-author
    └─ div.page-header-summary
       ├─ div.page-header-description > p
       └─ div.page-header-ctas > a.button
```

Generated HTML (Landing, simplified).

```
<div class="page-header block landing page-header-has-image">
  <div class="page-header-intro">
    <p class="page-header-eyebrow">Industry</p>
    <h1 class="page-header-title">Banking and financial services</h1>
    <div class="page-header-description"><p>Reimagine banking …</p></div>
    <div class="page-header-ctas">
      <a href="/en/contact" class="button primary">Contact us</a></div>
  </div>
  <div class="page-header-media"><picture>…<img loading="eager" fetchpriority="high"
    alt=""></picture></div>
</div>
```

Generated JSON: none. **Generated Metadata:** none (reads metadata only).

Analytics Tracking. Component state (lazy phase, after sections load):

```
{ "component": { "page-header-1": {
  "@type": "tcs-corp/components/page-header", "parentId": "page-1x2y3z",
  "dc:title": "Cloud transformation for banks" } } }
```

Clicks: CTAs (`interaction link`), LinkedIn (`link`, `xdm:linkType exit`), and the actions with their `data-cmp-interaction`:

```
{ "event": "cmp:click", "eventInfo": {
  "path": "component.page-header-1", "interaction": "share",
  "dc:title": "Share this page", "xdm:linkType": "other", "breakpoint": "desktop" } }
```

The listen link reports `interaction listen` with its `xdm:linkURL` (`xdm:linkType download` for audio files such as `.mp3`); download reports `download`.

Personalization Output: none. **Search Index Output:** none from the block; the query index reads the same metadata from the page head (title, description, tags, pageType, contentType, authorId, publishDate, readingTime, featuredImage).

6.4.11 Business Rules

1. Layout = the Layout select value (or a layout class on the block); else Landing when Page Type (`page-type`, else `template`) is `landing` or `listing`, else Article.
2. Title (H1) = the page's `og:title` (page Title) or the document title as delivered, read before `meta-title` is applied; no title → no H1.
3. Description = the page `description` meta as delivered (before `meta-description`); empty → no description.
4. Tags = `article:tag` metas, else the comma-separated `tags` meta, de-duplicated; labels from the locale taxonomy (language sheet), else derived from the tag ID (last segment, hyphens to spaces, first letter capitalised), else the authored text.
5. Eyebrow: Landing and Secondary show `content-type`, else the tags; Display always shows the tags; Article doesn't use the eyebrow (tags and content type are in the meta row).

6. Landing: eyebrow, title, description and CTAs in one column; the featured image (decorative) beside it from 900px.
7. Landing image: same-origin `/media_` URLs become an optimized `<picture>` (1200px from 900px, 750px below) with `loading="eager"` and `fetchpriority="high"`; other URLs a plain lazy `<img>`.
8. Secondary: eyebrow above the columns; title | description and CTAs.
9. Article: meta row (tags and `content-type` | Published date, reading time, actions) then title and author | description and CTAs.
10. Display: tags and title | description and CTAs; tags separated by "|".
11. Published date: "Published" + date as dd/mm/yy in a `<time datetime="YYYY-MM-DD">` for ISO dates (no time zone shift); other text shown as authored; `publication-date` is a fallback for `publish-date`.
12. Reading time: digits only → "{n} min read" (placeholder `minRead`); other text as authored; empty → nothing.
13. Actions (Article only): listen only when `listen-link` is set (a link, same tab); download (prints the page with `window.print()`, the visitor can save as PDF) and share always.
14. Share uses the device share dialog when available (title = page title, URL = current URL); otherwise copies the URL and shows "Link copied to clipboard" for 3 seconds; a dismissed dialog or blocked clipboard is ignored silently.
15. Author: the `author-id` row of the Authors sheet (case-insensitive), else the `author` metadata for the name; designation = "designation, region"; name linked to `linkedin` in a new tab with the "(opens in a new tab)" hint; image decorative; no name and no designation → no lockup (even with an image).
16. CTAs: items without a link are skipped; style primary / secondary, anything else tertiary (Text link); Open in New Window → new tab.
17. Empty parts (columns, meta row parts, CTAs) are left out; if nothing remains the block is empty.
18. Placeholders, taxonomy and the Authors sheet are loaded in parallel; missing sheets give empty results (no author, ID-derived tag labels, English text).
19. Any previous layout class is removed and the resolved one added; `page-header-has-image` marks Landing with an image.
20. The Article, Blog and Press Release templates add an empty Page Header as the first section when the page has none; an authored one wins.
21. Metadata changes are read when the page loads: reload the editor after changing Page Properties.
22. Additional tab options apply (anchor id, hide, spacing).

6.4.12 Author Best Practices

- **Recommended Content Length:** Title 40–70 characters (Landing / Display at 6rem: keep under about 40 characters); Description 120–200 characters; `content-type` one or two words; 1–4 tags; 1–2 CTAs with 2–4 words each.
- **Image Dimensions:** the featured image has no fixed aspect ratio in the CSS (full column width, height automatic); it is requested at 1200px wide (desktop) and 750px (mobile). Recommended: landscape 4:3 or 16:9 at least 1200 × 900px, uploaded to AEM Assets so it's delivered as site media (`/media_`) and loaded eagerly. Author photos: square, at least 160 × 160px (shown as a 2.5rem circle, `object-fit: cover`).
- **CTA Writing Guidelines:** start with a verb ("Talk to an expert", "Download the report"); one Primary Button; use Secondary or Text link for further actions; avoid new windows for site pages.
- **Accessibility Guidelines:** one H1 per page (don't add a Title H1 below); don't rely on the featured image to convey information (it has no alt text); keep tag labels meaningful; check colour contrast of text on any section background.
- **SEO Guidelines:** the H1 is the page Title: write it for users and search; keep `meta-title` (document title) consistent with it; keep Description consistent with `meta-description`.

- **Metadata Guidelines:** always use ISO dates (YYYY-MM-DD) for `publish-date`; `reading-time` as a number; maintain `author-id` values in the Authors sheet; set `featured-image` for landing pages; set `page-type` landing / listing in the sheet for landing defaults.
- **Taxonomy Guidelines:** use tags from the TCS taxonomy (`tcs:...`) and keep their titles translated in each locale's taxonomy; free-text tags show as typed.
- **Localization Guidelines:** each locale has its own metadata sheet, Authors sheet, taxonomy and placeholders; translate `published`, `minRead`, `listenToThisPage`, `downloadThisPage`, `shareThisPage`, `linkCopied`, `opensInNewTab`. The date format dd/mm/yy is the same in every locale (not localised).

6.4.13 Troubleshooting Guide

Issue 1 — Title or description is wrong or missing

Symptoms: The H1 shows an old title or nothing; no description.

Root Cause: Page Properties Title / Description not updated or the page not republished; the block reads the authored page values, not `meta-title` / `meta-description`.

Diagnosis Steps: View the page source: `og:title` and `description` metas.

Resolution Steps: Update Title / Description in Page Properties; publish.

Validation Steps: The H1 matches the Title.

Preventive Measures: Edit page text in Page Properties, not in the block.

Issue 2 — Wrong layout

Symptoms: An article shows the Landing layout or the reverse.

Root Cause: Layout left on Default and the `page-type` metadata resolves differently (for example a sheet row sets `landing` for a whole folder).

Diagnosis Steps: Check `<meta name="page-type">`; check the block's Layout.

Resolution Steps: Choose the Layout explicitly, or fix the metadata sheet row.

Validation Steps: The block has the expected layout class.

Preventive Measures: Use explicit layouts where the Page Type doesn't match the design.

Issue 3 — Author lockup missing or incomplete

Symptoms: No author, or a name without photo or designation.

Root Cause: `author-id` not in `/{locale}/data/authors`, the sheet isn't published, columns misnamed, or the layout isn't Article.

Diagnosis Steps: Check `<meta name="author-id">`; open `/{locale}/data/authors.json`.

Resolution Steps: Add or fix the author row (author-id, name, designation, image, linkedin, region); publish.

Validation Steps: Full lockup appears.

Preventive Measures: Maintain authors in every locale's sheet.

Issue 4 — Tags show technical names

Symptoms: Tags like "Artificial intelligence" derived from IDs, or English labels on a German page.

Root Cause: The tag is missing from `/{locale}/taxonomy.json` or its language sheet.

Diagnosis Steps: Open the taxonomy JSON and search for the tag ID.

Resolution Steps: Add or translate the tag title; publish the taxonomy.

Validation Steps: Proper labels show.

Preventive Measures: Use only taxonomy tags.

Issue 5 — Date shows in the wrong format
Symptoms: "Published March 12" instead of "Published 12/03/26".

Root Cause: `publish-date` isn't ISO (YYYY-MM-DD); non-ISO values are shown as typed.

Diagnosis Steps: Check the metadata value.

Resolution Steps: Use YYYY-MM-DD.

Validation Steps: The date shows as dd/mm/yy with `datetime`.

Preventive Measures: Validate dates in the metadata sheet.

Issue 6 — Landing image loads late (poor LCP)
Symptoms: PageSpeed Insights reports a slow LCP on the image.

Root Cause: `featured-image` isn't site media (`/media_` on the same host), so it's a lazy `<img>`; or the image is very large.

Diagnosis Steps: Inspect the image: `loading="lazy"` and no `<picture>`.

Resolution Steps: Use an image delivered as site media; development can apply eager loading to plain images too.

Validation Steps: `loading="eager"`, `fetchpriority="high"`; LCP improves.

Preventive Measures: Use site media for featured images.

Issue 7 — Share does nothing
Symptoms: Selecting share shows no dialog and no message.

Root Cause: No device share dialog and the clipboard is blocked (insecure context, permissions, embedded frame); errors are ignored.

Diagnosis Steps: Test on HTTPS in a normal browser window; check permissions.

Resolution Steps: Use HTTPS hosts (`.aem.page`, `.aem.live`, production domain); no author action.

Validation Steps: Dialog opens or the copied message shows.

Preventive Measures: Test share on mobile and desktop browsers.

Issue 8 — Two H1 headings
Symptoms: Accessibility checker reports multiple H1.

Root Cause: A Title / Hero with H1 or a second Page Header on the page; or an authored Page Header placed after the template's automatic one (the template adds one only if none exists, so check for a Hero H1).

Diagnosis Steps: List headings in developer tools.

Resolution Steps: Change other H1s to H2; keep one Page Header.

Validation Steps: One H1.

Preventive Measures: Follow the heading guidance.

6.4.14 Support Runbook
Monitoring Requirements. Recommended: PageSpeed Insights checks of key landing and article pages (LCP from the Page Header); periodic validation that each locale's `data/authors.json`, `taxonomy.json` and `placeholders.json` return 200; data layer monitoring of `share` / `download` / `listen` interactions once Adobe Tags is installed.

Logs.

- Browser console: the block emits no messages of its own; exceptions are logged by `aem.js` as `failed to load module for page-header`.
- Network: requests for `authors.json`, `taxonomy.json`, `placeholders.json` (a 404 means an unpublished sheet).
- Admin API status of the page and sheets: `https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`.
- Author logs: AEM publication of pages and sheets.

Common Errors.

Error	Likely Cause	Fix
Old title / description	Page Properties not published	Update and publish
Wrong layout	Page Type resolution	Choose Layout explicitly / fix sheet
No author	author-id missing in Authors sheet	Add row; publish sheet
ID-like tag labels	Taxonomy missing or untranslated	Update taxonomy
Date shown as typed	Non-ISO date	Use YYYY-MM-DD
Slow LCP	Non-site-media featured image	Use site media
Bulk metadata not applied	Sheets not in `metadata.source`	Admin configures `metadata.source`

Escalation Matrix.

Level	Team	Checks
L1	Author support	Page Properties, metadata sheet row, Authors sheet, taxonomy, Layout, CTA List, publication
L2	AMS / site admin	`metadata.source` configuration, `paths.json` mappings of sheets, Admin API status, CDN purge
L3	Development team	`blocks/page-header/*`, article-meta decorator, metadata / data / taxonomy utils, console errors, template insertion
L4	Adobe Support	Metadata publishing by Edge Delivery, AEM / Universal Editor defects

Recovery Procedures. Restore previous page or sheet versions and publish; on template pages, deleting a broken authored Page Header brings back the automatic one; revert faulty code on `main`.

Health Checks. Sample one page per layout and locale: one H1 equal to the Title; correct eyebrow / tags; Article meta row with date, reading time and working actions; author lockup; Landing image eager; no console errors.

Operational Procedures. New author: add the row to every locale's Authors sheet where they publish. New content type: agree the `content-type` term per locale. New layout: development adds it to `LAYOUTS` in `page-header.js`, the CSS and the Layout options in `_page-header.json`, then runs `npm run build:json`.

Known Limitations.

- Text content can't be edited in the block (metadata only); metadata changes need an editor reload.
- `landing` / `listing` page types aren't Page Type options in Page Properties (metadata sheet only).
- Non-site-media featured images are lazy-loaded; featured image and author photo have no alt text (decorative).
- Download prints the page (no PDF generation); listen is a link (no embedded player); no JSON-LD structured data (Not implemented as of this handover).
- Date format fixed to dd/mm/yy for all locales.

- New-window CTAs aren't announced to screen readers.
- The template-generated Page Header has no CTAs and can't be edited in Universal Editor.

6.5 Section Block

6.5.1 Component Purpose

Business Objective: Give every page a consistent rhythm of horizontal bands, with the design system's spacing, content widths and a small set of authorable backgrounds, so that authors can structure pages without custom code.

Business Use Cases:

- Splitting a page into topics (introduction, services, case studies, contact).
- Highlighting a band of content with a light grey background.
- Placing content over a full-cover background image (campaign bands, banners built from default content).
- Constraining a band to the narrower content width, optionally with rounded corners (a "card-like" band).
- Making a band the target of an Anchor List or a `#link` (chapter navigation).

When To Use: Always: every piece of content on a page lives in a section. Start a new section whenever the background, width or topic changes.

When Not To Use:

- For carousels: use **Gen Carousel** (a special section added at page level).
- For tabbed content spanning several components: use **Tabs Container** (a special section added at page level) or the **Tabs** block.
- For a dark or brand-coloured band: not authorable on sections (see Variation notes); use a block with a background colour option (for example Hero, Teaser BG Color).

Dependencies:

Type	Item
Model	`models/_section.json` (definition `section`, model `section`, filter `section`)
Code	`scripts/scripts.js` (`decorateSectionsWithOptions`, `readSectionOptions`, `buildSectionBackground`); `scripts/aem.js` (`decorateSections`, `loadSection`, `loadSections`, `waitForFirstImage`); `decorators/responsive-image-decorator.js` (`buildImage`); `scripts/editor-support.js` (re-decoration in the editor)
Styles	`styles/styles.css` (layout tokens `--page-max-width`, `--content-max-width`, `--section-padding-inline`, `--section-spacing`; section rules; Hide block wrapper rule)
Data sources / sheets / placeholders	None

Related Components: Gen Carousel and Tabs Container (section-based containers with their own models); Anchor List (links to section anchors); every block and default-content component (children of a section).

Applicable Templates: Any page (Default, Home, Article, Blog, Press Release). The article, blog and press-release templates add their own sections (Page Header, About the author, Media contacts) before or after the authored ones.

Supported Content Types: Default content (Text, Title, Image, Button) and all blocks of the section filter (see 6.5.9).

6.5.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A page at 1440 px with three sections: a default section (white), a Highlight section (light grey band) and a Fixed Width section with a background image and rounded corners.

Desktop (1200 px and above): each section's content is centred and capped at 120rem (Full Width) or 90rem (Fixed Width), with 5rem side padding (`--section-padding-inline: var(--spacing-l)`). Sections are separated by a vertical margin of 7.5rem (`--section-spacing: var(--space-3000)`); the first section has no top margin. Highlight and background sections have no outer margin but 7.5rem padding top and bottom, so their colour or image fills the band.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same page at 768 px: narrower side padding, Highlight band edge to edge, background image cropped to cover the band.

Tablet (600–1199 px): side padding 2.5rem (`--spacing-m`). The section spacing is 2.5rem up to 899 px and 5rem (`--spacing-l`) from 900 px. A Fixed Width section is narrower than 90rem on these screens anyway, so it looks like a Full Width section unless the viewport is wider than 90rem.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The same page at 375 px: 1.5rem side padding, 2.5rem section spacing, the background image cropped around its centre.

Mobile (below 600 px): side padding 1.5rem (`--spacing-s`), section spacing 2.5rem (`--spacing-m`). Background images use the 750 px-wide rendition and `object-fit: cover`, so the left and right edges of wide images are cut off.

6.5.3 Component Variations

Variation 1 — Default (Full Width)

Variation Name: Default section (Width Type: Full Width, no Style).

Business Purpose: The standard band for most content.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-default.png`
> **Purpose:** Default section

> **Description:** A white section with a Title, Text and a Teaser, content capped at the page width with standard side padding.

Use Cases: Body content, introductions, any block that brings its own background.

Authoring Guidelines: Add a Section; leave Style empty and Width Type on Full Width.

Business Rules: Full Width is the default and adds no class.

Rendering Rules: `main > .section` with `--section-max-width: var(--page-max-width)` (120rem); each wrapper (`main > .section > div`) is centred with `margin: auto` and `padding: 0 var(--section-padding-inline)`; vertical `margin: var(--section-spacing) 0`; first section `margin-top: 0`.

Responsive Behaviour: Desktop: 5rem side padding, 7.5rem spacing. Tablet: 2.5rem side padding, 2.5rem / 5rem spacing (below / from 900 px). Mobile: 1.5rem side padding, 2.5rem spacing.

SEO Impact: None by itself; sections are `div` elements. Use headings inside to structure the content.

Accessibility Compliance: No landmark or role is added; the reading order follows the content tree order.

Variation 2 — Highlight

Variation Name: Highlight (Style: Highlight).

Business Purpose: Draw attention to a band, or alternate backgrounds to separate topics.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/section-highlight.png`
> **Purpose:** Highlight section
> **Description:** A light grey band spanning the viewport width with a Title, Text and a Button, between two white sections.

Use Cases: Key messages, statistics, a call-to-action band, alternating content bands.

Authoring Guidelines: Select the section; in Style choose **Highlight**.

Business Rules: Style is a multi-select with a single option (`highlight`); the value is added as a class.

Rendering Rules: `main .section.highlight` (and the CSS-only alias `.light`): `background-color: var(--color-bg-secondary)` (grey-100), `margin: 0`, `padding: var(--section-spacing) 0`. Two consecutive Highlight sections touch without a gap.

Responsive Behaviour: The padding follows `--section-spacing`: 7.5rem desktop, 5rem from 900 px, 2.5rem below 900 px.

SEO Impact: None.

Accessibility Compliance: Text colour stays the default (`--color-text-default`, black) on grey-100, which keeps body text contrast well above 4.5:1. Do not rely on the background alone to convey meaning (WCAG 1.4.1).

Variation 3 — Fixed Width

Variation Name: Fixed Width (Width Type: Fixed Width).

Business Purpose: Constrain a band to the content width (90rem) on very wide screens, for a calmer, card-like layout.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-fixed-width.png`
> **Purpose:** Fixed Width section
> **Description:** On a 1920 px screen, a Highlight section with Fixed Width: the grey band is 1440 px wide and centred, with white space left and right.

Use Cases: Highlight or background bands that should not run edge to edge on large monitors; combined with Rounded edge for a framed band.

Authoring Guidelines: Set Width Type to **Fixed Width**.

Business Rules: The class `fixed-width` is added only when the value normalises to `fixed-width`.

Rendering Rules: `--section-max-width: var(--content-max-width)` (90rem) for the wrappers, and the section box itself gets `max-width: var(--content-max-width); margin-inline: auto` (after the Style and background rules, so a Highlight or background section is also centred at 90rem).

Responsive Behaviour: Visible only on viewports wider than 90rem (1440 px); below that it looks like Full Width, except that the wrapper width cap is 90rem instead of 120rem.

SEO Impact: None.

Accessibility Compliance: No effect; content reflows at 320 px width (WCAG 1.4.10).

Variation 4 — Rounded edge

Variation Name: Rounded edge (Rounded edge: on).

Business Purpose: Soften the corners of a coloured or image band.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-rounded-edge.png`
> **Purpose:** Rounded edge section
> **Description:** A Fixed Width Highlight section with 16 px rounded corners on a 1920 px screen.

Use Cases: Framed call-to-action bands, image bands inside a page with white margins.

Authoring Guidelines: Switch **Rounded edge** on; combine with Highlight or a Background Image, and with Fixed Width so the corners are visible.

Business Rules: The class `rounded` is added only when the value is `true` (case-insensitive).

Rendering Rules: `border-radius: var(--radius-l)` (1rem) and `overflow: hidden` (content and background are clipped to the corners).

Responsive Behaviour: Same radius at all sizes. A Full Width section spans the viewport, so its rounded corners sit at the screen edges; combine with Fixed Width for a visible frame on wide screens.

SEO Impact: None.

Accessibility Compliance: `overflow: hidden` can clip focus outlines of elements touching the edge; keep content inside the default padding.

Variation 5 — Background Image

Variation Name: Background image (Background Image set, optional Image alt text).

Business Purpose: Place content over a full-cover image for campaign or mood bands.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-background-image.png`
> **Purpose:** Section with background image
> **Description:** A section with a full-cover photograph behind a white-on-image Title and Button; the photograph fills the band edge to edge.

Use Cases: Campaign bands, event teasers, section openers built from default content.

Authoring Guidelines: Pick an image in **Background Image**; fill **Image alt text** only if the image conveys information, otherwise leave it empty.

Business Rules: The image comes from the field's picture / image, else from its link or URL text; site media (`/media_`) is optimised, other URLs are used as they are. The alt field wins over an alt that AEM merged into the image; empty means decorative (`alt=""`).

Rendering Rules: A `div.section-background` with the picture is prepended to the section; the section gets `has-background` (`position: relative; isolation: isolate; margin: 0; padding: var(--section-spacing) 0`). The background is absolutely positioned (`inset: 0; z-index: -1`) and the image fills it (`width/height: 100%; object-fit: cover`). No overlay is added and the text colour does not change.

Responsive Behaviour: Picture sources: 2000 px wide from 600 px viewport width, 750 px below. Cropped by `object-fit: cover` to the band's shape at each size; the band height is set by its content plus the section spacing padding.

SEO Impact: Background images are real `<img>` elements; with alt text they can be indexed as images. A decorative background (empty alt) has no SEO weight.

Accessibility Compliance: Authors must ensure the text over the image meets 4.5:1 contrast (WCAG 1.4.3); the code adds no scrim. Decorative images get `alt=""` so screen readers skip them.

Variation 6 — Anchored section

Variation Name: Anchored section (Anchor name set).

Business Purpose: Let Anchor List links and `#links` jump to a section.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-anchored.png`
> **Purpose:** Anchored section
> **Description:** An Anchor List at the top of the page and the browser scrolled to the "Overview" section after selecting its link, with `#overview` in the address bar.

Use Cases: Long pages with chapter navigation, deep links from campaigns.

Authoring Guidelines: Enter a short lower-case name such as `overview` in **Anchor name**; use the same value in the Anchor List item.

Business Rules: A leading `#` is removed and the value is normalised (lower case, other characters become hyphens). The id is set only if no element already has it; there is no suffix.

Rendering Rules: `section.id = {anchor}`. After loading, the page scrolls to the element named in the URL hash.

Responsive Behaviour: None.

SEO Impact: Fragment links (`#overview`) can appear as "jump to" links in search results; keep anchors stable.

Accessibility Compliance: Supports WCAG 2.4.1 (bypass blocks) when used with an Anchor List.

> **Note:** `styles/styles.css` also styles the section classes `light` (same as Highlight), `dark` (black background `--color-bg-neutral`, white text, brand-300 links, dark-variant buttons) and `brand` (`--color-bg-brand-secondary-hover`). The section model offers only **Highlight**, so `dark`, `brand` and `light` are **not authorable** on sections (as of this handover); the `gen-carousel-section` and `tabs-section` styles are set by the Gen Carousel and Tabs Container definitions, not by authors.

6.5.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-authoring-view.png`
> **Purpose:** Section on the Universal Editor canvas
> **Description:** A selected section on the canvas (blue selection frame around the band) with its background image, a Title and a Button inside, and the section highlighted in the content tree. Mark A to D.

Annotations:

- A = the section band (select it to edit Section Name, Style, Width Type, Rounded edge, Anchor name)
- B = the section background image (Background Image, Image alt text)
- C = default content inside the section (edited through its own component)
- D = the section entry in the content tree (shows the Section Name)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-properties-panel.png`
> **Purpose:** Section properties panel
> **Description:** The Properties panel of a section showing Section Name, Style, Width Type, Rounded edge, Background Image, Image alt text and Anchor name.

The section dialog has a single list of fields (no tabs): **Section Name** (text, content tree label), **Style** (multi-select: Highlight), **Width Type** (select: Full Width / Fixed Width), **Rounded edge** (switch), **Background Image** (Assets picker), **Image alt text** (text, with the help text "Describe the image for screen readers. Leave empty if the image is decorative.") and **Anchor name** (text, help text "Lets Anchor List links jump to this section, e.g. overview (lowercase, no spaces)."). There is no Additional tab on sections.

6.5.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In the Sites console select the page and select **Edit**.

Expected Result: The page opens on the canvas; the content tree shows Main and its sections.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-step-01-open-page.png`
> **Purpose:** Step 1
> **Description:** The page open in the Universal Editor with the content tree expanded to Main.

Step 2 — Add a section

Action: Select **Main** in the content tree, select **Add (+)** and choose **Section**.

Expected Result: An empty section appears at the end of the page (drag it to its position in the content tree if needed).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-step-02-add-section.png`
> **Purpose:** Step 2
> **Description:** The Add (+) picker at Main level showing Section, Gen Carousel and Tabs Container, with Section highlighted.

Step 3 — Name the section

Action: With the section selected, enter a **Section Name** such as "Services overview".

Expected Result: The content tree shows the new label; nothing changes on the page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-step-03-name.png`
> **Purpose:** Step 3
> **Description:** The Section Name field filled in and the content tree showing the label.

Step 4 — Choose the style and width

Action: Select **Highlight** in Style if the band should be grey; choose **Fixed Width** in Width Type if it should be limited to the content width; switch **Rounded edge** on if wanted.

Expected Result: The canvas updates with the grey band, the narrower box and / or rounded corners.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-step-04-style-width.png`
> **Purpose:** Step 4
> **Description:** Style Highlight, Width Type Fixed Width and Rounded edge on, with the canvas showing the framed grey band.

Step 5 — Add a background image (optional)

Action: Select **Background Image**, pick an image in AEM Assets, and fill **Image alt text** only if the image is informative.

Expected Result: The image covers the band behind the content.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-step-05-background.png`
> **Purpose:** Step 5
> **Description:** The Assets picker opened from Background Image and the canvas showing the image behind the section content.

Step 6 — Set the anchor (optional)

Action: Enter an **Anchor name**, for example `overview`.

Expected Result: The section gets the id `overview`; `#overview` links jump to it.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-step-06-anchor.png`
> **Purpose:** Step 6
> **Description:** The Anchor name field with `overview` and an Anchor List item pointing to it.

Step 7 — Add content

Action: Select the section, select **Add (+)** and insert Title, Text, Button, Image or blocks.

Expected Result: The components appear inside the band in content tree order.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-step-07-add-content.png`
> **Purpose:** Step 7
> **Description:** The section with a Title, a Text and a Cards block added, shown in the content tree.

Step 8 — Preview

Action: Switch on **Preview** mode and check the section at desktop, tablet and mobile sizes.

Expected Result: Spacing and background behave as described in 6.5.2.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-step-08-preview.png`
> **Purpose:** Step 8
> **Description:** The section in Preview mode at mobile size.

Step 9 — Validate and publish

Action: Run the checklist below, then **Publish** to Preview and to Live.

Expected Result: The section shows on `.aem.page` and then `.aem.live`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/section-step-09-publish.png`
> **Purpose:** Step 9
> **Description:** The Publish dialog listing the page and the background image asset.

Preview Process
Publish to Preview and open `https://main--tcs-corp--sradha-tcs.aem.page/{language}/{path}`. Check the section at 375 px, 768 px and 1440 px (browser developer tools device mode). For a Fixed Width section, also check a viewport wider than 1440 px.

Publishing Process
Publish the page to Live from the Universal Editor or with Quick Publish / Manage Publication in the Sites console. Include the background image asset in the publication (the dialog lists references).

Validation Process
- ☐ The section appears in the intended order.
- ☐ Highlight band is grey and spans the viewport (or 90rem when Fixed Width).
- ☐ Background image covers the band; the important part of the image is not cropped away on mobile.
- ☐ Text over the background image has sufficient contrast (4.5:1 for body text).
- ☐ Background alt text is filled only for informative images.
- ☐ The anchor id is present (`#overview` in the address bar scrolls to the section).
- ☐ Rounded corners are visible (Fixed Width on wide screens, or with a background).
- ☐ No unexpected empty section remains (for example when all its blocks are hidden).

Rollback Process
Undo the field change in the Properties panel, or restore the previous page version in the Sites console (Timeline > Versions > Revert to this Version), then republish. Deleting a section deletes its content: restore the page version to recover it.

6.5.6 Authorable Field Specification
Model `section` (single panel, no tabs)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Section Name (`name`)	Text	No	No	(empty)	Label in the content tree	Short, descriptive name of the band's topic	None	Services overview
Style (`style`)	Multi-select	No	Yes	(empty)	Background style of the band	Choose Highlight for a grey band	Enforced: options list (Highlight)	highlight
Width Type (`sectionWidth`)	Select	No	No	Full Width (`full-width`)	Width of the band	Fixed Width for a 90rem band on wide screens	Enforced: Full Width or Fixed Width	fixed-width
Rounded edge (`roundedEdge`)	Boolean	No	No	false	Rounded corners	Combine with Highlight / background and Fixed Width	None	true
Background Image (`backgroundImage`)	Reference (AEM Assets)	No	No	(empty)	Image behind the section	Wide, high-resolution image; important subject in the centre	None	/content/dam/tcs-corp/campaigns/ai-band.jpg
Image alt text (`backgroundImageAlt`)	Text	No	No	(empty)	Accessible description of the background	Empty if decorative	None	Engineers in a TCS innovation lab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor name (`anchorName`)	Text	No	No	(empty)	Section id for Anchor List and `#links`	Lower case, no spaces, unique on the page	None in the model; normalised by code	overview

Additional tab: Not applicable. Sections have no Additional tab (the Anchor ID, Hide block, padding and margin fields exist only on blocks).

Field details

- **Section Name** — Purpose: content tree label. Allowed Values: any text. Character Limits: Recommended up to 40 characters. Business Rules: not rendered on the site. Dependency Rules: none. Visibility Rules: always visible. Sample Content: "Case studies".
- **Style** — Purpose: background style. Allowed Values: `highlight`. Character Limits: not applicable. Business Rules: each selected value becomes a class (normalised). Dependency Rules: combine with Rounded edge and Fixed Width as needed. Visibility Rules: always visible. Sample Content: Highlight.
- **Width Type** — Purpose: band width. Allowed Values: `full-width`, `fixed-width`. Character Limits: not applicable. Business Rules: only `fixed-width` adds a class. Dependency Rules: affects Highlight, background and rounded corners (the box is centred at 90rem). Visibility Rules: always visible. Sample Content: Fixed Width.
- **Rounded edge** — Purpose: corner radius 1rem. Allowed Values: on / off. Character Limits: not applicable. Business Rules: applied when the value is `true`. Dependency Rules: visible only with a background (Highlight or image) or Fixed Width on wide screens. Visibility Rules: always visible. Sample Content: on.
- **Background Image** — Purpose: full-cover background. Allowed Values: an AEM Assets image (or a URL). Character Limits: not applicable. Business Rules: site media is optimised (2000 px from 600 px, 750 px below); the first section's background image is loaded eagerly as the LCP candidate, others lazily. Dependency Rules: Image alt text. Visibility Rules: always visible. Sample Content: a 2400 × 1200 px photograph.
- **Image alt text** — Purpose: background alt. Allowed Values: text. Character Limits: Recommended up to 125 characters. Business Rules: wins over an alt merged by AEM; empty becomes `alt=""`. Dependency Rules: used only with Background Image. Visibility Rules: always visible. Sample Content: "Aerial view of the TCS Siruseri campus".
- **Anchor name** — Purpose: section id. Allowed Values: text; normalised to lower case letters, digits and hyphens. Character Limits: Recommended up to 30 characters. Business Rules: leading `#` removed; not set if the id already exists. Dependency Rules: Anchor List items must use the same value. Visibility Rules: always visible. Sample Content: `overview`.

6.5.7 Field Annotation Mapping

Marker	Field	Purpose
A	Section Name; Style; Width Type; Rounded edge; Anchor name	Properties of the selected band
B	Background Image; Image alt text	Background image and its alt text
C	(child components)	Content of the section, edited through each component
D	Section Name	Label shown in the content tree

6.5.8 Component Specification

Attribute	Value
Component Name	Section (`section`, resource type `core/franklin/components/section/v1/section`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`models/_section.json`; `scripts/scripts.js`; `scripts/aem.js`; `decorators/responsive-image-decorator.js`; `styles/styles.css`
Component Type	Container (section)
Supported Variants	Default (Full Width); Highlight; Fixed Width; Rounded edge; Background Image; Anchored section (combinable)
Supported Templates	Any page
Supported Metadata	None (section properties only)
Supported Tags	None
Supported Personalization	None (as of this handover)
Caching Considerations	Part of the page HTML, cached by the Edge Delivery CDN and purged on publish; background images are served from the media store with long cache lifetimes
Performance Considerations	The first section is loaded in the eager phase (its first image, often the background, gets `loading="eager"` and LCP waits for it); later sections load in the lazy phase; each section is hidden (`display: none`) until its blocks are loaded
Accessibility Compliance	WCAG 2.1 AA: author-controlled contrast over backgrounds (1.4.3), decorative backgrounds `alt=""` (1.1.1), anchors support bypass navigation (2.4.1), reflow at 320 px (1.4.10)
SEO Compliance	No semantic element is added; headings inside sections carry the structure; informative backgrounds can carry alt text

6.5.9 Input Model

- **Content Inputs:** the components in the section (default content and blocks).
- **Metadata Inputs:** none.
- **Tags Inputs:** none.
- **Page Context Inputs:** position of the section (the first section is eager); existing ids on the page (anchor conflicts).
- **Query Inputs:** none.
- **Manual Inputs:** the seven section fields.

Allowed children (filter `section`): Text, Image, Button, Title, Page Header, Hero, Teaser, Quote, Testimonial, Anchor List, Accordion, Tabs, Table, Table List, List, Separator, Embed, Video, Modal, Form Variations, Filter List, Interactive Dashboard, Conversation Canvas, Cards, Columns, Fragment, Footer.

Authored HTML received from AEM (a section with options; the code reads the options either as data attributes of the section or as rows of a `section-metadata` block, then removes them):

```
<div>
  <h2>Our services</h2>
  <p>From cloud to AI, ...</p>
  <div class="teaser"><div><div>...</div></div></div>
  <div class="section-metadata">
    <div><div>style</div><div>highlight</div></div>
    <div><div>sectionWidth</div><div>fixed-width</div></div>
```

```
    <div><div>roundedEdge</div><div>true</div></div>
    <div><div>backgroundImage</div><div><picture>...</picture></div></div>
    <div><div>backgroundImageAlt</div><div>Engineers in a TCS lab</div></div>
    <div><div>anchorName</div><div>overview</div></div>
  </div>
</div>
```

6.5.10 Output Model

Rendered Output: `main > div.section` with the classes `highlight`, `fixed-width`, `rounded`, `has-background` as applicable, an optional `id`, a first child `div.section-background` (when a background is set), then one wrapper per run of default content (`div.default-content-wrapper`) or per block (`div.{block}-wrapper` containing `div.{block}.block`). Status attribute `data-section-status` (`initialized`, `loading`, `loaded`).

Generated HTML:

```
<div class="section highlight fixed-width rounded has-background" id="overview"
     data-section-status="loaded">
  <div class="section-background">
    <picture>
      <source type="image/webp" srcset="./media_1a2b.jpg?width=2000&format=webply..."
              media="(min-width: 600px)">
      <img loading="eager" alt="Engineers in a TCS lab" src="./media_1a2b.jpg?
width=750...">
    </picture>
  </div>
  <div class="default-content-wrapper"><h2 id="our-services">Our services</h2><p>...</p></
div>
  <div class="teaser-wrapper"><div class="teaser block" data-block-name="teaser">...</
div></div>
</div>
```

Generated JSON: none.

Generated Metadata: none.

Analytics Tracking: sections are not components in the data layer (only `.block` elements are). Clicks on links in a section's default content are pushed as `cmp:click` with the page as the component (`eventInfo.path` `component.page-{hash}`); blocks in the section are tracked as their own components with the page as `parentId` (Part 11).

Personalization Output: none.

Search Index Output: none (the query index reads page metadata, not sections).

6.5.11 Business Rules

1. Section options are read before the section is decorated, from the section's data attributes (`sectionwidth`, `roundededge`, `backgroundimage`, `backgroundimagealt`, `anchorname`) or from `section-metadata` rows whose normalised names match; matching rows are removed.
2. Style values are split at commas, normalised (`toClassName`) and added as classes; empty values are ignored.
3. Width Type adds `fixed-width` only when the value normalises to `fixed-width`; Full Width (or any other value) adds nothing.
4. Rounded edge adds `rounded` only when the value is `true` (case-insensitive).
5. Anchor name: a leading `#` is removed, the rest is normalised; the id is set only when no element with that id exists (no suffix is added; the anchor is silently skipped on a conflict).

6. Background image: the cell's `<picture>` or `<img>` is used; otherwise the cell link or the URL text is built into an image; same-origin `/media_` URLs become an optimised picture (2000 px from 600 px, 750 px below), other URLs a plain lazy `<img>`.
7. Background alt: the Image alt text wins; otherwise an alt already on the image is kept; otherwise `alt=""` (decorative).
8. When a background exists, `div.section-background` is prepended and the section gets `has-background` (no outer margin, section-spacing padding).
9. Highlight, background, and the CSS-only `light`, `dark`, `brand` styles remove the outer margin and add section-spacing padding.
10. Fixed Width also limits and centres the section box itself (after the Style and background rules).
11. The first section has no top margin.
12. A section whose Style is `gen-carousel-section` or `tabs-section` is turned into a Gen Carousel or Tabs container (set by those definitions, not by the Section model).
13. A wrapper whose block has Hide block on is not displayed outside edit mode; the section itself stays.
14. The first section is loaded in the eager phase and waits for its first image; all other sections load in the lazy phase.
15. In the Universal Editor, a changed section is re-decorated in place (`decorateSectionsWithOptions`, blocks, block options); if that fails the canvas reloads.

6.5.12 Author Best Practices

Recommended Content Length: one topic per section; typically one heading, up to three paragraphs and one block, or one block with its own heading.

Image Dimensions: background images at least 2000 px wide (served at 2000 px from 600 px viewport width and 750 px below); landscape 2:1 to 3:1; keep the subject in the centre third because `object-fit: cover` crops the edges.

CTA Writing Guidelines: a background or Highlight band should have at most one primary button; use action verbs ("Explore our services").

Accessibility Guidelines: check text contrast over images; leave the alt text empty for decorative backgrounds; give each section a heading so screen reader users can navigate by headings.

SEO Guidelines: structure sections with h2 headings; do not put important text only in background images.

Metadata Guidelines: not applicable (sections have no metadata).

Taxonomy Guidelines: not applicable.

Localization Guidelines: Section Name and Anchor name are not translated on the site (Anchor name is technical; keep it identical across locales if campaigns link to it); translate background alt text per locale.

6.5.13 Troubleshooting Guide

Issue 1 — Anchor link does not jump to the section

Symptoms: Selecting an Anchor List link or `#overview` URL stays at the top.

Root Cause: The anchor name differs (case, spaces, typo), or another element already has that id so the section was skipped.

Diagnosis Steps: In the browser developer tools, search the DOM for `id="overview"`; check which element has it.

Resolution Steps: Use the same normalised value in the section and the link; rename the conflicting block Anchor ID or heading.

Validation Steps: Open the page with `#overview`; the page scrolls to the section.

Preventive Measures: Use lower-case, hyphenated anchors unique on the page.

Issue 2 — Background image does not appear

Symptoms: The section has no image on preview or live.

Root Cause: The asset was not published, or the field holds a URL that is not reachable.

Diagnosis Steps: Inspect the section for `div.section-background`; check the image URL in the network panel.

Resolution Steps: Publish the asset (Manage Publication with references), re-pick the image, republish the page.

Validation Steps: The section has the class `has-background` and the image loads.

Preventive Measures: Always publish with references.

Issue 3 — Text is unreadable over the background image

Symptoms: Low contrast between text and image.

Root Cause: The code adds no overlay or text colour change.

Diagnosis Steps: Check contrast with a contrast checker at all three sizes.

Resolution Steps: Choose a calmer or darker / lighter image, or move the text into a block with its own background.

Validation Steps: Contrast 4.5:1 for body text, 3:1 for large text.

Preventive Measures: Agree background images with the design team. [To be confirmed: request for a dark section style or overlay — owner: product owner]

Issue 4 — Rounded corners are not visible

Symptoms: Rounded edge is on but the corners look square.

Root Cause: The section has no background and is full width, so the corners are at the screen edges and invisible on white.

Diagnosis Steps: Check that the section has `rounded` and either `highlight` or `has-background`.

Resolution Steps: Add Highlight or a background image and use Fixed Width.

Validation Steps: On a wide screen the framed band shows rounded corners.

Preventive Measures: Use Rounded edge only with a background.

Issue 5 — Two highlight bands merge

Symptoms: Two consecutive Highlight sections look like one band.

Root Cause: Highlight sections have no outer margin by design.

Diagnosis Steps: Check the content tree order.

Resolution Steps: Put a default section between them or merge them into one section.

Validation Steps: The bands are visually separated.

Preventive Measures: Alternate Highlight and default sections.

Issue 6 — Fixed Width has no visible effect

Symptoms: A Fixed Width section looks the same as Full Width.

Root Cause: The viewport is narrower than 90rem (1440 px).

Diagnosis Steps: Check the viewport width.

Resolution Steps: None needed; check on a wider screen.

Validation Steps: At 1920 px the band is centred at 1440 px.

Preventive Measures: Explain to reviewers that Fixed Width matters only on large screens.

Issue 7 — Dark or brand section requested

Symptoms: An author cannot find a Dark or Brand style.

Root Cause: Only Highlight is offered in the section model; `dark` and `brand` exist in CSS only.

Diagnosis Steps: Open the Style select (only Highlight).

Resolution Steps: Use a block with a background colour option, or request the style options from the development team.

Validation Steps: Not applicable.

Preventive Measures: Plan section styles with the design team.

Issue 8 — An empty gap remains where content was hidden

Symptoms: Extra white space in the page.

Root Cause: All blocks of the section have Hide block on; the section and its spacing remain.

Diagnosis Steps: In edit mode look for faded blocks in the section.

Resolution Steps: Hide or delete the whole section's content differently (for example move it to another page) or delete the section.

Validation Steps: The gap is gone on preview.

Preventive Measures: Hide whole sections by removing them, not by hiding every block.

6.5.14 Support Runbook

Monitoring Requirements: No section-specific monitoring. Core Web Vitals (LCP) of pages whose first section has a background image should be checked in PageSpeed Insights after large changes. [To be confirmed: RUM / monitoring tool — owner: site administrator]

Logs: Sections emit no console messages. Editor re-decoration errors show as a full canvas reload. Publish status can be checked with the Admin API (`https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`, authenticated) and in the AEM author replication log.

Common Errors:

Error	Cause	Fix
Anchor ignored	Id already used	Rename one of the ids
No background	Asset not published	Publish the asset
Background blurry	Small original image	Upload a 2000 px or wider original
Contrast failure	Busy image	Change the image
Unexpected grey band	Highlight selected	Clear Style

Escalation Matrix:

Level	Who	Checks
L1	Author support	Field values, publication of page and asset, browser cache
L2	AMS / site administrator	Publish status (Admin API), CDN cache, content source mapping (`paths.json`)
L3	Development team	`scripts/scripts.js` section code, `styles/styles.css` section rules

Level	Who	Checks
Adobe Support	Adobe	Edge Delivery Services or AEM author outages, Universal Editor defects

Recovery Procedures: Revert the section's field values or restore the page version and republish.

Health Checks: Open a page with each section variation in the locale's block library on author and on preview after each code release.

Operational Procedures: Section style options are changed by developers in `models/_section.json` and `styles/styles.css`, followed by `npm run build:json`.

Known Limitations: Only Highlight is authorable; no overlay or text colour for background images; no per-breakpoint background image; section anchors are not de-duplicated; sections have no Additional tab (no Hide, padding or margin options).

6.6 Text Component

6.6.1 Component Purpose

Business Objective: Let authors write and format running text (paragraphs, lists, links, emphasis) directly on the page, with the site's typography applied automatically.

Business Use Cases:

- Introductions and body copy on any page.
- Lists of benefits, steps or facts.
- Inline links to other pages, documents or external sites.
- Quick calls to action: a link on its own line becomes a link button, or a button when it is bold and / or italic.

When To Use: For any flowing text that does not need a block's layout.

When Not To Use:

- For a heading on its own: use **Title** (explicit heading level).
- For a call to action that needs a specific view, icon or alignment: use **Button**.
- For text with an image side by side: use **Teaser** or **Columns**.
- For tabular data: use **Table**.

Dependencies:

Type	Item
Model	`models/_text.json` (definition `text`, resource type `core/franklin/components/text/v1/text`; no model fields: the content is rich text)
Code	`decorators/content-buttons-decorator.js` (`decorateButtons`), `utils/buttons.js` (`readButtonOptions`), `decorators/button-decorator.js`; `scripts/aem.js` (`decorateIcons`); `scripts/editor-support-rte.js` (`decorateRichtext`, groups rich text for inline editing)
Styles	`styles/styles.css` (body typography, headings, lists, links, buttons)
Data sources / sheets / placeholders	None

Related Components: Title, Button, Image (other default content); Columns (Text inside a Column); Tabs Container (Text inside tab panels).

Applicable Templates: Any page.

Supported Content Types: Paragraphs, headings (when available in the rich text toolbar), bulleted and numbered lists, bold, italic, links (internal pages, assets, external URLs, `mailto:`), line breaks.

6.6.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A Text component at 1440 px with two paragraphs, a bulleted list, an inline link in brand blue and a standalone link rendered as a link button with a chevron.

Desktop (1200 px and above): body text in Inter (`--body-font-family`) at `--body-font-size-m` = 1.125rem (from 900 px), line height 1.6, letter-spacing –0.005 × font size. Paragraphs and lists have a top margin of 0.8 × and a bottom margin of 0.25 × the body size. Links are brand blue (`--color-text-brand`), not underlined, underlined and brand-600 on hover, with a 0.125rem brand outline on keyboard focus. Text spans the section content width (up to 120rem minus 5rem side padding).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Text at 768 px, wrapping within the 2.5rem section padding.

Tablet (600–1199 px): body text 1rem below 900 px and 1.125rem from 900 px; 2.5rem section side padding.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The same Text at 375 px with 1.5rem side padding; the link button on its own line.

Mobile (below 600 px): body text 1rem; 1.5rem side padding; long URLs wrap (`overflow-wrap: break-word` on links).

6.6.3 Component Variations

Variation 1 — Rich text paragraphs

Variation Name: Paragraphs (plain rich text).

Business Purpose: Readable body copy.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-paragraphs.png`
> **Purpose:** Paragraphs
> **Description:** Two paragraphs with a bold phrase and an italic phrase.

Use Cases: Introductions, descriptions, explanations.

Authoring Guidelines: Add a Text component and type on the canvas; use the toolbar for bold and italic.

Business Rules: Bold renders with weight 600 (`strong`, `b`). Bold or italic applied to a whole standalone link changes it into a button (Variations 5–7).

Rendering Rules: `<p>` elements in `div.default-content-wrapper`, styled by the global `p` rules.

Responsive Behaviour: Desktop and tablet from 900 px: 1.125rem; below 900 px: 1rem.

SEO Impact: Indexable text in the delivered HTML.

Accessibility Compliance: Line height 1.6 and paragraph spacing support WCAG 1.4.12 (text spacing); text colour black on white.

Variation 2 — Lists

Variation Name: Bulleted and numbered lists.

Business Purpose: Scannable enumerations.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-lists.png`
> **Purpose:** Lists
> **Description:** A bulleted list of four services and a numbered list of three steps.

Use Cases: Benefits, features, steps.

Authoring Guidelines: Use the list buttons of the rich text toolbar.

Business Rules: None beyond the browser defaults.

Rendering Rules: `<ul>` / `<ol>` with the same margins as paragraphs.

Responsive Behaviour: Same as paragraphs.

SEO Impact: Lists are semantic and can be used for featured snippets.

Accessibility Compliance: Screen readers announce the list and item count.

Variation 3 — Inline link

Variation Name: Link inside a sentence.

Business Purpose: Cross-linking in context.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-inline-link.png`
> **Purpose:** Inline link
> **Description:** A paragraph with “learn more about our cloud services” linked in brand blue, shown in its hover state (underlined).

Use Cases: References to other pages, documents and external sources.

Authoring Guidelines: Select words in a sentence and add a link with the toolbar; pick internal pages with the content picker.

Business Rules: A link is left as a plain link when its paragraph contains other text. Every link in a paragraph gets a `title` equal to its text when it has no title.

Rendering Rules: `a:any-link` brand colour, no underline; hover underline and brand-600.

Responsive Behaviour: Wraps with the text; long URLs break.

SEO Impact: Descriptive anchor text helps search engines understand the target.

Accessibility Compliance: Link purpose must be clear from the link text (WCAG 2.4.4); colour plus hover underline; keyboard focus outline.

Variation 4 — Standalone link (link button)

Variation Name: Link on its own line, no formatting.

Business Purpose: A low-emphasis call to action without a button container.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-link-button.png`
> **Purpose:** Link button from text
> **Description:** A paragraph that contains only the link "Read the report", rendered underlined and semibold with a trailing chevron.

Use Cases: "Read more", "View all", secondary navigation at the end of a text.

Authoring Guidelines: Put the link alone in its own paragraph without bold or italic.

Business Rules: Applies when the paragraph's text equals the link text, the link text is not its own URL, and the link does not wrap an image.

Rendering Rules: The paragraph becomes `p.button-wrapper` and the link `a.link-button` (semibold 1rem, underlined, default text colour, chevron-right icon 1rem; brand colour on hover).

Responsive Behaviour: Same at all sizes.

SEO Impact: Same as a link.

Accessibility Compliance: Underline and icon distinguish it from text (WCAG 1.4.1); the chevron is CSS-only and not announced.

Variation 5 — Bold standalone link (primary button)

Variation Name: Standalone link formatted bold.

Business Purpose: The main call to action of a text.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-primary-button.png`
> **Purpose:** Primary button from text
> **Description:** A black pill-shaped button "Contact us" with a trailing arrow, created from a bold link.

Use Cases: Contact, download, register.

Authoring Guidelines: Put the link alone in its paragraph and make the whole link bold.

Business Rules: Bold = primary; the bold formatting is removed after it has been read.

Rendering Rules: `a.button.primary`: black background and border (`--color-bg-neutral`), grey-300 text, medium weight, 3.125rem minimum height, full radius, padding 0.75rem 1rem 0.75rem 1.5rem, trailing arrow-right icon 1.5rem; hover text slate-400.

Responsive Behaviour: Same at all sizes; the label does not wrap (`white-space: nowrap`) and is clipped if wider than the container.

SEO Impact: Same as a link.

Accessibility Compliance: Keyboard focusable link with visible focus outline; label text is the accessible name.

Variation 6 — Italic standalone link (secondary button)

Variation Name: Standalone link formatted italic.

Business Purpose: A second, lower-emphasis call to action.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-secondary-button.png`
> **Purpose:** Secondary button from text
> **Description:** An outlined pill button "Explore services" with a trailing arrow, created from an italic link, next to its hover state with a blue border and text.

Use Cases: "Explore", "Learn more" next to a primary button.

Authoring Guidelines: Put the link alone in its paragraph and make it italic.

Business Rules: Italic = secondary.

Rendering Rules: `a.button.secondary`: transparent background, slate-900 border and text; hover white background, brand border and text.

Responsive Behaviour: As Variation 5.

SEO Impact: Same as a link.

Accessibility Compliance: As Variation 5.

Variation 7 — Bold and italic standalone link (tertiary button)

Variation Name: Standalone link formatted bold and italic.

Business Purpose: A text-only call to action with an arrow.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-tertiary-button.png`
> **Purpose:** Tertiary button from text
> **Description:** The text "See all case studies" with a trailing arrow and no container, created from a bold and italic link.

Use Cases: Inline-looking calls to action in dense layouts.

Authoring Guidelines: Put the link alone in its paragraph and make it bold and italic.

Business Rules: Bold + italic = tertiary.

Rendering Rules: `a.button.tertiary`: no border or padding, default text colour, arrow; hover brand colour and the arrow moves closer (gap 0).

Responsive Behaviour: As Variation 5.

SEO Impact: Same as a link.

Accessibility Compliance: As Variation 5.

6.6.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-authoring-view.png`
> **Purpose:** Text on the Universal Editor canvas
> **Description:** A Text component in inline-editing mode with the rich text toolbar visible, a paragraph, a list and a standalone bold link. Mark A to D.

Annotations:

- A = the rich text area (edited inline on the canvas)
- B = the rich text toolbar (bold, italic, lists, link)
- C = an inline link
- D = a standalone formatted link that becomes a button

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-properties-panel.png`
> **Purpose:** Text properties panel
> **Description:** The Properties panel with a Text component selected, showing the rich text field (no other fields, no tabs).

The Text component has no model fields (`models/_text.json` has an empty models list), so there are no tabs and no Additional tab. The content is the rich text itself, edited inline on the canvas (and, depending on the Universal Editor version, in a rich text field in the Properties panel). [To be confirmed: rich text toolbar options (headings, tables) enabled for this project — owner: AEM administrator]

6.6.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Sites console > select the page > **Edit**.

Expected Result: The page opens on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-step-01-open-page.png`
> **Purpose:** Step 1
> **Description:** The page open in the Universal Editor.

Step 2 — Add a Text component

Action: Select the section in the content tree, **Add (+)** > **Text**.

Expected Result: An empty Text component appears at the end of the section.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-step-02-add.png`
> **Purpose:** Step 2
> **Description:** The component picker with Text highlighted under Default Content.

Step 3 — Write the text

Action: Double-click the component on the canvas and type or paste the text. Use the toolbar for bold, italic and lists.

Expected Result: The text shows in the site's typography.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-step-03-write.png`
> **Purpose:** Step 3
> **Description:** Inline editing with the toolbar and two paragraphs typed.

Step 4 — Add links

Action: Select words and add a link; for internal pages pick the page with the content picker.

Expected Result: The words become a brand blue link.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-step-04-links.png`
> **Purpose:** Step 4
> **Description:** The link dialog with an internal page selected.

Step 5 — Add a call to action (optional)

Action: Put a link alone in a new paragraph. Leave it unformatted for a link button, make it bold for a primary button, italic for a secondary button, bold and italic for a tertiary button.

Expected Result: The link is shown as the chosen button style.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-step-05-cta.png`
> **Purpose:** Step 5
> **Description:** A bold standalone link rendered as a primary button on the canvas.

Step 6 — Preview

Action: Switch on Preview mode, test the links and check the three device sizes.

Expected Result: Links open their targets; buttons do not overflow on mobile.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-step-06-preview.png`

> **Purpose:** Step 6
> **Description:** Preview mode at mobile size.

Step 7 — Validate and publish

Action: Run the checklist, then publish to Preview and Live.

Expected Result: The text shows on `.aem.page` and `.aem.live`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/text-step-07-publish.png`
> **Purpose:** Step 7
> **Description:** The Publish dialog for the page.

Preview Process
Publish to Preview and open the `.aem.page` URL; click every link; check that links point to published URLs (`/en/...`), not to `/content/tcs-corp/...`.

Publishing Process
Publish the page to Live. Linked assets (PDFs) must be published too (include references in Manage Publication).

Validation Process
- ☐ Spelling and grammar checked; text matches the approved copy.
- ☐ Inline links have descriptive text (no "click here").
- ☐ Standalone links render as the intended style (link button, primary, secondary, tertiary).
- ☐ Button labels fit on one line on mobile.
- ☐ Lists use the list buttons (not typed dashes).
- ☐ No pasted formatting (fonts, colours) from other tools.
- ☐ External links point to the right site; document links open the published asset.

Rollback Process
Undo in the editor, or restore the page version in the Sites console and republish.

6.6.6 Authorable Field Specification
Model `text` (no model fields; rich text content)

Field Name	Field Type	Mandatory	Multi Value	De-fault Value	Business Purpose	Author Guidance	Valida-tion Rules	Example Value
Text (rich text content of the component)	Rich text	Recom-mended (an empty Text renders noth-ing)	No	(empty)	Body copy, lists, links	Short paragraphs; one idea per paragraph; standalone links for CTAs	None in the model	"TCS helps enter-prises … **Contact us**"

Additional tab: Not applicable. Default content has no Additional tab (no Anchor ID, Hide block, padding or margin).

Field details

- **Text** — Purpose: rich text body. Allowed Values: paragraphs, lists, bold, italic, links (and headings if enabled in the toolbar). Character Limits: none enforced; Recommended paragraphs of up to about 80 words, button labels up to about 25 characters (the button does not wrap). Business Rules: standalone links become buttons or link buttons by their formatting; every paragraph link without a title gets its text as title. Dependency Rules: none. Visibility Rules: always visible. Sample Content: "Our consultants help you modernise legacy systems." followed by a standalone bold link "Talk to an expert".

6.6.7 Field Annotation Mapping

Marker	Field	Purpose
A	Text	The rich text area
B	Text (toolbar)	Formatting of the rich text
C	Text (inline link)	Link in a sentence
D	Text (standalone formatted link)	Button created from a link

6.6.8 Component Specification

Attribute	Value
Component Name	Text (`text`, `core/franklin/components/text/v1/text`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`decorators/content-buttons-decorator.js`; `utils/buttons.js`; `decorators/button-decorator.js`; `scripts/editor-support-rte.js`; `styles/styles.css`
Component Type	Default content
Supported Variants	Paragraphs; Lists; Inline link; Link button; Primary, Secondary and Tertiary buttons from formatted links
Supported Templates	Any page
Supported Metadata	None
Supported Tags	None
Supported Personalization	None (as of this handover)
Caching Considerations	Part of the page HTML (Edge Delivery CDN, purged on publish)
Performance Considerations	No JavaScript of its own; buttons are decorated during the eager phase (`decorateMain`)
Accessibility Compliance	WCAG 2.1 AA: link purpose (2.4.4), focus visible (2.4.7), use of colour plus underline on hover (1.4.1), text spacing (1.4.12)
SEO Compliance	Server-rendered, indexable text and links

6.6.9 Input Model

- **Content Inputs:** the rich text.
- **Metadata Inputs:** none.
- **Tags Inputs:** none.
- **Page Context Inputs:** the section's style (link and button colours differ in the CSS-only `dark` section style).
- **Query Inputs:** none.
- **Manual Inputs:** the rich text.

Authored HTML received from AEM:

```
<p>TCS helps enterprises <a href="/en/services/cloud">modernise on the cloud</a>.</p>
<ul><li>Strategy</li><li>Migration</li></ul>
<p><strong><a href="/en/contact">Contact us</a></strong></p>
<p><a href="/en/insights">Read our insights</a></p>
```

6.6.10 Output Model

Rendered Output: the text in `div.default-content-wrapper`; standalone links turned into `p.button-wrapper > a.button.{type}` or `p.button-wrapper > a.link-button`.

Generated HTML:

```
<div class="default-content-wrapper">
  <p>TCS helps enterprises <a href="/en/services/cloud"
     title="modernise on the cloud">modernise on the cloud</a>.</p>
  <ul><li>Strategy</li><li>Migration</li></ul>
  <p class="button-wrapper"><a href="/en/contact" title="Contact us"
     class="button primary">Contact us</a></p>
  <p class="button-wrapper"><a href="/en/insights" title="Read our insights"
     class="link-button">Read our insights</a></p>
</div>
```

Generated JSON: none.

Generated Metadata: none.

Analytics Tracking: clicks on links in default content are pushed as `cmp:click` with the page as the component:

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.page-1x2y3z",
    "interaction": "link",
    "dc:title": "Contact us",
    "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/contact",
    "xdm:linkType": "other",
    "breakpoint": "desktop"
  }
}
```

The Text component itself is not a data layer component (default content is not a block).

Personalization Output: none.

Search Index Output: none directly (the query index uses page metadata; the text is indexed by search engines).

6.6.11 Business Rules

1. Only links inside paragraphs (`p a[href]`) are considered for buttons.
2. Every such link without a `title` gets its text as `title`.
3. A link stays a plain link when its paragraph contains any other text.
4. A link whose text is its own URL stays a plain link.
5. A link wrapping an image stays a plain link (unless it is an icon link).
6. Formatting decides the type: bold = primary, italic = secondary, bold + italic = tertiary; a type, `view-*` or `align-*` class or data value on the link or its paragraph also counts (a bare `button` class is not read as a view).

7. View: an explicit view wins; else an icon link becomes an icon button; else a link with a type becomes a button; else a link button.
8. An icon-only link without type or view stays plain.
9. The formatting element that encoded the type is removed after decoration.
10. The paragraph becomes `p.button-wrapper` (0.75rem vertical margin) and gets `button-align-{left|center|right}` when an alignment was read.
11. Buttons do not wrap their label (`white-space: nowrap; overflow: hidden; max-width: 100%`).
12. In the editor, after an inline edit the changed default content is replaced and buttons and icons are decorated again (`scripts/editor-support.js`).

6.6.12 Author Best Practices

Recommended Content Length: paragraphs of up to about 80 words; up to about 7 list items; button labels up to about 25 characters (Recommended; not enforced).

Image Dimensions: not applicable (use the Image component).

CTA Writing Guidelines: start with a verb ("Download the report"); one primary button per text; use secondary or link buttons for further options; avoid "Click here".

Accessibility Guidelines: descriptive link text; do not use bold or italic on standalone links unless you want a button; do not use colour alone for emphasis.

SEO Guidelines: use meaningful anchor text and link to related pages in the same locale.

Metadata Guidelines: not applicable.

Taxonomy Guidelines: not applicable.

Localization Guidelines: translate link text and button labels; link to the target locale's pages (`/de-de/...` from German pages); allow for text expansion (German and French labels can be 30 % longer), keeping button labels short.

6.6.13 Troubleshooting Guide

Issue 1 — A link did not become a button

Symptoms: A call-to-action link shows as a plain inline link.

Root Cause: The paragraph contains other text (even a space or a full stop), or several links share the paragraph.

Diagnosis Steps: Inspect the paragraph in developer tools; compare its text with the link text.

Resolution Steps: Put the link alone in its own paragraph.

Validation Steps: The paragraph has the class `button-wrapper`.

Preventive Measures: One CTA link per paragraph.

Issue 2 — A link became a button unintentionally

Symptoms: A bold link that should be inline shows as a black button.

Root Cause: It is alone in its paragraph and formatted bold.

Diagnosis Steps: Check the paragraph content.

Resolution Steps: Remove the bold formatting, or add the link to a sentence.

Validation Steps: It renders as a link or link button.

Preventive Measures: Use bold / italic on standalone links only for buttons.

Issue 3 — Button label is cut off on mobile

Symptoms: The end of a long button label is hidden.

Root Cause: Buttons do not wrap their text by design.

Diagnosis Steps: View at 375 px.

Resolution Steps: Shorten the label.

Validation Steps: Full label visible at 320 px width.

Preventive Measures: Keep labels short.

Issue 4 — Link shows the URL as text and is not a button

Symptoms: A standalone link with its URL as text is not decorated.

Root Cause: Links whose text is their URL are left alone on purpose.

Diagnosis Steps: Compare text and href.

Resolution Steps: Give the link a descriptive text.

Validation Steps: The link becomes a link button.

Preventive Measures: Never use raw URLs as link text.

Issue 5 — Internal link leads to a 404 on the live site

Symptoms: The link works on author but not on live.

Root Cause: The target page is not published, or the link points to an author path outside the locale mappings.

Diagnosis Steps: Check the href on the live page (`/en/...`) and open it.

Resolution Steps: Publish the target page; re-pick the link with the content picker.

Validation Steps: The link opens the page on live.

Preventive Measures: Publish linked pages first.

Issue 6 — Pasted text has odd formatting

Symptoms: Unexpected line breaks, empty paragraphs or spacing.

Root Cause: Formatting pasted from Word or e-mail.

Diagnosis Steps: Inspect for empty `<p>` or `<br>` elements.

Resolution Steps: Paste as plain text and reformat with the toolbar.

Validation Steps: Paragraph spacing is regular.

Preventive Measures: Paste as plain text.

Issue 7 — Text changes do not show on the canvas

Symptoms: The canvas keeps the old text after an edit.

Root Cause: The in-place update failed (the editor then reloads) or the session expired.

Diagnosis Steps: Reload the editor.

Resolution Steps: Re-enter the change; sign in again if needed.

Validation Steps: The canvas shows the new text after reload.

Preventive Measures: Avoid editing the same page in two tabs.

6.6.14 Support Runbook

Monitoring Requirements: None specific. Broken-link checks of published pages are recommended. [To be confirmed: link checking tool — owner: site administrator]

Logs: No console messages from the Text component. Publish status via the Admin API (`https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`) and AEM replication logs.

Common Errors:

Error	Cause	Fix
Link not a button	Other text in the paragraph	Separate paragraph
Unwanted button	Bold / italic standalone link	Remove formatting
Clipped label	Long label	Shorten
404 link	Target unpublished	Publish target
Author path in link	Link to `/content/...` outside the mappings	Re-pick with the content picker

Escalation Matrix:

Level	Who	Checks
L1	Author support	Paragraph structure, formatting, publication of targets
L2	AMS / site administrator	Publish status, `paths.json` mappings, cache
L3	Development team	`decorators/content-buttons-decorator.js`, `utils/buttons.js`, `styles/styles.css`
Adobe Support	Adobe	Universal Editor rich text defects, AEM rendering

Recovery Procedures: Restore the page version and republish.

Health Checks: After code releases, check the block library page with all four link styles on preview.

Operational Procedures: Changes to link and button styles are made by developers in `styles/styles.css` and the button decorators.

Known Limitations: No Additional tab; no text colour or size options; button labels never wrap; the button type depends on formatting, which is easy to apply by accident.

6.7 Title Component

6.7.1 Component Purpose

Business Objective: Provide headings with an explicit, author-chosen level so pages have a correct document outline for accessibility and SEO, with the design system's type scale.

Business Use Cases:

- The page heading (h1) on pages without a Page Header or Hero heading.
- Section headings (h2) and sub-headings (h3–h4).
- Small labels styled as headings (h5–h6) where the outline requires them.

When To Use: Whenever a heading stands on its own in default content.

When Not To Use:

- When a block already provides the heading (Page Header, Hero, Teaser, List and Cards headings).

- For visual emphasis of normal text (use bold in Text).

Dependencies:

Type	Item
Model	`models/_title.json` (definition `title`, resource type `core/franklin/components/title/v1/title`, model `title`)
Code	None specific (rendered by AEM / Edge Delivery; editor re-decoration via `scripts/editor-support.js`)
Styles	`styles/styles.css` (`h1`–`h6`, heading font size tokens)
Data sources / sheets / placeholders	None

Related Components: Text, Page Header, Hero, Anchor List (which can link to a section containing the heading).

Applicable Templates: Any page. On article, blog and press-release pages the template's Page Header already renders the h1.

Supported Content Types: Plain heading text.

6.7.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** Six Title components h1 to h6 stacked at 1440 px, showing the type scale.

Desktop (from 900 px): h1 3rem (line height 1.2, letter-spacing –0.03 × font size), h2 2.5rem, h3 1.75rem, h4 1.5rem, h5 1.25rem (light weight 300), h6 1rem (semibold 600); h1–h4 regular weight 400; line height 1.4 except h1. Each heading has a top margin of 0.8 × and a bottom margin of 0.25 × its font size.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same headings at 768 px.

Tablet: below 900 px the mobile sizes apply; from 900 px the desktop sizes.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The same headings at 375 px.

Mobile (below 900 px): h1 2rem, h2 1.75rem, h3 1.5rem, h4 1.125rem, h5 1rem, h6 0.875rem. Headings wrap; all have `scroll-margin: 2.5rem` so anchor jumps leave space above them.

6.7.3 Component Variations

Variation 1 — Heading level h1
Variation Name: Title Type h1.

Business Purpose: The main heading of the page.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/title-h1.png`
> **Purpose:** h1
> **Description:** An h1 “Cloud transformation services” at desktop size.

Use Cases: Page title on Default pages without a Page Header or a Hero with H1.

Authoring Guidelines: Title Type **h1**; exactly one h1 per page.

Business Rules: Recommended: one h1 per page (not enforced).

Rendering Rules: `h1`: `--heading-font-size-xxl`, line height 1.2, letter-spacing –0.03 × size.

Responsive Behaviour: 2rem below 900 px, 3rem from 900 px.

SEO Impact: Strongest on-page topic signal; should match the page title.

Accessibility Compliance: Screen reader users jump to the h1 to find the main content (WCAG 1.3.1, 2.4.6).

Variation 2 — Heading level h2
Variation Name: Title Type h2.

Business Purpose: Section headings.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/title-h2.png`
> **Purpose:** h2
> **Description:** An h2 “Our services” at desktop size.

Use Cases: One per section.

Authoring Guidelines: Title Type **h2**.

Business Rules: Follows the h1.

Rendering Rules: `--heading-font-size-xl`, letter-spacing –0.025 × size.

Responsive Behaviour: 1.75rem / 2.5rem.

SEO Impact: Structures the topics of the page.

Accessibility Compliance: Correct outline (WCAG 1.3.1).

Variation 3 — Heading level h3
Variation Name: Title Type h3.

Business Purpose: Sub-sections.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-h3.png`
> **Purpose:** h3
> **Description:** An h3 “Cloud migration” below an h2.

Use Cases: Topics within a section.

Authoring Guidelines: Title Type **h3** after an h2.

Business Rules: Do not skip levels (Recommended).

Rendering Rules: `--heading-font-size-l`, letter-spacing –0.02 × size.

Responsive Behaviour: 1.5rem / 1.75rem.

SEO Impact: Sub-topic structure.

Accessibility Compliance: As above.

Variation 4 — Heading level h4

Variation Name: Title Type h4.

Business Purpose: Third-level sub-sections.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-h4.png`
> **Purpose:** h4
> **Description:** An h4 below an h3.

Use Cases: Detailed documents, FAQs.

Authoring Guidelines: Title Type **h4** after an h3.

Business Rules: As above.

Rendering Rules: `--heading-font-size-m`, letter-spacing –0.015 × size.

Responsive Behaviour: 1.125rem / 1.5rem.

SEO Impact: Minor.

Accessibility Compliance: As above.

Variation 5 — Heading level h5

Variation Name: Title Type h5.

Business Purpose: Fourth-level headings.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-h5.png`
> **Purpose:** h5
> **Description:** An h5 in light weight.

Use Cases: Rare; deep document structures.

Authoring Guidelines: Title Type **h5** after an h4.

Business Rules: As above.

Rendering Rules: `--heading-font-size-s`, weight light (300), letter-spacing −0.01 × size.

Responsive Behaviour: 1rem / 1.25rem.

SEO Impact: Minor.

Accessibility Compliance: As above.

Variation 6 — Heading level h6

Variation Name: Title Type h6.

Business Purpose: Fifth-level headings or small heading-like labels.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-h6.png`
> **Purpose:** h6
> **Description:** An h6 in semibold.

Use Cases: Rare.

Authoring Guidelines: Title Type **h6** after an h5.

Business Rules: As above.

Rendering Rules: `--heading-font-size-xs`, weight semibold (600), no letter-spacing.

Responsive Behaviour: 0.875rem / 1rem.

SEO Impact: Minor.

Accessibility Compliance: As above.

6.7.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-authoring-view.png`
> **Purpose:** Title on the Universal Editor canvas
> **Description:** A selected Title component on the canvas with its heading text being edited inline. Mark A and B.

Annotations:

- A = the heading text (Title), editable inline
- B = the heading level as rendered (Title Type), visible through the size

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-properties-panel.png`
> **Purpose:** Title properties panel
> **Description:** The Properties panel with Title (text) and Title Type (select h1–h6).

The dialog has two fields and no tabs: **Title** (text) and **Title Type** (select with h1, h2, h3, h4, h5, h6). There is no Additional tab.

6.7.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Sites console > select the page > **Edit**.

Expected Result: The page opens on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-step-01-open-page.png`
> **Purpose:** Step 1
> **Description:** The page open in the Universal Editor.

Step 2 — Add a Title

Action: Select the section, **Add (+)** > **Title**.

Expected Result: A Title component appears in the section.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-step-02-add.png`
> **Purpose:** Step 2
> **Description:** The picker with Title highlighted.

Step 3 — Enter the heading text

Action: Type the text in **Title** (or inline on the canvas).

Expected Result: The heading shows on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-step-03-text.png`
> **Purpose:** Step 3
> **Description:** The Title field filled in.

Step 4 — Choose the level

Action: Select the **Title Type** that fits the outline (h1 for the page heading, h2 for sections, h3 below an h2, …).

Expected Result: The heading changes size.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-step-04-level.png`
> **Purpose:** Step 4
> **Description:** The Title Type select open with h2 highlighted.

Step 5 — Preview

Action: Switch on Preview mode; check the heading at the three sizes.

Expected Result: The heading wraps cleanly on mobile.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-step-05-preview.png`
> **Purpose:** Step 5
> **Description:** Preview mode at mobile size.

Step 6 — Validate and publish

Action: Check the heading outline (see Validation Process), then publish to Preview and Live.

Expected Result: The heading is live.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/title-step-06-publish.png`
> **Purpose:** Step 6
> **Description:** The Publish dialog.

Preview Process
On `.aem.page`, use a heading outline browser extension or the accessibility tree in developer tools to check the heading levels of the whole page.

Publishing Process
Publish the page to Live.

Validation Process
- ☐ Exactly one h1 on the page (including the Page Header / Hero heading).
- ☐ No skipped levels (h2 then h3, not h2 then h4).
- ☐ Heading text is short and descriptive.
- ☐ Heading text is not all upper case typed (use normal case).
- ☐ The heading wraps without breaking words badly on mobile.

Rollback Process
Change the text or level back, or restore the page version and republish.

6.7.6 Authorable Field Specification
Model `title`

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Title (`title`)	Text	Recommended (an empty Title renders an empty heading or nothing)	No	(empty)	The heading text	Short, descriptive, sentence case	None in the model	Our services
Title Type (`titleType`)	Select	Recommended	No	(none set in the model)	Heading level h1–h6	Follow the page outline	Enforced: h1, h2, h3, h4, h5, h6	h2

Additional tab: Not applicable (default content).

Field details

- **Title** — Purpose: heading text. Allowed Values: plain text. Character Limits: none enforced; Recommended up to 70 characters for h1 and up to 60 for h2–h3. Business Rules: Edge Delivery Services gives headings an `id` derived from their text (used by `#links`). Dependency Rules: none. Visibility Rules: always visible. Sample Content: "Why TCS for cloud".
- **Title Type** — Purpose: heading level. Allowed Values: h1–h6. Character Limits: not applicable. Business Rules: the model sets no default value; the level AEM renders when it is empty is the renderer's default. [To be confirmed: heading level rendered when Title Type is empty — expected h2] Dependency Rules: none. Visibility Rules: always visible. Sample Content: h2.

6.7.7 Field Annotation Mapping

Marker	Field	Purpose
A	Title	Heading text
B	Title Type	Heading level (size)

6.7.8 Component Specification

Attribute	Value
Component Name	Title (`title`, `core/franklin/components/title/v1/title`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`models/_title.json`; `styles/styles.css`
Component Type	Default content
Supported Variants	h1; h2; h3; h4; h5; h6
Supported Templates	Any page
Supported Metadata	None
Supported Tags	None
Supported Personalization	None (as of this handover)
Caching Considerations	Part of the page HTML
Performance Considerations	No JavaScript; an h1 in the first section can be the LCP element when there is no image
Accessibility Compliance	WCAG 2.1 AA: info and relationships (1.3.1), headings and labels (2.4.6), bypass blocks through heading navigation (2.4.1)
SEO Compliance	Server-rendered heading; level chosen by the author

6.7.9 Input Model

- **Content Inputs:** Title.
- **Metadata Inputs:** none.
- **Tags Inputs:** none.
- **Page Context Inputs:** none.
- **Query Inputs:** none.
- **Manual Inputs:** Title, Title Type.

Authored HTML received from AEM:

```
<h2 id="our-services">Our services</h2>
```

6.7.10 Output Model

Rendered Output: the heading element inside `div.default-content-wrapper`, unchanged by code.

Generated HTML:

```
<div class="default-content-wrapper">
  <h2 id="our-services">Our services</h2>
</div>
```

Generated JSON: none.

Generated Metadata: none.

Analytics Tracking: none for the Title itself. The first heading inside a block is used as that block's `dc:title` in the data layer; a Title is default content, so it is not used for any block.

Personalization Output: none.

Search Index Output: none (the query index uses page metadata).

6.7.11 Business Rules

1. The heading level is the Title Type value (h1–h6).
2. Heading ids are generated from the text by Edge Delivery Services; a block Anchor ID equal to a heading id gets a numeric suffix (5.6.3).
3. Heading sizes step at 900 px (mobile and desktop type scale).
4. All headings have `scroll-margin: 2.5rem` (anchor jumps leave space above).
5. h5 is light (300), h6 semibold (600), h1–h4 regular (400).
6. One h1 per page is Recommended, not enforced.
7. The article, blog and press-release templates add a Page Header with the page heading when none is authored; an extra h1 Title then duplicates it.

6.7.12 Author Best Practices

Recommended Content Length: h1 up to about 70 characters; h2–h3 up to about 60 characters; avoid headings longer than two lines on mobile.

Image Dimensions: not applicable.

CTA Writing Guidelines: not applicable (headings are not links).

Accessibility Guidelines: choose the level by structure, not size; never skip levels; no empty headings.

SEO Guidelines: the h1 should express the page topic and align with the page Title property.

Metadata Guidelines: keep the h1 consistent with the page Title and Description.

Taxonomy Guidelines: not applicable.

Localization Guidelines: translate per locale; allow for longer translations; heading ids change with translated text, so links to heading ids differ per locale (use Anchor names / IDs for stable links).

6.7.13 Troubleshooting Guide

Issue 1 — Two h1 headings on the page

Symptoms: Accessibility checker reports multiple h1.

Root Cause: A Title h1 plus a Page Header / Hero h1 (or a template-added Page Header).

Diagnosis Steps: Inspect the headings outline.

Resolution Steps: Change the Title to h2.

Validation Steps: One h1 remains.

Preventive Measures: On templated pages never add an h1 Title.

Issue 2 — Heading too large or too small

Symptoms: The heading size does not match the design.

Root Cause: The level was chosen for size.

Diagnosis Steps: Check Title Type against the outline.

Resolution Steps: Use the correct level; ask design if the scale does not fit.

Validation Steps: Outline correct.

Preventive Measures: Choose by structure.

Issue 3 — Anchor link to a heading breaks after translation

Symptoms: `#heading-id` link does not jump in another locale.

Root Cause: Heading ids are derived from the (translated) text.

Diagnosis Steps: Inspect the heading id on both locales.

Resolution Steps: Link to a section Anchor name or a block Anchor ID instead.

Validation Steps: The link jumps in every locale.

Preventive Measures: Use stable anchors.

Issue 4 — Empty heading in the outline

Symptoms: Screen reader announces an empty heading.

Root Cause: A Title without text.

Diagnosis Steps: Find empty `h1–h6` elements.

Resolution Steps: Fill or delete the Title.

Validation Steps: No empty headings.

Preventive Measures: Delete unused Titles.

Issue 5 — Heading level not as selected

Symptoms: The heading renders at another level.

Root Cause: Title Type empty (renderer default applies) or the change was not saved.

Diagnosis Steps: Check the Title Type field.

Resolution Steps: Select the level explicitly; republish.

Validation Steps: Correct element in the HTML.

Preventive Measures: Always set Title Type.

Issue 6 — Heading wraps awkwardly on mobile

Symptoms: A single word on the last line, or long words overflow.

Root Cause: Long heading text at 2rem on narrow screens.

Diagnosis Steps: View at 320 and 375 px.

Resolution Steps: Shorten the heading.

Validation Steps: Clean wrapping.

Preventive Measures: Keep headings short.

6.7.14 Support Runbook

Monitoring Requirements: Accessibility scans (heading order) after content releases are recommended. [To be confirmed: accessibility scanning tool — owner: site administrator]

Logs: None emitted.

Common Errors:

Error	Cause	Fix
Multiple h1	Title h1 on a templated page	Use h2
Skipped level	h2 followed by h4	Use h3
Empty heading	Empty Title	Delete or fill
Broken heading link	Translated heading id	Use anchors

Escalation Matrix:

Level	Who	Checks
L1	Author support	Title Type, outline, duplicates
L2	AMS / site administrator	Publish status, cache
L3	Development team	`styles/styles.css` heading rules and tokens
Adobe Support	Adobe	AEM title rendering defects

Recovery Procedures: Restore the page version and republish.

Health Checks: Block library page with h1–h6 on preview after releases.

Operational Procedures: Heading sizes are design tokens (`--heading-font-size-*`) changed by developers.

Known Limitations: No alignment, colour or size override; no Additional tab; no default Title Type in the model.

6.8 Image Component

6.8.1 Component Purpose

Business Objective: Place a single, automatically optimised image from AEM Assets in the flow of a page, with accessible alt text.

Business Use Cases:

- Illustrations, diagrams and photographs between paragraphs.
- Infographics with a text alternative.

- Images inside Columns or Tabs Container panels.

When To Use: For a standalone image in default content.

When Not To Use:

- For an image with text side by side: use **Teaser** or **Columns**.
- For a full-width banner with a heading and CTAs: use **Hero**.
- For a background behind content: use the section **Background Image**.
- For a clickable image: not supported by the Image component (no link field); use a block with a link (Teaser, Cards).
- For video or animation: use **Video**, **Embed** or the Hero Lottie option.

Dependencies:

Type	Item
Model	`models/_image.json` (definition `image`, resource type `core/franklin/components/image/v1/image`; model `image`)
Code	None specific; Edge Delivery Services delivers the responsive `<picture>`; `scripts/aem.js` `waitForFirstImage` makes the first image of the first section eager
Styles	`styles/styles.css` (`main img { max-width: 100%; width: auto; height: auto; }`)
Data sources	AEM Assets (`/content/dam/tcs-corp/`)

Related Components: Text, Title, Button; Columns; Teaser; Hero; section Background Image.

Applicable Templates: Any page.

Supported Content Types: Raster images from AEM Assets (JPEG, PNG, WebP, GIF); SVG as supported by AEM Assets and Edge Delivery. [To be confirmed: SVG handling for the Image component — owner: development team]

6.8.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** An Image component between two paragraphs at 1440 px, showing the image at its natural size up to the content width.

Desktop (1200 px and above): the image is shown at its intrinsic width, never wider than the content area (`max-width: 100%`), with its aspect ratio preserved (`height: auto`). It sits in a paragraph inside `div.default-content-wrapper`, so it is left-aligned and gets the paragraph margins.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same image at 768 px, scaled down to the content width.

Tablet (600–1199 px): the browser picks the large rendition (Edge Delivery typically serves 2000 px wide sources from 600 px) and scales it to the content width.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The same image at 375 px, full content width with 1.5rem side padding.

Mobile (below 600 px): the small rendition (typically 750 px wide) is loaded and scaled to the content width; no cropping.

6.8.3 Component Variations

Variation 1 — Informative image

Variation Name: Image with alt text.

Business Purpose: Convey information visually and to assistive technology.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-informative.png`
> **Purpose:** Informative image
> **Description:** A diagram of the TCS cloud migration approach with its alt text shown in the developer tools inspector.

Use Cases: Diagrams, product shots, photographs that support the text.

Authoring Guidelines: Pick the image in **Image**; describe it in **Alt Text**.

Business Rules: Alt Text is not enforced by the model (Recommended for every informative image).

Rendering Rules: `<picture>` with WebP and fallback sources and an `<img alt="...">`; lazy loading except for the first image of the first section.

Responsive Behaviour: Scales down with the container at all sizes; no crop.

SEO Impact: Alt text and the file context help image search.

Accessibility Compliance: WCAG 1.1.1 non-text content; complex diagrams also need the information in the text.

Variation 2 — Decorative image

Variation Name: Image without alt text.

Business Purpose: Visual atmosphere without information.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-decorative.png`
> **Purpose:** Decorative image
> **Description:** An abstract brand pattern image with an empty alt attribute in the inspector.

Use Cases: Mood images, separators made of images.

Authoring Guidelines: Leave **Alt Text** empty.

Business Rules: What AEM renders for an empty Alt Text (empty alt or a fallback from the asset metadata) depends on the AEM renderer. [To be confirmed: alt value rendered when Alt Text is empty]

Rendering Rules: As Variation 1.

Responsive Behaviour: As Variation 1.

SEO Impact: None.

Accessibility Compliance: An empty `alt=""` lets screen readers skip the image (WCAG 1.1.1).

6.8.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-authoring-view.png`
> **Purpose:** Image on the Universal Editor canvas
> **Description:** A selected Image component on the canvas and an asset being dragged onto it from the Assets panel. Mark A and B.

Annotations:

- A = the image (Image field; select it to open the Assets picker or drop an asset)
- B = the alt text, not visible on the canvas but shown in the Properties panel (Alt Text)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-properties-panel.png`
> **Purpose:** Image properties panel
> **Description:** The Properties panel with the Image reference field (thumbnail and picker button) and the Alt Text field.

The dialog has two fields and no tabs: **Image** (AEM Assets reference, single) and **Alt Text** (text). There is no Additional tab.

6.8.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Sites console > select the page > **Edit**.

Expected Result: The page opens on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-step-01-open-page.png`
> **Purpose:** Step 1
> **Description:** The page open in the Universal Editor.

Step 2 — Add an Image

Action: Select the section (or Column / tab panel), **Add (+)** > **Image**.

Expected Result: An empty Image component appears.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-step-02-add.png`

> **Purpose:** Step 2
> **Description:** The picker with Image highlighted.

Step 3 — Pick the image

Action: Select the **Image** field and choose the asset in the Assets picker (or drag it from the Assets panel).

Expected Result: The image shows on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-step-03-pick.png`
> **Purpose:** Step 3
> **Description:** The Assets picker with an image selected.

Step 4 — Write the alt text

Action: Fill **Alt Text** for informative images; leave it empty for decorative ones.

Expected Result: The alt text is saved (visible in the inspector).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-step-04-alt.png`
> **Purpose:** Step 4
> **Description:** The Alt Text field filled in.

Step 5 — Preview

Action: Switch on Preview mode and check the three device sizes.

Expected Result: The image scales down without distortion.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-step-05-preview.png`
> **Purpose:** Step 5
> **Description:** Preview at mobile size.

Step 6 — Validate and publish

Action: Run the checklist; publish the page with its references to Preview and Live.

Expected Result: The image is served from the site's media URL (`./media_...`).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/image-step-06-publish.png`
> **Purpose:** Step 6
> **Description:** The Publish dialog listing the page and the image asset.

Preview Process

On `.aem.page`, open developer tools > Network, filter on images and check that a WebP rendition of an appropriate width is loaded for the viewport.

Publishing Process
Publish the page; make sure the asset is included (references) or already published.

Validation Process
- ☐ The right image (version, crop) is used.
- ☐ Informative image: alt text present and meaningful; decorative: no alt text.
- ☐ The image is sharp on a desktop screen (original at least as wide as its display width, ideally 2×).
- ☐ No text baked into the image (or the text is repeated in alt text / the page).
- ☐ The image loads on the live site (asset published).
- ☐ File size reasonable (original under about 5 MB; Edge Delivery optimises renditions).

Rollback Process
Re-pick the previous asset, or restore the page version and republish.

6.8.6 Authorable Field Specification
Model `image`

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Image (`image`)	Reference (AEM Assets)	Recommended (without an image the component renders nothing)	No	(empty)	The image to show	High-quality original from `/content/dam/tcs-corp/`	None in the model	/content/dam/tcs-corp/insights/ai-lab.jpg
Alt Text (`imageAlt`)	Text	Recommended for informative images	No	(empty)	Text alternative	Describe content and purpose; empty if decorative	None in the model	Researchers testing a robotics prototype

Additional tab: Not applicable (default content).

Field details

- **Image** — Purpose: the asset. Allowed Values: one AEM Assets image. Character Limits: not applicable. Business Rules: delivered as a responsive picture; the first image of the first section is loaded eagerly (LCP), others lazily. Dependency Rules: Alt Text. Visibility Rules: always visible. Sample Content: a 2000 × 1125 px JPEG.
- **Alt Text** — Purpose: accessible description. Allowed Values: text. Character Limits: Recommended up to 125 characters (not enforced). Business Rules: rendered as the `alt` attribute. Dependency Rules: meaningful only with an image. Visibility Rules: always visible. Sample Content: “Map of TCS delivery centres in Europe”.

6.8.7 Field Annotation Mapping

Marker	Field	Purpose
A	Image	The asset shown
B	Alt Text	Text alternative (not visible)

6.8.8 Component Specification

Attribute	Value
Component Name	Image (`image`, `core/franklin/components/image/v1/image`)

Attribute	Value
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`models/_image.json`; AEM Assets; `styles/styles.css`
Component Type	Default content
Supported Variants	Informative image; Decorative image
Supported Templates	Any page
Supported Metadata	None
Supported Tags	None
Supported Personalization	None (as of this handover)
Caching Considerations	Images are copied to the site's media store on publish and served with content-hash URLs (`media_{hash}`), cacheable long-term; a changed asset gets a new hash after republishing the page
Performance Considerations	WebP renditions per breakpoint; lazy loading except the LCP candidate (first image of the first section, set to eager by `waitForFirstImage`)
Accessibility Compliance	WCAG 2.1 AA: text alternatives (1.1.1); images of text avoided (1.4.5)
SEO Compliance	Server-rendered `<img>` with alt; width and height attributes from the pipeline help layout stability

6.8.9 Input Model

- **Content Inputs:** Image, Alt Text.
- **Metadata Inputs:** none (asset metadata in AEM Assets is not read by the code).
- **Tags Inputs:** none.
- **Page Context Inputs:** position (first section's first image is eager).
- **Query Inputs:** none.
- **Manual Inputs:** Image, Alt Text.

Authored HTML received from AEM / Edge Delivery (typical; the widths and formats are chosen by Edge Delivery Services):

```
<p>
  <picture>
    <source type="image/webp" srcset="./media_3c4d.jpg?
width=2000&format=webply&optimize=medium"
            media="(min-width: 600px)">
    <source type="image/webp" srcset="./media_3c4d.jpg?
width=750&format=webply&optimize=medium">
    <source type="image/jpeg" srcset="./media_3c4d.jpg?
width=2000&format=jpeg&optimize=medium"
            media="(min-width: 600px)">
    <img loading="lazy" alt="Researchers testing a robotics prototype"
         src="./media_3c4d.jpg?width=750&format=jpeg&optimize=medium"
         width="2000" height="1125">
  </picture>
</p>
```

6.8.10 Output Model

Rendered Output: the picture is kept as delivered inside `div.default-content-wrapper`; the code does not change it (except `loading="eager"` for the LCP candidate).

Generated HTML: as in 6.8.9, inside `<div class="default-content-wrapper">`.

Generated JSON: none.

Generated Metadata: none (the page's `og-image` / `featured-image` come from the metadata sheet, not from Image components).

Analytics Tracking: none (an image without a link has no interaction; default content is not a data layer component).

Personalization Output: none.

Search Index Output: none (the query index's image columns come from metadata).

6.8.11 Business Rules

1. The image is shown at its natural size, never wider than its container; the aspect ratio is kept.
2. The first `img` of the first section is set to `loading="eager"`, and the first section is shown only after it has loaded (or failed).
3. Other images keep `loading="lazy"`.
4. There is no link, caption, alignment, crop or size option.
5. The alt text is the Alt Text field as rendered by AEM.
6. In the editor, a changed image is re-rendered in place.

6.8.12 Author Best Practices

Recommended Content Length: alt text up to about 125 characters; complex images (charts) explained in the surrounding text.

Image Dimensions: at least as wide as the largest display width (content area up to 110rem / 1760 px on wide screens with Full Width); 2000 px wide originals recommended; any aspect ratio (no crop is applied); the component does not stretch small images.

CTA Writing Guidelines: not applicable (no link).

Accessibility Guidelines: describe informative images; leave decorative images without alt text; avoid images of text; ensure charts have a text equivalent.

SEO Guidelines: descriptive file names and alt text; avoid very large originals.

Metadata Guidelines: set the page's social image in the metadata sheet (`og-image`), not by placing an Image.

Taxonomy Guidelines: not applicable.

Localization Guidelines: translate alt text per locale; use locale-specific images when they contain text or culture-specific content.

6.8.13 Troubleshooting Guide

Issue 1 — Image missing on the live site

Symptoms: A broken image or nothing where the image should be.

Root Cause: The asset is not published, or was moved / deleted in AEM Assets.

Diagnosis Steps: Check the image request in the network panel; check the asset in AEM Assets.

Resolution Steps: Publish the asset and republish the page.

Validation Steps: The image loads from `./media_...`.

Preventive Measures: Publish with references.

Issue 2 — Image looks blurry

Symptoms: Soft or pixelated image on desktop.

Root Cause: The original is smaller than its display size.

Diagnosis Steps: Check the asset's dimensions.

Resolution Steps: Upload a larger original and re-pick it.

Validation Steps: Sharp at 1440 px.

Preventive Measures: Use 2000 px originals.

Issue 3 — Old image still shows after replacing the asset

Symptoms: The site shows the previous version.

Root Cause: The page was not republished, so its media reference was not updated.

Diagnosis Steps: Compare the `media_` hash before and after.

Resolution Steps: Republish the page.

Validation Steps: New hash and new image.

Preventive Measures: Republish pages after replacing assets.

Issue 4 — Image is not full width

Symptoms: A small image does not fill the content width.

Root Cause: The CSS never enlarges images beyond their natural size.

Diagnosis Steps: Compare the natural width with the container.

Resolution Steps: Use a wider original, or a Hero / section background for full-width visuals.

Validation Steps: Image fills the area.

Preventive Measures: Choose the component by layout need.

Issue 5 — Accessibility checker reports missing alt text

Symptoms: "Image without alternative text" in scans.

Root Cause: Alt Text empty for an informative image, or the rendered alt attribute is missing.

Diagnosis Steps: Inspect the `img` alt attribute.

Resolution Steps: Fill Alt Text; republish.

Validation Steps: Scan passes.

Preventive Measures: Validation checklist before publishing.

Issue 6 — Slow page load (LCP)

Symptoms: PageSpeed Insights reports a slow LCP image.

Root Cause: A very large image in the first section, or many images above the fold.

Diagnosis Steps: Run PageSpeed Insights on the preview URL.

Resolution Steps: Use a smaller original or move the image below the first section.

Validation Steps: LCP under 2.5 s.

Preventive Measures: Keep the first section light.

Issue 7 — Image cannot be selected in the picker

Symptoms: The asset is not visible in the Assets picker.

Root Cause: Missing permissions on the DAM folder, or the asset is still processing.

Diagnosis Steps: Open the folder in AEM Assets.

Resolution Steps: Wait for processing; ask for access.

Validation Steps: Asset selectable.

Preventive Measures: Upload assets before editing.

6.8.14 Support Runbook

Monitoring Requirements: Core Web Vitals (LCP, CLS) for pages with large images. [To be confirmed: monitoring tool]

Logs: No console messages. Image failures show as failed requests in the browser network panel. Asset processing status in AEM Assets.

Common Errors:

Error	Cause	Fix
404 on `media_` URL	Page published before the asset was available	Republish the page
Broken image on author	Asset deleted	Re-pick
Blurry image	Small original	Larger original
Missing alt	Empty Alt Text	Fill Alt Text
Slow LCP	Huge first image	Optimise / move

Escalation Matrix:

Level	Who	Checks
L1	Author support	Asset publication, alt text, original size
L2	AMS / site administrator	Publish status, media delivery, CDN cache
L3	Development team	Image CSS, LCP handling in `scripts/aem.js`
Adobe Support	Adobe	AEM Assets processing, Edge Delivery media delivery

Recovery Procedures: Re-pick a valid asset or restore the page version; republish.

Health Checks: Block library image page on preview after releases.

Operational Procedures: DAM folder structure and asset governance. [To be confirmed: DAM governance — owner: brand / DAM team]

Known Limitations: No link, caption, alignment, crop, focal point or size options; no Additional tab; no art direction (different crops per device).

6.9 Button Component

6.9.1 Component Purpose

Business Objective: Give authors a consistent, accessible call to action with the TCS design system's button views, types, icons and alignment, without formatting tricks.

Business Use Cases:

- Primary call to action of a section ("Contact us", "Download the report").
- Secondary or tertiary actions next to a primary one.
- Icon-only actions (search, arrow, chevron).
- Text link calls to action ("View all insights").

When To Use: For any standalone call to action in default content, especially when you need a specific view, icon or alignment.

When Not To Use:

- Inside a sentence: use an inline link in **Text**.
- For CTAs that belong to a block (Hero, Teaser, Page Header, List, Cards have their own CTA fields).
- For navigation menus: use the navigation page (`references/nav`).

Dependencies:

Type	Item
Model	`models/_button.json` (definition `button`, resource type `core/franklin/components/button/v1/button`, model `button`)
Code	`decorators/content-buttons-decorator.js` (`decorateButtons`), `utils/buttons.js` (`BUTTON_VIEWS`, `BUTTON_TYPES`, `BUTTON_ALIGNMENTS`, `readButtonOptions`), `decorators/button-decorator.js` (`decorateButton`, `setButtonAlignment`); `decorators/data-layer-decorator.js` (click tracking)
Styles	`styles/styles.css` (`a.button`, `.primary`, `.secondary`, `.tertiary`, `.icon-button`, `a.link-button`, `.button-align-*`, `p.button-wrapper`)
Icons	`icons/arrow-right.svg`, `chevron-right.svg`, `chevron-down.svg`, `search.svg` (Icon options); `arrow-right.svg` (trailing icon of buttons); `chevron-right.svg` (link button)
Data sources / placeholders	None

Related Components: Text (formatted standalone links create the same buttons), Hero / Teaser / Page Header CTAs (same decorator), Columns (Button inside a Column), Tabs Container.

Applicable Templates: Any page.

Supported Content Types: A link to an AEM page, an asset or an external URL.

6.9.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A row of Button components at 1440 px: Primary, Secondary and Tertiary buttons, an Icon button (search) and a Link view, each in its own paragraph.

Desktop: buttons are pill-shaped (`--radius-full`), at least 3.125rem high, label 1rem (`--button-font-size`), padding 0.75rem 1rem 0.75rem 1.5rem, with a 1.5rem trailing arrow; icon buttons are 2.75rem circles with a 1.25rem icon; the link view is a semibold underlined text with a 1rem chevron. Each button sits in its own `p.button-wrapper` (0.75rem top and bottom margin), aligned left, centre or right.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same buttons at 768 px.

Tablet: identical sizes (button tokens do not change per breakpoint).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The same buttons at 375 px; a long label clipped at the content edge.

Mobile: identical sizes; labels never wrap (`white-space: nowrap`), and a label wider than the content area is clipped (`max-width: 100%; overflow: hidden`).

6.9.3 Component Variations

Variation 1 — Primary button

Variation Name: Views: Button, Type: Primary.

Business Purpose: The main action of a section.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-primary.png`
> **Purpose:** Primary button
> **Description:** Black pill button "Contact us" with a trailing arrow, default and hover states.

Use Cases: Contact, register, download.

Authoring Guidelines: Views **Button**, Type **Primary** (the defaults).

Business Rules: Primary is the default type; one primary per section is Recommended.

Rendering Rules: `a.button.primary`: black border and background (`--color-border-neutral`, `--color-bg-neutral`), grey-300 label (`--color-text-tertiary`), medium weight; hover / focus label slate-400.

Responsive Behaviour: Same at all sizes; label does not wrap.

SEO Impact: A normal link with its label as anchor text.

Accessibility Compliance: Keyboard focusable link with 0.125rem brand focus outline; the arrow is decorative CSS.

Variation 2 — Secondary button

Variation Name: Views: Button, Type: Secondary.

Business Purpose: An alternative action next to the primary.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-secondary.png`
> **Purpose:** Secondary button
> **Description:** Outlined pill button "Explore services", default and hover (blue border and label) states.

Use Cases: "Learn more", "Explore".

Authoring Guidelines: Views **Button**, Type **Secondary**.

Business Rules: None beyond the type.

Rendering Rules: `a.button.secondary`: transparent background, slate-900 border and label; hover white background, brand border and label. In the CSS-only `dark` section style: lighter border and white label.

Responsive Behaviour: As Variation 1.

SEO Impact: As Variation 1.

Accessibility Compliance: As Variation 1.

Variation 3 — Tertiary button

Variation Name: Views: Button, Type: Tertiary.

Business Purpose: A lightweight action with an arrow.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-tertiary.png`
> **Purpose:** Tertiary button
> **Description:** Text "See all case studies" with an arrow and no container, default and hover (blue) states.

Use Cases: Third-level actions, dense layouts.

Authoring Guidelines: Views **Button**, Type **Tertiary**.

Business Rules: None beyond the type.

Rendering Rules: `a.button.tertiary`: no border, radius or padding; default text colour; hover brand colour and gap 0.

Responsive Behaviour: As Variation 1.

SEO Impact: As Variation 1.

Accessibility Compliance: As Variation 1; minimum height 3.125rem keeps a large target.

Variation 4 — Icon button

Variation Name: Views: Icon (Type Primary / Secondary / Tertiary; Icon Arrow right / Chevron right / Chevron down / Search).

Business Purpose: A compact, icon-only action.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-icon.png`
> **Purpose:** Icon button
> **Description:** Three circular icon buttons with the search icon: primary (blue filled), secondary (blue outline) and tertiary (blue icon only).

Use Cases: Search entry points, "next" arrows, expand chevrons.

Authoring Guidelines: Views **Icon**; choose **Type** and **Icon**; fill **Text** with the action (it becomes the accessible name and tooltip, it is not shown).

Business Rules: The Icon field is shown only when Views is Icon. If no icon can be resolved, the link is rendered as a normal button of the chosen type.

Rendering Rules: `a.button.{type}.icon-button`: 2.75rem circle, padding 0.75rem, icon 1.25rem drawn from `/icons/{name}.svg` as a mask (follows the text colour); the text is wrapped in a hidden span; `aria-label` and `title` = the text (or the title, or the icon name). Primary: brand-500 background, white icon, brand-600 on hover. Secondary: brand border and icon. Tertiary: brand icon only.

Responsive Behaviour: Same at all sizes.

SEO Impact: The link has no visible text; search engines read the hidden text / `aria-label`.

Accessibility Compliance: Accessible name from the text (WCAG 4.1.2, 2.5.3); 2.75rem target (44 px) meets the recommended target size.

Variation 5 — Link view

Variation Name: Views: Link.

Business Purpose: A text call to action without a button container.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-link.png`
> **Purpose:** Link view
> **Description:** Underlined semibold "View all insights" with a trailing chevron, default and hover states.

Use Cases: "View all", "Read more" at the end of a section.

Authoring Guidelines: Views **Link** (Type is hidden for this view).

Business Rules: Type is ignored for the link view.

Rendering Rules: `a.link-button`: inline-flex, 1rem, semibold, underlined, default text colour, chevron-right 1rem; hover brand colour.

Responsive Behaviour: Same at all sizes.

SEO Impact: As a normal link.

Accessibility Compliance: Underline plus icon distinguish it from text.

Variation 6 — Centre alignment

Variation Name: Alignment: Center.

Business Purpose: Centre a button under centred content.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-align-center.png`
> **Purpose:** Centre-aligned button
> **Description:** A primary button centred in its section below a centred heading.

Use Cases: Call-to-action bands, end-of-page actions.

Authoring Guidelines: Alignment **Center**.

Business Rules: Alignment applies to the button's paragraph.

Rendering Rules: `p.button-wrapper.button-align-center { text-align: center; }`.

Responsive Behaviour: Centred at all sizes.

SEO Impact: None.

Accessibility Compliance: No effect on reading order.

Variation 7 — Right alignment

Variation Name: Alignment: Right.

Business Purpose: Align a button to the end of the line.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-align-right.png`
> **Purpose:** Right-aligned button
> **Description:** A secondary button aligned to the right edge of the content area.

Use Cases: "Next" actions, layouts mirroring a right-aligned element.

Authoring Guidelines: Alignment **Right**.

Business Rules: Uses `text-align: end`, so in a right-to-left context it aligns left.

Rendering Rules: `p.button-wrapper.button-align-right { text-align: end; }`.

Responsive Behaviour: Right-aligned at all sizes; consider Left on mobile-heavy pages (not configurable per breakpoint).

SEO Impact: None.

Accessibility Compliance: Screen magnifier users may miss right-aligned buttons; use sparingly.

> **Note:** Alignment **Left** (the default) adds `button-align-left` (`text-align: start`).

6.9.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-authoring-view.png`
> **Purpose:** Button on the Universal Editor canvas
> **Description:** A selected Button component on the canvas showing a primary button. Mark A to C.

Annotations:

- A = the button label (Text)
- B = the button's appearance (Views, Type, Icon)
- C = the button's position in its paragraph (Alignment)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-properties-panel.png`
> **Purpose:** Button properties panel
> **Description:** The Properties panel with Link, Text, Title, Views, Type, Icon (visible because Views is Icon) and Alignment.

The dialog has seven fields and no tabs: **Link** (AEM content picker or URL), **Text** (label), **Title** (tooltip), **Views** (Button / Icon / Link, default Button), **Type** (Primary / Secondary / Tertiary, default Primary; shown when Views is empty, Button or Icon), **Icon** (Arrow right / Chevron right / Chevron down / Search; shown only when Views is Icon) and **Alignment** (Left / Center / Right, default Left). There is no Additional tab.

6.9.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Sites console > select the page > **Edit**.

Expected Result: The page opens on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-step-01-open-page.png`
> **Purpose:** Step 1
> **Description:** The page open in the Universal Editor.

Step 2 — Add a Button

Action: Select the section (or Column / tab panel), **Add (+)** > **Button**.

Expected Result: A Button component appears.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-step-02-add.png`
> **Purpose:** Step 2
> **Description:** The picker with Button highlighted.

Step 3 — Set the link

Action: In **Link** pick a page (content picker) or paste a URL.

Expected Result: The link target is saved.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-step-03-link.png`
> **Purpose:** Step 3
> **Description:** The content picker selecting `/content/tcs-corp/us/en/contact`.

Step 4 — Write the label and title

Action: Enter the label in **Text** and, optionally, a tooltip in **Title**.

Expected Result: The canvas shows the label.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-step-04-text.png`
> **Purpose:** Step 4
> **Description:** Text “Contact us” and Title “Contact the TCS team”.

Step 5 — Choose the view and type

Action: Choose **Views** (Button, Icon or Link); for Button and Icon choose **Type**; for Icon choose **Icon**.

Expected Result: The button changes appearance.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-step-05-view-type.png`
> **Purpose:** Step 5
> **Description:** Views Icon, Type Secondary, Icon Search, and the canvas showing the outlined search icon button.

Step 6 — Choose the alignment

Action: Choose **Alignment** (Left, Center or Right).

Expected Result: The button moves within its line.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-step-06-alignment.png`
> **Purpose:** Step 6
> **Description:** Alignment Center with the button centred.

Step 7 — Preview

Action: Switch on Preview mode; click the button; check the three device sizes.

Expected Result: The link opens; the label fits on mobile.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-step-07-preview.png`
> **Purpose:** Step 7
> **Description:** Preview at mobile size.

Step 8 — Validate and publish

Action: Run the checklist; publish to Preview and Live.

Expected Result: The button is live.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/button-step-08-publish.png`
> **Purpose:** Step 8
> **Description:** The Publish dialog.

Preview Process
On `.aem.page`, click the button, check the target and inspect the classes (`button primary`, `icon-button`, `link-button`, `button-align-*`).

Publishing Process
Publish the page; publish the target page or asset if it is new.

Validation Process
- ☐ The link target is correct and published (internal links as `/en/...`).
- ☐ The label is action-oriented and fits on one line at 320 px.
- ☐ View, type and alignment as designed; at most one primary button per section.
- ☐ Icon buttons have a meaningful Text (accessible name) and the right icon.
- ☐ The tooltip (Title) adds information or repeats the label (it is set to the label when empty).
- ☐ `cmp:click` is pushed on click (console `adobeDataLayer`, see Part 11).

Rollback Process
Change the fields back, or restore the page version and republish.

6.9.6 Authorable Field Specification
Model `button`

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Link (`link`)	AEM content (picker or URL)	Recommended (without a link nothing is rendered as a button)	No	(empty)	Button target	Pick internal pages with the picker	None in the model	/content/tcs-corp/us/en/contact
Text (`linkText`)	Text	Recommended	No	(empty)	Visible label (accessible name for Icon view)	Verb first, up to about 25 characters	None in the model	Contact us
Title (`linkTitle`)	Text	No	No	(empty; the code sets the label as title)	Tooltip (`title` attribute)	Extra context only	None	Contact the TCS team
Views (`buttonView`)	Select	No	No	Button (`button`)	Button, icon-only or text link	Button for most CTAs	Enforced: Button, Icon, Link	button
Type (`linkType`)	Select	No	No	Primary (`primary`)	Emphasis of Button / Icon views	One primary per section	Enforced: Primary, Secondary, Tertiary	secondary
Icon (`buttonIcon`)	Select	Recommended	No	(empty)	Icon of the Icon view	Pick the icon matching the action	Enforced: Arrow right, Chevron	search

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
		when Views is Icon					right, Chevron down, Search	
Alignment (`buttonAlignment`)	Select	No	No	Left (`left`)	Horizontal position	Left unless the layout is centred	Enforced: Left, Center, Right	center

Additional tab: Not applicable (default content).

Field details

- **Link** — Purpose: target. Allowed Values: AEM page, asset or URL. Character Limits: not applicable. Business Rules: internal AEM paths are mapped to published paths by Edge Delivery (`paths.json`); downloads (`.pdf`, `.docx`, …) are tracked as `xdm:linkType download`, other origins as `exit`. Dependency Rules: none. Visibility Rules: always visible. Sample Content: `/content/tcs-corp/us/en/contact`.
- **Text** — Purpose: label. Allowed Values: text. Character Limits: Recommended up to about 25 characters (labels never wrap and are clipped). Business Rules: for the Icon view it is hidden and becomes `aria-label` and `title`. Dependency Rules: none. Visibility Rules: always visible. Sample Content: “Download the report”.
- **Title** — Purpose: tooltip. Allowed Values: text. Character Limits: Recommended up to 60 characters. Business Rules: when empty, the code sets the link text as title. Dependency Rules: none. Visibility Rules: always visible. Sample Content: “Annual report 2026 (PDF)”.
- **Views** — Purpose: view. Allowed Values: `button`, `icon`, `link`. Character Limits: not applicable. Business Rules: an explicit view wins; Link ignores Type. Dependency Rules: controls the visibility of Type and Icon. Visibility Rules: always visible. Sample Content: Button.
- **Type** — Purpose: emphasis. Allowed Values: `primary`, `secondary`, `tertiary`. Character Limits: not applicable. Business Rules: default primary. Dependency Rules: Views. Visibility Rules: shown when Views is empty, `button` or `icon` (model `condition`). Sample Content: Primary.
- **Icon** — Purpose: icon. Allowed Values: `arrow-right`, `chevron-right`, `chevron-down`, `search`. Character Limits: not applicable. Business Rules: drawn from `/icons/{name}.svg`; without an icon the Icon view falls back to a normal button. Dependency Rules: Views = Icon. Visibility Rules: shown only when Views is `icon` (model `condition`). Sample Content: Search.
- **Alignment** — Purpose: position. Allowed Values: `left`, `center`, `right`. Character Limits: not applicable. Business Rules: adds `button-align-{value}` to the paragraph. Dependency Rules: none. Visibility Rules: always visible. Sample Content: Center.

> **Note:** How AEM delivers the Views, Icon and Alignment values in the page markup (classes or data attributes of the link or its paragraph) could not be verified from the repository; `utils/buttons.js` reads both forms (type classes, `view-{view}` and `align-{alignment}` classes, and view / type / icon / alignment data values). [To be confirmed: delivered markup of the Button component’s Views, Icon and Alignment — owner: development team]

6.9.7 Field Annotation Mapping

Marker	Field	Purpose
A	Text; Title; Link	Label, tooltip and target
B	Views; Type; Icon	Appearance
C	Alignment	Position in the line

6.9.8 Component Specification

Attribute	Value
Component Name	Button (`button`, `core/franklin/components/button/v1/button`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`models/_button.json`; `decorators/content-buttons-decorator.js`; `decorators/button-decorator.js`; `utils/buttons.js`; `styles/styles.css`; `/icons`
Component Type	Default content
Supported Variants	Primary; Secondary; Tertiary; Icon (3 types × 4 icons); Link; Alignment Left / Center / Right
Supported Templates	Any page
Supported Metadata	None
Supported Tags	None
Supported Personalization	None (as of this handover); clicks are in the data layer for future Target / Journey Optimizer use
Caching Considerations	Part of the page HTML; icons are static files cached by the CDN
Performance Considerations	Decorated in the eager phase (`decorateMain`); icons are CSS masks (one small SVG request each)
Accessibility Compliance	WCAG 2.1 AA: focus visible (2.4.7), accessible names for icon buttons (4.1.2), target size 2.75rem / 3.125rem, colour contrast of the types
SEO Compliance	A real `<a href>` with label text (hidden text for icon buttons)

6.9.9 Input Model

- **Content Inputs:** Link, Text, Title.
- **Metadata Inputs:** none.
- **Tags Inputs:** none.
- **Page Context Inputs:** the section style (dark section overrides in CSS).
- **Query Inputs:** none.
- **Manual Inputs:** all seven fields.

Authored HTML received from AEM (typical; the standard button rendering encodes the type as formatting and the other options as classes or data values, see the note in 6.9.6):

```
<p class="button-container">
  <strong><a href="/en/contact" title="Contact the TCS team"
     class="button">Contact us</a></strong>
</p>
```

6.9.10 Output Model

Rendered Output: `p.button-wrapper.button-align-{alignment}` containing `a.button.{type}` (Button view), `a.button.{type}.icon-button` (Icon view, with `--icon-url`, `aria-label`, hidden text span) or `a.link-button` (Link view). The formatting element (`strong` / `em`) is removed.

Generated HTML:

```
<p class="button-wrapper button-align-center">
  <a href="/en/contact" title="Contact the TCS team" class="button primary">Contact us</a>
</p>
<p class="button-wrapper button-align-left">
  <a href="/en/search" title="Search" aria-label="Search" class="button secondary icon-
button"
     style="--icon-url: url('/icons/search.svg')"><span>Search</span></a>
</p>
```

Generated JSON: none.

Generated Metadata: none.

Analytics Tracking: a click pushes `cmp:click` with the page as the component (default content):

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.page-1x2y3z",
    "interaction": "link",
    "dc:title": "Contact us",
    "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/contact",
    "xdm:linkType": "other",
    "breakpoint": "mobile"
  }
}
```

For icon buttons `dc:title` is the `aria-label` (the Text). Links to other origins have `xdm:linkType` `exit`; links with `download` or a file extension (`pdf`, `doc(x)`, `xls(x)`, `ppt(x)`, `zip`, `csv`, `txt`, `mp3`, `mp4`) have `download`.

Personalization Output: none.

Search Index Output: none.

6.9.11 Business Rules

1. Only a link that is alone in its paragraph is decorated; a link sharing its paragraph with text stays a plain link.
2. A link whose text is its URL stays plain (unless it is an icon link).
3. View: the explicit Views value wins; else an icon link is an Icon button; else a link with a Type is a button; else a link button.
4. Type: an explicit type class or value wins; else bold = primary, italic = secondary, bold + italic = tertiary; default primary when a button is rendered without a type.
5. Type is ignored for the Link view (and hidden in the dialog).
6. The Icon field is shown only for the Icon view; an Icon view without a resolvable icon renders as a normal button.
7. An icon-only link without type or view stays plain.
8. Icon buttons: the text is moved into a hidden span; `aria-label` = existing aria-label, else text, else title, else the icon name with spaces; `title` set when empty.
9. Every decorated link gets `title` = its text when it has none.
10. Alignment adds `button-align-left|center|right` to the paragraph; unknown values are ignored.
11. Labels never wrap and are clipped at the container width.
12. Disabled states exist in the decorator and CSS but are not authorable for the Button component.
13. No open-in-new-tab option exists for the Button component (as of this handover).
14. In the editor, changed buttons are re-decorated in place (`decorateButtons` from `scripts/editor-support.js`).

6.9.12 Author Best Practices

Recommended Content Length: labels of 2–4 words, up to about 25 characters; Title up to 60 characters.

Image Dimensions: not applicable.

CTA Writing Guidelines: start with a verb; describe the outcome ("Download the 2026 report", not "Click here"); one primary per section; pair a primary with a secondary or link view; mention the format for downloads ("(PDF, 2 MB)") in the Title.

Accessibility Guidelines: always fill Text for icon buttons; do not rely on the icon alone; avoid right alignment for important actions.

SEO Guidelines: descriptive labels; link to the locale's own pages.

Metadata Guidelines: not applicable.

Taxonomy Guidelines: not applicable.

Localization Guidelines: translate Text and Title; check translated labels at 320 px (German and French expand); link to the target locale's pages.

6.9.13 Troubleshooting Guide

Issue 1 — Button shows as a plain link

Symptoms: No button styling.

Root Cause: No link set, the label equals the URL, or the paragraph contains other text.

Diagnosis Steps: Inspect the paragraph; check Link and Text.

Resolution Steps: Set a link and a descriptive Text.

Validation Steps: The link has the class `button` or `link-button`.

Preventive Measures: Always fill Link and Text.

Issue 2 — Wrong type shown

Symptoms: Primary instead of Secondary, or the reverse.

Root Cause: Type not saved, or an inconsistent value delivered by AEM.

Diagnosis Steps: Check the Type field and the classes in the HTML.

Resolution Steps: Re-select the type; republish.

Validation Steps: Correct class.

Preventive Measures: Check in Preview mode before publishing.

Issue 3 — Icon button shows a normal button with an arrow

Symptoms: Views is Icon but a pill button with text appears.

Root Cause: No icon resolved (Icon empty or not delivered).

Diagnosis Steps: Check the Icon field; inspect for `--icon-url`.

Resolution Steps: Choose an Icon; if set but ignored, escalate to development (see the note in 6.9.6).

Validation Steps: A circular icon button appears.

Preventive Measures: Always choose an Icon with the Icon view.

Issue 4 — Label clipped on mobile

Symptoms: The end of the label is cut off.

Root Cause: Labels never wrap.

Diagnosis Steps: View at 320 / 375 px.

Resolution Steps: Shorten the label.

Validation Steps: Full label visible.

Preventive Measures: Up to about 25 characters.

Issue 5 — Alignment has no effect

Symptoms: Centre / Right does not move the button.

Root Cause: The alignment value was not delivered, or the button is inside a block whose CSS overrides alignment.

Diagnosis Steps: Inspect for `button-align-center` on the paragraph.

Resolution Steps: Re-select the alignment; escalate if the class is missing.

Validation Steps: Class present and button moved.

Preventive Measures: Check in Preview mode.

Issue 6 — External link opens in the same tab

Symptoms: Visitors leave the site in the same tab.

Root Cause: The Button component has no new-tab option.

Diagnosis Steps: Inspect for `target`.

Resolution Steps: Accept, or use a block with an "open in new tab" option; request the feature from development.

Validation Steps: Not applicable.

Preventive Measures: Agree the external-link policy. [To be confirmed: external link policy — owner: product owner]

Issue 7 — Click not visible in analytics

Symptoms: No `cmp:click` in the data layer.

Root Cause: The button is inside an element with `data-analytics="off"`, or the check happened in the editor's edit mode (clicks select components).

Diagnosis Steps: On preview, run `adobeDataLayer.filter(e => e.event === 'cmp:click')` in the console after clicking (use a link that does not leave the page, or preserve the console log).

Resolution Steps: Test on preview / live; check ancestors for `data-analytics`.

Validation Steps: An entry with the button label appears.

Preventive Measures: Validate analytics on preview.

6.9.14 Support Runbook

Monitoring Requirements: Broken-link checks; click volumes once a tag manager is connected (Part 11).

Logs: No console messages. Clicks appear in `window.adobeDataLayer`.

Common Errors:

Error	Cause	Fix
Plain link	Missing link / text	Fill fields
Clipped label	Long label	Shorten

Error	Cause	Fix
Missing icon	Icon empty	Choose icon
Wrong target	Unpublished page	Publish target
No tracking	`data-analytics="off"` ancestor	Check container

Escalation Matrix:

Level	Who	Checks
L1	Author support	Field values, target publication, label length
L2	AMS / site administrator	Publish status, cache, `paths.json` mapping of the target
L3	Development team	`decorators/content-buttons-decorator.js`, `utils/buttons.js`, `decorators/button-decorator.js`, `styles/styles.css`
Adobe Support	Adobe	AEM button rendering, Universal Editor dialog defects

Recovery Procedures: Restore field values or the page version; republish.

Health Checks: After releases, check the locale block library page with every view, type and alignment.

Operational Procedures: New icons: add an SVG to `/icons` and an option to `models/_button.json`, then `npm run build:json` (development team).

Known Limitations: No new-tab option; no disabled state for authors; four icons only; labels never wrap; no Additional tab; alignment not configurable per breakpoint.

6.10 Hero Block

The Hero is the large, full-width opening banner of a page: an image or a Lottie animation fills (or sits contained inside) the banner, an overlay keeps the text readable, and a content block with an optional icon, eyebrow, title, description and a list of calls to action (CTAs) is placed at one of nine positions. It implements the TCS.com Design System Figma "L1 Header" variations.

Source files: `blocks/hero/hero.js`, `blocks/hero/hero.css`, `blocks/hero/_hero.json`.

6.10.1 Component Purpose

Business Objective

Give every key landing page a strong, brand-consistent first impression that states the page's message in one headline, supports it with imagery or motion, and routes visitors to the next step through one or more CTAs. As the first element of a page the Hero is usually the Largest Contentful Paint (LCP) element, so the block is optimised to load its image first.

Business Use Cases

- Home page opening banner of each of the 13 locale sites (`/en`, `/de-de`, …).
- Service, industry and capability landing pages ("L1" pages) with a headline and CTAs.
- Campaign pages with a Lottie animation as the background ("Lotto" image type).
- Illustrated banners where the artwork must be shown whole (Contained image view).
- Brand statements on a dark navy or black background with white text.
- A slide of a Gen Carousel section (the Gen Carousel filter allows the Hero as a slide).
- In-page "Discover more" banners whose CTA scrolls down to a section of the same page (CTA Arrow Down).

When To Use

- At the top of a page that needs a visual banner with a single, strong headline.
- When the message benefits from a full-bleed background image or animation.
- When the page needs up to a few prominent CTAs above the fold.

When Not To Use

- On article, blog and press-release pages: their templates add a Page Header block (with the page title as the H1) from the page metadata; use the Page Header block instead.
- For a smaller image-and-text promotion inside the page flow: use the Teaser block.
- For a list of pages or cards: use List, Cards or Filter List.
- For a background video: a video background is **Not implemented (as of this handover)** in the Hero; use the Video or Embed block in its own section instead.
- Several Heroes in a row: use a Gen Carousel section if several banners must rotate.

Dependencies

Type	Dependency	Used for
Code module	`scripts/scripts.js` `moveInstrumentation`	Keeps Universal Editor instrumentation on the rebuilt elements
Decorator	`decorators/animation-decorator.js` `createAnimation`	Lottie animation player and its pause / play button
Decorator	`decorators/button-decorator.js` `decorateButton`	CTA buttons (primary, secondary, tertiary) and new-window links
Decorator	`decorators/icon-button-decorator.js` (through the animation decorator)	Pause / play icon button
Decorator	`decorators/link-decorator.js` `setLinkTarget` (through the button decorator)	`target="_blank"` and `rel="noopener noreferrer"`
Util	`utils/buttons.js` `readButtonOptions`	Reads the CTA view (bold = primary, italic = secondary)
Util	`utils/links.js` `readLinkItems`, `NEW_WINDOW`	Reads the CTA List items, their Link Type and CTA Arrow values
Util	`utils/multifield.js` (through `readLinkItems`)	Splits the CTA List composite multifield into items
Util	`utils/environment.js` `isEditing`, `prefersReducedMotion`	Disables Motion Type in the editor and for reduced motion
Util	`utils/viewport.js` `onVisible`, `watchVisibility`	Starts motion and the animation when the hero is on screen
Util	`utils/media.js` `isAnimation`	Recognises a Lottie file (`.json` link)
Util	`utils/lottie.js` `loadLottie`	Loads the vendored player `scripts/lib/lottie/lottie_light.min.js` once per page
Util	`utils/placeholders.js` `getPlaceholders` (through the animation decorator)	Labels of the pause / play button
Global CSS	`styles/styles.css`	Button styles, design tokens, Additional tab classes
Data source	AEM Assets (DAM)	Hero image, icon and Lottie `.json` file
Sheet	`/{locale}/placeholders.json`	Keys `pauseAnimation`, `playAnimation` (English fallbacks “Pause animation”, “Play animation”)

Type	Dependency	Used for
Icons	`icons/arrow-right.svg`, `icons/arrow-down.svg`, `icons/pause.svg`, `icons/play.svg`	CTA arrows and the animation toggle

The Hero reads no page metadata, data sheet, taxonomy or query index.

Related Components

- Page Header block: the metadata-driven page title area of article, blog and press-release pages (use it instead of a Hero there).
- Teaser block: a smaller image-and-text or background-image promotion in the page flow; it shares the Lotto image type and the animation decorator.
- Gen Carousel section: rotates several Hero (or Teaser) blocks as slides.
- Video and Embed blocks: for video content (the Hero has no video background).
- Button component: the CTA buttons use the same button decorator and styles.

Applicable Templates

Any page. The Hero is allowed in every Section and in Gen Carousel sections (filters `section` and `gen-carousel`). On pages with Page Type `home`, the Home template removes the top margin and padding of the first section so that a Hero placed there meets the header. On `article`, `blog` and `press-release` pages a Page Header is added automatically; a Hero with an H1 there would create a second H1 (see the SEO guidance).

Supported Content Types

- Images (JPEG, PNG, WebP, SVG from AEM Assets), delivered as an optimised `<picture>`.
- Lottie animations: a Lottie JSON file (`.json`) from AEM Assets, with Hero Image type “Lotto”.
- Icon image (small square image or SVG above the eyebrow).
- Plain text (eyebrow, title), rich text (description), links (CTAs).
- Not supported: video files, YouTube / Vimeo, GIF animation as a “video” (a GIF can be used as an image), several images.

6.10.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A Hero at 1440 px wide with a full-bleed cover image under the default white overlay, centred content: icon, blue eyebrow, large title, grey description and a primary and a secondary CTA side by side.

Desktop (900 px and wider; the same rules apply at 1200 px and above)

- The Hero spans the full width of the browser: the section wrapper of a Hero has no maximum width and no padding (`.hero-container .hero-wrapper`).
- Minimum height 53.125rem (850 px); padding 7.5rem top and bottom, 2.75rem left and right.
- The image covers the whole hero (`object-fit: cover`), behind an overlay: white at 80 % (Default), none (Light, no overlay), navy `#01121f` at 80 % (Dark blue), navy at 60 % (Dark blue subtle) or black at 80 % (Black).
- Content block maximum width 84.5rem; when the Horizontal Alignment is Left or Right it is limited to 47.5rem so its position is visible.
- Title: display size 2 (4rem), medium weight, line height 1.2, letter spacing –0.025 em of the size, maximum width 72.5rem.

- Eyebrow: 1rem, semibold, brand blue; 2.5rem space below. Icon: 3rem × 3rem with 0.5rem below.
- Description: 1.5rem, line height 1.2, maximum width 53.125rem, 2.5rem above it.
- CTAs: in one row (wrapping when needed), 2.5rem between CTAs and 2.5rem above the row.
- Lottie: the pause / play button sits in the bottom-right corner of the hero, 1rem from the edges.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Hero at 768 px wide: minimum height 40rem, smaller title (2.5rem), description at 1rem, CTAs wrapping onto a second line when they do not fit.

Tablet (600 px to 1199 px)

- 600 px to 899 px: minimum height 40rem (640 px); padding 4rem top and bottom, 2.5rem left and right; title 2.5rem; description 1rem; eyebrow 1rem with 1.5rem below; icon 2.5rem; CTA gap 2rem; the content block uses the full width (Left / Right positions show only through the text alignment).
- 900 px to 1199 px: the desktop rules above apply (minimum height 53.125rem, 4rem title, 47.5rem content width for Left / Right).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Hero at 375 px wide: the cover image cropped to the centre, minimum height 35rem, stacked content with the title at 2.5rem and the CTAs wrapping one per line.

Mobile (below 600 px)

- Minimum height 35rem (560 px); padding 4rem top and bottom, 1.5rem left and right.
- The cover image is cropped around its centre to fill the taller, narrower box.
- Title 2.5rem; eyebrow and description 1rem; icon 2.5rem; gaps 0.75rem (icon), 1.5rem (eyebrow, description) and 2rem (CTAs).
- CTAs wrap; long CTA labels do not wrap inside a button (buttons use `white-space: nowrap`).
- The content block always uses the full width; Horizontal Alignment Left / Right moves the text start, not the block.

6.10.3 Component Variations

Each option of the Hero dialog that changes the presentation or behaviour is described below as its own variation. Options combine: for example Lotto + Dark blue + Bottom + Left + Fade in is one valid Hero. A video background is **Not implemented (as of this handover)**: the Hero Image Path accepts an image or a Lottie `.json` file only.

Variation 1 — Default Image Hero (cover image)

Variation Name: Hero Image type “Default” with Hero View “Default”.

Business Purpose: The standard brand banner: a photograph fills the whole hero behind the text.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-default-image.png`

> **Purpose:** Default image hero
> **Description:** A full-bleed photograph under the white 80 % overlay with centred dark title, description and CTAs.

Use Cases: Home page banner; service and industry landing pages.

Authoring Guidelines: Image tab: Hero Image type = Default, select an image in Hero Image Path, enter Hero Image alt text (or leave it empty if decorative), Hero View = Default.

Business Rules: Exactly one image is used; the image cell must contain only the picture. When the hero is in the first section of the page the image is loaded eagerly with high fetch priority (LCP).

Rendering Rules: The picture is moved into `div.hero-media` (absolute, behind the content); `img` uses `object-fit: cover`; the block gets `hero-has-media`, which draws the overlay (`::before`) in the BG Color.

Responsive Behaviour: Desktop: 53.125rem minimum height, image cropped to fill. Tablet: 40rem (600–899 px) or 53.125rem (900 px+). Mobile: 35rem, centre crop.

SEO Impact: The image is a real `<img>` with alt text (indexable); the eager, high-priority load supports a good LCP and Core Web Vitals score.

Accessibility Compliance: Meaningful images need alt text (WCAG 1.1.1); decorative images get an empty alt. The overlay keeps text contrast; check contrast against busy images (WCAG 1.4.3).

Variation 2 — Lotto (Lottie Animation) Hero

Variation Name: Hero Image type "Lotto" (block class `lottie`).

Business Purpose: A motion background for campaigns and innovation themes, using a light vector Lottie animation instead of a video.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-lotto-animation.png`
> **Purpose:** Lottie animation hero
> **Description:** A hero playing a Lottie animation behind the content, with the round pause button visible in the bottom-right corner.

Use Cases: Campaign launches, AI / innovation pages, animated brand moments.

Authoring Guidelines: Image tab: Hero Image type = Lotto; Hero Image Path = the Lottie `.json` file from AEM Assets; Hero Image alt text = a short description if the animation conveys information, or empty if decorative.

Business Rules: The animation plays only when the Hero Image type is Lotto **and** the file is a `.json` link. With Lotto and an image, the image is shown instead. With Default and a `.json` file, no media is shown at all. The animation loops; it starts paused on its first frame when the visitor prefers reduced motion; it pauses whenever it is off screen.

Rendering Rules: `createAnimation()` creates `div.hero-animation` inside `div.hero-media` (SVG renderer, `preserveAspectRatio` "xMidYMid meet"). The player (`/scripts/lib/lottie/lottie_light.min.js`) is loaded once per page when the hero comes within 200 px of the viewport and the browser is idle. A pause / play `<button class="animation-toggle">` is appended to the block (bottom-right, above the overlay and the content). With an alt text the animation has `role="img"` and `aria-label`; without, `aria-hidden="true"`.

Responsive Behaviour: The animation fills the hero box at every breakpoint and keeps its aspect ratio ("meet", so it may show bands at the sides on narrow screens); the toggle stays 1rem from the bottom-right corner.

SEO Impact: The animation is not indexable content and is not an LCP image; the title and description remain the indexable content. The player script is loaded after rendering, so it does not block the LCP.

Accessibility Compliance: WCAG 2.2.2 (Pause, Stop, Hide): the pause / play button (labels from placeholders `pauseAnimation` / `playAnimation`) lets visitors stop the loop. Reduced motion (WCAG 2.3.3) is respected: the animation starts paused.

Variation 3 — Contained Image View

Variation Name: Hero View "Contained image" (block class `media-contained`).

Business Purpose: Show an illustration or artwork whole, centred behind the content, instead of cropping it.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-contained-image.png`
> **Purpose:** Contained image hero
> **Description:** A hero with a transparent illustration shown whole and centred, with the content over it.

Use Cases: Illustrated hero artwork, product diagrams, logos or graphics that must not be cropped.

Authoring Guidelines: Image tab: Hero View = Contained image; choose an illustration (ideally a transparent PNG or SVG) in Hero Image Path.

Business Rules: Applies to images; a Lottie animation already keeps its aspect ratio.

Rendering Rules: `div.hero-media` becomes a centred flex box with 1.5rem padding; the image uses `object-fit: contain`, `max-width: 100%`, `max-height: 100%`, `width/height: auto`.

Responsive Behaviour: The whole image always shows; on mobile it is scaled down to the narrow width, leaving empty space above and below.

SEO Impact: As Variation 1.

Accessibility Compliance: As Variation 1; make sure the text placed over the artwork keeps enough contrast (use the overlay of the BG Color).

Variation 4 — Text-only Hero (no image)

Variation Name: Hero without Hero Image Path.

Business Purpose: A large typographic banner when no imagery is available or wanted.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-text-only.png`
> **Purpose:** Hero without media
> **Description:** A hero with only title, description and CTAs on a plain white (or navy / black) background, no overlay.

Use Cases: Statement pages, temporary pages before imagery is ready, dark brand banners.

Authoring Guidelines: Leave Hero Image Path empty; optionally choose a dark BG Color.

Business Rules: Without media the block does not get `hero-has-media` and no overlay is drawn; the background is white (Default, Light) or the BG Color's navy / black.

Rendering Rules: Only `div.hero-content` is rendered; minimum heights and paddings still apply.

Responsive Behaviour: As for the other variations (minimum heights 35 / 40 / 53.125rem).

SEO Impact: No image; the headline stays the main content signal.

Accessibility Compliance: Text colours follow the BG Color (dark text on white, white on navy / black), which meet WCAG AA contrast.

Variation 5 — BG Color "Default" (light overlay)

Variation Name: BG Color "Default" (no class).

Business Purpose: Light, airy banner with dark text over a softened photograph.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-bg-default.png`
> **Purpose:** Default light overlay
> **Description:** A photograph seen through an 80 % white overlay with black title and grey description.

Use Cases: Most corporate banners.

Authoring Guidelines: Style tab: BG Color = Default.

Business Rules: The overlay is drawn only when there is media.

Rendering Rules: `--hero-overlay: rgb(255 255 255 / 80%)`; title `--color-text-default`, eyebrow `--color-text-brand`, description `--color-text-secondary`; background `--color-bg-default` (white).

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Dark text on an 80 % white overlay keeps contrast above 4.5:1 in most cases; verify with very dark or busy images.

Variation 6 — BG Color "Light, no overlay"

Variation Name: BG Color "Light, no overlay" (class `bg-light`).

Business Purpose: Show the image exactly as designed (for example a light gradient artwork) with dark text.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-bg-light.png`
> **Purpose:** Light background without overlay
> **Description:** A hero with a light gradient image shown at full strength and dark text on it.

Use Cases: Pre-designed light backgrounds and gradients.

Authoring Guidelines: Style tab: BG Color = Light, no overlay; use a light image.

Business Rules: The overlay is transparent; the text stays dark, so the image must be light enough behind the text.

Rendering Rules: `--hero-overlay: transparent`; text colours as Default.

Responsive Behaviour: Same at every breakpoint; check the crop on mobile because the text may land on a darker area of the image.

SEO Impact: None.

Accessibility Compliance: The author is responsible for contrast (WCAG 1.4.3): only use light images.

Variation 7 — BG Color "Dark blue"

Variation Name: BG Color "Dark blue" (class `bg-dark`).

Business Purpose: A strong, premium brand look: white text on a navy overlay.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-bg-dark.png`
> **Purpose:** Dark blue overlay
> **Description:** A photograph under an 80 % navy overlay with white title, semi-transparent white description and white outline / text CTAs.

Use Cases: Brand statements, home page, technology themes.

Authoring Guidelines: Style tab: BG Color = Dark blue.

Business Rules: Secondary and Default CTAs switch to white; on hover they return to their normal hover colours (Default CTAs turn light blue `--brand-300`).

Rendering Rules: `--hero-overlay: rgb(1 18 31 / 80%)`, background `#01121f`; title `--color-text-on-neutral` (white); description white at 50 % with medium weight. The eyebrow keeps the brand blue.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: White text on 80 % navy meets AA. The eyebrow stays brand blue (`--brand-500 #4673db`) and the description is 50 % white: check their contrast (WCAG 1.4.3) in validation.

Variation 8 — BG Color "Dark blue (subtle)"

Variation Name: BG Color "Dark blue (subtle)" (class `bg-dark-subtle`).

Business Purpose: A dark look that lets more of the photograph show through.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-bg-dark-subtle.png`
> **Purpose:** Subtle dark blue overlay
> **Description:** A photograph under a 60 % navy overlay, the image more visible than in Dark blue, white text.

Use Cases: Dark banners with expressive imagery.

Authoring Guidelines: Style tab: BG Color = Dark blue (subtle).

Business Rules: As Dark blue; the lighter overlay needs a darker image.

Rendering Rules: `--hero-overlay: rgb(1 18 31 / 60%)`, background `#01121f`, white text and CTAs as Dark blue.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Verify contrast: with bright images 60 % navy may not be enough for the 50 % white description.

Variation 9 — BG Color "Black"

Variation Name: BG Color "Black" (class `bg-black`).

Business Purpose: A neutral, dramatic black banner.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-bg-black.png`
> **Purpose:** Black overlay
> **Description:** A photograph under an 80 % black overlay with white text and CTAs.

Use Cases: Events, campaigns, cinematic imagery.

Authoring Guidelines: Style tab: BG Color = Black.

Business Rules: As Dark blue.

Rendering Rules: `--hero-overlay: rgb(0 0 0 / 80%)`, background `--black-1000` (`#0c0c0d`), white text and CTAs.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: As Dark blue.

Variation 10 — Vertical Alignment "Top"

Variation Name: Vertical Alignment "Top" (class `align-top`).

Business Purpose: Place the content at the top of the banner, leaving the lower part of the image visible.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-align-top.png`
> **Purpose:** Content at the top
> **Description:** The content block starting right below the top padding, the image subject visible in the lower half.

Use Cases: Images with the subject at the bottom.

Authoring Guidelines: Style tab: Vertical Alignment = Top.

Business Rules: Only visible when the hero is taller than its content (minimum heights).

Rendering Rules: `justify-content: flex-start` on the hero (a column flex box).

Responsive Behaviour: Applies at every breakpoint.

SEO Impact: None.

Accessibility Compliance: No impact; reading order is unchanged.

Variation 11 — Vertical Alignment "Middle" (default)

Variation Name: Vertical Alignment "Middle" (class `align-middle`, set by the block template).

Business Purpose: The balanced default: the content in the vertical centre.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-align-middle.png`
> **Purpose:** Content in the middle
> **Description:** The content block centred vertically in the hero.

Use Cases: Most heroes.

Authoring Guidelines: Style tab: Vertical Alignment = Middle (pre-selected for new Heroes).

Business Rules: Default value of the model and of the block template.

Rendering Rules: `justify-content: center` (the base `.hero` rule).

Responsive Behaviour: Applies at every breakpoint.

SEO Impact: None.

Accessibility Compliance: No impact.

Variation 12 — Vertical Alignment "Bottom"

Variation Name: Vertical Alignment "Bottom" (class `align-bottom`).

Business Purpose: Anchor the content at the bottom, showing the top of the image.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-align-bottom.png`
> **Purpose:** Content at the bottom
> **Description:** The content block resting above the bottom padding, the sky / subject of the image visible above it.

Use Cases: Images with the subject at the top; editorial looks.

Authoring Guidelines: Style tab: Vertical Alignment = Bottom.

Business Rules: With a Lottie animation, the pause / play button is also at the bottom right; keep Right / Bottom content clear of it.

Rendering Rules: `justify-content: flex-end`.

Responsive Behaviour: Applies at every breakpoint.

SEO Impact: None.

Accessibility Compliance: No impact.

Variation 13 — Horizontal Alignment "Left"

Variation Name: Horizontal Alignment "Left" (class `align-left`).

Business Purpose: Place the content block on the left side of the banner, leaving the right side of the image free.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-align-left.png`
> **Purpose:** Content on the left
> **Description:** On desktop, a 47.5rem wide content block on the left of the hero, the image subject visible on the right.

Use Cases: Images with the subject on the right.

Authoring Guidelines: Style tab: Horizontal Alignment = Left; usually combined with Text Alignment = Left align.

Business Rules: If no Text Alignment is set (heroes authored before the option existed), the text follows the position (left aligned).

Rendering Rules: `--hero-position: flex-start`; at 900 px and wider the content block is limited to 47.5rem.

Responsive Behaviour: Desktop (900 px+): block on the left, 47.5rem wide. Tablet below 900 px and mobile: the block is full width, so only the text alignment shows.

SEO Impact: None.

Accessibility Compliance: Left-aligned text is the easiest to read for long descriptions (WCAG 1.4.8 advisory).

Variation 14 — Horizontal Alignment "Center" (default)

Variation Name: Horizontal Alignment "Center" (class `align-center`, set by the block template).

Business Purpose: The symmetrical default banner.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/hero-align-center.png`
> **Purpose:** Content in the centre
> **Description:** The content block centred horizontally with centred text.

Use Cases: Most heroes.

Authoring Guidelines: Style tab: Horizontal Alignment = Center (pre-selected).

Business Rules: Default value of the model and the block template.

Rendering Rules: `--hero-position: center`; content up to 84.5rem wide.

Responsive Behaviour: Centred at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Keep centred descriptions short; long centred paragraphs are harder to read.

Variation 15 — Horizontal Alignment "Right"

Variation Name: Horizontal Alignment "Right" (class `align-right`).

Business Purpose: Place the content block on the right, the image subject on the left.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/hero-align-right.png`
> **Purpose:** Content on the right
> **Description:** On desktop, the content block on the right half of the hero.

Use Cases: Images with the subject on the left.

Authoring Guidelines: Style tab: Horizontal Alignment = Right; choose a Text Alignment.

Business Rules: Without a Text Alignment the text is right aligned (`end`).

Rendering Rules: `--hero-position: flex-end`; 47.5rem content width at 900 px+.

Responsive Behaviour: As Variation 13, mirrored.

SEO Impact: None.

Accessibility Compliance: Avoid right-aligned long text for left-to-right languages.

Variation 16 — Text Alignment "Left align"

Variation Name: Text Alignment "Left align" (class `text-left`).

Business Purpose: Align the icon, eyebrow, title, description and CTAs to the start of the content block.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-text-left.png`
> **Purpose:** Left-aligned text
> **Description:** Icon, eyebrow, title, description and CTAs all starting at the left edge of the content block.

Use Cases: Left-positioned content; long descriptions.

Authoring Guidelines: Style tab: Text Alignment = Left align.

Business Rules: Overrides the text alignment implied by the Horizontal Alignment.

Rendering Rules: `--hero-text-align: start`, `--hero-align: flex-start` (icon and CTA row start).

Responsive Behaviour: Applies at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Recommended for readability.

Variation 17 — Text Alignment "Center align" (default)

Variation Name: Text Alignment "Center align" (class `text-center`, set by the block template).

Business Purpose: Centred text for symmetrical banners.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-text-center.png`
> **Purpose:** Centred text
> **Description:** Icon, title, description and the CTA row all centred inside the content block.

Use Cases: Default banners.

Authoring Guidelines: Style tab: Text Alignment = Center align (pre-selected).

Business Rules: Default of the model and the block template, so new Heroes are centred even when the Horizontal Alignment is changed to Left or Right (change Text Alignment too).

Rendering Rules: `--hero-text-align: center`, `--hero-align: center`.

Responsive Behaviour: Applies at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Keep centred text short.

Variation 18 — Text Alignment "Right align"

Variation Name: Text Alignment "Right align" (class `text-right`).

Business Purpose: Right-aligned content, for right-positioned blocks.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/hero-text-right.png`
> **Purpose:** Right-aligned text
> **Description:** Title, description and CTAs aligned to the right edge of the content block.

Use Cases: Right-positioned content blocks.

Authoring Guidelines: Style tab: Text Alignment = Right align.

Business Rules: Uses logical `end`, so it is right aligned in left-to-right languages.

Rendering Rules: `--hero-text-align: end`, `--hero-align: flex-end`.

Responsive Behaviour: Applies at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Avoid for long text.

Variation 19 — Motion Type "None" (default)

Variation Name: Motion Type "None" (no class).

Business Purpose: A static hero, shown immediately.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/hero-motion-none.png`
> **Purpose:** No motion
> **Description:** The hero fully visible on page load with no entrance animation.

Use Cases: Default; pages where motion is not wanted.

Authoring Guidelines: Links tab: Motion Type = None.

Business Rules: Default value.

Rendering Rules: No `motion-ready` / `is-visible` classes are added.

Responsive Behaviour: Not applicable.

SEO Impact: None.

Accessibility Compliance: No motion.

Variation 20 — Motion Type "Fade in"

Variation Name: Motion Type "Fade in" (class `motion-fade`).

Business Purpose: A subtle entrance: the content fades in when the hero comes into view.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-motion-fade.png`
> **Purpose:** Fade-in motion
> **Description:** A frame captured mid-animation: the content block half transparent over the image.

Use Cases: Heroes further down a page or in campaigns.

Authoring Guidelines: Links tab: Motion Type = Fade in.

Business Rules: Plays once, when 10 % of the hero is visible; never in Universal Editor edit mode or when the visitor prefers reduced motion (then the content is shown as is).

Rendering Rules: `hero.js` adds `motion-ready` (content at opacity 0) and, when visible, `is-visible`, which plays `hero-fade` (800 ms, ease) on `.hero-content`.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: The content is in the HTML from the start; search engines are not affected. For a first-section hero the content is briefly transparent, which can delay the visual LCP of the text slightly.

Accessibility Compliance: Respects `prefers-reduced-motion` (WCAG 2.3.3).

Variation 21 — Motion Type "Slide up"

Variation Name: Motion Type "Slide up" (class `motion-slide`).

Business Purpose: A more dynamic entrance: the content rises and fades in.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-motion-slide.png`
> **Purpose:** Slide-up motion
> **Description:** A frame captured mid-animation: the content block slightly below its final position and partly transparent.

Use Cases: Campaign and product launch heroes.

Authoring Guidelines: Links tab: Motion Type = Slide up.

Business Rules: As Fade in.

Rendering Rules: `hero-slide` (800 ms, ease) from opacity 0 and `translateY(2rem)`.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: As Fade in.

Accessibility Compliance: As Fade in.

Variation 22 — Motion Type "Zoom (image)"

Variation Name: Motion Type "Zoom (image)" (class `motion-zoom`).

Business Purpose: A cinematic effect: the background slowly settles from a slight zoom.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-motion-zoom.png`
> **Purpose:** Zoom motion
> **Description:** A frame captured mid-animation: the background image slightly enlarged while the text is static.

Use Cases: Photography-led banners.

Authoring Guidelines: Links tab: Motion Type = Zoom (image).

Business Rules: Animates the media only; the text is visible from the start. Not in the editor and not with reduced motion.

Rendering Rules: `hero-zoom` (1600 ms, ease-out) on `.hero-media` from `scale(1.08)`; the hero's `overflow: hidden` clips the enlarged image.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None (text is not hidden).

Accessibility Compliance: Respects reduced motion.

Variation 23 — CTA View "Default" (text link with arrow)

Variation Name: CTA view "Default" (tertiary button).

Business Purpose: A lightweight text CTA with an arrow, for secondary actions.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-cta-default.png`
> **Purpose:** Default CTA view
> **Description:** A CTA rendered as text with a trailing right arrow, no border or fill.

Use Cases: "Learn more", "Discover our story".

Authoring Guidelines: Links tab, CTA List item: CTA view = Default.

Business Rules: Any CTA that is not Primary or Secondary renders as tertiary.

Rendering Rules: `a.button.tertiary` with the arrow (`::after`, `arrow-right.svg`); white on dark BG Colors, light blue on hover there.

Responsive Behaviour: Wraps with the other CTAs.

SEO Impact: A crawlable link with descriptive text.

Accessibility Compliance: Use descriptive link text (WCAG 2.4.4); the arrow is decorative CSS.

Variation 24 — CTA View "Primary"

Variation Name: CTA view "Primary".

Business Purpose: The main action of the hero.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-cta-primary.png`

> **Purpose:** Primary CTA
> **Description:** A pill-shaped filled black button with white text and a right arrow.

Use Cases: "Contact us", "Get started".

Authoring Guidelines: CTA List item: CTA view = Primary (stored as bold formatting of the link).

Business Rules: Recommended: one Primary CTA per hero.

Rendering Rules: `a.button.primary` (neutral fill, rounded, arrow).

Responsive Behaviour: Does not wrap its text; wraps to a new line as a whole.

SEO Impact: As Variation 23.

Accessibility Compliance: Button colours meet AA; focus-visible styles come from the global button styles.

Variation 25 — CTA View "Secondary"
Variation Name: CTA view "Secondary".

Business Purpose: An alternative action next to the primary CTA.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-cta-secondary.png`
> **Purpose:** Secondary CTA
> **Description:** A pill-shaped outline button; white outline and text on dark BG Colors.

Use Cases: "View our services" next to "Contact us".

Authoring Guidelines: CTA List item: CTA view = Secondary (stored as italic formatting).

Business Rules: On dark BG Colors the border and text are white until hover / focus.

Rendering Rules: `a.button.secondary`.

Responsive Behaviour: As Variation 24.

SEO Impact: As Variation 23.

Accessibility Compliance: As Variation 24.

Variation 26 — CTA Arrow "Right" (default)
Variation Name: CTA Arrow "Right".

Business Purpose: Signals navigation to another page.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-cta-arrow-right.png`
> **Purpose:** Right arrow CTA
> **Description:** A CTA with the standard right-pointing arrow.

Use Cases: Links to other pages.

Authoring Guidelines: CTA List item: CTA Arrow = Right (pre-selected).

Business Rules: Default value. A link without an arrow value that starts with `#` points down automatically (see Variation 27).

Rendering Rules: Global `arrow-right.svg` mask on `a.button::after`.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Decorative icon only.

Variation 27 — CTA Arrow "Down" (in-page link)

Variation Name: CTA Arrow "Down" (class `hero-cta-down` on the link).

Business Purpose: Signals a jump to a section of the same page.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-cta-arrow-down.png`
> **Purpose:** Down arrow CTA
> **Description:** A "Discover our story" text CTA with a down-pointing arrow.

Use Cases: "Discover our story" linking to `#anchor-our-story`.

Authoring Guidelines: CTA List item: CTA Arrow = Down; CTA Link = the anchor of a block (Additional tab Anchor ID) or section on the page, e.g. `#anchor-our-story`.

Business Rules: The arrow points down when CTA Arrow = Down, or when no arrow value is stored and the link starts with `#`.

Rendering Rules: `a.button.hero-cta-down::after` uses `icons/arrow-down.svg`.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: In-page links do not affect crawling.

Accessibility Compliance: The link text must say where it goes; the arrow is decorative.

Variation 28 — CTA Link Type "Open in New Window"

Variation Name: CTA Link Type "Open in New Window".

Business Purpose: Open external or document links without leaving the TCS page.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-cta-new-window.png`
> **Purpose:** New-window CTA
> **Description:** A CTA to an external report; the browser status bar showing the link opens in a new tab.

Use Cases: External partner sites, PDF reports.

Authoring Guidelines: CTA List item: CTA Link Type = Open in New Window.

Business Rules: The item value `new-window` sets `target="_blank"` and `rel="noopener noreferrer"`. The Hero does not add a visual or screen-reader "opens in a new tab" hint (Not implemented (as of this handover) for the Hero; the Page Header and card lists have it).

Rendering Rules: `decorateButton(a, { newWindow: true })` → `setLinkTarget`.

Responsive Behaviour: Not applicable.

SEO Impact: None; `noopener noreferrer` is the safe default.

Accessibility Compliance: WCAG 3.2.5 (advisory): mention the new window in the CTA text, e.g. "Read the report (PDF)", because the hint is not added automatically.

Variation 29 — Title Heading Level "H2" (secondary hero)

Variation Name: Title heading level "H2" (default "H1").

Business Purpose: Use a Hero lower on a page, or in a Gen Carousel slide after the first, without creating several H1 headings.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-title-h2.png`
> **Purpose:** Hero with an H2 title
> **Description:** A hero that looks identical to the H1 version; the browser's accessibility tree shows the heading level 2.

Use Cases: Second and later heroes, carousel slides 2+, pages with a Page Header.

Authoring Guidelines: Content tab: Title heading level = H2.

Business Rules: The look is the same (`.hero-title` styles any level); only the semantics change. Default H1 (model and block template).

Rendering Rules: The heading is rendered as `<h1>` or `<h2>` with class `hero-title`.

Responsive Behaviour: Same as H1.

SEO Impact: Keep exactly one H1 per page; use H2 for every other hero.

Accessibility Compliance: Correct heading hierarchy (WCAG 1.3.1, 2.4.6).

6.10.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A Hero selected on the Universal Editor canvas with letters A–G marking the block, the background image, the icon, the eyebrow, the title, the description and the CTAs.

Annotations

- A = Hero block (select it to open its properties: Image, Content, Links, Style, Additional).
- B = Hero image or Lottie animation (Hero Image Path; the background behind the overlay).
- C = Icon (small image above the eyebrow).
- D = Eyebrow (short line above the title).
- E = Title (H1 or H2).
- F = Description (rich text below the title).
- G = CTA List (the row of CTAs; each CTA is one item of the list).

Notes on the canvas: Motion Type never plays in edit mode, so the content is always visible; the Lottie animation plays in the editor (with its pause / play button); a Hero with Hide block on is shown faded with a dashed outline. The text elements (eyebrow, title, description) can be edited in place on the canvas; images and CTAs are edited in the properties panel.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Hero properties panel with its five tabs (Image, Content, Links, Style, Additional), the Image tab open showing Icon, Icon Image Alt Text, Hero Image type, Hero Image Path, Hero Image alt text and Hero View.

Properties panel tabs and fields

- **Image** — Icon (asset picker), Icon Image Alt Text (text), Hero Image type (select: Default, Lotto), Hero Image Path (asset picker: an image, or a Lottie `.json` for Lotto), Hero Image alt text (text), Hero View (select: Default, Contained image).
- **Content** — Eyebrow (text), Title (text), Title heading level (select: H1, H2; default H1), Description (rich text).
- **Links** — CTA List (multifield; Add creates an item with CTA Title, CTA Link, CTA Link Type, CTA view, CTA Arrow), Motion Type (select: None, Fade in, Slide up, Zoom (image)).
- **Style** — Vertical Alignment (Top, Middle, Bottom; default Middle), Horizontal Alignment (Left, Center, Right; default Center), Text Alignment (Left align, Center align, Right align; default Center align), BG Color (Default, Light no overlay, Dark blue, Dark blue (subtle), Black).
- **Additional** — Anchor ID (validated text), Hide block (toggle), Padding left / right / top / bottom and Margin left / right / top / bottom (selects of the spacing tokens).

6.10.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM as a Cloud Service, open Sites, browse to the locale site (for example `/content/tcs-corp/us/en`), select the page and choose Edit; the page opens in Universal Editor.

Expected Result: The page renders on the canvas with the editing toolbar, the content tree and the properties panel.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** Universal Editor with the target page loaded and the content tree visible.

Step 2 — Add the Hero block

Action: Select the section that will hold the Hero (normally the first section; use a section of its own for the Hero), choose Add (+) and pick **Hero**.

Expected Result: A Hero is added with the defaults Vertical Alignment Middle, Horizontal Alignment Center, Text Alignment Center align and Title heading level H1; it shows as an empty banner until content is added.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-02-add-block.png`

> **Purpose:** Add the block
> **Description:** The component picker of a section with Hero highlighted.

Step 3 — Add the icon (optional)

Action: Select the Hero, open the **Image** tab, choose an asset in **Icon** and enter **Icon Image Alt Text** (leave empty if the icon is decorative).

Expected Result: A small icon appears above the eyebrow (2.5rem, 3rem on desktop).

> [Screenshot Placeholder]
> **File name:** `screenshots/hero-step-03-icon.png`
> **Purpose:** Add the icon
> **Description:** The Image tab with an icon selected and the icon visible above the title on the canvas.

Step 4 — Choose the Hero Image type and the image or animation

Action: In **Hero Image type** choose Default (image) or Lotto (Lottie animation). In **Hero Image Path** choose the image, or for Lotto the Lottie `.json` file, from AEM Assets.

Expected Result: The image fills the hero behind the content, or the animation starts playing with a pause button in the bottom-right corner.

> [Screenshot Placeholder]
> **File name:** `screenshots/hero-step-04-image.png`
> **Purpose:** Choose the image
> **Description:** The asset picker with a hero image selected and the Hero Image type set to Default.

Step 5 — Enter the alt text and choose the Hero View

Action: Enter **Hero Image alt text** (a short description of a meaningful image or animation; empty for decorative ones). Choose **Hero View**: Default (fills the hero) or Contained image (shown whole).

Expected Result: The image is cropped to fill the hero (Default) or shown entirely and centred (Contained image).

> [Screenshot Placeholder]
> **File name:** `screenshots/hero-step-05-alt-view.png`
> **Purpose:** Alt text and view
> **Description:** The Hero Image alt text filled in and Hero View set to Contained image, with the illustration shown whole.

Step 6 — Enter the eyebrow, title and heading level

Action: Open the **Content** tab. Enter **Eyebrow** and **Title**; choose **Title heading level** H1 for the first hero of the page, H2 for any other.

Expected Result: The eyebrow appears in brand blue above the large title.

> [Screenshot Placeholder]
> **File name:** `screenshots/hero-step-06-title.png`

> **Purpose:** Eyebrow and title
> **Description:** The Content tab with Eyebrow, Title and Title heading level filled in.

Step 7 — Enter the description

Action: Enter the **Description** in the rich-text editor (one or two short paragraphs).

Expected Result: The description appears below the title.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-07-description.png`
> **Purpose:** Description
> **Description:** The rich-text Description field and the description rendered below the title.

Step 8 — Add CTAs

Action: Open the **Links** tab. In **CTA List** choose Add for each CTA (recommended: one or two).

Expected Result: One empty item per CTA is created in the list.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-08-add-ctas.png`
> **Purpose:** Add CTA items
> **Description:** The CTA List multifield with two items added.

Step 9 — Configure each CTA

Action: For each item enter **CTA Title**, choose **CTA Link** (an AEM page or a URL), choose **CTA Link Type** (Default or Open in New Window), **CTA view** (Default, Primary, Secondary) and **CTA Arrow** (Right, Down).

Expected Result: The CTAs render as a text link with an arrow, a filled primary button or an outline secondary button, in a row below the description.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-09-configure-cta.png`
> **Purpose:** Configure a CTA
> **Description:** One CTA item with all five fields filled in and the resulting primary button on the canvas.

Step 10 — Choose the Motion Type

Action: In the **Links** tab choose **Motion Type**: None, Fade in, Slide up or Zoom (image).

Expected Result: No change in the editor (motion never plays in edit mode); the motion plays in Preview and on the published page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-10-motion.png`
> **Purpose:** Motion Type
> **Description:** The Motion Type select open with its four options.

Step 11 — Set the Style options

Action: Open the **Style** tab. Choose **Vertical Alignment**, **Horizontal Alignment**, **Text Alignment** and **BG Color**.

Expected Result: The content block moves to the chosen position, the text aligns and the overlay / text colours change.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-11-style.png`
> **Purpose:** Style options
> **Description:** The Style tab set to Bottom, Left, Left align and Dark blue, with the canvas showing white text at the bottom left.

Step 12 — Set the Additional options (optional)

Action: Open the **Additional** tab. Enter an **Anchor ID** if other links should jump to the Hero (for example `anchor-home-hero`), and adjust padding or margin only when the design requires it. Leave **Hide block** off.

Expected Result: The Anchor ID is accepted only in the form `anchor-` + lower-case letters, digits and hyphens; spacing changes show immediately.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-12-additional.png`
> **Purpose:** Additional tab
> **Description:** The Additional tab with an Anchor ID entered and the spacing selects at Default.

Step 13 — Preview

Action: Switch Universal Editor to Preview mode, then open the page preview (`https://main--tcs-corp--sradha-tcs.aem.page/{path}`) from AEM Sidekick or the Publish menu after a preview publish.

Expected Result: The Hero renders as on the live site, the motion plays once, the animation loops.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-13-preview.png`
> **Purpose:** Preview
> **Description:** The page on the .aem.page preview host with the Hero at desktop width.

Step 14 — Validate

Action: Run the Validation checklist below at desktop, tablet and mobile widths.

Expected Result: Every checklist item passes.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-14-validate.png`
> **Purpose:** Validate
> **Description:** The preview page in a browser's responsive mode at 375 px width with the Hero checked.

Step 15 — Publish

Action: In Universal Editor choose Publish and select Live (or use the AEM Sites console / AEM Sidekick Publish).

Expected Result: The page is available on `https://main--tcs-corp--sradha-tcs.aem.live/{path}` (and the production domain, [To be confirmed: production domain and CDN]).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/hero-step-15-publish.png`
> **Purpose:** Publish
> **Description:** The Publish dialog of Universal Editor with the page selected for Live.

Preview Process

1. Save the changes (Universal Editor saves automatically on each field change).
2. Use Preview mode in Universal Editor for a quick check (motion and links behave as on the site).
3. Publish to Preview to update `https://main--tcs-corp--sradha-tcs.aem.page/{path}`; review the page there with stakeholders.
4. Check desktop (1440 px), tablet (768 px and 1024 px) and mobile (375 px) widths.

Publishing Process

1. Confirm the Validation checklist below.
2. Publish the page to Live from Universal Editor or the Sites console.
3. If the Hero uses new assets (image, icon, Lottie `.json`), make sure the assets are published as well (Universal Editor publishes referenced assets with the page when selected in the publish dialog).
4. Open the live URL and reload with the browser cache disabled to confirm the change.

Validation Process

- ☐ Exactly one H1 on the page: the Hero title (heading level H1) or the Page Header, not both.
- ☐ The title is short and fits on at most two lines on desktop and four on mobile.
- ☐ The image is sharp at 1440 px and its subject stays visible in the 375 px centre crop.
- ☐ Meaningful images / animations have alt text; decorative ones have it empty.
- ☐ Text contrast is sufficient over the image (especially eyebrow and description on dark BG Colors and on Light, no overlay).
- ☐ Lotto: the animation plays, the pause / play button works and is reachable with the keyboard (Tab), and with reduced motion the animation is paused.
- ☐ Each CTA has a descriptive title, the correct link and view; in-page CTAs point down and jump to the right anchor.
- ☐ New-window CTAs open a new tab and their text says so where relevant.
- ☐ Motion Type plays once on the preview and the content is visible afterwards.
- ☐ The alignment (vertical, horizontal, text) matches the design at desktop width.
- ☐ Anchor ID (if any) is unique and links to it scroll to the Hero.
- ☐ Hide block is off.
- ☐ No errors in the browser console (for Lotto: no “failed to play the Lottie animation”).
- ☐ The hero image is the LCP element and loads quickly (PageSpeed Insights on the preview URL).

Rollback Process

1. In AEM Sites, open the page’s **Version history** (Timeline / Versions), compare and restore the previous version; or undo the field changes in Universal Editor.
2. Publish the restored page to Live.
3. If a new asset caused the problem, revert or replace the asset in AEM Assets and republish.

4. As an emergency measure, turn **Hide block** on (Additional tab) and publish: the Hero disappears from the page until it is fixed (remember that the page then has no H1 if the Hero held it).
5. Code problems are rolled back by the development team by reverting the commit on `main` (`blocks/hero/*`); Edge Delivery serves the reverted code after Code Sync.

6.10.6 Authorable Field Specification

The Hero model `hero` has 31 fields (excluding the five tab separators): 26 block fields and 5 fields per CTA List item. No field is marked required in the model, so every "Mandatory" value below is "No (not enforced)" unless stated otherwise; "Recommended" marks editorial guidance. Fields named `classes` or `classes_*` are stored as block classes (options), not as content.

Block model `hero` — Image tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Icon (`content_icon`)	reference (asset)	No	No	Empty	Small brand or topic icon above the eyebrow	Square icon, SVG or PNG; optional	None	`/content/dam/tcs/icons/ai.svg`
Icon Image Alt Text (`content_iconAlt`)	text	No	No	Empty	Text alternative of the icon	Leave empty if decorative	None	Artificial intelligence
Hero Image type (`classes_media`)	select	No	No	Default ("")	Image or Lottie animation	Lotto only with a `.json` file	Options: Default, Lotto	Lotto
Hero Image Path (`image`)	reference (asset)	No (Recommended)	No	Empty	The background image or animation	Landscape image, or Lottie `.json` for Lotto	None	`/content/dam/tcs/hero/home.jpg`
Hero Image alt text (`imageAlt`)	text	No	No	Empty	Text alternative of the image / animation	Describe meaningful media; empty if decorative	None	Engineers reviewing a digital twin
Hero View (`classes`)	select	No	No	Default ("")	Cover or contained image	Contained for illustrations	Options: Default, Contained image	Contained image

Block model `hero` — Content tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Eyebrow (`content_eyebrow`)	text	No	No	Empty	Short context line above the title	Max. about 40 characters (Recommended); only shown with a title	None	Enterprise AI
Title (`content_title`)	text	No (Recommended)	No	Empty	The headline of the hero and page	About 60 characters or less (Recommended)	None	Building on belief
Title heading level (`content_titleType`)	select	No	No	H1	Semantic heading level of the title	H1 for the first hero, H2 for others	Options: H1, H2	H1

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Description (`content_description`)	rich-text	No	No	Empty	Supporting text below the title	1–2 short paragraphs, about 200 characters (Recommended)	None	We help enterprises transform with AI-led services.

Block model `hero` — Links tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
CTA List (`ctas`)	container (composite multifield)	No	Yes	No items	The hero's calls to action	1–2 CTAs (Recommended); see the item table	None	Contact us; Our services
Motion Type (`classes_motion`)	select	No	No	None (“”)	Entrance animation	Use sparingly; none on most pages	Options: None, Fade in, Slide up, Zoom (image)	Fade in

Block model `hero` — Style tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Vertical Alignment (`classes_alignment`)	select	No	No	Middle (`align-middle`)	Vertical position of the content	Keep the image subject visible	Options: Top, Middle, Bottom	Bottom
Horizontal Alignment (`classes_horizontal`)	select	No	No	Center (`align-center`)	Horizontal position of the content (desktop)	Combine with a matching Text Alignment	Options: Left, Center, Right	Left
Text Alignment (`classes_textAlignment`)	select	No	No	Center align (`text-center`)	Alignment of icon, text and CTAs	Left for long descriptions	Options: Left align, Center align, Right align	Left align
BG Color (`classes_background`)	select	No	No	Default (“”)	Overlay and text colours	Dark options for white text	Options: Default, Light no overlay, Dark blue, Dark blue (subtle), Black	Dark blue

Block model `hero` — Additional tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	text	No	No	Empty	ID for in-page links to the block	Unique on the page, starts with `anchor-`	Enforced regExp `^anchor-[a-z0-9][a-z0-9-]*$` (“Start with anchor-, then lowercase letters, digits and hyphens.”)	anchor-home-hero
Hide block (`classes_hideblock`)	boolean	No	No	false	Temporarily hide the block	Turn off before go-live	None	false

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Padding left (`classes_paddingleft`)	select	No	No	Default (“”)	Extra left padding	Only when the design needs it	Options: Default, 0, space-100 … space-3000	pl-600
Padding right (`classes_paddingright`)	select	No	No	Default (“”)	Extra right padding	As above	Options: Default, 0, space-100 … space-3000	pr-600
Padding top (`classes_paddingtop`)	select	No	No	Default (“”)	Overrides the hero’s top padding	As above	Options: Default, 0, space-100 … space-3000	pt-1600
Padding bottom (`classes_paddingbottom`)	select	No	No	Default (“”)	Overrides the hero’s bottom padding	As above	Options: Default, 0, space-100 … space-3000	pb-1600
Margin left (`classes_marginleft`)	select	No	No	Default (“”)	Left margin of the block	Rarely needed for a full-width hero	Options: Default, 0, space-100 … space-3000	ml-0
Margin right (`classes_marginright`)	select	No	No	Default (“”)	Right margin of the block	As above	Options: Default, 0, space-100 … space-3000	mr-0
Margin top (`classes_margintop`)	select	No	No	Default (“”)	Space above the block	As above	Options: Default, 0, space-100 … space-3000	mt-0
Margin bottom (`classes_marginbottom`)	select	No	No	Default (“”)	Space below the block	As above	Options: Default, 0, space-100 … space-3000	mb-1000

Item model — CTA List item (fields of the `ctas` container)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
CTA Title (`ctaText`)	text	No (Recommended)	No	Empty	Visible text of the CTA	Verb first, 2–4 words, about 25 characters (Recommended)	None	Contact us
CTA Link (`cta`)	aem-content	Yes in practice (an item without a link is not rendered)	No	Empty	Target page or URL	AEM page, external URL or `#anchor-…`	None in the model; items without a link are dropped by the code	`/content/tcs-corp/us/en/contact-us`
CTA Link Type (`ctaTarget`)	select	No	No	Default (“”)	Same or new window	New window for external sites / documents	Options: Default, Open in New Window (`new-window`)	new-window
CTA view (`ctaType`)	select	No	No	Default (“”)	Text link, primary or secondary button	One Primary per hero	Options: Default, Primary, Secondary	primary
CTA Arrow (`ctaArrow`)	select	No	No	Right (`right`)	Arrow direction	Down for links within the page	Options: Right, Down	down

Field details

- **Icon (`content_icon`)** — Purpose: a small icon above the eyebrow. Allowed Values: one image asset (SVG, PNG, JPEG, WebP). Character Limits: not applicable. Business Rules: only the first picture before the title that stands alone (no text in the same paragraph) becomes the icon; it is shown at 2.5rem (3rem from 900 px) with `object-fit: contain`. Dependency Rules: part of the content group (`content_`), rendered in the content cell. Visibility Rules: always visible in the dialog. Sample Content: `/content/dam/tcs/icons/cloud.svg`.
- **Icon Image Alt Text (`content_iconAlt`)** — Purpose: alt text of the icon. Allowed Values: plain text. Character Limits: Recommended up to 60 characters. Business Rules: empty means a decorative icon. Dependency Rules: only meaningful with an Icon. Visibility Rules: always visible. Sample Content: Cloud services.
- **Hero Image type (`classes_media`)** — Purpose: chooses image or Lottie animation. Allowed Values: Default (“”), Lotto (`lottie`). Character Limits: not applicable. Business Rules: Lotto plays the animation only when Hero Image Path is a `.json` file; Default with a `.json` file shows no media. Dependency Rules: requires Hero Image Path. Visibility Rules: always visible. Sample Content: Lotto.
- **Hero Image Path (`image`)** — Purpose: the background image or animation. Allowed Values: one asset; an image, or a Lottie JSON file (`.json`) for Lotto. Character Limits: not applicable. Business Rules: the image of a first-section hero is loaded eagerly with `fetchpriority="high"`; without it the hero has no overlay. Dependency Rules: Hero Image type, Hero View, BG Color. Visibility Rules: always visible. Sample Content: `/content/dam/tcs/hero/home-hero.jpg` or `/content/dam/tcs/lottie/ai-waves.json`.
- **Hero Image alt text (`imageAlt`)** — Purpose: text alternative of the image or animation. Allowed Values: plain text. Character Limits: Recommended up to 125 characters. Business Rules: for an image it becomes the `alt`; for a Lottie animation the code reads it from the link’s `title` and, when present, gives the animation `role="img"` and `aria-label`; when empty the animation is `aria-hidden`. Dependency Rules: Hero Image Path. Visibility Rules: always visible. Sample Content: A data centre corridor with glowing servers.
- **Hero View (`classes`)** — Purpose: fill (cover) or contain the image. Allowed Values: Default (“”), Contained image (`media-contained`). Character Limits: not applicable. Business Rules: Contained image shows the whole image centred with 1.5rem padding. Dependency Rules: Hero Image Path. Visibility Rules: always visible. Sample Content: Contained image.
- **Eyebrow (`content_eyebrow`)** — Purpose: short kicker above the title. Allowed Values: plain text. Character Limits: Recommended up to 40 characters. Business Rules: the element before the title becomes the eyebrow (class `hero-eyebrow`); without a Title the eyebrow text is shown as part of the description instead. Dependency Rules: Title. Visibility Rules: always visible. Sample Content: Artificial Intelligence.
- **Title (`content_title`)** — Purpose: the headline. Allowed Values: plain text. Character Limits: Recommended up to 60 characters (display size 4rem on desktop). Business Rules: rendered as the heading of Title heading level with class `hero-title`; it is also the component title (`dc:title`) in the data layer. Dependency Rules: Title heading level. Visibility Rules: always visible. Sample Content: Building on belief.
- **Title heading level (`content_titleType`)** — Purpose: H1 or H2. Allowed Values: `h1`, `h2`. Character Limits: not applicable. Business Rules: default H1 (model and template); no visual difference. Dependency Rules: Title. Visibility Rules: always visible. Sample Content: h2.
- **Description (`content_description`)** — Purpose: supporting text. Allowed Values: rich text (paragraphs, bold, italic, links, lists). Character Limits: Recommended up to 200 characters. Business Rules: everything after the title (and anything before it that is not the icon or eyebrow) goes into `div.hero-description`; paragraphs are separated by 1rem. Dependency Rules: none. Visibility Rules: always visible. Sample Content: Our AI-led services help enterprises grow with resilience.
- **CTA List (`ctas`)** — Purpose: the calls to action. Allowed Values: any number of items. Character Limits: see the item fields. Business Rules: items without a link are dropped; the CTAs render in one wrapping row; if no item has a link the CTA row is omitted. Dependency Rules: none. Visibility Rules: always visible. Sample Content: two items “Contact us” (Primary) and “Our services” (Secondary).

- **Motion Type (`classes_motion`)** — Purpose: entrance animation. Allowed Values: None (“”), Fade in (`motion-fade`), Slide up (`motion-slide`), Zoom (image) (`motion-zoom`). Character Limits: not applicable. Business Rules: plays once when 10 % of the hero is visible; disabled in Universal Editor edit mode and for reduced motion. Dependency Rules: Zoom needs media. Visibility Rules: always visible. Sample Content: Fade in.
- **Vertical Alignment (`classes_alignment`)** — Purpose: vertical content position. Allowed Values: `align-top`, `align-middle`, `align-bottom`. Character Limits: not applicable. Business Rules: default Middle. Dependency Rules: none. Visibility Rules: always visible. Sample Content: align-bottom.
- **Horizontal Alignment (`classes_horizontal`)** — Purpose: horizontal content position. Allowed Values: `align-left`, `align-center`, `align-right`. Character Limits: not applicable. Business Rules: default Center; Left / Right limit the content to 47.5rem from 900 px; without a Text Alignment the text follows the position. Dependency Rules: Text Alignment. Visibility Rules: always visible. Sample Content: align-left.
- **Text Alignment (`classes_textAlignment`)** — Purpose: alignment inside the content block. Allowed Values: `text-left`, `text-center`, `text-right`. Character Limits: not applicable. Business Rules: default Center align; overrides the alignment implied by Horizontal Alignment. Dependency Rules: none. Visibility Rules: always visible. Sample Content: text-left.
- **BG Color (`classes_background`)** — Purpose: overlay and text colours. Allowed Values: Default (“”), `bg-light`, `bg-dark`, `bg-dark-subtle`, `bg-black`. Character Limits: not applicable. Business Rules: overlay only with media; dark options switch title, description and Default / Secondary CTAs to white. Dependency Rules: Hero Image Path. Visibility Rules: always visible. Sample Content: bg-dark.
- **Anchor ID (`classes_anchor`)** — Purpose: makes the block the target of `#anchor-…` links. Allowed Values: `anchor-` followed by lower-case letters, digits and hyphens. Character Limits: Enforced pattern; no length limit (Recommended up to 40 characters). Business Rules: becomes the block `id` (a duplicate on the same page gets `-2`, `-3`, …) and the component id in the data layer. Dependency Rules: none. Visibility Rules: always visible. Sample Content: anchor-home-hero.
- **Hide block (`classes_hideblock`)** — Purpose: hide the block without deleting it. Allowed Values: on / off. Character Limits: not applicable. Business Rules: hidden on preview and live (`display: none`, its wrapper too); faded with a dashed outline in edit mode. Dependency Rules: none. Visibility Rules: always visible. Sample Content: off.
- **Padding left (`classes_paddingleft`)** — Purpose: left padding from the spacing tokens. Allowed Values: Default, `pl-0`, `pl-100` (0.25rem), `pl-200` (0.5rem), `pl-300` (0.75rem), `pl-400` (1rem), `pl-500` (1.25rem), `pl-600` (1.5rem), `pl-800` (2rem), `pl-1000` (2.5rem), `pl-1200` (3rem), `pl-1600` (4rem), `pl-2000` (5rem), `pl-2400` (6rem), `pl-3000` (7.5rem). Character Limits: not applicable. Business Rules: overrides the hero’s own padding at every breakpoint. Dependency Rules: none. Visibility Rules: always visible. Sample Content: pl-1000.
- **Padding right (`classes_paddingright`)** — As Padding left with `pr-*` values. Sample Content: pr-1000.
- **Padding top (`classes_paddingtop`)** — As Padding left with `pt-*` values; replaces the hero’s 4rem / 7.5rem top padding. Sample Content: pt-2000.
- **Padding bottom (`classes_paddingbottom`)** — As Padding left with `pb-*` values. Sample Content: pb-2000.
- **Margin left (`classes_marginleft`)** — Purpose: left margin from the spacing tokens. Allowed Values: Default, `ml-0` … `ml-3000` (same 14 token steps). Business Rules: a margin makes the full-width hero narrower than the viewport. Dependency Rules: none. Visibility Rules: always visible. Sample Content: ml-0.
- **Margin right (`classes_marginright`)** — As Margin left with `mr-*` values.
- **Margin top (`classes_margintop`)** — As Margin left with `mt-*` values. Sample Content: mt-0.
- **Margin bottom (`classes_marginbottom`)** — As Margin left with `mb-*` values. Sample Content: mb-1000.
- **CTA Title (`ctaText`)** — Purpose: CTA text. Allowed Values: plain text. Character Limits: Recommended up to 25 characters (buttons do not wrap their text). Business Rules: becomes the link text. Dependency Rules: CTA Link. Visibility Rules: always visible. Sample Content: Contact us.
- **CTA Link (`cta`)** — Purpose: the target. Allowed Values: AEM page, external URL, `#anchor`. Character Limits: not applicable. Business Rules: an item without a link is not rendered; a link starting with `#` without an arrow

value points down. Dependency Rules: none. Visibility Rules: always visible. Sample Content: `/content/tcs-corp/us/en/contact-us`.

- **CTA Link Type (`ctaTarget`)** — Purpose: open in a new window. Allowed Values: Default (""), `new-window`. Business Rules: `new-window` sets `target="_blank"` and `rel="noopener noreferrer"`. Dependency Rules: CTA Link. Visibility Rules: always visible. Sample Content: new-window.
- **CTA view (`ctaType`)** — Purpose: CTA style. Allowed Values: Default (""), `primary`, `secondary`. Business Rules: delivered as bold (primary) or italic (secondary) formatting of the link; Default renders as a tertiary text link with an arrow. Dependency Rules: none. Visibility Rules: always visible. Sample Content: primary.
- **CTA Arrow (`ctaArrow`)** — Purpose: arrow direction. Allowed Values: `right`, `down`. Business Rules: default right; `down` adds `hero-cta-down`. Dependency Rules: none. Visibility Rules: always visible. Sample Content: down.

6.10.7 Field Annotation Mapping

Marker	Field	Purpose
A	Hero block (Style tab: Vertical Alignment, Horizontal Alignment, Text Alignment, BG Color; Links tab: Motion Type; Additional tab)	Block options that position and colour the hero
B	Hero Image type, Hero Image Path, Hero Image alt text, Hero View	The background image or Lottie animation
C	Icon, Icon Image Alt Text	The small icon above the eyebrow
D	Eyebrow	The short line above the title
E	Title, Title heading level	The headline (H1 / H2)
F	Description	The supporting rich text
G	CTA List (CTA Title, CTA Link, CTA Link Type, CTA view, CTA Arrow)	The calls to action

6.10.8 Component Specification

Attribute	Value
Component Name	Hero (`hero`; Universal Editor title "Hero")
Version	1.0
Owner	[To be confirmed: product owner — owner: Digital Marketing]; technical owner: development team
Dependencies	`decorators/animation-decorator.js`, `decorators/button-decorator.js`, `decorators/icon-button-decorator.js`, `decorators/link-decorator.js`, `utils/buttons.js`, `utils/links.js`, `utils/multifield.js`, `utils/environment.js`, `utils/viewport.js`, `utils/media.js`, `utils/lottie.js`, `utils/placeholders.js`, `scripts/scripts.js`, `scripts/lib/lottie/lottie_light.min.js`, `styles/styles.css`
Component Type	Block (resource type `core/franklin/components/block/v1/block`, model `hero`); the CTA List is a composite multifield inside the block, not a separate item component
Supported Variants	Image type Default / Lotto; Hero View Default / Contained image; BG Color Default / Light no overlay / Dark blue / Dark blue (subtle) / Black; Vertical Top / Middle / Bottom; Horizontal Left / Center / Right; Text Left / Center / Right; Motion None / Fade in / Slide up / Zoom; CTA view Default / Primary / Secondary; CTA Arrow Right / Down; CTA Link Type Default / New window; Title H1 / H2
Supported Templates	Any page; allowed in Section and Gen Carousel; Home template removes the first section's top spacing
Supported Metadata	None read by the block (page metadata is not used)
Supported Tags	None (the Hero has no tag field)

Attribute	Value
Supported Personalization	None in the block; the block is in the data layer (`tcs-corp/components/hero`) so Adobe Target could target it once Adobe Tags is added (Not implemented (as of this handover))
Caching Considerations	Rendered client-side from the page HTML cached by the Edge Delivery CDN; the Lottie player and `.json` file are cached as static resources; placeholders are fetched once per page (`/{locale}/placeholders.json`) only when a Lottie toggle is created
Performance Considerations	In the first section the block loads in the eager phase; its image gets `loading="eager"` and `fetchpriority="high"` (LCP). The Lottie player loads only when the hero is within 200 px of the viewport and the browser is idle; the animation pauses off screen. Motion uses CSS animations only
Accessibility Compliance	WCAG 2.1 AA: heading level selectable (1.3.1, 2.4.6); alt text for image, icon and animation (1.1.1); Lottie pause / play button with localised labels (2.2.2); reduced motion respected for motion and animation (2.3.3); white text and CTA colour switch on dark backgrounds (1.4.3); keyboard-reachable CTAs and toggle (2.1.1); no automatic new-window hint (author must state it)
SEO Compliance	One H1 per page when used as the first hero; title and description are server-delivered HTML (crawlable); image alt text indexable; optimised LCP image; CTAs are crawlable `<a href>` links

6.10.9 Input Model

Content Inputs: icon, eyebrow, title, title heading level, description (content group `content_`), hero image or Lottie file with its alt text, CTA List items.

Metadata Inputs: none.

Tags Inputs: none.

Page Context Inputs: whether the block is in the first section of `main` (eager image loading); Universal Editor edit mode (`html.adobe-ue-edit`, disables motion); the visitor's `prefers-reduced-motion` setting; the page's locale (placeholders for the animation toggle).

Query Inputs: none (no query index or sheet besides placeholders).

Manual Inputs: all fields are manual; the block options arrive as block classes.

Authored HTML received from AEM (rows can arrive in any order; the code sorts them into the content, media and CTA cells):

```
<div class="hero lottie align-bottom align-left text-left bg-dark motion-fade">
  <div><div>
    <p><picture><img src="./media_1a2b.svg" alt="Cloud services"></picture></p>
    <p>Enterprise AI</p>
    <h1>Building on belief</h1>
    <p>Our AI-led services help enterprises grow with resilience.</p>
  </div></div>
  <div><div>
    <p><a href="/content/dam/tcs/lottie/ai-waves.json"
      title="Animated waves of data">/content/dam/tcs/lottie/ai-waves.json</a></p>
  </div></div>
  <div><div>
    <p><strong><a href="/en/contact-us">Contact us</a></strong></p>
    <p>right</p>
    <hr>
    <p><em><a href="https://example.com/ai-report.pdf">AI report (PDF)</a></em></p>
    <p>new-window</p>
    <p>right</p>
    <hr>
    <p><a href="#anchor-our-story">Discover our story</a></p>
    <p>down</p>
```

```
  </div></div>
</div>
```

6.10.10 Output Model

Rendered Output

`hero.js` replaces the authored rows with at most two children; the block keeps its option classes and gets `hero-has-media` when there is media.

- `div.hero` (+ option classes, `hero-has-media`, and at runtime `motion-ready`, `is-visible`)
  - `div.hero-media` (only with media) containing the `<picture>`, or `div.hero-animation` (Lotto)
  - `div.hero-content`
    - `div.hero-icon` with the icon `<picture>` (optional)
    - `p.hero-eyebrow` (optional; only when there is a title)
    - `h1.hero-title` or `h2.hero-title` (optional)
    - `div.hero-description` (optional)
    - `div.hero-ctas` with `a.button.primary|secondary|tertiary` (optional; `hero-cta-down` on down-arrow links; `target="_blank"` on new-window links)
  - `button.animation-toggle.button.icon-button.secondary` (Lotto only; appended to the block)

Universal Editor instrumentation (`data-aue-*`, `data-richtext-*`) is moved from the authored cells to `hero-media`, `hero-content` and `hero-ctas`, so in-place editing keeps working.

Generated HTML

```
<div class="hero align-middle align-center text-center bg-dark hero-has-media block"
  data-block-name="hero" data-block-status="loaded" data-cmp-id="hero-1">
  <div class="hero-media">
    <picture><img src="./media_9f8e.jpg?width=2000" alt="" loading="eager"
      fetchpriority="high"></picture>
  </div>
  <div class="hero-content">
    <p class="hero-eyebrow">Enterprise AI</p>
    <h1 class="hero-title">Building on belief</h1>
    <div class="hero-description"><p>Our AI-led services help enterprises grow.</p></div>
    <div class="hero-ctas">
      <a href="/en/contact-us" title="Contact us" class="button primary">Contact us</a>
      <a href="#anchor-our-story" title="Discover our story"
        class="button tertiary hero-cta-down">Discover our story</a>
    </div>
  </div>
</div>
```

With Lotto, the media and the toggle look like this:

```
<div class="hero-media">
  <div class="hero-animation" role="img" aria-label="Animated waves of data">
    <svg><!-- rendered by the Lottie player --></svg>
  </div>
</div>
<button type="button" class="animation-toggle button secondary icon-button"
  aria-label="Pause animation" title="Pause animation"
  style="--icon-url: url('/icons/pause.svg')" data-cmp-interaction="animation"></button>
```

Generated JSON: none (the block produces no JSON).

Generated Metadata: none (the block does not write page metadata or `<head>` tags).

Analytics Tracking

The data layer decorator adds the block to `window.adobeDataLayer` once it is decorated, and tracks clicks on its links and buttons. The component id is the Anchor ID when set, else `hero-{n}` in page order.

```
{
  "component": {
    "hero-1": {
      "@type": "tcs-corp/components/hero",
      "parentId": "page-1x2y3z",
      "dc:title": "Building on belief"
    }
  }
}
```

A click on a CTA:

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.hero-1",
    "interaction": "link",
    "dc:title": "Contact us",
    "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/contact-us",
    "xdm:linkType": "other",
    "breakpoint": "desktop"
  }
}
```

`xdm:linkType` is `exit` for links to another origin and `download` for file links (PDF, DOCX, XLSX, PPTX, ZIP, CSV, TXT, MP3, MP4). A click on the Lottie pause / play button is a `cmp:click` with `"interaction": "animation"` and `dc:title` "Pause animation" (or "Play animation", the label at the time of the click). The Lottie animation is not an HTML media element, so no `cmp:media` events are sent. No tag manager reads the data layer yet (Adobe Tags / Web SDK are integration-ready, not implemented).

Personalization Output: none (no personalization marker or Target mbox).

Search Index Output: none. The query index (`helix-query.yaml`) only reads `<head>` metadata, so Hero content is not indexed in `/query-index.json`; the Hero HTML is crawlable by search engines as part of the page.

6.10.11 Business Rules

1. Rows are sorted by content, not position: a cell with only one picture (and no text), or only one `.json` link, is the media; a cell with only links (plus `new-window`, `right`, `down` values) and no heading is the CTA List; the first other cell is the content and any further cells are appended to it.
2. The media check runs first, so a cell with only a `.json` link is never read as a CTA List.
3. Lotto plays only when the block has class `lottie` and the media link ends in `.json` (query string or hash allowed).
4. With Lotto and an image, the image is shown (fallback).
5. With Default and a `.json` link, no media is rendered and the hero shows without media.
6. The Lottie label comes from the link's `title` when it differs from the link text; else the animation is decorative (`aria-hidden="true"`).
7. The Lottie toggle is appended to the block (outside `hero-media`) so it is above the overlay and the content; it is placed at the bottom right (`inset` 1rem).

8. Lottie: loop on; SVG renderer; `preserveAspectRatio` "xMidYMid meet"; starts paused on frame 0 for reduced motion; plays when visible and pauses when off screen; the player loads once per page, within 200 px of the viewport, in browser idle time (or after 200 ms where `requestIdleCallback` is missing).
9. A hero image in the first section of `main` gets `loading="eager"` and `fetchpriority="high"`.
10. Without media, `hero-has-media` is not set and no overlay is drawn.
11. Content order: icon, eyebrow, title, description, CTAs, regardless of the authored order of the description elements.
12. The icon is the first stand-alone picture before the heading (a paragraph with only a picture); only one icon is used; other pictures go to the description.
13. The eyebrow is the first non-icon element before the heading; it exists only when there is a heading; without a heading every element (except the icon) goes into the description.
14. The description container is omitted when empty.
15. CTA view: Primary and Secondary stay; anything else (Default) becomes tertiary (text + arrow).
16. CTA Arrow Down, or no arrow value and a link starting with `#`, adds `hero-cta-down`.
17. `new-window` sets `target="_blank"` and `rel="noopener noreferrer"`; no new-tab hint is added.
18. CTA items without a link are dropped; the CTA row is omitted when no CTA remains.
19. Motion Type applies only with a motion class, outside Universal Editor edit mode and without reduced motion; then `motion-ready` hides Fade / Slide content until 10 % of the hero is visible, when `is-visible` plays the animation once.
20. Motion CSS is also guarded by `@media (prefers-reduced-motion: no-preference)` and `:root:not(.adobe-ue-edit)`.
21. Horizontal Alignment Left / Right also aligns the text when no Text Alignment class is present (backwards compatibility); new Heroes get Text Alignment Center from the template.
22. From 900 px, Left / Right limit the content block to 47.5rem; below 900 px the content is full width.
23. Dark BG Colors switch the title to white, the description to 50 % white (medium weight), Secondary CTAs to white outline and Default CTAs to white (light blue on hover); the eyebrow stays brand blue.
24. The section wrapper of a Hero is full width without padding (`.hero-container .hero-wrapper`).
25. Additional tab: Anchor ID must match `^anchor-[a-z0-9][a-z0-9-]*$` (enforced in the dialog) and becomes the block id (unique, `-2`, `-3` suffixes); Hide block hides it outside edit mode; padding / margin classes override the hero's own spacing.
26. Error handling: a Lottie load failure logs `failed to play the Lottie animation` in the console and leaves the media box empty; the rest of the hero renders. Missing placeholders fall back to English labels.

6.10.12 Author Best Practices

Recommended Content Length

Element	Recommended	Why
Eyebrow	Up to 40 characters, one line	Kicker, not a sentence
Title	Up to 60 characters (2 lines on desktop)	Display size 4rem desktop, 2.5rem mobile
Description	Up to 200 characters, 1–2 paragraphs	Readability over images
CTA Title	2–4 words, up to 25 characters	Buttons do not wrap their text
Number of CTAs	1–2 (maximum 3)	Clear primary action
Alt text	Up to 125 characters	Concise screen-reader description

Image Dimensions

- Cover image (Hero View Default): landscape, at least 1920 × 1080 px (16:9), ideally 2400 px wide; the box is at least 850 px tall on desktop and 560 px on mobile, cropped around the centre (`object-fit: cover`), so keep the subject in the central area and away from where the text sits.
- Contained image: the whole image is shown inside the hero minus 1.5rem padding; a transparent PNG or SVG illustration, around 1600 px wide.
- Icon: square, at least 96 × 96 px (displayed at 40–48 px), SVG preferred.
- Lottie: a Lottie JSON exported for the SVG renderer, ideally under 500 KB; its own aspect ratio is kept ("meet"), so design it for a wide (16:9) frame.

CTA Writing Guidelines

- Start with a verb and say what happens: "Contact us", "Explore our services".
- One Primary CTA; a Secondary or Default CTA for the alternative.
- Use Arrow Down only for links to the same page (`#anchor-…`).
- For new-window links, say so in the text when it matters ("(PDF)", "(opens LinkedIn)").

Accessibility Guidelines

- Exactly one H1 per page; heading level H2 for every other Hero.
- Alt text for meaningful images and animations; empty for decorative ones.
- Check contrast: use a darker overlay (Dark blue instead of Dark blue (subtle)) when the image is bright; avoid Light, no overlay over busy images.
- Keep motion optional; avoid combining a Lottie background with Slide up / Zoom.
- Make link text meaningful out of context (no "Click here").

SEO Guidelines

- Put the main keyword in the H1 title; keep it consistent with the page's Title / meta title.
- Use real text, not text baked into the image.
- Optimise image weight in AEM Assets; Edge Delivery serves optimised renditions.

Metadata Guidelines: the Hero does not read page metadata. Set the page Title, Description and the SEO columns of the metadata sheet separately (they are not derived from the Hero).

Taxonomy Guidelines: the Hero has no tags; tag the page itself in its page properties.

Localization Guidelines

- Every locale site has its own Hero content; translate the eyebrow, title, description, CTA titles and alt texts, and link CTAs to the same-locale pages (`/de-de/...`).
- Allow 30 % more length for German, French and Dutch titles and CTAs; check that buttons still fit on mobile.
- The pause / play labels come from the locale's placeholders sheet (keys `pauseAnimation`, `playAnimation`); add translated texts to each locale's `placeholders`.
- Lottie animations with embedded text must be produced per language.

6.10.13 Troubleshooting Guide

Issue 1 — The Lottie animation does not play

Symptoms: The hero shows no background (or a static image) although a `.json` file was selected.

Root Cause: Hero Image type is Default (a `.json` file is then not rendered); or the file is not a valid Lottie JSON; or the player failed to load; or the visitor prefers reduced motion (the animation then starts paused on its first frame).

Diagnosis Steps: Check the Image tab (Hero Image type = Lotto?); inspect the block in the browser: does it have class `lottie` and a `div.hero-animation` with an `<svg>`? Look for `failed to play the Lottie animation` in the console; check the Network tab for `lottie_light.min.js` and the `.json` request (status 200?); check the OS reduced-motion setting.

Resolution Steps: Set Hero Image type to Lotto; re-export the animation as Lottie JSON for the SVG renderer (the light player does not support expressions); publish the asset; press Play on the toggle if reduced motion is on.

Validation Steps: Reload the preview page; the animation plays and the toggle switches between Pause and Play.

Preventive Measures: Test every new Lottie file on a block-library page first; keep the validation checklist.

Issue 2 — The title or CTAs are not visible on page load

Symptoms: The content appears only after scrolling or after a delay, or stays invisible.

Root Cause: Motion Type Fade in / Slide up hides the content (`motion-ready`) until 10 % of the hero is visible; if a script error stops the observer, it stays hidden.

Diagnosis Steps: Inspect the block classes: `motion-ready` without `is-visible`? Check the console for errors.

Resolution Steps: Set Motion Type to None for first-section heroes if the delay is not wanted; report script errors to L3.

Validation Steps: Reload the page; the content is visible immediately (None) or after the 800 ms animation.

Preventive Measures: Use motion only where it adds value; keep LCP heroes static.

Issue 3 — The image is cropped badly on mobile

Symptoms: On phones the subject of the image is cut off or hidden behind the text.

Root Cause: Hero View Default crops the image to fill a tall 35rem box around the centre.

Diagnosis Steps: View the page at 375 px; compare the image composition with the crop.

Resolution Steps: Choose an image with a centred subject; move the content with Vertical / Horizontal Alignment; or use Hero View Contained image for illustrations.

Validation Steps: Check 375 px, 768 px and 1440 px widths.

Preventive Measures: Brief photographers / designers on the safe central area.

Issue 4 — Two H1 headings on the page

Symptoms: SEO audit or accessibility tool reports multiple H1s.

Root Cause: The Hero title is H1 (default) while another Hero, a Page Header (article, blog, press-release templates) or an authored H1 also exists.

Diagnosis Steps: Run an accessibility / SEO checker or search the DOM for `h1`.

Resolution Steps: Set Title heading level to H2 on every Hero except the page's main one; on templated pages use the Page Header for the H1.

Validation Steps: Exactly one `h1` in the DOM.

Preventive Measures: Validation checklist item "exactly one H1".

Issue 5 — A CTA is missing or shows as plain text

Symptoms: Fewer CTAs than authored; or the CTA text appears inside the description.

Root Cause: The item has no CTA Link (dropped); or the CTA cell contains other text, so it is read as content.

Diagnosis Steps: Open each CTA List item and check CTA Link; inspect the authored HTML (`.plain.html` of the page) for the CTA row.

Resolution Steps: Fill CTA Link for every item; remove stray text from the CTA items.

Validation Steps: All CTAs show in the `hero-ctas` row with the right view.

Preventive Measures: Always fill CTA Title and CTA Link together.

Issue 6 — Text is hard to read over the image

Symptoms: Low contrast between text and background; accessibility checker flags contrast.

Root Cause: BG Color Light, no overlay over a dark / busy image; Dark blue (subtle) over a bright image; brand-blue eyebrow or 50 % white description on dark backgrounds.

Diagnosis Steps: Measure contrast with a contrast checker at the text position.

Resolution Steps: Choose Default (white overlay) or Dark blue / Black; replace the image; move the content to a calmer area.

Validation Steps: Contrast of at least 4.5:1 for body text, 3:1 for the large title.

Preventive Measures: Pair BG Colors with suitable images in the design system guidance.

Issue 7 — The CTA arrow points down for a normal link

Symptoms: A CTA to another page shows a down arrow.

Root Cause: CTA Arrow is Down, or the item has no arrow value and the link starts with `#`.

Diagnosis Steps: Open the item and check CTA Arrow and CTA Link.

Resolution Steps: Set CTA Arrow to Right.

Validation Steps: The arrow points right in preview.

Preventive Measures: Use Down only for in-page anchors.

Issue 8 — The hero is missing on the published page

Symptoms: The hero shows in the editor (faded) but not on preview / live.

Root Cause: Hide block is on.

Diagnosis Steps: Additional tab: Hide block; the block has class `hideblock`.

Resolution Steps: Turn Hide block off and publish.

Validation Steps: The hero renders on the live URL.

Preventive Measures: Review the Additional tab before publishing.

6.10.14 Support Runbook

Monitoring Requirements

- Core Web Vitals of pages that open with a Hero (LCP is usually the hero image): PageSpeed Insights or real-user monitoring ([To be confirmed: RUM / monitoring tool — owner: site admin]).
- Periodic visual checks of the home pages of all 13 locales.
- Availability of the Lottie player (`/scripts/lib/lottie/lottie_light.min.js`) and Lottie assets.

Logs

- Browser console: `failed to play the Lottie animation` (with the error) from `decorators/animation-decorator.js`; `failed to load template {name}` from `scripts.js` (not Hero-specific). The Hero itself logs nothing else.
- Admin API: preview / publish status of the page and assets (`https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`), visible in AEM Sidekick.
- AEM author: publication and replication logs in AEM as a Cloud Service (Cloud Manager log download) for publish failures.

Common Errors

Error	Likely cause	Fix
No background shown	Default type with a `.json` file; or asset not published	Set Lotto or choose an image; publish the asset
Console "failed to play the Lottie animation"	Invalid JSON, 404, unsupported Lottie features	Re-export / republish the file
Content invisible until scroll	Motion Type Fade / Slide	Use None for first-section heroes
Two H1s	Default H1 with a Page Header or second hero	Use H2
CTA missing	Item without CTA Link	Fill CTA Link
Anchor ID rejected	Does not match `anchor-…` pattern	Use lower-case `anchor-name`
Hero not visible live	Hide block on	Turn off and publish

Escalation Matrix

Level	Who	Checks
L1	Author support / content team lead	Dialog values (image type, alt text, CTAs, Hide block), asset publication, page preview vs live, checklist above
L2	AMS / site admin	Page and asset publish status in Sidekick / admin API, CDN cache (purge by republishing), placeholders sheet published, browser console errors
L3	Development team	`blocks/hero/*`, animation decorator, Lottie player, CSS regressions, Code Sync status of `main`, reproduction on a feature branch
L4	Adobe Support	Edge Delivery Services delivery or AEM as a Cloud Service / Universal Editor platform incidents

Recovery Procedures

1. Content error: restore the previous page version in AEM and publish.
2. Broken asset: replace or restore the asset in AEM Assets and republish the page.
3. Lottie failure: switch Hero Image type to Default and select a fallback image; publish.
4. Code regression: revert the offending commit on `main`; verify on `main--tcs-corp--sradha-tcs.aem.page`.
5. Emergency: Hide block on and publish while the fix is prepared.

Health Checks

- `curl -s https://main--tcs-corp--sradha-tcs.aem.live/en.plain.html` (or the page path) returns the hero rows.
- The home page of each locale renders the hero with no console errors.
- PageSpeed Insights LCP under 2.5 s for the home pages.
- The Lottie player URL returns 200.

Operational Procedures

- New hero design options are added by the development team in `_hero.json`, followed by `npm run build:json`, `npm run lint` and a pull request with a preview link.
- Placeholder labels (`pauseAnimation`, `playAnimation`) are maintained in each locale's placeholders sheet.
- Use the locale block-library pages to validate Hero variations after code changes.

Known Limitations

- No video background (Not implemented (as of this handover)).
- One media item only; no separate mobile / desktop image (art direction) field.
- No automatic "opens in a new tab" hint for new-window CTAs.
- The Lottie light player does not support After Effects expressions.

- Motion Type Fade / Slide briefly hides first-section content, which can affect perceived LCP.
- The eyebrow colour does not change on dark backgrounds.
- Horizontal Alignment only shows as a block position from 900 px.
- Without a title, the eyebrow is treated as description text.

6.11 Teaser Block

The Teaser is the site's most versatile content block: a title with optional eyebrow, tags, date, reading time, page actions, author, description, image or Lottie animation, statistics, links and a short description, presented in one of four layouts — Image & Text, Text split, Background image and Card — with many style options. It implements the TCS.com Design System Figma frames "Text & Image", "2 Slot Section", "Text Section - 2 Columns", "Secondary Page Header", "Careers", "Media Coverage", "Media Kit" (Card) and the article header elements of "L3 Header" (Text split).

Source files: `blocks/teaser/teaser.js`, `blocks/teaser/teaser.css`, `blocks/teaser/_teaser.json`. The Teaser is a **key-value block**: Universal Editor stores one row per property (property name | value), and the code reads the rows by name.

6.11.1 Component Purpose

Business Objective

Present a focused message — a service, an insight, a statistic-led story, a career call or a media resource — with a consistent, design-system look, and lead the visitor to the next page through clear links, while letting authors build many approved Figma variations from one component instead of maintaining several near-identical blocks.

Business Use Cases

- Image & Text promotion of a service, industry or capability, with the image left or right.
- Two-column text sections (Text split): title left, text, chart, stats and links right.
- Statistic-led sections ("40 % cost reduction") with the Stats tab.
- Secondary page headers of landing pages (Top spacing "Page header", large title).
- Article-style header (Figma "L3 Header"): eyebrow, tags, date, reading time, listen / download / share actions and author.
- Careers call to action without image (centred text).
- Full-width banners with a background image under a dark overlay (Background image).
- Media kit / media coverage cards with an image behind the text (Card).
- Link lists ("List" link style) for related resources.
- Animated (Lottie) illustrations beside the text (Image type Lotto).
- A slide of a Gen Carousel section.

When To Use

- For any single message that combines a heading with text, media, stats or links inside the page flow.
- When the design is one of the Teaser's Figma variations.

When Not To Use

- For the first, full-height banner of a landing page: use the Hero.
- For article, blog and press-release page headers built from page metadata: use the Page Header (added automatically by those templates).
- For lists of several items (cards, pages, people): use Cards, List or Filter List.
- For quotes and testimonials: use the Quote or Testimonial block.
- For video: use the Video or Embed block (the Teaser has no video media type; Not implemented (as of this handover)).

Dependencies

Type	Dependency	Used for
Core	`scripts/aem.js` `toClassName`; `scripts/scripts.js` `moveInstrumentation`	Normalising values; keeping Universal Editor instrumentation
Decorator	`decorators/a11y-decorator.js` `hideVisually`	Hide Title (visually hidden, kept for screen readers)
Decorator	`decorators/animation-decorator.js` `createAnimation`	Lotto image type and its pause / play button
Decorator	`decorators/article-meta-decorator.js` `buildPublished`, `buildReadingTime`, `buildPageActions`, `buildAuthorLockup`	Date, reading time, listen / download / share, author
Decorator	`decorators/date-decorator.js` (through article meta) and `utils/dates.js`	`<time>` element and date formats
Decorator	`decorators/button-decorator.js`, `decorators/icon-button-decorator.js`	Buttons, text links, icon buttons
Decorator	`decorators/content-decorator.js` `fromCell`	Moving eyebrow, description, author texts into the output
Decorator	`decorators/heading-decorator.js` `buildHeading`	Title with Title Type (H1–H6)
Decorator	`decorators/image-decorator.js` `pictureOf`	Image and author image with alt text
Decorator	`decorators/link-decorator.js` `setLinkTarget`	New-window links in List style
Decorator	`decorators/stats-decorator.js` `buildStats`	Stats as a description list
Decorator	`decorators/tag-decorator.js` `buildTagList`	Tag list
Util	`utils/properties.js` `readProperties`, `textOf`, `isOn`, `optionOf`	Reading the key-value rows
Util	`utils/links.js` `NEW_WINDOW`, `linkOf`, `readLinkItems`; `utils/multifield.js`	Links, Links multifield, Stats multifield
Util	`utils/media.js` `isAnimation`; `utils/lottie.js`	Lottie recognition and player
Util	`utils/placeholders.js` `getPlaceholders`	UI labels
Util	`utils/tags.js` `readTagValues`; `utils/taxonomy.js` `resolveTags`; `utils/site.js`	Tag IDs and their localised labels
Util	`utils/ids.js` `uniqueId` (through article meta)	Unique id of the share status message
Sheet	`/{locale}/placeholders.json`	Keys `minRead` (“{n} min read”), `listenToThisPage`, `downloadThisPage`, `shareThisPage`, `linkCopied`, `pauseAnimation`, `playAnimation`
Sheet	`/{locale}/taxonomy.json`	Tag titles (the locale’s language sheet when translated)
Data source	AEM Assets	Image, Lottie `.json`, author image
Data source	AEM tags	Tags field
Icons	`arrow-right.svg`, `arrow-up-right.svg`, `headphones.svg`, `download.svg`, `share.svg`, `pause.svg`, `play.svg`	Links, actions, animation toggle

Related Components

- Hero: full-height opening banner (shares the Lotto image type and animation decorator).

- Page Header: metadata-driven article header (shares the article meta decorator: date, reading time, actions, author).
- Cards and List: card collections (the List / Cards "Display as teaser" option is a card presentation, not this block).
- Testimonial: shares the Stats decorator.
- Gen Carousel section: can rotate Teasers as slides.
- Button component and Separator: often placed around Teasers.

Applicable Templates

Any page (home, article, blog, press-release or no template). Allowed in every Section and in Gen Carousel sections (filters `section` and `gen-carousel`).

Supported Content Types

- Text (eyebrow, title, labels), rich text (description, short description, stats).
- Image (JPEG, PNG, WebP, SVG) or Lottie JSON animation (Image type Lotto).
- Author image.
- Links: AEM pages, external URLs, anchors; a listen (audio) link.
- AEM tags.
- Date (date-time picker) and reading time.
- Not supported: video, several images, carousels inside the teaser.

6.11.2 Visual Overview

> [**Screenshot Placeholder**]
> **File name:** `screenshots/teaser-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** An Image & Text teaser at 1440 px: a rounded 10:9 image on the left, on the right a brand-blue eyebrow, a semibold title, grey description, a primary and a secondary button, vertically centred against the image.

Desktop (900 px and wider)

- Image & Text with an image: two equal columns (`repeat(2, minmax(0, 1fr))`) with a 5rem gap; the image left (or right), 10:9, 1rem rounded corners; the text vertically centred.
- Text split: the kicker (eyebrow, tags, date, actions) across the top, then two equal columns with a 1.5rem gap: title and author left, description, image, stats, links and short description right (3rem between them).
- Background image: at least 48.75rem tall, 7.5rem / 5rem padding, centred text (maximum 37.5rem wide) in white over the image and a 60 % dark overlay.
- Card: at least 35.625rem tall, 2.5rem padding, 1rem radius, header at the top and links at the bottom, white text over the image and a 70 % black overlay.
- With a BG Color (Grey, Dark, Dark green) the teaser becomes a panel with 7.5rem / 5rem padding.
- Type: title 2.5rem (Image & Text with image) or 3rem (other layouts); description 1.125rem (1.5rem in Text split); eyebrow 1rem; stat values 6rem.

> [**Screenshot Placeholder**]
> **File name:** `screenshots/teaser-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same teaser at 768 px: the image full width above the text, single column.

Tablet (600 px to 1199 px)

- 600 px to 899 px: the mobile layout (one column) with tablet section padding; type at mobile sizes.
- 900 px to 1199 px: the desktop layout (two columns, desktop type sizes).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The teaser at 375 px: the image on top (or at the bottom with Image position (mobile) Bottom), then eyebrow, title, description and stacked buttons.

Mobile (below 600 px)

- One column with a 2.5rem gap; Image & Text shows the image above the text (Image position (mobile) Bottom moves it below); Image position Right has no effect below 900 px.
- Text split: everything in one column in the order kicker, title, author, description, image, stats, links, short description (Image position (mobile) Top moves the image to the top of the text part, Bottom to the end).
- Background image: at least 35rem tall, 5rem / 1.5rem padding. Card: at least 25rem tall, 1.5rem padding.
- BG Color panels: 2.5rem / 1.5rem padding.
- Type: title 1.75rem (Image & Text with image) or 2rem; description 1rem; stat values 2.75rem.

6.11.3 Component Variations

All variations below combine: a teaser has one Layout, one BG Color, and any combination of the image, style and link options. Options that do not apply to the chosen Layout are hidden in the dialog (model `condition`) or have no effect, as noted.

Variation 1 — Layout "Image & Text", image left (default)

Variation Name: Layout "Image & Text" (no layout class) with Image position "Left".

Business Purpose: The classic promotion: image beside the text (Figma "Text & Image").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-image-text-left.png`
> **Purpose:** Image & Text, image left
> **Description:** A rounded image on the left half, eyebrow, title, description and buttons on the right half.

Use Cases: Service and industry promotions, "about us" sections.

Authoring Guidelines: Style tab: Layout = Image & Text; Image tab: choose an Image, Image position = Left (default).

Business Rules: The block gets `teaser-has-media` only when an image (or animation) is rendered; Hide Image or an empty Image gives Variation 3.

Rendering Rules: `div.teaser-media` is the first grid item, `div.teaser-content` the second; image 10:9, radius 1rem, `object-fit: cover`; grid `align-items: center`.

Responsive Behaviour: Desktop (900 px+): two equal columns, 5rem gap. Tablet below 900 px and mobile: one column, image above the text, 2.5rem gap.

SEO Impact: Title as a heading (default H2), image with alt text, crawlable links.

Accessibility Compliance: Reading order is image then text in the DOM; alt text per WCAG 1.1.1; heading level selectable (1.3.1).

Variation 2 — Layout "Image & Text", image right

Variation Name: Image position "Right" (class `image-right`).

Business Purpose: Alternate the image side between consecutive teasers for rhythm.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-image-text-right.png`
> **Purpose:** Image & Text, image right
> **Description:** The text on the left half and the rounded image on the right half.

Use Cases: Zig-zag sequences of teasers.

Authoring Guidelines: Layout = Image & Text; Image tab: Image position = Right (the field is shown only for the Image & Text layout).

Business Rules: Applied only for Image & Text with media; ignored for other layouts. An existing `image-right` block class is also honoured.

Rendering Rules: From 900 px `.teaser-has-media.image-right .teaser-media { order: 2 }`.

Responsive Behaviour: Desktop: image right. Tablet below 900 px / mobile: the image stays above the text (unless Image position (mobile) Bottom).

SEO Impact: None beyond Variation 1.

Accessibility Compliance: Visual order differs from the DOM order (image first) only visually; the reading order stays logical.

Variation 3 — Layout "Image & Text" without image (centred)

Variation Name: Image & Text with no Image or with Hide Image on (Figma "Careers").

Business Purpose: A centred text call to action.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-image-text-no-image.png`
> **Purpose:** Centred text teaser
> **Description:** Eyebrow, large title, description and buttons centred in a 37.5rem column.

Use Cases: Careers calls to action, newsletter sign-up prompts, section intros.

Authoring Guidelines: Layout = Image & Text; leave Image empty or turn Hide Image on.

Business Rules: Without media the content is centred (unless Text alignment Left or Right); the title uses the larger size (3rem desktop).

Rendering Rules: `.teaser:not(.no-image, .teaser-has-media, .card, .text-left, .text-right) .teaser-content` is 100 % wide, max 37.5rem, centred; kicker, tags, heading row and author are centred; links are centred.

Responsive Behaviour: Centred on all breakpoints.

SEO Impact: Text only.

Accessibility Compliance: Keep centred text short.

Variation 4 — Layout "Text split"

Variation Name: Layout "Text split (title left, text right)" (class `no-image`).

Business Purpose: A two-column text section: the title on the left, the content on the right (Figma "2 Slot Section", "Text Section - 2 Columns", "L3 Header").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-text-split.png`
> **Purpose:** Text split
> **Description:** The eyebrow and tags across the top, a large regular-weight title on the left, the description and text links on the right.

Use Cases: Section introductions, article headers with meta, statistics sections.

Authoring Guidelines: Style tab: Layout = Text split.

Business Rules: An image is placed in the right column (Variation 5), not beside the text. Image position is hidden; Image ratio and Image corners remain visible. Text alignment Center has no effect in this layout.

Rendering Rules: `.teaser.no-image`: block `align-items: start`; text colour default (black), title heading XXL at regular weight, description body L; from 900 px `.teaser-content` is a grid with areas `kicker kicker` / `heading body`; `.teaser-header` uses `display: contents`.

Responsive Behaviour: Desktop: two columns (1.5rem gap), kicker above both with 2.5rem below. Tablet below 900 px / mobile: single column.

SEO Impact: The title heading precedes the body in the DOM (good structure).

Accessibility Compliance: DOM order kicker → title → author → body is the reading order.

Variation 5 — Layout "Text split" with an image in the text column

Variation Name: Text split with an Image (e.g. a chart).

Business Purpose: Show a chart, diagram or picture inside the right-hand text column.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-text-split-image.png`
> **Purpose:** Text split with a chart
> **Description:** Title left; on the right the description followed by a chart image shown whole without rounded corners.

Use Cases: Research highlights with a chart, product diagrams.

Authoring Guidelines: Layout = Text split; Image tab: Image (and alt text); optional Image ratio.

Business Rules: The image comes right after the description (before stats and links); it is shown whole (no crop) unless Image ratio is Portrait or Square; corners are always square for images.

Rendering Rules: `div.teaser-media` inside `div.teaser-body`; `.teaser.no-image .teaser-media img { border-radius: 0 }` and `aspect-ratio: auto` unless `ratio-8-9` / `ratio-1-1`.

Responsive Behaviour: In the right column on desktop; in the single column on mobile, in the text order (or top / bottom with Image position (mobile)).

SEO Impact: Chart images need descriptive alt text (they carry information).

Accessibility Compliance: Complex charts need the key figures in the description too (WCAG 1.1.1).

Variation 6 — Layout "Background image"

Variation Name: Layout "Background image" (class `background`).

Business Purpose: A full-width banner inside the page with the image behind centred text.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-background-image.png`
> **Purpose:** Background image teaser
> **Description:** A wide image under a 60 % dark overlay with centred white eyebrow, title, description and white outline buttons.

Use Cases: Mid-page campaign banners, section breaks with a statement.

Authoring Guidelines: Style tab: Layout = Background image; Image tab: Image.

Business Rules: Image position, Image ratio and Image corners are hidden. Text colours are white / 70 % white whatever the BG Color. Top spacing has no effect (the layout's padding wins).

Rendering Rules: Flex column, centred, `min-height` 35rem (48.75rem from 900 px); media absolute behind (`z-index: -2`, cover, no radius); overlay `::before` `rgb(12 12 13 / 60%)` (80 % with BG Color Dark, dark green 80 % with Dark green).

Responsive Behaviour: Desktop: 48.75rem high, padding 7.5rem / 5rem. Tablet below 900 px / mobile: 35rem high, padding 5rem / 1.5rem.

SEO Impact: The image is still an `<img>` with alt text.

Accessibility Compliance: White text over the overlay; verify contrast with bright images.

Variation 7 — Layout "Card"

Variation Name: Layout "Card" (class `card`).

Business Purpose: A rounded card with an image behind its text and links at the bottom (Figma "Media Kit").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-card.png`
> **Purpose:** Card teaser
> **Description:** A rounded card with a dark 70 % overlay over an image, the title at the top left and the links at the bottom.

Use Cases: Media kit downloads, media coverage, side-by-side cards in Columns.

Authoring Guidelines: Style tab: Layout = Card; Image tab: Image.

Business Rules: Image position, Image ratio and Image corners are hidden. Text is left aligned by default; the overlay is always black 70 %; Top spacing has no effect.

Rendering Rules: Flex column, `min-height` 25rem (35.625rem from 900 px), radius 1rem, padding 1.5rem (2.5rem); `.teaser-content` stretches (`flex: 1`, `justify-content: space-between`) so the header is at the top and the body at the bottom; title max 31.25rem.

Responsive Behaviour: Desktop: 35.625rem minimum height. Tablet below 900 px / mobile: 25rem.

SEO Impact: As Variation 6.

Accessibility Compliance: As Variation 6.

Variation 8 — Image type "Lotto" (Lottie animation)

Variation Name: Image type "Lotto" (`imageFormat` = `lottie`).

Business Purpose: An animated illustration instead of a still image.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-lotto-animation.png`
> **Purpose:** Lottie teaser
> **Description:** An Image & Text teaser with a Lottie animation in the image column and the round pause button in its bottom-right corner.

Use Cases: Innovation stories, explainer graphics.

Authoring Guidelines: Image tab: Image type = Lotto; Image = the Lottie `.json` file; Image alt text = description (or empty if decorative).

Business Rules: Only with Lotto and a `.json` link; with Default and a `.json` file no media is shown. The animation loops, starts paused for reduced motion and pauses off screen. Behind the text (Background image, Card) it covers its box ("xMidYMid slice"); otherwise it is contained ("xMidYMid meet").

Rendering Rules: `div.teaser-media` holds `div.teaser-animation` and the `button.animation-toggle` (bottom right, 0.75rem from the edges); the animation uses the Image ratio (10:9 by default; Original does not change an animation) and rounded corners.

Responsive Behaviour: As the image of the chosen layout.

SEO Impact: Not indexable; use text for key messages.

Accessibility Compliance: Pause / play button (WCAG 2.2.2), labels from placeholders; `role="img"` + `aria-label` with alt text, else `aria-hidden`.

Variation 9 — Image position (mobile) "Default"

Variation Name: Image position (mobile) "Default" (no class).

Business Purpose: The natural order on small screens.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-mobile-image-default.png`
> **Purpose:** Default mobile image position
> **Description:** At 375 px, an Image & Text teaser with the image above the text.

Use Cases: Most teasers.

Authoring Guidelines: Image tab: Image position (mobile) = Default.

Business Rules: Image & Text: image above the text; Text split: image in the text order (after the description).

Rendering Rules: DOM order.

Responsive Behaviour: Applies below 900 px only.

SEO Impact: None.

Accessibility Compliance: Visual order equals DOM order.

Variation 10 — Image position (mobile) "Top"

Variation Name: Image position (mobile) "Top" (class `mobile-image-top`).

Business Purpose: Lead with the image on phones.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-mobile-image-top.png`
> **Purpose:** Mobile image on top
> **Description:** At 375 px, a Text split teaser with its chart moved to the top of the text part, above the description.

Use Cases: Text split teasers whose image should be seen first on mobile.

Authoring Guidelines: Image tab: Image position (mobile) = Top.

Business Rules: For Image & Text this equals Default (the image is already first); for Text split the image moves to the top of the body (below the title).

Rendering Rules: `.teaser.no-image.mobile-image-top .teaser-media { order: -1 }`; reset to `order: 0` from 900 px.

Responsive Behaviour: Below 900 px only.

SEO Impact: None.

Accessibility Compliance: Visual reordering only; DOM order unchanged.

Variation 11 — Image position (mobile) "Bottom"

Variation Name: Image position (mobile) "Bottom" (class `mobile-image-bottom`).

Business Purpose: Lead with the text on phones.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-mobile-image-bottom.png`
> **Purpose:** Mobile image at the bottom
> **Description:** At 375 px, an Image & Text teaser with the text first and the image below the buttons.

Use Cases: Teasers where the message matters more than the picture.

Authoring Guidelines: Image tab: Image position (mobile) = Bottom.

Business Rules: Image & Text: image after the text; Text split: image at the end of the body.

Rendering Rules: `.teaser.mobile-image-bottom .teaser-media { order: 2 }`; reset from 900 px.

Responsive Behaviour: Below 900 px only.

SEO Impact: None.

Accessibility Compliance: Visual reordering only.

Variation 12 — Image ratio "Default (10:9)"

Variation Name: Image ratio "Default (10:9)" (no class).

Business Purpose: The standard design-system image proportion.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-ratio-10-9.png`
> **Purpose:** 10:9 image
> **Description:** An Image & Text teaser with a slightly wider-than-tall rounded image.

Use Cases: Most Image & Text teasers.

Authoring Guidelines: Image tab: Image ratio = Default (shown for Image & Text and Text split).

Business Rules: In Text split, Default shows the image at its original proportion.

Rendering Rules: `aspect-ratio: 10 / 9`, `object-fit: cover` (crops).

Responsive Behaviour: Same proportion at every width.

SEO Impact: None.

Accessibility Compliance: Cropping must not remove information from the image.

Variation 13 — Image ratio "Portrait (8:9)"

Variation Name: Image ratio "Portrait (8:9)" (class `ratio-8-9`).

Business Purpose: A taller image, e.g. for people photography.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-ratio-8-9.png`
> **Purpose:** 8:9 image
> **Description:** An Image & Text teaser with a slightly taller-than-wide image.

Use Cases: Portraits, leadership teasers.

Authoring Guidelines: Image ratio = Portrait (8:9).

Business Rules: Applies to images and animations, also in Text split.

Rendering Rules: `aspect-ratio: 8 / 9`.

Responsive Behaviour: Same proportion at every width.

SEO Impact: None.

Accessibility Compliance: As Variation 12.

Variation 14 — Image ratio "Square (1:1)"

Variation Name: Image ratio "Square (1:1)" (class `ratio-1-1`).

Business Purpose: A square image.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-ratio-1-1.png`
> **Purpose:** Square image
> **Description:** An Image & Text teaser with a square image.

Use Cases: Logos, icons-as-illustration, social-style imagery.

Authoring Guidelines: Image ratio = Square (1:1).

Business Rules: Applies to images and animations, also in Text split.

Rendering Rules: `aspect-ratio: 1`.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: As Variation 12.

Variation 15 — Image ratio "Original"

Variation Name: Image ratio "Original" (class `ratio-original`).

Business Purpose: Show the image uncropped.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-ratio-original.png`
> **Purpose:** Original ratio
> **Description:** An Image & Text teaser showing a wide 16:9 image uncropped beside the text.

Use Cases: Diagrams, screenshots, charts.

Authoring Guidelines: Image ratio = Original.

Business Rules: Affects images only; an animation keeps 10:9.

Rendering Rules: `.teaser.ratio-original .teaser-media img { aspect-ratio: auto }`.

Responsive Behaviour: The height follows the image's proportion.

SEO Impact: None.

Accessibility Compliance: No information lost through cropping.

Variation 16 — Image corners "Rounded" (default)

Variation Name: Image corners "Rounded" (no class).

Business Purpose: The design system's soft image corners.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-corners-rounded.png`
> **Purpose:** Rounded corners
> **Description:** An image with 1rem rounded corners.

Use Cases: Default.

Authoring Guidelines: Image corners = Rounded (shown for Image & Text and Text split).

Business Rules: In Text split images are always square-cornered (the layout overrides it).

Rendering Rules: `border-radius: var(--radius-l)` (1rem).

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: None.

Variation 17 — Image corners "Square"

Variation Name: Image corners "Square" (class `square`).

Business Purpose: Sharp-edged images for technical or editorial looks.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-corners-square.png`
> **Purpose:** Square corners
> **Description:** An Image & Text teaser whose image has sharp 90-degree corners.

Use Cases: Diagrams, editorial photos.

Authoring Guidelines: Image corners = Square.

Business Rules: Applies to images and animations.

Rendering Rules: `.teaser.square .teaser-media :is(img, .teaser-animation) { border-radius: 0 }`.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: None.

Variation 18 — BG Color "Default"

Variation Name: BG Color "Default" (no class).

Business Purpose: The teaser sits directly on the section background without a panel.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-bg-default.png`
> **Purpose:** No background colour
> **Description:** A teaser on the white page background without extra padding.

Use Cases: Most teasers.

Authoring Guidelines: Style tab: BG Color = Default.

Business Rules: No panel padding; Background image and Card keep their own overlay and padding.

Rendering Rules: No `bg-*` class.

Responsive Behaviour: Not applicable.

SEO Impact: None.

Accessibility Compliance: Dark text on white.

Variation 19 — BG Color "Grey"

Variation Name: BG Color "Grey" (class `bg-grey`).

Business Purpose: A light grey panel to separate the teaser from neighbouring content.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-bg-grey.png`
> **Purpose:** Grey panel
> **Description:** A Text split teaser on a light grey panel with generous padding.

Use Cases: Alternating sections, stats panels.

Authoring Guidelines: BG Color = Grey.

Business Rules: Adds panel padding; text stays dark.

Rendering Rules: `background-color: var(--color-bg-secondary);` padding 2.5rem / 1.5rem (7.5rem / 5rem from 900 px).

Responsive Behaviour: Padding grows at 900 px.

SEO Impact: None.

Accessibility Compliance: Dark text on light grey meets AA.

Variation 20 — BG Color "Dark"

Variation Name: BG Color "Dark" (class `bg-dark`).

Business Purpose: A black panel with white text.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-bg-dark.png`
> **Purpose:** Dark panel
> **Description:** A teaser on a near-black panel with white title, 70 % white text, light-blue stat values and white buttons.

Use Cases: Statement sections, stats on dark.

Authoring Guidelines: BG Color = Dark.

Business Rules: Text white, body / meta text 70 % white, stat values light blue (`--brand-300`), list dividers `#767676`; Secondary buttons white outline; with Background image the overlay is 80 %.

Rendering Rules: `background-color: var(--color-bg-neutral)` (`#0c0c0d`), panel padding.

Responsive Behaviour: As Grey.

SEO Impact: None.

Accessibility Compliance: White on near-black meets AA; check 70 % white small text.

Variation 21 — BG Color "Dark green"

Variation Name: BG Color "Dark green" (class `bg-dark-green`).

Business Purpose: A dark green brand panel.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-bg-dark-green.png`
> **Purpose:** Dark green panel
> **Description:** A teaser on a dark green (#003b39) panel with white text.

Use Cases: Sustainability and ESG themes.

Authoring Guidelines: BG Color = Dark green.

Business Rules: As Dark; with Background image the overlay is green `rgb(0 59 57 / 80%)`.

Rendering Rules: `background-color: #003b39`, panel padding.

Responsive Behaviour: As Grey.

SEO Impact: None.

Accessibility Compliance: As Dark.

Variation 22 — Text alignment "Layout default"

Variation Name: Text alignment "Layout default" (no class).

Business Purpose: Each layout's designed alignment.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-text-layout-default.png`
> **Purpose:** Layout default alignment
> **Description:** Side by side: a left-aligned Image & Text teaser and a centred Background image teaser.

Use Cases: Default.

Authoring Guidelines: Style tab: Text alignment = Layout default.

Business Rules: Left for Image & Text with image, Text split and Card; centred for Background image and Image & Text without image.

Rendering Rules: No class.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: None.

Variation 23 — Text alignment "Left"

Variation Name: Text alignment "Left" (class `text-left`).

Business Purpose: Force left alignment in layouts that centre by default.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-text-left.png`
> **Purpose:** Left-aligned text
> **Description:** A Background image teaser with its text left aligned.

Use Cases: Long text in Background image or no-image teasers.

Authoring Guidelines: Text alignment = Left.

Business Rules: Removes the centred 37.5rem column of the centring layouts.

Rendering Rules: `text-align: start` on content and heading row.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: Improves readability of long text.

Variation 24 — Text alignment "Center"

Variation Name: Text alignment "Center" (class `text-center`).

Business Purpose: Centre the text, including in Image & Text and Card.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-text-center.png`
> **Purpose:** Centred text
> **Description:** An Image & Text teaser whose text column is centred next to the image.

Use Cases: Short promotional teasers.

Authoring Guidelines: Text alignment = Center.

Business Rules: No effect in Text split. With an image beside the text the column is not limited in width.

Rendering Rules: `.teaser.text-center:not(.no-image) .teaser-content` centred (max 37.5rem, none with media); kicker, tags, heading row, author and links centred.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: Keep centred text short.

Variation 25 — Text alignment "Right"

Variation Name: Text alignment "Right" (class `text-right`).

Business Purpose: Right-aligned text.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-text-right.png`
> **Purpose:** Right-aligned text
> **Description:** A Card teaser with title, text and links aligned to the right.

Use Cases: Special designs; rarely needed in left-to-right languages.

Authoring Guidelines: Text alignment = Right.

Business Rules: Applies to every layout, including Text split.

Rendering Rules: `text-align: end`; kicker, meta, tags, heading row and author `justify-content: flex-end`; links `align-items: flex-end`.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: Avoid for long text.

Variation 26 — Vertical alignment "Layout default"

Variation Name: Vertical alignment "Layout default" (no class).

Business Purpose: Each layout's designed vertical alignment.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-valign-default.png`
> **Purpose:** Layout default vertical alignment
> **Description:** An Image & Text teaser with the text centred vertically against the image.

Use Cases: Default.

Authoring Guidelines: Style tab: Vertical alignment = Layout default.

Business Rules: Centre for Image & Text and Background image; top for Text split; Card keeps the header at the top and the body at the bottom.

Rendering Rules: No class.

Responsive Behaviour: Visible where two columns exist (900 px+) and in Background image.

SEO Impact: None.

Accessibility Compliance: None.

Variation 27 — Vertical alignment "Top"

Variation Name: Vertical alignment "Top" (class `valign-top`).

Business Purpose: Align the text with the top of the image, or put Background image text at the top.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-valign-top.png`

> **Purpose:** Top alignment
> **Description:** An Image & Text teaser with the eyebrow level with the top edge of the image.

Use Cases: Long text next to a tall image.

Authoring Guidelines: Vertical alignment = Top.

Business Rules: Recommended: keep Layout default for Card (the option changes how the card content stretches).

Rendering Rules: `align-items: start`; Background image also `justify-content: flex-start`.

Responsive Behaviour: Desktop (900 px+) for Image & Text / Text split; all widths for Background image.

SEO Impact: None.

Accessibility Compliance: None.

Variation 28 — Vertical alignment "Center"

Variation Name: Vertical alignment "Center" (class `valign-middle`).

Business Purpose: Centre the columns against each other, e.g. in Text split.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-valign-middle.png`
> **Purpose:** Centre alignment
> **Description:** A Text split teaser whose short title is vertically centred against the longer right column.

Use Cases: Text split with a short title and a long body.

Authoring Guidelines: Vertical alignment = Center.

Business Rules: Equals the default for Image & Text and Background image.

Rendering Rules: `align-items: center`; in Text split from 900 px the content grid `align-items: center`.

Responsive Behaviour: Desktop for the two-column layouts.

SEO Impact: None.

Accessibility Compliance: None.

Variation 29 — Vertical alignment "Bottom"

Variation Name: Vertical alignment "Bottom" (class `valign-bottom`).

Business Purpose: Align the text to the bottom of the image, or put Background image text at the bottom.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-valign-bottom.png`
> **Purpose:** Bottom alignment
> **Description:** A Background image teaser with its text resting at the bottom of the banner.

Use Cases: Editorial banners.

Authoring Guidelines: Vertical alignment = Bottom.

Business Rules: As Top (avoid for Card).

Rendering Rules: `align-items: end`; Background image `justify-content: flex-end`; Text split grid `align-items: end` from 900 px.

Responsive Behaviour: As Top.

SEO Impact: None.

Accessibility Compliance: None.

Variation 30 — Title size "Layout default"

Variation Name: Title size "Layout default" (no class).

Business Purpose: The layout's designed title size.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-title-default.png`
> **Purpose:** Default title size
> **Description:** An Image & Text teaser with a 2.5rem title and a Text split teaser with a 3rem title.

Use Cases: Default.

Authoring Guidelines: Style tab: Title size = Layout default.

Business Rules: Heading XL (1.75rem mobile, 2.5rem desktop) for Image & Text with media; heading XXL (2rem, 3rem) for every other layout.

Rendering Rules: `.teaser-title` and the layout rules.

Responsive Behaviour: Steps at 900 px.

SEO Impact: None.

Accessibility Compliance: Text resizes with browser zoom (rem units).

Variation 31 — Title size "Medium (40px)"

Variation Name: Title size "Medium (40px)" (class `title-m`).

Business Purpose: A smaller title in layouts that default to XXL.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-title-m.png`
> **Purpose:** Medium title
> **Description:** A Background image teaser with a 2.5rem title.

Use Cases: Long titles, dense pages.

Authoring Guidelines: Title size = Medium (40px).

Business Rules: 40 px is the desktop size; mobile 1.75rem (28 px).

Rendering Rules: `font-size: var(--heading-font-size-xl)`, line height 1.4, letter spacing –0.025 of the size.

Responsive Behaviour: 1.75rem below 900 px, 2.5rem from 900 px.

SEO Impact: None (size only).

Accessibility Compliance: None.

Variation 32 — Title size "Large (48px)"
Variation Name: Title size "Large (48px)" (class `title-l`).

Business Purpose: A larger title for Image & Text teasers.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-title-l.png`
> **Purpose:** Large title
> **Description:** An Image & Text teaser with a 3rem title.

Use Cases: Key promotions.

Authoring Guidelines: Title size = Large (48px).

Business Rules: 48 px desktop, 2rem (32 px) mobile.

Rendering Rules: `--heading-font-size-xxl`, line height 1.2, letter spacing –0.03.

Responsive Behaviour: Steps at 900 px.

SEO Impact: None.

Accessibility Compliance: None.

Variation 33 — Title size "Extra large (64px)"
Variation Name: Title size "Extra large (64px)" (class `title-xl`).

Business Purpose: Hero-like titles inside the page, e.g. secondary page headers.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-title-xl.png`
> **Purpose:** Extra large title
> **Description:** A Text split teaser opening a page with a 4rem title.

Use Cases: Secondary page headers, campaign statements.

Authoring Guidelines: Title size = Extra large (64px).

Business Rules: 64 px desktop, 2.5rem (40 px) mobile.

Rendering Rules: `--display-font-size-2`, line height 1.2, letter spacing –0.03.

Responsive Behaviour: Steps at 900 px.

SEO Impact: Consider Title Type H1 when the teaser opens a page without a Hero / Page Header.

Accessibility Compliance: None.

Variation 34 — Title size "Display (96px)"
Variation Name: Title size "Display (96px)" (class `title-display`).

Business Purpose: The largest display title.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-title-display.png`
> **Purpose:** Display title
> **Description:** A teaser with a single short 6rem headline.

Use Cases: One- to three-word statements.

Authoring Guidelines: Title size = Display (96px); keep the title very short.

Business Rules: 96 px (6rem) from 900 px, 2.75rem (44 px) below.

Rendering Rules: `--display-font-size-1`, line height 1.2; from 900 px `font-size: 6rem`, `letter-spacing: -0.12rem`.

Responsive Behaviour: Steps at 900 px.

SEO Impact: None.

Accessibility Compliance: Long words may overflow narrow screens: test at 320 px.

Variation 35 — Title weight "Layout default"

Variation Name: Title weight "Layout default" (no class).

Business Purpose: The layout's designed weight.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-title-weight-default.png`
> **Purpose:** Default title weight
> **Description:** A semibold Image & Text title next to a regular-weight Text split title.

Use Cases: Default.

Authoring Guidelines: Title weight = Layout default.

Business Rules: Semibold (600) except Text split, which is regular (400).

Rendering Rules: `.teaser-title` / `.teaser.no-image .teaser-title`.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: None.

Variation 36 — Title weight "Regular"

Variation Name: Title weight "Regular" (class `title-regular`).

Business Purpose: A lighter, editorial title.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-title-regular.png`

> **Purpose:** Regular title
> **Description:** An Image & Text teaser with a regular-weight title.

Use Cases: Large titles (XL, Display).

Authoring Guidelines: Title weight = Regular.

Business Rules: Overrides the layout weight.

Rendering Rules: `font-weight: var(--font-weight-regular)`.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: None.

Variation 37 — Title weight "Semibold"

Variation Name: Title weight "Semibold" (class `title-semibold`).

Business Purpose: A stronger title, e.g. in Text split.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-title-semibold.png`
> **Purpose:** Semibold title
> **Description:** A Text split teaser with a semibold title.

Use Cases: Emphasis.

Authoring Guidelines: Title weight = Semibold.

Business Rules: Overrides the layout weight.

Rendering Rules: `font-weight: var(--font-weight-semibold)`.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: None.

Variation 38 — Description size "Layout default"

Variation Name: Description size "Layout default" (no class).

Business Purpose: The layout's designed text size.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-description-default.png`
> **Purpose:** Default description size
> **Description:** An Image & Text description at 1.125rem and a Text split description at 1.5rem.

Use Cases: Default.

Authoring Guidelines: Description size = Layout default.

Business Rules: Body M (1rem / 1.125rem, line height 1.39) for most layouts; Body L (1rem / 1.5rem, line height 1.417) for Text split. Applies to the description and the short description.

Rendering Rules: `.teaser-description`, `.teaser-short-description`.

Responsive Behaviour: Steps at 900 px.

SEO Impact: None.

Accessibility Compliance: None.

Variation 39 — Description size "Regular (18px)"

Variation Name: Description size "Regular (18px)" (class `text-regular`).

Business Purpose: Regular body text, e.g. long Text split copy.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-description-regular.png`
> **Purpose:** Regular description
> **Description:** A Text split teaser with 1.125rem body text.

Use Cases: Longer copy.

Authoring Guidelines: Description size = Regular (18px).

Business Rules: 18 px desktop, 16 px mobile.

Rendering Rules: `--body-font-size-m`, line height 1.61.

Responsive Behaviour: Steps at 900 px.

SEO Impact: None.

Accessibility Compliance: Comfortable line height for reading.

Variation 40 — Description size "Large (24px)"

Variation Name: Description size "Large (24px)" (class `text-large`).

Business Purpose: Lead-paragraph sized text.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-description-large.png`
> **Purpose:** Large description
> **Description:** An Image & Text teaser with a 1.5rem description.

Use Cases: Short, impactful copy.

Authoring Guidelines: Description size = Large (24px).

Business Rules: 24 px desktop, 16 px mobile.

Rendering Rules: `--body-font-size-l`, line height 1.417.

Responsive Behaviour: Steps at 900 px.

SEO Impact: None.

Accessibility Compliance: None.

Variation 41 — Eyebrow style "Brand"
Variation Name: Eyebrow style "Brand" (default, no class).

Business Purpose: The brand-blue kicker.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-eyebrow-brand.png`
> **Purpose:** Brand eyebrow
> **Description:** A semibold blue eyebrow above the title.

Use Cases: Default.

Authoring Guidelines: Eyebrow style = Brand.

Business Rules: Brand colour on every background (also dark ones).

Rendering Rules: `.teaser-eyebrow` `--color-text-brand`, Body L (1rem) / Body S (1rem) semibold; takes the full row of the kicker.

Responsive Behaviour: 1rem at all widths.

SEO Impact: None.

Accessibility Compliance: Check brand blue contrast on dark panels.

Variation 42 — Eyebrow style "Label (uppercase, grey)"
Variation Name: Eyebrow style "Label" (class `eyebrow-label`).

Business Purpose: A small uppercase label like the tags (article headers, media coverage).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-eyebrow-label.png`
> **Purpose:** Label eyebrow
> **Description:** A small grey uppercase "PRESS RELEASE" label above the title.

Use Cases: Content-type labels (Insight, Press release).

Authoring Guidelines: Eyebrow style = Label (uppercase, grey).

Business Rules: Uses the meta colour (grey; 70 % white on dark).

Rendering Rules: Body XS (0.75rem / 0.875rem), medium, uppercase, letter spacing 0.01.

Responsive Behaviour: Steps at 900 px.

SEO Impact: None (CSS uppercase; the text stays as authored).

Accessibility Compliance: Small text: check contrast.

Variation 43 — Top spacing "Default"
Variation Name: Top spacing "Default" (no class).

Business Purpose: Normal spacing in the page flow.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-top-spacing-default.png`
> **Purpose:** Default top spacing
> **Description:** A teaser in the middle of a page with the section's normal spacing.

Use Cases: Default.

Authoring Guidelines: Top spacing = Default.

Business Rules: No extra padding.

Rendering Rules: No class.

Responsive Behaviour: Not applicable.

SEO Impact: None.

Accessibility Compliance: None.

Variation 44 — Top spacing "Page header (extra space above)"

Variation Name: Top spacing "Page header" (class `spacing-header`).

Business Purpose: The larger space above a teaser that opens a page (Figma "Secondary Page Header").

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-spacing-header.png`
> **Purpose:** Page header spacing
> **Description:** A Text split teaser at the top of a page with 12.5rem space above it on desktop.

Use Cases: Landing pages without a Hero.

Authoring Guidelines: Style tab: Top spacing = Page header.

Business Rules: No effect with Background image or Card (their padding wins); Additional tab Padding top overrides it.

Rendering Rules: `padding-top: var(--space-1600)` (4rem); `var(--space-5000)` (12.5rem) from 900 px.

Responsive Behaviour: 4rem below 900 px, 12.5rem from 900 px.

SEO Impact: Consider Title Type H1 for a page-opening teaser.

Accessibility Compliance: None.

Variation 45 — Links "Primary and Secondary CTA" (Multiple links off)

Variation Name: Multiple links off (default).

Business Purpose: Up to two CTAs, each with its own style.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-links-primary-secondary.png`

> **Purpose:** Two CTAs
> **Description:** A filled primary button above an outline secondary button.

Use Cases: Most teasers.

Authoring Guidelines: Links tab: Multiple links off; fill Primary CTA Title / Link / Link Type / Link Style and the Secondary equivalents.

Business Rules: Missing style rows fall back to Primary Button and Secondary Button; an empty style is a Text link; CTAs without a link are skipped.

Rendering Rules: `div.teaser-links` (column, items start, 0.75rem gap).

Responsive Behaviour: Stacked at every width.

SEO Impact: Crawlable links.

Accessibility Compliance: Descriptive link texts.

Variation 46 — Links "Multiple links"

Variation Name: Multiple links on.

Business Purpose: Any number of links, each with its own style (e.g. a list of resources).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-links-multiple.png`
> **Purpose:** Multiple links
> **Description:** A Text split teaser with four links in List style below the description.

Use Cases: Related links, resource lists.

Authoring Guidelines: Links tab: Multiple links on; add items in Links (CTA Title, CTA Link, CTA Link Type, CTA Link Style).

Business Rules: Primary / Secondary CTA fields are hidden and ignored. Items without a link are skipped. Items without a style use the legacy Multi Link Style row if present, else Text link.

Rendering Rules: Links in authored order; consecutive List items share one list.

Responsive Behaviour: Stacked at every width.

SEO Impact: Crawlable links.

Accessibility Compliance: List-style links are in a `<ul>` (WCAG 1.3.1).

Variation 47 — Link style "Text link"

Variation Name: CTA Link Style "Text link" (value "").

Business Purpose: A light text CTA with an arrow.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-link-text.png`

> **Purpose:** Text link
> **Description:** "Learn more" in black text with a right arrow, turning blue on hover.

Use Cases: Secondary actions.

Authoring Guidelines: Any Link Style = Text link.

Business Rules: Rendered as a tertiary button.

Rendering Rules: `a.button.tertiary`; white on dark / background / card, light blue on hover there.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: Arrow is decorative.

Variation 48 — Link style "List"

Variation Name: CTA Link Style "List" (value `list`).

Business Purpose: Full-width link rows with dividers.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-link-list.png`
> **Purpose:** List links
> **Description:** Three full-width link rows separated by grey dividers, each with an arrow at the right end.

Use Cases: Related services, resource lists.

Authoring Guidelines: Link Style = List (Multiple links on, or for the Primary / Secondary CTA).

Business Rules: Consecutive List links form one `<ul>`; a new-window link shows an up-right arrow; the List arrow option changes the arrow of same-window links.

Rendering Rules: `ul.teaser-link-list > li > a.teaser-link-list-item`; rows 1.5rem apart, 0.75rem padding and a divider (`#b2b2b2`, `#767676` on dark) below all but the last; text Heading M (1.125rem / 1.5rem); hover: brand colour (light blue on dark).

Responsive Behaviour: Full width at every breakpoint.

SEO Impact: Crawlable list of links.

Accessibility Compliance: Semantic list; the arrow is a decorative pseudo-element.

Variation 49 — Link style "Primary Button"

Variation Name: CTA Link Style "Primary Button" (value `primary`).

Business Purpose: The main action.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-link-primary.png`
> **Purpose:** Primary button
> **Description:** A black pill button with white text and arrow.

Use Cases: “Contact us”, “Download the report”.

Authoring Guidelines: Link Style = Primary Button (default for Primary CTA).

Business Rules: Recommended: one per teaser.

Rendering Rules: `a.button.primary`.

Responsive Behaviour: Does not wrap its text.

SEO Impact: None.

Accessibility Compliance: Global button focus styles.

Variation 50 — Link style “Secondary Button”

Variation Name: CTA Link Style “Secondary Button” (value `secondary`).

Business Purpose: The alternative action.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-link-secondary.png`
> **Purpose:** Secondary button
> **Description:** An outline pill button; white outline on dark panels and images.

Use Cases: “Explore more”.

Authoring Guidelines: Link Style = Secondary Button (default for Secondary CTA).

Business Rules: White outline and text on Dark, Dark green, Background image and Card.

Rendering Rules: `a.button.secondary`.

Responsive Behaviour: Does not wrap its text.

SEO Impact: None.

Accessibility Compliance: As Primary.

Variation 51 — Mixed link styles

Variation Name: Links in different styles in one teaser.

Business Purpose: Combine a button with list rows or text links.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-links-mixed.png`
> **Purpose:** Mixed links
> **Description:** Two List rows followed by a primary button.

Use Cases: Resource lists with a final action.

Authoring Guidelines: Give the links different CTA Link Styles.

Business Rules: A non-List link ends the current list; the next List link starts a new one.

Rendering Rules: The container gets `teaser-links-mixed` (one style: `teaser-links-{style}`, `teaser-links-default` for Text links).

Responsive Behaviour: Stacked.

SEO Impact: None.

Accessibility Compliance: None.

Variation 52 — List arrow "Right"
Variation Name: List arrow "Right" (default, value "").

Business Purpose: The standard arrow of List rows.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-list-arrow-right.png`
> **Purpose:** Right arrow
> **Description:** List rows with right-pointing grey arrows.

Use Cases: Default.

Authoring Guidelines: Links tab: List arrow = Right.

Business Rules: New-window links always show the up-right arrow.

Rendering Rules: `arrow-right.svg` mask.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: Decorative.

Variation 53 — List arrow "Down right"
Variation Name: List arrow "Down right" (class `arrow-down-right`).

Business Purpose: The diagonal down-right arrow of some Figma frames.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-list-arrow-down-right.png`
> **Purpose:** Down-right arrow
> **Description:** List rows with diagonal down-right arrows.

Use Cases: Design-specific link lists.

Authoring Guidelines: List arrow = Down right (the field is always visible; it only affects List links).

Business Rules: Applies to same-window List links only.

Rendering Rules: `arrow-up-right.svg` rotated 90 degrees.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: Decorative.

Variation 54 — Stats
Variation Name: Teaser with Stats (Stats tab).

Business Purpose: Headline numbers that prove the message (Figma "Text Section - 2 Columns" with stats).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-stats.png`
> **Purpose:** Stats
> **Description:** A Text split teaser with three stats side by side: a label above a large brand-blue value such as "40%".

Use Cases: Case-study results, company facts.

Authoring Guidelines: Stats tab: add an item per stat with Stats Title and Stats Value (a single line).

Business Rules: In each item the last element is the value and the elements before it the title; stats appear after the description (and image) and before the links.

Rendering Rules: `buildStats` → `dl.teaser-stats > div.teaser-stat > dt.teaser-stat-label + dd.teaser-stat-value`; items flex at least 11.25rem wide, wrapping; value Display 1 (2.75rem; 6rem from 900 px), no wrap, brand blue (light blue on dark).

Responsive Behaviour: Several per row on desktop, wrapping to one per row on mobile.

SEO Impact: Numbers are text (indexable).

Accessibility Compliance: Description list associates each label with its value (WCAG 1.3.1).

Variation 55 — Date

Variation Name: Show Date with Date, Date Format and Date label.

Business Purpose: Publication date in article-style headers.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-date.png`
> **Purpose:** Date
> **Description:** "Published 13-10-2026" in small light text at the right of the kicker.

Use Cases: Insights, news, events.

Authoring Guidelines: Content tab: Show Date on; choose Date, Date Format and optional Date label.

Business Rules: Shown only with Show Date on and a Date; formats DD-MM-YYYY (default), MM-DD-YYYY, MMM d, yyyy (English month abbreviations), DD/MM/YY.

Rendering Rules: `p.teaser-date` with the label and `<time datetime="YYYY-MM-DD">`.

Responsive Behaviour: Wraps under the tags on narrow screens.

SEO Impact: Machine-readable `datetime`.

Accessibility Compliance: `<time>` element.

Variation 56 — Reading time

Variation Name: Reading time.

Business Purpose: Tell readers how long the content takes.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-reading-time.png`
> **Purpose:** Reading time
> **Description:** "14 min read" next to the date.

Use Cases: Articles, reports.

Authoring Guidelines: Content tab: Reading time = minutes (e.g. 14) or any text.

Business Rules: A whole number becomes the placeholder `minRead` ("{n} min read"); other text is shown as typed.

Rendering Rules: `p.teaser-reading-time` in `div.teaser-info`.

Responsive Behaviour: Wraps with the info row.

SEO Impact: None.

Accessibility Compliance: Plain text.

Variation 57 — Page actions (listen, download, share)

Variation Name: Listen link, Download and Share.

Business Purpose: Let visitors listen to, print / save or share the page.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-page-actions.png`
> **Purpose:** Page actions
> **Description:** Headphones, download and share icon buttons at the right of the kicker; after Share on desktop, "Link copied to clipboard" below them.

Use Cases: Article headers (Figma "L3 Header").

Authoring Guidelines: Content tab: Listen link (audio page or file), Download on, Share on.

Business Rules: Listen: the link becomes an icon button labelled `listenToThisPage` ("Listen to this page"). Download: `window.print()` (print or save as PDF), label `downloadThisPage`. Share: the device share dialog (title = teaser title or page title), else the URL is copied and "Link copied to clipboard" (`linkCopied`) shows for 3 seconds.

Rendering Rules: `div.teaser-actions` with tertiary icon buttons and `p.teaser-status` (`role="status"`).

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: Icon buttons have accessible names; the copy confirmation is announced through the live status region.

Variation 58 — Author lockup

Variation Name: Author name, designation and image.

Business Purpose: Credit the author of an insight.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-author.png`
> **Purpose:** Author
> **Description:** A round 2.5rem portrait next to the author's name and designation below the title.

Use Cases: Thought-leadership teasers and headers.

Authoring Guidelines: Content tab: Author name, Author designation, Author image, Author image alt text.

Business Rules: Shown only with a name or a designation (an image alone is not shown).

Rendering Rules: `div.teaser-author` (image `div.teaser-author-image`, round, cover) and `div.teaser-author-details` (`p.teaser-author-name` light, `p.teaser-author-designation` medium), below the title row; in Text split in the left column.

Responsive Behaviour: Same at every width.

SEO Impact: Author names are indexable text.

Accessibility Compliance: Author image alt text (usually the name, or empty when the name is next to it).

Variation 59 — Tags

Variation Name: Display Tags with Tags.

Business Purpose: Show topic labels (Figma "BANKING | AI NATIVE").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-tags.png`
> **Purpose:** Tags
> **Description:** Small grey uppercase tags separated by vertical bars below the eyebrow.

Use Cases: Insights, industry teasers.

Authoring Guidelines: Content tab: Display Tags on; choose Tags.

Business Rules: Labels come from the locale's taxonomy (`/{locale}/taxonomy.json`); tags missing there show a label derived from the tag ID.

Rendering Rules: `div.teaser-meta > ul.teaser-tags > li[data-tag]`.

Responsive Behaviour: Wraps.

SEO Impact: Display only; tags are not written to page metadata.

Accessibility Compliance: Semantic list.

Variation 60 — View All link

Variation Name: View All Text and View All Link.

Business Purpose: A "View all" button beside the title (e.g. above a related list).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-view-all.png`

> **Purpose:** View All
> **Description:** The title on the left and a primary "View all" button on the right of the same row.

Use Cases: Section headings for news or insights.

Authoring Guidelines: Content tab: View All Text and View All Link.

Business Rules: Always a primary button; no new-window option.

Rendering Rules: `a.teaser-view-all.button.primary` in `div.teaser-heading-row` (space-between, wrapping).

Responsive Behaviour: Moves below the title when space runs out.

SEO Impact: Crawlable link.

Accessibility Compliance: Use a specific text ("View all insights").

Variation 61 — Hide Title

Variation Name: Hide Title on.

Business Purpose: Keep a heading for structure and screen readers without showing it.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-hide-title.png`
> **Purpose:** Hidden title
> **Description:** A teaser that visually starts with the description; the accessibility tree still shows the heading.

Use Cases: Designs without a visible heading.

Authoring Guidelines: Content tab: Hide Title on.

Business Rules: The title stays in the DOM and in the data layer `dc:title`.

Rendering Rules: `hideVisually()` adds `visually-hidden`.

Responsive Behaviour: Hidden at every width.

SEO Impact: Visually hidden headings should match visible content; avoid stuffing keywords.

Accessibility Compliance: Preserves the heading structure (WCAG 1.3.1).

Variation 62 — Short description

Variation Name: Short Description.

Business Purpose: A note after the links (legal text, footnote).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-short-description.png`
> **Purpose:** Short description
> **Description:** A small paragraph shown below the buttons.

Use Cases: Footnotes, sources of statistics.

Authoring Guidelines: Content tab: Short Description.

Business Rules: Always shown when filled (no toggle).

Rendering Rules: `div.teaser-short-description`, styled like the description.

Responsive Behaviour: Same as description.

SEO Impact: Indexable text.

Accessibility Compliance: None.

Variation 63 — Show Eyebrow off

Variation Name: Show Eyebrow off.

Business Purpose: Hide an authored eyebrow without deleting it.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-eyebrow-hidden.png`
> **Purpose:** Hidden eyebrow
> **Description:** A teaser starting directly with its title.

Use Cases: Temporary changes, shared copy.

Authoring Guidelines: Content tab: Show Eyebrow off.

Business Rules: Default on; only an explicit off hides the eyebrow (removed from the page, not just visually).

Rendering Rules: No `p.teaser-eyebrow`.

Responsive Behaviour: Not applicable.

SEO Impact: The text is not output.

Accessibility Compliance: None.

Variation 64 — Show Description off

Variation Name: Show Description off.

Business Purpose: Hide the description without deleting it.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-description-hidden.png`
> **Purpose:** Hidden description
> **Description:** A teaser with only title and links.

Use Cases: Compact teasers.

Authoring Guidelines: Content tab: Show Description off.

Business Rules: Default on; the Short Description is not affected.

Rendering Rules: No `div.teaser-description`.

Responsive Behaviour: Not applicable.

SEO Impact: The text is not output.

Accessibility Compliance: None.

Variation 65 — Title Type (heading level H1–H6)

Variation Name: Title Type H1, H2 (default), H3, H4, H5 or H6.

Business Purpose: The correct heading level for the teaser's place in the page outline.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-title-type.png`
> **Purpose:** Heading level
> **Description:** Two identical-looking teasers; the accessibility tree shows H2 and H3.

Use Cases: H1 for a page-opening teaser without Hero / Page Header; H3 inside a section with its own H2.

Authoring Guidelines: Content tab: Title Type.

Business Rules: The look does not depend on the level (Title size controls it).

Rendering Rules: `buildHeading()` creates the chosen element with class `teaser-title`.

Responsive Behaviour: Not applicable.

SEO Impact: One H1 per page; logical hierarchy.

Accessibility Compliance: WCAG 1.3.1, 2.4.6.

Variation 66 — Personalization marker

Variation Name: Personalization on.

Business Purpose: Mark a teaser as a candidate for personalization (e.g. by Adobe Target).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-personalization.png`
> **Purpose:** Personalization marker
> **Description:** The browser's element inspector showing the teaser with class personalization and data-personalization="true"; the page looks unchanged.

Use Cases: Future Target activities.

Authoring Guidelines: Style tab: Personalization on.

Business Rules: No visible change; no personalization runs today (Adobe Target is Not implemented (as of this handover)).

Rendering Rules: Adds class `personalization` and `data-personalization="true"`.

Responsive Behaviour: Not applicable.

SEO Impact: None.

Accessibility Compliance: None.

6.11.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A fully populated Text split teaser selected on the Universal Editor canvas, with letters A–N marking the block, image, eyebrow, tags, date, reading time, page actions, title, View All, author, description, stats, links and short description.

Annotations

- A = Teaser block (select it to open its properties: Content, Image, Style, Links, Stats, Additional).
- B = Image or Lottie animation.
- C = Eyebrow.
- D = Tags.
- E = Date (with its label).
- F = Reading time.
- G = Page actions (listen, download, share icons).
- H = Title.
- I = View All button.
- J = Author (image, name, designation).
- K = Description.
- L = Stats.
- M = Links (Primary / Secondary CTA or the Links list).
- N = Short Description.

Notes on the canvas: the teaser is a key-value block, so its content is edited in the properties panel (texts that keep their editor instrumentation, such as the title, eyebrow and description, can also be selected on the canvas). Fields hidden by a `condition` (for example Date when Show Date is off) disappear from the panel until their toggle or Layout allows them. The Lottie animation plays in the editor. Personalization has no visible effect.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Teaser properties panel with its six tabs (Content, Image, Style, Links, Stats, Additional), the Content tab open with Show Date on so that Date, Date Format and Date label are visible.

Properties panel tabs and fields

- **Content** — Eyebrow, Show Eyebrow, Title, Title Type (H1–H6), Hide Title, Description, Show Description, Short Description, View All Text, View All Link, Show Date, Date, Date Format and Date label (the last three only with Show Date on), Reading time, Display Tags, Tags (only with Display Tags on), Listen link, Download, Share, Author name, Author designation, Author image, Author image alt text.
- **Image** — Image type (Default, Lotto), Image, Image alt text, Hide Image, Image position (only for Layout Image & Text), Image position (mobile), Image ratio and Image corners (only for Layouts Image & Text and Text split).
- **Style** — Layout (Image & Text, Text split, Background image, Card), BG Color (Default, Grey, Dark, Dark green), Text alignment, Vertical alignment, Title size, Title weight, Description size, Eyebrow style, Top spacing, Personalization.

- **Links** — Multiple links; Links multifield (only with Multiple links on: CTA Title, CTA Link, CTA Link Type, CTA Link Style); List arrow; Primary CTA Title / Link / Link Type / Link Style and Secondary CTA Title / Link / Link Type / Link Style (only with Multiple links off).
- **Stats** — Stats multifield (Stats Title, Stats Value).
- **Additional** — Anchor ID, Hide block, Padding left / right / top / bottom, Margin left / right / top / bottom.

6.11.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites browse to the locale site (for example `/content/tcs-corp/us/en`), select the page and choose Edit.

Expected Result: The page opens in Universal Editor.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** Universal Editor with the page loaded and the content tree visible.

Step 2 — Add the Teaser block

Action: Select the target section, choose Add (+) and pick **Teaser**.

Expected Result: An empty Teaser is added with the defaults Title Type H2, Image position Left, Show Eyebrow and Show Description on, Date Format DD-MM-YYYY, Primary CTA style Primary Button and Secondary CTA style Secondary Button.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** The component picker with Teaser highlighted.

Step 3 — Choose the Layout first

Action: Open the **Style** tab and choose **Layout** (Image & Text, Text split, Background image or Card), so that the dialog shows the fields that apply.

Expected Result: The Image tab shows or hides Image position, Image ratio and Image corners according to the Layout.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-03-layout.png`
> **Purpose:** Choose the layout
> **Description:** The Layout select open with its four options.

Step 4 — Enter the eyebrow and title

Action: Open the **Content** tab; enter **Eyebrow** (keep Show Eyebrow on), **Title**, choose **Title Type**, and leave Hide Title off unless the title must be hidden.

Expected Result: Eyebrow and title render on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-04-title.png`
> **Purpose:** Eyebrow and title
> **Description:** The Content tab with Eyebrow, Title and Title Type filled in.

Step 5 — Enter the description and short description

Action: Enter **Description** (keep Show Description on) and, if needed, **Short Description** (shown after the links).

Expected Result: The description appears under the title (in the right column for Text split).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-05-description.png`
> **Purpose:** Description
> **Description:** The rich-text Description field and the rendered text.

Step 6 — Add the article meta (optional)

Action: For article-style teasers: turn **Show Date** on and choose **Date**, **Date Format** and **Date label**; enter **Reading time**; turn **Display Tags** on and choose **Tags**; set **Listen link**, **Download** and **Share**; enter **View All Text** and **View All Link** if a View All button is needed.

Expected Result: Tags, date, reading time and action icons appear in the kicker row; View All appears beside the title.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-06-meta.png`
> **Purpose:** Article meta
> **Description:** The Content tab with Show Date and Display Tags on, and the kicker row on the canvas showing tags, date, reading time and icons.

Step 7 — Add the author (optional)

Action: Enter **Author name**, **Author designation**, choose **Author image** and enter **Author image alt text**.

Expected Result: The author lockup appears below the title.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-07-author.png`
> **Purpose:** Author
> **Description:** The author fields filled in and the round portrait with name and designation on the canvas.

Step 8 — Add the image or animation

Action: Open the **Image** tab: choose **Image type** (Default or Lotto), **Image** (a picture or a Lottie `.json`), **Image alt text**; then the layout options **Image position**, **Image position (mobile)**, **Image ratio**, **Image corners** as shown.

Expected Result: The image appears beside the text (Image & Text), in the text column (Text split) or behind the text (Background image, Card).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-08-image.png`
> **Purpose:** Image
> **Description:** The Image tab with an image selected, Image position Right and Image ratio Portrait.

Step 9 — Set the Style options

Action: In the **Style** tab choose **BG Color**, **Text alignment**, **Vertical alignment**, **Title size**, **Title weight**, **Description size**, **Eyebrow style** and **Top spacing**; leave **Personalization** off unless requested by the personalization team.

Expected Result: The canvas updates with the chosen colours, alignment and type sizes.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-09-style.png`
> **Purpose:** Style options
> **Description:** The Style tab with BG Color Dark and Title size Large, and the dark teaser on the canvas.

Step 10 — Add the links

Action: Open the **Links** tab. For one or two CTAs leave **Multiple links** off and fill the Primary and Secondary CTA fields (Title, Link, Link Type, Link Style). For more links turn **Multiple links** on and add items to **Links**, each with CTA Title, CTA Link, CTA Link Type and CTA Link Style. Choose **List arrow** if you use the List style.

Expected Result: Buttons, text links or list rows appear below the description and stats.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-10-links.png`
> **Purpose:** Links
> **Description:** The Links tab with Multiple links on and three List-style items, rendered as divided rows.

Step 11 — Add stats (optional)

Action: Open the **Stats** tab, add an item per stat and enter **Stats Title** and **Stats Value** (a single line such as `40%`).

Expected Result: Each stat shows its title above a large brand-blue value.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-11-stats.png`
> **Purpose:** Stats
> **Description:** The Stats tab with three items and the stats row on the canvas.

Step 12 — Set the Additional options (optional)

Action: Open the **Additional** tab; enter an **Anchor ID** (e.g. `anchor-our-services`) if the teaser is a link target; adjust padding / margin only when the design requires it; leave **Hide block** off.

Expected Result: A valid Anchor ID is accepted; spacing updates.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/teaser-step-12-additional.png`

> **Purpose:** Additional tab
> **Description:** The Additional tab with an Anchor ID entered.

Step 13 — Preview

Action: Switch to Preview mode, then publish to Preview and open `https://main--tcs-corp--sradha-tcs.aem.page/{path}`.

Expected Result: The teaser renders as on the live site; Share and Download work.

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-step-13-preview.png`
> **Purpose:** Preview
> **Description:** The teaser on the .aem.page preview host at desktop width.

Step 14 — Validate

Action: Run the Validation checklist below at desktop, tablet and mobile widths.

Expected Result: All items pass.

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-step-14-validate.png`
> **Purpose:** Validate
> **Description:** The teaser at 375 px width in the browser's responsive mode.

Step 15 — Publish

Action: Choose Publish → Live in Universal Editor (or publish from the Sites console).

Expected Result: The page is updated on `https://main--tcs-corp--sradha-tcs.aem.live/{path}`.

> [Screenshot Placeholder]
> **File name:** `screenshots/teaser-step-15-publish.png`
> **Purpose:** Publish
> **Description:** The Publish dialog with the page selected for Live.

Preview Process

1. Universal Editor saves each change automatically.
2. Use Preview mode for a first check, then publish to Preview (`.aem.page`) for review.
3. Check 1440 px, 1024 px, 768 px and 375 px widths; check light and dark BG Colors.
4. Check that the placeholders sheet and taxonomy of the locale are published if the teaser uses reading time, actions or tags.

Publishing Process

1. Complete the Validation checklist.
2. Publish the page (and newly used assets) to Live.
3. If tags are new, publish the locale's taxonomy page so that their labels resolve.
4. Reload the live page with the cache disabled to confirm.

Validation Process

- ☐ The Layout matches the design; options hidden for the layout are not relied on.
- ☐ Title present with the right Title Type (one H1 per page; H2 / H3 in the page flow).
- ☐ Image alt text set for meaningful images and charts; empty for decorative ones.
- ☐ Image crop (Image ratio) keeps the subject; Text split charts are fully visible.
- ☐ Image position (mobile) gives the intended order at 375 px.
- ☐ Background image / Card: text readable over the overlay; an image is selected.
- ☐ Links: each link has a title and a target; styles are as designed; List rows divide correctly; new-window links open a new tab.
- ☐ Multiple links: the Primary / Secondary CTA fields are not expected to show.
- ☐ Stats: each item shows a title and a single-line value.
- ☐ Date shows in the chosen format; reading time reads "{n} min read" (localised).
- ☐ Tags show localised labels (from the taxonomy), not raw IDs.
- ☐ Listen, Download and Share work; "Link copied to clipboard" appears where sharing is not supported.
- ☐ Author shows name and designation (an image alone is not shown).
- ☐ Lotto: animation plays, pause / play works.
- ☐ Anchor ID unique; Hide block off; Personalization off unless requested.
- ☐ No console errors.

Rollback Process

1. Restore the previous page version in AEM (Versions / Timeline) or undo in Universal Editor.
2. Publish the restored version.
3. Revert or replace changed assets in AEM Assets and republish.
4. Emergency: turn Hide block on (Additional tab) and publish.
5. Code regressions: the development team reverts the commit on `main` (`blocks/teaser/*` or the shared decorators).

6.11.6 Authorable Field Specification

The Teaser model `teaser` has 70 fields (excluding the six tab separators): 64 block fields, 4 fields per Links item and 2 per Stats item. The block definition sets `key-value: true`. Only the Anchor ID has an enforced validation; no field is marked required. Visibility conditions are listed in the Field details.

Block model `teaser` — Content tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Eyebrow (`eyebrow`)	text	No	No	Empty	Kicker above the title	Up to 40 characters (Recommended)	None	Insights
Show Eyebrow (`showEyebrow`)	boolean	No	No	true	Show / hide the eyebrow	Keep on	None	true
Title (`title`)	text	No (Recommended)	No	Empty	Heading of the teaser	Up to 70 characters (Recommended)	None	Banking on AI
Title Type (`titleType`)	select	No	No	H2	Heading level	H2 in the page flow; H1 only for a page-opening teaser	Options: H1, H2, H3, H4, H5, H6	h2

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Hide Title (`titleHidden`)	boolean	No	No	false	Visually hide the title	Use rarely	None	false
Description (`description`)	richtext	No	No	Empty	Main text	Up to 300 characters (Recommended)	None	How banks scale AI responsibly.
Show Description (`showDescription`)	boolean	No	No	true	Show / hide the description	Keep on	None	true
Short Description (`shortDescription`)	richtext	No	No	Empty	Note after the links	Footnotes, sources	None	Source: TCS AI study 2026.
View All Text (`viewAllText`)	text	No	No	Empty	View All button text	Specific text	None	View all insights
View All Link (`viewAll`)	aem-content	No	No	Empty	View All target	A listing page	None	`/content/tcs-corp/us/en/insights`
Show Date (`showDate`)	boolean	No	No	false	Show the date	On for dated content	None	true
Date (`date`)	date-time	No	No	Empty	Publication date	Pick a date	Date picker	2026-10-13
Date Format (`dateFormat`)	select	No	No	DD-MM-YYYY	Date display format	Match the locale's convention	Options: DD-MM-YYYY; MM-DD-YYYY; MMM d, yyyy; DD/MM/YY	dd/mm/yy
Date label (`dateLabel`)	text	No	No	Empty	Text before the date	Short word	None	Published
Reading time (`readingTime`)	text	No	No	Empty	Reading time	Minutes as a number	None	14
Display Tags (`displayTags`)	boolean	No	No	false	Show the tags	On to show Tags	None	true
Tags (`tags`)	aem-tag	No	Yes	Empty	Topic tags	1–3 tags (Recommended)	AEM tag picker	tcs:industries/banking
Listen link (`listenLink`)	aem-content	No	No	Empty	Audio version of the page	Audio page or file	None	`/content/dam/tcs/audio/ai.mp3`
Download (`download`)	boolean	No	No	false	Show the download (print) icon	For printable content	None	true
Share (`socialShare`)	boolean	No	No	false	Show the share icon	For shareable content	None	true
Author name (`authorName`)	text	No	No	Empty	Author's name	Full name	None	Priya Sharma

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Author designation (`authorDesignation`)	text	No	No	Empty	Author's role	e.g. TCS Role	None	Head of AI Research, TCS
Author image (`authorImage`)	reference (asset)	No	No	Empty	Author portrait	Square portrait	None	`/content/dam/tcs/people/priya.jpg`
Author image alt text (`authorImageAlt`)	text	No	No	Empty	Portrait alt text	Empty if decorative	None	Priya Sharma

Block model `teaser` — Image tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Image type (`imageFormat`)	select	No	No	Default (“”)	Picture or Lottie	Lotto only with `.json`	Options: Default, Lotto	lottie
Image (`image`)	reference (asset)	No	No	Empty	Image or animation	See Layout for placement	None	`/content/dam/tcs/teaser/cloud.jpg`
Image alt text (`imageAlt`)	text	No	No	Empty	Alt text / animation label	Empty if decorative	None	Cloud migration dashboard
Hide Image (`hideImage`)	boolean	No	No	false	Hide the image	Keeps the asset reference	None	false
Image position (`imagePosition`)	select	No	No	Left	Image side (desktop)	Alternate in sequences	Options: Left, Right; shown only for Layout Image & Text	right
Image position (mobile) (`imagePositionMobile`)	select	No	No	Default (“”)	Image order below 900 px	Bottom to lead with text	Options: Default, Top, Bottom	mobile-image-bottom
Image ratio (`imageRatio`)	select	No	No	Default (10:9)	Image proportion	Portrait for people	Options: Default (10:9), Portrait (8:9), Square (1:1), Original; shown for Image & Text and Text split	ratio-8-9
Image corners (`imageCorners`)	select	No	No	Rounded (“”)	Corner style	Square for diagrams	Options: Rounded, Square; shown for Image & Text and Text split	square

Block model `teaser` — Style tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Layout (`classes`)	select	No	No	Image & Text (“”)	Overall layout	Choose first	Options: Image & Text, Text split, Background image, Card	no-image
BG Color (`bgColor`)	select	No	No	Default (“”)	Panel colour	Dark for contrast sections	Options: Default, Grey, Dark, Dark green	dark

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Text alignment (`textAlignment`)	select	No	No	Layout default (“”)	Text alignment	Layout default in most cases	Options: Layout default, Left, Center, Right	text-left
Vertical alignment (`verticalAlignment`)	select	No	No	Layout default (“”)	Vertical alignment of columns	Top for long text	Options: Layout default, Top, Center, Bottom	valign-top
Title size (`titleSize`)	select	No	No	Layout default (“”)	Title size	Display only for 1–3 words	Options: Layout default, Medium (40px), Large (48px), Extra large (64px), Display (96px)	title-l
Title weight (`titleWeight`)	select	No	No	Layout default (“”)	Title weight	As designed	Options: Layout default, Regular, Semibold	title-regular
Description size (`descriptionSize`)	select	No	No	Layout default (“”)	Text size	Large for short copy	Options: Layout default, Regular (18px), Large (24px)	text-large
Eyebrow style (`eyebrowStyle`)	select	No	No	Brand (“”)	Eyebrow look	Label for content types	Options: Brand, Label (uppercase, grey)	eyebrow-label
Top spacing (`topSpacing`)	select	No	No	Default (“”)	Extra space above	Page header for page-opening teasers	Options: Default, Page header (extra space above)	spacing-header
Personalization (`personalization`)	boolean	No	No	false	Personalization marker	Only on request	None	false

Block model `teaser` — Links tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Multiple links (`multiLinks`)	boolean	No	No	false	List of links instead of two CTAs	On for 3+ links	None	true
Links (`links`)	container (composite multifield)	No	Yes	No items	The links list	See the Links item table	Shown only with Multiple links on	3 items
List arrow (`linkArrow`)	select	No	No	Right (“”)	Arrow of List-style links	Down right per design	Options: Right, Down right	arrow-down-right
Primary CTA Title (`primaryCtaText`)	text	No	No	Empty	Primary CTA text	Verb first	Shown only with Multiple links off	Contact us
Primary CTA Link (`primaryCta`)	aem-content	No	No	Empty	Primary CTA target	Page or URL	Shown only with Multiple links off	`/content/tcs-corp/us/en/contact-us`

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Primary CTA Link Type (`primaryCtaTarget`)	select	No	No	Default (“”)	Same / new window	New window for external	Options: Default, Open in New Window; shown only with Multiple links off	new-window
Primary CTA Link Style (`primaryCtaStyle`)	select	No	No	Primary Button	Primary CTA style	Keep Primary	Options: Text link, List, Primary Button, Secondary Button; shown only with Multiple links off	primary
Secondary CTA Title (`secondaryCtaText`)	text	No	No	Empty	Secondary CTA text	Verb first	Shown only with Multiple links off	Our services
Secondary CTA Link (`secondaryCta`)	aem-content	No	No	Empty	Secondary CTA target	Page or URL	Shown only with Multiple links off	`/content/tcs-corp/us/en/services`
Secondary CTA Link Type (`secondaryCtaTarget`)	select	No	No	Default (“”)	Same / new window	As above	Options: Default, Open in New Window; shown only with Multiple links off	(Default)
Secondary CTA Link Style (`secondaryCtaStyle`)	select	No	No	Secondary Button	Secondary CTA style	Secondary or Text link	Options: Text link, List, Primary Button, Secondary Button; shown only with Multiple links off	secondary

Block model `teaser` — Stats tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Stats (`stats`)	container (composite multifield)	No	Yes	No items	Headline statistics	2–4 stats (Recommended); see the Stats item table	None	3 items

Block model `teaser` — Additional tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	text	No	No	Empty	ID for in-page links	Unique, starts with `anchor-`	Enforced regExp `^anchor-[a-z0-9][a-z0-9-]*$` (“Start with anchor-, then lowercase letters, digits and hyphens.”)	anchor-our-services
Hide block (`classes_hideblock`)	boolean	No	No	false	Temporarily hide	Off before go-live	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default (“”)	Left padding	Only when the design needs it	Options: Default, 0, space-100 … space-3000	pl-600
Padding right (`classes_paddingright`)	select	No	No	Default (“”)	Right padding	As above	Options: Default, 0, space-100 … space-3000	pr-600

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Padding top (`classes_paddingtop`)	select	No	No	Default ("")	Top padding (overrides Top spacing and panel padding)	As above	Options: Default, 0, space-100 … space-3000	pt-1000
Padding bottom (`classes_paddingbottom`)	select	No	No	Default ("")	Bottom padding	As above	Options: Default, 0, space-100 … space-3000	pb-1000
Margin left (`classes_marginleft`)	select	No	No	Default ("")	Left margin	As above	Options: Default, 0, space-100 … space-3000	ml-0
Margin right (`classes_marginright`)	select	No	No	Default ("")	Right margin	As above	Options: Default, 0, space-100 … space-3000	mr-0
Margin top (`classes_margintop`)	select	No	No	Default ("")	Space above	As above	Options: Default, 0, space-100 … space-3000	mt-800
Margin bottom (`classes_marginbottom`)	select	No	No	Default ("")	Space below	As above	Options: Default, 0, space-100 … space-3000	mb-800

Item model — Links item (fields of the `links` container)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
CTA Title (`ctaText`)	text	No (Recommended)	No	Empty	Link text	Descriptive, up to 60 characters for List rows	None	Cloud migration services
CTA Link (`cta`)	aem-content	Yes in practice (items without a link are dropped)	No	Empty	Link target	Page, URL or anchor	None in the model	`/content/tcs-corp/us/en/services/cloud`
CTA Link Type (`ctaTarget`)	select	No	No	Default ("")	Same / new window	New window for external	Options: Default, Open in New Window	new-window
CTA Link Style (`ctaStyle`)	select	No	No	Text link ("")	Link style	List for resource lists	Options: Text link, List, Primary Button, Secondary Button	list

Item model — Stats item (fields of the `stats` container)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Stats Title (`statLabel`)	rich-text	No (Recommended)	No	Empty	What the number measures	Short label	None	Reduction in run costs

Field Name	Field Type	Mandatory	Multi Value	De-fault Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Stats Value (`statValue`)	rich-text	No (Rec-om-mended)	No	Empty	The num-ber	A single line, e.g. 40%	None (single line is guidance; the last element of the item is used as the value)	40%

Field details

Content tab:

- **Eyebrow (`eyebrow`)** — Purpose: kicker above the title. Allowed Values: plain text. Character Limits: Recommended up to 40 characters. Business Rules: shown only when filled and Show Eyebrow is not off; rendered as `p.teaser-eyebrow`, full width of the kicker row; style from Eyebrow style. Dependency Rules: Show Eyebrow, Eyebrow style. Visibility Rules: always visible. Sample Content: Insights.
- **Show Eyebrow (`showEyebrow`)** — Purpose: toggles the eyebrow. Allowed Values: on / off. Character Limits: not applicable. Business Rules: default on; a missing value counts as on; only an explicit off hides it. Dependency Rules: Eyebrow. Visibility Rules: always visible. Sample Content: on.
- **Title (`title`)** — Purpose: the teaser heading. Allowed Values: plain text. Character Limits: Recommended up to 70 characters (up to 25 for Display size). Business Rules: no title, no heading; it is also the share title and the data layer `dc:title`. Dependency Rules: Title Type, Hide Title, Title size, Title weight. Visibility Rules: always visible. Sample Content: Banking on AI.
- **Title Type (`titleType`)** — Purpose: heading level. Allowed Values: `h1`–`h6`. Business Rules: default H2 (model and template); an invalid value falls back to the authored heading level, else H2. Dependency Rules: Title. Visibility Rules: always visible. Sample Content: h3.
- **Hide Title (`titleHidden`)** — Purpose: hide the title visually. Allowed Values: on / off. Business Rules: the heading stays in the DOM with `visually-hidden`. Dependency Rules: Title. Visibility Rules: always visible. Sample Content: off.
- **Description (`description`)** — Purpose: main text. Allowed Values: rich text. Character Limits: Recommended up to 300 characters (Image & Text), 600 (Text split). Business Rules: shown when filled and Show Description is not off; paragraphs separated by 1.5rem. Dependency Rules: Show Description, Description size. Visibility Rules: always visible. Sample Content: How leading banks scale AI responsibly across the enterprise.
- **Show Description (`showDescription`)** — Purpose: toggles the description. Allowed Values: on / off. Business Rules: default on; a missing value counts as on. Dependency Rules: Description. Visibility Rules: always visible. Sample Content: on.
- **Short Description (`shortDescription`)** — Purpose: note after the links. Allowed Values: rich text. Character Limits: Recommended up to 200 characters. Business Rules: always shown when filled; last element of the body. Dependency Rules: Description size applies. Visibility Rules: always visible. Sample Content: Source: TCS Global AI Study 2026.
- **View All Text (`viewAllText`)** — Purpose: View All button text. Allowed Values: plain text. Character Limits: Recommended up to 25 characters. Business Rules: combined with View All Link into one link. Dependency Rules: View All Link. Visibility Rules: always visible. Sample Content: View all insights.
- **View All Link (`viewAll`)** — Purpose: View All target. Allowed Values: AEM page or URL. Business Rules: rendered as a primary button beside the title; no new-window option. Dependency Rules: View All Text. Visibility Rules: always visible. Sample Content: `/content/tcs-corp/us/en/insights`.
- **Show Date (`showDate`)** — Purpose: shows the date. Allowed Values: on / off. Business Rules: default off; the date is shown only when on and a Date is set. Dependency Rules: Date, Date Format, Date label. Visibility Rules: always visible; controls the visibility of the three date fields. Sample Content: on.
- **Date (`date`)** — Purpose: the date. Allowed Values: a date from the picker (stored as an ISO date / date-time). Business Rules: formatted from its calendar date (no time-zone shift); a value that is not a date is shown as

stored. Dependency Rules: Show Date. Visibility Rules: shown only when Show Date is on (`showDate === true`). Sample Content: 2026-10-13.

- **Date Format (`dateFormat`)** — Purpose: display format. Allowed Values: `dd-mm-yyyy` (13-10-2026), `mm-dd-yyyy` (10-13-2026), `mmm-d-yyyy` (Oct 13, 2026; English month abbreviations), `dd/mm/yy` (13/10/26). Business Rules: default `dd-mm-yyyy`. Dependency Rules: Date. Visibility Rules: shown only when Show Date is on. Sample Content: dd/mm/yy.
- **Date label (`dateLabel`)** — Purpose: text before the date. Allowed Values: plain text. Character Limits: Recommended up to 20 characters. Business Rules: followed by a space and the date. Dependency Rules: Date. Visibility Rules: shown only when Show Date is on. Sample Content: Published.
- **Reading time (`readingTime`)** — Purpose: reading time. Allowed Values: whole number of minutes or free text. Character Limits: Recommended up to 20 characters. Business Rules: a number becomes the localised placeholder `minRead` ("{n} min read"); other text is shown as typed. Dependency Rules: none. Visibility Rules: always visible. Sample Content: 14.
- **Display Tags (`displayTags`)** — Purpose: shows the tags. Allowed Values: on / off. Business Rules: default off. Dependency Rules: Tags. Visibility Rules: always visible; controls the visibility of Tags. Sample Content: on.
- **Tags (`tags`)** — Purpose: topic labels. Allowed Values: AEM tags (multi). Character Limits: Recommended 1–3 tags. Business Rules: labels resolved from `/{locale}/taxonomy.json` (the locale's language sheet when translated); unknown IDs are turned into a label from the last part of the ID (e.g. `tcs:topics/artificial-intelligence` → "Artificial intelligence"). Dependency Rules: Display Tags. Visibility Rules: shown only when Display Tags is on. Sample Content: tcs:industries/banking; tcs:topics/ai-native.
- **Listen link (`listenLink`)** — Purpose: audio version. Allowed Values: AEM page, asset or URL. Business Rules: the link text is replaced by the localised label "Listen to this page" and shown as a headphones icon button. Dependency Rules: none. Visibility Rules: always visible. Sample Content: `/content/dam/tcs/audio/banking-on-ai.mp3`.
- **Download (`download`)** — Purpose: print / save as PDF. Allowed Values: on / off. Business Rules: the icon button calls the browser's print dialog. Dependency Rules: none. Visibility Rules: always visible. Sample Content: on.
- **Share (`socialShare`)** — Purpose: share the page. Allowed Values: on / off. Business Rules: opens the device share sheet (Web Share API) with the teaser title (or page title) and URL; without it, copies the URL and shows "Link copied to clipboard" for 3 seconds; a dismissed dialog or blocked clipboard is ignored silently. Dependency Rules: none. Visibility Rules: always visible. Sample Content: on.
- **Author name (`authorName`)** — Purpose: author's name. Allowed Values: plain text. Character Limits: Recommended up to 50 characters. Business Rules: the author lockup is shown only with a name or designation. Dependency Rules: none. Visibility Rules: always visible. Sample Content: Priya Sharma.
- **Author designation (`authorDesignation`)** — Purpose: author's role. Allowed Values: plain text. Character Limits: Recommended up to 60 characters. Business Rules: as Author name. Visibility Rules: always visible. Sample Content: Head of AI Research, TCS.
- **Author image (`authorImage`)** — Purpose: portrait. Allowed Values: one image. Business Rules: shown round at 2.5rem, only together with a name or designation. Dependency Rules: Author name / designation. Visibility Rules: always visible. Sample Content: `/content/dam/tcs/people/priya-sharma.jpg`.
- **Author image alt text (`authorImageAlt`)** — Purpose: portrait alt text. Allowed Values: plain text. Business Rules: empty makes the image decorative (`alt=""`). Dependency Rules: Author image. Visibility Rules: always visible. Sample Content: Priya Sharma.

Image tab:

- **Image type (`imageFormat`)** — Purpose: picture or Lottie. Allowed Values: Default (""), Lotto (`lottie`). Business Rules: Lotto plays the animation only when Image is a `.json` file; Default with a `.json` file shows no media. Dependency Rules: Image. Visibility Rules: always visible. Sample Content: lottie.
- **Image (`image`)** — Purpose: the media. Allowed Values: one image or Lottie JSON asset. Business Rules: placement depends on Layout (beside, in the text column, or behind); ignored when Hide Image is on.

Dependency Rules: Image type, Layout. Visibility Rules: always visible. Sample Content: `/content/dam/tcs/teaser/cloud.jpg`.

- **Image alt text (`imageAlt`)** — Purpose: alt text or animation label. Allowed Values: plain text. Character Limits: Recommended up to 125 characters. Business Rules: empty makes the image decorative (`alt=""`) or the animation `aria-hidden`. Dependency Rules: Image. Visibility Rules: always visible. Sample Content: A cloud migration dashboard on a laptop.
- **Hide Image (`hideImage`)** — Purpose: hide the media. Allowed Values: on / off. Business Rules: Image & Text without media centres the content (Variation 3). Dependency Rules: Image. Visibility Rules: always visible. Sample Content: off.
- **Image position (`imagePosition`)** — Purpose: image side on desktop. Allowed Values: `left`, `right`. Business Rules: default Left; applied only to Image & Text with media, from 900 px. Dependency Rules: Layout. Visibility Rules: shown only when Layout is Image & Text (`classes` empty or unset). Sample Content: right.
- **Image position (mobile) (`imagePositionMobile`)** — Purpose: image order below 900 px. Allowed Values: Default (“”), `mobile-image-top`, `mobile-image-bottom`. Business Rules: no effect from 900 px or for Background image / Card. Dependency Rules: Image. Visibility Rules: always visible. Sample Content: mobile-image-bottom.
- **Image ratio (`imageRatio`)** — Purpose: image proportion. Allowed Values: Default (“”), `ratio-8-9`, `ratio-1-1`, `ratio-original`. Business Rules: default 10:9 (original proportion in Text split); Original affects images only. Dependency Rules: Layout. Visibility Rules: shown only when Layout is Image & Text or Text split. Sample Content: ratio-1-1.
- **Image corners (`imageCorners`)** — Purpose: corner style. Allowed Values: Rounded (“”), `square`. Business Rules: images in Text split are always square. Dependency Rules: Layout. Visibility Rules: shown only when Layout is Image & Text or Text split. Sample Content: square.

Style tab:

- **Layout (`classes`)** — Purpose: overall layout. Allowed Values: Image & Text (“”), `no-image` (Text split), `background`, `card`. Business Rules: stored as a block class (also read from a `classes` row); controls the visibility of Image position, Image ratio and Image corners. Dependency Rules: none. Visibility Rules: always visible. Sample Content: no-image.
- **BG Color (`bgColor`)** — Purpose: panel colour. Allowed Values: Default (“”), `grey`, `dark`, `dark-green` (rendered as `bg-grey`, `bg-dark`, `bg-dark-green`). Business Rules: adds panel padding; dark options switch text to white. Visibility Rules: always visible. Sample Content: dark.
- **Text alignment (`textAlignment`)** — Purpose: text alignment. Allowed Values: Layout default (“”), `text-left`, `text-center`, `text-right`. Business Rules: Center has no effect in Text split. Visibility Rules: always visible. Sample Content: text-center.
- **Vertical alignment (`verticalAlignment`)** — Purpose: vertical alignment. Allowed Values: Layout default (“”), `valign-top`, `valign-middle`, `valign-bottom`. Business Rules: affects the two-column layouts from 900 px and Background image at all widths. Visibility Rules: always visible. Sample Content: valign-top.
- **Title size (`titleSize`)** — Purpose: title size. Allowed Values: Layout default (“”), `title-m`, `title-l`, `title-xl`, `title-display`. Business Rules: the pixel sizes in the labels are desktop sizes; mobile sizes are smaller. Visibility Rules: always visible. Sample Content: title-xl.
- **Title weight (`titleWeight`)** — Purpose: title weight. Allowed Values: Layout default (“”), `title-regular`, `title-semibold`. Visibility Rules: always visible. Sample Content: title-regular.
- **Description size (`descriptionSize`)** — Purpose: description and short description size. Allowed Values: Layout default (“”), `text-regular`, `text-large`. Visibility Rules: always visible. Sample Content: text-large.
- **Eyebrow style (`eyebrowStyle`)** — Purpose: eyebrow look. Allowed Values: Brand (“”), `eyebrow-label`. Visibility Rules: always visible. Sample Content: eyebrow-label.

- **Top spacing (`topSpacing`)** — Purpose: page-header spacing. Allowed Values: Default (""), `spacing-header`. Business Rules: 4rem / 12.5rem top padding; no effect for Background image and Card; Additional Padding top overrides it. Visibility Rules: always visible. Sample Content: spacing-header.
- **Personalization (`personalization`)** — Purpose: personalization marker. Allowed Values: on / off. Business Rules: adds class `personalization` and `data-personalization="true"`; no visible change and no personalization engine connected. Visibility Rules: always visible. Sample Content: off.

Links tab:

- **Multiple links (`multiLinks`)** — Purpose: chooses the links mode. Allowed Values: on / off. Business Rules: on uses Links; off uses the Primary / Secondary CTA fields. Dependency Rules: controls visibility of the link fields. Visibility Rules: always visible. Sample Content: on.
- **Links (`links`)** — Purpose: list of links. Allowed Values: any number of items. Business Rules: items rendered in order in their own style; items without a link dropped; consecutive List items share one list. Visibility Rules: shown only when Multiple links is on. Sample Content: 4 List-style items.
- **List arrow (`linkArrow`)** — Purpose: arrow of List links. Allowed Values: Right (""), `arrow-down-right`. Business Rules: affects same-window List links only. Visibility Rules: always visible. Sample Content: arrow-down-right.
- **Primary CTA Title (`primaryCtaText`)** — Purpose: first CTA text. Allowed Values: plain text. Character Limits: Recommended up to 25 characters (buttons do not wrap). Visibility Rules: shown only when Multiple links is off. Sample Content: Contact us.
- **Primary CTA Link (`primaryCta`)** — Purpose: first CTA target. Allowed Values: page / URL / anchor. Business Rules: without a link the CTA is not rendered. Visibility Rules: Multiple links off. Sample Content: `/content/tcs-corp/us/en/contact-us`.
- **Primary CTA Link Type (`primaryCtaTarget`)** — Purpose: new window. Allowed Values: Default (""), `new-window`. Visibility Rules: Multiple links off. Sample Content: (Default).
- **Primary CTA Link Style (`primaryCtaStyle`)** — Purpose: style. Allowed Values: Text link (""), `list`, `primary`, `secondary`. Business Rules: default Primary Button; when the value is missing the code uses Primary Button, an empty value is Text link. Visibility Rules: Multiple links off. Sample Content: primary.
- **Secondary CTA Title (`secondaryCtaText`)** — As Primary CTA Title. Sample Content: Our services.
- **Secondary CTA Link (`secondaryCta`)** — As Primary CTA Link. Sample Content: `/content/tcs-corp/us/en/services`.
- **Secondary CTA Link Type (`secondaryCtaTarget`)** — As Primary CTA Link Type.
- **Secondary CTA Link Style (`secondaryCtaStyle`)** — As Primary CTA Link Style, default Secondary Button (fallback Secondary Button when missing). Sample Content: secondary.
- **CTA Title (`ctaText`, Links item)** — Purpose: link text. Character Limits: Recommended up to 60 characters for List rows, 25 for buttons. Visibility Rules: always visible in the item. Sample Content: Cloud migration services.
- **CTA Link (`cta`, Links item)** — Purpose: link target. Business Rules: item dropped without it. Sample Content: `/content/tcs-corp/us/en/services/cloud`.
- **CTA Link Type (`ctaTarget`, Links item)** — Allowed Values: Default (""), `new-window`. Business Rules: new-window List links show an up-right arrow. Sample Content: new-window.
- **CTA Link Style (`ctaStyle`, Links item)** — Allowed Values: Text link (""), `list`, `primary`, `secondary`. Business Rules: an item without a style uses the legacy Multi Link Style value of older teasers, else Text link. Sample Content: list.

Stats tab:

- **Stats (`stats`)** — Purpose: statistics. Allowed Values: any number of items. Character Limits: Recommended 2–4 items. Business Rules: rendered as a description list after the description / image and before the links; omitted when empty. Visibility Rules: always visible. Sample Content: 3 items.

- **Stats Title (`statLabel`, Stats item)** — Purpose: label. Allowed Values: rich text. Character Limits: Recommended up to 50 characters. Business Rules: all elements of the item except the last form the label. Sample Content: Reduction in run costs.
- **Stats Value (`statValue`, Stats item)** — Purpose: value. Allowed Values: rich text, a single line. Character Limits: Recommended up to 6 characters (no wrapping at 6rem). Business Rules: the last element of the item is the value; if the value is empty, the last line of the title becomes the value. Sample Content: 40%.

Additional tab:

- **Anchor ID (`classes_anchor`)** — Purpose: in-page link target. Allowed Values: `anchor-` + lower-case letters, digits, hyphens. Character Limits: Enforced pattern. Business Rules: becomes the block id (duplicates get `-2`, `-3`) and the data layer component id. Visibility Rules: always visible. Sample Content: anchor-our-services.
- **Hide block (`classes_hideblock`)** — Purpose: hide without deleting. Business Rules: hidden on preview / live, faded in the editor. Sample Content: off.
- **Padding left / right / top / bottom (`classes_paddingleft`, `classes_paddingright`, `classes_paddingtop`, `classes_paddingbottom`)** — Purpose: block padding. Allowed Values: Default, `p{l,r,t,b}-0`, `-100` (0.25rem), `-200` (0.5rem), `-300` (0.75rem), `-400` (1rem), `-500` (1.25rem), `-600` (1.5rem), `-800` (2rem), `-1000` (2.5rem), `-1200` (3rem), `-1600` (4rem), `-2000` (5rem), `-2400` (6rem), `-3000` (7.5rem). Business Rules: override the teaser's own padding (BG Color panels, Top spacing, Background image, Card) at every breakpoint. Visibility Rules: always visible. Sample Content: pt-1000.
- **Margin left / right / top / bottom (`classes_marginleft`, `classes_marginright`, `classes_margintop`, `classes_marginbottom`)** — Purpose: block margin. Allowed Values: Default, `m{l,r,t,b}-0` … `-3000` (same 14 steps). Business Rules: applied by `styles.css`. Visibility Rules: always visible. Sample Content: mb-800.

6.11.7 Field Annotation Mapping

Marker	Field	Purpose
A	Teaser block (Style tab: Layout, BG Color, alignments, sizes, Eyebrow style, Top spacing, Personalization; Additional tab)	Layout and style of the whole teaser
B	Image type, Image, Image alt text, Hide Image, Image position, Image position (mobile), Image ratio, Image corners	The image or Lottie animation
C	Eyebrow, Show Eyebrow	The kicker above the title
D	Display Tags, Tags	The topic tags
E	Show Date, Date, Date Format, Date label	The date
F	Reading time	The reading time
G	Listen link, Download, Share	The page action icons
H	Title, Title Type, Hide Title	The heading
I	View All Text, View All Link	The View All button
J	Author name, Author designation, Author image, Author image alt text	The author lockup
K	Description, Show Description	The main text
L	Stats (Stats Title, Stats Value)	The statistics
M	Multiple links, Links (CTA Title, CTA Link, CTA Link Type, CTA Link Style), List arrow, Primary / Secondary CTA fields	The links
N	Short Description	The note after the links

6.11.8 Component Specification

Attribute	Value
Component Name	Teaser (`teaser`; Universal Editor title "Teaser")
Version	1.0
Owner	[To be confirmed: product owner — owner: Digital Marketing]; technical owner: development team
Dependencies	`scripts/aem.js`, `scripts/scripts.js`; decorators `a11y`, `animation`, `article-meta`, `date`, `button`, `icon-button`, `content`, `heading`, `image`, `link`, `stats`, `tag`; utils `properties`, `links`, `multifield`, `media`, `lottie`, `placeholders`, `tags`, `taxonomy`, `site`, `dates`, `ids`, `environment`, `viewport`; `styles/styles.css`
Component Type	Key-value block (`key-value: true`, resource type `core/franklin/components/block/v1/block`, model `teaser`); Links and Stats are composite multifields inside the block
Supported Variants	Layouts Image & Text (left / right / no image), Text split (with / without image), Background image, Card; Image type Default / Lotto; mobile image position; 4 image ratios; 2 corner styles; 4 BG Colors; 4 text alignments; 4 vertical alignments; 5 title sizes; 3 title weights; 3 description sizes; 2 eyebrow styles; 2 top spacings; links mode (two CTAs / multiple); 4 link styles; 2 list arrows; content features (stats, date, reading time, actions, author, tags, View All, hidden title, short description, personalization marker)
Supported Templates	Any page; allowed in Section and Gen Carousel
Supported Metadata	None read from page metadata (date, reading time, author and listen link are authored in the block; the share fallback title is `document.title`)
Supported Tags	AEM tags in the Tags field; labels from the locale taxonomy (`/{locale}/taxonomy.json`)
Supported Personalization	Personalization marker only (`data-personalization="true"`); no Adobe Target integration (Not implemented (as of this handover)); the block is in the data layer as `tcs-corp/components/teaser`
Caching Considerations	Page HTML cached by the Edge Delivery CDN; placeholders and taxonomy fetched once per page (in-memory promise) and cached by the CDN as published sheets; republish the sheets after changes
Performance Considerations	Decorated in the eager phase when in the first section (its first image is the LCP candidate via `waitForFirstImage`), else lazily with its section; `await getPlaceholders()` (and the taxonomy when tags are shown) runs before rendering — one small JSON request each, cached per page; the Lottie player loads only near the viewport in idle time
Accessibility Compliance	WCAG 2.1 AA: selectable heading level and visually hidden title (1.3.1, 2.4.6); alt texts (1.1.1); stats as `<dl>`; links in `<ul>` for List style; icon buttons with localised accessible names; live status for "Link copied" (4.1.3); Lottie pause / play (2.2.2) and reduced motion; white text on dark panels and overlays (1.4.3); no automatic new-window hint
SEO Compliance	Server-delivered text content (crawlable); heading level control; `<time datetime>` for dates; image alt texts; crawlable links

6.11.9 Input Model

Content Inputs: eyebrow, title, description, short description, View All link, date and label, reading time, listen link, author name / designation / image, image or Lottie file with alt text, links (two CTAs or the Links multifield), stats.

Metadata Inputs: none from page metadata; `document.title` is used as the share title when the teaser has no title.

Tags Inputs: the Tags field (AEM tag IDs such as `tcs:industries/banking`), resolved through `/{locale}/taxonomy.json`.

Page Context Inputs: the locale (placeholders and taxonomy paths); the visitor's reduced motion preference (Lottie); Web Share API / clipboard availability (Share).

Query Inputs: none (no query index).

Manual Inputs: every field; Layout also from the block class `no-image` / `background` / `card`; style values from rows or matching block classes; a legacy `image-right` class and a legacy `multiLinkStyle` row are honoured.

Authored HTML received from AEM (one row per property: name | value; abridged):

```
<div class="teaser no-image">
  <div><div>eyebrow</div><div><p>Insights</p></div></div>
  <div><div>title</div><div><h2>Banking on AI</h2></div></div>
  <div><div>description</div><div><p>How leading banks scale AI.</p></div></div>
  <div><div>showDate</div><div><p>true</p></div></div>
  <div><div>date</div><div><p>2026-10-13T00:00:00.000Z</p></div></div>
  <div><div>dateFormat</div><div><p>dd-mm-yyyy</p></div></div>
  <div><div>dateLabel</div><div><p>Published</p></div></div>
  <div><div>readingTime</div><div><p>14</p></div></div>
  <div><div>displayTags</div><div><p>true</p></div></div>
  <div><div>tags</div><div><ul><li>tcs:industries/banking</li></ul></div></div>
  <div><div>socialShare</div><div><p>true</p></div></div>
  <div><div>multiLinks</div><div><p>true</p></div></div>
  <div><div>links</div><div>
    <p><a href="/en/services/cloud">Cloud services</a></p><p>list</p>
    <hr>
    <p><a href="https://example.com/report.pdf">AI report</a></p><p>new-window</
p><p>list</p>
  </div></div>
  <div><div>stats</div><div>
    <p>Reduction in run costs</p><p>40%</p>
    <hr>
    <p>Faster onboarding</p><p>3x</p>
  </div></div>
</div>
```

6.11.10 Output Model

Rendered Output

`teaser.js` reads the rows, then replaces them with at most two children. Classes added to the block: the Layout (`no-image`, `background`, `card`), `bg-{color}`, the style option classes (`text-*`, `valign-*`, `mobile-image-*`, `title-*`, `text-regular` / `text-large`, `eyebrow-label`, `spacing-header`, `ratio-*`, `square`, `arrow-down-right`), `image-right` and `teaser-has-media` (Image & Text with media only), and `personalization`.

- `div.teaser` (+ classes)
  - `div.teaser-media` (Image & Text, Background image, Card: first child) with `<picture>` or `div.teaser-animation` + `button.animation-toggle`
  - `div.teaser-content`
    - `div.teaser-header`
      - `div.teaser-kicker`: `p.teaser-eyebrow`; `div.teaser-meta > ul.teaser-tags > li[data-tag]`; `div.teaser-info` (`p.teaser-date` with `<time>`, `p.teaser-reading-time`, `div.teaser-actions` with icon buttons and `p.teaser-status[role=status]`)
      - `div.teaser-intro`: `div.teaser-heading-row` (`h2.teaser-title`, `a.teaser-view-all.button.primary`); `div.teaser-author` (`div.teaser-author-image`, `div.teaser-author-details` with `p.teaser-author-name`, `p.teaser-author-designation`)
    - `div.teaser-body`: `div.teaser-description`; `div.teaser-media` (Text split only); `dl.teaser-stats > div.teaser-stat > dt.teaser-stat-label + dd.teaser-stat-value`; `div.teaser-links.teaser-links-{default|list|primary|secondary|mixed}` with `a.button` and / or `ul.teaser-link-list > li > a.teaser-link-list-item`; `div.teaser-short-description`

Empty containers are not rendered. Universal Editor instrumentation is moved to the rendered elements (eyebrow, title, description, pictures, links, stats).

Generated HTML

```
<div class="teaser block no-image bg-grey" data-block-name="teaser" data-cmp-
id="teaser-1">
  <div class="teaser-content">
    <div class="teaser-header">
      <div class="teaser-kicker">
        <p class="teaser-eyebrow">Insights</p>
        <div class="teaser-meta"><ul class="teaser-tags">
          <li data-tag="tcs:industries/banking">Banking</li></ul></div>
        <div class="teaser-info">
          <p class="teaser-date">Published <time datetime="2026-10-13">13-10-2026</time></
p>
          <p class="teaser-reading-time">14 min read</p>
          <div class="teaser-actions">
            <button type="button" class="button tertiary icon-button"
              aria-label="Share this page" data-cmp-interaction="share"></button>
            <p class="teaser-status" role="status" id="teaser-status-1"></p>
          </div>
        </div>
      </div>
      <div class="teaser-intro"><div class="teaser-heading-row">
        <h2 class="teaser-title">Banking on AI</h2></div></div>
    </div>
    <div class="teaser-body">
      <div class="teaser-description"><p>How leading banks scale AI.</p></div>
      <dl class="teaser-stats"><div class="teaser-stat">
        <dt class="teaser-stat-label"><p>Reduction in run costs</p></dt>
        <dd class="teaser-stat-value"><p>40%</p></dd></div></dl>
      <div class="teaser-links teaser-links-list"><ul class="teaser-link-list">
        <li><a href="/en/services/cloud" class="teaser-link-list-item">Cloud services</
a></li>
      </ul></div>
    </div>
  </div>
</div>
```

Generated JSON: none.

Generated Metadata: none (no `<head>` changes).

Analytics Tracking

Component state (id = Anchor ID, else `teaser-{n}`; `dc:title` = the first heading, also when visually hidden):

```
{
  "component": {
    "teaser-1": {
      "@type": "tcs-corp/components/teaser",
      "parentId": "page-1x2y3z",
      "dc:title": "Banking on AI"
    }
  }
}
```

Click events (`cmp:click`) with `interaction`:

Element	`interaction`	Other eventInfo
CTA, link, List row, View All	`link`	`dc:title` (link text), `xdm:linkURL`, `xdm:linkType` (`other`, `exit`, `download`), `breakpoint`
Listen icon	`listen`	`xdm:linkURL` of the audio link
Download icon	`download`	`xdm:linkType` `other` (it is a button)
Share icon	`share`	—
Lottie pause / play	`animation`	`dc:title` "Pause animation" / "Play animation"

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.teaser-1",
    "interaction": "share",
    "dc:title": "Share this page",
    "xdm:linkType": "other",
    "breakpoint": "mobile"
  }
}
```

The share result (shared, copied, cancelled) is not tracked. No tag manager consumes the data layer yet (Adobe Tags / Web SDK integration-ready, not implemented).

Personalization Output: with Personalization on, `class="personalization"` and `data-personalization="true"` on the block, for a future personalization script; nothing else.

Search Index Output: none; the query index reads only `<head>` metadata. Teaser text is crawlable HTML.

6.11.11 Business Rules

1. Properties are read from two-cell rows by normalised name (lower case, no hyphens); rows with another number of cells are ignored.
2. Toggles are on only when the value is `true`; Show Eyebrow and Show Description are on unless explicitly `false` (a missing row means shown).
3. Select values are read from the row, else from a matching block class (so block options and rows both work).
4. Layout: `no-image`, `background`, `card` or Image & Text; layout and BG Color classes are removed and re-added from the resolved values.
5. BG Color `grey`, `dark`, `dark-green` becomes `bg-grey`, `bg-dark`, `bg-dark-green`.
6. The eyebrow renders only when filled and shown; the kicker renders only with an eyebrow, tags or info.
7. Tags render only with Display Tags on and at least one tag; labels from the locale taxonomy, else derived from the tag ID; authored titles (no `:` or `/`) are shown as typed.
8. Date renders only with Show Date on and a Date; default format `dd-mm-yyyy`; the `<time>` gets `datetime="YYYY-MM-DD"`; non-date text is shown as stored.
9. Reading time: digits only → `minRead` placeholder "{n} min read"; else as typed.
10. Page actions: Listen link (icon button, label `listenToThisPage`), Download (`window.print()`, `downloadThisPage`), Share (Web Share API with title = teaser title or `document.title`, else clipboard copy and `linkCopied` for 3 seconds; errors ignored). The actions container is omitted when none is set.
11. Title: built only when filled; level from Title Type (invalid → authored level → H2); Hide Title applies `visually-hidden`.
12. View All: a primary button in the heading row; requires a link.
13. Author lockup: requires a name or designation; the image is optional.

14. Description renders when filled and shown; Short Description whenever filled.
15. Media: none when Hide Image is on or no Image; Lotto needs a `.json` link (else no media unless a picture exists); animation fit "slice" for Background image / Card, "meet" otherwise; label from Image alt text.
16. Media placement: in the body after the description for Text split; first child of the block for the other layouts.
17. `teaser-has-media` and `image-right` are set only for Image & Text with media; Image position Right (or a legacy `image-right` class) moves the image right from 900 px.
18. Stats: items split by `<hr>` or list items; the last element is the value, the others the label; omitted when no item.
19. Links mode: Multiple links on → Links items with their CTA Link Style (fallback: legacy Multi Link Style, else Text link); off → Primary CTA (fallback style Primary Button) then Secondary CTA (fallback Secondary Button); an empty style row means Text link; CTAs without a link are skipped; the links container is omitted when empty.
20. List style: consecutive List links share one `<ul>`; any other style ends the list; new-window List links get `target="_blank"` / `rel="noopener noreferrer"` and an up-right arrow; List arrow Down right changes same-window arrows.
21. Text link renders as a tertiary button; Primary / Secondary as buttons; the links container gets `teaser-links-{style}` or `teaser-links-mixed`.
22. Layout-specific CSS: options without effect (Top spacing on Background image / Card, Text alignment Center on Text split, image corners in Text split, Image ratio Original on animations) are ignored silently.
23. Dialog conditions: Date / Date Format / Date label need Show Date; Tags needs Display Tags; Image position needs Layout Image & Text; Image ratio and Image corners need Image & Text or Text split; Links needs Multiple links; the eight Primary / Secondary CTA fields need Multiple links off. Hidden fields keep their stored values, which the code ignores where the controlling toggle is off.
24. Personalization adds a marker class and data attribute only.
25. Additional tab: Anchor ID pattern enforced and applied as the block id (unique); Hide block hides outside edit mode; padding / margin classes override the teaser's own spacing.
26. Error handling: missing placeholders or taxonomy fall back to English labels and ID-derived tag labels; Lottie failures log `failed to play the Lottie animation`; no other errors are logged by the block.

6.11.12 Author Best Practices

Recommended Content Length

Element	Recommended
Eyebrow	Up to 40 characters
Title	Up to 70 characters (Display size: 1–3 words)
Description	Up to 300 characters (Image & Text, Background image, Card); up to 600 (Text split)
Short Description	Up to 200 characters
CTA / button titles	Up to 25 characters
List link titles	Up to 60 characters
Stats	2–4 items; values up to 6 characters; labels up to 50 characters
Tags	1–3
Author designation	Up to 60 characters

Image Dimensions

Layout / option	Recommended image
Image & Text, Default ratio (10:9)	1200 × 1080 px
Image ratio Portrait (8:9)	1200 × 1350 px
Image ratio Square (1:1)	1200 × 1200 px
Image ratio Original / Text split charts	Native size, at least 1200 px wide; transparent PNG or SVG for charts
Background image	2400 × 1350 px (16:9) or wider; cropped to cover a 35rem–48.75rem high banner
Card	1200 × 1200 px or 4:3; cropped to cover the card (min. 25rem / 35.625rem high)
Author image	Square, at least 160 × 160 px (shown round at 40 px)
Lottie	Lottie JSON for the SVG renderer, ideally under 500 KB, designed at 10:9 (beside) or 16:9 (behind)

CTA Writing Guidelines

- Start with a verb; be specific ("Read the banking report").
- One Primary Button per teaser; Secondary Button or Text link for alternatives.
- Use List style for three or more related links; keep List titles parallel in form.
- Mention new windows or file types in the text ("(PDF)").
- View All text names what is listed ("View all insights").

Accessibility Guidelines

- Choose the Title Type that fits the page outline; never skip levels for looks — use Title size instead.
- Alt text for meaningful images and charts (state the key data in the description too).
- Check contrast for Background image and Card (white text) and brand-blue eyebrows on dark panels.
- Prefer Hide Title over deleting a title when the design has no visible heading.
- Keep stats values short so they do not overflow on mobile.

SEO Guidelines

- Write the key message in the title and description as text, not inside images.
- Link to canonical locale pages; avoid "Click here".
- Use one H1 per page (a page-opening teaser may use H1 when there is no Hero or Page Header).

Metadata Guidelines: the Teaser does not read page metadata; author dates, reading time and author details in the block and keep them consistent with the page's own metadata (`publish-date`, `reading-time`, `author-id`) where the teaser describes the same page.

Taxonomy Guidelines: use tags from the TCS taxonomy only; make sure each tag has a title in the locale's taxonomy (and its language sheet) so that a translated label is shown; publish the taxonomy after adding tags.

Localization Guidelines

- Translate every text field per locale; link to same-locale pages.
- Labels "{n} min read", "Listen to this page", "Download this page", "Share this page", "Link copied to clipboard", "Pause animation", "Play animation" come from each locale's placeholders (`minRead`, `listenToThisPage`, `downloadThisPage`, `shareThisPage`, `linkCopied`, `pauseAnimation`, `playAnimation`).
- Date Format "MMM d, yyyy" uses English month names in every locale; prefer the numeric formats on non-English sites.
- Allow for 30 % longer text in German, French and Dutch.

6.11.13 Troubleshooting Guide

Issue 1 — A field is missing from the dialog

Symptoms: The author cannot find Date, Tags, Image position, Image ratio, Image corners, Links or the Primary / Secondary CTA fields.

Root Cause: The field's `condition` hides it: Show Date / Display Tags off; Layout not Image & Text (Image position) or not Image & Text / Text split (ratio, corners); Multiple links off (Links) or on (Primary / Secondary CTA).

Diagnosis Steps: Check the controlling toggle or Layout in the dialog.

Resolution Steps: Turn the toggle on / off or change the Layout; the field appears.

Validation Steps: The field is visible and its value renders.

Preventive Measures: Choose the Layout and the link mode first (Steps 3 and 10).

Issue 2 — Links do not appear

Symptoms: No buttons or link rows render.

Root Cause: Multiple links is on but only the Primary / Secondary fields were filled (or the reverse); or the link fields are empty.

Diagnosis Steps: Check Multiple links and which fields hold links; inspect the `.plain.html` rows `multiLinks`, `links`, `primaryCta`.

Resolution Steps: Fill the fields of the active mode (or switch the toggle back).

Validation Steps: `div.teaser-links` renders with the expected styles.

Preventive Measures: Decide on two CTAs vs. a list before authoring.

Issue 3 — Tags show raw or English labels

Symptoms: Tags appear like "Ai native" or in English on a localised site.

Root Cause: The tag is missing from `/{locale}/taxonomy.json`, the taxonomy is not published, or the locale's language sheet lacks a translation.

Diagnosis Steps: Open `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/taxonomy.json` and look for the tag ID.

Resolution Steps: Add the tag and its title to the locale's taxonomy and publish it.

Validation Steps: Reload the page; the localised label shows.

Preventive Measures: Publish the taxonomy with every new tag.

Issue 4 — The Background image / Card teaser looks grey with unreadable text

Symptoms: White text on a grey box, no picture.

Root Cause: No Image (or Hide Image on, or a `.json` with Image type Default): the dark overlay is drawn over the white page.

Diagnosis Steps: Check Image, Hide Image and Image type.

Resolution Steps: Select an image (or set Lotto for a `.json`), or choose BG Color Dark / Dark green, or switch Layout to Image & Text.

Validation Steps: The image shows behind the overlay; contrast is sufficient.

Preventive Measures: Always pair Background image / Card with an image.

Issue 5 — The image is cropped or the chart is cut off
Symptoms: Important parts of the image are missing.

Root Cause: Image ratio 10:9 / 8:9 / 1:1 crops with `object-fit: cover`; Background image and Card always crop.

Diagnosis Steps: Check Image ratio and Layout.

Resolution Steps: Use Image ratio Original (Image & Text) or Default / Original in Text split; recompose the image.

Validation Steps: The whole chart is visible at all widths.

Preventive Measures: Use the Text split layout for charts.

Issue 6 — The date shows the wrong format or no date
Symptoms: No date, or a date like "2026-10-13T00:00:00.000Z".

Root Cause: Show Date off; Date empty; or a non-ISO value.

Diagnosis Steps: Check Show Date and the stored Date value.

Resolution Steps: Turn Show Date on and pick the date with the date picker; choose the Date Format.

Validation Steps: "Published 13-10-2026" (or the chosen format) shows.

Preventive Measures: Use the picker, not typed text.

Issue 7 — Share does nothing on desktop
Symptoms: Clicking Share shows no dialog.

Root Cause: Without the Web Share API the URL is copied to the clipboard and "Link copied to clipboard" shows for 3 seconds; if the clipboard is blocked (non-HTTPS or permissions), nothing happens.

Diagnosis Steps: Check for the status message; test on HTTPS; check browser permissions.

Resolution Steps: Test on the `.aem.page` / `.aem.live` hosts (HTTPS).

Validation Steps: The message appears and the URL is in the clipboard.

Preventive Measures: Document the expected behaviour for testers.

Issue 8 — Stats show the wrong value
Symptoms: The title text appears as the large value.

Root Cause: The Stats Value is empty or contains several lines: the last element of the item is taken as the value.

Diagnosis Steps: Open the Stats items.

Resolution Steps: Enter a single-line Stats Value in every item.

Validation Steps: Each stat shows its label above its value.

Preventive Measures: Keep values to one short line.

6.11.14 Support Runbook
Monitoring Requirements

- Visual checks of key pages with teasers in each locale after releases.
- Availability of `/{locale}/placeholders.json` and `/{locale}/taxonomy.json` (HTTP 200).
- Core Web Vitals of pages that open with a teaser.

Logs

- Browser console: `failed to play the Lottie animation` (animation decorator). The Teaser logs no other messages; missing placeholders / taxonomy fail silently (English fallbacks, ID-derived tag labels).
- Admin API / AEM Sidekick: preview and publish status of the page, assets, placeholders and taxonomy.

- AEM author: publish / replication logs (Cloud Manager).

Common Errors

Error	Likely cause	Fix
Field not in dialog	Hidden by a `condition`	Change the toggle / Layout
No links	Wrong link mode filled	Fill the active mode's fields
Raw tag labels	Taxonomy not published / tag missing	Update and publish the taxonomy
Grey Background / Card	No image	Add an image or change layout
English labels on a localised page	Placeholder keys missing in the locale sheet	Add the keys and publish
Console "failed to play the Lottie animation"	Bad / missing `.json`	Re-export / republish
Anchor ID rejected	Pattern not matched	Use `anchor-` + lower case
Teaser missing live	Hide block on	Turn off and publish

Escalation Matrix

Level	Who	Checks
L1	Author support / content team lead	Dialog values, conditions, toggles, link mode, assets published, Hide block, checklist
L2	AMS / site admin	Publish status of page, assets, placeholders and taxonomy (Sidekick / admin API); CDN cache; console errors
L3	Development team	`blocks/teaser/*` and the shared decorators / utils; CSS regressions; Code Sync of `main`
L4	Adobe Support	Edge Delivery Services or AEM as a Cloud Service / Universal Editor incidents

Recovery Procedures

1. Restore the previous page version and publish.
2. Republish the placeholders / taxonomy sheets if labels are wrong.
3. Replace broken assets and republish.
4. Hide block on as a temporary measure.
5. Revert the code commit on `main` for regressions.

Health Checks

- `curl -s https://main--tcs-corp--sradha-tcs.aem.live/{path}.plain.html` shows the teaser rows.
- `curl -s -o /dev/null -w "%{http_code}" https://main--tcs-corp--sradha-tcs.aem.live/en/placeholders.json` returns 200 (same for `taxonomy.json` and each locale).
- Block-library pages render every teaser variation without console errors.

Operational Procedures

- Dialog changes: edit `_teaser.json`, run `npm run build:json` and `npm run lint`, open a pull request with a preview link; keep old values working (the code already honours the legacy `image-right` class and `multiLinkStyle` row).
- Maintain the placeholder keys listed above in all 13 locale sheets.
- Publish the taxonomy whenever tags are added.

Known Limitations

- No video media type (Not implemented (as of this handover)).

- No automatic "opens in a new tab" hint for new-window links.
- Personalization is a marker only; no Target integration.
- Date Format "MMM d, yyyy" uses English month names.
- Text alignment Center has no effect in Text split; Top spacing has no effect in Background image and Card; Image ratio Original does not apply to animations.
- The Lottie light player does not support After Effects expressions.
- View All has no new-window option and is always a primary button.
- The share outcome is not tracked in the data layer.

6.12 Quote Block

The Quote block (`blocks/quote`) presents a single highlighted statement, for example a client or leadership quote, as a card with large brand-blue quotation marks that overlaps a portrait image. An optional eyebrow and section title sit above the card. It implements the TCS.com design system Figma component "Text & Image Overlap" (node 2590-300067).

6.12.1 Component Purpose

Business Objective: Give a single voice (a client, partner, analyst or TCS leader) visual prominence on a page, combining the statement with a portrait so that the quote is attributed to a real person and reads as a credible endorsement.

Business Use Cases:

- A client quote on a case study or industry page, with the client's portrait.
- A leadership statement (for example the CEO's view) on a corporate, investor or press page.
- A partner endorsement on a partnership or alliance page.
- An analyst or award jury statement on a recognition page.
- A pull quote in an article or blog post, with or without an image.

When To Use:

- One statement deserves emphasis on its own and should stand out from the body copy.
- A portrait (or a relevant image) is available and the overlapping card layout is wanted.
- The statement needs an attributed author (name, designation, organisation).

When Not To Use:

- For a testimonial that needs stats, a round author photo or a centred layout: use the Testimonial block (6.13).
- For several quotes side by side or a carousel of quotes: use Cards / List or a Gen Carousel section with several Quote blocks (one quote per block).
- For ordinary inline quotations inside running text: use a Text component (rich text).
- For a promotional message with a call to action: use the Teaser or Hero block. The Quote block has no CTA field.

Dependencies:

Type	Dependency	Use
Block code	`blocks/quote/quote.js`, `blocks/quote/quote.css`	Decoration and styles
Model	`blocks/quote/_quote.json` (aggregated in `component-definition.json`, `component-models.json`)	Universal Editor dialog

Type	Dependency	Use
Decorator	`decorators/content-decorator.js fromCell`	Moves eyebrow, quote and author content into new elements, keeping editor instrumentation
Decorator	`decorators/heading-decorator.js buildHeading`	Builds the title heading at the Title Type level
Decorator	`decorators/image-decorator.js pictureOf`	Takes the authored picture and applies the Image Alt Text field
Util	`utils/properties.js readProperties, optionOf, textOf`	Reads the key-value property rows and the select values
Global	`decorators/block-options-decorator.js, styles/styles.css`	Additional tab (anchor, hide, padding, margin)
Global	`decorators/data-layer-decorator.js, utils/analytics.js`	Adobe Client Data Layer component entry and click events
Global	`decorators/content-buttons-decorator.js`	A standalone link in the quote or author rich text becomes a link button
Data sources	None	The block reads no sheet, placeholder, taxonomy or query index
Placeholders	None	The block renders no fixed UI text

Related Components: Testimonial (6.13) for centred testimonials with stats; Teaser for image and text with CTAs; Columns (6.14) for free side-by-side layouts; Text (default content) for inline quotations.

Applicable Templates: Any page. The block is allowed in every section (section filter in `models/_section.json`), so it can be used on Home, Article, Blog, Press Release and pages without a Page Type. No template adds or requires it.

Supported Content Types: Client stories, case studies, leadership statements, partner and analyst endorsements, articles, blogs and press releases (as a pull quote).

6.12.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A Quote block at 1440px wide with eyebrow, brand-blue title, the white quote card with large blue opening and closing quote marks on the left, overlapping a tall portrait image on the right.

Desktop (900px and wider): The eyebrow (grey, semibold, 1.5rem) and the title (brand blue, medium weight, 3rem, tight letter-spacing) are stacked, 0.75rem apart, with 5rem space below them. The body is centred with a maximum width of 70rem (1120px). When an image is present, the body is a three-column grid in the proportions 498 : 204 : 418. The card spans the first two columns (702 of 1120, about 63%), the image the last two (622 of 1120, about 56%), so the card overlaps the image by 204/1120 (about 18%). The card sits above the image (z-index 1) and is vertically centred against it. The image has a 622 : 869 portrait aspect ratio (cropped with object-fit: cover). Inside the card, a three-column grid holds the opening quote mark (column 4.125rem), the text, and the closing quote mark (column 2.875rem, aligned to the bottom). The quote marks are 7.5rem high, in brand blue. The card padding is 1.5rem top and bottom and 2.5rem left and right. The quote text is 1.75rem (line height 1.39); the author is 1.5rem (line height 1.417); there is 2.5rem between quote and author.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Quote block at 768px wide: title above, the portrait image full width, the quote card overlapping the bottom edge of the image and inset from its sides.

Tablet (600px to 899px): The block uses the stacked mobile layout (the block's only breakpoint is 900px). The image comes first and fills the width with a 4 : 5 aspect ratio. The card follows and moves up over the bottom of the image by 3rem (`--space-1200`), inset 1rem (`--space-400`) on the left and right. Type uses the mobile sizes of the tokens (title 2rem, quote 1.5rem, author 1.125rem, eyebrow 1rem, quote marks 5rem in a 2.5rem column). From 600px the section's side padding grows to 2.5rem (`--section-padding-inline`), so the block is narrower than the screen by 5rem.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Quote block at 375px wide: eyebrow and title stacked, the image full width, the quote card with smaller quote marks overlapping the image's bottom edge.

Mobile (below 600px): As for tablet, with the section side padding of 1.5rem. The header has 0.75rem between eyebrow and title and 2.5rem below it. The card padding is 1.5rem on all sides and the quote marks are 5rem high. With Image Position (Mobile) = Bottom, the image is below the card and the card overlaps the top edge of the image instead. With BG Color = Grey, the whole block has a light grey background with 2.5rem top and bottom and 1.5rem side padding (7.5rem and 5rem from 900px).

6.12.3 Component Variations

The Style select (`classes`) has only the Default option, so there are no style classes. The variations come from Image Position (Desktop), Image Position (Mobile), BG Color and whether an image is set. These options combine freely (for example Image Left + Image Bottom + Grey).

Variation 1 — Default (Image Right, Image Top on Mobile)

Variation Name: Default — image on the right on desktop, above the quote on mobile and tablet.

Business Purpose: The standard Figma "Text & Image Overlap" presentation: the quote card reads first (left) on desktop and the portrait sits to the right.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-default-image-right.png`
> **Purpose:** Default variation
> **Description:** Desktop Quote block with the white card on the left overlapping a portrait on the right; default white background.

Use Cases: Client and leadership quotes on most pages; the first quote on a page.

Authoring Guidelines: Set Quote, Quote Author and Image (with Image Alt Text, or leave it empty for a decorative image). Leave Image Position (Desktop) = Right, Image Position (Mobile) = Top and BG Color = Default. These are the defaults of a new block.

Business Rules: Right and Top are the defaults of the model (`value`) and of the block template, and the code falls back to them when the rows are missing or hold an unknown value.

Rendering Rules: The block gets the class `quote-has-media`; neither `image-left` nor `image-bottom` is set. The desktop grid places the card in columns 1-2 and the image in columns 2-3.

Responsive Behaviour: Desktop: card left, image right, 18% overlap, card vertically centred. Tablet and mobile: image above (4 : 5), card overlapping its bottom edge by 3rem, inset 1rem.

SEO Impact: The title is a real heading (H2 by default) that contributes to the page outline. The quote is a `<blockquote>` and the author is a `<figcaption>`, which search engines can read as a quotation and its attribution. The image alt text is indexed for image search when set.

Accessibility Compliance: The quote marks are CSS-generated content with an empty alternative text (`content: '\201C' / ''`), so screen readers do not announce them. The image alt text comes from Image Alt Text; empty alt marks the image as decorative. DOM order is header, card, image, so screen readers read the quote before the image even on mobile, where the image is shown first (CSS `order: -1`). Text colours (black on white, brand blue for title) meet WCAG 2.1 AA contrast [To be confirmed: contrast measurement of brand-500 on white — owner: design lead].

Variation 2 — Image Left (Desktop)

Variation Name: Image Position (Desktop) = Left.

Business Purpose: A mirrored layout, so the portrait is on the left and the card on the right. Used to alternate the direction of several Quote blocks on one page, or when the person in the image faces right.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-image-left.png`
> **Purpose:** Image Left variation
> **Description:** Desktop Quote block with the portrait on the left and the quote card on the right, overlapping the image's right edge.

Use Cases: The second of two Quote blocks on a page (zig-zag rhythm); portraits whose subject looks to the right.

Authoring Guidelines: In the Content tab, set Image Position (Desktop) to Left. An Image must be set; without an image the option has no visible effect.

Business Rules: The value `left` is applied only when it is one of the allowed values (`right`, `left`); anything else falls back to Right.

Rendering Rules: The block gets the class `image-left`. From 900px the body grid becomes 418 : 204 : 498; the image spans columns 1-2 and the card columns 2-3. The rules only apply with `quote-has-media`.

Responsive Behaviour: Desktop: image left, card right, overlap 204/1120. Tablet and mobile: no effect; the mobile position (Top or Bottom) applies.

SEO Impact: None beyond Variation 1; DOM order is unchanged (card before image).

Accessibility Compliance: Only the visual order changes; the reading order stays card first, then image, which keeps the meaning (WCAG 1.3.2 Meaningful Sequence).

Variation 3 — Image Bottom (Mobile and Tablet)

Variation Name: Image Position (Mobile) = Bottom.

Business Purpose: On small screens, show the quote before the portrait, for example when the quote should be read before the reader scrolls past a tall image.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-image-bottom-mobile.png`
> **Purpose:** Image Bottom (mobile) variation
> **Description:** Mobile Quote block (375px) with the quote card first and the portrait below it; the card overlaps the top edge of the image.

Use Cases: Quotes high on a page on mobile; long portraits where the quote should stay above the fold.

Authoring Guidelines: In the Content tab, set Image Position (Mobile) to Bottom. An Image must be set.

Business Rules: The value `bottom` is applied only when it is one of the allowed values (`top`, `bottom`); anything else falls back to Top.

Rendering Rules: The block gets the class `image-bottom`. Below 900px the image gets `order: 1` (after the card) and the card gets a negative bottom margin of 3rem with 1rem side insets, so it overlaps the image's top edge.

Responsive Behaviour: Desktop: no effect (the desktop grid places card and image). Tablet and mobile: card first, image below, overlapping by 3rem.

SEO Impact: None.

Accessibility Compliance: Visual order matches the DOM order (card, then image).

Variation 4 — Grey Background

Variation Name: BG Color = Grey.

Business Purpose: Sets the quote apart from neighbouring white sections with a light grey panel, as in the Figma grey variant.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-grey-background.png`
> **Purpose:** Grey background variation
> **Description:** Desktop Quote block on a light grey panel with generous padding; the card is grey too and overlaps the portrait.

Use Cases: Pages with several white blocks in a row; emphasising a quote between two text sections.

Authoring Guidelines: In the Content tab, set BG Color to Grey.

Business Rules: Only `grey` is allowed; Default (empty) means the default white background.

Rendering Rules: The block gets the class `bg-grey` (any previous `bg-grey` class is removed first). `--quote-bg` becomes `--color-bg-secondary` (grey-100); the block and the card use it. Padding: 2.5rem top/bottom and 1.5rem left/right below 900px; 7.5rem and 5rem from 900px. The Additional tab padding selects override this padding.

Responsive Behaviour: Desktop: large padding (7.5rem / 5rem). Tablet and mobile: 2.5rem / 1.5rem.

SEO Impact: None.

Accessibility Compliance: Black text on grey-100 and brand blue on grey-100 must keep 4.5 : 1 contrast [To be confirmed: contrast measurement on grey-100 — owner: design lead].

Variation 5 — Text Only (No Image)

Variation Name: Quote without an image.

Business Purpose: A pull quote when no suitable image is available, or in articles where an image would compete with other media.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-text-only.png`
> **Purpose:** Text-only variation
> **Description:** Desktop Quote block without an image: the card with quote marks centred in a 70rem column under the title.

Use Cases: Article and blog pull quotes; quotes from people without an approved portrait.

Authoring Guidelines: Leave Image empty. Image Alt Text and both Image Position selects then have no effect.

Business Rules: The class `quote-has-media` is only set when the Image field holds a picture.

Rendering Rules: No `.quote-media` element is created. The card has no negative margin. From 900px the card uses the 4.125rem / 2.875rem quote mark columns and 1.5rem / 2.5rem padding inside the 70rem centred body.

Responsive Behaviour: Desktop: the card fills the centred body up to 70rem. Tablet and mobile: the card fills the block width with 1.5rem padding.

SEO Impact: As Variation 1, without an image.

Accessibility Compliance: As Variation 1.

6.12.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A Quote block selected in the Universal Editor canvas, with the letters A to E marking the eyebrow, title, quote text, quote author and image.

Annotations:

- A = Eyebrow (text above the title)
- B = Title (heading at the Title Type level)
- C = Quote (rich text in the card, editable in place)
- D = Quote Author (rich text below the quote, editable in place)
- E = Image (portrait, selected from AEM Assets)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Universal Editor properties panel of the Quote block showing the Content tab with all fields and the Additional tab header.

The properties panel has two tabs:

- **Content:** Eyebrow (text), Title (text), Title Type (select H1-H6, default H2), Quote (rich text), Quote Author (rich text; "Name on the first line, then designation and organisation."), Image (asset reference), Image Alt Text (text; "Leave empty if the image is decorative."), Image Position (Desktop) (Right / Left), Image Position (Mobile) (Top / Bottom), Style (Default only) and BG Color (Default / Grey).
- **Additional:** Anchor ID (validated text), Hide block (toggle), Padding left, Padding right, Padding top, Padding bottom, Margin left, Margin right, Margin top and Margin bottom (selects of the spacing tokens).

6.12.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites, open the locale's site (for example `/content/tcs-corp/us/en`), select the page and choose Edit, which opens it in Universal Editor.

Expected Result: The page is shown in the Universal Editor canvas in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** AEM Sites console with the page selected and the Edit action highlighted.

Step 2 — Select the section

Action: In the canvas or the content tree, select the section that will hold the quote.

Expected Result: The section is highlighted and the Add (+) action is available.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-02-select-section.png`
> **Purpose:** Select the section
> **Description:** A section selected in the content tree with the Add button visible.

Step 3 — Add the Quote block

Action: Select Add (+) and choose Quote from the component list.

Expected Result: An empty Quote block is added, with Title Type = H2, Image Position (Desktop) = Right and Image Position (Mobile) = Top preset. The block shows no content yet.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-03-add-block.png`
> **Purpose:** Add the block
> **Description:** The component picker with Quote highlighted.

Step 4 — Fill the header

Action: In the Content tab, enter the Eyebrow (optional), the Title (optional) and choose the Title Type that fits the page's heading outline.

Expected Result: The eyebrow and title appear above the card.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-04-header.png`
> **Purpose:** Eyebrow and title
> **Description:** Properties panel with Eyebrow, Title and Title Type filled; the canvas shows them above the card.

Step 5 — Write the quote and the author

Action: Enter the statement in Quote, without typing quotation marks (the block adds them). In Quote Author, put the name on the first line and the designation and organisation on the next line(s).

Expected Result: The card shows the quote between large blue quote marks, with the author below it.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-05-quote-author.png`
> **Purpose:** Quote and author
> **Description:** Rich text editor with the quote; the canvas showing the quote card with the author lines.

Step 6 — Add the image

Action: In Image, select a portrait from AEM Assets. Enter Image Alt Text describing it (for example "Portrait of Jane Doe"), or leave it empty if the image is decorative.

Expected Result: The image appears and the card overlaps it.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-06-image.png`
> **Purpose:** Image and alt text
> **Description:** Asset picker with a portrait selected, and the Image Alt Text field filled.

Step 7 — Set the layout options

Action: Choose Image Position (Desktop) (Right or Left), Image Position (Mobile) (Top or Bottom) and BG Color (Default or Grey). Leave Style = Default.

Expected Result: The canvas updates; desktop and mobile positions can be checked by resizing the canvas (emulation) in Universal Editor.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-07-layout.png`
> **Purpose:** Layout options
> **Description:** The Image Position and BG Color selects open in the properties panel.

Step 8 — Fill the Additional tab (optional)

Action: Open the Additional tab. Enter an Anchor ID if other content links to the quote (for example `anchor-client-voice`). Adjust padding and margin only if the design asks for it. Use Hide block to prepare the block without showing it on the page.

Expected Result: The Anchor ID is accepted only if it matches the pattern; spacing changes are visible on the canvas; a hidden block appears faded with a dashed outline in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-08-additional.png`
> **Purpose:** Additional tab
> **Description:** Additional tab with Anchor ID, Hide block and spacing selects.

Step 9 — Preview

Action: Switch Universal Editor to Preview mode and check the block at desktop, tablet and mobile widths.

Expected Result: The block renders as on the live site; Hide block blocks are not shown.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-09-preview.png`
> **Purpose:** Preview
> **Description:** Universal Editor in Preview mode with the Quote block rendered.

Step 10 — Validate

Action: Run the Validation Process checklist below.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-10-validate.png`
> **Purpose:** Validation
> **Description:** The block on the preview URL with the browser's accessibility tree showing the blockquote and figcaption.

Step 11 — Publish

Action: Select Publish in Universal Editor and publish the page (to preview first if your process requires it, then live).

Expected Result: The page is published; the change is visible on `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/{path}` after the CDN refresh.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/quote-step-11-publish.png`
> **Purpose:** Publish
> **Description:** The Universal Editor Publish dialog with the page selected.

Preview Process

1. In Universal Editor, use Preview mode for a quick check of the rendering.
2. Publish to preview (where available) and open `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}`.
3. Check the block at 375px, 768px and 1440px wide (browser developer tools device mode).
4. Check that the desktop image position switches at 900px.

Publishing Process

1. Publish the page from Universal Editor (or from the Sites console with Quick Publish / Manage Publication).

2. The Edge Delivery live URL updates once the content is published; the Edge Delivery CDN serves the new version after its cache is purged by the publish.
3. Referenced images from AEM Assets are published with the page.

Validation Process

- ☐ The title uses the right heading level for the page outline (only one H1 per page).
- ☐ The quote text contains no typed quotation marks (the block adds them).
- ☐ The author's name is on the first line and the designation and organisation follow.
- ☐ The image has meaningful alt text, or none if decorative.
- ☐ The portrait's subject is not cut off by the 4 : 5 (mobile) or 622 : 869 (desktop) crop.
- ☐ The card overlaps the image as expected at desktop and mobile widths.
- ☐ The chosen Image Position (Desktop / Mobile) and BG Color are as designed.
- ☐ Anchor ID (if set) is unique on the page and links to it with `#anchor-...` work.
- ☐ Hide block is off when the quote should be visible.
- ☐ Quote approval and usage rights of the quote and portrait are documented [To be confirmed: approval process for client quotes — owner: content lead].

Rollback Process

1. In AEM Sites, open the page's version history (Timeline or Versions) and restore the version before the change, or undo the change in Universal Editor.
2. To remove the quote quickly without deleting it, switch on Hide block (Additional tab).
3. Publish the page again; the Edge Delivery live page updates after publishing.
4. If the whole page must be withdrawn, unpublish it from the Sites console.

6.12.6 Authorable Field Specification

Model `quote` — Content tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Eyebrow (`eyebrow`)	text	No	No	empty	Short label above the title	A category or context, a few words	None	Client voice
Title (`title`)	text	No	No	empty	Section heading above the quote	Short, descriptive heading	None	What our clients say
Title Type (`titleType`)	select	No	No	h2	Heading level of the title	Choose the level that fits the outline	One of h1-h6 (select)	h2
Quote (`quote`)	rich-text	No (Recommended)	No	empty	The statement shown in the card	Do not type quote marks; keep it concise	None	TCS helped us transform our data estate in record time.
Quote Author (`author`)	rich-text	No (Recommended)	No	empty	Attribution below the quote	Name on line 1, then designation and organisation	None	Jane Doe; Chief Digital Officer, Example Bank
Image (`image`)	reference	No	No	empty	Portrait overlapped by the card	Portrait from AEM Assets	Asset reference (single)	/content/dam/tcs-corp/people/jane-doe.jpg
Image Alt Text (`imageAlt`)	text	No	No	empty	Alternative text of the image	Describe the image; empty if decorative	None	Portrait of Jane Doe

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Image Position (Desktop) (`imagePosition`)	select	No	No	right	Side of the image from 900px	Right or Left	right or left; other values fall back to right	left
Image Position (Mobile) (`imagePositionMobile`)	select	No	No	top	Image above or below the card below 900px	Top or Bottom	top or bottom; other values fall back to top	bottom
Style (`classes`)	select	No	No	Default (empty)	Reserved for style variants	Leave Default	Only Default exists	Default
BG Color (`bgColor`)	select	No	No	Default (empty)	Background of block and card	Default (white) or Grey	Only grey is applied	grey

Model `quote` — Additional tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	text	No	No	empty	Id for in-page links	Start with anchor-, lower case	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-client-voice
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide the block on the page	Use to prepare or temporarily remove	Toggle	false
Padding left (`classes_paddingleft`)	select	No	No	Default (empty)	Left padding from the spacing tokens	Only when the design asks	One of 15 options (pl-0 … pl-3000)	pl-400
Padding right (`classes_paddingright`)	select	No	No	Default (empty)	Right padding	Only when the design asks	One of 15 options (pr-0 … pr-3000)	pr-400
Padding top (`classes_paddingtop`)	select	No	No	Default (empty)	Top padding	Only when the design asks	One of 15 options (pt-0 … pt-3000)	pt-1000
Padding bottom (`classes_paddingbottom`)	select	No	No	Default (empty)	Bottom padding	Only when the design asks	One of 15 options (pb-0 … pb-3000)	pb-1000
Margin left (`classes_marginleft`)	select	No	No	Default (empty)	Left margin	Only when the design asks	One of 15 options (ml-0 … ml-3000)	ml-0
Margin right (`classes_marginright`)	select	No	No	Default (empty)	Right margin	Only when the design asks	One of 15 options (mr-0 … mr-3000)	mr-0
Margin top (`classes_margintop`)	select	No	No	Default (empty)	Top margin	Only when the design asks	One of 15 options (mt-0 … mt-3000)	mt-1600
Margin bottom (`classes_marginbottom`)	select	No	No	Default (empty)	Bottom margin	Only when the design asks	One of 15 options (mb-0 … mb-3000)	mb-1600

The 15 options of every padding and margin select are: Default (empty), 0 (none), 0.25rem (space-100), 0.5rem (space-200), 0.75rem (space-300), 1rem (space-400), 1.25rem (space-500), 1.5rem (space-600), 2rem (space-800), 2.5rem (space-1000), 3rem (space-1200), 4rem (space-1600), 5rem (space-2000), 6rem (space-2400) and 7.5rem

(space-3000). The values are the class prefix (`pl`, `pr`, `pt`, `pb`, `ml`, `mr`, `mt`, `mb`) plus the token number, for example `pt-1000`.

Field details

Eyebrow (`eyebrow`)

- Purpose: A short label above the title that gives context (for example the topic or the type of voice).
- Allowed Values: Plain text.
- Character Limits: None enforced. Recommended: up to 40 characters.
- Business Rules: Rendered as `<p class="quote-eyebrow">` only when it has text.
- Dependency Rules: Independent of Title; an eyebrow can be shown without a title.
- Visibility Rules: Always visible in the dialog (no `condition`).
- Sample Content: “Client voice”.

Title (`title`)

- Purpose: The heading of the quote section.
- Allowed Values: Plain text.
- Character Limits: None enforced. Recommended: up to 60 characters (one or two lines at 3rem).
- Business Rules: Rendered only when it has text, as the heading level of Title Type.
- Dependency Rules: Title Type sets the level; without a title, Title Type has no effect.
- Visibility Rules: Always visible.
- Sample Content: “What our clients say”.

Title Type (`titleType`)

- Purpose: Sets the HTML heading level of the title.
- Allowed Values: H1, H2, H3, H4, H5, H6 (values h1-h6). Default h2.
- Character Limits: Not applicable.
- Business Rules: An invalid or missing value falls back to a heading level found in the title cell, else H2 (`buildHeading`). The visual style is the same for every level.
- Dependency Rules: Only used when Title has text.
- Visibility Rules: Always visible.
- Sample Content: h2.

Quote (`quote`)

- Purpose: The statement shown in the card.
- Allowed Values: Rich text (paragraphs, bold, italic, links).
- Character Limits: None enforced. Recommended: up to 300 characters (about 3 to 6 lines on desktop).
- Business Rules: Rendered as `<blockquote class="quote-text">` when it has text. Paragraphs inside are spaced 0.5rem apart. A link alone in its own paragraph is turned into a link button by the content buttons decorator.
- Dependency Rules: The card is created when Quote or Quote Author has text.
- Visibility Rules: Always visible.
- Sample Content: “TCS helped us transform our data estate in record time.”

Quote Author (`author`)

- Purpose: Attribution of the quote.
- Allowed Values: Rich text; name on the first line, then designation and organisation.
- Character Limits: None enforced. Recommended: name up to 40 characters, designation and organisation up to 80 characters.
- Business Rules: Rendered as `<figcaption class="quote-author">` when it has text; all lines use the same style (1.125rem / 1.5rem), paragraphs 0.5rem apart.

- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: “Jane Doe” / “Chief Digital Officer, Example Bank”.

Image (`image`)

- Purpose: The portrait the card overlaps.
- Allowed Values: One image asset from AEM Assets.
- Character Limits: Not applicable.
- Business Rules: When the cell holds a picture, the block gets `quote-has-media` and the overlap layout. Cropped to 4 : 5 below 900px and 622 : 869 from 900px.
- Dependency Rules: Image Alt Text, Image Position (Desktop) and Image Position (Mobile) only have an effect with an image.
- Visibility Rules: Always visible.
- Sample Content: `/content/dam/tcs-corp/people/jane-doe.jpg`.

Image Alt Text (`imageAlt`)

- Purpose: Alternative text of the image for screen readers and search engines.
- Allowed Values: Plain text; empty for a decorative image.
- Character Limits: None enforced. Recommended: up to 125 characters.
- Business Rules: When set, it replaces the image’s alt; when empty and the image has no alt, `alt=""` is set (decorative).
- Dependency Rules: Requires Image.
- Visibility Rules: Always visible.
- Sample Content: “Portrait of Jane Doe”.

Image Position (Desktop) (`imagePosition`)

- Purpose: The side of the image from 900px.
- Allowed Values: Right (right, default), Left (left).
- Character Limits: Not applicable.
- Business Rules: Left adds `image-left`; any other value is treated as Right.
- Dependency Rules: Requires Image.
- Visibility Rules: Always visible.
- Sample Content: left.

Image Position (Mobile) (`imagePositionMobile`)

- Purpose: The image above or below the card below 900px.
- Allowed Values: Top (top, default), Bottom (bottom).
- Character Limits: Not applicable.
- Business Rules: Bottom adds `image-bottom`; any other value is treated as Top.
- Dependency Rules: Requires Image.
- Visibility Rules: Always visible.
- Sample Content: bottom.

Style (`classes`)

- Purpose: Reserved for future style variants of the block.
- Allowed Values: Default (empty) only.
- Character Limits: Not applicable.
- Business Rules: Adds no class today.
- Dependency Rules: None.
- Visibility Rules: Always visible.

- Sample Content: Default.

BG Color (`bgColor`)

- Purpose: The background colour of the block and the card.
- Allowed Values: Default (empty, white), Grey (grey).
- Character Limits: Not applicable.
- Business Rules: Grey adds `bg-grey` (grey-100 background and the block's own padding).
- Dependency Rules: The Additional tab padding overrides the grey padding.
- Visibility Rules: Always visible.
- Sample Content: grey.

Anchor ID (`classes_anchor`)

- Purpose: Gives the block an id for links like `#anchor-client-voice`.
- Allowed Values: `anchor-` followed by lower-case letters, digits and hyphens.
- Character Limits: None enforced. Recommended: up to 40 characters.
- Business Rules: Enforced by the model regExp `^anchor-[a-z0-9][a-z0-9-]*$` (error "Start with anchor-, then lower-case letters, digits and hyphens."). The block options decorator sets it as the block id; a duplicate on the page gets `-2`, `-3`, … appended.
- Dependency Rules: The id is also used as the block's data layer component id.
- Visibility Rules: Always visible.
- Sample Content: `anchor-client-voice`.

Hide block (`classes_hideblock`)

- Purpose: Hides the whole block on the page without deleting it.
- Allowed Values: On / Off (default Off).
- Character Limits: Not applicable.
- Business Rules: Adds the class `hideblock`; the block and its wrapper are not displayed outside edit mode; in Universal Editor edit mode the block is shown at 45% opacity with a dashed outline.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: Off.

Padding left (`classes_paddingleft`)

- Purpose: Left inner spacing of the block.
- Allowed Values: The 15 options listed above (class `pl-{token}`).
- Character Limits: Not applicable.
- Business Rules: Styled in `styles/styles.css` with a selector that outranks the block's own padding (including the Grey padding).
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 1rem (space-400).

Padding right (`classes_paddingright`)

- Purpose: Right inner spacing.
- Allowed Values: The 15 options (class `pr-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Padding left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 1rem (space-400).

Padding top (`classes_paddingtop`)

- Purpose: Top inner spacing.
- Allowed Values: The 15 options (class `pt-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Padding left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 2.5rem (space-1000).

Padding bottom (`classes_paddingbottom`)

- Purpose: Bottom inner spacing.
- Allowed Values: The 15 options (class `pb-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Padding left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 2.5rem (space-1000).

Margin left (`classes_marginleft`)

- Purpose: Left outer spacing of the block.
- Allowed Values: The 15 options (class `ml-{token}`).
- Character Limits: Not applicable.
- Business Rules: Styled in `styles/styles.css`.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 0 (none).

Margin right (`classes_marginright`)

- Purpose: Right outer spacing.
- Allowed Values: The 15 options (class `mr-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Margin left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 0 (none).

Margin top (`classes_margintop`)

- Purpose: Space above the block.
- Allowed Values: The 15 options (class `mt-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Margin left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 4rem (space-1600).

Margin bottom (`classes_marginbottom`)

- Purpose: Space below the block.
- Allowed Values: The 15 options (class `mb-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Margin left.

- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 4rem (space-1600).

6.12.7 Field Annotation Mapping

Marker	Field	Purpose
A	Eyebrow (`eyebrow`)	Context label above the title
B	Title (`title`) with Title Type (`titleType`)	Section heading
C	Quote (`quote`)	The statement in the card
D	Quote Author (`author`)	Name, designation and organisation
E	Image (`image`) with Image Alt Text (`imageAlt`)	Portrait overlapped by the card

6.12.8 Component Specification

Attribute	Value
Component Name	Quote (`quote`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`decorators/content-decorator.js`; `decorators/heading-decorator.js`; `decorators/image-decorator.js`; `utils/properties.js`; global block options, buttons and data layer decorators
Component Type	Block (key-value block, `core/franklin/components/block/v1/block`, no child items)
Supported Variants	Image Position (Desktop) Right / Left; Image Position (Mobile) Top / Bottom; BG Color Default / Grey; with or without image; Style: Default only
Supported Templates	Any page (allowed in every section); not added by any template
Supported Metadata	None read by the block
Supported Tags	None
Supported Personalization	None in the block; the data layer entry could be used by Adobe Target once Adobe Tags is integrated (not implemented)
Caching Considerations	Content is part of the page HTML cached by the Edge Delivery CDN and purged on publish; no sheet or query index is fetched
Performance Considerations	Decorated synchronously when its section loads; the image is the authored picture with Edge Delivery optimized sources; if the block is in the first section, its first image becomes the LCP candidate and is loaded eagerly (`waitForFirstImage`)
Accessibility Compliance	Semantic figure, blockquote and figcaption; heading level selectable; quote marks hidden from screen readers; alt text field; reading order card before image
SEO Compliance	Real heading for the title; quote text in the HTML (rendered client side from authored content); image alt text

6.12.9 Input Model

- **Content Inputs:** Eyebrow, Title, Title Type, Quote, Quote Author, Image, Image Alt Text, Image Position (Desktop), Image Position (Mobile), BG Color (property rows).
- **Metadata Inputs:** None.

- **Tags Inputs:** None.
- **Page Context Inputs:** None (the block does not read the page path, locale or Page Type).
- **Query Inputs:** None.
- **Manual Inputs:** All content is authored manually in the block; the Style and Additional tab fields arrive as block classes.

The block receives one row per property: the first cell holds the property name, the second its value. The code reads the names case-insensitively (`readProperties`).

```
<div class="quote image-left anchor-client-voice">
  <div><div>eyebrow</div><div><p>Client voice</p></div></div>
  <div><div>title</div><div><p>What our clients say</p></div></div>
  <div><div>titleType</div><div><p>h2</p></div></div>
  <div><div>quote</div><div><p>TCS helped us transform our data estate.</p></div></div>
  <div><div>author</div><div><p>Jane Doe</p><p>CDO, Example Bank</p></div></div>
  <div><div>image</div><div><picture><img src="./media_1a2b.jpeg" alt=""></picture></
div></div>
  <div><div>imageAlt</div><div><p>Portrait of Jane Doe</p></div></div>
  <div><div>imagePosition</div><div><p>left</p></div></div>
  <div><div>imagePositionMobile</div><div><p>top</p></div></div>
  <div><div>bgColor</div><div><p>grey</p></div></div>
</div>
```

6.12.10 Output Model

Rendered Output: The block's rows are replaced by up to two parts:

- `div.quote-header` with `p.quote-eyebrow` and the title heading `.quote-title` (only if one of them has text).
- `div.quote-body` with `figure.quote-card` (containing `div.quote-card-text` with `blockquote.quote-text` and `figcaption.quote-author`) and `div.quote-media` (the picture).

Block classes set by the code: `bg-grey`, `quote-has-media`, `image-left`, `image-bottom`. The block options decorator sets the `id` from the Anchor ID.

Generated HTML:

```
<div class="quote image-left bg-grey quote-has-media block" id="anchor-client-voice"
  data-block-name="quote" data-block-status="loaded">
  <div class="quote-header">
    <p class="quote-eyebrow">Client voice</p>
    <h2 class="quote-title">What our clients say</h2>
  </div>
  <div class="quote-body">
    <figure class="quote-card">
      <div class="quote-card-text">
        <blockquote class="quote-text"><p>TCS helped us transform our data estate.</p>
        </blockquote>
        <figcaption class="quote-author"><p>Jane Doe</p><p>CDO, Example Bank</p></
figcaption>
      </div>
    </figure>
    <div class="quote-media">
      <picture><img src="..." alt="Portrait of Jane Doe"></picture>
    </div>
  </div>
</div>
```

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking: After the sections load, the data layer decorator pushes the block's component state; its id is the block id (Anchor ID) or `quote-{n}` in page order, and its title is the first heading in the block. Clicks on links or buttons inside the quote or author text push `cmp:click`.

```
{
  "component": {
    "anchor-client-voice": {
      "@type": "tcs-corp/components/quote",
      "parentId": "page-1k3x9q",
      "dc:title": "What our clients say"
    }
  }
}
```

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.anchor-client-voice",
    "interaction": "link",
    "dc:title": "Read the case study",
    "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/case-studies/example",
    "xdm:linkType": "other",
    "breakpoint": "desktop"
  }
}
```

Personalization Output: None.

Search Index Output: None. The query index (`helix-query.yaml`) reads page metadata only, not block content. The quote text is part of the page HTML that search engines crawl.

6.12.11 Business Rules

1. The block is a key-value block: each authored row with exactly two cells is a property; rows with another number of cells are ignored.
2. The eyebrow is rendered only when it has text (after trimming).
3. The title is rendered only when it has text; its level is Title Type (h1-h6), else a heading level found in the title cell, else H2.
4. The header is left out when it has neither eyebrow nor title.
5. The quote is rendered as a blockquote only when it has text; the author as a figcaption only when it has text.
6. The card (`figure.quote-card`) is added only when it holds a quote or an author.
7. The image part is added only when the Image cell contains a picture; the block then gets `quote-has-media`.
8. Image Alt Text, when set, replaces the image's alt; otherwise an image without alt gets `alt=""` (decorative).
9. Image Position (Desktop) accepts `right` or `left`; a missing or unknown value means Right.
10. Image Position (Mobile) accepts `top` or `bottom`; a missing or unknown value means Top.
11. BG Color accepts only `grey`; any `bg-grey` class is removed before the current value is applied, so re-decoration in the editor never leaves a stale colour.
12. Select values are also read from the block classes as a fallback (`optionOf`), so a class `grey`, `left` or `bottom` on the block has the same effect.
13. The body is left out when it has neither card nor image; an empty block renders nothing.
14. The quote marks are always added by CSS; authors must not type them.
15. The desktop layout (overlap grid) starts at 900px; below 900px the stacked layout applies.

16. The Additional tab padding overrides the Grey padding (higher-specificity selectors).
17. Hide block hides the block on the published page and in preview; in edit mode it is shown faded.
18. Editor-only behaviour: editor instrumentation (`data-aue-*`, `data-richtext-*`) is moved to the rendered elements so in-place editing works; each change re-decorates the block.
19. Error handling: the block code has no try/catch of its own; an error is caught and logged by the block loader (`failed to load module for quote`), and the authored rows stay in place.

6.12.12 Author Best Practices

Recommended Content Length: Eyebrow up to 40 characters; Title up to 60 characters; Quote 100 to 300 characters; author name up to 40 characters and designation plus organisation up to 80. (All Recommended; nothing is enforced.)

Image Dimensions: Portrait images. The CSS crops them to 4 : 5 below 900px and to 622 : 869 (about 5 : 7) from 900px, with the image up to about 622px wide on desktop. Recommended: at least 1244 x 1738 px, with the face in the upper-middle area so that both crops keep it.

CTA Writing Guidelines: The block has no CTA field. If a link is needed (for example to the full case study), put it in its own paragraph at the end of the quote or author text, with a descriptive label ("Read the Example Bank case study"); it becomes a link button.

Accessibility Guidelines: Pick the heading level that fits the page outline; write alt text that identifies the person ("Portrait of Jane Doe"), or leave it empty if the name is in the caption and the image adds nothing; do not type quote marks; do not use colour or bold alone to convey meaning.

SEO Guidelines: Use a descriptive title; keep the exact wording approved by the person quoted; include the full name and organisation in the author text.

Metadata Guidelines: The block reads no metadata. Page-level metadata (title, description) is set in the page properties or the metadata sheet.

Taxonomy Guidelines: The block has no tags; tag the page itself if it should appear in List or Filter List results.

Localization Guidelines: Translate the eyebrow, title, quote and designation in each locale site; keep the person's name as approved; check that longer translations (for example German) still fit the card on mobile; translated quotes must be approved by the quoted person [To be confirmed: translation approval for quotes — owner: localisation lead].

6.12.13 Troubleshooting Guide

Issue 1 — The image does not overlap the card

Symptoms: The quote card and the image are shown one after the other without overlap.

Root Cause: The Image field is empty or holds no picture (for example a broken asset reference), so `quote-has-media` is not set.

Diagnosis Steps: Inspect the block in the browser; check whether it has the class `quote-has-media` and a `.quote-media` element.

Resolution Steps: Select the image again in the Image field from AEM Assets and publish the asset and the page.

Validation Steps: Reload the preview page; the block has `quote-has-media` and the card overlaps the image.

Preventive Measures: Publish referenced assets with the page; avoid deleting or moving assets that are in use.

Issue 2 — Image Position has no effect

Symptoms: Choosing Left (desktop) or Bottom (mobile) changes nothing.

Root Cause: The setting is checked at the wrong width (Left only applies from 900px, Bottom only below 900px), or there is no image.

Diagnosis Steps: Check the block classes (`image-left`, `image-bottom`) and the viewport width.

Resolution Steps: Resize the browser past 900px (Left) or below 900px (Bottom); make sure an image is set.

Validation Steps: The image moves to the chosen side at the right widths.

Preventive Measures: Validate both breakpoints in preview before publishing.

Issue 3 — Double quotation marks

Symptoms: The quote shows two sets of quote marks.

Root Cause: The author typed quotation marks in the Quote field; the block adds its own.

Diagnosis Steps: Look at the Quote rich text in the properties panel.

Resolution Steps: Remove the typed quotation marks at the start and end.

Validation Steps: Only the large blue marks remain.

Preventive Measures: Add the rule to the author training and the validation checklist.

Issue 4 — The face is cut off

Symptoms: The person's head is cropped on mobile or desktop.

Root Cause: The image is cropped to 4 : 5 (mobile) and 622 : 869 (desktop) with `object-fit: cover`, centred; a landscape or tightly framed image loses its edges.

Diagnosis Steps: Compare the source image's proportions with the two crops.

Resolution Steps: Use a portrait image with the face in the upper-middle area, or create a rendition with more space around the subject.

Validation Steps: Check at 375px, 768px and 1440px.

Preventive Measures: Brief photographers and DAM editors on the crop ratios.

Issue 5 — The anchor link does not jump to the quote

Symptoms: A link to `#anchor-...` does not scroll to the block.

Root Cause: The Anchor ID was rejected by validation, differs from the link, or another element already uses it (the block then gets `-2`).

Diagnosis Steps: Inspect the block's `id` attribute and compare it with the link's hash.

Resolution Steps: Use a unique Anchor ID and update the link.

Validation Steps: Open the page with the hash; it scrolls to the quote.

Preventive Measures: Keep a list of anchors per page; use descriptive, unique ids.

Issue 6 — The block is missing on the published page

Symptoms: The quote is visible in the editor (faded) but not on the page.

Root Cause: Hide block is on, or the page was not published after the change.

Diagnosis Steps: Check the Additional tab (Hide block) and the page's publication status.

Resolution Steps: Turn off Hide block and publish the page.

Validation Steps: The block appears on the live URL.

Preventive Measures: Use Hide block only on purpose and note it in the page's change log.

Issue 7 — The title has the wrong heading level

Symptoms: Accessibility checkers report skipped heading levels or several H1s.

Root Cause: Title Type is set to a level that does not fit the page outline.

Diagnosis Steps: Inspect the heading tag of `.quote-title`; review the page outline.

Resolution Steps: Change Title Type (for example to H2 or H3).

Validation Steps: Re-run the accessibility check.

Preventive Measures: Keep H1 for the page title (Page Header) only.

6.12.14 Support Runbook

Monitoring Requirements: No block-specific monitoring exists. Monitor the pages that use the block with Real User Monitoring (Edge Delivery operational telemetry) for LCP and layout shifts, and check browser console errors on key pages after code deployments.

Logs:

- Browser console: the block emits no messages of its own. A decoration error is logged by the block loader as `failed to load module for quote` (with the error); a CSS or network problem as `failed to load block quote`.
- Admin API: publish and preview status of the page (`https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`) shows whether the latest content is published.
- Author logs: AEM as a Cloud Service author logs and the page's Timeline (publication events).

Common Errors:

Error	Likely Cause	Action
No overlap, image missing	Image empty or asset unpublished	Re-select and publish the asset
Image Position ignored	Checked at the wrong width or no image	Check at the right breakpoint
Double quote marks	Marks typed in Quote	Remove them
Anchor ID rejected	Pattern not matched	Use `anchor-` plus lower case, digits, hyphens
Block not visible	Hide block on, or page not published	Turn off, publish
`failed to load module for quote`	Code error or unexpected markup	Escalate to L3 with the page URL

Escalation Matrix:

Level	Team	Checks
L1	Author support	Field values, Hide block, image and alt text, publication status, checklist
L2	AMS / site admin	Page and asset replication, Admin API status, CDN cache, console errors, recent deployments
L3	Development team	Block code, CSS, models, decorators; reproduce locally with `aem up`; fix and deploy
L4	Adobe Support	Platform issues (Universal Editor, Edge Delivery publishing, CDN) with the program and environment ids

Recovery Procedures: Restore the previous page version and republish; hide the block with Hide block while a fix is prepared; for code regressions, revert the commit on `main` (Code Sync deploys it).

Health Checks: Open a page with the block on the live host; check that the classes `quote-has-media` (if an image is set) and the expected position classes are present, that no console errors appear, and that the layout switches at 900px.

Operational Procedures: Content changes follow the standard publish workflow; code changes go through a feature branch, preview URL, pull request and merge to `main`. After `_quote.json` changes, run `npm run build:json` and `npm run lint`.

Known Limitations:

- Style has only the Default option; there are no other style variants.
- Only one quote per block; no carousel or multiple quotes.
- No CTA field; links can only be added in the rich text.
- The figcaption is inside a `div` within the figure rather than a direct child of it, which HTML validators report and which may weaken the figure/caption association for assistive technology.
- Image crops are fixed (4 : 5 and 622 : 869); there is no focal point setting.
- The layout switches at 900px only; tablet uses the mobile layout.

6.13 Testimonial Block

The Testimonial block (`blocks/testimonial`) shows a section title, then a centred testimonial: an eyebrow, the quote (Description) in large brand-blue type and the author (name, round photo, designation and location), followed by an optional row of stats (a title above a large value) under a divider. It implements the TCS.com design system Figma "Testimonial Block" (nodes 3291-268576 and 404-29822).

6.13.1 Component Purpose

Business Objective: Build trust by presenting a client's or partner's own words together with measurable results (stats), so a visitor sees both the endorsement and the business outcome.

Business Use Cases:

- A client testimonial with the outcomes of an engagement ("40% faster time to market").
- A case study summary: the client's statement and three or four key results.
- A results band without a quote (stats only), with the description and author hidden.
- A short endorsement with the author's name and role only (no photo).

When To Use:

- A single testimonial should be the focus of a section, centred and in large type.
- Quantified results support the testimonial.
- An author photo is small (a 3rem round avatar) rather than a large portrait.

When Not To Use:

- For a large portrait overlapped by the quote: use the Quote block (6.12).
- For several testimonials in a row or a slider: use Cards / List, or a Gen Carousel section with one Testimonial block per slide.
- For stats with images or CTAs per item: use Cards (Stat card style) or List.
- For a call to action: the block has no CTA field; use Teaser or Hero.

Dependencies:

Type	Dependency	Use
Block code	`blocks/testimonial/testimonial.js`, `blocks/testimonial/testimonial.css`	Decoration and styles

Type	Dependency	Use
Model	`blocks/testimonial/_testimonial.json` (aggregated in `component-definition.json`, `component-models.json`)	Universal Editor dialog
Decorator	`decorators/stats-decorator.js` `buildStats` (with `utils/multifield.js` `readMultifieldItems`)	Builds the stats description list from the Stats multifield
Decorator	`decorators/a11y-decorator.js` `hideVisually`	Hide Title keeps the title for screen readers
Decorator	`decorators/content-decorator.js` `fromCell`	Moves eyebrow, description, author and designation into new elements, keeping editor instrumentation
Decorator	`decorators/heading-decorator.js` `buildHeading`	Title at the Title Type level
Decorator	`decorators/image-decorator.js` `pictureOf`	Author image with its alt text
Util	`utils/properties.js` `readProperties`, `isOn`, `optionOf`, `textOf`	Property rows, toggles and selects
Global	`decorators/block-options-decorator.js`, `styles/styles.css` (`.visually-hidden`, block options)	Additional tab; visually hidden title
Global	`decorators/data-layer-decorator.js`, `utils/analytics.js`	Data layer component entry and click events
Global	`decorators/content-buttons-decorator.js`	A standalone link in rich text becomes a link button
Data sources	None	No sheet, query index or taxonomy is read
Placeholders	None	No fixed UI text

Related Components: Quote (6.12); Teaser (also has a Stats multifield built by the same stats decorator); Cards / List (Stat card style for stat grids); Page Header (author data from the Authors sheet, which the Testimonial does not use).

Applicable Templates: Any page; the block is allowed in every section (section filter). Not added by any template.

Supported Content Types: Client stories, case studies, industry and service pages, partner pages, results summaries.

6.13.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A Testimonial block at 1440px: left-aligned section title, centred eyebrow, large blue quote, author with round photo and designation, and four stats under a divider.

Desktop (900px and wider): The block is a vertical stack with 5rem between its parts (title, testimonial, stats). The section title is left aligned, 3rem, semibold, in the default text colour. The testimonial is centred (maximum width 59.375rem) with 0.75rem gaps: the eyebrow (1rem, regular, line height 1.625), the description (2.5rem, medium, brand blue, line height 1.4, maximum width 56.25rem; paragraphs 1rem apart) and the author 1.5rem lower (the 0.75rem gap plus the caption's 0.75rem top margin). The author is a centred row: a 3rem round photo (cropped

to a circle) and the text (name and designation, both 1rem medium, 0.25rem apart). The stats are a centred grid (maximum width 70rem) with a 1px grey divider on top and 2.5rem padding above the items. Each column is at least 10rem wide (`repeat(auto-fit, minmax(10rem, 1fr))`), so up to five stats fit in one row at 70rem; more wrap to a new row (gaps 2rem vertical, 2.5rem horizontal). Each stat shows its title (1rem, medium, grey) above its value (3rem, semibold, brand blue).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The Testimonial block at 768px: smaller type, the author row centred, three stats per row.

Tablet (600px to 899px): The block uses the base (mobile) sizes; the only breakpoint of the block is 900px. Gaps between parts are 2.5rem. Type: title 2rem, description 1.75rem, eyebrow and author 0.875rem, stat values 2rem. The stats grid fits as many 10rem columns as the width allows (typically three at 768px).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Testimonial block at 375px: title, centred quote and author, stats stacked one per row.

Mobile (below 600px): As tablet, with the section side padding of 1.5rem. The author's photo and text wrap onto separate lines when there is not enough room (the author row wraps, centred). The stats are usually one per row (two columns need about 22.5rem). With BG Color = Grey, the block has a light grey background and 2.5rem / 1.5rem padding (7.5rem / 5rem from 900px).

6.13.3 Component Variations

The Style select (`classes`) has only the Default option. The variations are BG Color, the author presentation, the stats and the Show / Hide toggles; they combine freely.

Variation 1 — Default Testimonial

Variation Name: Default — title, eyebrow, description and author on a white background.

Business Purpose: The standard Figma testimonial: one centred statement with attribution.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-default.png`
> **Purpose:** Default variation
> **Description:** Title, centred eyebrow, large blue quote and the author name with designation; white background, no stats.

Use Cases: Client testimonials on service, industry and case study pages.

Authoring Guidelines: Fill Eyebrow, Title, Description (Content tab), Author and Designation & Location (Author tab). Leave BG Color = Default and all Show / Hide toggles at their defaults (Show Description on, the others off).

Business Rules: Show Description is on by default (model value and block template); the description is shown only when it is on and has text.

Rendering Rules: Output order: title heading, `figure.testimonial-quote` (eyebrow, `blockquote.testimonial-description`, `figcaption` with the author), then the stats if any.

Responsive Behaviour: Desktop: 5rem gaps, large type. Tablet and mobile: 2.5rem gaps, smaller type, author row wraps if needed.

SEO Impact: The title is a real heading (H2 by default); the quote is a `<blockquote>` with a `<figcaption>` attribution, readable by search engines.

Accessibility Compliance: Semantic `figure` / `blockquote` / `figcaption`; selectable heading level; centred text is limited to 59.375rem for readability. Brand blue on white must keep 4.5 : 1 contrast for the 1.75rem+ description (large text needs 3 : 1) [To be confirmed: contrast measurement — owner: design lead].

Variation 2 — Grey Background

Variation Name: BG Color = Grey.

Business Purpose: Sets the testimonial apart from surrounding white content.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-grey-background.png`
> **Purpose:** Grey background variation
> **Description:** The full testimonial on a light grey panel with large padding on desktop.

Use Cases: Between white sections; to frame the testimonial and stats as one panel.

Authoring Guidelines: Style tab: BG Color = Grey.

Business Rules: Only `grey` is allowed; the class is reset on each decoration.

Rendering Rules: Class `bg-grey`: background `--color-bg-secondary` (grey-100), padding 2.5rem / 1.5rem below 900px and 7.5rem / 5rem from 900px. The Additional tab padding overrides it.

Responsive Behaviour: Padding changes at 900px as above.

SEO Impact: None.

Accessibility Compliance: Text colours must keep contrast on grey-100 [To be confirmed: contrast measurement on grey-100 — owner: design lead].

Variation 3 — Author with Photo

Variation Name: Author image set.

Business Purpose: Puts a face to the testimonial, as in Figma 404-29822.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-author-photo.png`
> **Purpose:** Author with photo
> **Description:** The author row with a 3rem round photo to the left of the name and designation.

Use Cases: Named client executives with an approved headshot.

Authoring Guidelines: Author tab: select Author Image from AEM Assets. Leave Author Image Alt Text empty when the author's name is shown next to the photo (the dialog's guidance), otherwise describe the person.

Business Rules: The image part is created only when the Author Image cell holds a picture; the alt text is applied by `pictureOf` (empty alt = decorative).

Rendering Rules: `div.testimonial-author-image` before `div.testimonial-author-text` inside `div.testimonial-author`; the image is 3rem x 3rem, `border-radius: full`, `object-fit: cover`.

Responsive Behaviour: Same size at every breakpoint; on narrow screens the text wraps below the photo.

SEO Impact: Minimal (decorative avatar).

Accessibility Compliance: A decorative photo has `alt=""` so the name is not announced twice.

Variation 4 — Author Text Only (No Photo)

Variation Name: Author without image.

Business Purpose: An attribution when no approved photo exists.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-author-text-only.png`
> **Purpose:** Author text only
> **Description:** The author name and the designation & location centred below the quote, without a photo.

Use Cases: Anonymous or role-only attributions ("CIO, a leading European bank").

Authoring Guidelines: Author tab: fill Author and/or Designation & Location, leave Author Image empty.

Business Rules: The author part is built from whichever of image, name and designation has content; with none, no figcaption is added.

Rendering Rules: `p.testimonial-author-name` (from Author) and `div.testimonial-author-details` (from Designation & Location, rich text) in `div.testimonial-author-text`.

Responsive Behaviour: Centred at every breakpoint.

SEO Impact: The attribution text is in the HTML.

Accessibility Compliance: As Variation 1.

Variation 5 — With Stats

Variation Name: Stats added (Stats tab).

Business Purpose: Shows measurable results of the engagement below the testimonial.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-with-stats.png`
> **Purpose:** Stats variation
> **Description:** Below the author, a divider and four stats, each with a grey title above a large blue value.

Use Cases: Case study results; programme outcomes; benefit summaries.

Authoring Guidelines: Stats tab: add one item per stat; enter Stats Title (what is measured) and Stats Value (a single line, for example "40%").

Business Rules: Each item's last element is the value and the elements before it are the title. A Stats Value of several paragraphs therefore moves all but the last into the title; an item with only one filled field shows that text as the value without a title. Items without content are skipped; with no items, no stats are shown.

Rendering Rules: `dl.testimonial-stats` with one `div.testimonial-stat` per item, containing `dt.testimonial-stat-label` (title) and `dd.testimonial-stat-value` (value), shown title first.

Responsive Behaviour: Auto-fit grid of columns at least 10rem wide: up to five per row on desktop, about three on tablet, one or two on mobile.

SEO Impact: The stats are text in the HTML (not images), so they can be indexed.

Accessibility Compliance: A description list associates each title (term) with its value (definition) for screen readers.

Variation 6 — Hide Eyebrow

Variation Name: Hide Eyebrow on.

Business Purpose: Removes the eyebrow without deleting its text (for example for a variant of the page).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-hide-eyebrow.png`
> **Purpose:** Hide Eyebrow variation
> **Description:** The testimonial without the eyebrow above the quote.

Use Cases: When the title already gives the context.

Authoring Guidelines: Show / Hide tab: Hide Eyebrow on.

Business Rules: The eyebrow is shown only when Hide Eyebrow is off and the Eyebrow has text.

Rendering Rules: No `p.testimonial-eyebrow` is created (it is not in the HTML).

Responsive Behaviour: No difference between breakpoints.

SEO Impact: The eyebrow text is not in the page.

Accessibility Compliance: Not applicable (content removed).

Variation 7 — Hide Title (Visually Hidden)

Variation Name: Hide Title on.

Business Purpose: Matches designs without a visible section title while keeping a heading for screen-reader navigation and the page outline.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-hide-title.png`
> **Purpose:** Hide Title variation
> **Description:** The testimonial starting with the centred eyebrow; no visible title above it.

Use Cases: Testimonials directly below a section heading in default content.

Authoring Guidelines: Fill Title anyway, then Show / Hide tab: Hide Title on.

Business Rules: The title stays in the HTML with the class `visually-hidden`; without Title text there is nothing to hide.

Rendering Rules: The heading is clipped to 1px (`.visually-hidden` in `styles/styles.css`).

Responsive Behaviour: Hidden at every breakpoint.

SEO Impact: The heading remains in the HTML and the outline.

Accessibility Compliance: Screen readers still announce the heading (WCAG 1.3.1, 2.4.6); it remains the block's `dc:title` in the data layer.

Variation 8 — Description Hidden (Show Description off)

Variation Name: Show Description off.

Business Purpose: Uses the block for the author and stats only, for example a results band.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-description-hidden.png`
> **Purpose:** Show Description off
> **Description:** The title, eyebrow, author and stats without the large quote.

Use Cases: Stats bands; reusing a block while the quote awaits approval.

Authoring Guidelines: Show / Hide tab: Show Description off.

Business Rules: The description is shown only when Show Description is explicitly on (`true`). A block whose Show Description row is missing (for example authored before the field existed) shows no description.

Rendering Rules: No `blockquote.testimonial-description` is created.

Responsive Behaviour: No difference.

SEO Impact: The quote text is not in the page.

Accessibility Compliance: Not applicable (content removed).

Variation 9 — Hide Author Details

Variation Name: Hide Author Details on.

Business Purpose: Hides the attribution (name, photo, designation and location) without deleting it, for example while approval is pending or for stats-only bands.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-hide-author.png`
> **Purpose:** Hide Author Details variation
> **Description:** The eyebrow and quote without the author row; stats below.

Use Cases: Anonymous testimonials; stats-only bands (together with Show Description off).

Authoring Guidelines: Show / Hide tab: Hide Author Details on.

Business Rules: No author part is built, whatever the Author tab contains.

Rendering Rules: No `figcaption`. If eyebrow and description are also absent, the whole figure is left out.

Responsive Behaviour: No difference.

SEO Impact: The attribution is not in the page.

Accessibility Compliance: A quote without attribution should say who is speaking in the text if it matters to the meaning.

6.13.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A Testimonial block selected in the canvas with letters A to H on the title, eyebrow, description, author image, author name, designation, stats titles and stats values.

Annotations:

- A = Title (heading, left aligned)
- B = Eyebrow (centred label)
- C = Description (the testimonial quote, rich text)
- D = Author Image (round photo)
- E = Author (name)
- F = Designation & Location (rich text)
- G = Stats Title (one per stat)
- H = Stats Value (one per stat)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Testimonial properties panel showing the six tabs (Content, Author, Style, Show / Hide, Stats, Additional) with the Stats tab open and two items.

The properties panel has six tabs:

- **Content:** Eyebrow (text), Title (text), Title Type (H1-H6, default H2), Description (rich text, "The testimonial quote.").
- **Author:** Author (text, for example "Ranil Boteju, Chief Data & Analytics Officer, Lloyds Banking Group"), Author Image (asset reference), Author Image Alt Text (text, "Leave empty if the name is next to the image."), Designation & Location (rich text, "e.g. name, designation, organisation.").
- **Style:** Style (Default only), BG Color (Default / Grey).
- **Show / Hide:** Hide Eyebrow (off), Hide Title (off; "The title stays available to screen readers."), Show Description (on), Hide Author Details (off; "Hides the author, author image and designation & location").
- **Stats:** Stats (multifield; each item has Stats Title and Stats Value, "A single line, e.g. 40%.").
- **Additional:** Anchor ID, Hide block, Padding left / right / top / bottom, Margin left / right / top / bottom.

6.13.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites, open the locale site, select the page and choose Edit.

Expected Result: The page opens in the Universal Editor canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** AEM Sites console with the page selected and Edit highlighted.

Step 2 — Add the Testimonial block

Action: Select the target section, select Add (+) and choose Testimonial.

Expected Result: An empty Testimonial block is added with Title Type = H2 and Show Description on.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** The component picker with Testimonial highlighted.

Step 3 — Fill the Content tab

Action: Enter Eyebrow, Title, choose Title Type and write the Description (the testimonial quote).

Expected Result: The title appears on the left; eyebrow and quote are centred below it.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-03-content.png`
> **Purpose:** Content tab
> **Description:** Content tab filled; canvas showing title, eyebrow and quote.

Step 4 — Fill the Author tab

Action: Enter Author (the name), select Author Image, set Author Image Alt Text (or leave it empty when the name is next to the photo) and enter Designation & Location.

Expected Result: The author row appears centred below the quote, with the round photo.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-04-author.png`
> **Purpose:** Author tab
> **Description:** Author tab filled; canvas showing the author row with photo.

Step 5 — Choose the style

Action: In the Style tab, choose BG Color (Default or Grey). Leave Style = Default.

Expected Result: The background changes accordingly.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-05-style.png`
> **Purpose:** Style tab
> **Description:** Style tab with BG Color = Grey and the grey block on the canvas.

Step 6 — Set Show / Hide

Action: In the Show / Hide tab, set Hide Eyebrow, Hide Title, Show Description and Hide Author Details as the design requires.

Expected Result: The hidden parts disappear from the canvas (the hidden title stays in the HTML for screen readers).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-06-show-hide.png`
> **Purpose:** Show / Hide tab
> **Description:** Show / Hide toggles with Hide Title on.

Step 7 — Add stats

Action: In the Stats tab, add an item for each stat; enter Stats Title (for example “Faster time to market”) and Stats Value (for example “40%”). Reorder items as needed.

Expected Result: The stats appear under a divider, title above value.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-07-stats.png`
> **Purpose:** Stats tab
> **Description:** Stats multifield with three items; canvas showing the stats row.

Step 8 — Fill the Additional tab (optional)

Action: Set an Anchor ID (for example `anchor-client-results`), Hide block, and spacing only when needed.

Expected Result: Invalid anchors are rejected; spacing changes are visible.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-08-additional.png`
> **Purpose:** Additional tab
> **Description:** Additional tab with Anchor ID filled.

Step 9 — Preview

Action: Switch to Preview mode and check desktop, tablet and mobile widths.

Expected Result: Layout and stats wrap as described in 6.13.2.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-09-preview.png`
> **Purpose:** Preview
> **Description:** Preview mode at mobile width with stats stacked.

Step 10 — Validate

Action: Run the Validation Process checklist.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-10-validate.png`
> **Purpose:** Validation
> **Description:** Accessibility tree showing the heading, blockquote, figcaption and description list.

Step 11 — Publish

Action: Publish the page from Universal Editor.

Expected Result: The live page shows the testimonial after publishing.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/testimonial-step-11-publish.png`
> **Purpose:** Publish
> **Description:** Universal Editor Publish dialog.

Preview Process

1. Use Universal Editor Preview mode for a quick check.
2. Open the preview URL `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}` after publishing to preview (where available).
3. Check at 375px, 768px and 1440px; check that the stats wrap cleanly.

Publishing Process

1. Publish the page from Universal Editor or the Sites console.
2. Referenced images (Author Image) are published with the page.
3. The Edge Delivery live page updates after publishing.

Validation Process

- ☐ The title's heading level fits the page outline (Hide Title still counts as a heading).
- ☐ Show Description is on when the quote should be shown.
- ☐ The author name and designation are correct and approved.
- ☐ The author photo is square, shows the face centred and has correct alt text (or none).
- ☐ Each Stats Value is a single line (for example "40%"), each Stats Title describes it.
- ☐ Stats are factual, sourced and approved [To be confirmed: approval process for published stats — owner: content lead].
- ☐ The stats row wraps well at mobile and tablet widths.
- ☐ Anchor ID (if any) is unique.
- ☐ Hide block is off when the block should be visible.

Rollback Process

1. Restore the previous page version from the page's version history, or undo the change.
2. Use Hide block (Additional tab) or Hide Author Details / Show Description off to withdraw a part quickly.
3. Publish the page again.

6.13.6 Authorable Field Specification

Model `testimonial` — Content, Author, Style, Show / Hide and Stats tabs

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Eyebrow (`eyebrow`)	text	No	No	empty	Centred label above the quote	A few words of context	None	Client success
Title (`title`)	text	No	No	empty	Section title	Short heading	None	What our clients say
Title Type (`titleType`)	select	No	No	h2	Heading level	Fit the outline	h1-h6	h2
Description (`description`)	richtext	No (Recommended)	No	empty	The testimonial quote	Concise, approved wording	None	TCS gave us a platform for growth.
Author (`authorName`)	text	No	No	empty	The author's name	Name (and role if no designation)	None	Ranil Boteju
Author Image (`authorImage`)	reference	No	No	empty	Round author photo	Square headshot	Single asset	/content/dam/tcs-corp/people/rb.jpg
Author Image Alt Text (`authorImageAlt`)	text	No	No	empty	Alt text of the photo	Empty if the name is next to it	None	Photo of Ranil Boteju
Designation & Location (`designationLocation`)	richtext	No	No	empty	Role, organisation, location	One or two lines	None	Chief Data & Analytics Officer, Lloyds Banking Group
Style (`classes`)	select	No	No	Default (empty)	Reserved for styles	Leave Default	Only Default	Default
BG Color (`bgColor`)	select	No	No	Default (empty)	Background	Default or Grey	Only grey applied	grey
Hide Eyebrow (`hideEyebrow`)	boolean	No	No	false	Hide the eyebrow	On to hide	Toggle	false
Hide Title (`titleHidden`)	boolean	No	No	false	Hide the title visually	Title stays for screen readers	Toggle	true
Show Description (`showDescription`)	boolean	No	No	true	Show the quote	Keep on to show the quote	Toggle	true
Hide Author Details (`hideAuthor`)	boolean	No	No	false	Hide name, photo, designation	On to hide	Toggle	false
Stats (`stats`)	container (multifield)	No	Yes	no items	Results below the testimonial	One item per stat	None	2 items

Model `testimonial` — Stats item (container `stats`, `multi: true`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Stats Title (`statLabel`)	richtext	No (Recommended)	No	empty	What the value measures	Short phrase	None	Faster time to market
Stats Value (`statValue`)	richtext	No (Recommended)	No	empty	The figure	A single line	None (single line Recommended)	40%

Model `testimonial` — Additional tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	text	No	No	empty	Id for in-page links	Start with anchor-	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-client-results
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide the block	Prepare or withdraw	Toggle	false
Padding left (`classes_paddingleft`)	select	No	No	Default (empty)	Left padding	Only when the design asks	15 options (pl-0 … pl-3000)	pl-400
Padding right (`classes_paddingright`)	select	No	No	Default (empty)	Right padding	Only when the design asks	15 options (pr-0 … pr-3000)	pr-400
Padding top (`classes_paddingtop`)	select	No	No	Default (empty)	Top padding	Only when the design asks	15 options (pt-0 … pt-3000)	pt-1000
Padding bottom (`classes_paddingbottom`)	select	No	No	Default (empty)	Bottom padding	Only when the design asks	15 options (pb-0 … pb-3000)	pb-1000
Margin left (`classes_marginleft`)	select	No	No	Default (empty)	Left margin	Only when the design asks	15 options (ml-0 … ml-3000)	ml-0
Margin right (`classes_marginright`)	select	No	No	Default (empty)	Right margin	Only when the design asks	15 options (mr-0 … mr-3000)	mr-0
Margin top (`classes_margintop`)	select	No	No	Default (empty)	Top margin	Only when the design asks	15 options (mt-0 … mt-3000)	mt-1600
Margin bottom (`classes_marginbottom`)	select	No	No	Default (empty)	Bottom margin	Only when the design asks	15 options (mb-0 … mb-3000)	mb-1600

The padding and margin selects offer the same 15 options as the Quote block (6.12.6): Default, 0 (none), 0.25rem (space-100), 0.5rem (space-200), 0.75rem (space-300), 1rem (space-400), 1.25rem (space-500), 1.5rem (space-600), 2rem (space-800), 2.5rem (space-1000), 3rem (space-1200), 4rem (space-1600), 5rem (space-2000), 6rem (space-2400) and 7.5rem (space-3000).

Field details

Eyebrow (`eyebrow`)

- Purpose: Context label, centred above the quote.
- Allowed Values: Plain text.

- Character Limits: None enforced. Recommended: up to 40 characters.
- Business Rules: Shown as `p.testimonial-eyebrow` when it has text and Hide Eyebrow is off.
- Dependency Rules: Hide Eyebrow.
- Visibility Rules: Always visible in the dialog (no `condition`).
- Sample Content: “Client success”.

Title (`title`)

- Purpose: Section title above the testimonial.
- Allowed Values: Plain text.
- Character Limits: None enforced. Recommended: up to 60 characters.
- Business Rules: Rendered as the heading of Title Type; visually hidden when Hide Title is on.
- Dependency Rules: Title Type, Hide Title.
- Visibility Rules: Always visible.
- Sample Content: “What our clients say”.

Title Type (`titleType`)

- Purpose: Heading level of the title.
- Allowed Values: H1-H6 (h1-h6), default h2.
- Character Limits: Not applicable.
- Business Rules: Invalid or missing: the heading level found in the cell, else H2.
- Dependency Rules: Only with Title text.
- Visibility Rules: Always visible.
- Sample Content: h2.

Description (`description`)

- Purpose: The testimonial quote.
- Allowed Values: Rich text.
- Character Limits: None enforced. Recommended: up to 250 characters (it is set at 1.75rem / 2.5rem).
- Business Rules: Shown as `blockquote.testimonial-description` only when Show Description is on and it has text; paragraphs 1rem apart.
- Dependency Rules: Show Description.
- Visibility Rules: Always visible.
- Sample Content: “TCS gave us a platform for growth.”

Author (`authorName`)

- Purpose: The author’s name (the dialog example also shows role and organisation in one line).
- Allowed Values: Plain text.
- Character Limits: None enforced. Recommended: up to 80 characters.
- Business Rules: Shown as `p.testimonial-author-name` when it has text and Hide Author Details is off.
- Dependency Rules: Hide Author Details.
- Visibility Rules: Always visible.
- Sample Content: “Ranil Boteju”.

Author Image (`authorImage`)

- Purpose: Round photo of the author.
- Allowed Values: One image asset.
- Character Limits: Not applicable.
- Business Rules: Shown 3rem x 3rem, cropped to a circle; only when the cell holds a picture and Hide Author Details is off.
- Dependency Rules: Author Image Alt Text; Hide Author Details.

- Visibility Rules: Always visible.
- Sample Content: `/content/dam/tcs-corp/people/ranil-boteju.jpg`.

Author Image Alt Text (`authorImageAlt`)

- Purpose: Alt text of the photo.
- Allowed Values: Plain text; empty when the name is next to the image.
- Character Limits: None enforced. Recommended: up to 100 characters.
- Business Rules: Replaces the image's alt when set; otherwise an image without alt gets `alt=""`.
- Dependency Rules: Requires Author Image.
- Visibility Rules: Always visible.
- Sample Content: empty (name shown next to it).

Designation & Location (`designationLocation`)

- Purpose: Role, organisation and location of the author.
- Allowed Values: Rich text.
- Character Limits: None enforced. Recommended: up to 100 characters.
- Business Rules: Shown as `div.testimonial-author-details` (rich text kept) below the name.
- Dependency Rules: Hide Author Details.
- Visibility Rules: Always visible.
- Sample Content: "Chief Data & Analytics Officer, Lloyds Banking Group".

Style (`classes`)

- Purpose: Reserved for style variants.
- Allowed Values: Default only.
- Character Limits: Not applicable.
- Business Rules: Adds no class.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: Default.

BG Color (`bgColor`)

- Purpose: Background colour.
- Allowed Values: Default (white), Grey (grey).
- Character Limits: Not applicable.
- Business Rules: Grey adds `bg-grey` with its own padding.
- Dependency Rules: Additional tab padding overrides it.
- Visibility Rules: Always visible.
- Sample Content: grey.

Hide Eyebrow (`hideEyebrow`)

- Purpose: Leaves out the eyebrow.
- Allowed Values: On / Off (default Off).
- Character Limits: Not applicable.
- Business Rules: Only an authored `true` turns it on.
- Dependency Rules: Eyebrow.
- Visibility Rules: Always visible.
- Sample Content: Off.

Hide Title (`titleHidden`)

- Purpose: Hides the title visually, keeping it for screen readers.
- Allowed Values: On / Off (default Off).

- Character Limits: Not applicable.
- Business Rules: Adds `visually-hidden` to the heading.
- Dependency Rules: Title must have text.
- Visibility Rules: Always visible.
- Sample Content: On.

Show Description (`showDescription`)

- Purpose: Shows the quote.
- Allowed Values: On / Off (default On).
- Character Limits: Not applicable.
- Business Rules: Only an authored `true` shows the description; a missing value hides it.
- Dependency Rules: Description.
- Visibility Rules: Always visible.
- Sample Content: On.

Hide Author Details (`hideAuthor`)

- Purpose: Leaves out the author, author image and designation & location.
- Allowed Values: On / Off (default Off).
- Character Limits: Not applicable.
- Business Rules: When on, no figcaption is built.
- Dependency Rules: Author tab fields.
- Visibility Rules: Always visible.
- Sample Content: Off.

Stats (`stats`)

- Purpose: A list of results shown under a divider.
- Allowed Values: Zero or more items, each with Stats Title and Stats Value.
- Character Limits: None enforced. Recommended: 2 to 5 items (five fit one desktop row).
- Business Rules: Empty items are skipped; with no items no stats are rendered.
- Dependency Rules: None (independent of the Show / Hide toggles).
- Visibility Rules: Always visible.
- Sample Content: “Faster time to market / 40%”, “Cost reduction / 25%”.

Stats Title (`statLabel`)

- Purpose: What the value measures, shown above it.
- Allowed Values: Rich text.
- Character Limits: None enforced. Recommended: up to 40 characters.
- Business Rules: All elements of an item before the last become the title (`dt`).
- Dependency Rules: Stats Value.
- Visibility Rules: Always visible within the item.
- Sample Content: “Faster time to market”.

Stats Value (`statValue`)

- Purpose: The figure.
- Allowed Values: Rich text; a single line (“A single line, e.g. 40%.”).
- Character Limits: None enforced. Recommended: up to 8 characters.
- Business Rules: The last element of an item is the value (`dd`); extra paragraphs move into the title.
- Dependency Rules: Stats Title.
- Visibility Rules: Always visible within the item.
- Sample Content: “40%”.

Anchor ID (`classes_anchor`)

- Purpose: Block id for `#anchor-...` links.
- Allowed Values: `anchor-` plus lower-case letters, digits and hyphens.
- Character Limits: None enforced. Recommended: up to 40 characters.
- Business Rules: Enforced by regExp `^anchor-[a-z0-9][a-z0-9-]*$`; duplicates get `-2`, `-3`.
- Dependency Rules: Becomes the data layer component id.
- Visibility Rules: Always visible.
- Sample Content: `anchor-client-results`.

Hide block (`classes_hideblock`)

- Purpose: Hides the whole block on the page.
- Allowed Values: On / Off (default Off).
- Character Limits: Not applicable.
- Business Rules: Not displayed outside edit mode; faded with a dashed outline in edit mode.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: Off.

Padding left (`classes_paddingleft`)

- Purpose: Left inner spacing.
- Allowed Values: The 15 options (class `pl-{token}`).
- Character Limits: Not applicable.
- Business Rules: Outranks the block's own padding (including Grey).
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 1rem (space-400).

Padding right (`classes_paddingright`)

- Purpose: Right inner spacing.
- Allowed Values: The 15 options (class `pr-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Padding left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 1rem (space-400).

Padding top (`classes_paddingtop`)

- Purpose: Top inner spacing.
- Allowed Values: The 15 options (class `pt-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Padding left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 2.5rem (space-1000).

Padding bottom (`classes_paddingbottom`)

- Purpose: Bottom inner spacing.
- Allowed Values: The 15 options (class `pb-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Padding left.

- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 2.5rem (space-1000).

Margin left (`classes_marginleft`)

- Purpose: Left outer spacing.
- Allowed Values: The 15 options (class `ml-{token}`).
- Character Limits: Not applicable.
- Business Rules: Styled in `styles/styles.css`.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 0 (none).

Margin right (`classes_marginright`)

- Purpose: Right outer spacing.
- Allowed Values: The 15 options (class `mr-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Margin left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 0 (none).

Margin top (`classes_margintop`)

- Purpose: Space above the block.
- Allowed Values: The 15 options (class `mt-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Margin left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 4rem (space-1600).

Margin bottom (`classes_marginbottom`)

- Purpose: Space below the block.
- Allowed Values: The 15 options (class `mb-{token}`).
- Character Limits: Not applicable.
- Business Rules: As Margin left.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: 4rem (space-1600).

6.13.7 Field Annotation Mapping

Marker	Field	Purpose
A	Title (`title`), Title Type (`titleType`)	Section heading
B	Eyebrow (`eyebrow`)	Context label above the quote
C	Description (`description`)	The testimonial quote
D	Author Image (`authorImage`), Author Image Alt Text (`authorImageAlt`)	Round author photo
E	Author (`authorName`)	Author's name

Marker	Field	Purpose
F	Designation & Location (`designationLocation`)	Role, organisation, location
G	Stats Title (`statLabel`)	What each stat measures
H	Stats Value (`statValue`)	The figure of each stat

6.13.8 Component Specification

Attribute	Value
Component Name	Testimonial (`testimonial`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`decorators/stats-decorator.js`; `utils/multifield.js`; `decorators/a11y-decorator.js`; `decorators/content-decorator.js`; `decorators/heading-decorator.js`; `decorators/image-decorator.js`; `utils/properties.js`; global block options, buttons and data layer decorators
Component Type	Block (key-value block with a composite multifield, no child components)
Supported Variants	BG Color Default / Grey; author with or without photo; with or without stats; Hide Eyebrow; Hide Title; Show Description; Hide Author Details; Style: Default only
Supported Templates	Any page (allowed in every section)
Supported Metadata	None read by the block
Supported Tags	None
Supported Personalization	None in the block; Adobe Target via the data layer once Adobe Tags is integrated (not implemented)
Caching Considerations	Part of the page HTML cached by the Edge Delivery CDN; no data fetched
Performance Considerations	Synchronous decoration; small author image (3rem) from the authored, optimized picture; no scripts or data requests
Accessibility Compliance	Figure / blockquote / figcaption; description list for stats; visually hidden title option; alt text field; selectable heading level
SEO Compliance	Real heading; quote and stats as text; title kept in the HTML when hidden visually

6.13.9 Input Model

- **Content Inputs:** Eyebrow, Title, Title Type, Description, Author, Author Image, Author Image Alt Text, Designation & Location, BG Color, the Show / Hide toggles, Stats.
- **Metadata Inputs:** None (the Authors data sheet is not used by this block).
- **Tags Inputs:** None.
- **Page Context Inputs:** None.
- **Query Inputs:** None.
- **Manual Inputs:** All content is authored in the block; Style and Additional tab values arrive as block classes.

Authored HTML (one row per property; the Stats cell holds the items as runs of elements separated by `<hr>`, or as list items when an item has a single element):

```
<div class="testimonial">
  <div><div>eyebrow</div><div><p>Client success</p></div></div>
  <div><div>title</div><div><p>What our clients say</p></div></div>
```

```
  <div><div>titleType</div><div><p>h2</p></div></div>
  <div><div>description</div><div><p>TCS gave us a platform for growth.</p></div></div>
  <div><div>authorName</div><div><p>Ranil Boteju</p></div></div>
  <div><div>authorImage</div><div><picture><img src="./media_9f.jpeg" alt=""></picture></
div>
  </div>
  <div><div>designationLocation</div><div><p>CDAO, Lloyds Banking Group</p></div></div>
  <div><div>bgColor</div><div><p>grey</p></div></div>
  <div><div>showDescription</div><div><p>true</p></div></div>
  <div><div>stats</div><div>
    <p>Faster time to market</p><p>40%</p><hr><p>Cost reduction</p><p>25%</p>
  </div></div>
</div>
```

6.13.10 Output Model

Rendered Output: The rows are replaced by, in order: the title heading `.testimonial-title` (with `visually-hidden` when Hide Title is on); `figure.testimonial-quote` with `p.testimonial-eyebrow`, `blockquote.testimonial-description` and `figcaption > div.testimonial-author` (`div.testimonial-author-image`, `div.testimonial-author-text` with `p.testimonial-author-name` and `div.testimonial-author-details`); and `dl.testimonial-stats` with `div.testimonial-stat > dt.testimonial-stat-label + dd.testimonial-stat-value`. Parts without content are left out. Block class from the code: `bg-grey`.

Generated HTML:

```
<div class="testimonial bg-grey block" data-block-name="testimonial">
  <h2 class="testimonial-title">What our clients say</h2>
  <figure class="testimonial-quote">
    <p class="testimonial-eyebrow">Client success</p>
    <blockquote class="testimonial-description"><p>TCS gave us a platform for growth.</p>
    </blockquote>
    <figcaption>
      <div class="testimonial-author">
        <div class="testimonial-author-image"><picture><img src="..." alt=""></picture></
div>
        <div class="testimonial-author-text">
          <p class="testimonial-author-name">Ranil Boteju</p>
          <div class="testimonial-author-details"><p>CDAO, Lloyds Banking Group</p></div>
        </div>
      </div>
    </figcaption>
  </figure>
  <dl class="testimonial-stats">
    <div class="testimonial-stat">
      <dt class="testimonial-stat-label"><p>Faster time to market</p></dt>
      <dd class="testimonial-stat-value"><p>40%</p></dd>
    </div>
  </dl>
</div>
```

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking: The block is added to the data layer after the sections load (id = Anchor ID, else `testimonial-{n}`; `dc:title` = the title heading, even when visually hidden). Links in the rich text push `cmp:click`.

```
{
  "component": {
    "testimonial-1": {
      "@type": "tcs-corp/components/testimonial",
      "parentId": "page-1k3x9q",
      "dc:title": "What our clients say"
    }
  }
}
```

Personalization Output: None.

Search Index Output: None (the query index reads page metadata only).

6.13.11 Business Rules

1. Key-value block: only rows with exactly two cells are properties; property names are matched case-insensitively.
2. The title is rendered only when it has text, at Title Type (h1-h6), else a heading level in the cell, else H2.
3. Hide Title adds `visually-hidden` to the title; the heading stays in the HTML.
4. The eyebrow is rendered only when Hide Eyebrow is off and the eyebrow has text.
5. The description is rendered only when Show Description is `true` and it has text; a missing Show Description value hides the description.
6. Toggles are on only when their value is exactly `true` (case-insensitive); a missing row means off.
7. Hide Author Details on: no author part, whatever the Author tab contains.
8. The author part is built from the parts with content (photo, name, designation); without any, no figcaption is added.
9. The author image's alt is Author Image Alt Text when set, else `alt=""` if the image has no alt.
10. The figure is left out when it has no eyebrow, description or author.
11. Stats: the multifield items are read as list items or as runs of elements between `<hr>`s; a list item's children are its elements; empty items are skipped.
12. In each stat, the last element is the value and all earlier elements are the title; a one-element item has a value and an empty title.
13. With no stat items, no stats list is rendered.
14. BG Color accepts only `grey`; the `bg-grey` class is reset on each decoration; a block class `grey` is a fallback.
15. The layout switches at 900px (gaps and Grey padding).
16. Stats columns are at least 10rem wide and wrap automatically; up to five fit one row at the 70rem maximum width.
17. The Additional tab padding overrides the Grey padding; Hide block hides the block outside edit mode.
18. Editor-only: instrumentation is moved to the rendered elements so in-place editing works; changes re-decorate the block.
19. Errors are caught by the block loader (`failed to load module for testimonial`).

6.13.12 Author Best Practices

Recommended Content Length: Eyebrow up to 40 characters; Title up to 60; Description up to 250 (large type); Author up to 80; Designation & Location up to 100; 2 to 5 stats with titles up to 40 characters and values up to 8 characters. (All Recommended.)

Image Dimensions: Author Image is shown 3rem x 3rem (48px) in a circle with `object-fit: cover`. Recommended: a square (1 : 1) headshot of at least 96 x 96 px, face centred.

CTA Writing Guidelines: No CTA field. If needed, add a link in its own paragraph in the description (it becomes a link button) with a descriptive label.

Accessibility Guidelines: Keep a title even when it is hidden (Hide Title) so the section has a heading; leave the alt text empty when the name is next to the photo; write stats so that the title explains the value ("Faster time to market" / "40%"); do not use only colour to convey meaning.

SEO Guidelines: Use a descriptive title; write numbers as text (not images); include the organisation name in the designation.

Metadata Guidelines: The block reads no page metadata.

Taxonomy Guidelines: No tags in the block; tag the page if it should appear in discovery lists.

Localization Guidelines: Translate eyebrow, title, description, designation and stat titles; localise number formats in values (for example "40 %" in French) [To be confirmed: number format rules per locale — owner: localisation lead]; keep names unchanged.

6.13.13 Troubleshooting Guide

Issue 1 — The quote (description) does not appear

Symptoms: Title and author are shown, but not the large quote.

Root Cause: Show Description is off, or its value is missing (blocks authored before the field existed), or the Description is empty.

Diagnosis Steps: Check the Show / Hide tab and the Description field.

Resolution Steps: Switch Show Description on (toggle it off and on again if it shows as on but the value was never saved) and fill the Description.

Validation Steps: The `blockquote.testimonial-description` appears.

Preventive Measures: Add the Show Description check to the validation checklist.

Issue 2 — A stat value appears in the title position or vice versa

Symptoms: A stat shows text where the value should be, or the value is missing.

Root Cause: The last element of an item is always the value; an empty Stats Value makes the title the value, and a multi-paragraph value moves its first paragraphs into the title.

Diagnosis Steps: Inspect the item's `dt` and `dd` in the HTML; open the item in the Stats tab.

Resolution Steps: Fill both fields; keep Stats Value on a single line.

Validation Steps: Title above, value below for each item.

Preventive Measures: Train authors on the single-line rule.

Issue 3 — The author row is missing

Symptoms: No name, photo or designation below the quote.

Root Cause: Hide Author Details is on, or all three author fields are empty.

Diagnosis Steps: Check the Show / Hide and Author tabs.

Resolution Steps: Turn Hide Author Details off and fill the fields.

Validation Steps: The `figcaption` with the author appears.

Preventive Measures: Use Hide Author Details only on purpose.

Issue 4 — The title is not visible
Symptoms: The section title is missing on the page but heard by screen readers.

Root Cause: Hide Title is on (visually hidden by design).

Diagnosis Steps: Check the `visually-hidden` class on `.testimonial-title`.

Resolution Steps: Turn Hide Title off if the title should be seen.

Validation Steps: The title is visible.

Preventive Measures: Document per page whether the title should be hidden.

Issue 5 — The author photo is distorted or cropped badly
Symptoms: The face is off-centre or cut in the circle.

Root Cause: A non-square or off-centre image is cropped to a 3rem circle.

Diagnosis Steps: Check the source image proportions.

Resolution Steps: Use a square, centred headshot.

Validation Steps: The face fills the circle.

Preventive Measures: Keep a library of approved square headshots.

Issue 6 — Stats wrap unevenly
Symptoms: For example four stats show as three plus one on tablet.

Root Cause: The grid fits as many 10rem columns as the width allows (`auto-fit`).

Diagnosis Steps: Check the viewport width and the number of items.

Resolution Steps: Use a number of stats that wraps well (for example 3 or 6), or shorten the values.

Validation Steps: Check at 375px, 768px and 1440px.

Preventive Measures: Plan the number of stats with the design team.

Issue 7 — The eyebrow is missing
Symptoms: The label above the quote is not shown.

Root Cause: Hide Eyebrow is on, or the Eyebrow is empty.

Diagnosis Steps: Check the Show / Hide tab.

Resolution Steps: Turn Hide Eyebrow off and fill the Eyebrow.

Validation Steps: `p.testimonial-eyebrow` appears.

Preventive Measures: Validate before publishing.

6.13.14 Support Runbook
Monitoring Requirements: No block-specific monitoring. Monitor key pages with Real User Monitoring (Edge Delivery operational telemetry) and check the console after deployments.

Logs:

- Browser console: no messages of its own; decoration errors appear as `failed to load module for testimonial`, CSS or loading errors as `failed to load block testimonial`.
- Admin API: page status at `https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`.
- Author logs: AEM author logs and the page Timeline.

Common Errors:

Error	Likely Cause	Action
Quote missing	Show Description off or missing	Turn it on
Author missing	Hide Author Details on or fields empty	Turn off, fill fields
Stat value and title swapped	Empty value or multi-line value	Fill both, single-line value
Title not visible	Hide Title on	Turn off if needed
Anchor ID rejected	Pattern not matched	Use `anchor-...`
`failed to load module for testimonial`	Code error	Escalate to L3

Escalation Matrix:

Level	Team	Checks
L1	Author support	Field values, Show / Hide toggles, stats items, images, publication
L2	AMS / site admin	Replication, Admin API status, CDN cache, console errors, deployments
L3	Development team	Block, stats decorator, multifield reading, CSS; local reproduction with `aem up`
L4	Adobe Support	Universal Editor, Edge Delivery publishing or CDN platform issues

Recovery Procedures: Restore the previous page version and republish; use Hide block or the Show / Hide toggles to withdraw parts; revert code regressions on `main`.

Health Checks: On a live page with the block, check that the figure, the author row and the stats list render, there are no console errors and the gaps change at 900px.

Operational Procedures: Standard publish workflow for content; feature branch, preview, pull request and merge for code. After `_testimonial.json` changes run `npm run build:json` and `npm run lint`. Changes to `decorators/stats-decorator.js` also affect the Teaser block.

Known Limitations:

- Style has only the Default option.
- One testimonial per block; no carousel.
- The block does not use the Authors data sheet; author data is entered per block.
- No CTA field.
- Stats are text only (no icons, no counting animation).
- Show Description must be explicitly on; blocks without the value hide the description.
- The layout switches at 900px only.

6.14 Columns Block

The Columns block (`blocks/columns`) places content side by side: a grid of rows and columns whose cells hold default content (Text, Image, Button, Title). It is the boilerplate Columns block of the AEM Edge Delivery Services xwalk project with the project's Additional tab. Columns are stacked on mobile and tablet and shown side by side, with equal widths, from 900px.

6.14.1 Component Purpose

Business Objective: Give authors a simple, flexible way to put text, images and buttons next to each other without a purpose-built block, for layouts not covered by the designed components.

Business Use Cases:

- Image and text side by side (for example a service description next to an illustration).
- Two or three text columns (for example “Our mission”, “Our vision”, “Our values”).
- A row of logos or images with captions.
- Contact or address blocks next to each other.
- Several rows of image and text pairs.

When To Use:

- The content is simple default content (text, headings, images, buttons) that should sit in columns.
- No designed component (Teaser, Cards, List, Quote, Testimonial) fits the need.

When Not To Use:

- For card grids with consistent card styling, links and responsive column control: use Cards or List (they have per-breakpoint column settings; Columns does not).
- For an image with text and CTAs in the design system style: use Teaser or Hero.
- For tabular data: use Table or Table List.
- For a quote with a portrait: use Quote.
- When columns of very different widths are needed: Columns always gives equal widths.

Dependencies:

Type	Dependency	Use
Block code	`blocks/columns/columns.js`, `blocks/columns/columns.css`	Column count class, image column detection, layout
Model	`blocks/columns/_columns.json` (definition, model `columns`, filters `columns` and `column`)	Universal Editor dialog and allowed content
Default content models	`models/_text.json`, `models/_title.json`, `models/_image.json`, `models/_button.json`	The components allowed in each column
Global	`decorators/content-buttons-decorator.js`	Links and Button components in the columns become buttons or link buttons
Global	`decorators/block-options-decorator.js`, `styles/styles.css`	Additional tab
Global	`decorators/data-layer-decorator.js`, `utils/analytics.js`	Data layer component entry and click events
Data sources / placeholders	None	The block reads no sheet, query index or placeholder

The block’s JavaScript imports no decorators or utils.

Related Components: Cards and List (structured card grids); Teaser (designed image and text); Table; the default content components Text, Title, Image and Button that go into the columns.

Applicable Templates: Any page. Columns is allowed in every section (section filter) and in Tabs sections (tabs filter). Not added by any template.

Supported Content Types: Any page content: corporate, service, industry, about, contact, article and press release pages.

6.14.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A two-column Columns block at 1440px: an image on the left and a heading, text and a button on the right, vertically centred.

Desktop (900px and wider): Each row is a horizontal flex row with a 1.5rem gap between columns. Every column takes an equal share of the width (`flex: 1`), whatever the number of columns, and the columns of a row are vertically centred against each other (`align-items: center`). Images fill their column's width (`width: 100%`, height automatic, no crop). The columns appear in authored order. There is no gap between rows other than the content's own margins.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same block at 768px: the image above, the text and button below, full width.

Tablet (600px to 899px): Each row is a vertical stack: the columns are full width, one under the other. A column that holds only an image is moved to the top of its row (CSS `order: 0`; other columns have `order: 1`). There is no gap between the stacked columns other than content margins. The section's side padding is 2.5rem from 600px.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The block at 375px: stacked columns, image first, text below.

Mobile (below 600px): As tablet, with the section side padding of 1.5rem.

6.14.3 Component Variations

Columns has no Style select and no style classes. The variations are the number of columns, the number of rows and image-only columns. The column count only changes how many equal columns share the row; the CSS has no rules per count.

Variation 1 — Two Columns

Variation Name: Two columns (the default of a new block: 2 columns, 1 row).

Business Purpose: The most common split: image and text, or two text blocks side by side.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-two-columns.png`
> **Purpose:** Two-column variation
> **Description:** Two equal columns with text in each, side by side on desktop.

Use Cases: Image and description; “Challenge” and “Solution”; two contact addresses.

Authoring Guidelines: Add the Columns block (it starts with Columns = 2 and Rows = 1); add Text, Title, Image or Button components to each column.

Business Rules: The block gets the class `columns-2-cols`, counted from the cells of the first row.

Rendering Rules: Each column is 50% of the row minus the 1.5rem gap on desktop.

Responsive Behaviour: Desktop: side by side, equal width, centred vertically. Tablet and mobile: stacked, image-only columns first.

SEO Impact: Content stays in the HTML in authored order; headings inside the columns contribute to the page outline.

Accessibility Compliance: The reading order is the authored order (row by row, column by column), except that on mobile an image-only column is shown before the text column visually; screen readers still follow the DOM order. Use headings inside columns so screen-reader users can navigate them.

Variation 2 — Three or More Columns

Variation Name: Three or more columns.

Business Purpose: Short parallel items: values, pillars, steps, logos.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-three-columns.png`
> **Purpose:** Three-column variation
> **Description:** Three equal columns with a heading and short text each on desktop.

Use Cases: “Mission / Vision / Values”; three key facts; a row of four logos.

Authoring Guidelines: Set Columns to 3 (or more) in the Content tab and fill each column.

Business Rules: The class is `columns-{n}-cols`, where n is the number of cells in the first row (for example `columns-3-cols`). No maximum is enforced.

Rendering Rules: n equal columns from 900px, 1.5rem apart. The class has no styles of its own.

Responsive Behaviour: Desktop: n equal columns (narrow with 4 or more). Tablet and mobile: stacked. There is no intermediate two-up tablet layout.

SEO Impact: As Variation 1.

Accessibility Compliance: Keep column text short; with many columns, text lines become very short on 900px to 1199px screens, which hurts readability.

Variation 3 — Image Column

Variation Name: A column whose only content is an image.

Business Purpose: Image and text layouts in which the image should appear above the text on small screens.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-image-column.png`

> **Purpose:** Image column variation
> **Description:** Desktop: text on the left, image on the right; mobile inset: the image moved above the text.

Use Cases: Text with an illustration on the right on desktop, image first on mobile.

Authoring Guidelines: Put only an Image component in the column (no text or button with it).

Business Rules: A column gets `columns-img-col` when it contains a picture and the element around it (the column cell) has exactly one child. A column with an image and text does not get the class.

Rendering Rules: `columns-img-col` has `order: 0` on mobile and tablet (other columns `order: 1`), so the image comes first; the image is a block element at full column width.

Responsive Behaviour: Desktop: authored position. Tablet and mobile: first in its row.

SEO Impact: Set Alt Text on the Image component for image search.

Accessibility Compliance: The visual reorder on mobile does not change the DOM order; if the image carries meaning, its alt text must make sense wherever it is read.

Variation 4 — Multiple Rows

Variation Name: Several rows.

Business Purpose: A repeated pattern, such as alternating image and text rows (zig-zag), or a grid of items.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/columns-multiple-rows.png`
> **Purpose:** Multiple rows variation
> **Description:** Three rows of image and text pairs, the image alternating between left and right on desktop.

Use Cases: Feature lists with images; step-by-step explanations; zig-zag layouts.

Authoring Guidelines: Set Rows to the number of rows; to alternate the image side, put the image in the first column of one row and the second column of the next.

Business Rules: The column count class is taken from the first row only; each row is laid out on its own.

Rendering Rules: Each row is its own flex container; rows with fewer cells share the width among their own cells.

Responsive Behaviour: Desktop: each row side by side. Tablet and mobile: all cells stacked row after row; image-only columns first within their row, so a zig-zag becomes image, text, image, text.

SEO Impact: As Variation 1.

Accessibility Compliance: Use a heading per row so each pair can be found by heading navigation.

6.14.4 Authoring Experience

> [Screenshot Placeholder]
> **File name:** `screenshots/columns-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A Columns block selected in the canvas with letters A to E on the block, a column, a Title, a Text and an Image component inside the columns.

Annotations:

- A = Columns block (select it to edit Columns, Rows and the Additional tab)
- B = A column (the container that accepts Text, Image, Button and Title)
- C = Title component inside a column
- D = Text component inside a column (rich text, editable in place)
- E = Image component inside a column (or a Button component)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Columns properties panel with the Content tab (Columns, Rows) and the Additional tab header.

The properties panel of the block has two tabs:

- **Content:** Columns (number, default 2 in a new block) and Rows (number, default 1). They set the grid of the block in AEM (one cell per column in each row); the block's code does not read them, it counts the cells of the first row.
- **Additional:** Anchor ID, Hide block, Padding left / right / top / bottom, Margin left / right / top / bottom.

The components inside the columns have their own panels: Title (Title, Title Type), Image (Image, Alt Text), Button (Link, Text, Title, Views, Type, Icon, Alignment) and Text (edited in place in the canvas).

6.14.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites, select the page in the locale site and choose Edit.

Expected Result: The page opens in the Universal Editor canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** Sites console with the page selected and Edit highlighted.

Step 2 — Add the Columns block

Action: Select the section (or a Tabs section), select Add (+) and choose Columns.

Expected Result: A Columns block with two empty columns in one row is added.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** The component picker with Columns highlighted; the new empty block on the canvas.

Step 3 — Set the number of columns and rows

Action: With the block selected, set Columns and Rows in the Content tab.

Expected Result: The grid shows the requested number of cells.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-03-columns-rows.png`
> **Purpose:** Columns and Rows
> **Description:** Content tab with Columns = 3 and Rows = 1, and the three empty columns on the canvas.

Step 4 — Add content to each column

Action: Select a column, select Add (+) and choose Title, Text, Image or Button. Repeat for each column.

Expected Result: The components appear in the column; only these four component types are offered.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-04-add-content.png`
> **Purpose:** Add column content
> **Description:** The Add menu of a column listing Text, Image, Button and Title.

Step 5 — Fill the components

Action: Enter the Title and its Title Type, write the Text in place, select the Image and its Alt Text, and set the Button's Link, Text, Views, Type and Alignment.

Expected Result: The columns show the content; on desktop they sit side by side.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-05-fill-content.png`
> **Purpose:** Fill the content
> **Description:** A filled two-column block: image left, title, text and button right.

Step 6 — Order the content

Action: Drag components within a column, or between columns, in the content tree to change their order.

Expected Result: The canvas reflects the new order.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-06-reorder.png`
> **Purpose:** Reorder
> **Description:** The content tree with a component being moved between columns.

Step 7 — Fill the Additional tab (optional)

Action: Select the block and set an Anchor ID, Hide block or spacing if required.

Expected Result: Anchor IDs that do not match the pattern are rejected.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-07-additional.png`
> **Purpose:** Additional tab
> **Description:** Additional tab of the Columns block.

Step 8 — Preview

Action: Use Preview mode; check the layout at desktop, tablet and mobile widths.

Expected Result: Columns side by side from 900px, stacked below, image-only columns first.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-08-preview.png`
> **Purpose:** Preview
> **Description:** Preview at mobile width with the image above the text.

Step 9 — Validate

Action: Run the Validation Process checklist.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-09-validate.png`
> **Purpose:** Validation
> **Description:** The page's heading outline and image alt texts checked in the browser tools.

Step 10 — Publish

Action: Publish the page from Universal Editor.

Expected Result: The live page shows the columns.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/columns-step-10-publish.png`
> **Purpose:** Publish
> **Description:** Universal Editor Publish dialog.

Preview Process

1. Check in Universal Editor Preview mode.
2. Open the preview URL `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}` after publishing to preview (where available).
3. Check at 375px, 768px, 900px and 1440px; the layout switches at 900px.

Publishing Process

1. Publish the page from Universal Editor or the Sites console.
2. Images used in the columns are published with the page.
3. The live page updates after publishing.

Validation Process

- ☐ Every row has the intended number of columns (the first row sets the column count class).
- ☐ No column is empty (an empty cell still takes an equal share of the width).
- ☐ Images have Alt Text (or are decorative on purpose).
- ☐ Image-only columns appear first on mobile as intended.
- ☐ Headings in the columns follow the page outline.

- ☐ Buttons have clear labels and correct links.
- ☐ Text stays readable at 900px to 1199px with the chosen number of columns.
- ☐ Anchor ID (if any) is unique; Hide block is off when the block should be visible.

Rollback Process

1. Restore the previous page version from the version history, or undo the change.
2. Use Hide block to withdraw the block quickly.
3. Publish the page again.

6.14.6 Authorable Field Specification

Model `columns` — Content and Additional tabs

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Columns (`columns`)	text (number)	No	No	empty in the model; 2 in a new block (template)	Number of columns of the grid	2 to 4	Number value type; no range enforced	3
Rows (`rows`)	text (number)	No	No	empty in the model; 1 in a new block (template)	Number of rows	As many as needed	Number value type; no range enforced	2
Anchor ID (`classes_anchor`)	text	No	No	empty	Id for in-page links	Start with anchor-	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-our-values
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide the block	Prepare or withdraw	Toggle	false
Padding left (`classes_paddingleft`)	select	No	No	Default (empty)	Left padding	Only when the design asks	15 options (pl-0 … pl-3000)	pl-400
Padding right (`classes_paddingright`)	select	No	No	Default (empty)	Right padding	Only when the design asks	15 options (pr-0 … pr-3000)	pr-400
Padding top (`classes_paddingtop`)	select	No	No	Default (empty)	Top padding	Only when the design asks	15 options (pt-0 … pt-3000)	pt-800
Padding bottom (`classes_paddingbottom`)	select	No	No	Default (empty)	Bottom padding	Only when the design asks	15 options (pb-0 … pb-3000)	pb-800
Margin left (`classes_marginleft`)	select	No	No	Default (empty)	Left margin	Only when the design asks	15 options (ml-0 … ml-3000)	ml-0
Margin right (`classes_marginright`)	select	No	No	Default (empty)	Right margin	Only when the design asks	15 options (mr-0 … mr-3000)	mr-0
Margin top (`classes_margintop`)	select	No	No	Default (empty)	Top margin	Only when the design asks	15 options (mt-0 … mt-3000)	mt-1600
Margin bottom (`classes_marginbottom`)	select	No	No	Default (empty)	Bottom margin	Only when the design asks	15 options (mb-0 … mb-3000)	mb-1600

The padding and margin selects have the same 15 options as in 6.12.6 (Default, 0 (none), and the tokens space-100 to space-3000).

Column item (`column`)

The column has no model of its own (no fields). Its filter allows four components: Text, Image, Button and Title. Their fields:

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Text (Text component)	rich text (edited in place)	No	No	empty	Paragraphs, lists, links	Keep it short in narrow columns	None	Our values guide every engagement.
Title (Title component)	text	No	No	empty	Heading in the column	Short heading	None	Our mission
Title Type (Title component)	select	No	No	none set	Heading level	h1-h6; fit the outline	h1-h6	h3
Image (Image component)	reference	No	No	empty	Image in the column	From AEM Assets	Single asset	/content/dam/tcs-corp/illustrations/cloud.png
Alt Text (Image component)	text	No	No	empty	Image alternative text	Describe the image	None	Cloud migration illustration
Link (Button component)	aem-content	No	No	empty	Button target	Page or URL	None	/content/tcs-corp/us/en/services/cloud
Text (Button component)	text	No	No	empty	Button label	Action-oriented	None	Explore cloud services
Title (Button component)	text	No	No	empty	Link title attribute	Optional extra description	None	Cloud services overview
Views (Button component)	select	No	No	button	Button, Icon or Link	Choose the view	button, icon, link	button
Type (Button component)	select	No	No	primary	Button style	Primary, Secondary, Tertiary	Shown when Views is empty, Button or Icon	secondary
Icon (Button component)	select	No	No	none	Icon of an icon button	Arrow right, Chevron right, Chevron down, Search	Shown when Views is Icon	arrow-right
Alignment (Button component)	select	No	No	left	Button alignment	Left, Center, Right	left, center, right	left

Field details

Columns (`columns`)

- Purpose: The number of columns of the grid.
- Allowed Values: A number (value type number); a new block starts with 2.
- Character Limits: Not applicable. No range is enforced. Recommended: 2 to 4.
- Business Rules: Used by AEM to build the grid; the code counts the first row's cells for the `columns-{n}-cols` class.
- Dependency Rules: Rows.

- Visibility Rules: Always visible (no `condition`).
- Sample Content: 3.

Rows (`rows`)

- Purpose: The number of rows.
- Allowed Values: A number; a new block starts with 1.
- Character Limits: Not applicable. No range is enforced.
- Business Rules: Each row is laid out on its own.
- Dependency Rules: Columns.
- Visibility Rules: Always visible.
- Sample Content: 2.

Anchor ID (`classes_anchor`)

- Purpose: Block id for `#anchor-...` links.
- Allowed Values: `anchor-` plus lower-case letters, digits and hyphens.
- Character Limits: None enforced. Recommended: up to 40 characters.
- Business Rules: Enforced by regExp `^anchor-[a-z0-9][a-z0-9-]*$`; duplicates get `-2`, `-3`.
- Dependency Rules: Becomes the data layer component id.
- Visibility Rules: Always visible.
- Sample Content: `anchor-our-values`.

Hide block (`classes_hideblock`)

- Purpose: Hides the block on the page.
- Allowed Values: On / Off (default Off).
- Character Limits: Not applicable.
- Business Rules: Not displayed outside edit mode; faded with a dashed outline in edit mode.
- Dependency Rules: None.
- Visibility Rules: Always visible.
- Sample Content: Off.

Padding left / right / top / bottom (`classes_paddingleft`, `classes_paddingright`, `classes_paddingtop`, `classes_paddingbottom`) — each field separately:

- Padding left — Purpose: left inner spacing; Allowed Values: the 15 options (`pl-{token}`); Character Limits: not applicable; Business Rules: styled in `styles/styles.css`; Dependency Rules: none; Visibility Rules: always visible; Sample Content: 1rem (space-400).
- Padding right — Purpose: right inner spacing; Allowed Values: the 15 options (`pr-{token}`); Character Limits: not applicable; Business Rules: as Padding left; Dependency Rules: none; Visibility Rules: always visible; Sample Content: 1rem (space-400).
- Padding top — Purpose: top inner spacing; Allowed Values: the 15 options (`pt-{token}`); Character Limits: not applicable; Business Rules: as Padding left; Dependency Rules: none; Visibility Rules: always visible; Sample Content: 2rem (space-800).
- Padding bottom — Purpose: bottom inner spacing; Allowed Values: the 15 options (`pb-{token}`); Character Limits: not applicable; Business Rules: as Padding left; Dependency Rules: none; Visibility Rules: always visible; Sample Content: 2rem (space-800).

Margin left / right / top / bottom (`classes_marginleft`, `classes_marginright`, `classes_margintop`, `classes_marginbottom`) — each field separately:

- Margin left — Purpose: left outer spacing; Allowed Values: the 15 options (`ml-{token}`); Character Limits: not applicable; Business Rules: styled in `styles/styles.css`; Dependency Rules: none; Visibility Rules: always visible; Sample Content: 0 (none).

- Margin right — Purpose: right outer spacing; Allowed Values: the 15 options (`mr-{token}`); Character Limits: not applicable; Business Rules: as Margin left; Dependency Rules: none; Visibility Rules: always visible; Sample Content: 0 (none).
- Margin top — Purpose: space above the block; Allowed Values: the 15 options (`mt-{token}`); Character Limits: not applicable; Business Rules: as Margin left; Dependency Rules: none; Visibility Rules: always visible; Sample Content: 4rem (space-1600).
- Margin bottom — Purpose: space below the block; Allowed Values: the 15 options (`mb-{token}`); Character Limits: not applicable; Business Rules: as Margin left; Dependency Rules: none; Visibility Rules: always visible; Sample Content: 4rem (space-1600).

Column content (Text, Title, Image, Button components)

- Purpose: The content of each column.
- Allowed Values: Only Text, Image, Button and Title can be added to a column (filter `column`).
- Character Limits: None enforced. Recommended: a heading up to 40 characters and up to 300 characters of text per column in three-column layouts.
- Business Rules: A standalone link in a paragraph and the Button component are turned into buttons or link buttons by the content buttons decorator; an image alone in a column makes it an image column.
- Dependency Rules: Button Type is shown only when Views is empty, Button or Icon; Button Icon only when Views is Icon (model `condition`).
- Visibility Rules: As above for the Button fields; all other fields always visible.
- Sample Content: Title “Our mission” (h3), Text “We help enterprises…”, Button “Learn more”.

6.14.7 Field Annotation Mapping

Marker	Field	Purpose
A	Columns block (Columns, Rows, Additional tab)	Grid size and block options
B	Column (container, no fields)	Holds the column’s components
C	Title component (Title, Title Type)	Heading in a column
D	Text component (rich text)	Body text in a column
E	Image component (Image, Alt Text) or Button component	Image or call to action in a column

6.14.8 Component Specification

Attribute	Value
Component Name	Columns (`columns`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	No imports in the block code; default content models (Text, Title, Image, Button); global buttons, block options and data layer decorators
Component Type	Container block (`core/franklin/components/columns/v1/columns`), with column items holding default content
Supported Variants	Any number of columns and rows; image-only columns; no style options
Supported Templates	Any page; allowed in sections and Tabs sections
Supported Metadata	None

Attribute	Value
Supported Tags	None
Supported Personalization	None; Adobe Target via the data layer once Adobe Tags is integrated (not implemented)
Caching Considerations	Part of the page HTML cached by the Edge Delivery CDN; nothing fetched
Performance Considerations	Very small synchronous decoration; images are the authored, optimized pictures; in the first section the first image is the LCP candidate and loaded eagerly
Accessibility Compliance	DOM order kept; image alt text from the Image component; headings from the Title component; visual reorder of image columns on mobile only
SEO Compliance	All content in the HTML; headings and alt texts authored per component

6.14.9 Input Model

- **Content Inputs:** The rows and cells of the grid, each cell holding Text, Title, Image and Button content.
- **Metadata Inputs:** None.
- **Tags Inputs:** None.
- **Page Context Inputs:** None.
- **Query Inputs:** None.
- **Manual Inputs:** Columns and Rows (grid size), the column content and the Additional tab values (as block classes).

Authored HTML (one `div` per row, one `div` per column):

```
<div class="columns anchor-our-values">
  <div>
    <div><picture><img src="./media_4c.png" alt="Cloud migration illustration"></
picture></div>
    <div>
      <h3>Our mission</h3>
      <p>We help enterprises grow with technology.</p>
      <p><a href="/en/about">Learn more</a></p>
    </div>
  </div>
</div>
```

6.14.10 Output Model

Rendered Output: The rows and cells keep their structure. The code adds the class `columns-{n}-cols` to the block (n = cells in the first row) and `columns-img-col` to each image-only column. Links and Button components were already turned into buttons (`p.button-wrapper`, `a.button`) by the content buttons decorator before the block loads. The block options decorator sets the `id` from the Anchor ID.

Generated HTML:

```
<div class="columns anchor-our-values columns-2-cols block" id="anchor-our-values"
  data-block-name="columns" data-block-status="loaded">
  <div>
    <div class="columns-img-col"><picture><img src="..." alt="Cloud migration
illustration">
    </picture></div>
    <div>
      <h3>Our mission</h3>
      <p>We help enterprises grow with technology.</p>
```

```
      <p class="button-wrapper">
        <a href="/en/about" title="Learn more" class="link-button">Learn more</a>
      </p>
    </div>
  </div>
</div>
```

An unformatted standalone link becomes a link button (`link-button`); a bold or italic link, or a Button component with Views = Button, becomes a button (`button primary`, `button secondary` or `button tertiary`); Views = Icon adds `icon-button`. See the Button chapter for details.

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking: The block is added to the data layer after the sections load (id = Anchor ID, else `columns-{n}`; `dc:title` = the first heading in any column). Clicks on links and buttons in the columns push `cmp:click` with the link URL and type (`other`, `exit` or `download`).

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.anchor-our-values",
    "interaction": "link",
    "dc:title": "Learn more",
    "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/about",
    "xdm:linkType": "other",
    "breakpoint": "tablet"
  }
}
```

Personalization Output: None.

Search Index Output: None (the query index reads page metadata only).

6.14.11 Business Rules

1. The column count class is `columns-{n}-cols`, with n the number of cells in the first row.
2. Columns and Rows are not read by the code; they define the grid authored in AEM.
3. A column gets `columns-img-col` when it contains a picture and the picture's nearest `div` (the column) has exactly one child.
4. A column with an image plus other content is a normal column.
5. Below 900px each row stacks its columns; image-only columns are shown first (`order: 0`), the others after (`order: 1`).
6. From 900px each row is horizontal, columns have equal widths (`flex: 1`), a 1.5rem gap and are vertically centred; the authored order applies.
7. Images fill their column's width; there is no crop or fixed aspect ratio.
8. Only Text, Image, Button and Title can be added to a column (filter `column`); only columns can be added to the block (filter `columns`).
9. Links and Button components are decorated as buttons or link buttons by the global content buttons decorator.
10. An empty Columns block (no rows) fails to decorate (no first row); the error is logged by the block loader as `failed to load module for columns` and nothing is rendered.
11. The Additional tab works as for every block: Anchor ID (validated), Hide block (hidden outside edit mode, faded in edit mode), padding and margin classes.

12. Editor-only: each change re-decorates the block.
13. There are no style options and no per-breakpoint column settings.

6.14.12 Author Best Practices

Recommended Content Length: Two columns: up to about 600 characters per column; three columns: up to about 300; four columns: up to about 150. A heading per column up to 40 characters. (Recommended.)

Image Dimensions: Images are shown at the full width of their column with their own aspect ratio. Use the same aspect ratio for images in the same row (for example 16 : 9 or 1 : 1) so the row looks even; width at least twice the displayed column width (for example 1200px for a two-column block at 1200px wide).

CTA Writing Guidelines: Use the Button component (or a standalone link in its own paragraph) with an action-oriented label ("Explore cloud services"); one primary button per column at most.

Accessibility Guidelines: Start each column with a Title at the right level; set Alt Text on every meaningful image; do not rely on the side-by-side position to convey meaning (on mobile the columns are stacked and image columns move up).

SEO Guidelines: Use real headings (Title component) rather than bold text; use descriptive link labels.

Metadata Guidelines: The block reads no metadata.

Taxonomy Guidelines: No tags in the block.

Localization Guidelines: Translations can be much longer (for example German); check that columns still balance on desktop and keep text short in three- and four-column layouts.

6.14.13 Troubleshooting Guide

Issue 1 — The block shows nothing

Symptoms: An empty space or nothing where the Columns block should be; the console shows `failed to load module for columns`.

Root Cause: The block has no rows (for example all rows were removed), so the code finds no first row.

Diagnosis Steps: Inspect the block in the browser; check for rows and the console error.

Resolution Steps: Set Rows to at least 1 and Columns to at least 1 and add content, or delete the block.

Validation Steps: The block renders and the error is gone.

Preventive Measures: Delete unused Columns blocks rather than emptying them.

Issue 2 — Columns have different widths than expected

Symptoms: A column is wider or narrower than intended, or a design asks for 1/3 and 2/3.

Root Cause: Columns always share the row equally; there is no width setting.

Diagnosis Steps: Check the number of cells in the row.

Resolution Steps: Use the equal split, or a designed block (Teaser) for asymmetric layouts; request a new variant from development if needed.

Validation Steps: Review at 1440px.

Preventive Measures: Agree layouts with the design team before authoring.

Issue 3 — The image is not first on mobile

Symptoms: On mobile the text appears before the image.

Root Cause: The image column also contains text or a button, so it is not an image column.

Diagnosis Steps: Check whether the column has the class `columns-img-col`.

Resolution Steps: Move the other content to another column, leaving only the Image.

Validation Steps: On mobile the image appears first.

Preventive Measures: Keep images in their own column.

Issue 4 — Columns are too narrow on small laptops

Symptoms: At 900px to 1199px the text in four or more columns wraps after a few words.

Root Cause: The layout becomes horizontal at 900px whatever the number of columns.

Diagnosis Steps: Check at 900px and 1024px.

Resolution Steps: Reduce the number of columns or shorten the text; use Cards or List for many items (they have per-breakpoint columns).

Validation Steps: Text is readable at 900px.

Preventive Measures: Prefer 2 to 3 columns.

Issue 5 — A link shows as a button (or not)

Symptoms: A text link in a column is shown as a link button, or a Button component looks different than expected.

Root Cause: A link alone in its paragraph is turned into a link button by the content buttons decorator; Button components are styled from their Views and Type.

Diagnosis Steps: Check whether the link is alone in its paragraph; check the Button's Views and Type.

Resolution Steps: Put the link inside a sentence to keep it inline, or set the Button's Views and Type.

Validation Steps: The link or button looks as intended.

Preventive Measures: Use the Button component for calls to action.

Issue 6 — Rows do not line up

Symptoms: Items in a multi-row block do not align in a grid.

Root Cause: Each row is a separate flex row; rows with fewer cells share the width among fewer columns, and images of different ratios have different heights.

Diagnosis Steps: Count the cells per row; compare image ratios.

Resolution Steps: Use the same number of cells in every row and images with the same aspect ratio.

Validation Steps: Rows align at 1440px.

Preventive Measures: Use Cards or List for regular grids.

6.14.14 Support Runbook

Monitoring Requirements: No block-specific monitoring; monitor key pages with Real User Monitoring and check the console after deployments.

Logs:

- Browser console: no messages of its own; decoration errors appear as `failed to load module for columns` and CSS or loading errors as `failed to load block columns`.
- Admin API: page status at `https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`.
- Author logs: AEM author logs and the page Timeline.

Common Errors:

Error	Likely Cause	Action
`failed to load module for columns`	Block without rows, or code error	Add a row or delete the block; else L3
Image not first on mobile	Image column has other content	Keep the image alone in its column
Unequal or unwanted widths	Equal split by design	Use another block or request a variant
Narrow text on small laptops	Many columns at 900px	Fewer columns or shorter text
Anchor ID rejected	Pattern not matched	Use `anchor-...`

Escalation Matrix:

Level	Team	Checks
L1	Author support	Rows and columns, column content, alt texts, Hide block, publication
L2	AMS / site admin	Replication, Admin API status, CDN cache, console errors, deployments
L3	Development team	`columns.js` / `columns.css`, filters, button decoration; local reproduction with `aem up`
L4	Adobe Support	Universal Editor columns component or Edge Delivery platform issues

Recovery Procedures: Restore the previous page version and republish; hide the block with Hide block; revert code regressions on `main`.

Health Checks: On a live page with the block, check that the block has `columns-{n}-cols`, image-only columns have `columns-img-col`, there are no console errors and the layout switches at 900px.

Operational Procedures: Standard publish workflow for content; feature branch, preview, pull request and merge for code. After `_columns.json` changes run `npm run build:json` and `npm run lint`.

Known Limitations:

- No style options, no column width ratios and no per-breakpoint column counts.
- The `columns-{n}-cols` class has no styles; the count only affects how many equal columns share the row.
- No tablet-specific layout: stacked until 900px.
- No gap between stacked columns or between rows other than content margins.
- An empty block (no rows) fails to decorate.
- Columns and Rows have no enforced range.

6.15 Cards Block

The Cards block shows **manually authored cards**: authors type each card's image, eyebrow, title, description, label, tags, date, link and call to action directly in the block. It is the authoring alternative to the List block (chapter 6.16), which builds its cards from the metadata of pages. Both blocks share one presentation, the TCS.com Design System (DS) Figma "List" variations 1–14, and one set of block settings, so a Cards block looks exactly like a List with the same settings: image, panel, report and statistic cards, a large feature card or rows of links, shown in a grid or a horizontal scroller chosen separately for desktop, tablet and mobile.

Source files: `blocks/cards/cards.js`, `blocks/cards/cards-data.js`, `blocks/cards/cards.css` (imports `styles/card-list.css`), `blocks/cards/_cards.json`, `blocks/cards/README.md`; shared with List: `decorators/card-list-decorator.js`, `utils/card-list-config.js`, `styles/card-list.css`, `models/_card-list.json`; unit tests: `test/cards/cards-data.test.mjs`.

6.15.1 Component Purpose

Business Objective

Let authors present curated, hand-written content as consistent, brand-compliant cards without creating a page for every card. Typical content is not a page of its own: statistics ("4bn revenue"), capabilities, services described in campaign wording, partner logos with a text, or a set of promotions pointing to pages or external sites. Because Cards uses the same design and settings as the List, a page can mix curated cards and page lists without visual differences.

Business Use Cases

- "At a glance" statistic rows on corporate, investor and about pages (big number, label and explanation, in brand or grey colours).
- Capability or service overviews with an image, a title, a rich text description and a link.
- Themed panels (dark, brand blue, image background) linking to topic hubs.
- A "featured" spotlight card followed by smaller cards (First card as feature).
- Report or download teasers (report cards with a "Download" call to action).
- Rows of links (title, eyebrow, description, date) for news-style link lists that are not driven by the query index.
- Horizontally scrolling card rails on mobile (or on every breakpoint) with a scroll indicator and previous / next buttons.
- Cards created with the earlier Block Collection Cards (image + rich text) keep working and render in the new design.

When To Use

- The cards' content is written for this placement and doesn't exist as page metadata.
- A card must link to an external site, to any URL, or to nothing at all.
- The description needs rich text (paragraphs, bold, bullet lists, inline links).
- Statistic cards with typed numbers and labels.
- The order of the cards is curated and must not change when pages are added.

When Not To Use

- The cards represent pages of the site that already have their metadata (card title, card description, card image): use the **List** block (6.16) so the cards update with the pages.
- The list must grow automatically with new child pages or tagged pages: use **List** (Child pages or Tagged pages source).
- Visitors must filter, sort or search the cards: use **Filter List**.
- A single image-and-text promotion: use **Teaser**.
- Slides that rotate: use the Gen Carousel block (6.24); Cards has no auto-rotation (by design).
- Tabular data: use **Table** or **Table List**.

Dependencies

Type	Dependency	Used for
Decorator	`decorators/card-list-decorator.js` (`buildCardListHeader`, `renderCardList`, `renderCardListEmpty`, `applyCardListLayout`, `cardListId`, `trackCardList`)	Header, cards, feature card, rows of links, grid / scroll classes, empty state, data layer state (shared with List)
Decorator	`decorators/scroller-decorator.js` `createScroller` (through the card list decorator)	Previous / next buttons, scroll indicator, mouse drag, keyboard, loop, initial item
Decorator	`decorators/button-decorator.js`, `decorators/icon-button-decorator.js`	Header CTA, card calls to action, previous / next icon buttons

Type	Dependency	Used for
Decorator	`decorators/link-decorator.js` (`setLinkTarget`, `announceNewTab`)	`target="_blank"`, `rel="noopener noreferrer"` and the "(opens in a new tab)" text
Decorator	`decorators/heading-decorator.js` `buildHeading`	The block title in the chosen heading level
Decorator	`decorators/responsive-image-decorator.js` `buildImage`	Card images (the authored picture is cloned)
Decorator	`decorators/data-layer-decorator.js`	Page entry (`trackPage`) and the click events (`cmp:click`)
Util	`utils/card-list-config.js` (`normalizePresentation`, `kindOf`, `isSafeUrl`, `toISODate`, …)	Settings, defaults, validation, card kind (shared with List)
Util	`blocks/cards/cards-data.js` (`buildManualCards`, `itemToCard`)	A Card's values as the card model (no DOM; unit tested)
Util	`utils/cells.js` `matchCells`	Reads Card cells when AEM leaves out fields that were never set
Util	`utils/properties.js`, `utils/tags.js`, `utils/taxonomy.js` `resolveTags`	Key-value rows, tag values, tag labels
Util	`utils/dates.js` `formatDate`	The card date in the block's Date format
Util	`utils/placeholders.js` `getPlaceholders`	UI text (see below)
Util	`utils/environment.js` `isEditing`	Editor hint instead of hiding an empty block
Util	`utils/analytics.js` `pushComponents`, `utils/ids.js` `uniqueId`	Data layer state; generated ids
Script	`scripts/scripts.js` `moveInstrumentation`	Keeps Universal Editor instrumentation on rebuilt cards and the rich text
Styles	`styles/card-list.css` (imported by `blocks/cards/cards.css`)	The whole presentation (shared with List)
Sheet	`/{locale}/placeholders.json`	`cards`, `cardsEditorHint`, `listPrevious`, `listNext`, `opensInNewTab`, `explore`, `download`, `listEmpty`, `minutesShort`
Sheet	`/{locale}/taxonomy.json`	Labels of the cards' tags (fallback: labels derived from the tag IDs)
Data sources	None	Cards doesn't read the query index, the redirects sheet or the Authors sheet

Related Components

- **List** (6.16): the same presentation for pages (child pages, fixed list, tagged pages).
- **Filter List** (6.17): page discovery with filter pills and sorting.
- **Teaser** (6.11) and **Hero** (6.10): single promotions.
- **Columns** (6.14): free side-by-side layouts without card styling.
- **Section** (6.5, with its style and width options): the container of the block.

Applicable Templates

Any page: the section filter allows Cards in every section, and no page template (home, article, blog, press-release) adds or removes it. Typical placements are home and landing pages (statistics, capabilities), article and blog pages (related content curated by hand) and press-release pages (downloads).

Supported Content Types

Manually entered content per card: image from AEM Assets (with alt text), plain text (eyebrow, title, label, content type, reading time), rich text (description), AEM tags (taxonomy), a date, an internal page link or an external URL (`https://…`, `mailto:`, `tel:`), and display options (card style, card colour). No page metadata, query index or content fragment is used.

6.15.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A Cards block at 1440 px wide with default settings: the header (title on the left, CTA text link with arrow on the right) above four image cards in one row (16:9 images with rounded corners, uppercase eyebrow and tags, title, description, content type and date).

Desktop (1200 px and wider). The block is a vertical stack with a 5rem gap between the header, the cards and the navigation row. The header puts the title slot (title and description) on the left and the CTA on the right, aligned to the bottom; the title is 2.5rem, semibold, line height 1.4, near-black (`--color-text-default`), the description 1.125rem medium grey (`--color-text-secondary`, #999). By default the cards are a 4-column grid (`--list-columns-desktop`, 1–6) with a 1.5rem gap (Desktop gap, 24 px). Image cards show the image in a 16:9 frame with a 1rem radius (`--radius-l`) and a light grey placeholder background (#d9d9d9), then the kicker (eyebrow, up to 3 tags and the reading time, uppercase, 0.875rem, separated by "|"), the card title (1.5rem, medium, line height 1.42), the description (1rem, 70 % black) and the meta row (content type and date, 1rem). Panel, report and statistic cards are coloured panels with a 1.5rem padding (from 900 px) and a 2:3 portrait shape by default. In horizontal scroll mode, 4.2 cards are visible by default (the fifth card is cut off to show that the row scrolls) and the row runs to the right edge of the section (edge bleed).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Cards block at 1024 px: the header with a 2rem title, four narrower image cards in one row (default tablet columns 4), smaller card titles (1.25rem).

Tablet (600–1199 px). The layout uses the tablet settings: grid with 4 columns by default (Tablet columns, 1–6) and a 1.5rem gap, or a horizontal scroller with 3.3 cards visible by default. The block gap stays 5rem. Type sizes step down: block title 2rem (line height 1.4), card titles 1.25rem, kicker and meta 0.875rem, panel titles 1.5rem, statistic numbers up to 4rem. The card padding of coloured cards is 1rem below 900 px and 1.5rem from 900 px. Rows of links lose their bottom border unless the Dividers style option is set. Because four columns can be narrow on a 600 px screen, authors often set Tablet columns to 2 or 3 or use horizontal scroll.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Cards block at 402 px: the header stacked (title, description, CTA below), one image card per row in a single column with a 1rem gap.

Mobile (below 600 px). One column by default (Mobile columns 1) with a 1rem gap (Mobile gap, 16 px); the block gap is 2.5rem. The header wraps: the title slot takes the full width and the CTA moves below it. Type: block title 1.75rem (line height 1.39), card titles 1.125rem, kicker, meta and description 0.75rem, statistic numbers up to 5rem (they shrink to fit narrow cards, at most 26 % of the card width). In horizontal scroll mode 1.2 cards are visible by default and the scroll indicator (a 0.25rem grey track with a darker thumb) is shown by default; the scroller bleeds to the right edge of the screen.

6.15.3 Component Variations

The Cards block has no "variant" select of its own: its presentation comes from the shared settings of `models/_card-list.json` (Cards, Layout, Scroll and Advanced tabs) and from each Card's own Card style and Card colour. Every distinct mode an author can choose is documented below as its own variation. The TCS DS Figma variations 1–14 are combinations of these modes; they are documented one by one for the List in 6.16.3 and are reproduced in Cards with the same settings (Cards README):

Figma variation	Cards settings
1 Themes	Card style Panel; each card's Card colour (dark, brand, image); Mobile display Horizontal scroll with indicator
2 Insight & Opinion	Card style Mixed; First card as feature; Image ratio 1:1; header CTA link
3 Related; 8 Media coverage	Mixed or Report; Horizontal scroll on every breakpoint
4 Intelligence in Action; 11 2-up	Image cards; Image ratio 1:1; 3 or 2 columns
5 News	Display as teaser off; Style option Dividers between rows
6, 7, 10, 12, 13, 14 Statistics	Card style Statistic (Title: the number; Label; Description); card colours; Compact statistic title

Variation 1 — Image Card (default)

Variation Name: Card style "Image card" (`image`, the default).

Business Purpose: The standard editorial card: a picture with a short text and a link, for capabilities, services and curated content.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-image-card.png`
> **Purpose:** Image card style
> **Description:** Four image cards in a row: 16:9 rounded image, uppercase grey kicker (eyebrow | tag | 6m), card title, two-line description, meta row "Article | 04 Sep 26".

Use Cases: Service and capability overviews; "Explore more" rows; related content.

Authoring Guidelines: Cards tab: Display as teaser on, Card style = Image card. Per Card: Image and Image alt text, Eyebrow, Title, Description, optional Tags, Date, Content type, Reading time, Link (and optionally Call to action label).

Business Rules: A card needs a Title, a Description or an Image, otherwise it isn't shown. The image shows only when Show image is on and the card has an image. Up to 3 tags are shown. Image cards have no default call to action: without a Call to action label the title is the link.

Rendering Rules: `article.list-card.list-card-image` with, in order: `div.list-card-media` (the authored `<picture>` cloned; `loading` from Image loading), `p.list-card-kicker` (eyebrow, tags, reading time — the

reading time only when Show tags is on), the title heading (`.list-card-title`, one level below the block title), the description (`div.list-card-body` rich text), `p.list-card-meta` (content type, `<time datetime>`), and `p.list-card-action` when a call to action is shown (tertiary button). The card's link covers the whole card (`.list-card-link::before`, `inset: 0`).

Responsive Behaviour: Desktop: title 1.5rem, kicker and description 0.875rem / 1rem; grid of 4 columns by default. Tablet: title 1.25rem, kicker / meta 0.875rem; 4 columns by default. Mobile: title 1.125rem, kicker / meta / description 0.75rem; 1 column. The image keeps its ratio (16:9 by default) at every breakpoint, so there is no layout shift.

SEO Impact: Real headings and links with the title as anchor text; images with alt text are indexable. Card titles are headings one level below the block heading, keeping the outline valid.

Accessibility Compliance: `<ul>`/`<li>` list in a labelled region; one link per card; decorative images `alt=""` when Image alt text is empty; visible focus ring around the whole card (WCAG 2.4.7). The grey kicker colour (#999) on white is about 2.8:1, below the 4.5:1 of WCAG 1.4.3 for small text — [To be confirmed: design sign-off of the Figma secondary text colour — owner: design lead].

Variation 2 — Panel Card

Variation Name: Card style "Panel" (`panel`).

Business Purpose: A bold coloured panel with a large title and an "Explore" call to action, for themes, topic hubs and navigation tiles (Figma variation 1 "Themes").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-panel-card.png`
> **Purpose:** Panel card style
> **Description:** Three tall (2:3) dark navy-grey panels (#20293c) with a small light kicker, a large light title, a short description and an "Explore" text link at the bottom.

Use Cases: Theme and topic tiles; industry entry points; campaign navigation.

Authoring Guidelines: Cards tab: Card style = Panel (or a Card's own Card style = Panel); optional Card colour; Panel height. Per Card: Eyebrow (or a Tag), Title, Description, Link; optionally a Call to action label (else "Explore").

Business Rules: The default call to action is the placeholder `explore` ("Explore"), shown when the card is linked, Link items is on and Show call to action is on. The kicker shows the eyebrow, else the first tag. Images aren't shown on panels unless Card colour is Image background (Variation 15).

Rendering Rules: `article.list-card.list-card-panel` (padding 1rem, 1.5rem from 900 px; radius 1rem; `overflow: hidden`), background `--list-card-bg` (dark #20293c by default), text #f3f3f3, muted kicker and description in 70 % white. Title 1.75rem / 1.5rem / 1.75rem (mobile / tablet / desktop), regular weight, line height 1.21. The action paragraph is pushed to the bottom (`margin-top: auto`); the tertiary button takes the card's light text colour.

Responsive Behaviour: Desktop: 2:3 portrait panels, description 1rem. Tablet: description 0.875rem, title 1.5rem. Mobile: description 0.75rem, title 1.75rem, 1 column or a scroller. With Panel height Portrait a long text is clipped (`overflow: hidden`) — keep texts short or use Panel height Content.

SEO Impact: As Variation 1; the call to action carries `aria-label="Explore: {title}"`, so the link's name includes the card title.

Accessibility Compliance: Light text on the dark panel meets WCAG 1.4.3; the call to action is the card's only link (no duplicate links); the accessible name joins the label and the title (WCAG 2.4.4).

Variation 3 — Report Card

Variation Name: Card style "Report" (`report`).

Business Purpose: A download-style teaser for reports, studies and white papers, with a "Download" call to action (Figma variations 8 and 9).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-report-card.png`
> **Purpose:** Report card style
> **Description:** Tall cards with the grey gradient background (light blue-grey to warm grey), a kicker "REPORT | 04 Sep 26", a large dark title, a description and a "Download" outlined button at the bottom.

Use Cases: Reports, research, annual report downloads, investor documents.

Authoring Guidelines: Card style = Report (block or Card). Per Card: Content type (e.g. Report; shown in the kicker), Date, Title, Description, Link (e.g. a PDF in AEM Assets or a page); optionally Call to action label (else "Download").

Business Rules: The kicker shows the content type, else the eyebrow, and the date. The default call to action is the placeholder `download` ("Download"). No image, no tags.

Rendering Rules: `article.list-card.list-card-report`, background `--list-gradient` (`linear-gradient(135deg, #cfd9e3, #e2e1dd)`), text #303030, muted text 70 % black; the call to action is a secondary (outlined) button. A link to a file (`.pdf`, `.docx`, `.zip`, …) is reported as `xdm:linkType: download` in the data layer.

Responsive Behaviour: As panels (2:3 by default; title 1.75rem / 1.5rem / 1.75rem; description 1rem / 0.875rem / 0.75rem).

SEO Impact: The title is a heading; the button's accessible name is "Download: {title}". File links aren't indexed as pages; describe the file in the title and description.

Accessibility Compliance: Dark text on the light gradient meets WCAG 1.4.3. State the file type and size in the description when linking to files (WCAG 2.4.4 recommended practice).

Variation 4 — Statistic Card

Variation Name: Card style "Statistic" (`stat`).

Business Purpose: Key figures with a large number, a label and an explanation ("At a glance", "Banking today"; Figma variations 6, 7, 10, 12, 13, 14).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-statistic-card.png`
> **Purpose:** Statistic card style
> **Description:** Four light grey (#d9d9d9) cards, each with a small kicker at the top, a very large number such as "4bn", and at the bottom a bold label "Revenue" and a short description.

Use Cases: Company facts, investor highlights, impact numbers, survey results.

Authoring Guidelines: Card style = Statistic. Per Card: Eyebrow (optional, small label at the top), Title = the number (e.g. `4bn`, `600,000+`), Label (`subtitle`, e.g. Revenue), Description; optional Link with a Call to action label.

Business Rules: Statistic cards have no default call to action: a link shows a button only with a Call to action label; otherwise the number (title) is the link. No image, tags or date are shown on statistic cards.

Rendering Rules: `article.list-card.list-card-stat` with `div.list-card-top` (kicker, number) and `div.list-card-bottom` (`p.list-card-subtitle`, description), spread top and bottom (`justify-content: space-between`). The number uses `min(var(--list-stat-number-size), 26cqi)` (6rem desktop, 4rem tablet, 5rem mobile at most; it shrinks with narrow cards) and wraps anywhere. Default background #d9d9d9, text #0c0c0d. The call to action is a secondary button.

Responsive Behaviour: Desktop: number up to 6rem, label and description 1.125rem. Tablet: number up to 4rem, texts 1rem. Mobile: number up to 5rem; usually a scroller (Figma 6, 10).

SEO Impact: The number is a heading; give the Label and Description the context (a number alone has little meaning for search engines and screen readers).

Accessibility Compliance: The label and description follow the number in reading order; avoid units that are only visual (use "4bn" rather than an image of a number).

Variation 5 — Mixed Card Style

Variation Name: Card style "Mixed (by page)" (`mixed`).

Business Purpose: One block that mixes image cards, panels and report cards automatically, as in Figma variations 2, 3, 8 and 9.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-mixed.png`
> **Purpose:** Mixed card style
> **Description:** A row with an image card, a dark panel (a card without an image) and a gradient report card (Content type "Report") side by side.

Use Cases: Insight rows that combine articles, opinions and reports.

Authoring Guidelines: Card style = Mixed. Per Card: set Content type `Report` for report cards, add an Image for image cards, leave the image empty for panels — or set the Card's own Card style, which always wins.

Business Rules: `kindOf()`: a Card's own Card style first; else a content type containing "report" (any case) gives a report card; else a card with an image is an image card; else a panel.

Rendering Rules: Each `<li>` holds the card kind's markup (Variations 1–4); the block has the class `style-mixed`; the data layer state reports `xdm:variant: "mixed"`.

Responsive Behaviour: As each card kind. Panels (2:3) and image cards (image ratio plus text) have different heights: keep Stretch cards to row height on (default) so a row lines up.

SEO Impact: As the card kinds.

Accessibility Compliance: As the card kinds; the mixed colours don't change the reading order.

Variation 6 — Feature Card (First card as feature)

Variation Name: First card as feature (`featureFirst` on).

Business Purpose: A large spotlight for the most important item, above the other cards (Figma variation 2 "Insight & Opinion").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-feature-card.png`
> **Purpose:** Feature card
> **Description:** A full-width card (about 35.6rem high on desktop) with the image filling it, a white gradient from the left, the tags and reading time and a large dark title on the left; below it a row of three smaller cards.

Use Cases: Lead story of an insights section; campaign spotlight.

Authoring Guidelines: Cards tab: Display as teaser on, First card as feature on. The first Card should have an Image, a Title, Tags and Reading time, and a Link.

Business Rules: Only with Display as teaser on and at least one card. The first shown card becomes the feature and is removed from the list. The feature always shows its image (even with Show image off). It shows tags (up to 3) and the reading time (with Show tags on) and the title — not the eyebrow, description, date or call to action. Cards has no author lockup on the feature card (the List takes it from the page's author).

Rendering Rules: `article.list-feature` before the list region: the image (`div.list-card-media`, absolute, cover) under a white gradient (`90deg`, 90 % → 70 % white → transparent) and `div.list-feature-content` (kicker, `.list-feature-title`, semibold, max width 31.25rem). The block has the class `feature-first`. Clicks on the feature card are reported with the block's data layer path (it isn't one of the list items).

Responsive Behaviour: Minimum height 35.625rem (desktop), 27.1875rem (tablet), 32.5rem (mobile); title 2.5rem (tablet and desktop) / 2rem (mobile); content padding 2.5rem (tablet+) / 1.5rem (mobile). Images 1600 px wide from 900 px, 750 px below (when built from a URL).

SEO Impact: The feature title is a heading at the card level; the image is a real `<img>` with its alt text.

Accessibility Compliance: Dark title on the white gradient; check legibility with busy images. The feature is a separate `<article>` outside the `<ul>`, so screen readers announce the list with one item less.

Variation 7 — Rows of Links (Display as teaser off)

Variation Name: Display as teaser off (`links`).

Business Purpose: A text-only list of links with large titles and arrows, for news and announcement lists (Figma variation 5 "News").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-rows-of-links.png`
> **Purpose:** Rows of links
> **Description:** Full-width rows, each with an uppercase eyebrow and a tag, a large title, a description, a date and a grey arrow on the right that turns blue and moves on hover; thin grey lines between rows.

Use Cases: News lists, announcements, press links, document lists.

Authoring Guidelines: Cards tab: Display as teaser off. Per Card: Eyebrow (or Content type), Tags (the first one is shown), Title, Description, Date, Link. Add the Style option Dividers between rows for lines on tablet and desktop.

Business Rules: Card style, Card colour, Image ratio, Panel height and First card as feature are hidden in the dialog and ignored. Images and calls to action aren't shown. The arrow shows only for a linked card with a title. Always one row per line (columns ignored).

Rendering Rules: Block classes `links`; `article.list-row` (grid: text column and arrow column; padding 2.5rem top and bottom) with `p.list-card-kicker` (eyebrow or content type, then the first tag in normal case), `.list-`

`row-title`, `div.list-row-description`, `p.list-card-meta` (date) and `span.list-row-arrow` (`aria-hidden`). Data layer variant `links`.

Responsive Behaviour: Desktop: title 2.5rem medium, arrow 3rem centred. Tablet: title 2rem semibold, arrow 3rem; no borders unless Dividers. Mobile: title 1.75rem, arrow 2rem at the top, grey border under every row.

SEO Impact: Text-rich rows with descriptive anchor texts are good for internal linking.

Accessibility Compliance: The arrow is decorative (`aria-hidden="true"`); one link per row (the title) covering the row; hover and focus both animate the arrow (`:focus-within`), with no animation for reduced motion.

Variation 8 — Card Colour "Dark"

Variation Name: Card colour "Dark" (`dark`).

Business Purpose: The dark navy-grey Figma panel colour (#20293c) for strong contrast tiles.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-colour-dark.png`
> **Purpose:** Dark card colour
> **Description:** A statistic card and a report card both rendered on #20293c with light text.

Use Cases: Themes; statistics on a light section; making one card stand out.

Authoring Guidelines: Cards tab: Card colour = Dark (all cards), or a Card's Display tab: Card colour = Dark (that card).

Business Rules: A Card's own colour wins over the block's. Colours change panel, report and statistic cards; on image cards they have no visible effect. Dark is the panel default.

Rendering Rules: Class `theme-dark` on the `article`: background #20293c, text #f3f3f3, muted text 70 % white, statistic kicker #f3f3f3; tertiary and secondary buttons switch to the light text colour.

Responsive Behaviour: Colour is the same on every breakpoint.

SEO Impact: None (presentation only).

Accessibility Compliance: Light text on #20293c exceeds 4.5:1 (WCAG 1.4.3).

Variation 9 — Card Colour "Navy"

Variation Name: Card colour "Navy" (`navy`).

Business Purpose: The brand's darkest blue (`--brand-600`, #011627), used for report cards in Figma variations 8 and 9.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-colour-navy.png`
> **Purpose:** Navy card colour
> **Description:** Report cards on a near-black navy background with light text and a light outlined Download button.

Use Cases: Media coverage and investor relations report cards; premium highlights.

Authoring Guidelines: Card colour = Navy on the block or on a Card.

Business Rules: As Variation 8.

Rendering Rules: `theme-navy`: background #011627, text #f3f3f3, muted 70 % white; buttons in the light colour.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Very high contrast for light text (WCAG 1.4.3 and 1.4.6).

Variation 10 — Card Colour "Brand blue"

Variation Name: Card colour "Brand blue" (`brand`).

Business Purpose: The TCS brand blue (#4673db) for highlighted statistics and themes (Figma variations 1, 10 and 13).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-colour-brand.png`
> **Purpose:** Brand blue card colour
> **Description:** A statistic card on brand blue with white number, label and description.

Use Cases: One highlighted statistic in a row; brand themes.

Authoring Guidelines: Card colour = Brand blue on the block or a Card.

Business Rules: As Variation 8.

Rendering Rules: `theme-brand`: background `--color-bg-brand` (#4673db), text white, muted text 60 % white, statistic kicker #f3f3f3; buttons in white.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: White on #4673db is about 4.2:1: it meets WCAG 1.4.3 for large text (numbers, panel titles) but not for small body text, and the 60 % white muted text is lower. Keep descriptions on brand blue short and large, or use Navy / Dark — [To be confirmed: design sign-off — owner: design lead].

Variation 11 — Card Colour "Light blue"

Variation Name: Card colour "Light blue" (`brand-light`).

Business Purpose: A soft brand tint (#a7c8f1) for friendly statistic tiles (Figma variation 13).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-colour-brand-light.png`
> **Purpose:** Light blue card colour
> **Description:** A statistic card on light blue with a black number and dark description.

Use Cases: Statistic rows with alternating colours.

Authoring Guidelines: Card colour = Light blue.

Business Rules: As Variation 8.

Rendering Rules: `theme-brand-light`: background `--brand-300`, text #0c0c0d, muted text 70 % black; buttons keep their default (dark) colours.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Dark text on light blue meets WCAG 1.4.3.

Variation 12 — Card Colour "Light grey"

Variation Name: Card colour "Light grey" (`grey`).

Business Purpose: The neutral grey (#d9d9d9); the statistic default, also usable for panels and reports (Figma variation 6).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-colour-grey.png`
> **Purpose:** Light grey card colour
> **Description:** Panel cards on light grey with black titles instead of the dark default.

Use Cases: Calm, neutral panels; "At a glance" statistics.

Authoring Guidelines: Card colour = Light grey.

Business Rules: As Variation 8. On a panel the tertiary "Explore" button keeps the panel's text colour (black on grey).

Rendering Rules: `theme-grey`: background `--color-bg-tertiary` (#d9d9d9), text #0c0c0d, muted 70 % black.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Meets WCAG 1.4.3 for all text.

Variation 13 — Card Colour "Mid grey"

Variation Name: Card colour "Mid grey" (`mid-grey`).

Business Purpose: A darker neutral (#999) for contrast in colour rows (Figma variation 13).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-colour-mid-grey.png`
> **Purpose:** Mid grey card colour
> **Description:** A statistic card on mid grey with a black number and dark grey description.

Use Cases: Four-up statistic rows with different colours.

Authoring Guidelines: Card colour = Mid grey.

Business Rules: As Variation 8.

Rendering Rules: `theme-mid-grey`: background `--grey-400` (#999), text #0c0c0d, muted text `--black-600`.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Black on #999 is about 6.9:1; check the muted description text (semi-transparent black) on #999 with a contrast checker.

Variation 14 — Card Colour "Grey gradient"

Variation Name: Card colour "Grey gradient" (`gradient`).

Business Purpose: The Figma grey gradient (the report default) for any card kind (Figma variation 13).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-colour-gradient.png`
> **Purpose:** Grey gradient card colour
> **Description:** A statistic card with the diagonal light blue-grey to warm grey gradient and dark text.

Use Cases: Statistic or panel cards that should match report cards.

Authoring Guidelines: Card colour = Grey gradient.

Business Rules: As Variation 8.

Rendering Rules: `theme-gradient`: background `linear-gradient(135deg, #cfd9e3, #e2e1dd)`, text #303030, muted 70 % black.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Dark text on the light gradient meets WCAG 1.4.3.

Variation 15 — Card Colour "Image background"

Variation Name: Card colour "Image background" (`image`).

Business Purpose: A panel with the card's image filling it behind the text (Figma variation 1 "Themes").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-colour-image.png`
> **Purpose:** Image background card colour
> **Description:** A tall panel whose image covers the whole card, with the kicker, title, description and Explore link on top of it.

Use Cases: Theme tiles with photography between solid-colour panels.

Authoring Guidelines: Card style = Panel (block or Card), Card colour = Image background, and an Image (with alt text if it carries meaning). Show image must be on.

Business Rules: Only panel cards render the image behind the text; report and statistic cards ignore the image (they only get the light background). Not intended for image cards (their image would be positioned behind the card's text without a panel background).

Rendering Rules: `theme-image`: background #f5f5f5, text #303030, muted #999; the `div.list-card-media` is absolute (`inset: 0`, cover), the texts are positioned above it. Images are requested 1600 px wide from 900 px and 750 px below (when built from a URL).

Responsive Behaviour: The image is cropped to the panel's shape (2:3 portrait, 9:10 compact or the content height) at every breakpoint.

SEO Impact: The image is a real `<img>` with its alt text.

Accessibility Compliance: There is no overlay behind the text: choose light, calm images so the dark text keeps a 4.5:1 contrast (WCAG 1.4.3); check every image.

Variation 16 — Image Ratio 1:1
Variation Name: Image ratio "1:1" (`1-1`).

Business Purpose: Square images, as in Figma variations 2, 4 and 11.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-ratio-1-1.png`
> **Purpose:** Square image ratio
> **Description:** Three image cards with square images above their texts.

Use Cases: Portrait-like photography; "Intelligence in Action" and "2-up" layouts.

Authoring Guidelines: Cards tab: Image ratio = 1:1 (Display as teaser on).

Business Rules: Applies to image cards (and the image of a Mixed image card); the feature card and image-background panels are not affected.

Rendering Rules: Block class `ratio-1-1`; `.list-card-media { aspect-ratio: 1 }`, the image is cropped with `object-fit: cover`.

Responsive Behaviour: Square at every breakpoint; the image height follows the column width.

SEO Impact: None (the same image, cropped by CSS).

Accessibility Compliance: Cropping must not cut off content that the alt text describes.

Variation 17 — Image Ratio 4:3
Variation Name: Image ratio "4:3" (`4-3`).

Business Purpose: A taller landscape image for product and people photography.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-ratio-4-3.png`
> **Purpose:** 4:3 image ratio
> **Description:** Image cards with 4:3 images.

Use Cases: Photography that would be cropped too much at 16:9.

Authoring Guidelines: Image ratio = 4:3.

Business Rules: As Variation 16.

Rendering Rules: Class `ratio-4-3`; `aspect-ratio: 4 / 3`, cover.

Responsive Behaviour: 4:3 at every breakpoint.

SEO Impact: None.

Accessibility Compliance: As Variation 16.

Variation 18 — Image Ratio 3:2
Variation Name: Image ratio "3:2" (`3-2`).

Business Purpose: The classic photo ratio between 16:9 and 4:3.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/cards-ratio-3-2.png`
> **Purpose:** 3:2 image ratio
> **Description:** Image cards with 3:2 images.

Use Cases: Camera photography without re-cropping.

Authoring Guidelines: Image ratio = 3:2.

Business Rules: As Variation 16. The default ratio is 16:9 (Variation 1).

Rendering Rules: Class `ratio-3-2`; `aspect-ratio: 3 / 2`, cover.

Responsive Behaviour: 3:2 at every breakpoint.

SEO Impact: None.

Accessibility Compliance: As Variation 16.

Variation 19 — Panel Height "Compact"

Variation Name: Panel height "Compact" (`compact`, 9:10).

Business Purpose: Shorter, almost square coloured cards for statistic rows and dense panels.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/cards-panel-compact.png`
> **Purpose:** Compact panel height
> **Description:** Four statistic cards that are nearly square (9:10) instead of tall portrait.

Use Cases: Statistic rows (Figma 6, 10, 13); panels with short texts.

Authoring Guidelines: Cards tab: Panel height = Compact.

Business Rules: Applies to panel, report and statistic cards; image cards and rows aren't affected. The default is Portrait (2:3).

Rendering Rules: Block class `panel-compact`; `aspect-ratio: 9 / 10` on the coloured cards; `overflow: hidden` clips text that doesn't fit.

Responsive Behaviour: The ratio is kept at every breakpoint; on narrow cards long texts may be clipped — use Content (Variation 20) for long texts.

SEO Impact: None (clipped text is still in the HTML).

Accessibility Compliance: Text clipped by `overflow: hidden` is still read by screen readers but invisible to sighted users (WCAG 1.4.10 risk): keep the texts short.

Variation 20 — Panel Height "Content"

Variation Name: Panel height "Content" (`auto`).

Business Purpose: Coloured cards as tall as their content, for texts of varying length.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-panel-content.png`
> **Purpose:** Content panel height
> **Description:** Dark panels whose height fits the text; the cards of a row share the tallest height.

Use Cases: Panels with long descriptions or rich text lists.

Authoring Guidelines: Panel height = Content.

Business Rules: No fixed shape; with Stretch cards to row height on (default) the cards of a row are as tall as the row.

Rendering Rules: Block class `panel-auto`; no aspect ratio on the coloured cards.

Responsive Behaviour: Heights follow the text at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Nothing is clipped; text reflows (WCAG 1.4.10).

Variation 21 — Style Option "Compact statistic title"

Variation Name: Style options: Compact statistic title (`stat-compact`).

Business Purpose: Statements and short phrases in statistic cards instead of a single number (Figma variations 7 and 12 "Statements").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-stat-compact.png`
> **Purpose:** Compact statistic title
> **Description:** Three statistic cards with a medium-size title such as "Net zero by 2030" instead of a huge number, a label and a description, and a call to action button.

Use Cases: Commitments, statements, short claims.

Authoring Guidelines: Card style = Statistic; Advanced tab: Style options = Compact statistic title. Often 3 columns and a Call to action label per Card.

Business Rules: Only affects statistic cards.

Rendering Rules: Block class `stat-compact`; the statistic title size becomes 0.42 × the number size (about 2.52rem desktop, 1.68rem tablet, 2.1rem mobile), without the 26 % width cap.

Responsive Behaviour: As statistic cards, with the smaller title.

SEO Impact: None.

Accessibility Compliance: As Variation 4.

Variation 22 — Style Option "Dividers between rows"

Variation Name: Style options: Dividers between rows (`dividers`).

Business Purpose: Grey lines between the rows of links on tablet and desktop (Figma variation 5).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-dividers.png`
> **Purpose:** Dividers between rows
> **Description:** Rows of links on desktop separated by thin grey (#b2b2b2) lines.

Use Cases: News and document lists that need visual separation.

Authoring Guidelines: Display as teaser off; Advanced tab: Style options = Dividers between rows.

Business Rules: Only affects rows of links. On mobile the rows always have dividers.

Rendering Rules: Block class `dividers`; from 600 px `.list-row` keeps its `border-bottom` (1 border width, `--slate-400`) instead of a transparent one.

Responsive Behaviour: Desktop and tablet: lines with the option, none without. Mobile: lines always.

SEO Impact: None.

Accessibility Compliance: Decorative borders only.

Variation 23 — Style Option "Header centred"

Variation Name: Style options: Header centred (`header-center`).

Business Purpose: A centred title, description and CTA above the cards.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-header-centred.png`
> **Purpose:** Centred header
> **Description:** The block title, description and CTA centred above a row of statistic cards.

Use Cases: Statistic rows and campaign sections with a centred heading.

Authoring Guidelines: Advanced tab: Style options = Header centred.

Business Rules: Only the header is centred; the cards keep their alignment.

Rendering Rules: Block class `header-center`: `.list-header` becomes a centred column (`text-align: center`), the CTA below the texts.

Responsive Behaviour: Centred on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Keep centred texts short (long centred paragraphs are harder to read).

Variation 24 — Header Call to Action

Variation Name: Header with a CTA (CTA link, CTA text, CTA style: Text link, Primary button, Secondary button).

Business Purpose: A "View all" or "Explore more" link next to the title.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-header-cta.png`
> **Purpose:** Header call to action

> **Description:** The block header with the title and description on the left and a "View all" text link with an arrow on the right; a second capture with a primary (filled) button.

Use Cases: Linking to an overview page of all items.

Authoring Guidelines: Content tab: CTA link (a page or URL), CTA text, CTA style (Text link = default, Primary button, Secondary button).

Business Rules: The CTA shows only with a CTA link; CTA text replaces the link text. Values other than primary / secondary give the text link (tertiary).

Rendering Rules: `div.list-cta-slot` with the link decorated by the button decorator (`button tertiary` / `primary` / `secondary`), `data-cmp-interaction="cta"` (data layer interaction `cta`).

Responsive Behaviour: Right of the title slot, aligned to its bottom, when it fits (the title slot is at least 20rem); otherwise it wraps below the title (mobile).

SEO Impact: A descriptive CTA text ("View all insights") helps internal linking; avoid "Click here".

Accessibility Compliance: A link with a visible name; the button decorator handles focus styles and new-window links.

Variation 25 — Grid on Every Breakpoint

Variation Name: Desktop, Tablet and Mobile display = Grid (the default).

Business Purpose: All cards visible at once in columns (Figma variations 2 and 11).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/cards-grid-all.png`
> **Purpose:** Grid on every breakpoint
> **Description:** Eight cards in two rows of four on desktop; the same cards in a single column on mobile.

Use Cases: Overviews where every card must be visible without interaction.

Authoring Guidelines: Layout tab: Desktop / Tablet / Mobile display = Grid; set Desktop / Tablet / Mobile columns (1–6; defaults 4 / 4 / 1) and gaps (0–120 px; defaults 24 / 24 / 16).

Business Rules: Columns are validated (1–6, rounded); gaps (0–120); invalid values fall back to the defaults. Cards wrap to new rows.

Rendering Rules: Block classes `desktop-grid tablet-grid mobile-grid`; custom properties `--list-columns-{bp}` and `--list-gap-{bp}` (px converted to rem); `grid-template-columns: repeat(var(--list-columns), minmax(0, 1fr))`. No previous / next buttons or indicator.

Responsive Behaviour: Desktop: Desktop columns; tablet: Tablet columns; mobile: Mobile columns, each with its gap.

SEO Impact: All cards are in the HTML and visible.

Accessibility Compliance: No horizontal scrolling; content reflows (WCAG 1.4.10).

Variation 26 — Horizontal Scroll on Every Breakpoint

Variation Name: Desktop, Tablet and Mobile display = Horizontal scroll.

Business Purpose: A card rail that scrolls sideways on all devices (Figma variations 3, 8 and 9).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-scroll-all.png`
> **Purpose:** Horizontal scroll on every breakpoint
> **Description:** A row of cards with 4.2 visible on desktop, the fifth card cut off at the right edge of the screen, and the previous / next buttons and a scroll indicator below.

Use Cases: Related content, media coverage, long sets of cards in little vertical space.

Authoring Guidelines: Layout tab: each Display = Horizontal scroll; set Items visible (1–8; defaults 4.2 / 3.3 / 1.2) or Card width (px); optionally Previous / next buttons and Scroll indicator per breakpoint; the Scroll tab for snap, drag, keyboard, step, loop and edge.

Business Rules: Items visible can be a decimal (the fraction shows part of the next card). The scroller is enhanced only when the cards don't fit (`data-overflow="true"`); otherwise it behaves like a row and the buttons and indicator are hidden.

Rendering Rules: Block classes `desktop-scroll tablet-scroll mobile-scroll`; `.list-scroller` has `overflow-x: auto` and `overscroll-behavior-x: contain`; the `<ul>` is a flex row; each item's width is the Card width or `(100cqi - bleed - (visible - 1) × gap) / visible`. Scroll snap (`x mandatory`, start) and hidden scrollbars by default. No carousel library and no auto-rotation.

Responsive Behaviour: Each breakpoint uses its own Items visible / Card width, gap, buttons and indicator; resizing or rotating recalculates (ResizeObserver).

SEO Impact: All cards are in the HTML (none are loaded later), so all links are crawlable.

Accessibility Compliance: While it scrolls the region is focusable (`tabindex="0"`) and scrolls with arrow keys, Home and End; links stay reachable with Tab; nothing traps focus; buttons are named and `disabled` at the ends.

Variation 27 — Scroll on Desktop Only

Variation Name: Desktop = Horizontal scroll; Tablet = Grid; Mobile = Grid.

Business Purpose: A single rail on wide screens, all cards stacked on smaller screens.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-scroll-desktop-only.png`
> **Purpose:** Desktop-only scroll
> **Description:** Desktop: a scrolling row of cards with buttons; tablet: a 2-column grid; mobile: one column.

Use Cases: Long sets on wide screens where a grid would be too tall.

Authoring Guidelines: Desktop display = Horizontal scroll (Items visible, buttons, indicator); Tablet and Mobile display = Grid (columns).

Business Rules: Breakpoints are independent: only desktop scrolls.

Rendering Rules: Classes `desktop-scroll tablet-grid mobile-grid`; the mobile scroll styles don't apply, and from 600 px the grid styles reset the scroller (`overflow-x: visible`).

Responsive Behaviour: Desktop (1200 px+) scrolls; tablet and mobile show grids.

SEO Impact: None.

Accessibility Compliance: As Variations 25 and 26 per breakpoint.

Variation 28 — Scroll on Tablet Only

Variation Name: Desktop = Grid; Tablet = Horizontal scroll; Mobile = Grid.

Business Purpose: Avoid cramped tablet grids while keeping desktop and mobile as grids.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/cards-scroll-tablet-only.png`
> **Purpose:** Tablet-only scroll
> **Description:** Desktop: a 4-column grid; tablet: a scrolling row with 2.5 cards visible; mobile: one column.

Use Cases: Many columns on desktop that don't fit tablets.

Authoring Guidelines: Tablet display = Horizontal scroll (Tablet items visible); the others Grid.

Business Rules: As Variation 27.

Rendering Rules: Classes `desktop-grid tablet-scroll mobile-grid`.

Responsive Behaviour: Only 600–1199 px scrolls.

SEO Impact: None.

Accessibility Compliance: As Variations 25 and 26.

Variation 29 — Scroll on Mobile Only

Variation Name: Desktop = Grid; Tablet = Grid; Mobile = Horizontal scroll.

Business Purpose: The most common Figma pattern (variations 1, 4, 6 and 10): a grid on large screens, a swipeable rail with an indicator on phones.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/cards-scroll-mobile-only.png`
> **Purpose:** Mobile-only scroll
> **Description:** Mobile: 1.2 cards visible with the next card peeking at the right edge and a grey scroll indicator below.

Use Cases: Statistic rows and themes that would be very long as a single column.

Authoring Guidelines: Mobile display = Horizontal scroll (Mobile items visible 1.2 default; Scroll indicator on mobile is on by default); Desktop and Tablet display = Grid.

Business Rules: As Variation 27.

Rendering Rules: Classes `desktop-grid tablet-grid mobile-scroll mobile-indicator`.

Responsive Behaviour: Below 600 px the row scrolls with touch swipe; from 600 px a grid.

SEO Impact: None.

Accessibility Compliance: As Variations 25 and 26.

Variation 30 — Scroll on Desktop and Tablet, Grid on Mobile

Variation Name: Desktop = Horizontal scroll; Tablet = Horizontal scroll; Mobile = Grid.

Business Purpose: Rails on large screens, a simple stacked list on phones.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-scroll-desktop-tablet.png`
> **Purpose:** Desktop and tablet scroll
> **Description:** Desktop and tablet: scrolling rows with buttons; mobile: a single column.

Use Cases: Content where swiping on phones isn't wanted.

Authoring Guidelines: Desktop and Tablet display = Horizontal scroll; Mobile display = Grid.

Business Rules: As Variation 27.

Rendering Rules: Classes `desktop-scroll tablet-scroll mobile-grid`.

Responsive Behaviour: 600 px and wider scroll; below, one column (or Mobile columns).

SEO Impact: None.

Accessibility Compliance: As Variations 25 and 26.

Variation 31 — Scroll on Desktop and Mobile, Grid on Tablet

Variation Name: Desktop = Horizontal scroll; Tablet = Grid; Mobile = Horizontal scroll.

Business Purpose: Rails where a tablet grid fits better (for example 2 columns).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-scroll-desktop-mobile.png`
> **Purpose:** Desktop and mobile scroll
> **Description:** Desktop and mobile: scrolling rows; tablet: a 2-column grid.

Use Cases: Card sets whose count suits a 2-column tablet grid.

Authoring Guidelines: Desktop and Mobile display = Horizontal scroll; Tablet display = Grid (Tablet columns e.g. 2).

Business Rules: As Variation 27.

Rendering Rules: Classes `desktop-scroll tablet-grid mobile-scroll`.

Responsive Behaviour: 1200 px+ and below 600 px scroll; 600–1199 px grid.

SEO Impact: None.

Accessibility Compliance: As Variations 25 and 26.

Variation 32 — Grid on Desktop, Scroll on Tablet and Mobile

Variation Name: Desktop = Grid; Tablet = Horizontal scroll; Mobile = Horizontal scroll.

Business Purpose: Everything visible on desktop, rails on touch devices.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-scroll-tablet-mobile.png`

> **Purpose:** Tablet and mobile scroll
> **Description:** Desktop: a 4-column grid; tablet and mobile: scrolling rows with an indicator.

Use Cases: Four or more cards that should be one row on desktop and swipeable elsewhere.

Authoring Guidelines: Desktop display = Grid; Tablet and Mobile display = Horizontal scroll.

Business Rules: As Variation 27.

Rendering Rules: Classes `desktop-grid tablet-scroll mobile-scroll`.

Responsive Behaviour: Below 1200 px the cards scroll.

SEO Impact: None.

Accessibility Compliance: As Variations 25 and 26.

Variation 33 — Previous / Next Buttons

Variation Name: Previous / next buttons on desktop, tablet or mobile (`{bp}Controls`).

Business Purpose: Visible controls for mouse and keyboard users to move through a rail.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-previous-next.png`
> **Purpose:** Previous and next buttons
> **Description:** Two round chevron icon buttons at the bottom right under a scrolling row; the previous button is disabled at the start.

Use Cases: Desktop rails; rails with many cards.

Authoring Guidelines: Layout tab, for a breakpoint with Display = Horizontal scroll: Previous / next buttons on {breakpoint} on. Scroll tab: Scroll step (One card or Viewport) and Loop.

Business Rules: Shown only where that breakpoint scrolls and the cards don't fit. Disabled at the start / end unless Loop is on. Default off on every breakpoint.

Rendering Rules: `div.list-nav > div.list-controls` with two icon buttons (`button.list-previous`, `button.list-next`, chevron-left / chevron-right) named by the placeholders `listPrevious` ("Previous items") and `listNext` ("Next items"), `aria-controls` the region, `data-cmp-interaction previous / next`. Block class `{bp}-controls`.

Responsive Behaviour: Each breakpoint independently; the nav row sits 1.5rem closer to the cards than the block gap.

SEO Impact: None (buttons, not links).

Accessibility Compliance: Real `<button>`s with names (WCAG 4.1.2), `disabled` state at the ends, keyboard operable; clicks never move focus.

Variation 34 — Scroll Indicator

Variation Name: Scroll indicator on desktop, tablet or mobile (`{bp}Indicator`).

Business Purpose: A thin progress bar that shows how far the rail is scrolled (Figma variations 1, 6 and 10 on mobile).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-scroll-indicator.png`
> **Purpose:** Scroll indicator
> **Description:** A 4-pixel light grey track under the cards with a darker grey thumb whose width shows the visible share.

Use Cases: Mobile rails (on by default for mobile).

Authoring Guidelines: Layout tab: Scroll indicator on {breakpoint} (with Display = Horizontal scroll). Default: off on desktop and tablet, on for mobile.

Business Rules: Shown only where the breakpoint scrolls and the cards don't fit.

Rendering Rules: `div.list-indicator` (`aria-hidden="true"`, 0.25rem high, track #e6e6e6) with `span.list-indicator-thumb` (#767676; width `max(5rem, visible share)`, offset by the scroll position via `--scroller-thumb-size` / `--scroller-thumb-offset`). Block class `{bp}-indicator`.

Responsive Behaviour: Per breakpoint; fills the nav row next to the buttons.

SEO Impact: None.

Accessibility Compliance: Decorative (hidden from assistive technology); not a control.

Variation 35 — Fixed Card Width

Variation Name: Card width (px) per breakpoint (`{bp}CardWidth` ≥ 120).

Business Purpose: Cards of a fixed width in a rail instead of a number of visible cards.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-fixed-card-width.png`
> **Purpose:** Fixed card width
> **Description:** A scrolling row of 320 px wide cards on desktop; the number of visible cards depends on the screen width.

Use Cases: Statistic cards or logos with a consistent width on all screens.

Authoring Guidelines: Layout tab, Display = Horizontal scroll: {Breakpoint} card width (px), 120–1200. 0 uses Items visible.

Business Rules: Values below 120 are treated as 0 (Items visible); above 1200 are capped.

Rendering Rules: `--list-card-width-{bp}` in rem (px / 16) replaces the Items visible calculation.

Responsive Behaviour: Per breakpoint.

SEO Impact: None.

Accessibility Compliance: As Variation 26.

Variation 36 — Edge "Inside the content width"

Variation Name: Scroll tab: Edge = Inside the content width (`none`); default Cards run to the edge (`bleed`).

Business Purpose: Choose whether a rail runs to the screen edge (Figma) or stops at the content width.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-edge-none.png`
> **Purpose:** Scroller inside the content width
> **Description:** Two captures: with bleed the last visible card is cut by the screen edge; with "Inside the content width" it is cut at the section's right padding.

Use Cases: Boxed sections or sections with a background where bleed looks wrong.

Authoring Guidelines: Scroll tab: Edge.

Business Rules: Only applies where the breakpoint scrolls.

Rendering Rules: Bleed: class `edge-bleed`; in scroll mode the scroller gets a negative right margin of the section padding (1.5rem mobile, 2.5rem tablet, 5rem desktop) and the list the same right padding; the visible-cards calculation subtracts it.

Responsive Behaviour: Per breakpoint in scroll mode; grids are never bled.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 37 — Snap Alignment "Center"

Variation Name: Scroll tab: Snap alignment = Center (Scroll snap on).

Business Purpose: Centre the snapped card instead of aligning it left.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-snap-center.png`
> **Purpose:** Centre snap alignment
> **Description:** On mobile, the current card centred with parts of the previous and next cards visible on both sides.

Use Cases: Mobile rails with 1.2 to 1.5 visible cards.

Authoring Guidelines: Scroll tab: Scroll snap on, Snap alignment = Center. Scroll snap off gives free scrolling.

Business Rules: The alignment field shows only when Scroll snap is on.

Rendering Rules: Classes `snap snap-center`; `scroll-snap-align: center`. Snap is suspended while dragging with a mouse.

Responsive Behaviour: Where a breakpoint scrolls.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 38 — Scroll Step "Viewport"

Variation Name: Scroll tab: Scroll step = Viewport (default One card).

Business Purpose: Page through a rail a screen at a time instead of card by card.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-viewport.png`
> **Purpose:** Viewport scroll step
> **Description:** Before and after pressing Next: the whole visible set of cards is replaced by the next set.

Use Cases: Long rails on desktop.

Authoring Guidelines: Scroll step = Viewport.

Business Rules: Applies to the buttons and the arrow keys.

Rendering Rules: `scrollBy` the scroller's width; One card scrolls to the next card's start.

Responsive Behaviour: Where a breakpoint scrolls.

SEO Impact: None.

Accessibility Compliance: Smooth scrolling is turned off for visitors who prefer reduced motion (Reduced motion support on).

Variation 39 — Loop

Variation Name: Scroll tab: Loop on.

Business Purpose: "Next" at the end returns to the start (and "Previous" at the start goes to the end).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-loop.png`
> **Purpose:** Loop
> **Description:** At the last card the Next button is still enabled; pressing it shows the first card again.

Use Cases: Short rails that should never "dead-end".

Authoring Guidelines: Loop on.

Business Rules: The buttons are never disabled at the ends. No auto-rotation.

Rendering Rules: The scroller jumps to 0 or to the end; the cards aren't cloned.

Responsive Behaviour: Where a breakpoint scrolls.

SEO Impact: None.

Accessibility Compliance: No automatic movement (WCAG 2.2.2 not triggered).

Variation 40 — Initial Item

Variation Name: Scroll tab: Initial item index > 0.

Business Purpose: Start a rail at a given card (for example the current year in a timeline-like row).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-initial-item.png`
> **Purpose:** Initial item
> **Description:** A rail that opens scrolled to its third card, with the Previous button enabled.

Use Cases: Rails where the most relevant card is not the first.

Authoring Guidelines: Initial item index (0 = the first card; 0–200).

Business Rules: Applied once, when the scroller first scrolls; focus isn't moved. Out-of-range values go to the last card.

Rendering Rules: `scrollToItem(index, instant)` without animation.

Responsive Behaviour: Where a breakpoint scrolls; grids ignore it.

SEO Impact: None.

Accessibility Compliance: Screen readers still read the cards from the first.

Variation 41 — Equal Card Height

Variation Name: Layout tab: Equal card height on.

Business Purpose: All grid rows as tall as the tallest card.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-equal-height.png`
> **Purpose:** Equal card height
> **Description:** Two grid rows of the same height although the second row's texts are shorter.

Use Cases: Grids with several rows of panels.

Authoring Guidelines: Equal card height on.

Business Rules: Default off.

Rendering Rules: Class `equal-height`; `grid-auto-rows: 1fr`.

Responsive Behaviour: In grid modes.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 42 — Cards Not Stretched

Variation Name: Layout tab: Stretch cards to row height off (default on).

Business Purpose: Let each card keep its own height in a row.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-no-stretch.png`
> **Purpose:** Cards not stretched
> **Description:** Image cards of different heights aligned to the top of the row.

Use Cases: Image cards with very different text lengths where empty space is unwanted.

Authoring Guidelines: Stretch cards to row height off.

Business Rules: With stretch on (default) the call to action of panels lines up at the bottom.

Rendering Rules: Without the `stretch` class, `.list-items` uses `align-items: start`.

Responsive Behaviour: In grid and scroll modes.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 43 — Unlinked Cards

Variation Name: Cards tab: Link items off, or Cards without a Link.

Business Purpose: Informational cards (for example statistics) that don't navigate.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/cards-unlinked.png`
> **Purpose:** Unlinked cards
> **Description:** Statistic cards without any link or button; hovering shows no pointer.

Use Cases: Facts and figures; capability lists without detail pages.

Authoring Guidelines: Link items off (all cards), or leave a Card's Link empty (that card).

Business Rules: Without a link there is no call to action button (even with a label) and the title is plain text. Links inside the rich text description still work.

Rendering Rules: No `a.list-card-link`; rows of links show no arrow.

Responsive Behaviour: Unchanged.

SEO Impact: No links are passed from the cards.

Accessibility Compliance: No empty or duplicate links (WCAG 2.4.4).

Variation 44 — Open Links in a New Tab

Variation Name: Cards tab: Open links in a new tab; Card: Open in a new tab (As the block, Yes, No).

Business Purpose: Open external destinations in a new browser tab.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/cards-new-tab.png`
> **Purpose:** New-tab links
> **Description:** A card linking to an external site; the browser's status bar shows the URL and the link opens a new tab.

Use Cases: Partner sites, external press articles.

Authoring Guidelines: Block default on the Cards tab; per Card on the Link tab.

Business Rules: A Card's Yes / No wins over the block's setting.

Rendering Rules: `target="_blank" rel="noopener noreferrer"`; a hidden "(opens in a new tab)" text (placeholder `opensInNewTab`) is added to title links. External links get `data-external="true"`; the data layer reports `xdm:linkType: exit`.

Responsive Behaviour: Unchanged.

SEO Impact: None.

Accessibility Compliance: The new-tab hint is announced on title links (WCAG 3.2.5 advisory). Known gap: on call to action buttons the generated `aria-label` ("Explore: {title}") replaces the hidden hint, so the new-tab warning isn't announced there unless an Accessibility label is set that mentions it.

Variation 45 — Cards Made Before (backward compatible)

Variation Name: Cards created with the earlier Block Collection Cards (image + rich text per card, no block settings).

Business Purpose: Existing pages keep working after the redesign.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-legacy.png`
> **Purpose:** Legacy cards in the new design
> **Description:** An older Cards block (image and rich text cells) rendered as four image cards with the heading as title and the trailing link as call to action.

Use Cases: Pages created before this version.

Authoring Guidelines: No action needed. To use the new settings, add a new Cards block and move the cards into it (old blocks are not key-value blocks).

Business Rules: A row with two cells or fewer is read as a card made before: the cell with an image is the image, the other the rich text. In the text the first heading (or a bold first paragraph) becomes the title, a link alone in the last paragraph becomes the call to action (its text the label), the rest is the description. Default settings apply (image cards, 4 / 4 / 1 columns).

Rendering Rules: As Variation 1.

Responsive Behaviour: As Variation 25.

SEO Impact: Headings may change level (card titles are one level below the block title).

Accessibility Compliance: As Variation 1.

Variation 46 — Empty State

Variation Name: Advanced tab: When there are no cards = Hide the block (default) or Show a message.

Business Purpose: Control what visitors see when no card has content.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-empty-state.png`
> **Purpose:** Empty state
> **Description:** In Universal Editor: the block header and a grey box "Add a Card to the block and give it a title, a description or an image."

Use Cases: Newly added blocks; blocks whose cards were removed.

Authoring Guidelines: Advanced tab: When there are no cards; Empty message (shown when Show a message is chosen).

Business Rules: In the editor (edit mode) the hint is always shown. On the page: Hide removes the block and hides its wrapper; Show a message shows the Empty message, else the placeholder `listEmpty` ("There are no items to show.").

Rendering Rules: `p.list-empty` (centred, 2.5rem padding, light grey background, radius 1rem) after the header.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: A hidden block outputs nothing.

Accessibility Compliance: The message is plain text in reading order.

6.15.4 Authoring Experience

Cards is a container block: the block holds the settings (key-value properties edited in the properties panel) and **Card** items (one per card) that are added, duplicated, deleted and reordered in the content tree. The decorated cards keep the Universal Editor instrumentation of their Card rows, so selecting a card on the canvas selects its Card item; the rich text Description can be edited in place on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A page in Universal Editor with a Cards block selected: header with title, description and CTA, three image cards with images, kickers, titles, descriptions and dates, and the content tree on the left listing Cards with three Card children and the add (+) button.

Annotations

- A = Block title (Content tab: Title, rendered in the Heading type)
- B = Block description (Content tab: Description)
- C = Header call to action (Content tab: CTA link, CTA text, CTA style)
- D = A Card item on the canvas (selecting it selects the Card in the content tree)
- E = Card image (Card, Content tab: Image, Image alt text)
- F = Card kicker: eyebrow, tags and reading time (Card: Eyebrow, Tags, Reading time)
- G = Card title (Card: Title; the link when there is no call to action)
- H = Card description, rich text editable in place (Card: Description)
- I = Card meta row: content type and date (Card: Content type, Date)
- J = Card call to action (Card, Link tab: Call to action label, Link)
- K = Previous / next buttons and scroll indicator (Layout tab, per breakpoint, in scroll mode)
- L = Content tree: the Cards block with its Card items (add, duplicate, delete, drag to reorder)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Universal Editor properties panel of a selected Cards block showing the tabs Content, Cards, Layout, Scroll, Advanced and Additional, with the Cards tab open (Display as teaser, Card style, Card colour, Image ratio, Panel height, First card as feature, the Show switches, Date format, Image loading).

Block tabs (Cards)

- **Content:** Title; Heading type (H2–H6, default H2); Description (rich text); CTA link; CTA text; CTA style (Text link, Primary button, Secondary button); Accessible list label; Component ID (validated: starts with a letter, then letters, digits, - and _).
- **Cards:** Display as teaser (on: cards; off: rows of links); Card style (Image card, Panel, Report, Statistic, Mixed); Card colour (Default for the style, Dark, Navy, Brand blue, Light blue, Light grey, Mid grey, Grey gradient, Image background); Image ratio (16:9, 1:1, 4:3, 3:2); Panel height (Portrait 2:3, Compact, Content); First card as feature — these five are shown only when Display as teaser is on; Link items; Open links in a new tab; Show image; Show eyebrow; Show tags; Show description; Show date; Date format (shown when Show date is on); Show call to action; Image loading (Lazy, Eager).
- **Layout:** for Desktop, Tablet and Mobile: display (Grid, Horizontal scroll); columns (grid only); gap (px); items visible, card width (px), previous / next buttons, scroll indicator (scroll only); then Equal card height and Stretch cards to row height.
- **Scroll:** Scroll snap; Snap alignment (when snap is on); Hide native scrollbar; Allow mouse drag; Allow touch swipe; Keyboard scrolling; Scroll step (One card, Viewport); Loop; Initial item index; Edge (Cards run to the edge, Inside the content width); Reduced motion support.
- **Advanced:** When there are no cards (Hide the block, Show a message); Empty message (when Show a message); Analytics; Style options (Compact statistic title, Dividers between rows, Header centred).
- **Additional:** Anchor ID; Hide block; Padding left / right / top / bottom; Margin left / right / top / bottom (spacing tokens 0 to 3000).

Item tabs (Card)

- **Content:** Image; Image alt text; Eyebrow; Title; Description (rich text); Label; Tags (multi); Date; Content type; Reading time.
- **Link:** Link; Link text; Call to action label; Open in a new tab (As the block, Yes, No).
- **Display:** Card style (As the block, Image card, Panel, Report, Statistic); Card colour (As the block, or a colour); Analytics label (at most 120 characters); Accessibility label.

6.15.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites, select the page (for example `/content/tcs-corp/us/en/...`) and choose Edit; the page opens in Universal Editor.

Expected Result: The page renders in the editor canvas with the content tree available.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** The Sites console with the page selected and the Edit action highlighted, then the page open in Universal Editor.

Step 2 — Add the Cards block to a section

Action: Select the section in the content tree (or on the canvas), choose Add (+) and pick **Cards** from the component list.

Expected Result: An empty Cards block is inserted with the default settings; in edit mode the hint "Add a Card to the block and give it a title, a description or an image." is shown.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** The add-component menu of a section with Cards highlighted, and the new empty Cards block with the editor hint.

Step 3 — Fill the Content tab

Action: With the block selected, open the properties panel, Content tab: enter the Title, choose the Heading type (H2 for a top-level section, H3 inside a section that already has an H2), enter the Description, and optionally the CTA link, CTA text and CTA style. If there is no title, enter an Accessible list label. Leave Component ID empty unless analytics or deep links need a fixed name.

Expected Result: The header shows the title, description and CTA.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-03-content-tab.png`
> **Purpose:** Content tab
> **Description:** The Content tab filled with title “Our services”, heading type H2, a description and a “View all services” CTA, and the header updated on the canvas.

Step 4 — Set the Cards tab

Action: Choose Display as teaser (cards) or turn it off (rows of links); choose the Card style and Card colour, Image ratio, Panel height and First card as feature; set Link items, Open links in a new tab, the Show switches, Date format and Image loading (Eager only when the block is on the first screen of the page).

Expected Result: The canvas re-renders with the chosen card style.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-04-cards-tab.png`
> **Purpose:** Cards tab
> **Description:** The Cards tab with Card style Statistic and Card colour Light grey selected.

Step 5 — Set the Layout tab

Action: For Desktop, Tablet and Mobile choose Grid (and the columns) or Horizontal scroll (Items visible or Card width, Previous / next buttons, Scroll indicator), and the gaps; set Equal card height and Stretch cards to row height.

Expected Result: The canvas shows the layout of the editor’s current width; resize the editor’s device preview to check the other breakpoints.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-05-layout-tab.png`
> **Purpose:** Layout tab
> **Description:** The Layout tab with Desktop display Grid (3 columns), Tablet display Grid (2 columns) and Mobile display Horizontal scroll (1.2 items visible, scroll indicator on).

Step 6 — Set the Scroll tab (only for scrolling breakpoints)

Action: Adjust Scroll snap, Snap alignment, Hide native scrollbar, Allow mouse drag, Allow touch swipe, Keyboard scrolling, Scroll step, Loop, Initial item index, Edge and Reduced motion support. The defaults suit most rails.

Expected Result: Scrolling behaviour changes where a breakpoint scrolls; grids are not affected.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-06-scroll-tab.png`
> **Purpose:** Scroll tab
> **Description:** The Scroll tab with the default values (snap on, start, scrollbar hidden, drag, swipe and keyboard on, one card, loop off, index 0, bleed, reduced motion on).

Step 7 — Set the Advanced tab

Action: Choose When there are no cards (Hide the block is the default) and the Empty message; keep Analytics on; add Style options (Compact statistic title, Dividers between rows, Header centred) if the design requires them.

Expected Result: The style options apply on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-07-advanced-tab.png`
> **Purpose:** Advanced tab
> **Description:** The Advanced tab with Style options "Header centred" selected.

Step 8 — Set the Additional tab

Action: Optionally enter an Anchor ID (`anchor-…`), and padding or margin values from the spacing scale. Leave Hide block off.

Expected Result: The block's spacing changes; the anchor makes `#anchor-…` links scroll to the block.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-08-additional-tab.png`
> **Purpose:** Additional tab
> **Description:** The Additional tab with Anchor ID "anchor-our-services" and Margin top "mt-1000".

Step 9 — Add a Card

Action: In the content tree, select the Cards block, choose Add (+) and pick **Card**.

Expected Result: A new Card item appears in the tree; it isn't shown on the canvas until it has a title, a description or an image.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-09-add-card.png`
> **Purpose:** Add a Card
> **Description:** The content tree with the Cards block expanded and the add menu offering Card.

Step 10 — Fill the Card's Content tab

Action: Select an Image from AEM Assets and enter the Image alt text (empty for a decorative image); enter Eyebrow, Title (for statistic cards: the number), Description (rich text), Label (statistic cards), Tags, Date, Content type (e.g. Report for report cards in Mixed) and Reading time (minutes, e.g. `6`).

Expected Result: The card renders with its content.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-10-card-content.png`
> **Purpose:** Card Content tab
> **Description:** The Card's Content tab filled in and the matching image card on the canvas.

Step 11 — Fill the Card's Link tab

Action: Choose the Link (a page of the site or an `https://…` URL; empty for an unlinked card), the Link text (only used as title when the card has none), the Call to action label (to show a button) and Open in a new tab.

Expected Result: The card's title (or its call to action) becomes the link.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-11-card-link.png`
> **Purpose:** Card Link tab
> **Description:** The Link tab with a page selected and Call to action label "Learn more", and the card showing the button.

Step 12 — Fill the Card's Display tab

Action: Optionally override the Card style and Card colour for this card, and enter an Analytics label and an Accessibility label.

Expected Result: The card uses its own style and colour.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-12-card-display.png`
> **Purpose:** Card Display tab
> **Description:** The Display tab with Card colour Brand blue; the card on the canvas turns brand blue.

Step 13 — Add, duplicate and reorder the cards

Action: Repeat Steps 9–12, or duplicate a Card in the content tree and change it; drag the Cards in the tree to reorder them.

Expected Result: The cards show in the authored order.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-13-reorder.png`
> **Purpose:** Reorder cards
> **Description:** A Card being dragged to a new position in the content tree.

Step 14 — Preview

Action: Switch Universal Editor to Preview mode, then use the device sizes (desktop, tablet, mobile); publish to Preview for a review on `aem.page`.

Expected Result: Links work, scrollers scroll, the layout matches the design at every breakpoint.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-14-preview.png`
> **Purpose:** Preview
> **Description:** The page in Universal Editor preview mode at mobile width with the cards in a scroller.

Step 15 — Validate

Action: Run the Validation Process checklist below.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-15-validate.png`
> **Purpose:** Validate
> **Description:** The preview page with the browser's accessibility inspector showing the cards list region and its name.

Step 16 — Publish

Action: Publish the page (and new assets) to Live from Universal Editor or the Sites console.

Expected Result: The cards are visible on `https://main--tcs-corp--sradha-tcs.aem.live/{path}`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/cards-step-16-publish.png`
> **Purpose:** Publish
> **Description:** The Universal Editor publish dialog with the page and its referenced images selected.

Preview Process

1. Universal Editor saves each field change automatically.
2. Use Preview mode in Universal Editor to try links, scrolling, buttons and keyboard (in edit mode clicks select content).
3. Publish to Preview and review `https://main--tcs-corp--sradha-tcs.aem.page/{path}`.
4. Check desktop (1440 px), tablet (1024 px and 768 px) and mobile (402 px or 375 px).

Publishing Process

1. Confirm the Validation Process checklist.
2. Publish the page to Live from Universal Editor or the Sites console, with the referenced images (and any linked PDF documents).
3. Pages linked from the cards must be published too (otherwise the links lead to the 404 page).
4. Open the live URL and reload without cache to confirm.

Validation Process

- ☐ Every card has a title (or a description / image) and shows as intended; no card is missing (cards without content or with a `javascript:` / `data:` link are dropped).
- ☐ The card style and colours match the design; text on coloured cards is readable.

- ☐ Statistic cards: Title holds the number, Label and Description explain it.
- ☐ Report cards: Content type and date in the kicker; the Download link points to the file.
- ☐ Each linked card has exactly one link; calls to action have meaningful labels.
- ☐ New-tab cards open a new tab; external links work.
- ☐ Images have alt text when they carry meaning; decorative ones have it empty.
- ☐ The block title's heading level fits the page outline (card titles are one level below).
- ☐ Desktop, tablet and mobile show the intended grid or scroller; scroll buttons and the indicator appear only where the cards don't fit.
- ☐ Panels with Portrait or Compact height don't clip text.
- ☐ Dates show in the chosen format; tags show their taxonomy labels (not IDs).
- ☐ Anchor ID (if any) is unique; Hide block is off.
- ☐ With Analytics on, `window.adobeDataLayer` contains the block and its cards (preview page, browser console).

Rollback Process

1. Restore the previous page version in AEM (Versions / Timeline) or undo in Universal Editor.
2. Publish the restored version.
3. Revert or replace changed assets in AEM Assets and republish them.
4. Emergency: turn Hide block on (Additional tab) and publish; the block disappears until fixed.
5. Code regressions: the development team reverts the commit on `main` (`blocks/cards/*` or the shared `decorators/card-list-decorator.js`, `utils/card-list-config.js`, `styles/card-list.css` — which also affect the List block).

6.15.6 Authorable Field Specification

The Cards model `cards` has 72 fields in six tabs; the Card model `card` has 18 fields in three tabs. The block definition sets `key-value: true` (new blocks store their settings as key | value rows). The Content, Cards, Layout, Scroll and Advanced groups (except the Cards-only "When there are no cards") are the shared groups of `models/_card-list.json`, spread into `blocks/cards/_cards.json`. No field is marked required. Enforced validation: Component ID and Anchor ID patterns, number ranges (`min` / `max` in the model, clamped again in code) and the Card's Analytics label (120 characters). Every value is validated again in `utils/card-list-config.js` `normalizePresentation()`: an invalid value falls back to its default.

Block model `cards` — Content tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Title (`title`)	text	No	No	Empty	Heading of the block	Short, descriptive; up to 60 characters (Recommended)	None	Our services
Heading type (`titleType`)	select	No	No	h2	Heading level of the title	H2 for a section heading; card titles are one level below	h2–h6 (h1 still read for older blocks)	h2
Description (`description`)	richtext	No	No	Empty	Intro text under the title	One or two sentences	None	What we do for our clients.
CTA link (`link`)	aem-content	No	No	Empty	Header call to action destination	A page or URL; leave empty for no CTA	None	/content/tcs-corp/us/en/services
CTA text (`linkText`)	text	No	No	Empty	Header CTA label	Action wording, 2–4 words	None	View all services

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
CTA style (`linkType`)	select	No	No	Text link (empty)	Look of the header CTA	Text link for secondary links; buttons sparingly	'', primary, secondary	primary
Accessible list label (`listLabel`)	text	No	No	Empty	Names the cards region for screen readers	Needed when there is no Title	None	Our services
Component ID (`componentId`)	text	No	No	Generated (`cards-1` …)	Stable id for deep links and analytics	Leave empty unless needed; don't change after go-live	Enforced regExp `^[A-Za-z][A-Za-z0-9_-]*$`; duplicates suffixed in code	our-services

Block model `cards` — Cards tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Display as teaser (`displayAsTeaser`)	boolean	No	No	true	Cards (on) or rows of links (off)	Off for news-style link lists	Boolean	true
Card style (`cardStyle`)	select	No	No	image	Kind of every card	Image, Panel, Report, Statistic or Mixed	image, panel, report, stat, mixed; shown when teaser on	stat
Card colour (`cardTheme`)	select	No	No	Default for the style (empty)	Colour of panel / report / statistic cards	A Card's own colour wins	'', dark, navy, brand, brand-light, grey, mid-grey, gradient, image; teaser on	grey
Image ratio (`imageRatio`)	select	No	No	16-9	Image shape of image cards	Match the photography	16-9, 1-1, 4-3, 3-2; teaser on	1-1
Panel height (`panelHeight`)	select	No	No	portrait	Shape of coloured cards	Content for long texts	portrait, compact, auto; teaser on	compact
First card as feature (`featureFirst`)	boolean	No	No	false	First card as a large spotlight	The first card needs an image	Boolean; teaser on	true
Link items (`linkItems`)	boolean	No	No	true	Link the cards	Off for informational cards	Boolean	true
Open links in a new tab (`openInNewTab`)	boolean	No	No	false	Default target of the card links	Only for external destinations	Boolean	false
Show image (`showImage`)	boolean	No	No	true	Show card images	Off for text-only cards	Boolean	true
Show eyebrow (`showEyebrow`)	boolean	No	No	true	Show the eyebrow	—	Boolean	true
Show tags (`showTags`)	boolean	No	No	true	Show tags and reading time	—	Boolean	true
Show description (`showDescription`)	boolean	No	No	true	Show the description	—	Boolean	true

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Show date (`showDate`)	boolean	No	No	true	Show the date	—	Boolean	true
Date format (`dateFormat`)	select	No	No	dd-mmm-yy (04 Sep 26)	Format of the dates	Use the locale's convention	dd-mmm-yy, dd-mm-yyyy, mm-dd-yyyy, mmm-d-yyyy, dd/mm/yy; shown when Show date on	dd-mm-yyyy
Show call to action (`showCta`)	boolean	No	No	true	Show call to action buttons	Off: the titles are the links	Boolean	true
Image loading (`imageLoading`)	select	No	No	lazy	Lazy or eager image loading	Eager only on the first screen	lazy, eager	lazy

Block model `cards` — Layout tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Desktop display (`desktopMode`)	select	No	No	grid	Grid or scroll at 1200 px+	—	grid, horizontal-scroll	grid
Desktop columns (`desktopColumns`)	number	No	No	4	Grid columns on desktop	2–4 typical	Enforced 1–6 (model and code, rounded); grid only	3
Desktop gap (px) (`desktopGap`)	number	No	No	24	Space between cards	Keep the default	Enforced 0–120	24
Desktop items visible (`desktopVisible`)	number	No	No	4.2	Cards in view when scrolling	A fraction shows part of the next	Enforced 1–8; scroll only	3.3
Desktop card width (px) (`desktopCardWidth`)	number	No	No	0	Fixed card width instead	0 uses Items visible	Enforced 0–1200; below 120 treated as 0; scroll only	320
Previous / next buttons on desktop (`desktopControls`)	boolean	No	No	false	Show the buttons	On for desktop rails	Boolean; scroll only	true
Scroll indicator on desktop (`desktopIndicator`)	boolean	No	No	false	Show the progress bar	—	Boolean; scroll only	false
Tablet display (`tabletMode`)	select	No	No	grid	Grid or scroll at 600–1199 px	—	grid, horizontal-scroll	horizontal-scroll
Tablet columns (`tabletColumns`)	number	No	No	4	Grid columns on tablet	2–3 often fit better	Enforced 1–6; grid only	2
Tablet gap (px) (`tabletGap`)	number	No	No	24	Space between cards	—	Enforced 0–120	24
Tablet items visible (`tabletVisible`)	number	No	No	3.3	Cards in view when scrolling	—	Enforced 1–8; scroll only	2.5
Tablet card width (px) (`tabletCardWidth`)	number	No	No	0	Fixed card width	—	Enforced 0–1200; below 120 = 0; scroll only	0

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Previous / next buttons on tablet (`tabletControls`)	boolean	No	No	false	Show the buttons	—	Boolean; scroll only	false
Scroll indicator on tablet (`tabletIndicator`)	boolean	No	No	false	Show the progress bar	—	Boolean; scroll only	true
Mobile display (`mobileMode`)	select	No	No	grid	Grid or scroll below 600 px	Scroll for long sets	grid, horizontal-scroll	horizontal-scroll
Mobile columns (`mobileColumns`)	number	No	No	1	Grid columns on mobile	1 (2 for small statistic tiles)	Enforced 1–6; grid only	1
Mobile gap (px) (`mobileGap`)	number	No	No	16	Space between cards	—	Enforced 0–120	16
Mobile items visible (`mobileVisible`)	number	No	No	1.2	Cards in view when scrolling	1.2–1.5	Enforced 1–8; scroll only	1.2
Mobile card width (px) (`mobileCardWidth`)	number	No	No	0	Fixed card width	—	Enforced 0–1200; below 120 = 0; scroll only	0
Previous / next buttons on mobile (`mobileControls`)	boolean	No	No	false	Show the buttons	Swipe is usually enough	Boolean; scroll only	false
Scroll indicator on mobile (`mobileIndicator`)	boolean	No	No	true	Show the progress bar	Keep on	Boolean; scroll only	true
Equal card height (`equalHeight`)	boolean	No	No	false	All rows equally tall	For multi-row grids	Boolean	false
Stretch cards to row height (`stretchCards`)	boolean	No	No	true	Cards fill the row height	Keep on	Boolean	true

Block model `cards` — Scroll tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Scroll snap (`scrollSnap`)	boolean	No	No	true	Cards snap into place	Keep on	Boolean	true
Snap alignment (`snapAlign`)	select	No	No	start	Where cards snap	Center for mobile peeking rails	start, center; shown when snap on	start
Hide native scrollbar (`hideScrollbar`)	boolean	No	No	true	Hide the browser scrollbar	Use the indicator instead	Boolean	true
Allow mouse drag (`mouseDrag`)	boolean	No	No	true	Drag with the mouse	Keep on	Boolean	true
Allow touch swipe (`touchSwipe`)	boolean	No	No	true	Swipe on touch screens	Off only with buttons	Boolean	true

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Keyboard scrolling (`keyboardScrolling`)	boolean	No	No	true	Arrow keys, Home, End	Keep on	Boolean	true
Scroll step (`scrollStep`)	select	No	No	one-card	Step of buttons and keys	—	one-card, viewport	one-card
Loop (`loop`)	boolean	No	No	false	Next at the end goes to the start	—	Boolean	false
Initial item index (`initialIndex`)	number	No	No	0	Card shown first	0 = first	Enforced 0–200	0
Edge (`edgePadding`)	select	No	No	bleed	Scroller to the screen edge or not	Bleed as in Figma	bleed, none	bleed
Reduced motion support (`reducedMotion`)	boolean	No	No	true	No smooth scrolling for reduced motion	Keep on	Boolean	true

Block model `cards` — Advanced tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
When there are no cards (`emptyBehavior`)	select	No	No	hide	What the page shows without cards	Hide in most cases	hide, message (code falls back to hide)	hide
Empty message (`emptyMessage`)	text	No	No	Empty (placeholder `listEmpty`)	Text of the empty state	Short sentence	Shown when behaviour = message	No services to show yet.
Analytics (`analyticsEnabled`)	boolean	No	No	true	Data layer state and events	Keep on	Boolean	true
Style options (`classes`)	multiselect	No	Yes	None	Extra styles	Only as designed	stat-compact, dividers, header-center	header-center

Block model `cards` — Additional tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	text	No	No	Empty	In-page link target (`#anchor-…`)	Unique on the page	Enforced regExp `^anchor-[a-z0-9][a-z0-9-]*$`; duplicates get –2, -3	anchor-our-services
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide the block on the page	Temporary use only	Boolean	false
Padding left (`classes_paddingleft`)	select	No	No	None	Inner spacing left	Spacing scale	pl-0, pl-100 … pl-3000	pl-0
Padding right (`classes_paddingright`)	select	No	No	None	Inner spacing right	Spacing scale	pr-0 … pr-3000	pr-0

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Padding top (`classes_paddingtop`)	select	No	No	None	Inner spacing top	Spacing scale	pt-0 … pt-3000	pt-600
Padding bottom (`classes_paddingbottom`)	select	No	No	None	Inner spacing bottom	Spacing scale	pb-0 … pb-3000	pb-600
Margin left (`classes_marginleft`)	select	No	No	None	Outer spacing left	Spacing scale	ml-0 … ml-3000	ml-0
Margin right (`classes_marginright`)	select	No	No	None	Outer spacing right	Spacing scale	mr-0 … mr-3000	mr-0
Margin top (`classes_margintop`)	select	No	No	None	Outer spacing top	Spacing scale	mt-0 … mt-3000	mt-1000
Margin bottom (`classes_marginbottom`)	select	No	No	None	Outer spacing bottom	Spacing scale	mb-0 … mb-3000	mb-1000

The padding and margin options are the spacing tokens 0, 100, 200, 300, 400, 500, 600, 800, 1000, 1200, 1600, 2000, 2400 and 3000 (`--space-*`, 0 to 7.5rem).

Item model `card` — Content, Link and Display tabs

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Image (`image`)	reference	No	No	Empty	Card image (also image-background panels and the feature)	From AEM Assets; landscape for 16:9	None	/content/dam/tcs/services/consulting.jpg
Image alt text (`imageAlt`)	text	No	No	Empty (decorative)	Text alternative	Describe meaning; empty if decorative	None	Consultants in a workshop
Eyebrow (`eyebrow`)	text	No	No	Empty	Label above the title	1–3 words	None	Perspective
Title (`title`)	text	No	No	Empty	Card heading; statistic number	Up to 60 characters (Recommended); stat: the number	None	4bn
Description (`text`)	richtext	No	No	Empty	Card text	1–3 sentences; lists and links allowed	None	Revenue in fiscal 2026.
Label (`subtitle`)	text	No	No	Empty	Statistic label above the description	1–3 words	None	Revenue
Tags (`tags`)	aem-tag	No	Yes	None	Labels in the kicker	Up to 3 shown	Tag IDs from the taxonomy	tcs:topic/ai
Date (`date`)	date-time	No	No	Empty	Card date	—	Read as YYYY-MM-DD	2026-09-04
Content type (`contentLabel`)	text	No	No	Empty	Meta label; Report in Mixed	Article, Report, …	None	Report
Reading time (`readingTime`)	text	No	No	Empty	Minutes after the tags	Digits only (6 shows 6m)	None	6

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Link (`link`)	aem-content	No	No	Empty (not linked)	Card destination	A page or `https://…`	Unsafe schemes (`javascript:`, `data:`) drop the card	/content/tcs-corp/us/en/services/consulting
Link text (`linkText`)	text	No	No	Empty	Title when the card has none	—	None	Consulting
Call to action label (`ctaLabel`)	text	No	No	Empty (panel: Explore; report: Download)	Shows a button as the card's link	1–3 words, a verb	None	Learn more
Open in a new tab (`newTab`)	select	No	No	As the block (empty)	New tab for this card	Yes for external sites	'', true, false	true
Card style (`kind`)	select	No	No	As the block (empty)	This card's kind	Overrides the block	'', image, panel, report, stat	panel
Card colour (`theme`)	select	No	No	As the block (empty)	This card's colour	Overrides the block	'', dark, navy, brand, brand-light, grey, mid-grey, gradient, image	brand
Analytics label (`analyticsLabel`)	text	No	No	Empty	`dc:title` of the card in the data layer	Stable, language-neutral name	Enforced maxLength 120	consulting-card
Accessibility label (`ariaLabel`)	text	No	No	Empty	Accessible name of the card link	When title / CTA isn't enough	None	Read about our consulting services

Field details

Format of each entry: Purpose; Allowed Values; Character Limits; Business Rules; Dependency Rules; Visibility Rules (model `condition`); Sample Content. Fields that differ only by breakpoint are described together.

Block — Content tab

- **Title (`title`)** — Purpose: the block heading and the name of the cards region (`aria-labelledby`). Allowed: plain text. Limits: none enforced; Recommended up to 60 characters (one line on desktop). Rules: an empty title gives no heading; the region is then named by the Accessible list label, else "Cards" (placeholder `cards`). Dependency: card titles use the next heading level. Visibility: always. Sample: "Our services".
- **Heading type (`titleType`)** — Purpose: the title's level. Allowed: h2–h6. Limits: n/a. Rules: card titles are one level lower (h3 under h2; at most h6). Dependency: Title. Visibility: always. Sample: h2.
- **Description (`description`)** — Purpose: intro under the title. Allowed: rich text (paragraphs, bold, links). Limits: Recommended up to 200 characters. Rules: empty text isn't rendered. Visibility: always. Sample: "From strategy to operations."
- **CTA link (`link`)** — Purpose: header call to action. Allowed: AEM page or URL. Rules: no link, no CTA. Dependency: CTA text, CTA style. Visibility: always. Sample: a services overview page.
- **CTA text (`linkText`)** — Purpose: the CTA label. Allowed: text. Limits: Recommended up to 25 characters. Rules: replaces the link's text. Dependency: CTA link. Visibility: always. Sample: "View all services".

- **CTA style (`linkType`)** — Purpose: CTA look. Allowed: Text link (empty), Primary button, Secondary button. Rules: anything else gives the text link (tertiary). Dependency: CTA link. Visibility: always. Sample: Primary button.
- **Accessible list label (`listLabel`)** — Purpose: names the region for screen readers. Allowed: text. Limits: Recommended up to 60 characters. Rules: when set it is the region's `aria-label` (also with a title); also the data layer `dc:title` when there is no title. Visibility: always. Sample: "Key facts".
- **Component ID (`componentId`)** — Purpose: the block id, the analytics id and the prefix of card ids (`{id}-item-{n}`). Allowed: letters, digits, `-`, `_`, starting with a letter (Enforced). Rules: empty: the Anchor ID, else `cards-1`, `cards-2`, … in page order; a duplicate gets a numeric suffix; values are lower-cased. Changing it breaks deep links and starts new analytics series. Visibility: always. Sample: "our-services".

Block — Cards tab

- **Display as teaser (`displayAsTeaser`)** — Purpose: cards or rows of links. Allowed: on / off. Rules: off ignores card style, colour, ratio, panel height and feature; data layer variant `links`. Visibility: always. Sample: on.
- **Card style (`cardStyle`)** — Purpose: card kind for all cards. Allowed: image, panel, report, stat, mixed. Rules: a Card's own style wins; Mixed picks per card (6.15.3 Variation 5); invalid values: image. Visibility: Display as teaser on. Sample: stat.
- **Card colour (`cardTheme`)** — Purpose: colour of coloured cards. Allowed: empty (default for the style: panel dark, report gradient, statistic grey) or one of 8 colours. Rules: a Card's colour wins; Image background renders the image only on panels. Visibility: teaser on. Sample: brand.
- **Image ratio (`imageRatio`)** — Purpose: image card image shape. Allowed: 16-9, 1-1, 4-3, 3-2. Rules: invalid: 16-9. Visibility: teaser on. Sample: 1-1.
- **Panel height (`panelHeight`)** — Purpose: shape of panel, report and statistic cards. Allowed: portrait (2:3), compact (9:10), auto (content). Rules: fixed shapes clip overflowing text. Visibility: teaser on. Sample: compact.
- **First card as feature (`featureFirst`)** — Purpose: spotlight the first card. Allowed: on / off. Rules: feature shows image, tags, reading time and title only. Visibility: teaser on. Sample: on.
- **Link items (`linkItems`)** — Purpose: make cards clickable. Allowed: on / off. Rules: off: no card links, no call to action buttons, no row arrows. Visibility: always. Sample: on.
- **Open links in a new tab (`openInNewTab`)** — Purpose: default target. Allowed: on / off. Rules: a Card's Yes / No wins. Visibility: always. Sample: off.
- **Show image / Show eyebrow / Show tags / Show description / Show date / Show call to action (`showImage`, `showEyebrow`, `showTags`, `showDescription`, `showDate`, `showCta`)** — Purpose: hide parts of all cards. Allowed: on / off (default on). Rules: Show tags also hides the reading time; Show call to action off makes titles the links; the feature card always shows its image. Visibility: always. Sample: Show date off.
- **Date format (`dateFormat`)** — Purpose: date display. Allowed: dd-mmm-yy (04 Sep 26), dd-mm-yyyy (04-09-2026), mm-dd-yyyy (09-04-2026), mmm-d-yyyy (Sep 4, 2026), dd/mm/yy (04/09/26). Rules: month names are English abbreviations in every locale; the machine date is in `<time datetime="YYYY-MM-DD">`. Visibility: Show date on. Sample: dd-mm-yyyy.
- **Image loading (`imageLoading`)** — Purpose: loading priority. Allowed: lazy, eager. Rules: eager also sets `fetchpriority="high"`. Visibility: always. Sample: lazy.

Block — Layout tab

- **Desktop / Tablet / Mobile display (`desktopMode`, `tabletMode`, `mobileMode`)** — Purpose: grid or horizontal scroll per breakpoint (desktop ≥ 1200 px, tablet 600–1199 px, mobile < 600 px). Allowed: grid, horizontal-scroll. Rules: independent per breakpoint (8 combinations); invalid: grid. Visibility: always. Sample: Mobile = horizontal-scroll.
- **Columns (`desktopColumns`, `tabletColumns`, `mobileColumns`)** — Purpose: grid columns. Allowed: 1–6 (Enforced; rounded). Defaults 4 / 4 / 1. Visibility: that breakpoint's display = grid. Sample: 3 / 2 / 1.

- **Gap (px) (`desktopGap`, `tabletGap`, `mobileGap`)** — Purpose: space between cards. Allowed: 0–120 (Enforced). Defaults 24 / 24 / 16; converted to rem. Visibility: always. Sample: 24.
- **Items visible (`desktopVisible`, `tabletVisible`, `mobileVisible`)** — Purpose: card width in a scroller as the number of visible cards. Allowed: 1–8, decimals (Enforced). Defaults 4.2 / 3.3 / 1.2. Dependency: ignored when Card width is set. Visibility: display = horizontal-scroll. Sample: 1.2.
- **Card width (px) (`desktopCardWidth`, `tabletCardWidth`, `mobileCardWidth`)** — Purpose: fixed card width. Allowed: 0 or 120–1200 (model 0–1200; values below 120 count as 0). Visibility: horizontal-scroll. Sample: 320.
- **Previous / next buttons on … (`desktopControls`, `tabletControls`, `mobileControls`)** — Purpose: buttons below the scroller. Default off. Rules: shown only when the cards overflow. Visibility: horizontal-scroll. Sample: on (desktop).
- **Scroll indicator on … (`desktopIndicator`, `tabletIndicator`, `mobileIndicator`)** — Purpose: progress bar. Defaults off / off / on. Rules: shown only when the cards overflow. Visibility: horizontal-scroll. Sample: on (mobile).
- **Equal card height (`equalHeight`)** — Purpose: equal grid rows. Default off. Visibility: always. Sample: off.
- **Stretch cards to row height (`stretchCards`)** — Purpose: cards fill their row. Default on. Visibility: always. Sample: on.

Block — Scroll tab (all: Visibility always, except Snap alignment)

- **Scroll snap (`scrollSnap`)** — Purpose: snapping. Default on. Rules: suspended while dragging. Sample: on.
- **Snap alignment (`snapAlign`)** — Purpose: snap position. Allowed: start, center. Visibility: Scroll snap on. Sample: center.
- **Hide native scrollbar (`hideScrollbar`)** — Purpose: hide the browser scrollbar. Default on. Sample: on.
- **Allow mouse drag (`mouseDrag`)** — Purpose: drag scrolling with a mouse (a drag doesn't follow the link under the pointer; 6 px threshold). Default on. Sample: on.
- **Allow touch swipe (`touchSwipe`)** — Purpose: swipe. Default on. Rules: off sets `touch-action: pan-y` (buttons still work). Sample: on.
- **Keyboard scrolling (`keyboardScrolling`)** — Purpose: arrow keys, Home, End on the focused scroller. Default on. Sample: on.
- **Scroll step (`scrollStep`)** — Purpose: step of buttons / keys. Allowed: one-card, viewport. Sample: one-card.
- **Loop (`loop`)** — Purpose: wrap around at the ends. Default off. Sample: off.
- **Initial item index (`initialIndex`)** — Purpose: card shown first. Allowed: 0–200 (Enforced). Sample: 0.
- **Edge (`edgePadding`)** — Purpose: bleed to the screen edge. Allowed: bleed, none. Sample: bleed.
- **Reduced motion support (`reducedMotion`)** — Purpose: instant instead of smooth scrolling for `prefers-reduced-motion`. Default on. Sample: on.

Block — Advanced tab

- **When there are no cards (`emptyBehavior`)** — Purpose: empty state. Allowed: hide, message (anything else: hide). Rules: in edit mode the editor hint is always shown. Visibility: always. Sample: hide.
- **Empty message (`emptyMessage`)** — Purpose: empty state text. Limits: Recommended up to 80 characters. Rules: empty: placeholder `listEmpty`. Visibility: behaviour = message. Sample: "No services to show yet."
- **Analytics (`analyticsEnabled`)** — Purpose: data layer. Default on. Rules: off sets `data-analytics="off"`: no state, no click events. Visibility: always. Sample: on.
- **Style options (`classes`)** — Purpose: block classes. Allowed: stat-compact, dividers, header-center (multi). Visibility: always. Sample: dividers.

Block — Additional tab

- **Anchor ID (`classes_anchor`)** — Purpose: in-page anchor. Allowed: `anchor-` + lower-case letters, digits, hyphens (Enforced). Rules: becomes the block id when there is no Component ID; duplicates get `-2`, `-3`. Visibility: always. Sample: anchor-key-facts.

- **Hide block (`classes_hideblock`)** — Purpose: temporarily hide. Rules: hidden on the page and in preview; shown faded in edit mode. Visibility: always. Sample: off.
- **Padding left / right / top / bottom (`classes_paddingleft`, `classes_paddingright`, `classes_paddingtop`, `classes_paddingbottom`)** and **Margin left / right / top / bottom (`classes_marginleft`, `classes_marginright`, `classes_margintop`, `classes_marginbottom`)** — Purpose: spacing around / inside the block. Allowed: empty or a token class (`pl-0` … `pl-3000`, etc.). Rules: styled in `styles/styles.css`. Visibility: always. Sample: mt-1000.

Card item

- **Image (`image`) and Image alt text (`imageAlt`)** — Purpose: the card picture. Allowed: an AEM Assets image; alt text. Limits: alt Recommended up to 125 characters. Rules: shown on image cards, image-background panels and the feature; empty alt = decorative (`alt=""`). Dependency: Show image. Visibility: always. Sample: a workshop photo, "Consultants in a workshop".
- **Eyebrow (`eyebrow`)** — Purpose: kicker label. Limits: Recommended up to 30 characters. Rules: panels show it, else the first tag; report cards show the content type first. Dependency: Show eyebrow. Sample: "Perspective".
- **Title (`title`)** — Purpose: card heading / statistic number. Limits: Recommended up to 60 characters (statistics: up to 8 characters). Rules: a card needs a title, a description or an image; a card without a title has no heading and is linked only through a call to action. Sample: "Consulting" / "4bn".
- **Description (`text`)** — Purpose: card text. Allowed: rich text (paragraphs, bold, lists, inline links). Limits: Recommended up to 160 characters. Rules: inline links stay clickable above the card link; also the data layer title of a card without a title (first 100 characters). Dependency: Show description. Sample: "We help you shape strategy."
- **Label (`subtitle`)** — Purpose: statistic label. Limits: Recommended up to 30 characters. Rules: only statistic cards show it. Sample: "Revenue".
- **Tags (`tags`)** — Purpose: kicker labels. Allowed: taxonomy tags (multi). Rules: up to 3 shown (image cards, feature), 1 on panels and rows; labels from the locale taxonomy, else derived from the ID. Dependency: Show tags. Sample: `tcs:topic/ai`.
- **Date (`date`)** — Purpose: card date. Rules: read as the first 10 characters (YYYY-MM-DD); other values show no date. Dependency: Show date, Date format. Sample: 2026-09-04.
- **Content type (`contentLabel`)** — Purpose: meta label; Report detection in Mixed. Limits: Recommended 1–2 words. Sample: "Report".
- **Reading time (`readingTime`)** — Purpose: minutes. Rules: digits show as `{n}m` (placeholder `minutesShort`); other text is shown as typed; hidden with Show tags off. Sample: "6".
- **Link (`link`)** — Purpose: destination. Allowed: http(s), mailto, tel, paths; never `javascript:` / `data:` (the card is dropped). Rules: empty: unlinked card. Internal links aren't resolved through the redirects sheet. Sample: a page.
- **Link text (`linkText`)** — Purpose: the title when the card has none. Rules: ignored when it equals the URL. Sample: "Consulting".
- **Call to action label (`ctaLabel`)** — Purpose: a button as the card's one link. Limits: Recommended up to 20 characters. Rules: needs a link, Link items on and Show call to action on; panels default "Explore", reports "Download". Sample: "Learn more".
- **Open in a new tab (`newTab`)** — Purpose: per-card target. Allowed: As the block, Yes, No. Sample: Yes.
- **Card style (`kind`)** — Purpose: per-card kind. Allowed: As the block, image, panel, report, stat. Rules: wins over the block's style and over Mixed. Visibility: always (applies with Display as teaser on). Sample: panel.
- **Card colour (`theme`)** — Purpose: per-card colour. Allowed: As the block or one of 8 colours. Sample: brand.
- **Analytics label (`analyticsLabel`)** — Purpose: data layer `dc:title` of the card and `data-analytics-label`. Limits: Enforced maxLength 120. Sample: "consulting-card".
- **Accessibility label (`ariaLabel`)** — Purpose: the link's accessible name. Limits: Recommended up to 100 characters. Rules: replaces the generated "{CTA}: {title}" name. Sample: "Read about our consulting services".

6.15.7 Field Annotation Mapping

Marker	Field	Purpose
A	Title (`title`), Heading type (`titleType`)	Block heading; names the cards region
B	Description (`description`)	Intro text under the title
C	CTA link (`link`), CTA text (`linkText`), CTA style (`linkType`)	Header call to action
D	Card item (model `card`); Card style (`kind`), Card colour (`theme`)	One card; its own kind and colour
E	Image (`image`), Image alt text (`imageAlt`)	Card image and its text alternative
F	Eyebrow (`eyebrow`), Tags (`tags`), Reading time (`readingTime`)	Kicker above the title
G	Title (`title`), Link (`link`), Link text (`linkText`)	Card heading and its link
H	Description (`text`)	Card text (rich text, editable in place)
I	Content type (`contentLabel`), Date (`date`)	Meta row of image cards (kicker of report cards)
J	Call to action label (`ctaLabel`), Link (`link`), Open in a new tab (`newTab`)	Card button
K	`{bp}Controls`, `{bp}Indicator` (Layout tab)	Previous / next buttons and scroll indicator
L	Content tree: Card items	Add, duplicate, delete, reorder cards

6.15.8 Component Specification

Attribute	Value
Component Name	Cards (`cards`), item Card (`card`)
Version	1.0
Owner	[To be confirmed: product owner — owner: TCS digital marketing]; technical owner: development team
Dependencies	`decorators/card-list-decorator.js`, `decorators/scroller-decorator.js`, `utils/card-list-config.js`, `blocks/cards/cards-data.js`, `utils/cells.js`, `utils/taxonomy.js`, `utils/placeholders.js`, `utils/analytics.js`, `styles/card-list.css`; placeholders and taxonomy sheets of the locale
Component Type	Container block (key-value settings) with item components (Card)
Supported Variants	Card styles image, panel, report, stat, mixed; feature card; rows of links; 8 card colours; 4 image ratios; 3 panel heights; style options stat-compact, dividers, header-center; grid / horizontal scroll per breakpoint (8 combinations) with buttons, indicator, card width, edge, snap, step, loop, initial item; Figma variations 1–14 by configuration
Supported Templates	Any page (home, article, blog, press-release and pages without a template)
Supported Metadata	None read from the page; the page's language comes from the page for the placeholders and taxonomy
Supported Tags	Card Tags (AEM tags of the locale taxonomy), labels from `/{locale}/taxonomy.json`
Supported Personalization	None. The data layer state and click events allow future Adobe Target / Journey Optimizer activities once Adobe Tags is added (not implemented as of this handover)
Caching Considerations	The page HTML (with the cards) is cached by the Edge Delivery CDN and refreshed on publish; the placeholders and taxonomy sheets are fetched once per page view and cached in memory; no query index is used
Performance Considerations	Loaded with the section (lazy phase unless the block is in the first section); no extra data requests except placeholders and taxonomy; images lazy by default (eager + high priority option for the first screen); im-

Attribute	Value
	age ratios reserve space (no layout shift); scrolling by CSS, the scroller script only adds controls; no carousel library
Accessibility Compliance	WCAG 2.1 AA target: `<ul>/<li>` in a named region; headings one level below the block title; one link per card; named buttons with `disabled` states and `aria-controls`; focusable scroller only while it scrolls; `<time datetime>`; decorative images `alt=""`; reduced motion; known gaps: #999 secondary text contrast, brand blue small-text contrast, new-tab hint on call to action buttons
SEO Compliance	Server-delivered card content in HTML (no client data fetch for the text); semantic headings and descriptive links; image alt text; no hidden duplicate links

6.15.9 Input Model

Content Inputs: the block settings (key | value rows) and one row per Card with its cells: image (picture with alt), eyebrow, title, description (rich text), label, tags, date, content type, reading time, link (anchor with the link text), call to action label, new tab, card style, card colour, analytics label, accessibility label.

Metadata Inputs: none from the page metadata.

Tags Inputs: the Card Tags (tag IDs such as `tcs:topic/ai`), resolved to labels with the locale's taxonomy (`utils/taxonomy.js resolveTags`).

Page Context Inputs: the page origin (to tell external links), the edit mode of Universal Editor (`html.adobe-ue-edit`, for the hint), the locale (placeholders and taxonomy), the viewport (layout by CSS media queries; breakpoint in click events).

Query Inputs: none (Cards doesn't read the query index or the redirects sheet).

Manual Inputs: everything above is typed by the author; the order is the content tree order.

Authored HTML received from AEM (simplified; in the editor the rows also carry `data-aue-*` instrumentation, e.g. `data-aue-model="card"`):

```
<div class="cards">
  <div><div><p>title</p></div><div><p>Our services</p></div></div>
  <div><div><p>cardStyle</p></div><div><p>image</p></div></div>
  <div><div><p>mobileMode</p></div><div><p>horizontal-scroll</p></div></div>
  <div>
    <div><picture><img src="./media_1a2b.jpg" alt="Consultants in a workshop"></picture></
div>
    <div><p>Perspective</p></div>
    <div><p>Consulting</p></div>
    <div><p>We help you shape <strong>strategy</strong>.</p></div>
    <div></div>
    <div><p>tcs:topic/consulting</p></div>
    <div><p>2026-09-04</p></div>
    <div><p>Article</p></div>
    <div><p>6</p></div>
    <div><p><a href="/en/services/consulting">Consulting</a></p></div>
    <div><p>Learn more</p></div>
  </div>
</div>
```

A row with exactly two cells whose first cell is a setting name is a setting; other rows are Cards (cells matched to fields by position and content when AEM leaves out unset fields); a row of at most two cells that isn't a setting is a card made before (image + rich text).

6.15.10 Output Model

Rendered Output: the block gets the classes `card-list`, `{desktop,tablet,mobile}-{grid, scroll}`, `{bp}-controls`, `{bp}-indicator`, `snap`, `snap-{start,center}`, `hide-scrollbar`, `no-swipe`, `edge-bleed`, `equal-height`, `stretch`, `feature-first`, `teaser` or `links`, `style-{cardStyle}`, `ratio-{ratio}`, `panel-{height}`, the style options and `source-manual`; the custom properties `--list-{columns,gap,visible,card-width}-{bp}`; the id (Component ID, Anchor ID or `cards-{n}`). Children: `div.list-header`, optional `article.list-feature`, `div.list-scroller[role=region]` with `ul.list-items > li.list-item > article.list-card.list-card-{kind}` (or `article.list-row`), and `div.list-nav` (indicator and buttons).

Generated HTML (one image card, shortened):

```
<div class="cards block card-list desktop-grid tablet-grid mobile-scroll mobile-indicator
  snap hide-scrollbar edge-bleed stretch snap-start teaser style-image ratio-16-9
  panel-portrait source-manual" id="cards-1" data-analytics="custom">
  <div class="list-header">
    <div class="list-title-slot">
      <h2 id="cards-1-title" class="list-title">Our services</h2>
    </div>
  </div>
  <div class="list-scroller" id="cards-1-items" role="region"
    aria-labelledby="cards-1-title" data-overflow="true" tabindex="0">
    <ul class="list-items">
      <li class="list-item" data-card-id="cards-1-item-1" data-cmp-id="cards-1-item-1"
        data-cmp-position="1">
        <article class="list-card list-card-image">
          <div class="list-card-media"><picture>…</picture></div>
          <p class="list-card-kicker"><span>Perspective</span><span>Consulting</span>
            <span>6m</span></p>
          <h3 class="list-card-title">Consulting</h3>
          <div class="list-card-description list-card-body"><p>We help you …</p></div>
          <p class="list-card-meta"><span>Article</span>
            <time datetime="2026-09-04">04 Sep 26</time></p>
          <p class="list-card-action"><a class="list-card-link button tertiary"
            href="/en/services/consulting" aria-label="Learn more: Consulting"
            data-cmp-interaction="card">Learn more</a></p>
        </article>
      </li>
    </ul>
  </div>
  <div class="list-nav">
    <div class="list-indicator" aria-hidden="true">
      <span class="list-indicator-thumb"></span>
    </div>
    <div class="list-controls">
      <button class="list-previous button secondary icon-button" aria-label="Previous
items"
        aria-controls="cards-1-items" data-cmp-interaction="previous"></button>
      <button class="list-next button secondary icon-button" aria-label="Next items"
        aria-controls="cards-1-items" data-cmp-interaction="next"></button>
    </div>
  </div>
</div>
```

Generated JSON: none (no files or API responses are produced).

Generated Metadata: none (no `<meta>` tags; the cards don't change the page metadata).

Analytics Tracking: with Analytics on, the block pushes its state to `window.adobeDataLayer` (Adobe Client Data Layer, AEM Core Components schema) and marks itself `data-analytics="custom"`:

```
{ component: {
  'cards-1': { '@type': 'tcs-corp/components/cards', parentId: 'page-1x2y3z',
    'dc:title': 'Our services', 'xdm:variant': 'image' },
  'cards-1-item-1': { '@type': 'tcs-corp/components/cards/item', parentId: 'cards-1',
    'dc:title': 'Consulting', 'xdm:linkURL': '/en/services/consulting',
    'xdm:position': 1, 'repo:modifyDate': '2026-09-04' },
} }
```

Clicks are pushed by the site-wide data layer decorator:

```
{ event: 'cmp:click', eventInfo: { path: 'component.cards-1-item-1', interaction: 'card',
  'dc:title': 'Learn more: Consulting',
  'xdm:linkURL': 'https://main--tcs-corp--sradha-tcs.aem.live/en/services/consulting',
  'xdm:linkType': 'other', breakpoint: 'desktop' } }
{ event: 'cmp:click', eventInfo: { path: 'component.cards-1', interaction: 'next',
  'dc:title': 'Next items', 'xdm:linkType': 'other', breakpoint: 'mobile' } }
```

`xdm:variant` is the card style, or `links` with Display as teaser off. `dc:title` of a card is the Analytics label, else the title, else the first 100 characters of the description. The header CTA reports `interaction: 'cta'`; external links `xdm:linkType: 'exit'`, file links `'download'`. The card's position is in the component state (`xdm:position`); the click event doesn't carry a `position` for card links (the position attribute is on the `<li>`, the event reads it from the clicked link), and clicks on the feature card are reported with the block's path. With Analytics off: `data-analytics="off"`, no state, no events.

Personalization Output: none.

Search Index Output: none of its own. The cards are part of the page HTML; the query index (`helix-query.yaml`) indexes page metadata, not block content.

6.15.11 Business Rules

1. The block is decorated once (`data-cards-decorated`); a second decoration call does nothing.
2. Rows with two cells whose first cell (normalized: lower case, no dashes or spaces) is a setting name are settings; rows instrumented as `card`, and other rows, are Cards.
3. A Card row with two cells or fewer is read as a card made before: the cell with an image is the image, the other the rich text; the first heading (or a bold-only first paragraph) is the title; a link alone in the last paragraph is the call to action and its link; the rest is the description.
4. Card cells are matched to the 16 model fields in order; when AEM leaves out unset fields, cells are matched by content (an image, a `YYYY-MM-DD` date, a link, `true` / `false`, a known style or colour value, single-line text).
5. When a Card has no Title, the Link text becomes the title (unless it equals the URL).
6. A card is dropped when its link is unsafe (any scheme except `http`, `https`, `mailto`, `tel`; e.g. `javascript:`, `data:`) or when it has no title, no description text and no image.
7. Cards keep the authored order: no sorting, no deduplication, no maximum.
8. Card ids are `{component id}-item-{n}`, numbered in the shown order (after dropped cards).
9. The component id is the Component ID (lower-cased, unsafe characters replaced; an id not starting with a letter gets the prefix `cards-`), else the block's id from the Anchor ID, else `cards-1`, `cards-2`, …; an authored id already used on the page gets a numeric suffix.
10. Every setting is validated: unknown select values, out-of-range numbers and non-numbers fall back to the defaults (image cards, 16:9, portrait, grid 4 / 4 / 1, gaps 24 / 24 / 16, visible 4.2 / 3.3 / 1.2, mobile indicator on, snap start, bleed, …). Booleans accept yes, true, 1, on, y; empty uses the default.
11. Columns are rounded and clamped to 1–6; gaps 0–120; items visible 1–8; card width 0–1200 with values below 120 treated as 0; initial index 0–200.

12. The card kind: the Card's own style; else the block's style; for Mixed: a content type containing "report" gives a report card, an image gives an image card, otherwise a panel.
13. The card colour: the Card's colour, else the block's; empty uses the style's default (panel dark, report gradient, statistic grey).
14. Tags: labels from the locale taxonomy; unknown IDs get a label derived from the ID; at most 3 tags on image cards and the feature card; the first tag only on panels (when there is no eyebrow) and rows.
15. Reading time made of digits shows as `{n}m` (placeholder `minutesShort`); it is part of the tags and hidden with Show tags off.
16. Dates: only `YYYY-MM-DD` values (or ISO date-times) are shown, in the Date format, with English month abbreviations, inside `<time datetime>`.
17. A card is linked only with a link and Link items on. Its one link is the call to action when shown (Show call to action on and a label: the Card's, or "Explore" for panels, "Download" for reports), else the title. A card without a title and without a call to action isn't linked.
18. The call to action's accessible name is "{label}: {title}" unless an Accessibility label is set; the Accessibility label is the `aria-label` of the card link.
19. Call to action types: secondary buttons on report and statistic cards; tertiary (text link) on image cards and panels.
20. Links open in a new tab when the Card says Yes, or the Card says As the block and the block's Open links in a new tab is on; they get `rel="noopener noreferrer"` and the hidden text "(opens in a new tab)".
21. Links to another origin are marked `data-external="true"`.
22. Images: the authored `<picture>` is cloned; the alt is the Image alt text (empty = decorative); `loading` is the Image loading value; eager images get `fetchpriority="high"`. Show image off hides images except on the feature card.
23. First card as feature (with Display as teaser on and at least one card): the first card is shown as the feature (image, tags, reading time, title) and removed from the list.
24. Display as teaser off: rows of links (kicker, title, description, date, arrow); no images, calls to action, styles, colours or feature.
25. The rich text description keeps its Universal Editor instrumentation (editable in place); links inside it stay clickable above the card link.
26. Card headings are one level below the block heading (h3 for an h2 title; h3 without a title; at most h6).
27. The cards region is named by the title (`aria-labelledby`), else the Accessible list label, else "Cards" (placeholder `cards`); an Accessible list label always wins as `aria-label`.
28. No cards: in Universal Editor edit mode the header and the hint (placeholder `cardsEditorHint`) are shown; on the page, Hide the block (default) empties and hides the block and its wrapper; Show a message shows the Empty message, else `listEmpty`.
29. Scrolling breakpoints: the scroller is enhanced only while the cards overflow (`data-overflow="true"`, focusable, buttons enabled); otherwise buttons and indicator are hidden. The state is recalculated on scroll and resize (ResizeObserver).
30. Buttons step one card or one viewport; with Loop they wrap; otherwise they are disabled at the start / end. Keys: Arrow Left / Right, Home, End on the focused scroller.
31. Mouse drag scrolls after 6 px of movement and cancels the click on the link under the pointer; touch scrolling is native (`touch-action: pan-y` with swipe off).
32. Smooth scrolling is replaced by instant scrolling for `prefers-reduced-motion` (with Reduced motion support on).
33. Analytics on: block and card state in the data layer (see 6.15.10); off: `data-analytics="off"` and no events. Analytics errors never break rendering.
34. Hide block (Additional tab) hides the block on the page and in preview, faded in edit mode; the Anchor ID becomes the block id (duplicates suffixed).

6.15.12 Author Best Practices

Recommended Content Length (Recommended, not enforced)

Element	Recommended length
Block title	Up to 60 characters, one line on desktop
Block description	Up to 200 characters
Eyebrow	1–3 words, up to 30 characters
Card title	Up to 60 characters (2 lines on a 4-column desktop grid)
Statistic title (number)	Up to 8 characters (e.g. `4bn`, `600k+`); longer phrases: Compact statistic title
Label	1–3 words
Description	Up to 160 characters; panels with Portrait / Compact height: up to 120
Call to action label	1–3 words, up to 20 characters
Cards per block	Grid: a multiple of the columns (e.g. 3, 4, 6, 8); scroller: 5–12

Image Dimensions

Use	Ratio (CSS)	Recommended source size
Image card 16:9 (default)	16 / 9	1600 × 900 px
Image card 1:1	1 / 1	1200 × 1200 px
Image card 4:3	4 / 3	1600 × 1200 px
Image card 3:2	3 / 2	1500 × 1000 px
Image background panel	2 / 3 (portrait), 9 / 10 (compact), content height	1200 × 1800 px, subject centred
Feature card	Full width, min. height 35.625rem desktop / 27.1875rem tablet / 32.5rem mobile	2400 × 1200 px, subject on the right (text sits on the left gradient)

Images are cropped with `object-fit: cover`; keep the subject in the centre. Use JPEG or WebP photography from AEM Assets; Edge Delivery delivers optimized renditions.

CTA Writing Guidelines

- Start with a verb (“Explore services”, “Download the report”, “Read the story”).
- Keep labels short and specific; the screen reader name adds the title (“Learn more: Consulting”), but the visible label should still make sense alone where possible.
- Use the header CTA for “View all” links to an overview page.
- Don’t put the same destination in the card link and in a description link.

Accessibility Guidelines

- Give every card a title; cards without a title have no heading and no link unless they have a call to action.
- Write alt text for informative images; leave it empty for decorative images next to a title.
- Use an Accessible list label when the block has no title.
- Choose the heading type that fits the page outline (H2 under the page H1).
- Check text contrast on Brand blue and Image background cards; prefer Dark or Navy for long text.
- For new-tab links on call to action buttons, mention the new tab in the Accessibility label.
- Don’t rely on colour alone to distinguish cards (add an eyebrow or label).

SEO Guidelines

- Use descriptive card titles; they are the link text of title-linked cards.

- Link to canonical published pages (not to redirecting URLs: Cards doesn't resolve the redirects sheet).
- Avoid duplicate cards linking to the same page in one block.

Metadata Guidelines

Cards doesn't read page metadata. If the cards represent pages that have card metadata (`card-title`, `card-description`, `card-image`), use the List block instead so the cards stay in sync with the pages.

Taxonomy Guidelines

- Use tags from the locale's taxonomy (`/{locale}/taxonomy`); the labels are translated per locale sheet.
- Add at most 3 tags per card (only 3 are shown; 1 on panels and rows).
- Prefer the eyebrow for a fixed label and tags for taxonomy-driven labels.

Localization Guidelines

- Each locale site has its own pages: translate every card's texts, alt texts and labels in the locale copy; links should point to the same locale.
- UI texts (Explore, Download, Previous items, Next items, "(opens in a new tab)", empty message, `{n}m`) come from the locale's placeholders sheet; keep the keys translated.
- Date formats: choose the format customary for the locale; month abbreviations are English in every locale (dd-mmm-yy and mmm-d-yyyy) — use a numeric format for non-English locales.
- Statistic numbers: use the locale's number style (e.g. `4 Mrd.` in German) and check that long values fit.

6.15.13 Troubleshooting Guide

Issue 1 — A card is missing on the page

Symptoms: A Card exists in the content tree but no card is rendered for it; the other cards show.

Root Cause: The card has no Title, no Description text and no Image, or its Link uses an unsafe scheme (`javascript:`, `data:` or another non-web scheme); such cards are dropped by `itemToCard()`.

Diagnosis Steps: Select the Card in the content tree and check Title, Description, Image and Link; check whether the link starts with `http`, `https`, `mailto:`, `tel:` or `/`.

Resolution Steps: Add a title (or description / image); replace the link with a page or an `https://` URL.

Validation Steps: Reload the editor; the card shows; publish and check the live page.

Preventive Measures: Fill the Title of every card; use the page picker for internal links.

Issue 2 — The whole Cards block is missing on the published page

Symptoms: The block shows in Universal Editor (edit mode) but not in preview or on the live page; the section may be empty.

Root Cause: No card has content and When there are no cards = Hide the block (default), so the block and its wrapper are hidden; or Hide block is on (Additional tab); or the block's JavaScript failed to load.

Diagnosis Steps: In edit mode check for the hint "Add a Card to the block …"; check the Additional tab's Hide block; open the browser console on the page and look for `failed to load block cards` or `failed to load module for cards`.

Resolution Steps: Add content to the cards; turn Hide block off; for a load error, escalate to L3 with the console message.

Validation Steps: The block is visible in preview and live.

Preventive Measures: Use Show a message during content build-up; review Hide block before publishing.

Issue 3 — Tags show technical IDs or wrong labels

Symptoms: The kicker shows labels such as "Genai" or "Cloud migration" (derived from the tag ID) instead of the business wording, or English labels on a translated locale.

Root Cause: The tag isn't in the locale's taxonomy (`/{locale}/taxonomy.json`), its translation sheet is missing, or the taxonomy page isn't published; the code then derives a label from the last part of the tag ID (hyphens become spaces, first letter upper case).

Diagnosis Steps: Open `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/taxonomy.json` and search for the tag ID; check the locale's language sheet.

Resolution Steps: Add or translate the tag in the locale taxonomy page and publish it.

Validation Steps: Reload the page (new session); the label is correct.

Preventive Measures: Pick tags only from the governed taxonomy (Part 7); publish taxonomy changes before pages that use new tags.

Issue 4 — Text is cut off in panel, report or statistic cards

Symptoms: The end of a description or the call to action is not visible in coloured cards, especially on tablet or mobile.

Root Cause: Panel height Portrait (2:3) or Compact (9:10) gives the card a fixed shape and `overflow: hidden` clips longer content.

Diagnosis Steps: Check the Panel height and the text length; resize the preview to the narrowest column width.

Resolution Steps: Shorten the texts, reduce the number of columns, or set Panel height = Content.

Validation Steps: All text and the call to action are visible at 1440, 1024, 768 and 375 px.

Preventive Measures: Respect the recommended lengths (6.15.12).

Issue 5 — Previous / next buttons or the scroll indicator don't appear

Symptoms: A scrolling row has no buttons or no indicator, or they appear on one device only.

Root Cause: Buttons and indicators are per breakpoint and only for breakpoints in Horizontal scroll mode; they are hidden when all cards fit (no overflow); the defaults are off (desktop, tablet) and indicator on (mobile).

Diagnosis Steps: Check the Layout tab for the breakpoint in question (display, buttons, indicator); count the cards against Items visible; inspect `.list-scroller[data-overflow]`.

Resolution Steps: Turn on Previous / next buttons on {breakpoint} / Scroll indicator on {breakpoint}; reduce Items visible or add cards if the row doesn't overflow.

Validation Steps: The controls show and work at the breakpoint.

Preventive Measures: Configure each breakpoint explicitly; test at all three widths.

Issue 6 — An older Cards block ignores the new settings, or a title shows as text

Symptoms: Changing Card style or layout has no effect on an existing block; or a card's title appears as plain description text.

Root Cause: Blocks created before this version aren't key-value blocks, so the new settings aren't read; card rows with two cells or fewer are read as cards made before (image + rich text), where only the first heading or a bold-only first paragraph becomes the title.

Diagnosis Steps: Check the page's HTML (`.plain.html` of the page) for setting rows (key | value) and count the cells of the card rows.

Resolution Steps: Add a new Cards block, set it up and move (or recreate) the cards; in an old card, format the title as a heading.

Validation Steps: The block responds to the settings; titles render as headings.

Preventive Measures: Migrate legacy Cards blocks page by page; fill more than one field of each new Card.

Issue 7 — The date or reading time doesn't show

Symptoms: A card has a Date or Reading time, but nothing is shown.

Root Cause: Show date or Show tags (which also controls the reading time) is off; the card kind doesn't show it (panels and statistic cards show no date; report cards show the date in the kicker); the date value isn't a valid date.

Diagnosis Steps: Check the Cards tab switches, the card kind and the Date field.

Resolution Steps: Turn the switches on, choose an image or report card, re-enter the date.

Validation Steps: The date shows in the chosen Date format.

Preventive Measures: Use image or report cards for dated content.

Issue 8 — The call to action button is missing

Symptoms: A Call to action label is set but no button shows; or panels show "Explore" unexpectedly.

Root Cause: The card has no Link, Link items is off or Show call to action is off; panels and reports have default labels ("Explore", "Download").

Diagnosis Steps: Check the Card's Link and the block's Link items / Show call to action.

Resolution Steps: Add a link and turn the switches on; to remove the default label on panels turn Show call to action off (the title becomes the link).

Validation Steps: The button shows with the right label and destination.

Preventive Measures: Decide per block whether cards use buttons or title links.

Issue 9 — Card clicks are missing in analytics

Symptoms: No `cmp:click` events or no `cards` component in `window.adobeDataLayer`.

Root Cause: Analytics is off for the block (`data-analytics="off"`); no tag manager is loaded yet (Adobe Tags isn't implemented, so nothing is sent to a reporting tool); the click was a mouse drag (cancelled).

Diagnosis Steps: In the browser console run `window.adobeDataLayer` and look for the block id; check the block's `data-analytics` attribute.

Resolution Steps: Turn Analytics on; for reporting, the Adobe Tags implementation is needed (Part 11).

Validation Steps: The state and a `cmp:click` entry appear after a click.

Preventive Measures: Keep Analytics on; set stable Component IDs for reported blocks.

Issue 10 — A card links to an old URL

Symptoms: Clicking a card goes through a redirect or to a 404 page.

Root Cause: Cards uses the authored link as is; it doesn't resolve the locale redirects sheet (the List does for pages); the target page was moved or unpublished.

Diagnosis Steps: Check the Card's Link and the locale's `/{locale}/redirects.json`.

Resolution Steps: Update the Link to the new page and republish.

Validation Steps: The card opens the target directly.

Preventive Measures: Check card links after page moves (AEM shows references of a moved page).

6.15.14 Support Runbook

Monitoring Requirements

- Real User Monitoring of Edge Delivery (Core Web Vitals) for pages with Cards above the fold (LCP when Image loading is eager).
- Periodic checks of key pages at desktop, tablet and mobile widths after code releases that touch `decorators/card-list-decorator.js`, `utils/card-list-config.js` or `styles/card-list.css` (shared with List).
- Data layer checks (block and card entries present) once Adobe Tags is implemented — [To be confirmed: analytics monitoring owner — owner: analytics team].

Logs

- Browser console: `failed to load block cards` / `failed to load module for cards` (from `scripts/aem.js`) when the block code can't load. The Cards block itself writes no console messages (there is no debug flag, unlike the List's `?debug=list`).
- Admin API status of a page: `https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}` (preview / live state and last modification).
- Author side: AEM as a Cloud Service logs in Cloud Manager (publish / replication errors); Universal Editor shows save errors in the editor.

Common Errors

Error	Likely cause	Action
Card not shown	No title / description / image, or unsafe link	Fill the card; fix the link
Block hidden on the page	No card content with Hide the block; Hide block on	Add content; turn Hide block off
Editor hint "Add a Card …"	No card with content	Add a Card and fill it
Tag labels look technical	Tag missing in the locale taxonomy	Add / translate the tag, publish the taxonomy
Text clipped	Portrait / Compact panel height	Shorter text or Panel height Content
No scroll controls	Breakpoint in grid mode, setting off, or no overflow	Configure the breakpoint
Settings have no effect	Block made before (not key-value)	Replace with a new Cards block
`failed to load block cards`	Code deployment or network problem	Escalate to L3

Escalation Matrix

Level	Who	Checks	Escalate when
L1	Author support / content team	Card fields, Show switches, Layout per breakpoint, Hide block, publish state of the page and assets	The configuration is right but the rendering is wrong
L2	AMS / site admin	Preview vs live, Admin API status, CDN cache (republish), placeholders and taxonomy sheets published, console errors	Code errors, or the same issue on several pages
L3	Development team	`blocks/cards/*`, `decorators/card-list-decorator.js`, `utils/card-list-config.js`, `styles/card-list.css`, unit tests (`npm test`), Code Sync	Platform problems (Edge Delivery, Universal Editor, AEM author)
L4	Adobe Support	Edge Delivery Services delivery, Universal Editor, AEM as a Cloud Service	—

[To be confirmed: names, contact channels and response times per level — owner: service manager].

Recovery Procedures

1. Content error: restore the previous page version and republish (6.15.5 Rollback Process).

2. Visual regression after a code release: revert the commit on `main`; Code Sync redeploys; both Cards and List are affected by shared files.
3. Temporary mitigation: Hide block on the affected block and publish.
4. Missing labels: republish the locale's placeholders or taxonomy sheet.

Health Checks

- The Cards blocks of the block library page (`/{locale}/block-library/...`, author only) render every style, colour and layout.
- `npm test` passes `test/cards/cards-data.test.mjs`; `npm run lint` passes.
- On a live page: `document.querySelectorAll('.cards.card-list .list-item').length` equals the number of filled cards; `window.adobeDataLayer` contains the `cards-...` component.

Operational Procedures

- Releases of the shared card list code are tested with both blocks (Cards and List) and all Figma variations.
- New placeholders keys must be added to all 13 locale placeholders sheets.
- Changes to the models (`_cards.json`, `models/_card-list.json`) require `npm run build:json` and a lint run.

Known Limitations

- No author lockup on the feature card (List only).
- Internal links aren't resolved through the redirects sheets.
- The card text fields are edited in the properties panel; only the rich text description can be edited in place.
- Blocks made before this version must be replaced to use the new settings.
- Card click events carry no `position` (it's in the component state), and feature card clicks are reported with the block's path.
- The new-tab hint isn't announced on call to action buttons without an Accessibility label.
- Month abbreviations in dates are English in every locale.
- No auto-rotation, no filtering or sorting (use Filter List).

6.16 List Block

The List block is the AEM Core Components **List** for Edge Delivery Services: a list of pages or links taken from **child pages** of a page, a **fixed list** authored in the block, or **tagged pages**, shown as cards or as rows of links, in a grid or a horizontal scroller chosen separately for desktop, tablet and mobile. Its cards are built from the listed pages' metadata (card title, card description, card image, eyebrow, tags, dates, content type, reading time, author), read from the published query index; a Fixed list item can override each value. The presentation follows the TCS DS Figma List variations 1–14 and is shared with the Cards block (6.15): the same settings, markup and styles.

Source files: `blocks/list/list.js`, `blocks/list/list-data.js`, `blocks/list/list.css` (imports `styles/card-list.css`), `blocks/list/_list.json`, `blocks/list/README.md`; shared: `decorators/card-list-decorator.js`, `decorators/scroller-decorator.js`, `utils/card-list-config.js`, `styles/card-list.css`, `models/_card-list.json`; data: `utils/query-index.js`, `utils/redirects.js`, `utils/data.js`, `helix-query.yaml`; unit tests: `test/list/list-data.test.mjs` (with `test/list/fixtures.mjs`).

6.16.1 Component Purpose

Business Objective

Keep listings of site content up to date without manual work: when an author publishes a new child page or tags a page, every List that selects it shows the new card, with the title, text and image the page defines in its metadata. Authors decide what is listed (source, tags, order, maximum) and how it looks; the page owners maintain the card content once, on the page.

Business Use Cases

- Section landing pages listing their child pages (services, industries, insights hubs).
- "Related insights" rows of pages tagged with the same topic (Tagged pages).
- News and press release lists ordered by last modified date (rows of links or cards).
- Media coverage and investor relations rails of report pages (Mixed, report cards).
- Curated editorial selections of specific pages (Fixed list) with per-item overrides of the title, description, image, eyebrow and style.
- Statistic rows (Fixed list with overrides: the number as Title, Label, Description).
- A featured spotlight with the author's photo, name and designation (First card as feature).
- External links mixed with site pages in one fixed list.

When To Use

- The cards represent pages of the same locale site that are published and indexed.
- The list should update itself (Child pages, Tagged pages).
- The cards should reuse the pages' own metadata (one source of truth).
- Ordering by title or last modified date, or a maximum number of items, is needed.

When Not To Use

- Content that isn't a page (statistics without detail pages, capabilities with their own wording, rich text descriptions, unlinked cards): use **Cards** (6.15).
- Visitors must filter or search: use **Filter List** (6.17).
- Content Fragment lists: not supported (by decision).
- Full-text search results: the Core List search source is not available.
- Pages of another locale: the List only reads the current locale's pages from the index.

Dependencies

Type	Dependency	Used for
Decorator	`decorators/card-list-decorator.js`	Header, cards, feature card (with author lockup), rows, grid / scroll classes, empty state, data layer state (shared with Cards)
Decorator	`decorators/scroller-decorator.js`	Previous / next, scroll indicator, drag, keyboard, loop, initial item
Decorator	`decorators/article-meta-decorator.js` `buildAuthorLockup`	Author lockup on the feature card
Decorator	`decorators/button-decorator.js`, `icon-button-decorator.js`, `link-decorator.js`, `heading-decorator.js`, `responsive-image-decorator.js`, `data-layer-decorator.js`	Buttons, links and new tab, title heading, images, page entry and clicks
Util	`blocks/list/list-data.js`	Sources, discovery, page metadata mapping (`PAGE_MAPPING`), overrides, deduplication, sorting, limits, validation (no DOM; unit tested)
Util	`utils/card-list-config.js`	Presentation settings, defaults, card kinds (shared with Cards)
Util	`utils/query-index.js` `getPages`, `columnOf`, `listOf`	The locale's pages from `/query-index.json` (500 rows per request, loaded once, cached in memory)
Util	`utils/redirects.js` `loadRedirects`, `resolveRedirect`	Card links to the redirect targets of `/{locale}/redirects.json` (up to 5 hops)
Util	`utils/data.js` `getAuthors`	Authors sheet for the feature card's author

Type	Dependency	Used for
Util	`utils/site.js` `toPublishedPath`	AEM paths (`/content/tcs-corp/us/en/…`) to published paths (`/en/…`)
Util	`utils/cells.js`, `utils/properties.js`, `utils/tags.js`, `utils/taxonomy.js`, `utils/dates.js`, `utils/placeholders.js`, `utils/environment.js`, `utils/analytics.js`	Item cells, settings, tags and labels, dates, UI text, editor detection, data layer
Styles	`styles/card-list.css` (imported by `blocks/list/list.css`)	The presentation
Index	`/query-index.json` (`helix-query.yaml`)	Pages and their card metadata columns
Sheets	`/{locale}/redirects.json`, `/{locale}/data/authors.json`, `/{locale}/taxonomy.json`, `/{locale}/placeholders.json`	Redirects, authors, tag labels, UI text
Placeholders	`listPrevious`, `listNext`, `opensInNewTab`, `explore`, `download`, `listEmpty`, `listEditorHint`, `listEditorPages`, `minutesShort`, `list`	UI text with English fallbacks
Page metadata	`card-title`, `card-description`, `card-image`, `card-image-alt`, `image-alt`, `eyebrow`, `category`, `thumbnail-image`, `featured-image`, `modified-date`, `publish-date`, `content-type`, `reading-time`, `author-id`, `hide-in-nav`, page tags	Card content of each listed page

Related Components

- **Cards** (6.15): the same presentation, content typed in the block.
- **Filter List** (6.17): query-index lists with filter pills, sorting and advanced filters.
- **Table List** (6.18): tabular listings.
- **Page Header** (6.4): uses the same page metadata (title, description, tags, dates, author).
- Page metadata and the bulk metadata sheet (Part 8) and the taxonomy (Part 7) supply the card content.

Applicable Templates

Any page; the section filter allows List in every section. Typical: home and landing pages (child pages, curated selections), article and blog pages (related tagged pages), press-release and news overview pages (rows of links ordered by date).

Supported Content Types

Pages of the current locale site that are published and in the query index (any page type: article, blog, press release, landing page, report pages, …), and for Fixed lists also paths that aren't indexed and external URLs (`https://…`). Card content: page metadata (text, image paths, tags, dates, numbers as text) with optional overrides (text, image from AEM Assets, tags, style, colour).

6.16.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A List at 1440 px with Child pages source and default settings: the header (title "Insights", description, "View all" text link on the right) and four image cards in a row built from the child pages' card images, eyebrows, titles, descriptions, content types and dates.

Desktop (1200 px and wider). Identical to the Cards block (6.15.2): a vertical stack with a 5rem gap; header with a 2.5rem semibold title, a 1.125rem grey description and the CTA on the right; a 4-column grid by default (Desktop columns 1–6, gap 1.5rem) or a horizontal scroller with 4.2 cards visible; image cards with 16:9 rounded images, uppercase kicker, 1.5rem titles, 1rem description and meta; panel, report and statistic cards as 2:3 coloured panels with a 1.5rem padding; the feature card at least 35.625rem high. Card images from page metadata are requested at 750 px width (from 600 px) and 600 px below; feature images at 1600 px (from 900 px) and 750 px.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same List at 1024 px: a 2rem title, four narrower image cards (default Tablet columns 4) with 1.25rem titles and 0.875rem kicker and meta.

Tablet (600–1199 px). The tablet settings apply: grid with 4 columns by default or a scroller with 3.3 cards visible; block title 2rem, card titles 1.25rem, kicker / meta 0.875rem, panel titles 1.5rem, statistic numbers up to 4rem; coloured card padding 1rem below 900 px and 1.5rem from 900 px; feature card at least 27.1875rem; rows of links without borders unless Dividers is set.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The List at 402 px: the header stacked (title, description, CTA below) and the cards in one column with a 1rem gap.

Mobile (below 600 px). One column by default with a 1rem gap and a 2.5rem block gap; the header wraps with the CTA below the title; block title 1.75rem, card titles 1.125rem, kicker, meta and description 0.75rem, statistic numbers up to 5rem (scaled to the card width); in scroll mode 1.2 cards visible, the scroll indicator on by default and the scroller bleeding to the right screen edge (1.5rem section padding).

6.16.3 Component Variations

The List's variations are its three **sources** (with their discovery and ordering modes), the shared **presentation** modes (card styles, feature card, rows of links, colours, ratios, panel heights, style options, grid or scroll per breakpoint and the scroll options — the same CSS and rules as the Cards block, 6.15.3) and the **TCS DS Figma variations 1–14**, each documented with the settings that reproduce it (Variations 56–69).

Variation 1 — Source "Child pages"

Variation Name: List source = Child pages (`parent`), direct children (default).

Business Purpose: List the pages below a section page automatically, like the AEM Core List "child pages" source.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-source-child-pages.png`
> **Purpose:** Child pages source
> **Description:** An "Our industries" landing page with a List of its child pages as image cards (card images, titles and descriptions from each child page's metadata).

Use Cases: Section landing pages; sub-navigation teasers; hub pages.

Authoring Guidelines: Source tab: List source = Child pages; Parent page (pick a page) or leave it empty with Use the current page when empty on (default); Direct children only on (default).

Business Rules: Lists published, indexed pages of the current locale directly below the parent (depth 1); pages with `hide-in-nav` = yes / true are left out; with an empty Parent page and Use the current page off, nothing is listed (editor hint "parent-missing"). Order by Manual = path order (the repository order isn't in the index).

Rendering Rules: Cards from `normalizePageToCard()` (6.16.9 metadata mapping); block class `source-parent`; data layer `xdm:listSource: "parent"`. Links go to the page path or its redirect target.

Responsive Behaviour: As the chosen presentation (default: grid 4 / 4 / 1).

SEO Impact: Automatic internal links from the parent to all its children (good for crawl depth); the card texts come from the pages' metadata, consistent with their titles.

Accessibility Compliance: As the presentation; card names come from the page titles, so write meaningful card titles in the page metadata.

Variation 2 — Child Pages with Depth

Variation Name: Child pages with Direct children only off and a Child depth (1–10, model default 2).

Business Purpose: Include grandchildren (and deeper pages) of a section.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-source-child-depth.png`
> **Purpose:** Child pages with depth
> **Description:** A List showing the child and grandchild pages of an insights section in one grid, ordered by title.

Use Cases: Hubs whose content sits in sub-folders (e.g. insights by year).

Authoring Guidelines: Source tab: Child pages; Direct children only off; Child depth (e.g. 2); usually Order by Title or Last modified and a Maximum items.

Business Rules: Pages at most Child depth levels below the parent; a depth outside 1–10 is clamped; an empty depth in code means 1 (the dialog pre-fills 2).

Rendering Rules: As Variation 1.

Responsive Behaviour: As the presentation.

SEO Impact: As Variation 1; deep lists can become long — set Maximum items.

Accessibility Compliance: As Variation 1.

Variation 3 — Source "Fixed list"

Variation Name: List source = Fixed list (`fixed`, the default) with List Items.

Business Purpose: A curated selection of pages and links in a chosen order, with per-item overrides of the card content.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-source-fixed.png`

> **Purpose:** Fixed list source
> **Description:** A List with four List Items: three site pages (cards from their metadata, one with an overridden title) and an external partner link with its own title and image.

Use Cases: Editor's picks; campaign selections; mixed internal and external links; statistic rows (overrides).

Authoring Guidelines: Source tab: List source = Fixed list. Add List Items to the block: Page or link (required), optionally Link text, Open in a new tab, Title, Description, Image, Image alt text, Eyebrow, Tags, Label, Call to action label, Card style, Card colour, Analytics label, Accessibility label.

Business Rules: Authored order (Order by Manual), or sorted by Title / Last modified. Precedence: item override → the page's metadata → defaults. An item without a link, with an unsafe link (`javascript:`, `data:`) or without any title (no override, no page title, no link text) is skipped. A link to an indexed page uses its metadata; another path or an external URL becomes a card with only the item's values. Duplicate destinations are kept once. A Fixed list without items shows the editor hint "items-missing".

Rendering Rules: Block class `source-fixed`; `xdm:listSource: "fixed"`; same-site absolute URLs are read as paths; external links get `data-external="true"`.

Responsive Behaviour: As the presentation.

SEO Impact: Curated internal links; external links are marked as exits in analytics.

Accessibility Compliance: Use the Accessibility label when several items have similar titles.

Variation 4 — Source "Tagged pages" (any tag)

Variation Name: List source = Tagged pages (`tags`) with Match = Any of the tags (default).

Business Purpose: List all pages below a root that carry at least one of the selected tags.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-source-tagged-any.png`
> **Purpose:** Tagged pages source (any)
> **Description:** A "Related to Agentic AI" rail of pages tagged with the AI topic (or its child tags), as mixed cards.

Use Cases: Related content; topic and industry pages; campaign pages pulling tagged insights.

Authoring Guidelines: Source tab: Tagged pages; Search root (a page) or Use the current page when empty (default on); Tags (one or more); Match = Any; Include descendant pages (default on) and Maximum search depth (0 = no limit).

Business Rules: A page matches when one of its tags equals a selected tag or is a child tag of it (`tcs:topic/ai` matches `tcs:topic/ai/genai`). Without tags nothing is listed (hint "tags-missing"); with an empty root and Use the current page off, hint "root-missing". Pages hidden from navigation are left out. Include descendant pages off: children of the root only.

Rendering Rules: Block class `source-tags`; `xdm:listSource: "tags"`.

Responsive Behaviour: As the presentation.

SEO Impact: Topic clusters link related pages (good internal linking).

Accessibility Compliance: As Variation 1.

Variation 5 — Tagged Pages with All Tags

Variation Name: Tagged pages with Match = All of the tags (`all`).

Business Purpose: Narrow lists to pages that carry every selected tag (e.g. industry Banking and topic AI).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-source-tagged-all.png`
> **Purpose:** Tagged pages source (all)
> **Description:** A list of only those insights tagged with both “Banking” and “AI”.

Use Cases: Intersections of industry, topic and content type.

Authoring Guidelines: As Variation 4 with Match = All of the tags.

Business Rules: Every selected tag (or one of its child tags) must be on the page.

Rendering Rules: As Variation 4.

Responsive Behaviour: As the presentation.

SEO Impact: As Variation 4.

Accessibility Compliance: As Variation 1.

Variation 6 — Order by Title

Variation Name: Order by = Title, Sort order Ascending or Descending.

Business Purpose: Alphabetical lists.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-order-title.png`
> **Purpose:** Order by title
> **Description:** Industry pages listed A to Z by their card titles.

Use Cases: Industries, services, locations.

Authoring Guidelines: Source tab: Order by = Title; Sort order (shown when Order by isn’t Manual).

Business Rules: Locale-aware, case-insensitive and numeric comparison (`Intl.Collator` with the page’s language; “Item 2” before “Item 10”); ties by path. Applies to all sources.

Rendering Rules: Sorting happens before Maximum items.

Responsive Behaviour: Unchanged.

SEO Impact: None.

Accessibility Compliance: A predictable order helps users scan the list.

Variation 7 — Order by Last Modified

Variation Name: Order by = Last modified (`lastModified`), usually Descending.

Business Purpose: Newest first lists (news, press releases, insights).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-order-last-modified.png`
> **Purpose:** Order by last modified
> **Description:** A news list with the most recently modified pages first, each with its date.

Use Cases: News, press releases, latest insights.

Authoring Guidelines: Order by = Last modified; Sort order = Descending; Maximum items.

Business Rules: The page's `modified-date` metadata, else when it was last published (index `lastModified`); external links count as 0 (last); ties by path. The shown date is `modified-date`, else `publish-date`, else the last publication.

Rendering Rules: As Variation 6.

Responsive Behaviour: Unchanged.

SEO Impact: Fresh content gets links from the list pages.

Accessibility Compliance: Dates are in `<time datetime>`.

Variation 8 — Maximum Items

Variation Name: Maximum items > 0.

Business Purpose: Limit long lists (e.g. "latest 4").

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-max-items.png`
> **Purpose:** Maximum items
> **Description:** A list of 40 tagged pages limited to the 4 newest.

Use Cases: Teaser rows of the latest content; home page sections.

Authoring Guidelines: Source tab: Maximum items (0–200; 0 = all).

Business Rules: Applied after discovery, filtering, deduplication and sorting; 0, empty, negative or invalid values list all.

Rendering Rules: Only the first N cards are rendered and tracked.

Responsive Behaviour: Unchanged.

SEO Impact: Fewer links per page; link to the full list with the header CTA.

Accessibility Compliance: None specific.

Variation 9 — Fallback Items

Variation Name: When there are no items = Show the fallback items (`fallback`).

Business Purpose: Show curated List Items when Child pages or Tagged pages find nothing.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-fallback-items.png`

> **Purpose:** Fallback items
> **Description:** A tagged-pages List that found no pages showing its two fallback List Items instead.

Use Cases: New topics without tagged pages yet; protecting important sections from empty lists.

Authoring Guidelines: Advanced tab: When there are no items = Show the fallback items; add List Items to the block (as for a Fixed list).

Business Rules: Only for Child pages and Tagged pages; the List Items are processed as a Fixed list (overrides, order, maximum). If the fallback is empty too, the empty message is shown.

Rendering Rules: As Variation 3, but the block class and data layer keep the original source.

Responsive Behaviour: As the presentation.

SEO Impact: None.

Accessibility Compliance: As Variation 3.

Variation 10 — Image Card (default)

Variation Name: Card style = Image card (`image`).

Business Purpose: The standard page teaser card with the page's card image.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-image-card.png`
> **Purpose:** Image card style
> **Description:** Four image cards with page images, kickers (eyebrow | tags | reading time), titles, descriptions and "Article | 04 Sep 26".

Use Cases: Insights, services, any page listing.

Authoring Guidelines: Cards tab: Card style = Image card. Content comes from page metadata (`card-image`, else `og:image` if it isn't the site's default social image, `thumbnail-image`, `featured-image`).

Business Rules: As Cards Variation 1 (6.15.3); pages without an image show no image area. Up to 3 tags; reading time from `reading-time`; content type from `content-type`.

Rendering Rules: `article.list-card.list-card-image`; images built with `createOptimizedPicture` for `/media_` images of the site (750 / 600 px), otherwise a plain `<img>`.

Responsive Behaviour: Title 1.5 / 1.25 / 1.125rem; grid 4 / 4 / 1 by default.

SEO Impact: Descriptive links with the page's card title.

Accessibility Compliance: Alt from `card-image-alt`, else `image-alt`, else decorative. Same contrast note as 6.15.3 Variation 1 (#999 kicker).

Variation 11 — Panel Card

Variation Name: Card style = Panel (`panel`).

Business Purpose: Dark coloured tiles with an "Explore" call to action (Figma 1).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-panel-card.png`
> **Purpose:** Panel card style
> **Description:** Tall #20293c panels with the page eyebrow, a large light title, the description and “Explore”.

Use Cases: Topic hubs, industries.

Authoring Guidelines: Card style = Panel (block or List Item).

Business Rules: As Cards Variation 2: “Explore” by default; kicker = eyebrow / category, else first tag; no image unless Card colour = Image background.

Rendering Rules: `list-card-panel`, colours as 6.15.3 Variation 2.

Responsive Behaviour: 2:3 portrait by default; titles 1.75 / 1.5 / 1.75rem.

SEO Impact: The link’s name is “Explore: {title}”.

Accessibility Compliance: As 6.15.3 Variation 2.

Variation 12 — Report Card

Variation Name: Card style = Report (`report`).

Business Purpose: Report teasers with the content type and date and a “Download” button.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-report-card.png`
> **Purpose:** Report card style
> **Description:** Gradient cards “REPORT | 12 Aug 26” with large titles and an outlined Download button.

Use Cases: Media coverage, investor relations, research pages.

Authoring Guidelines: Card style = Report; pages with `content-type` = Report.

Business Rules: As Cards Variation 3; the link goes to the page (not a file) unless a Fixed item links a file.

Rendering Rules: `list-card-report`, gradient background, secondary button.

Responsive Behaviour: As panels.

SEO Impact: As Variation 10.

Accessibility Compliance: “Download” leads to a page in most cases — consider a Call to action label such as “Read the report” on List Items.

Variation 13 — Statistic Card

Variation Name: Card style = Statistic (`stat`).

Business Purpose: Key figures; in a List usually a Fixed list whose items override the title with the number.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-statistic-card.png`

> **Purpose:** Statistic card style
> **Description:** Four grey statistic cards (number, label, description) each linking to a detail page.

Use Cases: Facts linked to their detail pages.

Authoring Guidelines: Fixed list; per List Item: Page or link (required), Title = the number, Label, Description; Card style = Statistic.

Business Rules: As Cards Variation 4. Because a List Item requires a link, List statistic cards are always linked (Link items off removes the links); for unlinked statistics use Cards.

Rendering Rules: `list-card-stat` with top (kicker, number) and bottom (label, description).

Responsive Behaviour: Number up to 6 / 4 / 5rem, scaled to the card width.

SEO Impact: As 6.15.3 Variation 4.

Accessibility Compliance: As 6.15.3 Variation 4.

Variation 14 — Mixed Card Style

Variation Name: Card style = Mixed (by page) (`mixed`).

Business Purpose: Each page gets the card that suits it (Figma 2, 3, 8, 9).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-mixed.png`
> **Purpose:** Mixed card style
> **Description:** A rail with image cards (pages with images), a panel (a page without an image) and report cards (pages of content type Report).

Use Cases: Insight rails mixing articles and reports.

Authoring Guidelines: Card style = Mixed; maintain `content-type` and card images on the pages.

Business Rules: An item's Card style wins; else `content-type` containing "report" → report; else image if the page has one; else panel.

Rendering Rules: `style-mixed`; data layer `xdm:variant: "mixed"`.

Responsive Behaviour: Keep Stretch cards on so rows line up.

SEO Impact: As the card kinds.

Accessibility Compliance: As the card kinds.

Variation 15 — Feature Card with Author

Variation Name: First card as feature on.

Business Purpose: Spotlight the first page with its image and author (Figma 2).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-feature-card.png`
> **Purpose:** Feature card

> **Description:** A full-width feature with the page image, tags and reading time, a large title and the author's round photo, name and designation, above three smaller cards.

Use Cases: Lead insight of a hub.

Authoring Guidelines: Cards tab: First card as feature on; the first page (after sorting) needs an image and an `author-id` matching the Authors sheet.

Business Rules: As Cards Variation 6, plus the author lockup: the Authors sheet (`/{locale}/data/authors.json`) is loaded only when the feature is on; the page's `author-id` must equal the sheet's id exactly; no match, no lockup.

Rendering Rules: `article.list-feature` with `div.list-author` (`list-author-image` 2.5rem round photo, `list-author-name`, `list-author-designation`).

Responsive Behaviour: Min. height 35.625 / 27.1875 / 32.5rem.

SEO Impact: As 6.15.3 Variation 6.

Accessibility Compliance: The author photo is decorative (`alt=""`); the name is text.

Variation 16 — Rows of Links (Display as teaser off)

Variation Name: Display as teaser off.

Business Purpose: Text rows with large titles and arrows (Figma 5 "News").

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-rows-of-links.png`
> **Purpose:** Rows of links
> **Description:** News rows with eyebrow and tag, a large title, description, date and an arrow on the right.

Use Cases: News, press releases, announcement lists.

Authoring Guidelines: Cards tab: Display as teaser off; usually Order by Last modified, descending.

Business Rules: As Cards Variation 7 (kicker: eyebrow or content type, then the first tag).

Rendering Rules: Block class `links`; `article.list-row`; data layer variant `links`.

Responsive Behaviour: Title 2.5 / 2 / 1.75rem; arrow 3rem (tablet+) / 2rem.

SEO Impact: Text-rich internal links.

Accessibility Compliance: As 6.15.3 Variation 7.

Variation 17 — Card Colour "Dark"

Variation Name: Card colour = Dark (`dark`), on the block or a List Item.

Business Purpose: Dark navy-grey (#20293c) coloured cards.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-colour-dark.png`
> **Purpose:** Dark card colour
> **Description:** Panel and statistic cards of a List on #20293c with light text.

Use Cases: Themes, highlighted items.

Authoring Guidelines: Cards tab: Card colour = Dark, or a List Item's Display tab.

Business Rules: A List Item's colour wins; colours affect panel, report and statistic cards (not image cards); Dark is the panel default.

Rendering Rules: `theme-dark`: background #20293c, text #f3f3f3, muted 70 % white; light buttons.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: High contrast (WCAG 1.4.3).

Variation 18 — Card Colour "Navy"

Variation Name: Card colour = Navy (`navy`).

Business Purpose: Darkest brand blue (#011627), the report colour of Figma 8 and 9.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-colour-navy.png`
> **Purpose:** Navy card colour
> **Description:** Report cards on #011627 with light text and light outlined buttons.

Use Cases: Media coverage, investor relations.

Authoring Guidelines: Card colour = Navy.

Business Rules: As Variation 17.

Rendering Rules: `theme-navy`: #011627, text #f3f3f3, muted 70 % white.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Very high contrast.

Variation 19 — Card Colour "Brand blue"

Variation Name: Card colour = Brand blue (`brand`).

Business Purpose: TCS brand blue (#4673db) cards (Figma 1, 10, 13).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-colour-brand.png`
> **Purpose:** Brand blue card colour
> **Description:** A brand blue statistic card with white text among grey ones.

Use Cases: Highlighting one item.

Authoring Guidelines: Card colour = Brand blue.

Business Rules: As Variation 17.

Rendering Rules: `theme-brand`: #4673db, text white, muted 60 % white.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: White on #4673db ≈ 4.2:1 — AA for large text only (see 6.15.3 Variation 10).

Variation 20 — Card Colour "Light blue"
Variation Name: Card colour = Light blue (`brand-light`).

Business Purpose: Brand tint #a7c8f1 (Figma 13).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-colour-brand-light.png`
> **Purpose:** Light blue card colour
> **Description:** A light blue statistic card with black text.

Use Cases: Colour rows of statistics.

Authoring Guidelines: Card colour = Light blue.

Business Rules: As Variation 17.

Rendering Rules: `theme-brand-light`: #a7c8f1, text #0c0c0d, muted 70 % black.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Meets WCAG 1.4.3.

Variation 21 — Card Colour "Light grey"
Variation Name: Card colour = Light grey (`grey`).

Business Purpose: Neutral #d9d9d9 (the statistic default; Figma 6).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-colour-grey.png`
> **Purpose:** Light grey card colour
> **Description:** Grey panels with black titles.

Use Cases: Calm panels and statistics.

Authoring Guidelines: Card colour = Light grey.

Business Rules: As Variation 17.

Rendering Rules: `theme-grey`: #d9d9d9, text #0c0c0d, muted 70 % black.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Meets WCAG 1.4.3.

Variation 22 — Card Colour "Mid grey"
Variation Name: Card colour = Mid grey (`mid-grey`).

Business Purpose: Darker neutral #999 (Figma 13).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-colour-mid-grey.png`
> **Purpose:** Mid grey card colour
> **Description:** A mid grey statistic card with black text.

Use Cases: Four-up colour rows.

Authoring Guidelines: Card colour = Mid grey.

Business Rules: As Variation 17.

Rendering Rules: `theme-mid-grey`: #999, text #0c0c0d, muted `--black-600`.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Black on #999 ≈ 6.9:1; check the muted text.

Variation 23 — Card Colour "Grey gradient"

Variation Name: Card colour = Grey gradient (`gradient`).

Business Purpose: The report gradient on any card kind (Figma 13).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-colour-gradient.png`
> **Purpose:** Grey gradient card colour
> **Description:** A statistic card with the diagonal grey gradient.

Use Cases: Matching statistics to report cards.

Authoring Guidelines: Card colour = Grey gradient.

Business Rules: As Variation 17.

Rendering Rules: `theme-gradient`: `linear-gradient(135deg, #cfd9e3, #e2e1dd)`, text #303030.

Responsive Behaviour: Same on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Meets WCAG 1.4.3.

Variation 24 — Card Colour "Image background"

Variation Name: Card colour = Image background (`image`).

Business Purpose: Panels with the page's (or item's) image behind the text (Figma 1).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-colour-image.png`

> **Purpose:** Image background card colour
> **Description:** A panel filled with the page image, the title and Explore link on top.

Use Cases: Theme tiles with photography.

Authoring Guidelines: Card style Panel; Card colour Image background (usually on one List Item); the page or item needs an image.

Business Rules: As 6.15.3 Variation 15: only panels render the image; report and statistic cards ignore it.

Rendering Rules: `theme-image`; the media absolute and cover; 1600 / 750 px images.

Responsive Behaviour: Cropped to the panel shape.

SEO Impact: Real `<img>` with alt.

Accessibility Compliance: No overlay: pick calm images for text contrast.

Variation 25 — Image Ratio 1:1

Variation Name: Image ratio = 1:1.

Business Purpose: Square card images (Figma 2, 4, 11).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-ratio-1-1.png`
> **Purpose:** Square image ratio
> **Description:** Image cards with square page images.

Use Cases: Insight grids.

Authoring Guidelines: Cards tab: Image ratio = 1:1.

Business Rules: Image cards only.

Rendering Rules: `ratio-1-1`; `aspect-ratio: 1`, cover.

Responsive Behaviour: Square on every breakpoint.

SEO Impact: None.

Accessibility Compliance: Cropping must keep the described content.

Variation 26 — Image Ratio 4:3

Variation Name: Image ratio = 4:3.

Business Purpose: Taller landscape images.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-ratio-4-3.png`
> **Purpose:** 4:3 image ratio
> **Description:** Image cards with 4:3 images.

Use Cases: People and product photography.

Authoring Guidelines: Image ratio = 4:3.

Business Rules: Image cards only.

Rendering Rules: `ratio-4-3`.

Responsive Behaviour: 4:3 on every breakpoint.

SEO Impact: None.

Accessibility Compliance: As Variation 25.

Variation 27 — Image Ratio 3:2

Variation Name: Image ratio = 3:2.

Business Purpose: Classic photo ratio (16:9 is the default, Variation 10).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-ratio-3-2.png`
> **Purpose:** 3:2 image ratio
> **Description:** Image cards with 3:2 images.

Use Cases: Camera photography.

Authoring Guidelines: Image ratio = 3:2.

Business Rules: Image cards only.

Rendering Rules: `ratio-3-2`.

Responsive Behaviour: 3:2 on every breakpoint.

SEO Impact: None.

Accessibility Compliance: As Variation 25.

Variation 28 — Panel Height "Compact"

Variation Name: Panel height = Compact (9:10).

Business Purpose: Shorter coloured cards.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-panel-compact.png`
> **Purpose:** Compact panel height
> **Description:** Nearly square statistic cards.

Use Cases: Statistic rows, short panels.

Authoring Guidelines: Panel height = Compact.

Business Rules: Panel, report, statistic cards; text that doesn't fit is clipped.

Rendering Rules: `panel-compact`; `aspect-ratio: 9 / 10`.

Responsive Behaviour: Same ratio everywhere.

SEO Impact: None.

Accessibility Compliance: Avoid clipped text (keep page descriptions short).

Variation 29 — Panel Height "Content"

Variation Name: Panel height = Content (`auto`).

Business Purpose: Coloured cards as tall as the text.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-panel-content.png`
> **Purpose:** Content panel height
> **Description:** Panels sized by their content, equal per row.

Use Cases: Pages with long card descriptions.

Authoring Guidelines: Panel height = Content.

Business Rules: No fixed shape.

Rendering Rules: `panel-auto`.

Responsive Behaviour: Heights follow the text.

SEO Impact: None.

Accessibility Compliance: Nothing clipped.

Variation 30 — Style Option "Compact statistic title"

Variation Name: Style options: Compact statistic title (`stat-compact`).

Business Purpose: Statements in statistic cards (Figma 7, 12).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-stat-compact.png`
> **Purpose:** Compact statistic title
> **Description:** Statistic cards with medium-size statement titles and a call to action.

Use Cases: Commitments and claims.

Authoring Guidelines: Card style Statistic; Advanced tab: Style options = Compact statistic title.

Business Rules: Statistic cards only.

Rendering Rules: Title size 0.42 × the number size.

Responsive Behaviour: About 2.52 / 1.68 / 2.1rem.

SEO Impact: None.

Accessibility Compliance: As Variation 13.

Variation 31 — Style Option "Dividers between rows"

Variation Name: Style options: Dividers between rows (`dividers`).

Business Purpose: Lines between rows of links on tablet and desktop (Figma 5).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-dividers.png`
> **Purpose:** Dividers between rows
> **Description:** News rows separated by thin grey lines on desktop.

Use Cases: News lists.

Authoring Guidelines: Display as teaser off; Style options = Dividers between rows.

Business Rules: Rows only; mobile rows always have lines.

Rendering Rules: `dividers`: `border-bottom` in #b2b2b2 from 600 px.

Responsive Behaviour: Lines on every breakpoint with the option.

SEO Impact: None.

Accessibility Compliance: Decorative.

Variation 32 — Style Option "Header centred"

Variation Name: Style options: Header centred (`header-center`).

Business Purpose: Centred title, description and CTA.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-header-centred.png`
> **Purpose:** Centred header
> **Description:** A centred list header above a grid of cards.

Use Cases: Campaign and statistic sections.

Authoring Guidelines: Style options = Header centred.

Business Rules: Header only.

Rendering Rules: `.list-header` as a centred column.

Responsive Behaviour: Centred everywhere.

SEO Impact: None.

Accessibility Compliance: Keep centred text short.

Variation 33 — Header Call to Action

Variation Name: CTA link with CTA text and CTA style (Text link, Primary button, Secondary button).

Business Purpose: "View all" link to the full listing (Figma 2: text link; Figma 5: primary button).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-header-cta.png`
> **Purpose:** Header call to action
> **Description:** The header with "View all news" as a primary button on the right.

Use Cases: Linking to an overview or archive page.

Authoring Guidelines: Content tab: CTA link, CTA text, CTA style.

Business Rules: As 6.15.3 Variation 24. Lists made before this version keep their authored CTA style (bold / italic).

Rendering Rules: `div.list-cta-slot`, button decorator, `data-cmp-interaction="cta"`.

Responsive Behaviour: Right of the title, below it on narrow screens.

SEO Impact: Descriptive CTA texts.

Accessibility Compliance: Named link with focus styles.

Variation 34 — Grid on Every Breakpoint

Variation Name: Desktop / Tablet / Mobile display = Grid (default).

Business Purpose: All cards visible in columns (Figma 2, 11).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-grid-all.png`
> **Purpose:** Grid on every breakpoint
> **Description:** A 4-column desktop grid and the same list in one column on mobile.

Use Cases: Overviews.

Authoring Guidelines: Layout tab: each display Grid; columns 1–6 (4 / 4 / 1); gaps.

Business Rules: As 6.15.3 Variation 25; lists start as grids unless authored otherwise.

Rendering Rules: `desktop-grid tablet-grid mobile-grid`.

Responsive Behaviour: Each breakpoint's columns and gap.

SEO Impact: All cards visible.

Accessibility Compliance: No horizontal scrolling.

Variation 35 — Horizontal Scroll on Every Breakpoint

Variation Name: Every display = Horizontal scroll.

Business Purpose: Rails on all devices (Figma 3, 8, 9).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-scroll-all.png`
> **Purpose:** Horizontal scroll on every breakpoint
> **Description:** A rail with 4.2 cards on desktop, bleeding to the right edge, with buttons and indicator.

Use Cases: Related content, media coverage.

Authoring Guidelines: Each display Horizontal scroll; Items visible or Card width; buttons; indicator; Scroll tab.

Business Rules: As 6.15.3 Variation 26.

Rendering Rules: `desktop-scroll tablet-scroll mobile-scroll`; CSS scrolling with snap.

Responsive Behaviour: Per breakpoint settings.

SEO Impact: All cards in the HTML.

Accessibility Compliance: Focusable region while scrolling; keys; named buttons.

Variation 36 — Scroll on Desktop Only
Variation Name: Desktop scroll; Tablet grid; Mobile grid.

Business Purpose: A rail only on wide screens.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-scroll-desktop-only.png`
> **Purpose:** Desktop-only scroll
> **Description:** Desktop rail; tablet and mobile grids.

Use Cases: Long lists on desktop.

Authoring Guidelines: Desktop display Horizontal scroll; others Grid.

Business Rules: Breakpoints are independent.

Rendering Rules: `desktop-scroll tablet-grid mobile-grid`.

Responsive Behaviour: Only 1200 px+ scrolls.

SEO Impact: None.

Accessibility Compliance: As Variations 34 and 35.

Variation 37 — Scroll on Tablet Only
Variation Name: Desktop grid; Tablet scroll; Mobile grid.

Business Purpose: Avoid cramped tablet grids.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-scroll-tablet-only.png`
> **Purpose:** Tablet-only scroll
> **Description:** Desktop grid; tablet rail; mobile column.

Use Cases: Many desktop columns.

Authoring Guidelines: Tablet display Horizontal scroll; others Grid.

Business Rules: As Variation 36.

Rendering Rules: `desktop-grid tablet-scroll mobile-grid`.

Responsive Behaviour: Only 600–1199 px scrolls.

SEO Impact: None.

Accessibility Compliance: As Variations 34 and 35.

Variation 38 — Scroll on Mobile Only
Variation Name: Desktop grid; Tablet grid; Mobile scroll.

Business Purpose: The most common Figma pattern (1, 4, 6, 10).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-scroll-mobile-only.png`
> **Purpose:** Mobile-only scroll
> **Description:** Mobile: 1.2 cards with the indicator below.

Use Cases: Themes, statistics, insight grids on phones.

Authoring Guidelines: Mobile display Horizontal scroll (indicator on by default); others Grid.

Business Rules: As Variation 36.

Rendering Rules: `desktop-grid tablet-grid mobile-scroll mobile-indicator`.

Responsive Behaviour: Below 600 px scrolls.

SEO Impact: None.

Accessibility Compliance: As Variations 34 and 35.

Variation 39 — Scroll on Desktop and Tablet

Variation Name: Desktop scroll; Tablet scroll; Mobile grid.

Business Purpose: Rails on large screens, stacked on phones.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-scroll-desktop-tablet.png`
> **Purpose:** Desktop and tablet scroll
> **Description:** Rails at 1440 and 1024 px, a column at 402 px.

Use Cases: Content where swiping isn't wanted on phones.

Authoring Guidelines: Desktop and Tablet Horizontal scroll; Mobile Grid.

Business Rules: As Variation 36.

Rendering Rules: `desktop-scroll tablet-scroll mobile-grid`.

Responsive Behaviour: 600 px+ scrolls.

SEO Impact: None.

Accessibility Compliance: As Variations 34 and 35.

Variation 40 — Scroll on Desktop and Mobile

Variation Name: Desktop scroll; Tablet grid; Mobile scroll.

Business Purpose: Rails where a tablet grid fits better.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-scroll-desktop-mobile.png`
> **Purpose:** Desktop and mobile scroll
> **Description:** Rails at 1440 and 402 px, a 2-column grid at 1024 px.

Use Cases: Even card counts on tablets.

Authoring Guidelines: Desktop and Mobile Horizontal scroll; Tablet Grid.

Business Rules: As Variation 36.

Rendering Rules: `desktop-scroll tablet-grid mobile-scroll`.

Responsive Behaviour: 1200 px+ and below 600 px scroll.

SEO Impact: None.

Accessibility Compliance: As Variations 34 and 35.

Variation 41 — Grid on Desktop, Scroll on Tablet and Mobile

Variation Name: Desktop grid; Tablet scroll; Mobile scroll.

Business Purpose: One desktop row, rails on touch devices.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-scroll-tablet-mobile.png`
> **Purpose:** Tablet and mobile scroll
> **Description:** A 4-column grid at 1440 px; rails at 1024 and 402 px.

Use Cases: Four-item rows.

Authoring Guidelines: Desktop Grid; Tablet and Mobile Horizontal scroll.

Business Rules: As Variation 36.

Rendering Rules: `desktop-grid tablet-scroll mobile-scroll`.

Responsive Behaviour: Below 1200 px scrolls.

SEO Impact: None.

Accessibility Compliance: As Variations 34 and 35.

Variation 42 — Previous / Next Buttons

Variation Name: Previous / next buttons on {desktop, tablet, mobile}.

Business Purpose: Visible rail controls.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-previous-next.png`
> **Purpose:** Previous and next buttons
> **Description:** Two chevron buttons below a rail, Previous disabled at the start.

Use Cases: Desktop rails.

Authoring Guidelines: Layout tab (breakpoint in Horizontal scroll): Previous / next buttons on; Scroll tab: Scroll step, Loop.

Business Rules: As 6.15.3 Variation 33: only when the cards overflow; disabled at the ends unless Loop.

Rendering Rules: `button.list-previous` / `button.list-next` (placeholders `listPrevious`, `listNext`), `aria-controls`; data layer `interaction: previous / next` with the list's path.

Responsive Behaviour: Per breakpoint.

SEO Impact: None.

Accessibility Compliance: Named buttons with `disabled` states (WCAG 4.1.2).

Variation 43 — Scroll Indicator

Variation Name: Scroll indicator on {desktop, tablet, mobile} (mobile on by default).

Business Purpose: Progress bar of the rail.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-scroll-indicator.png`
> **Purpose:** Scroll indicator
> **Description:** A thin grey track with a darker thumb below a mobile rail.

Use Cases: Mobile rails.

Authoring Guidelines: Layout tab: Scroll indicator on {breakpoint}.

Business Rules: Only when the breakpoint scrolls and the cards overflow.

Rendering Rules: `div.list-indicator` (`aria-hidden`), thumb sized by the visible share.

Responsive Behaviour: Per breakpoint.

SEO Impact: None.

Accessibility Compliance: Decorative.

Variation 44 — Fixed Card Width

Variation Name: {Breakpoint} card width (px) 120–1200.

Business Purpose: Fixed card widths in rails.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-fixed-card-width.png`
> **Purpose:** Fixed card width
> **Description:** A rail of 320 px cards.

Use Cases: Uniform report or statistic rails.

Authoring Guidelines: Layout tab: Card width (scroll mode); 0 = Items visible.

Business Rules: Below 120 = 0; capped at 1200.

Rendering Rules: `--list-card-width-{bp}` in rem.

Responsive Behaviour: Per breakpoint.

SEO Impact: None.

Accessibility Compliance: As Variation 35.

Variation 45 — Edge "Inside the content width"

Variation Name: Scroll tab: Edge = Inside the content width (`none`).

Business Purpose: Rails that stop at the content width instead of the screen edge (default bleed, as in Figma).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-edge-none.png`
> **Purpose:** Scroller inside the content width
> **Description:** The last visible card cut at the section padding instead of the screen edge.

Use Cases: Sections with backgrounds or boxed layouts.

Authoring Guidelines: Edge = Inside the content width.

Business Rules: Scroll modes only.

Rendering Rules: Without `edge-bleed` no negative right margin (bleed: 1.5 / 2.5 / 5rem).

Responsive Behaviour: Per scrolling breakpoint.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 46 — Snap Alignment "Center"

Variation Name: Scroll snap on, Snap alignment = Center.

Business Purpose: Centre the snapped card.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-snap-center.png`
> **Purpose:** Centre snap alignment
> **Description:** A mobile rail with the current card centred.

Use Cases: Mobile rails with peeking neighbours.

Authoring Guidelines: Scroll tab: Snap alignment = Center.

Business Rules: Shown only with snap on; snap is suspended while dragging.

Rendering Rules: `snap snap-center`.

Responsive Behaviour: Scrolling breakpoints.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 47 — Scroll Step "Viewport"

Variation Name: Scroll step = Viewport.

Business Purpose: Page by screen width.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-viewport.png`

> **Purpose:** Viewport scroll step
> **Description:** After Next, the next set of cards fills the rail.

Use Cases: Long desktop rails.

Authoring Guidelines: Scroll step = Viewport.

Business Rules: Buttons and arrow keys.

Rendering Rules: `scrollBy` the visible width.

Responsive Behaviour: Scrolling breakpoints.

SEO Impact: None.

Accessibility Compliance: Reduced motion respected.

Variation 48 — Loop

Variation Name: Loop on.

Business Purpose: Wrap from the end to the start.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-loop.png`
> **Purpose:** Loop
> **Description:** Next at the last card returns to the first.

Use Cases: Short rails.

Authoring Guidelines: Scroll tab: Loop.

Business Rules: Buttons never disabled; no auto-rotation.

Rendering Rules: Scroll to 0 / the end.

Responsive Behaviour: Scrolling breakpoints.

SEO Impact: None.

Accessibility Compliance: No automatic movement.

Variation 49 — Initial Item

Variation Name: Initial item index > 0.

Business Purpose: Open the rail at a given card.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-initial-item.png`
> **Purpose:** Initial item
> **Description:** A rail opened at its third card.

Use Cases: Timelines, ordered rails.

Authoring Guidelines: Initial item index (0–200).

Business Rules: Applied once, without moving focus; the index refers to the sorted, limited list (after the feature card is taken out).

Rendering Rules: Instant scroll to the item.

Responsive Behaviour: Scrolling breakpoints only.

SEO Impact: None.

Accessibility Compliance: Reading order unchanged.

Variation 50 — Equal Card Height

Variation Name: Equal card height on.

Business Purpose: Equal row heights in multi-row grids.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-equal-height.png`
> **Purpose:** Equal card height
> **Description:** Two grid rows of identical height.

Use Cases: Grids of panels.

Authoring Guidelines: Layout tab: Equal card height.

Business Rules: Default off.

Rendering Rules: `equal-height`; `grid-auto-rows: 1fr`.

Responsive Behaviour: Grid modes.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 51 — Cards Not Stretched

Variation Name: Stretch cards to row height off.

Business Purpose: Natural card heights.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-no-stretch.png`
> **Purpose:** Cards not stretched
> **Description:** Image cards of different heights aligned to the top.

Use Cases: Varying description lengths.

Authoring Guidelines: Layout tab: Stretch cards off.

Business Rules: Default on.

Rendering Rules: `align-items: start`.

Responsive Behaviour: All modes.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 52 — Unlinked Cards (Link items off)
Variation Name: Cards tab: Link items off.

Business Purpose: Show pages' cards without links (e.g. a visual overview).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-unlinked.png`
> **Purpose:** Unlinked cards
> **Description:** Statistic cards without links or buttons.

Use Cases: Statistics from a Fixed list (items require a link).

Authoring Guidelines: Link items off.

Business Rules: No card links, no call to action buttons, no row arrows; the data layer still lists the items with their URLs.

Rendering Rules: No `a.list-card-link`.

Responsive Behaviour: Unchanged.

SEO Impact: No internal links from the list.

Accessibility Compliance: No empty links.

Variation 53 — Open Links in a New Tab
Variation Name: Open links in a new tab (block) and Open in a new tab (List Item: As the list, Yes, No).

Business Purpose: External destinations in a new tab.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-new-tab.png`
> **Purpose:** New-tab links
> **Description:** A fixed list item linking to an external partner site opening in a new tab.

Use Cases: External links in Fixed lists.

Authoring Guidelines: Block default on the Cards tab; per List Item on the Link tab.

Business Rules: The item's Yes / No wins; pages from Child / Tagged sources use the block setting.

Rendering Rules: `target="_blank" rel="noopener noreferrer"`, hidden "(opens in a new tab)"; `xdm:linkType: exit` for external URLs.

Responsive Behaviour: Unchanged.

SEO Impact: None.

Accessibility Compliance: Same call-to-action gap as 6.15.3 Variation 44 (the generated `aria-label` hides the new-tab hint).

Variation 54 — Lists Made Before (legacy)
Variation Name: A List created before this version (a title and a CTA, positional rows, no key-value settings).

Business Purpose: Old pages keep rendering.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-legacy.png`
> **Purpose:** Legacy list
> **Description:** An older List rendered as its title (large heading) and CTA only.

Use Cases: Pages authored with the earlier List.

Authoring Guidelines: Replace with a new List to list pages.

Business Rules: Detected when there are no setting rows and every row has at most one cell; it renders the authored heading and link (CTA style from bold / italic) and no items.

Rendering Rules: Classes `card-list list-legacy`; the title uses `--heading-font-size-xxl`; only the generic block entry of the data layer decorator (no item state).

Responsive Behaviour: Header only.

SEO Impact: As before.

Accessibility Compliance: As before.

Variation 55 — Empty State and Editor Hints

Variation Name: When there are no items = Show a message (default) or Hide the list.

Business Purpose: Control the result when nothing is found and guide authors in the editor.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-empty-state.png`
> **Purpose:** Empty state
> **Description:** In Universal Editor a Tagged pages List showing "The pages show on the published page (from the query index).."; on the page a grey box "There are no items to show."

Use Cases: New sections, new tags, misconfigured lists.

Authoring Guidelines: Advanced tab: When there are no items; Empty message.

Business Rules: On the page: Hide the list empties and hides the block and its wrapper; Show a message shows the Empty message, else `listEmpty`; Show the fallback items: Variation 9. In edit mode a problem hint is shown instead ("Set up the list in its properties: tags-missing, …", placeholder `listEditorHint`), or, for Child / Tagged pages without index data, "The pages show on the published page (from the query index)." (`listEditorPages`).

Rendering Rules: `p.list-empty` after the header.

Responsive Behaviour: Same everywhere.

SEO Impact: A hidden list outputs nothing.

Accessibility Compliance: Plain text message.

Figma variations 1–14

The following variations reproduce the TCS DS Figma List frames (1440 / 1024 / 402 px). Each lists the settings to set; settings not listed keep their defaults. The source is the author's choice (any source works with every presentation); a typical source is suggested. In Cards (6.15) the same presentation settings apply, without the Source tab.

Variation 56 — Figma 1 "Themes"

Variation Name: Figma variation 1, Themes.

Business Purpose: Topic or theme tiles in brand colours and imagery.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-figma-01-themes.png`
> **Purpose:** Figma variation 1
> **Description:** A row of tall panels: a dark panel, a brand blue panel and an image-background panel, each with eyebrow, large title, description and Explore; on mobile a rail with the scroll indicator.

Use Cases: Home page themes; industry entry points.

Authoring Guidelines:

Setting	Value
List source	Fixed list (typical)
Card style	Panel
List Items: Card colour	Dark; Brand blue; Image background (with an image) — per item
Desktop / Tablet display	Grid (columns e.g. 3 or 4)
Mobile display	Horizontal scroll; Scroll indicator on mobile on (default)

Business Rules: Panel defaults: "Explore" call to action, eyebrow (or first tag) in the kicker; image only on the Image background item.

Rendering Rules: `style-panel desktop-grid tablet-grid mobile-scroll mobile-indicator`; items `theme-dark`, `theme-brand`, `theme-image`.

Responsive Behaviour: Desktop / tablet grid of 2:3 panels; mobile rail with 1.2 visible.

SEO Impact: Links to theme pages.

Accessibility Compliance: Check text on brand blue and on the image (no overlay).

Variation 57 — Figma 2 "Insight & Opinion"

Variation Name: Figma variation 2, Insight & Opinion.

Business Purpose: Lead insight with author plus a grid of insights and opinions.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/list-figma-02-insight-opinion.png`
> **Purpose:** Figma variation 2
> **Description:** A feature card with image, tags, reading time, title and author, followed by square-image cards and panels in a 4-column grid; a "View all" text link in the header.

Use Cases: Insights hubs; thought-leadership sections.

Authoring Guidelines:

Setting	Value
List source	Tagged pages or Child pages; Order by Last modified, Descending (typical)
Card style	Mixed
First card as feature	On
Image ratio	1:1
CTA link / CTA text / CTA style	Overview page / "View all" / Text link
Display	Grid 4 / 4 / 1 (default)

Business Rules: The first page becomes the feature (with its author from the Authors sheet); the others follow Mixed rules.

Rendering Rules: `feature-first style-mixed ratio-1-1`.

Responsive Behaviour: Feature full width at every breakpoint; grid 4 / 4 / 1.

SEO Impact: Fresh internal links to insights.

Accessibility Compliance: As Variations 14 and 15.

Variation 58 — Figma 3 "Related"

Variation Name: Figma variation 3, Related.

Business Purpose: "Related to …" rail at the end of articles.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-03-related.png`
> **Purpose:** Figma variation 3
> **Description:** A horizontally scrolling rail of mixed cards bleeding to the right edge on every breakpoint.

Use Cases: Related insights on article and blog pages.

Authoring Guidelines:

Setting	Value
List source	Tagged pages (Search root: the locale's insights; Tags: the article's topics)
Card style	Mixed
Desktop / Tablet / Mobile display	Horizontal scroll (Items visible 4.2 / 3.3 / 1.2 defaults)
Edge	Cards run to the edge (default)
Maximum items	e.g. 8

Business Rules: The current page is listed too if it matches (it isn't excluded automatically) — use a Search root that doesn't contain it, or accept it.

Rendering Rules: `style-mixed desktop-scroll tablet-scroll mobile-scroll edge-bleed`.

Responsive Behaviour: Rail everywhere.

SEO Impact: Topic cluster links.

Accessibility Compliance: As Variation 35.

Variation 59 — Figma 4 "Intelligence in Action"

Variation Name: Figma variation 4, Intelligence in Action.

Business Purpose: Case studies with square images and descriptions.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-04-intelligence-in-action.png`
> **Purpose:** Figma variation 4
> **Description:** Three image cards with square images, titles and descriptions; on mobile a rail.

Use Cases: Case studies, client stories.

Authoring Guidelines:

Setting	Value
List source	Fixed list or Tagged pages
Card style	Image card
Image ratio	1:1
Show description	On (default)
Desktop / Tablet columns	3 / 3
Mobile display	Horizontal scroll

Business Rules: As Variation 10.

Rendering Rules: `style-image ratio-1-1 desktop-grid tablet-grid mobile-scroll; --list-columns-desktop: 3.`

Responsive Behaviour: 3 columns on desktop and tablet; mobile rail.

SEO Impact: As Variation 10.

Accessibility Compliance: As Variation 10.

Variation 60 — Figma 5 "News"

Variation Name: Figma variation 5, News.

Business Purpose: News list with large titles, dates and arrows.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-05-news.png`
> **Purpose:** Figma variation 5
> **Description:** Rows with eyebrow and tag, large titles, descriptions and dates, separated by lines, with an arrow on the right; a primary "All news" button in the header.

Use Cases: Newsroom, press releases, announcements.

Authoring Guidelines:

Setting	Value
List source	Child pages of the news folder; Order by Last modified, Descending; Maximum items e.g. 5
Display as teaser	Off
CTA style	Primary button
Style options	Dividers between rows

Business Rules: As Variation 16.

Rendering Rules: `links dividers`.

Responsive Behaviour: Rows at every breakpoint; lines everywhere.

SEO Impact: Text-rich internal links to news.

Accessibility Compliance: As Variation 16.

Variation 61 — Figma 6 "At a glance"

Variation Name: Figma variation 6, At a glance.

Business Purpose: Key company figures in grey tiles.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-06-at-a-glance.png`
> **Purpose:** Figma variation 6
> **Description:** Four light grey statistic cards with large numbers, labels and descriptions; a rail on mobile.

Use Cases: About us, investor pages.

Authoring Guidelines:

Setting	Value
List source	Fixed list; per item: Page or link, Title (the number), Label, Description
Card style	Statistic
Card colour	Light grey (or default)
Mobile display	Horizontal scroll

Business Rules: As Variation 13; Figma's two-tone title isn't an option (one colour).

Rendering Rules: `style-stat`, `theme-grey` on the cards, `mobile-scroll`.

Responsive Behaviour: Grid 4 / 4 on desktop / tablet; mobile rail.

SEO Impact: As Variation 13.

Accessibility Compliance: As Variation 13.

Variation 62 — Figma 7 "Statements"

Variation Name: Figma variation 7, Statements.

Business Purpose: Short commitments with a call to action each.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-07-statements.png`
> **Purpose:** Figma variation 7
> **Description:** Three statistic cards with medium statement titles, labels, descriptions and a “Learn more” outlined button.

Use Cases: Sustainability and purpose statements.

Authoring Guidelines:

Setting	Value
List source	Fixed list; per item Title (statement), Label, Description, Call to action label
Card style	Statistic
Style options	Compact statistic title
Desktop / Tablet columns	3 / 3

Business Rules: The item’s Call to action label shows a secondary button.

Rendering Rules: `style-stat stat-compact`.

Responsive Behaviour: 3 columns; mobile 1 column (or a rail).

SEO Impact: Descriptive statements.

Accessibility Compliance: Button names “{label}: {title}”.

Variation 63 — Figma 8 “Media coverage”

Variation Name: Figma variation 8, Media coverage.

Business Purpose: Coverage and report items in a rail.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-08-media-coverage.png`
> **Purpose:** Figma variation 8
> **Description:** A rail mixing image cards and navy report cards on every breakpoint.

Use Cases: Newsroom media coverage.

Authoring Guidelines:

Setting	Value
List source	Tagged pages (e.g. a media coverage tag)
Card style	Mixed
Card colour	Navy (report cards)
Desktop / Tablet / Mobile display	Horizontal scroll

Business Rules: Report pages (`content-type` Report) become navy report cards; the block's Navy colour also applies to panels (pages without an image), while image cards are unaffected.

Rendering Rules: `style-mixed`, `theme-navy`, scroll on all breakpoints.

Responsive Behaviour: Rail everywhere.

SEO Impact: As Variation 14.

Accessibility Compliance: As Variations 14 and 18.

Variation 64 — Figma 9 "Investor relations"

Variation Name: Figma variation 9, Investor relations.

Business Purpose: Investor documents and pages in a rail.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-09-investor-relations.png`
> **Purpose:** Figma variation 9
> **Description:** A rail of navy report cards (quarterly results) and image cards.

Use Cases: Investor relations hub.

Authoring Guidelines: As Variation 63 (Mixed, Navy, scroll everywhere), typically Child pages of the investors section ordered by Last modified, Descending.

Business Rules: As Variation 63.

Rendering Rules: As Variation 63.

Responsive Behaviour: Rail everywhere.

SEO Impact: As Variation 14.

Accessibility Compliance: As Variation 63.

Variation 65 — Figma 10 "Banking today"

Variation Name: Figma variation 10, Banking today.

Business Purpose: Industry figures in brand blue.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-10-banking-today.png`
> **Purpose:** Figma variation 10
> **Description:** Brand blue statistic cards with white numbers; a rail on mobile.

Use Cases: Industry pages.

Authoring Guidelines:

Setting	Value
List source	Fixed list (statistics as item overrides)
Card style	Statistic

Setting	Value
Card colour	Brand blue
Mobile display	Horizontal scroll

Business Rules: As Variation 13.

Rendering Rules: `style-stat`, `theme-brand`, `mobile-scroll`.

Responsive Behaviour: Grid on desktop / tablet; mobile rail.

SEO Impact: As Variation 13.

Accessibility Compliance: Large numbers pass; keep descriptions short (Variation 19).

Variation 66 — Figma 11 "2-up"

Variation Name: Figma variation 11, 2-up.

Business Purpose: Two large image cards side by side.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-11-2-up.png`
> **Purpose:** Figma variation 11
> **Description:** Two image cards with square images on desktop and tablet, one column on mobile.

Use Cases: Two featured pages.

Authoring Guidelines:

Setting	Value
List source	Fixed list (2 items)
Card style	Image card
Image ratio	1:1
Desktop / Tablet / Mobile columns	2 / 2 / 1

Business Rules: As Variation 10.

Rendering Rules: `ratio-1-1`, `--list-columns-desktop: 2`.

Responsive Behaviour: 2 / 2 / 1 grid.

SEO Impact: As Variation 10.

Accessibility Compliance: As Variation 10.

Variation 67 — Figma 12 "Statements"

Variation Name: Figma variation 12, Statements (second layout).

Business Purpose: As Figma 7 in a second page context.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-12-statements.png`
> **Purpose:** Figma variation 12
> **Description:** Three compact statistic statements with calls to action.

Use Cases: Purpose and values sections.

Authoring Guidelines: As Variation 62 (Statistic, Compact statistic title, 3 columns, items' Call to action label).

Business Rules: As Variation 62.

Rendering Rules: As Variation 62.

Responsive Behaviour: As Variation 62.

SEO Impact: As Variation 62.

Accessibility Compliance: As Variation 62.

Variation 68 — Figma 13 "4-up colours"

Variation Name: Figma variation 13, 4-up colours.

Business Purpose: Four statistics in four different colours.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-13-4-up-colours.png`
> **Purpose:** Figma variation 13
> **Description:** Four statistic cards: brand blue, grey gradient, mid grey and light blue.

Use Cases: Highlight figures on campaign pages.

Authoring Guidelines:

Setting	Value
List source	Fixed list (4 items)
Card style	Statistic
List Items: Card colour	Brand blue; Grey gradient; Mid grey; Light blue
Desktop / Tablet columns	4 / 4 (default)

Business Rules: Item colours win over the block colour.

Rendering Rules: `theme-brand`, `theme-gradient`, `theme-mid-grey`, `theme-brand-light`.

Responsive Behaviour: 4 columns; mobile 1 column or a rail.

SEO Impact: As Variation 13.

Accessibility Compliance: See Variations 19–23.

Variation 69 — Figma 14 "Statistics row"

Variation Name: Figma variation 14, statistics row.

Business Purpose: A plain statistics row.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-figma-14-statistics.png`
> **Purpose:** Figma variation 14
> **Description:** A row of statistic cards within the content width.

Use Cases: Facts sections.

Authoring Guidelines: Fixed list; Card style Statistic; colours as designed.

Business Rules: The old design's edge-to-edge row isn't reproduced: the row stays within the content width.

Rendering Rules: `style-stat`.

Responsive Behaviour: Grid 4 / 4 / 1 by default.

SEO Impact: As Variation 13.

Accessibility Compliance: As Variation 13.

6.16.4 Authoring Experience

List is a container block with key-value settings and, for the Fixed list source (and for fallback items), **List Item** children added, duplicated, deleted and reordered in the content tree. Fields that don't apply to the chosen source or option are hidden in the dialog (model conditions). Child pages and Tagged pages are discovered from the published query index, which isn't available in the editor: in Universal Editor such lists show the hint "The pages show on the published page (from the query index)." — check them in preview on `aem.page`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A page in Universal Editor with a Fixed list List selected: the header (title, description, CTA), three cards and the content tree on the left listing the List with its List Item children.

Annotations

- A = List title (Content tab: Title, Heading type)
- B = List description (Content tab: Description)
- C = Header call to action (Content tab: CTA link, CTA text, CTA style)
- D = A card of a List Item (selecting it selects the List Item; Card style and Card colour)
- E = Card image (page metadata, or the List Item's Image / Image alt text)
- F = Card kicker: eyebrow and tags (page metadata or List Item Eyebrow / Tags) and reading time
- G = Card title and link (page title / card title, or List Item Title; Page or link)
- H = Card description (page description, or List Item Description)
- I = Card meta row: content type and date (page metadata)
- J = Call to action (List Item Call to action label; panel / report defaults)
- K = Previous / next buttons and scroll indicator (Layout tab)
- L = Content tree: List with its List Items (Fixed list and fallback items)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The properties panel of a selected List with the tabs Content, Source, Cards, Layout, Scroll, Advanced

and Additional; the Source tab open with List source = Tagged pages and the Search root, Use the current page when empty, Tags, Match, Include descendant pages and Maximum search depth fields, followed by Order by, Sort order and Maximum items.

Block tabs (List)

- **Content:** Title; Heading type (H2–H6); Description; CTA link; CTA text; CTA style (Text link, Primary button, Secondary button); Accessible list label; Component ID.
- **Source:** List source (Child pages, Fixed list, Tagged pages; required). Child pages: Parent page, Use the current page when empty, Direct children only, Child depth (when Direct children only is off). Tagged pages: Search root, Use the current page when empty, Tags, Match (Any of the tags, All of the tags), Include descendant pages, Maximum search depth (when descendants are included). Always: Order by (Manual, Title, Last modified), Sort order (when Order by isn't Manual), Maximum items.
- **Cards, Layout, Scroll:** as the Cards block (6.15.4): Display as teaser, Card style, Card colour, Image ratio, Panel height, First card as feature, Link items, Open links in a new tab, Show image / eyebrow / tags / description / date / call to action, Date format, Image loading; per breakpoint display, columns, gap, items visible, card width, buttons, indicator; Equal card height, Stretch cards; Scroll snap, Snap alignment, Hide native scrollbar, Allow mouse drag, Allow touch swipe, Keyboard scrolling, Scroll step, Loop, Initial item index, Edge, Reduced motion support.
- **Advanced:** When there are no items (Hide the list, Show a message, Show the fallback items); Empty message (when Show a message); Analytics; Style options.
- **Additional:** Anchor ID; Hide block; Padding and Margin left / right / top / bottom.

Item tabs (List Item)

- **Link:** Page or link (required); Link text; Open in a new tab (As the list, Yes, No).
- **Content:** Title; Description; Image; Image alt text; Eyebrow; Tags; Label; Call to action label.
- **Display:** Card style (As the list, Image card, Panel, Report, Statistic); Card colour (As the list or a colour); Analytics label; Accessibility label.

6.16.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites select the page and choose Edit.

Expected Result: The page opens in Universal Editor.

[Screenshot Placeholder]
File name: `screenshots/list-step-01-open-page.png`
Purpose: Open the page
Description: The Sites console with the page selected and Edit highlighted.

Step 2 — Add the List block

Action: Select a section, choose Add (+) and pick **List**.

Expected Result: A List with the default settings (Fixed list, image cards, grid 4 / 4 / 1) is inserted; the editor shows "Set up the list in its properties: items-missing."

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** The add menu with List highlighted and the new List showing the set-up hint.

Step 3 — Fill the Content tab

Action: Enter Title, Heading type, Description and optionally CTA link, CTA text, CTA style; an Accessible list label if there is no title; Component ID only when needed.

Expected Result: The header renders.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-03-content-tab.png`
> **Purpose:** Content tab
> **Description:** Content tab with title “Latest insights” and a “View all” CTA.

Step 4 — Choose the source (Source tab)

Action: Choose List source. Child pages: pick the Parent page (or leave it empty for the current page), Direct children only or a Child depth. Tagged pages: pick the Search root (or the current page), the Tags, Match, Include descendant pages and Maximum search depth. Fixed list: continue with Step 9. Then Order by, Sort order and Maximum items.

Expected Result: Fields of other sources are hidden. In the editor, Child / Tagged lists show the published-page hint; the cards appear in preview.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-04-source-tab.png`
> **Purpose:** Source tab
> **Description:** Source tab with Tagged pages, two tags selected, Match Any, Order by Last modified, Descending, Maximum items 8.

Step 5 — Set the Cards tab

Action: Choose Display as teaser, Card style, Card colour, Image ratio, Panel height, First card as feature, Link items, Open links in a new tab, the Show switches, Date format and Image loading.

Expected Result: The presentation changes (visible for Fixed lists in the editor).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-05-cards-tab.png`
> **Purpose:** Cards tab
> **Description:** Cards tab with Card style Mixed and First card as feature on.

Step 6 — Set the Layout tab

Action: Per breakpoint choose Grid (columns) or Horizontal scroll (items visible or card width, buttons, indicator) and the gaps; Equal card height; Stretch cards.

Expected Result: The layout follows the breakpoint of the editor preview width.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-06-layout-tab.png`
> **Purpose:** Layout tab
> **Description:** Layout tab with scroll on all breakpoints and desktop buttons on.

Step 7 — Set the Scroll and Advanced tabs

Action: Adjust the Scroll options if needed; Advanced: When there are no items, Empty message, Analytics, Style options.

Expected Result: Scroll behaviour and empty state as chosen.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-07-scroll-advanced.png`
> **Purpose:** Scroll and Advanced tabs
> **Description:** Advanced tab with “Show the fallback items” selected.

Step 8 — Set the Additional tab

Action: Optionally an Anchor ID, padding and margin; keep Hide block off.

Expected Result: Spacing and anchor applied.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-08-additional-tab.png`
> **Purpose:** Additional tab
> **Description:** Additional tab with Anchor ID “anchor-latest-insights”.

Step 9 — Add List Items (Fixed list or fallback)

Action: In the content tree select the List, Add (+) → **List Item**; on the Link tab pick the Page or link (a page of the site, a path or `https://…`), optionally Link text and Open in a new tab.

Expected Result: The item renders when it has a title (from an override or the Link text in the editor; the page’s metadata on the published page).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-09-add-item.png`
> **Purpose:** Add a List Item
> **Description:** The content tree add menu offering List Item, and the Link tab with a page selected.

Step 10 — Override the item content (optional)

Action: Content tab: Title, Description, Image and Image alt text, Eyebrow, Tags, Label, Call to action label — only where the page’s metadata shouldn’t be used.

Expected Result: The overrides replace the page values on the card.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-10-item-content.png`

> **Purpose:** Item overrides
> **Description:** A List Item with an overridden title and image and the card updated.

Step 11 — Set the item display (optional)

Action: Display tab: Card style, Card colour, Analytics label, Accessibility label.

Expected Result: The card uses its own style / colour.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-11-item-display.png`
> **Purpose:** Item display
> **Description:** A List Item with Card colour Image background in a panel list.

Step 12 — Reorder and duplicate items

Action: Drag List Items in the content tree; duplicate to create similar items.

Expected Result: With Order by Manual the cards follow the tree order.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-12-reorder.png`
> **Purpose:** Reorder items
> **Description:** A List Item dragged to the first position.

Step 13 — Check the listed pages' metadata

Action: For Child / Tagged lists (and page items without overrides), make sure the listed pages have Card title, Card description, Card image (or thumbnail / featured image), Eyebrow, Content type, dates and tags in their page metadata or the bulk metadata sheet, and that they are published.

Expected Result: The cards on the preview page show the intended content.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-13-page-metadata.png`
> **Purpose:** Page metadata
> **Description:** The bulk metadata sheet row of a listed page with card-title, card-description and card-image filled.

Step 14 — Preview

Action: Publish to Preview and open the page on `aem.page`; check every breakpoint.

Expected Result: The discovered (published and indexed) pages show, in the right order and number.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-14-preview.png`
> **Purpose:** Preview
> **Description:** The preview page with the tagged pages rendered as mixed cards.

Step 15 — Validate

Action: Run the Validation Process checklist (optionally with `?debug=list` to log the configuration and cards in the browser console).

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-15-validate.png`
> **Purpose:** Validate
> **Description:** The browser console with the `[list] config` and `[list] cards` debug output.

Step 16 — Publish

Action: Publish the page (and new assets; and the listed pages if not yet live) to Live.

Expected Result: The List shows on the live page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/list-step-16-publish.png`
> **Purpose:** Publish
> **Description:** The publish dialog with the page selected.

Preview Process

1. Universal Editor saves automatically; Fixed lists render in the editor with overrides and link texts (the index isn't available there).
2. Publish to Preview; on `https://main--tcs-corp--sradha-tcs.aem.page/{path}` the List reads `/query-index.json` of that host and shows Child / Tagged pages. Pages are listed once they are published and indexed (README: "a page shows once it is published") — [To be confirmed: whether pages that are only previewed appear in the preview index — owner: site admin].
3. Append `?debug=list` to log the normalized configuration, warnings and cards.
4. Check desktop (1440 px), tablet (1024 / 768 px) and mobile (402 / 375 px).

Publishing Process

1. Confirm the Validation checklist.
2. Make sure the listed pages are published (only published, indexed pages are listed; the index updates after publication).
3. Publish the page to Live, with new assets.
4. Open the live URL; if a new page is missing, wait for the index update and reload.

Validation Process

- ☐ The list shows the expected pages (count, order, Maximum items).
- ☐ No unwanted pages: pages hidden from navigation (`hide-in-nav`) and pages outside the Search root / Parent are not listed; duplicates appear once.
- ☐ Card titles, descriptions, images and alt texts come from the right metadata or overrides.
- ☐ Tags show their labels; dates in the chosen format.
- ☐ Links go to the right pages (redirect targets are used for redirected pages).
- ☐ Card style, colours, feature card (with author) as designed.
- ☐ Grid / scroll per breakpoint; buttons and indicator only where the cards overflow.
- ☐ Headings: the list title level fits the page; card titles one level below.
- ☐ Fixed list: every item has a link; external links open as intended.

- ☐ Empty behaviour configured (message, hide or fallback).
- ☐ With Analytics on, `window.adobeDataLayer` contains the list and its items.

Rollback Process

1. Restore the previous page version (Versions / Timeline) or undo in Universal Editor.
2. Republish the page; if listed pages' metadata changed, restore and republish those pages (or the bulk metadata sheet).
3. Emergency: Hide block on, publish.
4. Code regressions: the development team reverts the commit on `main` (`blocks/list/*` or the shared card list files, which also affect Cards).

6.16.6 Authorable Field Specification

The List model `list` has 86 fields in seven tabs; the List Item model `list-item` has 15 fields in three tabs. The block definition sets `key-value: true`. The Content, Cards, Layout, Scroll and Advanced groups (except the List's "When there are no items") are the shared groups of `models/_card-list.json`, spread into `blocks/list/_list.json`; the Source tab is List-only. Mandatory: List source (`required` in the model; the default Fixed list is pre-filled) and the List Item's Page or link. Enforced validation: Component ID and Anchor ID patterns and the number ranges (`min` / `max`, clamped again in `list-data.js` / `card-list-config.js`); every value is re-validated in code and invalid values fall back to the defaults.

Block model `list` — Content tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Title (`title`)	text	No	No	Empty	Heading of the list	Short and descriptive; up to 60 characters (Recommended)	None	Latest insights
Heading type (`titleType`)	select	No	No	h2	Heading level of the title	Card titles are one level below	h2–h6 (h1 kept for lists made before)	h2
Description (`description`)	richtext	No	No	Empty	Intro under the title	One or two sentences	None	Our latest thinking on AI.
CTA link (`link`)	aem-content	No	No	Empty	Header call to action	Overview / archive page	None	/content/tcs-corp/us/en/insights
CTA text (`linkText`)	text	No	No	Empty	CTA label	2–4 words	None	View all insights
CTA style (`linkType`)	select	No	No	Text link (empty)	CTA look	Text link or a button	'', primary, secondary	''
Accessible list label (`listLabel`)	text	No	No	Empty	Region name without a title	Needed without Title	None	Latest insights
Component ID (`componentId`)	text	No	No	Generated (`list-1`, …)	Deep links and analytics id	Don't change after go-live	Enforced regExp `^[A-Za-z][A-Za-z0-9_-]*$`; duplicates suffixed	latest-insights

Block model `list` — Source tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
List source (`source`)	select	Yes	No	fixed	What is listed	Child pages, Fixed list or Tagged pages	parent, fixed, tags (invalid: fixed)	tags
Parent page (`parentPath`)	aem-content	No	No	Empty	Parent of the child pages	A section page of the same locale	AEM or published path; shown when source = parent	/content/tcs-corp/us/en/industries
Use the current page when empty (`parentUseCurrentPage`)	boolean	No	No	true	Current page as parent	Keep on for section pages	Boolean; source = parent	true
Direct children only (`directChildrenOnly`)	boolean	No	No	true	Depth 1	Off to include deeper pages	Boolean; source = parent	true
Child depth (`childDepth`)	number	No	No	2 (code: 1 when empty)	Levels below the parent	2–3 typical	Enforced 1–10; source = parent and Direct children only off	2
Search root (`rootPath`)	aem-content	No	No	Empty	Root of the tagged pages	The locale or a section	Shown when source = tags	/content/tcs-corp/us/en/insights
Use the current page when empty (`tagsUseCurrentPage`)	boolean	No	No	true	Current page as root	Off with an explicit root	Boolean; source = tags	true
Tags (`tags`)	aem-tag	No (needed)	Yes	None	Tags to match	1–3 tags	Tag IDs; child tags match; source = tags	tcs:topic/ai
Match (`tagMatch`)	select	No	No	any	Any or all tags	All for intersections	any, all; source = tags	any
Include descendant pages (`includeDescendants`)	boolean	No	No	true	All levels below the root	Off: children only	Boolean; source = tags	true
Maximum search depth (`maxDepth`)	number	No	No	0 (no limit)	Levels below the root	0 for all	Enforced 0–10; tags and descendants on	0
Order by (`orderBy`)	select	No	No	manual	Sorting	Last modified for news	manual, title, lastModified	lastModified
Sort order (`sortOrder`)	select	No	No	ascending	Direction	Descending for newest first	ascending, descending; shown when Order by isn't manual	descending
Maximum items (`maxItems`)	number	No	No	0 (all)	Limit	e.g. 4, 8	Model 0–200; code: 0, empty, negative or invalid = all	8

Block model `list` — Cards tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Display as teaser (`displayAsTeaser`)	boolean	No	No	true	Cards (on) or rows of links (off)	Off for news-style link lists	Boolean	true

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Card style (`cardStyle`)	select	No	No	image	Kind of every card	Image, Panel, Report, Statistic or Mixed	image, panel, report, stat, mixed; shown when teaser on	stat
Card colour (`cardTheme`)	select	No	No	Default for the style (empty)	Colour of panel / report / statistic cards	A Card's own colour wins	'', dark, navy, brand, brand-light, grey, mid-grey, gradient, image; teaser on	grey
Image ratio (`imageRatio`)	select	No	No	16-9	Image shape of image cards	Match the photography	16-9, 1-1, 4-3, 3-2; teaser on	1-1
Panel height (`panelHeight`)	select	No	No	portrait	Shape of coloured cards	Content for long texts	portrait, compact, auto; teaser on	compact
First card as feature (`featureFirst`)	boolean	No	No	false	First card as a large spotlight with the author	The first page needs an image and author-id	Boolean; teaser on	true
Link items (`linkItems`)	boolean	No	No	true	Link the cards	Off for informational cards	Boolean	true
Open links in a new tab (`openInNewTab`)	boolean	No	No	false	Default target of the card links (a List Item's setting wins)	Only for external destinations	Boolean	false
Show image (`showImage`)	boolean	No	No	true	Show card images	Off for text-only cards	Boolean	true
Show eyebrow (`showEyebrow`)	boolean	No	No	true	Show the eyebrow	—	Boolean	true
Show tags (`showTags`)	boolean	No	No	true	Show tags and reading time	—	Boolean	true
Show description (`showDescription`)	boolean	No	No	true	Show the description	—	Boolean	true
Show date (`showDate`)	boolean	No	No	true	Show the date	—	Boolean	true
Date format (`dateFormat`)	select	No	No	dd-mmm-yy (04 Sep 26)	Format of the dates	Use the locale's convention	dd-mmm-yy, dd-mm-yyyy, mm-dd-yyyy, mmm-d-yyyy, dd/mm/yy; shown when Show date on	dd-mm-yyyy
Show call to action (`showCta`)	boolean	No	No	true	Show call to action buttons	Off: the titles are the links	Boolean	true
Image loading (`imageLoading`)	select	No	No	lazy	Lazy or eager image loading	Eager only on the first screen	lazy, eager	lazy

Block model `list` — Layout tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Desktop display (`desktopMode`)	select	No	No	grid	Grid or scroll at 1200 px+	—	grid, horizontal-scroll	grid

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Desktop columns (`desktopColumns`)	number	No	No	4	Grid columns on desktop	2–4 typical	Enforced 1–6 (model and code, rounded); grid only	3
Desktop gap (px) (`desktopGap`)	number	No	No	24	Space between cards	Keep the default	Enforced 0–120	24
Desktop items visible (`desktopVisible`)	number	No	No	4.2	Cards in view when scrolling	A fraction shows part of the next	Enforced 1–8; scroll only	3.3
Desktop card width (px) (`desktopCardWidth`)	number	No	No	0	Fixed card width instead	0 uses Items visible	Enforced 0–1200; below 120 treated as 0; scroll only	320
Previous / next buttons on desktop (`desktopControls`)	boolean	No	No	false	Show the buttons	On for desktop rails	Boolean; scroll only	true
Scroll indicator on desktop (`desktopIndicator`)	boolean	No	No	false	Show the progress bar	—	Boolean; scroll only	false
Tablet display (`tabletMode`)	select	No	No	grid	Grid or scroll at 600–1199 px	—	grid, horizontal-scroll	horizontal-scroll
Tablet columns (`tabletColumns`)	number	No	No	4	Grid columns on tablet	2–3 often fit better	Enforced 1–6; grid only	2
Tablet gap (px) (`tabletGap`)	number	No	No	24	Space between cards	—	Enforced 0–120	24
Tablet items visible (`tabletVisible`)	number	No	No	3.3	Cards in view when scrolling	—	Enforced 1–8; scroll only	2.5
Tablet card width (px) (`tabletCardWidth`)	number	No	No	0	Fixed card width	—	Enforced 0–1200; below 120 = 0; scroll only	0
Previous / next buttons on tablet (`tabletControls`)	boolean	No	No	false	Show the buttons	—	Boolean; scroll only	false
Scroll indicator on tablet (`tabletIndicator`)	boolean	No	No	false	Show the progress bar	—	Boolean; scroll only	true
Mobile display (`mobileMode`)	select	No	No	grid	Grid or scroll below 600 px	Scroll for long sets	grid, horizontal-scroll	horizontal-scroll
Mobile columns (`mobileColumns`)	number	No	No	1	Grid columns on mobile	1 (2 for small statistic tiles)	Enforced 1–6; grid only	1
Mobile gap (px) (`mobileGap`)	number	No	No	16	Space between cards	—	Enforced 0–120	16
Mobile items visible (`mobileVisible`)	number	No	No	1.2	Cards in view when scrolling	1.2–1.5	Enforced 1–8; scroll only	1.2
Mobile card width (px) (`mobileCardWidth`)	number	No	No	0	Fixed card width	—	Enforced 0–1200; below 120 = 0; scroll only	0

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Previous / next buttons on mobile (`mobileControls`)	boolean	No	No	false	Show the buttons	Swipe is usually enough	Boolean; scroll only	false
Scroll indicator on mobile (`mobileIndicator`)	boolean	No	No	true	Show the progress bar	Keep on	Boolean; scroll only	true
Equal card height (`equalHeight`)	boolean	No	No	false	All rows equally tall	For multi-row grids	Boolean	false
Stretch cards to row height (`stretchCards`)	boolean	No	No	true	Cards fill the row height	Keep on	Boolean	true

Block model `list` — Scroll tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Scroll snap (`scrollSnap`)	boolean	No	No	true	Cards snap into place	Keep on	Boolean	true
Snap alignment (`snapAlign`)	select	No	No	start	Where cards snap	Center for mobile peeking rails	start, center; shown when snap on	start
Hide native scrollbar (`hideScrollbar`)	boolean	No	No	true	Hide the browser scrollbar	Use the indicator instead	Boolean	true
Allow mouse drag (`mouseDrag`)	boolean	No	No	true	Drag with the mouse	Keep on	Boolean	true
Allow touch swipe (`touchSwipe`)	boolean	No	No	true	Swipe on touch screens	Off only with buttons	Boolean	true
Keyboard scrolling (`keyboardScrolling`)	boolean	No	No	true	Arrow keys, Home, End	Keep on	Boolean	true
Scroll step (`scrollStep`)	select	No	No	one-card	Step of buttons and keys	—	one-card, viewport	one-card
Loop (`loop`)	boolean	No	No	false	Next at the end goes to the start	—	Boolean	false
Initial item index (`initialIndex`)	number	No	No	0	Card shown first	0 = first	Enforced 0–200	0
Edge (`edgePadding`)	select	No	No	bleed	Scroller to the screen edge or not	Bleed as in Figma	bleed, none	bleed
Reduced motion support (`reducedMotion`)	boolean	No	No	true	No smooth scrolling for reduced motion	Keep on	Boolean	true

Block model `list` — Advanced tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
When there are no items (`emptyBehavior`)	select	No	No	message	Empty state	Fallback for important sections	hide, message, fallback (invalid: message)	fallback
Empty message (`emptyMessage`)	text	No	No	Empty (placeholder `listEmpty`)	Empty state text	Short sentence	Shown when behaviour = message	No insights on this topic yet.
Analytics (`analyticsEnabled`)	boolean	No	No	true	Data layer	Keep on	Boolean	true
Style options (`classes`)	multiselect	No	Yes	None	Extra styles	Only as designed	stat-compact, dividers, header-center	dividers

Block model `list` — Additional tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	text	No	No	Empty	In-page link target (`#anchor-…`)	Unique on the page	Enforced regExp `^anchor-[a-z0-9][a-z0-9-]*$`; duplicates get –2, -3	anchor-our-services
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide the block on the page	Temporary use only	Boolean	false
Padding left (`classes_paddingleft`)	select	No	No	None	Inner spacing left	Spacing scale	pl-0, pl-100 … pl-3000	pl-0
Padding right (`classes_paddingright`)	select	No	No	None	Inner spacing right	Spacing scale	pr-0 … pr-3000	pr-0
Padding top (`classes_paddingtop`)	select	No	No	None	Inner spacing top	Spacing scale	pt-0 … pt-3000	pt-600
Padding bottom (`classes_paddingbottom`)	select	No	No	None	Inner spacing bottom	Spacing scale	pb-0 … pb-3000	pb-600
Margin left (`classes_marginleft`)	select	No	No	None	Outer spacing left	Spacing scale	ml-0 … ml-3000	ml-0
Margin right (`classes_marginright`)	select	No	No	None	Outer spacing right	Spacing scale	mr-0 … mr-3000	mr-0
Margin top (`classes_margintop`)	select	No	No	None	Outer spacing top	Spacing scale	mt-0 … mt-3000	mt-1000
Margin bottom (`classes_marginbottom`)	select	No	No	None	Outer spacing bottom	Spacing scale	mb-0 … mb-3000	mb-1000

The padding and margin options are the spacing tokens 0, 100, 200, 300, 400, 500, 600, 800, 1000, 1200, 1600, 2000, 2400 and 3000 (`--space-*`, 0 to 7.5rem).

Item model `list-item` — Link, Content and Display tabs

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Page or link (`link`)	aem-content	Yes	No	Empty	The page (its metadata) or the URL	A page of the locale, a path or `https://…`	Required; unsafe schemes skip the item; items without a link are ignored	/content/tcs-corp/us/en/insights/agentic-bank
Link text (`linkText`)	text	No	No	Empty	Title when the page has none	For external links without a Title	Ignored when equal to the URL	Partner portal
Open in a new tab (`newTab`)	select	No	No	As the list (empty)	Per-item target	Yes for external sites	'', true, false	true
Title (`headline`)	text	No	No	Page title	Overrides the card title	Only when the page's card title doesn't fit	None	The agentic bank
Description (`summary`)	text	No	No	Page description	Overrides the description	Plain text, 1–2 sentences	None	How AI is changing banking.
Image (`image`)	reference	No	No	Page image	Overrides the image	From AEM Assets	None	/content/dam/tcs/insights/agentic.jpg
Image alt text (`imageAlt`)	text	No	No	Page alt (empty = decorative)	Text alternative	Describe meaningful images	None	Bank customer using an AI assistant
Eyebrow (`eyebrow`)	text	No	No	Page eyebrow / category	Overrides the eyebrow	1–3 words	None	Perspective
Tags (`tags`)	aem-tag	No	Yes	Page tags	Overrides the tags	Up to 3 shown	Tag IDs	tcs:industry/banking
Label (`subtitle`)	text	No	No	Empty	Statistic label	Statistic cards	None	Revenue
Call to action label (`ctaLabel`)	text	No	No	Empty (panel Explore, report Download)	Shows a button	1–3 words	None	Read the report
Card style (`kind`)	select	No	No	As the list (empty)	This item's kind	Overrides the list	'', image, panel, report, stat	report
Card colour (`theme`)	select	No	No	As the list (empty)	This item's colour	Overrides the list	'', dark, navy, brand, brand-light, grey, mid-grey, gradient, image	navy
Analytics label (`analyticsLabel`)	text	No	No	Empty	Data layer `dc:title`	Stable name	None (no maxLength in this model)	agentic-bank-card
Accessibility label (`ariaLabel`)	text	No	No	Empty	Link's accessible name	When the title isn't enough	None	Read the agentic bank insight

Field details

Format: Purpose; Allowed Values; Character Limits; Business Rules; Dependency Rules; Visibility Rules (model `condition`); Sample Content. Fields that differ only by breakpoint are described together. The Cards, Layout, Scroll and Additional fields are the same shared fields as in the Cards block; they are repeated here with the List's differences.

Block — Content tab

- **Title (`title`)** — Purpose: list heading, region name and data layer `dc:title`. Allowed: text. Limits: Recommended up to 60 characters. Rules: empty: no heading; the region is named by the Accessible list label, else "List" (placeholder `list`). Dependency: card heading levels. Visibility: always. Sample: "Latest insights".
- **Heading type (`titleType`)** — Purpose: title level. Allowed: h2–h6 (h1 kept in lists made before). Rules: card titles one level lower. Visibility: always. Sample: h2.
- **Description (`description`)** — Purpose: intro. Allowed: rich text. Limits: Recommended up to 200 characters. Visibility: always. Sample: "Our latest thinking."
- **CTA link (`link`)** — Purpose: header call to action destination. Allowed: page or URL. Rules: no link, no CTA (field name unchanged from lists made before). Dependency: CTA text, CTA style. Visibility: always. Sample: the insights overview page.
- **CTA text (`linkText`)** — Purpose: CTA label. Limits: Recommended up to 25 characters. Rules: replaces the link text. Visibility: always. Sample: "View all insights".
- **CTA style (`linkType`)** — Purpose: CTA look. Allowed: Text link (empty), Primary button, Secondary button. Rules: other values: text link; lists made before keep their bold / italic style. Visibility: always. Sample: Text link.
- **Accessible list label (`listLabel`)** — Purpose: region name. Limits: Recommended up to 60 characters. Rules: when set it is the region's `aria-label`; data layer title without a Title. Visibility: always. Sample: "Latest insights".
- **Component ID (`componentId`)** — Purpose: block id, analytics id, card id prefix. Allowed: Enforced pattern (letter first; letters, digits, `-`, `_`), lower-cased in code. Rules: empty: Anchor ID, else `list-1`, `list-2` in page order; duplicates suffixed (logged with `?debug=list`); card ids are `{id}-item-{hash of the destination}`, stable while the destination stays the same. Visibility: always. Sample: "latest-insights".

Block — Source tab

- **List source (`source`)** — Purpose: discovery. Allowed: parent, fixed, tags (Mandatory; invalid: fixed). Rules: decides which fields show and whether the index is loaded (always for Child / Tagged pages; for a Fixed list only when an item links a path). Visibility: always. Sample: tags.
- **Parent page (`parentPath`)** — Purpose: parent of the child pages. Allowed: an AEM path (`/content/tcs-corp/us/en/…`, converted to the published path) or a published path. Rules: only the current locale's pages are in the index. Dependency: Use the current page when empty. Visibility: source = parent. Sample: the industries page.
- **Use the current page when empty (`parentUseCurrentPage`)** — Purpose: current page as the parent. Rules: off without a Parent page: hint `parent-missing`, no items. Visibility: source = parent. Sample: on.
- **Direct children only (`directChildrenOnly`)** — Purpose: depth 1. Rules: on ignores Child depth. Visibility: source = parent. Sample: on.
- **Child depth (`childDepth`)** — Purpose: levels below the parent. Allowed: 1–10 (Enforced). Rules: empty in code = 1 (the dialog pre-fills 2). Visibility: source = parent and Direct children only off. Sample: 2.
- **Search root (`rootPath`)** — Purpose: root of the tagged pages. Allowed: AEM or published path. Rules: the root itself isn't listed. Dependency: Use the current page when empty. Visibility: source = tags. Sample: the insights page.
- **Use the current page when empty (`tagsUseCurrentPage`)** — Purpose: current page as the root. Rules: off without a root: hint `root-missing`. Visibility: source = tags. Sample: on.
- **Tags (`tags`)** — Purpose: tags to match. Allowed: taxonomy tags (multi). Rules: none selected: hint `tags-missing`, no items; child tags match. Visibility: source = tags. Sample: `tcs:topic/ai`.

- **Match (`tagMatch`)** — Purpose: any or all tags. Allowed: any, all. Visibility: source = tags. Sample: any.
- **Include descendant pages (`includeDescendants`)** — Purpose: all levels or children of the root only. Visibility: source = tags. Sample: on.
- **Maximum search depth (`maxDepth`)** — Purpose: depth limit. Allowed: 0–10 (Enforced; 0 = no limit). Visibility: source = tags and Include descendant pages on. Sample: 0.
- **Order by (`orderBy`)** — Purpose: sorting. Allowed: manual (authored order for Fixed, path order for pages), title, lastModified. Rules: ties by path. Visibility: always. Sample: lastModified.
- **Sort order (`sortOrder`)** — Purpose: direction. Allowed: ascending, descending. Visibility: Order by isn't manual. Sample: descending.
- **Maximum items (`maxItems`)** — Purpose: limit. Allowed: 0–200 in the dialog. Rules: applied after deduplication and sorting; 0, empty, negative or invalid = all (no upper limit in code). Visibility: always. Sample: 8.

Block — Cards tab

- **Display as teaser (`displayAsTeaser`)** — Purpose: cards or rows of links. Allowed: on / off. Rules: off ignores card style, colour, ratio, panel height and feature; data layer variant `links`. Visibility: always. Sample: on.
- **Card style (`cardStyle`)** — Purpose: card kind for all items. Allowed: image, panel, report, stat, mixed. Rules: a List Item's own style wins; Mixed picks per card (6.15.3 Variation 5); invalid values: image. Visibility: Display as teaser on. Sample: stat.
- **Card colour (`cardTheme`)** — Purpose: colour of coloured cards. Allowed: empty (default for the style: panel dark, report gradient, statistic grey) or one of 8 colours. Rules: a List Item's colour wins; Image background renders the image only on panels. Visibility: teaser on. Sample: brand.
- **Image ratio (`imageRatio`)** — Purpose: image card image shape. Allowed: 16-9, 1-1, 4-3, 3-2. Rules: invalid: 16-9. Visibility: teaser on. Sample: 1-1.
- **Panel height (`panelHeight`)** — Purpose: shape of panel, report and statistic cards. Allowed: portrait (2:3), compact (9:10), auto (content). Rules: fixed shapes clip overflowing text. Visibility: teaser on. Sample: compact.
- **First card as feature (`featureFirst`)** — Purpose: spotlight the first card. Allowed: on / off. Rules: the feature shows image, tags, reading time, title and the author lockup (Authors sheet, by the page's `author-id`). Visibility: teaser on. Sample: on.
- **Link items (`linkItems`)** — Purpose: make cards clickable. Allowed: on / off. Rules: off: no card links, no call to action buttons, no row arrows. Visibility: always. Sample: on.
- **Open links in a new tab (`openInNewTab`)** — Purpose: default target. Allowed: on / off. Rules: a List Item's Yes / No wins. Visibility: always. Sample: off.
- **Show image / Show eyebrow / Show tags / Show description / Show date / Show call to action (`showImage`, `showEyebrow`, `showTags`, `showDescription`, `showDate`, `showCta`)** — Purpose: hide parts of all cards. Allowed: on / off (default on). Rules: Show tags also hides the reading time; Show call to action off makes titles the links; the feature card always shows its image. Visibility: always. Sample: Show date off.
- **Date format (`dateFormat`)** — Purpose: date display. Allowed: dd-mmm-yy (04 Sep 26), dd-mm-yyyy (04-09-2026), mm-dd-yyyy (09-04-2026), mmm-d-yyyy (Sep 4, 2026), dd/mm/yy (04/09/26). Rules: month names are English abbreviations in every locale; the machine date is in `<time datetime="YYYY-MM-DD">`. Visibility: Show date on. Sample: dd-mm-yyyy.
- **Image loading (`imageLoading`)** — Purpose: loading priority. Allowed: lazy, eager. Rules: eager also sets `fetchpriority="high"`. Visibility: always. Sample: lazy.

Block — Layout tab

- **Desktop / Tablet / Mobile display (`desktopMode`, `tabletMode`, `mobileMode`)** — Purpose: grid or horizontal scroll per breakpoint (desktop ≥ 1200 px, tablet 600–1199 px, mobile < 600 px). Allowed: grid, horizontal-scroll. Rules: independent per breakpoint (8 combinations); invalid: grid. Visibility: always. Sample: Mobile = horizontal-scroll.

- **Columns (`desktopColumns`, `tabletColumns`, `mobileColumns`)** — Purpose: grid columns. Allowed: 1–6 (Enforced; rounded). Defaults 4 / 4 / 1. Visibility: that breakpoint's display = grid. Sample: 3 / 2 / 1.
- **Gap (px) (`desktopGap`, `tabletGap`, `mobileGap`)** — Purpose: space between cards. Allowed: 0–120 (Enforced). Defaults 24 / 24 / 16; converted to rem. Visibility: always. Sample: 24.
- **Items visible (`desktopVisible`, `tabletVisible`, `mobileVisible`)** — Purpose: card width in a scroller as the number of visible cards. Allowed: 1–8, decimals (Enforced). Defaults 4.2 / 3.3 / 1.2. Dependency: ignored when Card width is set. Visibility: display = horizontal-scroll. Sample: 1.2.
- **Card width (px) (`desktopCardWidth`, `tabletCardWidth`, `mobileCardWidth`)** — Purpose: fixed card width. Allowed: 0 or 120–1200 (model 0–1200; values below 120 count as 0). Visibility: horizontal-scroll. Sample: 320.
- **Previous / next buttons on … (`desktopControls`, `tabletControls`, `mobileControls`)** — Purpose: buttons below the scroller. Default off. Rules: shown only when the cards overflow. Visibility: horizontal-scroll. Sample: on (desktop).
- **Scroll indicator on … (`desktopIndicator`, `tabletIndicator`, `mobileIndicator`)** — Purpose: progress bar. Defaults off / off / on. Rules: shown only when the cards overflow. Visibility: horizontal-scroll. Sample: on (mobile).
- **Equal card height (`equalHeight`)** — Purpose: equal grid rows. Default off. Visibility: always. Sample: off.
- **Stretch cards to row height (`stretchCards`)** — Purpose: cards fill their row. Default on. Visibility: always. Sample: on.

Block — Scroll tab (all: Visibility always, except Snap alignment)

- **Scroll snap (`scrollSnap`)** — Purpose: snapping. Default on. Rules: suspended while dragging. Sample: on.
- **Snap alignment (`snapAlign`)** — Purpose: snap position. Allowed: start, center. Visibility: Scroll snap on. Sample: center.
- **Hide native scrollbar (`hideScrollbar`)** — Purpose: hide the browser scrollbar. Default on. Sample: on.
- **Allow mouse drag (`mouseDrag`)** — Purpose: drag scrolling with a mouse (a drag doesn't follow the link under the pointer; 6 px threshold). Default on. Sample: on.
- **Allow touch swipe (`touchSwipe`)** — Purpose: swipe. Default on. Rules: off sets `touch-action: pan-y` (buttons still work). Sample: on.
- **Keyboard scrolling (`keyboardScrolling`)** — Purpose: arrow keys, Home, End on the focused scroller. Default on. Sample: on.
- **Scroll step (`scrollStep`)** — Purpose: step of buttons / keys. Allowed: one-card, viewport. Sample: one-card.
- **Loop (`loop`)** — Purpose: wrap around at the ends. Default off. Sample: off.
- **Initial item index (`initialIndex`)** — Purpose: card shown first. Allowed: 0–200 (Enforced). Sample: 0.
- **Edge (`edgePadding`)** — Purpose: bleed to the screen edge. Allowed: bleed, none. Sample: bleed.
- **Reduced motion support (`reducedMotion`)** — Purpose: instant instead of smooth scrolling for `prefers-reduced-motion`. Default on. Sample: on.

Block — Advanced tab

- **When there are no items (`emptyBehavior`)** — Purpose: empty state. Allowed: hide, message, fallback (invalid: message). Rules: fallback renders the List Items (Child / Tagged pages only); in edit mode problem hints replace the message. Visibility: always. Sample: message.
- **Empty message (`emptyMessage`)** — Purpose: empty state text. Limits: Recommended up to 80 characters. Rules: empty: placeholder `listEmpty`. Visibility: behaviour = message. Sample: "No insights yet."
- **Analytics (`analyticsEnabled`)** — Purpose: data layer. Rules: off: `data-analytics="off"`, no state, no events. Visibility: always. Sample: on.
- **Style options (`classes`)** — Purpose: block classes. Allowed: stat-compact, dividers, header-center (multi). Visibility: always. Sample: dividers.

Block — Additional tab

- **Anchor ID (`classes_anchor`)** — Purpose: in-page anchor. Allowed: `anchor-` + lower-case letters, digits, hyphens (Enforced). Rules: becomes the block id when there is no Component ID; duplicates get `-2`, `-3`. Visibility: always. Sample: anchor-key-facts.
- **Hide block (`classes_hideblock`)** — Purpose: temporarily hide. Rules: hidden on the page and in preview; shown faded in edit mode. Visibility: always. Sample: off.
- **Padding left / right / top / bottom (`classes_paddingleft`, `classes_paddingright`, `classes_paddingtop`, `classes_paddingbottom`)** and **Margin left / right / top / bottom (`classes_marginleft`, `classes_marginright`, `classes_margintop`, `classes_marginbottom`)** — Purpose: spacing around / inside the block. Allowed: empty or a token class (`pl-0` … `pl-3000`, etc.). Rules: styled in `styles/styles.css`. Visibility: always. Sample: mt-1000.

List Item

- **Page or link (`link`)** — Purpose: the item's page (its metadata) or destination. Allowed: a page, a path, `https://`, `mailto:`, `tel:` (Mandatory). Rules: unsafe schemes and empty links skip the item; same-site absolute URLs are read as paths; a path that isn't an indexed page becomes a link card with only the item's values; internal links are replaced by their redirect targets from `/{locale}/redirects.json`. Visibility: always. Sample: an insight page.
- **Link text (`linkText`)** — Purpose: title fallback. Rules: used when there is no Title override and no page title (e.g. external links); ignored when it equals the URL. Sample: "Partner portal".
- **Open in a new tab (`newTab`)** — Purpose: per-item target. Allowed: As the list, Yes, No. Sample: Yes.
- **Title (`headline`)** — Purpose: overrides the card title (`card-title`, else the page title). Limits: Recommended up to 60 characters (statistics: the number). Rules: an item without any title is skipped. Sample: "The agentic bank".
- **Description (`summary`)** — Purpose: overrides the description (plain text). Limits: Recommended up to 160 characters. Sample: "How AI is changing banking."
- **Image (`image`) and Image alt text (`imageAlt`)** — Purpose: override the card image and its alt. Rules: an override image uses its own alt (empty = decorative); without an override image an alt override replaces the page's alt. Limits: alt Recommended up to 125 characters. Sample: an AEM Assets image.
- **Eyebrow (`eyebrow`)** — Purpose: overrides `eyebrow` / `category`. Limits: Recommended up to 30 characters. Sample: "Perspective".
- **Tags (`tags`)** — Purpose: replaces all the page tags. Rules: up to 3 shown. Sample: `tcs:industry/banking`.
- **Label (`subtitle`)** — Purpose: statistic label above the description. Sample: "Revenue".
- **Call to action label (`ctaLabel`)** — Purpose: a button as the card's one link. Limits: Recommended up to 20 characters. Rules: panels default "Explore", reports "Download". Sample: "Read the report".
- **Card style (`kind`)** — Purpose: per-item kind. Allowed: As the list, image, panel, report, stat. Rules: wins over the list's style and Mixed. Sample: report.
- **Card colour (`theme`)** — Purpose: per-item colour. Allowed: As the list or one of 8 colours. Sample: navy.
- **Analytics label (`analyticsLabel`)** — Purpose: data layer `dc:title` and `data-analytics-label`. Limits: not enforced in the List Item model (the Card model enforces 120); Recommended up to 120 characters. Sample: "agentic-bank-card".
- **Accessibility label (`ariaLabel`)** — Purpose: accessible name of the card link. Limits: Recommended up to 100 characters. Sample: "Read the agentic bank insight".

6.16.7 Field Annotation Mapping

Marker	Field	Purpose
A	Title (`title`), Heading type (`titleType`)	List heading; names the region
B	Description (`description`)	Intro text

Marker	Field	Purpose
C	CTA link (`link`), CTA text (`linkText`), CTA style (`linkType`)	Header call to action
D	List Item (model `list-item`); Card style (`kind`), Card colour (`theme`)	One card; its own kind and colour
E	Page metadata `card-image` (…); List Item Image (`image`), Image alt text (`imageAlt`)	Card image and alt
F	Page `eyebrow` / `category`, tags, `reading-time`; List Item Eyebrow (`eyebrow`), Tags (`tags`)	Kicker
G	Page `card-title` / title; List Item Page or link (`link`), Title (`headline`), Link text (`linkText`)	Card title and link
H	Page `card-description` / description; List Item Description (`summary`)	Card text
I	Page `content-type`, `modified-date` / `publish-date`	Meta row
J	List Item Call to action label (`ctaLabel`), Open in a new tab (`newTab`)	Card button
K	`{bp}Controls`, `{bp}Indicator` (Layout tab)	Previous / next and indicator
L	Content tree: List Items	Fixed list and fallback items

6.16.8 Component Specification

Attribute	Value
Component Name	List (`list`), item List Item (`list-item`)
Version	1.0
Owner	[To be confirmed: product owner — owner: TCS digital marketing]; technical owner: development team
Dependencies	`blocks/list/list-data.js`, `decorators/card-list-decorator.js`, `decorators/scroller-decorator.js`, `utils/card-list-config.js`, `utils/query-index.js`, `utils/redirects.js`, `utils/data.js`, `utils/site.js`, `utils/taxonomy.js`, `utils/placeholders.js`, `utils/analytics.js`, `styles/card-list.css`; `/query-index.json`, redirects, authors, taxonomy and placeholders sheets
Component Type	Container block (key-value settings) with item components (List Item)
Supported Variants	Sources Child pages (with depth), Fixed list, Tagged pages (any / all); order manual / title / last modified; maximum items; fallback items; the shared presentation (card styles, feature with author, rows of links, 8 colours, 4 ratios, 3 panel heights, 3 style options, grid / scroll per breakpoint with its options); Figma variations 1–14
Supported Templates	Any page
Supported Metadata	Of the listed pages: `card-title`, `title`, `card-description`, `description`, `card-image`, `og:image`, `thumbnail-image`, `featured-image`, `card-image-alt`, `image-alt`, `eyebrow`, `category`, tags, `modified-date`, `publish-date`, `content-type`, `reading-time`, `author-id`, `hide-in-nav` (from the query index)
Supported Tags	Tagged pages source (any / all, child tags included); card tags from page tags or item overrides; labels from the locale taxonomy
Supported Personalization	None. Data layer state and events are ready for Adobe Target / Journey Optimizer through Adobe Tags (not implemented as of this handover)
Caching Considerations	Page HTML cached by the Edge Delivery CDN; `/query-index.json` (500 rows per request) loaded once per page view and cached in memory (shared with other blocks); redirects, authors, taxonomy and placeholders sheets loaded once and cached; index updates after publication, so new pages appear after indexing

Attribute	Value
Performance Considerations	Index, redirects and authors (feature only) are fetched in parallel; the index is fetched only when needed (not for Fixed lists of external links only); images lazy by default, optimized `/media_` renditions (750 / 600 px; feature 1600 / 750 px); CSS-only layout and scrolling; the list renders after the data arrives (avoid a List as the LCP element)
Accessibility Compliance	WCAG 2.1 AA target, as Cards (6.15.8): named region, list semantics, heading levels, one link per card, named buttons, focus management, `<time>`, reduced motion; same known contrast and new-tab gaps
SEO Compliance	Cards are rendered in the browser from the index (the links are in the DOM after JavaScript); descriptive titles from the pages' metadata; links point to redirect targets (no redirect chains)

6.16.9 Input Model

Content Inputs: the block settings (key | value rows) and List Item rows (Page or link with link text, Title, Description, Image with alt, Eyebrow, Tags, Label, Call to action label, Open in a new tab, Card style, Card colour, Analytics label, Accessibility label).

Metadata Inputs (of the listed pages, in order of preference — `PAGE_MAPPING` in `list-data.js`):

Card part	Metadata (query index column)
Title	`card-title` (`cardTitle`), else the page title (`title`), else the last path segment
Description	`card-description`, else `description`
Image	`card-image`, else `og:image` (`image`, unless it's the site's `default-meta-image`), `thumbnail-image`, `featured-image`
Image alt	`card-image-alt`, else `image-alt`; else decorative
Eyebrow	`eyebrow`, else `category`
Tags	page tags (`tags`)
Date	`modified-date`, else `publish-date`, else the last publication (`lastModified`)
Content type / reading time / author	`content-type`, `reading-time`, `author-id`
Hidden	`hide-in-nav` = yes / true / 1 / on / y: not listed

Tags Inputs: the Tags of the Tagged pages source (IDs), the pages' tags, List Item tag overrides; labels from `/{locale}/taxonomy.json`.

Page Context Inputs: the current page's published path (the default parent / root), the locale (index rows of the locale only, locale sheets), the page language (`<html lang>`, for title sorting), the origin (external links), the editor mode.

Query Inputs: `/query-index.json?offset={n}&limit=500` (all rows, filtered in the browser by path, depth, tags and `hide-in-nav`); `/{locale}/redirects.json` (Source, Destination; up to 5 hops, loops stopped); `/{locale}/data/authors.json` when First card as feature is on. No repository (JCR) queries run in the browser. A query index row:

```
{
  "path": "/en/insights/agentic-bank",
  "title": "The agentic bank",
  "description": "Banking is entering a new era.",
  "image": "/default-meta-image.png",
  "tags": "[\"tcs:industry/banking\",\"tcs:topic/ai\"]",
  "contentType": "Article",
```

```
  "publishDate": "2026-09-04",
  "readingTime": "16",
  "thumbnailImage": "/en/media/agentic.jpg",
  "cardTitle": "", "cardDescription": "", "cardImage": "", "eyebrow": "",
  "hideInNav": "",
  "lastModified": "1790000300"
}
```


Manual Inputs: the Fixed list items and their overrides; the settings.

Authored HTML received from AEM (simplified; settings as key | value rows, then List Item rows):

```
<div class="list">
  <div><div><p>title</p></div><div><p>Latest insights</p></div></div>
  <div><div><p>source</p></div><div><p>fixed</p></div></div>
  <div><div><p>cardStyle</p></div><div><p>mixed</p></div></div>
  <div>
    <div><p><a href="/en/insights/agentic-bank">/en/insights/agentic-bank</a></p></div>
    <div><p>The agentic bank</p></div>
    <div><p>How AI is changing banking.</p></div>
    <div></div>
    <div><p>Perspective</p></div>
  </div>
</div>
```

A two-cell row whose first cell is a List setting is a setting; other rows are List Items (cells matched to the 13 item fields, also when AEM leaves out unset ones). A list with no setting rows and only single-cell rows is a list made before (title and CTA).

6.16.10 Output Model

Rendered Output: as Cards (6.15.10): `card-list` with the layout classes and custom properties, `source-{parent,fixed,tags}` instead of `source-manual`; `div.list-header`, optional `article.list-feature` (with `div.list-author`), `div.list-scroller[role=region] > ul.list-items > li.list-item > article.list-card`… or `article.list-row`, and `div.list-nav`. Lists made before: `card-list list-legacy` with the header only.

Generated HTML (one card of a Tagged pages list, shortened):

```
<div class="list block card-list desktop-scroll tablet-scroll mobile-scroll
  mobile-indicator snap hide-scrollbar edge-bleed stretch snap-start teaser style-mixed
  ratio-16-9 panel-portrait source-tags" id="list-1" data-analytics="custom">
  <div class="list-header">…</div>
  <div class="list-scroller" id="list-1-items" role="region" aria-labelledby="list-1-
title">
    <ul class="list-items">
      <li class="list-item" data-card-id="list-1-item-hxobwp" data-cmp-id="list-1-item-
hxobwp"
        data-cmp-position="1">
        <article class="list-card list-card-image">
          <div class="list-card-media"><picture>…</picture></div>
          <p class="list-card-kicker"><span>Banking</span><span>AI</span><span>16m</
span></p>
          <h3 class="list-card-title"><a class="list-card-link"
            href="/en/insights/agentic-bank" data-card-id="list-1-item-hxobwp"
            data-cmp-interaction="card">The agentic bank</a></h3>
          <p class="list-card-description">Banking is entering a new era.</p>
          <p class="list-card-meta"><span>Article</span>
```

```
            <time datetime="2026-09-04">04 Sep 26</time></p>
        </article>
      </li>
    </ul>
  </div>
  <div class="list-nav">…</div>
</div>
```

Generated JSON: none.

Generated Metadata: none.

Analytics Tracking: with Analytics on (Adobe Client Data Layer, Core Components schema):

```
{ component: {
  'list-1': { '@type': 'tcs-corp/components/list', parentId: 'page-1x2y3z',
    'dc:title': 'Latest insights', 'xdm:listSource': 'tags', 'xdm:variant': 'mixed' },
  'list-1-item-hxobwp': { '@type': 'tcs-corp/components/list/item', parentId: 'list-1',
    'dc:title': 'The agentic bank', 'xdm:linkURL': '/en/insights/agentic-bank',
    'xdm:position': 1, 'repo:modifyDate': '2026-09-04' },
} }
{ event: 'cmp:click', eventInfo: { path: 'component.list-1-item-hxobwp', interaction:
'card',
  'dc:title': 'The agentic bank',
  'xdm:linkURL': 'https://main--tcs-corp--sradha-tcs.aem.live/en/insights/agentic-bank',
  'xdm:linkType': 'other', breakpoint: 'desktop' } }
{ event: 'cmp:click', eventInfo: { path: 'component.list-1', interaction: 'next',
  'dc:title': 'Next items', 'xdm:linkType': 'other', breakpoint: 'desktop' } }
```

`xdm:variant` is the card style or `links`; `xdm:position` is the shown order (including the feature card as position 1); card ids are stable hashes of the destination. As for Cards, the click events carry no `position` for card links, feature card clicks use the list's path, the header CTA reports `interaction: 'cta'`, external links `exit`, files `download`. Analytics off: `data-analytics="off"`. A `?debug=list` URL parameter logs `[list] config` (with warnings) and `[list] cards` with `console.debug`.

Personalization Output: none.

Search Index Output: the List consumes the query index; it adds nothing to it. The page containing the List is indexed with its own metadata. Because the cards are rendered in the browser, crawlers that don't run JavaScript don't see the card links.

6.16.11 Business Rules

1. The block is decorated once (`data-list-decorated`).
2. Rows: two-cell rows whose normalized first cell is a List setting (the shared presentation settings plus source, parentpath, parentusecurrentpage, directchildrenonly, childdepth, rootpath, tagsusecurrentpage, tags, tag-match, includedescendants, maxdepth, orderby, sortorder, maxitems) are settings; rows instrumented as `list-item`, and other rows, are List Items.
3. A list without setting rows whose rows all have at most one cell is a list made before: only its heading and link are rendered (`list-legacy`), with the CTA style read from bold / italic formatting.
4. Paths (Parent page, Search root) are converted from AEM paths to published paths (`toPublishedPath`) and normalized (no `.html`, query, hash or trailing slash).
5. Defaults when missing or invalid: source fixed; parent / root = the current page (Use the current page on); direct children only; child depth 1 when empty (dialog 2), clamped 1–10; match any; descendants included; maximum depth 0 (no limit), clamped 0–10; order manual, ascending; maximum items all; empty behaviour message; presentation defaults as Cards (6.15.11 rule 10).

6. Editor warnings (shown only in edit mode when the list is empty): `tags-missing` (Tagged pages without tags), `parent-missing` (no parent and Use the current page off), `root-missing` (no root and Use the current page off), `items-missing` (Fixed list without items).
7. The query index is loaded only when needed: for Child / Tagged pages, and for a Fixed list with at least one item linking a path (an item written as an absolute URL of the site alone doesn't trigger it, so such items get no page metadata unless another item links a path). It is read in pages of 500 rows, once per page view, cached, and filtered to the current locale (rows below the locale root). A missing or failing index gives no pages (no error), and the empty behaviour applies.
8. The redirects sheet is loaded for every List; the Authors sheet only with First card as feature (a failure gives no author).
9. Listable pages: a path, and `hide-in-nav` not yes / true / 1 / on / y.
10. Child pages: pages below the parent (not the parent itself) at most Child depth levels down.
11. Tagged pages: pages below the root (not the root itself), all levels (or Maximum search depth levels) with Include descendant pages on, else children only; then tag matching — any or all selected tags; a page tag matches a selected tag when it equals it or is a child tag (`{tag}/…`). No tags: no pages.
12. Fixed list: items without a link, or with an unsafe link (not http(s), mailto, tel or a path), are skipped; a same-site absolute URL is read as its path; an indexed page gives its metadata card; another path or external URL gives a card with only the item's values; after overrides, an item without a title is skipped.
13. Overrides (Fixed): Title, Description, Eyebrow, Label, Call to action label, Analytics label, Accessibility label, Card style and Card colour replace the page values when set; the Link text is the title when there is still none; an override image replaces the page image and its alt; Tags replace all page tags; Open in a new tab Yes / No replaces the list default.
14. Card mapping from the index: see 6.16.9 (title fallback: the last path segment with hyphens as spaces; the site's default social image is never a card image).
15. Links of pages (and internal Fixed links) use the redirect target from `/{locale}/redirects.json`, followed up to 5 hops and stopped on loops; external targets end the chain.
16. Deduplication: by destination (path without `.html`, query, hash and trailing slash, or the full external URL, lower case); the first card with a key is kept; cards without a key are dropped.
17. Sorting: Manual keeps the authored order for Fixed lists and uses path order for pages; Title uses a locale-aware, case-insensitive, numeric collator; Last modified uses `modified-date`, else the last publication time (external links 0); Descending reverses; ties by path.
18. Maximum items is applied after deduplication and sorting; 0, empty, negative or invalid values list all.
19. Card ids: `{component id}-item-{hash}` (djb2 hash of the deduplication key).
20. Fallback: with Show the fallback items and no cards from Child / Tagged pages, the List Items are processed as a Fixed list (same rules 12–18).
21. Tag labels from the locale taxonomy (unknown IDs: derived labels).
22. Feature card: the first card after sorting and limiting; the author from the Authors sheet where `id` equals the page's `author-id` (exact match), with photo (decorative), name and designation.
23. Presentation rules 12–32 of Cards (6.15.11) apply unchanged (card kind, colours, tags limits, reading time, dates, linking, call to action defaults and types, new tab, images, feature, rows, heading levels, region name, scrolling, buttons, keys, drag, reduced motion). The region's fallback name is "List" (placeholder `list`).
24. Empty list: on the page, Hide the list empties and hides the block and its wrapper; Show a message shows the Empty message, else `listEmpty`. In edit mode: the warnings (rule 6) as "Set up the list in its properties: {problems}." (`listEditorHint`); for Child / Tagged pages without index data "The pages show on the published page (from the query index)." (`listEditorPages`); else the message.
25. Duplicate Component IDs are suffixed; `?debug=list` logs the duplicate, the configuration, warnings and the cards with `console.debug`.
26. Analytics on: state with `xdm:listSource` and `xdm:variant`; off: `data-analytics="off"`. Analytics errors never break rendering.

27. All texts are set with `textContent` (no HTML injection from metadata); one invalid item never breaks the others.

6.16.12 Author Best Practices

Recommended Content Length (Recommended; set on the listed pages as metadata or as item overrides)

Element	Recommended length
List title	Up to 60 characters
List description	Up to 200 characters
`card-title` / Title override	Up to 60 characters
`card-description` / Description override	Up to 160 characters (panels: 120)
`eyebrow` / `category`	1–3 words
Statistic Title override	Up to 8 characters (or Compact statistic title)
Call to action label	1–3 words
Items	Grid: a multiple of the columns; rails 5–12; set Maximum items for Child / Tagged lists

Image Dimensions

Use	Ratio (CSS)	Recommended source
`card-image` / item image, 16:9	16 / 9	1600 × 900 px
1:1, 4:3, 3:2 ratios	1 / 1, 4 / 3, 3 / 2	1200 × 1200, 1600 × 1200, 1500 × 1000 px
Image background panels	2 / 3, 9 / 10 or content	1200 × 1800 px
Feature card	Full width, min. 35.625rem high (desktop)	2400 × 1200 px, subject on the right
Author photo (Authors sheet)	Round, 2.5rem	Square, at least 160 × 160 px

Give pages a dedicated `card-image` when the page's social image doesn't crop well; the site's default social image (`default-meta-image`) is never used as a card image.

CTA Writing Guidelines

- Header CTA: "View all {topic}" to the overview or archive page.
- Item Call to action labels: verbs ("Read the report", "Explore banking").
- Report lists linking to pages: replace "Download" with a label that matches the destination.

Accessibility Guidelines

- Write meaningful `card-title` and `card-image-alt` metadata on the pages.
- Give lists without a title an Accessible list label.
- Choose the heading type that fits the page outline.
- Use Accessibility labels for items with similar titles; mention new tabs on call to action buttons.
- Check contrast on Brand blue and Image background cards.

SEO Guidelines

- Keep `card-title` close to the page title (consistent anchor texts).
- Use the List for internal linking of hubs (Child pages) and topic clusters (Tagged pages).
- Maintain the redirects sheet when pages move; the List links to the redirect target.
- The cards are rendered with JavaScript; important links should also exist in the page's navigation or text.

Metadata Guidelines

- Maintain per page (page metadata or the locale's bulk metadata sheet): `card-title`, `card-description`, `card-image`, `card-image-alt`, `eyebrow` or `category`, `content-type` (Article, Report, …), `publish-date` / `modified-date` (YYYY-MM-DD), `reading-time` (minutes), `author-id` (matching the Authors sheet), `hide-in-nav` (yes to keep a page out of lists).
- Use `content-type` = Report for pages that should become report cards in Mixed lists.
- After changing metadata, republish the page so the index updates.

Taxonomy Guidelines

- Tag pages with governed tags of the locale taxonomy (Part 7); Tagged pages lists match child tags automatically — select the parent tag to include its children.
- Use Match All only for intentional intersections; it narrows lists quickly.
- Keep tags consistent across locales so equivalent lists work in every locale.

Localization Guidelines

- A List only lists pages of its own locale (index rows below the locale root): configure the same List in each locale copy (Parent page / Search root of that locale).
- Card texts come from each locale's page metadata; translate `card-title` and `card-description` with the pages.
- UI texts come from the locale placeholders sheet (`listPrevious`, `listNext`, `opensInNewTab`, `explore`, `download`, `listEmpty`, `listEditorHint`, `listEditorPages`, `minutesShort`, `list`).
- Title sorting uses the page language (`<html lang>`); month abbreviations in dates are English — prefer numeric date formats in non-English locales.

6.16.13 Troubleshooting Guide

Issue 1 — In Universal Editor the List shows no cards

Symptoms: A Child pages or Tagged pages List shows "The pages show on the published page (from the query index)." in the editor; a Fixed list shows fewer items than authored.

Root Cause: The query index isn't available in the editor, so pages can't be discovered; Fixed list items linking pages are shown only when they have a Title override or a Link text (their metadata comes from the index).

Diagnosis Steps: Check the source; open the page on `aem.page` / `aem.live` to see the real result.

Resolution Steps: None needed for Child / Tagged lists (expected). For Fixed lists, add Title overrides if the editor view must show the items, or validate on preview.

Validation Steps: The preview / live page shows the cards.

Preventive Measures: Always validate Lists on the preview page, not only in the editor.

Issue 2 — A new or changed page doesn't appear in the List

Symptoms: A page that should be listed is missing, or old card content is shown.

Root Cause: The page isn't published or not yet indexed; it has `hide-in-nav` = yes; it is outside the Parent page / Search root or deeper than Child depth / Maximum search depth; it doesn't have the tags (or Match = All); it belongs to another locale; Maximum items cut it off; it is in a path excluded by `helix-query.yaml` (references, error pages, data, block library, templates).

Diagnosis Steps: Open `/query-index.json` on the live host and search for the path; check its columns (`tags`, `hideInNav`, `cardTitle`, …); open the page with `?debug=list` and read `[list] config` and `[list] cards` in the console.

Resolution Steps: Publish the page; fix its metadata or tags; adjust the List's root, depth, tags, match or maximum; re-index the page via the Admin API if needed (L2).

Validation Steps: The page is in the index and the List shows it.

Preventive Measures: Publish pages before linking them; maintain metadata with the bulk metadata sheet.

Issue 3 — Wrong, missing or generic card image

Symptoms: Cards without images, or with an unexpected image.

Root Cause: The page has no `card-image`, its social image is the site's default (skipped), and it has no `thumbnail-image` / `featured-image`; or `card-image` points to an unpublished asset; or Show image is off.

Diagnosis Steps: Check the index row (`cardImage`, `image`, `thumbnailImage`, `featuredImage`) and the page metadata.

Resolution Steps: Set `card-image` (and `card-image-alt`) on the page or bulk sheet and republish; or add an Image override on the List Item.

Validation Steps: The card shows the right image with alt text.

Preventive Measures: Define card images for every listable page type.

Issue 4 — A Tagged pages List is empty or lists too much

Symptoms: No cards (or the message), or unrelated pages.

Root Cause: No tags selected (`tags-missing`); Match = All with tags that no page has together; a too narrow or too broad Search root; parent tags include all child tags.

Diagnosis Steps: Check the Source tab; compare the pages' `tags` in the index; use `?debug=list`.

Resolution Steps: Select the right tags, change Match, set the Search root or Maximum search depth.

Validation Steps: The expected pages are listed.

Preventive Measures: Agree list tag strategies with the taxonomy owner (Part 7).

Issue 5 — A Fixed list item is missing

Symptoms: An authored List Item has no card.

Root Cause: No link or an unsafe link; no title (no override, no page title — e.g. an external link without Link text); a duplicate destination (kept once); Maximum items; a same-site absolute URL while no other item links a path (the index isn't loaded, so there is no page metadata).

Diagnosis Steps: Check the item's Page or link, Title, Link text; look for duplicates; `?debug=list`.

Resolution Steps: Pick the page with the page picker (a path), add a Title or Link text, remove duplicates.

Validation Steps: The item shows.

Preventive Measures: Use the page picker for internal pages; give external links a Title.

Issue 6 — The order of the cards is unexpected

Symptoms: Child or tagged pages aren't in the AEM tree order; "newest first" isn't right.

Root Cause: Order by Manual uses the path order for pages (the repository order isn't in the index); Last modified uses `modified-date` if set, else the last publication time (a republish moves a page up).

Diagnosis Steps: Check Order by / Sort order and the pages' `modified-date` values.

Resolution Steps: Use Title or Last modified with consistent `modified-date` metadata, or use a Fixed list for an editorial order.

Validation Steps: The order matches the expectation.

Preventive Measures: Maintain `modified-date` / `publish-date` for news and insights.

Issue 7 — A card links to a different URL than the page
Symptoms: The card goes to another page than the listed one.

Root Cause: The locale's redirects sheet has an entry for the page path; the List links to the final target (up to 5 hops).

Diagnosis Steps: Check `/{locale}/redirects.json` for the path.

Resolution Steps: Correct or remove the redirect; or hide the old page (`hide-in-nav`).

Validation Steps: The card links where expected.

Preventive Measures: Review redirects when moving pages.

Issue 8 — The feature card has no author
Symptoms: The feature card shows tags and title but no author photo / name.

Root Cause: The page has no `author-id`, or it doesn't match an `id` in `/{locale}/data/authors.json` exactly, or the Authors sheet isn't published.

Diagnosis Steps: Compare the index `authorId` with the Authors sheet ids.

Resolution Steps: Fix the `author-id` metadata or the Authors sheet and publish.

Validation Steps: The lockup shows name, designation and photo.

Preventive Measures: Use the Authors sheet ids consistently (Part 9).

Issue 9 — "There are no items to show." on the live page
Symptoms: The live List shows the empty message.

Root Cause: Nothing matches the source settings; the index request failed (network, CDN); the fallback items are empty.

Diagnosis Steps: Browser network tab: `/query-index.json` responses; `?debug=list`.

Resolution Steps: Fix the settings or content; if the index fails, escalate to L2 / L3.

Validation Steps: Cards show.

Preventive Measures: Use Show the fallback items for critical lists.

Issue 10 — An older List shows only a title and a link
Symptoms: A List created before this version renders a large title and a CTA without cards.

Root Cause: Lists made before (positional rows, no key-value settings) render as `list-legacy` (header only), by design.

Diagnosis Steps: The block has the class `list-legacy`.

Resolution Steps: Replace it with a new List and configure the source.

Validation Steps: The new List shows cards.

Preventive Measures: Inventory legacy Lists before go-live.

6.16.14 Support Runbook
Monitoring Requirements

- Availability and size of `/query-index.json` (the List and Filter List depend on it); watch for index growth (all rows are loaded in 500-row pages).
- Core Web Vitals of pages with Lists in the first screen (the cards appear after the index loads).

- After code releases touching the shared card list files, check Lists and Cards on key pages.
- Data layer checks once Adobe Tags is implemented — [To be confirmed: analytics monitoring owner — owner: analytics team].

Logs

- `?debug=list`: `[list] config` (normalized settings and warnings), `[list] cards`, and `component id "…" is used twice` (`console.debug`, visible with verbose console levels).
- `failed to load block list` / `failed to load module for list` from `scripts/aem.js`.
- Network: `/query-index.json?offset=…&limit=500`, `/{locale}/redirects.json`, `/{locale}/data/authors.json`, `/{locale}/taxonomy.json`, `/{locale}/placeholders.json`.
- Admin API: page status `https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`; indexing via the Admin API index endpoint (`/index/sradha-tcs/tcs-corp/main/{path}`).
- AEM author: publication / replication logs in Cloud Manager.

Common Errors

Error	Likely cause	Action
Editor: "The pages show on the published page"	No index in the editor	Validate on preview
Editor: "Set up the list in its properties: tags-missing"	Tagged pages without tags	Select tags
Editor: "… parent-missing" / "… root-missing"	No parent / root, current page off	Set the path or turn the option on
Editor: "… items-missing"	Fixed list without items	Add List Items
Page missing from the list	Not published / indexed, hidden, outside root, tags	See Issue 2
Empty message on live	No matches or index failure	See Issue 9
Wrong link target	Redirects sheet	Check redirects
No author on the feature	`author-id` mismatch	Fix metadata / Authors sheet

Escalation Matrix

Level	Who	Checks	Escalate when
L1	Author support / content team	Source settings, tags, page metadata, publish state, List Items, `?debug=list` output	Settings and content are right but the list is wrong
L2	AMS / site admin	Query index content and freshness, `helix-query.yaml` exclusions, redirects / authors / taxonomy sheets, Admin API status and re-indexing, CDN cache	Code or index configuration problems
L3	Development team	`blocks/list/*`, `list-data.js` unit tests (`npm test`), shared card list code, `helix-query.yaml`	Platform problems
L4	Adobe Support	Edge Delivery indexing and delivery, Universal Editor, AEM as a Cloud Service	—

[To be confirmed: names, contact channels and response times per level — owner: service manager].

Recovery Procedures

1. Content: restore and republish the page (and the listed pages' metadata if changed).
2. Index: republish affected pages or re-index them via the Admin API (L2).
3. Code: revert the commit on `main` (affects Cards too when shared files change).
4. Mitigation: Hide block, or switch the List temporarily to a Fixed list.

Health Checks

- `npm test` passes `test/list/list-data.test.mjs` (sources, metadata, redirects, deduplication, sorting, maximum items, ids, card kinds, the eight responsive combinations).
- `/query-index.json` returns rows for each locale with the card columns.
- Block library pages render every source and Figma variation.
- On a live page `?debug=list` shows no warnings for configured lists.

Operational Procedures

- New card metadata columns need `helix-query.yaml` and `PAGE_MAPPING` changes (development) and a re-index.
- New placeholders keys must be added to all 13 locale placeholders sheets.
- Model changes (`_list.json`, `models/_card-list.json`) require `npm run build:json` and lint.

Known Limitations

- Content Fragment lists aren't supported (by decision); no full-text search source.
- Child and tagged pages come from the query index: unpublished, excluded or not yet indexed pages aren't listed; Manual order for pages is the path order.
- Tag matching includes child tags (by design).
- In Universal Editor the index isn't available; pages show on the published page.
- Statistic cards don't style "bn"-style units separately; Figma's two-tone title (variation 6) isn't an option; Figma 14's edge-to-edge row isn't reproduced.
- Cards are rendered with JavaScript from the index (not in the delivered HTML).
- The current page isn't excluded from its own Tagged pages list.
- Card click events carry no `position`; feature card clicks use the list's path; the new-tab hint isn't announced on call to action buttons; English month abbreviations in dates.
- The List Item's Analytics label has no length limit in the model (the Card's has 120).

6.17 Filter List Block

The Filter List is the site's content discovery block. It lists the current locale's published pages from the query index (or a manual list of pages), builds a card for each page from the page's metadata, and lets visitors narrow the list with filter pills and an advanced filter panel. It implements the TCS DS Figma designs "Explore the Agentic bank" and "Explore insights & opinions".

Source files: `blocks/filter-list/filter-list.js` (595 lines), `blocks/filter-list/filter-list.css` (623 lines), `blocks/filter-list/_filter-list.json` (the Universal Editor definition and model). The dialog has 52 model entries: 7 tab separators and 45 fields in the tabs Content, Source, Filters, Advanced, Cards, Mapping and Additional, plus 3 fields inside the two composite multifields (Primary filter tags › Tag Id; Advanced filter categories › Category label, Taxonomy root).

6.17.1 Component Purpose

Business Objective

Give visitors one place to explore a collection of content (insights, articles, blogs, reports, case studies, news, events, whitepapers, podcasts, videos) and to narrow it by topic, industry or any other taxonomy, without authors maintaining the list by hand. New pages appear automatically once they are published with the right metadata, which keeps listing pages current and reduces authoring effort.

Business Use Cases

- An "Insights" hub page that lists every insight below `/en/insights`, newest first, with a large hero card for the most important piece.
- An industry landing page ("Explore the Agentic bank") that lists articles and reports tagged with banking topics, with pills for each sub-topic.
- A campaign page that shows a hand-picked, ordered set of pages (Manual List) with the same card design.
- A resource library with an advanced filter panel (for example Industry, Service, Content format categories) combined with AND or OR logic.

When To Use

- The list should be built from page metadata and stay up to date automatically.
- Visitors need to filter a large set of pages by tags.
- The page needs hero and featured spotlight cards chosen by rules (featured flag, priority, latest) rather than by hand.

When Not To Use

- For a short, fixed set of cards with authored text and images that are not pages of the site: use the Cards block (manual authoring) instead.
- For child pages of a page, a fixed list with overrides, or tagged pages without visitor filtering, in the List block's Figma presentations (grid or horizontal scroll per breakpoint): use the List block.
- For a numbered or divided list of services, solutions or products with arrows: use the Table List block (6.18).
- For in-page chapter navigation: use the Anchor List block (6.19).
- When the listed content is not published to Edge Delivery Services (the block only reads `/query-index.json`), or must be listed in Universal Editor itself (the editor shows a hint instead of the results, see 6.17.11).

Dependencies

Kind	Dependency	Used for
Block code	`blocks/filter-list/filter-list.js`, `filter-list.css`, `_filter-list.json`	Decoration, styles, dialog
Decorator	`decorators/masonry-decorator.js` (`layoutMasonry`)	Masonry layout of the card grid
Decorator	`decorators/button-decorator.js` (`decorateButton`)	Load more, Reset filters, Clear all and the card call-to-action buttons
Decorator	`decorators/icon-button-decorator.js` (`createIconButton`)	The advanced filter toggle (filter icon)
Decorator	`decorators/responsive-image-decorator.js` (`buildImage`)	Card, spotlight and author images (optimized when they are site media)
Decorator	`decorators/date-decorator.js` (`buildTime`)	The publish date as a `<time>` element (format `dd-mmm-yy`, e.g. 04 Sep 26)
Decorator	`decorators/article-meta-decorator.js` (`buildAuthorLockup`)	The author lockup on hero and featured spotlight cards
Utility	`utils/query-index.js` (`getPages`, `columnOf`, `listOf`)	The locale's pages and their metadata columns, loaded once per page
Utility	`utils/taxonomy.js` (`loadTaxonomy`, `resolveTags`)	Tag labels for pills, options and card tags
Utility	`utils/tags.js` (`readTagValues`)	Reading tag and multiselect values from the dialog
Utility	`utils/data.js` (`getAuthors`)	The Authors data sheet for spotlight cards
Utility	`utils/placeholders.js` (`getPlaceholders`)	Every user-facing label
Utility	`utils/properties.js` (`readProperties`, `textOf`, `isOn`)	Reading the key-value block properties

Kind	Dependency	Used for
Utility	`utils/multifield.js` (`readMultifieldItems`)	Reading Primary filter tags and Advanced filter categories
Utility	`utils/site.js` (`toPublishedPath`)	Converting AEM paths (`/content/tcs-corp/us/en/...`) to published paths (`/en/...`)
Utility	`utils/environment.js` (`isEditing`), `utils/ids.js` (`uniqueId`)	Editor hint; unique ids for ARIA relationships
Core	`scripts/aem.js` (`toClassName`, `createOptimizedPicture` via buildImage)	Value normalization, responsive images
Icon	`icons/filter.svg`	Advanced filter toggle
Data	`/query-index.json` (defined in `helix-query.yaml`)	The pages and their metadata (all locales; the block keeps the current locale's rows)
Data	`/{locale}/taxonomy.json`	Tag titles in the locale's language
Data	`/{locale}/data/authors.json`	Author name, designation and image for spotlight cards
Data	`/{locale}/placeholders.json`	UI text (keys listed below)
Page metadata	`title`, `description`, `tags` (article:tag), `page-type`, `content-type`, `publish-date`, `reading-time`, `featured-image`, `thumbnail-image`, `author`, `author-id`, `featured`, `priority` (fallbacks `feature-card`, `featured-priority`)	Card content, filtering, sorting, hero and featured rules

Placeholder keys used by the block (Key, English fallback):

Key	English fallback	Where it appears
`filters`	Filters	Label before the pills; accessible name of the pill group
`all`	All	The All pill
`advancedFilters`	Advanced filters	Toggle button name and tooltip; panel title and region name
`selectedCount`	{n} selected	Added to the toggle's accessible name when options are selected
`clearAll`	Clear all	Panel button that clears the selected options
`resultCountOne`	1 result	Result count for one result
`resultCount`	{n} results	Result count
`showingCount`	Showing {n} of {total}	Result count after Load more, while more remain
`loadMore`	Load more	Load more button
`noResources`	No resources found.	Empty state title
`noResourcesHint`	Try selecting another filter.	Empty state hint
`resetFilters`	Reset filters	Empty state button (only when a filter is selected)
`filterListEditor`	The results show on the published page, from the pages' metadata.	Editor hint when the index is not available
`minutesShort`	{n}m	Reading time on cards (a number, e.g. 5 becomes 5m)
`readMore`	Read more	Call to action of Featured style cards

Key	English fallback	Where it appears
`download`	Download	Call to action of report cards (Mixed style)
`explore`	Explore	Call to action of dark cards (Mixed style)

Related Components

- List block (`blocks/list`): child pages, fixed list or tagged pages from the same query index, without visitor filters; applies redirects and leaves out `hide-in-nav` pages (the Filter List does neither).
- Cards block (`blocks/cards`): manual cards in the List's presentation.
- Page Header block: reads the same page metadata (title, description, tags, content type, publish date, reading time, author) on the listed pages.
- Teaser block: a single promoted page.
- Table List (6.18) and Anchor List (6.19): other list-like blocks with authored content.

Applicable Templates

Any page. The block is allowed in every section (`models/_section.json` section filter). It is typically used on landing and listing pages (Page Type `landing` / `listing`) and on the home page; it is not intended for article, blog or press-release detail pages, although nothing prevents it.

Supported Content Types

The pages listed can be of any Page Type and Content Type. The dialog offers these filter values: Page types Article, Blog, Report, Case Study, News, Event; Content types Article, Blog, Report, Case Study, News, Event, Whitepaper, Podcast, Video. Content whose eyebrow (Content type by default) contains "report" gets the report card in the Mixed card style.

6.17.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A Filter List at 1440px wide with Title, Description, the Filters label, All and four tag pills, the round advanced filter button, the hero spotlight, a four-column masonry grid of Mixed cards (image cards, a report card and a dark card), a full-width featured spotlight and the Load more button.

Desktop (1200px and wider). The block is a vertical stack (gap 2.5rem, `--space-1000`): header, results, Load more. The title is a semibold display heading (`--display-font-size-1` from 900px) with a letter-spacing of –0.02 of its size; the description is limited to 50rem wide in the secondary text colour at `--body-font-size-l`. The toolbar shows the "Filters" label, the pills (outline pills with a full radius; the selected pill is filled with the neutral background and light text) and the advanced filter toggle on the same line. The advanced filter panel, when open, is a light grey box (`--color-bg-secondary`, radius l) with a header (title and Clear all) and one column per category (auto-fit, at least 12rem each). The grid has 4 columns for the Default, Compact and Mixed card styles and 3 columns for the Featured card style; the List layout always has one column. Column gap is 1.5rem and row gap 2.5rem. Hero and featured spotlights span the full width: the image covers the card behind a white gradient (from 80% white at the left to transparent at the right), the content has 3rem padding, the spotlight title is 3rem semibold (letter-spacing –0.06rem). The hero is at least 35rem high, a featured spotlight at least 30rem. Card images zoom to 104% on hover and the card title is underlined.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Filter List at 768px wide (two columns) and at 1024px wide (three columns), showing the pills on their own row below the Filters label and the filter button, the panel open with its categories side by side, and the hero spotlight.

Tablet (600px to 1199px). From 600px the grid has 2 columns (all layouts except List). From 900px it has 3 columns (2 for the Featured card style), the title grows to `--display-font-size-1`, the pills sit on the same line as the label (they no longer take a full row) and the toggle follows them; the spotlight content padding grows to 3rem and the spotlight title to 3rem. Between 600px and 899px the pills take a full row below the label and the toggle, and the toggle is pushed to the right edge. The panel shows its categories side by side (auto-fit columns of at least 12rem). In the List layout each card with an image becomes a two-column row: the image in the first third (spanning the rows of the text) and the text in the remaining two thirds, with a divider below each card.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Filter List at 375px wide: title, description, the Filters label with the filter button at the right, the pills wrapping below, one column of cards, a featured spotlight with the image behind its text, and the Load more button centred.

Mobile (below 600px). One column of cards in every layout. The title uses `--display-font-size-2`. The Filters label and the toggle share the first line (the toggle at the right), the pills wrap on the next lines (gap 0.75rem). Spotlight content padding is 2rem by 1.5rem and the spotlight title uses `--display-font-size-3`. List layout cards are stacked (image above the text). The Load more button is centred below the results. Report and dark cards keep a minimum height of 22.5rem.

6.17.3 Component Variations

The Filter List has no "variant" class field; its presentation and behaviour come from select and toggle fields. Each mode the author can choose is described separately below. Layout (3 modes) and Card style (4 styles, of which Mixed produces 2 additional card kinds) are presentation variations; Hero rule, Featured rule, Datasource, pill selection, Filter logic and Sort by are behaviour variations.

Variation 1 — Layout: Masonry

Variation Name: Masonry (Content tab › Default layout = Masonry; the default).

Business Purpose: A magazine-style wall of cards of different heights with no gaps, so mixed content (image cards, report and dark panels) reads as an editorial collection.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-layout-masonry.png`
> **Purpose:** Masonry layout
> **Description:** Four columns of cards of different heights packed below each other without row gaps, in reading order 1, 2, 3, 4 across the top, with a full-width featured spotlight starting below all columns.

Use Cases: Insights hubs, resource libraries with mixed card heights, Mixed card style pages.

Authoring Guidelines: Content tab › Default layout: Masonry. Works with every card style; best with Mixed or Default.

Business Rules: The block class is `layout-masonry` and the grid class `filter-list-grid layout-masonry`. Cards keep their DOM (reading) order across the columns; a full-width item (spotlight) starts below the longest column and the next card starts in the first column again.

Rendering Rules: `layoutMasonry(grid)` adds `masonry-native` in browsers that support `grid-template-rows: masonry` (the browser lays out the grid) and `masonry-spans` elsewhere. With `masonry-spans` the grid uses 0.25rem rows without a row gap; each card spans as many rows as its height plus its 2.5rem bottom margin need, placed by inline `grid-column` / `grid-row` styles. A `ResizeObserver` and `MutationObserver` re-run the layout on content, size and column changes and when Load more adds cards; it is skipped while the block is not rendered and re-runs when it shows. Without `ResizeObserver` the grid falls back to a normal grid.

Responsive Behaviour: Desktop: 4 columns (3 for the Featured card style). Tablet: 2 columns from 600px, 3 from 900px (2 for Featured). Mobile: 1 column (masonry placement is then a simple stack).

SEO Impact: None beyond the block's general impact (cards are rendered in the browser from the index, see 6.17.8). Reading order in the DOM equals the visual order per row.

Accessibility Compliance: DOM order is preserved, so keyboard and screen reader order follows the visual reading order (WCAG 1.3.2 Meaningful Sequence, 2.4.3 Focus Order).

Variation 2 — Layout: Grid

Variation Name: Grid (Content tab › Default layout = Grid).

Business Purpose: A regular, aligned grid where every row starts at the same line, for a tidy catalogue look.

Visual Example:

> [**Screenshot Placeholder**]
> **File name:** `screenshots/filter-list-layout-grid.png`
> **Purpose:** Grid layout
> **Description:** Four aligned columns of Default style cards in rows with a 2.5rem row gap; shorter cards leave space below them.

Use Cases: Catalogues of similar content (all with images and descriptions), event lists, Compact style lists.

Authoring Guidelines: Content tab › Default layout: Grid.

Business Rules: The block and grid get the class `layout-grid`; no masonry placement runs.

Rendering Rules: A CSS grid with `repeat(var(--filter-list-columns), minmax(0, 1fr))` columns, row gap 2.5rem, column gap 1.5rem. Spotlights span all columns (`grid-column: 1 / -1`).

Responsive Behaviour: Desktop: 4 columns (3 for Featured style). Tablet: 2 (600px+) then 3 (900px+; 2 for Featured). Mobile: 1.

SEO Impact: None specific.

Accessibility Compliance: Reading order follows rows left to right; cards are `<article>` elements with an `<h3>` title.

Variation 3 — Layout: List

Variation Name: List (Content tab › Default layout = List).

Business Purpose: A single-column list for scanning titles, such as news or press listings.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-layout-list.png`
> **Purpose:** List layout
> **Description:** One card per row at tablet and desktop width, the 4:3 image in the left third and the tags, title and date in the right two thirds, with a divider line below each card.

Use Cases: News, press releases, events, long text-led lists.

Authoring Guidelines: Content tab › Default layout: List. Combine with Default (shows descriptions) or Compact (no images).

Business Rules: The block gets `layout-list`; the column count stays 1 at every width (the column rules exclude `.layout-list`).

Rendering Rules: Each non-spotlight card has a bottom padding of 2.5rem and a 1px divider. From 600px, cards that have an image become a grid of `1fr 2fr`: the media in column 1 spanning 6 rows, every other part in column 2, column gap 1.5rem. Cards without an image (Compact, dark, report) stay single column.

Responsive Behaviour: Desktop and tablet: image beside the text. Mobile: image above the text.

SEO Impact: None specific.

Accessibility Compliance: Same structure as the other layouts; dividers are decorative borders.

Variation 4 — Card style: Default

Variation Name: Default (Cards tab › Card style = Default).

Business Purpose: The complete card: image, tags, title, description and meta line, for content where the summary helps visitors choose.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-card-default.png`
> **Purpose:** Default card style
> **Description:** A card with a 4:3 rounded image, an uppercase grey tag line (two tags and 5m separated by bars), a regular-weight title, a three-line description and a meta line "Article | 04 Sep 26".

Use Cases: Insights and article listings where descriptions are well written.

Authoring Guidelines: Cards tab › Card style: Default. Make sure the listed pages have a Description and a featured or thumbnail image.

Business Rules: Block class `card-default`. Every standard card is the `standard` kind.

Rendering Rules: `<article class="filter-list-card filter-list-card-standard">` with: media (if the page has an image; 4:3, radius l, `object-fit: cover`), tag row (up to 2 tag labels and the reading time), `<h3>` title with the card link, description (clamped to 3 lines), meta row (eyebrow and publish date). The title link's `::before` covers the whole card, so the whole card is clickable.

Responsive Behaviour: Desktop 4 columns; tablet 2 / 3; mobile 1.

SEO Impact: Title and description text are present in the rendered DOM.

Accessibility Compliance: One link per card (the title), so screen readers announce the title once; images have empty alt (decorative, the title names the link); focus ring drawn around the whole card (`:focus-visible::before` outline).

Variation 5 — Card style: Compact

Variation Name: Compact (Cards tab › Card style = Compact).

Business Purpose: A text-only, outlined card for dense lists where images add no value.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/filter-list-card-compact.png`
> **Purpose:** Compact card style
> **Description:** Outlined text cards with a tag line, a medium-size title and the meta line, no image and no description.

Use Cases: News, events, press, document libraries.

Authoring Guidelines: Cards tab › Card style: Compact.

Business Rules: Block class `card-compact`; cards are the `compact` kind and never show an image or description, even when the page has them.

Rendering Rules: Tag row, `<h3>` title link (`--heading-font-size-m`, letter-spacing –0.015 of its size), meta row; 1.5rem padding, 1px border, radius l.

Responsive Behaviour: Desktop 4 columns; tablet 2 / 3; mobile 1.

SEO Impact: Titles only; descriptions are not rendered.

Accessibility Compliance: As Default; no images.

Variation 6 — Card style: Featured

Variation Name: Featured (Cards tab › Card style = Featured).

Business Purpose: Larger cards with an explicit call to action, for fewer, more important items.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/filter-list-card-featured.png`
> **Purpose:** Featured card style
> **Description:** Three-column large cards with an image, tags, a semibold title, a description, the meta line and a "Read more" text button at the bottom.

Use Cases: Flagship reports, campaign pages, small curated collections.

Authoring Guidelines: Cards tab › Card style: Featured. Use with Grid or Masonry. Do not confuse with the featured spotlight cards (Enable featured cards), which are a separate setting.

Business Rules: Block class `card-featured`; cards are the `large` kind. The grid has one column fewer at large widths (2 from 900px, 3 from 1200px).

Rendering Rules: Media, tag row, `<h3>` title (semibold, not a link), description, meta row and a `Read more` tertiary button (`placeholder readMore`) that is the card's link; its accessible name is "Read more: {title}". The link covers the card.

Responsive Behaviour: Desktop 3 columns; tablet 2 (600px to 1199px); mobile 1.

SEO Impact: The link text is generic ("Read more"); the title is in the heading and the `aria-label`, but search engines read the visible link text. Prefer Default for SEO-sensitive listings.

Accessibility Compliance: The button's `aria-label` includes the title (WCAG 2.4.4 Link Purpose in context, 2.5.3 Label in Name: the visible "Read more" starts the label).

Variation 7 — Card style: Mixed (standard image cards)

Variation Name: Mixed (Cards tab › Card style = Mixed; the default style).

Business Purpose: The Figma "Explore insights & opinions" mix: image cards, report panels and dark panels in one wall, chosen per page from its metadata.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-card-mixed.png`
> **Purpose:** Mixed card style
> **Description:** A masonry wall where most cards are image cards without a description, one card is a light gradient report panel with a Download button and one is a dark blue panel with an Explore link.

Use Cases: Insights hubs with a mix of articles, reports and image-less pages.

Authoring Guidelines: Cards tab › Card style: Mixed. The card kind is chosen per page: pages whose eyebrow contains "report" become report cards (Variation 8), pages without an image become dark cards (Variation 9), all others are standard image cards without a description.

Business Rules: Block class `card-mixed`. Order of the checks: report first, then no image, then standard. A report page without an image is a report card.

Rendering Rules: Standard kind with media, tag row, `<h3>` title link and meta row; no description.

Responsive Behaviour: Desktop 4 columns; tablet 2 / 3; mobile 1.

SEO Impact: Descriptions only appear on report and dark cards.

Accessibility Compliance: As Default.

Variation 8 — Mixed card kind: Report card

Variation Name: Report card (Mixed style, automatic).

Business Purpose: Highlights research reports and whitepapers with a distinct panel and a Download call to action.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-card-report.png`
> **Purpose:** Report card in the Mixed style
> **Description:** A card with a light brand-to-white diagonal gradient, the line "REPORT | 12 Aug 26", a semibold title, a three-line description and a secondary "Download" button at the bottom.

Use Cases: Reports, whitepapers and other downloadable content in an insights hub.

Authoring Guidelines: On the listed page, set the metadata used as the eyebrow (Mapping tab › Eyebrow source, `content-type` by default) to a value that contains "report" (case-insensitive), e.g. `Report`. Card style must be Mixed.

Business Rules: Applies only in the Mixed style. The test is `/report/i` on the eyebrow text, so "Annual Report" and "Reports" also match. The Download button links to the page itself (the card URL), not to a file.

Rendering Rules: `filter-list-card-report`: tag row (eyebrow and date), `<h3>` title (not a link), description, `Download` secondary button (link to the page, accessible name "Download: {title}"). Padding 2rem by 1.5rem, minimum height 22.5rem, radius l, background `linear-gradient(160deg, var(--brand-200) 0%, var(--color-bg-default) 70%)`. No image.

Responsive Behaviour: Same size rules as the other cards in the grid at every breakpoint.

SEO Impact: The visible link text is "Download"; search engines may not associate it with the title.

Accessibility Compliance: The button's accessible name includes the title. Because the button opens the page, the label "Download" may be unexpected; consider this when writing report pages (see 6.17.12).

Variation 9 — Mixed card kind: Dark card

Variation Name: Dark card (Mixed style, automatic).

Business Purpose: Gives image-less pages a strong visual presence instead of an empty image frame.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-card-dark.png`
> **Purpose:** Dark card in the Mixed style
> **Description:** A dark brand-blue panel with one uppercase tag, a white semibold title, a light description and a white "Explore" text button at the bottom.

Use Cases: Opinion pieces, podcasts or pages that have no featured or thumbnail image.

Authoring Guidelines: Leave both image columns (Image source and Fallback image, by default `featured-image` and `thumbnail-image`) empty on the listed page. Card style must be Mixed.

Business Rules: Applies in the Mixed style to non-report pages without an image. Shows only the first tag (no reading time).

Rendering Rules: `filter-list-card-dark`: tag row (one tag), `<h3>` title, description, `Explore` tertiary button (link to the page, accessible name "Explore: {title}"). Background `--brand-600`, light text (`--color-text-on-neutral`, `--color-text-light-overlay`), padding 2rem by 1.5rem, minimum height 22.5rem.

Responsive Behaviour: As other cards.

SEO Impact: Generic link text "Explore".

Accessibility Compliance: Light text on `--brand-600`; [To be confirmed: colour contrast of `--color-text-light-overlay` on `--brand-600` measured against WCAG 1.4.3 — owner: design system].

Variation 10 — Hero spotlight: Featured rule

Variation Name: Hero card, Hero rule = Featured (Cards tab › Enable hero card = on, Hero rule = Featured; Featured is the default rule).

Business Purpose: Opens the list with one large spotlight for the most important featured page.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-hero-featured.png`
> **Purpose:** Hero spotlight (Featured rule)
> **Description:** A full-width card at least 35rem high above the grid: the page's image covers the card behind a white gradient; at the left the tags, a 3rem semibold title, the description and the author lockup with a round photo, name and designation.

Use Cases: Insights hub with a lead story.

Authoring Guidelines: Turn on Enable hero card; keep Hero rule = Featured. Mark the lead page with `featured` = `yes` (or `true`, `1`, `on`, `y`) and give it the lowest `priority` among featured pages.

Business Rules: The hero is the matching featured page with the lowest priority (pages without a priority last; ties by latest publish date). If no matching page is featured, the latest page (publish date, then last modified) is the hero. The hero is chosen again whenever the filters change, from the pages that match them; it is removed from the featured and standard cards.

Rendering Rules: `<article class="filter-list-card filter-list-card-spotlight filter-list-card-hero">` placed above the grid (inside `filter-list-results`, before the grid). Media is absolutely positioned behind the content, the image requested at 1600px (900px+) or 750px wide and loaded eagerly (`loading="eager"`). Content: tag row, `<h3>` title link, description, author lockup (when an author is found).

Responsive Behaviour: Desktop and tablet from 900px: content padding 3rem, title 3rem. Below 900px: padding 2rem by 1.5rem, title `--display-font-size-3`. Always full width, minimum height 35rem.

SEO Impact: The hero title is an `<h3>` (below the block's `<h2>` title).

Accessibility Compliance: Image alt is empty (decorative); text sits on a white gradient over the image for contrast; the card link covers the card with a visible focus outline.

Variation 11 — Hero spotlight: Latest rule

Variation Name: Hero card, Hero rule = Latest.

Business Purpose: Always promotes the newest page, without editorial flags.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-hero-latest.png`
> **Purpose:** Hero spotlight (Latest rule)
> **Description:** The hero spotlight showing the page with the most recent publish date of the list.

Use Cases: News or blog hubs.

Authoring Guidelines: Enable hero card = on; Hero rule = Latest. Give every listed page a `publish-date` (YYYY-MM-DD).

Business Rules: The page with the latest `publish-date` (compared as text, so the ISO format is required); ties and pages without a date by last modified (newest first). Pages without a publish date sort after dated ones. Independent of Sort by.

Rendering Rules: As Variation 10.

Responsive Behaviour: As Variation 10.

SEO Impact: As Variation 10.

Accessibility Compliance: As Variation 10.

Variation 12 — Hero spotlight: Manual rule

Variation Name: Hero card, Hero rule = Manual.

Business Purpose: The author chooses the hero page explicitly.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-hero-manual.png`
> **Purpose:** Hero spotlight (Manual rule)
> **Description:** The dialog's Cards tab with Hero rule = Manual and the Hero page picker filled, next to the resulting hero card.

Use Cases: Campaign launches; a specific flagship report.

Authoring Guidelines: Enable hero card = on; Hero rule = Manual; pick the Hero page (shown only for this rule).

Business Rules: The Hero page is used only when it is among the pages that match the source settings and the current filters; otherwise there is no hero (no fallback). When the visitor filters it out, the hero disappears until the filters match it again.

Rendering Rules: As Variation 10.

Responsive Behaviour: As Variation 10.

SEO Impact: As Variation 10.

Accessibility Compliance: As Variation 10.

Variation 13 — Featured spotlights: featured=true rule

Variation Name: Featured cards, Featured rule = featured=true (Cards tab › Enable featured cards = on, the default; Featured rule = featured=true, the default).

Business Purpose: Breaks up long lists with large spotlight cards for flagged pages.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-featured-flag.png`
> **Purpose:** Featured spotlight (featured=true)
> **Description:** A full-width spotlight card at least 30rem high inserted after the first page of nine cards, with tags, title and author lockup over the image; a second one after the next Load more.

Use Cases: Editorially promoted pages inside a hub.

Authoring Guidelines: Mark pages with `featured` = `yes` (or `feature-card`, the earlier name); order them with `priority` (1 first). Set Featured card count (default 2).

Business Rules: Featured pages are the matching pages (after the hero) with the featured flag, ordered by priority then latest. If none is flagged, pages with `priority` 3 or lower are used instead. At most Featured card count are taken; they are removed from the standard cards. One featured card is inserted after each page of cards (after the first Cards per page, after each Load more); when all standard cards are shown, all remaining featured cards follow at the end.

Rendering Rules: `filter-list-card-spotlight filter-list-card-featured` inside the grid, spanning all columns, minimum height 30rem; no description; lazy-loaded image.

Responsive Behaviour: Full width at every breakpoint; content padding 3rem from 900px.

SEO Impact: As other cards.

Accessibility Compliance: As hero; the spotlight is a normal grid item in DOM order.

Variation 14 — Featured spotlights: priority<=3 rule

Variation Name: Featured cards, Featured rule = priority<=3.

Business Purpose: Spotlights the top-ranked pages by a numeric priority instead of a flag.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/filter-list-featured-priority.png`
> **Purpose:** Featured spotlight (priority<=3)
> **Description:** A spotlight card for the page with priority 1 after the first page of cards.

Use Cases: Ranked content programmes.

Authoring Guidelines: Set `priority` (a number; `featured-priority` is read as a fallback) on the pages; values 1, 2, 3 qualify.

Business Rules: Pages with a numeric priority of 3 or less, ascending, ties by latest; at most Featured card count. Pages without a priority never qualify. Placement as Variation 13.

Rendering Rules: As Variation 13.

Responsive Behaviour: As Variation 13.

SEO Impact: As Variation 13.

Accessibility Compliance: As Variation 13.

Variation 15 — Featured spotlights: Manual rule

Variation Name: Featured cards, Featured rule = Manual.

Business Purpose: The author picks the spotlight pages and their order.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/filter-list-featured-manual.png`
> **Purpose:** Featured spotlight (Manual)
> **Description:** The Cards tab with Featured rule = Manual and three Featured pages picked, and the resulting spotlights in the authored order.

Use Cases: Campaign-driven promotion.

Authoring Guidelines: Featured rule = Manual; add pages in Featured pages (shown only for this rule) in the order wanted; set Featured card count to at least the number of pages.

Business Rules: Only Featured pages that are among the matching pages (and not the hero) are used, in the authored order, up to Featured card count. Pages filtered out by the source or the visitor's filters are skipped.

Rendering Rules: As Variation 13.

Responsive Behaviour: As Variation 13.

SEO Impact: As Variation 13.

Accessibility Compliance: As Variation 13.

Variation 16 — Datasource: Query Index

Variation Name: Query Index (Source tab › Datasource = Query Index; the default).

Business Purpose: Automatic lists that grow as pages are published.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-source-query-index.png`
> **Purpose:** Query Index source settings
> **Description:** The Source tab with Datasource = Query Index, Root path /en/insights, Page types Article and Report selected and Sort by Latest First.

Use Cases: Hubs, listing pages, topic pages.

Authoring Guidelines: Set Root path (optional), Page types and Content types (optional) and Sort by. The Pages field is hidden.

Business Rules: Candidates are the current locale's index rows below Root path (strictly below: the root page itself is not listed; empty Root path means the whole locale), excluding the current page, whose `page-type` is one of Page types (none selected: all) and whose `content-type` is one of Content types (none: all), compared after normalization (e.g. "Case Study" = `case-study`), sorted by Sort by.

Rendering Rules: No difference in markup.

Responsive Behaviour: Not applicable (data selection).

SEO Impact: Pages excluded from the index (`helix-query.yaml`: references, error pages, data, nav, footer, block libraries, templates, fragments, modals) are never listed.

Accessibility Compliance: Not applicable.

Variation 17 — Datasource: Manual List

Variation Name: Manual List (Source tab › Datasource = Manual List).

Business Purpose: A curated, ordered list with the automatic card design and filters.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-source-manual.png`
> **Purpose:** Manual List source settings
> **Description:** The Source tab with Datasource = Manual List and five pages added in the Pages multifield.

Use Cases: Campaign pages, "recommended reading" sets.

Authoring Guidelines: Add pages to Pages (shown only for Manual List) in the display order. Root path is hidden.

Business Rules: The cards are the Pages that exist in the query index of the current locale, in the authored order; pages not in the index (unpublished, excluded, another locale) are silently left out. Sort by, Page types and Content

types are not applied in this mode, and the current page is not excluded. Card content still comes from each page's metadata.

Rendering Rules: As other modes.

Responsive Behaviour: Not applicable.

SEO Impact: As Variation 16.

Accessibility Compliance: Not applicable.

Variation 18 — Filter pills: single select

Variation Name: Single select pills (Filters tab › Allow multi select = off; the default).

Business Purpose: One topic at a time, the simplest filtering.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-pills-single.png`
> **Purpose:** Single select pills
> **Description:** The pill row with "All" outlined and "Banking" filled dark (pressed), and the grid showing only banking pages.

Use Cases: Topic hubs with a handful of topics.

Authoring Guidelines: Set Primary filter tags (or a Primary filter category) and keep Allow multi select off. Show All filter adds the All pill (on by default).

Business Rules: Selecting a pill replaces the previous selection; selecting the pressed pill again clears it (all pages). All clears the selection. A pill matches pages that have its tag or any tag below it (e.g. `tcs:industry/banking` matches `tcs:industry/banking/retail`).

Rendering Rules: Pills are `<button class="filter-list-filter" aria-pressed>`; the All pill is pressed when nothing is selected.

Responsive Behaviour: Pills wrap; below 900px they take a full row.

SEO Impact: Filtering is client side; the URL does not change (no shareable filtered URL).

Accessibility Compliance: A `role="group"` labelled "Filters"; toggle buttons with `aria-pressed`; `aria-controls` points to the results; the result count is a `role="status"` live region announcing "{n} results" after each change.

Variation 19 — Filter pills: multi select

Variation Name: Multi select pills (Filters tab › Allow multi select = on).

Business Purpose: Narrow to pages that combine several topics.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-pills-multi.png`
> **Purpose:** Multi select pills
> **Description:** Two pills pressed ("Banking" and "AI") and the grid showing only pages tagged with both.

Use Cases: Large libraries where visitors combine dimensions.

Authoring Guidelines: Allow multi select = on.

Business Rules: Each pill toggles on and off independently; a page must match every selected pill (AND). All clears every selection.

Rendering Rules: As Variation 18, several pills may be pressed.

Responsive Behaviour: As Variation 18.

SEO Impact: As Variation 18.

Accessibility Compliance: As Variation 18.

Variation 20 — Advanced filter: AND logic

Variation Name: Advanced filter with Filter logic = AND (Filters tab › Enable advanced filter = on, the default; Advanced tab › Filter logic = AND, the default).

Business Purpose: A precise faceted search: pages must have every option chosen.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-advanced-and.png`
> **Purpose:** Advanced filter panel (AND)
> **Description:** The round filter button with a "2" badge and the open panel titled "Advanced filters" with Clear all, two category columns ("INDUSTRY", "SERVICE") of checkboxes, two of them checked.

Use Cases: Resource libraries with several taxonomies.

Authoring Guidelines: Add Advanced filter categories (Category label and Taxonomy root, e.g. Industry / `tcs:industry`); keep Filter logic = AND.

Business Rules: Each category's options are the tags below its Taxonomy root that the candidate pages use, sorted by label; categories without options are not shown; with no category shown there is no filter button. A page must have every checked option (or a tag below it). The options combine with the pills by AND.

Rendering Rules: Toggle: an icon button (`filter` icon, brand outline; filled when open) with `aria-expanded`, `aria-controls` and a count badge. Panel: `<div role="region" aria-label="Advanced filters" hidden>` with a header (title, Clear all, hidden while nothing is checked) and one `<fieldset>` with a `<legend>` per category and a list of checkboxes with labels.

Responsive Behaviour: Categories side by side from 600px (auto-fit columns of at least 12rem), stacked on mobile.

SEO Impact: None.

Accessibility Compliance: Opening the panel moves focus to the first checkbox; Escape closes it and returns focus to the toggle; Clear all returns focus to the first checkbox; the toggle's accessible name includes "{n} selected"; fieldset and legend group the options (WCAG 1.3.1, 2.1.1, 2.4.3, 4.1.2).

Variation 21 — Advanced filter: OR logic

Variation Name: Advanced filter with Filter logic = OR.

Business Purpose: Broad discovery: pages with any of the chosen options.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-advanced-or.png`
> **Purpose:** Advanced filter panel (OR)

> **Description:** The open panel with three options checked across two categories and the grid showing pages that have any of them.

Use Cases: Exploratory browsing.

Authoring Guidelines: Advanced tab › Filter logic = OR.

Business Rules: A page needs at least one of the checked options, across all categories (OR is not applied per category). Pills still combine by AND with the result.

Rendering Rules: As Variation 20.

Responsive Behaviour: As Variation 20.

SEO Impact: None.

Accessibility Compliance: As Variation 20.

Variation 22 — Sort by: Latest First

Variation Name: Latest First (Source tab › Sort by = Latest First; the default).

Business Purpose: Freshest content first.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-sort-latest.png`
> **Purpose:** Latest First order
> **Description:** Cards whose meta lines show dates descending from left to right, top to bottom.

Use Cases: News, blogs, insights.

Authoring Guidelines: Sort by = Latest First; give pages a `publish-date` in YYYY-MM-DD.

Business Rules: Descending `publish-date` (text comparison); ties and undated pages by index last modified (newest first); undated pages after dated ones. Query Index source only.

Rendering Rules: Order of cards in the DOM.

Responsive Behaviour: Not applicable.

SEO Impact: None.

Accessibility Compliance: Not applicable.

Variation 23 — Sort by: Featured First

Variation Name: Featured First.

Business Purpose: Promoted pages lead the standard list.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-sort-featured.png`
> **Purpose:** Featured First order
> **Description:** The first standard cards are the featured pages (when the spotlights do not already take them), then the others by priority and date.

Use Cases: Hubs where featured pages should come first even when spotlights are off.

Authoring Guidelines: Sort by = Featured First.

Business Rules: Featured pages first, then by priority ascending (no priority last), then latest. Pages taken for the hero or featured spotlights are removed from the standard cards.

Rendering Rules: Order only.

Responsive Behaviour: Not applicable.

SEO Impact: None.

Accessibility Compliance: Not applicable.

Variation 24 — Sort by: Priority

Variation Name: Priority.

Business Purpose: Editorially ranked lists.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-sort-priority.png`
> **Purpose:** Priority order
> **Description:** Cards ordered by their priority metadata 1, 2, 3, then pages without priority by date.

Use Cases: Ranked programmes, "top 10" collections.

Authoring Guidelines: Sort by = Priority; set `priority` numbers on the pages.

Business Rules: Priority ascending; pages without a numeric priority last; ties by latest.

Rendering Rules: Order only.

Responsive Behaviour: Not applicable.

SEO Impact: None.

Accessibility Compliance: Not applicable.

Variation 25 — Sort by: Alphabetical

Variation Name: Alphabetical.

Business Purpose: A–Z directories.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-sort-alphabetical.png`
> **Purpose:** Alphabetical order
> **Description:** Compact cards sorted A to Z by title.

Use Cases: Glossaries, case study directories, offices.

Authoring Guidelines: Sort by = Alphabetical.

Business Rules: By the page title from the index (`og:title`), using the page language's collation (`Intl.Collator` with `<html lang>`, base sensitivity: case and accents ignored). The Mapping tab's Title source does not change the sort key.

Rendering Rules: Order only.

Responsive Behaviour: Not applicable.

SEO Impact: None.

Accessibility Compliance: Not applicable.

6.17.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A page open in Universal Editor with the Filter List selected (blue outline): the title, the description, the Filters label with the All pill and three Primary filter tag pills, the result count and the grey editor hint box "The results show on the published page, from the pages' metadata."

Annotations

- A = Title (Content tab › Title), rendered as the block's `<h2>`.
- B = Description (Content tab › Description), rich text below the title.
- C = Filters label (placeholder `filters`), shown when there is at least one pill.
- D = Filter pills: All (Filters tab › Show All filter) and the Primary filter tags in their authored order (Filters tab › Primary filter tags). Pills from a Primary filter category only appear when the index is available, because they are the tags the listed pages use.
- E = Result count (Content tab › Show result count); visually hidden when the toggle is off but still announced to screen readers.
- F = Results area: in the editor the query index is normally not available (`utils/query-index.js` returns no pages there), so the empty-state box shows the editor hint (placeholder `filterListEditor`) instead of cards. On preview and live pages this area shows the hero spotlight, the cards, the featured spotlights and Load more.

> **Note:** The advanced filter button, cards, spotlights and Load more cannot be reviewed on the editor canvas. Review them on the preview URL (`https://main--tcs-corp--sradha-tcs.aem.page/...`) after previewing the page and the listed pages.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Universal Editor properties panel of the Filter List with the seven tabs Content, Source, Filters, Advanced, Cards, Mapping and Additional, the Content tab open.

The properties panel has seven tabs:

- **Content:** Title (text), Description (rich text), Show result count (toggle, off), Default layout (Masonry, Grid, List; Masonry), Cards per page (number, 9), Load more count (number, 6).
- **Source:** Datasource (Query Index, Manual List; Query Index), Root path (page picker; only for Query Index), Pages (page picker, multiple; only for Manual List), Page types (multiselect: Article, Blog, Report, Case Study, News, Event), Content types (multiselect: Article, Blog, Report, Case Study, News, Event, Whitepaper, Podcast, Video), Sort by (Latest First, Featured First, Priority, Alphabetical; Latest First).

- **Filters:** Primary filter category (tag picker), Primary filter tags (multifield of Tag Id tag pickers), Show All filter (toggle, on), Allow multi select (toggle, off), Enable advanced filter (toggle, on).
- **Advanced:** Advanced filter categories (multifield of Category label text and Taxonomy root tag picker), Filter logic (AND, OR; AND).
- **Cards:** Card style (Default, Compact, Featured, Mixed; Mixed), Enable featured cards (toggle, on), Featured rule (featured=true, priority<=3, Manual; only when featured cards are on), Featured pages (page picker, multiple; only for the Manual rule), Featured card count (number, 2; only when featured cards are on), Enable hero card (toggle, off), Hero rule (Latest, Featured, Manual; Featured; only when the hero is on), Hero page (page picker; only for the Manual hero rule).
- **Mapping:** Image source (`featured-image`), Fallback image (`thumbnail-image`), Eyebrow source (`content-type`), Title source (`title`), Description source (`description`), Tag source (`tags`), Url source (`path`), Author source (`author`): the metadata column used for each card part.
- **Additional:** Anchor ID, Hide block, Padding left / right / top / bottom, Margin left / right / top / bottom (common to every block).

6.17.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites (`https://author-p220207-e2271469.adobeaemcloud.com`), go to `/content/tcs-corp/{country}/{language}/...`, select the listing page and choose **Edit** to open it in Universal Editor.

Expected Result: The page opens in the editor canvas in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** The AEM Sites console with the listing page selected and the Edit action highlighted.

Step 2 — Add the Filter List to a section

Action: Select the section in the canvas or the content tree, choose **Add** (+) and select **Filter List**.

Expected Result: A Filter List with the default settings is added. The canvas shows the editor hint (no cards in the editor).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** The Add component menu of a section with Filter List highlighted, and the new block on the canvas with the editor hint.

Step 3 — Fill the Content tab

Action: Open the properties panel. Enter the Title (e.g. "Explore insights & opinions") and the Description. Turn on Show result count if visitors should see the count. Choose the Default layout, Cards per page and Load more count.

Expected Result: Title and description appear on the canvas immediately.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-03-content-tab.png`

Purpose: Content tab
Description: The Content tab filled with title, description, Show result count on, Default layout Masonry, Cards per page 9 and Load more count 6.

Step 4 — Choose the source

Action: In the Source tab choose Datasource. For Query Index, pick the Root path (the folder page whose children and descendants are listed; leave empty for the whole locale). For Manual List, add each page to Pages in the display order.

Expected Result: The field that does not apply is hidden (Root path for Manual List; Pages for Query Index).

[Screenshot Placeholder]
File name: `screenshots/filter-list-step-04-source.png`
Purpose: Source tab
Description: The Source tab with Datasource Query Index and the Root path picker showing /content/tcs-corp/us/en/insights.

Step 5 — Limit by page and content type, and choose the order

Action: (Query Index only) Select Page types and Content types to include (none selected means all). Choose Sort by.

Expected Result: The selections are saved; they take effect on the preview page.

[Screenshot Placeholder]
File name: `screenshots/filter-list-step-05-types-sort.png`
Purpose: Types and sort
Description: Page types with Article and Report checked, Content types empty and Sort by Latest First.

Step 6 — Set up the filter pills

Action: In the Filters tab either add Primary filter tags (one tag per item, in the order the pills should appear) or pick a Primary filter category (a tag below the namespace, e.g. `tcs:industry`) so the tags below it that the listed pages use become the pills. Set Show All filter and Allow multi select.

Expected Result: Primary filter tags appear as pills on the canvas right away; category pills appear on the preview page.

[Screenshot Placeholder]
File name: `screenshots/filter-list-step-06-pills.png`
Purpose: Filters tab
Description: The Filters tab with three Primary filter tags added, Show All filter on, Allow multi select off and Enable advanced filter on.

Step 7 — Set up the advanced filter

Action: Keep Enable advanced filter on. In the Advanced tab add one Advanced filter category per facet: Category label (e.g. "Industry") and Taxonomy root (e.g. `tcs:industry`). Choose Filter logic (AND or OR).

Expected Result: The categories are saved; the filter button and panel appear on the preview page when the listed pages use tags below the roots.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-07-advanced.png`
> **Purpose:** Advanced tab
> **Description:** Two Advanced filter categories (Industry / tcs:industry and Service / tcs:service) and Filter logic AND.

Step 8 — Choose the card style and spotlights

Action: In the Cards tab choose Card style. Decide on featured spotlight cards (Enable featured cards, Featured rule, Featured pages for Manual, Featured card count) and on the hero card (Enable hero card, Hero rule, Hero page for Manual).

Expected Result: Dependent fields show or hide as the toggles and rules change.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-08-cards.png`
> **Purpose:** Cards tab
> **Description:** Card style Mixed, featured cards on with rule featured=true and count 2, hero card on with rule Featured.

Step 9 — Review the Mapping tab

Action: Keep the defaults unless the development team has agreed other metadata columns. Each field holds the name of a metadata column (e.g. `card-title` instead of `title`).

Expected Result: The column names are saved as typed.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-09-mapping.png`
> **Purpose:** Mapping tab
> **Description:** The Mapping tab with the eight default column names.

Step 10 — Set the Additional options (optional)

Action: In the Additional tab set an Anchor ID (e.g. `anchor-insights`) to link to the block, and spacing if the design needs it. Leave Hide block off.

Expected Result: An invalid Anchor ID shows the message "Start with anchor-, then lower-case letters, digits and hyphens."

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-10-additional.png`
> **Purpose:** Additional tab
> **Description:** The Additional tab with Anchor ID anchor-insights and the padding and margin selects at Default.

Step 11 — Prepare the listed pages

Action: On each page that should appear, make sure the page properties (Title, Description, Tags, Page Type) and the bulk metadata sheet columns used by the cards are set: `content-type`, `publish-date` (YYYY-MM-DD), `reading-time`, `featured-image` or `thumbnail-image`, `author-id` or `author`, `featured`, `priority`. Publish (or preview) those pages so the query index contains them.

Expected Result: The pages appear in `/query-index.json` with their columns.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-11-page-metadata.png`
> **Purpose:** Listed page metadata
> **Description:** The locale's metadata sheet with rows for /en/insights/** showing content-type, publish-date, featured-image and featured columns.

Step 12 — Preview

Action: Use the editor's Preview mode, then open the page preview (AEM Sidekick › Preview, or the `.aem.page` URL).

Expected Result: The preview page shows the cards, pills, panel and Load more.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-12-preview.png`
> **Purpose:** Preview
> **Description:** The preview URL showing the rendered Filter List with the hero, the masonry grid and Load more.

Step 13 — Validate

Action: Work through the Validation Process checklist below at desktop, tablet and mobile widths.

Expected Result: Every checklist item passes.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-13-validate.png`
> **Purpose:** Validate
> **Description:** The preview page at 375px with a pill selected and the result count updated.

Step 14 — Publish

Action: In Universal Editor choose **Publish** (or use the AEM Sites console / AEM Sidekick › Publish) for the page. Make sure the listed pages are published too.

Expected Result: The live page (`.aem.live`) shows the Filter List with the published pages.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-step-14-publish.png`
> **Purpose:** Publish
> **Description:** The Publish dialog in Universal Editor with the listing page selected.

Preview Process

1. Save all dialog changes (Universal Editor saves automatically).

2. Preview the listing page and every page that should appear in it. The preview query index is updated when pages are previewed; the live index when they are published.
3. Open `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}` and check the result; reload with the browser cache disabled if a recently previewed page is missing.
4. If metadata comes from the bulk metadata sheet, preview the sheet as well (and see the metadata configuration note in AGENTS.md: the sheets apply only once listed in the site configuration).

Publishing Process

1. Publish the listed pages first (they must be in the live query index).
2. Publish the listing page.
3. Publish the locale's taxonomy, placeholders and Authors sheet if they changed.
4. Open the `.aem.live` URL and confirm the cards, labels and author lockups.

Validation Process

- ☐ Title is an `<h2>`, card titles are `<h3>`; no skipped heading level in the section.
- ☐ The listed pages are the expected ones (Root path, Page types, Content types; the current page is not listed).
- ☐ Sort order is as chosen.
- ☐ Pill labels are taxonomy titles in the locale's language (not tag IDs).
- ☐ Each pill filters as expected; All resets; multi select (if on) requires every selected tag.
- ☐ The filter button shows only when categories have options; Escape closes the panel; the badge shows the number of checked options; Clear all works.
- ☐ AND / OR logic behaves as configured.
- ☐ The empty state shows "No resources found." with Reset filters when a filter excludes everything.
- ☐ Hero and featured spotlights show the expected pages and are not repeated in the grid.
- ☐ Load more adds Load more count cards and disappears when everything is shown; the count reads "Showing n of total" then "n results".
- ☐ Images load (no broken images); report and dark cards appear where expected (Mixed style).
- ☐ Author lockups show name, designation and photo from the Authors sheet.
- ☐ Desktop 4 columns (3 for Featured style), tablet 2 / 3, mobile 1; List layout image beside text from 600px.
- ☐ Keyboard: Tab reaches every pill, the filter button, each card link and Load more, with a visible focus ring.

Rollback Process

1. To undo a dialog change, use Universal Editor undo, or restore the previous page version in AEM Sites (Timeline › Versions › Restore) and publish again.
2. To remove the block quickly from the live page without deleting it, turn on Additional › Hide block and publish (the block is hidden on the page, shown faded in the editor).
3. If a listed page is wrong, correct or unpublish that page; the index updates after (un)publishing.
4. If a code release broke the block, the development team reverts the commit on `main` (Code Sync deploys it); no content change is needed.

6.17.6 Authorable Field Specification

Filter List block model (`filter-list`)

Field Name	Field Type	Mandatory	Multi-Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Content tab › Title (`title`)	text	No	No	(empty)	Heading of the block	Short, descriptive heading	None	Explore the Agentic bank

Field Name	Field Type	Mandatory	Multi-Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Content tab › Description (`description`)	rich-text	No	No	(empty)	Introduction below the title	One or two sentences	None	Insights on AI in banking.
Content tab › Show result count (`showResultCount`)	boolean	No	No	false	Shows the result count visibly	Turn on for large libraries	None	true
Content tab › Default layout (`layout`)	select	No	No	masonry	Arrangement of the cards	Masonry for mixed heights, Grid for uniform cards, List for text-led lists	Options masonry, grid, list; other values fall back to masonry	masonry
Content tab › Cards per page (`cardsPerPage`)	number	No	No	9	Cards shown first	Multiple of the column count (4 or 3) looks best, e.g. 8 or 12	Model min 1, max 48; code: empty or 0 gives 9, negative gives 1	12
Content tab › Load more count (`loadMoreCount`)	number	No	No	6	Cards added per Load more	Same multiple as Cards per page	Model min 1, max 48; code: empty or 0 gives 6, negative gives 1	8
Source tab › Datasource (`source`)	select	No	No	query-index	Where the pages come from	Query Index for automatic lists, Manual List for curated ones	Options query-index, manual	query-index
Source tab › Root path (`rootPath`)	aem-content	No	No	(empty)	Lists the pages below this page	Pick the folder page, e.g. /en/insights	Shown when Datasource is not Manual List	/content/tcs-corp/us/en/insights
Source tab › Pages (`pages`)	aem-content	No	Yes	(empty)	Curated pages, in order	Add published pages of the same locale	Shown when Datasource = Manual List	/content/tcs-corp/us/en/insights/ai-banking
Source tab › Page types (`pageTypes`)	multiselect	No	Yes	(none)	Limits by page-type	Leave empty for all page types	Options article, blog, report, case-study, news, event	article, report
Source tab › Content types (`contentTypes`)	multiselect	No	Yes	(none)	Limits by content-type	Leave empty for all content types	Options article, blog, report, case-study, news, event, whitepaper, podcast, video	report, whitepaper
Source tab › Sort by (`sortBy`)	select	No	No	latest	Order of the cards	Latest First for news and insights	Options latest, featured, priority, alphabetical; not applied to Manual List	latest
Filters tab › Primary filter category (`primaryCategory`)	aem-tag	No	No	(empty)	Pills from the tags below this tag	Use when the pills should follow the content; ignored when Primary filter tags are set	A tag below a namespace	tcs:industry
Filters tab › Primary filter tags (`primaryTags`)	container (composite multi-	No	Yes	(none)	The pills, in order	Five to eight pills	Each item one Tag Id	see item table

Field Name	Field Type	Mandatory	Multi-Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
	ti-field)							
Filters tab › Show All filter (`showAll`)	boolean	No	No	true	Adds the All pill	Keep on in most cases	None	true
Filters tab › Allow multi select (`multiSelect`)	boolean	No	No	false	Several pills at once (AND)	Use only when tags combine meaningfully	None	false
Filters tab › Enable advanced filter (`advancedFilter`)	boolean	No	No	true	Shows the filter button and panel	Needs Advanced filter categories	None	true
Advanced tab › Advanced filter categories (`advancedCategories`)	container (composite multi-field)	No	Yes	(none)	Facets of the advanced panel	Two to four categories	Items without a Taxonomy root are ignored	see item table
Advanced tab › Filter logic (`filterLogic`)	select	No	No	and	How checked options combine	AND for precise, OR for broad	Options and, or	and
Cards tab › Card style (`cardStyle`)	select	No	No	mixed	Card presentation	Mixed for insight hubs	Options default, compact, featured, mixed	mixed
Cards tab › Enable featured cards (`enableFeatured`)	boolean	No	No	true	Large spotlight cards between pages of cards	Turn off for plain lists	None	true
Cards tab › Featured rule (`featuredRule`)	select	No	No	featured	Which pages are spotlighted	featured=true with priority ordering is the usual choice	Options featured, priority, manual; shown when Enable featured cards = true	featured
Cards tab › Featured pages (`featuredPages`)	aem-content	No	Yes	(empty)	Hand-picked spotlight pages	In display order	Shown when featured cards are on and rule = manual	/content/tcs-corp/us/en/insights/report-2026
Cards tab › Featured card count (`featuredCount`)	number	No	No	2	Maximum spotlight cards	1 to 3	Model min 1, max 12; shown when featured cards are on; code: empty or 0 gives 2	2
Cards tab › Enable hero card (`enableHero`)	boolean	No	No	false	A large spotlight above the grid	One hero per page	None	true
Cards tab › Hero rule (`heroRule`)	select	No	No	featured	Which page is the hero	Featured, or Latest for news	Options latest, featured, manual; shown when Enable hero card = true	featured
Cards tab › Hero page (`heroPage`)	aem-content	No	No	(empty)	The hero page (Manual)	Must also match the source	Shown when the hero is on and rule = manual	/content/tcs-corp/us/en/insights/

Field Name	Field Type	Mandatory	Multi-Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
								agentic-bank
Mapping tab › Image source (`imageSource`)	text	No	No	featured-image	Column of the card image	Keep the default	Free text; empty falls back to featured-image	featured-image
Mapping tab › Fallback image (`fallbackImage`)	text	No	No	thumbnail-image	Column used when the image source is empty	Keep the default	Free text; empty falls back to thumbnail-image	thumbnail-image
Mapping tab › Eyebrow source (`eyebrowSource`)	text	No	No	content-type	Column of the meta-line label (and the report test)	Keep content-type	Free text; empty falls back to content-type	content-type
Mapping tab › Title source (`titleSource`)	text	No	No	title	Column of the card title	Use card-title only if every page has it (else the page title is used)	Free text; empty falls back to title	title
Mapping tab › Description source (`descriptionSource`)	text	No	No	description	Column of the card description	Keep the default	Free text; empty falls back to description	description
Mapping tab › Tag source (`tagSource`)	text	No	No	tags	Column of the card tags	Keep tags	Free text; empty falls back to tags	tags
Mapping tab › Url source (`urlSource`)	text	No	No	path	Column of the card link	Keep path	Free text; empty falls back to path	path
Mapping tab › Author source (`authorSource`)	text	No	No	author	Column of the spotlight author	author or author-id	Free text; empty falls back to author	author-id
Additional tab › Anchor ID (`classes_anchor`)	text	No	No	(empty)	Block id for #links	anchor- plus a short slug	Regex `^anchor-[a-z0-9][a-z0-9-]*$` (Enforced, message “Start with anchor-, then lowercase letters, digits and hyphens.”)	anchor-insights
Additional tab › Hide block (`classes_hideblock`)	boolean	No	No	false	Hides the block on the page	Use temporarily	None	false
Additional tab › Padding left (`classes_paddingleft`)	select	No	No	Default (“”)	Extra left padding	Only when the design needs it	15 options (see Spacing values)	pl-400
Additional tab › Padding right (`classes_paddingright`)	select	No	No	Default (“”)	Extra right padding	As above	15 options	pr-400
Additional tab › Padding top (`classes_paddingtop`)	select	No	No	Default (“”)	Extra top padding	As above	15 options	pt-800

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Additional tab › Padding bottom (`classes_paddingbottom`)	select	No	No	Default (“”)	Extra bottom padding	As above	15 options	pb-800
Additional tab › Margin left (`classes_marginleft`)	select	No	No	Default (“”)	Left margin	As above	15 options	ml-0
Additional tab › Margin right (`classes_marginright`)	select	No	No	Default (“”)	Right margin	As above	15 options	mr-0
Additional tab › Margin top (`classes_margintop`)	select	No	No	Default (“”)	Top margin	As above	15 options	mt-1200
Additional tab › Margin bottom (`classes_marginbottom`)	select	No	No	Default (“”)	Bottom margin	As above	15 options	mb-1200

Primary filter tags item (composite multifield `primaryTags`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Tag Id (`tagId`)	aem-tag	No	No	(empty)	One pill	Pick a tag the listed pages use	Must be a tag ID (namespace:path); other values are ignored	tcs:industry/banking

Advanced filter categories item (composite multifield `advancedCategories`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Category label (`categoryLabel`)	text	No	No	(empty)	The category heading (legend) in the panel	Short noun, e.g. Industry	Empty: the taxonomy title of the root is used	Industry
Taxonomy root (`taxonomyRoot`)	aem-tag	No	No	(empty)	Options are the tags below it	A tag below the namespace	Without a root the item is ignored	tcs:industry

Spacing values (all eight padding and margin selects)

Option label	Value (padding left; the other selects use pr-, pt-, pb-, ml-, mr-, mt-, mb-)	Size
Default	(empty)	The block’s own spacing
0 (none)	pl-0	0
0.25rem (space-100)	pl-100	0.25rem
0.5rem (space-200)	pl-200	0.5rem
0.75rem (space-300)	pl-300	0.75rem
1rem (space-400)	pl-400	1rem
1.25rem (space-500)	pl-500	1.25rem

Option label	Value (padding left; the other selects use pr-, pt-, pb-, ml-, mr-, mt-, mb-)	Size
1.5rem (space-600)	pl-600	1.5rem
2rem (space-800)	pl-800	2rem
2.5rem (space-1000)	pl-1000	2.5rem
3rem (space-1200)	pl-1200	3rem
4rem (space-1600)	pl-1600	4rem
5rem (space-2000)	pl-2000	5rem
6rem (space-2400)	pl-2400	6rem
7.5rem (space-3000)	pl-3000	7.5rem

Note: None of the Filter List fields is marked `required` in the model, so Universal Editor does not enforce any field as mandatory. The only enforced pattern is the Anchor ID regular expression. The number fields declare `min` / `max` in the model; the code itself only guarantees a minimum of 1 and has no maximum.

Field details

- **Title** — Purpose: block heading. Allowed Values: plain text. Character Limits: Recommended 60 characters; none enforced. Business Rules: rendered as `<h2 class="filter-list-title">` only when not empty. Dependency Rules: none. Visibility Rules: always shown. Sample Content: “Explore insights & opinions”.
- **Description** — Purpose: introduction. Allowed Values: rich text (paragraphs, links, lists). Character Limits: Recommended 200 characters; none enforced. Business Rules: the authored nodes are moved into `filter-list-description` (max width 50rem) only when it has text. Dependency Rules: none. Visibility Rules: always shown. Sample Content: “Perspectives from our experts on AI-led transformation.”
- **Show result count** — Purpose: visible count. Allowed Values: on / off. Character Limits: not applicable. Business Rules: off adds `visually-hidden` to the count, which stays a `role="status"` live region. Dependency Rules: none. Visibility Rules: always shown. Sample Content: on.
- **Default layout** — Purpose: arrangement. Allowed Values: Masonry (`masonry`), Grid (`grid`), List (`list`). Character Limits: not applicable. Business Rules: any other value gives masonry; adds `layout-{value}` to block and grid. Dependency Rules: Featured card style reduces the columns of Masonry and Grid. Visibility Rules: always shown. Sample Content: Masonry.
- **Cards per page** — Purpose: first page size of standard cards (hero and featured spotlights are not counted). Allowed Values: whole numbers; model min 1 / max 48. Character Limits: not applicable. Business Rules: `Math.max(1, parseInt(value) || 9)`. Dependency Rules: a featured spotlight follows each page of cards. Visibility Rules: always shown. Sample Content: 12.
- **Load more count** — Purpose: cards added per Load more. Allowed Values: whole numbers; model min 1 / max 48. Business Rules: `Math.max(1, parseInt(value) || 6)`; Load more is hidden when every standard and featured card is shown. Dependency Rules: as above. Visibility Rules: always shown. Sample Content: 8.
- **Datasource** — Purpose: source of the pages. Allowed Values: Query Index (`query-index`), Manual List (`manual`). Business Rules: only the exact value `manual` selects the manual mode; anything else is Query Index. Dependency Rules: switches the visibility of Root path and Pages; Manual List ignores Page types, Content types and Sort by. Visibility Rules: always shown. Sample Content: Query Index.
- **Root path** — Purpose: listing scope. Allowed Values: a page of the same locale (AEM path or published path). Business Rules: converted to the published path (`/content/tcs-corp/us/en/insights.html` to `/en/insights`); only pages strictly below it are listed; empty lists the whole locale. Dependency Rules: Datasource

= Query Index. Visibility Rules: condition `source !== "manual"`. Sample Content: /content/tcs-corp/us/en/insights.

- **Pages** — Purpose: curated list. Allowed Values: pages of the same locale. Business Rules: listed in the authored order; pages missing from the locale's index are dropped. Dependency Rules: Datasource = Manual List. Visibility Rules: condition `source === "manual"`. Sample Content: three insight pages.
- **Page types** — Purpose: filter on `page-type`. Allowed Values: Article, Blog, Report, Case Study, News, Event. Business Rules: compared with the normalized `pageType` index column; none selected means every page type (including pages without one). Dependency Rules: Query Index only. Visibility Rules: always shown (no condition, although it has no effect for Manual List). Sample Content: Article.
- **Content types** — Purpose: filter on `content-type`. Allowed Values: Article, Blog, Report, Case Study, News, Event, Whitepaper, Podcast, Video. Business Rules: as Page types with the `contentType` column. Dependency Rules: Query Index only. Visibility Rules: always shown. Sample Content: Report, Whitepaper.
- **Sort by** — Purpose: order. Allowed Values: Latest First, Featured First, Priority, Alphabetical. Business Rules: see Variations 22 to 25. Dependency Rules: Query Index only. Visibility Rules: always shown. Sample Content: Latest First.
- **Primary filter category** — Purpose: automatic pills. Allowed Values: a tag ID. Business Rules: pills are the tags starting with `{category}/` that the candidate pages use, sorted by label; ignored when Primary filter tags has a valid tag. Dependency Rules: needs the index (no pills in the editor). Visibility Rules: always shown. Sample Content: tcs:topics.
- **Primary filter tags** — Purpose: curated pills. Allowed Values: items of one tag each. Character Limits: Recommended at most 8 items. Business Rules: only values that look like a tag ID (`namespace:path`) become pills; labels come from the locale's taxonomy, else from the tag ID. Dependency Rules: replaces Primary filter category. Visibility Rules: always shown. Sample Content: tcs:industry/banking, tcs:industry/insurance.
- **Tag Id** (item) — Purpose: one pill. Allowed Values: a tag ID. Business Rules: matches pages with the tag or a tag below it. Sample Content: tcs:industry/banking.
- **Show All filter** — Purpose: All pill. Allowed Values: on / off. Business Rules: a missing value counts as on (the code default is true when the property row is absent; an authored "false" turns it off). Visibility Rules: always shown. Sample Content: on.
- **Allow multi select** — Purpose: several pills. Allowed Values: on / off. Business Rules: on: pills toggle independently and a page needs every selected tag. Visibility Rules: always shown. Sample Content: off.
- **Enable advanced filter** — Purpose: filter button and panel. Allowed Values: on / off. Business Rules: a missing value counts as on; the button only appears when at least one category has options. Dependency Rules: needs Advanced filter categories. Visibility Rules: always shown (the Advanced tab is not hidden when it is off). Sample Content: on.
- **Advanced filter categories** — Purpose: facets. Allowed Values: items of Category label and Taxonomy root. Character Limits: Recommended 4 categories. Business Rules: options are the tags below the root used by the candidate pages, sorted by label; empty categories are not shown. Visibility Rules: always shown. Sample Content: Industry / tcs:industry.
- **Category label** (item) — Purpose: legend. Allowed Values: text that is not a tag ID. Character Limits: Recommended 20 characters. Business Rules: empty uses the taxonomy title of the root; displayed in upper case by CSS. Sample Content: Industry.
- **Taxonomy root** (item) — Purpose: options source. Allowed Values: a tag ID. Business Rules: required for the item to be used. Sample Content: tcs:industry.
- **Filter logic** — Purpose: option combination. Allowed Values: AND (`and`), OR (`or`). Business Rules: applies across all checked options; other values give AND. Visibility Rules: always shown. Sample Content: AND.
- **Card style** — Purpose: presentation. Allowed Values: Default, Compact, Featured, Mixed. Business Rules: other values give Mixed; adds `card-{value}` to the block. Visibility Rules: always shown. Sample Content: Mixed.

- **Enable featured cards** — Purpose: spotlights in the grid. Allowed Values: on / off. Business Rules: a missing value counts as on. Dependency Rules: controls the visibility of Featured rule, Featured pages and Featured card count. Visibility Rules: always shown. Sample Content: on.
- **Featured rule** — Purpose: spotlight selection. Allowed Values: featured=true (`featured`), priority<=3 (`priority`), Manual (`manual`). Business Rules: see Variations 13 to 15. Visibility Rules: condition `enableFeatured === true`. Sample Content: featured=true.
- **Featured pages** — Purpose: manual spotlights. Allowed Values: pages. Business Rules: only pages among the matches are used. Visibility Rules: condition `enableFeatured === true AND featuredRule === "manual"`. Sample Content: two report pages.
- **Featured card count** — Purpose: maximum spotlights. Allowed Values: model min 1 / max 12. Business Rules: `Math.max(1, parseInt(value) || 2)`. Visibility Rules: condition `enableFeatured === true`. Sample Content: 2.
- **Enable hero card** — Purpose: hero spotlight. Allowed Values: on / off. Business Rules: a missing value counts as off. Dependency Rules: controls Hero rule and Hero page. Visibility Rules: always shown. Sample Content: on.
- **Hero rule** — Purpose: hero selection. Allowed Values: Latest, Featured, Manual. Business Rules: see Variations 10 to 12. Visibility Rules: condition `enableHero === true`. Sample Content: Featured.
- **Hero page** — Purpose: manual hero. Allowed Values: a page. Business Rules: used only when it matches the source and filters. Visibility Rules: condition `enableHero === true AND heroRule === "manual"`. Sample Content: /content/tcs-corp/us/en/insights/agentic-bank.
- **Image source / Fallback image / Eyebrow source / Title source / Description source / Tag source / Url source / Author source** — Purpose: the metadata column read for each card part. Allowed Values: a metadata column name in kebab case (`featured-image`) or the index name (`featuredImage`); the code converts kebab case to the index's camel case and also tries the name as typed. Character Limits: not applicable. Business Rules: an empty field uses its default; a column that is not in the index (`helix-query.yaml`) is always empty, and the card then falls back (title to the page title, then the path; URL to the page path; image to the fallback column; author to the `author-id` lookup). Dependency Rules: Eyebrow source drives the report card test in the Mixed style. Visibility Rules: always shown. Sample Content: `card-title` in Title source.
- **Anchor ID** — Purpose: a stable id for links and the data layer. Allowed Values: `anchor-` followed by lower-case letters, digits and hyphens. Character Limits: Recommended 40 characters. Business Rules: becomes the block's `id` (a duplicate on the page gets `-2`, `-3`); also the data layer component id. Visibility Rules: always shown. Sample Content: anchor-insights.
- **Hide block** — Purpose: temporarily hide. Allowed Values: on / off. Business Rules: hidden on preview and live pages and in the editor's preview mode; shown at 45% opacity with a dashed outline in edit mode. Sample Content: off.
- **Padding left / right / top / bottom** and **Margin left / right / top / bottom** — Purpose: spacing around or inside the block from the spacing tokens. Allowed Values: the 15 options of the Spacing values table. Business Rules: applied by `styles/styles.css` (`main .section .block.pl-400` etc.), overriding the block's own spacing. Visibility Rules: always shown. Sample Content: Margin top 3rem (space-1200).

> **Note:** Two properties are read by the code but are not in the current dialog: `filters` (a multifield of a tag title and tags, from Filter Lists created before Primary filter tags existed; used only when Primary filter tags is empty) and `filtersLabel` (the text before the pills; the placeholder `filters` is used when absent). They only exist on legacy content.

6.17.7 Field Annotation Mapping

Marker	Field	Purpose
A	Content tab › Title	Block heading (`<h2>`)
B	Content tab › Description	Introduction text
C	Placeholder `filters` (legacy `filtersLabel`)	Label before the pills
D	Filters tab › Show All filter; Primary filter tags; Primary filter category	The filter pills
E	Content tab › Show result count	Visible result count
F	Source, Cards and Mapping tabs (results; editor hint in the editor)	Cards, spotlights, Load more; the editor hint where the index is not available

6.17.8 Component Specification

Attribute	Value
Component Name	Filter List (`filter-list`; Universal Editor component "Filter List", resource type `core/franklin/components/block/v1/block`, model `filter-list`, key-value block)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`decorators/masonry-decorator.js`, `button-decorator.js`, `icon-button-decorator.js`, `responsive-image-decorator.js`, `date-decorator.js`, `article-meta-decorator.js`; `utils/query-index.js`, `taxonomy.js`, `tags.js`, `data.js`, `placeholders.js`, `properties.js`, `multifield.js`, `site.js`, `environment.js`, `ids.js`; `scripts/aem.js`; `icons/filter.svg`; `/query-index.json`, `/{locale}/taxonomy.json`, `/{locale}/data/authors.json`, `/{locale}/placeholders.json`
Component Type	Block (no child items; two composite multifields in the dialog)
Supported Variants	Layout: Masonry, Grid, List; Card style: Default, Compact, Featured, Mixed (with report and dark card kinds); Hero rule: Latest, Featured, Manual; Featured rule: featured=true, priority<=3, Manual; Datasource: Query Index, Manual List; Pills: single or multi select; Filter logic: AND, OR; Sort: Latest First, Featured First, Priority, Alphabetical
Supported Templates	Any page (allowed in every section); typically landing / listing pages and the home page
Supported Metadata	Read from the listed pages via the index: title (og:title), description, image (og:image, not used by default), tags (article:tag), page-type, content-type, publish-date, reading-time, featured-image, thumbnail-image, author, author-id, featured (feature-card), priority (featured-priority), lastModified; any other indexed column through the Mapping tab
Supported Tags	AEM tags (tag IDs `namespace:path`) of the listed pages; labels from `/{locale}/taxonomy.json` in the locale's language, else derived from the tag ID; hierarchical matching (a tag matches its descendants)
Supported Personalization	None. Visitor filter choices are not stored. The interactions are in the Adobe Client Data Layer, so Adobe Target could use them once Adobe Tags is installed (not implemented)
Caching Considerations	`/query-index.json`, taxonomy, Authors sheet and placeholders are served from the Edge Delivery CDN and fetched once per page view (module-level promises); the index is read in pages of 500 rows until `total` is reached; the whole index (all 13 locales) is fetched and filtered to the current locale in the browser. Index content updates when pages are previewed (preview index) or published (live index); the CDN cache is purged by Edge Delivery on publish
Performance Considerations	Loaded like other blocks (lazy unless in the first section). Decoration waits for four parallel fetches (placeholders, index, taxonomy, Authors) plus tag label resolution, so the block appears after them. Card images are lazy (`createOptimizedPicture`, 600px from 900px wide, 750px below); spotlight images 1600px / 750px; the hero image is eager. Only Cards per page cards are rendered initially. Avoid

Attribute	Value
	placing the block in the first section (it would delay LCP while the index loads). Large indexes increase the number of 500-row requests
Accessibility Compliance	WCAG 2.1 AA intent: pills are toggle buttons with `aria-pressed` in a labelled `role="group"`; `aria-controls` links pills and toggle to the results and panel; `aria-expanded` on the toggle; Escape closes the panel and returns focus; focus moves to the first new card after Load more; result count is a `role="status"` live region; fieldsets with legends; one link per card with a visible focus outline around the card; reduced motion removes the image zoom transition; decorative images have empty alt
SEO Compliance	Cards are rendered in the browser from JSON; they are not in the server-delivered HTML. Search engines that render JavaScript can see them; listed pages are discoverable through the sitemap (`helix-sitemap.yaml`). Headings: block `<h2>`, card `<h3>`. Links are real `<a href>` links to the listed pages. Filtered states have no URL

6.17.9 Input Model

Content Inputs: Title (text) and Description (rich text) from the dialog.

Metadata Inputs: The listed pages' metadata, through the query index columns defined in `helix-query.yaml`: `path`, `title`, `description`, `image`, `tags`, `pageType`, `contentType`, `publishDate`, `readingTime`, `thumbnailImage`, `featuredImage`, `author`, `authorId`, `featured`, `featureCard`, `priority`, `featuredPriority`, `lastModified` (and any other column named in the Mapping tab, e.g. `cardTitle`, `cardDescription`, `cardImage`, `eyebrow`, `category`). The Authors sheet (`author-id`, `name`, `designation`, `image`).

Tags Inputs: Primary filter tags, Primary filter category and Taxonomy roots (tag IDs from the dialog); the `tags` column of the pages; the taxonomy (`tag`, `title` per language sheet).

Page Context Inputs: The current path (to exclude the page itself and to find the locale via `utils/site.js getLocale`), `<html lang>` (alphabetical collation), the Universal Editor edit-mode class `adobe-ue-edit` (editor hint).

Query Inputs: Datasource, Root path, Page types, Content types, Sort by, Featured and Hero rules.

Manual Inputs: Pages (Manual List), Featured pages, Hero page.

Authored HTML received from AEM (simplified, key-value rows):

```
<div class="filter-list">
  <div><div>title</div><div>Explore insights &amp;amp; opinions</div></div>
  <div><div>description</div><div><p>Perspectives from our experts.</p></div></div>
  <div><div>layout</div><div>masonry</div></div>
  <div><div>cardsPerPage</div><div>9</div></div>
  <div><div>source</div><div>query-index</div></div>
  <div><div>rootPath</div><div><a href="/content/tcs-corp/us/en/insights.html">
    /content/tcs-corp/us/en/insights.html</a></div></div>
  <div><div>pageTypes</div><div>article,report</div></div>
  <div><div>primaryTags</div><div><ul><li>tcs:industry/banking</li>
    <li>tcs:industry/insurance</li></ul></div></div>
  <div><div>advancedCategories</div><div><p>Industry</p><p>tcs:industry</p><hr>
    <p>Service</p><p>tcs:service</p></div></div>
  <div><div>cardStyle</div><div>mixed</div></div>
  <div><div>enableHero</div><div>true</div></div>
</div>
```

Keys are matched without case and hyphens (`cardsPerPage` and `cardsperpage` are the same). Composite multi-field items arrive as list items (single-field items) or as runs of elements separated by `<hr>` (several fields). The Additional tab values arrive as block classes (e.g. `anchor-insights`, `mt-1200`), not as rows.

A query index row the block reads (excerpt of `/query-index.json`):

```
{
  "path": "/en/insights/agentic-bank",
  "title": "The agentic bank",
  "description": "How AI agents change retail banking.",
  "tags": "[\"tcs:industry/banking\",\"tcs:topics/ai\"]",
  "pageType": "article",
  "contentType": "Article",
  "publishDate": "2026-09-04",
  "readingTime": "8",
  "featuredImage": "/en/insights/media_1a2b3c.jpeg",
  "authorId": "jdoe",
  "featured": "yes",
  "priority": "1",
  "lastModified": "1788000000"
}
```


6.17.10 Output Model

Rendered Output

- `div.filter-list.block.card-{style}.layout-{layout}` (plus Additional tab classes) containing:
- `div.filter-list-header`: `h2.filter-list-title`, `div.filter-list-description`, `div.filter-list-toolbar` (`p.filter-list-label`, `div.filter-list-filters[role=group]` with `button.filter-list-filter` pills, `button.filter-list-toggle` with `span.filter-list-toggle-count`), `div.filter-list-panel[role=region]` (`div.filter-list-panel-header`, `fieldset.filter-list-category` with `ul.filter-list-options`), `p.filter-list-count[role=status]`.
- `div.filter-list-results#filter-list-{n}-results`: the hero `article.filter-list-card-spotlight.filter-list-card-hero`, then `div.filter-list-grid.layout-{layout}` (plus `masonry-spans` or `masonry-native`) with `article.filter-list-card-{standard|compact|large|report|dark}` and `article.filter-list-card-spotlight.filter-list-card-featured`; or `div.filter-list-empty` (`p.filter-list-empty-title`, `p.filter-list-empty-hint`, Reset filters button).
- `p.filter-list-more` with the Load more button (hidden when nothing more).

Generated HTML (simplified)

```
<div class="filter-list block card-mixed layout-masonry">
  <div class="filter-list-header">
    <h2 class="filter-list-title">Explore insights &amp;amp; opinions</h2>
    <div class="filter-list-toolbar">
      <p class="filter-list-label">Filters</p>
      <div class="filter-list-filters" role="group" aria-label="Filters">
        <button class="filter-list-filter" type="button" data-filter=""
          data-cmp-interaction="filter" data-cmp-value="all"
          aria-controls="filter-list-1-results" aria-pressed="true">All</button>
        <button class="filter-list-filter" type="button"
          data-filter="tcs:industry/banking" data-cmp-interaction="filter"
          data-cmp-value="tcs:industry/banking" aria-pressed="false">Banking</button>
      </div>
      <button class="filter-list-toggle button secondary icon-button" type="button"
        aria-label="Advanced filters" aria-controls="filter-list-1-panel"
        aria-expanded="false"><span class="filter-list-toggle-count"
        aria-hidden="true"></span></button>
    </div>
    <p class="filter-list-count visually-hidden" role="status">24 results</p>
  </div>
  <div class="filter-list-results" id="filter-list-1-results">
```

```
    <div class="filter-list-grid layout-masonry masonry-spans">
      <article class="filter-list-card filter-list-card-standard">
        <div class="filter-list-card-media"><picture>...</picture></div>
        <p class="filter-list-card-tags"><span>Banking</span><span>8m</span></p>
        <h3 class="filter-list-card-title">
          <a class="filter-list-card-link"
            href="/en/insights/agentic-bank">The agentic bank</a>
        </h3>
        <p class="filter-list-card-meta"><span>Article</span>
          <time datetime="2026-09-04">04 Sep 26</time></p>
      </article>
    </div>
  </div>
  <p class="filter-list-more"><button type="button" class="button secondary"
    data-cmp-interaction="load-more">Load more</button></p>
</div>
```

Generated JSON: None. The block reads JSON (index, taxonomy, Authors, placeholders) but produces none.

Generated Metadata: None. The block does not change page metadata.

Analytics Tracking

The block is registered in the Adobe Client Data Layer by `decorators/data-layer-decorator.js` after the lazy phase:

```
{
  "component": {
    "anchor-insights": {
      "@type": "tcs-corp/components/filter-list",
      "parentId": "page-1x2y3z",
      "dc:title": "Explore insights & opinions"
    }
  }
}
```

The component id is the Anchor ID (block id), else `filter-list-{n}` in page order. Interactions:

Interaction	Event	eventInfo (besides `path: "component.{id}"`)
Pill click	`cmp:click`	`interaction: "filter"`, `dc:title`: pill label, `value`: tag ID or `all`, `xdm:linkType: "other"`, `breakpoint`
Advanced option checked / unchecked	`cmp:change`	`interaction: "filter-option"`, `value`: tag ID, `checked`: true / false
Filter toggle, Clear all, Reset filters	`cmp:click`	`interaction: "button"`, `dc:title`: the accessible name
Load more	`cmp:click`	`interaction: "load-more"`, `dc:title: "Load more"`
Card link	`cmp:click`	`interaction: "link"`, `dc:title`: link text or `aria-label`, `xdm:linkURL`: page URL, `xdm:linkType: "other"` (or `exit` / `download`)

Example:

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.filter-list-1",
```

```
    "interaction": "filter",
    "dc:title": "Banking",
    "xdm:linkType": "other",
    "value": "tcs:industry/banking",
    "breakpoint": "desktop"
  }
}
```

Personalization Output: None.

Search Index Output: None. The Filter List page itself is indexed like any page (its own metadata); the cards are not part of its indexed content.

6.17.11 Business Rules

1. The block reads its properties as key-value rows; unknown rows are ignored.
2. Defaults when a property is missing or invalid: layout masonry; cards per page 9; load more count 6; data-source Query Index; sort latest; Show All on; multi select off; advanced filter on; filter logic AND; card style mixed; featured cards on; featured rule featured; featured count 2; hero off; hero rule featured; mapping columns `featured-image`, `thumbnail-image`, `content-type`, `title`, `description`, `tags`, `path`, `author`.
3. Number fields: `Math.max(1, parseInt(value) || default)`; 0, empty or text gives the default, negatives give 1; there is no maximum in code.
4. Toggles: an absent property uses the default; a present property is on only when its text is `true`.
5. Paths (Root path, Pages, Featured pages, Hero page) are read from links or text, converted to published paths, `.html` removed and a trailing slash removed.
6. The index is the current locale's rows only (path equal to or below the locale root, e.g. `/de-de`).
7. Query Index source: pages strictly below Root path (not the root itself); the current page is excluded; Page types and Content types filter by normalized `page-type` / `content-type` (none selected: no filtering); sorted by Sort by.
8. Manual List source: the Pages in authored order, only those present in the locale's index; no sort, no type filters, the current page is not excluded.
9. Pages flagged `hide-in-nav` and redirect targets are not considered (unlike the List block).
10. Sorting: Latest = publish date descending (text comparison of YYYY-MM-DD), then last modified descending; Priority = numeric priority ascending (missing last), then latest; Featured First = featured first, then priority, then latest; Alphabetical = title with the page language's collator, ignoring case and accents.
11. `featured` is true when the `featured` or `feature-card` column is yes, true, 1, on or y (any case); `priority` is the number in `priority`, else `featured-priority`.
12. Pills: the Primary filter tags (valid tag IDs only, in order); else the tags below the Primary filter category used by the candidate pages, sorted by label; else no pills.
13. Pill labels and option labels come from the taxonomy titles of the locale's language; missing tags get a label derived from the ID (last segment, hyphens to spaces, first letter capitalised). Labels are never the raw IDs.
14. A pill or option matches a page that has the tag itself or any tag below it.
15. Single select: one pill at a time; clicking the pressed pill clears it. Multi select: pills toggle; a page must match every selected pill. All clears the selection and is pressed when nothing is selected.
16. The toolbar label and pill group appear only when there is at least one pill (including All).
17. Advanced categories need a Taxonomy root; options are the root's tags used by the candidate pages; categories without options are left out; with no categories (or Enable advanced filter off) there is no filter button or panel.
18. Advanced options combine by Filter logic across all categories; the result combines with the pills by AND.
19. Arrangement on every render: first the hero (if on), then the featured spotlights (if on), each removed from the remaining cards, so no page appears twice.

20. Hero: Manual = the Hero page if it matches, else none; Featured = the featured matching page with the lowest priority, else the latest page; Latest = the latest page.
21. Featured: Manual = the Featured pages that match, in authored order; featured=true = featured pages by priority, else (if none) pages with priority <= 3; priority<=3 = pages with priority <= 3 by priority. At most Featured card count.
22. Placement: the first render shows Cards per page standard cards, then one featured spotlight; each Load more adds Load more count cards, then the next featured spotlight; once every standard card is shown, all remaining featured spotlights follow.
23. Load more is hidden when every standard and featured card is shown; after a Load more the focus moves to the first new card's link and the count reads "Showing {n} of {total}" (or "{n} results" when everything is shown).
24. The result count counts the hero, all featured and all standard cards of the current filters ("1 result" / "{n} results").
25. Every filter change re-renders from the index already loaded (no new requests); the grid is rebuilt and the visible count resets to the first page.
26. Empty state: "No resources found." and "Try selecting another filter."; a Reset filters button appears when pills or options are selected; it clears both and moves focus to the first pill (or the first toolbar button).
27. Editor-only: when Universal Editor is in edit mode and the index returned no pages, the empty state shows only the editor hint ("The results show on the published page, from the pages' metadata.").
28. Card title: the Title source column, else the index title, else the path. Card URL: the Url source column, else the path. Image: the Image source column, else the Fallback image column; no image means no media (or a dark card in Mixed). Description: the Description source column.
29. Card tags: the first 2 tags of the Tag source column as labels, followed by the reading time; a whole number reading time is shown as "{n}m" (placeholder `minutesShort`), other text as typed.
30. Meta row: the eyebrow (Eyebrow source) and the publish date formatted `dd Mmm yy` (e.g. 04 Sep 26) in a `<time datetime>`; a non-ISO date is shown as typed without `datetime`.
31. Mixed style: report card when the eyebrow contains "report" (any case); else dark card when there is no image; else a standard image card without description.
32. Featured style: large cards with a Read more button; Compact: no image or description; Default: image, description and meta.
33. Spotlights: the image behind the content; tags, title link, description (hero only) and the author lockup.
34. Author lockup: the Author source value looked up as an Authors sheet `author-id` (exact match), else the page's `author-id`; if no sheet row matches, the Author source text is shown as the name (without designation or image); no value and no match means no lockup. A failing Authors sheet is treated as empty.
35. Images from site media (`/media_` on the same origin) are optimized responsive pictures; other URLs are plain lazy `<img>` elements. All card images have empty alt text.
36. The hero image loads eagerly; others lazily.
37. Fetch failures (index, taxonomy, placeholders, Authors) never throw: they give empty data, so the block shows the empty state or derived labels.
38. Masonry uses native CSS masonry when supported, else JavaScript placement that follows resizes and added cards.
39. Columns: 1 (mobile); 2 from 600px; 3 from 900px (Featured style 2); 4 from 1200px for Default, Compact and Mixed (Featured 3); List layout always 1.
40. Legacy content: the `filters` multifield (tag title and tags) is read when Primary filter tags is empty; `filtersLabel` replaces the "Filters" label.

6.17.12 Author Best Practices

Recommended Content Length

Element	Recommended	Enforced
Block title	Up to 60 characters	No
Block description	Up to 200 characters, 1 to 2 sentences	No
Card title (page title or card-title)	Up to 80 characters (titles wrap; long titles make masonry uneven)	No
Card description	Up to 160 characters (3 lines are shown, the rest is cut with an ellipsis)	3-line visual clamp (CSS)
Pills	3 to 8 pills, 1 to 3 words each	No
Categories	2 to 4 categories with 3 to 12 options each	No
Card tags	The first 2 tags of each page are shown; put the most relevant tags first	2 tags (code)

Image Dimensions

Image	CSS aspect / size	Requested width	Recommended source
Card media (Default, Mixed, Featured)	4:3, `object-fit: cover`	600px (900px+ viewports), 750px below	At least 1200 x 900px, subject centred
Hero spotlight	Covers a full-width card of at least 35rem (560px) height	1600px (900px+), 750px below	At least 2400 x 1200px landscape; keep the left half calm (text sits there on a white gradient)
Featured spotlight	Full-width card of at least 30rem (480px)	1600px / 750px	As hero
Author photo	3rem (48px) circle, `object-fit: cover`	750px	Square, at least 200 x 200px, face centred

Use images from AEM Assets published as site media, so they are optimized (`/media_` URLs); external image URLs are used as they are, without optimization.

CTA Writing Guidelines

The call-to-action texts come from placeholders (`readMore`, `download`, `explore`, `loadMore`, `resetFilters`) and are the same for all cards; they are maintained in the locale's placeholders sheet, not in the block. Because the report card's "Download" opens the report page, make sure the report page offers the download prominently. Write page titles that make sense out of context, because the title is the link text of most cards.

Accessibility Guidelines

- Always set a Title so the list has an `<h2>`; place the block in a section whose heading levels lead to it.
- Use meaningful taxonomy titles: they are the pill and option names.
- Keep pill counts small enough to scan by keyboard.
- Do not rely on the image to carry information (card images are decorative).
- Check the colour contrast of dark and report cards with real content.

SEO Guidelines

- The listing page's own Title and Description matter more than the cards for its ranking (cards are rendered in the browser).
- Keep listed pages in the sitemap and indexed; the Filter List does not replace internal linking from navigation.
- Prefer the Default style when descriptive link text matters (Featured, report and dark cards use generic button labels).

Metadata Guidelines

- Give every listed page: Title, Description, Tags, Page Type, `content-type`, `publish-date` (YYYY-MM-DD), `reading-time` (a number of minutes), `featured-image` or `thumbnail-image`, and `author-id` (Authors sheet) or `author`.
- Use `featured` = `yes` and `priority` = 1, 2, 3 consistently; avoid flagging many pages.
- Use the same Content type spelling everywhere (e.g. `Report`); it is the visible eyebrow and drives the report card.
- After changing metadata, preview and publish the page (and the metadata sheet) so the index updates.

Taxonomy Guidelines

- Use tags below a namespace (e.g. `tcs:industry/banking`) so categories and hierarchical matching work.
- Publish the locale's taxonomy with translated titles; missing tags fall back to labels derived from the ID.
- Taxonomy roots for categories should be the parent tags (e.g. `tcs:industry`), not the namespace alone.

Localization Guidelines

- Each locale's Filter List lists only that locale's pages; set Root path and pages within the same locale.
- Translate the placeholder keys listed in 6.17.1 in each locale's placeholders sheet.
- Translate the tag titles in the taxonomy (one sheet per language) and the Authors sheet designations.
- Alphabetical sorting follows the page language (`<html lang>`).

6.17.13 Troubleshooting Guide

Issue 1 — No cards in Universal Editor

Symptoms: The editor canvas shows "The results show on the published page, from the pages' metadata." and no cards.

Root Cause: The query index is not available in the editor; this is expected behaviour.

Diagnosis Steps: Confirm the canvas is in edit mode; open the preview URL of the page.

Resolution Steps: Review the cards on the preview (`.aem.page`) or live (`.aem.live`) URL.

Validation Steps: The preview page shows cards.

Preventive Measures: Tell authors that results are only visible outside the editor.

Issue 2 — "No resources found." on the preview or live page

Symptoms: The empty state shows without any filter selected.

Root Cause: No index row matches: Root path wrong or in another locale, Page types / Content types too narrow (values not matching the pages' metadata), the listed pages are not previewed / published, the pages are excluded by `helix-query.yaml`, or `/query-index.json` failed.

Diagnosis Steps: Open `https://main--tcs-corp--sradha-tcs.aem.page/query-index.json?limit=500` and search for the expected paths; compare their `pageType` / `contentType` with the selections; check the Root path; check the browser network panel for the index request.

Resolution Steps: Correct the Root path or type selections; preview / publish the pages; fix the metadata values.

Validation Steps: Reload the page; cards appear and the count is correct.

Preventive Measures: Use the standard Content type values; publish pages before the listing page.

Issue 3 — A new page does not appear

Symptoms: A recently published page is missing from the list.

Root Cause: The page is not yet in the (live) index, it is the current page, it is the Root path page itself, it is in a different locale, or its types do not match.

Diagnosis Steps: Check the page's row in `/query-index.json` on `.aem.live`; check its `path` against Root path; check the AEM Sidekick / admin status of the page.

Resolution Steps: Publish the page again; if needed ask L2 to re-index it with the Admin API (`POST https://admin.hlx.page/index/sradha-tcs/tcs-corp/main/{path}`).

Validation Steps: The row appears in the index and the card on the page.

Preventive Measures: Publish listed pages before announcing the listing.

Issue 4 — Pills show raw-looking labels or are missing

Symptoms: Pills read "Banking" derived from the ID instead of the translated title, or no pills appear.

Root Cause: The tag is missing from `/{locale}/taxonomy.json` or the language sheet; or Primary filter tags contain values that are not tag IDs; or a Primary filter category is used and no listed page has tags below it.

Diagnosis Steps: Open `/{locale}/taxonomy.json` and look for the tag; check the dialog values; check the pages' `tags` column in the index.

Resolution Steps: Publish the taxonomy with the tags and translations; select tags with the tag picker; tag the pages.

Validation Steps: Pills show the taxonomy titles.

Preventive Measures: Publish the taxonomy after tag changes.

Issue 5 — The advanced filter button does not appear

Symptoms: No round filter button next to the pills.

Root Cause: Enable advanced filter is off, no Advanced filter category has a Taxonomy root, or no candidate page uses tags below the roots (categories without options are hidden).

Diagnosis Steps: Check the Filters and Advanced tabs; check the pages' tags in the index.

Resolution Steps: Turn the toggle on; add categories with roots that the pages use.

Validation Steps: The button appears and the panel lists the options.

Preventive Measures: Choose taxonomy roots that the content is tagged with.

Issue 6 — Images missing or not optimized

Symptoms: Cards without images, dark cards where image cards were expected, or slow large images.

Root Cause: `featured-image` and `thumbnail-image` are empty for the page, the Mapping tab names a column that is not indexed, or the image URL is external (not optimized).

Diagnosis Steps: Inspect the page's `featuredImage` / `thumbnailImage` in the index; check the Mapping tab.

Resolution Steps: Set the metadata to an AEM Assets image published as site media; restore the default mapping.

Validation Steps: The card shows a `<picture>` with `/media_` sources.

Preventive Measures: Include images in the metadata checklist for every page.

Issue 7 — Wrong hero or featured cards

Symptoms: An unexpected page is the hero, or no featured spotlights appear.

Root Cause: The rules: Featured hero falls back to the latest page when no matching page is featured; featured=true falls back to priority <= 3; Manual pages that do not match the source or current filters are skipped; Featured card count limits the number; featured flags use values other than yes / true / 1 / on / y.

Diagnosis Steps: Check the pages' `featured` and `priority` columns in the index; check the rules and Featured pages.

Resolution Steps: Correct the metadata or rules; make sure manual pages are inside the Root path and types.

Validation Steps: The expected pages show as hero and spotlights.

Preventive Measures: Document which pages are featured per hub.

Issue 8 — Author lockup missing or shows only a name

Symptoms: Spotlight cards show no author, or a name without photo and designation.

Root Cause: The `author-id` of the page does not exactly match an `author-id` in `/{locale}/data/authors.json` (the match is case-sensitive in this block), or the Authors sheet is not published; with only an `author` text the name is shown alone.

Diagnosis Steps: Compare the page's `authorId` / `author` in the index with the sheet.

Resolution Steps: Correct the ID (same case) or add the author row; publish the sheet.

Validation Steps: The lockup shows the photo, name and designation.

Preventive Measures: Use lower-case author IDs consistently.

6.17.14 Support Runbook

Monitoring Requirements

- Availability of `/query-index.json`, `/{locale}/taxonomy.json`, `/{locale}/data/authors.json` and `/{locale}/placeholders.json` on `.aem.live` (HTTP 200, JSON).
- Index freshness: newly published pages appear in the live index.
- Real User Monitoring (Edge Delivery RUM) for the listing pages' performance ([To be confirmed: monitoring tool and alerting — owner: AMS]).

Logs

- Browser console: the block emits no console messages; fetch failures are swallowed and lead to empty data. Use the network panel to see failing requests (index, taxonomy, Authors, placeholders).
- Admin API: page status `GET https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}` (preview, live and index state); re-index `POST https://admin.hlx.page/index/sradha-tcs/tcs-corp/main/{path}`.
- AEM author: replication / publish history of the listed pages (Sites console Timeline) and Universal Editor errors.

Common Errors

Error	Likely cause	First action
Editor hint instead of cards	Index not available in the editor (expected)	Check the preview URL
No resources found.	Source settings or unpublished pages	Check the index rows and dialog
Pills not translated	Taxonomy missing the tag or language sheet	Publish the taxonomy
No filter button	No category options	Check roots and page tags
Dark cards everywhere (Mixed)	Image metadata missing	Fill featured-image / thumbnail-image
English labels on a localized page	Placeholder keys missing in the locale sheet	Add the keys to the placeholders sheet
Duplicate-looking cards	Same content published at two paths	Unpublish one or exclude it
Very slow block	Large index (many 500-row requests) or block in the first section	Move the block below the fold; L3 to review index size

Escalation Matrix

Level	Who	What they check
L1	Author support	Dialog settings, metadata of listed pages, preview / publish state, taxonomy and placeholders sheets, the expected editor hint
L2	AMS / site admin	Index rows in `/query-index.json`, Admin API status and re-index, site configuration (metadata sources), CDN and cache
L3	Development team	`filter-list.js` rules, `utils/query-index.js`, `helix-query.yaml` columns, masonry and layout defects, code releases
L4	Adobe Support	Edge Delivery indexing or CDN incidents, Universal Editor platform defects

Recovery Procedures

1. Content problem: correct the dialog or page metadata, preview and publish.
2. Index problem: re-publish the affected pages or re-index them via the Admin API; if `helix-query.yaml` changed, the development team triggers a re-index of the site.
3. Block defect after a release: revert the code change on `main` (Code Sync deploys); clear nothing manually (Edge Delivery purges on deploy).
4. Emergency: turn on Hide block and publish the listing page.

Health Checks

- The listing page shows cards and the expected count on `.aem.live`.
- `/query-index.json?limit=1` returns `total` greater than 0.
- A pill click changes the count; the panel opens and closes with Escape.
- Load more adds cards.

Operational Procedures

- When adding a new metadata column for cards, the development team adds it to `helix-query.yaml` and re-indexes; authors then use it in the Mapping tab.
- When adding tags, publish the taxonomy for every locale.
- When adding a locale, translate the placeholder keys and publish the locale's Authors sheet.

Known Limitations

- Results are not visible in Universal Editor.
- Filters are not reflected in the URL (no deep links or back-button state for filters).
- The whole index (all locales) is downloaded in the browser; very large sites increase load time.
- `hide-in-nav` pages and redirect targets are not handled (unlike the List block).
- The Authors lookup is case-sensitive.
- Card call-to-action labels are global placeholders, not per card; the report "Download" opens the page, not a file.
- Number limits (`min` / `max`) are declared in the model but the code only enforces a minimum of 1.
- No personalization; no server-side rendering of cards.

6.18 Table List Block

The Table List is a divided list of items (industries, solutions, products) under a heading with an optional call to action. Each item has a title, an optional number and trademark sign, a description and a link shown as an arrow, a diagonal arrow or a button. It implements the TCS DS Figma "List" variations 1 to 3.

Source files: `blocks/table-list/table-list.js` (333 lines), `blocks/table-list/table-list.css` (453 lines), `blocks/table-list/_table-list.json`. The dialog has a block model (`table-list`: 4 tabs, 10 fields plus the 10

Additional tab fields), an item model (`table-list-card`, "Table List Item": 6 fields) and a legacy header item model (`table-list-header`, "Table List Header": 2 fields, no longer addable).

6.18.1 Component Purpose

Business Objective

Present a short, scannable list of offerings (industries served, transformation solutions, products and platforms) with a strong typographic style and a clear path to each detail page, without images.

Business Use Cases

- Home page "Industries" list: title, short description and an arrow per industry, with a "View all industries" button.
- "Our transformation solutions": description on the left, a large solution name on the right and an outline "Explore" button.
- "Products & Platforms": description, a large product name with a number (01, 02 …) and a trademark sign (e.g. TwinX™), and a diagonal arrow.

When To Use

- Up to 10 items that each link to a page.
- Text-led lists where typography, not imagery, carries the design.

When Not To Use

- More than 10 items, or lists that should update automatically from page metadata: use the List or Filter List block.
- Image cards: use Cards or List.
- Tabular data with columns: use the Table block.
- Expandable question-and-answer content: use the Accordion block.

Dependencies

Kind	Dependency	Used for
Block code	`blocks/table-list/table-list.js`, `table-list.css`, `_table-list.json`	Decoration, styles, dialog
Decorator	`decorators/button-decorator.js` (`decorateButton`)	The heading call to action and item buttons
Utility	`utils/buttons.js` (`pick`)	Normalizing select values
Utility	`utils/cells.js` (`matchCells`)	Matching item cells (and unkeyed property rows) to fields when AEM leaves out unset fields
Utility	`utils/properties.js` (`normalizeKey`, `textOf`)	Reading property rows
Utility	`utils/urls.js` (`toSitePath`)	Converting AEM paths to published paths
Utility	`utils/environment.js` (`isEditing`, `prefersReducedMotion`)	Editor previews of unlinked buttons; motion off
Utility	`utils/viewport.js` (`onVisible`)	Motion when items scroll into view
Utility	`utils/placeholders.js` (`getPlaceholders`, `fillPlaceholder`)	UI text
Core	`scripts/scripts.js` (`moveInstrumentation`)	Keeps Universal Editor instrumentation on rebuilt elements

Kind	Dependency	Used for
Icons	`icons/arrow-right.svg`, `icons/arrow-up-right.svg`	Arrow masks
Data	`/{locale}/placeholders.json`	UI text

Placeholder keys:

Key	English fallback	Where it appears
`viewAll`	View all	Heading call to action text when a Link is set without Text
`explore`	Explore	Item button text when neither the item nor the block sets one
`itemLinkLabel`	{text}: {title}	Accessible name of item buttons, e.g. “Explore: TwinX”
`exploreItem`	Explore {title}	Accessible name of item arrows, e.g. “Explore TwinX”

Related Components

Cards and List (image cards), Filter List (6.17, metadata-driven lists), Table (tabular data), Accordion, Anchor List (6.19).

Applicable Templates

Any page (allowed in every section); typically the home page and landing pages. Not specific to article, blog or press-release templates.

Supported Content Types

Authored text only: titles, rich-text descriptions, links to site pages, anchors or external URLs. No images or media.

6.18.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A Table List (Variation 1) at 1440px: the 3rem heading at the left and a filled “View all” button at the right, then four rows, each with a semibold title, a description in the middle column and a grey arrow at the right, separated by thin dividers; the block inset 5rem from the section edges.

Desktop (1200px and wider). The header is a flex row: the heading (3rem, semibold, letter-spacing –0.09rem, line-height 1.2) and the call to action (button font at `--button-font-size`) at the right; it wraps under the heading when there is not enough space. Header margin below: 5rem (`--space-2000`). Items are grid rows `main | description | action` with a column gap of 1.5rem and an item gap of 1.5rem (2.5rem for Variation 3), each with 2.5rem bottom padding and a divider (1px `--slate-400`; 2px `--grey-200` for Variation 3). Variation-specific sizes are listed per variation in 6.18.3.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Table List at 768px: the heading at 2.5rem with the button beside it, items in three columns (title, description, arrow), no numbers in Variation 1.

Tablet (600px to 1199px). The heading is 2.5rem (letter-spacing -0.075rem); header margin 5rem. Items switch to three columns: two equal text columns and the action (`minmax(0, 1fr) minmax(0, 1fr) auto`, column gap 1rem); Variation 2 and 3 put the description first and the title second, vertically centred, with the title right-aligned. The action has a start margin of 1.5rem. "Show on mobile only" numbers are hidden from 600px. The call-to-action button text uses `--body-font-size-xs`.

> [**Screenshot Placeholder**]
> **File name:** `screenshots/table-list-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Table List at 375px: a 2rem heading with the button wrapping below it, then items with the number and title on the first line, the description below and the arrow at the top right.

Mobile (below 600px). Heading 2rem (letter-spacing –0.06rem, line-height 1.19), header margin 2.5rem, call to action below the heading when it does not fit. Items are stacked grids: Variation 1 `main` over `description` with the arrow at the right; Variation 2 description, title and button stacked; Variation 3 description over the number and title with the arrow at the right. Item gap 1rem (2.5rem in Variation 3).

6.18.3 Component Variations

The Variation select sets the layout and the defaults of the call-to-action style, numbers and item link; those three can be changed independently ("Variation default" keeps the variation's choice). Motion and the item Trademark are further options.

Variation 1 — Industries (Variation 1: title, description, arrow)

Variation Name: Variation 1 - Title, description, arrow (`industries`; the default).

Business Purpose: A list of industries or capabilities with a short explanation each.

Visual Example:

> [**Screenshot Placeholder**]
> **File name:** `screenshots/table-list-industries.png`
> **Purpose:** Variation 1 (industries)
> **Description:** Rows with a semibold title ("Banking"), a description in the middle column and a grey right arrow; the last row has no divider; numbers visible only on mobile.

Use Cases: Industries, capabilities, service lines.

Authoring Guidelines: Content tab › Variation: Variation 1. Give each item a Title, a one-line Description and a Link.

Business Rules: Defaults: call to action Filled button (`primary`), Numbers Show on mobile only, Item link Arrow. Block classes `industries numbers-{value} motion-{value}`.

Rendering Rules: Mobile title line: `--body-font-size-m`, medium weight, line-height 1.375; description 1.5rem semibold (line-height 1.42). Arrow 3rem. The last item has no divider and no bottom padding. From 600px the description is 1.25rem (line-height 1.4) and the arrow `--slate-400`. From 1200px the block has 5rem inline padding, items are vertically centred, the title line is 1.75rem semibold (letter-spacing –0.035rem, line-height 1.39) and the description regular weight.

Responsive Behaviour: Desktop: title | description | arrow, inset 5rem. Tablet: three columns, numbers hidden (default). Mobile: number and title over the description, arrow at the right.

SEO Impact: Item titles are headings one level below the block heading (e.g. `<h3>`); links point to the detail pages.

Accessibility Compliance: The arrow link has the accessible name "Explore {title}" and covers the whole row; focus outline on the arrow (2px brand, 0.25rem offset); links inside the description stay separately clickable.

Variation 2 — Solutions (Variation 2: description, large title, button)

Variation Name: Variation 2 - Description, large title, button (`solutions`).

Business Purpose: Emphasizes solution names in large type with an explicit button.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-solutions.png`
> **Purpose:** Variation 2 (solutions)
> **Description:** Rows with a short description at the left, a 4rem semibold solution name right-aligned in the middle and an outline "Explore" button at the right.

Use Cases: Transformation solutions, offerings, programmes.

Authoring Guidelines: Variation: Variation 2. Set the block Button text (Items tab, shown for this variation) or each item's Button text.

Business Rules: Defaults: call to action Outline button (`secondary`), Numbers Hide, Item link Button.

Rendering Rules: Mobile: description (body-m, `--grey-900`, line-height 1.39), then the title line (2.5rem semibold, line-height 1.2, letter-spacing –0.025rem), then the button. From 600px: description | title | button, centred, title right-aligned, description 1.125rem. From 1200px the title is 4rem (line-height 1.4, letter-spacing -0.04rem).

Responsive Behaviour: Desktop and tablet: three columns. Mobile: stacked.

SEO Impact: Visible button text is generic ("Explore"); the accessible name adds the title.

Accessibility Compliance: Buttons have the accessible name "{text}: {title}" (placeholder `itemLinkLabel`); the visible text starts the name (WCAG 2.5.3). The button does not cover the row.

Variation 3 — Products (Variation 3: description, number, large title, arrow)

Variation Name: Variation 3 - Description, number, large title, arrow (`products`).

Business Purpose: A numbered showcase of products and platforms with very large names.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-products.png`
> **Purpose:** Variation 3 (products)
> **Description:** Rows with a description at the left, then a light grey number "01" and a 5.625rem product name "TwinX™" spread across the right column, and a diagonal arrow; 2px light grey dividers; a "All Products & Platforms" text link next to the regular-weight heading.

Use Cases: Products and platforms, numbered programmes.

Authoring Guidelines: Variation: Variation 3. Leave Number empty for automatic numbering. Use Trademark for ™ / ®.

Business Rules: Defaults: call to action Text link (`tertiary`), Numbers Show, Item link Diagonal arrow. Item gap 2.5rem; divider 2px `--grey-200`; text colour `--grey-900`.

Rendering Rules: Heading regular weight (line-height 1.4); a tertiary call to action uses `--body-font-size-l`. Mobile: description over the number and title (3rem, line-height 1.17, letter-spacing -0.075rem; number `--slate-300` 1.5rem), arrow `--icon-size-m`, centred vertically. From 600px: columns `2fr 3fr auto`, title 3.5rem with the number at the left and the title at the right (`space-between`), description 1.125rem. From 1200px: columns `476fr 695fr auto` with a 2.5rem gap, title 5.625rem (letter-spacing –0.1406rem, line-height 1.4), number 2.4975rem, description 1.5rem, arrow `--icon-size-l`. The trademark gets a start margin (0.45rem, 0.525rem, 0.8438rem by breakpoint).

Responsive Behaviour: Desktop and tablet: description | number and title | arrow. Mobile: description above, title row below, arrow at the right.

SEO Impact: As Variation 1. Very long product names wrap anywhere (`overflow-wrap: anywhere`).

Accessibility Compliance: As Variation 1; the number is text, read before the title.

Variation 4 — Call to action style: Filled button

Variation Name: Call to action › Style = Filled button (`primary`; Variation 1 default).

Business Purpose: A prominent "View all" next to the heading.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-cta-filled.png`
> **Purpose:** Filled button call to action
> **Description:** The heading with a filled brand button "View all industries" at the right.

Use Cases: Home page sections leading to a hub.

Authoring Guidelines: Call to action tab: Text, Link, Style = Filled button (or Variation default with Variation 1).

Business Rules: Shown only when a Link is set (on the page); without Text the label is "View all" (placeholder `viewAll`). In the editor, a Text without a Link shows a non-focusable preview.

Rendering Rules: `<p class="table-list-cta"><a class="button primary">`; font `--body-font-size-xs` below 1200px, `--button-font-size` from 1200px.

Responsive Behaviour: Beside the heading; wraps below it when space is short.

SEO Impact: A crawlable link with descriptive text when Text is written well.

Accessibility Compliance: Standard button styles and focus states from `styles/styles.css`.

Variation 5 — Call to action style: Outline button

Variation Name: Style = Outline button (`secondary`; Variation 2 default).

Business Purpose: A quieter button, matching outline item buttons.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-cta-outline.png`
> **Purpose:** Outline button call to action
> **Description:** The heading with an outline button "All solutions".

Use Cases: Solutions lists.

Authoring Guidelines: Style = Outline button.

Business Rules: As Variation 4.

Rendering Rules: `a.button.secondary`; same font sizes as Variation 4.

Responsive Behaviour: As Variation 4.

SEO Impact: As Variation 4.

Accessibility Compliance: As Variation 4.

Variation 6 — Call to action style: Text link

Variation Name: Style = Text link (`tertiary`; Variation 3 default).

Business Purpose: A lightweight link for editorial layouts.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-list-cta-text-link.png`
> **Purpose:** Text link call to action
> **Description:** The heading with a text link "All Products & Platforms".

Use Cases: Product lists.

Authoring Guidelines: Style = Text link.

Business Rules: As Variation 4.

Rendering Rules: `a.button.tertiary`; keeps the tertiary button size (in Variation 3 `--body-font-size-l`).

Responsive Behaviour: As Variation 4.

SEO Impact: As Variation 4.

Accessibility Compliance: As Variation 4.

Variation 7 — Numbers: Show

Variation Name: Items › Numbers = Show (`show`; Variation 3 default).

Business Purpose: Numbered items at every width.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-list-numbers-show.png`
> **Purpose:** Numbers shown
> **Description:** Items with "01", "02", "03" before the titles at desktop width.

Use Cases: Ordered steps, ranked products.

Authoring Guidelines: Numbers = Show; leave item Number empty for 01, 02, … or type a number.

Business Rules: The item's Number is used when it is 1 to 3 digits (optionally followed by "." or ")"); otherwise the position is used, zero-padded to two digits (01, 02 … 10). Positions count the displayed items.

Rendering Rules: `span.table-list-number` before the title in `div.table-list-main`; class `numbers-show`.

Responsive Behaviour: Visible at every breakpoint.

SEO Impact: Numbers are text in the item.

Accessibility Compliance: Read as part of the item content.

Variation 8 — Numbers: Show on mobile only

Variation Name: Numbers = Show on mobile only (`mobile`; Variation 1 default).

Business Purpose: Numbers help on stacked mobile layouts but are omitted on wider layouts.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-list-numbers-mobile.png`
> **Purpose:** Numbers on mobile only
> **Description:** Side-by-side views: at 375px the items show "01 Banking"; at 1024px only "Banking".

Use Cases: Industries lists.

Authoring Guidelines: Numbers = Show on mobile only (or Variation default with Variation 1).

Business Rules: As Variation 7.

Rendering Rules: Class `numbers-mobile`; CSS hides `.table-list-number` from 600px.

Responsive Behaviour: Visible below 600px, hidden from 600px.

SEO Impact: The number stays in the DOM (hidden with `display: none` from 600px).

Accessibility Compliance: Hidden numbers are not announced at those widths.

Variation 9 — Numbers: Hide

Variation Name: Numbers = Hide (`hide`; Variation 2 default).

Business Purpose: Unordered lists.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-list-numbers-hide.png`
> **Purpose:** Numbers hidden
> **Description:** Items without numbers at any width.

Use Cases: Solutions, unordered offerings.

Authoring Guidelines: Numbers = Hide.

Business Rules: No number element is rendered; authored item Numbers are ignored.

Rendering Rules: Class `numbers-hide`; no `span.table-list-number`.

Responsive Behaviour: Same at every width.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 10 — Item link: Arrow

Variation Name: Items › Item link = Arrow (`arrow`; Variation 1 default).

Business Purpose: The whole row is a link, signalled by a right arrow.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-item-arrow.png`
> **Purpose:** Arrow item link
> **Description:** A row hovered: the arrow turns brand blue and moves slightly right.

Use Cases: Navigation lists.

Authoring Guidelines: Item link = Arrow; set each item's Link.

Business Rules: With a Link: `<a class="table-list-arrow arrow">` with the accessible name "Explore {title}" (placeholder `exploreItem`), covering the item (`::after`, inset 0). Without a Link: a decorative `<span aria-hidden="true">` arrow; the item is not clickable.

Rendering Rules: Arrow drawn with a CSS mask of `icons/arrow-right.svg`, 3rem, `--color-icon-default`; on hover of the item or focus: `--color-text-brand` and `translateX(0.25rem)` (no transition with reduced motion).

Responsive Behaviour: Always in the action column at the right.

SEO Impact: The link's text is its `aria-label` only (no visible text); search engines rely on the item title for context.

Accessibility Compliance: Accessible name from `aria-label`; visible focus outline; one link per item.

Variation 11 — Item link: Diagonal arrow

Variation Name: Item link = Diagonal arrow (`arrow-up-right`; Variation 3 default).

Business Purpose: Product-style link affordance.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-item-diagonal-arrow.png`
> **Purpose:** Diagonal arrow item link
> **Description:** A product row with an up-right arrow at the right.

Use Cases: Products and platforms.

Authoring Guidelines: Item link = Diagonal arrow.

Business Rules: As Variation 10, class `arrow-up-right`.

Rendering Rules: Mask `icons/arrow-up-right.svg`; size 3rem, or `--icon-size-m` / `--icon-size-l` (1200px+) in Variation 3.

Responsive Behaviour: As Variation 10.

SEO Impact: As Variation 10.

Accessibility Compliance: As Variation 10.

Variation 12 — Item link: Button

Variation Name: Item link = Button (`button`; Variation 2 default).

Business Purpose: An explicit, labelled action per item.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-item-button.png`
> **Purpose:** Button item link
> **Description:** A row with an outline “Explore” button at the right.

Use Cases: Solutions, offerings needing a verb.

Authoring Guidelines: Item link = Button; set Button text on the block (Items tab) and optionally per item.

Business Rules: Text: the item’s Button text, else the block’s Button text, else “Explore” (placeholder `explore`). With a Link: an outline (`secondary`) button with the accessible name “{text}: {title}”. Without a Link: nothing on the page; a non-focusable preview in the editor.

Rendering Rules: `div.table-list-action > a.button.secondary`; the row is not covered by the link.

Responsive Behaviour: At the right (tablet and desktop) or below the title (Variation 2 mobile).

SEO Impact: Generic visible text.

Accessibility Compliance: As Variation 2.

Variation 13 — Motion: None

Variation Name: Motion Type = None (`none`; the default).

Business Purpose: Static rendering.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-motion-none.png`
> **Purpose:** No motion
> **Description:** All items visible immediately.

Use Cases: Most pages.

Authoring Guidelines: Motion Type = None.

Business Rules: No observers are created.

Rendering Rules: Class `motion-none`.

Responsive Behaviour: Same everywhere.

SEO Impact: None.

Accessibility Compliance: No motion.

Variation 14 — Motion: Fade

Variation Name: Motion Type = Fade (`fade`).

Business Purpose: Items fade in as they scroll into view.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-motion-fade.png`

> **Purpose:** Fade motion
> **Description:** A sequence: items below the fold invisible, then fading in as they enter the viewport.

Use Cases: Storytelling sections.

Authoring Guidelines: Motion Type = Fade.

Business Rules: Runs only when `IntersectionObserver` exists, not in Universal Editor edit mode, and not when the visitor prefers reduced motion. Each item animates once, when 15% of it is visible.

Rendering Rules: The block gets `motion-ready`; items without `is-visible` have opacity 0; on view they get `is-visible` and the 500ms `table-list-fade` animation (opacity from 0).

Responsive Behaviour: Same at every width.

SEO Impact: Content is in the DOM; hidden only visually until scrolled.

Accessibility Compliance: Respects `prefers-reduced-motion` (WCAG 2.3.3 intent).

Variation 15 — Motion: Slide
Variation Name: Motion Type = Slide (`slide`).

Business Purpose: Items rise slightly while fading in.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-list-motion-slide.png`
> **Purpose:** Slide motion
> **Description:** Items moving up 1.25rem while fading in.

Use Cases: Landing pages.

Authoring Guidelines: Motion Type = Slide.

Business Rules: As Variation 14.

Rendering Rules: `table-list-slide`: from opacity 0 and `translateY(var(--space-500))` (1.25rem), 500ms ease.

Responsive Behaviour: As Variation 14.

SEO Impact: As Variation 14.

Accessibility Compliance: As Variation 14.

Variation 16 — Motion: Reveal
Variation Name: Motion Type = Reveal (`reveal`).

Business Purpose: Items are revealed from the top down.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-list-motion-reveal.png`
> **Purpose:** Reveal motion
> **Description:** An item partially revealed by a clip from top to bottom.

Use Cases: Product showcases.

Authoring Guidelines: Motion Type = Reveal.

Business Rules: As Variation 14.

Rendering Rules: `table-list-reveal`: from opacity 0 and `clip-path: inset(0 0 100% 0)`, 500ms ease.

Responsive Behaviour: As Variation 14.

SEO Impact: As Variation 14.

Accessibility Compliance: As Variation 14.

Variation 17 — Item Trademark sign

Variation Name: Table List Item › Trademark = ™ Trademark (`tm`) or ® Registered (`r`).

Business Purpose: Correct legal marking of product names.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-trademark.png`
> **Purpose:** Trademark sign
> **Description:** "TwinX™" and "ignio®" product titles with the raised signs at title size.

Use Cases: Product and platform names.

Authoring Guidelines: Select Trademark on the item; do not type the sign in the Title.

Business Rules: `tm` adds ™, `r` adds ®; None adds nothing.

Rendering Rules: `<sup class="table-list-trademark">` appended to the title, at the title's font size with line-height 0 (the glyph is raised itself); extra start margin in Variation 3.

Responsive Behaviour: Scales with the title.

SEO Impact: The sign is part of the heading text.

Accessibility Compliance: Screen readers announce the sign as "trademark" or "registered" depending on the reader.

6.18.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A Table List selected in Universal Editor (Variation 3) with the heading, the call to action, three items (number, title with trademark, description, arrow) and the content tree showing the Table List with three Table List Item children.

Annotations

- A = Heading (Content tab › Heading, level from Heading Type).
- B = Call to action (Call to action tab › Text, Link, Style).
- C = Item number (Table List Item › Number, or the automatic 01, 02 …; shown per the Numbers setting).
- D = Item title (Table List Item › Title) with the Trademark sign.

- E = Item description (Table List Item › Description).
- F = Item link (Table List Item › Link and Button text; arrow, diagonal arrow or button per Item link).
- G = Items in the content tree (add, reorder and delete Table List Items; at most Maximum items are shown).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The properties panel of the Table List with the tabs Content, Call to action, Items and Additional, and, for a selected item, the Table List Item panel with Number, Title, Trademark, Description, Link and Button text.

Block tabs:

- **Content:** Variation (Variation 1 / 2 / 3), Heading (text), Heading Type (H2, H3, H4), Maximum items (number, 4; 1 to 10).
- **Call to action:** Text, Link (page picker or URL), Style (Variation default, Filled button, Outline button, Text link).
- **Items:** Numbers (Variation default, Show, Show on mobile only, Hide), Item link (Variation default, Arrow, Diagonal arrow, Button), Button text (shown for Button item links, including Variation 2 with Variation default), Motion Type (None, Fade, Slide, Reveal).
- **Additional:** Anchor ID, Hide block, the 4 padding and 4 margin selects.

Item panel (Table List Item): Number, Title, Trademark (None, ™ Trademark, ® Registered), Description (rich text), Link, Button text.

6.18.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites select the page under `/content/tcs-corp/{country}/{language}/` and choose **Edit**.

Expected Result: The page opens in Universal Editor.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** The Sites console with the page selected and Edit highlighted.

Step 2 — Add the Table List

Action: Select a section, choose **Add** (+) and select **Table List**.

Expected Result: A Table List (Variation 1, Maximum items 4, Button text "Explore") appears, without items.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** The section's Add menu with Table List highlighted.

Step 3 — Choose the variation and heading

Action: In the Content tab choose Variation, type the Heading, choose Heading Type (H2 unless the section already has an H2 the list belongs under) and set Maximum items.

Expected Result: The canvas shows the heading at the chosen level and the variation's layout.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-03-content-tab.png`
> **Purpose:** Content tab
> **Description:** Variation 3 selected, Heading "Products & Platforms", Heading Type H2, Maximum items 6.

Step 4 — Add the call to action

Action: In the Call to action tab enter Text, pick or type the Link and choose a Style (or keep Variation default).

Expected Result: The call to action appears next to the heading.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-04-cta-tab.png`
> **Purpose:** Call to action tab
> **Description:** Text "All Products & Platforms", Link /content/tcs-corp/us/en/products, Style Variation default.

Step 5 — Set the item options

Action: In the Items tab choose Numbers, Item link, Button text (if buttons) and Motion Type.

Expected Result: The canvas updates the numbers and links (motion is not played in the editor).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-05-items-tab.png`
> **Purpose:** Items tab
> **Description:** Numbers Variation default, Item link Variation default, Motion Type Fade.

Step 6 — Add items

Action: Select the block in the content tree, choose **Add** and select **Table List Item**. Repeat for each item.

Expected Result: Each new item shows the sample title "TwinX" and description.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-06-add-item.png`
> **Purpose:** Add items
> **Description:** The content tree Add menu of the Table List with Table List Item.

Step 7 — Fill each item

Action: Select the item and fill Number (optional), Title, Trademark, Description, Link and Button text (optional, button item links only).

Expected Result: The item renders with its number, title, sign, description and link.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-07-item-fields.png`

> **Purpose:** Item fields
> **Description:** The item panel with Title “TwinX”, Trademark ™, Description and Link filled.

Step 8 — Order the items

Action: Drag items in the content tree into the display order; delete unneeded ones.

Expected Result: The canvas follows the order; only the first Maximum items are shown.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-08-reorder.png`
> **Purpose:** Reorder items
> **Description:** An item being dragged in the content tree.

Step 9 — Set the Additional options (optional)

Action: Set an Anchor ID and spacing if needed.

Expected Result: Invalid Anchor IDs are rejected with the validation message.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-09-additional.png`
> **Purpose:** Additional tab
> **Description:** Anchor ID anchor-products and Margin top 3rem.

Step 10 — Preview

Action: Switch to Preview mode, then preview the page (AEM Sidekick or the `.aem.page` URL).

Expected Result: The list renders with motion (if chosen) and working links.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-10-preview.png`
> **Purpose:** Preview
> **Description:** The preview page with the Table List at desktop width.

Step 11 — Validate

Action: Run the Validation Process checklist at the three breakpoints.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-11-validate.png`
> **Purpose:** Validate
> **Description:** The mobile view with numbers and stacked items.

Step 12 — Publish

Action: Publish the page from Universal Editor or the Sites console.

Expected Result: The live page shows the Table List.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-list-step-12-publish.png`
> **Purpose:** Publish
> **Description:** The Publish dialog with the page selected.

Preview Process

1. Changes save automatically in Universal Editor; Preview mode shows the page without the editor outlines (motion runs in Preview mode, not in edit mode).
2. Preview the page and open `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}`.
3. Check every link destination on the preview.

Publishing Process

1. Publish the target pages of the item links first (avoid links to unpublished pages).
2. Publish the page.
3. Check the live URL.

Validation Process

- ☐ The heading level fits the page outline; item titles are one level below it.
- ☐ The call to action shows only when a Link is set, with the right style and text.
- ☐ Every item has a Link; arrows are clickable across the whole row; buttons show the right text.
- ☐ Numbers follow the Numbers setting (mobile-only numbers disappear from 600px).
- ☐ Trademark signs are correct and not typed twice.
- ☐ No more than Maximum items are shown, and the right ones.
- ☐ Desktop / tablet / mobile layouts match the variation.
- ☐ Motion plays once on scroll and not with reduced motion turned on in the operating system.
- ☐ Keyboard: each item link or button reachable with a visible focus ring.

Rollback Process

1. Undo in Universal Editor, or restore a page version (Sites console › Timeline) and publish again.
2. Turn on Hide block and publish to remove the list temporarily.
3. Code defects: the development team reverts the change on `main`.

6.18.6 Authorable Field Specification

Table List block model (`table-list`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Content tab › Variation (`variations`)	select	No	No	industries	Layout and defaults	Pick by content	Options industries, solutions, products; others give industries	products
Content tab › Heading (`listHeading`)	text	No	No	(empty)	List heading	Short heading	None	Products & Platforms
Content tab › Heading Type (`listHeadingType`)	select	No	No	h2	Heading level	H2 normally	Options h2, h3, h4	h2

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Content tab › Maximum items (`maxCards`)	number	No	No	4	Items shown	Match the number of items	validation numberMin 1, numberMax 10, message “Maximum items must be between 1 and 10.” (Enforced); code clamps to 1 to 10, empty gives 4	6
Call to action tab › Text (`viewAllLabel`)	text	No	No	(empty)	Call to action text	Descriptive, e.g. All Products & Platforms	None; empty gives “View all”	All Products & Platforms
Call to action tab › Link (`viewAllLink`)	aem-content	No	No	(empty)	Call to action destination	Pick a page or type a URL	Must start with /, #, http:// or https:// to be used	/content/tcs-corp/us/en/products
Call to action tab › Style (`ctaStyle`)	select	No	No	“” (Variation default)	Call to action style	Keep Variation default	Options “”, primary, secondary, tertiary	“”
Items tab › Numbers (`numbers`)	select	No	No	“” (Variation default)	Item numbering	Keep Variation default	Options “”, show, mobile, hide	show
Items tab › Item link (`itemAction`)	select	No	No	“” (Variation default)	Item link style	Keep Variation default	Options “”, arrow, arrow-up-right, button	“”
Items tab › Button text (`itemButtonLabel`)	text	No	No	Explore	Item button text	A verb, 1 to 2 words	Shown when itemAction = button, or itemAction empty and variation = solutions	Explore
Items tab › Motion Type (`motion`)	select	No	No	none	Scroll animation	None unless the page design asks for it	Options none, fade, slide, reveal	fade
Additional tab › Anchor ID (`classes_anchor`)	text	No	No	(empty)	Block id for links	anchor-slug	Regex `^anchor-[a-z0-9][a-z0-9-]*$` (Enforced)	anchor-products
Additional tab › Hide block (`classes_hideblock`)	boolean	No	No	false	Hide the block	Temporary use	None	false
Additional tab › Padding left (`classes_paddingleft`)	select	No	No	Default (“”)	Left padding	Design needs only	15 options (see 6.17.6 Spacing values)	pl-0
Additional tab › Padding right (`classes_paddingright`)	select	No	No	Default (“”)	Right padding	As above	15 options	pr-0

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Additional tab › Padding top (`classes_paddingtop`)	select	No	No	Default (“”)	Top padding	As above	15 options	pt-800
Additional tab › Padding bottom (`classes_paddingbottom`)	select	No	No	Default (“”)	Bottom padding	As above	15 options	pb-800
Additional tab › Margin left (`classes_marginleft`)	select	No	No	Default (“”)	Left margin	As above	15 options	ml-0
Additional tab › Margin right (`classes_marginright`)	select	No	No	Default (“”)	Right margin	As above	15 options	mr-0
Additional tab › Margin top (`classes_margintop`)	select	No	No	Default (“”)	Top margin	As above	15 options	mt-1200
Additional tab › Margin bottom (`classes_marginbottom`)	select	No	No	Default (“”)	Bottom margin	As above	15 options	mb-1200

Table List Item model (`table-list-card`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Number (`number`)	text	No	No	(empty)	Own item number	Leave empty for automatic numbering	Used only when 1 to 3 digits, optionally with . or)	01
Title (`title`)	text	No	No	TwinX	Item name	Short name, no trademark sign	None	TwinX
Trademark (`trademark`)	select	No	No	“” (None)	Sign after the title	Use for registered names	Options “”, tm, r	tm
Description (`description`)	richtext	No	No	Risk free business experimentation and decision intelligence	Item explanation	One sentence	None	Risk free business experimentation and decision intelligence
Link (`link`)	aem-content	No	No	(empty)	Item destination	Pick a page	Must start with /, #, http:// or https://	/content/tcs-corp/us/en/products/twinx
Button text (`linkText`)	text	No	No	(empty)	Item button text override	Only for button item links	None	Discover TwinX

Table List Header model (`table-list-header`, legacy)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Heading (`heading`)	text	No	No	Our transformation solutions	Legacy heading	Move the text to the block's Heading	None	Our transformation solutions
Explore All Link (`exploreLink`)	rich-text	No	No	(empty)	Legacy heading link	The block's Call to action takes precedence	None	a link "Explore all"

> **Note:** The Table List Header cannot be added any more (the block's filter only allows Table List Item). Existing headers are still read: their heading is used when the block Heading is empty, their link when the Call to action Link is empty. Prefer the block fields.

Field details

- **Variation** — Purpose: layout and defaults. Allowed Values: Variation 1 - Title, description, arrow (`industries`); Variation 2 - Description, large title, button (`solutions`); Variation 3 - Description, number, large title, arrow (`products`). Character Limits: not applicable. Business Rules: sets the block class and the defaults of Style, Numbers and Item link. Dependency Rules: Button text visibility. Visibility Rules: always. Sample Content: Variation 1.
- **Heading** — Purpose: list heading. Allowed Values: plain text. Character Limits: Recommended 50 characters. Business Rules: rendered as `h2`, `h3` or `h4.table-list-heading` with the text only. Dependency Rules: when set, item titles are one level below it; without a heading (and no legacy header) item titles are `h2`. Visibility Rules: always. Sample Content: Industries.
- **Heading Type** — Purpose: heading level. Allowed Values: H2, H3, H4. Business Rules: the code reads the level from the heading element AEM renders for the Heading; anything other than h2 to h4 gives h2. Visibility Rules: always. Sample Content: H2.
- **Maximum items** — Purpose: limit. Allowed Values: 1 to 10 (Enforced by the model validation and clamped by code). Business Rules: only the first N items (in content tree order) are rendered; the others are removed from the page. Visibility Rules: always. Sample Content: 4.
- **Text** (call to action) — Purpose: link text. Character Limits: Recommended 30 characters. Business Rules: used with the Link; empty gives "View all". Without a Link nothing shows on the page (an editor preview only). Visibility Rules: always. Sample Content: View all industries.
- **Link** (call to action) — Purpose: destination. Allowed Values: site pages (AEM paths are converted to published paths), `#anchors`, `http(s)` URLs. Business Rules: other schemes (mailto:, tel:) and text without `/` or `http` are ignored. Visibility Rules: always. Sample Content: /content/tcs-corp/us/en/industries.
- **Style** — Purpose: call-to-action style. Allowed Values: Variation default, Filled button, Outline button, Text link. Business Rules: Variation default = primary / secondary / tertiary for Variation 1 / 2 / 3. Visibility Rules: always. Sample Content: Variation default.
- **Numbers** — Purpose: numbering. Allowed Values: Variation default, Show, Show on mobile only, Hide. Business Rules: Variation default = mobile / hide / show for Variation 1 / 2 / 3. Visibility Rules: always. Sample Content: Show.
- **Item link** — Purpose: link style. Allowed Values: Variation default, Arrow, Diagonal arrow, Button. Business Rules: Variation default = arrow / button / arrow-up-right. Dependency Rules: Button controls the Button text field. Visibility Rules: always. Sample Content: Variation default.
- **Button text** (block) — Purpose: default item button text. Character Limits: Recommended 15 characters. Business Rules: used when an item has no Button text; empty gives "Explore". Visibility Rules: condition

`itemAction === "button" OR (itemAction empty AND variations === "solutions")`. Sample Content: Explore.

- **Motion Type** — Purpose: entrance animation. Allowed Values: None, Fade, Slide, Reveal. Business Rules: not in edit mode, not with reduced motion, only with IntersectionObserver. Visibility Rules: always. Sample Content: None.
- **Anchor ID, Hide block, Padding and Margin selects** — as in 6.17.6 (same model fields, same values and rules).
- **Number** (item) — Purpose: custom number. Allowed Values: 1 to 3 digits, optionally followed by "." or ")". Business Rules: other text is not recognised as a number (it may be read as another field); empty gives the position, zero-padded (01). Visibility Rules: always. Sample Content: 07.
- **Title** (item) — Purpose: item name. Character Limits: Recommended 30 characters for Variations 2 and 3 (very large type), 40 for Variation 1. Business Rules: text only (formatting is not kept); heading one level below the block heading. Sample Content: Banking.
- **Trademark** (item) — Purpose: legal sign. Allowed Values: None, ™ Trademark (`tm`), ® Registered (`r`). Business Rules: appended as `<sup>`. Sample Content: ™.
- **Description** (item) — Purpose: explanation. Allowed Values: rich text (paragraphs, lists, links). Character Limits: Recommended 120 characters. Business Rules: paragraphs and lists are kept; plain text is wrapped in a paragraph; links inside stay clickable above the row link. Sample Content: "Risk free business experimentation and decision intelligence".
- **Link** (item) — Purpose: destination. Allowed Values: as the call-to-action Link. Business Rules: without a Link, an arrow is decorative and a button is not rendered on the page. Sample Content: /content/tcs-corp/us/en/industries/banking.
- **Button text** (item) — Purpose: per-item button text. Character Limits: Recommended 20 characters. Business Rules: only used for Button item links; AEM delivers it as the Link's text, and a text equal to the URL is ignored. Sample Content: Discover TwinX.

6.18.7 Field Annotation Mapping

Marker	Field	Purpose
A	Content tab › Heading, Heading Type	List heading
B	Call to action tab › Text, Link, Style	Link next to the heading
C	Table List Item › Number; Items tab › Numbers	Item number
D	Table List Item › Title, Trademark	Item name
E	Table List Item › Description	Item explanation
F	Table List Item › Link, Button text; Items tab › Item link, Button text	Item link
G	Content tree items; Content tab › Maximum items	Number and order of items

6.18.8 Component Specification

Attribute	Value
Component Name	Table List (`table-list`; Universal Editor components "Table List", "Table List Item" (`table-list-card`) and legacy "Table List Header" (`table-list-header`))
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`decorators/button-decorator.js`; `utils/buttons.js`, `cells.js`, `properties.js`, `urls.js`, `environment.js`, `viewport.js`, `placeholders.js`; `scripts/scripts.js`

Attribute	Value
	(`moveInstrumentation`); `icons/arrow-right.svg`, `icons/arrow-up-right.svg`; `/{locale}/placeholders.json`
Component Type	Container block (key-value properties plus Table List Item children; filter `table-list` allows only Table List Item)
Supported Variants	Variation 1 (industries), Variation 2 (solutions), Variation 3 (products); call-to-action style primary / secondary / tertiary; numbers show / mobile / hide; item link arrow / arrow-up-right / button; motion none / fade / slide / reveal; item trademark ™ / ®
Supported Templates	Any page (allowed in every section)
Supported Metadata	None (the block does not read page metadata)
Supported Tags	None
Supported Personalization	None; clicks are in the Adobe Client Data Layer for future use with Adobe Target (not implemented)
Caching Considerations	Authored content is part of the page HTML cached by the Edge Delivery CDN; the only fetch is the placeholders sheet (once per page, shared by all blocks)
Performance Considerations	No images; light DOM work. Decoration waits for the placeholders sheet. Motion uses one IntersectionObserver per item (disconnected after the first view). Items beyond Maximum items are removed. Suitable for the first section
Accessibility Compliance	`role="list"` / `role="listitem"`; item titles as headings one level below the list heading; arrows with `aria-label` "Explore {title}"; buttons with "{text}: {title}"; unlinked arrows `aria-hidden`; visible focus; description links remain operable; motion honours `prefers-reduced-motion` and is off in the editor
SEO Compliance	Server-delivered authored content (headings, text, links are in the HTML before decoration); heading hierarchy follows Heading Type; internal links converted to published paths

6.18.9 Input Model

Content Inputs: Heading, call-to-action text and link, and per item Number, Title, Trademark, Description, Link and Button text.

Metadata Inputs: None.

Tags Inputs: None.

Page Context Inputs: Universal Editor edit mode (unlinked button previews, no motion), the visitor's reduced-motion preference, the site's locale mapping for AEM paths (via `utils/urls.js` `toSitePath`).

Query Inputs: None.

Manual Inputs: All content is manual.

Authored HTML received from AEM (simplified):

```
<div class="table-list">
  <div><div>variations</div><div>products</div></div>
  <div><div>listHeading</div><div><h2>Products &amp;amp; Platforms</h2></div></div>
  <div><div>maxCards</div><div>4</div></div>
  <div><div>viewAllLabel</div><div>All Products &amp;amp; Platforms</div></div>
  <div><div>viewAllLink</div><div><a href="/content/tcs-corp/us/en/products.html">
    /content/tcs-corp/us/en/products.html</a></div></div>
  <div><div>itemButtonLabel</div><div>Explore</div></div>
  <div><div>motion</div><div>fade</div></div>
  <div>
    <div></div>
    <div>TwinX</div>
```

```
    <div>tm</div>
    <div><p>Risk free business experimentation and decision intelligence</p></div>
    <div><a href="/content/tcs-corp/us/en/products/twinx.html">Discover TwinX</a></div>
  </div>
</div>
```

How rows are read:

- In Universal Editor, rows carry `data-aue-model`: rows of the `table-list` model are properties, `table-list-header` rows the legacy header and `table-list-card` rows items.
- On the published page: a row with two cells whose first cell is a known property name (`variations`, `listheading`, `maxcards`, `viewalllabel`, `viewalllink`, `ctastyle`, `numbers`, `itemaction`, `itembuttonlabel`, `motion`, compared without case and hyphens) is a property; a row with more than two cells is an item; a row with exactly two cells that is not a property is read as a legacy header (heading | link).
- When no property row is keyed, single-cell rows are matched to the fields in model order by what their values look like (`utils/cells.js matchCells`).
- Item cells: number, title, trademark, description, link (the Button text is the link's text). When AEM leaves out never-set fields, cells are matched by content (a 1 to 3 digit number, a tm / r value, a link).

6.18.10 Output Model

Rendered Output

- `div.table-list.block.{variation}.numbers-{value}.motion-{value}` (plus `motion-ready` when animating and Additional tab classes) containing:
- `div.table-list-header`: `h2|h3|h4.table-list-heading` and `p.table-list-cta > a.button.{primary|secondary|tertiary}` (each only when present; the header is omitted when empty).
- `div.table-list-items[role=list]` with `div.table-list-item[role=listitem]` (the authored row element, kept for editor instrumentation; `is-visible` after the motion): `div.table-list-main` (`span.table-list-number`, `hN.table-list-title` with `sup.table-list-trademark`), `div.table-list-description`, `div.table-list-action` (`a.table-list-arrow.{arrow|arrow-up-right}`, `span.table-list-arrow` or `a.button.secondary`).

Generated HTML (simplified)

```
<div class="table-list block products numbers-show motion-fade motion-ready">
  <div class="table-list-header">
    <h2 class="table-list-heading">Products &amp;amp; Platforms</h2>
    <p class="table-list-cta"><a href="/en/products" class="button tertiary">
      All Products &amp;amp; Platforms</a></p>
  </div>
  <div class="table-list-items" role="list">
    <div class="table-list-item is-visible" role="listitem">
      <div class="table-list-main">
        <span class="table-list-number">01</span>
        <h3 class="table-list-title">TwinX<sup class="table-list-trademark">™</sup></h3>
      </div>
      <div class="table-list-description"><p>Risk free business experimentation ...</p></
div>
      <div class="table-list-action">
        <a class="table-list-arrow arrow-up-right" href="/en/products/twinx"
          aria-label="Explore TwinX"></a>
      </div>
    </div>
```

```
  </div>
</div>
```

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking

Component state (after the lazy phase):

```
{
  "component": {
    "table-list-1": {
      "@type": "tcs-corp/components/table-list",
      "parentId": "page-1x2y3z",
      "dc:title": "Products & Platforms"
    }
  }
}
```

Interaction	Event	eventInfo
Item arrow	`cmp:click`	`interaction: "link"`, `dc:title`: "Explore {title}", `xdm:linkURL`, `xdm:linkType` (`other` for site pages, `exit` for other origins, `download` for files), `breakpoint`
Item button	`cmp:click`	`interaction: "link"`, `dc:title`: "{text}: {title}", `xdm:linkURL`, `xdm:linkType`, `breakpoint`
Call to action	`cmp:click`	`interaction: "link"`, `dc:title`: the link text, `xdm:linkURL`, `xdm:linkType`, `breakpoint`
Link inside a description	`cmp:click`	`interaction: "link"`, as above

Example:

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.table-list-1",
    "interaction": "link",
    "dc:title": "Explore TwinX",
    "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/products/twinx",
    "xdm:linkType": "other",
    "breakpoint": "desktop"
  }
}
```

The component id is the Anchor ID if set, else `table-list-{n}`. Items have no component id of their own.

Personalization Output: None.

Search Index Output: None specific; the page's text (including the list) is part of the page, but `helix-query.yaml` indexes only head metadata.

6.18.11 Business Rules

1. Variation: `industries`, `solutions` or `products`; any other value gives `industries`.

2. Variation defaults: industries = primary call to action, numbers on mobile only, arrow; solutions = secondary, hidden, button; products = tertiary, shown, diagonal arrow. An explicit Style, Numbers or Item link overrides them.
3. Maximum items: whole number clamped to 1 to 10; missing or not a number gives 4; only the first items in order are rendered, the rest are removed.
4. Heading: rendered with the level of the authored heading element if it is h2, h3 or h4, else h2; only its text is kept.
5. Item title level: one below the block heading when a heading (block or legacy header) exists; `h2` otherwise.
6. Legacy header: used for the heading when the block Heading is empty and for the call to action when the block Link (and Text) are empty.
7. Call to action: rendered when there is a Link or Text; on the page only when there is a Link; text defaults to "View all".
8. Link values: `#anchor` (not `#` alone), `/path` or `http(s)://...`; AEM content paths are converted to published paths (`/content/tcs-corp/us/en/x.html` to `/en/x`); other values are ignored.
9. Link text: a link's text is used only when it differs from its URL and does not look like a path or URL.
10. Without a Link, a button is shown only in the editor (non-focusable, `aria-hidden`, click prevented); an arrow becomes a decorative `span`.
11. Numbers: Hide renders none; Show and Show on mobile only render the item Number or the zero-padded position; mobile-only numbers are hidden from 600px by CSS.
12. Trademark: `tm` gives ™, `r` gives ®, appended to the title.
13. Description: rich text is kept when it has paragraphs or lists; plain text becomes one paragraph; missing description renders nothing.
14. Button item links: text = item Button text, else block Button text, else "Explore"; accessible name "{text}: {title}".
15. Arrow item links: accessible name "Explore {title}"; the arrow link covers the whole item; description links stay clickable above it.
16. Motion: only for fade / slide / reveal, with IntersectionObserver, outside the editor's edit mode and without reduced motion; items are hidden until 15% visible, then animate once for 500ms.
17. A row with two cells whose first cell is not a property name, without editor instrumentation, is read as the legacy header, not as an item (see 6.18.13 Issue 6).
18. The authored item rows are reused (classes and children replaced), so Universal Editor selection keeps working.
19. All texts in the UI come from placeholders with English fallbacks.

6.18.12 Author Best Practices

Recommended Content Length

Element	Variation 1	Variation 2	Variation 3	Enforced
Heading	50 characters	50 characters	50 characters	No
Item title	40 characters	25 characters (4rem type on desktop)	15 characters (5.625rem type on desktop)	No
Item description	120 characters	120 characters	160 characters	No
Button text	not used	15 characters	not used	No
Items	3 to 8	3 to 6	3 to 6	1 to 10 (Enforced)

Image Dimensions: Not applicable; the block has no images. Arrows are SVG icons from `icons/`.

CTA Writing Guidelines

- Make the heading call to action specific ("View all industries", "All Products & Platforms"), not just "View all".
- Item buttons: a short verb ("Explore", "Discover"); the title is added to the accessible name automatically.
- Link every item; unlinked arrows and buttons give no destination.

Accessibility Guidelines

- Choose Heading Type so the page outline has no skipped levels.
- Do not type ™ or ® into the title; use Trademark.
- Keep descriptions short and meaningful without the title.
- Use Motion sparingly; it is disabled automatically for visitors who prefer reduced motion.

SEO Guidelines

- Item titles are headings; use real product or industry names.
- Link to canonical published pages (pick pages with the page picker).

Metadata Guidelines: Not applicable (the block uses no page metadata).

Taxonomy Guidelines: Not applicable.

Localization Guidelines

- Translate the Heading, call-to-action Text, Button texts, titles and descriptions per locale.
- Link to pages of the same locale.
- The placeholder keys `viewAll`, `explore`, `itemLinkLabel` and `exploreItem` must be translated in each locale's placeholders sheet (they include the `{text}` / `{title}` variables).
- Expect longer titles in German, French and other languages; check Variation 2 and 3 sizes.

6.18.13 Troubleshooting Guide

Issue 1 — Some items are missing

Symptoms: Fewer items on the page than in the content tree.

Root Cause: Maximum items (default 4) is lower than the number of items.

Diagnosis Steps: Check Content tab › Maximum items.

Resolution Steps: Raise Maximum items (up to 10) or remove extra items.

Validation Steps: All items appear.

Preventive Measures: Set Maximum items when adding items.

Issue 2 — The call to action does not appear

Symptoms: The heading shows without its button or link on the page (it may show in the editor).

Root Cause: The Link is empty or not a supported value (e.g. `mailto:` or text without `/` or `http`).

Diagnosis Steps: Check Call to action › Link.

Resolution Steps: Pick a page or enter a full URL.

Validation Steps: The call to action renders and opens the destination.

Preventive Measures: Always set Link with Text.

Issue 3 — Item buttons or arrows do nothing

Symptoms: No button on an item (Button link) or an arrow that is not clickable.

Root Cause: The item's Link is empty or unsupported.

Diagnosis Steps: Select the item and check Link.

Resolution Steps: Set the Link.

Validation Steps: The arrow or button navigates.

Preventive Measures: Validation checklist: every item has a Link.

Issue 4 — Wrong button text

Symptoms: Buttons show "Explore" or the URL instead of the expected text.

Root Cause: Item Button text empty (block Button text or "Explore" used); a Button text equal to the URL is ignored; Item link is not Button.

Diagnosis Steps: Check the item Button text, block Button text and Item link.

Resolution Steps: Set the texts; set Item link to Button.

Validation Steps: The expected text appears.

Preventive Measures: Use the block Button text for a common label.

Issue 5 — Heading Type has no effect

Symptoms: The heading stays `h2` although H3 or H4 was chosen.

Root Cause: The level is read from the heading element AEM renders for the Heading field; if the Heading is delivered as plain text (no heading element), `h2` is used.

Diagnosis Steps: Inspect `https://main--tcs-corp--sradha-tcs.aem.page/{path}.plain.html` and check whether the listHeading cell contains `<h3>` / `<h4>`.

Resolution Steps: Re-select Heading Type and re-enter the Heading; if the markup still has no heading element, escalate to L3.

Validation Steps: The rendered heading has the chosen level and item titles are one level lower.

Preventive Measures: Check heading levels in the validation checklist.

Issue 6 — An item visible in the editor is missing on the published page

Symptoms: An item shows in Universal Editor but not on preview / live, and the heading or call to action may change.

Root Cause: On the published page, a row with exactly two cells that is not a property row is read as the legacy Table List Header (code rule). This can happen if an item's markup arrives with only two cells. [To be confirmed: whether AEM delivers item rows with fewer than five cells — owner: development team.]

Diagnosis Steps: Inspect the `.plain.html` of the page and count the cells of the item's row.

Resolution Steps: Fill the item's other fields (e.g. Number or Link) so all cells are delivered; escalate to L3 for a code fix.

Validation Steps: The item renders on the preview page.

Preventive Measures: Give every item a Link (needed anyway).

Issue 7 — Items invisible until scrolled, or never shown

Symptoms: Items appear late, or stay invisible in a screenshot tool.

Root Cause: Motion is on; items are hidden until 15% of them enters the viewport. Tools that capture without scrolling may not trigger it.

Diagnosis Steps: Check Motion Type; scroll the page.

Resolution Steps: Set Motion Type to None if needed.

Validation Steps: Items are visible without scrolling.

Preventive Measures: Use motion only where the design asks for it.

Issue 8 — Unexpected numbers

Symptoms: Numbers missing at tablet / desktop, or custom numbers ignored.

Root Cause: Numbers is Show on mobile only (Variation 1 default) or Hide (Variation 2 default); item Numbers that are not 1 to 3 digits are not used as numbers.

Diagnosis Steps: Check Items › Numbers and the item Number values.

Resolution Steps: Set Numbers to Show; use numeric values like `01`.

Validation Steps: Numbers appear as intended at each width.

Preventive Measures: Leave Number empty for automatic numbering.

6.18.14 Support Runbook

Monitoring Requirements: No block-specific monitoring; general page availability and Core Web Vitals (RUM). Link checks for the item destinations ([To be confirmed: link checking tool — owner: AMS]).

Logs

- Browser console: the block emits no console messages.
- Admin API: page status `GET https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`.
- AEM author: page versions and publish history.

Common Errors

Error	Likely cause	First action
Items missing	Maximum items	Raise the limit
No call to action	Empty or unsupported Link	Set Link
No item button	Empty Link	Set Link
Heading level wrong	Heading markup without heading element	Check `.plain.html`; L3
English text on a localized page	Placeholder keys missing	Add keys to the locale's placeholders
Item missing only on the published page	Two-cell row read as legacy header	Fill the item's fields; L3

Escalation Matrix

Level	Who	What they check
L1	Author support	Dialog settings, item fields, links, Maximum items, publish state
L2	AMS / site admin	Delivered markup (`.plain.html`), placeholders sheet, CDN cache, page status
L3	Development team	`table-list.js` row reading, CSS per variation, defects after releases
L4	Adobe Support	Universal Editor or Edge Delivery platform issues

Recovery Procedures: Restore a previous page version and publish; hide the block with Hide block; revert code on `main` for code defects.

Health Checks: The list renders with the expected number of items; arrows and buttons navigate; the call to action opens its destination; layouts at 375px, 768px and 1440px match the variation.

Operational Procedures: Maintain the placeholder keys in each locale; review product names and trademarks with the legal team ([To be confirmed: trademark approval process — owner: brand / legal]).

Known Limitations

- At most 10 items.
- Titles and headings are plain text (formatting is dropped).
- Only `/`, `#` and `http(s)` links are supported (no `mailto:` / `tel:`).
- No images.
- The legacy header item cannot be added any more but is still read.
- Motion is not previewed in the editor's edit mode.

6.19 Anchor List Block

The Anchor List is an in-page chapter navigation ("Chapter Nav" in the TCS DS Figma, node 404-15450). It renders a horizontal row of links; each link scrolls to the section whose **Anchor name** (Section properties) matches, or to any element with that id (a block's Anchor ID, a heading id). The link of the chapter currently in view is marked as the current location, and with **Sticky** the row stays pinned to the top of the screen while the visitor scrolls through the page.

Source files: `blocks/anchor-list/anchor-list.js` (181 lines), `blocks/anchor-list/anchor-list.css` (80 lines), `blocks/anchor-list/_anchor-list.json`. The dialog has 2 tabs besides Additional: Content (the Links multifield with Link Title and Anchor Name, and the Sticky toggle) and the common Additional tab.

6.19.1 Component Purpose

Business Objective

Help visitors navigate long pages (reports, solution pages, case studies, annual-report style articles) by showing the page's chapters, jumping straight to one, and always showing where they are.

Business Use Cases

- A long report page with chapters "CH 1: Overview", "CH 2: Findings", "CH 3: Outlook" below the page header, pinned while reading.
- A service page with "Overview", "Capabilities", "Case studies", "Contact" links to its sections.
- A landing page that links to blocks by their Anchor IDs.

When To Use

- Pages with three or more substantial sections.
- When visitors benefit from knowing their position (current chapter highlight).

When Not To Use

- Navigation between pages: use the header navigation, the List or Teaser blocks or links.
- Switching content in place without scrolling: use the Tabs block.
- Short pages with one or two sections.
- More than about seven links (they overflow and scroll sideways, see 6.19.13).

Dependencies

Kind	Dependency	Used for
Block code	`blocks/anchor-list/anchor-list.js`, `anchor-list.css`, `_anchor-list.json`	Decoration, styles, dialog
Section model	`models/_section.json` › Anchor name (`anchorName`)	The ids the links jump to (applied in `scripts/scripts.js` `decorateSectionsWithOptions`)
Decorator	`decorators/block-options-decorator.js`	Block Anchor IDs (`anchor-...`) that links can also target

Kind	Dependency	Used for
Utility	`utils/multifield.js` (`readMultifieldItems`)	Reading the Links multifield
Utility	`utils/properties.js` (`readProperties`, `isOn`)	Reading the properties
Utility	`utils/environment.js` (`isEditing`, `scrollBehavior`)	No pinning while editing; smooth or instant scrolling (reduced motion)
Utility	`utils/placeholders.js` (`getPlaceholders`)	The navigation's accessible name
Core	`scripts/aem.js` (`toClassName`), `scripts/scripts.js` (`moveInstrumentation`)	Anchor normalization; editor instrumentation on the list
Header	`blocks/header` (`header .nav-wrapper`)	Offset below the fixed header (mobile and tablet below 900px)
Data	`/{locale}/placeholders.json`	Key `onThisPage` (fallback "On this page")

Related Components

Sections (Anchor name), the Additional tab Anchor ID of every block, Tabs (in-place switching), Page Header (usually directly above), Table List (6.18) and Filter List (6.17) as typical chapter targets.

Applicable Templates

Any page (allowed in every section); most useful on article, blog, press-release and long landing pages.

Supported Content Types

Links to sections, blocks or headings on the same page. Links to other pages are not supported (every link is `#anchor`).

6.19.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A centred row of four chapter links at 1440px below the page header: grey regular text, the current chapter "CH 2: Findings" dark, semibold, with a 3px brand-blue underline.

Desktop (1200px and wider). A centred row of links (the list is as wide as its links, at most the full width). Each link has 0.75rem vertical and 2.5rem horizontal padding (from 900px), `--body-font-size-m` text, line-height 1.61, in `--color-text-secondary`, with a transparent 3px (0.1875rem) bottom border. The current link (`aria-current="location"`) is `--color-text-default`, semibold, with a `--color-border-brand` underline. Hover and keyboard focus turn the text dark; focus also shows a 2px brand outline inset by 3px. Colour and border changes transition over 0.2s (none with reduced motion). With Sticky the row has the default background colour and, once scrolled to, is fixed at the top of the screen (the site header is not fixed from 900px, so the row pins at the very top).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The Anchor List at 768px pinned directly below the fixed site header, the current link underlined, the row scrolled sideways to keep the current link centred.

Tablet (600px to 1199px). Same design. From 900px the horizontal link padding is 2.5rem; below 900px it is 1.5rem. Below 900px the site header is fixed, so a sticky Anchor List pins directly below the header and clicked chapters scroll to just below both.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Anchor List at 375px: the links in one line overflowing to the right, scrollable sideways without a visible scrollbar, the current link brought into view.

Mobile (below 600px). Links never wrap (`white-space: nowrap`); when they do not fit, the row scrolls horizontally (scrollbar hidden) and the current link is scrolled to the centre of the row automatically (smoothly, or instantly with reduced motion).

6.19.3 Component Variations

Variation 1 — Default (in place)

Variation Name: Default (Content tab › Sticky = off; the default).

Business Purpose: A chapter index at its place in the page, typically below the page header.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-default.png`
> **Purpose:** Default Anchor List
> **Description:** The row of chapter links in the page flow below the page header; after scrolling down it is no longer visible.

Use Cases: Medium-length pages, a table of contents at the top of an article.

Authoring Guidelines: Add the links (Link Title and Anchor Name) and leave Sticky off. Set the Anchor name of each target section in its Section properties.

Business Rules: Clicking a link scrolls to its target with 16px space above it (plus the fixed header height below 900px); the current-chapter highlight still follows the scroll position (useful when the list is visible, e.g. at the top).

Rendering Rules: `div.anchor-list` (no `sticky` class) containing `nav.anchor-list-nav > ul.anchor-list-items > li > a.anchor-list-link`.

Responsive Behaviour: Desktop and tablet (900px+): links padded 2.5rem; below 900px 1.5rem; on narrow screens the row scrolls sideways.

SEO Impact: Same-page fragment links (`#overview`); they do not add crawlable pages but expose the page structure.

Accessibility Compliance: A `<nav>` landmark labelled "On this page" (placeholder `onThisPage`); `aria-current="location"` on the current link; after a click keyboard focus moves to the target section (made focusable with `tabindex="-1"`); smooth scrolling is replaced by instant scrolling with reduced motion (WCAG 2.4.1, 2.4.3, 2.4.8 intent, 2.3.3 intent).

Variation 2 — Sticky

Variation Name: Sticky (Content tab › Sticky = on).

Business Purpose: Keeps the chapters available and the reading position visible throughout a long page.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-sticky.png`
> **Purpose:** Sticky Anchor List
> **Description:** The page scrolled into chapter 3: the Anchor List fixed at the top of the screen across the full width with a white background, "CH 3: Outlook" underlined.

Use Cases: Reports, long solution pages, case studies.

Authoring Guidelines: Turn Sticky on. Place the block above the first chapter section.

Business Rules: Once the block's place scrolls up to the top of the screen (or to the bottom of the fixed header below 900px), the block is pinned for the rest of the page and released when the visitor scrolls back above its place. The block's wrapper keeps the block's height while it is pinned, so the page does not jump. Pinning is disabled while editing in Universal Editor (it follows the edit / preview switch). Clicked chapters scroll to just below the pinned list (header height + list height + 16px).

Rendering Rules: Class `sticky` (background `--color-bg-default`); when pinned the class `anchor-list-pinned` adds `position: fixed`, `left: 0`, `right: 0`, `z-index: 10`, the section's inline padding (`--section-padding-inline`) and an inline `top` equal to the header height; the wrapper gets an inline `min-height`. A `position: sticky` element is deliberately not used, because it would only stay within its own section.

Responsive Behaviour: Desktop (900px+): pinned at the top of the screen (the header scrolls away). Tablet below 900px and mobile: pinned below the fixed header.

SEO Impact: None beyond Variation 1.

Accessibility Compliance: As Variation 1. The pinned list covers the top of the content, which is compensated by the scroll margin on targets; [To be confirmed: WCAG 2.2 "Focus Not Obscured" behaviour when tabbing through content under the pinned list — owner: development team].

6.19.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A page in Universal Editor with the Anchor List selected below the page header showing four links, and a section selected further down with its Anchor name visible in the section properties.

Annotations

- A = Link Title of each link (Links multifield › Link Title), the visible text.
- B = Current-location underline (automatic; on the canvas the first link, or the chapter in view).
- C = Anchor Name (Links multifield › Anchor Name), not visible on the canvas; it is the `#anchor` of the link.
- D = Sticky (Content tab), visible only on preview and live pages (not pinned in edit mode).
- E = The target section's Anchor name (Section properties), which must match C.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Anchor List properties panel: the Content tab with the Links multifield (two items with Link Title and Anchor Name) and the Sticky toggle, and the Additional tab.

Tabs:

- **Content:** Links (composite multifield; each item: Link Title, Anchor Name), Sticky (toggle, off).
- **Additional:** Anchor ID, Hide block, Padding left / right / top / bottom, Margin left / right / top / bottom.

6.19.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites select the page and choose **Edit**.

Expected Result: The page opens in Universal Editor.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** The Sites console with the page selected and Edit highlighted.

Step 2 — Name the target sections

Action: Select each chapter section, open its properties and enter an **Anchor name** (lower case, no spaces, e.g. `overview`). To target a block instead, set the block's Additional › Anchor ID (e.g. `anchor-findings`).

Expected Result: Each section has a unique anchor name.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-step-02-section-anchor.png`
> **Purpose:** Section Anchor name
> **Description:** The section properties panel with Anchor name "overview".

Step 3 — Add the Anchor List

Action: Select the section above the chapters (e.g. the one with the page header), choose **Add** (+) and select **Anchor List**.

Expected Result: An empty Anchor List is added (no links yet).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-step-03-add-block.png`
> **Purpose:** Add the block
> **Description:** The Add menu with Anchor List highlighted.

Step 4 — Add the links

Action: In the Content tab, add an item to Links for each chapter: Link Title (e.g. "CH 1: Overview") and Anchor Name (e.g. `overview`, with or without `#`). Order the items as the chapters appear.

Expected Result: The links appear on the canvas in order.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-step-04-links.png`
> **Purpose:** Links multifield
> **Description:** Three Links items filled with titles and anchor names.

Step 5 — Choose Sticky

Action: Turn on Sticky if the list should stay at the top while scrolling.

Expected Result: No change on the edit canvas (pinning is off while editing).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-step-05-sticky.png`
> **Purpose:** Sticky toggle
> **Description:** The Sticky toggle switched on.

Step 6 — Set the Additional options (optional)

Action: Set spacing if needed; an Anchor ID is rarely needed for this block.

Expected Result: Saved options.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-step-06-additional.png`
> **Purpose:** Additional tab
> **Description:** The Additional tab with Margin bottom 2rem.

Step 7 — Preview and test

Action: Switch to Preview mode (or open the `.aem.page` URL), click each link, scroll the page.

Expected Result: Each link scrolls to its chapter; the current link follows the scroll position; Sticky pins the list.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-step-07-preview.png`
> **Purpose:** Preview
> **Description:** The preview page scrolled to chapter 2 with the pinned list and “CH 2” underlined.

Step 8 — Validate

Action: Run the Validation Process checklist.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-step-08-validate.png`
> **Purpose:** Validate
> **Description:** The mobile view with the list below the fixed header.

Step 9 — Publish

Action: Publish the page.

Expected Result: The live page shows the working chapter navigation.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/anchor-list-step-09-publish.png`
> **Purpose:** Publish
> **Description:** The Publish dialog with the page selected.

Preview Process

1. Use Preview mode in Universal Editor to test clicks and the current-link highlight (Sticky pins in Preview mode too).
2. Preview the page and test on `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}`, including opening the URL with `#anchor` appended.

Publishing Process

1. Publish the page (the anchors are part of the same page).
2. Test the live URL with a hash, e.g. `/en/insights/report#findings`.

Validation Process

- ☐ Every link scrolls to the right chapter, with the chapter heading visible (not hidden under the header or the pinned list).
- ☐ The current link changes while scrolling and the last link is current at the end of the page.
- ☐ Opening the page with `#anchor` marks that link as current.
- ☐ Sticky pins below the header on mobile and at the top on desktop, without the page jumping.
- ☐ On mobile the row scrolls sideways and keeps the current link in view.
- ☐ Keyboard: Tab reaches each link; Enter moves focus to the chapter.
- ☐ Anchor names are unique on the page.

Rollback Process

1. Undo in Universal Editor or restore a page version, then publish.
2. Turn on Hide block to remove the list temporarily.
3. Code defects: revert on `main`.

6.19.6 Authorable Field Specification

Anchor List block model (`anchor-list`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Content tab › Links (`anchors`)	container (composite multi-field)	No	Yes	(none)	The chapter links	One item per chapter, in page order	Items without Link Title and Anchor Name are ignored	see item table
Content tab › Sticky (`sticky`)	boolean	No	No	false	Pins the list while scrolling	On for long pages	None	true

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Additional tab › Anchor ID (`classes_anchor`)	text	No	No	(empty)	Block id	Rarely needed	Regex `^anchor-[a-z0-9][a-z0-9-]*$` (Enforced)	anchor-chapters
Additional tab › Hide block (`classes_hideblock`)	boolean	No	No	false	Hide the block	Temporary use	None	false
Additional tab › Padding left (`classes_paddingleft`)	select	No	No	Default ("")	Left padding	Design needs only	15 options (see 6.17.6 Spacing values)	pl-0
Additional tab › Padding right (`classes_paddingright`)	select	No	No	Default ("")	Right padding	As above	15 options	pr-0
Additional tab › Padding top (`classes_paddingtop`)	select	No	No	Default ("")	Top padding	As above	15 options	pt-400
Additional tab › Padding bottom (`classes_paddingbottom`)	select	No	No	Default ("")	Bottom padding	As above	15 options	pb-400
Additional tab › Margin left (`classes_marginleft`)	select	No	No	Default ("")	Left margin	As above	15 options	ml-0
Additional tab › Margin right (`classes_marginright`)	select	No	No	Default ("")	Right margin	As above	15 options	mr-0
Additional tab › Margin top (`classes_margintop`)	select	No	No	Default ("")	Top margin	As above	15 options	mt-800
Additional tab › Margin bottom (`classes_marginbottom`)	select	No	No	Default ("")	Bottom margin	As above	15 options	mb-800

Links item (composite multifield `anchors`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Link Title (`label`)	text	No	No	(empty)	Visible link text	Short chapter name	None	CH 1: Overview
Anchor Name (`anchorName`)	text	No	No	(empty)	Target id	The section's Anchor name, a block Anchor ID or a heading id	Normalized (lower case, other characters to hyphens, leading # removed)	overview

Field details

- **Links** — Purpose: the chapters. Allowed Values: items of Link Title and Anchor Name. Character Limits: Recommended at most 7 items. Business Rules: items keep their order; an item is used when it has at least one non-empty value; if only one value is filled, it is used as both the text and the anchor. Dependency Rules: each Anchor Name must match an id on the page. Visibility Rules: always shown. Sample Content: three chapters.

- **Link Title** — Purpose: link text. Allowed Values: plain text. Character Limits: Recommended 25 characters (links do not wrap). Business Rules: empty Link Title with an Anchor Name shows the Anchor Name as the text. Sample Content: CH 2: Findings.
- **Anchor Name** — Purpose: the target. Allowed Values: text; normalized with `toClassName` (e.g. "Our Findings" becomes `our-findings`, `#overview` becomes `overview`). Character Limits: Recommended 30 characters. Business Rules: empty Anchor Name uses the normalized Link Title (e.g. "CH 1: Overview" becomes `ch-1-overview`). Dependency Rules: the section Anchor name is normalized the same way, so case and spaces do not matter as long as the words match. Sample Content: findings.
- **Sticky** — Purpose: pinning. Allowed Values: on / off. Business Rules: on only when the property text is `true`; adds the `sticky` class; pinning needs at least one link and is off in edit mode. Visibility Rules: always shown. Sample Content: on.
- **Anchor ID, Hide block, Padding and Margin selects** — as in 6.17.6. A block Anchor ID of another block is a valid Anchor Name target.

6.19.7 Field Annotation Mapping

Marker	Field	Purpose
A	Links › Link Title	Visible chapter link text
B	(automatic) current location	Shows the chapter in view
C	Links › Anchor Name	The `#anchor` the link jumps to
D	Content tab › Sticky	Pins the list while scrolling
E	Section properties › Anchor name (or a block's Anchor ID)	The id of the target

6.19.8 Component Specification

Attribute	Value
Component Name	Anchor List (`anchor-list`; resource type `core/franklin/components/block/v1/block`, model `anchor-list`, key-value block)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`utils/multifield.js`, `properties.js`, `environment.js`, `placeholders.js`; `scripts/aem.js`, `scripts/scripts.js`; section Anchor name (`models/_section.json`, `scripts/scripts.js`); block Anchor IDs (`decorators/block-options-decorator.js`); header `.nav-wrapper`
Component Type	Block (no child items; one composite multifield)
Supported Variants	Default; Sticky
Supported Templates	Any page
Supported Metadata	None
Supported Tags	None
Supported Personalization	None
Caching Considerations	Authored content in the page HTML (CDN cached); only the placeholders sheet is fetched (once per page)
Performance Considerations	Small DOM; scroll, resize and `ResizeObserver` (on `document.body`) handlers are throttled with `requestAnimationFrame`; passive scroll listeners. Two independent update loops (pinning and current link) when Sticky is on

Attribute	Value
Accessibility Compliance	<nav> with aria-label "On this page"; aria-current="location"; focus moved to the target after a click; visible focus outline; reduced motion gives instant scrolling and no transitions; links are real #anchor links
SEO Compliance	Fragment links within the page; no impact on indexing; section ids give stable deep links

6.19.9 Input Model

Content Inputs: The Links (Link Title, Anchor Name) and Sticky.

Metadata Inputs: None.

Tags Inputs: None.

Page Context Inputs: The ids on the page (section Anchor names, block Anchor IDs, heading ids), the site header's `.nav-wrapper` position and height, the URL hash, the scroll position and viewport height, the editor's edit-mode class on `<html>`, the reduced-motion preference.

Query Inputs: None.

Manual Inputs: All links are manual.

Authored HTML received from AEM (simplified):

```
<div class="anchor-list">
  <div><div>anchors</div><div>
    <p>CH 1: Overview</p><p>overview</p><hr>
    <p>CH 2: Findings</p><p>findings</p><hr>
    <p>CH 3: Outlook</p><p>outlook</p>
  </div></div>
  <div><div>sticky</div><div>true</div></div>
</div>
```

The target section (AEM delivers section properties as data attributes; `scripts.js` turns the Anchor name into the section id):

```
<div class="section" data-anchorname="overview">...</div>
<!-- after decoration -->
<div class="section" id="overview">...</div>
```

6.19.10 Output Model

Rendered Output

- `div.anchor-list.block` (plus `sticky`, `anchor-list-pinned` while pinned, Additional tab classes) containing `nav.anchor-list-nav[aria-label] > ul.anchor-list-items > li > a.anchor-list-link[href="#anchor"]` (one has `aria-current="location"`).

Generated HTML

```
<div class="anchor-list block sticky anchor-list-pinned" style="top: 64px;">
  <nav class="anchor-list-nav" aria-label="On this page">
    <ul class="anchor-list-items">
      <li><a class="anchor-list-link" href="#overview">CH 1: Overview</a></li>
      <li><a class="anchor-list-link" href="#findings"
        aria-current="location">CH 2: Findings</a></li>
      <li><a class="anchor-list-link" href="#outlook">CH 3: Outlook</a></li>
```

```
    </ul>
  </nav>
</div>
```

While pinned, the block's wrapper (`div.anchor-list-wrapper`) has an inline `min-height` equal to the block's height. After a click the target gets an inline `scroll-margin-top` and, if it had none, `tabindex="-1"`; the URL hash is replaced (`history.replaceState`, no new history entry).

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking

Component state: `{ "component": { "anchor-list-1": { "@type": "tcs-corp/components/anchor-list", "parentId": "page-..." } } }` (no `dc:title`, the block has no heading). Each link click:

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.anchor-list-1",
    "interaction": "link",
    "dc:title": "CH 2: Findings",
    "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/insights/report#
findings",
    "xdm:linkType": "other",
    "breakpoint": "mobile"
  }
}
```

Scroll-driven changes of the current link are not tracked.

Personalization Output: None.

Search Index Output: None.

6.19.11 Business Rules

1. Links are read from the composite multifield in order; each item's non-empty texts are [Link Title, Anchor Name].
2. An item with neither value is dropped; with one value, that value is the text and (normalized) the anchor.
3. Anchor names are normalized with `toClassName` after removing a leading `#`; section Anchor names are normalized the same way.
4. A section gets its Anchor name as id only if no element on the page has that id yet (the first one wins).
5. Any element id on the page is a valid target, including block Anchor IDs (`anchor-...`) and heading ids.
6. Click: the default jump is prevented; the link becomes current immediately; the target is scrolled into view (smooth, instant with reduced motion) with a top margin of header height (when the header is fixed or sticky, i.e. below 900px) + list height (when Sticky) + 16px; the URL hash is replaced; focus moves to the target without a second scroll.
7. A link whose target is not on the page (e.g. on the block library page) only becomes current.
8. While a clicked chapter scrolls into view, scroll updates do not change the current link, until the `scrollend` event or 3 seconds.
9. Current link on scroll: the last target whose top is at or above the offset; at the very end of the page (after scrolling) the last link; before any target the first link; hidden targets (still loading) are skipped.
10. Without any visible target on the page, the clicked link, else the first link, stays current.

11. Opening the page with a hash that matches a link, or a hash change, marks that link current (and locks for 3 seconds), even when the chapter is not on the page.
12. When the current link changes, the row scrolls sideways to centre it.
13. Sticky: on only for the text `true`; the block is pinned (fixed) once its wrapper's top reaches the header bottom (0 from 900px), with the wrapper keeping its height; unpinned above that; never pinned in Universal Editor edit mode; re-evaluated on scroll, resize, body size changes and editor mode switches.
14. With no links the block renders an empty navigation and adds no listeners (Sticky has no effect).
15. The navigation's accessible name comes from the placeholder `onThisPage` ("On this page").

6.19.12 Author Best Practices

Recommended Content Length

Element	Recommended	Enforced
Number of links	3 to 7	No
Link Title	Up to 25 characters	No
Anchor Name	1 to 3 words, lower case, hyphens	No (normalized by code)

Image Dimensions: Not applicable (no images).

CTA Writing Guidelines: Use the chapter names exactly as the section headings read (or a consistent "CH n: Name" pattern), so visitors recognise where they land.

Accessibility Guidelines

- Give each target section a visible heading that matches the link text.
- Keep the number of links small so they fit without sideways scrolling on most screens.
- Do not use the Anchor List as the only navigation to important content.

SEO Guidelines

- Stable, meaningful anchor names (`findings`, not `section-2`) make shareable deep links.
- Do not change anchor names after publishing pages that others link to.

Metadata Guidelines: Not applicable.

Taxonomy Guidelines: Not applicable.

Localization Guidelines

- Translate Link Titles per locale.
- Anchor names may stay in English across locales (they are not visible), or be translated consistently with the section Anchor names of that locale; they must match on each page.
- Translate the placeholder `onThisPage` in each locale.

6.19.13 Troubleshooting Guide

Issue 1 — A link does not scroll

Symptoms: Clicking a link only underlines it; the page does not move.

Root Cause: No element on the page has the link's (normalized) anchor: the section's Anchor name is missing, different, or used twice (the second section gets no id).

Diagnosis Steps: In the browser, run `document.getElementById('findings')` in the console; check the section's Anchor name and the link's Anchor Name.

Resolution Steps: Make the names match (case and spaces do not matter, the words do); make section Anchor names unique.

Validation Steps: The link scrolls to the section.

Preventive Measures: Set section Anchor names before adding the Anchor List.

Issue 2 — The chapter heading is hidden under the header or the list

Symptoms: After a click the start of the chapter is covered.

Root Cause: The offset is computed from the header `.nav-wrapper` (fixed below 900px) and the list height (Sticky); other fixed elements (e.g. a banner) are not included.

Diagnosis Steps: Check for other fixed elements; compare at 375px and 1440px.

Resolution Steps: Remove the extra fixed element or escalate to L3.

Validation Steps: The heading is visible below the list after a click.

Preventive Measures: Avoid additional fixed elements on pages with a sticky Anchor List.

Issue 3 — Sticky does not pin in the editor

Symptoms: The list scrolls away in Universal Editor.

Root Cause: Pinning is disabled in edit mode by design (so it does not cover the content being edited).

Diagnosis Steps: Switch to Preview mode.

Resolution Steps: Test in Preview mode or on the preview URL.

Validation Steps: The list pins in Preview mode.

Preventive Measures: Inform authors.

Issue 4 — The wrong link is highlighted

Symptoms: The current link lags or jumps to a later chapter.

Root Cause: Short final chapters never reach the top (the last link becomes current only at the page end); after a click the highlight is locked until the scroll ends (or 3 seconds); links not in page order.

Diagnosis Steps: Check the order of the links against the section order; scroll slowly.

Resolution Steps: Order links as the sections; make the last chapter long enough or accept the end-of-page rule.

Validation Steps: The highlight follows the scroll.

Preventive Measures: Keep links in page order.

Issue 5 — Links overflow on mobile

Symptoms: Some links are off screen on mobile; no scrollbar is visible.

Root Cause: Links never wrap; the row scrolls sideways with a hidden scrollbar.

Diagnosis Steps: Check at 375px.

Resolution Steps: Shorten Link Titles or reduce the number of links.

Validation Steps: All links fit, or the current one is scrolled into view.

Preventive Measures: Recommended limits in 6.19.12.

Issue 6 — Link text shows the anchor name

Symptoms: A link reads “overview” instead of “CH 1: Overview”.

Root Cause: The item’s Link Title is empty, so its only value (the Anchor Name) is used as text.

Diagnosis Steps: Check the Links item.

Resolution Steps: Fill Link Title.

Validation Steps: The link shows the title.

Preventive Measures: Fill both fields for every item.

Issue 7 — The page jumps when the list pins

Symptoms: Content shifts when the list becomes fixed.

Root Cause: The wrapper's height is kept with an inline `min-height`; additional custom spacing on the block (Additional tab margins) is outside the measured height.

Diagnosis Steps: Check the block's Additional margins.

Resolution Steps: Use padding instead of margin, or no extra spacing.

Validation Steps: No visible jump.

Preventive Measures: Keep spacing defaults on sticky Anchor Lists.

6.19.14 Support Runbook

Monitoring Requirements: No block-specific monitoring; page availability and Core Web Vitals (layout shift when pinning should be zero).

Logs

- Browser console: the block emits no messages.
- Use `document.getElementById('<anchor>')` and the Elements panel (`aria-current`, `anchor-list-pinned`) to diagnose.
- Admin API page status: `GET https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`.

Common Errors

Error	Likely cause	First action
Link does not scroll	Missing or duplicate section Anchor name	Match and de-duplicate names
Heading covered after click	Extra fixed element	Remove it; L3
No pinning in editor	Edit mode (by design)	Use Preview mode
Link shows anchor name	Empty Link Title	Fill Link Title
English "On this page" read by screen readers on a localized page	Placeholder `onThisPage` missing	Add it to the locale's placeholders

Escalation Matrix

Level	Who	What they check
L1	Author support	Links items, section Anchor names, Sticky, preview vs edit mode
L2	AMS / site admin	Delivered markup (`data-anchorname`, section ids), placeholders, CDN cache
L3	Development team	`anchor-list.js` offsets and pinning, header interplay, `scripts.js` section anchors
L4	Adobe Support	Universal Editor or Edge Delivery platform issues

Recovery Procedures: Correct names and publish; restore a page version; hide the block; revert code on `main`.

Health Checks: Each link scrolls to its chapter; the highlight follows scrolling; Sticky pins at 375px (below the header) and 1440px (top); a URL with `#anchor` highlights the right link.

Operational Procedures: When restructuring a page, update the section Anchor names and the Anchor List together; avoid renaming anchors that external links use.

Known Limitations

- Same-page links only.
- No visible scrollbar or scroll buttons when links overflow.
- Scroll offset does not account for fixed elements other than the site header and the list itself.
- The highlight at the end of the page jumps to the last link.
- Scroll-based highlight changes are not tracked in the data layer.
- An empty Anchor List renders an empty, labelled navigation landmark.

6.20 Accordion Block

The Accordion block (`blocks/accordion`) shows a list of collapsible items. Each item has a summary line (the visible question or label) and a body (rich text) that opens and closes when the visitor selects the summary. It is the Block Collection accordion of Edge Delivery Services, built on the native HTML `<details>` / `<summary>` elements and styled for the TCS Corporate website.

6.20.1 Component Purpose

Business Objective

Present secondary or reference information (answers, specifications, terms, process details) in a compact way, so that visitors scan the summaries and open only the items they need, while the full text remains part of the page for search engines and assistive technology.

Business Use Cases

- Frequently asked questions (FAQ) on service, careers, investor or contact pages.
- Eligibility criteria, process steps or programme details on careers and education pages.
- Legal, compliance or policy clarifications that most visitors do not need to read.
- Long lists of capabilities or offerings that would otherwise make the page very long.

When To Use

- The content is a list of independent items, each with a short label and a longer answer.
- Visitors are expected to read one or a few items, not all of them.
- The page would otherwise be too long to scan.

When Not To Use

- The content is essential for every visitor (key messages, calls to action, prices, legal notices that must be seen): put it in default content (Text) instead, because collapsed content is easily overlooked.
- The items are alternative views of one topic that should be compared side by side: use the Tabs block (6.21) instead.
- There is only one item: use a Title and Text instead.
- The content needs images, videos or other blocks inside each item: the Accordion Item body is a rich text field only (no nested blocks).

Dependencies

Dependency	Type	Use
`blocks/accordion/accordion.js`	Block code	Turns each item row into `<details>` / `<summary>`
`blocks/accordion/accordion.css`	Block styles	Dividers, chevron, typography, focus styles
`blocks/accordion/_accordion.json`	Universal Editor models	Accordion and Accordion Item definitions, models and filter
`scripts/scripts.js moveInstrumentation`	Shared code	Keeps the Universal Editor instrumentation on the rendered item
`decorators/data-layer-decorator.js`	Shared code	Pushes `cmp:show` / `cmp:hide` when an item is opened or closed
`decorators/block-options-decorator.js` and `styles/styles.css`	Shared code	Additional tab (Anchor ID, Hide block, padding, margin)
`icons/chevron-down.svg`	Icon	The open / close indicator (CSS mask)
Placeholders sheet	Data	Not used: the block has no UI text of its own
Data sheets, query index, taxonomy	Data	Not used

Related Components

- Tabs Block (6.21): alternative for content that should be compared or switched between.
- Text (default content): for content that must always be visible.
- Fragment block: to reuse the same Accordion on several pages, author it on a fragment page.

Applicable Templates

Any page. The Accordion is allowed in every standard section (`section` filter), so it can be used on home, article, blog and press-release pages and on any page without a template.

Supported Content Types

Plain-text summaries and rich-text bodies (paragraphs, bold, italic, links, lists and other formatting offered by the rich text editor).

6.20.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** An Accordion with four items at a desktop width (1200px or wider), the second item open, showing the top divider, the dividers between items, the chevron on the right of each summary and the open item's chevron rotated upwards.

Desktop (1200px and wider). The block spans the width of its section content area. A 1px (0.0625rem) divider in the default border colour runs above the first item and below every item. Each summary is a full-width row with the label on the left and a 1.5rem chevron on the right, separated by at least 1.5rem; the row has 1.5rem padding above and below. Summary text uses the medium heading size (1.5rem from 900px), medium font weight, line height 1.417 and slightly negative letter-spacing. The open item's body appears below its summary in the secondary neutral text colour, body size (1.125rem from 900px), line height 1.6, with 1.5rem padding below. Hovering or keyboard-focusing a summary changes its text (and chevron) to the brand colour; keyboard focus also shows a 2px brand outline.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Accordion at a tablet width (600px to 1199px), showing that the layout is identical to desktop and that summaries wrap onto two lines when long.

Tablet (600px to 1199px). The block has no breakpoints of its own; the layout is the same as on desktop. The type tokens step at 900px: below 900px summaries use 1.125rem and bodies 1rem; from 900px they use 1.5rem and 1.125rem. Long summaries wrap; the chevron stays on the right, vertically centred.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Accordion at a mobile width (below 600px), first item open, showing the summary text at 1.125rem, the body text at 1rem and the chevron aligned to the right of wrapped summaries.

Mobile (below 600px). Same single-column structure at the smaller type sizes (summary 1.125rem, body 1rem). The section's own horizontal padding (1.5rem on mobile, 2.5rem from 600px) frames the block. The whole summary row is the touch target.

6.20.3 Component Variations

The Accordion has no Style, Layout or Variant field: its model contains only the Additional tab. There is therefore exactly one presentation. Spacing and visibility options of the Additional tab are block options, not variations (see 6.20.6).

Variation 1 — Default Accordion

Variation Name

Default Accordion (the only variation).

Business Purpose

Show several question / answer or label / detail pairs compactly, with every item closed when the page loads.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-default.png`
> **Purpose:** Default Accordion variation
> **Description:** An Accordion with five FAQ items, all closed, with dividers between the items and a downward chevron on each summary; one summary in the hover state (brand colour).

Use Cases

FAQs; programme details; policy clarifications; feature lists with explanations.

Authoring Guidelines

Add an Accordion block to a section, then add one Accordion Item per entry. In each item fill **Summary** (the always-visible label) and **Text** (the body that opens).

Business Rules

- All items are closed on page load; there is no "open by default" option.
- Several items can be open at the same time; opening one does not close the others.
- An item row without a summary cell is not converted (it is left as it is).
- An item without a body gets an empty body: opening it shows nothing.

Rendering Rules

- Each item row becomes `<details class="accordion-item">` with `<summary class="accordion-item-label">` (the Summary content) and `<div class="accordion-item-body">` (the Text content).
- The native disclosure marker is hidden; the chevron is drawn with `::after` and rotates 180 degrees when the item is open.
- The first and last child of the body lose their top and bottom margins.

Responsive Behaviour

- Desktop: full-width rows, summary 1.5rem, body 1.125rem.
- Tablet: same layout; sizes switch at 900px (below: 1.125rem / 1rem).
- Mobile: same layout at 1.125rem / 1rem; long summaries wrap.

SEO Impact

The body text is in the HTML of the page even when closed, so it is indexed. The summary is not a heading (it is a `<summary>` element), so it does not contribute to the heading outline; add a heading (Title) above the Accordion to introduce the topic. No structured data (for example FAQPage JSON-LD) is generated: Not implemented (as of this handover).

Accessibility Compliance

- Native `<details>` / `<summary>`: keyboard operable (Tab to the summary, Enter or Space to toggle) and announced as expandable / collapsed by screen readers without extra ARIA.
- Visible focus indicator: 2px brand outline with 0.125rem offset (`:focus-visible`).
- The chevron is decorative (a CSS mask on a pseudo-element), so it is not announced.
- Transitions (colour, chevron rotation) are disabled when the visitor prefers reduced motion.

6.20.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** The Universal Editor canvas with an Accordion selected in a section, showing the block outline, two Accordion Items (one selected), the item Summary and the open item Text, with letters A to D placed on the elements.

Annotations

- A = Accordion block (select it to see the block properties: Additional tab).
- B = Accordion Item (one per row; select it to edit Summary and Text).
- C = Summary of the item (plain text, the visible label).
- D = Text of the item (rich text body, editable in place when the item is open).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-properties-panel.png`
> **Purpose:** Properties panel

Description: The Universal Editor properties panel for the Accordion block (Additional tab with Anchor ID, Hide block and the eight spacing selects) and, side by side, for an Accordion Item (Summary and Text).

Properties panel, Accordion block

- **Additional** tab (the only tab): Anchor ID; Hide block; Padding left, Padding right, Padding top, Padding bottom; Margin left, Margin right, Margin top, Margin bottom.

Properties panel, Accordion Item

- No tabs; two fields: **Summary** (text) and **Text** (rich text).

Note: The Accordion block itself has no content fields. Its content is the list of Accordion Items, which you add with the + (Add) button of the block in the content tree or on the canvas. Only Accordion Items can be added inside an Accordion (filter `accordion`).

6.20.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites, select the page and choose **Edit** (or open it from the Universal Editor start screen).

Expected Result: The page opens in the Universal Editor canvas in edit mode.

[Screenshot Placeholder]
File name: `screenshots/accordion-step-01-open-page.png`
Purpose: Step 1
Description: The page open in Universal Editor with the content tree on the left and the canvas in edit mode.

Step 2 — Select the target section

Action: In the content tree or on the canvas, select the section where the Accordion should appear.

Expected Result: The section is outlined and its + (Add) button is available.

[Screenshot Placeholder]
File name: `screenshots/accordion-step-02-select-section.png`
Purpose: Step 2
Description: A section selected in the content tree, with the Add button highlighted.

Step 3 — Add the Accordion block

Action: Select + (Add) and choose **Accordion** from the component list.

Expected Result: An Accordion block is added to the section. In edit mode it is empty until items are added.

[Screenshot Placeholder]
File name: `screenshots/accordion-step-03-add-block.png`
Purpose: Step 3
Description: The component picker with Accordion selected and the new empty Accordion in the content tree.

Step 4 — Add the first Accordion Item

Action: Select the Accordion, select **+** (Add) and choose **Accordion Item**.

Expected Result: An item appears with the default content "Lorem Ipsum" (Summary) and a placeholder paragraph (Text).

> [**Screenshot Placeholder**]
> **File name:** `screenshots/accordion-step-04-add-item.png`
> **Purpose:** Step 4
> **Description:** The new Accordion Item with the default Lorem Ipsum summary on the canvas.

Step 5 — Fill in the Summary

Action: With the item selected, type the label or question in **Summary** in the properties panel, e.g. "How do I apply for a role at TCS?".

Expected Result: The summary on the canvas updates after the item is re-rendered.

> [**Screenshot Placeholder**]
> **File name:** `screenshots/accordion-step-05-summary.png`
> **Purpose:** Step 5
> **Description:** The Summary field filled in the properties panel and the updated summary on the canvas.

Step 6 — Fill in the Text

Action: Open the item on the canvas (select its summary) and edit **Text** in place or in the properties panel. Use paragraphs, lists and links as needed.

Expected Result: The body text is shown below the summary while the item is open.

> [**Screenshot Placeholder**]
> **File name:** `screenshots/accordion-step-06-text.png`
> **Purpose:** Step 6
> **Description:** The rich text editor of the Text field with a paragraph and a bulleted list.

Step 7 — Add, reorder and remove items

Action: Repeat Steps 4 to 6 for each entry. Drag items in the content tree to reorder them; use the delete action to remove an item.

Expected Result: The canvas shows the items in the content tree order.

> [**Screenshot Placeholder**]
> **File name:** `screenshots/accordion-step-07-items.png`
> **Purpose:** Step 7
> **Description:** The content tree with five Accordion Items under the Accordion, one being dragged to a new position.

Step 8 — Set the Additional options (optional)

Action: Select the Accordion block, open the **Additional** tab and, if needed, enter an **Anchor ID** (e.g. `anchor-careers-faq`), switch **Hide block** on or off, and choose padding or margin tokens.

Expected Result: The options are applied as block classes; a hidden block is shown faded and outlined in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-step-08-additional.png`
> **Purpose:** Step 8
> **Description:** The Additional tab with an Anchor ID entered and Margin bottom set to 2.5rem (space-1000).

Step 9 — Preview the page

Action: Switch Universal Editor to **Preview** mode and open and close each item.

Expected Result: Items open and close; the chevron rotates; links in the body work.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-step-09-preview.png`
> **Purpose:** Step 9
> **Description:** Preview mode with one item open and the rotated chevron.

Step 10 — Validate

Action: Run the validation checklist below (keyboard, text, links, mobile width).

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-step-10-validate.png`
> **Purpose:** Step 10
> **Description:** Keyboard focus on a summary showing the brand focus outline.

Step 11 — Publish

Action: Select **Publish** in Universal Editor (or use Manage Publication in AEM Sites) and publish the page.

Expected Result: The page is published to Edge Delivery Services and the Accordion appears on the live site.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/accordion-step-11-publish.png`
> **Purpose:** Step 11
> **Description:** The Publish dialog of Universal Editor with the page selected.

Preview Process

1. Use Universal Editor **Preview** mode to interact with the items as a visitor would.
2. For the Edge Delivery preview, open the page on `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}` (for example with AEM Sidekick **Preview**). Feature branches use `https://{branch}--tcs-corp--sradha-tcs.aem.page/`.
3. Check the page at mobile, tablet and desktop widths.

Publishing Process

1. Publish the page from Universal Editor or AEM Sites.
2. The live page is served from `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/{path}` ([To be confirmed: production domain and CDN]).
3. If the Accordion is on a fragment page, publish the fragment page as well.

Validation Process

- ☐ Every item has a Summary and a Text.
- ☐ Summaries are short and unique within the Accordion.
- ☐ All items are closed on load and open / close with mouse, touch, Enter and Space.
- ☐ The focus outline is visible on each summary when tabbing.
- ☐ Links in bodies work and have meaningful link text.
- ☐ A heading (Title) introduces the Accordion.
- ☐ The Anchor ID (if set) starts with `anchor-` and `#anchor-…` links reach the block.
- ☐ Hide block is off (unless intentionally hidden).
- ☐ The layout is correct at below 600px, 600 to 1199px and 1200px and wider.
- ☐ In the browser console, `window.adobeDataLayer` shows `cmp:show` / `cmp:hide` events when items are opened / closed.

Rollback Process

1. In AEM Sites, open the page's version history (**Timeline** / **Versions**) and restore the previous version, or edit the Accordion back to its previous content.
2. Publish the page again.
3. If the block must disappear immediately, switch **Hide block** on and publish, or unpublish the page.
4. For a code regression, the development team reverts the change in GitHub (`main` branch); AEM Code Sync deploys the revert automatically.

6.20.6 Authorable Field Specification

Model `accordion` (Accordion block)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Tab: Additional	tab	—	—	—	Groups the block options	—	—	—
Anchor ID (`classes_anchor`)	text	No	No	empty	Gives the block an id for # links	Start with `anchor-`, lower-case letters, digits, hyphens	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-careers-faq
Hide block (`classes_hideblock`)	boolean	No	No	false	Hides the block on the page	Use to prepare content before release	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default (empty)	Inner spacing on the left	Choose a spacing token only if the design needs it	One of 15 options	pl-400
Padding right (`classes_paddingright`)	select	No	No	Default (empty)	Inner spacing on the right	As above	One of 15 options	pr-400
Padding top (`classes_paddingtop`)	select	No	No	Default (empty)	Inner spacing above	As above	One of 15 options	pt-600

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Padding bottom (`classes_paddingbottom`)	select	No	No	Default (empty)	Inner spacing below	As above	One of 15 options	pb-600
Margin left (`classes_marginleft`)	select	No	No	Default (empty)	Outer spacing on the left	As above	One of 15 options	ml-0
Margin right (`classes_marginright`)	select	No	No	Default (empty)	Outer spacing on the right	As above	One of 15 options	mr-0
Margin top (`classes_margintop`)	select	No	No	Default (empty)	Outer spacing above	As above	One of 15 options	mt-800
Margin bottom (`classes_marginbottom`)	select	No	No	Default (empty)	Outer spacing below	As above	One of 15 options	mb-1000

Model `accordion-item` (Accordion Item)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Summary (`summary`)	text	No (not enforced)	No	"Lorem Ipsum" (new item template)	The visible label / question	Short, specific, unique; a question for FAQs	None in model or code	How do I apply for a role at TCS?
Text (`text`)	richtext	No (not enforced)	No	Lorem ipsum paragraph (new item template)	The body shown when the item is open	Short paragraphs, lists, descriptive links	None in model or code	Visit the Careers portal, choose a role and submit your profile.

Field details

- **Anchor ID**
  - Purpose: makes the block addressable by `#anchor-…` links (for example from an Anchor List or a CTA). `decorators/block-options-decorator.js` sets the class as the block's `id`.
  - Allowed Values: `anchor-` followed by lower-case letters, digits and hyphens.
  - Character Limits: none enforced; Recommended: under 40 characters.
  - Business Rules: if another element already has the id, a suffix `-2`, `-3`, … is added.
  - Dependency Rules: none. Visibility Rules: always visible (no `condition`).
  - Sample Content: `anchor-careers-faq`.
- **Hide block**
  - Purpose: hides the block on the page and in the editor's preview; in edit mode it is shown at 45% opacity with a dashed outline so it can still be selected.
  - Allowed Values: on / off. Character Limits: not applicable.
  - Business Rules: the block's HTML is still delivered (hidden with CSS), so hidden text is in the page source. Do not use it to hide confidential content.
  - Dependency Rules: none. Visibility Rules: always visible. Sample Content: off.
- **Padding left / right / top / bottom** and **Margin left / right / top / bottom**
  - Purpose: adjust the spacing around or inside the block with the design system's spacing tokens.
  - Allowed Values (each select, 15 options): Default (no class); 0 (none); 0.25rem (space-100); 0.5rem (space-200); 0.75rem (space-300); 1rem (space-400); 1.25rem (space-500); 1.5rem (space-600); 2rem (space-800); 2.5rem (space-1000); 3rem (space-1200); 4rem (space-1600); 5rem (space-2000); 6rem (space-2400); 7.5rem (space-3000). Values are classes with the prefixes `pl-`, `pr-`, `pt-`, `pb-`, `ml-`, `mr-`, `mt-`, `mb-` (e.g. `pt-600`).

    - Character Limits: not applicable. Business Rules: the section selector in `styles/styles.css` outranks the block's own spacing. Dependency Rules: none.
    - Visibility Rules: always visible. Sample Content: Margin bottom = 2.5rem (space-1000).
- **Summary**
    - Purpose: the always-visible label of the item; moved into the `<summary>` element.
    - Allowed Values: plain text.
    - Character Limits: none enforced; Recommended: up to 100 characters (one or two lines on mobile).
    - Business Rules: an item row with no summary cell is not converted into an accordion item.
    - Dependency Rules: none. Visibility Rules: always visible.
    - Sample Content: "What industries does TCS serve?"
- **Text**
    - Purpose: the content revealed when the item is opened.
    - Allowed Values: rich text (paragraphs, headings, lists, bold, italic, links).
    - Character Limits: none enforced; Recommended: up to about 150 words per item.
    - Business Rules: without Text the item opens to an empty body. Headings inside the body are allowed but should follow the page's heading hierarchy.
    - Dependency Rules: none. Visibility Rules: always visible.
    - Sample Content: "TCS serves banking, insurance, retail, manufacturing, life sciences and more."

6.20.7 Field Annotation Mapping

Marker	Field	Purpose
A	Accordion block (Additional tab: Anchor ID, Hide block, padding, margin)	Selects the block and its options
B	Accordion Item	One collapsible entry; added with + (Add)
C	Summary	The visible label of the item
D	Text	The body revealed when the item is open

6.20.8 Component Specification

Attribute	Value
Component Name	Accordion (`accordion`), item: Accordion Item (`accordion-item`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`scripts/scripts.js` (`moveInstrumentation`); `decorators/data-layer-decorator.js`; `decorators/block-options-decorator.js`; `styles/styles.css`; `icons/chevron-down.svg`
Component Type	Block (container block with items)
Supported Variants	One (Default); Additional tab options
Supported Templates	Any page (home, article, blog, press-release, no template)
Supported Metadata	None read by the block
Supported Tags	None
Supported Personalization	None in the block; open / close events are in the Adobe Client Data Layer for future use by Adobe Target / Adobe Journey Optimizer (not implemented)

Attribute	Value
Caching Considerations	Delivered in the page HTML, cached by the Edge Delivery CDN; republish the page to update; no sheet or query index is read
Performance Considerations	Loaded with its section (eager if in the first section, else lazy); no images, no network requests; about 30 lines of JS; CSS-only animation
Accessibility Compliance	WCAG 2.1 AA: native disclosure semantics (4.1.2); keyboard operable (2.1.1); visible focus (2.4.7); decorative chevron; reduced-motion support (2.3.3 good practice); colour contrast from design tokens
SEO Compliance	Closed content is in the HTML and indexable; no FAQ structured data (not implemented); summaries are not headings

6.20.9 Input Model

- **Content Inputs:** one row per Accordion Item, each with a Summary cell and a Text cell.
- **Metadata Inputs:** none.
- **Tags Inputs:** none.
- **Page Context Inputs:** none (the block does not read the locale, page type or URL).
- **Query Inputs:** none.
- **Manual Inputs:** all content is entered manually per item.

Authored HTML received from AEM (illustrative; instrumentation attributes shortened):

```
<div class="accordion anchor-careers-faq mb-1000" data-aue-resource="urn:...">
  <div data-aue-resource="urn:.../item0" data-aue-model="accordion-item">
    <div><p>How do I apply for a role at TCS?</p></div>
    <div><p>Visit the Careers portal and submit your profile.</p></div>
  </div>
  <div data-aue-resource="urn:.../item1" data-aue-model="accordion-item">
    <div><p>Can I apply for several roles?</p></div>
    <div><p>Yes, apply separately for each role.</p></div>
  </div>
</div>
```

6.20.10 Output Model

Rendered Output

- `div.accordion.block` (with Additional classes and, if set, the Anchor ID as `id`)
  - `details.accordion-item` (one per item; carries the item's Universal Editor instrumentation)
    - `summary.accordion-item-label` (the Summary content)
    - `div.accordion-item-body` (the Text content)

Generated HTML

```
<div class="accordion anchor-careers-faq mb-1000 block" id="anchor-careers-faq"
  data-block-name="accordion" data-block-status="loaded">
  <details class="accordion-item">
    <summary class="accordion-item-label"><p>How do I apply for a role at TCS?</p></
summary>
    <div class="accordion-item-body"><p>Visit the Careers portal ...</p></div>
  </details>
</div>
```

Generated JSON: none.

Generated Metadata: none.

Analytics Tracking

- Component state (pushed by `trackBlocks` after the sections load):

```
{
  "component": {
    "anchor-careers-faq": {
      "@type": "tcs-corp/components/accordion",
      "parentId": "page-1x2y3z"
    }
  }
}
```

The id is the block's id (Anchor ID), else `accordion-{n}` in page order. `dc:title` (the first heading inside the block) is omitted when the block has no heading, which is the normal case for an Accordion. - Events: opening an item pushes `cmp:show`, closing it pushes `cmp:hide` (captured from the `toggle` event of `<details>`):

```
{
  "event": "cmp:show",
  "eventInfo": {
    "path": "component.anchor-careers-faq",
    "dc:title": "How do I apply for a role at TCS?"
  }
}
```

- Clicks on the summary are deliberately not tracked as `cmp:click` (the toggle event is the interaction). Clicks on links inside the body push `cmp:click` with `xdm:linkURL`, `xdm:linkType` (`other`, `exit` or `download`) and `breakpoint`.

Personalization Output: none.

Search Index Output: the block does not write to the query index; the page text (including closed bodies) is available to external search engines.

6.20.11 Business Rules

1. Every direct child row of the block is treated as one item.
2. The first cell of a row is the summary, the second the body.
3. A row with no first cell is skipped and left unchanged.
4. A row with no second cell gets an empty body element.
5. All items render closed; no item is open by default.
6. Items open and close independently; several can be open at once.
7. The Universal Editor instrumentation of each row moves to its `<details>` element, so items stay selectable and editable on the canvas.
8. Only Accordion Items can be added inside an Accordion (filter `accordion`).
9. The Summary and Text fields have no `required` flag and no validation; empty values are accepted.
10. Opening / closing an item pushes `cmp:show` / `cmp:hide` to `window.adobeDataLayer`, unless the block is inside an element with `data-analytics="off"`.
11. Summary clicks are excluded from `cmp:click`.
12. Hide block hides the block on the page and in Preview; in edit mode it is faded and outlined.
13. The Anchor ID becomes the block's id only if it matches `^anchor-[a-z0-9][a-z0-9-]*$`; duplicates get `-2`, `-3`, ….
14. Reduced motion: colour and chevron transitions are disabled.

15. The block emits no console messages and has no error state; a failure leaves the authored rows visible as plain text.

6.20.12 Author Best Practices

- **Recommended Content Length:** 3 to 10 items per Accordion; Summary up to 100 characters; Text up to about 150 words per item (Recommended, not enforced).
- **Image Dimensions:** not applicable (the Text field can hold images only if the rich text editor offers them; the block CSS has no image rules, so avoid images in items).
- **CTA Writing Guidelines:** put calls to action outside the Accordion so they are always visible; inside bodies use descriptive link text ("Read the Careers FAQ") rather than "Click here".
- **Accessibility Guidelines:** make each Summary self-explanatory out of context; do not start all summaries with the same words; keep heading levels inside bodies consistent with the page.
- **SEO Guidelines:** introduce the Accordion with a heading containing the topic; write full answers (they are indexed); avoid duplicating the same FAQ on many pages (use a fragment).
- **Metadata Guidelines:** not applicable; the block reads no metadata.
- **Taxonomy Guidelines:** not applicable; the block reads no tags.
- **Localization Guidelines:** translate summaries and texts per locale site; keep the same Anchor ID across locales so links and analytics stay comparable; check that longer translations (for example German) still read well on mobile.

6.20.13 Troubleshooting Guide

Issue 1 — Items do not open or close

Symptoms: Selecting a summary does nothing; no chevron is shown; items show as plain rows.

Root Cause: The block JavaScript did not run (script error, block not decorated) or the block is not an Accordion (wrong component added).

Diagnosis Steps: Inspect the block in the browser developer tools: decorated items are `<details class="accordion-item">`. Check the console for errors and the Network tab for `blocks/accordion/accordion.js`.

Resolution Steps: Reload the page; if the files fail to load, escalate to development. If the wrong block was used, replace it with an Accordion and re-enter the items.

Validation Steps: Items open and close and the chevron rotates.

Preventive Measures: Validate in Preview before publishing; keep code changes tested on a feature branch.

Issue 2 — An item appears as plain text above the Accordion items

Symptoms: One row shows its text without a summary row and chevron.

Root Cause: The row has no first cell (an empty or malformed item), so the code skips it.

Diagnosis Steps: In the content tree, look for an Accordion Item with an empty Summary; inspect the row in the developer tools (it is a plain `div`, not `details`).

Resolution Steps: Fill the Summary of the item or delete the item.

Validation Steps: All rows render as accordion items.

Preventive Measures: Always fill Summary and Text when adding an item.

Issue 3 — An item opens but shows nothing

Symptoms: The chevron rotates but no text appears.

Root Cause: The item's Text is empty.

Diagnosis Steps: Select the item and check the Text field.

Resolution Steps: Enter the body text, or remove the item.

Validation Steps: The body shows when the item is open.

Preventive Measures: Use the validation checklist before publishing.

Issue 4 — The Accordion is missing on the live page

Symptoms: The block is visible in edit mode (faded) but not in Preview or on the live page.

Root Cause: Hide block is on, or the page (or the fragment that contains it) was not published.

Diagnosis Steps: Check the Additional tab; check the publication status of the page and any fragment.

Resolution Steps: Switch Hide block off and publish; publish the fragment page.

Validation Steps: The block appears on the `.aem.page` / `.aem.live` URL.

Preventive Measures: Record intentionally hidden blocks in the editorial plan.

Issue 5 — A link to the Accordion does not scroll to it

Symptoms: `#anchor-…` links do not jump to the block.

Root Cause: The Anchor ID was not saved (validation error), differs from the link, or was de-duplicated to `-2` because another element already uses it.

Diagnosis Steps: Inspect the block's `id` in the developer tools; compare with the link's `#` value.

Resolution Steps: Correct the Anchor ID or the link; make the id unique on the page.

Validation Steps: Selecting the link scrolls to the Accordion.

Preventive Measures: Keep a list of anchors per page; never reuse an Anchor ID on one page.

Issue 6 — No cmp:show / cmp:hide events in the data layer

Symptoms: Analytics shows no accordion interactions.

Root Cause: The block is inside an element with `data-analytics="off"`; the events are pushed but no tag manager reads them (Adobe Tags is not loaded yet); or the toggle happened before the data layer listeners were attached.

Diagnosis Steps: In the console, open an item and run `window.adobeDataLayer.slice(-3)`.

Resolution Steps: If events are present, configure the Adobe Tags rule (once Adobe Tags is embedded in `scripts/delayed.js`). If missing, escalate to development.

Validation Steps: `cmp:show` with `eventInfo.path` `component.{id}` appears on open.

Preventive Measures: Include data layer checks in release testing.

Issue 7 — Summary text is too large or wraps badly on mobile

Symptoms: Long summaries wrap over three or more lines on phones.

Root Cause: Summary text is too long for the 1.125rem heading size on small screens.

Diagnosis Steps: Check the page at 375px width.

Resolution Steps: Shorten the Summary; move details into the Text.

Validation Steps: Summaries fit on one or two lines on mobile.

Preventive Measures: Follow the Recommended 100-character limit.

6.20.14 Support Runbook

Monitoring Requirements

- Real User Monitoring (RUM) of Edge Delivery Services for page performance and errors of pages that contain the Accordion.
- Once Adobe Tags is embedded, monitor `cmp:show` / `cmp:hide` volumes per page.
- [To be confirmed: monitoring tool and alerting thresholds — owner: AMS lead].

Logs

- Browser console: the block emits no messages of its own; errors would come from block loading in `scripts/aem.js`: `failed to load module for accordion` or `failed to load block accordion`.
- Admin API: check the preview / live status of the page (`https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`).
- Author logs: AEM as a Cloud Service author logs (Cloud Manager) for publishing errors.

Common Errors

Error	Likely cause	First action
Items show as plain text	JS not loaded or failed	Reload; check console and Network
One row not converted	Empty Summary cell	Fill or delete the item
Empty body	Text not filled	Fill Text
Block missing on live	Hide block on, page not published	Check Additional tab; publish
Anchor link fails	Anchor ID mismatch or duplicate	Inspect the block id
No analytics	Adobe Tags not embedded; `data-analytics="off"`	Inspect `window.adobeDataLayer`

Escalation Matrix

Level	Who	What they check
L1	Author support	Item content, Hide block, Anchor ID, publication status, browser cache
L2	AMS / site admin	Preview / live status in the Admin API, CDN cache, code sync status, RUM
L3	Development team	`accordion.js` / `.css` regressions, console errors, data layer events
L4	Adobe Support	Platform issues (Edge Delivery delivery, Universal Editor, AEM author)

Recovery Procedures

- Content problem: restore the previous page version and republish.
- Code problem: revert the commit on `main`; Code Sync redeploys; purge is automatic on code sync.
- Urgent removal: Hide block on, publish.

Health Checks

- Open a page with an Accordion: items open with mouse and keyboard; the chevron rotates.
- `document.querySelectorAll('.accordion details').length` equals the number of items.
- Opening an item adds a `cmp:show` entry to `window.adobeDataLayer`.

Operational Procedures

- Changes to content: author, preview, validate, publish (6.20.5).
- Changes to code: feature branch, PR with a preview link, lint (`npm run lint`), review, merge.

Known Limitations

- No “open by default” or “only one open at a time” option.
- No FAQ structured data (JSON-LD).
- Summaries are not headings; no heading level option.
- Bodies are rich text only; no nested blocks or images styling.
- No expand-all / collapse-all control.

6.21 Tabs Block

The Tabs component (`blocks/tabs`) shows content in tabbed panels: a tab list (one button per tab) and one panel per tab, of which one is shown at a time. It behaves like the Tabs core component of AEM and is available in two authoring modes that render the same way:

- **Tabs Container** (model `tabs`): a special section. Inside it, **Tab** items (6.22) start the tabs, and the components placed after a Tab (Columns, Title, Text, Image, Button), up to the next Tab, are that tab’s content. `scripts/scripts.js` (`buildTabsSections`) wraps the section content in a synthetic `tabs` block.
- **Tabs** block (model `tabs-block`): a block placed in a normal section, with one **Tab** item (model `tabs-item`) per tab. Each Tab item has a Title, ID, Active tab and a **Fragment**: a page whose content is loaded into the panel.

Both modes support the Style (Underline, Boxed, Pills), Alignment (Left, Center, Full width), Orientation (Horizontal, Vertical), an accessible Label, deep links to a tab (`#id`) and the `cmp:show` data layer event.

6.21.1 Component Purpose

Business Objective

Let visitors switch between related views of one topic (for example services, industries, regions or product editions) without leaving the page, keeping the page short while every view stays one selection away and linkable.

Business Use Cases

- Services or offerings grouped by category (“Consulting”, “Cloud”, “Cybersecurity”).
- Industry or region views of the same content (“Americas”, “Europe”, “Asia Pacific”).
- Investor information by period (“Quarterly results”, “Annual reports”, “Presentations”).
- Careers content by audience (“Graduates”, “Experienced professionals”, “Interns”).
- Reusing the same tab content on several pages (Tabs block with fragment pages).

When To Use

- Two to about seven parallel, mutually exclusive views with short, clear labels.
- Visitors usually need only one view at a time.
- Each view needs rich layout (Columns, images, buttons): use the Tabs Container; when the views are shared between pages, use the Tabs block with fragments.

When Not To Use

- Content that visitors must compare side by side or read in sequence: use default content or Columns.
- Many short question / answer items: use the Accordion (6.20).
- Only one tab: use a normal section.
- Primary navigation between pages: use the header navigation or an Anchor List.
- Critical content (legal notices, main CTAs) that must be visible without interaction.

Dependencies

Dependency	Type	Use
`blocks/tabs/tabs.js`, `tabs.css`, `_tabs.json`	Block	Tab list, panels, keyboard, deep links, data layer; models `tabs`, `tabs-block`, `tabs-item`

Dependency	Type	Use
`blocks/tab/tab.js` (`readTab`)	Block	Reads the Tab items of a Tabs Container (6.22)
`blocks/fragment/fragment.js` (`loadFragment`)	Block	Loads a Tab item's Fragment page (Tabs block mode)
`scripts/scripts.js` (`buildTabsSections`, `wrapSectionContent`, `moveInstrumentation`)	Shared code	Builds the synthetic block for Tabs Container sections; keeps editor instrumentation
`scripts/aem.js` (`toClassName`)	Core library	Normalises IDs and option values
`utils/analytics.js` (`pushComponentEvent`)	Util	`cmp:show` when another tab is shown
`utils/buttons.js` (`pick`)	Util	Accepts only the allowed Style, Alignment and Orientation values (block mode)
`utils/environment.js` (`isEditing`)	Util	Empty-state hint in Universal Editor edit mode
`utils/ids.js` (`availableId`, `uniqueId`)	Util	Unique panel / button ids
`utils/placeholders.js` (`getPlaceholders`)	Util	UI text: `tabs`, `tabNumber`, `tabsEmpty`, `tabsBlockEmpty`
`utils/properties.js` (`isOn`, `normalizeKey`, `textOf`)	Util	Reads the block property rows
`decorators/data-layer-decorator.js`	Decorator	Component state and `cmp:click` on tab buttons
`decorators/block-options-decorator.js`, `styles/styles.css`	Shared code	Additional tab (Tabs block mode)
Placeholders sheet `/{locale}/placeholders.json`	Data	Keys `tabs` (default "Tabs"), `tabNumber` ("Tab {n}"), `tabsEmpty`, `tabsBlockEmpty`
Fragment pages	Content	Tabs block mode: one page per tab, e.g. under the locale's `references/`

Related Components

- Tab Block (6.22): the tab marker inside a Tabs Container.
- Accordion Block (6.20): collapsible alternative for many short items.
- Columns, Title, Text, Image, Button: the components allowed inside a Tabs Container.
- Fragment block: the mechanism behind the Tabs block's Fragment field.
- Anchor List: in-page navigation; links to a tab's ID open that tab.

Applicable Templates

Any page (home, article, blog, press-release, no template). The Tabs Container is added at the page level (the `main` filter allows `section`, `gen-carousel` and `tabs`); the Tabs block is added inside any standard section (`section` filter, as `tabs-block`).

Supported Content Types

- Tabs Container: Columns, Title, Text, Image and Button components (filter `tabs`), plus the Tab markers.
- Tabs block: any content of the referenced fragment pages (any sections and blocks the fragment page contains).

6.21.2 Visual Overview

> [Screenshot Placeholder]
> **File name:** `screenshots/tabs-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A horizontal Underline Tabs at 1200px or wider with four tabs, the second selected (semibold label, 3px brand underline), the grey divider under the tab list and the selected panel's Columns content 2.5rem below the list.

Desktop (1200px and wider). The tab list is a single row of buttons above the panels, with a 1px divider under the whole list. Each tab button has 0.75rem vertical and 2.5rem horizontal padding (from 900px), body text size (1.125rem from 900px), line height 1.61, labels on one line (`white-space: nowrap`). Unselected tabs use the secondary text colour and turn to the default text colour on hover; the selected tab is semibold in the default text colour with a 3px (0.1875rem) brand-coloured bottom border that overlaps the divider. The visible panel starts 2.5rem below the list; the other panels are hidden. With **Orientation: Vertical**, from 900px the block becomes a two-column grid: the tab list on the left (at least 12.5rem, one quarter of the width), the panel on the right (three quarters), 3rem apart; the list has a 1px left border and the selected tab a 3px brand left border; labels may wrap.

> [Screenshot Placeholder]
> **File name:** `screenshots/tabs-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Tabs at about 768px, showing the horizontal tab list; when the labels are wider than the screen, the list scrolls horizontally without a visible scrollbar.

Tablet (600px to 1199px). Between 900px and 1199px the layout is the desktop layout. Between 600px and 899px tabs use the mobile padding (0.75rem by 1.5rem) and 1rem text, and Vertical tabs are shown horizontally above the panel (the vertical layout starts at 900px). If the tabs do not fit, the tab list scrolls horizontally (the scrollbar is hidden; swipe or use the arrow keys).

> [Screenshot Placeholder]
> **File name:** `screenshots/tabs-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Tabs at 375px: the tab list cut off at the right edge, indicating that it scrolls sideways, and the selected panel stacked below with its content in one column.

Mobile (below 600px). The tab list stays a single horizontal row that scrolls sideways; tab labels do not wrap. Panel content follows the responsive rules of the components inside it (for example Columns stack). Vertical tabs are horizontal on mobile. Pills tabs use 0.5rem by 1.5rem padding.

6.21.3 Component Variations

The variations below can be combined: one authoring mode (1 or 2), one Style (3 to 5), one Alignment (6 to 8), one Orientation (9 or 10) and, for the Tabs Container only, one Width Type (11 or 12). The definitions start new Tabs with Underline, Left and Horizontal.

Variation 1 — Tabs Container (section mode)

Variation Name

Tabs Container (component title “Tabs Container”, model `tabs`, section style `tabs-section`).

Business Purpose

Author the tab content directly on the page, with full layout components, without creating separate fragment pages.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-container-mode.png`
> **Purpose:** Tabs Container variation
> **Description:** A Tabs Container with three tabs whose panels contain Columns with an image and text, a Title and a Button, as rendered on the page.

Use Cases

Page-specific tabbed content: services overview, region views, programme details.

Authoring Guidelines

At the page level (content tree, `main`), add **Tabs Container**. Inside it, add a **Tab**, then the components for that tab (Columns, Title, Text, Image, Button), then the next **Tab** and its components, and so on. Set the Tabs Container’s ID, Width Type, Label, Style, Alignment and Orientation in its properties.

Business Rules

- Each Tab starts a new tab; the components after it, up to the next Tab, are its content.
- Content placed before the first Tab forms a first, untitled tab labelled “Tab 1” (placeholder `tabNumber`).
- Allowed components: Tab, Columns, Title, Text, Image, Button (filter `tabs`).
- The Tabs Container’s ID (field `anchorName`) becomes the section’s id and the base for generated panel ids (`{id}-item-{n}`) when it is not already used on the page.
- The **Component** select has one option (Tabs = `tabs-section`); this class is what turns the section into tabs.

Rendering Rules

- `buildTabsSections` wraps the section content in `div.tabs` with `data-source="section"`; Style, Alignment and Orientation values become classes; the Label becomes `data-label`.
- Each component keeps its own markup, decoration and editor instrumentation inside the panel.
- The Tab markers’ wrappers stay inside the panels but are hidden on the page (shown in edit mode as a dashed label chip).

Responsive Behaviour

- Desktop / tablet / mobile: as described in 6.21.2; components inside the panels follow their own responsive rules (panel inner wrappers have no extra padding or max width).

SEO Impact

All panel content is in the page HTML (hidden panels use the `hidden` attribute), so it is indexable. Headings inside panels should continue the page’s heading outline.

Accessibility Compliance

WAI-ARIA tabs pattern: `role="tablist"` with `aria-label`, `role="tab"` buttons with `aria-selected`, `aria-controls` and a roving `tabindex`, `role="tabpanel"` panels with `aria-labelledby` and `tabindex="0"`; arrow keys, Home and End move between tabs.

Variation 2 — Tabs block with fragment items (block mode)

Variation Name

Tabs block (component title "Tabs", model `tabs-block`, items `tabs-item`).

Business Purpose

Reuse tab content maintained once on fragment pages across several pages and locales' pages, and keep complex panel content (any blocks) in its own page.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-block-fragments.png`
> **Purpose:** Tabs block variation
> **Description:** A Tabs block in a standard section with three Tab items, the selected panel showing the content of a fragment page (two sections with a Title, Text and Cards).

Use Cases

Shared product or service descriptions; the same "Our locations" tabs on several pages; panels that need blocks not allowed in a Tabs Container (e.g. Cards, List, Video).

Authoring Guidelines

In a standard section add **Tabs**. Set ID, Label, Style, Alignment and Orientation. Add one **Tab** item per tab; in each, set Title (required), ID, Active tab, and in the **Fragment** tab pick the fragment page.

Business Rules

- Each Tab item is one tab; its panel shows the content of the Fragment page.
- Fragments are loaded in parallel when the block is decorated; a missing or unloadable fragment leaves the panel empty (no error message).
- The Fragment path must be a site path starting with `/` (external URLs are ignored).
- Only Tab items can be added inside the block (filter `tabs-block`).
- Without any Tab item the block renders nothing on the page; in edit mode it shows the hint "Add a Tab, then choose the Fragment it shows in the Tab properties." (`tabsBlockEmpty`).

Rendering Rules

- The block's property rows (ID, Label, Style, Alignment, Orientation) are read and removed; every other row is a Tab item.
- The fragment content is wrapped in `div.tabs-fragment`; consecutive sections inside it are 2.5rem apart.
- Each Tab item's editor instrumentation moves to its panel, so selecting a panel in Universal Editor selects the Tab item.
- Items authored with an earlier model that had more cells use the last cell as the Fragment.

Responsive Behaviour

As 6.21.2; the fragment content follows the responsive rules of its own blocks.

SEO Impact

Fragment content is fetched by JavaScript after the page loads; it is not in the page's delivered HTML. Search engines that render JavaScript may see it; the fragment pages are separate URLs. Keep fragments under the locale's `references/` folder, which is excluded from the query index.

Accessibility Compliance

Same tabs pattern as Variation 1. Fragment content must carry its own accessible headings and alternative texts.

Variation 3 — Style: Underline

Variation Name

Underline (`tabsStyle` = `underline`, default).

Business Purpose

The standard, light tab style, matching the Anchor List chapter navigation.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-style-underline.png`
> **Purpose:** Underline style
> **Description:** Underline tabs: plain text labels, a grey divider under the list and a 3px brand underline under the selected, semibold label.

Use Cases

Most content tabs; tabs directly below a heading.

Authoring Guidelines

Styles tab, **Style** = Underline (the default).

Business Rules

Default for new Tabs; also applied when the value is empty or unknown (no extra class needed).

Rendering Rules

Class `underline` (no own CSS; the base styles are the underline style): transparent tab background, 3px transparent bottom border, brand bottom border on the selected tab.

Responsive Behaviour

Desktop / tablet from 900px: 0.75rem by 2.5rem padding; below 900px 0.75rem by 1.5rem; list scrolls sideways when it overflows.

SEO Impact

None (presentation only).

Accessibility Compliance

Selected state is shown by weight, colour and underline (not colour alone); focus outline 2px brand, inset 0.125rem.

Variation 4 — Style: Boxed

Variation Name

Boxed (`tabsStyle` = `boxed`).

Business Purpose

A stronger, folder-like look that ties the selected tab visually to its panel.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-style-boxed.png`
> **Purpose:** Boxed style

> **Description:** Boxed tabs with grey (secondary background) tab boxes with rounded top corners, 0.25rem apart, the selected tab white with a border and brand-coloured text, open towards the panel.

Use Cases

Data-rich panels (tables, figures); tabs in light-background sections.

Authoring Guidelines

Styles tab, **Style** = Boxed.

Business Rules

None beyond the style.

Rendering Rules

Class `boxed`: list gap 0.25rem; tabs with a 1px transparent border, 0.5rem top radius and the secondary background (hover: secondary hover background); selected tab: default border colour, bottom border in the page background colour, default background and brand text. Vertical (from 900px): all corners rounded, no list left border, selected tab fully bordered.

Responsive Behaviour

Same breakpoints as Underline; boxes keep their style in the horizontally scrolling list.

SEO Impact

None.

Accessibility Compliance

The selected tab is distinguished by background, border and text colour; focus outline as above.

Variation 5 — Style: Pills

Variation Name

Pills (`tabsStyle` = `pills`).

Business Purpose

A compact, button-like filter look for short labels.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-style-pills.png`
> **Purpose:** Pills style
> **Description:** Pill tabs: rounded, outlined labels 0.5rem apart without a divider under the list, the selected pill filled with the brand colour and white text.

Use Cases

Year or category switchers (“2024”, “2025”, “2026”); short audience labels.

Authoring Guidelines

Styles tab, **Style** = Pills.

Business Rules

None beyond the style.

Rendering Rules

Class `pills`: list gap 0.5rem and no bottom divider; tabs with a 1px default border, full radius, padding 0.5rem by 1.5rem (0.5rem by 2rem from 900px); hover: default hover background; selected: brand border and background, on-brand text. Vertical (from 900px): no list left border; each pill keeps its border.

Responsive Behaviour

Pills stay in one scrolling row on mobile and tablet; vertical stack from 900px if Vertical.

SEO Impact

None.

Accessibility Compliance

Selected pill uses fill and text colour; focus outline as above.

Variation 6 — Alignment: Left

Variation Name

Left (`tabsAlignment` = `left`, default).

Business Purpose

Aligns the tab list with left-aligned headings and text.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-alignment-left.png`
> **Purpose:** Left alignment
> **Description:** A tab list starting at the left edge of the content area with free space on the right.

Use Cases

Default for content pages.

Authoring Guidelines

Styles tab, **Alignment** = Left.

Business Rules

Default; no own CSS (flex start).

Rendering Rules

Class `left`; tabs size to their labels and start on the left.

Responsive Behaviour

Same on all breakpoints; the list scrolls when it overflows.

SEO Impact

None.

Accessibility Compliance

No effect on semantics or reading order.

Variation 7 — Alignment: Center

Variation Name

Center (`tabsAlignment` = `center`).

Business Purpose

Centres the tab list under a centred heading.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-alignment-center.png`
> **Purpose:** Center alignment
> **Description:** A tab list centred in the content area under a centred Title.

Use Cases

Landing and home page sections with centred layouts.

Authoring Guidelines

Styles tab, **Alignment** = Center.

Business Rules

None.

Rendering Rules

Class `center`: `justify-content: safe center`, so a list wider than the screen falls back to start alignment and stays scrollable (the first tab is never cut off).

Responsive Behaviour

Centred when the tabs fit; start-aligned and scrollable when they do not (typically mobile). With Vertical orientation from 900px the list is a column, so centring applies only below 900px.

SEO Impact

None.

Accessibility Compliance

No effect on semantics.

Variation 8 — Alignment: Full width (stretch)

Variation Name

Full width (stretch) (`tabsAlignment` = `stretch`).

Business Purpose

Makes the tabs share the full width of the content area, like a segmented control.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-alignment-stretch.png`
> **Purpose:** Full width alignment
> **Description:** Three tabs that together fill the full width of the content area, each at least as wide as its label.

Use Cases

Two to four tabs of similar label length.

Authoring Guidelines

Styles tab, **Alignment** = Full width (stretch).

Business Rules

None.

Rendering Rules

Class `stretch`: each tab `flex: 1 0 auto` (grows to share the free space, never shrinks below its label).

Responsive Behaviour

Fills the width when the tabs fit; scrolls when they do not. In the Vertical layout (from 900px) the tabs are full-width rows of the left column.

SEO Impact

None.

Accessibility Compliance

Larger targets (helps 2.5.5 Target Size, AAA); no semantic change.

Variation 9 — Orientation: Horizontal

Variation Name

Horizontal (`tabsOrientation` = `horizontal`, default).

Business Purpose

Standard layout: tab list above the panel.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-orientation-horizontal.png`
> **Purpose:** Horizontal orientation
> **Description:** The tab list in a row above the panel, at desktop width.

Use Cases

Most tabs; short labels.

Authoring Guidelines

Styles tab, **Orientation** = Horizontal.

Business Rules

Default; `aria-orientation` is not set (horizontal is the ARIA default).

Rendering Rules

Class `horizontal`; no own CSS.

Responsive Behaviour

Same on all breakpoints; tab padding grows from 900px.

SEO Impact

None.

Accessibility Compliance

All four arrow keys, Home and End move between tabs; selection follows focus.

Variation 10 — Orientation: Vertical
Variation Name

Vertical (`tabsOrientation` = `vertical`).

Business Purpose

Long or many tab labels on the left of the content, like a side menu, on desktop.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-orientation-vertical.png`
> **Purpose:** Vertical orientation
> **Description:** At 1200px: the tab list as a left column with a thin left border and the selected tab marked by a 3px brand left border; the panel in the right column, 3rem away.

Use Cases

Five or more tabs; long labels ("Banking, Financial Services and Insurance").

Authoring Guidelines

Styles tab, **Orientation** = Vertical. Combine with any Style and Alignment.

Business Rules

The tab list gets `aria-orientation="vertical"` at all widths (even where it is displayed horizontally below 900px).

Rendering Rules

From 900px: block grid `minmax(12.5rem, 1fr) 3fr` with a 3rem column gap; tab list as a column with a 1px left border; tabs with a 3px left border (brand when selected), 0.75rem by 1.5rem padding, start-aligned text that may wrap; panel without top padding. Boxed and Pills keep their boxes / pills without the list border.

Responsive Behaviour

Desktop and 900 to 1199px: two columns. Below 900px (mobile, small tablet): horizontal list above the panel, as Variation 9.

SEO Impact

None.

Accessibility Compliance

Arrow Up / Down (and Left / Right) move between tabs; reading order is tab list, then panel.

Variation 11 — Width Type: Full Width (Tabs Container only)
Variation Name

Full Width (`sectionWidth` = `full-width`, default).

Business Purpose

The Tabs Container section uses the page's maximum width (120rem).

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-width-full.png`
> **Purpose:** Full Width section
> **Description:** A Tabs Container spanning the full page width on a wide screen, with the section's inline padding.

Use Cases

Default; tabs with wide Columns content.

Authoring Guidelines

Tabs Container, Properties tab, **Width Type** = Full Width.

Business Rules

Default; no class is added.

Rendering Rules

Section content capped at `--page-max-width` (120rem) with the section inline padding (1.5rem mobile, 2.5rem from 600px).

Responsive Behaviour

Fluid at all widths up to 120rem.

SEO Impact

None.

Accessibility Compliance

Very long line lengths can reduce readability; keep text in Columns.

Variation 12 — Width Type: Fixed Width (Tabs Container only)

Variation Name

Fixed Width (`sectionWidth` = `fixed-width`).

Business Purpose

Limits the Tabs Container to the content width (90rem), centred.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-width-fixed.png`
> **Purpose:** Fixed Width section
> **Description:** A Tabs Container centred at the content width with empty space on both sides on a 1920px screen.

Use Cases

Text-heavy tabs; alignment with other fixed-width sections.

Authoring Guidelines

Tabs Container, Properties tab, **Width Type** = Fixed Width.

Business Rules

Adds the `fixed-width` section class.

Rendering Rules

Section and its content capped at `--content-max-width` (90rem), centred.

Responsive Behaviour

Below 90rem identical to Full Width; wider screens show side margins.

SEO Impact

None.

Accessibility Compliance

Shorter line lengths support readability.

6.21.4 Authoring Experience

> [Screenshot Placeholder]
> **File name:** `screenshots/tabs-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** Universal Editor edit mode with a Tabs Container selected: the tab list, the selected panel with its dashed Tab label chip, a Columns component in the panel, and in the content tree the Tabs Container with Tab, Columns, Tab, Text children; letters A to G placed on the elements.

Annotations

- A = Tabs Container (or Tabs block) — select it for ID, Width Type, Label, Style, Alignment, Orientation and Additional.
- B = Tab list (generated; shows the Tab titles; select a tab button to switch panels in the canvas).
- C = Tab marker (Tabs Container: the dashed label chip at the top of the panel, shown only in edit mode) / Tab item (Tabs block: the panel itself) — select it for Title, ID, Active tab (and Fragment).
- D = Tab Title (the label shown in the tab button).
- E = Panel content (Tabs Container: Columns, Title, Text, Image, Button components; Tabs block: the fragment page content, not editable in place).
- F = Add (+) button of the container / block, to add a Tab or a component.
- G = Empty-state hint (only when there is no tab yet, edit mode only).

> [Screenshot Placeholder]
> **File name:** `screenshots/tabs-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The properties panel of a Tabs Container with its four tabs (Properties, Accessibility, Styles, Additional), next to the properties of a Tabs block Tab item with its Tab and Fragment tabs.

Properties panel, Tabs Container (model `tabs`)

- **Properties:** Component (select, only "Tabs"); ID; Width Type (Full Width / Fixed Width).
- **Accessibility:** Label (aria-label of the tab list).
- **Styles:** Style (Underline / Boxed / Pills); Alignment (Left / Center / Full width (stretch)); Orientation (Horizontal / Vertical).
- **Additional:** Anchor ID; Hide block; Padding left / right / top / bottom; Margin left / right / top / bottom (see the note in 6.21.6: not applied to this section-based component).

Properties panel, Tabs block (model `tabs-block`)

- **Properties:** ID.

- **Accessibility:** Label.
- **Styles:** Style; Alignment; Orientation.
- **Additional:** Anchor ID; Hide block; 4 padding and 4 margin selects.

Properties panel, Tab item of the Tabs block (model `tabs-item`)

- **Tab:** Title (required); ID; Active tab.
- **Fragment:** Fragment (content picker for a page).

The Tab marker of the Tabs Container has its own model (`tab`), described in 6.22.

6.21.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites select the page and choose **Edit**.

Expected Result: The page opens in Universal Editor in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-01-open-page.png`
> **Purpose:** Step 1
> **Description:** The page open in Universal Editor with the content tree visible.

Step 2 — Choose the mode

Action: Decide: Tabs Container (content authored on this page) or Tabs block (content from fragment pages). For the Tabs block, create and publish the fragment pages first (one per tab, e.g. `/content/tcs-corp/us/en/references/tabs/cloud-overview`).

Expected Result: You know which component to add; fragment pages exist if needed.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-02-choose-mode.png`
> **Purpose:** Step 2
> **Description:** The Sites console with three fragment pages under references/tabs.

Step 3 — Add the component

Action: Tabs Container: select the page root (`main`) in the content tree, select **+** (Add) and choose **Tabs Container**. Tabs block: select a section, **+** (Add), **Tabs**.

Expected Result: The component is added with Style Underline, Alignment Left and Orientation Horizontal. With no tabs, edit mode shows the hint "Add a Tab, then the components of the tab after it." (container) or "Add a Tab, then choose the Fragment it shows in the Tab properties." (block).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-03-add-component.png`
> **Purpose:** Step 3
> **Description:** The new empty Tabs Container showing the dashed empty-state hint.

Step 4 — Fill the Properties tab

Action: Enter the **ID** (e.g. `our-services`, lower case, no spaces). Tabs Container only: choose **Width Type**.

Expected Result: Generated tab ids will be `our-services-item-1`, `-2`, … for tabs without their own ID.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-04-properties.png`
> **Purpose:** Step 4
> **Description:** Properties tab with ID our-services and Width Type Full Width.

Step 5 — Fill the Accessibility tab

Action: Enter the **Label**, e.g. "Our services".

Expected Result: Screen readers announce the tab list as "Our services, tab list"; without a Label it is "Tabs" (placeholder `tabs`).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-05-accessibility.png`
> **Purpose:** Step 5
> **Description:** Accessibility tab with the Label field filled.

Step 6 — Choose the Styles

Action: Choose **Style**, **Alignment** and **Orientation**.

Expected Result: The canvas re-renders with the chosen look.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-06-styles.png`
> **Purpose:** Step 6
> **Description:** Styles tab with Pills, Center and Horizontal selected and the re-rendered tabs.

Step 7 — Add the tabs

Action: Tabs Container: select the container, **+** (Add), **Tab**; enter its Title (and optionally ID and Active tab, see 6.22); then add the tab's components (Columns, Title, Text, Image, Button) after it; repeat for each tab. Tabs block: select the block, **+** (Add), **Tab**; enter Title, ID, Active tab; open the **Fragment** tab and pick the fragment page; repeat.

Expected Result: One tab button per Tab; selecting a component in the content tree shows its tab on the canvas.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-07-add-tabs.png`
> **Purpose:** Step 7
> **Description:** Content tree with Tab, Columns, Tab, Text, Tab, Image under the Tabs Container and three tab buttons on the canvas.

Step 8 — Order and check the tabs

Action: Drag Tabs and components in the content tree to reorder. Make sure every component is after the Tab it belongs to.

Expected Result: Tab order on the canvas follows the content tree; no "Tab 1" untitled tab appears (which would mean content is placed before the first Tab).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-08-order.png`
> **Purpose:** Step 8
> **Description:** A Tab being dragged in the content tree.

Step 9 — Additional options (Tabs block)

Action: On the Tabs block, set Anchor ID, Hide block, padding and margin if needed.

Expected Result: The options apply to the block.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-09-additional.png`
> **Purpose:** Step 9
> **Description:** Additional tab of the Tabs block with Margin top 3rem (space-1200).

Step 10 — Preview and test deep links

Action: Switch to **Preview**, select each tab, use the arrow keys, and open the page with `#{tab-id}` appended (e.g. `.../services#cloud`).

Expected Result: The tab named in the address is shown and scrolled to (below the fixed header).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-10-preview.png`
> **Purpose:** Step 10
> **Description:** Preview with the URL ending in #cloud and the Cloud tab selected.

Step 11 — Validate

Action: Run the validation checklist below.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-11-validate.png`
> **Purpose:** Step 11
> **Description:** Keyboard focus outline on a tab button and the data layer console output.

Step 12 — Publish

Action: Publish the page; for the Tabs block also publish every fragment page.

Expected Result: The tabs and their content appear on the live site.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tabs-step-12-publish.png`
> **Purpose:** Step 12
> **Description:** Publish dialog listing the page and its referenced fragment pages.

Preview Process

1. Universal Editor Preview mode: switch tabs with mouse and keyboard.
2. Edge Delivery preview: `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}` (AEM Sidekick **Preview**); fragment pages must be previewed too for the Tabs block.
3. Test `#id` deep links and the Vertical layout at 900px and wider.

Publishing Process

1. Publish fragment pages (Tabs block) before or together with the page.
2. Publish the page; the live URL is `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/{path}` ([To be confirmed: production domain and CDN]).

Validation Process

- ☐ Every tab has a meaningful, short Title; no “Tab 1” / “Tab 2” fallback labels appear.
- ☐ In a Tabs Container, no content sits before the first Tab.
- ☐ Exactly one tab (or none) has Active tab switched on.
- ☐ Tab IDs are lower case, unique on the page and deep links (`#id`) open the right tab.
- ☐ The Label (Accessibility) describes the tab group.
- ☐ Tabs block: every Tab item has a Fragment; every fragment page is published and shows content.
- ☐ Arrow keys, Home and End move between tabs; focus is visible.
- ☐ The layout is correct below 600px, 600 to 899px, 900 to 1199px and 1200px and wider (Vertical only from 900px).
- ☐ Switching tabs adds `cmp:show` entries to `window.adobeDataLayer`.

Rollback Process

1. Restore the previous page version in AEM Sites (Timeline / Versions) and republish.
2. Tabs block: if a fragment page caused the problem, restore and republish that fragment page (every page using it is affected).
3. Urgent removal of a Tabs block: Hide block on and publish. A Tabs Container cannot be hidden with Hide block (see 6.21.6); delete it or unpublish the page.
4. Code regressions: the development team reverts the commit on `main`; Code Sync redeploys.

6.21.6 Authorable Field Specification

Model `tabs` (Tabs Container, a section)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Tab: Properties	tab	—	—	—	Main properties	—	—	—
Component (`style`)	select	No (preset)	No	Tabs (`tabs-section`)	Marks the section as a Tabs Container	Leave on Tabs	One option: Tabs = `tabs-section`	tabs-section
ID (`anchorName`)	text	No	No	empty	Section id and base of generated tab ids	Lower case, hyphens, no spaces	None in model; normalised by code (`toClassName`, leading # removed)	our-services
Width Type (`sectionWidth`)	select	No	No	Full Width (`full-width`)	Page or content width	Fixed Width for	Full Width / Fixed Width	full-width

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
						text-heavy tabs		
Tab: Accessibility	tab	—	—	—	Accessible name	—	—	—
Label (`tabsAriaLabel`)	text	No	No	empty (code fallback "Tabs")	aria-label of the tab list	Describe the group	None (trimmed)	Our services
Tab: Styles	tab	—	—	—	Presentation	—	—	—
Style (`tabsStyle`)	select	No	No	Underline	Tab look	—	Underline / Boxed / Pills	underline
Alignment (`tabsAlignment`)	select	No	No	Left	Tab list alignment	—	Left / Center / Full width (stretch)	left
Orientation (`tabsOrientation`)	select	No	No	Horizontal	List above or beside panels	Vertical for many / long labels	Horizontal / Vertical	horizontal
Tab: Additional	tab	—	—	—	Block options	See note below	—	—
Anchor ID (`classes_anchor`)	text	No	No	empty	Block anchor (see note)	Use ID (Properties) instead	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-our-services
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide (see note)	Not effective here	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	pl-400
Padding right (`classes_paddingright`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	pr-400
Padding top (`classes_paddingtop`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	pt-400
Padding bottom (`classes_paddingbottom`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	pb-400
Margin left (`classes_marginleft`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	ml-0
Margin right (`classes_marginright`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	mr-0
Margin top (`classes_margintop`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	mt-800
Margin bottom (`classes_marginbottom`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	mb-800

Important: The Tabs Container is a section, not a block. `decorators/block-options-decorator.js` and the spacing / Hide block rules in `styles/styles.css` apply only to `.block` elements, and `buildTabsSections` copies only Style, Alignment, Orientation and Label to the synthetic `tabs` block. The Additional tab fields of the

> Tabs Container therefore have no effect on the page (as of this handover) [To be confirmed: whether AEM delivers `classes_*` section fields at all — owner: development team]. Use the section **ID** for anchors.

Model `tabs-block` (Tabs block)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Tab: Properties	tab	—	—	—	Main properties	—	—	—
ID (`tabsId`)	text	No	No	empty (generated `tabs-{n}`)	Base of generated tab ids	Lower case, hyphens	None in model; normalised by `toClassName`	our-services
Tab: Accessibility	tab	—	—	—	Accessible name	—	—	—
Label (`tabsAriaLabel`)	text	No	No	empty (fallback “Tabs”)	aria-label of the tab list	Describe the group	None	Our services
Tab: Styles	tab	—	—	—	Presentation	—	—	—
Style (`tabsStyle`)	select	No	No	Underline	Tab look	—	Enforced in code: only underline / boxed / pills become classes	boxed
Alignment (`tabsAlignment`)	select	No	No	Left	Alignment	—	Enforced in code: left / center / stretch	center
Orientation (`tabsOrientation`)	select	No	No	Horizontal	Orientation	—	Enforced in code: horizontal / vertical	vertical
Tab: Additional	tab	—	—	—	Block options	—	—	—
Anchor ID (`classes_anchor`)	text	No	No	empty	Block id for # links	Start with `anchor-`	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-our-services
Hide block (`classes_hideblock`)	boolean	No	No	false	Hides the block	Prepare before release	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default	Inner spacing left	Only if the design needs it	One of 15 options	pl-0
Padding right (`classes_paddingright`)	select	No	No	Default	Inner spacing right	As above	One of 15 options	pr-0
Padding top (`classes_paddingtop`)	select	No	No	Default	Inner spacing top	As above	One of 15 options	pt-600
Padding bottom (`classes_paddingbottom`)	select	No	No	Default	Inner spacing bottom	As above	One of 15 options	pb-600
Margin left (`classes_marginleft`)	select	No	No	Default	Outer spacing left	As above	One of 15 options	ml-0

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Margin right (`classes_marginright`)	select	No	No	Default	Outer spacing right	As above	One of 15 options	mr-0
Margin top (`classes_margintop`)	select	No	No	Default	Outer spacing top	As above	One of 15 options	mt-1200
Margin bottom (`classes_marginbottom`)	select	No	No	Default	Outer spacing bottom	As above	One of 15 options	mb-1200

Model `tabs-item` (Tab item of the Tabs block)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Tab: Tab	tab	—	—	—	Tab properties	—	—	—
Title (`title`)	text	Yes (`required`)	No	"Tab" (new item template)	Tab button label	1 to 3 words	Enforced: required in the model; code falls back to "Tab {n}" if empty	Cloud
ID (`tabId`)	text	No	No	empty (generated `{tabs-id}-item-{n}`)	Panel id for deep links	Lower case, no spaces	Normalised by `toClassName`; de-duplicated with `-2`, `-3`	cloud
Active tab (`activeTab`)	boolean	No	No	off	Tab shown on load	One tab at most	Only the value "true" counts	off
Tab: Fragment	tab	—	—	—	Content source	—	—	—
Fragment (`fragment`)	aem-content	No (code tolerates empty)	No	empty	Page whose content fills the panel	Pick a page under `references/`	Code: path must start with `/`; otherwise ignored	/content/tcs-corp/us/en/references/tabs/cloud

Field details

- **Component (`style`, Tabs Container)**
  - Purpose: the section style class `tabs-section` that `buildTabsSections` looks for.
  - Allowed Values: Tabs (`tabs-section`). Character Limits: not applicable.
  - Business Rules: without this class the section renders as a normal section (no tabs).
  - Dependency Rules: required for every other Tabs Container field to take effect.
  - Visibility Rules: always visible. Sample Content: Tabs.
- **ID (`anchorName` / `tabsId`)**
  - Purpose: Tabs Container: the section's `id` (anchor) and the base for generated tab ids; Tabs block: the base for generated tab ids (`{id}-item-{n}`); the block itself does not get this id (use the Anchor ID).
  - Allowed Values: any text; normalised to lower case with hyphens (`toClassName`).
  - Character Limits: none enforced; Recommended: under 30 characters.
  - Business Rules: Tabs Container: applied only if no element on the page already has that id. Empty: ids are generated (`tabs-1`, `tabs-2`, … in page order, then `tabs-1-item-1`, …).
  - Dependency Rules: a Tab's own ID takes priority for its panel.
  - Visibility Rules: always visible. Sample Content: `our-services`.
- **Width Type (`sectionWidth`, Tabs Container)**

    - Purpose: section width; Allowed Values: Full Width (`full-width`), Fixed Width (`fixed-width`).
    - Character Limits: not applicable. Business Rules: only `fixed-width` adds a class.
    - Dependency Rules: none. Visibility Rules: always visible. Sample Content: Full Width.
- **Label (`tabsAriaLabel`)**
    - Purpose: the tab list's `aria-label`.
    - Allowed Values: plain text. Character Limits: none enforced; Recommended: under 40 characters.
    - Business Rules: empty → the placeholder `tabs` ("Tabs").
    - Dependency Rules: none. Visibility Rules: always visible. Sample Content: "Our services".
- **Style (`tabsStyle`)**
    - Purpose: tab look. Allowed Values: Underline (`underline`), Boxed (`boxed`), Pills (`pills`).
    - Character Limits: not applicable. Business Rules: Tabs block: unknown values are ignored (`pick`); Tabs Container: the value is added as a class without checking.
    - Dependency Rules: combines with Alignment and Orientation. Visibility Rules: always visible. Sample Content: Underline.
- **Alignment (`tabsAlignment`)**
    - Purpose: horizontal alignment of the tab list. Allowed Values: Left (`left`), Center (`center`), Full width (stretch) (`stretch`).
    - Character Limits / Business Rules: as Style. Dependency Rules: in the Vertical layout (from 900px) Center has no visible effect. Visibility Rules: always visible. Sample Content: Left.
- **Orientation (`tabsOrientation`)**
    - Purpose: tab list above (Horizontal) or left of the panel (Vertical, from 900px).
    - Allowed Values: Horizontal (`horizontal`), Vertical (`vertical`).
    - Business Rules: Vertical sets `aria-orientation="vertical"`. Dependency Rules: none.
    - Visibility Rules: always visible. Sample Content: Horizontal.
- **Additional tab fields** (Anchor ID, Hide block, 4 padding and 4 margin selects)
    - Purpose and Allowed Values: as for every block: Anchor ID `anchor-...`; Hide block on / off; spacing selects with 15 options each: Default; 0 (none); 0.25rem (space-100); 0.5rem (space-200); 0.75rem (space-300); 1rem (space-400); 1.25rem (space-500); 1.5rem (space-600); 2rem (space-800); 2.5rem (space-1000); 3rem (space-1200); 4rem (space-1600); 5rem (space-2000); 6rem (space-2400); 7.5rem (space-3000), as classes `pl-`, `pr-`, `pt-`, `pb-`, `ml-`, `mr-`, `mt-`, `mb-` + token number.
    - Character Limits: Anchor ID: none enforced; Recommended under 40 characters.
    - Business Rules: effective on the Tabs block; not effective on the Tabs Container (see the note above). Anchor ID duplicates get `-2`, `-3`, ….
    - Dependency Rules: none. Visibility Rules: always visible (no `condition`).
    - Sample Content: Margin top = 3rem (space-1200).
- **Title (`title`, Tab item)**
    - Purpose: the tab button text and the accessible name of the panel.
    - Allowed Values: plain text. Character Limits: none enforced; Recommended: up to 25 characters (labels never wrap in the horizontal list).
    - Business Rules: required in the model; empty in the content → "Tab {n}" (placeholder `tabNumber`). Also sent as `dc:title` in `cmp:show`.
    - Dependency Rules: none. Visibility Rules: always visible. Sample Content: "Cloud".
- **ID (`tabId`, Tab item)**
    - Purpose: the panel's id, so `#id` links open the tab.
    - Allowed Values: text normalised by `toClassName`. Character Limits: none enforced; Recommended: under 30 characters.
    - Business Rules: empty → `{Tabs ID}-item-{n}`; an id already used on the page gets `-2`, `-3`, …. The tab button's id is `{panel id}-tab`.
    - Dependency Rules: unique per page. Visibility Rules: always visible.

    - Sample Content: `cloud`.
- **Active tab (`activeTab`, Tab item)**
    - Purpose: the tab shown when the page loads.
    - Allowed Values: on / off. Business Rules: if several are on, the first wins; if none, the first tab is shown; a deep link (`#id`) overrides it.
    - Dependency Rules: none. Visibility Rules: always visible. Sample Content: on for “Cloud”.
- **Fragment (`fragment`, Tab item)**
    - Purpose: the page whose content is shown in the panel.
    - Allowed Values: a site page path (picker). Character Limits: not applicable.
    - Business Rules: loaded with `loadFragment` (`{path}.plain.html`); paths not starting with `/` and failed requests leave the panel empty without a message; the fragment’s sections are fully decorated (blocks, buttons, block options).
    - Dependency Rules: the fragment page must be published (and previewed for the preview site). Visibility Rules: always visible.
    - Sample Content: `/content/tcs-corp/us/en/references/tabs/cloud`.

6.21.7 Field Annotation Mapping

Marker	Field	Purpose
A	Tabs Container / Tabs block (ID, Width Type, Label, Style, Alignment, Orientation, Additional)	Component properties
B	Tab list (generated from Titles)	Switches panels
C	Tab marker (Tabs Container) / Tab item (Tabs block)	Starts a tab; holds Title, ID, Active tab (and Fragment)
D	Title	Tab button label
E	Panel content (components / Fragment)	Content of the tab
F	Add (+)	Adds Tabs or components
G	Empty-state hint	Edit-mode guidance when there are no tabs

6.21.8 Component Specification

Attribute	Value
Component Name	Tabs: Tabs Container (`tabs`, section), Tabs (`tabs-block`), Tab item (`tabs-item`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`blocks/tab`, `blocks/fragment`, `scripts/scripts.js`, `scripts/aem.js`, `utils/analytics.js`, `utils/buttons.js`, `utils/environment.js`, `utils/ids.js`, `utils/placeholders.js`, `utils/properties.js`, `decorators/data-layer-decorator.js`
Component Type	Container section (Tabs Container) and container block with items (Tabs block)
Supported Variants	Modes: Container, Block; Style: Underline, Boxed, Pills; Alignment: Left, Center, Stretch; Orientation: Horizontal, Vertical; Width Type (Container): Full, Fixed
Supported Templates	Any page
Supported Metadata	None
Supported Tags	None

Attribute	Value
Supported Personalization	None; `cmp:show` per tab and the `tabs:change` DOM event are available for future Adobe Target / Journey Optimizer use (not implemented)
Caching Considerations	Page HTML and fragment `.plain.html` are cached by the Edge Delivery CDN; a change in a fragment page needs that page republished; the shown tab is remembered only in memory during Universal Editor re-renders
Performance Considerations	Loaded with its section; Tabs block fetches all fragments in parallel during decoration (in the first section this delays LCP); images in hidden panels are not displayed, so lazy images load only when their tab is shown
Accessibility Compliance	WCAG 2.1 AA: WAI-ARIA tabs pattern (4.1.2), keyboard (2.1.1: arrows, Home, End, roving tabindex), visible focus (2.4.7), named tab list, focusable panels, reduced motion
SEO Compliance	Container content in HTML (indexable, hidden until selected); fragment content loaded client-side; deep-linkable tab ids

6.21.9 Input Model

- **Content Inputs:** Tabs Container: Tab markers and the components after them. Tabs block: property rows and Tab item rows (Title, ID, Active tab, Fragment).
- **Metadata Inputs:** none.
- **Tags Inputs:** none.
- **Page Context Inputs:** the URL hash (`#id`, deep link); Universal Editor edit mode (`html.adobe-ue-edit`); Universal Editor selection events (`aue:ui-select`); the locale (for placeholders).
- **Query Inputs:** none (no query index).
- **Manual Inputs:** all fields; fragment pages chosen per tab.

Authored HTML of a Tabs Container section (illustrative; the section properties arrive as data attributes):

```
<div data-aue-resource="urn:.../section_1" data-aue-model="tabs" data-style="tabs-section"
  data-anchorname="our-services" data-tabsstyle="pills" data-tabsalignment="center"
  data-tabsorientation="horizontal" data-tabsarialabel="Our services">
  <div class="tab">
    <div><div>title</div><div>Consulting</div></div>
    <div><div>tabId</div><div>consulting</div></div>
    <div><div>activeTab</div><div>true</div></div>
  </div>
  <div class="columns">...</div>
  <div class="tab">...</div>
  <p>Text of the second tab.</p>
</div>
```

Authored HTML of a Tabs block (key-value property rows, then item rows with four cells):

```
<div class="tabs mt-1200">
  <div><div>tabsId</div><div>our-services</div></div>
  <div><div>tabsAriaLabel</div><div>Our services</div></div>
  <div><div>tabsStyle</div><div>boxed</div></div>
  <div><div>tabsAlignment</div><div>left</div></div>
  <div><div>tabsOrientation</div><div>vertical</div></div>
  <div data-aue-model="tabs-item">
    <div>Cloud</div><div>cloud</div><div>true</div>
    <div><a href="/en/references/tabs/cloud">/en/references/tabs/cloud</a></div>
  </div>
</div>
```

6.21.10 Output Model

Rendered Output

- Tabs Container: `div.section.tabs-section.tabs-container` (with the section `id`) > `div.tabs-wrapper` > `div.tabs.block` (`data-source="section"`, style classes).
- Tabs block: `div.tabs-wrapper` > `div.tabs.block` (style and Additional classes).
- Inside the block, in both modes:
  - `div.tabs-list[role=tablist]` with `aria-label` (and `aria-orientation="vertical"` when Vertical), containing one `button.tabs-tab[role=tab]` per tab with `id="{panel}-tab"`, `aria-controls`, `aria-selected` and `tabindex` (0 on the selected tab, –1 on the others).
  - One `div.tabs-panel[role=tabpanel]` per tab with `id`, `aria-labelledby`, `tabindex="0"` and `hidden` on the panels not shown. Container panels contain the component wrappers (including the hidden Tab marker wrapper); block panels contain `div.tabs-fragment`.
  - Edit mode with no tabs: `p.tabs-empty` (hint text).

Generated HTML

```
<div class="tabs pills center horizontal block" data-source="section"
  data-label="Our services" data-block-name="tabs">
  <div class="tabs-list" role="tablist" aria-label="Our services">
    <button type="button" class="tabs-tab" id="consulting-tab" role="tab"
      aria-controls="consulting" aria-selected="true" tabindex="0">Consulting</button>
    <button type="button" class="tabs-tab" id="our-services-item-2-tab" role="tab"
      aria-controls="our-services-item-2" aria-selected="false" tabindex="-1">Cloud</
button>
  </div>
  <div class="tabs-panel" id="consulting" role="tabpanel" tabindex="0"
    aria-labelledby="consulting-tab">
    <div class="tab-wrapper"><div class="tab block">...</div></div>
    <div class="columns-wrapper">...</div>
  </div>
  <div class="tabs-panel" id="our-services-item-2" role="tabpanel" tabindex="0"
    aria-labelledby="our-services-item-2-tab" hidden>...</div>
</div>
```

Generated JSON: none.

Generated Metadata: none.

DOM events: `tabs:change` (bubbles) on the block when the visitor shows another tab, with `detail: { index, id, title }`; not fired for the tab shown on load.

Analytics Tracking

- Component state (`trackBlocks`): the tabs block as `tcs-corp/components/tabs`, id = block id (Anchor ID, Tabs block) else `tabs-{n}`. Blocks inside panels (Columns, Tab, fragment blocks) are registered with the tabs block as `parentId`.
- Showing another tab (click, keyboard, deep link change) pushes `cmp:show` with the panel id as the component path (panels are not registered as components themselves):

```
{
  "event": "cmp:show",
  "eventInfo": { "path": "component.cloud", "dc:title": "Cloud" }
}
```

- A click on a tab button also pushes `cmp:click` (from `data-layer-decorator.js`, `[role=tab]`):

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.tabs-1",
    "interaction": "button",
    "dc:title": "Cloud",
    "xdm:linkType": "other",
    "breakpoint": "desktop"
  }
}
```

- Not pushed: the tab shown on load; selecting the already shown tab; re-renders in Universal Editor. Opening a tab through a deep link after load clicks its button (so `cmp:click` appears; `cmp:show` only if it differs from the tab already shown).

Personalization Output: none.

Search Index Output: none written by the block; Tabs Container text is part of the page HTML; fragment content belongs to the fragment pages.

6.21.11 Business Rules

1. Mode: a block with `data-source="section"` (built by `buildTabsSections`) is a Tabs Container; otherwise it is a Tabs block.
2. A section becomes a Tabs Container only with the style class `tabs-section` and only once (a section already containing `.tabs-wrapper` is skipped).
3. Tabs Container: each wrapper containing a `.tab` block starts a new tab; content before the first Tab starts an untitled tab.
4. Tabs block: rows with exactly two cells whose first cell (normalised) is `tabsid`, `tabsarialabel`, `tabsstyle`, `tabsalignment` or `tabsorientation` are properties; all other rows are Tab items.
5. Tabs block: Style, Alignment and Orientation are accepted only from their allowed lists; Tabs Container: the values are added as classes after `toClassName`.
6. Tab item cells: Title, ID, Active tab, then the Fragment as the last cell when the row has more than three cells; with three or fewer cells there is no fragment.
7. Fragment paths must start with `/` (not `//`); `.html` / `.plain.html` is removed and `{path}.plain.html` is fetched; failures give an empty panel without a message.
8. All fragments are requested in parallel; decoration waits for all of them.
9. Tab label: the Title, else “Tab {n}” (placeholder `tabNumber`).
10. Panel id: the Tab’s ID (normalised), else `{base}-item-{n}`; base = Tabs Container section id / Tabs block ID, else `tabs-{n}`; ids already on the page get `-2`, `-3`, ….
11. Tab button id: `{panel id}-tab`.
12. Tab list name: the Label, else “Tabs” (placeholder `tabs`).
13. Tab shown first: the tab shown before a Universal Editor re-render, else the tab named by the page address `#id`, else the first Active tab, else the first tab.
14. Active tab counts only when its value is exactly “true”.
15. No tabs: the block is emptied; in edit mode a hint is shown (`tabsEmpty` / `tabsBlockEmpty` placeholders); on the page nothing is shown.
16. Click selects a tab; ArrowLeft / ArrowUp select the previous tab, ArrowRight / ArrowDown the next (wrapping around), Home the first, End the last; the selected tab gets focus (automatic activation).
17. Deep links: on `hashchange`, and once after the page `load` event for the initial address, a hash naming a tab panel clicks its tab and scrolls the Tabs into view (`scroll-margin-top` = navigation height, 4rem).

18. Only panel ids open tabs; links to a tab button id (…`-tab`) or to the Tabs Container section id scroll but do not change the tab.
19. A tab change fires `tabs:change` and `cmp:show`; the initial selection does not.
20. Universal Editor: selecting a component inside a panel shows its tab (`aue:ui-select`); a Tab marker edited in the properties updates its tab label (`tabs:tab-update`); Tab markers are visible only in edit mode.
21. Tabs block items: each item's editor instrumentation moves to its panel.
22. Vertical layout applies from 900px; `aria-orientation="vertical"` is set at all widths.
23. Additional tab options apply to the Tabs block but not to the Tabs Container.
24. Reduced motion disables the tab colour / border transitions.

6.21.12 Author Best Practices

- **Recommended Content Length:** 2 to 7 tabs; Titles of 1 to 3 words (up to about 25 characters); keep panel content similar in length so the page does not jump (Recommended).
- **Image Dimensions:** images in panels follow their component (Image, Columns); they are limited to the panel width (`max-width: 100%`). Use at least 1200px wide images for full-width panels; 2000px for wide hero-like images.
- **CTA Writing Guidelines:** each panel may end with one clear Button ("Explore cloud services"); do not put the page's main CTA only in a non-default tab.
- **Accessibility Guidelines:** always fill the Label; use unique Titles; start each panel with a heading or text that matches the Title; avoid nesting Tabs inside Tabs.
- **SEO Guidelines:** put the most important content in the first (or Active) tab; for content that must rank on its own, prefer a separate page over a fragment-only tab.
- **Metadata Guidelines:** not applicable.
- **Taxonomy Guidelines:** not applicable.
- **Localization Guidelines:** translate Titles and Labels per locale; keep tab IDs identical across locales (they are used in links and in `cmp:show` paths); use locale-specific fragment pages (under each locale's `references/`); check that translated labels still fit (German and Dutch labels are longer; consider Vertical).

6.21.13 Troubleshooting Guide

Issue 1 — The section shows all content without tabs

Symptoms: Tab markers' content and components appear one below another; no tab list.

Root Cause: The section is a normal section, or its Component (style) is not `tabs-section`.

Diagnosis Steps: Inspect the section: it must have the class `tabs-section` and contain `div.tabs-wrapper`.

Resolution Steps: Add a Tabs Container (not a Section) and move the components into it; set Component = Tabs.

Validation Steps: The tab list appears.

Preventive Measures: Always start from the Tabs Container component.

Issue 2 — An extra tab "Tab 1" appears first

Symptoms: The first tab has a generic label and contains a stray component.

Root Cause: A component was placed before the first Tab in the Tabs Container.

Diagnosis Steps: Check the content tree order of the container.

Resolution Steps: Move the component after the right Tab, or add a Tab before it.

Validation Steps: Only titled tabs remain.

Preventive Measures: Add the Tab first, then its components.

Issue 3 — A Tabs block panel is empty
Symptoms: A tab shows nothing.

Root Cause: No Fragment chosen; the fragment page is not published / previewed; the path is not a site path; or the item was authored with fewer cells.

Diagnosis Steps: Check the Tab item's Fragment field; open `{fragment path}.plain.html` on the `.aem.page` / `.aem.live` host; check the Network tab for a 404.

Resolution Steps: Choose the fragment; publish it; re-add the item if its structure is outdated.

Validation Steps: The panel shows the fragment content.

Preventive Measures: Publish fragments with the page (publish references).

Issue 4 — A deep link does not open the tab
Symptoms: `page#cloud` opens the page on the first tab.

Root Cause: The tab's ID differs (normalised to lower case with hyphens), was de-duplicated (`cloud-2`), or the link targets the button id or the section id.

Diagnosis Steps: Inspect the panel `id`s in the developer tools.

Resolution Steps: Set the Tab's ID and use exactly that value after `#`.

Validation Steps: The tab opens and scrolls into view below the header.

Preventive Measures: Use unique, simple ids (`cloud`, `consulting`).

Issue 5 — The wrong tab is shown on load
Symptoms: A different tab than expected is selected.

Root Cause: Several tabs have Active tab on (the first wins); the URL contains a tab `#id`; in Universal Editor the tab shown before the re-render is kept.

Diagnosis Steps: Check Active tab on each Tab; check the address.

Resolution Steps: Keep at most one Active tab; reload without the hash.

Validation Steps: The intended tab shows on a fresh load.

Preventive Measures: Validation checklist.

Issue 6 — Vertical tabs show horizontally
Symptoms: Orientation is Vertical but the list is above the panel.

Root Cause: By design below 900px; or, on desktop, the value did not reach the block.

Diagnosis Steps: Check the viewport width; inspect the block for the `vertical` class.

Resolution Steps: Re-select Orientation = Vertical and save; reload.

Validation Steps: From 900px the list is in a left column.

Preventive Measures: Test at 900px and wider.

Issue 7 — Spacing or Hide block has no effect on a Tabs Container
Symptoms: Padding / margin / Hide block set in the Additional tab change nothing.

Root Cause: These options are applied only to blocks; the Tabs Container is a section.

Diagnosis Steps: Inspect the section and the `tabs` block for `pt-…` / `hideblock` classes.

Resolution Steps: Use a Tabs block (which supports them) or remove / unpublish the container; adjust spacing on the components inside.

Validation Steps: Expected spacing / visibility.

Preventive Measures: Follow 6.21.6.

Issue 8 — The tab label does not update after editing

Symptoms: Changing a Title in Universal Editor leaves the old label.

Root Cause: The re-render did not complete (editor event not applied) or ID / Active tab changes, which are only read when the Tabs are rendered again.

Diagnosis Steps: Check the console for errors; reload the editor.

Resolution Steps: Reload the page in Universal Editor.

Validation Steps: The label and behaviour match the properties.

Preventive Measures: Reload after structural changes.

6.21.14 Support Runbook

Monitoring Requirements

- RUM / Core Web Vitals for pages with Tabs blocks in the first section (fragment fetches can affect LCP).
- 404 monitoring of fragment `.plain.html` requests.
- Once Adobe Tags is embedded, `cmp:show` volumes per tab id.
- [To be confirmed: monitoring tool and thresholds — owner: AMS lead].

Logs

- Browser console: the block logs nothing itself; module errors appear as `failed to load module for tabs` / `failed to load block tabs` (`scripts/aem.js`). Failed fragments show only as failed requests in the Network tab.
- Admin API status of the page and fragment pages (`https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`).
- AEM author / Cloud Manager logs for publication errors.

Common Errors

Error	Likely cause	First action
No tab list	Not a Tabs Container / style missing	Check section class `tabs-section`
“Tab 1” untitled tab	Content before the first Tab	Reorder in the content tree
Empty panel	Fragment missing / unpublished / 404	Open the fragment `.plain.html`
Deep link ignored	Id mismatch or duplicate	Inspect panel ids
Wrong initial tab	Several Active tabs / hash in URL	Fix Active tab
Vertical not applied	Below 900px / class missing	Check width and class
Additional options ignored	Tabs Container is a section	Use Tabs block

Escalation Matrix

Level	Who	What they check
L1	Author support	Mode, tab order, Titles, IDs, Active tab, Fragments, publication of page and fragments
L2	AMS / site admin	Admin API preview / live status, fragment URLs (404), CDN cache, Code Sync, RUM
L3	Development team	`tabs.js` / `tab.js` / `buildTabsSections` regressions, console errors, data layer

Level	Who	What they check
L4	Adobe Support	Universal Editor rendering, Edge Delivery delivery, AEM author issues

Recovery Procedures

- Restore and republish the page or fragment version.
- Tabs block: Hide block and publish for an urgent removal.
- Code: revert on `main` (Code Sync deploys automatically).

Health Checks

- `document.querySelectorAll('.tabs [role=tab]').length` equals the number of tabs.
- Exactly one panel per Tabs has no `hidden` attribute.
- Arrow keys move the selection; `#id` links open tabs.
- Switching a tab adds `cmp:show` to `window.adobeDataLayer`.

Operational Procedures

- Fragment changes: edit, preview and publish the fragment page; every page using it shows the change after the CDN serves the new version.
- Code changes: feature branch, PR with preview link, `npm run lint`, review, merge.

Known Limitations

- Tabs Container allows only Tab, Columns, Title, Text, Image and Button.
- Additional tab options have no effect on the Tabs Container.
- No manual activation mode (selection follows focus).
- Fragment content is loaded client-side (not in the page HTML) and failures are silent.
- `aria-orientation="vertical"` is set even where the list is displayed horizontally (below 900px).
- Tab ID / Active tab changes in Universal Editor apply after the Tabs are rendered again.
- No URL update when a visitor changes tab (the hash is read, not written).

6.22 Tab Block

The Tab block (`blocks/tab`, model `tab`) is the tab marker inside a **Tabs Container** (6.21, Variation 1). It marks where a tab starts and holds that tab's **Title**, **ID** and **Active tab** setting; the components placed after it, up to the next Tab, are the tab's content. The Tabs block builds the tab list from the Tab markers, so on the page a Tab shows nothing itself; in Universal Editor edit mode it shows its title as a dashed label chip so authors can select it and edit its properties.

> **Note:** The Tab items of the **Tabs block** (fragment mode) are a different model (`tabs-item`, with a Fragment field) and are described in 6.21.6. This chapter covers the Tab marker (`tab`) used in the Tabs Container.

6.22.1 Component Purpose

Business Objective

Give authors a simple, visible marker for "a new tab starts here" so that tabbed content can be built directly on the page with standard components.

Business Use Cases

- Starting each tab of a services, industries or region Tabs Container.
- Naming a tab for visitors (Title) and for links and analytics (ID).

- Choosing which tab is shown when the page opens (Active tab).

When To Use

- Inside a Tabs Container, once at the start of every tab.

When Not To Use

- Outside a Tabs Container (it is only allowed there by the `tabs` filter; elsewhere it would render as a visible label).
- In a Tabs block (fragment mode): use its Tab items instead.

Dependencies

Dependency	Type	Use
`blocks/tab/tab.js`	Block	`readTab` (read and cache Title, ID, Active tab) and `decorate` (label chip, `tabs:tab-update`)
`blocks/tab/tab.css`	Block styles	Dashed label chip (edit mode)
`blocks/tab/_tab.json`	Model	Definition `tab` (key-value), model `tab`
`blocks/tabs/tabs.js`, `tabs.css`	Block	Builds tabs from the markers; hides the marker on the page
`scripts/aem.js` (`toClassName`)	Core library	Normalises the ID
`utils/properties.js` (`readProperties`, `textOf`, `isOn`)	Util	Reads the key-value rows
`decorators/block-options-decorator.js`, `styles/styles.css`	Shared code	Additional tab
Placeholders	Data	None directly; an empty Title becomes "Tab {n}" in the Tabs block (`tabNumber`)

Related Components

Tabs Block (6.21); the components allowed after a Tab: Columns, Title, Text, Image, Button.

Applicable Templates

Any page that contains a Tabs Container.

Supported Content Types

Plain-text Title and ID, a boolean Active tab. The Tab has no content of its own.

6.22.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A Tabs Container at 1200px on the published page, with the Tab markers' titles shown only as the tab buttons (Consulting, Cloud, Cybersecurity) and no marker visible inside the panel.

Desktop (1200px and wider). On the page the Tab marker is not visible: its wrapper inside the panel is hidden (`display: none`), and its Title appears as the label of its tab button in the tab list (see 6.21.2 for the tab list styling). In Universal Editor edit mode the marker is shown at the top of its panel as an inline chip: dashed 1px brand border, 0.25rem radius, 0.25rem by 0.75rem padding, brand-coloured semibold extra-small text (0.875rem from 900px), 1rem space below.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Tabs Container at about 768px in Universal Editor edit mode, showing the dashed Tab chip above the panel content.

Tablet (600px to 1199px). Same behaviour: hidden on the page, chip in edit mode. The chip text uses 0.75rem below 900px and 0.875rem from 900px.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** The Tabs Container at 375px on the page: the tab titles in the scrolling tab list, no marker visible in the panel.

Mobile (below 600px). Same; the chip (edit mode) is 0.75rem text. The tab button labels (Titles) never wrap in the horizontal tab list.

6.22.3 Component Variations

The Tab has no style options. The behaviour modes an author can choose are the three below.

Variation 1 — Standard Tab

Variation Name

Standard Tab (Title only).

Business Purpose

Start a tab with a visible label; its id is generated.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-standard.png`
> **Purpose:** Standard Tab variation
> **Description:** Edit mode: a Tab chip “Cloud” at the top of a panel, and the “Cloud” tab button in the tab list.

Use Cases

Every tab that does not need a stable link or initial selection.

Authoring Guidelines

Add a Tab inside the Tabs Container; enter **Title**; leave ID empty and Active tab off.

Business Rules

Panel id generated as `{Tabs Container ID}-item-{n}` (or `tabs-{n}-item-{n}`); the tab is shown first only if it is the first tab and no other tab is Active or deep-linked.

Rendering Rules

The Tab block becomes `<p class="tab-title">Cloud</p>`; its wrapper is hidden on the page.

Responsive Behaviour

Desktop / tablet / mobile: not visible on the page; chip in edit mode at all widths.

SEO Impact

The Title is the text of a `<button>`; it is not a heading. The marker's own text is hidden on the page (`display: none`).

Accessibility Compliance

The Title is the accessible name of the tab and, through `aria-labelledby`, of its panel. The hidden marker is not exposed to assistive technology.

Variation 2 — Active Tab

Variation Name

Active Tab (Active tab = on).

Business Purpose

Show a specific tab, not the first one, when the page opens.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-active.png`
> **Purpose:** Active Tab variation
> **Description:** A Tabs Container loaded with its third tab selected because that Tab has Active tab switched on.

Use Cases

Promote the most relevant view (e.g. the current year's results).

Authoring Guidelines

Select the Tab; switch **Active tab** on. Keep it off on all other Tabs of the container.

Business Rules

If several Tabs are Active, the first one wins. A deep link `#id` and (in Universal Editor) the tab shown before a re-render take priority over Active tab.

Rendering Rules

Stored as `data-tab-active="true"`; the Tabs block selects it on load (no `cmp:show` for the initial selection).

Responsive Behaviour

Same at all widths.

SEO Impact

The active panel is the one visible without interaction.

Accessibility Compliance

Its tab button has `aria-selected="true"` and `tabindex="0"` on load.

Variation 3 — Tab with ID (deep link)

Variation Name

Tab with ID.

Business Purpose

Give a tab a stable id so links (`page#cloud`), Anchor Lists and analytics can address it.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-deep-link.png`
> **Purpose:** Tab with ID variation
> **Description:** A browser address ending in #cloud and the page scrolled to the Tabs with the Cloud tab selected.

Use Cases

Linking from navigation, campaigns, emails or other pages directly to a tab.

Authoring Guidelines

Enter **ID**, e.g. `cloud` (lower case, no spaces).

Business Rules

The ID is normalised (`toClassName`: lower case, non-alphanumerics to hyphens); an id already used on the page gets `-2`, `-3`, ... The tab button id is `{id}-tab`.

Rendering Rules

Stored as `data-tab-id`; used as the panel `id`.

Responsive Behaviour

Same at all widths; the page scrolls the Tabs below the fixed header (4rem).

SEO Impact

Fragment links (`#cloud`) are not separate URLs for search engines.

Accessibility Compliance

Deep links move the view to the Tabs and select the tab; focus is not moved.

6.22.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** Edit mode with a Tab chip selected at the top of a panel of a Tabs Container, the corresponding tab button highlighted, and the Tab selected in the content tree; letters A to C placed on the elements.

Annotations

- A = Tab chip (the Tab block in edit mode; select it to edit its properties).
- B = Tab button in the tab list (shows the Title).
- C = The components after the Tab (its content, up to the next Tab).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The properties panel of a Tab with the Content tab (Title, ID, Active tab) and the Additional tab.

Properties panel, Tab (model `tab`)

- **Content:** Title (required); ID; Active tab.
- **Additional:** Anchor ID; Hide block; Padding left / right / top / bottom; Margin left / right / top / bottom (no visible effect on the page, see 6.22.6).

6.22.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Open the page that has (or will have) a Tabs Container in Universal Editor.

Expected Result: Edit mode; the Tabs Container is visible.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-01-open-page.png`
> **Purpose:** Step 1
> **Description:** The page in Universal Editor with a Tabs Container.

Step 2 — Select the Tabs Container

Action: Select the Tabs Container in the content tree.

Expected Result: Its + (Add) button offers Tab, Columns, Title, Text, Image, Button.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-02-select-container.png`
> **Purpose:** Step 2
> **Description:** The Add menu of the Tabs Container listing Tab and the content components.

Step 3 — Add a Tab

Action: Choose **Tab**.

Expected Result: A Tab with the default Title "Tab" is added; a new tab button appears.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-03-add-tab.png`
> **Purpose:** Step 3
> **Description:** The new Tab chip "Tab" and its tab button.

Step 4 — Enter the Title

Action: In the Content tab, replace the Title with the tab label, e.g. "Consulting".

Expected Result: The chip and the tab button show the new Title.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-04-title.png`
> **Purpose:** Step 4
> **Description:** Title field "Consulting" and the updated tab button.

Step 5 — Enter the ID (optional)

Action: Enter an ID, e.g. `consulting`.

Expected Result: After the Tabs are rendered again (e.g. reload), the panel id is `consulting`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-05-id.png`
> **Purpose:** Step 5
> **Description:** ID field filled with consulting.

Step 6 — Set Active tab (optional)

Action: Switch **Active tab** on for the tab to show on load (one Tab only).

Expected Result: On the next load the tab is selected.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-06-active.png`
> **Purpose:** Step 6
> **Description:** Active tab switched on in the properties panel.

Step 7 — Add the tab's content

Action: With the Tab (or the container) selected, add Columns, Title, Text, Image or Button components and make sure they are placed after this Tab and before the next one.

Expected Result: The components appear in this tab's panel.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-07-content.png`
> **Purpose:** Step 7
> **Description:** Content tree: Tab "Consulting", Columns, Text, then Tab "Cloud".

Step 8 — Repeat and reorder

Action: Add the next Tab and its components; drag Tabs (with their components) to reorder.

Expected Result: The tab order follows the content tree.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-08-reorder.png`
> **Purpose:** Step 8
> **Description:** A Tab dragged above another Tab in the content tree.

Step 9 — Preview

Action: Switch to **Preview**.

Expected Result: The chips disappear; the tabs work as on the live page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-09-preview.png`
> **Purpose:** Step 9
> **Description:** Preview mode without chips.

Step 10 — Validate and publish

Action: Run the checklist below, then publish the page.

Expected Result: The live page shows the tabs with the right labels and initial tab.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tab-step-10-publish.png`
> **Purpose:** Step 10
> **Description:** The Publish dialog for the page.

Preview Process
Universal Editor Preview (no chips); Edge Delivery preview on `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}`; test `#id` links.

Publishing Process
Publish the page (the Tab is part of the page; nothing else to publish). Live: `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/{path}` ([To be confirmed: production domain and CDN]).

Validation Process
- ☐ Every Tab has a Title (no "Tab {n}" fallback labels).
- ☐ The first item in the Tabs Container is a Tab.
- ☐ At most one Tab has Active tab on.
- ☐ IDs are unique on the page, lower case, and `#id` links open the tab.
- ☐ Each Tab is followed by at least one component (no empty panels).
- ☐ In Preview, no chips are visible.

Rollback Process
Restore the previous page version and republish; or correct / delete the Tab and republish.

6.22.6 Authorable Field Specification
Model `tab` (Tab marker)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Tab: Content	tab	—	—	—	Tab properties	—	—	—
Title (`title`)	text	Yes (`required`)	No	"Tab" (new component template)	Tab button label	1 to 3 words	Enforced: required in the model; empty in content → "Tab {n}"	Consulting

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
ID (`tabId`)	text	No	No	empty (generated)	Panel id for deep links	Lower case, no spaces	Normalised with `toClassName`; de-duplicated `-2`, `-3`	consulting
Active tab (`activeTab`)	boolean	No	No	off	Shown when the page loads	At most one per container	Only “true” counts	on
Tab: Additional	tab	—	—	—	Block options	Not needed on a Tab	—	—
Anchor ID (`classes_anchor`)	text	No	No	empty	Id of the (hidden) Tab block	Use ID (Content) for deep links	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-consulting
Hide block (`classes_hideblock`)	boolean	No	No	false	Hides the Tab block	Does not remove the tab	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default	Spacing of the Tab block	No visible effect on the page	One of 15 options	pl-0
Padding right (`classes_paddingright`)	select	No	No	Default	As above	As above	One of 15 options	pr-0
Padding top (`classes_paddingtop`)	select	No	No	Default	As above	As above	One of 15 options	pt-0
Padding bottom (`classes_paddingbottom`)	select	No	No	Default	As above	As above	One of 15 options	pb-0
Margin left (`classes_marginleft`)	select	No	No	Default	As above	As above	One of 15 options	ml-0
Margin right (`classes_marginright`)	select	No	No	Default	As above	As above	One of 15 options	mr-0
Margin top (`classes_margintop`)	select	No	No	Default	As above	As above	One of 15 options	mt-0
Margin bottom (`classes_marginbottom`)	select	No	No	Default	As above	As above	One of 15 options	mb-0

Field details

- **Title**
  - Purpose: the tab button text; also `dc:title` of `cmp:show` and the chip text in edit mode.
  - Allowed Values: plain text. Character Limits: none enforced; Recommended: up to 25 characters (tab labels do not wrap in the horizontal list).
  - Business Rules: required in the model; if the content is empty anyway, the Tabs block uses “Tab {n}” (placeholder `tabNumber`). Edits update the tab button immediately in Universal Editor (`tabs:tab-update`).
  - Dependency Rules: none. Visibility Rules: always visible. Sample Content: “Consulting”.
- **ID**
  - Purpose: the panel `id`; `#id` links open the tab.
  - Allowed Values: any text, normalised to lower case with hyphens. Character Limits: none enforced; Recommended: under 30 characters.

    - Business Rules: empty → `{Tabs Container ID}-item-{n}`, else `tabs-{n}-item-{n}`; duplicates on the page get `-2`, `-3`. Changes apply when the Tabs are rendered again.
    - Dependency Rules: unique on the page. Visibility Rules: always visible.
    - Sample Content: `consulting`.
- **Active tab**
    - Purpose: initial tab. Allowed Values: on / off.
    - Business Rules: first Active Tab wins; overridden by a deep link and, in Universal Editor, by the tab shown before a re-render. Changes apply when the Tabs are rendered again.
    - Dependency Rules: none. Visibility Rules: always visible. Sample Content: on.
- **Additional tab fields**
    - Purpose: the common block options (Anchor ID, Hide block, 15-option padding / margin selects: Default; 0 (none); 0.25rem (space-100); 0.5rem (space-200); 0.75rem (space-300); 1rem (space-400); 1.25rem (space-500); 1.5rem (space-600); 2rem (space-800); 2.5rem (space-1000); 3rem (space-1200); 4rem (space-1600); 5rem (space-2000); 6rem (space-2400); 7.5rem (space-3000)).
    - Business Rules: they apply to the Tab block, which is hidden on the page, so they have no visible effect there. Hide block does not remove the tab from the tab list (the Tabs block still creates a tab for it). Anchor ID gives the hidden Tab block an id; links to it do not open the tab (only panel ids do). Padding / margin affect only the chip in edit mode.
    - Character Limits: Anchor ID: none enforced. Dependency Rules: none. Visibility Rules: always visible. Sample Content: leave all at their defaults.

No field of the Tab model has a `condition`.

6.22.7 Field Annotation Mapping

Marker	Field	Purpose
A	Tab chip (Title, ID, Active tab, Additional)	Selects the Tab and its properties
B	Title (shown on the tab button)	Tab label
C	Following components (not Tab fields)	The tab's content

6.22.8 Component Specification

Attribute	Value
Component Name	Tab (`tab`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`blocks/tabs`, `scripts/aem.js` (`toClassName`), `utils/properties.js`, block options
Component Type	Block used as an item marker inside the Tabs Container section (key-value block)
Supported Variants	Standard, Active, with ID (behaviour modes; no styles)
Supported Templates	Any page with a Tabs Container
Supported Metadata	None
Supported Tags	None
Supported Personalization	None

Attribute	Value
Caching Considerations	Part of the page HTML (Edge Delivery CDN)
Performance Considerations	Tiny script; no requests; loaded with its section after the Tabs block (which reads the Tab first)
Accessibility Compliance	Hidden on the page (not exposed); its Title names the tab and panel (WCAG 4.1.2, 2.4.6)
SEO Compliance	Marker text hidden; Title is button text, not a heading

6.22.9 Input Model

- **Content Inputs:** Title, ID, Active tab (key-value rows `title`, `tabId`, `activeTab`; positional rows, one per field in dialog order, are also accepted).
- **Metadata Inputs:** none. **Tags Inputs:** none.
- **Page Context Inputs:** its position in the Tabs Container (which components follow it).
- **Query Inputs:** none. **Manual Inputs:** all fields.

Authored HTML (illustrative):

```
<div class="tab" data-aue-resource="urn:.../tab_1" data-aue-model="tab">
  <div><div>title</div><div>Consulting</div></div>
  <div><div>tabId</div><div>consulting</div></div>
  <div><div>activeTab</div><div>true</div></div>
</div>
```

6.22.10 Output Model

Rendered Output: `div.tab.block` with data attributes `data-tab-title`, `data-tab-id`, `data-tab-active` (cached values) containing `p.tab-title`; inside the Tabs panel its wrapper `div.tab-wrapper` is hidden except in edit mode.

Generated HTML

```
<div class="tab-wrapper">
  <div class="tab block" data-block-name="tab" data-tab-title="Consulting"
    data-tab-id="consulting" data-tab-active="true">
    <p class="tab-title">Consulting</p>
  </div>
</div>
```

Generated JSON: none. **Generated Metadata:** none.

DOM events: `tabs:tab-update` (bubbles) after each decoration, `detail: { title, id, active }`; the enclosing Tabs updates the tab label.

Analytics Tracking: the Tab block is registered in the data layer component state as `tcs-corp/components/tab` (id `tab-{n}`, or its Anchor ID) with the Tabs block as `parentId`. It pushes no events; the Tabs block pushes `cmp:show` with the Tab's panel id and Title (see 6.21.10).

Personalization Output: none. **Search Index Output:** none.

6.22.11 Business Rules

1. Properties are read once and cached on the block as data attributes, so they can be read again after the block is decorated.
2. Key-value rows (`title`, `tabid`, `activetab` after normalisation) are used if present; otherwise the last cell of each row in dialog order (Title, ID, Active tab).
3. The ID is normalised with `toClassName`.

4. Active tab is on only for the text “true”.
5. Decoration replaces the content with `p.tab-title` showing the Title.
6. After decoration a `tabs:tab-update` event tells the Tabs block to update the label.
7. The Tab’s wrapper is hidden in Tabs panels except in Universal Editor edit mode.
8. The Tab is allowed only in the Tabs Container (`tabs` filter).
9. A Tab outside a Tabs Container would show its Title as a visible chip (the hiding rule is scoped to `.tabs .tabs-panel`).
10. Hide block on a Tab does not remove the tab from the tab list.
11. The Tab emits no console messages.

6.22.12 Author Best Practices

- **Recommended Content Length:** Title 1 to 3 words, up to 25 characters; ID under 30 characters (Recommended).
- **Image Dimensions:** not applicable.
- **CTA Writing Guidelines:** not applicable (no CTA on the Tab; put Buttons after it).
- **Accessibility Guidelines:** unique, descriptive Titles; avoid “Tab 1” style labels.
- **SEO Guidelines:** start the tab’s content with a Title (heading) that repeats or extends the tab label.
- **Metadata Guidelines:** not applicable.
- **Taxonomy Guidelines:** not applicable.
- **Localization Guidelines:** translate Titles; keep IDs the same across locales.

6.22.13 Troubleshooting Guide

Issue 1 — The tab label shows “Tab 2” instead of the Title

Symptoms: A generic label in the tab list.

Root Cause: The Title is empty, or the content before the first Tab created an untitled tab.

Diagnosis Steps: Select the Tab; check Title; check the content tree order.

Resolution Steps: Enter the Title; move stray components after a Tab.

Validation Steps: All tab labels are the Titles.

Preventive Measures: Add the Tab before its components.

Issue 2 — The Tab chip is visible on the live page

Symptoms: A dashed label appears on the published page.

Root Cause: The Tab is not inside a Tabs Container (e.g. moved to a normal section by copy / paste), so the hiding rule does not apply.

Diagnosis Steps: Check whether the section has the class `tabs-section`.

Resolution Steps: Move the Tab into a Tabs Container or delete it.

Validation Steps: No chip on the page.

Preventive Measures: Only add Tabs inside Tabs Containers.

Issue 3 — The ID does not work in links

Symptoms: `#My Tab` or `#my_tab` links do not open the tab.

Root Cause: The ID is normalised (`my-tab`); or duplicated and suffixed (`my-tab-2`).

Diagnosis Steps: Inspect the panel id.

Resolution Steps: Use lower-case, hyphenated IDs and the same value in links.

Validation Steps: The link opens the tab.

Preventive Measures: Simple, unique IDs.

Issue 4 — Active tab is ignored
Symptoms: Another tab is shown on load.

Root Cause: An earlier Tab is also Active; a `#id` in the address; Universal Editor keeps the tab shown before a re-render; or the change was not yet re-rendered.

Diagnosis Steps: Check all Tabs; load the page without a hash.

Resolution Steps: Leave Active on one Tab only; reload.

Validation Steps: The Active tab shows on a fresh load.

Preventive Measures: Checklist.

Issue 5 — Hiding a Tab does not hide the tab
Symptoms: Hide block is on but the tab button and panel remain.

Root Cause: Hide block hides the Tab block only, not the tab created from it.

Diagnosis Steps: Check the Additional tab.

Resolution Steps: Delete the Tab and its components (or move them out) to remove a tab.

Validation Steps: The tab is gone.

Preventive Measures: Do not use Hide block on Tabs.

Issue 6 — A component appears in the wrong tab
Symptoms: Content shows under the previous or next tab.

Root Cause: The component's position in the content tree is before / after the wrong Tab.

Diagnosis Steps: Check the order in the content tree.

Resolution Steps: Drag the component after its Tab.

Validation Steps: Content in the correct panel.

Preventive Measures: Review the content tree before publishing.

6.22.14 Support Runbook
Monitoring Requirements: covered by the Tabs monitoring (6.21.14); no Tab-specific monitoring.

Logs: no console messages from the Tab; module errors `failed to load block tab` (`scripts/aem.js`); Admin API page status; AEM author logs.

Common Errors

Error	Likely cause	First action
Generic "Tab n" label	Empty Title / content before first Tab	Fill Title / reorder
Chip visible on page	Tab outside a Tabs Container	Move or delete
Deep link fails	Normalised / duplicated ID	Inspect panel id
Active ignored	Several Active / hash / no re-render	Fix and reload
Hide block ineffective	By design	Delete the Tab instead

Escalation Matrix

Level	Who	What they check
L1	Author support	Title, ID, Active tab, content tree order, publication
L2	AMS / site admin	Page preview / live status, CDN cache
L3	Development team	`tab.js` / `tabs.js` behaviour, console errors
L4	Adobe Support	Universal Editor and Edge Delivery platform issues

Recovery Procedures: restore the page version and republish; code fixes via revert on `main`.

Health Checks: on the page, `document.querySelectorAll('.tabs .tab')` exist but are not visible; each Title appears as a tab button.

Operational Procedures: as for the Tabs (6.21.14).

Known Limitations: ID and Active tab changes apply only after the Tabs are rendered again; Hide block does not hide a tab; Additional options have no visible effect; no icon or badge on tabs.

6.23 Modal Block

The Modal block (`blocks/modal`) shows the content of a fragment page in a dialog over the page. It is the Block Collection modal of Edge Delivery Services, adapted for the TCS Corporate website: an authored **Modal** block renders a button that opens the dialog; in addition, any link to a page whose path contains `/modals/` opens as a modal, and other blocks (the Video block's **Play in a Pop-up**) use its `createModal()` / `openModal()` functions. The dialog is a native `<dialog>` opened with `showModal()`, named by its first heading, closable with the close button, the Escape key or a click on the backdrop, and tracked in the Adobe Client Data Layer (`cmp:show` / `cmp:hide`).

6.23.1 Component Purpose

Business Objective

Show supplementary content (details, forms, disclaimers, videos) without leaving the page, so visitors keep their context and return to the same place when they close the dialog.

Business Use Cases

- "Read more" details of a service, offer or award.
- Terms, disclaimers or privacy notices linked from a CTA.
- A short sign-up or contact form (Form Variations on the fragment page).
- A video played in a pop-up (Video block, Play in a Pop-up).
- Biographies of leaders opened from a profile listing.

When To Use

- The content is short, secondary and self-contained.
- The visitor should return to the same place on the page.

When Not To Use

- Primary content or content that must be indexed with the page (the dialog content is loaded from another page when opened).
- Long articles (use a page); multi-step processes; content that must be shared by URL.
- Automatic pop-ups on page load: Not implemented (as of this handover); the modal opens only on a visitor action.

Dependencies

Dependency	Type	Use
`blocks/modal/modal.js`	Block	`decorate` (opener button), `openModal(url)`, `createModal(nodes)`, visible-area tracking in frames
`blocks/modal/modal.css`	Block styles	Dialog, backdrop, sticky close button, page scroll lock
`blocks/modal/_modal.json`	Model	Definition and model `modal`
`blocks/fragment/fragment.js` (`loadFragment`)	Block	Loads and decorates the fragment page
`scripts/aem.js` (`buildBlock`, `decorateBlock`, `loadBlock`, `loadCSS`)	Core library	Creates the dialog's block and loads its CSS
`scripts/scripts.js` (`moveInstrumentation`, `autolinkModals`)	Shared code	Editor instrumentation; links to `/modals/` pages open as a modal
`decorators/button-decorator.js` (`decorateButton`)	Decorator	Opener as a primary button
`decorators/icon-button-decorator.js` (`createIconButton`)	Decorator	Close icon button
`decorators/data-layer-decorator.js` (`componentIdOf`)	Decorator	Component id; `cmp:hide` on close
`utils/analytics.js` (`pushComponentEvent`)	Util	`cmp:show` on open
`utils/ids.js` (`uniqueId`)	Util	Id for the heading that names the dialog
`utils/placeholders.js`	Util	`open` ("Open"), `close` ("Close"), `dialog` ("Dialog")
`icons/close.svg`	Icon	Close button
Fragment pages	Content	The dialog content, e.g. under the locale's `modals/` or `references/` folder

Related Components

- Fragment block (same loading mechanism, inline instead of a dialog).
- Video block (Play in a Pop-up uses `createModal`).
- Button (default content): a Button linking to a `/modals/` page also opens a modal.
- Form Variations: a form can be placed on a modal fragment page.

Applicable Templates

Any page; the Modal block is allowed in every standard section (`section` filter). Links to `/modals/` pages work anywhere on the page (including header and footer links).

Supported Content Types

Any content of the fragment page (sections, default content and blocks), decorated like a page.

6.23.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** At 1200px: the page dimmed by the dark 75% backdrop, a white dialog with 1rem rounded corners

> centred vertically, at most 56.25rem wide, the close (X) icon button at the top right, a heading and text inside with 2rem padding.

Desktop (1200px and wider). The opener is a primary button where the block is placed. When opened, the page behind is covered with a backdrop of near-black at 75% opacity and stops scrolling (`body.modal-open`). The dialog is centred vertically in the visible area, `calc(100vw - 4rem)` wide up to 56.25rem (900px), 1rem corner radius, default background and text colours, and never taller than the visible area minus 2 × 2.5rem; longer content scrolls inside the dialog. The close button (tertiary icon button) sticks to the top right (0.5rem from the top) while the content scrolls. Content padding is 2rem on the sides and bottom (from 900px).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** At about 768px: the dialog nearly full width with 1.5rem margins left and right, content padding 1.5rem, the sticky close button at the top right of a long, scrolling content.

Tablet (600px to 1199px). From 900px as desktop. Between 600px and 899px the dialog is `calc(100vw - 3rem)` wide, the vertical gap is 1.5rem and the content padding 1.5rem.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** At 375px: the dialog 1.5rem from each screen edge, as tall as the content up to the screen height minus 3rem, with the close button at the top right and the content scrolling inside.

Mobile (below 600px). Same as small tablet: full width minus 3rem, 1.5rem padding, height limited to the dynamic viewport height (`100dvh`) minus 3rem; the content scrolls inside the dialog; overscroll does not scroll the page.

6.23.3 Component Variations

Variation 1 — Modal block (button opener)

Variation Name

Modal block.

Business Purpose

Give authors an explicit button that opens a fragment page in a dialog.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-block-button.png`
> **Purpose:** Modal block variation
> **Description:** A primary button “View our awards” on the page and, next to it, the dialog it opens.

Use Cases

“Learn more”, “View terms”, “Register interest” buttons.

Authoring Guidelines

Add **Modal** to a section; in the Content tab choose **Modal Content** (the fragment page) and enter **Button Text**.

Business Rules

- The block's first link becomes the button; without a link the block is left unchanged.
- Button text: the Button Text; if it is empty or equals the URL, "Open" (placeholder `open`).
- The opener is a `<button>` (not a link), so it also opens the dialog in Universal Editor's preview (the editor navigates to the targets of clicked links).
- A failed fragment (missing, not published, path not starting with `/`) opens nothing.

Rendering Rules

- `<button type="button" aria-haspopup="dialog" class="button primary">` replaces the block content; the link's editor instrumentation moves to the button; a link title different from the text and URL becomes the button's `title`.
- On click: the fragment is loaded and decorated, a new dialog is created and shown.

Responsive Behaviour

The button follows the site button styles at all widths; the dialog as described in 6.23.2.

SEO Impact

The dialog content is not in the page HTML (fetched on click); the opener is a button, so crawlers do not follow it to the fragment page. Modal fragment pages under `modals/` or `references/` are excluded from the query index.

Accessibility Compliance

Native modal dialog (focus moves into it, the page behind is inert, Escape closes); `aria-haspopup="dialog"` on the opener; accessible name from the first heading (`aria-labelledby`) or "Dialog"; close button labelled "Close"; focus returns to the opener on close.

Variation 2 — Link to a /modals/ page (auto-linked modal)

Variation Name

Auto-linked modal.

Business Purpose

Open dialogs from ordinary links (text links, Buttons, navigation links) without adding a Modal block.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-auto-link.png`
> **Purpose:** Auto-linked modal variation
> **Description:** A text link "privacy notice" in a paragraph and the dialog showing the /en/modals/privacy-notice fragment.

Use Cases

Inline links to terms or disclaimers; Buttons in Hero / Teaser CTAs; footer links.

Authoring Guidelines

Create the fragment page in a folder named `modals` (e.g. `/content/tcs-corp/us/en/modals/privacy-notice`, published as `/en/modals/privacy-notice`) and link to it from any text, Button or CTA.

Business Rules

- Any click on a link whose URL path contains `/modals/` opens it in a modal, unless another script already handled the click (`defaultPrevented`).
- The listener is attached in the lazy phase; clicks before that follow the link normally.
- In Universal Editor, link clicks may be handled by the editor (navigation) first.

Rendering Rules

`scripts/scripts.js` imports `blocks/modal/modal.js` on the first such click and calls `openModal(href)` (absolute URLs are reduced to their path).

Responsive Behaviour

As Variation 1.

SEO Impact

The link is a normal `<a href>`, so crawlers can follow it to the fragment page; `modals/` pages are excluded from the query index (`/**/modals/**`) and therefore from the sitemap.

Accessibility Compliance

As Variation 1, except the opener is a link: it has no `aria-haspopup`; write link text that says a dialog opens if that matters ("View the privacy notice").

Variation 3 — Dialog opened by another component (Video: Play in a Pop-up)

Variation Name

Component dialog (`createModal`).

Business Purpose

Reuse the same accessible dialog for component features, currently the Video block's **Play in a Pop-up** option.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-video-popup.png`
> **Purpose:** Component dialog variation
> **Description:** A video playing in the modal dialog over the dimmed page after the visitor selected the play button of a Video block with Play in a Pop-up on.

Use Cases

Videos in a pop-up.

Authoring Guidelines

Configured in the Video block (Play in a Pop-up = on); no Modal block needed.

Business Rules

The calling block supplies the content (no fragment); naming, close behaviour and data layer events are the same as for the other variations.

Rendering Rules

`createModal([content])` then `showModal()`; Video adds `.video-modal-player` styles inside `.modal`.

Responsive Behaviour

As 6.23.2; the video keeps its aspect ratio inside the dialog.

SEO Impact

None beyond the Video block.

Accessibility Compliance

Named "Dialog" (the content has no heading) unless the content provides a heading; focus returns to the play button on close.

6.23.4 Authoring Experience

> [Screenshot Placeholder]
> **File name:** `screenshots/modal-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** Edit mode with a Modal block selected in a section, showing the opener button labelled with the Button Text and the block outline; letters A and B on the elements.

Annotations

- A = Modal block (select it for Modal Content, Button Text and Additional).
- B = Opener button (shows the Button Text; in Preview mode it opens the dialog).

> [Screenshot Placeholder]
> **File name:** `screenshots/modal-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Modal properties panel with the Content tab (Modal Content picker and Button Text) and the Additional tab.

Properties panel, Modal (model `modal`)

- **Content:** Modal Content (content picker for the fragment page; description "A fragment page, e.g. under /modals/. Opens in a dialog."); Button Text.
- **Additional:** Anchor ID; Hide block; Padding left / right / top / bottom; Margin left / right / top / bottom.

> **Note:** The dialog content is edited on the fragment page itself, not in the Modal block.

6.23.5 Authoring Guide

Step 1 — Create the fragment page

Action: In AEM Sites, create a page for the dialog content in the locale, preferably in a `modals` folder (e.g. `/content/tcs-corp/us/en/modals/awards`). Start it with a heading (it names the dialog), add the content and publish it.

Expected Result: The fragment page exists and is published.

> [Screenshot Placeholder]
> **File name:** `screenshots/modal-step-01-fragment-page.png`
> **Purpose:** Step 1
> **Description:** The Sites console with the modals folder and the awards page.

Step 2 — Open the target page in Universal Editor

Action: Open the page where the button should appear.

Expected Result: Edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-step-02-open-page.png`
> **Purpose:** Step 2
> **Description:** The page open in Universal Editor.

Step 3 — Add the Modal block

Action: Select a section, **+** (Add), **Modal**.

Expected Result: An empty Modal block is added (nothing visible until a link is set).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-step-03-add-block.png`
> **Purpose:** Step 3
> **Description:** The component picker with Modal and the new block in the content tree.

Step 4 — Choose the Modal Content

Action: In the Content tab, use the **Modal Content** picker to select the fragment page.

Expected Result: A button appears (labelled with the Button Text, or "Open").

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-step-04-content.png`
> **Purpose:** Step 4
> **Description:** The content picker with the awards page selected.

Step 5 — Enter the Button Text

Action: Enter an action-oriented label, e.g. "View our awards".

Expected Result: The button shows the text.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-step-05-button-text.png`
> **Purpose:** Step 5
> **Description:** Button Text field and the updated button.

Step 6 — Set the Additional options (optional)

Action: Set Anchor ID, Hide block, padding or margin.

Expected Result: Options applied to the opener block.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-step-06-additional.png`

> **Purpose:** Step 6
> **Description:** Additional tab with Margin top set.

Step 7 — Preview and test the dialog

Action: Switch to **Preview**, select the button, scroll the content, close the dialog with the close button, Escape and a click on the backdrop.

Expected Result: The dialog opens with the fragment content, closes in all three ways, and focus returns to the button.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-step-07-preview.png`
> **Purpose:** Step 7
> **Description:** The open dialog in Universal Editor preview, centred in the visible area of the editor frame.

Step 8 — Validate

Action: Run the checklist below.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-step-08-validate.png`
> **Purpose:** Step 8
> **Description:** Browser developer tools showing the dialog's aria-labelledby pointing to its heading.

Step 9 — Publish

Action: Publish the fragment page (if changed) and the page.

Expected Result: The button opens the dialog on the live site.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/modal-step-09-publish.png`
> **Purpose:** Step 9
> **Description:** The Publish dialog listing the page and the referenced fragment page.

Preview Process

1. Universal Editor Preview: the button opens the dialog (the dialog is kept in the visible part of the editor frame).
2. Edge Delivery preview: `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}`; the fragment must be previewed as well (`/{locale}/modals/{name}.plain.html` must load).
3. Test at mobile, tablet and desktop widths with short and long content.

Publishing Process

1. Publish the fragment page first, then the page.
2. Live: `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/{path}` ([To be confirmed: production domain and CDN]).

Validation Process

- ☐ Modal Content points to an existing, published page in the same locale.
- ☐ Button Text is set and describes the action.
- ☐ The fragment page starts with a heading (dialog name).
- ☐ The dialog opens, scrolls inside when long, and closes with the close button, Escape and a backdrop click; focus returns to the button.
- ☐ The page behind does not scroll while the dialog is open.
- ☐ Links in the dialog work.
- ☐ `window.adobeDataLayer` shows `cmp:show` on open and `cmp:hide` on close.
- ☐ Correct layout at below 600px, 600 to 899px and 900px and wider.

Rollback Process

1. Restore the previous version of the page and / or the fragment page and republish.
2. Urgent removal of the opener: Hide block on and publish; to stop auto-linked modals, change or remove the links, or unpublish the fragment (the dialog then no longer opens).
3. Code regressions: revert on `main`.

6.23.6 Authorable Field Specification

Model `modal` (Modal block)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Tab: Content	tab	—	—	—	Content	—	—	—
Modal Content (`link`)	aem-content	No (not enforced; required in practice)	No	empty	The fragment page shown in the dialog	Pick a page, e.g. under `modals/`	None in model; code: must resolve to a site path starting with `/`	/content/tcs-corp/us/en/modals/awards
Button Text (`linkText`)	text	No	No	empty (fallback “Open”)	Label of the opener button	Verb + object	None in model; empty or equal to the URL → “Open”	View our awards
Tab: Additional	tab	—	—	—	Block options	—	—	—
Anchor ID (`classes_anchor`)	text	No	No	empty	Block id for # links	Start with `anchor-`	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-awards
Hide block (`classes_hideblock`)	boolean	No	No	false	Hides the opener	Prepare before release	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default	Inner spacing left	Only if needed	One of 15 options	pl-0
Padding right (`classes_paddingright`)	select	No	No	Default	Inner spacing right	As above	One of 15 options	pr-0
Padding top (`classes_paddingtop`)	select	No	No	Default	Inner spacing top	As above	One of 15 options	pt-400
Padding bottom (`classes_paddingbottom`)	select	No	No	Default	Inner spacing bottom	As above	One of 15 options	pb-400

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Margin left (`classes_marginleft`)	select	No	No	Default	Outer spacing left	As above	One of 15 options	ml-0
Margin right (`classes_marginright`)	select	No	No	Default	Outer spacing right	As above	One of 15 options	mr-0
Margin top (`classes_margintop`)	select	No	No	Default	Outer spacing top	As above	One of 15 options	mt-600
Margin bottom (`classes_marginbottom`)	select	No	No	Default	Outer spacing bottom	As above	One of 15 options	mb-600

Field details

- **Modal Content**
  - Purpose: the page whose content fills the dialog.
  - Allowed Values: a site page chosen with the content picker. The model description suggests a page under `/modals/`; any page path works for the Modal block.
  - Character Limits: not applicable.
  - Business Rules: the block's first link is used; absolute URLs are reduced to their path; the path must start with `/` (not `//`); `.html` / `.plain.html` is stripped and `{path}.plain.html` is fetched each time the dialog opens; a failed request opens nothing and shows no message.
  - Dependency Rules: the fragment page must be published (live) or previewed (preview site).
  - Visibility Rules: always visible (no `condition`).
  - Sample Content: `/content/tcs-corp/us/en/modals/awards` (published `/en/modals/awards`).
- **Button Text**
  - Purpose: the visible label of the opener.
  - Allowed Values: plain text. Character Limits: none enforced; Recommended: up to 30 characters.
  - Business Rules: empty or identical to the URL → "Open" (placeholder `open`). With the link it forms one link (field collapsing of `link` + `linkText`) that the block converts into a button.
  - Dependency Rules: needs Modal Content (without a link no button is rendered).
  - Visibility Rules: always visible. Sample Content: "View our awards".
- **Anchor ID / Hide block / Padding (4) / Margin (4)**
  - Purpose: the common block options; Allowed Values: Anchor ID `anchor-…`; Hide block on / off; each spacing select: Default; 0 (none); 0.25rem (space-100); 0.5rem (space-200); 0.75rem (space-300); 1rem (space-400); 1.25rem (space-500); 1.5rem (space-600); 2rem (space-800); 2.5rem (space-1000); 3rem (space-1200); 4rem (space-1600); 5rem (space-2000); 6rem (space-2400); 7.5rem (space-3000).
  - Character Limits: Anchor ID none enforced (Recommended under 40 characters).
  - Business Rules: they apply to the opener block on the page, not to the dialog. Anchor ID becomes the block id (also used as the data layer component id of the opener); duplicates get `-2`, `-3`.
  - Dependency Rules: none. Visibility Rules: always visible. Sample Content: Margin top = 1.5rem (space-600).

6.23.7 Field Annotation Mapping

Marker	Field	Purpose
A	Modal block (Modal Content, Button Text, Additional)	Block properties
B	Button Text (on the opener button)	Opens the dialog

6.23.8 Component Specification

Attribute	Value
Component Name	Modal (`modal`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`blocks/fragment`, `scripts/aem.js`, `scripts/scripts.js`, `decorators/button-decorator.js`, `decorators/icon-button-decorator.js`, `decorators/data-layer-decorator.js`, `utils/analytics.js`, `utils/ids.js`, `utils/placeholders.js`, `icons/close.svg`
Component Type	Block (opener) plus a dialog block created at runtime
Supported Variants	Modal block (button); auto-linked `/modals/` links; component dialogs (Video Play in a Pop-up)
Supported Templates	Any page
Supported Metadata	None
Supported Tags	None
Supported Personalization	None; `cmp:show` / `cmp:hide` are available for future Adobe Target / Journey Optimizer use (not implemented)
Caching Considerations	Fragment `.plain.html` is cached by the Edge Delivery CDN and the browser; republish the fragment to update every opener
Performance Considerations	No cost until opened: the fragment, modal CSS and the auto-link module load on click; the fragment is fetched again at each opening
Accessibility Compliance	WCAG 2.1 AA: native modal dialog (focus containment, Escape), accessible name (4.1.2), focus return (2.4.3), labelled icon close button (1.1.1, 4.1.2), `aria-haspopup` on the opener
SEO Compliance	Content not part of the page HTML; `modals/` and `references/` pages excluded from the query index and sitemap

6.23.9 Input Model

- **Content Inputs:** the Modal Content link and the Button Text (one link); the fragment page's content.
- **Metadata Inputs:** none. **Tags Inputs:** none.
- **Page Context Inputs:** whether the page runs in a frame (Universal Editor): then the visible area is measured; the locale (placeholders).
- **Query Inputs:** none.
- **Manual Inputs:** both fields; the fragment page content.

Authored HTML received by the block (illustrative; the AEM path is mapped to the published path on Edge Delivery):

```
<div class="modal mt-600">
  <div>
    <div>
      <p class="button-container">
        <a href="/en/modals/awards" title="View our awards">View our awards</a>
      </p>
    </div>
  </div>
</div>
```

6.23.10 Output Model

Rendered Output

- On the page: `div.modal.block` containing `button.button.primary[aria-haspopup=dialog]`.
- When open: a new `div.modal.block` appended to `main`, containing `dialog[aria-labelledby | aria-label] > button.close-button` (tertiary icon button, `aria-label` "Close") and `div.modal-content` (the fragment's sections); `body` gets `modal-open`. The dialog block is removed when the dialog closes.

Generated HTML

```
<div class="modal block" data-block-name="modal" data-block-status="loaded">
  <dialog aria-labelledby="modal-title-1" open>
    <button type="button" class="close-button button tertiary" aria-label="Close"
      title="Close"></button>
    <div class="modal-content">
      <div class="section">
        <div class="default-content-wrapper"><h2 id="modal-title-1">Our awards</h2>...</
div>
      </div>
    </div>
  </dialog>
</div>
```

Generated JSON: none. **Generated Metadata:** none.

Analytics Tracking

- Opening pushes `cmp:show` with the dialog block's component id (`modal-{n}`; a new id at every opening because a new block is created):

```
{ "event": "cmp:show", "eventInfo": { "path": "component.modal-2", "interaction":
"dialog" } }
```

- Closing (any way) pushes `cmp:hide` (captured `close` event):

```
{ "event": "cmp:hide", "eventInfo": { "path": "component.modal-2", "interaction":
"dialog" } }
```

- The opener button and the close button push `cmp:click` (`interaction` "button", `dc:title` the button text / "Close", `xdm:linkType` "other", `breakpoint`); links inside the dialog push `cmp:click` with `xdm:linkURL` and `xdm:linkType`.
- Component state: the opener block is registered as `tcs-corp/components/modal`; the dialog block may be registered by the delayed `trackBlocks` pass if it is open at that time.

Personalization Output: none. **Search Index Output:** none.

6.23.11 Business Rules

1. Modal block without a link: nothing is changed.
2. Opener text: link text, else "Open" when the text is empty or equals the URL.
3. The link `title` is copied to the button only if it differs from the text and the URL.
4. The opener is a `<button type="button">` with `aria-haspopup="dialog"`, styled as a primary button; the link's editor instrumentation moves to it.
5. Absolute URLs are reduced to their path; fragment paths must start with `/` and not `//`.
6. The fragment is fetched (`{path}.plain.html`) and fully decorated each time the dialog opens; a failed fetch silently does nothing.

7. The dialog's name: its first heading (h1 to h6; an id `modal-title-{n}` is added if it has none), else `aria-label` "Dialog" (placeholder `dialog`).
8. The close button is labelled "Close" (placeholder `close`).
9. A click on the backdrop (outside the dialog's box) closes the dialog; clicks inside do not.
10. Escape closes the dialog (native `<dialog>`).
11. On close: the page scroll lock is removed, the dialog block is removed from the page and focus returns to the element that opened it.
12. On open: the page is locked (`body.modal-open { overflow: hidden }`) and the dialog scrolls to the top.
13. In a frame (Universal Editor) the dialog is hidden until the visible area is measured, is centred in that area, and wheel scrolling outside it or past its ends is blocked.
14. Links whose URL path contains `/modals/` open as a modal, unless the click was already handled; this listener starts in the lazy phase.
15. `cmp:show` on open and `cmp:hide` on close (interaction `dialog`).
16. The dialog is at most 56.25rem wide and never taller than the visible area minus twice the gap (1.5rem below 900px, 2.5rem from 900px).

6.23.12 Author Best Practices

- **Recommended Content Length:** a heading and up to about 300 words, or a short form; longer content belongs on a page (Recommended).
- **Image Dimensions:** the content width is at most about 52rem (56.25rem minus padding); images of 1200px width are sufficient; use landscape images (16:9 or 3:2).
- **CTA Writing Guidelines:** Button Text with a verb and object ("View our awards", "Read the terms"); avoid "Click here" and "Open".
- **Accessibility Guidelines:** start every modal page with a heading; keep a single purpose per dialog; do not open a modal from inside a modal.
- **SEO Guidelines:** keep indexable content on regular pages; use modals for supplementary content only.
- **Metadata Guidelines:** modal fragment pages need no SEO metadata (excluded from the index); give them a clear Title for authors.
- **Taxonomy Guidelines:** not applicable.
- **Localization Guidelines:** create the modal page in each locale (`/de-de/modals/...`) and link the locale's own page; the "Open", "Close" and "Dialog" texts come from the locale's placeholders sheet.

6.23.13 Troubleshooting Guide

Issue 1 — The button does nothing

Symptoms: Selecting the opener shows no dialog and no error.

Root Cause: The fragment page is not published / previewed, the path is wrong, or the link is not a site path.

Diagnosis Steps: Inspect the button's original link in the authored content; open `{path}.plain.html` on the same host; check the Network tab for a 404.

Resolution Steps: Publish the fragment; correct Modal Content.

Validation Steps: The dialog opens.

Preventive Measures: Publish fragments with their referencing pages.

Issue 2 — No button appears

Symptoms: The Modal block is empty on the page.

Root Cause: Modal Content is not set (no link), or Hide block is on.

Diagnosis Steps: Check both fields.

Resolution Steps: Choose the fragment; switch Hide block off.

Validation Steps: The button is visible.

Preventive Measures: Checklist.

Issue 3 — The button says "Open"

Symptoms: Generic label.

Root Cause: Button Text is empty or the same as the URL.

Diagnosis Steps: Check Button Text.

Resolution Steps: Enter a descriptive text.

Validation Steps: The label is correct.

Preventive Measures: Always fill Button Text.

Issue 4 — A link opens a dialog instead of navigating (or the reverse)

Symptoms: A link unexpectedly opens a modal, or a modal link navigates to the page.

Root Cause: Its path contains `/modals/` (auto-link), or the click happened before the lazy phase / in Universal Editor (editor navigation).

Diagnosis Steps: Check the link URL; retest on the preview site after the page loads.

Resolution Steps: Move content pages out of `modals` folders; for dialogs use a Modal block in the editor.

Validation Steps: Behaviour as intended on `.aem.page`.

Preventive Measures: Reserve `modals` folders for dialog content.

Issue 5 — Screen readers announce "Dialog" only

Symptoms: Generic dialog name.

Root Cause: The fragment page has no heading.

Diagnosis Steps: Inspect the dialog for `aria-label="Dialog"`.

Resolution Steps: Add a heading at the top of the fragment page.

Validation Steps: `aria-labelledby` points to the heading.

Preventive Measures: Template for modal pages starting with a heading.

Issue 6 — Dialog content looks unstyled or blocks are missing

Symptoms: Blocks in the dialog show raw rows.

Root Cause: A block on the fragment page failed to load, or a block's styles depend on its page context.

Diagnosis Steps: Console errors `failed to load block …`; open the fragment page on its own.

Resolution Steps: Fix the block content; escalate block errors to development.

Validation Steps: Content renders like on the fragment page.

Preventive Measures: Preview fragment pages before linking them.

Issue 7 — No cmp:show / cmp:hide events

Symptoms: Analytics shows no dialog interactions.

Root Cause: Adobe Tags not yet embedded (events exist but are not sent), or the dialog did not open.

Diagnosis Steps: `window.adobeDataLayer.filter((e) => e.eventInfo?.interaction === 'dialog')`.

Resolution Steps: Configure the Adobe Tags rule once embedded; otherwise fix Issue 1.

Validation Steps: Events present with `component.modal-{n}` paths.

Preventive Measures: Note that the component id changes at every opening; build rules on `interaction: dialog`.

6.23.14 Support Runbook

Monitoring Requirements

- 404 monitoring for `/{locale}/modals/*.plain.html` and other fragment requests.
- RUM for pages with modals (INP of the opener).
- [To be confirmed: monitoring tool and thresholds — owner: AMS lead].

Logs

- Browser console: the block logs nothing itself; block load errors (`failed to load block modal`, or of blocks inside the fragment) come from `scripts/aem.js`.
- Network tab: the `.plain.html` request of the fragment and `blocks/modal/modal.css`.
- Admin API status of the page and the fragment page; AEM author logs for publication errors.

Common Errors

Error	Likely cause	First action
Nothing opens	Fragment 404 / wrong path	Open `{path}.plain.html`
No button	No link / Hide block	Check fields
“Open” label	Empty Button Text	Fill it
Unexpected modal	Link path contains `/modals/`	Move the page
“Dialog” name	No heading in fragment	Add a heading
Page scrolls behind	`modal-open` missing (script error)	Check console

Escalation Matrix

Level	Who	What they check
L1	Author support	Modal Content, Button Text, fragment publication, heading
L2	AMS / site admin	Fragment URL status (Admin API), CDN cache, redirects, Code Sync
L3	Development team	`modal.js` / `modal.css`, `autolinkModals`, blocks inside fragments, data layer
L4	Adobe Support	Edge Delivery delivery, Universal Editor frame behaviour

Recovery Procedures: restore and republish the fragment or page; Hide block for the opener; revert code on `main`.

Health Checks: opener opens the dialog; Escape, close button and backdrop close it; focus returns; `document.querySelector('main > .modal')` exists only while open.

Operational Procedures: maintain modal pages per locale in `modals/` folders; republish them after edits; check all openers after renaming or moving a modal page.

Known Limitations

- No automatic opening (on load, timer or exit intent); no size or style options.
- The fragment is fetched again at every opening; failures are silent.
- A new dialog block (and data layer id) is created at every opening.
- Dialog content is not part of the page HTML (not indexed with the page).

- Auto-linking depends on `/modals/` in the URL path and starts in the lazy phase.

6.24 Gen Carousel Block

The Gen Carousel (`blocks/gen-carousel`) shows a sequence of slides, one at a time, with previous / next buttons, dots, swipe and keyboard navigation and an optional autoplay with a pause / play button. It is a **section-based** component: authors add a **Gen Carousel** section at the page level and place **Hero** or **Teaser** blocks in it; `scripts/scripts.js` (`buildCarouselSections`) wraps the section content in a synthetic `gen-carousel` block, and every child of that block becomes a slide, whatever it contains. The carousel therefore never depends on the markup of its slides; each Hero or Teaser keeps its own design and authoring. It follows the WAI-ARIA carousel pattern (region and slide role descriptions from the placeholders sheet) and pushes a `cmp:show` data layer event for every slide change.

6.24.1 Component Purpose

Business Objective

Feature several campaigns, stories or offers in the same prominent space (typically the top of the home page) while keeping each message a full-size Hero or Teaser.

Business Use Cases

- Home page hero rotation (brand campaign, latest results, major event).
- Rotating success stories or case studies (Teasers).
- Event or campaign highlights on landing pages.

When To Use

- Two to five messages of similar importance that share one space.
- Each message is a complete Hero or Teaser with its own CTA.

When Not To Use

- One message only (a carousel with one slide shows no controls; use the Hero or Teaser directly).
- Critical information every visitor must see (later slides are often not viewed).
- Lists of many items (cards, news): use the List or Cards block (horizontal scroll layouts).
- Content other than Hero / Teaser: the section only allows these two blocks (filter `gen-carousel`).

Dependencies

Dependency	Type	Use
`blocks/gen-carousel/gen-carousel.js`	Block	Slides, controls, autoplay, keyboard, data layer, editor selection
`blocks/gen-carousel/gen-carousel.css`	Block styles	Scroll-snap track, slide sizing, controls and dots
`blocks/gen-carousel/_gen-carousel.json`	Model	Section definition `gen-carousel`, model, filter (hero, teaser)
`scripts/scripts.js` (`buildCarouselSections`, `wrapSectionContent`)	Shared code	Builds the synthetic block from the section and its options
`decorators/button-decorator.js` (`decorateButton`)	Decorator	Pause / play icon switching
`decorators/icon-button-decorator.js` (`createIconButton`)	Decorator	Previous, next, pause / play icon buttons

Dependency	Type	Use
`utils/analytics.js` (`pushComponentEvent`)	Util	`cmp:` show per slide change
`utils/environment.js` (`isEditing`, `prefersReducedMotion`, `scrollBehavior`)	Util	Editor behaviour; reduced motion
`utils/viewport.js` (`watchVisibility`)	Util	Pauses autoplay off screen
`utils/ids.js` (`uniqueId`)	Util	Ids `gen-carousel-{n}`, `…-slides`, `…-slide-{i}`
`utils/placeholders.js`	Util	`carouselRole`, `carousel`, `slideRole`, `slideOf`, `showSlide`, `pauseSlides`, `playSlides`, `previousSlide`, `nextSlide`
`decorators/data-layer-decorator.js`	Decorator	`cmp:click` for controls (`data-cmp-interaction`)
`icons/chevron-left.svg`, `chevron-right.svg`, `pause.svg`, `play.svg`	Icons	Control icons
Hero and Teaser blocks	Blocks	The slides

Related Components

Hero Block and Teaser Block (the slides); List / Cards (horizontal scroll card lists); Tabs (switchable content without rotation).

Applicable Templates

Any page; added at the page level (the `main` filter allows `section`, `gen-carousel` and `tabs`). Typical: home template, landing pages.

Supported Content Types

Hero and Teaser blocks with their own images, headings, text and CTAs.

6.24.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** At 1200px: a Gen Carousel of three Hero slides showing slide 1 at full content width, and centred below it 2rem lower the pause button, previous arrow, three dots (the first a wider brand-coloured pill) and next arrow.

Desktop (1200px and wider). One slide fills the full width of the section content area; slides of different blocks are stretched to the same height. Below the slides, the controls are centred in one row, 2rem below the slides (from 900px), 1rem apart: pause / play (Auto modes), previous (Manual modes), the dots, next (Manual modes). Each dot has a 1.5rem round hit area with a 0.5rem dot in the tertiary neutral border colour; the current slide's dot is a 1.5rem wide brand-coloured pill; hovering a dot turns it to the brand hover colour. Slides move with a smooth horizontal scroll and snap into place.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** At about 768px: a Teaser slide filling the width, with the controls centred below it 1.5rem lower.

Tablet (600px to 1199px). Same layout. Below 900px the controls are 1.5rem below the slides. In Manual modes the slides can be swiped horizontally (scroll snap, one slide at a time).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** At 375px: a Hero slide in its mobile layout, and below it the controls wrapped onto two lines if there are many dots.

Mobile (below 600px). One slide at full width in its own mobile layout (from the Hero / Teaser CSS); swipe in Manual modes; the controls and dots wrap onto several lines when they do not fit.

6.24.3 Component Variations

Variation 1 — Navigation: Manual

Variation Name

Manual (`carouselNavigation` = `manual`, default).

Business Purpose

The visitor controls the slides; nothing moves on its own (the most accessible option).

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-manual.png`
> **Purpose:** Manual navigation
> **Description:** Controls with previous arrow, dots and next arrow, no pause button.

Use Cases

Stories the visitor reads at their own pace; most carousels.

Authoring Guidelines

Gen Carousel section, Content tab, **Navigation** = Manual. The Autoplay interval field is hidden.

Business Rules

No autoplay; previous / next wrap around (from the last slide, next goes to the first).

Rendering Rules

Class `manual`; controls: previous, dots, next; swipe enabled (horizontal scroll with snap); ArrowLeft / ArrowRight work while focus is on the controls.

Responsive Behaviour

Same controls at all widths; swipe on touch devices.

SEO Impact

All slides are in the page HTML and indexable; each Hero / Teaser keeps its own headings.

Accessibility Compliance

No moving content (WCAG 2.2.2 not triggered); slide changes announced politely (`aria-live="polite"`); off-screen slides are `inert`.

Variation 2 — Navigation: Auto

Variation Name

Auto (`carouselNavigation` = `auto`).

Business Purpose

Slides advance on their own, like a rotating banner, with a pause button.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-auto.png`
> **Purpose:** Auto navigation
> **Description:** Controls with only the pause button and the dots; no previous / next arrows.

Use Cases

Ambient brand rotation where interaction is not expected.

Authoring Guidelines

Navigation = Auto; set **Autoplay interval (seconds)** (2 to 30, default 5).

Business Rules

- Slides advance every interval; the rotation pauses while the pointer is over the slides, while keyboard focus is inside the carousel (except on the pause / play button), while less than 25% of the carousel is on screen, while the browser tab is hidden, in Universal Editor edit mode, and when the visitor selects Pause.
- With "reduce motion" set in the operating system, the carousel starts paused (Play shown).
- Swiping is disabled (the slides cannot be scrolled by hand); dots remain clickable.

Rendering Rules

Class `auto`; controls: pause / play button and dots; the track has `overflow-x: hidden`; no arrow-key navigation.

Responsive Behaviour

Same at all widths; no swipe on touch devices.

SEO Impact

As Variation 1.

Accessibility Compliance

Pause control for moving content (WCAG 2.2.2); while playing, `aria-live` is "off" so automatic changes are not announced; button label switches between "Pause slides" and "Play slides".

Variation 3 — Navigation: Manual and Auto

Variation Name

Manual and Auto (`carouselNavigation` = `manual-auto`).

Business Purpose

Rotating slides that the visitor can also control.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-manual-auto.png`
> **Purpose:** Manual and Auto navigation
> **Description:** Controls with the pause button, previous arrow, dots and next arrow.

Use Cases

Home page hero rotation.

Authoring Guidelines

Navigation = Manual and Auto; set the Autoplay interval.

Business Rules

Autoplay as Variation 2; previous / next / dot clicks move the slide and restart the interval; swipe enabled.

Rendering Rules

Class `manual-auto`; controls: pause / play, previous, dots, next; arrow keys on the controls.

Responsive Behaviour

All controls at all widths, wrapping on narrow screens; swipe on touch.

SEO Impact

As Variation 1.

Accessibility Compliance

As Variations 1 and 2 combined.

Variation 4 — Single slide (static)

Variation Name

Single slide.

Business Purpose

A carousel prepared with one slide (for example while other slides are being produced) shows as a static block.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-single-slide.png`
> **Purpose:** Single slide
> **Description:** A Gen Carousel with one Hero and no controls below it.

Use Cases

Temporary state; seasonal reductions to one campaign.

Authoring Guidelines

Place only one Hero or Teaser in the section.

Business Rules

With fewer than two slides there are no controls, no autoplay, no keyboard handling and no slide events; the slide is marked active.

Rendering Rules

Region and slide roles are still set; no `div.gen-carousel-controls`.

Responsive Behaviour

The slide's own responsive layout.

SEO Impact

As Variation 1.

Accessibility Compliance

Still announced as a carousel region with "1 of 1"; consider using the Hero directly instead.

Variation 5 — Hero slides

Variation Name

Hero slides.

Business Purpose

Full-impact slides with background image, heading, text and CTAs.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-hero-slides.png`
> **Purpose:** Hero slides
> **Description:** A carousel of Hero blocks, each with a full-width image, a heading and a primary CTA.

Use Cases

Home page top banner.

Authoring Guidelines

Add **Hero** blocks to the Gen Carousel section; author each Hero as described in the Hero Block chapter.

Business Rules

Each Hero is one slide; its own options (layout, image, CTAs, Additional) apply inside the slide.

Rendering Rules

The Hero's wrapper becomes the slide content and is stretched to the slide height so all slides line up.

Responsive Behaviour

Each Hero follows its own breakpoints within the slide width.

SEO Impact

Use one page-level h1 only: if the first Hero has an h1, later Heroes should use h2.

Accessibility Compliance

Hero images need alternative text (or be marked decorative); CTAs in hidden slides are not focusable until their slide is shown (`inert`).

Variation 6 — Teaser slides

Variation Name

Teaser slides.

Business Purpose

Lighter slides (image and text side by side, story teasers) in the same rotation.

Visual Example

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-teaser-slides.png`
> **Purpose:** Teaser slides
> **Description:** A carousel of Teaser blocks with image and text, the dots below.

Use Cases

Case study or story rotation further down a page.

Authoring Guidelines

Add **Teaser** blocks to the section; Heroes and Teasers can be mixed.

Business Rules

As Variation 5; slides of different heights are aligned to the tallest one.

Rendering Rules

As Variation 5.

Responsive Behaviour

The Teaser's own breakpoints within the slide.

SEO Impact

As Variation 5.

Accessibility Compliance

As Variation 5.

6.24.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** Edit mode with a Gen Carousel section selected, the second slide (a Teaser) shown because it is selected in the content tree, and the controls below; letters A to E placed on the elements.

Annotations

- A = Gen Carousel section (select it for Style, Navigation, Autoplay interval, Additional).
- B = Slide (a Hero or Teaser block; select it to edit it with its own properties).
- C = Previous / next buttons (Manual modes).
- D = Dots (one per slide).
- E = Pause / play button (Auto modes; autoplay never runs in edit mode).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-properties-panel.png`
> **Purpose:** Properties panel

> **Description:** The Gen Carousel properties panel with the Content tab (Style, Navigation, Autoplay interval) and the Additional tab.

Properties panel, Gen Carousel (model `gen-carousel`)

- **Content:** Style (select, only "Default"); Navigation (Manual / Auto / Manual and Auto); Autoplay interval (seconds) (number 2 to 30; shown only when Navigation is not Manual).
- **Additional:** Anchor ID; Hide block; Padding left / right / top / bottom; Margin left / right / top / bottom (see the note in 6.24.6: not applied to this section-based component).

In edit mode, selecting a slide (or a component inside it) in the content tree scrolls the carousel to that slide, and all slides stay focusable (no `inert`).

6.24.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Open the page in Universal Editor.

Expected Result: Edit mode with the content tree.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-01-open-page.png`
> **Purpose:** Step 1
> **Description:** The page open in Universal Editor.

Step 2 — Add the Gen Carousel section

Action: Select the page root (`main`) in the content tree, + (Add), **Gen Carousel**.

Expected Result: A new section with Style Default, Navigation Manual and interval 5 is added.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-02-add-section.png`
> **Purpose:** Step 2
> **Description:** The Add menu at page level with Gen Carousel highlighted.

Step 3 — Choose the Navigation

Action: In the Content tab, choose Navigation: Manual, Auto or Manual and Auto.

Expected Result: For Auto modes the Autoplay interval field appears.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-03-navigation.png`
> **Purpose:** Step 3
> **Description:** The Navigation select open with its three options.

Step 4 — Set the Autoplay interval (Auto modes)

Action: Enter the seconds per slide (2 to 30), e.g. 7.

Expected Result: The value is accepted only within 2 to 30.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-04-interval.png`
> **Purpose:** Step 4
> **Description:** Autoplay interval field with the value 7.

Step 5 — Add the slides

Action: Select the Gen Carousel section, **+** (Add), choose **Hero** or **Teaser**; repeat for each slide.

Expected Result: Each block is a slide; dots appear from two slides.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-05-add-slides.png`
> **Purpose:** Step 5
> **Description:** Content tree with three Heroes under the Gen Carousel section.

Step 6 — Author each slide

Action: Select each Hero / Teaser and fill its fields (image with alternative text, heading, text, CTAs) as described in its chapter.

Expected Result: The selected slide is shown on the canvas while you edit it.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-06-author-slide.png`
> **Purpose:** Step 6
> **Description:** The second Hero selected and displayed, with its properties panel.

Step 7 — Order the slides

Action: Drag the blocks in the content tree.

Expected Result: The slide and dot order follows the content tree.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-07-order.png`
> **Purpose:** Step 7
> **Description:** A Hero dragged to the first position.

Step 8 — Preview

Action: Switch to **Preview**; use the arrows, dots, swipe, keyboard and pause / play.

Expected Result: Navigation works; in Auto modes the slides advance after the interval (the rotation runs only outside edit mode).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-08-preview.png`
> **Purpose:** Step 8
> **Description:** Preview mode with the second dot active after selecting next.

Step 9 — Validate

Action: Run the checklist below.

Expected Result: All checks pass.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-09-validate.png`
> **Purpose:** Step 9
> **Description:** The data layer in the console showing cmp:show entries for slides 2 and 3.

Step 10 — Publish

Action: Publish the page.

Expected Result: The carousel appears on the live site.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/gen-carousel-step-10-publish.png`
> **Purpose:** Step 10
> **Description:** The Publish dialog for the page.

Preview Process

1. Universal Editor Preview for interaction (autoplay runs in Preview, not in edit mode).
2. Edge Delivery preview: `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}`.
3. Test touch swipe on a phone or device emulation, and reduced motion (the carousel should start paused).

Publishing Process

Publish the page (slides are part of it). Live: `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/{path}` ([To be confirmed: production domain and CDN]).

Validation Process

- ☐ Two to five slides; each a Hero or Teaser with heading, image alternative text and CTA.
- ☐ Only one h1 on the page (first slide at most).
- ☐ Navigation mode as agreed; for Auto modes the interval gives enough reading time (at least 5 seconds Recommended).
- ☐ Previous / next / dots / swipe / arrow keys (on the controls) work; wrap-around works.
- ☐ Pause stops the rotation and Play resumes it; hovering and keyboard focus pause it.
- ☐ Hidden slides' links cannot be reached with Tab.
- ☐ `window.adobeDataLayer` shows `cmp:show` with `position` on each slide change.
- ☐ Correct at below 600px, 600 to 899px, 900px and wider.

Rollback Process

1. Restore the previous page version and republish.
2. To stop a rotation quickly: set Navigation to Manual and publish.
3. To remove a slide quickly: delete or move it (Hide block on a Hero / Teaser hides its content, but the empty slide and its dot remain).
4. Code regressions: revert on `main`.

6.24.6 Authorable Field Specification

Model `gen-carousel` (Gen Carousel section)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Tab: Content	tab	—	—	—	Carousel options	—	—	—
Style (`style`)	select	No (preset)	No	Default (`gen-carousel-section`)	Marks the section as a carousel	Leave on Default	One option: Default = `gen-carousel-section`	gen-carousel-section
Navigation (`carouselNavigation`)	select	No	No	Manual (`manual`)	How slides change	Manual unless rotation is required	Manual / Auto / Manual and Auto; unknown values → Manual (code)	manual-auto
Autoplay interval (seconds) (`carouselInterval`)	number	No	No	5	Seconds per slide	5 to 10 seconds for reading time	Enforced in the dialog: min 2, max 30; code: minimum 2, empty / invalid → 5	7
Tab: Additional	tab	—	—	—	Block options	See note	—	—
Anchor ID (`classes_anchor`)	text	No	No	empty	Anchor (see note)	Not effective here	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-highlights
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide (see note)	Not effective here	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	pl-0
Padding right (`classes_paddingright`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	pr-0
Padding top (`classes_paddingtop`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	pt-0
Padding bottom (`classes_paddingbottom`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	pb-0
Margin left (`classes_marginleft`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	ml-0
Margin right (`classes_marginright`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	mr-0
Margin top (`classes_margintop`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	mt-0
Margin bottom (`classes_marginbottom`)	select	No	No	Default	Spacing (see note)	Not effective here	One of 15 options	mb-0

Important: The Gen Carousel is a section. The Additional options are applied only to `.block` elements (`decorators/block-options-decorator.js`, `styles/styles.css`), and `buildCarouselSections` copies only Navigation and Autoplay interval to the synthetic `gen-carousel` block. The Additional tab of the Gen Carousel therefore has no effect on the page (as of this handover) [To be confirmed: whether AEM delivers `classes_*` section fields at all — owner: development team]. The Hero and Teaser slides have their own Additional tab, which does apply to them.

The slide items are Hero and Teaser blocks with their own models (see their chapters).

Field details

- **Style**
  - Purpose: the section class `gen-carousel-section` that `buildCarouselSections` looks for.
  - Allowed Values: Default (`gen-carousel-section`). Character Limits: not applicable.
  - Business Rules: without it the section shows its Heroes / Teasers one below another.
  - Dependency Rules: required for Navigation and interval to take effect.
  - Visibility Rules: always visible. Sample Content: Default.
- **Navigation**
  - Purpose: Manual: previous / next buttons, dots and swipe; Auto: slides advance on their own with a pause button; Manual and Auto: both (model description).
  - Allowed Values: `manual`, `auto`, `manual-auto`. Character Limits: not applicable.
  - Business Rules: becomes a class on the carousel block; a missing or unknown value is treated as Manual.
  - Dependency Rules: controls the visibility of Autoplay interval.
  - Visibility Rules: always visible. Sample Content: Manual and Auto.
- **Autoplay interval (seconds)**
  - Purpose: time each slide is shown before the next one appears.
  - Allowed Values: numbers 2 to 30 (dialog). Character Limits: not applicable.
  - Business Rules: read from the section's `data-carouselinterval`; `Number(value) || 5`; values below 2 are raised to 2 in code; there is no upper limit in code (30 is enforced by the dialog only).
  - Dependency Rules: used only when Navigation is Auto or Manual and Auto.
  - Visibility Rules: `condition {"!==": [{"var": "carouselNavigation"}, "manual"]}` — shown only when Navigation is not Manual.
  - Sample Content: 7.
- **Anchor ID / Hide block / Padding (4) / Margin (4)**
  - Purpose and Allowed Values: the common block options (Anchor ID `anchor-…`; Hide block on / off; spacing selects with 15 options: Default; 0 (none); 0.25rem (space-100); 0.5rem (space-200); 0.75rem (space-300); 1rem (space-400); 1.25rem (space-500); 1.5rem (space-600); 2rem (space-800); 2.5rem (space-1000); 3rem (space-1200); 4rem (space-1600); 5rem (space-2000); 6rem (space-2400); 7.5rem (space-3000)).
  - Character Limits: Anchor ID none enforced.
  - Business Rules: not effective on this section (see the note above).
  - Dependency Rules: none. Visibility Rules: always visible. Sample Content: defaults.

6.24.7 Field Annotation Mapping

Marker	Field	Purpose
A	Gen Carousel section (Style, Navigation, Autoplay interval, Additional)	Carousel options
B	Slide (Hero / Teaser block and its own fields)	Slide content
C	Previous / next (generated, Manual modes)	Navigation
D	Dots (generated)	Direct slide selection and position
E	Pause / play (generated, Auto modes)	Stops / resumes autoplay

6.24.8 Component Specification

Attribute	Value
Component Name	Gen Carousel (`gen-carousel`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`scripts/scripts.js`, `decorators/button-decorator.js`, `decorators/icon-button-decorator.js`, `decorators/data-layer-decorator.js`, `utils/analytics.js`, `utils/environment.js`, `utils/viewport.js`, `utils/ids.js`, `utils/placeholders.js`, Hero, Teaser, icons
Component Type	Container section (synthetic block whose children are slides)
Supported Variants	Navigation: Manual, Auto, Manual and Auto; single slide; Hero / Teaser slides
Supported Templates	Any page (typically home and landing pages)
Supported Metadata	None
Supported Tags	None
Supported Personalization	None; `cmp:show` per slide (with position) is available for future Adobe Target / Journey Optimizer use (not implemented)
Caching Considerations	Part of the page HTML (Edge Delivery CDN); no sheets or index read; placeholders sheet loaded once per page
Performance Considerations	Loaded with its section (often the first: the first slide's image is the LCP candidate); all slide blocks are decorated with the section; no extra requests apart from placeholders and icons; autoplay timers stop off screen and in hidden tabs
Accessibility Compliance	WCAG 2.1 AA: region with `aria-roledescription` "carousel" and label; slides as groups "slide" labelled "n of total"; pause control (2.2.2); labelled controls (4.1.2); `aria-current` on dots; `inert` hidden slides (2.4.3); polite live region when not playing; reduced motion (starts paused, no smooth scroll); 1.5rem dot hit area
SEO Compliance	All slides in the page HTML; slide headings follow the Hero / Teaser models

6.24.9 Input Model

- **Content Inputs:** the section's children (Hero / Teaser blocks), each a slide.
- **Metadata Inputs:** none. **Tags Inputs:** none.
- **Page Context Inputs:** Universal Editor edit mode; reduced-motion preference; visibility of the carousel and of the browser tab; pointer and focus position; editor selection (`aue:ui-select`); locale (placeholders).
- **Query Inputs:** none.
- **Manual Inputs:** Navigation, Autoplay interval; slide content.

Authored HTML of the section (illustrative; the section properties arrive as data attributes):

```
<div data-aue-model="gen-carousel" data-style="gen-carousel-section"
  data-carouselnavigation="manual-auto" data-carouselinterval="7">
  <div class="hero">...</div>
  <div class="hero">...</div>
  <div class="teaser">...</div>
</div>
```

6.24.10 Output Model

Rendered Output

- `div.section.gen-carousel-section.gen-carousel-container` > `div.gen-carousel-wrapper` > `div.gen-carousel.block.{navigation}` (`role="region"`, `aria-roledescription`, `aria-label`, `data-interval`)
  - `div.gen-carousel-slides` (`id="gen-carousel-{n}-slides"`, `aria-live`)
    - `div.gen-carousel-slide` per slide (`id`, `data-cmp-id`, `role="group"`, `aria-roledescription="slide"`, `aria-label="{n} of {total}"`, `inert` when not current outside edit mode; `gen-carousel-slide-active` on the current one) containing the Hero / Teaser wrapper.
  - `div.gen-carousel-controls` (from two slides): `button.gen-carousel-toggle` (Auto modes), `button.gen-carousel-prev`, `ul.gen-carousel-dots` > `li` > `button.gen-carousel-dot` (`aria-label`, `aria-controls`, `aria-current`), `button.gen-carousel-next` (Manual modes).

Generated HTML

```
<div class="gen-carousel manual-auto block" data-interval="7" role="region"
  aria-roledescription="carousel" aria-label="Carousel">
  <div class="gen-carousel-slides" id="gen-carousel-1-slides" aria-live="off">
    <div class="gen-carousel-slide gen-carousel-slide-active" id="gen-carousel-1-slide-1"
      data-cmp-id="gen-carousel-1-slide-1" role="group" aria-roledescription="slide"
      aria-label="1 of 3"><div class="hero-wrapper">...</div></div>
    <div class="gen-carousel-slide" id="gen-carousel-1-slide-2" inert ...>...</div>
  </div>
  <div class="gen-carousel-controls">
    <button class="gen-carousel-toggle button secondary" aria-label="Pause slides"
      data-cmp-interaction="pause" aria-controls="gen-carousel-1-slides"></button>
    <button class="gen-carousel-prev button secondary" aria-label="Previous slide"
      data-cmp-interaction="previous"></button>
    <ul class="gen-carousel-dots">
      <li><button class="gen-carousel-dot" aria-label="Show slide 1 of 3"
        aria-controls="gen-carousel-1-slide-1" aria-current="true"></button></li>
    </ul>
    <button class="gen-carousel-next button secondary" aria-label="Next slide"
      data-cmp-interaction="next"></button>
  </div>
</div>
```

Generated JSON: none. **Generated Metadata:** none.

Analytics Tracking

- Component state: the carousel block as `tcs-corp/components/gen-carousel` (id `gen-carousel-{n}`); the Hero / Teaser blocks with the carousel as `parentId`.
- Every change of slide (buttons, dots, keyboard, swipe, autoplay, editor selection) pushes `cmp:show` for the slide (not for the first slide on load):

```
{
  "event": "cmp:show",
  "eventInfo": { "path": "component.gen-carousel-1-slide-2", "position": 2 }
}
```

- Control clicks push `cmp:click` on the carousel component with `interaction` `previous`, `next` or `pause` (the pause / play button, both states), `button` for dots, and `dc:title` from the `aria-label`:

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.gen-carousel-1",
    "interaction": "next",
```

```
    "dc:title": "Next slide",
    "xdm:linkType": "other",
    "breakpoint": "mobile"
  }
}
```

- CTA clicks inside a slide are attributed to the Hero / Teaser block.

Personalization Output: none. **Search Index Output:** none.

6.24.11 Business Rules

1. A section becomes a carousel only with the style `gen-carousel-section`, once per section.
2. Navigation and Autoplay interval move from the section data attributes to the carousel block (class and `data-interval`).
3. Every child of the carousel block is a slide, whatever it contains.
4. Navigation: the first of `manual`, `auto`, `manual-auto` found in the classes, else Manual; Manual modes have previous / next and swipe, Auto modes autoplay and the pause button.
5. Interval: `data-interval` seconds, else 5; at least 2 seconds.
6. Fewer than two slides: no controls, no autoplay, the slide is active.
7. Dots are always shown from two slides; in Auto mode they are the only navigation besides pause / play.
8. Previous / next wrap around; a button or dot click restarts the autoplay interval.
9. ArrowLeft / ArrowRight move the slide only when focus is on the controls (widgets inside slides keep their keys); focus then moves to the current dot.
10. Swipe / scroll: the slide at least 60% visible in the track becomes current.
11. Autoplay runs only when Navigation includes Auto and the visitor has not paused it, and pauses while: the pointer is over the slides; keyboard focus (`:focus-visible`) is inside the carousel, except on the pause / play button; less than 25% of the carousel is visible; the browser tab is hidden; Universal Editor is in edit mode; the block is removed.
12. Reduced motion: autoplay starts paused; slide scrolling is instant instead of smooth; dot transitions are off.
13. Pause / play label and icon switch between “Pause slides” (pause icon) and “Play slides” (play icon).
14. Live region: `aria-live="off"` while playing, `"polite"` otherwise.
15. Non-current slides are `inert` (not focusable) except in edit mode.
16. Universal Editor: selecting a component inside a slide shows that slide.
17. A slide change pushes `cmp:show` with the slide id and its 1-based position; the initial slide does not.
18. Auto-only carousels cannot be swiped (`overflow-x: hidden`).
19. Region label: an existing `aria-label`, else “Carousel” (no authorable label field).
20. Allowed slide blocks: Hero and Teaser (filter `gen-carousel`).
21. Additional tab options of the section are not applied.
22. The block logs nothing to the console.

6.24.12 Author Best Practices

- **Recommended Content Length:** 2 to 5 slides; one message per slide with a short heading (up to about 60 characters) and one primary CTA (Recommended).
- **Image Dimensions:** follow the Hero / Teaser image rules; slides are full content width (up to 120rem in full-width sections), so use landscape images of at least 2000px width for Heroes; keep important image areas away from the edges for mobile cropping; use the same aspect ratio on all slides so heights stay consistent.
- **CTA Writing Guidelines:** one action per slide, verb first (“Explore our AI services”); distinct CTAs across slides.
- **Accessibility Guidelines:** prefer Manual; for Auto use at least 5 seconds; provide image alternative texts; one h1 on the page.

- **SEO Guidelines:** put the most important message on slide 1 (it is the only one most visitors see and the LCP element).
- **Metadata Guidelines:** not applicable.
- **Taxonomy Guidelines:** not applicable.
- **Localization Guidelines:** translate each slide; check that translated headings fit the Hero layout on mobile; control labels ("Previous slide", "Show slide {n} of {total}", …) and role descriptions ("carousel", "slide") are translated in the locale's placeholders sheet (keys `carouselRole`, `carousel`, `slideRole`, `slideOf`, `showSlide`, `pauseSlides`, `playSlides`, `previousSlide`, `nextSlide`; keep the `{n}` and `{total}` variables).

6.24.13 Troubleshooting Guide

Issue 1 — Slides appear one below another

Symptoms: No carousel; Heroes are stacked.

Root Cause: The section is not a Gen Carousel (style `gen-carousel-section` missing), or the script failed.

Diagnosis Steps: Inspect the section class and look for `div.gen-carousel-wrapper`; check the console.

Resolution Steps: Add a Gen Carousel section and move the blocks into it.

Validation Steps: One slide visible with controls.

Preventive Measures: Always start from the Gen Carousel component.

Issue 2 — No controls are shown

Symptoms: A single slide without arrows or dots.

Root Cause: Only one slide in the section.

Diagnosis Steps: Count the blocks in the section.

Resolution Steps: Add slides or replace the carousel with the Hero itself.

Validation Steps: Controls appear from two slides.

Preventive Measures: Plan at least two slides.

Issue 3 — Autoplay does not run

Symptoms: Slides do not advance.

Root Cause: Navigation is Manual; the visitor paused it; reduced motion is on; the pointer is over the slides; keyboard focus is inside; the carousel is mostly off screen; edit mode in Universal Editor.

Diagnosis Steps: Check the block class (`auto` / `manual-auto`) and the pause / play label; check the OS reduced-motion setting; test in Preview or on `.aem.page`.

Resolution Steps: Set Navigation to Auto or Manual and Auto; select Play.

Validation Steps: Slides advance after the interval.

Preventive Measures: Test autoplay outside edit mode.

Issue 4 — The interval is not the configured value

Symptoms: Slides change faster or slower than expected.

Root Cause: The value is missing or invalid (5 seconds used) or below 2 (2 seconds used); user interaction restarts the interval.

Diagnosis Steps: Inspect `data-interval` on the carousel block.

Resolution Steps: Enter a value between 2 and 30 and publish.

Validation Steps: Measured interval matches.

Preventive Measures: Use whole seconds within the range.

Issue 5 — Swipe does not work on mobile
Symptoms: Touch swipe does nothing.

Root Cause: Navigation is Auto (swipe disabled by design).

Diagnosis Steps: Check the `auto` class.

Resolution Steps: Use Manual and Auto if swipe is required.

Validation Steps: Swipe changes the slide.

Preventive Measures: Choose the mode with the design team.

Issue 6 — An empty slide / dot appears
Symptoms: A dot leads to a blank slide.

Root Cause: A slide block has Hide block on (its content is hidden but it still counts as a slide), or a block in the section has no content.

Diagnosis Steps: Check each Hero / Teaser's Additional tab and content.

Resolution Steps: Delete or move the block instead of hiding it; complete its content.

Validation Steps: Every dot shows content.

Preventive Measures: Do not use Hide block for slides.

Issue 7 — Spacing / Hide block of the carousel has no effect
Symptoms: Additional options on the Gen Carousel do nothing.

Root Cause: The options are not applied to sections.

Diagnosis Steps: Inspect the section and carousel block for the classes.

Resolution Steps: Adjust the slides' own options, or delete / move the carousel.

Validation Steps: Expected result on the page.

Preventive Measures: Follow 6.24.6.

Issue 8 — Too many cmp:show events
Symptoms: Analytics records many slide views.

Root Cause: Autoplay pushes `cmp:show` at every automatic change; swipes or smooth scrolls across several slides may report intermediate slides.

Diagnosis Steps: Inspect `window.adobeDataLayer` while autoplay runs.

Resolution Steps: Filter on interactions (`cmp:click` with `previous` / `next`) or on time in Adobe Tags rules once implemented.

Validation Steps: Reports match the intended definition of a slide view.

Preventive Measures: Agree the analytics definition with the analytics team [To be confirmed: analytics owner].

6.24.14 Support Runbook
Monitoring Requirements

- Core Web Vitals (LCP of the first slide, CLS) for pages with a carousel in the first section.
- Once Adobe Tags is embedded, `cmp:show` per slide position and `cmp:click` per control.

- [To be confirmed: monitoring tool and thresholds — owner: AMS lead].

Logs

- Browser console: no messages from the block; module errors `failed to load block gen-carousel` or of the slide blocks (`scripts/aem.js`).
- Admin API status of the page; AEM author logs for publication errors.
- Placeholders: `/{locale}/placeholders.json` request (labels fall back to English if it fails).

Common Errors

Error	Likely cause	First action
Stacked slides	Not a Gen Carousel section	Check `gen-carousel-section`
No controls	One slide	Add slides
No autoplay	Manual / paused / reduced motion / edit mode	Check mode and Play
No swipe	Auto mode	Use Manual and Auto
Blank slide	Hidden or empty block	Remove / fill it
English labels on a localised page	Placeholder keys missing	Add keys to the locale's sheet
Additional options ignored	Section-based component	Use slide options

Escalation Matrix

Level	Who	What they check
L1	Author support	Section type, Navigation, interval, slide count and content, publication
L2	AMS / site admin	Page status (Admin API), placeholders sheet publication, CDN cache, RUM
L3	Development team	`gen-carousel.js` / `.css`, `buildCarouselSections`, Hero / Teaser compatibility, data layer
L4	Adobe Support	Universal Editor and Edge Delivery platform issues

Recovery Procedures: restore and republish the page; switch Navigation to Manual to stop rotation; revert code on `main`.

Health Checks: `document.querySelectorAll('.gen-carousel-slide').length` equals the number of slides; exactly one slide has `gen-carousel-slide-active`; next / previous and dots change `aria-current`; `cmp:show` appears on change.

Operational Procedures: content changes via author → preview → publish; translation of control labels via the placeholders sheet per locale; code changes via PR with preview link and `npm run lint`.

Known Limitations

- Only Hero and Teaser slides; one slide visible at a time (no multi-item view).
- No authorable carousel label (always "Carousel" from the placeholders sheet).
- No Width Type, background or Additional options on the section.
- Auto-only carousels cannot be swiped.
- Autoplay pushes a `cmp:show` per automatic change.
- No slide counter text or thumbnails; no per-slide interval.

6.25 Table Block

The Table block presents structured, comparable information (rows and columns) as a native, accessible HTML `<table>`. It is based on the Edge Delivery Services Block Collection table (`blocks/table/table.js`, `table.css`, `_table.json`) and adapted for TCS: the first row is the column header row unless **No Header** is chosen, the first column can be row headers (**First Column Header**), the heading directly before the table names it, and a table wider than its container scrolls inside a labelled region that keyboard users can focus.

6.25.1 Component Purpose

Business Objective: Give authors a simple, accessible way to publish tabular data (facts, comparisons, schedules, specifications) that is readable on every device, understood by screen readers and indexable by search engines, without custom code or images of tables.

Business Use Cases:

- Service or offering comparisons (for example capability by region, feature by plan).
- Financial or results summaries in press releases and investor-related pages (quarter, revenue, growth).
- Event agendas and schedules (time, session, speaker).
- Office or contact directories (city, address, phone), when a Location List is not required.
- Specifications, eligibility criteria and glossaries (term in the first column, definition in the others).

When To Use:

- The content is genuinely two-dimensional: each value is meaningful in relation to a row and a column.
- Up to five columns of short values (the model offers 1 to 5 columns per row).
- The data must remain text (searchable, translatable, accessible), not an image.

When Not To Use:

- For page layout or to place content side by side (use Columns or the section layout).
- For a single list of items (use a bulleted list in default content, or the List / Cards blocks).
- For more than five columns: the model has no 6+ column row type (Not implemented as of this handover).
- For data that needs sorting, filtering, pagination or merged cells (colspan / rowspan): Not implemented (as of this handover).
- For content that is maintained in a spreadsheet and changes often: consider the Table List block where applicable, or a data sheet, rather than retyping values.

Dependencies:

Type	Dependency	Use
Code	`scripts/scripts.js` `moveInstrumentation`	Keeps Universal Editor instrumentation on the rendered rows and cells
Code	`utils/ids.js` `uniqueId`	Generates an id for the heading before the table (`table-title-1`, …) when it has none
Code	`utils/placeholders.js` `getPlaceholders`	Loads the fallback region label `table` ("Table")
Code	`decorators/block-options-decorator.js`	Additional tab (Anchor ID, Hide block, padding, margin)
Code	`decorators/data-layer-decorator.js`	Component state and `cmp:click` events for links in cells
Code	`decorators/content-buttons-decorator.js`	Standalone links in a cell paragraph become link buttons (runs on the whole page before blocks)

Type	Dependency	Use
Browser API	`ResizeObserver`	Detects when the table is wider than its container
Sheet	`/{locale}/placeholders.json`	Key `table` (scroll region label when there is no heading before the table)
Model	`models/_section.json`	Allows the Table in sections (filter `section`)

Related Components: Table List block (`blocks/table-list`, a different block for list-style tabular data), Separator (to divide a table from following content), Columns (layout, not data), List / Cards (collections of items).

Applicable Templates: Any page. The block is allowed in every section (`section` filter), so it can be used on home, article, blog, press-release and standard content pages.

Supported Content Types: Rich text in every cell: paragraphs, bold / italic, links, lists and line breaks. Images in cells are not part of the design (Recommended: text only).

6.25.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** A 4-column Table with a header row below a "Quarterly results" heading, at a viewport of 1440px; header row in semibold with a dark 2px rule below it, body rows separated by light 1px rules.

Desktop (≥ 1200px): The table spans the full width of its section content (`width: 100%`). From 900px the text size is `--body-font-size-m` (1.125rem). Cells have 0.75rem vertical and 1rem horizontal padding (`--space-300` / `--space-400`), text is left aligned and top aligned. Header cells (`th`) are semibold. The header row has a 2px bottom border in `--color-border-neutral` (slate 900, #303030); each body row has a 1px bottom border in `--color-border-default` (grey 200, #e6e6e6). Paragraphs inside cells have no margin; a second paragraph in a data cell is spaced 0.25rem below the first.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** The same Table at 768px and at 1024px, showing the smaller text below 900px and, at 768px, the table still fitting the content width.

Tablet (600–1199px): Same layout. Between 600px and 899px the text size is `--body-font-size-s` (0.75rem); from 900px it is 1.125rem. Columns shrink to fit; long words wrap. If the content cannot shrink further (long unbreakable values, many columns), the table becomes horizontally scrollable inside the block (`overflow-x: auto`) and the block becomes a focusable region (see Variation 12).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** A 5-column Table at 390px that is wider than the screen: the visible part of the table with a horizontal scroll, and the 2px brand-blue focus outline around the block after tabbing to it.

Mobile (< 600px): No stacking or card layout: the table keeps its rows and columns. Text is 0.75rem. Tables with three or more columns usually exceed the viewport and scroll horizontally within the block; the page itself never scrolls sideways. When the table scrolls, the block gets `tabindex="0"`, `role="region"` and an `aria-label`, and shows a 2px `--color-border-brand` outline (offset 0.25rem) when focused with the keyboard.

6.25.3 Component Variations

The Table has two option groups in the **Options** multiselect (Border: Striped, Bordered; Header: No Header, First Column Header), a **Columns** select (1–5 columns, which decides the row type authors can add) and one automatic behaviour (scroll region). Options in the multiselect can be combined.

Variation 1 — Standard (header row)

Variation Name: Standard table with a column header row (no option selected).

Business Purpose: The default for most data: the first row names the columns.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-standard.png`
> **Purpose:** Variation example
> **Description:** A 3-column table "Service / Region / Availability" with a semibold header row, a 2px dark rule below it and three body rows with light separators.

Use Cases: Comparisons, results tables, schedules.

Authoring Guidelines: Leave **Options** empty. Select the **Columns** count, then add Rows; the first Row holds the column titles.

Business Rules: The first row is always treated as the header row unless No Header is selected. Every cell of the first row becomes a column header.

Rendering Rules: Row 1 is rendered in `<thead>` with `<th scope="col">` cells; all other rows in `<tbody>` with `<td>` cells.

Responsive Behaviour: Desktop: full width, 1.125rem text. Tablet: 0.75rem text below 900px, 1.125rem from 900px; scrolls if too wide. Mobile: 0.75rem text; scrolls horizontally if too wide.

SEO Impact: A native `<table>` with `<th>` header cells is understood by search engines as structured tabular content.

Accessibility Compliance: WCAG 1.3.1 (Info and Relationships): `scope="col"` associates every data cell with its column header; screen readers announce the header when moving across cells.

Variation 2 — No Header

Variation Name: No Header (`no-header` class).

Business Purpose: Tables whose first row is data, not titles (for example a simple list of key facts in two columns).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-no-header.png`
> **Purpose:** Variation example

> **Description:** A 2-column table without a header row; the first body row has a 2px dark rule above it, all rows have regular weight text.

Use Cases: Fact sheets, key / value lists where a header would repeat the obvious.

Authoring Guidelines: In **Options**, select **No Header** (Header group).

Business Rules: No row is treated as a header; the table has no `<thead>`.

Rendering Rules: All rows render in `<tbody>` with `<td>` cells. CSS draws a 2px `--color-border-neutral` top border on the first body row, so the table still has a clear start.

Responsive Behaviour: Same as Standard on all breakpoints.

SEO Impact: Neutral; content is plain table data.

Accessibility Compliance: Without headers, screen readers cannot announce column context. Use only when the data is self-explanatory; prefer First Column Header with No Header (Variation 4) for key / value tables (WCAG 1.3.1).

Variation 3 — First Column Header

Variation Name: First Column Header (`row-header` class) with the header row.

Business Purpose: Matrices in which both the column titles and the first column (row titles) give context, for example "Capability × Region".

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-first-column-header.png`
> **Purpose:** Variation example
> **Description:** A 4-column matrix: header row in semibold, and in each body row the first cell in semibold as the row title.

Use Cases: Feature matrices, specification tables, pricing or plan comparisons.

Authoring Guidelines: In **Options**, select **First Column Header** (Header group). Put the row titles in the first column (Column1).

Business Rules: The first cell of every body row becomes a row header. The top-left cell of the header row stays a column header.

Rendering Rules: Header row: `<th scope="col">` cells. Body rows: first cell `<th scope="row">` (semibold, `text-align: start`), other cells `<td>`.

Responsive Behaviour: Same as Standard; when the table scrolls on small screens the row header scrolls with the row (it is not sticky: Not implemented as of this handover).

SEO Impact: Positive: explicit row and column headers describe the data relationships.

Accessibility Compliance: WCAG 1.3.1: screen readers announce both the row header and the column header for each data cell.

Variation 4 — Row headers only (No Header + First Column Header)

Variation Name: No Header combined with First Column Header (`no-header row-header`).

Business Purpose: Key / value tables where the first column is the label and there is no column title row (for example "Founded | 1968", "Headquarters | Mumbai").

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-row-headers-only.png`
> **Purpose:** Variation example
> **Description:** A 2-column fact table with a 2px dark rule above the first row; the first column (labels) in semibold, values in regular weight.

Use Cases: Company facts, contact details, document properties.

Authoring Guidelines: In **Options**, select both **No Header** and **First Column Header**.

Business Rules: No column header row; the first cell of every row is a row header.

Rendering Rules: No `<thead>`; every row in `<tbody>` with `<th scope="row">` as first cell; top border of the first row from the No Header style.

Responsive Behaviour: Same as Standard.

SEO Impact: Positive: labelled values.

Accessibility Compliance: WCAG 1.3.1: each value is associated with its row label.

Variation 5 — Striped

Variation Name: Striped (`striped` class).

Business Purpose: Improves readability of long or wide tables by shading alternate rows.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-striped.png`
> **Purpose:** Variation example
> **Description:** A 3-column table with 8 body rows in which the 1st, 3rd, 5th and 7th body rows have a light grey background.

Use Cases: Long schedules, results tables, directories.

Authoring Guidelines: In **Options**, select **Striped** (Border group). Can be combined with Bordered and with either Header option.

Business Rules: Only body rows are striped; the header row is never shaded.

Rendering Rules: `tbody tr:nth-child(odd)` gets `--color-bg-secondary` (grey 100, #f5f5f5). With No Header the first row of the table is shaded.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None (visual only).

Accessibility Compliance: Stripes are a visual aid only; the text contrast stays that of `--color-text-default` on a very light background (WCAG 1.4.3 met by the token colours).

Variation 6 — Bordered

Variation Name: Bordered (`bordered` class).

Business Purpose: A grid look for dense numeric data where each cell must be visually separated.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-bordered.png`
> **Purpose:** Variation example
> **Description:** A 5-column numeric table with a 1px light grey border around every cell.

Use Cases: Financial figures, specification grids.

Authoring Guidelines: In **Options**, select **Bordered**. Can be combined with Striped.

Business Rules: Applies to header and body cells.

Rendering Rules: Every `th` and `td` gets a 1px `--color-border-default` border (grey 200). The row borders of the standard style remain.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Visual only; borders are not needed to understand the data.

Variation 7 — 1 Column

Variation Name: Columns: 1 Column (filter `table`, row model `table-row`).

Business Purpose: A single-column list with a header (rarely needed).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-1-column.png`
> **Purpose:** Variation example
> **Description:** A one-column table with a header cell and four rows.

Use Cases: A titled list where table semantics are wanted (Recommended: prefer a list in default content).

Authoring Guidelines: Set **Columns** to **1 Column** (the default) before adding Rows. Each Row has one field, Column1.

Business Rules: The block accepts only `Row` items of model `table-row` while this option is selected.

Rendering Rules: Each row renders one cell.

Responsive Behaviour: Never scrolls in practice; full width.

SEO Impact: Neutral.

Accessibility Compliance: As Standard.

Variation 8 — 2 Columns

Variation Name: Columns: 2 Columns (filter `table-2-columns`, row model `table-col-2`).

Business Purpose: Key / value or term / definition tables.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-2-columns.png`
> **Purpose:** Variation example
> **Description:** A two-column "Term / Definition" table.

Use Cases: Glossaries, facts, contact lists.

Authoring Guidelines: Set **Columns** to **2 Columns** before adding Rows; each Row has Column1 and Column2.

Business Rules: Only `table-col-2` rows can be added while this option is selected.

Rendering Rules: Two cells per row.

Responsive Behaviour: Fits on all breakpoints unless values are very long.

SEO Impact: Neutral.

Accessibility Compliance: As Standard; combine with First Column Header for key / value data.

Variation 9 — 3 Columns

Variation Name: Columns: 3 Columns (filter `table-3-columns`, row model `table-col-3`).

Business Purpose: Typical comparisons and schedules.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-3-columns.png`
> **Purpose:** Variation example
> **Description:** A three-column agenda "Time / Session / Speaker".

Use Cases: Agendas, simple comparisons.

Authoring Guidelines: Set **Columns** to **3 Columns**; each Row has Column1–Column3.

Business Rules: Only `table-col-3` rows can be added while selected.

Rendering Rules: Three cells per row.

Responsive Behaviour: Usually fits on tablet and desktop; may scroll on mobile with long values.

SEO Impact: Neutral.

Accessibility Compliance: As Standard.

Variation 10 — 4 Columns

Variation Name: Columns: 4 Columns (filter `table-4-columns`, row model `table-col-4`).

Business Purpose: Results and multi-attribute comparisons.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-4-columns.png`
> **Purpose:** Variation example
> **Description:** A four-column results table "Quarter / Revenue / Growth / Margin".

Use Cases: Quarterly figures, office directories.

Authoring Guidelines: Set **Columns** to **4 Columns**; each Row has Column1–Column4.

Business Rules: Only `table-col-4` rows can be added while selected.

Rendering Rules: Four cells per row.

Responsive Behaviour: Expect horizontal scrolling on mobile.

SEO Impact: Neutral.

Accessibility Compliance: As Standard; the scroll region (Variation 12) keeps it operable by keyboard.

Variation 11 — 5 Columns

Variation Name: Columns: 5 Columns (filter `table-5-columns`, row model `table-col-5`).

Business Purpose: The widest supported table.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-5-columns.png`
> **Purpose:** Variation example
> **Description:** A five-column comparison table on desktop, and the same table scrolling horizontally on mobile.

Use Cases: Plan or capability comparisons with five attributes.

Authoring Guidelines: Set **Columns** to **5 Columns**; each Row has Column1–Column5.

Business Rules: Five is the maximum: there is no wider row model.

Rendering Rules: Five cells per row.

Responsive Behaviour: Scrolls horizontally on mobile and possibly on tablet.

SEO Impact: Neutral.

Accessibility Compliance: As Standard; keep values short to limit scrolling.

Variation 12 — Scroll region (automatic)

Variation Name: Scrollable table region (applied automatically, not an author option).

Business Purpose: Tables wider than the screen remain usable without breaking the page layout.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/table-scroll-region.png`
> **Purpose:** Variation example
> **Description:** A 5-column table on a 390px screen with a horizontal scrollbar and the focus outline drawn around the block after pressing Tab.

Use Cases: Any table whose content is wider than its container.

Authoring Guidelines: Nothing to set. Place a heading (H2–H4) directly before the table so the region gets a meaningful name.

Business Rules: The region is turned on only while `scrollWidth > clientWidth + 1`; it is turned off again when the table fits (for example after rotating the device).

Rendering Rules: A `ResizeObserver` on the block sets `tabindex="0"`, `role="region"` and `aria-label` (the text of the heading before the table, else the placeholder `table`, default "Table") while the table overflows, and removes them when it fits.

Responsive Behaviour: Typically active on mobile for 3+ columns; rarely on desktop.

SEO Impact: None.

Accessibility Compliance: WCAG 2.1.1 (Keyboard): keyboard users can focus the region and scroll it with the arrow keys; WCAG 2.4.7 (Focus Visible): 2px brand outline; WCAG 4.1.2: the region has a name; WCAG 1.4.10 (Reflow) exception for data tables that need two-dimensional layout.

6.25.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** Universal Editor in edit mode showing a heading followed by a 3-column Table; the Table block is selected (blue outline) and one Row is hovered; the content tree on the left lists the Table with its Rows.

Annotations:

- A = Heading in default content directly before the block (names the table; not part of the block)
- B = The Table block (select it to open the block properties: Options, Columns, Additional tab)
- C = The first Row (header row unless No Header is selected)
- D = A cell of a Row (Column1 … Column5, rich text, editable in place)
- E = A body Row (select to edit its columns in the properties panel)
- F = The "Add" (+) control of the block in the content tree, which adds a Row of the type chosen in Columns

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The properties panel of the selected Table showing the Content tab (Options multiselect with Border and Header groups, Columns select) and the Additional tab.

Properties panel, block (model `table`):

- **Content tab: Options** (multiselect: Border group Striped, Bordered; Header group No Header, First Column Header) and **Columns** (select: 1 Column, 2 Columns, 3 Columns, 4 Columns, 5 Columns), which decides which Row type the block accepts.
- **Additional tab:** Anchor ID, Hide block, Padding left / right / top / bottom, Margin left / right / top / bottom.

Properties panel, Row (models `table-row`, `table-col-2` … `table-col-5`): one rich-text field per column, labelled Column1 … Column5. A new Row is pre-filled with the text "Column1", "Column2", … (the template defaults), which authors replace.

6.25.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites, select the page and choose **Edit** (Universal Editor opens), or open the page from the Universal Editor start screen.

Expected Result: The page renders in the editor canvas in edit mode.

> [Screenshot Placeholder]
> **File name:** `screenshots/table-step-01-open-page.png`
> **Purpose:** Authoring step
> **Description:** The page opened in Universal Editor in edit mode.

Step 2 — Add a heading for the table

Action: In the section where the table goes, add a Text / Title component with a heading (for example H2 "Quarterly results") as the last default content element before the table.

Expected Result: The heading shows directly above the place where the table will be.

> [Screenshot Placeholder]
> **File name:** `screenshots/table-step-02-heading.png`
> **Purpose:** Authoring step
> **Description:** A heading added in the section, selected in the canvas.

Step 3 — Add the Table block

Action: Select the section in the content tree, choose **Add** (+) and select **Table**.

Expected Result: An empty Table block appears below the heading.

> [Screenshot Placeholder]
> **File name:** `screenshots/table-step-03-add-block.png`
> **Purpose:** Authoring step
> **Description:** The Add component menu with Table highlighted, and the new empty Table in the content tree.

Step 4 — Choose the number of columns

Action: With the Table selected, in the properties panel **Content** tab choose **Columns** (for example "3 Columns"). Do this before adding Rows.

Expected Result: The block now accepts Rows with three fields.

> [Screenshot Placeholder]
> **File name:** `screenshots/table-step-04-columns.png`
> **Purpose:** Authoring step
> **Description:** The Columns select open with "3 Columns" chosen.

Step 5 — Choose the options

Action: In **Options**, select the styles you need: Striped and / or Bordered, and No Header and / or First Column Header.

Expected Result: The canvas updates with the selected styles once rows exist.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-step-05-options.png`
> **Purpose:** Authoring step
> **Description:** The Options multiselect with Striped and First Column Header selected.

Step 6 — Add the header row

Action: Select the Table in the content tree, choose **Add** (+) and select **Row**. Replace the pre-filled "Column1", "Column2", "Column3" with the column titles.

Expected Result: The first row renders as the semibold header row (unless No Header).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-step-06-header-row.png`
> **Purpose:** Authoring step
> **Description:** The first Row edited in place with three column titles.

Step 7 — Add the data rows

Action: Add one Row per line of data and type the values in each column, in place or in the properties panel. Use **Duplicate** on a Row to copy one, and drag Rows in the content tree to reorder them.

Expected Result: Rows appear in order; with Striped, odd body rows are shaded.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-step-07-data-rows.png`
> **Purpose:** Authoring step
> **Description:** Several data Rows in the content tree and rendered in the canvas.

Step 8 — Fill the Additional tab (optional)

Action: On the **Additional** tab set an Anchor ID (for example `anchor-results`) if other pages link to the table, and spacing if the design requires it.

Expected Result: The values are accepted; an invalid Anchor ID shows "Start with anchor-, then lower-case letters, digits and hyphens."

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-step-08-additional.png`
> **Purpose:** Authoring step
> **Description:** The Additional tab with Anchor ID anchor-results and Margin bottom 2rem.

Step 9 — Preview

Action: Switch Universal Editor to **Preview** mode, then resize the editor / use the device emulation to check mobile width.

Expected Result: The table renders without editor outlines; on narrow widths it scrolls horizontally inside the block.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-step-09-preview.png`
> **Purpose:** Authoring step
> **Description:** The table in preview mode at mobile width with a horizontal scroll.

Step 10 — Validate and publish

Action: Run the Validation Process checklist below, then choose **Publish** in Universal Editor (or Quick Publish in AEM Sites).

Expected Result: The table is visible on the live site after publishing.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/table-step-10-publish.png`
> **Purpose:** Authoring step
> **Description:** The Publish dialog of Universal Editor for the page.

Preview Process

1. In Universal Editor use **Preview** mode to see the page without editing outlines.
2. To check the Edge Delivery preview, open the page on `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}` after it has been previewed (publishing to preview from AEM, or AEM Sidekick **Preview**).
3. Check the table at desktop, tablet (768px) and mobile (390px) widths.

Publishing Process

1. Publish the page from Universal Editor (**Publish**) or AEM Sites (Quick Publish / Manage Publication).
2. The live page `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/{path}` updates once the Edge Delivery CDN cache is refreshed (normally immediately after publishing).
3. A table has no separate assets; only the page needs publishing.

Validation Process

- ☐ A heading directly precedes the table (last element of the default content above it).
- ☐ The first row contains column titles (or No Header is selected deliberately).
- ☐ No Row still contains the template text “Column1”, “Column2”, …
- ☐ Every Row has the same number of columns (rows of different types are not mixed).
- ☐ With First Column Header, the first column holds the row labels.
- ☐ No empty header cells.
- ☐ At 390px width the table scrolls inside the block and the page does not scroll sideways.
- ☐ Tab key focuses the scrolling table (outline visible) and arrow keys scroll it.
- ☐ Links in cells work and have meaningful text.
- ☐ Anchor ID (if set) starts with `anchor-` and `#anchor-…` links jump to the table.
- ☐ Hide block is off (unless intended).

Rollback Process

1. In AEM Sites open the page’s **Version History** (Timeline / Versions) and restore the previous version, or undo the changes in Universal Editor before publishing.
2. Republish the page.
3. To remove the table temporarily without deleting it, turn on **Hide block** (Additional tab) and publish; turn it off later to restore it.

6.25.6 Authorable Field Specification

Model `table` (the block)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Options (`classes`) — Content tab	Multiselect (grouped)	No	Yes	None	Visual and header options	Border group: Striped, Bordered; Header group: No Header, First Column Header	Enforced: option list	Striped; First Column Header
Columns (`filter`) — Content tab	Select	No	No	1 Column (`table`)	Decides which Row type can be added	Choose before adding Rows	Enforced: option list (1–5 Columns)	3 Columns
Anchor ID (`classes_anchor`) — Additional tab	Text	No	No	Empty	Target for `#anchor-…` links	Start with `anchor-`; unique on the page	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`; duplicates get `-2`, `-3` in code	anchor-results
Hide block (`classes_hideblock`) — Additional tab	Boolean	No	No	Off	Hide the block without deleting it	Use temporarily	—	Off
Padding left (`classes_paddingleft`) — Additional tab	Select	No	No	Default	Extra inner space on the left	Use spacing tokens only	Enforced: option list	1rem (space-400)
Padding right (`classes_paddingright`) — Additional tab	Select	No	No	Default	Extra inner space on the right	As above	Enforced: option list	1rem (space-400)
Padding top (`classes_paddingtop`) — Additional tab	Select	No	No	Default	Extra inner space above	As above	Enforced: option list	0 (none)
Padding bottom (`classes_paddingbottom`) — Additional tab	Select	No	No	Default	Extra inner space below	As above	Enforced: option list	0 (none)
Margin left (`classes_marginleft`) — Additional tab	Select	No	No	Default	Outer space on the left	As above	Enforced: option list	Default
Margin right (`classes_marginright`) — Additional tab	Select	No	No	Default	Outer space on the right	As above	Enforced: option list	Default
Margin top (`classes_margintop`) — Additional tab	Select	No	No	Default	Outer space above	As above	Enforced: option list	1.5rem (space-600)
Margin bottom (`classes_marginbottom`) — Additional tab	Select	No	No	Default	Outer space below	As above	Enforced: option list	2rem (space-800)

Model `table-row` (Row, 1 column)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Column1 (`column1text`)	Rich text	No (Recommended: fill)	No	“Column1” (template text of a new Row)	The cell value	Replace the template text	None in the model	Cloud services

Model `table-col-2` (Row, 2 columns)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Column1 (`column1text`)	Rich text	No (Recommended: fill)	No	“Column1”	First cell (row label with First Column Header)	Short label	None	Founded
Column2 (`column2text`)	Rich text	No (Recommended: fill)	No	“Column2”	Second cell	Value	None	1968

Model `table-col-3` (Row, 3 columns)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Column1 (`column1text`)	Rich text	No (Recommended: fill)	No	“Column1”	First cell	Short label or value	None	09:00
Column2 (`column2text`)	Rich text	No (Recommended: fill)	No	“Column2”	Second cell	Value	None	Keynote
Column3 (`column3text`)	Rich text	No (Recommended: fill)	No	“Column3”	Third cell	Value	None	CEO

Model `table-col-4` (Row, 4 columns)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Column1 (`column1text`)	Rich text	No (Recommended: fill)	No	“Column1”	First cell	Short label	None	Q1 FY27
Column2 (`column2text`)	Rich text	No (Recommended: fill)	No	“Column2”	Second cell	Value	None	[To be confirmed: figure — owner: content owner]
Column3 (`column3text`)	Rich text	No (Recommended: fill)	No	“Column3”	Third cell	Value	None	5.2%
Column4 (`column4text`)	Rich text	No (Recommended: fill)	No	“Column4”	Fourth cell	Value	None	24.1%

Model `table-col-5` (Row, 5 columns)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Column1 (`column1text`)	Rich text	No (Recommended: fill)	No	“Column1”	First cell	Short label	None	Mumbai
Column2 (`column2text`)	Rich text	No (Recommended: fill)	No	“Column2”	Second cell	Value	None	India

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Column3 (`column3text`)	Rich text	No (Recommended: fill)	No	"Column3"	Third cell	Value	None	Headquarters
Column4 (`column4text`)	Rich text	No (Recommended: fill)	No	"Column4"	Fourth cell	Value	None	+91 …
Column5 (`column5text`)	Rich text	No (Recommended: fill)	No	"Column5"	Fifth cell	Value	None	Open

Field details

- **Options (`classes`)** — Purpose: style and header behaviour. Allowed Values: `striped`, `bordered`, `no-header`, `row-header` (any combination). Character Limits: n/a. Business Rules: options are added as block classes; No Header removes the `<thead>`; First Column Header makes the first cell of each body row a row header. Dependency Rules: Striped with No Header shades the first row; none. Visibility Rules: always visible (no `condition`). Sample Content: Striped + First Column Header.
- **Columns (`filter`)** — Purpose: chooses the Row type the block accepts (`table` → Row with 1 column, `table-2-columns` … `table-5-columns`). Allowed Values: 1–5 Columns. Character Limits: n/a. Business Rules: changing it later does not convert existing Rows; Rows already added keep their column count. Dependency Rules: decides the item model for new Rows. Visibility Rules: always visible. Sample Content: 4 Columns.
- **Anchor ID (`classes_anchor`)** — Purpose: a stable id for links (`#anchor-results`). Allowed Values: `anchor-` followed by lower-case letters, digits and hyphens. Character Limits: none enforced (Recommended: under 40 characters). Business Rules: becomes the block's `id`; if the id is already used on the page, `-2`, `-3`, … is appended; also used as the data layer component id. Dependency Rules: none. Visibility Rules: always visible. Sample Content: `anchor-results`.
- **Hide block (`classes_hideblock`)** — Purpose: hide the block on the published page. Allowed Values: on / off. Business Rules: hidden with `display: none` (also its wrapper) outside edit mode; shown at 45% opacity with a dashed outline in Universal Editor edit mode. Visibility Rules: always visible. Sample Content: off.
- **Padding left / right / top / bottom (`classes_padding*`)** — Purpose: inner spacing from the spacing tokens. Allowed Values: Default (none), 0 (none), 0.25rem (space-100), 0.5rem (space-200), 0.75rem (space-300), 1rem (space-400), 1.25rem (space-500), 1.5rem (space-600), 2rem (space-800), 2.5rem (space-1000), 3rem (space-1200), 4rem (space-1600), 5rem (space-2000), 6rem (space-2400), 7.5rem (space-3000); classes `pl-*`, `pr-*`, `pt-*`, `pb-*`. Business Rules: the same value on every breakpoint; styled by `styles/styles.css` (`main .section .block.pl-400`, …). Visibility Rules: always visible. Sample Content: Default.
- **Margin left / right / top / bottom (`classes_margin*`)** — Purpose: outer spacing. Allowed Values: as padding; classes `ml-*`, `mr-*`, `mt-*`, `mb-*`. Business Rules: same on every breakpoint. Visibility Rules: always visible. Sample Content: Margin bottom 2rem.
- **Column1 … Column5 (`column{n}text`)** — Purpose: the cell content. Allowed Values: rich text. Character Limits: none enforced; Recommended: header cells up to 30 characters, data cells up to 80 characters. Business Rules: content is moved into the `<th>` / `<td>` as authored; a paragraph containing only a link is turned into a link button by the content buttons decorator (bold link: primary button). Dependency Rules: the cell type (header / data) depends on the row position and the Options. Visibility Rules: always visible. Sample Content: "Revenue (USD bn)".

6.25.7 Field Annotation Mapping

Marker	Field	Purpose
A	Heading (default content, outside the block)	Names the table (`aria-labelledby`) and the scroll region

Marker	Field	Purpose
B	Table block: Options, Columns, Additional tab fields	Block-level styles, row type and spacing
C	First Row: Column1 … ColumnN	Column headers (unless No Header)
D	Cell: Column{n} (`column{n}text`)	The value of one cell
E	Body Row: Column1 … ColumnN	One line of data (first cell a row header with First Column Header)
F	Add (+) Row	Adds a Row of the type selected in Columns

6.25.8 Component Specification

Attribute	Value
Component Name	Table (`blocks/table`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`scripts/scripts.js` (moveInstrumentation); `utils/ids.js`; `utils/placeholders.js`; block options, data layer and content buttons decorators; ResizeObserver
Component Type	Container block (block with Row items)
Supported Variants	Standard; No Header; First Column Header; No Header + First Column Header; Striped; Bordered; 1–5 Columns; automatic scroll region
Supported Templates	Any page (all templates; any section)
Supported Metadata	None (the block reads no page metadata)
Supported Tags	None
Supported Personalization	None in code; future personalisation via the data layer and Adobe Target (not implemented)
Caching Considerations	Rendered from the page HTML, cached by the Edge Delivery CDN with the page; the placeholders sheet is fetched once per page and cached in memory; no other data is fetched
Performance Considerations	Small JS and CSS loaded only when the block is on the page; eager if in the first section, else lazy; no images; one ResizeObserver per table; no layout shift (rendered once)
Accessibility Compliance	Native table semantics; `scope="col"` / `scope="row"`; `aria-labelledby` to the heading before it; focusable named scroll region with visible focus (WCAG 1.3.1, 2.1.1, 2.4.7, 4.1.2)
SEO Compliance	Real `<table>`, `<thead>`, `<th>` markup in the DOM; the text stays crawlable; no images of text

6.25.9 Input Model

Content Inputs: The block's Rows, each with 1–5 rich-text cells.

Metadata Inputs: None.

Tags Inputs: None.

Page Context Inputs: The heading directly before the block (the last child of the preceding `.default-content-wrapper`, or a heading that is the preceding sibling of the block wrapper); the locale (for the placeholders sheet).

Query Inputs: None (no query index or sheet besides placeholders).

Manual Inputs: Options (classes), Columns (filter), Additional tab classes.

Authored HTML received from AEM (example, 3 columns, Striped + First Column Header):

```
<div class="table striped row-header">
  <div>
    <div><p>Office</p></div>
    <div><p>Country</p></div>
    <div><p>Type</p></div>
  </div>
  <div>
    <div><p>Mumbai</p></div>
    <div><p>India</p></div>
    <div><p>Headquarters</p></div>
  </div>
</div>
```

6.25.10 Output Model

Rendered Output: The block content is replaced by one `<table>`: optional `<thead>` with the first row (`<th scope="col">`), a `<tbody>` with the other rows (`<td>`, or `<th scope="row">` as first cell with First Column Header). The table gets `aria-labelledby` pointing to the heading before it (which receives an id `table-title-N` if it has none). When the table overflows, the block (`div.table.block`) gets `tabindex="0"`, `role="region"` and `aria-label`.

Generated HTML (example):

```
<h2 id="table-title-1">Our offices</h2>
<div class="table-wrapper">
  <div class="table striped row-header block" data-block-name="table"
       tabindex="0" role="region" aria-label="Our offices">
    <table aria-labelledby="table-title-1">
      <thead><tr><th scope="col">Office</th><th scope="col">Country</th>
        <th scope="col">Type</th></tr></thead>
      <tbody><tr><th scope="row">Mumbai</th><td>India</td><td>Headquarters</td></tr></
tbody>
    </table>
  </div>
</div>
```

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking: Generic data layer handling (`decorators/data-layer-decorator.js`):

```
{ "component": { "table-1": { "@type": "tcs-corp/components/table",
  "parentId": "page-1x2y3z" } } }
```

`dc:title` is added only if a heading is inside the block (normally it is left out). The component id is the Anchor ID when set, else `table-{n}`. Clicks on links in cells push:

```
{ "event": "cmp:click", "eventInfo": { "path": "component.table-1", "interaction": "link",
  "dc:title": "Annual report", "xdm:linkURL": "https://…/report.pdf",
  "xdm:linkType": "download", "breakpoint": "mobile" } }
```

Personalization Output: None.

Search Index Output: The query index (`helix-query.yaml`) indexes page metadata columns, not table content; the table text is part of the page HTML for external search engines.

6.25.11 Business Rules

1. Every child row of the block becomes a table row, in authored order.
2. The first row is the header row (`<thead>`, `<th scope="col">`) unless the `no-header` class is set.
3. With `row-header`, the first cell of every body row is `<th scope="row">`; in the header row the first cell stays a column header.
4. `no-header` and `row-header` can be combined; Striped and Bordered can be combined with everything.
5. Without a header row no `<thead>` is output; the first body row gets a 2px top border.
6. The number of cells per row is whatever the Row type delivers; the code does not pad rows or add colspans, so rows of different Row types produce ragged rows.
7. Cell content is moved as authored (rich text kept); editor instrumentation is moved to the new `<tr>` / `<td>` / `<th>` so in-place editing keeps working.
8. The heading before the table is found only if it is the last element of the immediately preceding default content wrapper (or the immediately preceding sibling of the block wrapper). If found and without an id, it gets `table-title-N`; the table gets `aria-labelledby`.
9. The scroll region is applied only while the content overflows (`scrollWidth > clientWidth + 1`) and removed when it fits; its label is the heading text, else the placeholder `table` (fallback "Table").
10. Empty block (no Rows): an empty `<table>` with an empty `<tbody>` is rendered (nothing visible).
11. Columns is limited to 1–5; a new Row is pre-filled with "Column1" … "ColumnN".
12. Changing Columns affects only Rows added afterwards.
13. Links alone in a cell paragraph are converted to link buttons by the content buttons decorator before the block is decorated (bold → primary button, italic → secondary, per the button options).
14. Additional tab: Anchor ID becomes the id (de-duplicated); Hide block hides it outside edit mode; padding / margin classes apply the spacing tokens on all breakpoints.
15. Text size: `--body-font-size-s` below 900px, `--body-font-size-m` from 900px.

6.25.12 Author Best Practices

Recommended Content Length: Up to 5 columns (maximum) and Recommended up to about 20 rows per table; header cells Recommended ≤ 30 characters; data cells Recommended ≤ 80 characters; split long tables by topic under separate headings.

Image Dimensions: Not applicable; images are not part of the table design (Recommended: no images in cells).

CTA Writing Guidelines: Links in cells should describe the target ("Download the annual report (PDF)"), not "Click here". Avoid buttons in cells; keep links inline in text where possible.

Accessibility Guidelines: Always place a descriptive heading directly before the table; use a header row (avoid No Header unless the data is obvious); use First Column Header for row labels; never leave header cells empty; don't use the table for layout; avoid merged meaning across cells; write units in the header ("Revenue (USD bn)").

SEO Guidelines: Use real text, not images of tables; give the heading before the table the key topic words; keep numbers and labels consistent with the page text.

Metadata Guidelines: No block metadata; ensure the page's Title and Description describe the data's topic.

Taxonomy Guidelines: Not applicable to the block; tag the page as usual.

Localization Guidelines: Translate all cell text in each locale; check that translated text (German, French) does not make the table much wider; adapt number and date formats to the locale (the block doesn't format them); the region label fallback comes from each locale's placeholders sheet (key `table`).

6.25.13 Troubleshooting Guide

Issue 1 — The first row of data is shown as a header

Symptoms: The first data row appears semibold with a thick rule below it.

Root Cause: The first row is always the header unless No Header is selected.

Diagnosis Steps: Select the block; check Options for No Header.

Resolution Steps: Either add a header Row at the top (and move it first in the content tree) or select No Header.

Validation Steps: Preview: the first row has normal weight (No Header) or column titles.

Preventive Measures: Always author the header Row first.

Issue 2 — "Add" offers only one-column Rows

Symptoms: New Rows have only Column1.

Root Cause: Columns is set to 1 Column (default filter `table`).

Diagnosis Steps: Select the block; check the Columns value.

Resolution Steps: Set Columns to the needed count; delete the one-column Rows and add new ones.

Validation Steps: A new Row shows Column1 … ColumnN in the properties panel.

Preventive Measures: Choose Columns before adding Rows.

Issue 3 — Ragged rows (some rows have fewer cells)

Symptoms: Some rows end early; borders and stripes don't line up.

Root Cause: Rows of different Row types (column counts) are mixed after Columns was changed.

Diagnosis Steps: Select each Row and count its fields in the properties panel.

Resolution Steps: Recreate the short Rows with the correct type and copy the values over.

Validation Steps: All rows have the same number of cells in preview.

Preventive Measures: Don't change Columns once Rows exist.

Issue 4 — Text "Column1", "Column2" visible on the page

Symptoms: Template text appears in published cells.

Root Cause: A new Row is pre-filled with the template values and wasn't edited.

Diagnosis Steps: Search the page preview for "Column".

Resolution Steps: Replace or clear the text in those cells.

Validation Steps: No template text in preview.

Preventive Measures: Validation checklist before publishing.

Issue 5 — The page scrolls sideways on mobile

Symptoms: The whole page can be dragged horizontally.

Root Cause: Normally prevented by `overflow-x: auto` on the block; a wide element inside a cell (for example a very long URL in a link button, or a margin from the Additional tab) or custom spacing can push beyond the section.

Diagnosis Steps: Inspect the page in the browser's device mode; find the element exceeding the viewport.

Resolution Steps: Shorten long unbreakable text (use link text instead of URLs), remove large left / right margins from the Additional tab.

Validation Steps: At 390px only the table scrolls.

Preventive Measures: Use descriptive link text; keep cell values short.

Issue 6 — Screen readers announce "Table" instead of the table's title

Symptoms: The scroll region or the table is announced without the expected name.

Root Cause: No heading is directly before the block (another element, or a heading in a different section), so no `aria-labelledby` and the fallback label "Table" are used.

Diagnosis Steps: In DevTools, check that the `<table>` has `aria-labelledby`; check what precedes the block wrapper.

Resolution Steps: Put the heading as the last default content element right before the table in the same section.

Validation Steps: `aria-labelledby` present; region label equals the heading text.

Preventive Measures: Always add a heading before a table.

Issue 7 — A link in a cell looks like a button

Symptoms: A cell link renders as a link button or a filled button.

Root Cause: The content buttons decorator turns a link that is alone in its paragraph into a link button (bold / italic links into buttons).

Diagnosis Steps: Check the cell's rich text: is the link the only content of the paragraph, or bold / italic?

Resolution Steps: Add text around the link in the same paragraph, or remove bold / italic.

Validation Steps: The link renders inline.

Preventive Measures: Write links inside sentences in table cells.

6.25.14 Support Runbook

Monitoring Requirements

- No runtime service; monitor page availability and Core Web Vitals (RUM / PageSpeed Insights) of pages with large tables.
- Check the placeholders sheet of each locale is published (`/{locale}/placeholders.json`).

Logs

- Browser console: the block emits no messages of its own. A failure would appear as an error in `table.js` (for example in very old browsers without `ResizeObserver`).
- Admin API / AEM Sidekick: preview and publish status of the page (`admin.hlx.page` status).
- AEM author: replication / publication logs for the page.

Common Errors

Error	Likely cause	Fix
Header row missing	No Header selected	Remove No Header
Ragged rows	Mixed Row types	Recreate Rows with one type
Template text published	New Rows not edited	Replace "ColumnN"
No table name	No heading before the block	Add a heading directly before
Table not visible	Hide block on	Turn off Hide block
Anchor link not jumping	Anchor ID invalid or duplicated (−2 suffix)	Fix the ID; update links

Escalation Matrix

Level	Who	Checks
L1	Author support	Options, Columns, Row types, template text, heading before table, Hide block, page published
L2	AMS / site admin	Preview vs live HTML (`.plain.html`), CDN cache, placeholders sheet published, publication logs
L3	Development team	`table.js` / `table.css` behaviour, console errors, browser support, code changes in `main`
L4	Adobe Support	Edge Delivery delivery or Universal Editor platform issues (rendering of Rows, instrumentation)

Recovery Procedures

1. Restore the previous page version in AEM and republish.
2. If a code change broke tables, revert the commit on `main` (code sync redeploys automatically) and verify on `main--tcs-corp--sradha-tcs.aem.page`.
3. Hide the block temporarily (Hide block) while fixing the content.

Health Checks

- Open a page with a table on desktop and mobile; check header semantics in the accessibility tree (DevTools > Accessibility).
- Confirm `aria-labelledby` and the region attributes on overflow.

Operational Procedures

- New column counts (6+) require a new Row model and filter in `_table.json` and `npm run build:json` (development task).
- Placeholder text for `table` is maintained per locale in the placeholders sheet.

Known Limitations

- Maximum 5 columns; no merged cells (colspan / rowspan), no sorting, filtering, captions (`<caption>`), column alignment options or sticky headers / first column.
- No responsive stacked layout: wide tables scroll horizontally.
- Changing Columns does not convert existing Rows.
- Only the heading immediately before the table names it.

6.26 Separator Block

The Separator is the Edge Delivery Services implementation of the AEM Core Components Separator: a horizontal division between pieces of content, rendered as one native `<hr>` element, with design-system variants, themes, widths, alignment, decorations, spacing, per-breakpoint (desktop / tablet / mobile) settings, an optional animation, print control and optional Adobe Client Data Layer state. Files: `blocks/separator/separator.js`, `separator-config.js` (normalisation of the settings, unit tested in `test/separator/separator-config.test.mjs`), `separator.css`, `_separator.json` and `README.md`. It is a key-value block: every dialog field arrives as a `name | value` row.

6.26.1 Component Purpose

Business Objective: Structure long pages visually and semantically with a consistent, design-system-controlled divider that authors can configure without CSS, that is accessible (semantic thematic break or purely decorative) and that never breaks the layout on any device.

Business Use Cases:

- Separate two stories, chapters or topics on a long article page (a semantic thematic break).
- Add a light visual rule below a section introduction (decorative).
- Add a short brand-coloured accent under a title (Branded, 25%, Start).
- Divide columns or content areas on desktop only (hidden on tablet / mobile).
- Add controlled vertical space only (Line style None), without a visible line.
- Create edge-to-edge dividers in a section (Full width / Full bleed).

When To Use:

- The topic changes within a section (Accessibility mode **Semantic**).
- A visual division between related content helps scanning (**Decorative**, the default).

When Not To Use:

- Only to add space when section spacing is enough (or use Line style None deliberately).
- To separate items of a list, cards or table rows (those blocks have their own dividers).
- Several separators in a row.
- As a heading replacement: a separator does not name the next section.

Dependencies:

Type	Dependency	Use
Code	`blocks/separator/separator-config.js`	Normalises properties (controlled lists, defaults, presets, legacy values) and builds the block classes; validation warnings; data layer entry
Code	`utils/properties.js` (`readProperties`, `textOf`)	Reads the key-value rows
Code	`utils/environment.js` (`isEditing`, `prefersReducedMotion`)	Editor notes; reduced-motion handling of the animation
Code	`utils/ids.js` (`uniqueId`)	Generated ids `separator-1`, `separator-2`, …; duplicate suffixes
Code	`utils/placeholders.js`	Editor note texts
Code	`utils/analytics.js` (`pushComponents`)	Data layer state when Analytics is on
Code	`decorators/block-options-decorator.js`	Additional tab (Anchor ID, Hide block, padding, margin)
Browser API	`IntersectionObserver`	Animate on view (only when an animation is set)
Sheet	`/{locale}/placeholders.json`	Keys `separatorHidden`, `separatorInvisible` (editor only)
Styles	`styles/styles.css` tokens	`--color-border-*`, `--slate-*`, `--brand-*`, `--white-*`, `--black-*`, `--border-width-*`, `--space-*`, `--content-max-width`, `--section-padding-inline`
Model	`models/_section.json`	Allows the Separator in sections

Related Components: Section (spacing, and the `dark` / `brand` section styles that Theme Auto follows), Table, Cards / List (own dividers), Anchor List (in-page navigation by Anchor ID).

Applicable Templates: Any page (home, article, blog, press-release, standard pages), in any section.

Supported Content Types: No content of its own: the separator renders no text, links or images on the published page. An `<hr>` authored inside the block is reused.

6.26.2 Visual Overview

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** At 1440px: a Default separator (1px, slate 400 line, 100% of the content width) between two text paragraphs, with 1.5rem space above and below; next to it a Branded 25% Start separator under a heading.

Desktop (≥ 1200px): The block is a flex row; the line (`.separator-line` containing the `<hr class="separator-rule">`) is centred by default (`justify-content` from Alignment). Default look: a solid 1px (`--border-width-100`) line in the default palette colour (`--color-border-neutral-tertiary`, slate 400 #b2b2b2) at 100% of the content width, capped at `--content-max-width` (90rem) because Respect content maximum width is on by default. Space above and below is padding of 1.5rem (`--space-600`, "24px"). Settings prefixed "Desktop" apply only here.

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** At 1024px: the same separators; one configured with Tablet width 50% shows at half width; a separator with Show on tablet off is absent.

Tablet (600–1199px): Same rendering; Tablet width / thickness / alignment / spacing overrides and Show on tablet apply only in this range (`@media (600px <= width < 1200px)`). Full bleed adds 2 × `--section-padding-inline` (2.5rem each side at this size).

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** At 390px: a separator at 100% width with Mobile top spacing None directly under a paragraph; a full-bleed separator touching both screen edges.

Mobile (< 600px): Same rendering with Mobile overrides and Show on mobile (`@media (width < 600px)`). Full bleed extends by the mobile section padding (1.5rem each side) so it reaches the screen edges without causing horizontal scrolling. All responsive behaviour is CSS; the script reads the settings once and adds no resize listeners.

6.26.3 Component Variations

Each variation below is a distinct presentation or behaviour the author can select. Settings combine: Variant sets line style, thickness, colour and width together, and any Appearance field other than "As the variant" overrides that part of the preset.

Variation 1 — Decorative (default accessibility mode)

Variation Name: Accessibility mode: Decorative (visual only) — class `decorative`.

Business Purpose: A visual division that assistive technologies ignore.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-decorative.png`
> **Purpose:** Variation example
> **Description:** A default thin grey line between two paragraphs; the DevTools accessibility tree beside it shows the line as ignored (aria-hidden).

Use Cases: Rules between related content blocks, visual rhythm in a section.

Authoring Guidelines: General tab > Accessibility mode = Decorative (visual only) (the default for a new block).

Business Rules: Default when the field is empty or unknown. Accessible label is ignored (and hidden in the dialog).

Rendering Rules: `<div class="separator-line" aria-hidden="true"><hr class="separator-rule"></div>`.

Responsive Behaviour: Same markup on all breakpoints; appearance per the other settings.

SEO Impact: None; an `<hr>` carries no ranking value.

Accessibility Compliance: WCAG 1.3.1: purely visual content is hidden from assistive technology so it does not add noise.

Variation 2 — Semantic (thematic break)

Variation Name: Accessibility mode: Semantic (a thematic break) — class `semantic`.

Business Purpose: Marks a real change of topic for screen-reader users, as the `<hr>` element intends.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-semantic.png`
> **Purpose:** Variation example
> **Description:** A separator between two stories; the accessibility tree shows a “separator” role named “End of the story”.

Use Cases: Between two stories, before an appendix, between chapters.

Authoring Guidelines: General tab > Accessibility mode = Semantic; optionally fill Accessible label (appears only in Semantic mode), for example “End of the case study”.

Business Rules: The Accessible label is trimmed and cut to 120 characters (Enforced in code); an empty label means an unnamed separator.

Rendering Rules: `<div class="separator-line"><hr class="separator-rule" aria-label="…"></div>` (no `aria-hidden`; `aria-label` only when a label is set).

Responsive Behaviour: Same on all breakpoints; if hidden on a breakpoint (Show on … off) it is removed from the accessibility tree there too (`display: none`).

SEO Impact: Neutral; semantically correct HTML.

Accessibility Compliance: WCAG 1.3.1 (the structure is conveyed programmatically); the label names the break; most screen readers read it, not all (README known limitation).

Variation 3 — Spacing only (Decorative + Line style None)

Variation Name: Spacing only — Decorative with Line style None (class `line-none`).

Business Purpose: Adds controlled vertical space with no visible line.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-spacing-only.png`
> **Purpose:** Variation example
> **Description:** Two paragraphs with a 4rem gap and no line; in Universal Editor the empty block area is selectable.

Use Cases: Extra breathing room before a call to action where section spacing is too small.

Authoring Guidelines: Accessibility mode = Decorative; Appearance > Line style = None (spacing only); set the top / bottom spacing on the Spacing tab (for example Desktop top spacing XL).

Business Rules: No `<hr>` is rendered for a decorative separator with Line style None; decorations are not rendered either. Semantic + None renders an invisible `<hr>` and shows the editor warning "Semantic separator without a line: screen readers announce a break nobody sees" (placeholder `separatorInvisible`).

Rendering Rules: The block keeps only its padding (default 1.5rem top and bottom); no child elements (except the editor note in edit mode, for semantic + none).

Responsive Behaviour: Spacing per breakpoint from the Spacing tab.

SEO Impact: None.

Accessibility Compliance: Nothing is exposed to assistive technology (decorative). Avoid Semantic + None (an announced break that nobody sees).

Variation 4 — Variant: Default

Variation Name: Separator variant: Default (`variant-default`).

Business Purpose: The standard neutral rule.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-variant-default.png`
> **Purpose:** Variation example
> **Description:** A solid 1px slate-400 line at 100% width.

Use Cases: Most dividers.

Authoring Guidelines: General tab > Separator variant = Default; leave Appearance fields on "As the variant".

Business Rules: Preset: line style solid, thickness thin, colour default, width 100%. Also the fallback for an empty or unknown variant.

Rendering Rules: Classes `variant-default line-solid thickness-thin color-default width-100-percent`; colour `--separator-default` of the active palette.

Responsive Behaviour: Identical on all breakpoints unless overridden on the Responsive tab.

SEO Impact: None.

Accessibility Compliance: As per the chosen Accessibility mode. The default colour on white is a light grey; it is decorative, so WCAG 1.4.11 (non-text contrast) does not require 3:1 for it; use Strong or High contrast where the line must be perceivable.

Variation 5 — Variant: Subtle

Variation Name: Separator variant: Subtle (`variant-subtle`).

Business Purpose: A very light rule that barely interrupts the reading flow.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-variant-subtle.png`
> **Purpose:** Variation example
> **Description:** A solid 1px very light grey (grey 200) line under a section intro.

Use Cases: Under introductions, between closely related paragraphs.

Authoring Guidelines: Separator variant = Subtle.

Business Rules: Preset: solid, thin, colour subtle, 100%.

Rendering Rules: `color-subtle` → `--separator-subtle` (light palette: `--color-border-default`, grey 200 #e6e6e6; dark: slate 800; brand: brand 300).

Responsive Behaviour: Identical on all breakpoints unless overridden.

SEO Impact: None.

Accessibility Compliance: Low contrast by design; use only decoratively.

Variation 6 — Variant: Strong

Variation Name: Separator variant: Strong (`variant-strong`).

Business Purpose: A clearly visible, heavier rule for major breaks.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-variant-strong.png`
> **Purpose:** Variation example
> **Description:** A solid 2px near-black (slate 900) line between two chapters.

Use Cases: Chapter breaks, the end of a main article before related content.

Authoring Guidelines: Separator variant = Strong (often with Semantic mode).

Business Rules: Preset: solid, medium (2px), colour strong, 100%.

Rendering Rules: `thickness-medium` → `--border-width-200` (0.125rem); `color-strong` → `--color-border-neutral` (slate 900 #303030) on light, slate 200 on dark, brand 600 on brand.

Responsive Behaviour: Identical unless overridden (for example Mobile thickness Thin).

SEO Impact: None.

Accessibility Compliance: High contrast line (meets 3:1 non-text contrast on white).

Variation 7 — Variant: Inset

Variation Name: Separator variant: Inset (`variant-inset`).

Business Purpose: A rule that stops short of both edges of its area.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-variant-inset.png`
> **Purpose:** Variation example
> **Description:** A thin grey line inset 2.5rem from the left and right content edges.

Use Cases: Dividers inside content areas that should not touch the edges.

Authoring Guidelines: Separator variant = Inset.

Business Rules: Preset: solid, thin, default colour, 100%; plus an inline inset.

Rendering Rules: `variant-inset` sets `--separator-inset: var(--space-1000)` (2.5rem) as inline padding of `.separator-line`; start / end decorations are positioned at the inset.

Responsive Behaviour: The 2.5rem inset is the same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: As per mode.

Variation 8 — Variant: Full width

Variation Name: Separator variant: Full width (`variant-full-width`).

Business Purpose: An edge-to-edge divider across the whole section.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-variant-full-width.png`
> **Purpose:** Variation example
> **Description:** A thin line spanning the full section width including the section's side padding.

Use Cases: Strong page structure, dividing full-width sections' content.

Authoring Guidelines: Separator variant = Full width. To let it reach beyond 90rem on very wide screens, also turn off Advanced > Respect content maximum width.

Business Rules: Preset: solid, thin, default colour, width Full bleed.

Rendering Rules: `width-full-bleed` sets `--separator-bleed: 1`: the line's width is `100% + 2 × --section-padding-inline` with matching negative inline margins, so it covers the section padding but never overflows the page.

Responsive Behaviour: The bleed follows the section padding: 1.5rem (mobile), 2.5rem (tablet), 5rem (desktop) each side.

SEO Impact: None.

Accessibility Compliance: As per mode.

Variation 9 — Variant: Dotted

Variation Name: Separator variant: Dotted (`variant-dotted`).

Business Purpose: A lighter, informal break.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-variant-dotted.png`
> **Purpose:** Variation example
> **Description:** A 2px dotted grey line.

Use Cases: Between informal content items, footnotes.

Authoring Guidelines: Separator variant = Dotted.

Business Rules: Preset: dotted, medium, default colour, 100%.

Rendering Rules: `line-dotted` sets `--separator-style: dotted` on the `border-block-start` of the `<hr>`.

Responsive Behaviour: Identical unless overridden.

SEO Impact: None.

Accessibility Compliance: As per mode.

Variation 10 — Variant: Dashed

Variation Name: Separator variant: Dashed (`variant-dashed`).

Business Purpose: A dashed break, e.g. for "continued" or optional content.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-variant-dashed.png`
> **Purpose:** Variation example
> **Description:** A 1px dashed grey line.

Use Cases: Before optional / supplementary content.

Authoring Guidelines: Separator variant = Dashed.

Business Rules: Preset: dashed, thin, default colour, 100%.

Rendering Rules: `line-dashed` → `--separator-style: dashed`.

Responsive Behaviour: Identical unless overridden.

SEO Impact: None.

Accessibility Compliance: As per mode.

Variation 11 — Variant: Gradient

Variation Name: Separator variant: Gradient (`variant-gradient`).

Business Purpose: A soft brand line that fades out at both ends.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-variant-gradient.png`
> **Purpose:** Variation example
> **Description:** A 2px brand-blue line fading to transparent at the left and right ends.

Use Cases: Premium / campaign pages, below hero-like intros.

Authoring Guidelines: Separator variant = Gradient.

Business Rules: Preset: gradient, medium, colour primary, 100%.

Rendering Rules: `line-gradient`: the `<hr>` has no border; its height is the thickness and its background is `linear-gradient(90deg, transparent, colour 20%, colour 80%, transparent)`. In forced-colours mode the background becomes `CanvasText`.

Responsive Behaviour: Identical unless overridden.

SEO Impact: None.

Accessibility Compliance: Visible in Windows high contrast (forced colours) thanks to the `CanvasText` fallback.

Variation 12 — Variant: Branded

Variation Name: Separator variant: Branded (`variant-branded`).

Business Purpose: A solid brand-colour accent.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-variant-branded.png`
> **Purpose:** Variation example
> **Description:** A 2px brand-blue line at 25% width aligned to the start under a heading.

Use Cases: Accent under titles (with 25%, Start), branded section breaks.

Authoring Guidelines: Separator variant = Branded; typically Width 25% and Alignment Start.

Business Rules: Preset: solid, medium, colour primary, 100%.

Rendering Rules: `color-primary` → `--color-border-brand` (brand 500 #4673db) on light; brand 300 on dark; brand 500 on brand.

Responsive Behaviour: Identical unless overridden.

SEO Impact: None.

Accessibility Compliance: Decorative accent normally (Decorative mode).

Variation 13 — Variant: Custom

Variation Name: Separator variant: Custom (only the Appearance settings) (`variant-custom-design`).

Business Purpose: Full control over the look via the Appearance tab.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-variant-custom.png`
> **Purpose:** Variation example
> **Description:** A custom separator: dashed, thick, accent (yellow) line at 75% width, end aligned.

Use Cases: Cases not covered by a preset.

Authoring Guidelines: Separator variant = Custom; set Line style, Line thickness, Line colour and Separator width explicitly.

Business Rules: Its preset equals Default (solid, thin, default, 100%) for any Appearance field left on "As the variant".

Rendering Rules: Classes from the Appearance values.

Responsive Behaviour: As set.

SEO Impact: None.

Accessibility Compliance: Author responsibility for contrast (prefer Strong / High contrast for meaningful lines).

Variation 14 — Theme: Auto (as the section)

Variation Name: Theme Auto (`theme-auto`, default).

Business Purpose: The palette follows the section background automatically.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-theme-auto.png`
> **Purpose:** Variation example
> **Description:** The same Default separator in a white section (slate 400 line) and in a Highlight (light grey) section.

Use Cases: All separators on standard sections.

Authoring Guidelines: General tab > Theme = Auto (default).

Business Rules: Uses the dark palette inside `.section.dark` and the brand palette inside `.section.brand`, else the light palette. Note: the Section dialog's Style currently offers only **Highlight**; the `dark` and `brand` section classes exist in the site CSS but are not selectable in the section model as of this handover, so Auto resolves to the light palette in authored sections ([To be confirmed: whether dark / brand section styles will be added — owner: product owner]).

Rendering Rules: `.section.dark .separator.theme-auto` / `.section.brand .separator.theme-auto` redefine the palette variables.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Keeps the line visible against dark / brand backgrounds.

Variation 15 — Theme: Light background

Variation Name: Theme Light (`theme-light`).

Business Purpose: Forces the light palette, for example on a light background image.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-theme-light.png`
> **Purpose:** Variation example
> **Description:** A separator on a light section background image using the light palette.

Use Cases: Sections whose background image is light.

Authoring Guidelines: Theme = Light background.

Business Rules: Never switches to the dark / brand palette.

Rendering Rules: Light palette: default slate 400, subtle grey 200, neutral slate 600, strong slate 900, primary brand 500, secondary brand 300, accent warning 500 (#e8b931), inverse white 70%.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Choose a colour with enough contrast against the image.

Variation 16 — Theme: Dark background

Variation Name: Theme Dark (`theme-dark`).

Business Purpose: Palette for dark backgrounds (images or dark sections).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-theme-dark.png`
> **Purpose:** Variation example
> **Description:** A Strong separator (slate 200) on a dark background image.

Use Cases: Sections with a dark background image.

Authoring Guidelines: Theme = Dark background.

Business Rules: Uses the dark palette regardless of the section.

Rendering Rules: Dark palette: default slate 600, subtle slate 800, neutral slate 500, strong slate 200, primary brand 300, secondary brand 400, inverse black 70%; accent unchanged.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Lighter lines for dark backgrounds.

Variation 17 — Theme: Brand background

Variation Name: Theme Brand (`theme-brand`).

Business Purpose: Palette for brand-coloured backgrounds.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-theme-brand.png`
> **Purpose:** Variation example
> **Description:** A Default separator (brand 500) on a light-blue brand background.

Use Cases: Campaign sections with brand backgrounds.

Authoring Guidelines: Theme = Brand background.

Business Rules: Uses the brand palette regardless of the section.

Rendering Rules: Brand palette: default brand 500, subtle brand 300, neutral slate 600, strong brand 600 (#011627), primary brand 500, secondary brand 400, inverse white.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Choose Strong for meaningful lines on brand backgrounds.

Variation 18 — High background contrast

Variation Name: Background contrast: High (stronger line) (`contrast-high`).

Business Purpose: Keeps the line visible on busy or tinted backgrounds.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-contrast-high.png`
> **Purpose:** Variation example
> **Description:** A separator on a tinted background using the strong palette colour.

Use Cases: Background images, Highlight sections.

Authoring Guidelines: Appearance > Background contrast = High (stronger line).

Business Rules: Overrides the Line colour: the line uses the palette's strong colour.

Rendering Rules: `.separator.contrast-high { --separator-color: var(--separator-strong) }` (declared after the colour classes, so it wins).

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Supports WCAG 1.4.11 for meaningful (semantic) lines.

Variation 19 — Line opacity

Variation Name: Line opacity 25% / 50% / 75% (`opacity-25`, `opacity-50`, `opacity-75`; 100% default).

Business Purpose: Softens any line colour.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-opacity.png`
> **Purpose:** Variation example
> **Description:** Four Strong separators at 100%, 75%, 50% and 25% opacity.

Use Cases: Fine-tuning of brand or strong lines.

Authoring Guidelines: Appearance > Line opacity.

Business Rules: Applies to the whole line including decorations; unknown values fall back to 100.

Rendering Rules: `--separator-opacity` on `.separator-line` (0.25 / 0.5 / 0.75 / 1). Print always uses opacity 1.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Lower opacity reduces contrast; keep meaningful lines at 100%.

Variation 20 — Partial width (25% / 50% / 75%)

Variation Name: Separator width 25%, 50% or 75% (`width-25-percent`, …).

Business Purpose: Short accent lines or centred dividers.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-partial-width.png`
> **Purpose:** Variation example
> **Description:** Three separators at 25%, 50% and 75% width, centred.

Use Cases: Accent under a title (25%), light division under an intro (50%).

Authoring Guidelines: Appearance > Separator width = 25%, 50% or 75%; set Alignment (Start, Center, End).

Business Rules: The share is of the content width of the block's container. Alignment applies to partial widths; Start and End follow the reading direction (right-to-left languages too).

Rendering Rules: `--separator-width: 25% / 50% / 75%`; `align-start | center | end` sets `justify-content` (logical values).

Responsive Behaviour: Can be changed per breakpoint on the Responsive tab.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 21 — Content width

Variation Name: Separator width: Content width (`width-content-width`).

Business Purpose: Full width, but never wider than the site's content width.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-content-width.png`
> **Purpose:** Variation example
> **Description:** On a 2560px screen, a line capped at 90rem in a full-width section.

Use Cases: Full-width sections on very wide screens.

Authoring Guidelines: Appearance > Separator width = Content width.

Business Rules: 100% width capped at `--content-max-width` (90rem).

Rendering Rules: `--separator-width: 100%`, `--separator-cap: var(--content-max-width)`.

Responsive Behaviour: Identical below 90rem of available width.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 22 — Full bleed width

Variation Name: Separator width: Full bleed (section edge to edge) (`width-full-bleed`).

Business Purpose: Edge-to-edge line with any variant.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-full-bleed.png`
> **Purpose:** Variation example
> **Description:** A Branded line covering the section padding on both sides at mobile width.

Use Cases: Strong section divisions on mobile and desktop.

Authoring Guidelines: Appearance > Separator width = Full bleed; turn off Respect content maximum width to exceed 90rem on very wide screens.

Business Rules: Same mechanics as the Full width variant (Variation 8).

Rendering Rules: `--separator-bleed: 1` (width plus the section's inline padding on both sides, negative margins).

Responsive Behaviour: Bleed 1.5rem / 2.5rem / 5rem per side on mobile / tablet / desktop.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 23 — Maximum width

Variation Name: Maximum width Small / Medium / Large / Extra large / Content container (`max-small` 20rem, `max-medium` 40rem, `max-large` 60rem, `max-extra-large` 80rem, `max-content-container` 90rem).

Business Purpose: Caps the line length regardless of the percentage width.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-max-width.png`
> **Purpose:** Variation example
> **Description:** A 100% separator capped at Medium (40rem) on desktop, centred.

Use Cases: Consistent short dividers on wide screens.

Authoring Guidelines: Appearance > Maximum width (labels show px: 320 / 640 / 960 / 1280px).

Business Rules: The line's max-width is the minimum of Maximum width, the Content width cap and the Respect content maximum width limit.

Rendering Rules: `max-width: min(--separator-max, --separator-cap, --separator-limit)`.

Responsive Behaviour: The cap is the same on all breakpoints; on small screens the percentage width usually wins.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 24 — Stretch to container width

Variation Name: Stretch to container width (`stretch`).

Business Purpose: Forces the line to fill its container on every breakpoint.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-stretch.png`
> **Purpose:** Variation example
> **Description:** A separator configured at 25% that still fills its container because Stretch is on.

Use Cases: Overriding widths inherited from a variant without changing each width field.

Authoring Guidelines: Advanced > Stretch to container width = on.

Business Rules: Ignores width, per-breakpoint widths and full bleed (width 100%, no negative margins). The maximum width caps still apply.

Rendering Rules: `.separator.stretch .separator-line { width: 100%; margin-inline: 0 }`.

Responsive Behaviour: 100% on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 25 — Dot decoration

Variation Name: Decoration: Dot (start, center and / or end) (`separator-decoration-dot`).

Business Purpose: A round ornament on the line.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-decoration-dot.png`
> **Purpose:** Variation example
> **Description:** A thin line with a dot at its centre.

Use Cases: Decorative chapter breaks.

Authoring Guidelines: Appearance > Start / Center / End decoration = Dot.

Business Rules: Decorations render only when the line is visible (not with Line style None); the block gets the `decorated` class.

Rendering Rules: `<span class="separator-decoration separator-decoration-{position} separator-decoration-dot" aria-hidden="true">`, a circle in the line colour, size `thickness × 3 + 0.375rem` (thin: 0.5625rem; medium: 0.75rem; thick: 0.9375rem), vertically centred on the line.

Responsive Behaviour: Scales with the (per-breakpoint) thickness.

SEO Impact: None.

Accessibility Compliance: Hidden from assistive technology; forced colours: `CanvasText`.

Variation 26 — Square decoration

Variation Name: Decoration: Square (`separator-decoration-square`).

Business Purpose: A square ornament.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-decoration-square.png`
> **Purpose:** Variation example
> **Description:** A line with squares at the start and the end.

Use Cases: Technical / structured design accents.

Authoring Guidelines: Start / Center / End decoration = Square.

Business Rules: As Dot.

Rendering Rules: Same span with the square shape (no radius).

Responsive Behaviour: Scales with the thickness.

SEO Impact: None.

Accessibility Compliance: `aria-hidden="true"`.

Variation 27 — Diamond decoration

Variation Name: Decoration: Diamond (`separator-decoration-diamond`).

Business Purpose: A diamond ornament (for example a decorated chapter break).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-decoration-diamond.png`
> **Purpose:** Variation example
> **Description:** A medium line at 50% width with a diamond at the centre.

Use Cases: "A decorated chapter break: Semantic, Medium, 50%, Center decoration Diamond" (README example).

Authoring Guidelines: Center decoration = Diamond (or start / end).

Business Rules: As Dot.

Rendering Rules: Square rotated 45° and scaled to 0.8.

Responsive Behaviour: Scales with the thickness.

SEO Impact: None.

Accessibility Compliance: `aria-hidden="true"`.

Variation 28 — Hidden on desktop

Variation Name: Show on desktop off (`hide-desktop`).

Business Purpose: A divider needed only on smaller screens (for example where columns stack).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-hidden-desktop.png`
> **Purpose:** Variation example
> **Description:** Mobile view with the separator between stacked content; desktop view of the same page without it.

Use Cases: Separating stacked columns on mobile / tablet.

Authoring Guidelines: Responsive > Show on desktop = off.

Business Rules: Hidden from 1200px; the spacing disappears too.

Rendering Rules: `@media (width >= 1200px) { .separator.hide-desktop { display: none } }`.

Responsive Behaviour: Visible on tablet and mobile only.

SEO Impact: None.

Accessibility Compliance: Removed from the accessibility tree where hidden.

Variation 29 — Hidden on tablet

Variation Name: Show on tablet off (`hide-tablet`).

Business Purpose: Remove a divider in the tablet range.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-hidden-tablet.png`
> **Purpose:** Variation example
> **Description:** The page at 1024px without the separator, and at 1440px with it.

Use Cases: Layouts that look crowded on tablet.

Authoring Guidelines: Responsive > Show on tablet = off.

Business Rules: Hidden from 600px to 1199px.

Rendering Rules: `display: none` in `@media (600px <= width < 1200px)`.

Responsive Behaviour: Visible on desktop and mobile.

SEO Impact: None.

Accessibility Compliance: Removed where hidden.

Variation 30 — Hidden on mobile

Variation Name: Show on mobile off (`hide-mobile`).

Business Purpose: A desktop / tablet-only divider (for example between two columns).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-hidden-mobile.png`
> **Purpose:** Variation example
> **Description:** Desktop view with the separator; mobile view without it.

Use Cases: "A desktop-only divider of two columns: Show on tablet / mobile off" (README).

Authoring Guidelines: Responsive > Show on mobile = off.

Business Rules: Hidden below 600px.

Rendering Rules: `display: none` in `@media (width < 600px)`.

Responsive Behaviour: Visible on desktop and tablet.

SEO Impact: None.

Accessibility Compliance: Removed where hidden.

Variation 31 — Hidden on all devices

Variation Name: Show on desktop, tablet and mobile all off (`hide-desktop hide-tablet hide-mobile hidden-all`).

Business Purpose: Temporarily removing a separator while keeping its settings (normally not recommended; Hide block on the Additional tab does the same more clearly).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-hidden-all.png`
> **Purpose:** Variation example
> **Description:** Universal Editor edit mode: a dashed placeholder box with the note "Separator hidden on all devices"; the published page shows nothing.

Use Cases: Temporary removal.

Authoring Guidelines: Turn off all three Show on … toggles.

Business Rules: Nothing on the page; in Universal Editor edit mode the block stays visible with a dashed outline and the note from placeholder `separatorHidden` ("Separator hidden on all devices").

Rendering Rules: `.separator.hidden-all { display: none }; .adobe-ue-edit .separator.hidden-all { display: block; outline: 1px dashed }`; the note is `<p class="separator-editor-note">`.

Responsive Behaviour: Hidden everywhere.

SEO Impact: None.

Accessibility Compliance: Not exposed.

Variation 32 — Desktop overrides (width, thickness, alignment)

Variation Name: Desktop width / Desktop thickness / Desktop alignment (`desktop-width-*`, `desktop-thickness-*`, `desktop-align-*`).

Business Purpose: A different look on large screens.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-desktop-overrides.png`
> **Purpose:** Variation example
> **Description:** A separator that is 100% thin on mobile and 50% medium, start-aligned on desktop.

Use Cases: Shorter accent lines on wide screens.

Authoring Guidelines: Responsive tab: Desktop width (As Appearance, Auto, 25%, 50%, 75%, 100%, Content width, Full bleed), Desktop thickness (As Appearance, Thin, Medium, Thick), Desktop alignment (As Appearance, Start, Center, End).

Business Rules: "As Appearance" (`inherit`) keeps the Appearance value; a breakpoint width other than Full bleed removes the bleed.

Rendering Rules: Classes set the custom properties inside `@media (width >= 1200px)`.

Responsive Behaviour: Applies from 1200px only.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 33 — Tablet overrides (width, thickness, alignment)
Variation Name: Tablet width / thickness / alignment (`tablet-width-*`, `tablet-thickness-*`, `tablet-align-*`).

Business Purpose: A different look in the tablet range.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-tablet-overrides.png`
> **Purpose:** Variation example
> **Description:** At 1024px a separator at 75% width, centred, while desktop shows 50%.

Use Cases: Intermediate widths for tablet layouts.

Authoring Guidelines: Responsive tab: Tablet width, Tablet thickness, Tablet alignment.

Business Rules: As desktop overrides.

Rendering Rules: Inside `@media (600px <= width < 1200px)`.

Responsive Behaviour: 600–1199px only.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 34 — Mobile overrides (width, thickness, alignment)
Variation Name: Mobile width / thickness / alignment (`mobile-width-*`, `mobile-thickness-*`, `mobile-align-*`).

Business Purpose: A different look on phones.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/separator-mobile-overrides.png`
> **Purpose:** Variation example
> **Description:** At 390px a separator at 100% thin while desktop shows a 25% medium accent.

Use Cases: Full-width lines on small screens.

Authoring Guidelines: Responsive tab: Mobile width, Mobile thickness, Mobile alignment.

Business Rules: As desktop overrides.

Rendering Rules: Inside `@media (width < 600px)`.

Responsive Behaviour: Below 600px only.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 35 — Desktop spacing
Variation Name: Desktop top / bottom spacing (`desktop-top-*`, `desktop-bottom-*`).

Business Purpose: Vertical space around the line on large screens.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-desktop-spacing.png`
> **Purpose:** Variation example
> **Description:** A separator with Desktop top spacing None and bottom spacing Large under a heading.

Use Cases: "A brand accent under a title: Desktop top spacing None" (README).

Authoring Guidelines: Spacing tab: Desktop top spacing, Desktop bottom spacing (Default, None, 2XS 4px, XS 8px, Small 16px, Medium 24px, Large 40px, XL 64px, 2XL 80px).

Business Rules: Default = 1.5rem (24px); values map to `--space-100` (0.25rem), `--space-200` (0.5rem), `--space-400` (1rem), `--space-600` (1.5rem), `--space-1000` (2.5rem), `--space-1600` (4rem), `--space-2000` (5rem); None = 0. Spacing is padding, so it does not collapse with neighbouring margins.

Rendering Rules: `--separator-space-top` / `--separator-space-bottom` inside `@media (width >= 1200px)`.

Responsive Behaviour: Desktop only; tablet and mobile keep their own values.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 36 — Tablet spacing

Variation Name: Tablet top / bottom spacing (`tablet-top-*`, `tablet-bottom-*`).

Business Purpose: Vertical space in the tablet range.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-tablet-spacing.png`
> **Purpose:** Variation example
> **Description:** At 1024px a separator with Small spacing above and below.

Use Cases: Tighter rhythm on tablet.

Authoring Guidelines: Spacing tab: Tablet top spacing, Tablet bottom spacing.

Business Rules: As desktop spacing.

Rendering Rules: Inside `@media (600px <= width < 1200px)`.

Responsive Behaviour: 600–1199px only.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 37 — Mobile spacing

Variation Name: Mobile top / bottom spacing (`mobile-top-*`, `mobile-bottom-*`).

Business Purpose: Vertical space on phones.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-mobile-spacing.png`
> **Purpose:** Variation example
> **Description:** At 390px a separator with XS spacing above and below.

Use Cases: Less whitespace on small screens.

Authoring Guidelines: Spacing tab: Mobile top spacing, Mobile bottom spacing.

Business Rules: As desktop spacing.

Rendering Rules: Inside `@media (width < 600px)`.

Responsive Behaviour: Below 600px only.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 38 — Use section padding

Variation Name: Use section padding (`section-padding`).

Business Purpose: Avoids doubled spacing when the section already provides it.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-section-padding.png`
> **Purpose:** Variation example
> **Description:** A separator at the top of a Highlight section with no space of its own.

Use Cases: A separator as the first or last element of a padded section.

Authoring Guidelines: Spacing tab > Use section padding = on.

Business Rules: Removes the separator's own top and bottom spacing on all breakpoints; all six spacing selects are then ignored.

Rendering Rules: `.separator.section-padding { padding-block: 0 }`.

Responsive Behaviour: All breakpoints.

SEO Impact: None.

Accessibility Compliance: Visual only.

Variation 39 — Animate on view: Fade

Variation Name: Animate on view = Fade (`animate-fade`).

Business Purpose: A gentle reveal as the user scrolls.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-animate-fade.png`
> **Purpose:** Variation example
> **Description:** Two frames: the line invisible before it enters the viewport, then fully visible.

Use Cases: Storytelling pages.

Authoring Guidelines: Advanced > Animate on view = Fade; Reduced motion fallback (shown only when an animation is set): Show immediately or Short fade.

Business Rules: Runs once, when 10% of the block is in view; for visitors who prefer reduced motion: Show immediately = no animation, Short fade = fade only. The hidden start state only exists after the script added `animate-ready`, so a script failure never hides the line.

Rendering Rules: `animate-ready` → line opacity 0; `is-visible` added by an IntersectionObserver (threshold 0.1) → opacity 1 with a 0.6s ease-out transition.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Respects the visitor's reduced-motion preference (good practice aligned with WCAG 2.3.3, an AAA criterion); the one-off 0.6s animation is not affected by WCAG 2.2.2. Implemented via `prefers-reduced-motion`.

Variation 40 — Animate on view: Draw from start

Variation Name: Animate on view = Draw from start (`animate-draw-from-start`).

Business Purpose: The line "draws" itself from the start edge.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-animate-draw-start.png`
> **Purpose:** Variation example
> **Description:** Three frames of the line growing from the left edge.

Use Cases: Emphasis at chapter starts.

Authoring Guidelines: Animate on view = Draw from start; set the Reduced motion fallback.

Business Rules: As Fade; with reduced motion and Short fade, it only fades.

Rendering Rules: Start state `transform: scaleX(0)`, origin left (right in right-to-left pages, `:dir(rtl)`), animated to full width in 0.6s.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Respects reduced motion.

Variation 41 — Animate on view: Draw from center

Variation Name: Animate on view = Draw from center (`animate-draw-from-center`).

Business Purpose: The line grows outwards from the centre.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-animate-draw-center.png`
> **Purpose:** Variation example
> **Description:** Three frames of the line growing from its centre to both ends.

Use Cases: Centred, symmetrical layouts.

Authoring Guidelines: Animate on view = Draw from center.

Business Rules: As Fade.

Rendering Rules: `scaleX(0)` with the default centre origin, animated to full width.

Responsive Behaviour: Same on all breakpoints.

SEO Impact: None.

Accessibility Compliance: Respects reduced motion.

Variation 42 — Hide when printed

Variation Name: Print = Hide when printed (`print-hide`).

Business Purpose: Screen-only dividers don't waste paper.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-print-hide.png`
> **Purpose:** Variation example
> **Description:** The browser print preview of a page without the separator.

Use Cases: Decorative separators on pages that are often printed (press releases).

Authoring Guidelines: Advanced > Print = Hide when printed.

Business Rules: Default is Show when printed; printed lines are always fully opaque and not animated.

Rendering Rules: `@media print { .separator.print-hide { display: none } }`.

Responsive Behaviour: Print only.

SEO Impact: None.

Accessibility Compliance: Not applicable.

Variation 43 — Analytics enabled

Variation Name: Analytics on (data attribute `data-analytics="custom"`).

Business Purpose: Lets analytics see that a specific separator was rendered (rarely needed).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-analytics.png`
> **Purpose:** Variation example
> **Description:** The browser console showing window.adobeDataLayer with the separator's component entry.

Use Cases: A/B or design tests that need the separator's configuration in the data layer.

Authoring Guidelines: Advanced > Analytics = on; optionally Analytics label (shown only when Analytics is on).

Business Rules: Off by default (`data-analytics="off"`: no tracking). On: one component state entry; no events (a separator has no interactions). The label is trimmed and cut to 120 characters (Enforced in code).

Rendering Rules: No visual change.

Responsive Behaviour: Not applicable; `xdm:visibility` lists the breakpoints it shows on.

SEO Impact: None.

Accessibility Compliance: Not applicable.

6.26.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** Universal Editor edit mode with a Separator selected between two text components; the blue selection outline around the thin line and its spacing; the content tree shows "Separator" in the section.

Annotations:

- A = The Separator block (the line and its top / bottom spacing; select it to open its properties)
- B = The rule (`<hr>`), styled by the variant, theme and Appearance settings
- C = A decoration (start / center / end), when set
- D = The editor note ("Separator hidden on all devices" or "Semantic separator without a line: …"), shown only in edit mode when a warning applies
- E = The block entry "Separator" in the content tree

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The properties panel of a selected Separator with its tabs General, Appearance, Responsive, Spacing, Advanced and Additional; the General tab open.

Properties panel tabs:

- **General:** Component ID; Accessibility mode; Accessible label (only when Semantic); Separator variant; Theme.
- **Appearance:** Line style; Line thickness; Line colour; Line opacity; Separator width; Maximum width; Alignment; Start decoration; Center decoration; End decoration; Background contrast.
- **Responsive:** Show on desktop; Show on tablet; Show on mobile; Desktop width (help "1200px and wider"); Desktop thickness; Desktop alignment; Tablet width ("600 to 1199px"); Tablet thickness; Tablet alignment; Mobile width ("up to 599px"); Mobile thickness; Mobile alignment.
- **Spacing:** Desktop / Tablet / Mobile top spacing and bottom spacing; Use section padding.
- **Advanced:** Stretch to container width; Respect content maximum width; Animate on view; Reduced motion fallback (only when an animation is set); Print; Analytics; Analytics label (only when Analytics is on).
- **Additional:** Anchor ID; Hide block; Padding left / right / top / bottom; Margin left / right / top / bottom.

6.26.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Open the page from AEM Sites with **Edit**.

Expected Result: The page is shown in the editor in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-01-open-page.png`
> **Purpose:** Authoring step
> **Description:** The page open in Universal Editor.

Step 2 — Add the Separator

Action: Select the section (or the component after which the separator goes), choose **Add** (+) and select **Separator**.

Expected Result: A Default, Decorative, centred, thin full-width line appears with 1.5rem space above and below.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-02-add-block.png`
> **Purpose:** Authoring step
> **Description:** The Add menu with Separator, and the new separator in the canvas.

Step 3 — General tab

Action: Choose the Accessibility mode (Semantic for a change of topic, else Decorative); for Semantic, optionally type an Accessible label. Choose a Separator variant and a Theme (Auto unless the background is an image). Leave Component ID empty unless analytics, tests or links need a fixed id.

Expected Result: The canvas updates with the preset look.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-03-general.png`
> **Purpose:** Authoring step
> **Description:** The General tab with Semantic mode and the Accessible label field visible.

Step 4 — Appearance tab (optional)

Action: Override parts of the preset: Line style, thickness, colour, opacity, width, maximum width, alignment, decorations and background contrast. Keep "As the variant" for what the preset should decide.

Expected Result: The line changes accordingly.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-04-appearance.png`
> **Purpose:** Authoring step
> **Description:** The Appearance tab with Width 25%, Alignment Start and Center decoration Diamond.

Step 5 — Responsive tab (optional)

Action: Turn devices on / off and set per-device width, thickness and alignment ("As Appearance" keeps the Appearance value).

Expected Result: The editor shows the setting of the current editor width; check other widths in Preview.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-05-responsive.png`

> **Purpose:** Authoring step
> **Description:** The Responsive tab with Show on mobile off and Desktop width 50%.

Step 6 — Spacing tab (optional)

Action: Choose top and bottom spacing per device, or turn on Use section padding.

Expected Result: The space around the line changes.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-06-spacing.png`
> **Purpose:** Authoring step
> **Description:** The Spacing tab with Desktop top spacing None.

Step 7 — Advanced tab (optional)

Action: Set Stretch to container width, Respect content maximum width, Animate on view and its Reduced motion fallback, Print, and Analytics (with an Analytics label).

Expected Result: Settings saved; animation is visible only when scrolling the page in Preview / on the site.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-07-advanced.png`
> **Purpose:** Authoring step
> **Description:** The Advanced tab with Animate on view Draw from start and the Reduced motion fallback field visible.

Step 8 — Additional tab (optional)

Action: Set an Anchor ID (only if there is no Component ID; Component ID takes precedence), Hide block, padding or margin.

Expected Result: Values accepted; an invalid Anchor ID shows the validation message.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-08-additional.png`
> **Purpose:** Authoring step
> **Description:** The Additional tab of the separator.

Step 9 — Preview

Action: Switch to **Preview** and check desktop, tablet and mobile widths, and the animation by scrolling.

Expected Result: The separator renders per breakpoint; editor notes are not shown in Preview.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-09-preview.png`
> **Purpose:** Authoring step
> **Description:** Preview mode at mobile width.

Step 10 — Validate and publish

Action: Run the Validation Process checklist, then **Publish**.

Expected Result: The separator is live.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/separator-step-10-publish.png`
> **Purpose:** Authoring step
> **Description:** The Publish dialog.

Preview Process

1. Universal Editor Preview mode (no editor notes, no hidden-everywhere placeholder).
2. Edge Delivery preview `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}`; append `?debug=separator` to log each separator's configuration and warnings to the browser console (`[separator]` debug messages).
3. Check at 390px, 1024px and 1440px; check print preview if Print is set.

Publishing Process

Publish the page from Universal Editor or AEM Sites; the live URL `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/{path}` updates after publication. The block has no assets of its own.

Validation Process

- ☐ Semantic only where the topic changes; Decorative otherwise.
- ☐ No Semantic separator with Line style None (no editor warning visible).
- ☐ Not hidden on all devices (no "Separator hidden on all devices" note).
- ☐ Theme matches the background (Auto for standard sections; Light / Dark / Brand on images).
- ☐ The line is visible enough where it matters (Strong or High contrast).
- ☐ Spacing is not doubled with section padding (use Use section padding where needed).
- ☐ No horizontal page scrolling at 390px (Full bleed is designed not to overflow).
- ☐ Animation (if any) plays once; with reduced motion the line is visible immediately.
- ☐ Component ID (if set) is unique and lower-case (the code lower-cases it).
- ☐ Not several separators in a row.

Rollback Process

1. Restore the previous page version in AEM (Version History) and republish.
2. Or delete the separator / turn on Hide block and publish.
3. If analytics rules depend on the Component ID, restore the previous ID.

6.26.6 Authorable Field Specification

Model `separator` (key-value block)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Component ID (`componentId`) — General	Text	No	No	Empty (generated `separator-N`)	Stable id for links, CSS, analytics, tests	Leave empty unless needed	Enforced: regExp `^[A-Za-z][A-Za-z0-9_-]*$`; code lower-cases and sanitises; duplicates suffixed	story-end

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Accessibility mode (`accessibilityMode`) — General	Select	No	No	Decorative	Semantic break or visual only	Semantic for topic changes	Enforced: option list	semantic
Accessible label (`accessibleLabel`) — General	Text	No	No	Empty	Names a semantic break	Short phrase	Enforced in code: trimmed, max 120 characters; shown only when Semantic	End of the case study
Separator variant (`variant`) — General	Select	No	No	Default	Preset of style, thickness, colour, width	Pick the closest preset	Enforced: option list	branded
Theme (`theme`) — General	Select	No	No	Auto (as the section)	Palette for the background	Auto unless on an image	Enforced: option list	auto
Line style (`lineStyle`) — Appearance	Select	No	No	As the variant	Solid, dashed, dotted, gradient, none	Override only if needed	Enforced: option list	dashed
Line thickness (`lineThickness`) — Appearance	Select	No	No	As the variant	1px / 2px / 3px	—	Enforced: option list	medium
Line colour (`lineColor`) — Appearance	Select	No	No	As the variant	Palette colour	—	Enforced: option list	primary
Line opacity (`lineOpacity`) — Appearance	Select	No	No	100%	Softens the line	Keep 100% for meaningful lines	Enforced: option list	50
Separator width (`width`) — Appearance	Select	No	No	As the variant	Share of the content width	Use with Alignment	Enforced: option list	25-percent
Maximum width (`maxWidth`) — Appearance	Select	No	No	None	Caps the line length	—	Enforced: option list	medium
Alignment (`alignment`) — Appearance	Select	No	No	Center	Position of partial widths	Start / End follow reading direction	Enforced: option list	start
Start decoration (`startDecoration`) — Appearance	Select	No	No	None	Shape at the start	Visual only	Enforced: option list	dot
Center decoration (`centerDecoration`) — Appearance	Select	No	No	None	Shape at the centre	Visual only	Enforced: option list	diamond
End decoration (`endDecoration`) — Appearance	Select	No	No	None	Shape at the end	Visual only	Enforced: option list	dot
Background contrast (`contrastMode`) — Appearance	Select	No	No	Standard	Stronger line on busy backgrounds	High on images / tints	Enforced: option list	high
Show on desktop (`showDesktop`) — Responsive	Boolean	No	No	On	Visibility ≥ 1200px	—	—	on

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Show on tablet (`showTablet`) — Responsive	Boolean	No	No	On	Visibility 600–1199px	—	—	on
Show on mobile (`showMobile`) — Responsive	Boolean	No	No	On	Visibility < 600px	—	—	off
Desktop width (`desktopWidth`) — Responsive	Select	No	No	As Appearance	Width ≥ 1200px	—	Enforced: option list	50-percent
Desktop thickness (`desktopThickness`) — Responsive	Select	No	No	As Appearance	Thickness ≥ 1200px	—	Enforced: option list	medium
Desktop alignment (`desktopAlignment`) — Responsive	Select	No	No	As Appearance	Alignment ≥ 1200px	—	Enforced: option list	start
Tablet width (`tabletWidth`) — Responsive	Select	No	No	As Appearance	Width 600–1199px	—	Enforced: option list	75-percent
Tablet thickness (`tabletThickness`) — Responsive	Select	No	No	As Appearance	Thickness 600–1199px	—	Enforced: option list	thin
Tablet alignment (`tabletAlignment`) — Responsive	Select	No	No	As Appearance	Alignment 600–1199px	—	Enforced: option list	center
Mobile width (`mobileWidth`) — Responsive	Select	No	No	As Appearance	Width < 600px	—	Enforced: option list	100-percent
Mobile thickness (`mobileThickness`) — Responsive	Select	No	No	As Appearance	Thickness < 600px	—	Enforced: option list	thin
Mobile alignment (`mobileAlignment`) — Responsive	Select	No	No	As Appearance	Alignment < 600px	—	Enforced: option list	center
Desktop top spacing (`desktopTopSpacing`) — Spacing	Select	No	No	Default (1.5rem)	Space above ≥ 1200px	—	Enforced: option list	none
Desktop bottom spacing (`desktopBottomSpacing`) — Spacing	Select	No	No	Default (1.5rem)	Space below ≥ 1200px	—	Enforced: option list	large
Tablet top spacing (`tabletTopSpacing`) — Spacing	Select	No	No	Default (1.5rem)	Space above 600–1199px	—	Enforced: option list	small
Tablet bottom spacing (`tabletBottomSpacing`) — Spacing	Select	No	No	Default (1.5rem)	Space below 600–1199px	—	Enforced: option list	small
Mobile top spacing (`mobileTopSpacing`) — Spacing	Select	No	No	Default (1.5rem)	Space above < 600px	—	Enforced: option list	xs
Mobile bottom spacing (`mobileBottomSpacing`) — Spacing	Select	No	No	Default (1.5rem)	Space below < 600px	—	Enforced: option list	xs

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Use section padding (`useSectionPadding`) — Spacing	Boolean	No	No	Off	No own spacing	On at the edges of padded sections	—	off
Stretch to container width (`stretchToContainer`) — Advanced	Boolean	No	No	Off	Fill the container always	—	—	off
Respect content maximum width (`respectMaxWidth`) — Advanced	Boolean	No	No	On	Cap at 90rem	Off only for true edge-to-edge on wide screens	—	on
Animate on view (`animateOnView`) — Advanced	Select	No	No	None	One-off reveal	Use sparingly	Enforced: option list	fade
Reduced motion fallback (`reducedMotionFallback`) — Advanced	Select	No	No	Show immediately	Behaviour for reduced motion	Show immediately recommended	Enforced: option list; shown only when Animate on view ≠ None	static
Print (`printVisibility`) — Advanced	Select	No	No	Show when printed	Print visibility	Hide for screen-only dividers	Enforced: option list	hide
Analytics (`analyticsEnabled`) — Advanced	Boolean	No	No	Off	Data layer state	Only when analysts ask	—	off
Analytics label (`analyticsLabel`) — Advanced	Text	No	No	Empty	`dc:title` in the data layer	Short, English, stable	Enforced in code: trimmed, max 120 characters; shown only when Analytics on	Story break
Anchor ID (`classes_anchor`) — Additional	Text	No	No	Empty	`#anchor-…` target (if no Component ID)	Start with anchor-	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-chapter-2
Hide block (`classes_hideblock`) — Additional	Boolean	No	No	Off	Hide without deleting	—	—	off
Padding left (`classes_paddingleft`) — Additional	Select	No	No	Default	Inner space left	Tokens only	Enforced: option list	Default
Padding right (`classes_paddingright`) — Additional	Select	No	No	Default	Inner space right	Tokens only	Enforced: option list	Default
Padding top (`classes_paddingtop`) — Additional	Select	No	No	Default	Inner space top (overrides the Spacing tab top value)	Prefer the Spacing tab	Enforced: option list	Default
Padding bottom (`classes_paddingbottom`) — Additional	Select	No	No	Default	Inner space bottom (overrides the Spacing tab bottom value)	Prefer the Spacing tab	Enforced: option list	Default

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Margin left (`classes_marginleft`) — Additional	Select	No	No	Default	Outer space left	—	Enforced: option list	Default
Margin right (`classes_marginright`) — Additional	Select	No	No	Default	Outer space right	—	Enforced: option list	Default
Margin top (`classes_margintop`) — Additional	Select	No	No	Default	Outer space top	—	Enforced: option list	Default
Margin bottom (`classes_marginbottom`) — Additional	Select	No	No	Default	Outer space bottom	—	Enforced: option list	Default

Field details

- **Component ID** — Purpose: the block's `id`. Allowed Values: starts with a letter; letters, digits, `-`, `_`. Character Limits: none enforced (Recommended ≤ 40). Business Rules: the code lower-cases it, replaces other characters with `-`, trims leading / trailing `-` / `_`, and prefixes `separator-` if it does not start with a letter; if the id is already on the page, a suffix is added (`story-end-1`, …). Empty: the Anchor ID if set, else `separator-1`, `separator-2`, … in page order. Dependency Rules: takes precedence over the Anchor ID. Visibility Rules: always. Sample: `story-end`.
- **Accessibility mode** — Allowed Values: `decorative`, `semantic`. Business Rules: decorative hides the line from assistive technology; semantic exposes the `<hr>`. Visibility: always. Sample: Semantic.
- **Accessible label** — Character Limits: 120 (Enforced by truncation in code). Visibility Rules: `condition accessibilityMode === "semantic"`. Business Rules: ignored in Decorative mode. Sample: "End of the case study".
- **Separator variant** — Allowed Values: default, subtle, strong, inset, full-width, dotted, dashed, gradient, branded, custom-design. Business Rules: presets — Default solid/thin/default/ 100%; Subtle solid/thin/subtle/100%; Strong solid/medium/strong/100%; Inset solid/thin/default/ 100% + 2.5rem inset; Full width solid/thin/default/full bleed; Dotted dotted/medium/default/100%; Dashed dashed/thin/default/100%; Gradient gradient/medium/primary/100%; Branded solid/medium/ primary/100%; Custom solid/thin/default/100%. Unknown values → Default.
- **Theme** — Allowed Values: auto, light, dark, brand. Business Rules: Auto follows `.section.dark` / `.section.brand` (not selectable in the section dialog as of this handover). Sample: Auto.
- **Line style** — Allowed Values: As the variant (empty), solid, dashed, dotted, gradient, none. Business Rules: None + Decorative renders no `<hr>` (spacing only); None + Semantic triggers the editor warning. The dialog help text explains this. Sample: Dashed.
- **Line thickness** — Allowed Values: As the variant, thin (1px, 0.0625rem), medium (2px, 0.125rem), thick (3px, 0.1875rem).
- **Line colour** — Allowed Values: As the variant, default, subtle, neutral, strong, primary (brand), secondary (light brand), accent, inverse (on dark). Business Rules: resolved through the active theme palette; overridden by High contrast.
- **Line opacity** — Allowed Values: 25, 50, 75, 100 (%). Default 100.
- **Separator width** — Allowed Values: As the variant, auto, 25-percent, 50-percent, 75-percent, 100-percent, content-width, full-bleed. Business Rules: legacy values such as "50%" are accepted by the code. Note: "Auto (as the variant)" renders as 100% of the content width (class `width-auto`); to keep the variant's own width (for example Full width's full bleed) choose "As the variant".

- **Maximum width** — Allowed Values: none, small (20rem / 320px), medium (40rem / 640px), large (60rem / 960px), extra-large (80rem / 1280px), content-container (90rem).
- **Alignment** — Allowed Values: start, center, end. Default center. Logical (RTL aware).
- **Start / Center / End decoration** — Allowed Values: none, dot, square, diamond. Business Rules: not rendered without a visible line.
- **Background contrast** — Allowed Values: standard, high. High uses the palette's strong colour.
- **Show on desktop / tablet / mobile** — Booleans, default on (template and code default). Business Rules: all three off = hidden everywhere (editor placeholder and note).
- **Desktop / Tablet / Mobile width** — Allowed Values: As Appearance (`inherit`) plus the widths. Business Rules: apply only in their range; help text shows the range.
- **Desktop / Tablet / Mobile thickness** — Allowed Values: As Appearance, thin, medium, thick.
- **Desktop / Tablet / Mobile alignment** — Allowed Values: As Appearance, start, center, end.
- **Top / bottom spacing (6 fields)** — Allowed Values: default (1.5rem), none (0), 2xs (0.25rem), xs (0.5rem), small (1rem), medium (1.5rem), large (2.5rem), xl (4rem), 2xl (5rem). Labels show px equivalents. Business Rules: padding of the block; ignored when Use section padding is on; the Additional tab's Padding top / bottom override them.
- **Use section padding** — Boolean, default off. Removes all own vertical spacing.
- **Stretch to container width** — Boolean, default off. Line always 100%, no bleed.
- **Respect content maximum width** — Boolean, default on. Caps at `--content-max-width` (90rem), also for full bleed.
- **Animate on view** — Allowed Values: none, fade, draw-from-start, draw-from-center. Once per page view, threshold 10% visible.
- **Reduced motion fallback** — Allowed Values: static (Show immediately), fade (Short fade). Visibility Rules: `condition animateOnView !== "none"`.
- **Print** — Allowed Values: show, hide.
- **Analytics** — Boolean, default off. On: data layer state entry; off: `data-analytics="off"`.
- **Analytics label** — Character Limits: 120 (Enforced by truncation). Visibility Rules: `condition analyticsEnabled === true`.
- **Additional tab fields** — as described for every block: Anchor ID regExp `^anchor-[a-z0-9][a-z0-9-]*$` (message "Start with anchor-, then lower-case letters, digits and hyphens."), de-duplicated with `-2`, `-3`; Hide block hides outside edit mode; padding / margin options Default, 0, 0.25rem, 0.5rem, 0.75rem, 1rem, 1.25rem, 1.5rem, 2rem, 2.5rem, 3rem, 4rem, 5rem, 6rem, 7.5rem (`--space-*` tokens). Business Rule specific to the Separator: Additional Padding top / bottom (`pt-*`, `pb-*`, higher specificity) replace the separator's own spacing on that side.

6.26.7 Field Annotation Mapping

Marker	Field	Purpose
A	Separator block (all tabs)	Select to configure; spacing from the Spacing tab
B	Variant, Theme, Line style / thickness / colour / opacity, Width, Maximum width, Alignment, Background contrast; Responsive overrides	The look of the rule
C	Start / Center / End decoration	Ornaments on the line
D	Show on desktop / tablet / mobile; Accessibility mode + Line style	Trigger the editor notes (hidden everywhere; invisible semantic break)
E	Content tree entry	Selecting, moving, deleting the block

6.26.8 Component Specification

Attribute	Value
Component Name	Separator (`blocks/separator`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	`separator-config.js`; utils properties, environment, ids, placeholders, analytics; block options decorator; IntersectionObserver (animation only)
Component Type	Block (key-value, no items)
Supported Variants	10 variants; 4 themes; 2 accessibility modes; line styles incl. none; widths, max widths, alignment, decorations, contrast, opacity; per-breakpoint visibility, width, thickness, alignment and spacing; animations; print; analytics
Supported Templates	Any page, any section
Supported Metadata	None
Supported Tags	None
Supported Personalization	None; its state can be in the data layer (Analytics on) for future Adobe Target / Analytics use
Caching Considerations	Rendered from page HTML cached by the Edge Delivery CDN; placeholders fetched only in Universal Editor edit mode when a warning exists
Performance Considerations	No images or network calls on the live site; no resize listeners (CSS only); IntersectionObserver only with an animation (not for reduced motion with Show immediately); eager in the first section, else lazy; no layout shift (padding fixed by classes)
Accessibility Compliance	Semantic `<hr>` or `aria-hidden` decoration; optional accessible name; decorations hidden; reduced motion respected; forced-colours support (`CanvasText`); RTL-aware alignment and draw origin
SEO Compliance	Native `<hr>`; no hidden text; no impact on ranking

6.26.9 Input Model

Content Inputs: None besides the settings; an `<hr>` authored in the block is reused (other rows ignored).

Metadata Inputs: None.

Tags Inputs: None.

Page Context Inputs: The section classes (`dark`, `brand`) for Theme Auto; the reading direction (`:dir(rtl)`) for alignment and draw-from-start; the viewport width (CSS media queries); `prefers-reduced-motion`; Universal Editor edit mode (`html.adobe-ue-edit`); the URL query `?debug=separator`.

Query Inputs: None.

Manual Inputs: All dialog fields, delivered as key-value rows; the Additional tab fields as block classes.

Authored HTML received from AEM (example, abridged):

```
<div class="separator anchor-chapter-2">
  <div><div>accessibilityMode</div><div>semantic</div></div>
  <div><div>accessibleLabel</div><div>End of the case study</div></div>
  <div><div>variant</div><div>branded</div></div>
  <div><div>width</div><div>25-percent</div></div>
  <div><div>alignment</div><div>start</div></div>
  <div><div>showMobile</div><div>false</div></div>
</div>
```

Keys are matched case- and separator-insensitively (`accessibilityMode` → `accessibilitymode`); values are matched case-insensitively, spaces / underscores become hyphens, "50%" becomes `50-percent`, and booleans accept true / yes / on / 1 / y and false / no / off / 0 / n. Any unknown value falls back to its default.

6.26.10 Output Model

Rendered Output: The rows are replaced by `div.separator-line` containing `hr.separator-rule` and optional `span.separator-decoration` elements, plus (edit mode only, with a warning) `p.separator-editor-note`. The block gets its id and classes: mode, `variant-*`, `theme-*`, `line-*`, `thickness-*`, `color-*`, `opacity-*`, `width-*`, `max-*`, `align-*`, and when applicable `contrast-high`, `section-padding`, `stretch`, `respect-max`, `print-hide`, `decorated`, `hide-{bp}`, `{bp}-width-*`, `{bp}-thickness-*`, `{bp}-align-*`, `{bp}-top-*`, `{bp}-bottom-*`, `hidden-all`, `animate-*`, `motion-*`, and at runtime `animate-ready`, `reduced-motion`, `is-visible`. Data attributes: `data-separator-ready="true"`, `data-analytics="off|custom"`.

Generated HTML (example: Semantic, Branded, 25%, Start, hidden on mobile):

```
<div class="separator anchor-chapter-2 block semantic variant-branded theme-auto line-
solid
  thickness-medium color-primary opacity-100 width-25-percent max-none align-start
  respect-max hide-mobile" id="anchor-chapter-2" data-separator-ready="true"
  data-analytics="off">
  <div class="separator-line">
    <hr class="separator-rule" aria-label="End of the case study">
  </div>
</div>
```

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking: The generic data layer decorator skips the separator (it always has a `data-analytics` attribute). With Analytics on, the separator pushes its own state:

```
{ "component": { "story-end": {
  "@type": "tcs-corp/components/separator", "dc:title": "Story break",
  "xdm:variant": "branded", "xdm:accessibilityMode": "semantic", "xdm:theme": "auto",
  "xdm:lineStyle": "solid", "xdm:visibility": "desktop,tablet" } } }
```

`xdm:visibility` is `none` when hidden everywhere. No events are pushed (no interactions).

Personalization Output: None.

Search Index Output: None (not indexed; no text).

6.26.11 Business Rules

1. The block decorates once (`data-separator-ready`).
2. Only two-cell rows are read as properties; an authored `<hr>` is reused (its attributes other than class removed); extra `<hr>` elements and other content are dropped.
3. Every value comes from a controlled list; unknown values fall back to defaults: mode decorative, variant default, theme auto, opacity 100, max width none, alignment center, decorations none, contrast standard, show on all devices, responsive values inherit, spacing default, use section padding off, stretch off, respect max width on, animation none, reduced motion fallback static, print show, analytics off.
4. Line style, thickness, colour and width fall back to the variant preset when empty ("As the variant").
5. Accessible label is used only in Semantic mode and cut to 120 characters; Analytics label is cut to 120 characters.

6. Id: sanitised Component ID; if already used on the page, a numeric suffix; empty → the Anchor ID (Additional tab) if present, else `separator-N`. A Component ID replaces the Anchor ID.
7. Decorative + Line style None → no `<hr>` (spacing only). Semantic always renders an `<hr>`.
8. Decorations render only when the line style is not None.
9. Warnings (edit mode only, from placeholders): hidden on all devices (`separatorHidden`); semantic without a line (`separatorInvisible`). Never on the published page.
10. Hidden on all devices: `display: none` on the site, dashed placeholder in the editor.
11. Responsive values apply only within their breakpoint range; "inherit" keeps the Appearance value; breakpoint widths other than Full bleed reset the bleed.
12. High contrast overrides the Line colour.
13. Stretch to container overrides all widths and bleed; maximum width caps still apply.
14. Use section padding removes all own vertical spacing (all breakpoints).
15. Respect content maximum width (default on) caps the line at 90rem including full bleed.
16. Animation: once, at 10% visibility; not prepared at all if IntersectionObserver is missing or for reduced motion with Show immediately; with reduced motion and Short fade only opacity animates; errors remove `animate-ready` so the line is never left hidden.
17. Print: shown (opacity 1, no transform) unless Hide when printed.
18. Forced colours: line, gradient and decorations use `CanvasText`.
19. Analytics: off → `data-analytics="off"` (no tracking, not even generic state); on → `data-analytics="custom"` and one state entry; errors never affect rendering.
20. `?debug=separator` logs the configuration, warnings and duplicate Component IDs via `console.debug`.

6.26.12 Author Best Practices

Recommended Content Length: Accessible label Recommended ≤ 60 characters (Enforced maximum 120); Analytics label Recommended ≤ 40 characters (Enforced maximum 120).

Image Dimensions: Not applicable (no images).

CTA Writing Guidelines: Not applicable (no links or buttons).

Accessibility Guidelines: Use Semantic only for real topic changes; give important breaks an Accessible label; never use Semantic with Line style None; keep meaningful lines at Strong colour or High contrast and 100% opacity; prefer Show immediately as the reduced motion fallback; do not use separators instead of headings.

SEO Guidelines: No SEO effect; structure the page with headings, not separators.

Metadata Guidelines: None.

Taxonomy Guidelines: None.

Localization Guidelines: Translate the Accessible label per locale; keep the Analytics label and Component ID identical across locales (stable for reporting); Start / End alignment adapts automatically to right-to-left languages; editor note texts come from each locale's placeholders sheet.

6.26.13 Troubleshooting Guide

Issue 1 — The separator doesn't appear on the page

Symptoms: Nothing visible on the published page.

Root Cause: All Show on … toggles off, Hide block on, Decorative + Line style None, or very low opacity / subtle colour on a light background.

Diagnosis Steps: Open the block in Universal Editor edit mode: a dashed placeholder with "Separator hidden on all devices" means all devices are off; check Hide block; check Line style; use `?debug=separator` on preview.

Resolution Steps: Turn on the needed devices, turn off Hide block, choose a visible line style / colour.

Validation Steps: The line shows in Preview at all intended widths.

Preventive Measures: Validation checklist.

Issue 2 — Editor warning "Semantic separator without a line"

Symptoms: A note under the separator in edit mode.

Root Cause: Accessibility mode Semantic with Line style None (screen readers announce an invisible break).

Diagnosis Steps: Check General > Accessibility mode and Appearance > Line style.

Resolution Steps: Switch to Decorative (spacing only) or choose a visible line style.

Validation Steps: The note disappears.

Preventive Measures: Use Decorative for spacing-only separators.

Issue 3 — Theme Auto doesn't adapt to a dark background

Symptoms: A light-palette line on a dark background image.

Root Cause: Auto follows only `.section.dark` / `.section.brand`; a background image doesn't change the palette, and the section dialog has no dark / brand style as of this handover.

Diagnosis Steps: Inspect the section's classes.

Resolution Steps: Set Theme to Dark (or Brand) explicitly; consider High contrast.

Validation Steps: The line is visible on the image.

Preventive Measures: Choose the theme explicitly on image backgrounds.

Issue 4 — Double spacing around the separator

Symptoms: Large gaps above / below.

Root Cause: The separator's 1.5rem default spacing adds to section spacing or block margins.

Diagnosis Steps: Inspect padding of `.separator` and neighbouring margins.

Resolution Steps: Turn on Use section padding, or set top / bottom spacing to None / smaller per device.

Validation Steps: Gaps match the design at each breakpoint.

Preventive Measures: Use section spacing as the primary rhythm.

Issue 5 — Spacing settings have no effect

Symptoms: Changing Desktop top spacing doesn't change the gap.

Root Cause: Use section padding is on, or the Additional tab Padding top / bottom override the separator's spacing, or the editor is not at desktop width (desktop values apply ≥ 1200px).

Diagnosis Steps: Check Spacing > Use section padding, Additional > Padding top / bottom, and the editor width.

Resolution Steps: Turn off Use section padding / reset the Additional padding to Default; preview at the right width.

Validation Steps: The gap changes at the intended breakpoint.

Preventive Measures: Use only the Spacing tab for vertical space of separators.

Issue 6 — Width "Auto" doesn't keep the Full width look

Symptoms: A Full width variant becomes content-width after choosing "Auto (as the variant)".

Root Cause: `auto` renders as 100% (class `width-auto`), not the variant's preset; only the empty value "As the variant" uses the preset.

Diagnosis Steps: Check Appearance > Separator width and the block class `width-auto`.

Resolution Steps: Choose "As the variant" or "Full bleed".

Validation Steps: The line bleeds to the section edges.

Preventive Measures: Use "As the variant" to keep presets.

Issue 7 — Anchor links to the separator break

Symptoms: `#anchor-…` links don't jump.

Root Cause: A Component ID is set (it replaces the Anchor ID), or the Component ID was lower-cased / suffixed because of a duplicate.

Diagnosis Steps: Inspect the block's `id`; check `?debug=separator` for "component id … is used twice".

Resolution Steps: Link to the actual id, or clear the Component ID to use the Anchor ID; make ids unique.

Validation Steps: The link scrolls to the separator.

Preventive Measures: Use one id mechanism; keep ids lower-case and unique.

Issue 8 — Animation doesn't play

Symptoms: The line just appears.

Root Cause: The visitor prefers reduced motion (fallback Show immediately), the separator was already in view when loaded (it animates immediately once), or the browser lacks IntersectionObserver.

Diagnosis Steps: Check the OS reduced-motion setting; check classes `animate-ready` / `is-visible`.

Resolution Steps: Expected behaviour for reduced motion; test with reduced motion off and the separator below the fold.

Validation Steps: The animation plays when scrolling into view.

Preventive Measures: Don't rely on animation to convey meaning.

6.26.14 Support Runbook

Monitoring Requirements

- No runtime service. Monitor visual regressions after CSS / token changes and the unit tests (`npm test`, `test/separator/separator-config.test.mjs`) in CI.
- If Analytics is used, verify the data layer entries in Adobe Tags once Tags is implemented.

Logs

- Browser console: `[separator]` debug messages only with `?debug=separator` (configuration, warnings, duplicate Component ID). No errors are logged in normal operation; analytics errors are swallowed.
- Admin API / AEM Sidekick: page preview / publish status.
- AEM author: publication logs.

Common Errors

Error	Likely cause	Fix
Invisible separator	All devices off / Hide block / Line style None	Re-enable / choose a line
Editor note "Semantic separator without a line"	Semantic + None	Decorative or visible line
Wrong colour on dark image	Theme Auto	Theme Dark / Brand, High contrast

Error	Likely cause	Fix
Double spacing	Own spacing + section spacing	Use section padding / None
Id differs from authored	Lower-casing, sanitising, duplicate suffix	Use lower-case unique ids
Not in data layer	Analytics off (default)	Turn on Analytics
Validation error on Component ID	Starts with a digit or has spaces	Start with a letter; use - / _

Escalation Matrix

Level	Who	Checks
L1	Author support	Mode, line style, devices, theme, spacing, Hide block, Component / Anchor ID; editor notes
L2	AMS / site admin	`?debug=separator` output, preview vs live HTML, CDN cache, placeholders sheet
L3	Development team	`separator-config.js` normalisation, CSS classes, unit tests, browser support
L4	Adobe Support	Universal Editor key-value rendering, Edge Delivery delivery issues

Recovery Procedures

1. Restore the previous page version and republish.
2. For code regressions, revert the commit on `main`; run `npm test` to confirm the configuration logic.
3. Temporarily hide the separator with Hide block.

Health Checks

- Check a page with separators at 390 / 1024 / 1440px; verify no horizontal page scroll.
- Check the accessibility tree: decorative separators hidden, semantic ones exposed with label.
- Check forced-colours (Windows high contrast) and print preview.

Operational Procedures

- A new variant is an extension: add a preset to `VARIANT_PRESETS` in `separator-config.js`, the option in `_separator.json`, its CSS class, run `npm run build:json` and `npm test`.
- Maintain `separatorHidden` and `separatorInvisible` in each locale's placeholders sheet.

Known Limitations

- Visual presentation uses site tokens until the Figma designs are applied (README).
- No brand-icon decoration (needs an approved icon asset); dot, square and diamond only.
- No free-form CSS classes or colours (controlled options only).
- An Accessible label is read as the separator's name by most screen readers, not all.
- Theme Auto depends on `dark` / `brand` section classes that the section dialog doesn't offer as of this handover.
- Width "Auto (as the variant)" renders 100%, not the variant's preset width.
- Dialog labels show px values (for example "XL (64px)"); the CSS uses the equivalent rem tokens.

6.27 Embed Block

The Embed block places third-party content from a URL on the page: YouTube and Vimeo videos, Twitter / X posts, or any other URL as a generic iframe. It is based on the Edge Delivery Services Block Collection embed (`blocks/embed/embed.js`, `embed.css`, `_embed.json`), adapted so that players are built with DOM APIs by the media decorator (no authored URL in HTML strings), the Play button comes from the button decorator, frames are named from the placeholders, and keyboard focus moves into the player after Play. Without a placeholder image

the embed loads when it scrolls into view; with a placeholder image it loads only when the visitor selects Play (click-to-load).

6.27.1 Component Purpose

Business Objective: Let authors include video and social content hosted on external platforms with minimal performance cost (lazy or click-to-load) and accessible naming.

Business Use Cases:

- A YouTube video of a keynote, customer story or product demo on a campaign or article page.
- A Vimeo-hosted brand film.
- An official TCS post on Twitter / X embedded in a press release or news page.
- An approved third-party widget or page shown in an iframe (for example an interactive report), where the provider allows framing.

When To Use:

- The content is hosted on YouTube, Vimeo or Twitter / X, or is an iframe-able URL.
- A simple embed without playback options is enough.

When Not To Use:

- For videos stored in AEM Assets (MP4 / WebM), background videos, captions files, autoplay, pop-up playback or privacy-enhanced YouTube: use the **Video** block (6.28), which supports these options. The Embed block always uses `www.youtube.com` (not the privacy-enhanced `youtube-nocookie.com` host).
- For providers not listed (Instagram, LinkedIn, Facebook posts): no dedicated handler (Not implemented as of this handover); a generic iframe of such URLs is usually blocked by the provider.
- For forms or content requiring interaction tracking (interactions inside iframes are not visible to the site's data layer).

Dependencies:

Type	Dependency	Use
Code	`decorators/media-decorator.js` (`createFrame`, `createYoutubeFrame`, `createVimeoFrame`)	Builds the iframes
Code	`utils/media.js` (`isYoutube`, `isVimeo`, `youtubeEmbedUrl`, `vimeoEmbedUrl`)	Recognises providers and builds player URLs
Code	`decorators/icon-button-decorator.js` (`createIconButton`)	Play button (primary icon button, icon `icons/play.svg`)
Code	`utils/viewport.js` (`onVisible`)	Loads the embed when it enters the viewport
Code	`utils/placeholders.js`	Labels `play`, `contentFrom`
Code	`scripts/aem.js` (`loadScript`)	Loads the Twitter widget script
Code	`decorators/block-options-decorator.js`, `decorators/data-layer-decorator.js`	Additional tab; data layer
Third party	`https://platform.twitter.com/widgets.js`	Renders Twitter / X posts
Third party	`www.youtube.com/embed/…`, `player.vimeo.com/video/…`	Video players
Sheet	`/{locale}/placeholders.json`	Keys `play` ("Play"), `contentFrom` ("Content from {source}")
Model	`models/_section.json`	Allows the Embed in sections

Related Components: Video block (richer video options), Modal (used by the Video block's pop-up), Hero / Teaser (Lottie animations, not embeds).

Applicable Templates: Any page (home, article, blog, press-release, standard pages), in any section.

Supported Content Types: YouTube (watch, `youtu.be`, `/embed/`, `/shorts/`, `/live/` URLs); Vimeo (`vimeo.com/{id}`); Twitter / X post URLs (`twitter.com`, `x.com`, with or without `www.`); any other `https` URL as an iframe; an optional placeholder image from AEM Assets.

6.27.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** At 1440px: a YouTube embed with a placeholder image and a large round brand-blue Play button in the centre, 900px wide, centred in the section, with 1rem rounded corners.

Desktop (≥ 1200px): The block is centred with a maximum width of 56.25rem (900px). Video and iframe embeds are 16:9 (`aspect-ratio: 16 / 9`) with 1rem rounded corners (`--radius-l`) and the iframe filling the frame without a border. With a placeholder image, the image covers the 16:9 area (`object-fit: cover`) and the Play button (a primary icon button, 2.75rem, scaled ×1.4) is centred on it. Twitter / X posts are centred and take the height the widget renders.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** At 768px: a Vimeo player in a 16:9 frame filling the content width.

Tablet (600–1199px): Same layout; below 900px of available width the embed fills the content width and keeps 16:9. There are no breakpoint-specific rules in `embed.css`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** At 390px: an embedded X post centred, and below it a YouTube placeholder image at full width with the Play button.

Mobile (< 600px): Same layout at full content width. Before an embed without placeholder loads, the block reserves a 16:9 area to avoid layout shift (`.embed:not(.embed-is-loaded, :has(.embed-placeholder))`).

6.27.3 Component Variations

The provider is chosen automatically from the URI; the placeholder image switches the block to click-to-load.

Variation 1 — YouTube

Variation Name: YouTube embed (`embed-youtube` class after loading).

Business Purpose: Show a YouTube video in the page.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/embed-youtube.png`
> **Purpose:** Variation example
> **Description:** A YouTube player in a 16:9 rounded frame, centred, 900px wide.

Use Cases: Keynotes, customer stories, product videos published on the TCS YouTube channel.

Authoring Guidelines: Paste a YouTube URL into **URI** (for example `https://www.youtube.com/watch?v=VIDEO_ID` or `https://youtu.be/VIDEO_ID`); leave Placeholder Image empty for lazy loading, or set it for click-to-load (Variation 5).

Business Rules: Recognised when the URL contains `youtube.com` or `youtu.be`. The video id comes from `youtu.be/{id}`, `/embed/{id}`, `/shorts/{id}`, `/live/{id}` or `?v={id}`. Other URL parameters (for example a start time `t=`) are not carried over.

Rendering Rules: `div.embed-frame > iframe` with `src` `https://www.youtube.com/embed/{id}?rel=0`, `title` "Content from youtube.com" (placeholder `contentFrom`), `loading="lazy"`, `allow="autoplay; fullscreen; picture-in-picture; encrypted-media; accelerometer; gyroscope"`, `allowfullscreen`. Loaded when the block enters the viewport.

Responsive Behaviour: 16:9 at max 900px on desktop; full content width on tablet / mobile.

SEO Impact: The video is in a third-party iframe; its content isn't part of the page's text. Describe the video in the surrounding text for search engines.

Accessibility Compliance: The iframe has a title (WCAG 4.1.2); captions and keyboard control come from the YouTube player (WCAG 1.2.2 depends on captions on YouTube).

Variation 2 — Vimeo

Variation Name: Vimeo embed (`embed-vimeo`).

Business Purpose: Show a Vimeo-hosted video.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/embed-vimeo.png`
> **Purpose:** Variation example
> **Description:** A Vimeo player in a 16:9 rounded frame.

Use Cases: Brand films hosted on Vimeo.

Authoring Guidelines: Paste `https://vimeo.com/{id}` into URI.

Business Rules: Recognised when the URL contains `vimeo.com`. The video id is the first path segment of the URL, so only `vimeo.com/{id}` URLs work; channel, showcase, private-hash or `player.vimeo.com/video/{id}` URLs produce a wrong player URL.

Rendering Rules: iframe `src` `https://player.vimeo.com/video/{id}`, title "Content from vimeo.com", `allow="autoplay; fullscreen; picture-in-picture; encrypted-media"`, lazy.

Responsive Behaviour: As YouTube.

SEO Impact: As YouTube.

Accessibility Compliance: Iframe title; captions and controls from the Vimeo player.

Variation 3 — Twitter / X post

Variation Name: Twitter / X embed (`embed-twitter`).

Business Purpose: Show an official social post with its native rendering.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-twitter.png`
> **Purpose:** Variation example
> **Description:** An embedded X post centred in the content column, with the post's own card design.

Use Cases: Press releases and news quoting an official post.

Authoring Guidelines: Paste the post URL (`https://x.com/{account}/status/{id}` or `https://twitter.com/...`) into URI.

Business Rules: Recognised when the host is `twitter.com`, `x.com`, `www.twitter.com` or `www.x.com`. `x.com` URLs are rewritten to `twitter.com` for the widget. The widget script `https://platform.twitter.com/widgets.js` is loaded (once per page) when the embed loads.

Rendering Rules: `<blockquote class="twitter-tweet"><a href="…">…URL…</a></blockquote>`; the widget replaces it with its own iframe (`.twitter-tweet-rendered`, centred). If the script does not load (blocked, offline), the post's URL remains as a link.

Responsive Behaviour: Width and height from the widget; centred on all breakpoints.

SEO Impact: The fallback link is in the HTML; the rendered post is third-party content.

Accessibility Compliance: The fallback is a real link; the widget's accessibility is provided by the platform. Third-party cookies may be set by the platform (see Known Limitations).

Variation 4 — Generic iframe

Variation Name: Generic iframe (no provider class).

Business Purpose: Embed any other URL that allows framing.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-generic-iframe.png`
> **Purpose:** Variation example
> **Description:** A third-party interactive page displayed in a 16:9 rounded frame.

Use Cases: Approved interactive content, maps or reports from partners.

Authoring Guidelines: Paste the URL into URI. Confirm with the provider that the page may be framed.

Business Rules: Any URL not recognised as YouTube, Vimeo or Twitter / X. There is no allow-list of domains in code: governance of which URLs are embedded is editorial ([To be confirmed: approved embed domains — owner: site admin / security]).

Rendering Rules: `div.embed-frame > iframe` with `src` = the URL, `title` "Content from {host}", `loading="lazy"`, `allow="encrypted-media"`, `allowfullscreen`. The frame is always 16:9.

Responsive Behaviour: 16:9 frame; the framed page must be responsive itself.

SEO Impact: The framed content is not part of the page.

Accessibility Compliance: Iframe title from the host name; the framed page's own accessibility is the provider's responsibility. Content that needs another aspect ratio is cropped or letterboxed.

Variation 5 — Placeholder image with Play (click-to-load)

Variation Name: Placeholder image with Play button (`.embed-placeholder`).

Business Purpose: Better performance and privacy: nothing from the provider loads until the visitor chooses Play; a branded image replaces the provider's preview.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-placeholder-play.png`
> **Purpose:** Variation example
> **Description:** A 16:9 placeholder image with the round brand-blue Play button centred; second frame: the YouTube player playing after the click.

Use Cases: Above-the-fold videos, pages with several videos, campaign pages with branded thumbnails.

Authoring Guidelines: Set **Placeholder Image** (from AEM Assets) and **Placeholder Image Alt Text** (describe the image, or leave empty if it is decorative), plus the URI.

Business Rules: The provider content loads only on Play; YouTube and Vimeo are requested with `autoplay=1` and `muted=1` (Vimeo's mute parameter; YouTube's own parameter is `mute`, so YouTube ignores it, and browsers that block autoplay with sound will show the YouTube play button instead). Twitter / X and generic iframes ignore autoplay. After loading, keyboard focus moves to the iframe (or the post link).

Rendering Rules: `div.embed-placeholder` (grid, 16:9, rounded) with the `<picture>` covering it and `button.embed-placeholder-play` (primary icon button, `aria-label` "Play" from placeholder `play`, `data-cmp-interaction="play"`). On click, the placeholder is replaced by the embed and the block gets `embed-is-loaded`.

Responsive Behaviour: 16:9 image at max 900px; full width on tablet / mobile; the Play button stays centred.

SEO Impact: The image is in the page HTML (with alt text if set); faster pages benefit Core Web Vitals.

Accessibility Compliance: The Play button is a native `<button>` with an accessible name (WCAG 4.1.2, 2.1.1); focus moves into the player after loading (WCAG 2.4.3); alt text on the image (WCAG 1.1.1).

6.27.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** Universal Editor edit mode with an Embed block selected that has a placeholder image and Play button; the properties panel open on the Content tab.

Annotations:

- A = The Embed block (select it to edit its fields)
- B = The placeholder image (Placeholder Image, with its alt text)
- C = The Play button (generated, not authorable; label from the placeholders sheet)
- D = The player area (the embed loaded from the URI after Play, or on scroll without a placeholder)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Embed properties panel: Content tab with Placeholder Image, Placeholder Image Alt Text and URI; the Additional tab.

Properties panel tabs:

- **Content:** Placeholder Image (asset picker, single image); Placeholder Image Alt Text (text); URI (text: the URL of the video, post or page).
- **Additional:** Anchor ID; Hide block; Padding left / right / top / bottom; Margin left / right / top / bottom.

6.27.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Open the page with **Edit** from AEM Sites.

Expected Result: The page is in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-step-01-open-page.png`
> **Purpose:** Authoring step
> **Description:** The page open in Universal Editor.

Step 2 — Add the Embed block

Action: Select the section, choose **Add** (+), select **Embed**.

Expected Result: An empty Embed block is added (nothing renders until a URI is set).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-step-02-add-block.png`
> **Purpose:** Authoring step
> **Description:** The Add menu with Embed, and the empty block in the content tree.

Step 3 — Enter the URI

Action: In the Content tab, paste the full URL (starting with `https://`) into **URI**.

Expected Result: After saving, the embed renders in the canvas when visible (or the placeholder shows if an image is set).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-step-03-uri.png`
> **Purpose:** Authoring step
> **Description:** The URI field with a YouTube URL.

Step 4 — Add a placeholder image (optional, recommended for videos)

Action: Choose **Placeholder Image** from AEM Assets (16:9 recommended) and type the **Placeholder Image Alt Text** (or leave it empty if the image is decorative and the surrounding text names the video).

Expected Result: The image shows with the Play button.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-step-04-placeholder.png`
> **Purpose:** Authoring step
> **Description:** The asset picker and the rendered placeholder with the Play button.

Step 5 — Additional tab (optional)

Action: Set an Anchor ID, Hide block or spacing if required.

Expected Result: Values accepted.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-step-05-additional.png`
> **Purpose:** Authoring step
> **Description:** The Additional tab of the Embed block.

Step 6 — Preview and test playback

Action: Switch to **Preview**; select Play (if a placeholder is used) and check playback; check mobile width.

Expected Result: The player loads and plays; the page doesn't shift while loading.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-step-06-preview.png`
> **Purpose:** Authoring step
> **Description:** The video playing in preview mode after Play.

Step 7 — Validate and publish

Action: Run the Validation Process checklist, then **Publish** the page (and make sure the placeholder image asset is published with it).

Expected Result: The embed works on the live site.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/embed-step-07-publish.png`
> **Purpose:** Authoring step
> **Description:** The Publish dialog listing the page and its referenced image.

Preview Process

1. Universal Editor **Preview** mode; test Play and keyboard use (Tab to Play, Enter).
2. Edge Delivery preview `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}`.
3. Check `{path}.plain.html` if the block renders nothing: the URI must arrive as a link (`<a>`).

Publishing Process

Publish the page; the placeholder image is delivered through Edge Delivery media optimisation. Third-party content is fetched by the visitor's browser from the provider.

Validation Process

- ☐ URI is a full `https://` URL of the intended provider.
- ☐ YouTube URL contains the video id (`watch?v=`, `youtu.be/`, `/shorts/`, `/live/`, `/embed/`).
- ☐ Vimeo URL has the form `vimeo.com/{id}`.
- ☐ The video has captions on its platform (YouTube / Vimeo).
- ☐ Placeholder image is 16:9 and has alt text (or is intentionally decorative).
- ☐ Play button works with mouse and keyboard; focus moves into the player.
- ☐ Generic iframe URL is approved and allows framing (no "refused to connect").
- ☐ Surrounding text names / describes the embedded content.
- ☐ No horizontal page scroll at 390px.

Rollback Process

1. Restore the previous page version and republish, or
2. turn on Hide block and publish, or
3. replace the URI with the previous URL.

6.27.6 Authorable Field Specification

Model `embed`

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Placeholder Image (`embed_placeholder`) — Content	Reference (image)	No	No	Empty	Click-to-load preview image	16:9, from AEM Assets	None in the model	/content/dam/tcs-corp/video-thumb.jpg
Placeholder Image Alt Text (`embed_placeholderAlt`) — Content	Text	No	No	Empty	Alt text of the placeholder	Describe the image	None	Keynote speaker on stage
URI (`embed_uri`) — Content	Text	No (block renders nothing without it)	No	Empty	Address of the content	Full https URL	None in the model; the code needs a link	https://www.youtube.com/watch?v=VIDEO_ID
Anchor ID (`classes_anchor`) — Additional	Text	No	No	Empty	`#anchor-…` target	Start with anchor-	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-keynote
Hide block (`classes_hideblock`) — Additional	Boolean	No	No	Off	Hide without deleting	—	—	off
Padding left (`classes_paddingleft`) — Additional	Select	No	No	Default	Inner space left	Tokens	Enforced: option list	Default
Padding right (`classes_paddingright`) — Additional	Select	No	No	Default	Inner space right	Tokens	Enforced: option list	Default

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Padding top (`classes_paddingtop`) — Additional	Select	No	No	Default	Inner space top	Tokens	Enforced: option list	Default
Padding bottom (`classes_paddingbottom`) — Additional	Select	No	No	Default	Inner space bottom	Tokens	Enforced: option list	Default
Margin left (`classes_marginleft`) — Additional	Select	No	No	Default	Outer space left (replaces the centring auto margin)	Avoid unless intended	Enforced: option list	Default
Margin right (`classes_marginright`) — Additional	Select	No	No	Default	Outer space right (replaces the centring auto margin)	Avoid unless intended	Enforced: option list	Default
Margin top (`classes_margintop`) — Additional	Select	No	No	Default	Outer space top	Tokens	Enforced: option list	2rem (space-800)
Margin bottom (`classes_marginbottom`) — Additional	Select	No	No	Default	Outer space bottom	Tokens	Enforced: option list	2rem (space-800)

Field details

- **Placeholder Image** — Purpose: switches the block to click-to-load and shows a branded image. Allowed Values: one image from AEM Assets. Character Limits: n/a. Business Rules: the first `<picture>` in the block is used; it is displayed with `object-fit: cover` in 16:9. Dependency Rules: requires a URI (without URI nothing renders). Visibility Rules: always. Sample: a 1920 × 1080 still of the video.
- **Placeholder Image Alt Text** — Purpose: describes the image. Allowed Values: text. Character Limits: none enforced; Recommended ≤ 125 characters. Business Rules: written into the image's alt by AEM's field collapsing (`…Alt` suffix); the block code doesn't change it. Dependency: only meaningful with a Placeholder Image. Visibility: always. Sample: "TCS CEO presenting at the AI summit".
- **URI** — Purpose: the content to embed. Allowed Values: a URL; the block reads the first link (`<a href>`) in the block, so the value must be delivered as a link (paste a full URL). Character Limits: none. Business Rules: provider detection — YouTube (`youtube.com`, `youtu.be` in the URL), Vimeo (`vimeo.com` in the URL), Twitter / X (host `twitter.com` / `x.com`, optionally `www.`), otherwise generic iframe. Visibility: always. Sample: `https://vimeo.com/123456789`.
- **Additional tab fields** — Anchor ID (regExp `^anchor-[a-z0-9][a-z0-9-]*$`, de-duplicated `-2`, `-3`), Hide block (hidden outside edit mode, faded in edit mode), Padding / Margin (Default, 0, 0.25rem … 7.5rem `--space-*` tokens, same on all breakpoints). Note: the block is centred with `margin: 0 auto`; Margin left / right replace that side's auto margin, so the block is no longer centred.

6.27.7 Field Annotation Mapping

Marker	Field	Purpose
A	Embed block (all fields)	Selection and Additional tab
B	Placeholder Image + Placeholder Image Alt Text	Click-to-load image
C	(generated) Play button	Starts loading; label from placeholder `play`

Marker	Field	Purpose
D	URI	The embedded content

6.27.8 Component Specification

Attribute	Value
Component Name	Embed (`blocks/embed`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	Media decorator, media utils, icon button decorator, viewport utils, placeholders, `loadScript`; third-party YouTube, Vimeo, Twitter / X
Component Type	Block (single cell of grouped fields)
Supported Variants	YouTube; Vimeo; Twitter / X; generic iframe; placeholder with Play
Supported Templates	Any page, any section
Supported Metadata	None
Supported Tags	None
Supported Personalization	None; Play clicks are in the data layer for future Adobe Analytics / Target use
Caching Considerations	Page HTML and placeholder image cached by the Edge Delivery CDN; third-party players are cached by their providers; the Twitter script is loaded once per page
Performance Considerations	No provider request until the block is in the viewport (no placeholder) or until Play (placeholder); iframes `loading="lazy"`; 16:9 reserved space avoids layout shift; a placeholder image in the first section can be the LCP element
Accessibility Compliance	Iframe `title` “Content from {host}”; native Play `<button>` with accessible name; focus moved into the player; alt text for the placeholder; captions depend on the platform
SEO Compliance	Embedded content isn’t page text; provide a textual description near the embed

6.27.9 Input Model

Content Inputs: URI (link), optional placeholder image (picture with alt).

Metadata Inputs: None.

Tags Inputs: None.

Page Context Inputs: The locale (placeholders), visibility in the viewport.

Query Inputs: None.

Manual Inputs: Additional tab classes.

Authored HTML received from AEM (example):

```
<div class="embed">
  <div>
    <div>
      <picture><img src="./media_1a2b.jpg?width=750" alt="Keynote speaker on stage"></
picture>
      <p><a href="https://www.youtube.com/watch?v=VIDEO_ID">
        https://www.youtube.com/watch?v=VIDEO_ID</a></p>
    </div>
```

```
  </div>
</div>
```

6.27.10 Output Model

Rendered Output: Before loading: `div.embed-placeholder` (picture + `button.embed-placeholder-play`) or an empty block reserving 16:9. After loading: `div.embed-frame > iframe` (YouTube, Vimeo, generic) or `blockquote.twitter-tweet` (Twitter / X); the block gets `embed-is-loaded` and `embed-youtube` / `embed-vimeo` / `embed-twitter`.

Generated HTML (YouTube after Play):

```
<div class="embed block embed-youtube embed-is-loaded" data-block-name="embed">
  <div class="embed-frame">
    <iframe src="https://www.youtube.com/embed/VIDEO_ID?rel=0&amp;muted=1&amp;autoplay=1"
      title="Content from youtube.com" loading="lazy" allowfullscreen
      allow="autoplay; fullscreen; picture-in-picture; encrypted-media; accelerometer;
      gyroscope">
    </iframe>
  </div>
</div>
```

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking: Component state (generic): `{ "component": { "embed-1": { "@type": "tcs-corp/components/embed", "parentId": "page-…" } } }` (id = Anchor ID if set). Play click:

```
{ "event": "cmp:click", "eventInfo": { "path": "component.embed-1",
  "interaction": "play", "dc:title": "Play", "xdm:linkType": "other",
  "breakpoint": "desktop" } }
```

Interactions inside the third-party iframes (play / pause in the YouTube player, clicks in a post) are not visible to the page and are not tracked. A click on the Twitter fallback link (before the widget renders) is tracked as `cmp:click` with `xdm:linkType exit`.

Personalization Output: None.

Search Index Output: None (the query index stores page metadata, not embeds).

6.27.11 Business Rules

1. The block reads the first `<picture>` (placeholder) and the first `<a>` (URI) and clears its content.
2. No link → the block stays empty (nothing renders, also in the editor).
3. Provider detection order: YouTube (`youtube.com` / `youtu.be` anywhere in the URL), Vimeo (`vimeo.com` anywhere in the URL), Twitter / X (host match), else generic iframe.
4. With a placeholder: the embed loads only after Play, with autoplay parameters for YouTube / Vimeo (`muted=1`, `autoplay=1`), and focus moves into the loaded iframe / link.
5. Without a placeholder: the embed loads once the block enters the viewport (IntersectionObserver, no margin), without autoplay.
6. The embed loads only once (`embed-is-loaded` guard).
7. Iframe title: placeholder `contentFrom` "Content from {source}", source = host name without `www.`.
8. YouTube: `https://www.youtube.com/embed/{id}?rel=0` (+ autoplay parameters); no privacy host; other URL parameters dropped; unrecognised URL shapes use the URL path on www.youtube.com.

9. Vimeo: `https://player.vimeo.com/video/{first path segment}`.
10. Twitter / X: `x.com` rewritten to `twitter.com`; widget script loaded; the link stays as the fallback.
11. Generic iframe: `allow="encrypted-media"`, `allowfullscreen`, lazy, 16:9.
12. Before loading (no placeholder), the block reserves 16:9 space; after loading, Twitter posts take the widget's height.
13. The block is centred with max width 56.25rem; Additional margin left / right override the centring.
14. Additional tab rules (Anchor ID, Hide block, spacing) as for every block.

6.27.12 Author Best Practices

Recommended Content Length: Placeholder alt text Recommended ≤ 125 characters; add a heading and a one- or two-sentence description near the embed.

Image Dimensions: Placeholder image 16:9 (the area is 16:9, `object-fit: cover`), Recommended at least 1800 × 1012px for the 900px maximum width on high-density screens (for example 1920 × 1080); keep the subject centred because edges may be cropped.

CTA Writing Guidelines: The Play label is generic ("Play"); name the video in the heading or text before the block so the context is clear.

Accessibility Guidelines: Ensure captions on YouTube / Vimeo; provide a transcript link in the text where the video carries important information (or use the Video block's Transcript field); write alt text describing the placeholder image; don't embed autoplaying third-party content without a placeholder.

SEO Guidelines: Describe the video in page text (title, key points); use the official video URL; avoid embedding the same video several times on a page.

Metadata Guidelines: None at block level.

Taxonomy Guidelines: None.

Localization Guidelines: Use locale-appropriate videos or videos with captions in the locale's language; translate the alt text; "Play" and "Content from {source}" are translated in each locale's placeholders sheet (keep the `{source}` variable).

6.27.13 Troubleshooting Guide

Issue 1 — The block renders nothing

Symptoms: Empty space or nothing where the embed should be.

Root Cause: No URI, or the URI is not delivered as a link (`<a>`) in the block HTML.

Diagnosis Steps: Check the URI field; open `{page}.plain.html` on the preview host and look for an `<a href>` inside `div.embed`.

Resolution Steps: Paste the complete URL starting with `https://`; republish.

Validation Steps: The embed or placeholder renders.

Preventive Measures: Always paste full URLs.

Issue 2 — "Video unavailable" in the YouTube player

Symptoms: The YouTube frame loads with an error.

Root Cause: The video id could not be extracted (unsupported URL shape, playlist URL), the video is private, or embedding is disabled by the owner.

Diagnosis Steps: Inspect the iframe `src`; open the URL `https://www.youtube.com/embed/{id}` in a browser.

Resolution Steps: Use a `watch?v=` or `youtu.be/` URL; ask the channel owner to allow embedding.

Validation Steps: The video plays.

Preventive Measures: Test embeds in preview before publishing.

Issue 3 — Vimeo shows "Sorry, this video does not exist"

Symptoms: Vimeo error inside the frame.

Root Cause: The URL is not `vimeo.com/{id}` (for example a channel, showcase, `player.vimeo.com` or private-hash URL), so the first path segment is not the id.

Diagnosis Steps: Inspect the iframe `src` (`player.vimeo.com/video/{segment}`).

Resolution Steps: Use `https://vimeo.com/{numeric id}`; for unlisted videos, consider the Video block or ask development (private hash not supported).

Validation Steps: The video plays.

Preventive Measures: Use canonical Vimeo URLs.

Issue 4 — Twitter / X post shows only a link

Symptoms: A plain URL instead of the post card.

Root Cause: `platform.twitter.com/widgets.js` blocked (content blocker, consent tool, network policy) or the post was deleted / is protected.

Diagnosis Steps: Browser DevTools > Network: check the widgets.js request; open the post URL.

Resolution Steps: Confirm the post is public; check consent / CSP settings with the site admin.

Validation Steps: The post card renders.

Preventive Measures: Use screenshots plus link for critical posts if the widget may be blocked.

Issue 5 — Generic iframe "refused to connect"

Symptoms: Browser error page inside the frame.

Root Cause: The target site forbids framing (`X-Frame-Options` / `frame-ancestors`) or the URL is `http://` on an `https` page (mixed content).

Diagnosis Steps: Console messages about X-Frame-Options / Content-Security-Policy / mixed content.

Resolution Steps: Use an embeddable URL from the provider (often an "embed" link) over https, or link to the content instead.

Validation Steps: The framed page displays.

Preventive Measures: Check embeddability before authoring.

Issue 6 — YouTube video starts with sound or doesn't start after Play

Symptoms: After Play, the player shows its own play button, or plays with sound.

Root Cause: The block requests autoplay with `muted=1`, which YouTube ignores (its parameter is `mute`); browser autoplay policies then decide.

Diagnosis Steps: Inspect the iframe `src` parameters; test in different browsers.

Resolution Steps: Expected behaviour; for precise control (muted, controls, start / end), use the Video block.

Validation Steps: Video plays after one or two clicks.

Preventive Measures: Use the Video block for videos needing playback control.

Issue 7 — Placeholder image cropped badly

Symptoms: Faces or text cut off.

Root Cause: The image isn't 16:9; it is cropped by `object-fit: cover`.

Diagnosis Steps: Check the asset's dimensions.

Resolution Steps: Use a 16:9 rendition with the subject centred.

Validation Steps: The image displays fully at all widths.

Preventive Measures: Prepare 16:9 thumbnails.

6.27.14 Support Runbook

Monitoring Requirements

- Availability of third-party providers (YouTube, Vimeo, X) affects the block; no internal monitoring exists.
- Core Web Vitals of pages with embeds (CLS should be ~0 thanks to the reserved area).

Logs

- Browser console: the block logs nothing itself; errors come from third-party frames (CSP, X-Frame-Options, mixed content, widget errors).
- Network panel: requests to `youtube.com`, `player.vimeo.com`, `platform.twitter.com`.
- Admin API / Sidekick: page status; AEM author: publication logs of the page and placeholder image.

Common Errors

Error	Likely cause	Fix
Empty block	No / non-link URI	Full https URL
YouTube "Video unavailable"	Wrong id / embedding disabled	Correct URL / owner setting
Vimeo "does not exist"	Non-canonical Vimeo URL	`vimeo.com/{id}`
Post shows link only	Widget script blocked	Check consent / CSP
Refused to connect	Framing forbidden / http	Embeddable https URL
Broken placeholder image	Asset not published	Publish the asset

Escalation Matrix

Level	Who	Checks
L1	Author support	URI format, provider permissions, placeholder image and alt, Hide block, page / asset published
L2	AMS / site admin	`.plain.html` markup, CDN cache, CSP / consent tooling, blocked third-party domains
L3	Development team	`embed.js`, media utils (id parsing), media decorator, console errors
L4	Adobe Support	Edge Delivery delivery or Universal Editor issues; third-party provider support for their platforms

Recovery Procedures

1. Replace or remove the URI; restore the previous page version.
2. Hide the block while a provider is unavailable.
3. Revert code changes on `main` if a regression is identified.

Health Checks

- Load a page with each provider type; check the iframe `title`, the Play button and focus behaviour.
- Check that no provider request happens before Play (placeholder) in the Network panel.

Operational Procedures

- New providers require a new entry in the `EMBEDS` list of `embed.js` (development task).
- Translate `play` and `contentFrom` in each locale's placeholders sheet.

Known Limitations

- No privacy-enhanced YouTube host and no consent integration (third-party cookies may be set when the embed loads) — [To be confirmed: consent management requirement — owner: privacy / legal].
- No start time, playlist or other YouTube URL parameters; Vimeo only `vimeo.com/{id}`.
- No dedicated handlers for Instagram, LinkedIn, Facebook, TikTok (Not implemented as of this handover).
- Generic iframes are always 16:9; no height or aspect ratio option.
- No domain allow-list for generic iframes.
- Interactions inside iframes are not tracked in the data layer.

6.28 Video Block

The Video block plays a video file from AEM Assets (MP4, WebM) in a native HTML5 `<video>` player, or a YouTube or Vimeo video in the provider's player, with authorable playback (play button on a poster image, autoplay when in view, background video), sound, loop, controls, start and end times, pop-up playback, pause when out of view, WebVTT captions, caption text, transcript link, aspect ratio, size, alignment, rounded corners and a privacy-enhanced mode. It is based on the Edge Delivery Services Block Collection video block (`blocks/video/video.js`, `video.css`, `_video.json`) and is a key-value block. The model has 38 entries: 6 tabs (Video, Playback, Captions, Display, Privacy, Additional) and 32 fields. Videos never autoplay for visitors who prefer reduced motion; a video that plays on its own without controls always gets a pause / stop button (WCAG 2.2.2).

6.28.1 Component Purpose

Business Objective: Present brand, product and thought-leadership videos with full control over playback, accessibility (captions, transcript, names, pause control), privacy and performance (poster first, load on demand), from AEM Assets or YouTube / Vimeo.

Business Use Cases:

- A product or service explainer video with a branded poster and play button.
- A decorative, silent, looping background video (for example an ambient office scene).
- A keynote from YouTube played in a pop-up dialog over the page.
- A muted autoplay teaser that starts when scrolled into view.
- A vertical (9:16) social-style clip or a square (1:1) video.
- A captioned corporate film hosted in AEM Assets with a transcript link.

When To Use:

- Any video that needs more than a simple embed: AEM Assets files, captions, transcript, pop-up, autoplay / background, start / end times, privacy mode, sizes and aspect ratios.

When Not To Use:

- Social posts (Twitter / X) or arbitrary iframes: use the Embed block (6.27).
- Lottie animations: use the Lotto (Lottie) image type of Hero / Teaser.
- Very large uncompressed video files: provide web-optimised MP4 / WebM renditions (see 6.28.12).

Dependencies:

Type	Dependency	Use
Code	`decorators/media-decorator.js` (`createVideoElement`, `createYoutubeFrame`, `createVimeoFrame`)	Builds the players
Code	`utils/media.js` (`isYoutube`, `isVimeo`, `youtubeId`, `videoType`)	Provider detection, ids, MIME type
Code	`decorators/icon-button-decorator.js`, `decorators/button-decorator.js`	Play button, pause / stop toggle
Code	`decorators/content-decorator.js` (`fromCell`)	Caption with editor instrumentation
Code	`decorators/image-decorator.js` (`pictureOf`)	Poster image with alt text
Code	`utils/properties.js`, `utils/links.js`	Reading key-value fields and links
Code	`utils/environment.js` (`prefersReducedMotion`)	No autoplay for reduced motion
Code	`utils/viewport.js` (`onVisible`, `watchVisibility`)	Load when visible; pause off screen
Code	`utils/site.js` (`getLocale`)	Default captions language
Code	`utils/placeholders.js`	UI labels
Code	`blocks/modal/modal.js` (`createModal`, loaded on demand)	Play in a Pop-up
Code	Block options and data layer decorators	Additional tab; `cmp:media`, `cmp:click`
Content	AEM Assets (`/content/dam/tcs-corp/...`, published via `paths.json` includes)	Video files, poster, captions (.vtt), transcript documents
Third party	`www.youtube-nocookie.com` / `www.youtube.com`, `player.vimeo.com`	Provider players
Sheet	`/{locale}/placeholders.json`	`video`, `playVideo`, `playVideoTitle`, `pauseVideo`, `stopVideo`, `readTheTranscript`; modal: `dialog`, `close`
Model	`models/_section.json`	Allows the Video in sections

Related Components: Embed (simple third-party embeds), Modal (pop-up), Hero / Teaser (media in larger compositions), Separator.

Applicable Templates: Any page (home, article, blog, press-release, standard), any section.

Supported Content Types: MP4 and WebM files (any file URL; the MIME type is derived from the extension), YouTube URLs (watch, `youtu.be`, `/embed/`, `/shorts/`, `/live/`), Vimeo URLs (`vimeo.com/{id}`), poster image, WebVTT captions, transcript page or document, rich-text caption.

6.28.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** At 1440px: a Large (900px) 16:9 video, centred, with 1rem rounded corners, a poster image and a round brand-blue Play button in the centre; a grey caption and a “Read the transcript” link below.

Desktop (≥ 1200px): The block's maximum width depends on Size (Large 56.25rem default, Medium 40rem, Small 26.25rem, Full width none) and its position on Alignment (centre default; left / right for sizes other than full width). The media area keeps the chosen aspect ratio (default 16:9), has a black background (`--color-bg-neutral`), clips its content and has rounded corners (1rem, `--radius-l`) unless Rounded Corners is off. The poster covers the area with the Play button (primary icon button, scaled ×1.4) centred. Video files fill the area (`object-fit: cover`). Below: caption (0.75rem top margin, `--color-text-secondary`, small body size) and the transcript link (0.5rem top margin, small body size).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** At 768px: the same video filling the content width with the aspect ratio kept; a background video with the pause button in the bottom-right corner.

Tablet (600–1199px): Same rules; when the content width is less than the Size maximum, the video fills the content width. There are no breakpoint-specific rules in `video.css`; the body font size of caption and transcript steps up at 900px (global tokens).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** At 390px: a 9:16 vertical video at full width, and a 16:9 video with poster and Play button; caption below.

Mobile (< 600px): Full content width for all sizes; aspect ratio kept (a 9:16 video becomes tall: limit its Size to Small or Medium). Video files play inline (`playsinline`), not in the phone's full-screen player, unless the visitor chooses full screen.

6.28.3 Component Variations

Variation 1 — Video file from AEM Assets

Variation Name: Video source: file (native `<video>` player).

Business Purpose: Self-hosted brand videos without third-party players or cookies.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-file.png`
> **Purpose:** Variation example
> **Description:** A native HTML5 video player with browser controls (play, timeline, volume, captions, full screen) inside the rounded 16:9 area.

Use Cases: Corporate films, product demos, captioned content.

Authoring Guidelines: Video tab > **Video**: select an MP4 or WebM file from AEM Assets.

Business Rules: Any URL not recognised as YouTube or Vimeo is played as a file; the MIME type is `video/{extension}` (`video/mp4`, `video/webm`). The asset must be published (AEM Assets under `/content/dam/tcs-corp/` is included in `paths.json`).

Rendering Rules: `<video>` with `controls` (unless off / background), `controlslist="nodownload"`, right-click menu suppressed, `playsinline`, `preload="metadata"` (or `auto` when autoplaying), `poster` from the Poster Image, `aria-label` = Title, `<source src="…#t=start,end" type="video/mp4">` and an optional captions `<track>`.

Responsive Behaviour: Fills the media area (`object-fit: cover`) on all breakpoints.

SEO Impact: The file URL is in the DOM after decoration; no video structured data is generated (Not implemented as of this handover).

Accessibility Compliance: Native keyboard-operable controls; accessible name from Title; captions track (WCAG 1.2.2) when provided; Download option hidden.

Variation 2 — YouTube

Variation Name: Video source: YouTube (`iframe` in `div.video-frame`).

Business Purpose: Play videos published on the TCS YouTube channel.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-youtube.png`
> **Purpose:** Variation example
> **Description:** The YouTube player inside the rounded 16:9 area, privacy-enhanced (youtube-nocookie.com).

Use Cases: Keynotes, campaigns, interviews.

Authoring Guidelines: Video tab > **Video**: paste the YouTube URL.

Business Rules: Recognised when the URL contains `youtube.com` or `youtu.be`. Player parameters: `rel=0`, `autoplay` 0 / 1, `mute` 0 / 1, `controls` 0 / 1, `playsinline=1`; Loop adds `loop=1&playlist={id}`; background adds `disablekb=1`; Start At / End At add `start` / `end` (whole seconds). Privacy-enhanced Mode (default on) uses `www.youtube-nocookie.com`. Captions file fields are ignored (use YouTube's captions).

Rendering Rules: `div.video-frame > iframe` with `title` = Title, lazy, `allow="autoplay; fullscreen; picture-in-picture; encrypted-media; accelerometer; gyroscope"`, `allowfullscreen`.

Responsive Behaviour: Frame fills the media area; YouTube letterboxes inside it.

SEO Impact: Third-party iframe; describe the video in text.

Accessibility Compliance: Iframe title; captions and controls from YouTube.

Variation 3 — Vimeo

Variation Name: Video source: Vimeo.

Business Purpose: Play Vimeo-hosted videos.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-vimeo.png`
> **Purpose:** Variation example
> **Description:** The Vimeo player in the rounded media area.

Use Cases: Brand films hosted on Vimeo.

Authoring Guidelines: Paste `https://vimeo.com/{id}` into **Video**.

Business Rules: Recognised when the URL contains `vimeo.com`; the id is the first path segment. Parameters: `autoplay`, `muted`, `loop`, `controls` (0 / 1), `playsinline=1`; `background=1` for background videos; `dnt=1` (do not track) with Privacy-enhanced Mode; Start At as `#t={seconds}s`. End At is not applied to Vimeo (Not implemented as of this handover).

Rendering Rules: `div.video-frame > iframe` (`player.vimeo.com/video/{id}`), title = Title, `allow="autoplay; fullscreen; picture-in-picture; encrypted-media"`.

Responsive Behaviour: As YouTube.

SEO Impact: As YouTube.

Accessibility Compliance: Iframe title; captions and controls from Vimeo.

Variation 4 — Play button on the poster image (inline)

Variation Name: Playback = Play button (on the poster image), with a Poster Image (`placeholder` class).

Business Purpose: Fast page (no player until requested) with a branded first impression.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-play-button-poster.png`
> **Purpose:** Variation example
> **Description:** A poster image with the centred Play button; second frame: the video playing in place.

Use Cases: Most content videos.

Authoring Guidelines: Playback = Play button (default); set Poster Image and its alt text.

Business Rules: On Play, the player is built with autoplay; it plays with sound unless Muted is on (the visitor started it). Keyboard focus moves into the player once it can play / has loaded.

Rendering Rules: `div.video-placeholder` (poster picture + `button.video-placeholder-play`, label "Play video: {Title}" from `playVideoTitle`, or "Play video" without a Title). The player is appended behind the poster and hidden until it can play (`data-embed-loaded="false"`), then the poster is hidden (`data-embed-loaded="true"`).

Responsive Behaviour: Same on all breakpoints.

SEO Impact: Poster image in the page; good for Core Web Vitals.

Accessibility Compliance: Native button with a descriptive name; focus management (WCAG 2.4.3); poster alt text (WCAG 1.1.1).

Variation 5 — Click to play without a poster

Variation Name: Playback = Play button, no Poster Image, no pop-up.

Business Purpose: The provider's / browser's own player with its first frame and play button.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-no-poster.png`

> **Purpose:** Variation example
> **Description:** A native video player showing the first frame and its own play control (or the YouTube thumbnail).

Use Cases: Quick additions where no poster exists.

Authoring Guidelines: Playback = Play button; leave Poster Image empty; Play in a Pop-up off.

Business Rules: The player loads when the block enters the viewport, without autoplay; Muted, Loop, controls and times apply.

Rendering Rules: `video-media` contains only the player (no `placeholder` class).

Responsive Behaviour: Same on all breakpoints.

SEO Impact: Neutral.

Accessibility Compliance: Player's native controls.

Variation 6 — Play in a Pop-up

Variation Name: Play in a Pop-up (modal dialog).

Business Purpose: Large, focused viewing without leaving the page.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-popup.png`
> **Purpose:** Variation example
> **Description:** The page dimmed behind a dialog containing the playing video and a Close button at the top.

Use Cases: Teaser images that open a keynote; compact layouts.

Authoring Guidelines: Playback = Play button; Playback tab > **Play in a Pop-up** = on; recommended with a Poster Image.

Business Rules: Only with Playback = Play button (ignored for autoplay / background). The play button opens a dialog (the Modal block is loaded on demand) with the player autoplaying (sound unless Muted). Closing the dialog (Close button, Esc, click on the backdrop) removes it and stops the video; focus returns to the Play button. Without a poster, the black media area shows only the Play button.

Rendering Rules: `dialog` with `aria-label` "Dialog" (placeholder `dialog`), Close button (placeholder `close`), `div.video-modal-player` sized `min(90vw, 75rem, 80vh × aspect ratio)` with the chosen aspect ratio.

Responsive Behaviour: The player scales to the viewport (90% width, max 75rem, height ≤ 80vh).

SEO Impact: Neutral.

Accessibility Compliance: Native `<dialog>` (focus trapped by the browser, Esc closes); focus returned on close; data layer `cmp:show` / `cmp:hide`.

Variation 7 — Autoplay when in view (with controls)

Variation Name: Playback = Autoplay when in view (muted), Show Player Controls on.

Business Purpose: Attract attention with motion while letting visitors control playback.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-autoplay.png`
> **Purpose:** Variation example
> **Description:** A muted video playing as the section scrolls into view, with native controls visible.

Use Cases: Product teasers, highlight reels.

Authoring Guidelines: Playback = Autoplay when in view; keep Show Player Controls on.

Business Rules: Starts muted (always, regardless of Muted) when the block enters the viewport; the poster (if any) shows without a play button until the video can play. For video files, Pause When Out of View (default on) pauses it off screen and resumes it when back (unless the visitor paused it). Reduced motion → behaves as Play button.

Rendering Rules: Player with autoplay, muted, controls; no toggle (controls allow pausing).

Responsive Behaviour: Same on all breakpoints.

SEO Impact: Neutral; preload `auto` for files increases data usage.

Accessibility Compliance: WCAG 2.2.2: visible controls allow pause; no sound without user action (WCAG 1.4.2).

Variation 8 — Autoplay without controls (with playback toggle)

Variation Name: Playback = Autoplay when in view, Show Player Controls off.

Business Purpose: A clean, controls-free autoplaying video that still can be paused.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-autoplay-no-controls.png`
> **Purpose:** Variation example
> **Description:** A playing video without controls and a round “Pause video” button in the bottom-right corner.

Use Cases: Visual storytelling sections.

Authoring Guidelines: Playback = Autoplay when in view; Show Player Controls = off.

Business Rules: A pause / play toggle is added (files: “Pause video” / “Play video”; YouTube / Vimeo: “Stop video” removes the frame, “Play video” rebuilds it and restarts from the start time).

Rendering Rules: `button.video-toggle` (secondary icon button, `pause` / `play` icon, `data-cmp-interaction="playback"`), positioned 0.75rem from the bottom and inline end.

Responsive Behaviour: Toggle in the same corner on all breakpoints (left corner in right-to-left pages, logical positioning).

SEO Impact: Neutral.

Accessibility Compliance: WCAG 2.2.2 (Pause, Stop, Hide) met by the toggle; its label updates with the state.

Variation 9 — Background video

Variation Name: Playback = Background video (autoplay, muted, loop, no controls).

Business Purpose: A decorative, ambient moving image.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-background.png`
> **Purpose:** Variation example
> **Description:** A full-width looping silent video with only the small pause button in the corner.

Use Cases: Atmosphere at the top of campaign pages.

Authoring Guidelines: Playback = Background video; usually Size Full width; set a Poster Image.

Business Rules: Always muted, looping, without controls (Show Player Controls and Muted / Loop are overridden); the player is hidden from assistive technology (`aria-hidden="true"`; iframes `tabindex="-1"`); a pause / stop toggle is always added; YouTube adds `disablekb=1`, Vimeo `background=1`. Files pause off screen with Pause When Out of View. Reduced motion → behaves as Play button.

Rendering Rules: As Variation 8 plus `aria-hidden`.

Responsive Behaviour: Same on all breakpoints; consider data usage on mobile.

SEO Impact: None (decorative).

Accessibility Compliance: Decorative (hidden), pausable (WCAG 2.2.2), no sound.

Variation 10 — Muted

Variation Name: Muted = on.

Business Purpose: Start without sound even when the visitor presses Play.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-muted.png`
> **Purpose:** Variation example
> **Description:** A video playing after Play with the volume control showing muted.

Use Cases: Videos without important audio; office environments.

Authoring Guidelines: Playback tab > Muted = on.

Business Rules: Default off. Always on for autoplay and background (help text: "always on for autoplay and background videos").

Rendering Rules: Files: `muted` property and attribute; YouTube `mute=1`; Vimeo `muted=1`.

Responsive Behaviour: All breakpoints.

SEO Impact: None.

Accessibility Compliance: Visitors can unmute via controls (if shown).

Variation 11 — Loop

Variation Name: Loop = on.

Business Purpose: Repeat short clips.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-loop.png`
> **Purpose:** Variation example
> **Description:** A short clip restarting at its end (two frames).

Use Cases: Short product loops.

Authoring Guidelines: Playback tab > Loop = on.

Business Rules: Default off; always on for background.

Rendering Rules: Files `loop`; YouTube `loop=1&playlist={id}`; Vimeo `loop=1`.

Responsive Behaviour: All breakpoints.

SEO Impact: None.

Accessibility Compliance: Looping content that plays on its own must be pausable (controls or toggle are provided).

Variation 12 — Start At / End At

Variation Name: Start At (seconds) / End At (seconds).

Business Purpose: Play an excerpt of a longer video.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-start-end.png`
> **Purpose:** Variation example
> **Description:** A player whose timeline starts at 0:30 after Play.

Use Cases: Highlighting one answer in an interview.

Authoring Guidelines: Playback tab > Start At (e.g. 30), End At (e.g. 90).

Business Rules: Empty → 0 (from the start / to the end); negative or non-numeric values → 0. YouTube: whole seconds (`start`, `end`); Vimeo: start only (`#t=30s`); files: media fragment `#t=30,90`. End before Start is not validated.

Rendering Rules: As above.

Responsive Behaviour: All breakpoints.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 13 — Pause When Out of View

Variation Name: Pause When Out of View (default on).

Business Purpose: Saves CPU / battery and avoids distraction.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-pause-offscreen.png`

> **Purpose:** Variation example
> **Description:** An autoplaying video paused after scrolling away, resuming when scrolled back.

Use Cases: Autoplay and background video files.

Authoring Guidelines: Playback tab > Pause When Out of View on (default) / off.

Business Rules: Only for video files in Autoplay or Background mode; resumes only if it was paused by scrolling (not by the visitor). Not applied to YouTube / Vimeo.

Rendering Rules: `watchVisibility` IntersectionObserver on the `<video>`.

Responsive Behaviour: All breakpoints.

SEO Impact: None.

Accessibility Compliance: Reduces motion off screen.

Variation 14 — Captions (WebVTT)

Variation Name: Captions File (WebVTT) with language and name.

Business Purpose: Accessible, translatable subtitles for self-hosted videos.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-captions.png`
> **Purpose:** Variation example
> **Description:** A native player showing subtitles and the captions menu listing “English”.

Use Cases: Every video file with speech.

Authoring Guidelines: Captions tab: Captions File (WebVTT) (a `.vtt` from AEM Assets), Captions Language (e.g. `en`, `de`, `ja`), Captions Name (e.g. “English”).

Business Rules: Only for video files (YouTube / Vimeo use their own captions). Language defaults to the page’s language (for example `de` on `/de-de`); name defaults to the language code. One track; it is the default track.

Rendering Rules: `<track kind="captions" src="…" srclang="en" label="English" default>`.

Responsive Behaviour: Rendered by the browser.

SEO Impact: None directly (captions aren’t indexed from the track); provide a transcript.

Accessibility Compliance: WCAG 1.2.2 (Captions, Prerecorded).

Variation 15 — Caption text and Transcript link

Variation Name: Caption (rich text) and Transcript (link) below the video.

Business Purpose: Context and a text alternative.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-caption-transcript.png`
> **Purpose:** Variation example
> **Description:** A video with a caption line below and a “Read the transcript” link.

Use Cases: Interviews, keynotes, any video with information.

Authoring Guidelines: Video tab: Caption (rich text); Transcript (select a page or document).

Business Rules: Caption shown only if it has text. Transcript link text: the authored link text, unless it looks like a URL or path, then "Read the transcript" (placeholder `readTheTranscript`).

Rendering Rules: `div.video-caption` after the media; `<p><a class="video-transcript">` after the caption.

Responsive Behaviour: Text wraps; small body size (steps up at 900px).

SEO Impact: Positive: caption and transcript are indexable text / links.

Accessibility Compliance: WCAG 1.2.1 / 1.2.3 (text alternative / media alternative) via transcript.

Variation 16 — Aspect ratio 16:9 (widescreen)

Variation Name: Aspect Ratio = 16:9 (default).

Business Purpose: Standard widescreen video.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-ratio-16-9.png`
> **Purpose:** Variation example
> **Description:** A 16:9 media area.

Use Cases: Most videos.

Authoring Guidelines: Display tab > Aspect Ratio = 16:9 (widescreen).

Business Rules: Also the fallback for unknown values.

Rendering Rules: `--video-aspect-ratio: 16 / 9` on the block; media area `aspect-ratio`.

Responsive Behaviour: Kept at all widths.

SEO Impact: None.

Accessibility Compliance: Reserved space prevents layout shift.

Variation 17 — Aspect ratio 4:3

Variation Name: Aspect Ratio = 4:3.

Business Purpose: Legacy or presentation-format videos.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-ratio-4-3.png`
> **Purpose:** Variation example
> **Description:** A 4:3 media area.

Use Cases: Archive footage, slide recordings.

Authoring Guidelines: Aspect Ratio = 4:3.

Business Rules: Choose the ratio of the video itself: files are cropped to fill the area (`object-fit: cover`).

Rendering Rules: `--video-aspect-ratio: 4 / 3`.

Responsive Behaviour: Kept at all widths.

SEO Impact: None.

Accessibility Compliance: As 16:9.

Variation 18 — Aspect ratio 1:1 (square)
Variation Name: Aspect Ratio = 1:1 (square).

Business Purpose: Square social-style videos.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-ratio-1-1.png`
> **Purpose:** Variation example
> **Description:** A square video, Size Medium, centred.

Use Cases: Social cut-downs.

Authoring Guidelines: Aspect Ratio = 1:1; Size Medium or Small recommended.

Business Rules: As 4:3.

Rendering Rules: `--video-aspect-ratio: 1 / 1`.

Responsive Behaviour: Kept; becomes tall on mobile at full width.

SEO Impact: None.

Accessibility Compliance: As 16:9.

Variation 19 — Aspect ratio 21:9 (cinema)
Variation Name: Aspect Ratio = 21:9 (cinema).

Business Purpose: Cinematic, wide banners.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-ratio-21-9.png`
> **Purpose:** Variation example
> **Description:** A 21:9 full-width background video.

Use Cases: Background videos at the top of a page.

Authoring Guidelines: Aspect Ratio = 21:9; often Size Full width.

Business Rules: As 4:3.

Rendering Rules: `--video-aspect-ratio: 21 / 9`.

Responsive Behaviour: Kept; shallow on mobile.

SEO Impact: None.

Accessibility Compliance: As 16:9.

Variation 20 — Aspect ratio 9:16 (vertical)
Variation Name: Aspect Ratio = 9:16 (vertical).

Business Purpose: Vertical / short-form videos.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-ratio-9-16.png`
> **Purpose:** Variation example
> **Description:** A vertical video, Size Small, centred on desktop; full width on mobile.

Use Cases: YouTube Shorts, social clips.

Authoring Guidelines: Aspect Ratio = 9:16; Size Small recommended.

Business Rules: As 4:3.

Rendering Rules: `--video-aspect-ratio: 9 / 16`.

Responsive Behaviour: Very tall at full width; limit with Size Small (26.25rem).

SEO Impact: None.

Accessibility Compliance: As 16:9.

Variation 21 — Size Full width

Variation Name: Size = Full width (`video-size-full`).

Business Purpose: Maximum impact.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-size-full.png`
> **Purpose:** Variation example
> **Description:** A video spanning the full content width of the section.

Use Cases: Background videos, hero-like videos.

Authoring Guidelines: Display tab > Size = Full width.

Business Rules: Alignment has no effect.

Rendering Rules: `--video-max-width: none`.

Responsive Behaviour: Always the full content width.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 22 — Size Large (900px)

Variation Name: Size = Large (`video-size-large`, default).

Business Purpose: Standard content video.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-size-large.png`

> **Purpose:** Variation example
> **Description:** A 900px-wide video centred on desktop.

Use Cases: Most videos.

Authoring Guidelines: Size = Large (900px).

Business Rules: Default and fallback.

Rendering Rules: Max width 56.25rem.

Responsive Behaviour: Full width below 900px of content width.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 23 — Size Medium (640px)

Variation Name: Size = Medium (`video-size-medium`).

Business Purpose: Secondary videos.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-size-medium.png`
> **Purpose:** Variation example
> **Description:** A 640px-wide video.

Use Cases: Supporting clips in articles.

Authoring Guidelines: Size = Medium (640px).

Business Rules: Works with Alignment.

Rendering Rules: Max width 40rem.

Responsive Behaviour: Full width on mobile.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 24 — Size Small (420px)

Variation Name: Size = Small (`video-size-small`).

Business Purpose: Compact or vertical videos.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-size-small.png`
> **Purpose:** Variation example
> **Description:** A 420px-wide vertical video aligned left.

Use Cases: 9:16 and 1:1 clips.

Authoring Guidelines: Size = Small (420px).

Business Rules: Works with Alignment.

Rendering Rules: Max width 26.25rem.

Responsive Behaviour: Full width only below 420px of content width.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 25 — Alignment Center

Variation Name: Alignment = Center (`video-align-center`, default).

Business Purpose: Balanced layout.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-align-center.png`
> **Purpose:** Variation example
> **Description:** A Medium video centred in the section.

Use Cases: Default.

Authoring Guidelines: Display tab > Alignment = Center.

Business Rules: For sizes other than Full width.

Rendering Rules: `margin: 0 auto`.

Responsive Behaviour: Visible only when the video is narrower than the content.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 26 — Alignment Left

Variation Name: Alignment = Left (`video-align-left`).

Business Purpose: Align with left-aligned text.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-align-left.png`
> **Purpose:** Variation example
> **Description:** A Small video aligned with the start of the text column.

Use Cases: Editorial layouts.

Authoring Guidelines: Alignment = Left.

Business Rules: For sizes other than Full width; uses logical margins, so in right-to-left pages "Left" aligns to the start (right) side.

Rendering Rules: `margin-inline: 0 auto`.

Responsive Behaviour: As above.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 27 — Alignment Right
Variation Name: Alignment = Right (`video-align-right`).

Business Purpose: Align to the end of the column.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-align-right.png`
> **Purpose:** Variation example
> **Description:** A Small video aligned to the right edge.

Use Cases: Editorial layouts.

Authoring Guidelines: Alignment = Right.

Business Rules: As Left (logical: end side).

Rendering Rules: `margin-inline: auto 0`.

Responsive Behaviour: As above.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 28 — Rounded corners
Variation Name: Rounded Corners = on (`video-rounded`, default).

Business Purpose: Site design language (1rem radius).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-rounded.png`
> **Purpose:** Variation example
> **Description:** A video with 1rem rounded corners.

Use Cases: Default.

Authoring Guidelines: Display tab > Rounded Corners on.

Business Rules: Default on.

Rendering Rules: `--video-radius: var(--radius-l)` on the media area.

Responsive Behaviour: Same radius on all breakpoints.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 29 — Square corners
Variation Name: Rounded Corners = off.

Business Purpose: Edge-to-edge or technical layouts.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-square-corners.png`
> **Purpose:** Variation example
> **Description:** A full-width background video with square corners.

Use Cases: Full-width background videos.

Authoring Guidelines: Rounded Corners off.

Business Rules: —

Rendering Rules: `--video-radius: 0`.

Responsive Behaviour: All breakpoints.

SEO Impact: None.

Accessibility Compliance: None specific.

Variation 30 — Privacy-enhanced mode on

Variation Name: Privacy-enhanced Mode = on (default).

Business Purpose: Reduce tracking by video providers.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-privacy-on.png`
> **Purpose:** Variation example
> **Description:** DevTools showing the iframe src on www.youtube-nocookie.com.

Use Cases: Default for all YouTube / Vimeo videos.

Authoring Guidelines: Privacy tab > Privacy-enhanced Mode on.

Business Rules: YouTube: `www.youtube-nocookie.com` (no cookies until the video plays, per the help text); Vimeo: `dnt=1`. No effect on files.

Rendering Rules: Player URL host / parameter.

Responsive Behaviour: n/a.

SEO Impact: None.

Accessibility Compliance: n/a. Privacy: supports data-protection requirements; consent management is not implemented ([To be confirmed: consent requirements — owner: privacy / legal]).

Variation 31 — Privacy-enhanced mode off

Variation Name: Privacy-enhanced Mode = off.

Business Purpose: Standard player behaviour (for example when YouTube features need the regular host).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-privacy-off.png`

> **Purpose:** Variation example
> **Description:** DevTools showing the iframe src on www.youtube.com.

Use Cases: Exceptions approved by privacy owners.

Authoring Guidelines: Privacy-enhanced Mode off.

Business Rules: YouTube `www.youtube.com`; Vimeo without `dnt`.

Rendering Rules: As above.

Responsive Behaviour: n/a.

SEO Impact: None.

Accessibility Compliance: n/a.

Variation 32 — Reduced motion (automatic)

Variation Name: Reduced motion fallback (not an author option).

Business Purpose: Respect visitors who ask the system to reduce motion.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-reduced-motion.png`
> **Purpose:** Variation example
> **Description:** A background video configured block shown with its poster and a Play button because the OS reduced-motion setting is on.

Use Cases: All autoplay and background videos.

Authoring Guidelines: Nothing to set; always provide a Poster Image for autoplay / background videos so the reduced-motion view looks good.

Business Rules: With `prefers-reduced-motion: reduce`, Playback is treated as Play button (with a poster or pop-up: poster + Play; else the player without autoplay).

Rendering Rules: As Variations 4–6.

Responsive Behaviour: All breakpoints.

SEO Impact: None.

Accessibility Compliance: WCAG 2.2.2 / 2.3.3 good practice.

Variation 33 — Blocks authored with an earlier model (compatibility)

Variation Name: Legacy Video blocks (link + picture + "autoplay" option, or one row per field).

Business Purpose: Existing pages keep working after the model became key-value.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/video-legacy.png`
> **Purpose:** Variation example
> **Description:** A Video block created with the earlier model rendering correctly.

Use Cases: Pages created before the current model.

Authoring Guidelines: Re-open and save the block to move to the current fields when convenient.

Business Rules: If no `uri` property exists: the first link is the video, the first picture the poster, and an `autoplay` block class means Background video (else Play button). Blocks delivered as one row per field (in dialog order) are read positionally when the Playback row holds click / autoplay / background.

Rendering Rules: As the current model.

Responsive Behaviour: As the current model.

SEO Impact: None.

Accessibility Compliance: As the current model (Title defaults to "Video").

6.28.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** Universal Editor edit mode with a Video block selected: poster image with Play button, caption and transcript link below; the properties panel open on the Video tab.

Annotations:

- A = The Video block (select to open its properties)
- B = The poster image (Poster Image + Poster Image Alt Text)
- C = The Play button (generated; its label uses the Title)
- D = The media area (shows the player after Play / when in view; aspect ratio, size, alignment and corners from the Display tab)
- E = The caption (Caption, rich text)
- F = The transcript link (Transcript)
- G = The pause / stop toggle (generated for autoplay without controls and background videos)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The Video properties panel with its tabs Video, Playback, Captions, Display, Privacy and Additional.

Properties panel tabs:

- **Video:** Video (required: asset or link); Title; Caption (rich text); Transcript (page or document); Poster Image; Poster Image Alt Text.
- **Playback:** Playback (Play button (on the poster image) / Autoplay when in view (muted) / Background video (autoplay, muted, loop, no controls)); Muted; Loop; Show Player Controls; Start At (seconds); End At (seconds); Play in a Pop-up; Pause When Out of View.
- **Captions:** Captions File (WebVTT); Captions Language; Captions Name.
- **Display:** Aspect Ratio (16:9, 4:3, 1:1, 21:9, 9:16); Size (Full width, Large (900px), Medium (640px), Small (420px)); Alignment (Center, Left, Right); Rounded Corners.
- **Privacy:** Privacy-enhanced Mode.
- **Additional:** Anchor ID; Hide block; Padding left / right / top / bottom; Margin left / right / top / bottom.

None of the fields has a `condition`: all are always visible; fields that don't apply to the chosen source or playback are ignored by the code (see Field details).

6.28.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Open the page with **Edit** from AEM Sites.

Expected Result: The page is in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-step-01-open-page.png`
> **Purpose:** Authoring step
> **Description:** The page open in Universal Editor.

Step 2 — Upload or locate the assets

Action: In AEM Assets, upload the web-optimised video (MP4 / WebM), the poster image and, for files, the `.vtt` captions under `/content/dam/tcs-corp/…` (skip for YouTube / Vimeo except the poster).

Expected Result: The assets are available in the asset picker.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-step-02-assets.png`
> **Purpose:** Authoring step
> **Description:** AEM Assets folder with the MP4, the poster JPG and the VTT file.

Step 3 — Add the Video block

Action: Select the section, choose **Add** (+), select **Video**.

Expected Result: A Video block with defaults (Play button, controls on, 16:9, Large, Center, rounded, privacy on) is added; nothing renders until Video is set.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-step-03-add-block.png`
> **Purpose:** Authoring step
> **Description:** The Add menu with Video and the new block in the content tree.

Step 4 — Video tab

Action: Set **Video** (pick the asset or paste the YouTube / Vimeo URL); type a **Title** (names the video for screen readers); optionally a **Caption** and a **Transcript**; choose a **Poster Image** and its **Alt Text**.

Expected Result: The poster with the Play button renders (or the player, without poster).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-step-04-video-tab.png`
> **Purpose:** Authoring step
> **Description:** The Video tab filled in.

Step 5 — Playback tab

Action: Choose **Playback**; set Muted, Loop, Show Player Controls, Start At / End At, Play in a Pop-up and Pause When Out of View as needed.

Expected Result: Settings saved; autoplay behaviour is visible in Preview.

> [Screenshot Placeholder]
> **File name:** `screenshots/video-step-05-playback-tab.png`
> **Purpose:** Authoring step
> **Description:** The Playback tab with Playback set to Background video.

Step 6 — Captions tab (video files)

Action: Select the **Captions File (WebVTT)**; enter **Captions Language** (e.g. `en`) and **Captions Name** (e.g. "English").

Expected Result: The captions option appears in the native player.

> [Screenshot Placeholder]
> **File name:** `screenshots/video-step-06-captions-tab.png`
> **Purpose:** Authoring step
> **Description:** The Captions tab filled in.

Step 7 — Display tab

Action: Choose **Aspect Ratio** (match the video), **Size**, **Alignment** and **Rounded Corners**.

Expected Result: The media area changes shape and size.

> [Screenshot Placeholder]
> **File name:** `screenshots/video-step-07-display-tab.png`
> **Purpose:** Authoring step
> **Description:** The Display tab with 9:16 and Small.

Step 8 — Privacy tab

Action: Keep **Privacy-enhanced Mode** on unless an exception is approved.

Expected Result: YouTube uses youtube-nocookie.com; Vimeo gets do-not-track.

> [Screenshot Placeholder]
> **File name:** `screenshots/video-step-08-privacy-tab.png`
> **Purpose:** Authoring step
> **Description:** The Privacy tab.

Step 9 — Additional tab (optional)

Action: Set Anchor ID, Hide block, padding or margin.

Expected Result: Values accepted.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-step-09-additional-tab.png`
> **Purpose:** Authoring step
> **Description:** The Additional tab.

Step 10 — Preview and test

Action: Switch to **Preview**; play the video with mouse and keyboard; test the pop-up, the pause toggle, captions; check mobile width.

Expected Result: Playback works as configured.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-step-10-preview.png`
> **Purpose:** Authoring step
> **Description:** The video playing in preview with captions on.

Step 11 — Validate and publish

Action: Run the Validation Process checklist; **Publish** the page together with the referenced assets (video, poster, captions, transcript).

Expected Result: The video plays on the live site.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/video-step-11-publish.png`
> **Purpose:** Authoring step
> **Description:** The Publish dialog listing the page and its referenced assets.

Preview Process

1. Universal Editor **Preview** mode.
2. Edge Delivery preview `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{path}`; check that the video file URL loads (Network panel) — assets must be published / previewed with the page.
3. Test with the OS reduced-motion setting on (autoplay / background become Play button).

Publishing Process

Publish the page and its referenced assets from Universal Editor / AEM Sites (Manage Publication lists references). AEM Assets under `/content/dam/tcs-corp/` are delivered at their own path (included in `paths.json`).

Validation Process

- ☐ Video set (required) and playing in preview on desktop and mobile.
- ☐ Title set (descriptive, not “Video”).
- ☐ Poster image set (mandatory in practice for autoplay / background and pop-up) with alt text or intentionally decorative.
- ☐ Aspect Ratio matches the video (no unwanted cropping of files).
- ☐ Video files have a WebVTT captions file with correct language; YouTube / Vimeo have captions on the platform.
- ☐ Transcript provided for videos with important information.
- ☐ Autoplay / background videos are muted and show the pause / stop toggle (or controls).

- ☐ Start At / End At correct (End after Start; End not available for Vimeo).
- ☐ Privacy-enhanced Mode on (unless an approved exception).
- ☐ Pop-up opens, plays, closes with Esc / Close, and focus returns.
- ☐ Assets published.

Rollback Process

1. Restore the previous page version and republish.
2. Turn on Hide block while fixing.
3. Replace the asset with the previous version in AEM Assets (asset versioning) and republish the asset.

6.28.6 Authorable Field Specification

Model `video` (key-value block)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Video (`uri`) — Video	AEM content (asset / link)	Yes	No	Empty	The video	MP4 / WebM asset, or YouTube / Vimeo URL	Enforced: `required` in the model; nothing renders without it	/content/dam/tcs-corp/videos/ai-lab.mp4
Title (`label`) — Video	Text	No (Recommended)	No	Empty → "Video"	Accessible name; Play label	Describe the video	None	TCS AI lab tour
Caption (`caption`) — Video	Rich text	No	No	Empty	Text below the video	One or two sentences	None	Inside the TCS AI lab in Pune.
Transcript (`transcript`) — Video	AEM content (page / document)	No	No	Empty	Link to a transcript	Page or PDF	None	/content/tcs-corp/us/en/ai-lab-transcript
Poster Image (`placeholder_image`) — Video	Reference (image)	No (Recommended)	No	Empty	Image before playback	Same ratio as the video	None	/content/dam/tcs-corp/videos/ai-lab-poster.jpg
Poster Image Alt Text (`placeholder_imageAlt`) — Video	Text	No	No	Empty	Alt text of the poster	Describe, or leave empty if decorative	None	Engineers at a robotics bench
Playback (`playback`) — Playback	Select	No	No	Play button (`click`)	How the video starts	Play button for most videos	Enforced: option list	background
Muted (`muted`) — Playback	Boolean	No	No	Off	Start without sound	Always on for autoplay / background	—	off
Loop (`loop`) — Playback	Boolean	No	No	Off	Repeat	Short clips only	—	off
Show Player Controls (`controls`) — Playback	Boolean	No	No	On	Player controls	Keep on except for de-	—	on

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
						sign-led autoplay		
Start At (seconds) (`startTime`) — Playback	Number	No	No	Empty (0)	Start position	Whole seconds	Model: number; code: negative / invalid → 0	30
End At (seconds) (`endTime`) — Playback	Number	No	No	Empty (to the end)	End position	After Start; not for Vimeo	Model: number; code: negative / invalid → 0 (no end)	90
Play in a Pop-up (`openInModal`) — Playback	Boolean	No	No	Off	Play in a dialog	With Play button playback	—	on
Pause When Out of View (`pauseOffscreen`) — Playback	Boolean	No	No	On	Pause autoplay / background files off screen	Keep on	—	on
Captions File (WebVTT) (`captions`) — Captions	AEM content (.vtt)	No (Recommended for files with speech)	No	Empty	Subtitles for files	`.vtt` from AEM Assets	None	/content/dam/tcs-corp/videos/ai-lab-en.vtt
Captions Language (`captionsLanguage`) — Captions	Text	No	No	Empty → page language	`srclang` of the track	ISO code (en, de, ja)	None	en
Captions Name (`captionsLabel`) — Captions	Text	No	No	Empty → the language code	Name in the captions menu	Language name in that language	None	English
Aspect Ratio (`aspectRatio`) — Display	Select	No	No	16:9	Shape of the media area	Match the video	Enforced: option list	9-16
Size (`size`) — Display	Select	No	No	Large (900px)	Maximum width	—	Enforced: option list	medium
Alignment (`alignment`) — Display	Select	No	No	Center	Horizontal position	For sizes other than Full width	Enforced: option list	left
Rounded Corners (`rounded`) — Display	Boolean	No	No	On	1rem radius	—	—	on
Privacy-enhanced Mode (`privacyMode`) — Privacy	Boolean	No	No	On	youtube-nocookie / Vimeo dnt	Keep on	—	on
Anchor ID (`classes_anchor`) — Additional	Text	No	No	Empty	`#anchor-…` target	Start with anchor-	Enforced: regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-lab-tour

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Hide block (`classes_hideblock`) — Additional	Boolean	No	No	Off	Hide without deleting	—	—	off
Padding left (`classes_paddingleft`) — Additional	Select	No	No	Default	Inner space left	Tokens	Enforced: option list	Default
Padding right (`classes_paddingright`) — Additional	Select	No	No	Default	Inner space right	Tokens	Enforced: option list	Default
Padding top (`classes_paddingtop`) — Additional	Select	No	No	Default	Inner space top	Tokens	Enforced: option list	Default
Padding bottom (`classes_paddingbottom`) — Additional	Select	No	No	Default	Inner space bottom	Tokens	Enforced: option list	Default
Margin left (`classes_marginleft`) — Additional	Select	No	No	Default	Outer space left (overrides alignment margins)	Avoid; use Alignment	Enforced: option list	Default
Margin right (`classes_marginright`) — Additional	Select	No	No	Default	Outer space right (overrides alignment margins)	Avoid; use Alignment	Enforced: option list	Default
Margin top (`classes_margintop`) — Additional	Select	No	No	Default	Outer space top	Tokens	Enforced: option list	2rem (space-800)
Margin bottom (`classes_marginbottom`) — Additional	Select	No	No	Default	Outer space bottom	Tokens	Enforced: option list	2rem (space-800)

Field details

- **Video (`uri`)** — Purpose: the video source. Allowed Values: an AEM Assets video (MP4 / WebM) or a YouTube / Vimeo URL. Character Limits: n/a. Business Rules: YouTube if the URL contains `youtube.com` / `youtu.be`; Vimeo if it contains `vimeo.com`; otherwise a file. Dependency Rules: decides which other fields apply (captions only for files; End At not for Vimeo). Visibility: always; `required: true` (Enforced). Sample: `https://youtu.be/VIDEO_ID`.
- **Title (`label`)** — Purpose: accessible name (`aria-label` on `<video>`, `title` on iframes) and the Play button label ("Play video: {Title}"). Not shown on the page. Character Limits: none enforced; Recommended ≤ 80. Business Rules: empty → placeholder `video` ("Video") and Play label "Play video". Sample: "TCS AI lab tour".
- **Caption (`caption`)** — Purpose: visible text below the video. Allowed Values: rich text. Character Limits: none; Recommended ≤ 200. Business Rules: rendered only if it has text; keeps editor instrumentation. Sample: "Inside the TCS AI lab."
- **Transcript (`transcript`)** — Purpose: text alternative. Allowed Values: page or document link. Business Rules: link text = the authored text unless it starts with `http(s):` or `/`, then "Read the transcript" (placeholder `readTheTranscript`). Sample: a transcript page.
- **Poster Image (`placeholder_image`)** — Purpose: image before playback; for files also the `<video poster>`. Business Rules: with Play button playback it shows the Play button; with autoplay / background it shows (without button) until the video can play; empty alt text marks it decorative. Sample: a 1920 × 1080 still.

- **Poster Image Alt Text (`placeholder_imageAlt`)** — Purpose: alt text. Character Limits: Recommended ≤ 125. Business Rules: applied by the image decorator (`pictureOf`); empty → `alt=""`.
- **Playback (`playback`)** — Allowed Values: `click`, `autoplay`, `background`. Business Rules: reduced motion forces `click`; also read from block classes (legacy). Default click.
- **Muted** — Boolean, default off; forced on for autoplay and background.
- **Loop** — Boolean, default off; forced on for background.
- **Show Player Controls** — Boolean, default on; forced off for background; off with autoplay adds the pause / stop toggle.
- **Start At / End At** — Numbers (seconds), parsed with `parseFloat`, minimum 0; YouTube uses whole seconds; Vimeo start only; files as media fragment. No validation that End > Start.
- **Play in a Pop-up** — Boolean, default off; only with Play button playback.
- **Pause When Out of View** — Boolean, default on; only video files in autoplay / background.
- **Captions File / Language / Name** — files only; language default = page language (from the locale: `en`, `de`, `fr`, `es`, `pt`, `nl`, `sv`, `da`, `no`, `zh`, `ja`, `ko`); name default = the language code; one default captions track.
- **Aspect Ratio** — `16-9`, `4-3`, `1-1`, `21-9`, `9-16`; unknown → 16:9; also read from block classes (legacy).
- **Size** — `full`, `large`, `medium`, `small`; unknown → large.
- **Alignment** — `center`, `left`, `right`; unknown → center; help text “For sizes other than full width”.
- **Rounded Corners** — Boolean, default on.
- **Privacy-enhanced Mode** — Boolean, default on.
- **Additional tab fields** — Anchor ID (regExp, de-duplicated), Hide block, Padding / Margin (`--space-*` tokens). Margin left / right replace the alignment margins of the block.

6.28.7 Field Annotation Mapping

Marker	Field	Purpose
A	Video block (all tabs)	Selection and configuration
B	Poster Image + Poster Image Alt Text	Image before playback
C	(generated) Play button; Title	Starts playback; label “Play video: {Title}”
D	Video (`uri`); Display tab; Playback tab	The player, its shape and behaviour
E	Caption	Text below the video
F	Transcript	Transcript link
G	(generated) pause / stop toggle; Playback + Show Player Controls	WCAG 2.2.2 control

6.28.8 Component Specification

Attribute	Value
Component Name	Video (`blocks/video`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	Media, icon button, button, content and image decorators; properties, links, media, environment, viewport, site and placeholders utils; Modal block (on demand); YouTube / Vimeo players
Component Type	Block (key-value)

Attribute	Value
Supported Variants	File / YouTube / Vimeo; Play button (with / without poster), pop-up, autoplay (with / without controls), background; muted, loop, start / end, pause off screen; captions; caption and transcript; 5 aspect ratios; 4 sizes; 3 alignments; rounded / square; privacy on / off
Supported Templates	Any page, any section
Supported Metadata	None (page language used as the captions default)
Supported Tags	None
Supported Personalization	None; `cmp:media` / `cmp:click` data layer events available for future Adobe Analytics / Target
Caching Considerations	Page HTML and assets cached by the Edge Delivery CDN; video files served from AEM Assets delivery path; placeholders loaded once per page
Performance Considerations	No player until in view or until Play; `preload="metadata"` for click-to-play files; reserved aspect-ratio area (no CLS); Modal code loaded only on demand; a poster in the first section may be the LCP image; background / autoplay files use `preload="auto"` (bandwidth)
Accessibility Compliance	Accessible names (Title); native button controls with labels; focus management; WCAG 2.2.2 pause / stop toggle; background videos hidden from assistive technology; captions track; transcript link; reduced motion respected; no download control
SEO Compliance	Caption and transcript are indexable; no VideoObject structured data (Not implemented as of this handover)

6.28.9 Input Model

Content Inputs: Video URL (asset path or link), Title, Caption, Transcript, Poster Image + alt, Captions file, language and name.

Metadata Inputs: None directly; the locale (from the URL) provides the default captions language.

Tags Inputs: None.

Page Context Inputs: Viewport visibility; `prefers-reduced-motion`; page locale.

Query Inputs: None.

Manual Inputs: Playback, Display and Privacy settings; Additional tab classes.

Authored HTML received from AEM (example, key-value rows, abridged):

```
<div class="video">
  <div><div>uri</div>
    <div><a href="/content/dam/tcs-corp/videos/ai-lab.mp4">ai-lab.mp4</a></div></div>
  <div><div>label</div><div>TCS AI lab tour</div></div>
  <div><div>placeholder_image</div>
    <div><picture><img src="./media_9f.jpg" alt="Lab"></picture></div></div>
  <div><div>playback</div><div>click</div></div>
  <div><div>aspectRatio</div><div>16-9</div></div>
  <div><div>size</div><div>large</div></div>
</div>
```

Keys are matched case- and separator-insensitively. Blocks authored with the earlier model (a link and a picture) or delivered as one row per field are also read (Variation 33).

6.28.10 Output Model

Rendered Output: The block gets `video-size-{size}`, `video-align-{alignment}`, optionally `video-rounded` and `placeholder`, the custom property `--video-aspect-ratio` and `data-embed-loaded` (`false` → `true` when the player can play). Content: `div.video-media` (poster `div.video-placeholder` with `button.video-placeholder-`

`play`, the player `<video>` or `div.video-frame > iframe`, and optionally `button.video-toggle`), then `div.video-caption` and `<p><a class="video-transcript"></a></p>`.

Generated HTML (file, Play button, after Play):

```
<div class="video block video-size-large video-align-center video-rounded placeholder"
  style="--video-aspect-ratio: 16 / 9;" data-embed-loaded="true">
  <div class="video-media">
    <div class="video-placeholder"><picture>…</picture>
      <button class="video-placeholder-play button primary icon-button"
        aria-label="Play video: TCS AI lab tour" title="Play video: TCS AI lab tour"></
button>
    </div>
    <video controls controlslist="nodownload" playsinline preload="auto" autoplay
      aria-label="TCS AI lab tour" poster="…/media_9f.jpg">
      <source src="/content/dam/tcs-corp/videos/ai-lab.mp4" type="video/mp4">
      <track kind="captions" src="…/ai-lab-en.vtt" srclang="en" label="English" default>
    </video>
  </div>
  <div class="video-caption">Inside the TCS AI lab.</div>
  <p><a class="video-transcript" href="/en/ai-lab-transcript">Read the transcript</a></p>
</div>
```

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking:

- Component state (generic): `{ "component": { "video-1": { "@type": "tcs-corp/components/video", "parentId": "page-…" } } }` (id = Anchor ID if set).
- Play button: `cmp:click` with `interaction` `button` and `dc:title` “Play video: …”.
- Pause / stop toggle: `cmp:click` with `interaction` `playback`.
- Video files: `cmp:media` on play, pause and ended (not for muted, looping videos without controls, i.e. background videos):

```
{ "event": "cmp:media", "eventInfo": { "path": "component.video-1",
  "interaction": "play", "xdm:mediaURL": "https://…/ai-lab.mp4", "position": 0 } }
```

- Pop-up: `cmp:show` (`interaction` `dialog`) when opened and `cmp:hide` when closed, on the modal component; media events inside the dialog are attributed to the modal block.
- YouTube / Vimeo: player interactions inside the iframe are not tracked.

Personalization Output: None.

Search Index Output: None (the query index indexes page metadata only).

6.28.11 Business Rules

1. No Video (`uri`, or a link for legacy blocks) → the block is emptied and nothing renders.
2. Source detection: YouTube (`youtube.com` / `youtu.be`), Vimeo (`vimeo.com`), else a file with MIME type `video/{extension}`.
3. Title empty → “Video” (placeholder `video`); Play label “Play video” without a Title, else “Play video: {Title}” (`playVideoTitle`).
4. Reduced motion → Playback treated as Play button.
5. Play button with a poster or the pop-up → poster with Play button; click builds the player with autoplay (sound unless Muted) or opens the pop-up.

6. Play button without poster and without pop-up → the player loads when visible, without autoplay.
7. Autoplay / background → the poster (if any) without button until the video can play; the player loads when visible, always muted, autoplaying.
8. Background → loop, muted, no controls, `aria-hidden`, iframe not focusable, toggle always.
9. Autoplay with controls off → toggle; with controls on → no toggle.
10. Toggle: files pause / play; YouTube / Vimeo frames are removed (stop) and rebuilt (play, restarting at Start At); labels `pauseVideo`, `stopVideo`, `playVideo`.
11. Pause When Out of View (default on) only for files in autoplay / background; resumes only if paused by scrolling.
12. Start / End: `max(0, parseFloat(value))`; YouTube whole seconds (`start`, `end`); Vimeo start only (`#t=Ns`); files media fragment `#t=start,end`.
13. YouTube Loop needs `playlist={id}` (added automatically).
14. Privacy (default on): YouTube `youtube-nocookie.com`; Vimeo `dnt=1`.
15. Captions only for files; one default track; language default from the page locale; label default = language.
16. Transcript text replaced by "Read the transcript" when it looks like a URL / path.
17. Caption rendered only if it has text.
18. Poster stays visible until the player can play (`canplay` for files, iframe `load`), then is hidden; until then the player is hidden.
19. After Play, focus moves into the player.
20. Pop-up: Modal loaded on demand; closing removes the dialog (and the video); focus returns.
21. Files: no download control (`controlslist="nodownload"`), context menu suppressed, `playsinline`, `preload metadata` / `auto`.
22. Aspect ratio unknown → 16:9; size unknown → large; alignment unknown → center.
23. Legacy model: no `uri` property → first link and picture; `autoplay` class → background.
24. Additional tab rules as for every block; Margin left / right override Alignment.

6.28.12 Author Best Practices

Recommended Content Length: Title ≤ 80 characters; Caption ≤ 200 characters; poster alt text ≤ 125 characters; background videos Recommended ≤ 30 seconds and loop seamlessly.

Image Dimensions: Poster in the same aspect ratio as the chosen Aspect Ratio (it is cropped with `object-fit: cover`): 16:9 e.g. 1920 × 1080; 4:3 e.g. 1600 × 1200; 1:1 e.g. 1200 × 1200; 21:9 e.g. 2520 × 1080; 9:16 e.g. 1080 × 1920. Video files: web-optimised H.264 MP4 (or WebM), resolution matched to the Size (Large 900px → 1080p is ample); keep files small ([To be confirmed: maximum asset size for delivery — owner: AEM administrator]).

CTA Writing Guidelines: The Play button is generated; give the video a clear Title so the button reads "Play video: {Title}". Transcript link text: "Read the transcript" or a specific text (not a URL).

Accessibility Guidelines: Always set a Title; provide captions (WebVTT for files, platform captions for YouTube / Vimeo); provide a transcript for informative videos; avoid autoplay with sound (not possible by design); keep controls on unless the toggle is enough; set posters for autoplay / background videos so reduced-motion visitors see a meaningful image; note the caption text colour (`--color-text-secondary`, #999 on white, about 2.8:1 contrast) is below the WCAG AA 4.5:1 minimum for small text ([To be confirmed: design review of the caption colour — owner: design]).

SEO Guidelines: Describe the video in the page text and Caption; link a transcript page; use descriptive file names for assets.

Metadata Guidelines: None at block level.

Taxonomy Guidelines: Tag the video assets in AEM Assets as per DAM governance; the block does not read tags.

Localization Guidelines: Provide a captions file per locale (Captions Language and Name in that language), localised poster images if they contain text, translated Title, Caption and transcript; UI labels come from each locale's placeholders sheet (`video`, `playVideo`, `playVideoTitle` with `{title}`, `pauseVideo`, `stopVideo`, `readTheTranscript`, `dialog`, `close`).

6.28.13 Troubleshooting Guide

Issue 1 — Nothing renders
Symptoms: The block area is empty.

Root Cause: Video (`uri`) empty or not delivered as a link / path.

Diagnosis Steps: Check the Video field; open `{page}.plain.html` and look for the `uri` row.

Resolution Steps: Select the asset or paste the URL; republish.

Validation Steps: Poster or player renders.

Preventive Measures: Video is required in the dialog; fill it first.

Issue 2 — Video file doesn't play on the live site (404)
Symptoms: Black area; Network panel shows 404 for the MP4.

Root Cause: The asset isn't published, or was moved / renamed after authoring.

Diagnosis Steps: Open the file URL directly; check its publication status in AEM Assets.

Resolution Steps: Publish the asset (Manage Publication with references); re-select moved assets.

Validation Steps: The URL returns the video; playback works.

Preventive Measures: Publish pages with their references.

Issue 3 — Autoplay video doesn't start
Symptoms: Poster stays; no motion.

Root Cause: The visitor prefers reduced motion (by design → Play button), the block isn't in view yet, or the browser blocked autoplay (power saving).

Diagnosis Steps: Check the OS reduced-motion setting; scroll the block into view; check the console for play() promise rejections.

Resolution Steps: Expected for reduced motion; ensure the poster is meaningful.

Validation Steps: Plays with reduced motion off.

Preventive Measures: Never rely on autoplay for essential information.

Issue 4 — The video is cropped
Symptoms: Edges of the picture are cut off.

Root Cause: Aspect Ratio doesn't match the file (files use `object-fit: cover`).

Diagnosis Steps: Compare the file's dimensions with the Aspect Ratio.

Resolution Steps: Set the matching Aspect Ratio.

Validation Steps: The full frame is visible.

Preventive Measures: Check the ratio when authoring.

Issue 5 — Captions don't show
Symptoms: No CC option in the native player.

Root Cause: The source is YouTube / Vimeo (captions file ignored), the VTT isn't published, or the file isn't valid WebVTT.

Diagnosis Steps: Check the source type; open the `.vtt` URL; validate it starts with `WEBVTT`.

Resolution Steps: Upload captions to YouTube / Vimeo, or publish / fix the VTT.

Validation Steps: Captions appear and are labelled correctly.

Preventive Measures: Test captions in preview.

Issue 6 — End At doesn't stop a Vimeo video

Symptoms: The Vimeo video plays to the end.

Root Cause: End At is not applied to Vimeo (only YouTube and files).

Diagnosis Steps: Check the source.

Resolution Steps: Edit the video on Vimeo, or host the excerpt as a file.

Validation Steps: Playback stops at the intended time.

Preventive Measures: Use End At only with YouTube or files.

Issue 7 — Pop-up doesn't open

Symptoms: Play does nothing, or the dialog is empty.

Root Cause: Playback isn't Play button (pop-up ignored for autoplay / background), the Modal code failed to load, or the source URL is wrong.

Diagnosis Steps: Check Playback; console / Network for `blocks/modal/modal.js` and `modal.css`.

Resolution Steps: Set Playback to Play button; escalate loading errors to development.

Validation Steps: Dialog opens and plays; Esc closes it.

Preventive Measures: Test pop-ups in preview.

Issue 8 — Vimeo / YouTube shows an error

Symptoms: "Video unavailable" / "does not exist".

Root Cause: Unsupported URL shape (Vimeo id must be the first path segment), private video, embedding disabled.

Diagnosis Steps: Inspect the iframe `src`.

Resolution Steps: Use `vimeo.com/{id}` / a YouTube watch or share URL; allow embedding.

Validation Steps: The player loads.

Preventive Measures: Use canonical URLs.

6.28.14 Support Runbook

Monitoring Requirements

- Asset availability (video, poster, VTT) on the live host; Core Web Vitals (LCP of posters, data usage of autoplay / background videos).
- Once Adobe Tags is implemented, `cmp:media` events can feed video reporting.

Logs

- Browser console: the block logs nothing itself; autoplay rejections are caught silently; errors may come from the Modal import or third-party players.

- Network panel: video file, VTT, poster requests; youtube-nocookie.com / player.vimeo.com.
- Admin API / Sidekick: page and asset status; AEM author: publication logs, asset processing.

Common Errors

Error	Likely cause	Fix
Empty block	No Video	Fill Video
404 on MP4 / VTT	Asset not published	Publish references
No autoplay	Reduced motion / browser policy	Expected; set a poster
Cropped video	Wrong Aspect Ratio	Match the ratio
No captions	YouTube / Vimeo source or bad VTT	Platform captions / fix VTT
Vimeo end ignored	Not supported	Edit on Vimeo / use file
Pop-up not opening	Playback not Play button	Set Play button

Escalation Matrix

Level	Who	Checks
L1	Author support	Video, Title, poster, playback settings, aspect ratio, captions, publication of page and assets, Hide block
L2	AMS / site admin	Asset delivery (`/content/dam/tcs-corp/` in `paths.json` includes), CDN cache, file sizes and formats, CSP for third-party players
L3	Development team	`video.js`, media decorator / utils, Modal block, console errors
L4	Adobe Support	AEM Assets delivery, Edge Delivery, Universal Editor issues

Recovery Procedures

1. Restore the previous page version; republish with references.
2. Restore the previous asset version in AEM Assets and republish it.
3. Hide the block (Hide block) during an outage of a provider.
4. Revert code changes on `main` if a regression is identified.

Health Checks

- Play one file, one YouTube and one Vimeo video on a test page (desktop and mobile).
- Check the pause toggle and `aria-hidden` on a background video.
- Check the pop-up opens, plays and closes with focus returned.
- With reduced motion on, check autoplay videos show the Play button.

Operational Procedures

- Maintain the video placeholders (`video`, `playVideo`, `playVideoTitle`, `pauseVideo`, `stopVideo`, `readTheTranscript`) and modal placeholders (`dialog`, `close`) in each locale's placeholders sheet.
- Follow DAM governance for video renditions and captions files.

Known Limitations

- One captions track per video; no audio description track.
- End At not supported for Vimeo; Vimeo URLs must be `vimeo.com/{id}`.
- The YouTube / Vimeo stop toggle restarts the video when played again.
- No VideoObject structured data, no chapters, no playlists.

- No consent management for third-party players ([To be confirmed: consent requirements — owner: privacy / legal]).
- Files are cropped to the chosen aspect ratio (`object-fit: cover`).
- Caption text colour contrast below AA (see 6.28.12).
- Interactions inside YouTube / Vimeo players aren't tracked.

6.29 Form Variations Block

The Form Variations block builds an accessible, validated web form from child field items that authors add in Universal Editor: text inputs (Text, Email, Tel), dropdowns, a textarea with a character counter, a consent checkbox, contact cards and one or more submit buttons. Fields, contact cards and buttons can be made conditional ("Show only if dropdown value equals"), so one form can route visitors, for example by region, to the right contact. The block lives in `blocks/form-variations/` (`form-variations.js`, `form-variations.css`, `_form-variations.json`).

> **Important:** there is **no submit endpoint**. A valid submit dispatches a bubbling `form-variations:submit` DOM event on the block with the form data, resets the form and shows the success message "Thank you, your details were sent." The data is **not** sent anywhere unless a script on the page listens for that event and forwards it. Any connection to Salesforce, Marketo, an email service or an API is Not implemented (as of this handover). [To be confirmed: form back end — owner: product owner / solution architect]

6.29.1 Component Purpose

Business Objective: Let authors create contact, enquiry and feedback forms without developer involvement, with consistent TCS styling, built-in client-side validation, WCAG 2.1 AA accessibility (labels, inline errors, live status message, input purpose) and conditional content that routes visitors to the right regional contact.

Business Use Cases:

- "Contact us" / "Talk to an expert" enquiry forms (name, email, company, message, consent).
- Regional enquiry routing: a Region dropdown that reveals the matching regional contact card (name, emails, phone numbers) and, optionally, region-specific fields or submit buttons.
- Sector- or topic-specific questions: a dropdown that reveals extra fields only for one value (for example "Other" reveals a free-text field).
- Feedback forms (textarea with a character limit and live counter).
- Newsletter or event-interest capture (email plus a mandatory consent checkbox) — subject to the back end being connected.

When To Use:

- A form with up to a few dozen simple fields (text, email, phone, dropdown, long text, single consent checkbox).
- When contact details must change with a visitor's choice (contact cards).
- When the data is handled by a script listening for `form-variations:submit` (once a back end integration exists).

When Not To Use:

- When the data must actually reach a system today: Not implemented (as of this handover); use a plain email link or contact card until the back end is connected.
- File uploads, date pickers, radio-button groups, multi-select lists, number fields, multiple checkboxes as a group, multi-step (wizard) forms, CAPTCHA, server-side validation: Not implemented (as of this handover).
- For site search or "Ask us a question" experiences (use the Conversation Canvas, 6.30).

- For content that needs no input (use Teaser, Cards or default content).

Dependencies:

Type	Dependency	Use
Code	`scripts/scripts.js moveInstrumentation`	Keeps Universal Editor instrumentation on the rendered fields, labels, options and buttons
Code	`decorators/button-decorator.js decorateButton`	Styles the submit button as a primary button (view `button`, type `primary`)
Code	`decorators/content-decorator.js fromCell`	Moves the rich-text Form Description into `.form-variations-description`
Code	`decorators/heading-decorator.js buildHeading`	Builds the form title as an h2 (or the heading level found in the cell)
Code	`utils/environment.js isInEditor`	Detects Universal Editor (edit and preview mode): conditional items stay visible
Code	`utils/properties.js textOf`	Reads trimmed cell text
Code	`utils/ids.js createIdFactory, uniqueId`	Unique ids per form instance (`form-variations-1-firstname`, ...)
Code	`utils/placeholders.js getPlaceholders, fillPlaceholder`	UI text (errors, success, required note, select prompt) with English fallbacks
Code	`decorators/block-options-decorator.js`	Additional tab: Anchor ID becomes the block id; Hide block, padding and margin classes
Code	`decorators/data-layer-decorator.js`	Component state, `cmp:click` on the submit button, `cmp:submit` with field names
Styles	`styles/styles.css`	Design tokens, primary button styles, `.hideblock`, spacing classes
Data	Locale placeholders sheet / `{locale}/placeholders.json`	Keys `fieldRequired`, `fieldEmail`, `fieldFormat`, `fieldTooLong`, `selectAnOption`, `send`, `shownWhenValue`, `requiredNote`, `formSuccess`
Data	None other	The block reads no data sheet, query index, metadata or taxonomy
Back end	None	No endpoint; `form-variations:submit` event only

Related Components: Modal (can show a page that contains a form in a pop-up), Fragment (reuse one form on many pages by placing it on a reference page), Conversation Canvas (6.30, the search / question entry point, not a data-capture form), Page Header and Teaser (CTAs that link to the page or anchor holding the form; use the form's Anchor ID).

Applicable Templates: Any page. The block is in the section filter (`models/_section.json`), so it can be added to any section of home, article, blog, press-release or generic pages. No template adds it automatically.

Supported Content Types: Contact and enquiry pages, landing pages, campaign pages, event pages, career or partner interest pages. The block does not read the page's `content-type` metadata; it works the same on every content type.

6.29.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-desktop-view.png`

> **Purpose:** Desktop rendering
> **Description:** Form Variations at 1280 px wide, Default theme: title "Contact us", description, "* Required fields" note, First name and Last name side by side, full-width Email, Region dropdown, Message textarea with "0/1500" counter, consent checkbox, black primary "Send" button with arrow.

Desktop (>= 900 px, including >= 1200 px): The block is centred with a maximum width of 45rem (720 px) in the Default theme. A header (title, `h2`, and the description in the muted text colour) sits above the form with 2rem space below it. The form is a two-column CSS grid (gap 1.5rem vertical, 1rem horizontal). Text input fields whose **Field Name / ID** contains "name" (`half-width`) take one column, so "firstName" and "lastName" sit side by side; all other items (`full-width`) span both columns. Labels use the label font size, medium weight, secondary neutral colour; a red asterisk follows mandatory labels. Inputs are full width, 0.75rem/1rem padding, 1 px tertiary border, 0.5rem radius; focus shows a 2 px brand outline. The textarea is at least 7.5rem tall and resizable vertically, with a right-aligned "n/limit" counter. The consent checkbox (1.25rem) sits left of its small consent text. Contact cards have a top divider, the contact name in the brand colour (heading size M) and a vertical list of mail and phone links. The submit button is the site primary button (black, white text, trailing arrow), left-aligned. The status message appears below the form in the brand colour after a successful submit.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** Form Variations at 768 px wide, Default theme: every field stacked in one column, including First name and Last name, counter under the textarea, button left-aligned.

Tablet (600–899 px): One column: every item, including half-width name fields, is full width. The block is still limited to 45rem and centred. In the Dark theme the panel padding grows to 3rem top/bottom and 2.5rem left/right. Between 900 and 1199 px the desktop two-column grid already applies (the form's only breakpoints are 600 px and 900 px).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** Form Variations at 375 px wide: single column, full-width inputs and dropdown, consent checkbox with wrapped text, primary Send button, success message under the form after submitting.

Mobile (< 600 px): One column (`grid-template-columns: minmax(0, 1fr)`), full-width controls, the same font sizes as on tablet. In the Dark theme the panel has 2.5rem top/bottom and 1.5rem left/right padding with a 1rem radius.

6.29.3 Component Variations

The block has one author-selectable style option (**Theme**: Default / Dark). The other variations below are layout and behaviour modes produced by the item types and their settings. Each is described separately.

Variation 1 — Default theme

Variation Name: Default (Theme = Default, no class).

Business Purpose: The standard light form for content pages, sitting on the page background.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-default-theme.png`
> **Purpose:** Default theme
> **Description:** Light form on a white section: dark text, grey-bordered inputs, black primary Send button, brand-coloured status message.

Use Cases: Contact pages, enquiry sections in articles and landing pages.

Authoring Guidelines: In the block's **Content** tab leave **Theme** on **Default**.

Business Rules: Default is the model default (`value: ""`); no class is added.

Rendering Rules: Block max width 45rem, centred (`margin-inline: auto`); text colour `--color-text-default`; inputs on `--color-bg-default` with `--color-border-tertiary` borders; error colour `#c8102e` (hard-coded, there is no error colour token yet).

Responsive Behaviour: Desktop (>= 900 px): two-column grid with half-width name fields. Tablet (600–899 px): one column. Mobile (< 600 px): one column.

SEO Impact: The form title is an `h2`, contributing to the page outline; form controls are not indexed content. No structured data is generated.

Accessibility Compliance: Labels linked with `for`/`id`; required marked by the native `required` attribute (the visual asterisk is `aria-hidden` and explained once by "* Required fields"); the red error colour (`#c8102e` on white) is used together with error text, never colour alone (WCAG 1.4.1).

Variation 2 — Dark theme

Variation Name: Dark (Theme = Dark, class `dark`).

Business Purpose: A high-contrast, framed form panel that stands out on a light page, for campaign or conversion sections.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-dark-theme.png`
> **Purpose:** Dark theme
> **Description:** Black rounded panel spanning the section width, white text, transparent inputs with light borders, white primary Send button, contact name in the light brand colour.

Use Cases: "Get in touch" end-of-page panels, campaign landing pages, event registration interest sections.

Authoring Guidelines: In the **Content** tab set **Theme** to **Dark**.

Business Rules: Class `dark` on the block. The block's width limit is removed (`max-width: none`) and moved to its children (each direct child max 45rem, centred).

Rendering Rules: Background `--color-bg-neutral` (black), text `--color-text-on-neutral`, 1rem radius (`--radius-l`). Labels and muted text use `--color-text-light-overlay`; inputs have a transparent background with `--color-border-neutral-secondary` borders; dropdown options use a `--grey-900` background. Error colour `#ff6b6b`. The primary button is inverted (white background, dark text; hover `--color-bg-secondary-hover`). Contact name and status message use `--color-bg-brand-secondary`.

Responsive Behaviour: Desktop (>= 900 px): panel padding 4rem on all sides, two-column grid. Tablet (600–899 px): padding 3rem / 2.5rem, one column. Mobile (< 600 px): padding 2.5rem / 1.5rem, one column.

SEO Impact: None beyond Variation 1.

Accessibility Compliance: Same structure as Default. Contrast of the light overlay text and the `#ff6b6b` error colour on black should be verified with the final design tokens during the accessibility audit ([To be confirmed: contrast audit of the Dark theme — owner: accessibility lead]).

Variation 3 — Two-column layout with half-width name fields

Variation Name: Half-width fields (automatic, class `half-width`).

Business Purpose: Shorter, more scannable forms on desktop by placing First name and Last name side by side, matching the TCS design.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-half-width.png`
> **Purpose:** Half-width name fields
> **Description:** Desktop form where "First name" and "Last name" share one row and Email spans the full width below.

Use Cases: First / last name pairs; any pair of short text inputs.

Authoring Guidelines: Add two **Text Input Field** items whose **Field Name / ID** contains the letters "name" (any case), for example `firstName` and `lastName`, one after the other. There is no explicit width field.

Business Rules: Enforced by code: a Text Input Field whose name includes "name" (case-insensitive) gets `half-width`; every other item gets `full-width`. This also applies to names such as `companyName` or `username`, which therefore become half width too. Dropdowns, textareas, checkboxes, contact cards and buttons are always full width.

Rendering Rules: At >= 900 px `.form-variations-group.full-width` spans `1 / -1`; half-width groups take one of the two grid columns in source order.

Responsive Behaviour: Desktop (>= 900 px): side by side. Tablet and mobile: stacked, full width.

SEO Impact: None.

Accessibility Compliance: Reading and tab order follow the source order (left, then right), so the visual order and the focus order match (WCAG 1.3.2, 2.4.3).

> **Note:** when the form has any mandatory field, the "* Required fields" note is the first grid item and, at >= 900 px, occupies one column. A half-width field that directly follows it therefore sits next to the note, and its partner field moves to the next row (checked with the repository CSS at 1280 px). Start the form with a full-width item (for example Email or a dropdown) or ask the development team to make the note span both columns.

Variation 4 — Single-column (full-width) layout

Variation Name: Full-width fields (automatic, class `full-width`).

Business Purpose: A simple single-column form, easiest to complete on every device.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-full-width.png`
> **Purpose:** Single-column form
> **Description:** Desktop form where every field (Full name, Email, Phone, Message) spans the full 45rem width.

Use Cases: Short forms, feedback forms, forms with a single "fullName" field.

Authoring Guidelines: Use field names that do not contain "name" (for example `email`, `phone`, `message`), or a single text field such as `fullname` placed where it can stand alone.

Business Rules: Every non-"name" item is `full-width`. A lone half-width field still takes only one column on desktop, leaving the other column empty.

Rendering Rules: Grid item spanning both columns at >= 900 px.

Responsive Behaviour: Full width at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Single-column forms are the most robust for zoom (WCAG 1.4.10 reflow) and screen magnifiers.

Variation 5 — Text input, input type Text

Variation Name: Text Input Field, Input Type = Text (`type="text"`).

Business Purpose: Free short text such as names, company, job title, city.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-input-text.png`
> **Purpose:** Text input
> **Description:** "Company" label with asterisk above an empty single-line input.

Use Cases: First name, last name, company, job title, country, city, postal code.

Authoring Guidelines: Add **Text Input Field**; set **Field Name / ID** (for example `company`), **Field Label** ("Company"), **Input Type** = Text, and **Mandatory Field** as needed.

Business Rules: Input Type defaults to Text (template `inputKind: text`). Any unknown value also falls back to `text`. The browser autocomplete hint is derived from the name (see 6.29.11).

Rendering Rules: `<input class="form-variations-control" type="text">` inside a `.form-variations-group.field-text` with a `<label>`; `autocomplete` set when the name matches first/given, last/surname/family, name/full name, company/organisation, job/designation/ position, country, city/town, zip/postal/post code.

Responsive Behaviour: Half or full width on desktop (Variation 3/4); full width on tablet and mobile.

SEO Impact: None.

Accessibility Compliance: Visible label linked to the input; `autocomplete` identifies the input purpose (WCAG 1.3.5); required state exposed by the native attribute.

Variation 6 — Text input, input type Email

Variation Name: Text Input Field, Input Type = Email (`type="email"`).

Business Purpose: Collect a valid email address with browser and code validation.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-input-email.png`
> **Purpose:** Email input with error
> **Description:** "Business email" input with a red border and the inline error "Enter a valid email address." below it.

Use Cases: Business email, contact email.

Authoring Guidelines: Set **Input Type** = Email and a name such as `email`.

Business Rules: Enforced on submit: an invalid address shows "Enter a valid email address." (placeholder `fieldEmail`); empty and mandatory shows "This field is required." (`fieldRequired`). Autocomplete is always `email`. The format check is the browser's standard email check (it does not verify that the mailbox exists or block free-mail domains).

Rendering Rules: `<input type="email" autocomplete="email">`; mobile keyboards show the email layout.

Responsive Behaviour: As Variation 5.

SEO Impact: None.

Accessibility Compliance: Error text is linked with `aria-describedby` and the field gets `aria-invalid="true"` (WCAG 3.3.1); focus moves to the first invalid field.

Variation 7 — Text input, input type Tel

Variation Name: Text Input Field, Input Type = Tel (`type="tel"`).

Business Purpose: Collect a phone number with the telephone keypad on mobile devices.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-input-tel.png`
> **Purpose:** Phone input
> **Description:** "Phone number" input on a mobile screen with the numeric telephone keypad open.

Use Cases: Phone or mobile number.

Authoring Guidelines: Set **Input Type** = Tel and a name such as `phone`.

Business Rules: No format or length validation is applied to phone numbers (browsers do not validate `type="tel"`, and the block sets no pattern); only Mandatory is enforced. Autocomplete is `tel`.

Rendering Rules: `<input type="tel" autocomplete="tel">`.

Responsive Behaviour: As Variation 5.

SEO Impact: None.

Accessibility Compliance: Same as Variation 5.

Variation 8 — Dropdown / Select field

Variation Name: Dropdown / Select Field (`field-select`).

Business Purpose: Let the visitor choose one value from a fixed list; the chosen value also drives conditional fields and contact cards.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-dropdown.png`
> **Purpose:** Dropdown field
> **Description:** "Region" dropdown open, showing the disabled prompt "Select an option" and the options "Asia Pacific", "Europe", "India", "North America".

Use Cases: Sector, region, country, topic of enquiry, "How did you hear about us?".

Authoring Guidelines: Add **Dropdown / Select Field**; set **Dropdown Name / ID** (`region`), **Dropdown Label** ("Region"), **Options** as a comma-separated list ("Asia Pacific, Europe, India, North America"), **Mandatory Field** as needed.

Business Rules: Options are split on commas and trimmed; empty entries are dropped; an option cannot contain a comma. The option text is also the submitted value. A disabled, pre-selected first option "Select an option" (placeholder `selectAnOption`) is always added; it has an empty value, so a mandatory dropdown left on it fails with "This field is required.". The dropdown has no Show-only-if limit on how many conditions refer to it.

Rendering Rules: Native `<select class="form-variations-control">` with the browser's own arrow (`appearance: auto`), full width.

Responsive Behaviour: Full width at all breakpoints; the native picker is used on mobile.

SEO Impact: None.

Accessibility Compliance: Native select (fully keyboard and screen reader accessible), visible label; the prompt option cannot be re-selected.

Variation 9 — Textarea with character counter

Variation Name: Textarea Field (`field-textarea`).

Business Purpose: Longer free-text messages with a visible character limit.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-textarea.png`
> **Purpose:** Textarea with counter
> **Description:** "How can we help you?" textarea four rows tall with some typed text and the counter "124/1500" right-aligned below it.

Use Cases: Message, feedback, project description.

Authoring Guidelines: Add **Textarea Field**; set **Field Name / ID** (`message`), **Field Label**, **Max Character Limit** (default 1500), **Mandatory Field** as needed.

Business Rules: Enforced: the browser stops typing at the limit (`maxlength`). The limit is the authored number when it is a positive whole number, else 1500. The counter shows "{typed}/{limit}" and updates on every input and after a reset.

Rendering Rules: `<textarea rows="4" maxlength="…">`, minimum height 7.5rem, vertical resize only; counter `.form-variations-counter` (extra small, muted) linked with `aria-describedby`.

Responsive Behaviour: Full width at every breakpoint.

SEO Impact: None.

Accessibility Compliance: The counter is the textarea's description, so screen readers announce the limit; it is not a live region, so the count is not re-announced on every keystroke.

Variation 10 — Consent checkbox

Variation Name: Consent Checkbox Field (`field-checkbox`).

Business Purpose: Capture explicit consent (privacy notice, marketing opt-in) with authorable rich text including links.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-consent-checkbox.png`
> **Purpose:** Consent checkbox
> **Description:** Checkbox followed by small text "I agree to the processing of my personal data as described in the Privacy Notice" with "Privacy Notice" underlined as a link, and a red asterisk.

Use Cases: Privacy notice acknowledgement, marketing opt-in, terms acceptance.

Authoring Guidelines: Add **Consent Checkbox Field**; set **Checkbox Name / ID** (`consent`), **Consent Text / Disclaimer** (rich text, links allowed), **Mandatory Field** on for a required acknowledgement.

Business Rules: Submitted value is `"true"` when checked; an unchecked box is not included in the data at all. Mandatory and unchecked shows "This field is required.". If the consent text is empty, the name is shown as the label.

Rendering Rules: Row layout: 1.25rem checkbox (brand accent colour) then the label with the rich text kept (`.form-variations-consent`, last paragraph inline so the asterisk follows the text); links in the text are underlined and inherit the text colour.

Responsive Behaviour: Same row layout at all breakpoints; the text wraps.

SEO Impact: Links in the consent text are normal links (crawlable).

Accessibility Compliance: The whole text is the checkbox's label (clicking it toggles the box). Placing links inside a label is allowed but means a click on the link navigates rather than toggling; keep link text descriptive.

> **Note:** an inline error on the checkbox is placed in the same row, to the right of the consent text, not below it (the row has no wrapping; checked with the repository CSS). It is still announced through `aria-describedby`.

Variation 11 — Conditional fields ("Show only if dropdown value equals")

Variation Name: Conditional item (`data-depends-on`).

Business Purpose: Ask only relevant questions: show extra fields, buttons or text for one dropdown answer.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-conditional-fields.png`
> **Purpose:** Conditional field shown
> **Description:** Sector dropdown set to "Other" and the newly visible "Please specify your sector" text input below it.

Use Cases: "Other — please specify"; region-specific consent texts; a different submit button per region; extra questions for one sector.

Authoring Guidelines: On a Text, Dropdown, Textarea, Checkbox or Submit item, type the exact dropdown option text in **Show only if dropdown value equals** (for example `Other`). Leave it empty to always show the item.

Business Rules: Enforced by code: the item is visible when **any** dropdown of the same form currently has that value (compared trimmed and case-insensitively). One value per item only (a comma-separated list is treated as one value and will not match). Hidden items are disabled: they are not validated, not required and not submitted. Conditions can chain (a conditional dropdown can control further items); the code repeats the check until nothing changes. A hidden dropdown's value does not count. In Universal Editor (edit and preview mode) every conditional item stays visible with an italic note "Shown when a dropdown value is"{value}"" and a dashed border.

Rendering Rules: `hidden` attribute on the group (`display: none`); evaluated on load, on every dropdown `change` and after a successful submit (reset).

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Hidden items are removed from the accessibility tree; newly shown items appear after the dropdown in reading order. No announcement is made when items appear (Recommended: place conditional items directly after their dropdown).

Variation 12 — Contact card output (region routing)

Variation Name: Contact Card Output (`contact-card`).

Business Purpose: Show the right contact person and their email addresses and phone numbers for the visitor's selection, without submitting anything.

Visual Example:

> [**Screenshot Placeholder**]
> **File name:** `screenshots/form-variations-contact-card.png`
> **Purpose:** Contact card shown for a region
> **Description:** Region dropdown set to "India"; below it, after a thin divider line, the contact name in brand blue and two mailto links and one phone link listed vertically.

Use Cases: Regional media, sales or partner contacts; office contacts per country.

Authoring Guidelines: Add **Contact Card Output**; set **Show when dropdown value equals** (mandatory, e.g. `India`), **Contact Name**, **Emails** and **Phones** (comma-separated).

Business Rules: A contact card is always conditional (its region is its condition); with no matching dropdown value it is hidden. Emails become `mailto:` links; phones become `tel:` links with every character except digits and `+` removed from the link (the visible text is kept as typed). No email or phone format validation. Several cards can share a value (all are shown).

Rendering Rules: `.form-variations-group.form-variations-contact-card.full-width` with a top divider, `p.form-variations-contact-name` (brand colour, heading size M, medium weight), and a `ul.form-variations-contact-links` of `li.form-variations-contact-email` / `-phone` links.

Responsive Behaviour: Full width at every breakpoint.

SEO Impact: Contact names, emails and phone numbers are in the page HTML (hidden until selected); they can be crawled. Publish only business contact data approved for the web.

Accessibility Compliance: Real links (keyboard operable); the name is a paragraph, not a heading. The card has no content of its own to submit.

Variation 13 — Submit button (always shown or conditional)

Variation Name: Submit Button (`field-submit`).

Business Purpose: Submits the form; can be shown for one dropdown value only (for example a region-specific call to action).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-submit-button.png`
> **Purpose:** Submit button and success message
> **Description:** Black primary "Send" button with trailing arrow, and below the form the brand-coloured message "Thank you, your details were sent.".

Use Cases: "Send", "Submit enquiry", "Request a call back".

Authoring Guidelines: Add **Submit Button**; set **Button Label** (default "Send"); optionally **Show only if dropdown value equals**.

Business Rules: Empty label falls back to the placeholder `send` ("Send"). Several submit buttons are allowed; each submits the whole form. If every submit button is conditional and no matching value is selected, the visitor has no button to submit with. After a valid submit the form is reset, so a conditional button may disappear; focus then moves to the status message.

Rendering Rules: `<button type="submit" class="button primary">` from the button decorator (trailing arrow icon); in the Dark theme inverted to white. Left-aligned (`align-items: flex-start`).

Responsive Behaviour: Same at every breakpoint; the button keeps its content width.

SEO Impact: None.

Accessibility Compliance: Native button; the success message is a `role="status"` live region that also receives focus (`tabindex="-1"`), so it is announced (WCAG 4.1.3).

6.29.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A Form Variations block selected in Universal Editor with the content tree showing its items; all conditional items visible with dashed borders and italic condition notes; letters A–J marking the elements listed below.

Annotations:

- A = Form Title / Heading (block property, rendered as the `h2`)
- B = Form Description (rich text)
- C = "* Required fields" note (generated when any field is mandatory; not editable)
- D = Text Input Field item (label, input; half width when the name contains "name")
- E = Dropdown / Select Field item (label, options)

- F = Textarea Field item (label, textarea, counter)
- G = Consent Checkbox Field item (consent rich text)
- H = Contact Card Output item (contact name, emails, phones)
- I = Condition note on a conditional item ("Shown when a dropdown value is"India""; editor only)
- J = Submit Button item (button label)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** The properties panel of the Form Variations block with the Content and Additional tabs, and, side by side, the panel of a Text Input Field item.

Block properties panel:

- **Content** tab: **Form Title / Heading** (text), **Form Description** (rich text), **Theme** (select: Default, Dark).
- **Additional** tab: **Anchor ID** (text, validated `^anchor-[a-z0-9][a-z0-9-]*$`), **Hide block** (toggle), **Padding left / right / top / bottom** and **Margin left / right / top / bottom** (selects: Default, 0 (none), 0.25rem … 7.5rem from the spacing tokens).

Item properties panels (no tabs):

- **Text Input Field:** Field type (read-only `field-text`), Field Name / ID, *Field Label*, Input Type (Text, Email, Tel), Mandatory Field (*), Show only if dropdown value equals.
- **Dropdown / Select Field:** Field type (read-only `field-select`), Dropdown Name / ID, *Dropdown Label*, Mandatory Field (*), Options, Show only if dropdown value equals.
- **Textarea Field:** Field type (read-only `field-textarea`), Field Name / ID, *Field Label*, Max Character Limit (number, 1500), Mandatory Field (*), Show only if dropdown value equals.
- **Consent Checkbox Field:** Field type (read-only `field-checkbox`), Checkbox Name / ID, *Consent Text / Disclaimer* (rich text), Mandatory Field (*), Show only if dropdown value equals.
- **Contact Card Output:** Field type (read-only `contact-card`), Show when dropdown value equals, *Contact Name*, Emails, Phones.
- **Submit Button:** Field type (read-only `field-submit`), Button Label ("Send"), Show only if dropdown value equals.

(* = required in the dialog.) The **Add** menu of the block offers exactly these six item types (filter `form-variations`).

6.29.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites, open the locale site (for example `/content/tcs-corp/us/en`), select the page and choose **Edit** to open it in Universal Editor.

Expected Result: The page opens in the Universal Editor canvas in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** AEM Sites console with the page selected and the Edit action highlighted.

Step 2 — Add the Form Variations block

Action: Select the section that should hold the form, choose **Add** (+) and pick **Form Variations**.

Expected Result: An empty Form Variations block appears; the properties panel shows the Content tab.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** The Add menu of a section with "Form Variations" highlighted.

Step 3 — Fill the Content tab

Action: Type the **Form Title / Heading** (for example "Contact us"), add an optional **Form Description** (rich text; links allowed), and choose the **Theme** (Default or Dark).

Expected Result: The title appears as a heading and the description below it; Dark shows the black panel.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-03-content-tab.png`
> **Purpose:** Content tab
> **Description:** Content tab with Form Title "Contact us", a description and Theme = Dark.

Step 4 — Fill the Additional tab (optional)

Action: Open **Additional**. Set an **Anchor ID** such as `anchor-contact-form` if CTAs should jump to the form; adjust padding or margin only when the design requires it; leave **Hide block** off.

Expected Result: The block gets the id `anchor-contact-form` on the page (links `#anchor-contact-form` scroll to it); spacing classes are applied.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-04-additional-tab.png`
> **Purpose:** Additional tab
> **Description:** Additional tab with Anchor ID "anchor-contact-form" and default spacing.

Step 5 — Add text input fields

Action: With the block selected, choose **Add** > **Text Input Field**. Set **Field Name / ID** (`firstName`), **Field Label** ("First name"), **Input Type** (Text), **Mandatory Field** on. Repeat for `lastName`, `email` (Input Type Email), `company`, `phone` (Input Type Tel).

Expected Result: Each field appears with its label; mandatory fields show a red asterisk; "* Required fields" appears at the top of the form; first and last name sit side by side on desktop.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-05-text-fields.png`
> **Purpose:** Text fields added
> **Description:** Canvas with First name, Last name, Email and Phone fields and the Text Input Field panel open.

Step 6 — Add a dropdown

Action: Add > **Dropdown / Select Field**. Set **Dropdown Name / ID** (`region`), **Dropdown Label** ("Region"), **Options** ("Asia Pacific, Europe, India, North America"), and **Mandatory Field** as needed.

Expected Result: A dropdown with the prompt “Select an option” and the four options.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-06-dropdown.png`
> **Purpose:** Dropdown added
> **Description:** Dropdown item panel with the comma-separated Options field filled.

Step 7 — Add a textarea

Action: Add > **Textarea Field**. Set **Field Name / ID** (`message`), **Field Label** (“How can we help you?”), **Max Character Limit** (keep 1500 or set another positive number).

Expected Result: A four-row textarea with the counter “0/1500”.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-07-textarea.png`
> **Purpose:** Textarea added
> **Description:** Textarea with its counter and the panel showing Max Character Limit 1500.

Step 8 — Add the consent checkbox

Action: Add > **Consent Checkbox Field**. Set **Checkbox Name / ID** (`consent`), write the **Consent Text / Disclaimer** with a link to the locale’s privacy notice, and turn **Mandatory Field** on if consent is required.

Expected Result: A checkbox with the consent text and link.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-08-consent.png`
> **Purpose:** Consent added
> **Description:** Consent checkbox with rich-text disclaimer and a linked Privacy Notice.

Step 9 — Add contact cards (optional)

Action: Add > **Contact Card Output** once per dropdown value. Set **Show when dropdown value equals** to the exact option text (`India`), **Contact Name**, **Emails** and **Phones** (comma-separated).

Expected Result: In the editor every card is visible with the note “Shown when a dropdown value is”India”“; on the published page it appears only when India is selected.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-09-contact-cards.png`
> **Purpose:** Contact cards added
> **Description:** Two contact cards with dashed borders and their condition notes.

Step 10 — Make fields conditional (optional)

Action: On any Text, Dropdown, Textarea, Checkbox or Submit item, type the dropdown value in **Show only if dropdown value equals** (for example `Other`).

Expected Result: The item shows a dashed border and condition note in the editor.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-10-conditions.png`
> **Purpose:** Conditional item
> **Description:** A text field panel with "Show only if dropdown value equals" set to "Other".

Step 11 — Add the submit button

Action: Add > **Submit Button**. Keep or change the **Button Label** ("Send").

Expected Result: A black primary button with an arrow at the end of the form.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-11-submit.png`
> **Purpose:** Submit button added
> **Description:** The Submit Button item selected with its Button Label panel.

Step 12 — Order the items

Action: In the content tree, drag the items into the visible order: dropdowns before the items that depend on them, the submit button last. Start with a full-width item if the first fields are half-width name fields (see the note in Variation 3).

Expected Result: The canvas shows the items in the new order.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-12-order-items.png`
> **Purpose:** Reorder items
> **Description:** Content tree with the form items being dragged into order.

Step 13 — Preview

Action: Switch Universal Editor to **Preview**, then open the page preview (`https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{page}`) with AEM Sidekick or the publish-to-preview action.

Expected Result: In the Universal Editor preview, conditional items are still visible (the page is still instrumented). On the `.aem.page` preview, conditional items are hidden until their value is selected.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-13-preview.png`
> **Purpose:** Preview
> **Description:** The form on the aem.page preview with conditional fields hidden.

Step 14 — Test the form

Action: On the `.aem.page` preview: submit empty; enter an invalid email; select each dropdown value; type up to the textarea limit; finally submit a complete form.

Expected Result: Inline errors appear under the invalid fields and focus moves to the first; conditional items and contact cards appear per value; a valid submit resets the form and shows "Thank you, your details were sent.".

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-14-test.png`
> **Purpose:** Test validation
> **Description:** Form with inline errors under First name and Email after an empty submit.

Step 15 — Publish

Action: In Universal Editor choose **Publish** (or use AEM Sites / Sidekick) to publish the page to live.

Expected Result: The form is live at `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/ {page}` (production domain: [To be confirmed: production domain and CDN]).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/form-variations-step-15-publish.png`
> **Purpose:** Publish
> **Description:** The Universal Editor Publish dialog with the page selected.

Preview Process

1. Save all item changes in Universal Editor (changes save automatically on field blur).
2. Publish to preview and open the `.aem.page` URL; hard-refresh to bypass the browser cache.
3. Check the form on desktop (>= 900 px), tablet (600–899 px) and mobile (< 600 px) widths.
4. Remember that conditional items are always visible inside Universal Editor; judge the conditional behaviour only on `.aem.page` / `.aem.live`.

Publishing Process

1. Confirm with the development team that a listener for `form-variations:submit` exists on the page or site, or that the business accepts that submissions are not stored ([To be confirmed: form back end — owner: product owner]).
2. Publish the page; if the form sits on a fragment / reference page, publish that page too.
3. Publish the locale's placeholders sheet if UI texts were translated or changed.
4. Verify the live URL once the CDN cache has refreshed.

Validation Process

- ☐ Every item has a unique, meaningful **Name / ID** (camelCase, no spaces), e.g. `email`.
- ☐ Every field has a visible label; consent text is complete and links to the locale's privacy notice.
- ☐ Mandatory fields are marked and an empty submit shows "This field is required." under each.
- ☐ Email fields use Input Type Email; an invalid address shows "Enter a valid email address.".
- ☐ Dropdown options are spelled exactly like the conditions that refer to them.
- ☐ Each condition shows its items on `.aem.page` only for its value; hidden fields are not required.
- ☐ Contact cards show correct names, working `mailto:` and `tel:` links.
- ☐ Textarea counter shows the intended limit.
- ☐ At least one submit button is visible in every state of the form.
- ☐ A valid submit shows the success message and resets the form.
- ☐ Half-width name fields line up as intended on desktop (see Variation 3 note).
- ☐ UI texts appear in the locale's language (placeholders sheet).
- ☐ The business has confirmed where submissions go (no back end by default).

Rollback Process

1. To remove a faulty form quickly, turn on **Hide block** (Additional tab) and publish: the block and its wrapper are hidden on the site and shown faded in the editor.
2. To restore an earlier version, use AEM **Timeline / Versions** on the page, restore the version and republish.
3. To take the page offline, **Unpublish** it in AEM Sites (consider a redirect in the locale's redirects sheet and the redirect mechanism, see Part 4).
4. Code issues are rolled back by the development team by reverting the commit on `main` (Edge Delivery Code Sync deploys the revert).

6.29.6 Authorable Field Specification

Block model `form-variations` — Content tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Form Title / Heading (`formHeading`)	text	No	No	(empty)	Names the form (h2) and labels it for screen readers	Short, action-oriented title	None in the model; no title means no header and no accessible name for the form	Contact us
Form Description (`formDescription`)	rich-text	No	No	(empty)	Explains the purpose of the form	One or two sentences; links allowed	None; shown only when it has text	Tell us about your project and our experts will get back to you.
Theme (`classes`)	select	No	No	Default (`""`)	Light or dark presentation	Dark for framed conversion panels	Options: Default (`""`), Dark (`dark`)	Dark

Block model `form-variations` — Additional tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	text	No	No	(empty)	Makes the block a jump target (`#anchor-…`)	Unique per page, starts with `anchor-`	Enforced regExp `^anchor-[a-z0-9][a-z0-9-]*$`; message "Start with anchor-, then lower-case letters, digits and hyphens."; duplicates get `-2`, `-3` in code	anchor-contact-form
Hide block (`classes_hideblock`)	boolean	No	No	false	Temporarily hides the block	Use for staged launches or quick rollback	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default (`""`)	Extra inner space on the left	Only when the design requires it	15 options: Default, `pl-0`, `pl-100` … `pl-3000`	pl-400
Padding right (`classes_paddingright`)	select	No	No	Default (`""`)	Extra inner space on the right	As above	Default, `pr-0` … `pr-3000`	pr-400
Padding top (`classes_paddingtop`)	select	No	No	Default (`""`)	Extra inner space above	As above	Default, `pt-0` … `pt-3000`	pt-800
Padding bottom (`classes_paddingbottom`)	select	No	No	Default (`""`)	Extra inner space below	As above	Default, `pb-0` … `pb-3000`	pb-800

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Margin left (`classes_marginleft`)	select	No	No	Default (`""`)	Outer space on the left	Rarely needed (the block is centred)	Default, `ml-0` … `ml-3000`	ml-0
Margin right (`classes_marginright`)	select	No	No	Default (`""`)	Outer space on the right	Rarely needed	Default, `mr-0` … `mr-3000`	mr-0
Margin top (`classes_margintop`)	select	No	No	Default (`""`)	Outer space above	To separate from previous content	Default, `mt-0` … `mt-3000`	mt-1200
Margin bottom (`classes_marginbottom`)	select	No	No	Default (`""`)	Outer space below	To separate from following content	Default, `mb-0` … `mb-3000`	mb-1200

The 15 spacing options of every padding / margin select are: Default (`""`), 0 (none) (`-0`), 0.25rem space-100 (`-100`), 0.5rem space-200 (`-200`), 0.75rem space-300 (`-300`), 1rem space-400 (`-400`), 1.25rem space-500 (`-500`), 1.5rem space-600 (`-600`), 2rem space-800 (`-800`), 2.5rem space-1000 (`-1000`), 3rem space-1200 (`-1200`), 4rem space-1600 (`-1600`), 5rem space-2000 (`-2000`), 6rem space-2400 (`-2400`), 7.5rem space-3000 (`-3000`), each prefixed with `pl`, `pr`, `pt`, `pb`, `ml`, `mr`, `mt` or `mb`.

Item model `field-text` — Text Input Field

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Field type (`classes`)	text, read-only	—	No	`field-text`	Identifies the item type (first cell)	Do not change	Read-only	field-text
Field Name / ID (`name`)	text	Yes (dialog)	No	(empty)	Key of the value in the submitted data; drives width and autocomplete	camelCase, unique in the form; contains "name" = half width	Required in the dialog; code fallback `text`	firstName
Field Label (`label`)	text	Yes (dialog)	No	(empty)	Visible label	Short noun phrase	Required in the dialog; code fallback = the name	First name
Input Type (`inputKind`)	select	No	No	Text (`text`)	Input type and keyboard	Email for emails, Tel for phones	Options: Text (`text`), Email (`email`), Tel (`tel`); unknown = text	email
Mandatory Field (*) (`required`)	boolean	No	No	false	Makes the field required	Only what is truly needed	Only `true` turns it on	true
Show only if dropdown value equals (`dependsOn`)	text	No	No	(empty)	Conditional display	Exact option text; empty = always shown	One value, case-insensitive match	Other

Item model `field-select` — Dropdown / Select Field

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Field type (`classes`)	text, read-only	—	No	`field-select`	Item type	Do not change	Read-only	field-select
Dropdown Name / ID (`name`)	text	Yes (dialog)	No	(empty)	Data key	camelCase, unique	Required in the dialog; fallback `select`	region
Dropdown Label (`label`)	text	Yes (dialog)	No	(empty)	Visible label	Short noun	Required in the dialog; fallback = name	Region
Mandatory Field (*) (`required`)	boolean	No	No	false	Requires a choice	On when the choice routes the enquiry	Only `true` turns it on	true
Options (`options`)	text	No	Comma-separated list	(empty)	The choices (text = value)	Separate with commas; no commas inside an option	Split on commas, trimmed, empty entries dropped	Asia Pacific, Europe, India, North America
Show only if dropdown value equals (`dependsOn`)	text	No	No	(empty)	Conditional display (chained dropdowns)	Exact value of another dropdown	One value	Other

Item model `field-textarea` — Textarea Field

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Field type (`classes`)	text, read-only	—	No	`field-textarea`	Item type	Do not change	Read-only	field-textarea
Field Name / ID (`name`)	text	Yes (dialog)	No	(empty)	Data key	camelCase	Required in the dialog; fallback `message`	message
Field Label (`label`)	text	Yes (dialog)	No	(empty)	Visible label	A question works well	Required in the dialog; fallback = name	How can we help you?
Max Character Limit (`maxLength`)	number	No	No	1500	Maximum characters	Keep 1500 unless the receiving system needs less	Positive whole number, else 1500 (code); enforced in the browser via `maxlength`	1000
Mandatory Field (*) (`required`)	boolean	No	No	false	Requires text	—	Only `true`	false
Show only if dropdown value equals (`dependsOn`)	text	No	No	(empty)	Conditional display	Exact option text	One value	Other

Item model `field-checkbox` — Consent Checkbox Field

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Field type (`classes`)	text, read-only	—	No	`field-checkbox`	Item type	Do not change	Read-only	field-checkbox
Checkbox Name / ID (`name`)	text	Yes (dialog)	No	(empty)	Data key	e.g. consent, marketingOptIn	Required in the dialog; fallback `consent`	consent
Consent Text / Disclaimer (`label`)	rich-text	Yes (dialog)	No	(empty)	The consent wording, with links	Approved legal wording per locale	Required in the dialog; fallback = name	I agree to the processing of my data as described in the Privacy Notice.
Mandatory Field (*) (`required`)	boolean	No	No	false	Requires the box to be checked	On for required acknowledgements; off for optional marketing opt-ins	Only `true`	true
Show only if dropdown value equals (`dependsOn`)	text	No	No	(empty)	Region-specific consent	Exact option text	One value	Europe

Item model `contact-card` — Contact Card Output

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Field type (`classes`)	text, read-only	—	No	`contact-card`	Item type	Do not change	Read-only	contact-card
Show when dropdown value equals (`region`)	text	Yes (dialog)	No	(empty)	The value that shows the card	Exact option text	Required in the dialog; one value	India
Contact Name (`contactName`)	text	Yes (dialog)	No	(empty)	Name shown in brand colour	Person or team name	Required in the dialog	Iris Tham
Emails (`emails`)	text	No	Comma-separated list	(empty)	`mailto:` links	Business addresses only	Split on commas; no format check	media.india@example.com
Phones (`phones`)	text	No	Comma-separated list	(empty)	`tel:` links	International format with +	Split on commas; link keeps digits and + only	+91 00 0000 0000

Item model `field-submit` — Submit Button

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Field type (`classes`)	text, read-only	—	No	`field-submit`	Item type	Do not change	Read-only	field-submit
Button Label (`label`)	text	No	No	Send	Button text	Verb, 1–3 words	Empty = placeholder	Submit enquiry

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
							`send` ("Send")	
Show only if dropdown value equals (`dependsOn`)	text	No	No	(empty)	Conditional button	Ensure some button is always visible	One value	North America

Note: the example email and phone values above are illustrative formats, not verified TCS contacts.

Field details

- **Form Title / Heading** — Purpose: the form's heading and accessible name (`aria-labelledby`). Allowed values: plain text. Character limits: none enforced; Recommended up to 60 characters. Business rules: rendered as `h2` (or the heading level present in the cell); id `form-variations-{n}-title`. Dependency rules: none. Visibility rules: no model condition; the header is omitted when title and description are empty. Sample: "Contact us".
- **Form Description** — Purpose: context and expectations (response time, privacy). Allowed values: rich text (paragraphs, bold, italic, links, lists). Character limits: none enforced; Recommended up to 300 characters. Business rules: shown only when it contains text; rich text kept. Dependency rules: none. Visibility rules: none. Sample: "Tell us about your project."
- **Theme** — Purpose: Default or Dark presentation. Allowed values: Default (`""`), Dark (`dark`). Character limits: n/a. Business rules: becomes a block class. Dependency rules: none. Visibility rules: none. Sample: Dark.
- **Anchor ID** — Purpose: jump target. Allowed values: `anchor-` followed by lower-case letters, digits and hyphens. Character limits: none enforced; Recommended up to 40. Business rules: Enforced pattern in the dialog; the block-options decorator sets it as the block id, adding `-2`, `-3` … if the id is already used on the page. Dependency rules: none. Visibility rules: none. Sample: `anchor-contact-form`.
- **Hide block** — Purpose: hide without deleting. Allowed values: on / off. Business rules: hidden on the site and in Universal Editor preview; shown at 45% opacity with a dashed outline in edit mode; applies inside `main` only. Dependency / visibility rules: none. Sample: off.
- **Padding left / right / top / bottom** — Purpose: extra inner spacing from the spacing tokens. Allowed values: the 15 options listed above. Business rules: classes `main .section .block.p?-{token}` outrank the block's own spacing; apply inside `main` only. Dependency / visibility rules: none. Sample: Padding top 2rem (`pt-800`).
- **Margin left / right / top / bottom** — Purpose: outer spacing from the tokens. Allowed values: the 15 options. Business rules: as padding; a left / right margin overrides the block's automatic centring (`margin-inline: auto`). Dependency / visibility rules: none. Sample: Margin bottom 3rem (`mb-1200`).
- **Field type (all items)** — Purpose: the item type the code reads from the first cell (or the Universal Editor model). Allowed values: fixed per item. Business rules: read-only; rows whose type is not one of the six types are ignored (before the first item they are block properties). Dependency / visibility rules: none.
- **Field Name / ID (text, textarea, checkbox) and Dropdown Name / ID** — Purpose: the key in the submitted data, the base of the element id and, for text inputs, the width and autocomplete hint. Allowed values: any text; Recommended camelCase letters and digits. Character limits: Recommended up to 40. Business rules: two fields with the same name are submitted as an array; for Text Input Fields "name" anywhere in the name gives half width. Dependency rules: none. Visibility rules: none. Sample: `email`.
- **Field Label / Dropdown Label** — Purpose: visible label. Allowed values: plain text. Character limits: Recommended up to 40. Business rules: a red `*` is appended when mandatory. Sample: "Business email".

- **Input Type** — Purpose: `text`, `email` or `tel` input. Allowed values: Text, Email, Tel. Business rules: Email adds format validation and `autocomplete="email"`; Tel adds `autocomplete="tel"` and no validation. Sample: Email.
- **Mandatory Field (*)** — Purpose: required input. Allowed values: on / off. Business rules: only the value `true` counts; a hidden conditional field is never required. Sample: on.
- **Show only if dropdown value equals / Show when dropdown value equals** — Purpose: conditional display. Allowed values: one dropdown option text. Character limits: n/a. Business rules: case-insensitive, trimmed exact match against the current value of any enabled dropdown in the same form. Dependency rules: requires a dropdown with that option in the same block. Visibility rules (model): none — the field is always shown in the dialog; the condition is evaluated by the code at runtime. Sample: "India".
- **Options** — Purpose: dropdown choices. Allowed values: comma-separated text. Character limits: none enforced. Business rules: the text is the value; a disabled "Select an option" prompt is added first. Sample: "Technology, Sport, Asia Pacific, India" (the dialog's own example).
- **Max Character Limit** — Purpose: textarea limit. Allowed values: whole numbers > 0. Business rules: invalid or empty values fall back to 1500; enforced by `maxlength`. Sample: 1500.
- **Consent Text / Disclaimer** — Purpose: consent wording. Allowed values: rich text with links. Character limits: Recommended up to 400. Business rules: markup kept; last paragraph inline. Sample: "I agree to the Privacy Notice."
- **Contact Name** — Purpose: name on the card. Character limits: Recommended up to 50. Sample: "Iris Tham" (the dialog's example).
- **Emails / Phones** — Purpose: contact links. Allowed values: comma-separated addresses / numbers. Business rules: no validation; `tel:` links keep digits and `+`. Sample: "+65 0000 0000" (illustrative).
- **Button Label** — Purpose: submit text. Character limits: Recommended up to 25. Business rules: empty = "Send" (placeholder `send`). Sample: "Send".

6.29.7 Field Annotation Mapping

Marker	Field	Purpose
A	Form Title / Heading (`formHeading`)	Form heading (h2) and accessible name of the form
B	Form Description (`formDescription`)	Introductory rich text under the title
C	(generated) "* Required fields" note	Explains the asterisk; placeholder `requiredNote`; not authorable in the block
D	Text Input Field: Field Label, Field Name / ID, Input Type, Mandatory	Single-line input
E	Dropdown / Select Field: Dropdown Label, Options	Choice list that also drives conditions
F	Textarea Field: Field Label, Max Character Limit	Long text with counter
G	Consent Checkbox Field: Consent Text / Disclaimer	Consent checkbox label
H	Contact Card Output: Contact Name, Emails, Phones, Show when dropdown value equals	Conditional contact details
I	Show only if dropdown value equals (any item)	Editor-only condition note; condition evaluated on the site
J	Submit Button: Button Label	Submits the form

6.29.8 Component Specification

Attribute	Value
Component Name	Form Variations (`form-variations`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	scripts.js `moveInstrumentation`; button, content, heading, block-options and data-layer decorators; utils environment, properties, ids, placeholders; locale placeholders sheet
Component Type	Container block with six item types (`field-text`, `field-select`, `field-textarea`, `field-checkbox`, `contact-card`, `field-submit`)
Supported Variants	Theme Default / Dark; half-width and full-width layout; input types Text / Email / Tel; conditional items; contact cards
Supported Templates	Any page (allowed in every section)
Supported Metadata	None read
Supported Tags	None
Supported Personalization	None. Future: data layer (`cmp:submit`) can feed Adobe Target / Journey Optimizer once Adobe Tags is installed (not implemented)
Caching Considerations	Page HTML cached by the Edge Delivery CDN; the placeholders sheet is fetched once per page view (in-memory promise); no form data is cached
Performance Considerations	Block JS/CSS loaded when its section loads (eager only if in the first section); no images, no third-party scripts; waits for the placeholders sheet before rendering
Accessibility Compliance	Labels with `for`; native `required`; `aria-invalid` and `aria-describedby` errors; focus to first error; `role="status"` success message that receives focus; `autocomplete` per input purpose (1.3.5); asterisk `aria-hidden` and explained once; visible focus outline
SEO Compliance	One h2; no structured data; contact card text present in HTML; form submits nowhere (no crawlable endpoint)

6.29.9 Input Model

Content Inputs: Form Title, Form Description (block properties: the first rows before any item), and the item rows: each starts with its type cell followed by its fields in model order.

Metadata Inputs: None.

Tags Inputs: None.

Page Context Inputs: Universal Editor detection (`data-aue-resource` on an ancestor) keeps conditional items visible; the locale (from the URL) selects the placeholders sheet.

Query Inputs: None (no query index, no sheets other than placeholders).

Manual Inputs: Everything is authored manually in the dialog.

Authored HTML received from AEM (simplified):

```
<div class="form-variations dark">
  <div><div><p>Contact us</p></div></div>
  <div><div><p>Tell us about your project.</p></div></div>
  <div>
    <div>field-text</div><div>firstName</div><div>First name</div>
    <div>text</div><div>true</div><div></div>
  </div>
```

```
  <div>
    <div>field-select</div><div>region</div><div>Region</div>
    <div>true</div><div>Asia Pacific, Europe, India</div><div></div>
  </div>
  <div>
    <div>contact-card</div><div>India</div><div>Iris Tham</div>
    <div>contact@example.com</div><div>+91 00 0000 0000</div>
  </div>
  <div><div>field-submit</div><div>Send</div><div></div></div>
</div>
```

> **Note:** the code reads item cells **by position** (it does not use `utils/cells.js`). If a field in the middle of an item was never saved and AEM leaves its cell out, the following values shift (for example a Dropdown's condition could be read as its options). Open each item and set its fields, including the toggles, before publishing. [To be confirmed: whether AEM omits never-set item fields for this model — owner: development team]

6.29.10 Output Model

Rendered Output: The block is replaced by up to three children: `.form-variations-header` (title and description, omitted when both are empty), the `form.form-variations-form` (`novalidate`, `aria-labelledby` = title id) and `p.form-variations-status` (`role="status"`, `tabindex="-1"`). In the form: the optional `p.form-variations-required-note`, then one `.form-variations-group` per item with the type class (`field-text`, `field-select`, `field-textarea`, `field-checkbox`, `form-variations-contact-card`, `field-submit`) and `half-width` / `full-width`; conditional groups carry `data-depends-on` and `hidden` when not matched. Errors are added on submit as `p.form-variations-error` inside the group.

Generated HTML (simplified, after an invalid submit):

```
<div class="form-variations block" data-block-name="form-variations">
  <div class="form-variations-header">
    <h2 class="form-variations-title" id="form-variations-1-title">Contact us</h2>
  </div>
  <form class="form-variations-form" novalidate aria-labelledby="form-variations-1-title">
    <p class="form-variations-required-note">* Required fields</p>
    <div class="form-variations-group field-text full-width">
      <label class="form-variations-label" for="form-variations-1-email">
        <span>Business email</span>
        <span class="form-variations-required" aria-hidden="true">*</span>
      </label>
      <input class="form-variations-control" type="email" id="form-variations-1-email"
        name="email" autocomplete="email" required data-required="true"
        aria-invalid="true" aria-describedby="form-variations-1-email-error">
      <p class="form-variations-error" id="form-variations-1-email-error">
        Enter a valid email address.</p>
    </div>
    <div class="form-variations-group form-variations-contact-card full-width"
      data-depends-on="India" hidden>
      <p class="form-variations-contact-name">Iris Tham</p>
      <ul class="form-variations-contact-links">
        <li class="form-variations-contact-email">
          <a href="mailto:contact@example.com">contact@example.com</a></li>
      </ul>
    </div>
    <div class="form-variations-group field-submit full-width">
```

```
      <button type="submit" class="button primary">Send</button>
    </div>
  </form>
  <p class="form-variations-status" role="status" tabindex="-1"></p>
</div>
```

Generated JSON: No JSON is sent to a server. A valid submit dispatches:

```
// fired on the block element, bubbles to document
new CustomEvent('form-variations:submit', {
  bubbles: true,
  detail: {
    data: {
      firstName: 'Asha', lastName: 'Rao', email: 'asha@example.com',
      region: 'India', message: 'We need help with...', consent: 'true',
    },
  },
});
```

Values are strings; repeated names become arrays; hidden (disabled) conditional fields and unchecked checkboxes are absent. A future integration listens for it, for example:

```
document.addEventListener('form-variations:submit', (e) => {
  // e.target is the .form-variations block; e.detail.data holds the values
  // send to the agreed back end here (not implemented as of this handover)
});
```

Generated Metadata: None.

Analytics Tracking (Adobe Client Data Layer):

- Component state (lazy phase, once): `{ component: { 'form-variations-1': { '@type': 'tcs-corp/components/form-variations', parentId: 'page-…', 'dc:title': 'Contact us' } } }` (the id is the Anchor ID when set).
- `cmp:click` on the submit button (and on contact-card links: `xdm:linkType` `exit` for `mailto:` / `tel:` links, because their origin differs from the site):

```
{ "event": "cmp:click", "eventInfo": { "path": "component.form-variations-1",
  "interaction": "button", "dc:title": "Send", "xdm:linkType": "other",
  "breakpoint": "desktop" } }
```

- `cmp:submit` with the field **names only** (never values):

```
{ "event": "cmp:submit", "eventInfo": { "path": "component.form-variations-1",
  "dc:title": "form-variations",
  "fields": ["firstName", "lastName", "email", "region", "message", "consent"] } }
```

> **Important:** the data layer listens to `submit` in the capture phase, before the block's own validation. `cmp:submit` is therefore pushed on **every submit attempt, including invalid ones**, and `fields` lists all named controls, including hidden conditional fields. `dc:title` is the block name because the form has no `aria-label` or id. Analytics rules for "successful leads" must not rely on `cmp:submit` alone; a success signal (for example a data-layer push from the future `form-variations:submit` handler) is Not implemented (as of this handover).

Personalization Output: None.

Search Index Output: None. The query index (`helix-query.yaml`) indexes page metadata only; form fields are not indexed.

6.29.11 Business Rules

1. Rows are read in order; rows before the first item are block properties: the first is the Form Title, the second the Form Description. Rows after the first item that are not items are ignored.
2. An item's type is the text of its first cell when it is one of the six types, else its Universal Editor model (`data-aue-model`); rows of unknown type are not rendered.
3. Item cells are read by position in model order (type cell excluded).
4. Name fallbacks when empty: Text `text`, Dropdown `select`, Textarea `message`, Checkbox `consent`. Label fallback: the name.
5. Element ids are `form-variations-{n}-{name}` (lower-cased, hyphenated, de-duplicated with `-2`, `-3`); `{n}` is the form's instance number on the page.
6. Text Input Field width: half width when the name contains "name" (any case), else full width; all other items are full width. Two columns only at >= 900 px.
7. Input Type: `text`, `email` or `tel`; anything else becomes `text`.
8. Autocomplete: Email type `email`; Tel type `tel`; otherwise by name pattern — first/given `given-name`; last/surname/family `family-name`; `name` / `fullname` / `full-name` / `full_name` / `full name` `name`; e-mail/ email `email`; phone/mobile/tel `tel`; company/organisation/organization `organization`; job/designation/ position `organization-title`; country `country-name`; city/town `address-level2`; zip/postal/ postcode `postal-code`. First match wins in that order; no match = no attribute.
9. Mandatory: only the value `true` (case-insensitive) makes a field required.
10. Dropdown options: comma-separated, trimmed, empty entries dropped; a disabled, pre-selected "Select an option" prompt with an empty value comes first.
11. Textarea limit: the authored number if > 0, else 1500; enforced with `maxlength`; counter "{length}/{limit}" updates on input and after reset.
12. Checkbox value is `"true"`; unchecked boxes are not submitted. Empty consent text shows the name.
13. Contact cards: the region value is the condition; emails `mailto:`, phones `tel:` with only digits and `+` in the link; no list when both are empty.
14. Submit label fallback: placeholder `send` ("Send"); primary button style.
15. Conditional items: shown when any enabled dropdown of the same form has the item's value (trimmed, case-insensitive); hidden items are `hidden`, their controls `disabled` and not required; re-evaluated on dropdown change and after reset; repeated until stable for chained conditions.
16. Editor-only: inside Universal Editor (edit and preview), all conditional items stay visible and enabled with the note "Shown when a dropdown value is"{value}"" (placeholder `shownWhenValue`) and a dashed border.
17. "* Required fields" (placeholder `requiredNote`) is added at the top of the form when any field is mandatory (including conditional ones).
18. Validation runs on submit only (the native validation UI is turned off with `novalidate`): every enabled input, select and textarea is checked; errors appear in the field's group; the first invalid field receives focus.
19. Error messages: missing value `fieldRequired` "This field is required."; invalid email `fieldEmail` "Enter a valid email address."; other type / pattern mismatch `fieldFormat` "Enter a valid value."; too long `fieldTooLong` "This text is too long."; anything else the browser's own message.
20. An error is removed as soon as the field becomes valid while typing (`input` event), or on the next submit.
21. On a valid submit: the `form-variations:submit` event is dispatched with `{ data }`, the form is reset, counters and conditional visibility are updated, the status shows `formSuccess` "Thank you, your details were sent." and receives focus. The message is shown whether or not any listener handled the data.
22. The status message is cleared on the next input or submit.

23. No submit endpoint, no network request, no storage of form data, no spam protection.
24. Placeholders: texts come from the locale's placeholders sheet; missing or empty keys use the English fallbacks above; the block waits for the sheet before rendering.
25. The form is labelled by the title only when a title exists.
26. Additional tab: Anchor ID becomes the block id (unique on the page); Hide block hides the block and its wrapper outside edit mode; spacing classes apply inside `main`.

6.29.12 Author Best Practices

Recommended Content Length: Title up to 60 characters; description up to 300; labels up to 40; consent text up to 400; button label 1–3 words (up to 25 characters); 3–8 fields for lead forms (each extra field lowers completion). All Recommended, not enforced.

Image Dimensions: Not applicable — the block has no image fields.

CTA Writing Guidelines: Use a specific verb on the submit button ("Send enquiry", "Request a call back"), not "Click here". Keep one primary action; avoid several visible submit buttons unless each is conditional.

Accessibility Guidelines: Give every field a clear visible label (do not rely on the description); mark only genuinely required fields; use Input Type Email / Tel so the right keyboard and autocomplete apply; use names that match the autocomplete patterns (`firstName`, `lastName`, `email`, `phone`, `company`, `jobTitle`, `country`, `city`, `postalCode`); place conditional items directly after the dropdown that controls them; write descriptive link text in the consent text.

SEO Guidelines: Use a meaningful title (it is an `h2` in the page outline); keep the page's main `h1` in the Page Header; do not put essential content only inside contact cards (hidden until selected).

Metadata Guidelines: The block reads no metadata. Set the page's own title and description for the contact page as usual.

Taxonomy Guidelines: Not applicable; dropdown options are free text, not tags. Keep option wording consistent with the taxonomy where the values must match a downstream system.

Localization Guidelines: Author labels, options, consent text and contact cards in each locale's copy of the page. Translate the UI keys (`fieldRequired`, `fieldEmail`, `fieldFormat`, `fieldTooLong`, `selectAnOption`, `send`, `shownWhenValue`, `requiredNote`, `formSuccess`) in each locale's placeholders sheet. Keep **Field Name / ID** values identical across locales (they are data keys, not visible text), and remember that conditions must match the **translated** option text of that locale.

6.29.13 Troubleshooting Guide

Issue 1 — Visitors see "Thank you" but nobody receives the submission

Symptoms: The success message appears; no email, CRM record or log entry exists.

Root Cause: By design there is no submit endpoint; the block only dispatches `form-variations:submit`.

Diagnosis Steps: In the browser console run `document.addEventListener('form-variations:submit', (e) => console.log(e.detail))` and submit; the data is logged. Check `scripts/delayed.js` and the site code for a listener (none exists as of this handover).

Resolution Steps: Do not use the form for live data capture until the development team adds the agreed integration ([To be confirmed: form back end — owner: solution architect]). Meanwhile show a contact card or an email link.

Validation Steps: After an integration, submit a test entry on `.aem.page` and confirm it in the target system.

Preventive Measures: Include "back end connected?" in the publishing checklist.

Issue 2 — A conditional field or contact card never appears

Symptoms: Selecting the dropdown value does not show the item on `.aem.page`.

Root Cause: The condition text differs from the option text (spelling, extra words, a comma-separated list), or the controlling dropdown is itself hidden.

Diagnosis Steps: Compare **Show only if dropdown value equals** with the option exactly; inspect the group's `data-depends-on` attribute in the browser's developer tools.

Resolution Steps: Copy the option text into the condition; use one value per item (add a second item for a second value).

Validation Steps: Re-test on `.aem.page` for every option.

Preventive Measures: Keep a list of options and their dependent items per form.

Issue 3 — All conditional fields are visible

Symptoms: Every contact card and conditional field shows at once.

Root Cause: The page is viewed in Universal Editor (edit or preview mode), where conditional items stay visible on purpose.

Diagnosis Steps: Check whether the URL is the AEM author / Universal Editor URL; look for the italic condition notes.

Resolution Steps: Test on `https://main--tcs-corp--sradha-tcs.aem.page/…` instead.

Validation Steps: On `.aem.page` only unconditional items show initially.

Preventive Measures: Train authors that conditional behaviour is visible only outside the editor.

Issue 4 — The form cannot be submitted (no button)

Symptoms: No Send button is visible.

Root Cause: Every Submit Button item has a condition and no matching value is selected, or no Submit Button item exists.

Diagnosis Steps: Check the submit items for **Show only if dropdown value equals**.

Resolution Steps: Add an unconditional Submit Button or clear the condition.

Validation Steps: A button is visible before and after selecting each value.

Preventive Measures: Validation checklist item "at least one button always visible".

Issue 5 — Fields are misaligned on desktop

Symptoms: "First name" sits next to "* Required fields" and "Last name" drops to the next row; or a lone field takes only half the width; or a "companyName" field is half width.

Root Cause: Width is automatic: Text Input names containing "name" are half width, and the required note occupies one grid column at >= 900 px.

Diagnosis Steps: Check the field names; check whether the first item is half width.

Resolution Steps: Put a full-width item first or pair half-width fields; rename fields that should be full width (for example `company` instead of `companyName`). A permanent fix for the note is a CSS change by the development team.

Validation Steps: View at 1280 px and 900 px.

Preventive Measures: Follow the naming guidance in 6.29.12.

Issue 6 — Values appear in the wrong field (options show the condition text, etc.)

Symptoms: A dropdown shows its condition as an option, or a checkbox is unexpectedly mandatory.

Root Cause: Item cells are read by position; a missing cell for a never-saved field shifts the later values (see the note in 6.29.9).

Diagnosis Steps: Open `{page}.plain.html` on `.aem.page` and count the cells of the item row.

Resolution Steps: Open the item in Universal Editor, set every field (toggle Mandatory on and off if needed) and republish.

Validation Steps: The item row has one cell per field; the field renders correctly.

Preventive Measures: Report to the development team so the block can use `utils/cells.js`.

Issue 7 — Error or button texts are in English on a translated site

Symptoms: "This field is required." on `/de-de/`.

Root Cause: The key is missing or empty in the locale's placeholders sheet, or the sheet is not published.

Diagnosis Steps: Open `/{locale}/placeholders.json` and look for the key.

Resolution Steps: Add the key and translation; publish the sheet.

Validation Steps: Reload the page; the translated text appears.

Preventive Measures: Copy all form keys when creating a new locale's sheet.

Issue 8 — Analytics shows more "submits" than real enquiries

Symptoms: `cmp:submit` counts exceed successful submissions.

Root Cause: `cmp:submit` is pushed on every submit attempt, including invalid ones (capture listener before validation).

Diagnosis Steps: In the console, inspect `window.adobeDataLayer` after an empty submit.

Resolution Steps: Analytics rules should combine with a success signal once implemented; the development team can add one in the submit handler.

Validation Steps: An invalid submit no longer counts as a lead in reports.

Preventive Measures: Document the event semantics in the analytics solution design.

6.29.14 Support Runbook

Monitoring Requirements

- No server-side monitoring exists (no endpoint). Monitor page availability of the form pages (`.aem.live` URL returns 200) and, once Adobe Tags is installed, `cmp:submit` volumes.
- After any integration is built: monitor the integration's error rate and delivery ([To be confirmed: monitoring tool and SLA — owner: AMS lead]).

Logs

- Browser console: the block emits **no** console messages of its own. JavaScript errors in the console (for example a failed `form-variations.js` load) indicate code problems.
- Network: `/{locale}/placeholders.json` (200 expected; a 404 only means English fallbacks).
- Admin API: preview / publish status of the page via AEM Sidekick or `https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`.
- Author logs: AEM author publication logs (replication) for the page.

Common Errors

Error	Cause	Fix
Success message but no data received	No back end (by design)	Connect an integration; meanwhile use contact cards
"This field is required." on a hidden field	Not possible on the site (hidden fields are disabled); in the editor all fields are visible and enabled	Test on `.aem.page`
Conditional item never shows	Condition text does not match option	Match spelling exactly, one value per item
No submit button	All buttons conditional	Add an unconditional button
English UI texts on a locale	Placeholder key missing	Add keys to the locale's placeholders sheet
Form empty on the site but fine in the editor	Item rows lack their type cell (type read from the editor model only)	Re-save items; escalate to development
Duplicate id warnings in audits	Two fields with the same name in one form	Use unique names

Escalation Matrix

Level	Who	Checks
L1	Author support / content lead	Field names, labels, conditions, options spelling, button presence, placeholders translations, publish status
L2	AMS / site admin	Page and placeholders sheet published, `.aem.live` reachable, CDN cache, Sidekick status, user permissions
L3	Development team	Console errors, `form-variations.js` behaviour, cell order in `.plain.html`, data-layer events, any back-end listener
Adobe Support	Adobe	Edge Delivery delivery or Universal Editor platform issues (rendering of item rows, publishing failures)

Recovery Procedures

1. Faulty form on a live page: turn on **Hide block**, publish (immediate mitigation).
2. Wrong content: restore the previous page version in AEM and republish.
3. Code regression: development team reverts the commit on `main`; verify on `.aem.page`.
4. After recovery, purge or wait for the CDN cache and re-run the validation checklist.

Health Checks

- Load each form page on `.aem.live`: form renders, no console errors.
- Submit empty: inline errors appear; submit valid: success message appears.
- Select each dropdown value: correct conditional items and contact cards.
- Check `window.adobeDataLayer` contains the component entry and `cmp:submit`.

Operational Procedures

- New form: follow 6.29.5; get legal approval of consent text per locale.
- New locale: copy the form keys to the new placeholders sheet.
- Back-end integration (future): the development team adds a listener for `form-variations:submit` (for example in a block-independent script) and a success data-layer event; update this runbook.

Known Limitations

- No submit endpoint, storage, email notification, CRM/marketing automation connection, spam protection or server-side validation (Not implemented as of this handover).

- The success message is shown even if nothing handled the data.
- `cmp:submit` fires on invalid attempts and lists hidden field names.
- No radio groups, multi-select, file upload, date, number or multi-step forms.
- One value per condition; conditions match any dropdown in the form, not a specific one.
- No phone format validation.
- Width is derived from field names (no width field); the required note takes one grid column at >= 900 px; the checkbox error sits beside the consent text.
- Item cells are read by position.
- Error colours are hard-coded (`#c8102e`, `#ff6b6b`), not tokens.

6.30 Conversation Canvas Block

The Conversation Canvas is the TCS "ask us a question" entry point: an optional icon, eyebrow, title and text, a search row (round menu button with a pop-up site menu, a search field with voice input and a submit icon button) and an optional list of buttons. It has three styles: **Default**, **Secondary** (with a filter-tag list filled by script) and **Primary (sticky)**, a dock that floats at the bottom of the viewport with the search pill and, on desktop, a pill-based site navigation built from the query index. The block lives in `blocks/conversation-canvas/` (`conversation-canvas.js`, `.css`, `_conversation-canvas.json`) and is also used inside the site footer (the Footer's **Conversation Canvas Page**).

> **Important:** no search or conversational (AI) service is connected. Submitting a query dispatches a bubbling `conversation-canvas:search` DOM event with `{ detail: { query } }` on the block; nothing else happens unless a script on the page listens for it. Suggestions and filter tags can be supplied by scripts through `setConversationCanvasSuggestions()` and `setConversationCanvasFilters()`; no such script exists as of this handover. [To be confirmed: search / conversational back end — owner: product owner / solution architect]

6.30.1 Component Purpose

Business Objective: Give every page a consistent, prominent invitation to ask a question or search, plus quick access to the site structure, ready to be connected to a search or conversational service without changing the authored content.

Business Use Cases:

- Hero-like "How can we help?" panel on home and landing pages (Default style).
- Topic exploration panel with filter tags supplied by a personalisation or search service (Secondary style).
- A persistent floating search and section navigation dock that follows the visitor down the page until the footer (Primary (sticky) style).
- The "Ask us a question" area at the top of the site footer (footer canvas page).
- Entry point for a future AI assistant: suggestions and the search event are the integration points.

When To Use:

- When the page should invite free-text questions or searches.
- When visitors need quick navigation through a section of the site (menu from the query index).
- At most one Primary (sticky) canvas per page.

When Not To Use:

- To collect personal data (use Form Variations, 6.29).
- As the only route to content while no search service is connected: the query goes nowhere.

- As a replacement for the site header navigation (the header remains the main navigation; on mobile the Primary dock shows only the search pill).
- Several canvases on one page (only one can float; repeated search rows confuse visitors).

Dependencies:

Type	Dependency	Use
Code	`scripts/aem.js` `toClassName`	Normalises the BG Style value into a class
Code	`scripts/scripts.js` `moveInstrumentation`	Keeps editor instrumentation on button items
Code	`decorators/responsive-image-decorator.js` `buildImage`	Builds background, icon and audio images from URLs (site media optimized)
Code	`decorators/a11y-decorator.js` `hideVisually`	Visually hidden label of the search input
Code	`decorators/button-decorator.js` `createButton`	Button List items
Code	`decorators/icon-button-decorator.js` `createIconButton`	Search submit icon button (arrow-right)
Code	`utils/buttons.js`	Button views, types, alignments and `readButtonOptions`
Code	`utils/properties.js` `normalizeKey`, `textOf`	Reads the key-value property rows
Code	`utils/query-index.js` `getPages`	Pages of the current locale for the menu and dock navigation
Code	`utils/site.js` `getLocale`	Default menu root (the locale root, e.g. `/de-de`)
Code	`utils/urls.js` `toSitePath`	Converts the Menu root path (AEM path) to the published path
Code	`utils/metadata.js` `getPageTags`	The page's tags as default suggestions (Primary)
Code	`utils/taxonomy.js` `resolveTags`	Tag labels for the suggestions from the locale's taxonomy
Code	`utils/ids.js` `uniqueId`	Ids per instance (`conversation-canvas-1-q`, `-menu`, `-nav`, `-panel`)
Code	`utils/placeholders.js`	UI text with English fallbacks
Code	`decorators/block-options-decorator.js`, `decorators/data-layer-decorator.js`	Additional tab; data layer (`cmp:click`, `cmp:search`)
Data	`/query-index.json` (`helix-query.yaml`)	`path` and `title` of pages for the menu / navigation
Data	`/{locale}/taxonomy.json`	Suggestion labels
Data	`/{locale}/placeholders.json`	Keys `menu`, `back`, `noPagesFound`, `siteNavigation`, `suggestions`, `searchPlaceholder`, `searchByVoice`, `submit`
Browser	Web Speech API (`SpeechRecognition` / `webkitSpeechRecognition`)	Voice input, only where supported
Icons	`icons/menu.svg`, `mic.svg`, `arrow-right.svg`, `arrow-left.svg`, `chevron-down.svg`	CSS masks and the submit icon
Item model	`models/_button.json` `button`	Conversation Canvas Button items

Related Components: Footer (see the Header / Footer chapter: the Footer block's Conversation Canvas Page loads a page with this block as a fragment), Header (main navigation), List and Filter List (also use the query index), Form Variations (data capture), Teaser / Hero (other hero-like entry points).

Applicable Templates: Any page (the block is in the section filter). Typical: home and landing pages (Default / Secondary), any page (Primary), and the footer reference page (`references/footer-conversation-canvas` in the English sample content).

Supported Content Types: All; the block does not read `content-type`. The Primary dock reads the page's tags (suggestions) and its position in the site tree.

6.30.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** Default style at 1440 px: rounded white panel with soft blue and yellow glows at the bottom, centred icon, eyebrow, large title, intro text, the search row (round menu button, pill search field with microphone and blue arrow button) and two buttons.

Desktop (>= 1200 px): A rounded (1.25rem) panel with centred text and 7.5rem top / 5rem side and bottom padding (Secondary: 10rem / 5rem). Without a background image two soft radial glows (blue and yellow) are drawn at the bottom (`::before`). Content is a centred column (max 80rem, gap 5rem; Secondary 2.5rem): icon 2.5rem, eyebrow 1rem semibold in brand colour, title 4rem medium (Bigger 5rem, Medium 2.5rem), text 1.125rem medium (max 50.75rem). The search row is up to 42.875rem wide: a 3.6875rem round outlined menu button and a pill-shaped field (2 px dark outline, translucent white, brand border on focus) containing the input, the microphone button (where supported) and the arrow submit icon button. The pop-up menu opens **above** the search row (min 15rem, max 22.5rem wide, max 60 % of the viewport height, scrolls). Buttons are centred, wrapping, 1.5rem apart. The Primary dock is described in Variation 3.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** Default style at 768 px: same layout with smaller title (3.125rem), panel padding 5rem / 2.5rem, search row full width up to 42.875rem.

Tablet (600–1199 px): Padding 5rem top/bottom, 2.5rem sides; content gap 2.5rem; eyebrow 0.875rem; title 3.125rem (Bigger 4rem, Medium 2rem); text 1.125rem; search row gap 1.25rem; the field has 0.75rem / 1.5rem padding. The Primary dock shows its desktop form from 900 px.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** Default style at 375 px: compact panel, 2.5rem title, search row with menu button and narrow search pill, buttons wrapping below.

Mobile (< 600 px): Padding 4rem top/bottom, 1.5rem sides; content gap 2.5rem; eyebrow 0.75rem; title 2.5rem (Bigger 2.75rem, Medium 1.75rem); text 1rem; search row gap 0.75rem. The menu button stays visible in the block (it is hidden on mobile only in the footer canvas). The Primary dock shows only the search pill (no menu button, no navigation).

6.30.3 Component Variations

Variation 1 — Default style

Variation Name: Default (Canvas Style = Default, no style class).

Business Purpose: A prominent in-page "ask or search" panel.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/conversation-canvas-default.png`
> **Purpose:** Default style
> **Description:** White rounded panel with blue / yellow glows, title "How can we help you today?", search row and buttons.

Use Cases: Home page, landing pages, help / contact hubs.

Authoring Guidelines: Leave **Canvas Style** on **Default**; fill Eyebrow, Title, Text, Menu and Search bar fields; add buttons with **Add** > **Conversation Canvas Button**.

Business Rules: Intro elements render only when they have a value. The search row is always rendered. The menu loads the site tree the first time it is opened. When the search row is scrolled out of view above the viewport, it floats at the bottom (Variation 17).

Rendering Rules: `.conversation-canvas.conversation-canvas-bg-default` with `.conversation-canvas-content > .conversation-canvas-intro`, `.conversation-canvas-search-slot > .conversation-canvas-search`, `ul.conversation-canvas-buttons`.

Responsive Behaviour: Desktop (>= 1200 px): 7.5rem / 5rem padding, 4rem title. Tablet (600–1199 px): 5rem / 2.5rem, 3.125rem title. Mobile (< 600 px): 4rem / 1.5rem, 2.5rem title.

SEO Impact: The title is an `h2`; text and button links are crawlable. The menu links are built only when opened (not in the initial HTML).

Accessibility Compliance: Search input with a visually hidden label (the placeholder text); menu button with `aria-label`, `aria-controls`, `aria-expanded`; menu is a `nav` with `aria-label`; Escape closes the menu and returns focus to the button; clicking or tabbing outside closes it.

Variation 2 — Secondary style

Variation Name: Secondary (Canvas Style = Secondary, class `secondary`).

Business Purpose: An exploration panel with filter tags that a script can supply (for example popular topics from a personalisation or search service).

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/conversation-canvas-secondary.png`
> **Purpose:** Secondary style

> **Description:** Panel with yellow glow on the left and blue on the right, larger top padding, search row and a row of pill filter tags (one active in brand blue).

Use Cases: Topic or industry hubs, insights landing pages.

Authoring Guidelines: Set **Canvas Style** = **Secondary**. Filter tags cannot be authored; they are supplied by code.

Business Rules: An empty `ul.conversation-canvas-filters` (`aria-label` "Suggestions") is added after the search row and hidden while empty. `setConversationCanvasFilters(block, tags)` fills it with `{ label, href?, active? }` items: with `href` a link `a.tag.tag-filter` (`aria-current="true"` when active), without `href` a `button` with `aria-pressed`. The buttons get no click behaviour from the block; the calling script must add it. Each tag carries `data-cmp-interaction="filter"` and `data-cmp-value` (label) for analytics.

Rendering Rules: Different glow positions; intro gap 2.5rem; at >= 1200 px padding 10rem / 5rem and content gap 2.5rem. Filters wrap, centred, gap 0.75rem, max 56.25rem.

Responsive Behaviour: Desktop (>= 1200 px): 10rem / 5rem padding. Tablet: 5rem / 2.5rem. Mobile: 4rem / 1.5rem.

SEO Impact: Filter tags are added by script after load (not in the initial HTML).

Accessibility Compliance: Filter list labelled; active state conveyed by `aria-current` / `aria-pressed` and colour.

```
import { setConversationCanvasFilters }
  from '/blocks/conversation-canvas/conversation-canvas.js';

const block = document.querySelector('.conversation-canvas.secondary');
setConversationCanvasFilters(block, [
  { label: 'Cloud', href: '/en/what-we-do/services/cloud', active: true },
  { label: 'AI' },
]);
```

Variation 3 — Primary (sticky) dock

Variation Name: Primary (sticky) (Canvas Style = Primary (sticky), class `primary`).

Business Purpose: A persistent floating search pill and, on desktop, contextual section navigation that stays available while the visitor scrolls.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-primary-dock.png`
> **Purpose:** Primary (sticky) dock on desktop
> **Description:** Bottom of a desktop page with a white fade: a round menu button, a search pill "Ask us a question" and, above it, a row of navigation pills (Back arrow, "Services", "Industries" with chevron, current page outlined in blue).

Use Cases: Long content pages, section landing pages, articles that should keep search and navigation in reach.

Authoring Guidelines: Set **Canvas Style** = **Primary (sticky)**. Set **Menu root path** (default: the locale's Home) and **Menu depth** (1–3) and the **Search bar** fields. Eyebrow, Title, Text, Icon, Background image and buttons are **not rendered** in this style. Place it in its own section (that section then gets no margin or padding).

Business Rules:

- Only the dock is rendered; it is fixed to the bottom of the viewport and is visible (and reachable by keyboard) only while floating: when no footer is in view, no Default / Secondary canvas's search row is in view in its place, and no canvas registered before it (higher on the page) is already floating. Otherwise it is hidden and `inert`.
- The site tree loads immediately from the query index (pages below the root, up to the depth), together with the page's tag labels (suggestions).
- The navigation shows one of three views, chosen from the current page's place in the tree:

Current page	Default view	Pills
The menu root (Home) or a page not in the tree	Root	Links to the top-level sections
A page with sub-pages	Section	Its sub-pages; sub-pages that have their own children are toggle pills that open a panel
A page without sub-pages, with suggestions	Suggestions	One button per suggestion (page tags or script-supplied); clicking one searches for it
A page without sub-pages, no suggestions	Parent's section (root if the parent is Home)	Its siblings

- A **Back** button (arrow-left, `aria-label` "Back") appears in every view except Root and goes up one level (suggestions to the parent's section, a section to its parent's section, then Root).
- The **menu button**: when the default view is Root, it shows / hides the navigation (`aria-expanded`); otherwise it switches between the full menu (Root) and the default view (`aria-pressed`).
- Toggle pills (section view) open a panel listing the sub-pages **in place of the search row**; 6 or more sub-pages use two balanced columns (wider panel). Escape or a click outside closes the panel and returns focus to its pill.
- Pills for the current page or its ancestors are outlined in blue (`aria-current="page"` for the page itself, `"true"` for ancestors).
- Page titles in pills and panel drop anything after a `|` (e.g. "Cloud | TCS" shows "Cloud").
- `setConversationCanvasSuggestions(block, [...])` replaces the suggestions (for example prompts from a personalisation service) and re-renders; suggestion text has surrounding quotes stripped before it is submitted.

Rendering Rules: `.conversation-canvas.primary > .conversation-canvas-dock` containing `.conversation-canvas-nav-panel`, `.conversation-canvas-search` (menu toggle and form) and `nav.conversation-canvas-nav` (`aria-label` "Site"). The block wrapper is `display: contents`. The dock fades and slides in (0.3 s; no transition with reduced motion).

Responsive Behaviour: Desktop and tablet >= 900 px: 5rem top padding over a white gradient fade, search row up to 37.9375rem with the menu button, navigation pills (3rem tall, 1.125rem text, blue `#4e84ff` active border) and the panel. Below 900 px: only the search pill (up to 21.25rem wide, grey translucent, blurred), no menu button, navigation or panel (the site header provides navigation).

SEO Impact: Navigation links are created by script after the query index loads; they are not in the initial HTML. No heading is rendered.

Accessibility Compliance: Hidden dock is `inert` (not focusable, not announced); `nav` with label; current page marked with `aria-current`; panel toggles with `aria-expanded` and `aria-controls`; Escape closes the panel. The panel container has an `aria-label` but no role (Known limitation).

```
import { setConversationCanvasSuggestions }
  from '/blocks/conversation-canvas/conversation-canvas.js';
```

```
const dock = document.querySelector('.conversation-canvas.primary');
setConversationCanvasSuggestions(dock, ['"What is agentic AI?"', 'Cloud migration']);
```

Variation 4 — Title Type: Default

Variation Name: Title Type = Default (no class).

Business Purpose: Standard title size for the canvas.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-title-default.png`
> **Purpose:** Default title size
> **Description:** Canvas title at 4rem on desktop.

Use Cases: Most canvases.

Authoring Guidelines: Leave **Title Type** on **Default**.

Business Rules: The title is always an `h2`; Title Type changes only its size.

Rendering Rules: 2.5rem / line height 1.2 / –0.05rem letter-spacing (mobile); 3.125rem / –0.0625rem (>= 600 px); 4rem / -0.08rem (>= 1200 px). 1.5rem extra space after an icon or eyebrow.

Responsive Behaviour: As above per breakpoint.

SEO Impact: One `h2`.

Accessibility Compliance: Heading level is fixed at 2; check the page outline (Page Header `h1` above it).

Variation 5 — Title Type: Bigger

Variation Name: Title Type = Bigger Title (class `conversation-canvas-title-bigger`).

Business Purpose: A hero-scale statement title.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-title-bigger.png`
> **Purpose:** Bigger title
> **Description:** Canvas with a 5rem title on desktop.

Use Cases: Home page canvas, campaign pages.

Authoring Guidelines: Set **Title Type** = **Bigger Title**; keep the title short (up to about 40 characters).

Business Rules: Class only; still an `h2`.

Rendering Rules: 2.75rem / –0.0825rem (mobile); 4rem / –0.12rem (>= 600 px); 5rem / –0.15rem (>= 1200 px).

Responsive Behaviour: As above.

SEO Impact: None beyond the `h2`.

Accessibility Compliance: Large text; ensure it reflows at 400 % zoom (short titles).

Variation 6 — Title Type: Medium

Variation Name: Title Type = Medium Title (class `conversation-canvas-title-medium`).

Business Purpose: A smaller title for secondary placements (for example lower on the page).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-title-medium.png`
> **Purpose:** Medium title
> **Description:** Canvas with a 2.5rem title on desktop and line height 1.4.

Use Cases: Mid-page canvases, footer-adjacent panels.

Authoring Guidelines: Set **Title Type** = **Medium Title**.

Business Rules: Class only; still an `h2`.

Rendering Rules: 1.75rem / line height 1.4 / –0.0438rem (mobile); 2rem / –0.05rem (>= 600 px); 2.5rem / –0.0625rem (>= 1200 px).

Responsive Behaviour: As above.

SEO Impact: None beyond the `h2`.

Accessibility Compliance: As Variation 4.

Variation 7 — BG Style: Default (glow)

Variation Name: BG Style = Default (class `conversation-canvas-bg-default`), no background image.

Business Purpose: The brand's soft gradient look without an image.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-bg-default.png`
> **Purpose:** Default background glow
> **Description:** White panel with blue (left) and yellow (right) radial glows rising from the bottom edge.

Use Cases: Any canvas without photography.

Authoring Guidelines: Leave **BG Style** = **Default** (the only option) and **BG Background Image** empty.

Business Rules: BG Style has one option; the class `conversation-canvas-bg-{value}` is always added (`default` when empty) but has no CSS of its own today — it is a hook for future styles.

Rendering Rules: `::before` with two radial gradients (blue `rgb(70 115 219 / 22%)`, yellow `rgb(232 185 49 / 22%)`; Secondary 25 % with swapped positions) on `--color-bg-default`.

Responsive Behaviour: The glow scales with the panel at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Decorative; text contrast is on the white background.

Variation 8 — Background image

Variation Name: BG Background Image set (class `conversation-canvas-has-background`).

Business Purpose: A photographic or illustrated canvas.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-background-image.png`
> **Purpose:** Background image
> **Description:** Canvas with a light abstract image filling the rounded panel behind the centred title and search row.

Use Cases: Campaign or event canvases.

Authoring Guidelines: Pick an image in **BG Background Image** from AEM Assets. Choose a light, low-contrast image (the text colour stays dark).

Business Rules: The image is decorative (`alt=""` unless the asset already has one); the glow is removed. Site media (`/media_` URLs) is optimized to a responsive `<picture>`; other URLs are used as is with `loading="lazy"`. Not rendered in the Primary style.

Rendering Rules: `.conversation-canvas-background` absolutely positioned behind the content, `object-fit: cover`, clipped to the 1.25rem radius.

Responsive Behaviour: Covers the panel at every breakpoint; cropping changes with the panel height.

SEO Impact: Decorative image, no alt text value for SEO.

Accessibility Compliance: Verify text contrast (WCAG 1.4.3, 4.5:1) over the image; the code does not add an overlay.

Variation 9 — Search button icon type: Primary

Variation Name: Search bar button icon type = Primary.

Business Purpose: The default, most prominent submit button.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-search-primary.png`
> **Purpose:** Primary search button
> **Description:** Filled brand-blue round arrow button inside the search pill.

Use Cases: Most canvases.

Authoring Guidelines: **Search bar** > **Search bar button icon type** = **Primary** (default).

Business Rules: Icon button (arrow-right) of type primary; its accessible name and tooltip is **Search bar button icon type hover text**, else "Submit".

Rendering Rules: `button.button.icon-button.primary` (2.75rem, brand background, white icon; hover brand-hover).

Responsive Behaviour: Same size at every breakpoint.

SEO Impact: None.

Accessibility Compliance: `aria-label` and `title` from the hover text; visible focus style.

Variation 10 — Search button icon type: Secondary

Variation Name: Search bar button icon type = Secondary.

Business Purpose: A lighter, outlined submit button.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/conversation-canvas-search-secondary.png`
> **Purpose:** Secondary search button
> **Description:** Round arrow button with a brand-blue outline and transparent fill.

Use Cases: Canvases on image backgrounds where a filled button is too heavy.

Authoring Guidelines: Set **Search bar button icon type** = **Secondary**.

Business Rules: As Variation 9 with type secondary.

Rendering Rules: `icon-button.secondary`: brand border, transparent background, brand icon; hover secondary brand colours.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: As Variation 9.

Variation 11 — Search button icon type: Tertiary

Variation Name: Search bar button icon type = Tertiary.

Business Purpose: A minimal icon-only submit.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/conversation-canvas-search-tertiary.png`
> **Purpose:** Tertiary search button
> **Description:** Brand-blue arrow icon without fill or border inside the search pill.

Use Cases: Minimal designs.

Authoring Guidelines: Set **Search bar button icon type** = **Tertiary**.

Business Rules: As Variation 9 with type tertiary; any unknown value falls back to primary.

Rendering Rules: `icon-button.tertiary`: brand icon colour only.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: As Variation 9; the button boundary is less visible, so the focus outline is important.

Variation 12 — Voice input with the default microphone

Variation Name: Audio button, default icon (class `conversation-canvas-audio-default`).

Business Purpose: Let visitors dictate their question.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/conversation-canvas-voice-default.png`

> **Purpose:** Voice input
> **Description:** Search pill with the microphone icon highlighted in brand blue with an outline while listening.

Use Cases: All canvases (automatic).

Authoring Guidelines: Leave **Search bar audio icon** empty; optionally set **Search bar audio icon alt text** (the button's name, default "Search by voice").

Business Rules: Shown only where the browser supports the Web Speech API (for example Chromium-based browsers and Safari); hidden elsewhere (for example Firefox). A click starts recognition in the page language (`<html lang>`, else `en`); the first result is written into the field and the field is focused — the query is **not** submitted automatically. A second click stops listening. Errors or the end of speech reset the button. Browsers may process the audio with the vendor's online speech service (privacy statement review recommended: [To be confirmed: privacy review of voice input — owner: data protection officer]).

Rendering Rules: 2rem round button with a 1.5rem `mic.svg` mask; `aria-pressed="true"` while listening (brand colour and outline).

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: Toggle button with `aria-label` and `aria-pressed`; the browser asks for microphone permission.

Variation 13 — Voice input with a custom audio icon

Variation Name: Audio button with **Search bar audio icon** set.

Business Purpose: Brand-specific microphone artwork.

Visual Example:

> [Screenshot Placeholder]
> **File name:** `screenshots/conversation-canvas-voice-custom-icon.png`
> **Purpose:** Custom audio icon
> **Description:** Search pill with a custom 1.5rem audio icon image in place of the default microphone.

Use Cases: Campaigns with their own icon set.

Authoring Guidelines: Pick an SVG or PNG in **Search bar audio icon**; set **Search bar audio icon alt text** to describe the action ("Search by voice").

Business Rules: The image is decorative (`alt=""`); the button's name is the alt text field. Same browser support rules as Variation 12.

Rendering Rules: `<img>` 1.5rem inside the audio button.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: None.

Accessibility Compliance: As Variation 12; the icon colour does not follow the pressed state (only the outline does).

Variation 14 — Canvas button, Views = Button

Variation Name: Conversation Canvas Button with **Views** = Button.

Business Purpose: Calls to action below the search row.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-button-view.png`
> **Purpose:** Button items
> **Description:** Two centred pill buttons below the search row: a black primary "Contact us" and an outlined secondary "Explore services", each with a trailing arrow.

Use Cases: "Contact us", "Explore our services".

Authoring Guidelines: Add > **Conversation Canvas Button**; set **Link**, **Text**, optional **Title**, **Views** = Button, **Type** (Primary / Secondary / Tertiary).

Business Rules: The Type field is shown when Views is Button or Icon (model condition). **Alignment** has no effect here (buttons are always centred). A Title equal to the text is dropped. Rows without a link are ignored. Not rendered in the Primary style.

Rendering Rules: `ul.conversation-canvas-buttons > li > a.button.{type}` from the button decorator.

Responsive Behaviour: Wrap and centre at every breakpoint (in the footer canvas they stack in a column on mobile).

SEO Impact: Crawlable links; use descriptive text.

Accessibility Compliance: Real links; the title attribute is a tooltip only.

Variation 15 — Canvas button, Views = Icon

Variation Name: Conversation Canvas Button with **Views** = Icon.

Business Purpose: Compact icon-only action (for example a search or arrow shortcut).

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-icon-view.png`
> **Purpose:** Icon button item
> **Description:** A round brand-blue icon button with a chevron-right icon below the search row.

Use Cases: Secondary shortcuts.

Authoring Guidelines: Set **Views** = Icon, choose **Icon** (Arrow right, Chevron right, Chevron down, Search) and **Type**; the **Text** becomes the accessible name.

Business Rules: The Icon field is shown only when Views is Icon (model condition).

Rendering Rules: `a.button.icon-button.{type}` with the icon as a CSS mask; the text is kept for screen readers.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: The link text stays in the HTML (hidden with CSS) and becomes the `aria-label`.

Accessibility Compliance: `aria-label` and title from the text.

Variation 16 — Canvas button, Views = Link

Variation Name: Conversation Canvas Button with **Views** = Link.

Business Purpose: A low-emphasis text link.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-link-view.png`
> **Purpose:** Link item
> **Description:** A text link "View all FAQs" with a trailing chevron below the search row.

Use Cases: "View all FAQs", "Browse topics".

Authoring Guidelines: Set **Views** = Link (Type and Icon are hidden).

Business Rules: Rendered as a link button (no type).

Rendering Rules: `a.link-button`.

Responsive Behaviour: Same at every breakpoint.

SEO Impact: Crawlable link.

Accessibility Compliance: Underline / chevron convey a link.

Variation 17 — Floating search row (Default and Secondary)

Variation Name: Floating search row (class `conversation-canvas-floating` on the row).

Business Purpose: Keep the search reachable after the visitor scrolls past the canvas.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-floating-row.png`
> **Purpose:** Floating search row
> **Description:** Mid-page scroll position with the menu button and search pill floating centred 1.5rem above the bottom of the viewport, with shadow and blur.

Use Cases: Automatic for every Default / Secondary canvas.

Authoring Guidelines: No setting. To avoid it, do not place a canvas high on very long pages or use only one canvas per page.

Business Rules: Across all canvases on a page at most one floats: the first registered canvas that qualifies. A Default / Secondary row floats when its place has scrolled above the viewport and no other Default / Secondary row is visible in place (for example the footer canvas). The row's place keeps its height, so the page does not jump. Positions are updated on scroll, resize and body size changes.

Rendering Rules: `position: fixed; bottom: 1.5rem;` centred, width `min(42.875rem, 100vw - 3rem)`, 90 % white with blur and shadow, 0.3 s float-in animation (off with reduced motion).

Responsive Behaviour: Same at every breakpoint (narrower on mobile).

SEO Impact: None.

Accessibility Compliance: The floating row keeps its focus order (it is the same element); the menu opens above it.

Variation 18 — Footer canvas placement

Variation Name: Conversation Canvas in the footer (inside `.footer-canvas`).

Business Purpose: The "Ask us a question" area at the top of the site footer, shared by all pages that use the footer.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-footer.png`
> **Purpose:** Footer canvas
> **Description:** Site footer with a centred 3.25rem balanced title, the search row 38.0625rem wide and stacked buttons on mobile, above the copyright and footer links bar.

Use Cases: Site-wide footer entry point.

Authoring Guidelines: Create a page in the locale's `references/` folder (the English sample is `references/footer-conversation-canvas`) with a Default-style canvas; select it in the Footer block's **Conversation Canvas Page** on the footer page; publish both.

Business Rules: The footer loads the page as a fragment and removes the canvas surface, glow and padding; the menu button is hidden below 600 px (`--footer-menu-display`). Additional tab spacing and Hide block do not apply here (they are scoped to `main`). While the footer canvas is in view, no other canvas floats.

Rendering Rules: Footer CSS overrides title size (3.25rem, max 20rem, balanced), gaps and search widths (see the Footer chapter).

Responsive Behaviour: Mobile: no menu button, buttons stacked; >= 600 px: menu button shown; >= 1200 px: footer desktop sizes.

SEO Impact: The footer canvas text appears on every page using that footer.

Accessibility Compliance: As Default; the title is an `h2` inside `footer`.

6.30.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** A Default-style Conversation Canvas selected in Universal Editor with the letters A–I marking the elements listed below.

Annotations:

- A = Background (BG Background Image or the default glow)
- B = Icon (with Icon Alt Text)
- C = Eyebrow
- D = Title (size from Title Type)
- E = Text (rich text)
- F = Menu button (menu from Menu root path / Menu depth)
- G = Search field (Search bar water mark)
- H = Audio button (Search bar audio icon / alt text) and submit icon button (button icon type / hover text)
- I = Conversation Canvas Button items

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-properties-panel.png`

> **Purpose:** Properties panel
> **Description:** Properties panel with the Content tab (Canvas Style, BG Style, images, texts, collapsible Menu and Search bar groups) and the Additional tab.

Content tab: Canvas Style (Default, Primary (sticky), Secondary); BG Style (Default); BG Background Image (asset picker); Icon (asset picker); Icon Alt Text; Eyebrow (placeholder text "Infrastructure to Intelligence"); Title (placeholder "Statistic"); Title Type (Default, Bigger Title, Medium Title); Text (rich text); **Menu** group (collapsible): Menu root path (page picker), Menu depth (number 1–3, default 1); **Search bar** group (collapsible): Search bar water mark (placeholder "Ask us a question"), Search bar audio icon (asset picker), Search bar audio icon alt text, Search bar button icon type (Primary, Secondary, Tertiary), Search bar button icon type hover text (placeholder "Submit").

Additional tab: Anchor ID, Hide block, Padding left / right / top / bottom, Margin left / right / top / bottom.

Conversation Canvas Button item (model `button`): Link (page picker / URL), Text, Title, Views (Button, Icon, Link), Type (Primary, Secondary, Tertiary; shown for Button and Icon), Icon (Arrow right, Chevron right, Chevron down, Search; shown for Icon), Alignment (Left, Center, Right; not used by this block).

> **Note:** the dialog shows every field for every Canvas Style (there are no model `conditions` on the block fields); fields not used by the chosen style are ignored (see 6.30.6).

6.30.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: In AEM Sites open the page (for example under `/content/tcs-corp/us/en`) and choose **Edit**.

Expected Result: The page opens in Universal Editor.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** AEM Sites console with the page selected and Edit highlighted.

Step 2 — Add the block

Action: Select a section, choose **Add** (+) > **Conversation Canvas**. For the Primary style, add a new section for it.

Expected Result: A canvas with an empty search row appears.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** Section Add menu with "Conversation Canvas" highlighted.

Step 3 — Choose the style and background

Action: In **Content** choose **Canvas Style**; keep **BG Style** = Default; optionally pick a **BG Background Image**.

Expected Result: The panel changes to the chosen style; Primary shows only the dock (visible when scrolling, see Step 9).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-03-style.png`
> **Purpose:** Style
> **Description:** Canvas Style select open with Default, Primary (sticky), Secondary.

Step 4 — Fill the intro

Action: Pick an **Icon** and enter **Icon Alt Text** (empty for a decorative icon); enter **Eyebrow**, **Title**, **Title Type** and **Text**.

Expected Result: The intro appears centred above the search row.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-04-intro.png`
> **Purpose:** Intro fields
> **Description:** Canvas with icon, eyebrow, title and text filled.

Step 5 — Configure the menu

Action: Open the **Menu** group; select the **Menu root path** page (leave empty for the locale Home) and set **Menu depth** (1–3).

Expected Result: On the published page the menu lists the pages below the root.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-05-menu.png`
> **Purpose:** Menu settings
> **Description:** Menu group with root path /content/tcs-corp/us/en and depth 2.

Step 6 — Configure the search bar

Action: Open **Search bar**: water mark text, optional audio icon and its alt text, button icon type and hover text.

Expected Result: The field shows the water mark; the arrow button's tooltip shows the hover text.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-06-search-bar.png`
> **Purpose:** Search bar settings
> **Description:** Search bar group with water mark "Ask us a question" and hover text "Submit".

Step 7 — Add buttons

Action: With the block selected, **Add** > **Conversation Canvas Button**; set Link, Text, Views, Type (and Icon for the Icon view). Repeat as needed.

Expected Result: Buttons appear centred below the search row (not in Primary).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-07-buttons.png`
> **Purpose:** Buttons
> **Description:** Two button items in the content tree and on the canvas.

Step 8 — Fill the Additional tab (optional)

Action: Set an **Anchor ID** if links should jump to the canvas; leave spacing on Default for the Primary style.

Expected Result: The block id is set on the page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-08-additional.png`
> **Purpose:** Additional tab
> **Description:** Additional tab with Anchor ID "anchor-ask-us".

Step 9 — Preview

Action: Publish to preview and open `https://main--tcs-corp--sradha-tcs.aem.page/{path}`. Open the menu, scroll down and up, and resize below and above 900 px.

Expected Result: The menu lists pages (the query index is not available inside the editor, where it may show "No pages found"); the search row floats after scrolling past it; a Primary dock appears while the footer is out of view.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-09-preview.png`
> **Purpose:** Preview
> **Description:** Preview page with the menu open above the search row.

Step 10 — Validate the search hand-off

Action: Type a question and press Enter; in the browser console, a listener for `conversation-canvas:search` (development team) receives the query.

Expected Result: No navigation happens (no search service); the event carries the query.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-10-search-event.png`
> **Purpose:** Search event
> **Description:** Browser console showing the conversation-canvas:search event detail.

Step 11 — Publish

Action: Publish the page (and the footer and canvas reference pages if changed).

Expected Result: The canvas is live on `.aem.live`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/conversation-canvas-step-11-publish.png`
> **Purpose:** Publish
> **Description:** Universal Editor Publish dialog.

Preview Process

1. Publish to preview; open the `.aem.page` URL (the menu and dock need `/query-index.json`, which is not served inside Universal Editor).

2. Check desktop (>= 1200 px), tablet (600–1199 px; the Primary dock switches at 900 px) and mobile (< 600 px).
3. Scroll to the footer: the floating row or dock must hide when the footer appears.

Publishing Process

1. Publish the page; newly published pages appear in the menu only after the query index has been updated for them (automatic on publish).
2. For the footer canvas, publish the canvas reference page and the footer page.
3. Publish the taxonomy and placeholders sheets if suggestion labels or UI texts changed.

Validation Process

- ☐ Style is correct; at most one Primary canvas on the page.
- ☐ Title is concise; Title Type suits the placement; only one `h1` on the page (the canvas title is an `h2`).
- ☐ Icon alt text set only if the icon conveys meaning.
- ☐ Background image keeps the dark text readable (4.5:1).
- ☐ Menu root path is in the same locale; Menu depth 1–3; the menu lists the expected pages.
- ☐ Water mark and hover text are meaningful and localised (no test values).
- ☐ Buttons have descriptive text and working links.
- ☐ Primary: dock appears on scroll, hides at the footer; desktop pills reflect the page's place; Back works; panels open and close with Escape.
- ☐ Voice button appears in supporting browsers and fills the field.
- ☐ Stakeholders know that queries are not answered until a search service is connected.

Rollback Process

1. Turn on **Hide block** and publish (inside `main`; for the footer canvas, clear the Footer's Conversation Canvas Page or unpublish the canvas page).
2. Restore a previous page version in AEM and republish.
3. Code issues: the development team reverts the commit on `main`.

6.30.6 Authorable Field Specification

Block model `conversation-canvas` — Content tab (a key-value block: each field is one `name | value` row)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Canvas Style (`classes`)	select	No	No	Default (`""`)	Chooses Default, Primary (sticky) or Secondary	One Primary per page	Options: Default (`""`), Primary (sticky) (`primary`), Secondary (`secondary`)	primary
BG Style (`bgStyle`)	select	No	No	Default (`default`)	Background style hook	Leave Default	One option: Default (`default`); becomes class `conversation-canvas-bg-default`	default
BG Background Image (`backgroundImage`)	reference (asset)	No	No	(empty)	Decorative background image	Light, low-detail image	Single asset; not used by Primary	/content/dam/tcs/canvas-bg.jpg
Icon (`icon`)	reference (asset)	No	No	(empty)	Small icon above the eyebrow	SVG or PNG, square	Single asset; shown at 2.5rem; not used by Primary	/content/dam/tcs/icons/spark.svg

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Icon Alt Text (`iconAlt`)	text	No	No	(empty)	Alternative text of the icon	Empty when decorative	None	(empty)
Eyebrow (`eyebrow`)	text	No	No	(empty)	Small label above the title	1–4 words	Plain text; dialog placeholder “Infrastructure to Intelligence”; not used by Primary	Infrastructure to Intelligence
Title (`title`)	text	No	No	(empty)	Main heading (h2)	Short question or promise	Plain text; dialog placeholder “Statistic”; not used by Primary	How can we help you today?
Title Type (`titleSize`)	select	No	No	Default (`""`)	Title size	Bigger for hero use, Medium lower on the page	Options: Default (`""`), Bigger Title (`bigger`), Medium Title (`medium`)	bigger
Text (`text`)	rich-text	No	No	(empty)	Supporting text	1–2 sentences	Rich text kept; not used by Primary	Ask about our services, industries or insights.
Menu root path (`menuRootPath`, group Menu)	aem-content	No	No	(empty = locale Home)	Root of the menu / navigation tree	A page in the same locale	Converted to the published path; empty = `getLocale().root` (e.g. `/en`)	/content/tcs-corp/us/en
Menu depth (`menuDepth`, group Menu)	number	No	No	1	Levels below the root	1 for sections only, 2–3 for deeper menus	Enforced in the dialog: min 1, max 3; code: invalid = 1, minimum 1	2
Search bar watermark (`searchPlaceholder`, group Search bar)	text	No	No	(empty = placeholder `searchPlaceholder` label)	Placeholder and hidden label of the search field	A short invitation	Plain text; dialog placeholder “Ask us a question”	Ask us a question
Search bar audio icon (`searchAudioIcon`)	reference (asset)	No	No	(empty = mic icon)	Custom microphone image	24 x 24 icon	Single asset	/content/dam/tcs/icons/mic.svg
Search bar audio icon alt text (`searchAudioIconAlt`)	text	No	No	(empty = placeholder `searchByVoice`)	Accessible name of the voice button	Describe the action	Plain text	Search by voice
Search bar button icon type (`searchButtonStyle`)	select	No	No	Primary (`primary`)	Submit button style	Primary unless the design asks otherwise	Options: Primary, Secondary, Tertiary; unknown = primary	primary
Search bar button icon type hover text (`searchButtonLabel`)	text	No	No	(empty = placeholder `submit`)	Accessible name and tooltip of the submit button	A verb such as “Submit” or “Ask”	Plain text; dialog placeholder “Submit”	Submit

Block model `conversation-canvas` — Additional tab

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	text	No	No	(empty)	Jump target	Unique, starts with `anchor-`	Enforced regExp `^anchor-[a-z0-9][a-z0-9-]*$`	anchor-ask-us
Hide block (`classes_hideblock`)	boolean	No	No	false	Hide without deleting	Quick rollback	None; applies inside `main`	false
Padding left (`classes_paddingleft`)	select	No	No	Default	Inner space left	Avoid on Primary	Default, `pl-0` … `pl-3000` (15 options)	pl-0
Padding right (`classes_paddingright`)	select	No	No	Default	Inner space right	Avoid on Primary	Default, `pr-0` … `pr-3000`	pr-0
Padding top (`classes_paddingtop`)	select	No	No	Default	Inner space top	Avoid on Primary	Default, `pt-0` … `pt-3000`	pt-1600
Padding bottom (`classes_paddingbottom`)	select	No	No	Default	Inner space bottom	Avoid on Primary	Default, `pb-0` … `pb-3000`	pb-1600
Margin left (`classes_marginleft`)	select	No	No	Default	Outer space left	Rarely needed	Default, `ml-0` … `ml-3000`	ml-0
Margin right (`classes_marginright`)	select	No	No	Default	Outer space right	Rarely needed	Default, `mr-0` … `mr-3000`	mr-0
Margin top (`classes_margintop`)	select	No	No	Default	Outer space top	Separate from previous block	Default, `mt-0` … `mt-3000`	mt-1200
Margin bottom (`classes_marginbottom`)	select	No	No	Default	Outer space bottom	Separate from next block	Default, `mb-0` … `mb-3000`	mb-1200

The spacing options are the same 15 values as in 6.29.6 (Default, 0, 0.25rem … 7.5rem from `--space-100` … `--space-3000`).

Item model `button` — Conversation Canvas Button (`models/_button.json`)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Link (`link`)	aem-content	No (a row without a link is ignored)	No	(empty)	Destination	Page picker or URL	Required in practice: rows without a `[href]` are skipped	/content/tcs-corp/us/en/contact-us
Text (`linkText`)	text	No	No	(empty = the URL)	Visible text / accessible name	Action verb + object	None	Contact us
Title (`linkTitle`)	text	No	No	(empty)	Tooltip	Only if it adds information	Dropped when equal to the text	Talk to our experts

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Views (`buttonView`)	select	No	No	Button (`button`)	Button, icon-only or text link	Button for CTAs	Options: Button, Icon, Link	button
Type (`linkType`)	select	No	No	Primary (`primary`)	Emphasis	One primary per canvas	Options: Primary, Secondary, Tertiary; condition: shown when Views is empty, Button or Icon	secondary
Icon (`buttonIcon`)	select	No	No	(none)	Icon of the Icon view	—	Options: Arrow right, Chevron right, Chevron down, Search; condition: shown when Views = Icon	arrow-right
Alignment (`buttonAlignment`)	select	No	No	Left (`left`)	Alignment (not used by this block)	Leave as is	Options: Left, Center, Right; ignored, buttons are centred	left

Fields used per style

Field	Default	Secondary	Primary (sticky)
BG Background Image, Icon, Icon Alt Text, Eyebrow, Title, Title Type, Text	Used	Used	Ignored
Menu root path, Menu depth	Pop-up menu	Pop-up menu	Dock navigation
Search bar fields	Used	Used	Used
Conversation Canvas Button items	Used	Used	Ignored
Filter tags (script only)	—	`setConversationCanvasFilters`	
Suggestions (page tags or script)	—	—	`setConversationCanvasSuggestions`

Field details

- **Canvas Style** — Purpose: behaviour and layout mode. Allowed values: Default, Primary (sticky), Secondary. Character limits: n/a. Business rules: Primary renders only the dock; Secondary adds the filter list. Dependency rules: Primary ignores intro and buttons. Visibility rules: no model condition. Sample: Default.
- **BG Style** — Purpose: future background variants. Allowed values: Default only. Business rules: class `conversation-canvas-bg-default` (no CSS yet). Sample: default.
- **BG Background Image** — Purpose: decorative background. Allowed values: one image asset (or a URL). Business rules: replaces the glow; optimized when served as site media; `alt=""`. Dependency: not used in Primary. Sample: a light abstract image.
- **Icon** — Purpose: small visual cue. Allowed values: one image. Business rules: 2.5rem, `object-fit: contain`. Dependency: alt from Icon Alt Text. Sample: a spark icon.
- **Icon Alt Text** — Purpose: text alternative. Character limits: Recommended up to 80. Business rules: applied only when not empty; else the icon is decorative. Sample: empty.
- **Eyebrow** — Purpose: context label. Character limits: Recommended up to 40. Business rules: plain text paragraph in brand colour. Sample: “Infrastructure to Intelligence”.
- **Title** — Purpose: main heading. Character limits: Recommended up to 60 (Bigger: up to 40). Business rules: always `h2`, plain text (formatting is not kept). Sample: “How can we help you today?”.
- **Title Type** — Purpose: size. Allowed values: Default, Bigger Title, Medium Title. Business rules: class `conversation-canvas-title-{bigger|medium}`. Sample: Default.

- **Text** — Purpose: supporting copy. Character limits: Recommended up to 200. Business rules: rich text moved as is, margins removed, max width 50.75rem. Sample: one sentence.
- **Menu root path** — Purpose: tree root. Allowed values: a page of the same locale. Business rules: AEM path converted with `toSitePath` (for example `/content/tcs-corp/us/en` to `/en`); pages are read from the query index filtered to the **current** locale, so a root in another locale gives "No pages found". Sample: `/content/tcs-corp/us/en/what-we-do`.
- **Menu depth** — Purpose: levels below the root. Allowed values: 1–3 (Enforced in the dialog). Business rules: pages deeper than the depth are not listed; a page whose parent is not in the index is skipped. Sample: 2.
- **Search bar water mark** — Purpose: input placeholder and its hidden label. Character limits: Recommended up to 40 (the field is narrow on mobile). Business rules: empty = "Ask us a question" (`searchPlaceholder`). Sample: "Ask us a question".
- **Search bar audio icon / alt text** — Purpose: voice button artwork and name. Business rules: alt text empty = "Search by voice" (`searchByVoice`); the image itself is decorative. Sample: empty (default mic).
- **Search bar button icon type** — Purpose: submit style. Allowed values: Primary, Secondary, Tertiary. Sample: Primary.
- **Search bar button icon type hover text** — Purpose: submit accessible name and tooltip. Character limits: Recommended up to 20. Business rules: empty = "Submit" (`submit`). Sample: "Submit".
- **Anchor ID, Hide block, Padding, Margin** — as described in 6.29.6; in the Primary style, padding or margin only add empty space (the dock is fixed and takes no room) and should stay Default; inside the footer these options have no effect.
- **Button item fields** — Link (destination; required in practice), Text (Recommended up to 25 characters), Title (tooltip), Views, Type (condition: Views empty / Button / Icon), Icon (condition: Views = Icon), Alignment (no effect in this block).

6.30.7 Field Annotation Mapping

Marker	Field	Purpose
A	BG Background Image (`backgroundImage`) / BG Style	Background image or default glow
B	Icon (`icon`), Icon Alt Text (`iconAlt`)	Small icon above the eyebrow
C	Eyebrow (`eyebrow`)	Context label
D	Title (`title`), Title Type (`titleSize`)	Main h2 heading and its size
E	Text (`text`)	Supporting rich text
F	Menu root path (`menuRootPath`), Menu depth (`menuDepth`)	Menu button and pop-up menu / dock navigation
G	Search bar water mark (`searchPlaceholder`)	Search field placeholder and label
H	Search bar audio icon / alt text; button icon type / hover text	Voice and submit buttons
I	Conversation Canvas Button items (`button` model)	Calls to action

6.30.8 Component Specification

Attribute	Value
Component Name	Conversation Canvas (`conversation-canvas`)
Version	1.0
Owner	[To be confirmed: product owner]; technical owner: development team

Attribute	Value
Dependencies	aem.js, scripts.js; responsive-image, a11y, button, icon-button, block-options, data-layer decorators; utils buttons, properties, query-index, site, urls, metadata, taxonomy, ids, placeholders; query index, taxonomy and placeholders sheets; Web Speech API
Component Type	Container block (key-value properties) with `button` items (Conversation Canvas Button)
Supported Variants	Canvas Style Default / Secondary / Primary (sticky); Title Type Default / Bigger / Medium; BG glow or image; search button Primary / Secondary / Tertiary; voice input; button views Button / Icon / Link; footer placement
Supported Templates	Any page; the footer canvas reference page
Supported Metadata	Page tags (`article:tag` meta, else `tags` metadata) for Primary suggestions; query index `path` and `title` (og:title)
Supported Tags	Page tags, labelled from `/{locale}/taxonomy.json` (Primary suggestions)
Supported Personalization	Script APIs `setConversationCanvasSuggestions` and `setConversationCanvasFilters` for a personalisation service (none connected); data layer `cmp:search` for Adobe Target / Journey Optimizer in future (Adobe Tags not installed)
Caching Considerations	Page HTML cached by the CDN; `/query-index.json` fetched in pages of 500 rows once per page view and cached in memory; taxonomy and placeholders loaded once per page view; menu changes appear after the query index updates on publish
Performance Considerations	Default / Secondary: the query index loads only when the menu is first opened; Primary: loads the index and taxonomy at decoration; images use `createOptimizedPicture` with lazy loading (a background image in the first section is not eager); listeners on scroll (passive, one animation frame per update), resize and a body ResizeObserver, shared by all canvases
Accessibility Compliance	Hidden label on the search input; labelled menu toggle with `aria-expanded` / `aria-controls`; labelled `nav`; Escape and click-outside close; `aria-pressed` voice and menu toggles; `aria-current` on the current page; hidden dock `inert`; reduced-motion support; icon buttons with `aria-label` and title
SEO Compliance	One h2 (not in Primary); menu and dock links are created by script; buttons are crawlable links; no structured data

6.30.9 Input Model

Content Inputs: Key-value property rows (`name | value`) for every set field (empty fields are omitted), plus one row per Conversation Canvas Button (a link and its option cells). If no row has a known key, the code falls back to one property per single-cell row in model order.

Metadata Inputs: The page's tags (`meta[property="article:tag"]`, else the `tags` meta) for Primary suggestions.

Tags Inputs: Tag IDs resolved to labels via `/{locale}/taxonomy.json`; unknown IDs are labelled from the ID.

Page Context Inputs: `window.location.pathname` (locale, current page in the tree), the document language (voice recognition), viewport position of the footer and other canvases, and the editor context for the query index (empty inside Universal Editor).

Query Inputs: `/query-index.json` rows of the current locale (`path`, `title`), filtered to the menu root and depth, sorted by level and path.

Manual Inputs: All dialog fields; scripts may add suggestions and filter tags.

Authored HTML received from AEM (the English footer canvas sample, `content/en/references/footer-conversation-canvas.plain.html`, simplified):

```
<div class="conversation-canvas">
  <div><div>bgStyle</div><div>default</div></div>
```

```
  <div><div>menuRootPath</div><div><a href="/en/">/content/tcs-corp/us/en</a></div></div>
  <div><div>menuDepth</div><div>2</div></div>
  <div><div>searchPlaceholder</div><div>Ask us a question</div></div>
  <div><div>searchButtonStyle</div><div>primary</div></div>
  <div><div>searchButtonLabel</div><div>sradha</div></div>
</div>
```

> **Warning:** in this sample the **Search bar button icon type hover text** is "sradha", a test value that becomes the submit button's accessible name and tooltip. [To be confirmed: the live English footer canvas uses a real label such as "Submit" — owner: content lead]

6.30.10 Output Model

Rendered Output (Default / Secondary): The block keeps its classes and gets `conversation-canvas-bg-{style}`, optional `conversation-canvas-title-{size}` and `conversation-canvas-has-background`. Children: optional `.conversation-canvas-background`, then `.conversation-canvas-content` with `.conversation-canvas-intro` (`.conversation-canvas-icon`, `p.conversation-canvas-eyebrow`, `h2.conversation-canvas-title`, `.conversation-canvas-text`), `.conversation-canvas-search-slot` > `.conversation-canvas-search` (menu toggle, `nav.conversation-canvas-menu`, `form.conversation-canvas-search-form`), `ul.conversation-canvas-filters` (Secondary) and `ul.conversation-canvas-buttons`.

Rendered Output (Primary): Only `.conversation-canvas-dock` with `.conversation-canvas-nav-panel`, `.conversation-canvas-search` (menu toggle, form) and `nav.conversation-canvas-nav` > `ul.conversation-canvas-nav-list`.

Generated HTML (Default, simplified, menu opened):

```
<div class="conversation-canvas block conversation-canvas-bg-default"
  data-block-name="conversation-canvas">
  <div class="conversation-canvas-content">
    <div class="conversation-canvas-intro">
      <p class="conversation-canvas-eyebrow">Infrastructure to Intelligence</p>
      <h2 class="conversation-canvas-title">How can we help you today?</h2>
    </div>
    <div class="conversation-canvas-search-slot">
      <div class="conversation-canvas-search">
        <button type="button" class="conversation-canvas-menu-toggle" aria-label="Menu"
          aria-controls="conversation-canvas-1-menu" aria-expanded="true"></button>
        <nav class="conversation-canvas-menu" id="conversation-canvas-1-menu"
          aria-label="Menu">
          <ul><li><a href="/en/what-we-do">What we do</a><ul>…</ul></li></ul>
        </nav>
        <form class="conversation-canvas-search-form">
          <label class="visually-hidden" for="conversation-canvas-1-q">
            Ask us a question</label>
          <input type="search" name="q" id="conversation-canvas-1-q"
            placeholder="Ask us a question" autocomplete="off">
          <button type="button"
            class="conversation-canvas-audio conversation-canvas-audio-default"
            aria-label="Search by voice" aria-pressed="false"></button>
          <button type="submit" class="button primary icon-button" aria-label="Submit"
            title="Submit"></button>
        </form>
      </div>
```

```
    </div>
    <ul class="conversation-canvas-buttons">
      <li><a href="/en/contact-us" class="button primary">Contact us</a></li>
    </ul>
  </div>
</div>
```

Generated HTML (Primary, section view, simplified):

```
<div class="conversation-canvas block primary conversation-canvas-bg-default">
  <div class="conversation-canvas-dock conversation-canvas-floating">
    <div class="conversation-canvas-nav-panel" id="conversation-canvas-1-panel" hidden></
div>
    <div class="conversation-canvas-search">
      <button type="button" class="conversation-canvas-menu-toggle" aria-label="Menu"
        aria-controls="conversation-canvas-1-nav" aria-pressed="false"></button>
      <form class="conversation-canvas-search-form">…</form>
    </div>
    <nav class="conversation-canvas-nav" id="conversation-canvas-1-nav" aria-label="Site">
      <ul class="conversation-canvas-nav-list">
        <li><button type="button" class="conversation-canvas-nav-back"
          aria-label="Back"></button></li>
        <li><button type="button" class="conversation-canvas-nav-item
          conversation-canvas-nav-toggle" aria-expanded="false"
          aria-controls="conversation-canvas-1-panel">Services</button></li>
        <li><a class="conversation-canvas-nav-item conversation-canvas-nav-current"
          href="/en/what-we-do/industries" aria-current="page">Industries</a></li>
      </ul>
    </nav>
  </div>
</div>
```

Generated JSON / events: No JSON is requested from a service. A non-empty submit (Enter, the arrow button, or a suggestion pill) dispatches:

```
new CustomEvent('conversation-canvas:search', {
  bubbles: true,
  detail: { query: 'How do you support cloud migration?' },
});
```

A future search integration listens for it:

```
document.addEventListener('conversation-canvas:search', (e) => {
  const { query } = e.detail; // e.target is the canvas block
  // open the search / assistant experience here (not implemented as of this handover)
});
```

Public functions exported by `conversation-canvas.js`:

Function	Arguments	Effect
`setConversationCanvasSuggestions(block, suggestions)`	Primary block; array of strings	Replaces the suggestion pills (shown on pages without sub-pages) and re-renders
`setConversationCanvasFilters(block, tags)`	Secondary block; array of `{ label, href?, active? }`	Renders filter tags (links or toggle buttons); no-op on other styles

Generated Metadata: None.

Analytics Tracking (Adobe Client Data Layer):

- Component state (once): `{ component: { 'conversation-canvas-1': { '@type': 'tcs-corp/components/conversation-canvas', parentId: 'page-…', 'dc:title': 'How can we help you today?' } } }` (no `dc:title` for Primary, which has no heading). Footer canvases are added in the delayed phase.
- `cmp:click` for the menu toggle ("Menu"), menu and pill links (`interaction: link`, `xdm:linkURL`), Back, toggles, voice, submit and suggestion buttons (`interaction: button`), canvas buttons, and filter tags (`interaction: filter`, `value` = label).
- `cmp:search` on every submit of the search form:

```
{ "event": "cmp:search", "eventInfo": { "path": "component.conversation-canvas-1",
  "xdm:searchTerm": "How do you support cloud migration?" } }
```

> **Note:** `cmp:search` is pushed even when the field is empty (`xdm:searchTerm: ""`), because the data layer listens before the block ignores empty queries; the term is trimmed and cut to 100 characters. Visitors may type personal data into a free-text question; the analytics design should review whether search terms may be collected ([To be confirmed: privacy review of search term tracking — owner: data protection officer]).

Personalization Output: None by default. Script-supplied suggestions and filters are the intended personalization outputs.

Search Index Output: The block is not indexed; it reads `/query-index.json`. Its own text (title, text, buttons) is part of the page HTML.

6.30.11 Business Rules

1. Properties are read from two-cell rows whose key (normalised: lower-case, no separators) is a known field; other rows with a link are button items; if no keyed row exists, single-cell rows are mapped to the fields in model order (Canvas Style excluded).
2. Canvas Style: `primary` or `secondary` (from the row or the block classes); anything else is Default.
3. BG Style adds `conversation-canvas-bg-{value}` (`default` when empty).
4. Title Type `bigger` / `medium` adds a size class; anything else is Default.
5. Primary renders only the dock; intro, background, buttons and filters are not rendered.
6. Images: `<picture>` / `<img>` in the cell is used, else the cell's link or text is built into an image (site `/media_` URLs optimized, others lazy `<img>`); the background is decorative; the icon takes Icon Alt Text when set.
7. Eyebrow, title (always `h2`) and text render only when not empty; the intro is omitted when empty.
8. Search: input `type="search"`, `name="q"`, `autocomplete="off"`; placeholder and hidden label = water mark, else `searchPlaceholder` ("Ask us a question").
9. Submit icon button: arrow-right, type from Search bar button icon type (unknown = primary), label = hover text, else `submit` ("Submit").
10. A submit with an empty (trimmed) query does nothing in the block; a non-empty one dispatches `conversation-canvas:search` with `{ query }`. No navigation or request.
11. Voice: shown only if `SpeechRecognition` / `webkitSpeechRecognition` exists; language = `<html lang>` or `en`; first result fills the field and focuses it; click again to stop; `aria-pressed` reflects listening; start errors reset it.
12. Menu root: the Menu root path converted to a published path, else the locale root; depth = the Menu depth, minimum 1 (dialog 1–3).

13. Tree: query-index pages of the current locale under the root, levels 1..depth, sorted by level then path; pages whose parent is not in the index are skipped; titles fall back to the last path segment with hyphens as spaces; an index error gives an empty tree.
14. Default / Secondary menu: loaded on first open; nested lists by level; empty tree shows `noPagesFound` ("No pages found"); closes on Escape (focus back to the button), outside click or focus leaving the row.
15. The pop-up menu titles are shown in full; dock pills and panel links drop text after `|`.
16. Primary views: Root on the menu root or when the page is not in the tree; Section for a page with sub-pages; Suggestions for a leaf page when suggestions exist; otherwise the parent's section (Root if the parent is the root).
17. Primary suggestions: the page's tag labels, unless a script set suggestions first; `setConversationCanvasSuggestions` replaces them. Clicking a suggestion strips surrounding straight or curly quotes, fills the field and submits.
18. Primary toggles: only in Section view; sub-pages with children become toggles; the panel replaces the search row on desktop; 6+ items use two balanced columns.
19. Primary menu button: on Root-default pages it shows / hides the navigation (`aria-expanded`); elsewhere it switches between Root and the default view (`aria-pressed`).
20. Primary Back: suggestions to parent section; section to its parent section; then Root.
21. Floating: at most one canvas floats; Primary floats while no footer and no in-place Default / Secondary search row is in view; Default / Secondary rows float after scrolling past their place; the first registered qualifying canvas wins; a non-floating dock is `inert`.
22. Primary on < 900 px: search pill only; >= 900 px: menu button, navigation and panel.
23. Secondary filter list: empty and hidden until `setConversationCanvasFilters` fills it; links when `href`, otherwise toggle buttons without behaviour.
24. Button items: link required; view / type / icon from the item's option cells or link formatting, defaults button / primary; alignment ignored; title dropped when equal to the text.
25. Placeholders keys with English fallbacks: `menu`, `back`, `noPagesFound`, `siteNavigation` ("Site"), `suggestions`, `searchPlaceholder`, `searchByVoice`, `submit`.
26. Additional tab applies in `main` only (not in the footer); spacing on Primary only adds empty space.

6.30.12 Author Best Practices

Recommended Content Length: Eyebrow up to 40 characters; title up to 60 (Bigger up to 40); text up to 200; water mark up to 40; hover text up to 20; button text up to 25; 1–3 buttons. All Recommended.

Image Dimensions: Background: landscape, at least 1920 x 1080 for full-width panels (rendered with `object-fit: cover`, so keep important content centred; the panel height varies with the content). Icon: square (rendered 2.5rem / 40 px; supply at least 80 x 80 or SVG). Audio icon: square, rendered 1.5rem / 24 px (SVG preferred). There is no fixed aspect ratio in the CSS.

CTA Writing Guidelines: Use clear verbs ("Contact us", "Explore services"); one primary button; use the Title only to add information; keep the water mark inviting ("Ask us a question").

Accessibility Guidelines: Leave Icon Alt Text empty for decorative icons; keep text readable over background images; make the hover text meaningful (it is the submit button's name); avoid more than one canvas per page; check that the canvas title fits the page's heading order.

SEO Guidelines: The title is an `h2`; keep the page `h1` in the Page Header. Do not rely on the menu or dock for internal linking (script-generated); keep important links in content or buttons.

Metadata Guidelines: For good Primary pills, give pages concise `og:title` values (anything after `|` is dropped in the dock) and correct tags (suggestions).

Taxonomy Guidelines: Tags used as suggestions should have translated titles in each locale's taxonomy; otherwise ID-derived labels are shown.

Localization Guidelines: Author the water mark, alt texts, hover text and buttons per locale; point the Menu root path to a page of the same locale; translate the canvas placeholder keys in each locale's placeholders sheet; voice input uses the page language (`<html lang>`).

6.30.13 Troubleshooting Guide

Issue 1 — Pressing Enter or the arrow does nothing

Symptoms: The query stays in the field; no results.

Root Cause: By design: no search service; only `conversation-canvas:search` is dispatched.

Diagnosis Steps: In the console add a listener for `conversation-canvas:search` and submit; check `window.adobeDataLayer` for `cmp:search`.

Resolution Steps: Connect a search or assistant integration (development, [To be confirmed: search back end]).

Validation Steps: After integration, a query opens the results experience.

Preventive Measures: Communicate the limitation to stakeholders before launch.

Issue 2 — The menu shows "No pages found"

Symptoms: The pop-up menu or dock navigation is empty.

Root Cause: Viewed inside Universal Editor (no query index), Menu root path in another locale, pages not yet indexed, or intermediate pages missing from the index (their children are skipped).

Diagnosis Steps: Open `/query-index.json` on `.aem.page` and look for the root's child paths; check the root path's locale.

Resolution Steps: Test on `.aem.page`; use a root in the same locale; publish missing parent pages; check the `helix-query.yaml` exclusions.

Validation Steps: The menu lists the expected pages.

Preventive Measures: Keep a published page at every folder level.

Issue 3 — The Primary dock never appears

Symptoms: No floating bar while scrolling.

Root Cause: The footer is always in view (short page); another canvas's search row is in view or floating first; Hide block is on; the page is narrow and the dock shows only the search pill.

Diagnosis Steps: Scroll on a long page; check for other canvases; inspect the dock for `conversation-canvas-floating`.

Resolution Steps: Use Primary on long pages only; keep one canvas per page.

Validation Steps: The dock fades in after scrolling and hides at the footer.

Preventive Measures: Validation checklist (one Primary per page).

Issue 4 — Dock pills show the wrong level or titles with "| TCS"

Symptoms: Unexpected pills, or long titles.

Root Cause: The current page is not in the tree (deeper than the Menu depth or outside the root), so Root is shown; titles come from `og:title`; only text after `|` is removed.

Diagnosis Steps: Compare the URL with the root and depth; check the page's `og:title`.

Resolution Steps: Increase Menu depth (max 3) or change the root; shorten titles.

Validation Steps: Reload: the expected view and labels appear.

Preventive Measures: Plan root / depth with the information architecture.

Issue 5 — The microphone button is missing
Symptoms: No voice button.

Root Cause: The browser does not support the Web Speech API (for example Firefox).

Diagnosis Steps: Check `window.SpeechRecognition || window.webkitSpeechRecognition` in the console.

Resolution Steps: None needed; test in a supporting browser.

Validation Steps: The button appears in Chrome or Edge.

Preventive Measures: Document supported browsers for stakeholders.

Issue 6 — Voice input fills the field but does not search
Symptoms: The spoken text appears; nothing else happens.

Root Cause: By design the result is not auto-submitted; the visitor presses Enter or the arrow.

Diagnosis Steps: Observe that the field receives focus after speaking.

Resolution Steps: None (or change the behaviour in code on request).

Validation Steps: Pressing Enter dispatches the search event.

Preventive Measures: Mention in user guidance if needed.

Issue 7 — The submit button tooltip shows an odd word
Symptoms: Hovering the arrow shows a test word (for example "sradha" in the English footer sample).

Root Cause: Search bar button icon type hover text contains a test value.

Diagnosis Steps: Open the canvas (or the footer canvas reference page) and check the field.

Resolution Steps: Set a real label ("Submit") or clear it to use the placeholder; publish.

Validation Steps: Tooltip and screen reader name read "Submit".

Preventive Measures: Review test values before go-live.

Issue 8 — Filter tags or suggestions never show
Symptoms: Secondary shows no filters; Primary shows no suggestion pills.

Root Cause: Filters are script-only (no script supplies them); suggestions need page tags or a script, and only appear on pages without sub-pages.

Diagnosis Steps: Check the page's tags; check whether the page has sub-pages in the tree.

Resolution Steps: Tag the page or implement the supplying script (development).

Validation Steps: Pills appear on a tagged leaf page.

Preventive Measures: Agree the personalisation integration before relying on these features.

6.30.14 Support Runbook

Monitoring Requirements
- Availability of `/query-index.json` (the menu and dock depend on it) and of the locale's `taxonomy.json` and `placeholders.json`.
- Once Adobe Tags is installed: `cmp:search` volumes and top terms (subject to privacy review).

Logs

- Browser console: the block emits no messages of its own; index or taxonomy failures are swallowed (empty menu / labels). Check the Network tab for `/query-index.json?offset=0&limit=500` (200 expected) and the taxonomy / placeholders sheets.
- Admin API / Sidekick: preview and publish status of the page, the footer and the canvas reference page; index status of pages.
- AEM author logs: publication of the pages.

Common Errors

Error	Cause	Fix
Search does nothing	No search service (by design)	Integrate a listener for `conversation-canvas:search`
"No pages found"	Editor context, wrong locale root, unindexed pages	Test on `.aem.page`; fix root; publish pages
Dock not visible	Short page, other canvas, Hide block	Use on long pages; one canvas
Intro or buttons missing	Canvas Style is Primary	Use Default or Secondary
Voice button missing	Unsupported browser	Use Chrome / Edge / Safari
English UI labels on a locale	Placeholder keys missing	Add keys to the locale sheet
Test word as button tooltip	Hover text test value	Correct the field

Escalation Matrix

Level	Who	Checks
L1	Author support / content lead	Style, fields, root path locale, depth, test values, tags, publish status
L2	AMS / site admin	Query index content and freshness, sheet publication, CDN cache, footer configuration
L3	Development team	Console / network errors, floating logic, tree building, data layer, integrations for search, suggestions and filters
Adobe Support	Adobe	Query index generation, Edge Delivery delivery or Universal Editor platform issues

Recovery Procedures

1. Faulty canvas in a page: Hide block and publish; or restore a page version.
2. Faulty footer canvas: clear the Footer's Conversation Canvas Page (or fix the reference page) and publish the footer page.
3. Empty menus across the site: check and re-index (`/query-index.json`) via the Admin API with the site admin.
4. Code regression: revert on `main`.

Health Checks

- On `.aem.live`: canvas renders; menu lists pages; search submit dispatches the event; dock appears on scroll on a long page and hides at the footer; voice button works in Chrome.
- `window.adobeDataLayer` contains the component entry and `cmp:search` after a query.

Operational Procedures

- New section of the site: publish it so it appears in menus and dock pills after indexing.
- New locale: translate the canvas placeholder keys; create the footer canvas reference page.

- Search integration (future): add a listener for `conversation-canvas:search`, optionally call the two exported functions; update this runbook.

Known Limitations

- No search or conversational service; empty queries are ignored by the block but still tracked as `cmp:search` with an empty term.
- Filters and suggestions require scripts (none exist); filter buttons have no behaviour.
- Menu and dock exclude nothing beyond the query index exclusions (for example `hide-in-nav` and `robots` are not checked); depth max 3 in the dialog.
- Menus are built from the current locale's index only; the index is not available in the editor.
- BG Style has one option and no styles.
- Button Alignment is ignored.
- Hide block and spacing do not apply in the footer; spacing on Primary adds empty space.
- The pop-up menu titles keep the "| …" suffix (only the dock removes it).
- The dock panel has an `aria-label` without a landmark role.
- With unkeyed (single-cell) property rows, a row containing a link (for example the menu root) is read as a button item; authored content from Universal Editor uses keyed rows, so this affects only hand-made markup.
- Background images load lazily even in the first section (possible LCP impact).

6.31 Interactive Dashboard Block

The Interactive Dashboard block is a **placeholder** reserved for a future interactive data dashboard. Its design, content model and logic have not been defined yet. As of this handover the block renders **nothing** on the site; in Universal Editor edit mode it shows a dashed, selectable box labelled "Interactive Dashboard (placeholder)". The block lives in `blocks/interactive-dashboard/` (`interactive-dashboard.js`, 13 lines; `interactive-dashboard.css`, 19 lines; `_interactive-dashboard.json`, which has only the Additional tab).

> **Important:** do not use this block on live pages. Charts, data sources, filters, data refresh and any interactivity are Not implemented (as of this handover). [To be confirmed: dashboard requirements, data source and design — owner: product owner]

6.31.1 Component Purpose

Business Objective: Reserve a component name and a place in the section filter for a future dashboard (for example interactive figures or KPIs), so that the authoring palette, the data layer and the Additional options are already in place when it is built. [To be confirmed: business objective and KPIs of the dashboard — owner: product owner]

Business Use Cases: None today. Possible future use cases are not specified in the code or the repository ([To be confirmed: use cases — owner: product owner]).

When To Use: Only on test or block-library pages, to reserve a position for the future dashboard in a draft layout.

When Not To Use: On any published page: it shows nothing to visitors and may leave an empty section.

Dependencies:

Type	Dependency	Use
Code	`blocks/interactive-dashboard/interactive-dashboard.js`	Empties the block and sets `data-placeholder="Interactive Dashboard"`

Type	Dependency	Use
Code	`blocks/interactive-dashboard/interactive-dashboard.css`	Hides the empty block; shows the labelled box in edit mode
Code	`decorators/block-options-decorator.js`	Additional tab: Anchor ID becomes the block id; Hide block / spacing classes
Code	`decorators/data-layer-decorator.js`	Adds the block to the data layer like any block
Styles	`styles/styles.css`	Tokens used by the placeholder box; `.hideblock` and spacing classes
Data	None	No sheets, query index, metadata, taxonomy or placeholders

Related Components: Table and Table List (tabular data available today), Cards / List (statistic-style cards), Embed (an external dashboard could be embedded where the Embed block supports the provider).

Applicable Templates: Technically any page (the block is in the section filter); not used by any template.

Supported Content Types: Not applicable (placeholder).

6.31.2 Visual Overview

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-desktop-view.png`
> **Purpose:** Desktop rendering
> **Description:** Desktop page on `.aem.page` where the section containing the Interactive Dashboard shows no dashboard content (empty space only from the section's own spacing).

Desktop (>= 1200 px): Nothing is rendered: the block is empty and `display: none` (`.interactive-dashboard:empty`). Its wrapper and section remain, so a section containing only this block appears as empty space (the section's padding / margin).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-tablet-view.png`
> **Purpose:** Tablet rendering
> **Description:** Tablet page with no visible dashboard.

Tablet (600–1199 px): Same as desktop: nothing rendered; there are no breakpoints in the block CSS.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-mobile-view.png`
> **Purpose:** Mobile rendering
> **Description:** Mobile page with no visible dashboard.

Mobile (< 600 px): Same: nothing rendered.

6.31.3 Component Variations

The block has no style, layout or content options. Its two rendering modes depend on where it is viewed.

Variation 1 — Editor placeholder (Universal Editor edit mode)

Variation Name: Placeholder box (`html.adobe-ue-edit`).

Business Purpose: Let authors see, select, move or delete the reserved block.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-editor-placeholder.png`
> **Purpose:** Editor placeholder
> **Description:** Universal Editor canvas with a dashed, rounded grey box at least 10rem tall, centred text "Interactive Dashboard (placeholder)".

Use Cases: Draft layouts, block library pages.

Authoring Guidelines: Add the block; nothing can be configured except the Additional tab.

Business Rules: Shown only in Universal Editor **edit** mode (class `adobe-ue-edit` on `<html>`). Any authored content in the block is removed by the code.

Rendering Rules: `display: grid; place-items: center; min-height: 10rem`; dashed 2 px border (`--color-border-secondary`), 1rem radius, secondary text colour, body font size M; text from `attr(data-placeholder)` + " (placeholder)" via `::before`.

Responsive Behaviour: The box fills the section width at every breakpoint; no breakpoints.

SEO Impact: None (editor only).

Accessibility Compliance: Editor-only visual; the label is CSS-generated content (not part of the published DOM text).

Variation 2 — Published and preview rendering (empty)

Variation Name: Empty block (`.interactive-dashboard:empty { display: none; }`).

Business Purpose: Ensure visitors never see an unfinished component.

Visual Example:

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-published-empty.png`
> **Purpose:** Published rendering
> **Description:** The `.aem.page` page with developer tools open, showing the empty `div.interactive-dashboard` with `display: none` and `data-placeholder`.

Use Cases: Applies automatically on `.aem.page`, `.aem.live` and in Universal Editor preview mode.

Authoring Guidelines: None.

Business Rules: The block is emptied and hidden. Hide block, padding and margin have no visible effect (the block is not displayed).

Rendering Rules: Empty `div` with `data-placeholder="Interactive Dashboard"`.

Responsive Behaviour: Hidden at every breakpoint.

SEO Impact: None (no content).

Accessibility Compliance: Not exposed to assistive technology (`display: none`).

6.31.4 Authoring Experience

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-authoring-view.png`
> **Purpose:** Universal Editor canvas
> **Description:** The placeholder box selected in Universal Editor with marker A on the box.

Annotations:

- A = The placeholder box (the block itself; it has no authorable elements on the canvas)

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-properties-panel.png`
> **Purpose:** Properties panel
> **Description:** Properties panel of the block showing only the Additional tab.

Properties panel: One tab, **Additional**: Anchor ID (text, validated `^anchor-[a-z0-9][a-z0-9-]*$`), Hide block (toggle), Padding left / right / top / bottom and Margin left / right / top / bottom (selects with the 15 spacing options). There is no Content tab and no **Add** menu for child items (the block has no filter).

6.31.5 Authoring Guide

Step 1 — Open the page in Universal Editor

Action: Open a test or block-library page in Universal Editor.

Expected Result: The page opens in edit mode.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-step-01-open-page.png`
> **Purpose:** Open the page
> **Description:** AEM Sites console with a block-library page selected and Edit highlighted.

Step 2 — Add the block

Action: Select a section, **Add** (+) > **Interactive Dashboard**.

Expected Result: The dashed "Interactive Dashboard (placeholder)" box appears.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-step-02-add-block.png`
> **Purpose:** Add the block
> **Description:** Section Add menu with "Interactive Dashboard" highlighted.

Step 3 — Fill the Additional tab (optional)

Action: Set an Anchor ID if the reserved position must be addressable later; leave the other options on their defaults.

Expected Result: The block id is set; no visible change on the site.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-step-03-additional.png`
> **Purpose:** Additional tab
> **Description:** Additional tab with Anchor ID “anchor-dashboard”.

Step 4 — Preview

Action: Switch to Preview or open the `.aem.page` URL.

Expected Result: Nothing is shown where the block is.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-step-04-preview.png`
> **Purpose:** Preview
> **Description:** The preview page without any dashboard content.

Step 5 — Do not publish to live

Action: Remove the block (or keep it only on unpublished block-library pages) before publishing a live page.

Expected Result: No empty section on the live page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/interactive-dashboard-step-05-remove.png`
> **Purpose:** Remove before publishing
> **Description:** The block selected with the Delete action highlighted.

Preview Process
Preview shows nothing for this block; confirm only that the surrounding layout has no unwanted gap.

Publishing Process
Not recommended. If a page containing the block must be published, confirm that the section does not leave visible empty space.

Validation Process
- ☐ The block is not on any live page (search the locale sites for it before go-live).
- ☐ If present on a draft, its section has other content or no visible gap.
- ☐ No Anchor ID links point to it from live content.

Rollback Process
Delete the block from the page and republish, or restore the previous page version.

6.31.6 Authorable Field Specification
Block model `interactive-dashboard` — Additional tab (the only tab)

Field Name	Field Type	Mandatory	Multi Value	Default Value	Business Purpose	Author Guidance	Validation Rules	Example Value
Anchor ID (`classes_anchor`)	text	No	No	(empty)	Id for future # links	Starts with `anchor-`	Enforced regExp `^anchor-[a-z0-9][a-z0-9-]*$`; message "Start with anchor-, then lowercase letters, digits and hyphens."; duplicates get `-2`, `-3`	anchor-dashboard
Hide block (`classes_hideblock`)	boolean	No	No	false	Hides the block and its wrapper	Not needed (already hidden)	None	false
Padding left (`classes_paddingleft`)	select	No	No	Default	Inner space left	Leave Default	Default, `pl-0` … `pl-3000` (15 options)	(Default)
Padding right (`classes_paddingright`)	select	No	No	Default	Inner space right	Leave Default	Default, `pr-0` … `pr-3000`	(Default)
Padding top (`classes_paddingtop`)	select	No	No	Default	Inner space top	Leave Default	Default, `pt-0` … `pt-3000`	(Default)
Padding bottom (`classes_paddingbottom`)	select	No	No	Default	Inner space bottom	Leave Default	Default, `pb-0` … `pb-3000`	(Default)
Margin left (`classes_marginleft`)	select	No	No	Default	Outer space left	Leave Default	Default, `ml-0` … `ml-3000`	(Default)
Margin right (`classes_marginright`)	select	No	No	Default	Outer space right	Leave Default	Default, `mr-0` … `mr-3000`	(Default)
Margin top (`classes_margintop`)	select	No	No	Default	Outer space top	Leave Default	Default, `mt-0` … `mt-3000`	(Default)
Margin bottom (`classes_marginbottom`)	select	No	No	Default	Outer space bottom	Leave Default	Default, `mb-0` … `mb-3000`	(Default)

The spacing options are the 15 values listed in 6.29.6.

Field details

- **Anchor ID** — Purpose: block id. Allowed values: `anchor-` + lower-case letters, digits, hyphens. Character limits: none enforced (Recommended up to 40). Business rules: set as the id by the block-options decorator; a link to it does not scroll, because the block is hidden. Dependency / visibility rules: none. Sample: `anchor-dashboard`.
- **Hide block** — Purpose: hide. Business rules: on the site the block is already hidden; the option additionally hides the wrapper (removing a possible wrapper gap) outside edit mode; in edit mode the placeholder is shown faded. Sample: off.
- **Padding left / right / top / bottom, Margin left / right / top / bottom** — Purpose: spacing. Allowed values: the 15 options. Business rules: visible only on the editor placeholder (the published block is `display: none`). Sample: Default.

There are no content fields and no item models.

6.31.7 Field Annotation Mapping

Marker	Field	Purpose
A	(block, no field)	The editor placeholder box; select it to reach the Additional tab

6.31.8 Component Specification

Attribute	Value
Component Name	Interactive Dashboard (`interactive-dashboard`)
Version	1.0 (placeholder)
Owner	[To be confirmed: product owner]; technical owner: development team
Dependencies	block-options and data-layer decorators; `styles/styles.css` tokens
Component Type	Block (placeholder; no items)
Supported Variants	None (editor placeholder; empty on the site)
Supported Templates	Any page technically; not recommended on live pages
Supported Metadata	None
Supported Tags	None
Supported Personalization	None
Caching Considerations	Only the page HTML (CDN); no data is loaded
Performance Considerations	Negligible: the block's JS and CSS are loaded when its section loads (two small files)
Accessibility Compliance	Not exposed on the site (`display: none`); future implementation must meet WCAG 2.1 AA (text alternatives for charts, keyboard operation, non-colour encoding)
SEO Compliance	No content

6.31.9 Input Model

Content Inputs: None (any rows are discarded).

Metadata Inputs: None.

Tags Inputs: None.

Page Context Inputs: Universal Editor edit mode (`html.adobe-ue-edit`) for the placeholder.

Query Inputs: None.

Manual Inputs: Additional tab only (as block classes).

Authored HTML received from AEM (simplified):

```
<div class="interactive-dashboard anchor-dashboard"></div>
```

6.31.10 Output Model

Rendered Output: The same `div`, emptied, with `data-placeholder="Interactive Dashboard"`; the block-options decorator turns an `anchor-…` class into its id.

Generated HTML:

```
<div class="interactive-dashboard anchor-dashboard block" id="anchor-dashboard"
  data-block-name="interactive-dashboard" data-block-status="loaded"
  data-placeholder="Interactive Dashboard"></div>
```

Generated JSON: None.

Generated Metadata: None.

Analytics Tracking: The data layer still adds a component entry (lazy phase):

```
{ "component": { "anchor-dashboard": {
  "@type": "tcs-corp/components/interactive-dashboard", "parentId": "page-…" } } }
```

No events (nothing to click).

Personalization Output: None.

Search Index Output: None.

6.31.11 Business Rules

1. The block removes all of its children (`block.replaceChildren()`); authored content is never shown.
2. It sets `data-placeholder="Interactive Dashboard"`.
3. On the site and in Universal Editor preview mode the empty block is `display: none`.
4. In Universal Editor edit mode it is a dashed box, at least 10rem tall, labelled "Interactive Dashboard (placeholder)" (the label text is fixed in CSS / JS, not a placeholder key, and is not translated).
5. The model has only the Additional tab; there is no filter, so no child items can be added.
6. The Anchor ID becomes the block id (unique on the page); links to it do not scroll.
7. Hide block hides the block's wrapper too (outside edit mode); in edit mode the placeholder is faded with a dashed outline.
8. The block is added to the data layer like any other block.

6.31.12 Author Best Practices

Recommended Content Length: Not applicable.

Image Dimensions: Not applicable.

CTA Writing Guidelines: Not applicable.

Accessibility Guidelines: Do not reference the block from live content (for example "see the dashboard below").

SEO Guidelines: Do not publish pages that rely on the dashboard for content.

Metadata Guidelines: Not applicable.

Taxonomy Guidelines: Not applicable.

Localization Guidelines: Not applicable; the placeholder label is English and editor-only.

6.31.13 Troubleshooting Guide

Issue 1 — The dashboard does not appear on the published page

Symptoms: Nothing is visible where the block was added.

Root Cause: By design: the block is a placeholder and renders nothing.

Diagnosis Steps: Inspect the page: an empty `div.interactive-dashboard` with `data-placeholder`.

Resolution Steps: Remove the block; use Table, Cards or Embed for current needs.

Validation Steps: The page shows the intended alternative content.

Preventive Measures: Communicate the block's status to authors (the Authoring Guide lists it as a placeholder).

Issue 2 — The placeholder box is not visible in Universal Editor

Symptoms: The block cannot be found on the canvas.

Root Cause: Universal Editor is in Preview mode (the box is shown only in edit mode).

Diagnosis Steps: Check the editor mode toggle.

Resolution Steps: Switch to edit mode, or select the block in the content tree.

Validation Steps: The dashed box appears.

Preventive Measures: Use the content tree to locate hidden or empty blocks.

Issue 3 — Authored text disappears

Symptoms: Content pasted into the block is gone.

Root Cause: The code empties the block; the model has no content fields.

Diagnosis Steps: Check the model: only the Additional tab.

Resolution Steps: Place content in other blocks or default content.

Validation Steps: Content shows in the chosen block.

Preventive Measures: Do not use the placeholder for content.

Issue 4 — An empty gap appears on the page

Symptoms: Extra white space in a section.

Root Cause: The section (or its wrapper) remains with its own spacing although the block is hidden.

Diagnosis Steps: Inspect the section containing the block.

Resolution Steps: Delete the block, or turn on Hide block (hides the wrapper), or remove the empty section.

Validation Steps: The gap is gone on `.aem.page`.

Preventive Measures: Validation checklist in 6.31.5.

Issue 5 — A link to the dashboard anchor does not scroll

Symptoms: `#anchor-dashboard` links do nothing.

Root Cause: The target exists but is `display: none`.

Diagnosis Steps: Check the target id in the page.

Resolution Steps: Remove such links until the dashboard exists.

Validation Steps: No links to the anchor remain.

Preventive Measures: Do not link to placeholder blocks.

Issue 6 — Analytics reports an “interactive-dashboard” component

Symptoms: The data layer contains a component entry for the dashboard.

Root Cause: Every block is tracked, including empty ones.

Diagnosis Steps: Inspect `window.adobeDataLayer`.

Resolution Steps: Remove the block from the page; or ignore the component type in reports.

Validation Steps: The entry no longer appears.

Preventive Measures: Keep the placeholder off live pages.

6.31.14 Support Runbook

Monitoring Requirements

None. Optionally check before releases that no live page contains the block (search the AEM content or the published HTML for `interactive-dashboard`).

Logs
The block emits no console messages and makes no requests. Admin API / Sidekick status and AEM author publication logs apply as for any page.

Common Errors

Error	Cause	Fix
Nothing on the page	Placeholder by design	Remove the block
Box missing in the editor	Preview mode	Switch to edit mode
Empty gap	Section remains	Delete the block or section; Hide block
Anchor link does nothing	Target hidden	Remove the link

Escalation Matrix

Level	Who	Checks
L1	Author support / content lead	Block present on live pages? Remove it
L2	AMS / site admin	Publish status; content audit for the block
L3	Development team	Future implementation (requirements, data source, design)
Adobe Support	Adobe	Not expected (no platform features used beyond standard blocks)

Recovery Procedures
Delete the block and republish, or restore a previous page version.

Health Checks
Not applicable beyond confirming that live pages do not include the block.

Operational Procedures
When the dashboard is commissioned: define requirements, data source and design ([To be confirmed — owner: product owner]); the development team replaces the placeholder code and model and updates this chapter.

Known Limitations
- Placeholder only: no content model, no rendering, no data, no interactivity (Not implemented as of this handover).
- The editor label is fixed English text.
- The block is still tracked in the data layer and its section may leave empty space.

Part 7 — Taxonomy Documentation

This part describes how tags (the taxonomy) are structured, published, resolved into labels and used by the TCS Corporate Website. It covers the AEM side (the Tagging console, the taxonomy page per locale) and the Edge Delivery side (the published `taxonomy.json`, the query index, the blocks that read tags and the Adobe Client Data Layer).

> **Note:** Everything in this part was checked against the code in the `main` branch (`utils/taxonomy.js`, `utils/tags.js`, `utils/metadata.js`, `utils/query-index.js`, `blocks/list/list-data.js`, `blocks/filter-list/filter-list.js`, `blocks/teaser/teaser.js`, `blocks/page-header/page-header.js`, `blocks/cards/cards.js`, `blocks/conversation-canvas/conversation-canvas.js`, `decorators/tag-decorator.js`, `decorators/card-list-decorator.js`, `decorators/data-layer-decorator.js`, `helix-query.yaml`, `paths.json`) and, where stated, against the published site on 13 October 2026.

7.1 Taxonomy Structure

7.1.1 AEM Tags and Tag IDs

Tags are created and maintained in AEM as a Cloud Service, in the **Tagging** console (Tools → General → Tagging). Every tag has:

Property	Description	Example
Namespace	The top level of a tag tree. Its ID ends with a colon.	`tcs-corp:`
Tag ID	The namespace, a colon and the tag's path below the namespace. This is the value stored on pages and in block fields.	`tcs-corp:industry/banking`
Title	The label shown to visitors. Can be translated per language in AEM (tag localisation).	`Banking`
Path	The repository location of the tag (below `/content/cq:tags`).	`/content/cq:tags/tcs-corp/industry/banking`
Description	Optional text for authors (not used by the site code).	`Banking and financial services`

The website code works **only with tag IDs** and turns them into labels at runtime. Universal Editor also stores tag IDs: every tag field in the block models is an `aem-tag` field, which saves the selected tag IDs (for example `tcs-corp:region/europe`) in the component's properties.

Tag fields that exist in the component models (verified in `blocks/*/_*.json`):

Component	Field (name)	Model file	Multi value	Visible when
Teaser	Tags (`tags`)	`blocks/teaser/_teaser.json`	Yes	Display Tags is on
List	Tags (`tags`)	`blocks/list/_list.json`	Yes	List source = Tagged pages (`tags`)
List Item	Tags (`tags`), "Overrides the page tags."	`blocks/list/_list.json`	Yes	Always

Component	Field (name)	Model file	Multi value	Visible when
Card (Cards block)	Tags (`tags`)	`blocks/cards/_cards.json`	Yes	Always
Filter List	Primary filter category (`primaryCategory`)	`blocks/filter-list/_filter-list.json`	No	Always (Filters tab)
Filter List	Primary filter tags → Tag Id (`tagId`)	`blocks/filter-list/_filter-list.json`	Container, multi	Always (Filters tab)
Filter List	Advanced filter categories → Taxonomy root (`taxonomyRoot`)	`blocks/filter-list/_filter-list.json`	Container, multi	Always (Advanced tab)

> **Note:** None of the `aem-tag` fields restricts the tag picker to a namespace or root path: authors can pick any tag of any namespace. Keeping to the agreed namespaces is a governance rule (see 7.4), not something the editor enforces.

The **page tags** are not a field of the Universal Editor page properties model (`models/_page.json` has Title, Description, Keywords, Page Type, Navigation, Footer and 404 Page only). Page tags are set either in the standard AEM page properties (Sites console → Properties → Basic → Tags, stored as `cq:tags`) or in a `tags` column of the locale's bulk metadata sheet. See 7.8.1 for how the code reads them and an open point about the published meta name.

7.1.2 Namespaces in Use

Two sets of tag IDs appear in the project, and they are not the same:

Where	Namespace and tags	Status
Published taxonomy `/en/taxonomy.json` (checked 13 October 2026)	`tcs-corp:` (title "TCS Corp") with three trees: `tcs-corp:content-type` (Case Study, Article, Blog, Press Release, Whitepaper, Research, Webinar, Event), `tcs-corp:industry` (no child tags yet), `tcs-corp:region` (Americas, Europe, Asia Pacific, Middle East Africa)	The real taxonomy in AEM
Code comments, field descriptions and unit tests	`tcs:` — `tcs:industry`, `tcs:industry/banking`, `tcs:topic/ai`, `tcs:topic/ai/genai`, `tcs:topic/cloud`, `tcs:topics/cloud`, `tcs:topics/artificial-intelligence`	Illustrative examples only (test fixtures in `test/list/fixtures.mjs`, descriptions such as "A tag root, e.g. tcs:industry")

The published `tcs-corp:` tree (16 rows) is:

```
tcs-corp:                              TCS Corp
├── tcs-corp:content-type              Content Type
│   ├── tcs-corp:content-type/article        Article
│   ├── tcs-corp:content-type/blog           Blog
│   ├── tcs-corp:content-type/case-study     Case Study
│   ├── tcs-corp:content-type/event          Event
│   ├── tcs-corp:content-type/press-release  Press Release
│   ├── tcs-corp:content-type/research       Research
│   ├── tcs-corp:content-type/webinar        Webinar
│   └── tcs-corp:content-type/whitepaper     Whitepaper
├── tcs-corp:industry                  Industry          (no child tags yet)
└── tcs-corp:region                    Region
    ├── tcs-corp:region/americas             Americas
```

```
    ├── tcs-corp:region/asia-pacific          Asia Pacific
    ├── tcs-corp:region/europe                Europe
    └── tcs-corp:region/middle-east-africa    Middle East Africa
```

> **Important:** When an author follows a field description literally (for example types `tcs:industry` as a Filter List Primary filter category), nothing matches, because the real namespace is `tcs-corp:`. Always pick tags with the tag picker instead of typing IDs.

The code is namespace-agnostic. A value is treated as a tag ID when:

- `utils/taxonomy.js` `resolveTags`: it contains a colon or a slash (`/[:/]/`);
- `blocks/filter-list/filter-list.js` `isTagId`: it matches `^[\w-]+:[\w/-]*$` (namespace of letters, digits, _ or -, a colon, then an optional path), so `tcs-corp:industry/banking` and the bare namespace `tcs-corp:` both qualify.

7.1.3 The Taxonomy Page and /{locale}/taxonomy.json

AEM exposes tag titles to Edge Delivery Services through a **taxonomy page**: a page created from the AEM *Taxonomy* template whose page properties select the tags or namespaces to include. The page content is generated by AEM from the selected tags and is read-only. When published, it is delivered as a spreadsheet JSON with the columns `tag` and `title`.

Each of the 13 locale sites has its own taxonomy page at `/content/tcs-corp/{country}/{language}/taxonomy`. `paths.json` (generated by `npm run build:paths` from `LOCALES` in `utils/site.js`) maps each of them to a JSON resource:

AEM taxonomy page	Published as
/content/tcs-corp/us/en/taxonomy	/en/taxonomy.json
/content/tcs-corp/de/de-de/taxonomy	/de-de/taxonomy.json
/content/tcs-corp/fr/fr-fr/taxonomy	/fr-fr/taxonomy.json
/content/tcs-corp/ca/fr-ca/taxonomy	/fr-ca/taxonomy.json
/content/tcs-corp/es/es-es/taxonomy	/es-es/taxonomy.json
/content/tcs-corp/pt/pt-pt/taxonomy	/pt-pt/taxonomy.json
/content/tcs-corp/nl/nl-nl/taxonomy	/nl-nl/taxonomy.json
/content/tcs-corp/se/sv-se/taxonomy	/sv-se/taxonomy.json
/content/tcs-corp/dk/da-dk/taxonomy	/da-dk/taxonomy.json
/content/tcs-corp/no/no-no/taxonomy	/no-no/taxonomy.json
/content/tcs-corp/cn/zh-cn/taxonomy	/zh-cn/taxonomy.json
/content/tcs-corp/jp/ja-jp/taxonomy	/ja-jp/taxonomy.json
/content/tcs-corp/kr/ko-kr/taxonomy	/ko-kr/taxonomy.json

A published taxonomy row looks like this (from `/en/taxonomy.json`):

```
{
  "total": 16,
```

```
  "offset": 0,
  "limit": 16,
  "data": [
    { "tag": "tcs-corp:", "title": "TCS Corp" },
    { "tag": "tcs-corp:region", "title": "Region" },
    { "tag": "tcs-corp:region/europe", "title": "Europe" }
  ],
  "columns": ["tag", "title"],
  ":type": "sheet"
}
```

> **Note:** On 13 October 2026 only `/en/taxonomy.json` was published; the other 12 locale taxonomy URLs returned 404. Until a locale's taxonomy page is published, its pages show labels derived from the tag IDs (see 7.1.4). [To be confirmed: publication of the 12 remaining locale taxonomy pages — owner: content operations lead]

> **[Screenshot Placeholder]**
> **File name:** `screenshots/taxonomy-page-properties.png`
> **Purpose:** Configuring a locale taxonomy page
> **Description:** AEM Sites console, page properties of `/content/tcs-corp/us/en/taxonomy`, with the Tags field listing the `tcs-corp:` namespace (or the `content-type`, `industry` and `region` trees) that the taxonomy publishes.

7.1.4 How Tag Labels Are Resolved

Labels are resolved in the browser by `utils/taxonomy.js` and `utils/tags.js`:

```
tag values (IDs or titles)
   |  e.g. from the page head, a block field or the query index
   v
resolveTags(values)                                  utils/taxonomy.js
   |
   +-- loadTaxonomy()  -> fetch {locale root}/taxonomy.json   (once per page view)
   |      rows of the language sheet (multi-sheet) or the only sheet
   |      Map: tag ID (trimmed) -> title (trimmed)
   |
   +-- for each value:
         1. Map has a non-empty title       -> { id, label: title }
         2. value contains ":" or "/"       -> parseTag(value)        utils/tags.js
              last path segment, "-" and "_" become spaces, first letter upper case
              tcs-corp:industry/financial-services -> "Financial services"
         3. otherwise (an authored title)   -> { id: value, label: value }
```

Precise behaviour (from the code):

1. `loadTaxonomy()` fetches `localeSheetPath('taxonomy')`, that is `{root}/taxonomy.json` of the current locale (`/de-de/taxonomy.json` on a German page; `/en/taxonomy.json` on pages outside a locale, such as drafts). The request is made **once per page view**: the promise is kept in a module variable and shared by every block on the page.
2. Rows without a `tag` value are ignored. Tag IDs and titles are trimmed.
3. A non-OK response (404, 500) or a network error gives an **empty map** — the page still renders, with ID-derived labels. No error is shown to the visitor.

4. A taxonomy row with an empty `title` counts as missing (the ID-derived label is used).
5. `parseTag` uses only the **last segment** of the ID: `tcs-corp:content-type/case-study` becomes "Case study" (sentence case, not the AEM title "Case Study"). A bare namespace (`tcs-corp:`) gives an empty label.
6. Values that contain neither `:` nor `/` are treated as titles typed by an author and shown as typed.
7. The Filter List keeps a second fallback: `labels.get(tag) || taxonomy.get(tag) || tag`. The List and Cards use `labels.get(tag) || tag`. Because `resolveTags` always returns a label, these fallbacks only apply in edge cases (for example an empty ID-derived label).

7.1.5 Language Sheets and Fallbacks

When tag titles are translated in AEM, the taxonomy JSON can contain one sheet per language (a multi-sheet JSON, `":type": "multi-sheet"`). `utils/site.js` `sheetRows(json, sheet)` picks:

1. the sheet named after the locale's **ISO 639-1 language** (`getLocale().language`: `en`, `de`, `fr`, `es`, `pt`, `nl`, `sv`, `da`, `no`, `zh`, `ja`, `ko`) — `fr-fr` and `fr-ca` both use `fr`;
2. else the sheet named `default`;
3. else the first sheet listed in `:names`.

For a single-sheet JSON (the current state of `/en/taxonomy.json`) its `data` rows are used.

Situation	Label shown
Tag in the locale's taxonomy, title present (language sheet or single sheet)	The taxonomy title, e.g. "Asia Pacific"
Multi-sheet taxonomy without the language sheet	The `default` sheet's title, else the first sheet's
Tag not in the taxonomy (not included in the taxonomy page, or deleted)	ID-derived label, e.g. "Middle east africa"
Taxonomy JSON not published (404) or not reachable	ID-derived label for every tag
Authored text without `:` or `/`	The text as authored

> **Note:** The code requests `taxonomy.json` without a `sheet` parameter and relies on the response containing the language sheets. The aem.live documentation describes reading a language with `?sheet={language}`. When translated tag titles are introduced, check the published JSON of a locale taxonomy page: if it returns only one sheet, the translated titles will not be picked up. [To be confirmed: shape of a translated taxonomy JSON — owner: development team]

> **Tip:** Because each locale has its own taxonomy page, a locale can also show translated labels by publishing a taxonomy whose single sheet already holds the titles of its language. [To be confirmed: which approach AEM produces for this site — owner: development team]

7.1.6 Reading Tag Values from Fields

`utils/tags.js` `readTagValues(cell)` reads a multi-value tag property as Universal Editor renders it: the text of each `<li>` when the values are a list, else the cell text split on commas. Values are trimmed and empty values dropped. The result is the list of tag IDs **as authored** (no validation, no de-duplication). `readTags(cell)` returns them with ID-derived labels (no taxonomy lookup); it is exported but not used by any block today.

Page tags are read by `utils/metadata.js` `getPageTags()`: the `content` of every `<meta property="article:tag">` in the page head, else a comma-separated `<meta name="tags">`; trimmed, empty values removed, **de-duplicated**.

7.2 Tag Hierarchy

7.2.1 The Hierarchy Model

AEM tags form trees: a namespace contains tags, which can contain child tags to any depth. The tag ID encodes the hierarchy as a path: `tcs-corp:region/europe` is a child of `tcs-corp:region`, which is a child of the namespace `tcs-corp:`. The code derives parent-child relationships **only from the ID string** — it never reads the AEM tag tree, the taxonomy order or tag properties.

7.2.2 Child Tags Match Parent Filters (hasTag)

The List and the Filter List both use the same rule (two identical implementations):

```
// blocks/list/list-data.js (exported, unit tested)
export const hasTag = (tags, tag) => tags.some((id) => id === tag ||
id.startsWith(`${tag}/`));

// blocks/filter-list/filter-list.js
const hasTag = (page, tag) => page.tags.some((id) => id === tag || id.startsWith(`${tag}/
`));
```

A page matches a filter tag when it carries the tag itself **or any descendant** of it, at any depth. The comparison is exact and case-sensitive on the ID; the slash boundary prevents partial matches (`tcs-corp:region/europe` does not match `tcs-corp:region/euro`).

Filter tag	Page tag	Match?	Why
`tcs-corp:region`	`tcs-corp:region/europe`	Yes	Child tag
`tcs-corp:industry`	`tcs-corp:industry/banking/retail`	Yes	Grandchild tag (any depth)
`tcs-corp:region/europe`	`tcs-corp:region`	No	A parent does not match a child filter
`tcs-corp:region/europe`	`tcs-corp:region/europe`	Yes	Same tag
`tcs-corp:region/euro`	`tcs-corp:region/europe`	No	Not a path boundary
`tcs-corp:content-type`	`tcs-corp:region/europe`	No	Different tree
`tcs-corp:`	`tcs-corp:region/europe`	No	`tcs-corp:/` is not a prefix of `tcs-corp:region/…`; a namespace alone matches only pages tagged with the bare namespace

> **Warning:** Selecting a bare namespace (for example `tcs-corp:`) as a List tag or a Filter List Primary filter tag does **not** match the tags below it, because the code looks for `tcs-corp:/`. Use a tag tree root such as `tcs-corp:region` instead.

7.2.3 Where the Hierarchy Is Used

Feature	Rule	Code
List, source Tagged pages, Match = Any	A page matches when it has at least one of the List's tags or a descendant of one	`applyTagMatching` in `blocks/list/list-data.js`
List, source Tagged pages, Match = All	A page matches when, for every List tag, it has that tag or a descendant	`applyTagMatching`
Filter List pills (single select)	Pages with the selected pill's tag or a descendant	`matches()` in `filter-list.js`
Filter List pills (Allow multi select on)	Pages that match **every** selected pill (AND)	`matches()`
Filter List Advanced filter options, Filter logic AND / OR	Pages with every / any checked option tag or a descendant	`matches()`
Filter List pills from Primary filter category	The pills are all descendants (any depth) of the category used by the listed pages: tags that start with `{category}/`	`used(root)` in `filter-list.js`
Filter List Advanced filter category options	The options are the descendants of the Taxonomy root used by the listed pages, sorted by label	`used(root)`

The unit tests (`test/list/list-data.test.mjs`, fixtures in `test/list/fixtures.mjs`) confirm the behaviour: `tcs:topic/ai` matches a page tagged `tcs:topic/ai/genai`; "all" of `tcs:topic/ai` and `tcs:topic/cloud` matches only the page carrying both; a page with `hide-in-nav = yes` is never listed even if its tags match; no tags, or tags with no matches, give no cards.

7.3 Business Taxonomy

7.3.1 Current State

The only namespace published is `tcs-corp:` with the trees `content-type`, `industry` (empty) and `region` (see 7.1.2). No topic, service or audience tags exist yet.

> **Note:** "Content type" exists twice in the implementation: as a **tag tree** (`tcs-corp:content-type/*`, usable in the List and the Filter List tag filters and shown as tag labels) and as the **`content-type` metadata** (free text, shown as the card eyebrow / meta and the Page Header eyebrow, filtered by the Filter List *Content types* field). The two are not connected by code. Keep them aligned: use the taxonomy title (for example `Case Study`) as the `content-type` metadata value and tag the page with the matching `tcs-corp:content-type/…` tag.

7.3.2 Proposed Business Taxonomy

[To be confirmed: business taxonomy (namespaces, trees and tag lists) — owner: digital marketing / content strategy lead]

The following structure is **proposed**. It stays inside the existing `tcs-corp:` namespace, uses lower-case, hyphenated tag names (as the published tags do), and keeps every tree two or three levels deep so that the parent-child matching in 7.2 is useful.

Tree (tag root)	Purpose	Proposed child tags (examples)	Used by
`tcs-corp:industry`	The industries TCS serves	`banking`, `financial-services`, `insurance`, `capital-markets`, `retail`, `consumer-packaged-goods`,	Filter List (Primary filter category, Advanced category "Industry"); List tag source; Teaser / Page Header labels

Tree (tag root)	Purpose	Proposed child tags (examples)	Used by
		`manufacturing`, `life-sciences`, `healthcare`, `communications-media`, `energy-utilities`, `travel-hospitality`, `public-services`	
`tcs-corp:service`	Service lines and offerings	`consulting`, `cloud`, `cybersecurity`, `data-analytics`, `artificial-intelligence`, `enterprise-solutions`, `iot-digital-engineering`, `cognitive-business-operations`	Filter List Advanced category "Service"; List tag source
`tcs-corp:topic`	Editorial themes across industries	`agentic-ai`, `generative-ai`, `sustainability`, `future-of-work`, `customer-experience`; children allowed, e.g. `artificial-intelligence/generative-ai`	Filter List pills; List "related insights" by topic
`tcs-corp:content-type`	The kind of content (exists)	`article`, `blog`, `case-study`, `event`, `press-release`, `research`, `webinar`, `whitepaper`; proposed additions `report`, `podcast`, `video`, news (to align with the Filter List Content types options)	Filter List Advanced category "Content type"; kept aligned with the `content-type` metadata
`tcs-corp:region`	Geography of the content (exists)	`americas`, `europe`, `asia-pacific`, `middle-east-africa`; children for countries if needed, e.g. `europe/uk`	Filter List Advanced category "Region"; List by region
`tcs-corp:audience`	Target audience (optional)	`cxo`, `it-leaders`, `investors`, `analysts`, `job-seekers`, `media`	Future personalisation; List / Filter List
`tcs-corp:campaign`	Marketing campaigns (optional, time-bound)	One tag per campaign, e.g. `campaign/agentic-bank-2026`	Campaign landing lists; to be retired after the campaign (see 7.6)

Note: The `region` tag tree and the `region` **metadata** (free text used by the press-release template to filter Media Contacts) are different things. If both are used, keep their values consistent (see 8.2, `region`).

7.3.3 Tags Versus Metadata

Need	Use a tag	Use metadata
Filter or list pages by a classification	Yes (List Tagged pages, Filter List pills and Advanced filter)	Only page type and content type (Filter List Page types / Content types)
Translated label shown to visitors	Yes (taxonomy titles per locale)	No (metadata is shown as authored)
Hierarchical matching (parent matches children)	Yes	No
Single value per page that drives layout or code	No	Yes (`page-type`, `content-type`, `featured`, `priority`, `hide-in-nav`)
Values read by templates (author, region of media contacts)	No	Yes (`author-id`, `region`)

7.4 Tag Governance

Tag governance keeps the taxonomy small, consistent across the 13 locales and safe to change. Nothing below is enforced by the code or the models unless marked **Enforced**.

7.4.1 Governance Rules

No.	Rule	Type	Reason
1	Create tags only in the `tcs-corp:` namespace (no new namespaces without approval).	Recommended	The code and the taxonomy pages are set up for one namespace; the tag picker does not restrict namespaces.
2	Tag names (the last ID segment) are lower case, words separated by hyphens, ASCII only, no spaces.	Recommended	`parseTag` builds fallback labels from the name; Filter List `isTagId` accepts only `[\w-]` in the namespace and `[\w/-]` in the path.
3	Every tag has a Title in English; titles are translated for every locale language before the tag is used on that locale.	Recommended	Without a title, visitors see the ID-derived label.
4	Pick tags with the tag picker; never type IDs into fields.	Recommended	Typed IDs (for example the `tcs:` examples in field descriptions) match nothing.
5	A page carries the most specific tags that apply, not their parents as well.	Recommended	Parents match their children automatically (7.2); duplicates add noise to card labels.
6	At most 3 tags per page should be "display" tags (the first ones in the tag list).	Recommended	List and Cards show up to 3 tags on image and feature cards; the Filter List shows up to 2 (`MAX_CARD_TAGS`). **Enforced** truncation in code.
7	Tag trees stay at most 3 levels below the namespace.	Recommended	Deep trees are hard to curate and to translate; matching works at any depth.
8	Changes to tag IDs (move, rename of the name, merge, delete) go through a change request (7.6).	Recommended	IDs are stored on pages, in block fields and in the query index.
9	After any tag or title change, republish the taxonomy page of every affected locale.	Recommended	The taxonomy JSON only changes when its page is published.
10	The taxonomy page of each locale includes the whole `tcs-corp:` namespace (or every tree in use).	Recommended	A tag missing from the taxonomy page has no title on the site.
11	`content-type` metadata values match the `tcs-corp:content-type` titles.	Recommended	Keeps eyebrows, filters and tag labels consistent (7.3.1).
12	Deprecated tags are not removed until no page, block field or index row uses them.	Recommended	See 7.6.4.

7.4.2 Review Cadence

Activity	Frequency	Responsible
Review new tag requests	[To be confirmed: frequency — owner: taxonomy owner]	Taxonomy owner
Check for unused and duplicate tags (AEM Tagging console usage count, query index)	[To be confirmed: frequency — owner: taxonomy owner]	Taxonomy owner with content operations
Translation check of tag titles for 13 locales	Before a tag is used on a locale	Localisation team
Republish all locale taxonomy pages	After every taxonomy change	Content operations

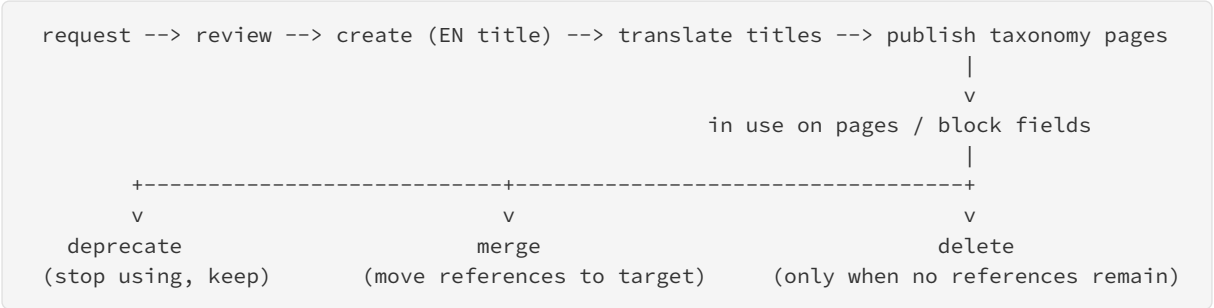
7.5 Tag Ownership

Role	Responsibilities	Named owner
Taxonomy owner	Approves new tags, structure changes, merges and deletions; keeps the business taxonomy (7.3)	[To be confirmed: taxonomy owner — owner: digital marketing lead]

Role	Responsibilities	Named owner
Tree owners (industry, service, topic, content type, region)	Curate the tags of their tree; propose changes	[To be confirmed: tree owners per business unit — owner: taxonomy owner]
Localisation team	Translates tag titles for each locale language in AEM	[To be confirmed: localisation contact — owner: content operations lead]
Content operations	Creates tags in the Tagging console after approval; publishes the 13 taxonomy pages; checks the published JSON	[To be confirmed: content operations contact — owner: content operations lead]
Authors	Tag pages and blocks with approved tags; request new tags	All authors
Development team	Code that reads tags (`utils/taxonomy.js`, `utils/tags.js`, List, Filter List), the query index definition, `paths.json` mappings	Development team
AEM administrators	Permissions on `/content/cq:tags` (who can create, move, delete tags)	[To be confirmed: AEM administrator — owner: platform owner]

> **Tip:** Restrict create / move / delete permissions on `/content/cq:tags/tcs-corp` to the taxonomy owner and content operations through an AEM user group, so authors can only apply tags. [To be confirmed: tag permission groups — owner: AEM administrator]

7.6 Tag Lifecycle



7.6.1 Create

1. The author or tree owner requests the tag (name, English title, parent, reason).
2. The taxonomy owner approves it (checks for duplicates and the naming rules in 7.4.1).
3. Content operations creates the tag in AEM: Tools → General → Tagging → select the parent → Create Tag; enter the Title and the Name (lower case, hyphenated).
4. Republish the taxonomy page of every locale (`/content/tcs-corp/{country}/{language}/taxonomy`) so the title is in `/{locale}/taxonomy.json`.
5. Check the published JSON, for example `https://main--tcs-corp--sradha-tcs.aem.live/en/taxonomy.json`.

Impact: none on existing pages. Until step 4, the tag shows an ID-derived label.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/aem-tagging-console.png`

Purpose: Managing tags in AEM
Description: The AEM Tagging console (Tools → General → Tagging) in column view with the `TCS Corp` namespace open, showing the Content Type, Industry and Region trees and the toolbar actions Create, Edit, Move, Merge, Delete and Publish.

[Screenshot Placeholder]
File name: `screenshots/aem-tagging-create-tag.png`
Purpose: Creating a tag
Description: The Create Tag dialog below `TCS Corp > Industry`, with Title "Banking", Name "banking" and an optional Description filled in.

7.6.2 Translate

1. In the Tagging console select the tag → Edit (or Properties) → Localization: add the title for each language used by the locales (de, fr, es, pt, nl, sv, da, no, zh, ja, ko).
2. Republish the affected locale taxonomy pages.
3. Check a page of that locale: the tag label is translated.

Impact: labels only. Tag IDs, matching, the query index and the data layer (`xdm:tags` holds IDs) are unchanged. See the open point in 7.1.5 about how translated titles are published.

[Screenshot Placeholder]
File name: `screenshots/aem-tagging-localization.png`
Purpose: Translating tag titles
Description: The tag edit dialog of `tcs-corp:region/europe` with the Localization section listing translated titles (German "Europa", French "Europe", Japanese "□□□□□").

7.6.3 Deprecate

1. Agree a replacement tag (if any) and a removal date.
2. Prefix the English title with a marker for authors, for example "(Deprecated) …", **or** keep the title and announce the deprecation. [To be confirmed: deprecation marker convention — owner: taxonomy owner]
3. Remove the tag from pages and block fields over time (the Tagging console shows where a tag is used through its references; the published `/query-index.json tags` column shows published pages that still carry it).

Impact: none until pages change. Note that a title marker would also be visible to visitors (the title is the label), so a separate communication is usually preferable.

7.6.4 Merge

1. In the Tagging console select the tag to remove → Merge → choose the target tag.
2. AEM replaces the references in page content with the target tag on author. [To be confirmed: whether tag references stored in Universal Editor block properties (List Tags, List Item Tags, Card Tags, Teaser Tags, Filter List tag fields) are updated by a merge — owner: development team]
3. Republish every page whose tags changed (page tags drive the query index) and every page whose blocks reference the tag; republish the taxonomy pages.

Impact:

Area	Until pages are republished	After republishing
Page head (`article:tag`), Page Header and Conversation Canvas labels	Old ID; label falls back to the ID-derived label once the taxonomy no longer contains it	New ID and title
Query index `tags` column	Old ID on the published row	New ID
List (Tagged pages) / Filter List filters on the new tag	Pages with the old ID are not found	Found
Filter List pills defined with the old tag	Match only pages not yet republished	Must be re-selected if the block property was not updated
Data layer `xdm:tags`	Old ID	New ID; analytics reports split across both IDs for the transition period

7.6.5 Move or Rename

Moving a tag or changing its **name** changes its ID (for example `tcs-corp:topic/ai` → `tcs-corp:topic/artificial-intelligence`), with the same impact as a merge. Changing only the **title** keeps the ID: republish the taxonomy pages and nothing else is needed.

7.6.6 Delete

1. Make sure no page, block field or query index row uses the tag (see 7.6.3 step 3).
2. Delete the tag in the Tagging console and republish the taxonomy pages.

Impact if references remain: published pages keep the ID; the label becomes ID-derived; List / Filter List filters that use the tag keep working on the remaining pages; the data layer keeps sending the ID.

7.6.7 Lifecycle Impact Summary

Change	Republish taxonomy pages	Republish tagged pages	Update block fields	Query index	Analytics
Create tag	Yes	No	No	No change	No change
Change title / translate	Yes	No	No	No change	No change (IDs)
Move / rename name	Yes	Yes	Check and re-select	Rows change on republish	New ID from then on
Merge	Yes	Yes	Check and re-select	Rows change on republish	New ID from then on
Deprecate	Optional	Over time	Over time	Changes as pages are edited	Gradual
Delete	Yes	Before deleting	Before deleting	Must contain no rows with the ID	Old ID stops

7.7 Parent-Child Relationships

7.7.1 How Relationships Are Derived

- A tag's parents are the prefixes of its ID at `/` boundaries: `tcs-corp:industry/banking/retail` → `tcs-corp:industry/banking` → `tcs-corp:industry`.
- The bare namespace (`tcs-corp:`) is not treated as a parent by the matching code (7.2.2).

- No code walks the tree downwards from the AEM tag tree; the Filter List finds children by looking at the tags **used by the listed pages** (`used(root)`): unused child tags never appear as pills or options.

7.7.2 Rules for Authors

Situation	What to do	Result
A page is about retail banking	Tag `tcs-corp:industry/banking/retail` only	Found by filters on …/`banking/retail`, …/`banking` and `tcs-corp:industry`
A List should show all industry pages	List Tags = `tcs-corp:industry`	Every page with any industry tag
A Filter List should offer one pill per industry	Primary filter category = `tcs-corp:industry` (no Primary filter tags)	One pill per industry tag used by the listed pages, sorted by label
A Filter List should offer exactly three pills	Primary filter tags = the three tags, in order	Exactly those pills, in the authored order
A parent tag should be shown as a card label	Add the parent tag to the page as well	The parent label appears (if within the first 2 or 3 tags)

7.7.3 Label Order and Display

Tag labels are shown in the order the tags are stored on the page or in the field (the order of the `article:tag` elements or of the query index array). Parents are not shown automatically, and labels are not sorted except for Filter List pills and options built from a category or taxonomy root (sorted with `localeCompare` on the label).

7.8 Usage Examples

7.8.1 Page Tags

Where they are set: AEM page properties (Basic → Tags) or a `tags` column in the locale's bulk metadata sheet (one row per URL pattern; several tags separated by commas).

What the code reads: `getPageTags()` reads `<meta property="article:tag">` (Edge Delivery publishes one such element per value of a `tags` metadata property), else `<meta name="tags">`.

```
<meta property="article:tag" content="tcs-corp:industry/banking">
<meta property="article:tag" content="tcs-corp:region/europe">
```

Used by: Page Header (tag list), Conversation Canvas (suggestion pills), the data layer page entry (`xdm:tags`) and, through the query index `tags` column, the List and the Filter List.

> **Important:** The aem.live documentation states that AEM publishes the page property Tags (`cq:tags`) as `<meta name="cq-tags">` (a comma-separated list of tag IDs). The code and `helix-query.yaml` do not read `cq-tags`. Verify on a published page with tags which meta elements are present. If only `cq-tags` is present, page tags set in page properties will not reach the Page Header, the query index (List, Filter List) or the data layer; use the bulk metadata `tags` column, or extend `getPageTags()` and `helix-query.yaml` to read `cq-tags`. [To be confirmed: published meta name of page tags set in AEM page properties — owner: development team]

> **[Screenshot Placeholder]**
> **File name:** `screenshots/aem-page-properties-tags.png`
> **Purpose:** Tagging a page
> **Description:** AEM Sites console page properties of an insights article, Basic tab, with the Tags field showing "TCS Corp : Industry / Banking" and "TCS Corp : Region / Europe".

7.8.2 List Block — Tagged Pages

Field	Value	Effect
List source	Tagged pages (`tags`)	Pages from the query index
Search root	`/en/insights` (empty with "Use the current page when empty" on: the current page)	Only pages below the root
Tags	`tcs-corp:industry/banking`, `tcs-corp:region/europe`	The tags to match
Match	Any of the tags / All of the tags	OR / AND (each with child tags)
Include descendant pages / Maximum search depth	On / 0	All levels below the root

Pages with `hide-in-nav = yes` are skipped. In the editor, a Tagged pages list without tags shows the hint `tags-missing` (placeholder `listEditorHint`). Card kickers show the tag labels (see 7.8.6). List Items (Fixed list) can override a page's tags with their own Tags field.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/ue-tag-picker-list.png`
> **Purpose:** Tag picker in Universal Editor
> **Description:** Universal Editor properties panel of a List block with List source "Tagged pages"; the Tags field's tag picker dialog is open, browsing TCS Corp > Industry with "Banking" selected.

7.8.3 Filter List Block — Filters

Field (tab)	Example	Effect
Primary filter category (Filters)	`tcs-corp:industry`	Without Primary filter tags: one pill per industry tag used by the listed pages
Primary filter tags → Tag Id (Filters)	`tcs-corp:industry/banking`, `tcs-corp:industry/insurance`	Exactly these pills, in this order; labels from the taxonomy
Show All filter (Filters)	On (default)	An "All" pill (placeholder `all`)
Allow multi select (Filters)	Off (default)	One pill at a time; on: pages with every selected pill
Enable advanced filter (Filters)	On (default)	The filter button opens the Advanced filter panel
Advanced filter categories (Advanced)	Category label "Region", Taxonomy root `tcs-corp:region`	A fieldset "Region" with one checkbox per region tag used by the pages
Filter logic (Advanced)	AND (default) / OR	Every / any checked option
Tag source (Mapping)	`tags` (default)	The index column whose first 2 tags are shown on cards

Filtering always uses the pages' `tags` index column (whatever the Tag source mapping); the Tag source only changes which column's tags are shown on the cards. Lists made before the Primary filter tags field (legacy "filters" with a Tag title and Tags) are still read.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/ue-filter-list-filters-tab.png`
> **Purpose:** Configuring tag filters
> **Description:** Universal Editor properties panel of a Filter List, Filters tab, with Primary filter category `TCS Corp : Industry`, two Primary filter tags, Show All filter on and Allow multi select off.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/filter-list-tag-pills.png`
> **Purpose:** Tag filters on the page
> **Description:** A published Filter List showing the pills "All", "Banking", "Insurance" with "Banking" pressed, the filter button with a count badge and the open Advanced filters panel with a "Region" fieldset of checkboxes.

7.8.4 Teaser Block — Tag List

Turn on **Display Tags**; the **Tags** field appears (model `condition`). The Teaser renders `<ul class="teaser-tags">` with one `<li data-tag="{ID}">{label}</li>` per tag, labels from the taxonomy, in the authored order (no maximum).

7.8.5 Page Header Block — Tag List

The Page Header has no tag field: it shows the **page tags** (7.8.1) as `<ul class="page-header-tags">`, labels from the taxonomy:

Layout	Where the tags appear
Article (default for most page types)	In the meta row, before the content type
Display	As the eyebrow above the title
Landing / Secondary	As the eyebrow only when the page has no `content-type` metadata

7.8.6 Card Kickers (List and Cards)

The List and the Cards blocks share `decorators/card-list-decorator.js`. With **Show tags** on (default):

Card	Tags shown
Image card	Eyebrow, up to 3 tag labels, then the reading time (`{n}m`, placeholder `minutesShort`)
Feature card (First card as feature)	Up to 3 tag labels and the reading time
Panel card	The eyebrow, else the first tag label
Row (Display as teaser off)	Eyebrow (or content type) and the first tag label
Report and Statistic cards	No tags

In the Cards block each Card has its own Tags field; in the List the tags come from the page's index row or the List Item's Tags override.

7.8.7 Filter List Cards

Standard, compact and featured cards show up to **2** tag labels of the Tag source column plus the reading time; the dark (image-less) card of the Mixed style shows the first tag label only.

7.8.8 Conversation Canvas

The canvas uses the labels of the page tags as its suggestion pills when no other suggestions have been supplied (`setConversationCanvasSuggestions`).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/tag-labels-rendered.png`
> **Purpose:** Tag labels on the site
> **Description:** A collage of rendered tag labels: a Page Header meta row with “Banking” and “Europe”, a List image card kicker with eyebrow, two tags and “16m”, and a Teaser tag list.

7.9 Tags in the Query Index and Analytics

7.9.1 Query Index

`helix-query.yaml` indexes the page tags as a multi-value column:

```
tags:
  select: head > meta[property="article:tag"]
  values: attribute(el, "content")
```

- The index stores tag **IDs**, never titles; a page’s row changes only when the page is republished (and re-indexed).
- `utils/query-index.js getPages()` loads `/query-index.json` once per page view, 500 rows per request, keeps the rows of the current locale (paths below `/{locale}`), and turns `tags` into an array with `listOf` (a JSON array, a JSON string or a comma-separated list).
- Excluded from the index: each locale’s `references/`, `error-pages/`, `data/`, `nav`, `footer`, block libraries, `site-global-templates`, `fragments` and `modals` paths and JSON resources.
- The List and the Filter List filter the index in the browser; no repository query runs.

> **Warning:** On 13 October 2026 the published `/query-index.json` listed columns that differ from `helix-query.yaml` in the repository (it contained earlier columns such as `pageCategory`, `storyTags`, `timeToRead`, and lacked `author`, `cardTitle`, `cardDescription`, `cardImage`, `cardImageAlt`, `imageAlt`, `eyebrow`, `category`, `hideInNav`, `featureCard`, `featuredPriority`). The `tags` column was present. Check which index definition is active in the site configuration and re-index after aligning it. [To be confirmed: active query index definition — owner: development team / site administrator]

7.9.2 Data Layer Page Entry

`decorators/data-layer-decorator.js` `trackPage()` pushes the page once per page view (on load, in the eager phase) to `window.adobeDataLayer`:

```
{
  "page": {
```

```
    "page-1x2y3z": {
      "@type": "tcs-corp/components/page",
      "dc:title": "The agentic bank",
      "xdm:language": "en-US",
      "xdm:template": "article",
      "xdm:tags": ["tcs-corp:industry/banking", "tcs-corp:region/europe"],
      "repo:path": "/en/insights/agentic-bank"
    }
  }
}
```

`xdm:tags` is `getPageTags()`: the page tag **IDs** (not labels), trimmed and de-duplicated, in page order. IDs are language-independent, so reports aggregate the same tag across all 13 locales; map IDs to labels in the analytics tool (for example a classification based on `/en/taxonomy.json`). [To be confirmed: analytics classification of tag IDs — owner: analytics lead]

7.9.3 Tag Interactions

Interaction	Event	Values
Filter List pill click	`cmp:click`	`interaction: "filter"`, `value`: the pill's tag ID or `all`, `dc:title`: the label
Filter List Advanced filter check-box	`cmp:change`	`interaction: "filter-option"`, `value`: the tag ID, `checked`: true / false
List / Cards card click	`cmp:click`	The card's component id (`data-cmp-id`), link URL and type; tags are not sent

> **Note:** No tag manager is loaded yet (`scripts/delayed.js` is empty), so these data layer entries are collected in `window.adobeDataLayer` but not sent to Adobe Analytics, Customer Journey Analytics or Adobe Experience Platform. Personalisation by tag (for example Adobe Target audiences) is not implemented (as of this handover).

Part 8 — Metadata Documentation

This part explains where page metadata comes from, how it reaches the published HTML, which code reads each value, and how metadata feeds search (query index, sitemap), analytics (data layer) and future personalisation. Section 8.2 is the field-by-field reference; 8.3 is a one-page cheat sheet.

8.1 How Metadata Works

8.1.1 Sources of Metadata

Source	Where authors set it	Fields	Published as
Universal Editor page properties (model `page-metadata` in `models/_page.json`)	Universal Editor → page properties panel (also the AEM Sites console page properties)	Title (`jcr:title`), Description (`jcr:description`), Keywords (`keywords`, multi), Page Type (`page-type`), Navigation (`nav`), Footer (`footer`), 404 Page (`error-page`)	`<title>` / `og:title` / `twitter:title`, `description` / `og:description` / `twitter:description`, `keywords`, `<meta name="page-type">`, `nav`, `footer`, `error-page`
Standard AEM page properties	AEM Sites console → Properties	Tags (`cq:tags`); other standard properties (for example robots, canonical) as supported by AEM	Per the aem.live documentation: `cq-tags` (comma-separated IDs); see 7.8.1
Values added by AEM / Edge Delivery automatically	—	Last modification and publication times, the page image	`modified-time`, `published-time`, `og:image`, `og:image:secure_url`, `twitter:image`, `twitter:card` (default `summary_large_image`), `og:url`
Bulk metadata sheet of the locale	AEM spreadsheet `/content/tcs-corp/{country}/{language}/metadata`, published as `/{locale}/metadata.json`	One row per URL pattern (`url` column); one column per metadata name (39 columns in AGENTS.md, see 8.2)	`<meta name="{column}">` for every non-empty cell of the matching rows (`tags` as `article:tag`)

Important: The bulk metadata sheets only apply once **all 13** are listed in the site configuration's `metadata.source` (`POST https://admin.hlx.page/config/{org}/sites/{site}/metadata.json` with `{"source": ["/en/metadata.json", "/de-de/metadata.json", …]}`). Until then no bulk metadata applies. Each sheet's URL patterns must start with its locale (for example `/de-de/**`). [To be confirmed: metadata.source configured for the 13 locales — owner: site administrator]

Note: On 13 October 2026 `/en/metadata.json` was published with 29 columns (`url` to `priority`) and no rows; the columns `listen-link`, `author`, `card-title`, `card-description`, `card-image`, `card-image-alt`, `image-alt`, `eyebrow`, `category` and `hide-in-nav` were not yet in the sheet, and `/de-de/metadata.json` returned 404. Add the missing columns to the sheets before using them in bulk. [To be confirmed: column set of the 13 bulk metadata sheets — owner: content operations lead]

> **[Screenshot Placeholder]**
> **File name:** `screenshots/ue-page-properties-panel.png`
> **Purpose:** Page properties in Universal Editor
> **Description:** Universal Editor with the page selected and the properties panel showing Title, Description, Keywords, Page Type (select: Default, Home, Article, Blog, Press Release), Navigation, Footer and 404 Page, each with its help text.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/aem-bulk-metadata-sheet.png`
> **Purpose:** The bulk metadata sheet
> **Description:** The AEM spreadsheet editor for `/content/tcs-corp/us/en/metadata` showing the header row (url, page-type, meta-title, …, hide-in-nav) and example rows `/en/newsroom/**` → `press-release` and `/en/insights/**` → `article`.

8.1.2 Precedence

1. **Page-level values win over bulk metadata.** A page property (for example Page Type) overrides the bulk sheet's column for that page (the Page Type help text states this; page-level metadata taking precedence is standard Edge Delivery behaviour).
2. **Bulk rows are evaluated top to bottom.** Put site-wide rows (`/en/**`) before more specific rows (`/en/insights/**`); a later matching row overrides values of an earlier one.
3. **Code fallbacks** apply on top (see each field in 8.2): for example `og-title` → `meta-title` → Title; `card-title` → Title; `page-type` → `template`; `publish-date` → `publication-date`.

8.1.3 From AEM to the Browser

```
AEM author                  Edge Delivery Services          Browser (code)
----------                  ----------------------          --------------
page properties --publish-> page HTML <head>:               scripts.js loadEager:
(Title, Page Type, nav)     <title>, <meta name="page-type">  - decorateHead() (SEO)
                            + bulk metadata of matching rows  - trackInteractions()
metadata sheet  --publish-> <meta name="{column}">              -> trackPage() (ACDL)
(/{locale}/metadata.json)                                     - decoratePageType()
                            query index (helix-query.yaml)    - loadTemplate()
                            /query-index.json <- the <head> blocks: getPageMetadata(),
                            sitemap.xml <- the query index  getPages() (index)
```

Every metadata value is a `<meta>` element in the head of the published HTML. Code reads it with `getMetadata(name)` from `scripts/aem.js` (meta `name`, or `property` for names with a colon). The query index reads the **server-delivered** head, so changes made by JavaScript (for example the SEO decorator) are **not** in the index.

8.1.4 Inherited Properties: Navigation, Footer and 404 Page

`utils/inherited-metadata.js` `getInheritedMetadata(name)` emulates AEM's inherited page properties for `nav`, `footer` and `error-page` (`INHERITED_METADATA`):

1. The page's own value (page property, or the `nav` / `footer` / `error-page` column of the bulk sheet).
2. Else the nearest parent page's value, up to the locale's Home page (`utils/site.js` `getParentPages`: for `/de-de/insights/blogs/x` → `/de-de/insights/blogs`, `/de-de/insights`, `/de-de/`). The parents' published HTML is fetched **in parallel** and their `nav`, `footer` and `error-page` metas are read. Values are cached in

`sessionStorage` (key `tcs-inherited-metadata`) for the browser session: a changed parent value shows in a new session.

3. Else the default: `references/nav`, `references/footer` of the locale (header and footer), and for the 404 page `/{locale}/error-pages/404`, then `/en/error-pages/404`, then built-in content (`scripts.js loadErrorPage`).

Values can be AEM paths, published paths or paths relative to the locale root (`references/nav-careers`); `utils/site.js resolvePagePath` normalises them. On the published Home page `/en/` (checked 13 October 2026) the head contains `<meta name="nav" content="/en/references/nav">`, `footer /en/references/footer` and `error-page /en/error-pages/404`.

8.1.5 SEO Decorator Mapping

`decorators/seo-decorator.js decorateHead()` runs first in `loadEager` and writes the bulk metadata SEO and social columns to the standard head tags. It only sets a tag when a value exists (otherwise the tag delivered by Edge Delivery stays).

Column(s)	Head element set	Fallback chain
`language` (via `getPageLanguage`)	`<html lang>`	`language` if it is a language code (`^[a-z]{2,3}(-[a-z0-9]{2,8})*$`), else the locale's tag (`en-US`, `de-DE`, ...)
`meta-title`	`document.title`	Not set: the Title stays
`meta-description`	`<meta name="description">`	Not set: the Description stays
`meta-keywords`	`<meta name="keywords">`	Not set: the Keywords stay
`og-title`	`<meta property="og:title">`	`og-title` → `meta-title` → (unchanged)
`og-description`	`og:description`	`og-description` → `meta-description` → (unchanged)
`og-image`	`og:image` (made absolute)	(unchanged)
`og-type`	`og:type`	(not set)
`twitter-card`	`<meta name="twitter:card">`	(unchanged: `summary_large_image`)
`twitter-title`	`twitter:title`	`twitter-title` → `og-title` → `meta-title`
`twitter-description`	`twitter:description`	`twitter-description` → `og-description` → `meta-description`
`twitter-image`	`twitter:image` (absolute)	`twitter-image` → `og-image`
`canonical-url`	`<link rel="canonical">` and `og:url` (absolute)	(unchanged)
`robots`	— (standard name, applied by Edge Delivery in the delivered HTML)	—

> **Warning:** The SEO decorator changes the head **in the browser**. Search engines that render JavaScript see the result; social network crawlers and other clients that do not run JavaScript see only the values delivered by Edge Delivery (from Title, Description, Keywords and the page image). `og:image:secure_url` is not updated by the decorator. For values that must be present without JavaScript, prefer the standard names Edge Delivery handles on the server (`title`, `description`, `image`, `canonical`, `robots`).

8.1.6 Page Type and Templates

- `scripts.js` `decoratePageType()` adds the class-name form of `page-type` to `<body>` (for example `press-release`).
- `loadTemplate()` uses `page-type`, else the standard `template` metadata; when the value is one of `TEMPLATES` (`home`, `article`, `blog`, `press-release`), `templates/{name}/{name}.js` and `.css` load before the page is decorated (see templates/README.md).
- The Page Header uses Page Type for its default layout: `landing` for page types `landing` and `listing`, else `article` (an authored Layout wins).
- The data layer page entry sends it as `xdm:template`.

8.1.7 getPageMetadata() Naming

`utils/metadata.js` `getPageMetadata()` returns typed values with camelCase keys:

Key	Read from (first non-empty)	Type / conversion
`title`	`og:title` meta at module load (the authored Title), else `document.title`	string; read before the SEO decorator runs, so it is never `meta-title`
`description`	`description` meta at module load	string; never `meta-description`
`image`	`featured-image`, `og:image`	string (not used by blocks)
`pageType`	`page-type`, `template`	string as authored
`contentType`	`content-type`	string
`category`	`category`	string (not used by blocks)
`tags`	`getPageTags()`	string[] of IDs
`authorId`	`author-id`	string
`author`	`author`	string
`publishDate`	`publish-date`, `publication-date`	string (YYYY-MM-DD expected)
`modifiedDate`	`modified-date`	string (not used by blocks)
`readingTime`	`reading-time`	string
`listenLink`	`listen-link`	string (URL)
`thumbnailImage`	`thumbnail-image`	string (not used by blocks)
`featuredImage`	`featured-image`	string
`region`	`region`	string
`country`	`country`, else the locale's country folder (`us`, `de`, ...)	string (not used by blocks)
`language`	`getPageLanguage()`	string
`persona`	`persona`	string (not used by blocks)
`campaignId`	`campaign-id`	string (not used by blocks)
`featured`	`featured`	boolean: true for `yes`, `true`, `1`, `on`, `y` (case-insensitive)
`priority`	`priority`	number (`parseFloat`), else `null`

Users of `getPageMetadata()`: Page Header (`blocks/page-header/page-header.js`), the article template (`authorId`), the press-release template (`region`), the data layer page entry (`title`, `pageType`).

8.1.8 Query Index Columns

`helix-query.yaml` builds `/query-index.json` from the published heads. The index columns use camelCase names; `utils/query-index.js` `columnOf(page, name)` converts metadata names (`featured-image`) to index names (`featuredImage`), so block Mapping fields accept either.

Index column	Selector (server head)	Read by
`path`	(page path)	List, Filter List, Conversation Canvas
`lastModified`	`last-modified` response header	List (date and sort fallback), Filter List (sort tie-break)
`robots`	`meta[name="robots"]`	Sitemap (noindex pages excluded)
`title`	`meta[property="og:title"]`	List, Filter List, Conversation Canvas menu
`description`	`meta[name="description"]`	List, Filter List
`image`	`meta[property="og:image"]` (path only)	List (card image, 2nd choice; the site's default image is ignored)
`tags`	`meta[property="article:tag"]` (multi-value)	List, Filter List
`metaTitle`, `metaDescription`	`meta-title`, `meta-description`	Not read by code (available to mappings)
`pageType`	`page-type`	Filter List (Page types)
`contentType`	`content-type`	List, Filter List, card kind (report)
`author`, `authorId`	`author`, `author-id`	Filter List (author lockup), List (feature card author)
`publishDate`, `modifiedDate`	`publish-date`, `modified-date`	Filter List (date, Latest sort), List (date, last modified)
`readingTime`	`reading-time`	List, Filter List
`thumbnailImage`, `featuredImage`	`thumbnail-image`, `featured-image`	List (image 3rd / 4th choice), Filter List (image source and fallback)
`region`, `country`, `language`, `persona`, `campaignId`	same names	Not read by code
`featured`, `priority`	`featured`, `priority`	Filter List (featured / hero cards, sorts)
`featureCard`, `featuredPriority`	`feature-card`, `featured-priority`	Filter List fallbacks for `featured` / `priority`
`cardTitle`, `cardDescription`, `cardImage`, `cardImageAlt`, `imageAlt`	`card-title`, `card-description`, `card-image`, `card-image-alt`, `image-alt`	List
`eyebrow`, `category`	`eyebrow`, `category`	List (eyebrow, then category)
`hideInNav`	`hide-in-nav`	List (page skipped when yes)

Warning: See 7.9.1: on 13 October 2026 the published index did not match this definition (no `author`, card, `eyebrow`, `category`, `hideInNav`, `featureCard`, `featuredPriority` columns). Until it does, those values have no effect on the List and the Filter List.

8.1.9 Sitemap

`helix-sitemap.yaml` generates `/sitemap.xml` from `/query-index.json` with `lastmod` (`YYYY-MM-DD`, from `lastModified`), origin `https://main--tcs-corp--sradha-tcs.aem.live` until a production domain is configured. Because the index includes `robots`, pages with `robots: noindex` are left out of the sitemap. Pages excluded from the index (references, error pages, data, block libraries, templates) are not in the sitemap. No other metadata affects the sitemap. [To be confirmed: production domain and CDN]

8.1.10 Data Layer Page Entry

Data layer field	Metadata source
`dc:title`	Title (authored, `getPageMetadata().title`), else `document.title`
`xdm:language`	`language` metadata if a language code, else the locale's tag
`xdm:template`	`page-type`, else `template`; omitted when empty
`xdm:tags`	page tags (IDs)
`repo:path`	the page path (not metadata)

No other metadata (author, content type, region, persona, campaign, dates) is in the data layer today. Adding them is a change to `trackPage()` in `decorators/data-layer-decorator.js`.

8.1.11 Personalisation

No personalisation is implemented (as of this handover): no Adobe Target or Adobe Journey Optimizer is loaded, and `scripts/delayed.js` is empty. Metadata that could feed audiences later (`persona`, `campaign-id`, `region`, `country`, `content-type`, tags) is published in the head, partly indexed, and available through `getPageMetadata()`; only page type, language and tags reach the data layer. The Teaser's **Personalization** toggle only adds a marker class and `data-personalization="true"` for a future script.

8.2 Metadata Field Reference

Each field below lists where it is set, what it does and which code reads it. "Used By" entries were verified by searching the code for the meta name and its camelCase index name.

> **Note:** No metadata field is validated by the models or the code (there are no `required` or `validation` rules in `models/_page.json`, and the bulk sheet is free text). "Mandatory" therefore means **Recommended** unless marked otherwise. Yes / no fields accept `yes`, `true`, `1`, `on`, `y` (case-insensitive) as yes; anything else is no.

Fields 8.2.1 to 8.2.8 are page properties (the Universal Editor page properties model, and Tags from the standard AEM page properties). Fields 8.2.9 to 8.2.46 are the bulk metadata columns in the order of AGENTS.md (page-type is described once, in 8.2.5). Fields 8.2.47 and 8.2.48 are legacy names read as fallbacks.

8.2.1 title — Title

- **Field Name:** Page property **Title** (`jcr:title`, `models/_page.json`); published as `<title>`, `og:title` and `twitter:title`. Can also be set as bulk metadata `title`.
- **Purpose:** The page's name: browser tab title, the Page Header heading (`h1`), the default card title in lists and the title sent to analytics.
- **Mandatory:** Recommended for every page (not enforced). Without it the List derives a title from the last path segment (`agentic-bank` → "agentic bank").

- **Allowed Values:** Plain text.
- **Business Rules:** `meta-title` replaces the browser title in the browser but **not** the Page Header heading, the index `title` or `dc:title` (those keep the authored Title). The List prefers `card-title`. Recommended: unique per locale, up to about 60 characters.
- **Used By Components:** Page Header (`getPageMetadata().title` → `h1.page-header-title`; also the title of the share action); List (index `title`, after `card-title`; Order by Title); Filter List (Title source default `title`; Alphabetical sort); Conversation Canvas (menu labels from the index `title`).
- **Used By Search:** Query index `title` (from `og:title`); search result title for engines (unless `meta-title` is set and the engine renders JavaScript).
- **Used By Personalization:** None (not implemented).
- **Used By Analytics:** `dc:title` of the page entry.
- **Example Values:** `The agentic bank`, `Banking and financial services`.

8.2.2 description — Description

- **Field Name:** Page property **Description** (`jcr:description`); published as `<meta name="description">`, `og:description`, `twitter:description`.
- **Purpose:** Summary of the page for search results, social previews, the Page Header and cards.
- **Mandatory:** Recommended (not enforced).
- **Allowed Values:** Plain text.
- **Business Rules:** `meta-description` replaces the head description in the browser; the Page Header, the index and the cards keep the authored Description. The List prefers `card-description`. Recommended: 120–160 characters.
- **Used By Components:** Page Header (`.page-header-description`); List (index `description`, after `card-description`); Filter List (Description source default `description`; shown on default, featured, report, dark and hero cards).
- **Used By Search:** Query index `description` (from the server `description` meta).
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `Banking is entering a new era of autonomous, AI-driven operations.`

8.2.3 keywords — Keywords

- **Field Name:** Page property **Keywords** (`keywords`, multi value); published as `<meta name="keywords">`.
- **Purpose:** Keyword list in the page head.
- **Mandatory:** No.
- **Allowed Values:** One or more words or phrases.
- **Business Rules:** `meta-keywords` overrides it in the browser. Major search engines ignore the keywords meta; keep it short or empty.
- **Used By Components:** None.
- **Used By Search:** Not indexed (no `keywords` column in `helix-query.yaml`).
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `agentic AI, banking`.

8.2.4 tags — Tags

- **Field Name:** Standard AEM page property **Tags** (`cq:tags`, Sites console → Properties → Basic) or bulk metadata column `tags` (comma-separated). Read from `<meta property="article:tag">`, else `<meta name="tags">`.
- **Purpose:** Classifies the page with taxonomy tags (Part 7).
- **Mandatory:** Recommended for content pages that appear in lists or filters.

- **Allowed Values:** Tag IDs from the `tcs-corp:` namespace, e.g. `tcs-corp:region/europe`. Titles without `:` or `/` are accepted and shown as typed (not recommended).
- **Business Rules:** De-duplicated and trimmed by `getPageTags()`; order kept. Child tags match parent filters (7.2). Up to 3 labels on List / Cards image and feature cards, 2 on Filter List cards. See the open point on `cq-tags` in 7.8.1.
- **Used By Components:** Page Header (tag list), Conversation Canvas (suggestion pills), List and Filter List (index `tags`).
- **Used By Search:** Query index `tags` (multi-value, from `article:tag`).
- **Used By Personalization:** None (not implemented).
- **Used By Analytics:** `xdm:tags` of the page entry (IDs).
- **Example Values:** `tcs-corp:industry/banking`, `tcs-corp:content-type/article`.

8.2.5 page-type — Page Type

- **Field Name:** Page property **Page Type** (`page-type`, select) and bulk column `page-type`. Fallback: the standard `template` metadata.
- **Purpose:** Selects the Site Global Template (layout and template code) and classifies the page.
- **Mandatory:** No (Default = empty: no template).
- **Allowed Values:** Page property select: Default (empty), `home`, `article`, `blog`, `press-release`. Bulk sheet: any value; values with code effects are `home`, `article`, `blog`, `press-release` (templates), `landing` and `listing` (Page Header Landing layout) and the Filter List Page types options `article`, `blog`, `report`, `case-study`, `news`, `event`.
- **Business Rules:** The page property overrides the sheet. The class-name form is added to `<body>`. Only `home`, `article`, `blog`, `press-release` load template code (`TEMPLATES` in `scripts/scripts.js`); the blog template reuses the article template. The Filter List compares class-name forms (`Case Study` matches `case-study`). Note the mismatch: the Page Type select cannot set `report`, `case-study`, `news`, `event`, `landing` or `listing` (sheet only), and the Filter List Page types field cannot select `press-release` or `home`.
- **Used By Components:** `scripts.js` (`decoratePageType`, `loadTemplate`); templates `home`, `article`, `blog`, `press-release`; Page Header (default layout); Filter List (Page types).
- **Used By Search:** Query index `pageType`.
- **Used By Personalization:** None.
- **Used By Analytics:** `xdm:template` of the page entry.
- **Example Values:** `article`, `press-release`; sheet row `/en/newsroom/**` → `press-release`.

8.2.6 nav — Navigation

- **Field Name:** Page property **Navigation** (`nav`, aem-content) and bulk column `nav`.
- **Purpose:** The header (nav) fragment of the page and, unless they set their own, its child pages.
- **Mandatory:** No. Empty: inherited (8.1.4), else `/{locale}/references/nav`.
- **Allowed Values:** A page path: AEM path, published path or a path relative to the locale root (`references/nav-careers`).
- **Business Rules:** Inherited from the nearest parent up to the locale Home page; parent values cached per browser session.
- **Used By Components:** Header (`blocks/header/header.js` via `getInheritedMetadata('nav')`).
- **Used By Search:** Not indexed; `/*/nav` and `/*/references/**` pages are excluded from the index and sitemap.
- **Used By Personalization:** None (a different nav per section is possible by authoring).
- **Used By Analytics:** None.
- **Example Values:** `/content/tcs-corp/us/en/references/nav`, `references/nav-careers`.

8.2.7 footer — Footer

- **Field Name:** Page property **Footer** (`footer`, aem-content) and bulk column `footer`.
- **Purpose:** The footer fragment of the page and its child pages.
- **Mandatory:** No. Empty: inherited, else `/{locale}/references/footer`.
- **Allowed Values:** As Navigation.
- **Business Rules:** As Navigation.
- **Used By Components:** Footer (`blocks/footer/footer.js` via `getInheritedMetadata('footer')`).
- **Used By Search:** Not indexed; footer pages excluded.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `/content/tcs-corp/de/de-de/references/footer`.

8.2.8 error-page — 404 Page

- **Field Name:** Page property **404 Page** (`error-page`, aem-content) and bulk column `error-page`.
- **Purpose:** The page shown when a page below this one is not found.
- **Mandatory:** No. Empty: inherited, else `/{locale}/error-pages/404`, else `/en/error-pages/404`, else built-in content.
- **Allowed Values:** As Navigation.
- **Business Rules:** For a missing page the lookup starts at its nearest existing parent; a configured page that fails to load falls through to the next option; content from another locale gets that locale's `lang`.
- **Used By Components:** `404.html` → `scripts.js` `loadErrorPage`.
- **Used By Search:** Error pages are excluded from the index and sitemap.
- **Used By Personalization:** None.
- **Used By Analytics:** None specific (the 404 page sends its own page entry).
- **Example Values:** `/content/tcs-corp/us/en/error-pages/404`.

8.2.9 url — URL Pattern (bulk sheet only)

- **Field Name:** Bulk metadata column `url`.
- **Purpose:** The pages a row applies to.
- **Mandatory:** Yes, for every sheet row (a row without a pattern applies to nothing).
- **Allowed Values:** A published path or pattern with `*` / `**` wildcards, starting with the locale: `/en/**`, `/de-de/insights/**`, `/en/newsroom/2026/*`.
- **Business Rules:** Rows are evaluated top to bottom; general rows first, specific rows later. Patterns use published paths (`/de-de/…`), not AEM paths. The `url` value is the row key, not page metadata used by code.
- **Used By Components:** None (the platform applies the rows).
- **Used By Search:** None directly.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `/en/insights/**`.

8.2.10 meta-title — SEO Title

- **Field Name:** Bulk column `meta-title`.
- **Purpose:** A search-optimised browser title different from the Title.
- **Mandatory:** No.
- **Allowed Values:** Plain text; Recommended up to about 60 characters.
- **Business Rules:** Applied in the browser by the SEO decorator (`document.title`); fallback for `og:title` and `twitter:title`. Not used for the Page Header, cards or `dc:title`.
- **Used By Components:** SEO decorator (`decorators/seo-decorator.js`).

- **Used By Search:** Query index `metaTitle` (not read by code); search engines that render JavaScript.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `Agentic AI in Banking – TCS`.

8.2.11 meta-description — SEO Description

- **Field Name:** Bulk column `meta-description`.
- **Purpose:** A search-optimised description.
- **Mandatory:** No.
- **Allowed Values:** Plain text; Recommended 120–160 characters.
- **Business Rules:** Replaces `<meta name="description">` in the browser; fallback for `og:description` and `twitter:description`.
- **Used By Components:** SEO decorator.
- **Used By Search:** Query index `metaDescription` (not read by code).
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `How agentic AI changes operations, risk and customer service in banking.`

8.2.12 meta-keywords — SEO Keywords

- **Field Name:** Bulk column `meta-keywords`.
- **Purpose:** Replaces the keywords meta.
- **Mandatory:** No.
- **Allowed Values:** Comma-separated words.
- **Business Rules:** Applied in the browser; ignored by major search engines.
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `agentic AI, banking, automation`.

8.2.13 author-id — Author ID

- **Field Name:** Bulk column `author-id` (or page metadata `author-id`).
- **Purpose:** Links the page to an entry of the locale's Authors sheet (`/{locale}/data/authors.json`: author-id, name, designation, bio, image, linkedin, region).
- **Mandatory:** Recommended for articles and blogs.
- **Allowed Values:** An `author-id` value of the Authors sheet.
- **Business Rules:** The Page Header and article template match the ID case-insensitively (`getAuthor`); the List and Filter List match it exactly. When no Authors row matches, the Page Header shows the `author` metadata as the name (no image or designation) and the article template adds no "About the author".
- **Used By Components:** Page Header (author lockup: name linked to LinkedIn, image, designation and region); article and blog templates ("About the author", placeholder `aboutTheAuthor`); List (feature card author, when First card as feature is on); Filter List (hero and featured spotlight cards, when the Author source column has no match).
- **Used By Search:** Query index `authorId`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `jdoe`, `author-0042`.

8.2.14 publish-date — Publish Date

- **Field Name:** Bulk column `publish-date`; fallback `publication-date`.
- **Purpose:** The publication date shown to visitors and used for "latest" ordering.
- **Mandatory:** Recommended for articles, blogs, press releases and insights.
- **Allowed Values:** `YYYY-MM-DD` (an ISO date-time also works: the first 10 characters are used).
- **Business Rules:** Other formats are shown as typed, without a machine-readable `datetime`, and sort incorrectly in the Filter List (string comparison). Not the AEM `published-time`.
- **Used By Components:** Page Header (article layout: "Published dd/mm/yy", placeholder `published`); Filter List (card meta `dd Mmm yy`; Latest sort, then last modified); List (card date when `modified-date` is empty; Order by last modified uses `modified-date`, else the publication time).
- **Used By Search:** Query index `publishDate`.
- **Used By Personalization:** None.
- **Used By Analytics:** None (List / Cards items send `repo:modifyDate` from the card date).
- **Example Values:** `2026-09-04`.

8.2.15 modified-date — Modified Date

- **Field Name:** Bulk column `modified-date`.
- **Purpose:** An editorial "last updated" date.
- **Mandatory:** No.
- **Allowed Values:** `YYYY-MM-DD`.
- **Business Rules:** In the List it is the **first** choice for the card date (before `publish-date`) and the value of Order by last modified (else the page's last publication). Not shown by the Page Header.
- **Used By Components:** List (`PAGE_MAPPING.date`, `lastModified`).
- **Used By Search:** Query index `modifiedDate`.
- **Used By Personalization:** None.
- **Used By Analytics:** Indirectly: `repo:modifyDate` of List items (the card date).
- **Example Values:** `2026-10-01`.

8.2.16 reading-time — Reading Time

- **Field Name:** Bulk column `reading-time`.
- **Purpose:** Estimated reading time.
- **Mandatory:** No.
- **Allowed Values:** Whole minutes (`16`) or any text (`2 hours`).
- **Business Rules:** A number is formatted: Page Header "16 min read" (placeholder `minRead`), cards "16m" (placeholder `minutesShort`); text is shown as typed.
- **Used By Components:** Page Header (article layout); List and Cards (kicker of image and feature cards, with Show tags on); Filter List (tag row of cards).
- **Used By Search:** Query index `readingTime`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `16`, `6`.

8.2.17 og-title — Social Title

- **Field Name:** Bulk column `og-title`.
- **Purpose:** The title of social link previews.
- **Mandatory:** No.
- **Allowed Values:** Plain text.

- **Business Rules:** Sets `og:title` in the browser; fallback `meta-title`; also the fallback of `twitter:title`. The server-delivered `og:title` (from the Title) is what the query index and non-JavaScript crawlers read.
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed under this name (the index `title` reads the server `og:title`).
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `The agentic bank — TCS perspective`.

8.2.18 og-description — Social Description

- **Field Name:** Bulk column `og-description`.
- **Purpose:** The description of social link previews.
- **Mandatory:** No.
- **Allowed Values:** Plain text.
- **Business Rules:** Sets `og:description`; fallback `meta-description`; fallback of `twitter:description`.
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `Five shifts every bank leader should plan for.`

8.2.19 og-image — Social Image

- **Field Name:** Bulk column `og-image`.
- **Purpose:** The image of social link previews.
- **Mandatory:** No.
- **Allowed Values:** An image URL or site path (made absolute).
- **Business Rules:** Sets `og:image` in the browser (not `og:image:secure_url`); fallback of `twitter:image`. An invalid URL is ignored. The index `image` column is the server `og:image` (the page image), not this column.
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `/en/media/agentic-bank-social.jpg`.

8.2.20 og-type — Open Graph Type

- **Field Name:** Bulk column `og-type`.
- **Purpose:** The Open Graph object type.
- **Mandatory:** No.
- **Allowed Values:** Open Graph types, e.g. `website`, `article`.
- **Business Rules:** Set only when present; no default.
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `article`.

8.2.21 twitter-card — Twitter / X Card Type

- **Field Name:** Bulk column `twitter-card`.
- **Purpose:** The card layout on X (Twitter).

- **Mandatory:** No (Edge Delivery delivers `summary_large_image`).
- **Allowed Values:** `summary`, `summary_large_image`, `app`, `player`.
- **Business Rules:** Overrides the delivered value in the browser.
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `summary`.

8.2.22 twitter-title — Twitter / X Title

- **Field Name:** Bulk column `twitter-title`.
- **Purpose:** The title on X cards.
- **Mandatory:** No.
- **Allowed Values:** Plain text.
- **Business Rules:** Fallbacks `og-title`, then `meta-title`.
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `The agentic bank`.

8.2.23 twitter-description — Twitter / X Description

- **Field Name:** Bulk column `twitter-description`.
- **Purpose:** The description on X cards.
- **Mandatory:** No.
- **Allowed Values:** Plain text.
- **Business Rules:** Fallbacks `og-description`, then `meta-description`.
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `Five shifts every bank leader should plan for.`

8.2.24 twitter-image — Twitter / X Image

- **Field Name:** Bulk column `twitter-image`.
- **Purpose:** The image on X cards.
- **Mandatory:** No.
- **Allowed Values:** An image URL or site path (made absolute).
- **Business Rules:** Fallback `og-image`.
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `/en/media/agentic-bank-x.jpg`.

8.2.25 canonical-url — Canonical URL

- **Field Name:** Bulk column `canonical-url`.
- **Purpose:** The preferred URL of duplicate or syndicated content.

- **Mandatory:** No.
- **Allowed Values:** An absolute URL or a site path (made absolute).
- **Business Rules:** Sets `<link rel="canonical">` and `og:url` in the browser; an invalid URL is ignored. Edge Delivery also supports the standard `canonical` metadata on the server, which is visible without JavaScript (Recommended for critical cases).
- **Used By Components:** SEO decorator.
- **Used By Search:** Not indexed; affects search engines that render JavaScript.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `https://[To be confirmed: production domain]/en/insights/agentic-bank`.

8.2.26 robots — Robots

- **Field Name:** Bulk column `robots` (standard metadata).
- **Purpose:** Instructs search engines whether to index the page and follow links.
- **Mandatory:** No (no value: indexable).
- **Allowed Values:** `index`, `noindex`, `follow`, `nofollow` and combinations, e.g. `noindex, nofollow`.
- **Business Rules:** Delivered by Edge Delivery in the HTML (no code needed). Pages with `noindex` are left out of the sitemap because the index includes `robots`; they still appear in the List and the Filter List (no code checks `robots`).
- **Used By Components:** None.
- **Used By Search:** Query index `robots`; sitemap exclusion; search engines.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `noindex, nofollow` (campaign test pages).

8.2.27 region — Region

- **Field Name:** Bulk column `region`.
- **Purpose:** The region the page belongs to; selects the Media Contacts of press releases.
- **Mandatory:** Recommended for press releases.
- **Allowed Values:** A value of the `region` column of the locale's Media Contacts sheet (`/{locale}/data/media-contacts.json`). [To be confirmed: list of region values — owner: corporate communications]
- **Business Rules:** The press-release template shows the contacts whose region matches (case-insensitive); when none match, or the page has no region, all contacts are shown. Free text, not a tag: keep it consistent with the `tcs-corp:region` titles if both are used.
- **Used By Components:** Press-release template (`templates/press-release/press-release.js`, `inRegion`).
- **Used By Search:** Query index `region` (not read by code).
- **Used By Personalization:** None (candidate for future regional targeting).
- **Used By Analytics:** None.
- **Example Values:** `Europe`, `Americas`.

8.2.28 country — Country

- **Field Name:** Bulk column `country`.
- **Purpose:** The country of the page's content.
- **Mandatory:** No.
- **Allowed Values:** Free text; Recommended ISO 3166-1 alpha-2 codes in lower case (as the locale folders: `us`, `de`, `fr`, …).
- **Business Rules:** `getPageMetadata().country` falls back to the locale's country folder; no block reads it.
- **Used By Components:** None.

- **Used By Search:** Query index `country` (not read by code).
- **Used By Personalization:** None (available for the future).
- **Used By Analytics:** None.
- **Example Values:** `de`, `jp`.

8.2.29 language — Language

- **Field Name:** Bulk column `language`.
- **Purpose:** Overrides the page language (`<html lang>`).
- **Mandatory:** No (default: the locale's language tag, e.g. `de-DE`).
- **Allowed Values:** A BCP 47 language code matching `^[a-z]{2,3}(-[a-z0-9]{2,8})*$` (case-insensitive), e.g. `en`, `en-GB`. Other values are ignored.
- **Business Rules:** Use only for pages whose content differs from the locale language (for example an English page on `/ja-jp/`). It also changes the sort collation of the List and the Filter List (`document.documentElement.lang`).
- **Used By Components:** SEO decorator (`<html lang>`); `getPageLanguage()`; data layer.
- **Used By Search:** Query index `language`; search engines (language detection).
- **Used By Personalization:** None.
- **Used By Analytics:** `xdm:language` of the page entry.
- **Example Values:** `en-GB`.

8.2.30 persona — Persona

- **Field Name:** Bulk column `persona`.
- **Purpose:** The target persona of the page.
- **Mandatory:** No.
- **Allowed Values:** Free text. [To be confirmed: persona list — owner: digital marketing lead]
- **Business Rules:** Published and indexed; no code uses it.
- **Used By Components:** None (`getPageMetadata().persona` exists but is not read).
- **Used By Search:** Query index `persona`.
- **Used By Personalization:** None today; intended for future audiences (Adobe Target / Journey Optimizer, not implemented).
- **Used By Analytics:** None (not in the data layer).
- **Example Values:** `cxo`, `it-leader`.

8.2.31 thumbnail-image — Thumbnail Image

- **Field Name:** Bulk column `thumbnail-image`.
- **Purpose:** A small image for cards.
- **Mandatory:** No.
- **Allowed Values:** An image path or URL (AEM Assets / site media).
- **Business Rules:** List card image order: `card-image`, page image (`og:image`, except the site's default image), `thumbnail-image`, `featured-image`. Filter List: the Fallback image column (default `thumbnail-image`) when the Image source (default `featured-image`) is empty.
- **Used By Components:** List; Filter List.
- **Used By Search:** Query index `thumbnailImage`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `/en/media/agentic-bank-thumb.jpg`.

8.2.32 featured-image — Featured Image

- **Field Name:** Bulk column `featured-image`.
- **Purpose:** The large image of the page: the Page Header Landing image and the default Filter List card image.
- **Mandatory:** Recommended for landing pages and for pages listed by Filter Lists.
- **Allowed Values:** An image path or URL.
- **Business Rules:** Page Header Landing layout loads it eagerly with high fetch priority (the page's LCP candidate), widths 1200 (≥ 900px) / 750. Filter List Image source default; List fourth choice.
- **Used By Components:** Page Header (Landing); Filter List (cards, hero and featured spotlights); List.
- **Used By Search:** Query index `featuredImage`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `/en/media/agentic-bank-hero.jpg`.

8.2.33 campaign-id — Campaign ID

- **Field Name:** Bulk column `campaign-id`.
- **Purpose:** The marketing campaign the page belongs to.
- **Mandatory:** No.
- **Allowed Values:** Campaign codes. [To be confirmed: campaign code format — owner: marketing operations]
- **Business Rules:** Published and indexed; no code uses it.
- **Used By Components:** None (`getPageMetadata().campaignId` is not read).
- **Used By Search:** Query index `campaignId`.
- **Used By Personalization:** None today (future campaign targeting).
- **Used By Analytics:** None today (not in the data layer; would need `trackPage()` changes).
- **Example Values:** `cmp-2026-agentic-bank`.

8.2.34 content-type — Content Type

- **Field Name:** Bulk column `content-type`.
- **Purpose:** The kind of content (Article, Case Study, Report …) shown as a label.
- **Mandatory:** Recommended for every page listed by a List or Filter List.
- **Allowed Values:** Free text; Recommended: the titles of the `tcs-corp:content-type` tags (7.3.1). The Filter List Content types options are `article`, `blog`, `report`, `case-study`, `news`, `event`, `whitepaper`, `podcast`, `video`.
- **Business Rules:** Filter List Content types compares class-name forms (`Case Study` = `case-study`). A value containing "report" makes the card a **report card** in the Mixed card style (List / Cards `kindOf`; Filter List when the eyebrow column contains "report").
- **Used By Components:** Page Header (eyebrow in Landing and Secondary; label in the Article meta row); List (image card meta, report kicker, row eyebrow fallback, card kind); Filter List (Eyebrow source default; Content types filter; report cards).
- **Used By Search:** Query index `contentType`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `Article`, `Case Study`, `Report`.

8.2.35 featured — Featured

- **Field Name:** Bulk column `featured`; fallback `feature-card` (8.2.47).
- **Purpose:** Marks a page for Filter List spotlight cards.
- **Mandatory:** No.
- **Allowed Values:** Yes / no (yes: `yes`, `true`, `1`, `on`, `y`).

- **Business Rules:** Filter List: Featured rule `featured` takes featured pages sorted by priority (if there are none, pages with priority ≤ 3); Hero rule `featured` takes the top featured page (else the latest); Sort by Featured puts featured pages first.
- **Used By Components:** Filter List.
- **Used By Search:** Query index `featured`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `yes`.

8.2.36 priority — Priority

- **Field Name:** Bulk column `priority`; fallback `featured-priority` (8.2.48).
- **Purpose:** Ranks pages in the Filter List.
- **Mandatory:** No.
- **Allowed Values:** A number; lower is more important. Empty = lowest.
- **Business Rules:** Sort by Priority (ascending, ties: latest); Featured rule `priority`: pages with priority ≤ 3 (`FEATURED_PRIORITY`), up to the Featured card count.
- **Used By Components:** Filter List.
- **Used By Search:** Query index `priority`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `1`, `2.5`.

8.2.37 listen-link — Listen Link

- **Field Name:** Bulk column `listen-link`.
- **Purpose:** An audio version of the page.
- **Mandatory:** No.
- **Allowed Values:** A URL of an audio file or player page.
- **Business Rules:** Adds the Listen action to the Page Header (Article layout only); without it only Download and Share are shown.
- **Used By Components:** Page Header.
- **Used By Search:** Not indexed.
- **Used By Personalization:** None.
- **Used By Analytics:** The Listen link click is a `cmp:click` of the Page Header.
- **Example Values:** `/en/media/agentic-bank.mp3`.

8.2.38 author — Author

- **Field Name:** Bulk column `author` (standard Author metadata).
- **Purpose:** The author's name for pages without an Authors sheet entry, and the default Filter List Author source.
- **Mandatory:** No.
- **Allowed Values:** A name, or an author ID of the Authors sheet (Filter List).
- **Business Rules:** Page Header: shown when `author-id` has no Authors entry. Filter List: the Author source value (default column `author`) is looked up in the Authors sheet, else `author-id`, else shown as a name.
- **Used By Components:** Page Header; Filter List (hero and featured cards).
- **Used By Search:** Query index `author`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `Jane Doe`.

8.2.39 card-title — Card Title

- **Field Name:** Bulk column `card-title`.
- **Purpose:** A shorter title for List cards.
- **Mandatory:** No.
- **Allowed Values:** Plain text.
- **Business Rules:** List first choice (then Title); a List Item's Headline overrides it.
- **Used By Components:** List (`PAGE_MAPPING.title`).
- **Used By Search:** Query index `cardTitle`.
- **Used By Personalization:** None.
- **Used By Analytics:** `dc:title` of List items (unless an Analytics label is set).
- **Example Values:** `Cloud, at scale`.

8.2.40 card-description — Card Description

- **Field Name:** Bulk column `card-description`.
- **Purpose:** A shorter description for List cards.
- **Mandatory:** No.
- **Allowed Values:** Plain text.
- **Business Rules:** List first choice (then Description).
- **Used By Components:** List.
- **Used By Search:** Query index `cardDescription`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `How the cloud scales.`

8.2.41 card-image — Card Image

- **Field Name:** Bulk column `card-image`.
- **Purpose:** A dedicated image for List cards.
- **Mandatory:** No.
- **Allowed Values:** An image path or URL.
- **Business Rules:** List first choice (then the page image, `thumbnail-image`, `featured-image`).
- **Used By Components:** List.
- **Used By Search:** Query index `cardImage`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `/en/media/cloud-card.jpg`.

8.2.42 card-image-alt — Card Image Alt Text

- **Field Name:** Bulk column `card-image-alt`.
- **Purpose:** Alternative text of the List card image.
- **Mandatory:** Recommended when the image conveys information.
- **Allowed Values:** Plain text.
- **Business Rules:** List first choice, then `image-alt`; empty: the image is decorative (`alt=""`, the title is next to it).
- **Used By Components:** List.
- **Used By Search:** Query index `cardImageAlt`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `A data centre`.

8.2.43 image-alt — Image Alt Text

- **Field Name:** Bulk column `image-alt`.
- **Purpose:** Alternative text of the page image used on cards.
- **Mandatory:** No.
- **Allowed Values:** Plain text.
- **Business Rules:** List fallback after `card-image-alt`.
- **Used By Components:** List.
- **Used By Search:** Query index `imageAlt`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `Bank employees reviewing an AI dashboard`.

8.2.44 eyebrow — Eyebrow

- **Field Name:** Bulk column `eyebrow`.
- **Purpose:** The small label above the List card title.
- **Mandatory:** No.
- **Allowed Values:** Plain text.
- **Business Rules:** List first choice, then `category`; shown with Show eyebrow on. The Filter List uses it only if its Eyebrow source is set to `eyebrow`.
- **Used By Components:** List (image card kicker, panel card, rows, statistic); Filter List (optional mapping).
- **Used By Search:** Query index `eyebrow`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `Perspective`.

8.2.45 category — Category

- **Field Name:** Bulk column `category`.
- **Purpose:** A category label, the fallback eyebrow of List cards.
- **Mandatory:** No.
- **Allowed Values:** Plain text.
- **Business Rules:** Used only when `eyebrow` is empty. Not the Filter List “Primary filter category” (a tag field of the block).
- **Used By Components:** List.
- **Used By Search:** Query index `category`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `Banking`.

8.2.46 hide-in-nav — Hide in Navigation

- **Field Name:** Bulk column `hide-in-nav`.
- **Purpose:** Keeps a page out of automatic lists.
- **Mandatory:** No.
- **Allowed Values:** Yes / no.
- **Business Rules:** The List (Child pages and Tagged pages sources) skips the page. It does **not** hide the page from the header navigation (authored), the Filter List, the Conversation Canvas menu, a Fixed list item, the sitemap or search engines.
- **Used By Components:** List (`isListable`).
- **Used By Search:** Query index `hideInNav`.

- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `yes`.

8.2.47 feature-card — Feature Card (legacy)

- **Field Name:** Bulk column `feature-card` (an earlier name of `featured`).
- **Purpose:** Read as a fallback so pages tagged with the earlier name stay featured.
- **Mandatory:** No. Not in the AGENTS.md column list; prefer `featured`.
- **Allowed Values:** Yes / no.
- **Business Rules:** A page is featured when `featured` **or** `feature-card` is yes.
- **Used By Components:** Filter List (through `utils/query-index.js`).
- **Used By Search:** Query index `featureCard`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `true`.

8.2.48 featured-priority — Featured Priority (legacy)

- **Field Name:** Bulk column `featured-priority` (an earlier name of `priority`).
- **Purpose:** Read as a fallback for `priority`.
- **Mandatory:** No; prefer `priority`.
- **Allowed Values:** A number.
- **Business Rules:** Used only when `priority` is empty or not a number.
- **Used By Components:** Filter List (through `utils/query-index.js`).
- **Used By Search:** Query index `featuredPriority`.
- **Used By Personalization:** None.
- **Used By Analytics:** None.
- **Example Values:** `2`.

8.3 Metadata Cheat Sheet

Field	Where to set	Used by	Example
`title`	Page property Title	Page Header h1; List / Filter List titles; index `title`; `dc:title`	`The agentic bank`
`description`	Page property Description	Page Header; List / Filter List; index `description`	`Banking is entering a new era.`
`keywords`	Page property Keywords	Head only	`agentic AI, banking`
`tags`	AEM page properties Tags, or sheet `tags`	Page Header; Conversation Canvas; List / Filter List (index); `xdm:tags`	`tcs-corp:industry/banking`
`page-type`	Page property Page Type, or sheet	Body class; templates; Page Header layout; Filter List Page types; `xdm:template`	`article`
`nav`	Page property Navigation, or sheet	Header (inherited)	`references/nav-careers`
`footer`	Page property Footer, or sheet	Footer (inherited)	`/en/references/footer`

Field	Where to set	Used by	Example
error-page	Page property 404 Page, or sheet	404 page (inherited)	/en/error-pages/404
url	Sheet	Row pattern	/en/insights/**
meta-title	Sheet	SEO decorator (title, og / twitter fallback)	Agentic AI in Banking – TCS
meta-description	Sheet	SEO decorator	How agentic AI changes banking.
meta-keywords	Sheet	SEO decorator	agentic AI, banking
author-id	Sheet	Page Header author; article / blog "About the author"; List feature card; Filter List spotlights	jdoe
publish-date	Sheet	Page Header date; Filter List date and Latest sort; List date fallback	2026-09-04
modified-date	Sheet	List date and Order by last modified	2026-10-01
reading-time	Sheet	Page Header; List / Cards and Filter List kickers	16
og-title	Sheet	SEO decorator og:title	The agentic bank — TCS
og-description	Sheet	SEO decorator og:description	Five shifts for bank leaders.
og-image	Sheet	SEO decorator og:image	/en/media/social.jpg
og-type	Sheet	SEO decorator og:type	article
twitter-card	Sheet	SEO decorator twitter:card	summary
twitter-title	Sheet	SEO decorator twitter:title	The agentic bank
twitter-description	Sheet	SEO decorator twitter:description	Five shifts for bank leaders.
twitter-image	Sheet	SEO decorator twitter:image	/en/media/x.jpg
canonical-url	Sheet	SEO decorator canonical link and og:url	https://…/en/insights/agentic-bank
robots	Sheet	Search engines; index robots; sitemap exclusion	noindex, nofollow
region	Sheet	Press-release Media Contacts; index	Europe
country	Sheet	Index only	de
language	Sheet	<html lang>; xdm:language; sort collation	en-GB
persona	Sheet	Index only (future personalisation)	cxo
thumbnail-image	Sheet	List image; Filter List fallback image	/en/media/thumb.jpg
featured-image	Sheet	Page Header Landing image (LCP); Filter List image; List image	/en/media/hero.jpg
campaign-id	Sheet	Index only	cmp-2026-agentic-bank
content-type	Sheet	Page Header eyebrow / label; List meta and report cards; Filter List eyebrow and Content types	Case Study

Field	Where to set	Used by	Example
`featured`	Sheet	Filter List featured / hero cards, Featured sort	`yes`
`priority`	Sheet	Filter List Priority sort, featured rule (≤ 3)	`1`
`listen-link`	Sheet	Page Header Listen action (Article layout)	`/en/media/page.mp3`
`author`	Sheet	Page Header name fallback; Filter List Author source	`Jane Doe`
`card-title`	Sheet	List card title	`Cloud, at scale`
`card-description`	Sheet	List card description	`How the cloud scales.`
`card-image`	Sheet	List card image	`/en/media/card.jpg`
`card-image-alt`	Sheet	List card image alt	`A data centre`
`image-alt`	Sheet	List image alt fallback	`Bank employees`
`eyebrow`	Sheet	List eyebrow; Filter List (if mapped)	`Perspective`
`category`	Sheet	List eyebrow fallback	`Banking`
`hide-in-nav`	Sheet	List: page not listed (Child pages, Tagged pages)	`yes`
`feature-card`	Sheet (legacy)	Filter List fallback for `featured`	`true`
`featured-priority`	Sheet (legacy)	Filter List fallback for `priority`	`2`

> **Tip:** Changes to page properties take effect when the page is published; changes to the bulk metadata sheet when the sheet is published (and only once `metadata.source` lists it). Lists and filters reflect metadata only after the affected pages are republished and re-indexed, because they read `/query-index.json`.

Part 9 — Data Sheets Documentation

This part documents every spreadsheet (and the taxonomy page) that the TCS Corporate Website reads. All sheets exist **once per locale**: they are authored in AEM as a Cloud Service under the locale folder and published as JSON at the locale's language-only path, mapped in `paths.json`.

Locale (AEM folder)	Published root	Locale (AEM folder)	Published root
us/en	/en	se/sv-se	/sv-se
de/de-de	/de-de	dk/da-dk	/da-dk
fr/fr-fr	/fr-fr	no/no-no	/no-no
ca/fr-ca	/fr-ca	cn/zh-cn	/zh-cn
es/es-es	/es-es	jp/ja-jp	/ja-jp
pt/pt-pt	/pt-pt	kr/ko-kr	/ko-kr
nl/nl-nl	/nl-nl		

Sheet	AEM path (below **/content/tcs-corp/{country}/{language}/**)	Published URL	**paths.json** mapping (example de-de)
Placeholders	placeholders	/{locale}/placeholders.json	/content/tcs-corp/de/de-de/placeholders:/de-de/placeholders
Metadata	metadata	/{locale}/metadata.json	/content/tcs-corp/de/de-de/metadata:/de-de/metadata
Taxonomy	taxonomy (taxonomy page)	/{locale}/taxonomy.json	/content/tcs-corp/de/de-de/taxonomy:/de-de/taxonomy.json
Redirects	redirects	/{locale}/redirects.json	/content/tcs-corp/de/de-de/redirects:/de-de/redirects
Authors	data/authors	/{locale}/data/authors.json	/content/tcs-corp/de/de-de/data/authors:/de-de/data/authors
Locations	data/locations	/{locale}/data/locations.json	.../data/locations:/de-de/data/locations
Partners	data/partners	/{locale}/data/partners.json	.../data/partners:/de-de/data/partners
Social Links	data/social-links	/{locale}/data/social-links.json	.../data/social-links:/de-de/data/social-links
Media Contacts	data/media-contacts	/{locale}/data/media-contacts.json	.../data/media-contacts:/de-de/data/media-contacts

Spreadsheet mappings get `.json` appended by the platform; the taxonomy page mapping carries `.json` explicitly.

Status on 13 October 2026 (preview `main--tcs-corp--sradha-tcs.aem.page`): for `/en/`, all nine are published: the placeholders sheet has one row, the taxonomy 16 tags, and the other sheets have their column headers but no rows. None of the other 12 locales has published sheets yet (404).

Common rules for every sheet (enforced by code):

- The code finds a sheet by its **exact page name and location**; a sheet with another name, or in another folder, is ignored.

- Each sheet is loaded **at most once per page view** (the request is shared by all components; there is no cross-page cache in code). The browser and the Edge Delivery CDN cache the JSON; publishing a sheet updates it.
- A missing, unpublished or invalid sheet never breaks a page: the code continues with an empty list (or empty map) and its fallbacks.
- Multi-sheet JSON (`:type` = `multi-sheet`) is supported by `sheetRows()`: the requested sheet, else `default`, else the first sheet. Only the taxonomy uses a requested sheet (the language); other sheets read `default` / the first sheet.
- Values are read as text and trimmed by `utils/data.js`; column headers must be written exactly as documented (lower case with hyphens, except Placeholders `Key` / `Text` and Redirects `Source` / `Destination`, which accept both cases).

Common authoring procedure (AEM as a Cloud Service with Edge Delivery Services):

1. In AEM, open **Sites** and navigate to the locale folder (for example `TCS Corp > us > en`, or `... > en > data` for a data sheet; create the `data` folder page first, left empty).
2. Select **Create > Page** and choose the spreadsheet template for the sheet (Placeholders, Metadata, Redirects, Taxonomy, or the general Spreadsheet template for the data sheets), as listed in each chapter.
3. Enter the **Name** exactly as documented (lower case, hyphens). The Title is free.
4. Define the columns (headers) exactly as documented, then open the sheet and add the rows.
5. Publish the sheet: to **Preview** first (Manage Publication, destination Preview), check it, then **Publish** (live).
6. Verify the JSON at `https://main--tcs-corp--sradha-tcs.aem.page/{locale}/{sheet}.json` (preview) and `https://main--tcs-corp--sradha-tcs.aem.live/{locale}/{sheet}.json` (live): the `columns` array must list the headers and `data` the rows.

> **Note:** The names of the AEM spreadsheet page templates (Placeholders, Metadata, Redirects, Taxonomy, Spreadsheet) are those offered by the site's AEM templates. [To be confirmed: template names as shown in the AEM Create Page wizard of this environment — owner: AEM administrator]

9.1 Placeholders

9.1.1 Purpose

The user-interface text of all components (labels, accessible names, messages, counters) for the locale, so that every visible or announced text can be translated without a code change. See 4.5 for the full list of keys.

9.1.2 Ownership

[To be confirmed: business owner of UI text — owner: Localization / content operations lead]. Each locale's content manager or translator maintains that locale's sheet; the development team adds new keys (with an English fallback in code) and documents them.

9.1.3 Published path (per locale)

`/{locale}/placeholders.json`: `/en/placeholders.json`; `/de-de/placeholders.json`; `/fr-fr/placeholders.json`; `/fr-ca/placeholders.json`; `/es-es/placeholders.json`; `/pt-pt/placeholders.json`; `/nl-nl/placeholders.json`; `/sv-se/placeholders.json`; `/da-dk/placeholders.json`; `/no-no/placeholders.json`; `/zh-cn/placeholders.json`; `/ja-jp/placeholders.json`; `/ko-kr/placeholders.json`. AEM: `/content/tcs-corp/{country}/{language}/placeholders`.

9.1.4 Columns

Column	Type	Mandatory	Description	Example
`Key`	text	Yes (rows without a Key are ignored)	The placeholder key. Matched without case and separators (`read-more` = `readMore` = `Read More`). `key` (lower case header) is accepted.	`nextSlide`
`Text`	text	Recommended (empty Text uses the English fallback)	The translated text. May contain `{variables}` that the code fills; keep them as written. `text` (lower case header) is accepted.	`Nächste Folie`
`Description`	text	No (not read by code)	Where and how the text appears, for translators.	`Accessible name of the carousel's next button`
`Category`	text	No (not read by code)	A grouping for authors (for example CTA, Navigation, Carousel, Forms, Filter List, Editor).	`Carousel`

9.1.5 Relationships

- Keys correspond one-to-one to the `t('key', 'fallback')` calls in the code (4.5.3).
- Each locale's sheet is independent; there is no inheritance from the English sheet.

9.1.6 Dependencies

`utils/placeholders.js` (`loadPlaceholders`, `getPlaceholders`, `fillPlaceholder`), used by: Anchor List, Cards, Conversation Canvas, Embed, Filter List, Footer, Form Variations, Gen Carousel, Header, List, Modal, Page Header, Separator, Table, Table List, Tabs, Teaser, Video, Hero (Lottie toggle), `decorators/animation-decorator.js`, `decorators/article-meta-decorator.js`, `decorators/card-list-decorator.js`, the Article / Blog and Press Release templates.

9.1.7 Examples

Key	Text	Description	Category
`read-more`	Read More	CTA label	CTA
`nextSlide`	Next slide	Accessible name of the carousel's next button	Carousel
`showingCount`	Showing {n} of {total}	Filter List status under the results	Filter List
`followUsOn`	Follow us on {platform}	Accessible name of each footer social icon	Footer
`fieldRequired`	This field is required.	Form validation message	Forms

The first row is the row currently published in `/en/placeholders.json`; the others are illustrative.

9.1.8 Business Rules

1. Missing sheet, missing key or empty Text: the English fallback from the code is shown (enforced).
2. Key matching ignores case and every character other than letters and digits (enforced).
3. Duplicate keys (after normalisation): the last row wins (enforced).
4. `{name}` tokens are replaced only with the values the code passes; unknown tokens stay visible (enforced).
5. Loaded once per page view and shared by all components (enforced).
6. Recommended: the English sheet lists every key of 4.5.3 with Description and Category; other locales copy it and translate the Text column.
7. Recommended: keep translations short enough for the component (button labels, pills); there is no enforced length.

9.1.9 Authoring steps

Step 1 — Create the sheet. In Sites, open the locale folder (for example `/content/tcs-corp/de/de-de`), select **Create** > **Page**, choose the **Placeholders** template, set the Name `placeholders` and create it.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-placeholders-create.png`
> **Purpose:** Creating the placeholders sheet
> **Description:** The AEM Create Page wizard in /content/tcs-corp/de/de-de with the Placeholders template selected and the Name field set to `placeholders`.

Step 2 — Add the rows. Open the sheet, check the headers `Key`, `Text`, `Description`, `Category`, and add one row per key (copy the English sheet and translate the Text column).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-placeholders-rows.png`
> **Purpose:** Editing placeholders
> **Description:** The placeholders sheet editor showing the four columns and rows such as `nextSlide` / Nächste Folie / Accessible name of the carousel's next button / Carousel.

Step 3 — Publish. Publish to Preview, check, then publish to live.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-placeholders-publish.png`
> **Purpose:** Publishing the sheet
> **Description:** The Sites console with the `placeholders` sheet selected and the Manage Publication dialog open with the Preview destination.

Step 4 — Verify. Open `https://main--tcs-corp--sradha-tcs.aem.page/de-de/placeholders.json`: the JSON lists `"columns":["Key","Text","Description","Category"]` and the rows in `data`. Then open a German page with a carousel and check the button labels (browser developer tools, accessibility tree).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-placeholders-json.png`
> **Purpose:** Verifying the published JSON
> **Description:** The browser showing /de-de/placeholders.json with the columns array and the translated rows.

9.2 Metadata

9.2.1 Purpose

The **bulk metadata** sheet of a locale: page-level values (page type, SEO and social overrides, author, dates, card overrides, featured flags, region, inherited Navigation / Footer / 404 Page) applied to all pages whose URL matches a row's pattern. It is the only place for page-level values other than the seven page properties (4.3).

9.2.2 Ownership

[To be confirmed: owner of the metadata sheets — owner: SEO lead with the web content lead]. The site administrator lists the sheets in the site configuration (`metadata.source`).

9.2.3 Published path (per locale)

`/{locale}/metadata.json`: `/en/metadata.json`; `/de-de/metadata.json`; `/fr-fr/metadata.json`; `/fr-ca/metadata.json`; `/es-es/metadata.json`; `/pt-pt/metadata.json`; `/nl-nl/metadata.json`; `/sv-se/metadata.json`; `/da-dk/metadata.json`; `/no-no/metadata.json`; `/zh-cn/metadata.json`; `/ja-jp/metadata.json`; `/ko-kr/metadata.json`. AEM: `/content/tcs-corp/{country}/{language}/metadata`.

> **Important:** The platform applies these sheets only when **all 13** are listed in the site configuration's `metadata.source` (`POST https://admin.hlx.page/config/sradha-tcs/sites/tcs-corp/metadata.json` with `{"source": ["/en/metadata.json", "/de-de/metadata.json", ...]}`). Until then no bulk metadata applies. There is no root `/metadata.json`. [To be confirmed: metadata.source configured — owner: site administrator]

9.2.4 Columns

Every column is published on the matching pages as `<meta name="{column}" content="...">`. "Read by" names the code that uses the value. Dates are `YYYY-MM-DD`. "yes" values accept yes / true / 1 / on / y (case-insensitive).

Column	Type	Mandatory	Description	Example
`url`	URL pattern	Yes	The pages the row applies to; must start with the locale root; `**` matches any depth.	`/en/newsroom/**`
`page-type`	text (class name)	No	Page Type when the page property is empty: template (`home`, `article`, `blog`, `press-release`), Page Header default layout (`landing`, `listing`), Filter List page type filter; body class; index `pageType`; data layer `xdm:template`.	`press-release`
`meta-title`	text	No	`<title>` in the browser (SEO decorator); also `og:title` / `twitter:title` fallback; index `metaTitle`.	`Q3 results 2026 – TCS`
`meta-description`	text	No	`<meta name="description">` (SEO decorator); `og:description` / `twitter:description` fallback; index `metaDescription`.	`Highlights of the third quarter.`
`meta-keywords`	text (comma-separated)	No	`<meta name="keywords">` (SEO decorator).	`results, finance`
`author-id`	text (ID)	Recommended for article, blog	The Authors sheet entry (9.5): Page Header author, "About the author", List / Filter List author lockups; index `authorId`.	`example-author-1`
`publish-date`	date	Recommended for article, blog, press-release	Page Header "Published" date; Filter List sort (Latest) and card date; List / Cards date (when no `modified-date`); index `publishDate`.	`2026-09-04`
`modified-date`	date	No	List / Cards card date (preferred over `publish-date`); index `modifiedDate`.	`2026-09-10`
`reading-time`	number (minutes) or text	No	Page Header "{n} min read"; List / Cards / Filter List "{n} m"; non-numeric text shown as written; index `readingTime`.	`6`

Column	Type	Mandatory	Description	Example
og-title	text	No	og:title (SEO decorator); twitter:title fallback.	Q3 results 2026
og-description	text	No	og:description; twitter:description fallback.	Revenue and highlights.
og-image	image URL	No	og:image (made absolute); twitter:image fallback.	/en/newsroom/media_1a2b.jpg
og-type	text	No	og:type.	article
twitter-card	text	No	twitter:card.	summary_large_image
twitter-title	text	No	twitter:title.	Q3 results 2026
twitter-description	text	No	twitter:description.	Revenue and highlights.
twitter-image	image URL	No	twitter:image (made absolute).	/en/newsroom/media_1a2b.jpg
canonical-url	URL	No	<link rel="canonical"> and og:url (made absolute).	https://main--tcs-corp--sradha-tcs.aem.live/en/newsroom/q3
robots	text	No	Standard robots meta, applied server side; index robots.	noindex, nofollow
region	text	No	Press Release media contacts by region; getPageMetadata().region; index region.	Europe
country	text	No	getPageMetadata().country (else the locale's country); index country. No component reads it (as of this handover).	de
language	language code	No	<html lang> when a valid code (else the locale's tag); data layer xdm:language; index language.	en-GB
persona	text	No	getPageMetadata().persona; index persona. No component reads it (as of this handover).	investor
thumbnail-image	image URL	No	List card image fallback; Filter List fallback image; index thumbnailImage.	/en/insights/media_3c4d.jpg
featured-image	image URL	No	Page Header Landing image; Filter List card image (default Image source); List image fallback; index featuredImage.	/en/insights/media_5e6f.jpg
campaign-id	text	No	getPageMetadata().campaignId; index campaignId. No component reads it (as of this handover).	cmp-2026-q4
content-type	text	Recommended for listed pages	The editorial label (4.2): Page Header eyebrow / meta row; List / Cards label and report cards; Filter List Content types filter and eyebrow; index contentType.	Report
featured	yes / no	No	Filter List hero (Featured rule) and featured cards; Sort "Featured First"; index featured.	yes
priority	number	No	Filter List order (lower first); featured cards fallback (priority 3 or lower); index priority.	1

Column	Type	Mandatory	Description	Example
`listen-link`	URL	No	Page Header Listen action (audio version). Not indexed.	`/content/dam/tcs-corp/audio/q3.mp3`
`author`	text	No	Author name when there is no `author-id` (Page Header); Filter List author source (matched to an Authors ID, else shown as the name); index `author`.	`TCS Newsroom`
`card-title`	text	No	List card title (else the page title); index `cardTitle`.	`Q3 at a glance`
`card-description`	text	No	List card description (else the page description); index `cardDescription`.	`Revenue up, margins stable.`
`card-image`	image URL	No	List card image (first choice); index `cardImage`.	`/en/newsroom/media_7a8b.jpg`
`card-image-alt`	text	No	List card image alt text (first choice); index `cardImageAlt`.	`Chart of quarterly revenue`
`image-alt`	text	No	List card image alt text (fallback); index `imageAlt`.	`Office building`
`eyebrow`	text	No	List card eyebrow (first choice); index `eyebrow`.	`Investor relations`
`category`	text	No	List card eyebrow fallback; `getPageMetadata().category`; index `category`.	`Financials`
`hide-in-nav`	yes / no	No	`yes`: the page is left out of List results; index `hideInNav`.	`yes`

Additional metadata names read by the code (not in the standard column list; can be added as columns):

Column	Type	Mandatory	Description	Example
`nav`	page path	No	Navigation (inherited by child pages); AEM path, published path or relative to the locale root.	`references/nav-careers`
`footer`	page path	No	Footer (inherited).	`references/footer-campaign`
`error-page`	page path	No	404 Page (inherited).	`error-pages/404-careers`
`tags`	text (comma-separated)	No	Page tags fallback when the page has no `article:tag` tags.	`tcs-corp:region/europe, tcs-corp:industry`
`template`	text	No	Page Type fallback for template loading (standard metadata).	`article`
`publication-date`	date	No	Publish date fallback.	`2026-09-04`
`feature-card`	yes / no	No	Earlier name of `featured` (index fallback `featureCard`).	`yes`

Column	Type	Mandatory	Description	Example
`featured-priority`	number	No	Earlier name of `priority` (index fallback `featuredPriority`).	2

> **Note:** The published `/en/metadata.json` (13 October 2026) has the columns `url` to `priority` (29 columns); `listen-link`, `author`, `card-title`, `card-description`, `card-image`, `card-image-alt`, `image-alt`, `eyebrow`, `category` and `hide-in-nav` still need to be added as columns where they are used. Every field is specified in detail in Part 8.

9.2.5 Relationships

- `author-id` references `author-id` in the locale's Authors sheet (9.5).
- `region` should use the same values as `region` in the Media Contacts (9.9) and Authors sheets, and the titles of the `tcs-corp:region/*` tags.
- `content-type` values should align with the Filter List options and the `tcs-corp:content-type/*` tags (4.2).
- `nav`, `footer` and `error-page` reference pages under the locale's `references/` and `error-pages/`.
- The query index (`helix-query.yaml`) reads the published meta tags; listings (List, Filter List) depend on it.

9.2.6 Dependencies

The platform (applies the rows to pages); `decorators/seo-decorator.js`; `utils/metadata.js`; `utils/inherited-metadata.js`; `scripts/scripts.js` (Page Type, templates); Page Header; templates; List; Filter List (the Cards block is authored manually and reads no page metadata); `decorators/data-layer-decorator.js`; `helix-query.yaml`; `helix-sitemap.yaml` (via the index and `robots`).

9.2.7 Examples

url	page-type	content-type	author-id	publish-date	reading-time	region	robots
`/en/references/**`							noindex, nofollow
`/en/error-pages/**`							noindex, nofollow
`/en/data/**`							noindex, nofollow
`/en/newsroom/**`	press-release	Press Release					
`/en/insights/cloud-outlook-2026`	article	Report	example-author-1	2026-09-04	6	Europe	

Illustrative values; the `/en/` sheet currently has no rows.

9.2.8 Business Rules

1. The sheets apply only after all 13 are listed in `metadata.source` (platform).
2. A row's pattern must start with the locale's root; a row whose pattern points to another locale's pages has no effect in practice and must be maintained in that locale's sheet instead (platform: only pages whose URL matches the pattern receive values).

3. Page properties win over sheet values (platform).
4. Several rows can match a page; each contributes its non-empty columns (platform behaviour for bulk metadata). [To be confirmed: precedence between overlapping rows in this configuration — owner: development team]
5. SEO / social columns are applied in the browser by the SEO decorator; `robots` works server side (code).
6. Page Type values outside `home`, `article`, `blog`, `press-release` load no template (code).
7. `featured` accepts yes / true / 1 / on / y; `priority` must be a number (else ignored) (code).
8. `hide-in-nav` = yes removes the page from List results (code); it does not hide the page from the header navigation, which is authored.
9. Changes need a republish of the sheet; pages show the new values after the CDN refresh, and listings after the page is re-indexed (republish the affected pages).

9.2.9 Authoring steps

Step 1 — Create the sheet. In the locale folder select **Create** > **Page**, choose the **Metadata** template, Name `metadata`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-metadata-create.png`
> **Purpose:** Creating the metadata sheet
> **Description:** The AEM Create Page wizard in /content/tcs-corp/us/en with the Metadata template selected and the Name `metadata`.

Step 2 — Add the columns and rows. Add `url` and the needed columns (9.2.4), then one row per URL pattern; start every pattern with the locale root (`/en/`).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-metadata-rows.png`
> **Purpose:** Editing bulk metadata
> **Description:** The metadata sheet editor showing the url, page-type, content-type, author-id, publish-date and robots columns with the example rows of 9.2.7.

Step 3 — Publish. Publish to Preview, check, then publish to live. Ask the site administrator to confirm the sheet is listed in `metadata.source`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-metadata-publish.png`
> **Purpose:** Publishing the sheet
> **Description:** The Manage Publication dialog for the `metadata` sheet with Preview selected.

Step 4 — Verify. Open `/en/metadata.json` on `.aem.page`: the `columns` and `data` arrays show the headers and rows. Then open a matching page, view the page source and check the `<meta name="...">` tags (for example `<meta name="page-type" content="press-release">`).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-metadata-verify-head.png`

Purpose: Verifying the applied metadata
Description: The page source of /en/newsroom/q3-results with the `page-type`, `content-type` and `robots` meta tags highlighted.

9.3 Taxonomy

9.3.1 Purpose

The labels of the AEM tags for the locale: maps each tag ID (`tcs-corp:region/europe`) to its title (Europe), in the locale's language. Components show these labels instead of tag IDs.

9.3.2 Ownership

[To be confirmed: taxonomy owner — owner: Content strategy / taxonomy manager]. Tags are created and translated in the AEM Tagging console; the taxonomy page of each locale is republished after every change.

9.3.3 Published path (per locale)

`/{locale}/taxonomy.json`: `/en/taxonomy.json`; `/de-de/taxonomy.json`; `/fr-fr/taxonomy.json`; `/fr-ca/taxonomy.json`; `/es-es/taxonomy.json`; `/pt-pt/taxonomy.json`; `/nl-nl/taxonomy.json`; `/sv-se/taxonomy.json`; `/da-dk/taxonomy.json`; `/no-no/taxonomy.json`; `/zh-cn/taxonomy.json`; `/ja-jp/taxonomy.json`; `/ko-kr/taxonomy.json`. AEM: `/content/tcs-corp/{country}/{language}/taxonomy` (a taxonomy page, not a spreadsheet).

9.3.4 Columns

Column	Type	Mandatory	Description	Example
`tag`	tag ID	Yes (rows without `tag` are ignored)	The AEM tag ID (`namespace:path`); generated from the tags selected for the taxonomy page.	`tcs-corp:region/europe`
`title`	text	Recommended (an empty title falls back to a label derived from the ID)	The tag title in the sheet's language.	`Europe`

With translated titles the JSON is a multi-sheet: one sheet per language, named by the ISO 639-1 code (`de`, `fr`, `ja`, ...), plus `default`. The code reads `{language}`, else `default`, else the first sheet.

9.3.5 Relationships

- Page tags (`<meta property="article:tag">`, index column `tags`), Teaser and List / Cards tag fields and Filter List filter tags all hold tag IDs that are resolved here.
- Filter List: **Primary filter category** and **Advanced filter categories** are tag roots (for example `tcs-corp:industry`); their child tags used by the pages become the filter options, labelled from this sheet.
- `tcs-corp:content-type/*` and `tcs-corp:region/*` tags relate to the `content-type` and `region` metadata values (4.2, 9.2.5), but the code does not link them.

9.3.6 Dependencies

`utils/taxonomy.js` (`loadTaxonomy`, `resolveTags`), `utils/tags.js` (`parseTag`, `readTagValues`), `decorators/tag-decorator.js`; used by Page Header, Teaser, List, Cards, Filter List, Conversation Canvas.

9.3.7 Examples

Rows of the published `/en/taxonomy.json` (13 October 2026, 16 rows in total):

tag	title
`tcs-corp:`	TCS Corp
`tcs-corp:content-type`	Content Type
`tcs-corp:content-type/case-study`	Case Study
`tcs-corp:industry`	Industry
`tcs-corp:region/asia-pacific`	Asia Pacific

The full current list: `tcs-corp:` (TCS Corp); `tcs-corp:content-type` (Content Type) with `article` (Article), `blog` (Blog), `case-study` (Case Study), `event` (Event), `press-release` (Press Release), `research` (Research), `webinar` (Webinar), `whitepaper` (Whitepaper); `tcs-corp:industry` (Industry, no child tags yet); `tcs-corp:region` (Region) with `americas` (Americas), `asia-pacific` (Asia Pacific), `europe` (Europe), `middle-east-africa` (Middle East Africa).

9.3.8 Business Rules

1. Loaded once per page view; a missing taxonomy gives an empty map (code).
2. A tag ID in the taxonomy shows its title; a tag ID not in the taxonomy shows a label derived from its last segment (hyphens and underscores become spaces, first letter upper case: `tcs-corp:industry/banking-financial` gives "Banking financial"); a value without `:` or `/` is shown as written (code).
3. The language sheet is chosen by the locale's ISO 639-1 language; `fr-fr` and `fr-ca` share `fr` (code).
4. Matching is exact on the tag ID (after trimming) (code).
5. Recommended: republish the taxonomy page of every locale after creating, renaming or translating tags.

9.3.9 Authoring steps

Step 1 — Create or update the tags. In AEM **Tools > General > Tagging**, create the tags in the `tcs-corp` namespace and add their translated titles.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/taxonomy-tagging-console.png`
> **Purpose:** Managing tags
> **Description:** The AEM Tagging console with the TCS Corp namespace open, showing Content Type, Industry and Region and the child tags of Region.

Step 2 — Create the taxonomy page. In the locale folder select **Create > Page**, choose the **Taxonomy** template, Name `taxonomy`, and select the `tcs-corp` namespace (or the tags to include).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/taxonomy-page-create.png`
> **Purpose:** Creating the taxonomy page
> **Description:** The Create Page wizard with the Taxonomy template, the Name `taxonomy` and the tag picker with the TCS Corp namespace selected.

Step 3 — Publish. Publish the taxonomy page (Preview, then live).

Step 4 — Verify. Open `/en/taxonomy.json` on `.aem.page`: `"columns":["tag","title"]` and one row per tag (or one sheet per language). Then check a tagged page: the Page Header shows the titles, not the IDs.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/taxonomy-json-verify.png`
> **Purpose:** Verifying the taxonomy JSON
> **Description:** The browser showing /en/taxonomy.json with the 16 tag rows and the columns tag and title.

9.4 Redirects

9.4.1 Purpose

The locale's list of moved pages (source path to destination). In this implementation the sheet is **not applied by the platform**: Edge Delivery applies only a root `/redirects.json`, and there is none. The code uses the sheet in one place: the List block links its cards to the final destination of a redirected page instead of the old URL.

9.4.2 Ownership

[To be confirmed: owner of redirects — owner: SEO lead with the site administrator]. Applying the redirects (CDN rules) is an infrastructure task: [To be confirmed: production domain and CDN].

9.4.3 Published path (per locale)

`/{locale}/redirects.json`: `/en/redirects.json`; `/de-de/redirects.json`; `/fr-fr/redirects.json`; `/fr-ca/redirects.json`; `/es-es/redirects.json`; `/pt-pt/redirects.json`; `/nl-nl/redirects.json`; `/sv-se/redirects.json`; `/da-dk/redirects.json`; `/no-no/redirects.json`; `/zh-cn/redirects.json`; `/ja-jp/redirects.json`; `/ko-kr/redirects.json`. AEM: `/content/tcs-corp/{country}/{language}/redirects`.

9.4.4 Columns

Column	Type	Mandatory	Description	Example
`source` (or `Source`)	path	Yes	The old path, starting with `/` (rows not starting with `/` are ignored). Normalised: query, hash, `.html` and a trailing slash are removed.	`/en/insights/old-report`
`destination` (or `Destination`)	path or URL	Yes (rows without one are ignored)	The new path (internal, `/...`) or an absolute URL.	`/en/insights/cloud-outlook-2026`
`description`	text	No (not read by code)	Why the redirect exists (the published `/en/redirects.json` has this column).	`Report renamed in 2026`

9.4.5 Relationships

- Sources and destinations are published paths of the same locale (`/en/...` in the `/en/` sheet).
- The List block resolves card links (query index pages and fixed items) through this sheet.

9.4.6 Dependencies

`utils/redirects.js` (`loadRedirects`, `resolveRedirect`, `normalizeRedirectPath`), used by `blocks/list/list.js`. A CDN rule set ([To be confirmed: CDN]) would be the consumer that actually redirects visitors.

9.4.7 Examples

source	destination	description
`/en/insights/old-report`	`/en/insights/cloud-outlook-2026`	Report renamed
`/en/about-us`	`/en/about`	Folder restructure
`/en/careers/jobs`	`https://www.example.com/careers`	Moved to the careers site

Illustrative values; the `/en/` sheet currently has no rows. The redirect `/` to `/en/` cannot be expressed in a locale sheet (its source is outside the locale) and must be configured at the CDN.

9.4.8 Business Rules

1. Not applied to visitors by Edge Delivery (platform: only the root `/redirects.json` is applied server side).
2. List block: a link to an internal path follows the redirects up to 5 hops, stops at loops and at an external destination (code, `MAX_HOPS = 5`).
3. Paths are compared after normalisation (no query, hash, `.html` or trailing slash) (code).
4. Rows without a source starting with `/` or without a destination are ignored (code).
5. Loaded once per page view; a missing sheet means no redirects (code).
6. Recommended: when a page is moved or deleted, add a row in the locale's sheet and ask the site administrator to update the CDN rules.

9.4.9 Authoring steps

Step 1 — Create the sheet. In the locale folder select **Create** > **Page**, choose the **Redirects** template, Name `redirects`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-redirects-create.png`
> **Purpose:** Creating the redirects sheet
> **Description:** The Create Page wizard in /content/tcs-corp/us/en with the Redirects template and the Name `redirects`.

Step 2 — Add rows. Fill `source`, `destination` and `description`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-redirects-rows.png`
> **Purpose:** Editing redirects
> **Description:** The redirects sheet editor with the three columns and the example rows of 9.4.7.

Step 3 — Publish and verify. Publish (Preview, then live) and open `/en/redirects.json`; check a List block that links to a redirected page: the card links to the destination.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-redirects-json.png`
> **Purpose:** Verifying the published JSON
> **Description:** The browser showing /en/redirects.json with the columns source, destination and description and the rows.

> **Warning:** Publishing the sheet alone does not redirect visitors. Until CDN rules apply the sheets, old URLs return 404.

9.5 Authors

9.5.1 Purpose

The author profiles of the locale: name, designation, bio, image, LinkedIn profile and region, referenced from pages by their `author-id` metadata. One profile is maintained once and shown consistently in page headers, author boxes and listing cards.

9.5.2 Ownership

[To be confirmed: owner of author profiles — owner: Editorial / content strategy lead]. Authors' consent for name, image and LinkedIn URL: [To be confirmed: consent process — owner: Legal / privacy].

9.5.3 Published path (per locale)

`/{locale}/data/authors.json`: `/en/data/authors.json`; `/de-de/data/authors.json`; `/fr-fr/data/authors.json`; `/fr-ca/data/authors.json`; `/es-es/data/authors.json`; `/pt-pt/data/authors.json`; `/nl-nl/data/authors.json`; `/sv-se/data/authors.json`; `/da-dk/data/authors.json`; `/no-no/data/authors.json`; `/zh-cn/data/authors.json`; `/ja-jp/data/authors.json`; `/ko-kr/data/authors.json`. AEM: `/content/tcs-corp/{country}/{language}/data/authors`.

9.5.4 Columns

`getAuthors()` maps each row to `{ id, name, designation, bio, image, linkedin, region }` (trimmed text) and drops rows without `author-id`.

Column	Type	Mandatory	Description	Example
`author-id`	text (ID)	Yes (rows without it are dropped)	The ID used in the pages' `author-id` metadata. `getAuthor()` (Page Header, templates) matches it case-insensitively; List and Filter List match it exactly.	`example-author-1`
`name`	text	Yes (the "About the author" box needs it)	The display name. Without it the author box is not shown; the Page Header falls back to the page's `author` metadata.	`Sample Author A`
`designation`	text	No	Job title; shown as "designation, region" in the Page Header and author box; alone on List / Filter List cards.	`Head of Cloud Research`
`bio`	text	No	Short biography; shown only in the "About the author" box (Article / Blog).	`Writes about cloud strategy.`
`image`	image URL or path	No	Author photo. Same-origin `/media_` URLs are optimised (responsive `<picture>`); other URLs are used as they are (lazy-loaded `<img>`). Treated as decorative (empty alt).	`/en/data/media_9f8e7d.jpg`
`linkedin`	URL	No	LinkedIn profile: the author name becomes a link in the Page Header (new tab, announced); the "View LinkedIn profile" link in the author box.	`https://www.linkedin.com/in/example`
`region`	text	No	Appended to the designation in the Page Header and the author box.	`Europe`

9.5.5 Relationships

- Pages reference authors through the `author-id` metadata (bulk metadata sheet, 9.2); the query index column `authorId` carries it to listings.
- Filter List: the **Author source** column (default `author`) is matched against `author-id` values first, then the page's `authorId`.
- `region` uses the same vocabulary as the `region` metadata and the Media Contacts sheet (recommended).

9.5.6 Dependencies

`utils/data.js` (`getAuthors`, `getAuthor`); `blocks/page-header` (author lockup in the Article layout); `templates/article` and `templates/blog` (`decorators/author-bio-decorator.js`); `blocks/list` (feature card author, only when "first card as feature" is on); `blocks/filter-list` (hero and featured card author lockups); `decorators/article-meta-decorator.js` (`buildAuthorLockup`).

9.5.7 Examples

author-id	name	designation	bio	image	linkedin	region
`example-author-1`	Sample Author A	Head of Cloud Research	Writes about cloud strategy.	`/en/data/media_1a.jpg`	`https://www.linkedin.com/in/example-a`	Europe
`example-author-2`	Sample Author B	Chief Economist	Covers markets and growth.		`https://www.linkedin.com/in/example-b`	Americas
`example-author-3`	Sample Author C	Industry Advisor, Banking		`/en/data/media_3c.jpg`		Asia Pacific

Illustrative values; the `/en/` sheet has the headers and no rows (13 October 2026).

9.5.8 Business Rules

1. Rows without `author-id` are ignored (code).
2. Page Header and templates: case-insensitive ID match after trimming; List and Filter List: exact match (code). Recommended: lower-case IDs with hyphens everywhere.
3. "About the author" appears only when the page has `author-id`, the ID is found and the entry has a `name`; otherwise its section is removed (code).
4. Page Header author name: the sheet's `name`, else the page's `author` metadata; no name, no author lockup (code).
5. The sheet is loaded once per page view; the List loads it only when its first card is a feature card; a failure gives an empty list (code).
6. Author images are decorative (empty alt); the name is always shown as text next to them (code).
7. Recommended: square images of at least 192 x 192 px (shown at 6rem, 96 px, in the author box; cropped to a circle).

9.5.9 Authoring steps

Step 1 — Create the data folder and the sheet. In the locale folder create the `data` folder page (empty) if needed; inside it select **Create > Page**, choose the **Spreadsheet** template, Name `authors`, columns `author-id`, `name`, `designation`, `bio`, `image`, `linkedin`, `region`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-authors-create.png`

> **Purpose:** Creating the Authors sheet
> **Description:** The Create Page wizard in /content/tcs-corp/us/en/data with the Spreadsheet template, the Name `authors` and the seven column names entered.

Step 2 — Add authors. One row per author; use the same `author-id` in the pages' metadata.

> [**Screenshot Placeholder**]
> **File name:** `screenshots/sheet-authors-rows.png`
> **Purpose:** Editing authors
> **Description:** The authors sheet editor with the seven columns and two example rows.

Step 3 — Publish and verify. Publish (Preview, then live), open `/en/data/authors.json` and check that `data` lists the rows; open an article with that `author-id` and check the Page Header author and the "About the author" box.

> [**Screenshot Placeholder**]
> **File name:** `screenshots/sheet-authors-verify.png`
> **Purpose:** Verifying the author on a page
> **Description:** An article page with the Page Header author lockup (image, linked name, designation and region) and the "About the author" box at the end, side by side with /en/data/authors.json in a second browser tab.

9.6 Locations

9.6.1 Purpose

The office or delivery-centre locations of the locale with coordinates, prepared for location components (for example a map or a contact page). **No component reads this sheet yet (as of this handover)**; the data is available to code through `getLocations()`.

9.6.2 Ownership

[To be confirmed: owner of the location list — owner: Corporate communications / facilities].

9.6.3 Published path (per locale)

`/{locale}/data/locations.json`: `/en/data/locations.json`; `/de-de/data/locations.json`; `/fr-fr/data/locations.json`; `/fr-ca/data/locations.json`; `/es-es/data/locations.json`; `/pt-pt/data/locations.json`; `/nl-nl/data/locations.json`; `/sv-se/data/locations.json`; `/da-dk/data/locations.json`; `/no-no/data/locations.json`; `/zh-cn/data/locations.json`; `/ja-jp/data/locations.json`; `/ko-kr/data/locations.json`. AEM: `/content/tcs-corp/{country}/{language}/data/locations`.

9.6.4 Columns

`getLocations()` returns `{ id, name, country, region, latitude, longitude }` and drops rows without both `location-id` and `location-name`.

Column	Type	Mandatory	Description	Example
`location-id`	text (ID)	Yes, or `location-name`	Stable ID of the location.	`loc-example-1`
`location-name`	text	Yes, or `location-id`	Display name.	`Example City Office`
`country`	text	No	Country (name or code; no format enforced).	`India`
`region`	text	No	Region; can be filtered with `inRegion()` (case-insensitive).	`Asia Pacific`
`latitude`	number (decimal degrees)	No	Converted to a number; empty or non-numeric becomes null.	`19.0760`
`longitude`	number (decimal degrees)	No	Converted to a number; empty or non-numeric becomes null.	`72.8777`

9.6.5 Relationships

`region` should use the same values as the `region` metadata and the other data sheets. No page metadata references locations.

9.6.6 Dependencies

`utils/data.js` (`getLocations`, `inRegion`). No block, decorator or template uses it (as of this handover).

9.6.7 Examples

location-id	location-name	country	region	latitude	longitude
`loc-example-1`	Example City Office	India	Asia Pacific	19.0760	72.8777
`loc-example-2`	Example Delivery Centre	Germany	Europe	50.1109	8.6821
`loc-example-3`	Example Innovation Hub	United States	Americas	40.7128	–74.0060

Illustrative values (coordinates of public city centres); the `/en/` sheet has the headers and no rows.

9.6.8 Business Rules

1. Rows without both ID and name are dropped (code).
2. Latitude and longitude must be decimal numbers (dot as decimal separator); other values become null (code).
3. Loaded once per page view by a consumer; missing sheet gives an empty list (code).
4. Recommended: keep the sheet up to date now so a future location component can use it without a data migration.

9.6.9 Authoring steps

Step 1 — Create the sheet. In `data/`, **Create** > **Page**, **Spreadsheet** template, Name `locations`, columns `location-id`, `location-name`, `country`, `region`, `latitude`, `longitude`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-locations-create.png`
> **Purpose:** Creating the Locations sheet

> **Description:** The Create Page wizard in /content/tcs-corp/us/en/data with the Spreadsheet template, the Name `locations` and the six column names.

Step 2 — Add rows, publish and verify. Add one row per location, publish (Preview, then live) and open `/en/data/locations.json`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-locations-json.png`
> **Purpose:** Verifying the published JSON
> **Description:** The browser showing /en/data/locations.json with the six columns and the rows.

9.7 Partners

9.7.1 Purpose

The technology and alliance partners of the locale (name, logo, website, tier, description), prepared for partner components (for example a partner logo wall). **No component reads this sheet yet (as of this handover)**; the data is available through `getPartners()`.

9.7.2 Ownership

[To be confirmed: owner of the partner list — owner: Alliances / partnerships team]. Logo usage rights: [To be confirmed: brand approval process — owner: Brand / legal].

9.7.3 Published path (per locale)

`/{locale}/data/partners.json`: `/en/data/partners.json`; `/de-de/data/partners.json`; `/fr-fr/data/partners.json`; `/fr-ca/data/partners.json`; `/es-es/data/partners.json`; `/pt-pt/data/partners.json`; `/nl-nl/data/partners.json`; `/sv-se/data/partners.json`; `/da-dk/data/partners.json`; `/no-no/data/partners.json`; `/zh-cn/data/partners.json`; `/ja-jp/data/partners.json`; `/ko-kr/data/partners.json`. AEM: `/content/tcs-corp/{country}/{language}/data/partners`.

9.7.4 Columns

`getPartners()` returns `{ id, name, logo, website, tier, description }` and drops rows without both `partner-id` and `partner-name`.

Column	Type	Mandatory	Description	Example
`partner-id`	text (ID)	Yes, or `partner-name`	Stable ID.	`partner-example-1`
`partner-name`	text	Yes, or `partner-id`	Display name.	`Example Cloud Partner`
`logo`	image URL or path	No	Partner logo.	`/en/data/media_4b.png`
`website`	URL	No	Partner website.	`https://www.example.com`
`tier`	text	No	Partnership level (free text; no values enforced).	`Strategic`

Column	Type	Mandatory	Description	Example
`description`	text	No	Short description.	`Joint cloud migration offerings.`

9.7.5 Relationships
None in code. [To be confirmed: partner tiers vocabulary — owner: Alliances team].

9.7.6 Dependencies
`utils/data.js` (`getPartners`). No block, decorator or template uses it (as of this handover).

9.7.7 Examples

partner-id	partner-name	logo	website	tier	description
`partner-example-1`	Example Cloud Partner	`/en/data/media_4b.png`	`https://www.example.com`	Strategic	Joint cloud migration offerings.
`partner-example-2`	Example Data Partner	`/en/data/media_5c.png`	`https://www.example.org`	Premier	Analytics platforms.
`partner-example-3`	Example Security Partner		`https://www.example.net`	Select	Managed security services.

Illustrative values; the `/en/` sheet has the headers and no rows.

9.7.8 Business Rules
1. Rows without both ID and name are dropped (code).
2. Loaded once per page view by a consumer; missing sheet gives an empty list (code).
3. Recommended: logos as SVG or PNG with a transparent background, uploaded to AEM Assets.

9.7.9 Authoring steps
Step 1 — Create the sheet. In `data/`, **Create > Page**, **Spreadsheet** template, Name `partners`, columns `partner-id`, `partner-name`, `logo`, `website`, `tier`, `description`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-partners-create.png`
> **Purpose:** Creating the Partners sheet
> **Description:** The Create Page wizard in /content/tcs-corp/us/en/data with the Spreadsheet template, the Name `partners` and the six column names.

Step 2 — Add rows, publish and verify. Add one row per partner, publish (Preview, then live) and open `/en/data/partners.json`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-partners-json.png`
> **Purpose:** Verifying the published JSON
> **Description:** The browser showing /en/data/partners.json with the six columns and the rows.

9.8 Social Links

9.8.1 Purpose
The social media profiles of the locale, shown as icon links in the page footer of every page.

9.8.2 Ownership
[To be confirmed: owner of the social profiles — owner: Social media / digital marketing team].

9.8.3 Published path (per locale)
`/{locale}/data/social-links.json`: `/en/data/social-links.json`; `/de-de/data/social-links.json`; `/fr-fr/data/social-links.json`; `/fr-ca/data/social-links.json`; `/es-es/data/social-links.json`; `/pt-pt/data/social-links.json`; `/nl-nl/data/social-links.json`; `/sv-se/data/social-links.json`; `/da-dk/data/social-links.json`; `/no-no/data/social-links.json`; `/zh-cn/data/social-links.json`; `/ja-jp/data/social-links.json`; `/ko-kr/data/social-links.json`. AEM: `/content/tcs-corp/{country}/{language}/data/social-links`.

9.8.4 Columns
`getSocialLinks()` returns `{ platform, url, icon, sortOrder }`, drops rows without `url`, and sorts by `sort-order`.

Column	Type	Mandatory	Description	Example
`platform`	text	Recommended	The platform name: used in the accessible name "Follow us on {platform}" and, without an Icon, as the icon name. `Twitter` uses the `x` icon.	`LinkedIn`
`url`	URL	Yes (rows without it are dropped)	The profile URL. External URLs open in a new tab.	`https://www.linkedin.com/company/example`
`icon`	icon name or image URL	No	An icon of `/icons` (`linkedin`, `x`, `facebook`, `instagram`, `youtube`) or an image URL / path (contains `.` or `/`). Empty: the platform name is used when `/icons` has it; otherwise the platform name is shown as text.	`linkedin`
`sort-order`	number	No	Display order, ascending; rows without a number come last, in sheet order.	`1`

9.8.5 Relationships
Displayed by the page footer of every page of the locale, whichever footer page (`references/footer` or an alternative) the page uses.

9.8.6 Dependencies
`utils/data.js` (`getSocialLinks`), `decorators/social-links-decorator.js` (`buildSocialLinks`), `blocks/footer/footer.js` (page footer role, list class `footer-social`), placeholder `followUsOn`, icons in `/icons/`.

9.8.7 Examples

platform	url	icon	sort-order
LinkedIn	`https://www.linkedin.com/company/example`	linkedin	1
X	`https://x.com/example`	x	2

platform	url	icon	sort-order
YouTube	`https://www.youtube.com/@example`	youtube	3
Instagram	`https://www.instagram.com/example`	instagram	4
Facebook	`https://www.facebook.com/example`	facebook	5

Illustrative URLs; [To be confirmed: official TCS social profile URLs per locale — owner: Social media team]. The `/en/` sheet has the headers and no rows, so no social icons are shown yet.

9.8.8 Business Rules

1. Rows without `url` are dropped (code).
2. Sort: ascending `sort-order`; rows without a valid number after the numbered ones; ties keep the sheet order (code).
3. Icon resolution: an Icon containing `.` or `/` is an image (decorative); else the Icon name (class-name form) or, when empty, the platform name (with `twitter` mapped to `x`); an empty Icon with a platform that has no icon in `/icons` shows the platform name as text (code).
4. With an icon, the link text "Follow us on {platform}" is visually hidden but read by screen readers (code).
5. Links to another origin open in a new tab (code).
6. No rows, or a missing sheet: no social list in the footer (code).
7. Recommended: one row per platform; use the five icon names available in `/icons` (adding a platform icon is a development task: add `icons/{name}.svg`).

9.8.9 Authoring steps

Step 1 — Create the sheet. In `data/`, **Create > Page**, **Spreadsheet** template, Name `social-links`, columns `platform`, `url`, `icon`, `sort-order`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-social-links-create.png`
> **Purpose:** Creating the Social Links sheet
> **Description:** The Create Page wizard in /content/tcs-corp/us/en/data with the Spreadsheet template, the Name `social-links` and the four column names.

Step 2 — Add rows. One row per profile, with `sort-order` 1, 2, 3, …

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-social-links-rows.png`
> **Purpose:** Editing social links
> **Description:** The social-links sheet editor with five rows (LinkedIn, X, YouTube, Instagram, Facebook) and their sort order.

Step 3 — Publish and verify. Publish (Preview, then live), open `/en/data/social-links.json`, then any `/en/` page: the footer shows the icons in sort order; inspecting the first link (developer tools or a screen reader) shows "Follow us on LinkedIn".

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-social-links-footer.png`
> **Purpose:** Verifying the footer icons
> **Description:** The page footer at desktop width with the five social icons in sort order, and the developer tools showing the hidden “Follow us on LinkedIn” text of the first link.

9.9 Media Contacts

9.9.1 Purpose

The press and media relations contacts of the locale, shown at the end of every press release (“Media contacts”), filtered by the release’s region.

9.9.2 Ownership

[To be confirmed: owner of the media contacts — owner: Corporate communications]. Contacts’ consent to publish name, email and phone: [To be confirmed: consent process — owner: Legal / privacy].

9.9.3 Published path (per locale)

`/{locale}/data/media-contacts.json`: `/en/data/media-contacts.json`; `/de-de/data/media-contacts.json`; `/fr-fr/data/media-contacts.json`; `/fr-ca/data/media-contacts.json`; `/es-es/data/media-contacts.json`; `/pt-pt/data/media-contacts.json`; `/nl-nl/data/media-contacts.json`; `/sv-se/data/media-contacts.json`; `/da-dk/data/media-contacts.json`; `/no-no/data/media-contacts.json`; `/zh-cn/data/media-contacts.json`; `/ja-jp/data/media-contacts.json`; `/ko-kr/data/media-contacts.json`. AEM: `/content/tcs-corp/{country}/{language}/data/media-contacts`.

9.9.4 Columns

`getMediaContacts()` returns `{ id, name, designation, email, phone, region }` and drops rows without both `id` and `name`.

Column	Type	Mandatory	Description	Example
`id`	text (ID)	Yes, or `name`	Stable ID of the contact (not displayed).	`media-europe-1`
`name`	text	Recommended (a row with only an ID shows no name)	Contact name.	`Sample Contact A`
`designation`	text	No	Role; shown as “designation, region”.	`Media Relations Manager`
`email`	email address	No	Shown as a `mailto:` link.	`media.europe@example.com`
`phone`	phone number	No	Shown as written; linked as `tel:` with only digits and +.	`+44 20 7946 0000`
`region`	text	No	Matched (case-insensitive) with the press release’s `region` metadata; shown after the designation.	`Europe`

9.9.5 Relationships

- Press releases (Page Type `press-release`) select contacts by their `region` metadata (9.2).

- `region` uses the same vocabulary as the metadata `region` and the `tcs-corp:region/*` tag titles (recommended).

9.9.6 Dependencies

`utils/data.js` (`getMediaContacts`, `inRegion`), `templates/press-release/press-release.js`, `decorators/contact-list-decorator.js` (`buildContactList`), `decorators/template-decorator.js` (`appendLoadedSection`), placeholder `mediaContacts`, `templates/press-release/press-release.css`.

9.9.7 Examples

id	name	designation	email	phone	region
`media-global-1`	Sample Contact A	Global Head of Media Relations	`media@example.com`	+1 555 0100	
`media-europe-1`	Sample Contact B	Media Relations Manager	`media.europe@example.com`	+44 20 7946 0000	Europe
`media-americas-1`	Sample Contact C	Media Relations Lead	`media.americas@example.com`	+1 555 0123	Americas
`media-apac-1`	Sample Contact D	Media Relations Lead	`media.apac@example.com`	+61 2 5550 0000	Asia Pacific

Illustrative values (reserved example numbers and domains); the `/en/` sheet has the headers and no rows.

9.9.8 Business Rules

1. Rows without both `id` and `name` are dropped (code).
2. A press release with a `region` shows the contacts of that region (case-insensitive exact match); if none match, or the page has no region, all contacts are shown (code).
3. Contacts without a region (for example global contacts) are shown only when all contacts are shown (code). Recommended: give global contacts a row per region if they must appear on every release.
4. No contacts: the "Media contacts" section is removed (code).
5. The section is loaded after the page content, without delaying it (code).
6. Order: sheet order (code).
7. Layout: one column on mobile, two from 600 px, three from 900 px (CSS).

9.9.9 Authoring steps

Step 1 — Create the sheet. In `data/`, **Create > Page**, **Spreadsheet** template, Name `media-contacts`, columns `id`, `name`, `designation`, `email`, phone, `region`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-media-contacts-create.png`
> **Purpose:** Creating the Media Contacts sheet
> **Description:** The Create Page wizard in /content/tcs-corp/us/en/data with the Spreadsheet template, the Name `media-contacts` and the six column names.

Step 2 — Add contacts. One row per contact, with the region values used in the press releases' `region` metadata.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/sheet-media-contacts-rows.png`

Purpose: Editing media contacts
Description: The media-contacts sheet editor with the six columns and four example rows with different regions.

Step 3 — Publish and verify. Publish (Preview, then live), open `/en/data/media-contacts.json`, then a press release with `region` = Europe: the “Media contacts” section lists only the Europe contacts; a release without a region lists all.

[**Screenshot Placeholder**]
File name: `screenshots/sheet-media-contacts-press-release.png`
Purpose: Verifying the contacts on a press release
Description: The end of a press release at desktop width with the “Media contacts” heading and the regional contacts in a three-column grid with email and phone links.

Part 10 — Error Pages

10.1 Overview

Error	HTTP status	Served by	Authorable	Locale-specific	Status
Not Found	404	Edge Delivery Services (`404.html` in the repository)	Yes: `error-pages/404` per locale and the inherited **404 Page** page property	Yes	Implemented
Forbidden	403	Not served by Edge Delivery Services	Authored `error-pages/403` per locale, for a CDN	Yes (by CDN rule)	Content only; CDN not configured
Server Error	500	Not served by Edge Delivery Services	Authored `error-pages/500` per locale, for a CDN	Yes (by CDN rule)	Content only; CDN not configured
Maintenance	503 (recommended)	—	—	—	Not implemented

Error pages live in every locale site at `/content/tcs-corp/{country}/{code}/error-pages/` and are published at `/{code}/error-pages/{404|403|500}`. They are excluded from the query index (`/*/error-pages/**` in `helix-query.yaml`) and therefore from the sitemap and from List / Filter List results.

```
Request /de-de/insights/missing-page
       |
       v
Edge Delivery Services: no such page -> HTTP 404 + 404.html
       |
       v
404.html: window.isErrorPage = true; window.errorCode = '404'; scripts.js
       |
       v
loadEager -> loadErrorPage(main)
  1. 404 Page property of the nearest parent (/de-de/insights, then /de-de/) -> load it
  2. else /de-de/error-pages/404
  3. else /en/error-pages/404   (marked lang="en-US")
  4. else keep the built-in "404 / Page Not Found / Go home"
       |
       v
decorateMain -> first section -> header (inherited nav) -> footer (inherited footer)
       |
       v
window load: "Go back" button (same-origin referrer); RUM checkpoint '404'
```

10.2 404 Not Found

10.2.1 How Edge Delivery Serves the 404

When a requested path has no published page, Edge Delivery Services responds with HTTP status **404** and the repository's `404.html`. The status code stays 404 regardless of the content loaded afterwards, so search engines don't index the missing URL.

`404.html` contains:

Part	Content
CSP	Same policy as `head.html` (`script-src 'nonce-aem' 'strict-dynamic' …; base-uri 'self'; object-src 'none';`, moved to the HTTP header)
Title	`<title>Page not found</title>`, `<meta property="og:title" content="Page not found">` (English, not localised)
Flags	Inline script: `window.isErrorPage = true; window.errorCode = '404';`
Scripts	`/scripts/scripts.js` (which imports `aem.js`); a "Go back" script; a RUM script `sampleRUM('404', { source: document.referrer })`
Styles	`styles.css`, `lazy-styles.css`, inline styles: `main.error` fills the viewport below the header (`min-height: calc(100vh - var(--nav-height))`, flex column, centred); `.error-number` SVG text in monospace
Built-in content	`<main class="error">` with one section: an SVG "404", `<h2 class="error-message">Page Not Found</h2>`, `<p class="button-container"><a href="/" class="button secondary error-button-home">Go home</a></p>`
Header / footer	Empty `<header>` and `<footer>`, filled by the lazy phase like any page

10.2.2 Loading the Authored 404 (`loadErrorPage` in `scripts/scripts.js`)

1. The built-in **Go home** button is pointed to the locale's Home: `localePagePath('')`, e.g. `/de-de/` for a missing `/de-de/…` URL (`/en/` for paths outside a locale).
2. If `window.errorCode` is set and isn't `404`, nothing else happens (reserved for other codes).
3. `utils/inherited-metadata.js` is imported on demand and `getInheritedMetadata('error-page')` looks up the **404 Page** value:
   - the page's own `error-page` metadata (not present on `404.html`), then
   - the nearest parent page of the missing URL up to the locale Home (`getParentPages`: for `/de-de/insights/blogs/x` → `/de-de/insights/blogs`, `/de-de/insights`, `/de-de/`). The parents' published HTML is fetched in parallel and `<meta name="error-page">` is read. Missing parents (their own 404) give empty values. Values are cached in `sessionStorage` (`tcs-inherited-metadata`) for the session.
   - The value comes from the 404 Page page property or the bulk metadata sheet's `error-page` column of that parent.
4. The candidate list is built and de-duplicated:
   1. the configured 404 Page, resolved with `resolvePagePath(value, 'error-pages/404')` (AEM path, published path or a path relative to the locale root are accepted; an invalid value falls back to the locale's 404);
   2. the locale's `error-pages/404` (`localePagePath('error-pages/404')`);
   3. the default locale's `/en/error-pages/404`.
5. Each candidate is tried in order with `loadPageContent`: `fetch('{path}.plain.html')`; on a 200 response `main.innerHTML` is replaced, and relative media (`./media_…` in `img src` / `source srcset`) is rewritten relative to the error page. Network errors or non-200 responses move to the next candidate.
6. On success `main` gets the class `error-authored` (in addition to `error`). If the loaded page belongs to another locale than the requested URL (e.g. the `/en` fallback for a `/de-de` URL), `main.lang` is set to that locale's language (e.g. `en-US`) so screen readers pronounce it correctly.
7. If no candidate loads, the built-in content stays.
8. `decorateMain(main)` then decorates the content like a normal page (sections, blocks, buttons, Additional tab options), the first section is loaded (LCP), and the lazy phase loads the remaining sections, the header and the footer.

10.2.3 Header, Footer and Language on the 404 Page

- `<html lang>` is the locale of the missing URL (`decorateHead` → `getPageLanguage`; `404.html` has no `language` metadata).
- The header and footer use the **inherited** Navigation and Footer of the missing URL's nearest parent pages, else the locale's `references/nav` / `references/footer` — so a missing Careers page shows the Careers header if the Careers area sets one.
- The data layer receives a page entry for the missing URL (`repo:path` = requested path; `dc:title` "Page not found").

10.2.4 "Go back" Button

On `window.load`, if `document.referrer` is from the same origin, an `<a class="button error-button-back" href="{referrer path}">Go back</a>` (title "Go back") is appended to the first `main .button-container`. This works for the built-in content and for an authored 404 page that contains a button (a paragraph with a link decorated as a button). The label is English and hard-coded in `404.html` (not from placeholders).

> **Note:** The built-in texts ("Page Not Found", "Go home", "Go back", the `<title>`) are English. Localised text comes from the authored 404 pages, so every locale should publish its `error-pages/404`.

10.2.5 Inheritance of the 404 Page Property

Page tree	404 Page set on	Missing URL	Page shown
`/en/` → `/en/careers/` → `/en/careers/jobs/`	`/en/careers/` = `/content/tcs-corp/us/en/error-pages/404-careers`	`/en/careers/jobs/old-role`	`/en/error-pages/404-careers`
same	same	`/en/about/missing`	`/en/error-pages/404` (no parent sets a value)
`/de-de/` (no `error-pages/404` published)	—	`/de-de/missing`	`/en/error-pages/404` with `lang="en-US"` on `main`
—	—	`/unknown-path` (outside any locale)	`/en/error-pages/404`
nothing published	—	any	Built-in 404 content

10.2.6 Authoring the 404 Page per Locale

Step 1 — Open the locale's error pages folder

Action: In AEM Sites, open `TCS Corporate` → the country → the locale (e.g. `de` → `de-de`) → `error-pages`.

Expected Result: The folder lists `404`, `403` and `500`.

> [Screenshot Placeholder]
> **File name:** `screenshots/error-pages-sites-console.png`
> **Purpose:** Locating the error pages
> **Description:** AEM Sites console with /content/tcs-corp/de/de-de/error-pages selected, showing the 404, 403 and 500 pages.

Step 2 — Open the 404 page in Universal Editor

Action: Select `404` and select **Edit** (opens the Universal Editor).

Expected Result: The 404 page opens on the canvas with its sections.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/error-pages-404-universal-editor.png`
> **Purpose:** Editing the authored 404 page
> **Description:** Universal Editor canvas with the de-de 404 page: a heading, a short explanation, a button to the Home page and a List or Cards block with popular pages.

Step 3 — Write the content in the locale's language

Action: Add a heading (Title, e.g. Title Type H1), a short Text explaining that the page wasn't found, and a **Button** linking to the locale Home (e.g. `/content/tcs-corp/de/de-de`). Optionally add a List (child pages of a hub) or Cards with popular destinations, or a Conversation Canvas for search.

Expected Result: The canvas shows the content; the Button paragraph is rendered as a button (the "Go back" button will be added next to it on the site when the visitor came from the site).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/error-pages-404-content.png`
> **Purpose:** Recommended 404 content
> **Description:** The 404 page canvas with the heading, explanation, Home button and a list of popular pages, all in German.

Step 4 — Set page properties

Action: Open the page properties: set **Title** (e.g. "Seite nicht gefunden") and, in the locale's metadata sheet, `robots` = `noindex` for `/de-de/error-pages/**` (Recommended, so the directly reachable error page URLs aren't indexed).

Expected Result: The properties are saved.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/error-pages-404-properties.png`
> **Purpose:** Error page properties
> **Description:** Page properties dialog of the 404 page with Title filled in and Navigation, Footer and 404 Page left empty.

Step 5 — Preview and test

Action: Preview the page, then open `https://main--tcs-corp--sradha-tcs.aem.page/de-de/error-pages/404` and a missing URL such as `https://main--tcs-corp--sradha-tcs.aem.page/de-de/this-page-does-not-exist`.

Expected Result: The missing URL shows the authored German 404 content with the German header and footer; the browser's developer tools show HTTP status 404.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/error-pages-404-preview.png`
> **Purpose:** Verifying the authored 404 on preview

> **Description:** Browser on a missing /de-de URL of the aem.page host showing the authored German 404 content, the Go back button and the locale header and footer.

Step 6 — Publish

Action: Publish the 404 page (Universal Editor **Publish**, or Quick Publish in the Sites console).

Expected Result: Missing URLs on the live host (`*.aem.live`) show the authored page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/error-pages-404-publish.png`
> **Purpose:** Publishing the error page
> **Description:** Universal Editor Publish dialog for the de-de 404 page with Live selected.

10.2.7 Authoring an Area-Specific 404 Page

Step 1 — Create the page

Action: In the locale's `error-pages` folder, create a page (e.g. `404-careers`) and author it like the 404 page.

Expected Result: The page exists at `/content/tcs-corp/us/en/error-pages/404-careers`.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/error-pages-create-area-404.png`
> **Purpose:** Creating an area-specific 404 page
> **Description:** AEM Sites Create Page wizard in /content/tcs-corp/us/en/error-pages with the name 404-careers.

Step 2 — Set the 404 Page property on the area's top page

Action: Open the page properties of the area's top page (e.g. `/content/tcs-corp/us/en/careers`) and select the new page in **404 Page**.

Expected Result: All missing URLs below `/en/careers/` show `404-careers`, unless a deeper page sets its own 404 Page.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/error-pages-404-page-property.png`
> **Purpose:** Setting the inherited 404 Page
> **Description:** Page properties of /content/tcs-corp/us/en/careers with the 404 Page field set to /content/tcs-corp/us/en/error-pages/404-careers.

Step 3 — Publish both pages and test

Action: Publish `404-careers` and the careers page; open a missing URL below `/en/careers/` in a **new browser session** (private window).

Expected Result: The careers 404 content is shown.

> **[Screenshot Placeholder]**
> **File name:** `screenshots/error-pages-area-404-result.png`

Purpose: Verifying the area-specific 404
Description: Browser on a missing /en/careers/ URL showing the careers 404 content with the careers navigation.

Tip: Inherited values are cached per browser session; use a new private window to see a changed 404 Page, Navigation or Footer immediately.

The same value can be set in bulk with the metadata sheet's `error-page` column (e.g. URL `/en/careers/**`, `error-page` = `/en/error-pages/404-careers`).

10.2.8 404 Validation Checklist

- ☐ Every locale publishes `error-pages/404` with content in its language.
- ☐ A missing URL in each locale returns HTTP 404 and shows the locale's content, header and footer.
- ☐ The Go home / Home button points to the locale Home.
- ☐ Area-specific 404 pages are selected in the 404 Page property of the area's top page.
- ☐ The error page has a single H1 and meaningful link texts.
- ☐ `robots: noindex` is applied to `/{locale}/error-pages/**` (Recommended).

10.3 403 Forbidden

- Edge Delivery Services doesn't serve a custom 403 page; the public site has no access-restricted pages. If site authentication is enabled for preview / live hosts, the platform returns its own response [To be confirmed: site authentication — owner: Site Administrator].
- Each locale has an authored `error-pages/403` page (`/{code}/error-pages/403`), not used by code. It's intended for the production CDN (e.g. WAF blocks, geo restrictions) [To be confirmed: CDN 403 configuration — owner: CDN Owner].
- **Recommended:** the CDN responds with status 403 and the body of the locale's `/{code}/error-pages/403` (cached copy of the published page, chosen by the first path segment, default `/en/`).
- Authoring: same steps as 10.2.6 on the `403` page (heading, explanation, contact / Home link).

[Screenshot Placeholder]
File name: `screenshots/error-pages-403-universal-editor.png`
Purpose: Editing the 403 page
Description: Universal Editor canvas with the en 403 page: heading "Access denied", explanation and a Home button.

10.4 500 Server Error

- Edge Delivery Services doesn't serve a custom 5xx page. Origin errors are rare (static delivery, no server-side project code); most "errors" are client-side (block failures, see 10.6), which never produce a 500.
- Each locale has an authored `error-pages/500` page for the production CDN.
- **Recommended:** the CDN stores a static copy of each locale's `/{code}/error-pages/500` (refreshed on publish) and serves it with status 500 / 502 / 503 / 504 when the origin fails — a copy, because fetching it from the failing origin would fail too.

- Authoring: same steps as 10.2.6 on the `500` page; keep it self-contained (no List / Filter List, which depend on the query index; no fragments).

> **[Screenshot Placeholder]**
> **File name:** `screenshots/error-pages-500-universal-editor.png`
> **Purpose:** Editing the 500 page
> **Description:** Universal Editor canvas with the en 500 page: heading "Something went wrong", a short apology and a link to the Home page, without dynamic blocks.

10.5 Maintenance Page

Status: Not implemented (as of this handover).

When it's needed: Code and content deployments on Edge Delivery Services don't need downtime (code is live per branch on merge; content on publish). An AEM author outage doesn't affect the live site. A maintenance page is therefore only needed for business reasons (e.g. legal hold) or a planned CDN / domain change.

Recommended approach:

1. Author a page per locale, e.g. `error-pages/maintenance` (or one global English page), self-contained (no dynamic blocks).
2. Store a static copy at the CDN.
3. Add a CDN rule, switchable without deployment, that returns **503 Service Unavailable** with a `Retry-After` header and the maintenance body for all paths except an allow-list (internal IPs, `/.well-known/`).
4. Add `robots: noindex` to the maintenance page; 503 tells crawlers to retry later.
5. Document the switch in the AMS runbook [To be confirmed: maintenance procedure — owner: CDN Owner].

10.6 Custom Error Handling

10.6.1 Block and Template Load Failures

Failure	Behaviour	Console message
Block JS fails to import or its `decorate` throws	Caught in `loadBlock` (`scripts/aem.js`); the block is marked `data-block-status="loaded"` and stays undecorated (authored rows visible, possibly unstyled); the section is still shown; other blocks continue	`failed to load module for {name}` / `failed to load block {name}`
Block CSS missing	The block renders unstyled	Network 404 in DevTools
Template fails (`templates/{name}`)	Caught in `loadTemplate`; the page renders without the template structure	`failed to load template {name}`
Auto blocking fails	Caught	`Auto Blocking failed`
Lottie animation fails	Caught in the animation decorator	`failed to play the Lottie animation`
Universal Editor change can't be applied	Page reload	—

10.6.2 Fragment Failures (`loadFragment`)

`loadFragment(path)` returns `null` for a path that doesn't start with `/` (or starts with `//`) and for a non-200 response.

Consumer	Behaviour when the fragment is missing
Fragment block	Keeps its authored content (the link / path)
Header (Navigation page)	Header left empty
Footer (footer page)	Only the Social Links sheet (if any) is shown
Footer block (Conversation Canvas Page)	The canvas is omitted; the copyright / links bar still renders
Modal (link to `/modals/...` or Modal block)	Nothing opens (for auto-linked `/modals/` links the navigation is already prevented)
Tabs (Fragment of a Tab item)	The tab's panel stays empty
Authored 404	Next candidate, then built-in content

10.6.3 Site Data Failures

Resource	Fallback
`/{locale}/placeholders.json`	English fallback texts passed by the code
`/{locale}/taxonomy.json`	Labels derived from the tag IDs (`utils/tags.js` `parseTag`) or the authored titles
`/{locale}/data/*.json`	Empty lists: no social links in the footer; Page Header uses the `author` metadata when the Authors sheet has no row; "About the author" (article / blog) and "Media contacts" (press release) sections are removed (`appendLoadedSection`)
`/query-index.json`	Empty page list (List, Filter List, Conversation Canvas menu) — see empty states
`/{locale}/redirects.json`	Links unchanged
Parent page heads (inherited metadata)	Treated as empty: defaults `references/nav`, `references/footer`, `error-pages/404`
Images	`waitForFirstImage` resolves on error, so the page isn't blocked
Data layer	`pushToDataLayer` never throws

10.6.4 Empty States

Block	Condition	Site behaviour	Universal Editor behaviour
List	No cards	**When there are no items** = Hide the list: block and its wrapper hidden; Show a message: the **Empty message** or placeholder `listEmpty` ("There are no items to show."); Show the fallback items: the block's List Items (Child pages / Tagged pages)	Hint `listEditorHint` ("Set up the list in its properties: {problems}.") or `listEditorPages` ("The pages show on the published page (from the query index).")
Cards	No cards	**When there are no cards** = hide or message (same placeholder `listEmpty`)	Hint `cardsEditorHint` ("Add a Card to the block and give it a title, a description or an image.")
Filter List	No results	`noResources` ("No resources found."), `noResourcesHint` ("Try selecting another filter.") and a reset button when filters are selected	`filterListEditor` ("The results show on the published page, from the pages' metadata.")
Tabs	No tabs	—	`tabsEmpty` / `tabsBlockEmpty` hints
Separator	Hidden on every breakpoint, or a semantic separator without a line	Nothing visible on the page	Note `separatorHidden` ("Separator hidden on all devices") / `separatorInvisible`

Block	Condition	Site behaviour	Universal Editor behaviour
Interactive Dashboard	Always	Renders nothing	Labelled placeholder
Conversation Canvas menu	No pages below the Menu root path	Empty menu item (`conversation-canvas-menu-empty`)	Same
Header	No Navigation page	Empty header	Same

10.6.5 Form Validation Errors

Form Variations validates on submit: invalid fields get `aria-invalid="true"` and inline error messages; an error clears when the field becomes valid on input. A valid submit dispatches `form-variations:submit` and shows `formSuccess` ("Thank you, your details were sent."). No network request is made (no back end).

10.7 Redirection Rules

Rule	Where	Status
`/` → `/en/`	CDN rule or root `/redirects.json`	Not implemented
Per-locale redirects (`/{locale}/redirects.json`: Source, Destination)	Authored per locale	Published; not applied by the platform; used by the List block to link to final targets (up to 5 hops, loop-safe)
Root `/redirects.json`	Edge Delivery (the only sheet the platform applies, server side, 301)	Not present
CDN redirects from the locale sheets	Production CDN	Not implemented [To be confirmed: production domain and CDN — owner: Site Administrator]
Error page links	Authored	The built-in Go home link points to the locale Home; Go back to the same-origin referrer

Rules for redirect sheet entries (Recommended):

1. Sources start with `/` and with the sheet's locale (`/de-de/…`); paths without `.html`, without trailing slash.
2. Destinations are final targets (no chains); external URLs allowed.
3. Remove redirects whose source page has been re-created.
4. Don't redirect error pages.

10.8 Recovery Behaviour

Situation	Automatic recovery	Manual recovery
Locale 404 page not published	`/en/error-pages/404` shown (marked `lang`)	Publish `/{locale}/error-pages/404`
`/en/error-pages/404` also missing	Built-in English content	Publish the English 404

Situation	Automatic recovery	Manual recovery
404 Page property points to a deleted page	Next candidate (locale 404)	Correct or clear the property; republish the parent page
Changed 404 Page / Navigation / Footer not visible	Takes effect in a new browser session (session cache)	Test in a private window; republish the parent page if the value isn't in its head
Block fails to load	Page continues; block undecorated	Check the console; revert the code change on `main`
Fragment missing	Graceful omission (see 10.6.2)	Publish the fragment / fix the path
Sheets missing	Fallbacks (see 10.6.3)	Publish the sheet (preview and publish the spreadsheet in AEM)
Origin error (5xx)	None today (no CDN error page)	Escalate to Adobe Support; with the recommended CDN, the static 500 page is served
Bad code deployment	—	Revert the commit on `main`; Code Sync deploys within seconds
Wrong content published	—	Restore a previous page version in AEM and republish, or unpublish

[Screenshot Placeholder]
File name: `screenshots/error-pages-404-fallback-en.png`
Purpose: Fallback behaviour
Description: Browser on a missing /sv-se URL where the Swedish 404 isn't published, showing the English 404 content inside the Swedish header and footer.

Part 11 — Analytics & Data Layer

This part describes the analytics foundation of the TCS Corporate Website: the **Adobe Client Data Layer** that every page and block fills, the exact entries and events it contains, how blocks refine it, and how to connect **Adobe Tags** and the **Adobe Experience Platform Web SDK** to it. It is written for analytics engineers, developers and the AMS team; authors only need 11.5 (what the analytics-related fields do) and 11.9 (privacy).

11.1 Overview

11.1.1 What is implemented

The site pushes structured information about the page, its components and visitor interactions to `window.adobeDataLayer`, following the **AEM Core Components data layer** schema (the same structure AEM Sites Core Components produce). Adobe Tags rules and the Web SDK can read it without block-specific code.

Item	Status (as of this handover)	Where
Data layer array `window.adobeDataLayer`	Implemented (created on the first push, `window.adobeDataLayer || []`)	`utils/analytics.js pushToDataLayer`
Page state	Implemented, once per page	`decorators/data-layer-decorator.js trackPage`
Component state for every block	Implemented, once per block	`trackBlocks`
Interaction events `cmp:click`, `cmp:show`, `cmp:hide`, `cmp:media`, `cmp:search`, `cmp:submit`, `cmp:change`	Implemented, document-wide listeners	`trackInteractions`
Block-specific state (List, Cards, Separator) and events (Tabs, Gen Carousel, Modal)	Implemented	the blocks and `decorators/card-list-decorator.js`
Adobe Client Data Layer **library** (`adobe-client-data-layer.min.js`: `getState()`, `addEventListener()`)	**Not loaded** — the array is a plain JavaScript array until a library (for example the Tags ACDL extension) processes it	—
Adobe Tags (Launch) embed	**Not implemented** — `scripts/delayed.js` contains only a comment	`scripts/delayed.js`
Adobe Experience Platform Web SDK, Adobe Analytics, Customer Journey Analytics, Adobe Target, Adobe Journey Optimizer	**Integration-ready, not implemented**	—
Consent management	**Not implemented**	—

> **Important:** Nothing is sent to any analytics server today. The data layer only exists in the visitor's browser until a tag manager is added (11.7).

11.1.2 Code modules

Module	Exports	Role
`utils/analytics.js`	`pushToDataLayer(entry)`, `pushComponents(components)`, `pushComponentEvent(event, id, info)`	Low-level pushes; every push is wrapped in `try/catch` so analytics never breaks rendering or links
`decorators/data-layer-decorator.js`	`trackPage()`, `trackBlocks(container)`, `trackInteractions()`, `componentIdOf(block)`	Page and component state, document-wide event listeners, component ids
`decorators/card-list-decorator.js`	`trackCardList(block, options)`	State of List and Cards and their cards
`blocks/separator/separator.js`, `separator-config.js`	`analyticsEntry(config, id)`	Separator state (opt-in)
`blocks/tabs/tabs.js`, `blocks/gen-carousel/gen-carousel.js`, `blocks/modal/modal.js`	—	`cmp:` show pushes for tab panels, slides and dialogs
`utils/viewport.js`	`currentBreakpoint()`	`breakpoint` of clicks (mobile < 600 px, tablet 600–1199 px, desktop >= 1200 px)
`utils/metadata.js`	`getPageMetadata()`, `getPageLanguage()`, `getPageTags()`	Page state values

11.1.3 Entry shapes

```
State entries                                Event entries
{ page:      { 'page-{hash}': {...} } }      { event: 'cmp:click',
{ component: { '{id}': {...}, ... } }          eventInfo: { path: 'component.{id}', ... } }
```

- **State** entries describe what is on the page; with the ACDL library they are merged into one computed state (`adobeDataLayer.getState()`).
- **Event** entries describe an interaction; `eventInfo.path` points to the component in the state (`component.{id}`).

11.1.4 When entries are pushed

```
Phase    When                               What is pushed
-------  ---------------------------------
-------------------------------------------------
Eager    loadEager: trackInteractions()     page state (first entry); listeners start
Eager+   while blocks decorate              List, Cards, Separator state (their own
entries)
Lazy     after loadSections(main)           state of all blocks in main: trackBlocks(main)
Delayed  3 s after load: trackBlocks(doc)   header, footer, blocks added later (e.g.
modals)
Delayed  then import('./delayed.js')        nothing today (the Tags embed goes here, 11.7)
Any      on interaction                     cmp:* events
```

> **Note:** The same code runs in the Universal Editor, so entries are also pushed while authors edit. No data leaves the browser today; when a tag manager is added it must not be loaded in the editor (11.7.3).

11.2 Page state

`trackPage()` runs once, at the start of the eager phase, and pushes one page entry. The page id is `page-` followed by a hash of `window.location.pathname` (a djb2-style 32-bit hash written in base 36), so the same path always has the same id.

Field	Value	Source
(key)	`page-{hash}`	Hash of the URL path
`@type`	`tcs-corp/components/page`	Constant
`dc:title`	The authored page title: the `og:title` metadata, else `document.title`, read before the SEO decorator applies `meta-title`	`getPageMetadata().title`
`xdm:language`	The `language` metadata when it is a language code (for example `en`, `en-US`), else the locale's language tag (`en-US`, `de-DE`, `fr-FR`, `fr-CA`, `es-ES`, `pt-PT`, `nl-NL`, `sv-SE`, `da-DK`, `no-NO`, `zh-CN`, `ja-JP`, `ko-KR`)	`getPageLanguage()`
`xdm:template`	The Page Type (`page-type` metadata, else `template`); omitted when empty	`getPageMetadata().pageType`
`xdm:tags`	Array of the page tags: the `article:tag` meta values (tag IDs or titles as authored), else the comma-separated `tags` meta; empty array when none	`getPageTags()`
`repo:path`	`window.location.pathname` (the published path, for example `/de-de/newsroom/q3-results`)	Location

Example:

```
{
  "page": {
    "page-1k9x0qz": {
      "@type": "tcs-corp/components/page",
      "dc:title": "TCS reports Q3 results",
      "xdm:language": "en-US",
      "xdm:template": "press-release",
      "xdm:tags": ["tcs:topics/financial-results", "tcs:regions/global"],
      "repo:path": "/en/newsroom/q3-results"
    }
  }
}
```

> **Note:** Region, country, persona, content type, author and campaign ID are available as page metadata (`utils/metadata.js`) but are **not** in the page state today. Add them in `trackPage()` if the measurement plan needs them (11.7.4).

11.3 Component state

11.3.1 Generic component entries (trackBlocks)

`trackBlocks(container)` adds every decorated block (`.block`) once. It runs for `main` after the sections are loaded and for the whole document 3 seconds later (header, footer, modals).

Field	Value
(key)	The component id from `componentIdOf(block)`
`@type`	`tcs-corp/components/{block}` where `{block}` is the block's `data-block-name`, else its first class (for example `teaser`, `hero`, `header`, `footer`)
`parentId`	The id of the nearest enclosing block (for example a Hero inside a Gen Carousel, a block inside a Fragment), else the page id
`dc:title`	The text of the first heading (`h1`–`h6`) inside the block: its `aria-label` or text, whitespace collapsed, up to 100 characters; omitted when there is no heading

Skipped blocks: a block is not added when it was already tracked (`data-cmp-tracked`), or when it or an ancestor has a `data-analytics` attribute (`off` or `custom`; see 11.5). Hidden blocks (Additional tab **Hide block**) are still added.

11.3.2 Component ids (componentIdOf)

1. If the block already has `data-cmp-id`, that value is used.
2. Else the block's `id` (the **Anchor ID** from the Additional tab, or the **Component ID** of List, Cards and Separator, or an id a block sets itself).
3. Else `{block}-{n}`: the block name and a counter per block name, in the order the ids are first requested (normally page order), for example `teaser-1`, `teaser-2`.

The id is stored as `data-cmp-id` on the block.

> **Tip:** `{block}-{n}` ids change when blocks are added, removed or reordered. Give blocks that matter for reporting a stable **Anchor ID** (or Component ID) so reports stay comparable over time.

Example (a Hero inside a Gen Carousel, and a Teaser):

```
{
  "component": {
    "gen-carousel-1": {
      "@type": "tcs-corp/components/gen-carousel",
      "parentId": "page-1k9x0qz"
    },
    "hero-1": {
      "@type": "tcs-corp/components/hero",
      "parentId": "gen-carousel-1",
      "dc:title": "Building on belief"
    },
    "anchor-ai-services": {
      "@type": "tcs-corp/components/teaser",
      "parentId": "page-1k9x0qz",
      "dc:title": "AI services for the enterprise"
    }
  }
}
```

11.3.3 Custom component entries

Blocks marked `data-analytics="custom"` push their own, richer state (11.6): List, Cards (with one entry per card) and Separator (when its Analytics option is on).

11.4 Events

All events share this envelope (`pushComponentEvent`):

```
{ "event": "cmp:{name}", "eventInfo": { "path": "component.{id}", "...": "..." } }
```

Which component an event belongs to (`componentOf`): the nearest element with `data-cmp-id` inside the same block (for example a card or a carousel slide), else the block's id, else the page id (`component.page-{hash}` for links in default content).

> **Note:** For links in default content the path is `component.page-{hash}`, while the page state is stored under `page.page-{hash}`. With the ACDL library, `getState('component.page-…')` therefore returns nothing for these clicks; read the page state with `getState('page')` instead. (`decorators/README.md` describes the path as `page.{id}`; the code produces `component.page-{hash}`.)

The listeners are registered once on `document` in the capture phase, so they see events before block code handles them.

11.4.1 cmp:click

Trigger: a click on a link (`a[href]`), a `button`, an element with `role="tab"` or `role="button"`. Excluded: `summary` elements (accordion items push `cmp:show` / `cmp:hide` instead), elements inside `data-analytics="off"`, and elements outside any block unless they are links inside `main` (so a click on a link in the page's text counts, a click outside `main` and outside header / footer blocks does not).

Field	Value
`path`	`component.{id}` (see above)
`interaction`	The element's `data-cmp-interaction` (for example `card`, `cta`, `next`, `filter`), else `link` for links and `button` for other elements
`dc:title`	The element's `aria-label`, else its text; whitespace collapsed; up to 100 characters (localised: it is the visible or accessible label in the page language)
`xdm:linkURL`	The absolute URL of the link (omitted for buttons)
`xdm:linkType`	`download` when the link has a `download` attribute or its URL ends in `.pdf`, `.doc`, `.docx`, `.xls`, `.xlsx`, `.ppt`, `.pptx`, `.zip`, `.csv`, `.txt`, `.mp3`, `.mp4` (before ?, # or the end); `exit` when the link goes to another origin; otherwise `other` (also for buttons)
`value`	The element's `data-cmp-value` (omitted when absent)
`position`	The element's `data-cmp-position` as a number (omitted when absent)
`breakpoint`	`mobile`, `tablet` or `desktop` at the time of the click

Example (a card in a List):

```
{
  "event": "cmp:click",
```

```
  "eventInfo": {
    "path": "component.insights-list-item-3f9a2k",
    "interaction": "card",
    "dc:title": "How cloud platforms change banking operations",
    "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/insights/cloud-
banking",
    "xdm:linkType": "other",
    "breakpoint": "desktop"
  }
}
```

Example (a PDF download link in default content):

```
{
  "event": "cmp:click",
  "eventInfo": {
    "path": "component.page-1k9x0qz",
    "interaction": "link",
    "dc:title": "Download the annual report",
    "xdm:linkURL": "https://main--tcs-corp--sradha-tcs.aem.live/en/reports/ar-2026.pdf",
    "xdm:linkType": "download",
    "breakpoint": "mobile"
  }
}
```

11.4.2 cmp:show

Triggers and fields:

Source	When	Path	Extra fields
Accordion item (`<details>` in any block)	The item is opened (`toggle` event, captured)	the item's nearest `data-cmp-id`, else the block	`dc:title`: the item label (`summary` text)
Tabs (block and Tabs Container)	The user selects another tab (not on the initial selection)	`component.{panel id}` (the Tab ID, else `{tabs id}-item-{n}`)	`dc:title`: the tab label
Gen Carousel	The active slide changes (previous / next, dots, keyboard, swipe / scroll, or autoplay), not for the first slide on load	`component.{carousel}-slide-{n}`	`position`: the slide number (1-based)
Modal	The dialog opens	`component.{modal block id}`	`interaction`: `dialog`

Example (accordion item opened):

```
{
  "event": "cmp:show",
  "eventInfo": { "path": "component.accordion-1", "dc:title": "What is TCS Cognix?" }
}
```

Example (tab selected):

```
{
  "event": "cmp:show",
```

```
  "eventInfo": { "path": "component.tabs-1-item-2", "dc:title": "Banking" }
}
```

Example (carousel slide):

```
{
  "event": "cmp:show",
  "eventInfo": { "path": "component.gen-carousel-1-slide-3", "position": 3 }
}
```

Example (modal opened):

```
{
  "event": "cmp:show",
  "eventInfo": { "path": "component.modal-1", "interaction": "dialog" }
}
```

Note: Tab panels and carousel slides are not separate entries in the component state; their ids appear only in event paths. Autoplay slide changes are pushed like user changes; filter them in Tags if only user interactions should count (a preceding `cmp:click` with `interaction` `next` / `previous` identifies user navigation by buttons).

11.4.3 cmp:hide

Source	When	Path	Extra fields
Accordion item (`<details>` in a block)	The item is closed	the item's nearest `data-cmp-id`, else the block	`dc:title`: the item label
Any `<dialog>` (Modal)	The dialog closes (close button, Escape, backdrop click)	the dialog's block	`interaction`: `dialog`

Example:

```
{
  "event": "cmp:hide",
  "eventInfo": { "path": "component.modal-1", "interaction": "dialog" }
}
```

11.4.4 cmp:media

Trigger: `play`, `pause` and `ended` events of an HTML `<video>` or `<audio>` element inside a block (captured). Background videos (no controls, looping and muted) are ignored. Embedded YouTube or Vimeo players (iframes) do not expose these events and are not tracked, except the click on the Embed placeholder's play button (`cmp:click`, `interaction` `play`).

Field	Value
`path`	`component.{id}` of the media's component
`interaction`	`play`, `pause` or `ended`
`xdm:mediaURL`	The media's `currentSrc` (omitted when empty)

Field	Value
position	The playback position in whole seconds

Example:

```
{
  "event": "cmp:media",
  "eventInfo": {
    "path": "component.video-1",
    "interaction": "pause",
    "xdm:mediaURL": "https://main--tcs-corp--sradha-tcs.aem.live/media_7e1f.mp4",
    "position": 42
  }
}
```

11.4.5 cmp:search

Trigger: the `submit` event of a form inside a block that has `role="search"` or contains an `input[type="search"]`. Today this is the Conversation Canvas search form (the only search form; its query is also dispatched as the DOM event `conversation-canvas:search` for a future search service).

Field	Value
path	`component.{id}` of the block (for example `conversation-canvas-1`)
xdm:searchTerm	The value of the search input (`input[type="search"]`, else `input[name="q"]`, else the first text input), trimmed, up to 100 characters

Example:

```
{
  "event": "cmp:search",
  "eventInfo": {
    "path": "component.conversation-canvas-1",
    "xdm:searchTerm": "cloud migration"
  }
}
```

> **Note:** The event is pushed in the capture phase, before the block checks the query, so an empty submit pushes `"xdm:searchTerm": ""`. Exclude empty terms in the Tags rule.

11.4.6 cmp:submit

Trigger: the `submit` event of any other form inside a block (today the Form Variations form).

Field	Value
path	`component.{id}` of the block (for example `form-variations-1`)
dc:title	The form's `aria-label`, else its `id`, else the block name (the Form Variations form is labelled with `aria-labelledby` and has no id, so the value is `form-variations`)
fields	The unique, non-empty `name` attributes of the form's controls — names only, never values

Example:

```
{
  "event": "cmp:submit",
  "eventInfo": {
    "path": "component.form-variations-1",
    "dc:title": "form-variations",
    "fields": ["first-name", "last-name", "email", "country", "message", "consent"]
  }
}
```

Important: `cmp:submit` is pushed for every submit **attempt**, before the form validates itself (Form Variations validates in its own submit handler and blocks invalid submissions). It is not a success signal. A valid submission additionally dispatches the DOM event `form-variations:submit` on the block (with the data in `detail.data`); that event is **not** in the data layer. There is no form back end ([To be confirmed: form back end — owner: product owner]). To measure successful submissions, add a data layer push in the success branch of `blocks/form-variations/form-variations.js` (recommended next step), never including the field values.

11.4.7 cmp:change

Trigger: the `change` event of an `input` or `select` inside a block that carries `data-cmp-interaction` (or whose ancestor inside the control carries it). Form fields without that attribute (all Form Variations fields) are not tracked.

Field	Value
`path`	`component.{id}` of the block
`interaction`	The `data-cmp-interaction` value (today `filter-option`)
`value`	`data-cmp-value`, else the control's value (for Filter List options: the tag ID)
`checked`	`true` / `false` for checkboxes; omitted otherwise

Example (Filter List advanced filter option):

```
{
  "event": "cmp:change",
  "eventInfo": {
    "path": "component.filter-list-1",
    "interaction": "filter-option",
    "value": "tcs:industries/banking",
    "checked": true
  }
}
```

11.5 data-cmp-* attributes and data-analytics

These attributes let blocks refine the generic tracking without their own analytics code. Developers set them in block JavaScript; authors influence them only through the fields listed in the last table.

Attribute	Set on	Read by	Effect
`data-cmp-id`	A block (set by `componentIdOf`) or a	`componentOf`	The component id of events inside it

Attribute	Set on	Read by	Effect
	child component (card `li`, carousel slide)		
`data-cmp-interaction`	Links, buttons, inputs	`cmp:click` (`interaction`), `cmp:change` (trigger and `interaction`)	Names the interaction (`card`, `cta`, `next`, `previous`, `pause`, `filter`, `filter-option`, `load-more`, `play`, `playback`, `listen`, `download`, `share`, `animation`)
`data-cmp-value`	Links, buttons, inputs	`cmp:click` / `cmp:change` (`value`)	A value for the interaction (for example the filter tag ID, `all`, or a tag label)
`data-cmp-position`	The clicked element	`cmp:click` (`position`)	A 1-based position
`data-cmp-tracked`	A block (set by `trackBlocks`)	`trackBlocks`	Prevents duplicate state entries
`data-analytics="off"`	A block	`trackBlocks` (skip), all listeners (`tracked()` is false inside it)	No state and no events for the block and everything inside it
`data-analytics="custom"`	A block	`trackBlocks` (skip)	The block pushes its own state; its clicks and other events are still tracked
`data-analytics-label`	Card links (List, Cards)	Not read by the click listener	Informational; the card's Analytics label is used as `dc:title` of the card's **state** entry

Author-facing fields that affect analytics:

Block	Field	Effect
Every block with the Additional tab	Anchor ID	Becomes the component id (stable ids in reports)
List, Cards (`models/_card-list.json`)	Component ID	The component id of the list and the prefix of its card ids
List, Cards	Analytics (`analyticsEnabled`, default on)	On: `data-analytics="custom"` and the list / card state; Off: `data-analytics="off"` — no state and no click events for the block
List items, Cards items	Analytics label	`dc:title` of the card's state entry (Cards: up to 120 characters, enforced by the model)
Separator	Component ID	The separator's id
Separator	Analytics (default off), Analytics label	On: `data-analytics="custom"` and a state entry (label as `dc:title`); Off: `data-analytics="off"`

11.5.1 Adding tracking to a new block (developer checklist)

1. Do nothing for the basics: state, clicks, accordion toggles, dialogs, media and forms are tracked automatically.
2. Name important controls with `data-cmp-interaction` (and `data-cmp-value` / `data-cmp-position` when useful).
3. Mark child components with `data-cmp-id` (unique on the page) when clicks should be attributed to them.
4. For state richer than `@type` / `parentId` / `dc:title`, set `data-analytics="custom"` and push with `pushComponents()` (see `trackCardList`).
5. For show / hide of custom widgets, call `pushComponentEvent('cmp:show', id, {...})`.
6. Never push form values or other personal data.
7. Document the new interactions in `decorators/README.md`.

11.6 Block-specific tracking

11.6.1 List

Source: `blocks/list/list.js`, `decorators/card-list-decorator.js` (`trackCardList`, `renderCardList`).

Aspect	Detail
Analytics option	`analyticsEnabled` (label "Analytics", default on, description "Card clicks and scrolling in the Adobe Data Layer.")
Component id	Component ID, else the block id (Anchor ID), else `list-{n}`; a Component ID already used by another element gets a suffix
List state	`@type` `tcs-corp/components/list`, `parentId` = page id (always, even when nested), `dc:title` = the list heading text or the list label, `xdm:listSource` = the configured source: `parent` (child pages), `fixed` (List Items) or `tags` (tagged pages), `xdm:variant` = the card style when shown as teasers, else `links`
Card state (one per card)	Key `{list id}-item-{hash}`; `@type` `tcs-corp/components/list/item`, `parentId` = list id, `dc:title` = Analytics label, else title, else the first 100 characters of the description, `xdm:linkURL`, `xdm:position` (1-based in the data), `repo:modifyDate` (the card date, ISO)
Card markup	`li[data-cmp-id="{card id}"][data-cmp-position="{n}"]`; card link `data-cmp-interaction="card"`
Header CTA	`data-cmp-interaction="cta"`
Scroll controls	Previous / next buttons `data-cmp-interaction="previous"` / `"next"`
Events	`cmp:click` with `interaction` `card`, `cta`, `previous`, `next`; path = the card id for card clicks, the list id otherwise
Analytics off	`data-analytics="off"`: no state and no events

> **Note:** `data-cmp-position` is on the card's `li`, but the click listener reads `position` from the clicked link, so card clicks carry no `position` field. Use the card's state entry (`xdm:position`, via `getState(eventInfo.path)`) to get the position, or move the attribute to the link (development backlog item).

Example state:

```
{
  "component": {
    "insights-list": {
      "@type": "tcs-corp/components/list",
      "parentId": "page-1k9x0qz",
      "dc:title": "Latest insights",
      "xdm:listSource": "tags",
      "xdm:variant": "image"
    },
    "insights-list-item-3f9a2k": {
      "@type": "tcs-corp/components/list/item",
      "parentId": "insights-list",
      "dc:title": "How cloud platforms change banking operations",
      "xdm:linkURL": "/en/insights/cloud-banking-operations",
      "xdm:position": 1,
      "repo:modifyDate": "2026-09-30"
    }
```

```
    }
  }
```

11.6.2 Cards

Source: `blocks/cards/cards.js`, `blocks/cards/cards-data.js`, `decorators/card-list-decorator.js`.

Aspect	Detail
Analytics option	`analyticsEnabled` (shared model `models/_card-list.json`, default on)
Component id	Component ID, else Anchor ID, else `cards-{n}`
Block state	`@type` `tcs-corp/components/cards`, `parentId` = page id, `dc:title` = heading text or list label, `xdm:variant` = card style or `links`
Card state	Key `{cards id}-item-{n}` (1-based authoring order); `@type` `tcs-corp/components/cards/item`; `dc:title` = Analytics label (max 120 characters, enforced), else title, else the start of the text; `xdm:linkURL`; `xdm:position`; `repo:modifyDate`
Events	As List: `card`, `cta`, `previous`, `next`
Analytics off	`data-analytics="off"`

11.6.3 Separator

Source: `blocks/separator/separator.js`, `separator-config.js` (`analyticsEntry`).

Aspect	Detail
Analytics option	`analyticsEnabled` (label "Analytics", default **off**, description "Adds the separator to the Adobe Data Layer (no interaction events).")
Component id	Component ID (validated `^[A-Za-z][A-Za-z0-9_-]*$`), else generated `separator-{n}`
State (when on)	`@type` `tcs-corp/components/separator`, `dc:title` = Analytics label (up to 120 characters in code), `xdm:variant`, `xdm:accessibilityMode`, `xdm:theme`, `xdm:lineStyle`, `xdm:visibility` (the breakpoints where it is shown, comma-separated, or `none`); no `parentId`
Events	None (a separator has no interactions)
Off	`data-analytics="off"`

Example:

```
{
  "component": {
    "separator-1": {
      "@type": "tcs-corp/components/separator",
      "dc:title": "Before case studies",
      "xdm:variant": "default",
      "xdm:accessibilityMode": "decorative",
      "xdm:theme": "auto",
      "xdm:lineStyle": "solid",
      "xdm:visibility": "mobile,tablet,desktop"
    }
  }
}
```

The values of `xdm:variant`, `xdm:accessibilityMode`, `xdm:theme` and `xdm:lineStyle` are the Separator's configured options (see the Separator chapter); the example values are illustrative.

11.6.4 Tabs (block and Tabs Container)

Source: `blocks/tabs/tabs.js`.

Aspect	Detail
State	Generic (`tcs-corp/components/tabs`)
`cmp:show`	When the user selects another tab: `path` `component.{panel id}`, `dc:title` the tab label; not pushed for the initial tab
`cmp:click`	Tab buttons (`role="tab"`) also push `cmp:click` with `interaction` `button` and the tab label
DOM event	`tabs:change` on the block (`detail`: `index`, `id`, `title`) for other scripts

11.6.5 Gen Carousel

Source: `blocks/gen-carousel/gen-carousel.js`.

Aspect	Detail
State	Generic for the carousel (`tcs-corp/components/gen-carousel`); its Hero / Teaser slides are blocks with `parentId` = the carousel
Slide ids	`gen-carousel-{n}-slide-{i}` as `data-cmp-id` on each slide
`cmp:show`	Every change of the active slide (not the first slide on load), with `position`
`cmp:click`	Pause / play (`pause`), previous (`previous`), next (`next`); dots push `interaction` `button` with their label ("Show slide 2 of 4")
Clicks in slides	CTAs inside a Hero / Teaser slide are attributed to that Hero / Teaser block

11.6.6 Modal

Source: `blocks/modal/modal.js`, `scripts/scripts.js` (`autolinkModals`: links to `/modals/` pages open a modal).

Aspect	Detail
State	Generic (`tcs-corp/components/modal`), added by the delayed `trackBlocks(document)` or on first use
`cmp:show`	When the dialog opens, `interaction` `dialog`
`cmp:hide`	When the dialog closes (close button, Escape, backdrop), `interaction` `dialog`
`cmp:click`	The link that opened the modal (`interaction` `link`); the close button (`interaction` `button`, `dc:title` "Close" in the page language)

11.6.7 Filter List

Source: `blocks/filter-list/filter-list.js`.

Aspect	Detail
State	Generic (`tcs-corp/components/filter-list`)
Filter pills	`cmp:click` with `interaction` `filter`, `value` the tag ID or `all`
Advanced filter options	`cmp:change` with `interaction` `filter-option`, `value` the tag ID, `checked`
Load more	`cmp:click` with `interaction` `load-more`
Other controls	Advanced filter toggle, Clear all: `cmp:click` with `interaction` `button`

Aspect	Detail
Result cards	`cmp:click` with `interaction` `link` (no per-card component ids)

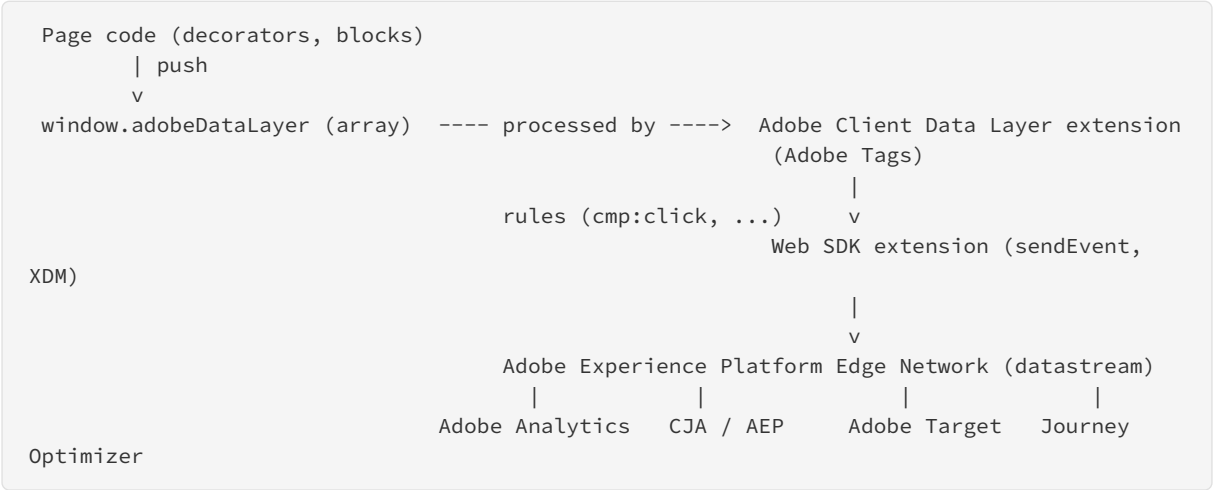
11.6.8 Other interactions

Block / decorator	Element	Event	`interaction` / `value`
Accordion	Item open / close	`cmp:show` / `cmp:hide`	`dc:title` = item label
Conversation Canvas	Tag filters (links or buttons)	`cmp:click`	`filter`, `value` = the tag label
Conversation Canvas	Search form	`cmp:search`	`xdm:searchTerm`
Embed	Placeholder play button	`cmp:click`	`play`
Video	Playback toggle	`cmp:click`	`playback`
Video (file)	Native video play / pause / ended	`cmp:media`	`play` / `pause` / `ended`
Page Header (article meta decorator)	Listen, Download (print), Share	`cmp:click`	`listen`, `download`, `share`
Hero / Teaser (Lottie animation)	Pause / play animation	`cmp:click`	`animation`
Form Variations	Submit attempt	`cmp:submit`	`fields` (names only)
Header, Footer	Navigation and footer links	`cmp:click`	`link` (component `header-1` / `footer-1`)

11.7 Integrating Adobe Tags and the Web SDK (recommended next step)

> **Important:** Everything in this chapter is a **recommendation**. None of it is implemented as of this handover. [To be confirmed: Adobe Tags property, Web SDK datastreams, Adobe Analytics / CJA report suites or data views — owner: analytics lead]

11.7.1 Target architecture



11.7.2 Where the embed goes

The Tags embed belongs in `scripts/delayed.js`, which `scripts/scripts.js` imports 3 seconds after the page has loaded (delayed phase), so analytics never competes with the Largest Contentful Paint. Entries pushed before Tags loads stay in the array and are processed by the ACDL extension when it initialises.

```
// scripts/delayed.js (recommended, not implemented)
const TAGS = {
  // [To be confirmed: Tags library URLs per environment]
  prod: 'https://assets.adobedtm.com/{company}/{property}/launch-{prod-id}.min.js',
  stage: 'https://assets.adobedtm.com/{company}/{property}/launch-{stage-id}-
staging.min.js',
};

const { hostname } = window.location;
// never in the Universal Editor or on the author environment
const inEditor = document.querySelector('[data-aue-resource]')
  || hostname.endsWith('adobeaemcloud.com');
if (!inEditor) {
  const live = hostname.endsWith('.aem.live'); // add the production domain when
configured
  const script = document.createElement('script');
  script.src = live ? TAGS.prod : TAGS.stage;
  script.async = true;
  document.head.append(script);
}
```

Notes on the embed:

- **Content Security Policy:** `head.html` sets `script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:`. With `strict-dynamic`, scripts created by the trusted module scripts (such as `delayed.js`) are allowed, and so are the scripts Tags loads itself. Verify the Tags library, its extensions and any custom code under this policy on the feature preview.
- **Environments:** use the staging library on `.aem.page` and feature previews, the production library on `.aem.live` and the production domain ([To be confirmed: production domain and CDN]).
- **Timing trade-off:** loading in the delayed phase means visits shorter than about 3 seconds send no page view. If the measurement plan requires them, load Tags at the start of the lazy phase instead and re-check Core Web Vitals.
- **Single loader:** do not add the Tags script to `head.html` (it would block rendering and run in the editor).

11.7.3 Tags property set-up

Item	Recommendation
Extensions	Core; Adobe Client Data Layer (data layer name `adobeDataLayer`; it loads the ACDL library and processes the existing array); Adobe Experience Platform Web SDK (one datastream per environment); consent extension of the chosen consent platform
Data elements	`page` — Data Layer Computed State path `page`, with custom code `Object.values(state)[0]` to get the single page object; `eventInfo` — the event's `eventInfo` (ACDL extension event object); `component` — computed state at `event.eventInfo.path`; `breakpoint`, `linkType`, `linkURL`, `interaction` from `eventInfo`
Environments	Development (feature previews), Staging (`main--tcs-corp--sradha-tcs.aem.page`), Production (`.aem.live` and the production domain)
Editor exclusion	Already handled by the loader; optionally add a rule condition "hostname does not end with adobeaemcloud.com"

11.7.4 Mapping the data layer to XDM

Data layer value	XDM field (Web SDK `sendEvent`)	Notes
page `dc:title`	`web.webPageDetails.name`	Authored title
page `repo:path`	`web.webPageDetails.URL` (with origin) or a custom page path field	
page `xdm:language`	Custom field, for example `_{tenant}.page.language`	Or `environment.language` per the schema
page `xdm:template`	Custom field `_{tenant}.page.pageType`; also usable as `web.webPageDetails.siteSection`	
page `xdm:tags`	Custom field `_{tenant}.page.tags` (array)	
page view	`eventType: 'web.webpagedetails.pageViews', web.webPageDetails.pageViews.value: 1`	Send once per page
`cmp:click` `dc:title`	`web.webInteraction.name`	Localised label; prefer `interaction` + component id for language-neutral reports
`cmp:click` `xdm:linkURL`	`web.webInteraction.URL`	
`cmp:click` `xdm:linkType`	`web.webInteraction.type` (`other`, `exit`, `download`)	Values already match the XDM enumeration
`cmp:click`	`eventType: 'web.webinteraction.linkClicks', web.webInteraction.linkClicks.value: 1`	
`eventInfo.path`, `interaction`, `value`, `position`, `breakpoint`	Custom fields `_{tenant}.interaction.*`	
component state `@type`, `dc:title`, `parentId`	Custom fields `_{tenant}.component.*`	Read with `getState(eventInfo.path)`
`cmp:search` `xdm:searchTerm`	Custom field `_{tenant}.search.term` (or the schema's search field group)	Review for personal data (11.9)
`cmp:media`	Custom fields, or the Media Analytics / Media Edge integration	Only native video / audio
`cmp:submit` `fields`	Custom field `_{tenant}.form.fields`	Attempt, not success

11.7.5 Sample Tags rules

Rule	Event	Conditions	Actions
Page view	Core: Library Loaded (Page Top)	Not in the editor	Web SDK Send event: `web.webpagedetails.pageViews`, XDM from the `page` data element
Link click	ACDL: specific event `cmp:click`	`xdm:linkType` = `other`	Send event `web.webinteraction.linkClicks` with name, URL, type, component and interaction fields
Exit link	ACDL: `cmp:click`	`xdm:linkType` = `exit`	Send event (link type `exit`)
Download	ACDL: `cmp:click`	`xdm:linkType` = `download`	Send event (link type `download`)
Card click	ACDL: `cmp:click`	`interaction` = `card`	Send event with the card's state (`getState(eventInfo.path)`: title, position, URL)

Rule	Event	Conditions	Actions
Carousel / tab / accordion view	ACDL: `cmp:show`	—	Send event (custom interaction `show`, component id, position / title)
Dialog closed / item collapsed	ACDL: `cmp:hide`	—	Send event (custom interaction `hide`)
Video	ACDL: `cmp:media`	`interaction` in `play`, `pause`, `ended`	Send event (media milestone)
Internal search	ACDL: `cmp:search`	`xdm:searchTerm` not empty	Send event with the search term
Form submit attempt	ACDL: `cmp:submit`	—	Send event (form name, field names)
Filter used	ACDL: `cmp:change`	`interaction` = `filter-option`	Send event (filter value, checked)

11.7.6 Consent

No consent banner or consent management platform is implemented. Before any data is sent:

1. Choose and implement a consent platform (Adobe-integrated or third party). [To be confirmed: consent management platform — owner: privacy / legal]
2. Configure the Web SDK with `defaultConsent: 'pending'` (or `out` where required) and call `setConsent` from the consent platform's callback; or load Tags only after consent.
3. Apply the rules of each locale's jurisdiction (for example GDPR for the EU and EEA locales `de-de`, `fr-fr`, `es-es`, `pt-pt`, `nl-nl`, `sv-se`, `da-dk`, `no-no`; PIPL for `zh-cn`; PIPA for `ko-kr`; APPI for `ja-jp`; Quebec Law 25 / PIPEDA for `fr-ca`). [To be confirmed: legal requirements per locale — owner: legal]
4. Keep the data layer itself (local, no network) as it is; consent governs sending.
5. Translate the consent banner texts per locale (placeholders or the consent platform's own localisation).

11.7.7 Personalisation (future)

Adobe Target and Adobe Journey Optimizer can use the same Web SDK connection (decision scopes, offers). Nothing is prepared in the blocks beyond the data layer (for example the Teaser's Personalization flag is an authored field; see its chapter). Personalising content above the fold affects LCP and flicker; plan it with the development team. [To be confirmed: personalisation roadmap — owner: product owner]

11.8 Validation

11.8.1 Browser console (works today)

Open the page on the preview or live site (not in the Universal Editor's edit mode, where clicks select components), open the developer tools console and use:

Check	Console command
All entries	`window.adobeDataLayer`
Page entry	`adobeDataLayer.find((e) => e.page)`
All component ids and types	`adobeDataLayer.filter((e) => e.component).flatMap((e) => Object.entries(e.component)).map(([id, c]) => [id, c['@type'], c.parentId])`

Check	Console command
Events only	`adobeDataLayer.filter((e) => e.event)`
Last event	`adobeDataLayer.filter((e) => e.event).at(-1)`
Clicks of a component	`adobeDataLayer.filter((e) => e.event === 'cmp:click' && e.eventInfo.path === 'component.teaser-1')`
Copy for a ticket	`copy(JSON.stringify(adobeDataLayer, null, 2))`

Tip: Clicking a link navigates away and clears the console. Enable **Preserve log** in the console settings, or check with Ctrl / Cmd + click (new tab keeps the current page), or use links to `#` anchors.

Note: `adobeDataLayer.getState()` and `adobeDataLayer.addEventListener()` exist only after the ACDL library is loaded (normally by the Tags ACDL extension). Until then `adobeDataLayer` is a plain array.

11.8.2 Test cases

#	Action	Expected entry
1	Load any page	First entry is `{ page: { 'page-…': { '@type': 'tcs-corp/components/page', … } } }` with the right language and Page Type
2	Wait 3 seconds	Component entries for all blocks, including `header` and `footer`
3	Click a link in a Text	`cmp:click`, path `component.page-…`, `interaction` `link`
4	Click a PDF link	`xdm:linkType` `download`
5	Click an external link	`xdm:linkType` `exit`
6	Click a List / Cards card	`cmp:click`, `interaction` `card`, path = the card id; the card id exists in a component entry
7	Open and close an accordion item	`cmp:show`, then `cmp:hide`, with the item label
8	Select the second tab	`cmp:show`, path `component.{panel id}`
9	Move a carousel to slide 2	`cmp:show` with `position` 2 (and `cmp:click` `next`)
10	Open and close a modal	`cmp:show` and `cmp:hide`, `interaction` `dialog`
11	Play and pause a file video	`cmp:media` `play`, `pause`, with position
12	Search in the Conversation Canvas	`cmp:search` with the term
13	Submit the Form Variations form	`cmp:submit` with field names, **no values**
14	Tick a Filter List advanced option	`cmp:change`, `filter-option`, tag ID, `checked` true
15	Turn List Analytics off and republish	No list or card entries and no clicks from the list
16	Resize to 375 px and click	`breakpoint` `mobile`

11.8.3 Adobe Experience Platform Debugger (after Tags is integrated)

1. Install the Adobe Experience Platform Debugger browser extension.
2. Open the page on preview; open the Debugger; check **Summary** (Tags property and environment, Web SDK version, datastream).

3. **Experience Platform Web SDK:** inspect each `sendEvent` request and its XDM (page view, link clicks).
4. **Tags > Logs** (or `_satellite.setDebug(true)` in the console): see which rules fired for which `cmp:*` event.
5. **Edge Trace / Adobe Experience Platform Assurance:** verify the data reaches Analytics / AEP.
6. Check consent: with consent pending or denied, no `sendEvent` calls should leave the browser.

11.8.4 Regression checks after releases

Run test cases 1, 2, 3, 6 and 13 on the feature preview of every release that changes blocks or decorators. The data layer decorator has no unit tests today (the `test/` folder covers List, Cards and Separator data and configuration); adding them is recommended.

11.9 Privacy

Rule	How the code enforces it
No form values in the data layer	`cmp:submit` sends only the field **names**; Form Variations fields have no `data-cmp-interaction`, so typing or changing them pushes nothing
No personal data in state	Page and component state contain titles, types, paths, tags and ids only
Labels are truncated	`dc:title` values are limited to 100 characters (Cards analytics labels to 120 by the model, Separator labels to 120 by code)
Analytics never breaks the page	All pushes are wrapped in `try/catch`
Opt-out per block	`data-analytics="off"` (List, Cards and Separator Analytics fields) stops state and events for that block
Nothing is sent today	No tag manager or network call exists (11.1.1)

Residual risks and recommendations:

1. **Search terms** (`cmp:search` `xdm:searchTerm`) are free text typed by visitors and can contain personal data (names, e-mail addresses). Before sending them, filter patterns such as e-mail addresses and numbers in the Tags rule, or send only whether a search happened. [To be confirmed: search term policy — owner: privacy / legal]
2. **Analytics labels, Anchor IDs and Component IDs** are authored; authors must not put personal data into them.
3. **Link URLs** of `mailto:` and `tel:` links (for example Media Contacts) are pushed as `xdm:linkURL` and contain e-mail addresses or phone numbers of named contacts; they are classified as `exit` links. Mask or drop them in Tags before sending. [To be confirmed: handling of contact links in analytics — owner: privacy / legal]
4. **URLs with query parameters** (campaign links) are pushed as they are; strip personal parameters in Tags.
5. **Consent** must be in place before data is sent (11.7.6).
6. **Retention and access** of analytics data are governed in the Adobe products, not in this code. [To be confirmed: data retention policy — owner: privacy / legal]

Part 12 — Publishing, Deployment & Release

This part describes how content and code reach the TCS Corporate website, how configuration is changed, how a release is prepared and verified, and how each kind of change is rolled back. It covers the three independent delivery streams of an Edge Delivery Services site:

Stream	Source of truth	Delivered by	Who changes it
Content (pages, fragments, sheets, assets)	AEM as a Cloud Service author `https://author-p220207-e2271469.adobeaemcloud.com` (`/content/tcs-corp`, `/content/dam/tcs-corp`)	AEM publication to Edge Delivery Services (preview and live tiers)	Content authors and publishers
Code (blocks, styles, scripts, models, index and sitemap definitions)	GitHub `sradha-tcs/tcs-corp`, branch `main`	AEM Code Sync (GitHub app) to the Edge Delivery code bus	Development team
Configuration (site configuration service: paths, metadata sources, CDN, robots, access)	Edge Delivery configuration service for org `sradha-tcs`, site `tcs-corp`	Admin API (`admin.hlx.page`)	Site administrator / AMS

> **Important:** The three streams are released independently. A code merge does not publish content, and publishing content does not deploy code. A release that needs both (for example a new block and the pages that use it) must sequence them: code first, then content.

12.1 Content Publishing Lifecycle

12.1.1 Environments and Tiers

Environment	URL	Content shown	Code shown	Typical use
AEM author (Universal Editor)	`https://author-p220207-e2271469.adobeaemcloud.com` (pages under `/content/tcs-corp/...`)	Current, unpublished authoring state	Code of `main` (the fstab mount `bin/franklin.delivery/sradha-tcs/tcs-corp/main`)	Authoring, in-context editing
Local development	`http://localhost:3000` (`aem up`)	The preview tier (content previewed by authors)	Local working copy, including uncommitted code	Development and debugging
Feature preview	`https://{branch}--tcs-corp--sradha-tcs.aem.page/`	Preview tier	The feature branch	Pull request review, PageSpeed check
Preview (main)	`https://main--tcs-corp--sradha-tcs.aem.page/`	Preview tier	`main`	Content review before go-live
Live (main)	`https://main--tcs-corp--sradha-tcs.aem.live/`	Live tier (published content)	`main`	Production origin
Production domain / CDN	[To be confirmed: production domain and CDN — owner: site administrator]	Live tier through the customer CDN	`main`	Public website

Published paths are language-only (`/en/`, `/de-de/`, `/fr-fr/`, `/fr-ca/`, `/es-es/`, `/pt-pt/`, `/nl-nl/`, `/sv-se/`, `/da-dk/`, `/no-no/`, `/zh-cn/`, `/ja-jp/`, `/ko-kr/`), mapped from the AEM paths `/content/tcs-corp/{country}/{language}` by `paths.json` (143 mappings, 14 includes, 28 excludes; see 12.5).

12.1.2 Lifecycle Overview

```
 AUTHOR (AEM as a Cloud Service)                EDGE DELIVERY SERVICES
 -----------------------------------
-----------------------------------------------
 1. Create / edit page in Universal Editor
 2. Review in Universal Editor (Preview mode)
 3. Publish  ---------- preview tier ---------> main--tcs-corp--sradha-tcs.aem.page/en/...
           (optional review step)                  (content review; Sidekick)
 4. Publish  ---------- live tier ------------> main--tcs-corp--sradha-tcs.aem.live/en/...
                                                   - CDN cache purged for the path
                                                   - query index row updated (/query-
index.json)
                                                   - sitemap regenerated (/sitemap.xml)
                                                   - production CDN [To be confirmed]
 5. Unpublish ---------------------------------> path removed from live, index row removed
```

Content states:

State	Where it is visible	How it is reached
Draft	AEM author / Universal Editor only	Saving edits in Universal Editor (edits save automatically)
Previewed	`*.aem.page` and `localhost:3000`	Publishing to the preview tier (where the Publish dialog offers Preview) or an Admin API preview operation
Published (live)	`*.aem.live` and the production domain	Publish in Universal Editor or the AEM Sites console
Unpublished	Nowhere on Edge Delivery (404 page shown)	Unpublish / Manage Publication → Unpublish
Excluded	Never published	Paths in `paths.json` `excludes`: the 13 `block-library` folders and `site-global-templates`

12.1.3 Publishing from Universal Editor

1. Open the page in Universal Editor (AEM Sites console → select the page → **Edit**).
2. Make and check the changes; switch to **Preview** mode to see the page as visitors will (in Preview mode hidden blocks are hidden, carousels and motion run; see 13.18).
3. Select **Publish** in the Universal Editor header.
4. In the Publish dialog, review the list of items: the page and the references it uses (for example fragments and AEM Assets). Keep referenced items selected if they changed.
5. Choose the destination (Live, or Preview where the dialog offers it) and confirm.

[Screenshot Placeholder]
File name: `screenshots/publishing-ue-publish-dialog.png`
Purpose: Universal Editor publishing
Description: The Universal Editor Publish dialog for a `us/en` page, showing the page and its referenced items (a fragment and an image) selected, and the destination choice.

6. Open the published URL (for example `https://main--tcs-corp--sradha-tcs.aem.live/en/about`) and verify the change. Use a private window or a cache-busting query (`?nocache=1`) to bypass the browser cache.

> **Note:** Pages under `block-library/` and `site-global-templates/` cannot be published: they are in the `excludes` list of `paths.json`. This is intentional (they are author-only).

12.1.4 Publishing from the AEM Sites Console (single, bulk, tree, scheduled)

Task	Steps in the AEM Sites console	Notes
Quick publish one or more pages	Select the pages → **Quick Publish** → confirm	Publishes the selected pages and their changed references
Bulk publish with options	Select pages → **Manage Publication** → Action: Publish → Scheduling: Now or Later → include children if needed → Publish	Use "Include children" to publish a section of the site tree
Scheduled publication	Manage Publication → Scheduling: **Later** → date and time	The publication runs as an AEM workflow at the given time
Publish a whole locale	Select the locale page (for example `/content/tcs-corp/de/de-de`) → Manage Publication → include children	Excluded paths (block library) are skipped by the path mapping
Unpublish	Select pages → **Manage Publication** → Action: Unpublish	The live URL returns the 404 page; the query index row is removed
Publish sheets	Select the spreadsheet (for example `/content/tcs-corp/us/en/placeholders`) → Quick Publish	Sheets are served as JSON (see 12.1.6)
Publish assets	AEM Assets → select assets → Quick Publish	Assets referenced by pages are also offered in the page's Publish dialog; `/content/dam/tcs-corp/` is in `paths.json` `includes`

> **[Screenshot Placeholder]**
> **File name:** `screenshots/publishing-manage-publication.png`
> **Purpose:** Bulk publication in the Sites console
> **Description:** The AEM Sites console Manage Publication wizard for `/content/tcs-corp/us/en`, with Action Publish, Scheduling Now and the "Include children" option visible.

> **Tip:** After a bulk publication, check a sample of URLs on `*.aem.live` and the query index (`/query-index.json`) for the new or changed pages. Large publications are processed asynchronously; allow a few minutes before concluding that a page is missing.

12.1.5 AEM Sidekick

AEM Sidekick is the browser extension used on the Edge Delivery URLs. For this AEM-authored (xwalk) site its main uses are:

Sidekick function	Use on this project
Environment switcher	Switch between preview (`.aem.page`) and live (`.aem.live`) for the current path

Sidekick function	Use on this project
Edit	Opens the page in the AEM editor. The project configuration `tools/sidekick/config.json` sets `editUrlLabel: "AEM Editor"` and `editUrlPattern: "{{contentSourceUrl}}{{pathname}}?cmd=open"`
Status	Shows when the path was last previewed and published
Preview / Publish (Reload)	Can re-preview or re-publish a path from the content source; the authoring workflow of record remains publishing in AEM

Note: `tools/sidekick/config.json` still carries the boilerplate project name `"AEM XWalk Boilerplate"`. It is a label only; rename it to the TCS project name in a future code change ([To be confirmed: preferred Sidekick project label — owner: product owner]).

12.1.6 Publishing Sheets and Site Data

Every locale has its own sheets. They are AEM spreadsheet pages published as JSON at the locale's published path (mapped one by one in `paths.json`; the system appends `.json` to spreadsheets; the taxonomy page is mapped explicitly to `taxonomy.json`).

Sheet (AEM path below `/content/tcs-corp/{country}/{language}/`)	Published URL (example `/de-de`)	Read by (code)	Effect of publishing
`placeholders`	`/de-de/placeholders.json`	`utils/placeholders.js` (all blocks with UI text, header, footer)	New UI text on the next page load (loaded once per page)
`metadata`	`/de-de/metadata.json`	Edge Delivery (bulk metadata), applied only when listed in the site configuration's `metadata.source`	Page `<meta>` values change on pages matching the URL patterns
`taxonomy`	`/de-de/taxonomy.json`	`utils/taxonomy.js` (teaser, page-header, filter-list, conversation-canvas, list, cards)	Tag labels change on the next page load
`redirects`	`/de-de/redirects.json`	`utils/redirects.js` (List card links only)	Not applied by the platform (see 13.12)
`data/authors`	`/de-de/data/authors.json`	`utils/data.js` `getAuthor` / `getAuthors` (page-header, filter-list, list feature card, article template)	Author lockups change
`data/locations`	`/de-de/data/locations.json`	`utils/data.js` `getLocations`	Loader exists; no block uses it yet
`data/partners`	`/de-de/data/partners.json`	`utils/data.js` `getPartners`	Loader exists; no block uses it yet
`data/social-links`	`/de-de/data/social-links.json`	`utils/data.js` `getSocialLinks` (footer)	Footer social icons change
`data/media-contacts`	`/de-de/data/media-contacts.json`	`utils/data.js` `getMediaContacts` (press-release template)	Media contacts change

Important: A sheet change is visible only after the sheet itself is published. Editing the sheet in AEM without publishing it has no effect on preview or live.

12.1.7 Query Index and Sitemap Updates

Query index. `helix-query.yaml` defines one index (`pages`, target `/query-index.json`) over `/**`, excluding `/**.json`, each locale's `references/**`, `error-pages/**`, `data/**`, `nav`, `footer`, `block-library` and `site-global-templates`, and any `fragments/` or `modals/` folder. It indexes `lastModified`, `robots`, `title`, `description`, `image`, `tags` and the metadata columns (metaTitle, metaDescription, pageType, contentType, author, authorId, publishDate, modifiedDate, readingTime, thumbnailImage, featuredImage, region, country, language, persona, campaignId, featured, priority, featureCard, featuredPriority, cardTitle, cardDescription, cardImage, cardImageAlt, imageAlt, eyebrow, category, hideInNav).

- Publishing a page adds or updates its row; unpublishing removes it. This is automatic.
- One index serves all 13 locales. `utils/query-index.js` loads it once per page in pages of 500 rows (`?offset=…&limit=500`) and filters the rows to the current locale's root (for example `/de-de/`).
- Consumers: List (child pages, tagged pages), Filter List, Conversation Canvas.
- After a change to `helix-query.yaml` (new column, changed exclude), existing rows are not rewritten until the pages are published again or a re-index is triggered (Admin API index operation, see 12.1.9).

Sitemap. `helix-sitemap.yaml` defines one sitemap (`default`) built from `/query-index.json` into `/sitemap.xml`, with `lastmod: YYYY-MM-DD` and `origin: https://main--tcs-corp--sradha-tcs.aem.live`. The sitemap is regenerated when the index changes. Change the `origin` to the production domain when it is configured (see 12.5 and 13.25).

12.1.8 Caching

Cache	What it holds	Lifetime / invalidation	Notes
Edge Delivery CDN (`*.aem.page`, `*.aem.live`)	HTML, `.plain.html`, JSON sheets, media, code	Purged automatically for the affected path when content is previewed or published and when code is synced	Manual purge: Admin API cache operation
Production CDN	Live responses	[To be confirmed: production CDN (Adobe-managed or customer CDN) and its purge integration — owner: site administrator]	A customer-managed CDN needs push invalidation configured in the site configuration
Browser HTTP cache	Any response	Short TTLs set by Edge Delivery	Use a private window or `?nocache=…` to verify a change
In-page memory caches (code)	Query index, placeholders, taxonomy, redirects, data sheets	One page view (fetched once per page)	`utils/query-index.js`, `utils/placeholders.js`, `utils/taxonomy.js`, `utils/redirects.js`, `utils/data.js`
Session cache `tcs-inherited-metadata` (`sessionStorage`)	The Navigation, Footer and 404 Page values of parent pages	Browser session	`utils/inherited-metadata.js`: a changed parent property shows in a new session (new tab or window after closing the browser, or after clearing site data)
Session flag `fonts-loaded` (`sessionStorage`)	Whether web fonts were loaded	Browser session	`scripts/scripts.js` (not set on `localhost`)

12.1.9 Admin API Operations (conceptual)

The Edge Delivery Admin API (`https://admin.hlx.page`) performs the operations behind publishing. Calls need an authenticated user or an API key with the right role in the site configuration; no credentials are recorded in this document. Path segments are `{org}` = `sradha-tcs`, `{site}` = `tcs-corp`, `{ref}` = `main` (or a branch), `{path}` = the published path (for example `/de-de/about`).

Operation	Endpoint pattern	Purpose	When AMS uses it
Status	`GET /status/{org}/{site}/{ref}/{path}`	Shows the preview, live and edit (source) state and last-modified times of a path	First diagnostic step for "page not updating"
Preview	`POST /preview/{org}/{site}/{ref}/{path}`	Fetches the path from the content source into the preview tier	Re-preview a path when AEM publication did not reach preview
Live (publish)	`POST /live/{org}/{site}/{ref}/{path}`	Promotes the previewed resource to live	Re-publish a path
Unpublish	`DELETE /live/{org}/{site}/{ref}/{path}`	Removes the path from live	Emergency takedown when AEM unpublication is not possible
Cache	`POST /cache/{org}/{site}/{ref}/{path}`	Purges the CDN cache for the path	Stale content after a confirmed publish
Index	`POST /index/{org}/{site}/{ref}/{path}`	Re-indexes the path in `/query-index.json`	Page missing from the index or stale columns
Sitemap	`POST /sitemap/{org}/{site}/{ref}/{path}`	Regenerates a sitemap (for example `/sitemap.xml`)	Sitemap outdated after configuration changes
Code	`POST /code/{org}/{site}/{ref}/{path}`	Re-syncs code from GitHub	Code Sync did not run after a merge
Log	`GET /log/{org}/{site}/{ref}`	Audit log of preview, publish and code operations	Who published what and when
Configuration	`/config/{org}/sites/{site}.json` and sub-resources (for example `public.json`, `metadata.json`)	Reads and updates the site configuration	Configuration changes (see 12.5)

Warning: Admin API operations act immediately on the public site. Bulk operations (for example `/*` paths) must be agreed with the site owner and logged in the change record.

12.2 Code Deployment

12.2.1 Code Repository and Delivery

Item	Value
Repository	`https://github.com/sradha-tcs/tcs-corp` (based on `adobe-rnd/aem-boilerplate-xwalk`)
Default branch	`main` (production)
Build step	None: vanilla JavaScript (ES modules) and CSS are served as committed
Delivery	AEM Code Sync GitHub app pushes every commit of every branch to the code bus; branch code is served at `https://{branch}--tcs-corp--sradha-tcs.aem.page/`
Node.js	`README.md` states Node.js 20 or newer; the GitHub workflow installs Node 24
Dependencies	Development only (ESLint 8 with Airbnb base and `eslint-plugin-xwalk`, Stylelint, Husky, `merge-json-cli`, `npm-run-all`); nothing is bundled into the site

12.2.2 Developer Workflow

Step	Action	Command / location
1	Update `main` and create a feature branch named `feature/tcs-<topic>`	`git checkout main && git pull && git checkout -b feature/tcs-<topic>`
2	Install dependencies (also installs the Husky pre-commit hook through `prepare`)	`npm install`
3	Run the local server against preview content	`npx -y @adobe/aem-cli up --no-open --forward-browser-logs` (or `aem up`)
4	Implement the change following AGENTS.md (blocks, decorators, utils, rem units, mobile-first breakpoints 600 / 900 / 1200 px)	`blocks/`, `decorators/`, `utils/`, `styles/`
5	If models changed (`_*.json`), rebuild the aggregated component files	`npm run build:json` (the pre-commit hook does it automatically for staged `_*.json` files)
6	If `utils/site.js` LOCALES changed, regenerate `paths.json`	`npm run build:paths`
7	Lint and run unit tests	`npm run lint` and `npm test`
8	Commit and push the branch	`git push -u origin feature/tcs-<topic>`
9	Verify on the feature preview	`https://feature-tcs-<topic>--tcs-corp--sradha-tcs.aem.page/{path}` (slashes in branch names become hyphens in the host name)
10	Run PageSpeed Insights on the feature preview URL; target a score of 100	`https://developers.google.com/speed/pagespeed/insights/?url=<feature URL>`
11	Open a pull request to `main` with the preview link	See 12.2.3
12	Review, approve, merge	GitHub

> **Note:** Branch names must be valid in a host name. Keep them short and lower case: the branch, repository and owner together form the `aem.page` sub-domain.

12.2.3 Pull Request Requirements

- Title: what the change does, prefixed with the area (the history uses `Cards: …`, `List: …`, `Footer: …`, `Styles: …`).
- Description (AGENTS.md "Publishing Process"): a link to `https://{branch}--tcs-corp--sradha-tcs.aem.page/{path}` showing the change. **Without it the PR is rejected.** If no page exists yet, create test content (static HTML in `drafts/`, served with `--html-folder drafts`) and ask an author to create a CMS page to link.
- Use `.github/pull_request_template.md` (GitHub issue reference, Before / After test URLs).

> **Warning:** `.github/pull_request_template.md` still contains the unreplaced boilerplate placeholders `{repo}` and `{owner}`. Replace them manually with `tcs-corp` and `sradha-tcs` in each PR, or fix the template in a code change.

12.2.4 Automated Checks

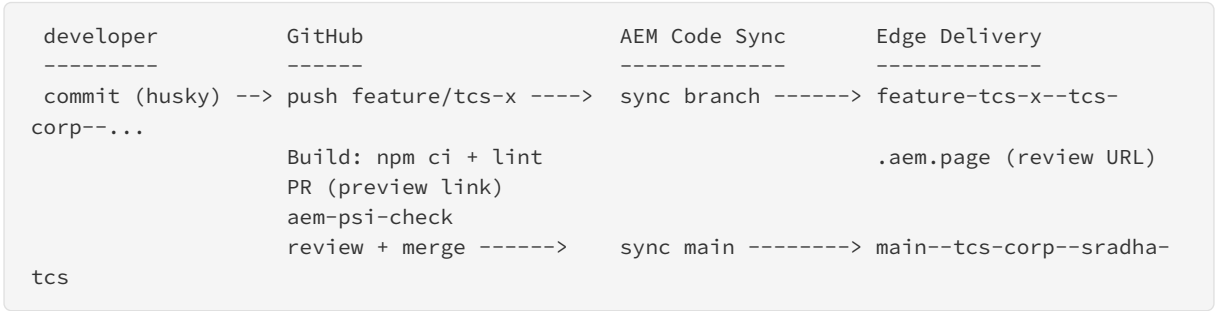
Check	Where defined	When	What it verifies	On failure
Husky pre-commit	`.husky/pre-commit` → `.husky/pre-commit.mjs`	Every local commit	If staged files include a model partial (`_*.json`), runs `npm run build:json` and stages `component-models.json`, `component-definition.json`, `component-filters.json`	Commit continues; the aggregated files are added
Build (GitHub Actions)	`.github/workflows/main.yaml`	Every push (`on: [push]`), every branch	`npm ci` then `npm run lint` (ESLint for `.json`, `.js`, `.mjs` including the xwalk model rules, and Stylelint for `blocks/**/*.css` and `styles/*.css`)	Red check on the commit / PR; fix and push again
AEM Code Sync	GitHub app installed on the repository	Every push	Syncs the branch to the code bus; reports a check on the commit	Code not visible on `aem.page`; see 13.22
PageSpeed (aem-psi-check)	AEM GitHub app check (not a workflow file in the repository)	Pull requests	PageSpeed Insights on the URLs in the PR description	Fix performance regressions (13.15)
Renovate	`.renovaterc.json`	Scheduled	Dependency update PRs; devDependency updates are labelled `ignore-psi-check` and auto-merged when tests pass; ESLint pinned to `^8.0.0`	Review the PR
Unit tests	`package.json` `test` (`node --test 'test/**/*.test.mjs'`)	**Manual only**	`test/list/list-data.test.mjs`, `test/cards/cards-data.test.mjs`, `test/separator/separator-config.test.mjs`	Not run by CI; run `npm test` before every PR

> **Important:** The GitHub workflow runs lint only. Unit tests are not part of CI as of this handover; reviewers must confirm `npm test` passed. Adding `npm test` to `main.yaml` is a recommended improvement.

`.github/workflows/cleanup-on-create.yaml` is a boilerplate workflow that runs only on repository creation or manual dispatch; it has no role in releases.

12.2.5 Merge and AEM Code Sync

1. A reviewer checks the code, opens the preview link and confirms the checks are green.
2. The PR is merged to `main` (the history uses merge commits: "Merge pull request #NN from sradha-tcs/feature/tcs-…").
3. AEM Code Sync pushes `main` to the code bus within seconds to a few minutes.
4. `https://main--tcs-corp--sradha-tcs.aem.page/` and `.aem.live` serve the new code; the CDN is purged for changed code files automatically.
5. Universal Editor uses the new `component-definition.json`, `component-models.json` and `component-filters.json` after the author reloads the editor.



```
                                                        .aem.page / .aem.live
                                                        Universal Editor
(reload)
```

12.3 Branching and Versioning

Topic	Current practice (from the repository history)	Recommendation
Branch model	Trunk-based: short-lived feature branches merged into `main` through PRs (#1–#62 at handover)	Keep; one PR per change
Branch naming	`feature/tcs-<topic>` (for example `feature/tcs-cards-manual-list`, `feature/tcs-separator-block`)	Also use `fix/tcs-<topic>` for defects and `hotfix/tcs-<topic>` for urgent production fixes ([To be confirmed: naming convention for fixes — owner: technical lead])
Protected branch	[To be confirmed: branch protection rules on `main` (required reviews, required checks) — owner: repository administrator]	Require one review and the Build and Code Sync checks
Versions / tags	No Git tags exist; `package.json` keeps the boilerplate version `1.3.0`	Tag each production release (for example `release-2026.10.13`) so rollbacks can target a known state
Changelog	None (no `CHANGELOG.md`)	Use PR titles and GitHub releases as the changelog
Document version	This handover: version 1.0, 13 October 2026	Update when the release process changes
Content versions	AEM page versions (created on publication and on demand)	Create a named version before large content changes

12.4 Release Checklist

12.4.1 Pre-release

#	Check	Owner	Done
1	All PRs of the release are merged to `main`; Build check green on `main`	Development team	[ ]
2	`npm run lint` and `npm test` pass on `main` locally	Development team	[ ]
3	`component-models.json`, `component-definition.json`, `component-filters.json` are rebuilt and committed (no diff after `npm run build:json`)	Development team	[ ]
4	If locales changed: `paths.json` regenerated (`npm run build:paths`) and the site configuration's `public.json` paths updated to match	Development team / site administrator	[ ]
5	Feature preview URLs reviewed by the product owner at 1440, 1024 and 402 px widths	Product owner	[ ]
6	PageSpeed Insights score checked for the changed templates (mobile and desktop)	Development team	[ ]
7	Accessibility spot check (keyboard, focus, screen reader labels) on changed blocks	QA	[ ]
8	New placeholders keys added to all 13 `placeholders` sheets (English fallbacks exist in code)	Content team	[ ]

#	Check	Owner	Done
9	New tags added to the AEM tag namespace and translated in the taxonomy	Content team	[ ]
10	Content dependent on new code is prepared in AEM but not yet published	Content team	[ ]
11	Change record raised with scope, rollback plan and time window	Release manager [To be confirmed: change management process — owner: AMS lead]	[ ]
12	AEM page versions created for pages that will be changed significantly	Content team	[ ]

12.4.2 Release

#	Step	Owner	Done
1	Confirm Code Sync processed the `main` merge (check on the merge commit; new code visible on `main--tcs-corp--sradha-tcs.aem.page`)	Development team	[ ]
2	Apply configuration changes (site configuration, metadata sources, sitemap origin) if part of the release	Site administrator	[ ]
3	Reload Universal Editor and confirm new fields or components appear	Content team	[ ]
4	Publish dependent sheets first (placeholders, taxonomy, data, metadata), then fragments (nav, footer), then pages	Content team	[ ]
5	Publish the pages of the release (Manage Publication, children where needed)	Content team	[ ]
6	Tag the release commit (recommended)	Development team	[ ]

12.4.3 Post-release

#	Check	Owner	Done
1	Changed pages render on `*.aem.live` in all affected locales (no console errors such as `failed to load module for …`)	QA	[ ]
2	Header and footer correct per locale (Navigation / Footer inheritance)	QA	[ ]
3	A missing URL in each changed locale shows that locale's 404 page	QA	[ ]
4	`/query-index.json` contains the new pages with the expected columns	Site administrator	[ ]
5	`/sitemap.xml` lists the new pages with the expected host	Site administrator	[ ]
6	Sheets JSON reachable (`/{locale}/placeholders.json`, `/{locale}/taxonomy.json`, `/{locale}/data/*.json`)	Site administrator	[ ]
7	Data layer: `window.adobeDataLayer` contains the page entry and the blocks; clicks push `cmp:click`	Analytics owner	[ ]
8	PageSpeed Insights on the live URLs; RUM data checked the following day	Development team	[ ]
9	Change record closed with the verification evidence	Release manager	[ ]

12.5 Configuration Changes

Configuration	Location	Applied by	Change procedure	Verification
Locale path mapping	`paths.json` (generated by `npm run build:paths` from `utils/site.js` `LOCALES`; includes `/content/dam/tcs-corp/`)	The site configuration's `public.json` → `paths` (AEM uses it instead of the repository file)	Change `LOCALES`, run `npm run build:paths`, commit both, then update `public.json` paths through the configuration Admin API with the same content	Publish a page of the locale; the published path appears at `/{code}/...`; the Home page resolves at `/{code}/`
Bulk metadata sources	Site configuration `metadata.json`	Admin API (POST `https://admin.hlx.page/config/{org}/sites/{site}/metadata.json` with `{"source": ["/en/metadata.json", "/de-de/metadata.json", …]}`)	List all 13 locale sheets; each sheet's URL patterns must start with its locale (for example `/de-de/**`)	A page matching a pattern shows the column as `<meta name="…">` in its head
Component models, definitions, filters	`blocks/*/_*.json`, `models/_*.json` → `component-models.json`, `component-definition.json`, `component-filters.json`	`npm run build:json` (`merge-json-cli`), then Code Sync	Edit the partials only; never edit the aggregated files by hand; lint with `npm run lint`	Reload Universal Editor; the dialog shows the fields
Query index	`helix-query.yaml`	Code Sync; rows update on publish or re-index	Add the property with `select` and `value`; keep the excludes for references, error pages, data, nav, footer	`/query-index.json` shows the column after pages are re-published or re-indexed
Sitemap	`helix-sitemap.yaml` (`origin`, `source`, `destination`, `lastmod`)	Code Sync; regenerated from the index	Change `origin` to the production domain once configured (or remove it)	`/sitemap.xml` URLs use the right host
Files not served	`.hlxignore` (`.*`, `*.md`, `karma.config.js`, `LICENSE`, `package.json`, `package-lock.json`, `test/*`, `_*`, `docs/*`)	Code Sync	Add patterns for any repository file that must not be public	Requesting the file on `aem.page` returns 404
Content Security Policy	`head.html` and `404.html`: `script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:; base-uri 'self'; object-src 'none';` with `move-to-http-header="true"`	Delivered as an HTTP header	Scripts must be added through code (`loadScript` in `delayed.js`) or carry `nonce="aem"`; change the policy in both files together	Browser console has no CSP violation messages
Content source	`fstab.yaml` (mount `/` to the AEM author `bin/franklin.delivery/sradha-tcs/tcs-corp/main`, type `markup`, suffix `.html`)	Code Sync / site configuration	Only when the AEM environment changes	Preview of a page succeeds
Universal Edi-	`xwalk.json` (`multi-field` enabled)	Site configuration	Rarely changed	Composite multifields (CTA List, Links) available in the editor

Config-uration	Location	Applied by	Change procedure	Verification
tor fea-tures				
Sidekick	`tools/sidekick/config.json`	Code Sync	Edit labels or the edit URL pattern	Sidekick Edit opens AEM
robots.txt, CDN, access control, API keys	Site configuration (not in the repository)	Admin API	[To be confirmed: production robots.txt, CDN and access settings — owner: site administrator]	Fetch `/robots.txt` on the live host

Important: `paths.json` in the repository and the site configuration's `public.json` paths must stay identical. Changing only one of them causes pages to be published at unexpected paths or not at all (see 13.4).

12.6 Rollback

12.6.1 Code Rollback

Situation	Procedure	Time to effect
A merged PR causes a defect	In GitHub, open the merged PR → **Revert** → creates a revert PR → checks → merge. Code Sync deploys the previous code	Minutes after merge
Several PRs must be undone	`git revert -m 1 <merge-commit>` for each merge, newest first, on a branch; PR; merge	Minutes
Urgent (outage)	Revert directly with a `hotfix/tcs-…` branch and an expedited review ([To be confirmed: emergency change approval — owner: AMS lead])	Minutes
Model change reverted	Revert restores the partials and the aggregated `component-*.json`; authors reload Universal Editor. Content already saved with new fields keeps the values but they are no longer shown	Minutes

Warning: Never force-push or rewrite `main`. Always revert with a new commit so the history and Code Sync stay consistent.

12.6.2 Content Rollback

1. In the AEM Sites console select the page → **Timeline** (or Versions) → choose the version before the change → **Revert to this Version** (or **Restore** → Restore Version for deleted pages).
2. Check the restored page in Universal Editor.
3. Publish the page again (Quick Publish). The live tier is updated and the CDN purged.
4. For sheets (placeholders, metadata, taxonomy, data) restore the sheet version the same way and publish the sheet.
5. For an unwanted publication with no good version, **Unpublish** the page (Manage Publication) and republish when corrected.
6. For removed assets, restore the asset version in AEM Assets and publish it.

12.6.3 Configuration Rollback

Configuration	Rollback
`public.json` paths	Re-apply the previous `paths.json` content from Git history (`git show <commit>:paths.json`) through the configuration API
Metadata sources	Re-POST the previous `source` list (keep a copy of the JSON before each change)
Repository configuration files (`helix-query.yaml`, `helix-sitemap.yaml`, `head.html`, `.hlxignore`, `fstab.yaml`)	Revert the PR; re-index or regenerate the sitemap if the index or sitemap changed
CDN / robots / access	[To be confirmed: backup of the site configuration before changes — owner: site administrator]

12.6.4 Rollback Decision Table

Symptom after release	Likely stream	First action
Blocks broken on every page	Code	Revert the last merge
One page wrong, others fine	Content	Restore the page version and republish
A whole locale missing or at wrong paths	Configuration (paths)	Re-apply the previous paths configuration
Metadata missing site-wide	Configuration (metadata sources)	Re-POST the previous metadata source list
Index or sitemap wrong	Code (YAML) or index state	Revert the YAML change; re-index

Part 13 — Global Troubleshooting Guide

This guide covers problems that affect the whole site or many components. Component-specific issues are in each component chapter (section 6.X.13). Every issue lists the **Symptoms**, the **Root Cause** as implemented in this code base, **Diagnosis Steps**, **Resolution Steps**, **Validation Steps** and **Preventive Measures**.

13.1 Triage Approach and Diagnostic Toolkit

13.1.1 First Questions

Question	Why it matters
Does the problem appear in Universal Editor, on preview (`.aem.page`), on live (`.aem.live`) or on the production domain?	Separates authoring state, publication state and CDN caching
Which locale(s)? (`/en/`, `/de-de/`, ...)	Most site data (nav, footer, placeholders, taxonomy, data sheets, 404) is per locale
One page, one block type, or every page?	One page: content; one block type: code or block content; every page: code, configuration or platform
When did it start, and what was released or published just before?	Links the issue to a merge, a publication or a configuration change (Admin API log)
Which breakpoint? (mobile < 600 px, tablet 600–1199 px, desktop >= 1200 px; header switches at 900 px)	Layout issues are often breakpoint-specific

13.1.2 Diagnostic Toolkit

Tool	How to use it
Browser developer tools, Console	Look for `failed to load module for <block>`, `failed to load block <block>`, `failed to load template <name>`, `failed to play the Lottie animation`, CSP violations, 404s of JSON sheets
Browser developer tools, Network	Check the requests to `/{locale}/placeholders.json`, `/{locale}/taxonomy.json`, `/{locale}/data/*.json`, `/query-index.json?offset=0&limit=500`, `*.plain.html` (fragments, nav, footer, error pages) and block JS / CSS
`curl` the delivered markup	`curl https://main--tcs-corp--sradha-tcs.aem.page/en/page.plain.html` shows the authored rows and cells a block receives; `curl -I` shows the status and cache headers
Debug switches	`?debug=list` logs the List configuration and cards; `?debug=separator` logs the Separator configuration (`console.debug`, enable the Verbose level)
RUM switch	`?rum=on` forces Real Use Monitoring sampling for the view (the console shows `ping:<checkpoint>` debug lines)
Data layer	In the console: `window.adobeDataLayer` (array of page, component and event entries)
Admin API status	`GET https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}` (preview, live and source times)
AEM Sidekick	Status, environment switch, open in editor
Local server	`aem up` against preview content with the local code, to isolate code from content

13.2 Page Not Updating After Publish

Symptoms - The live URL still shows the old text or images after the author published. - Preview (`.aem.page`) shows the change, live (`.aem.live`) does not, or the reverse. - Some visitors see the new version, others the old one.

Root Cause - The page was published to preview only, or the publication is still being processed. - A referenced item (fragment, nav or footer page, asset, sheet) changed but was not published; the page itself is up to date but the referenced content is not. - Browser cache, or a production CDN without purge integration ([To be confirmed: production CDN]). - The Navigation / Footer / 404 Page value of a parent page changed: the inherited values are cached in the browser session (`sessionStorage` key `tcs-inherited-metadata`).

Diagnosis Steps 1. Open the page on `.aem.page` and `.aem.live` in a private window and compare. 2. Call the Admin API status for the path and compare the preview, live and source modification times. 3. Check the AEM Sites console for the publication status of the page and of its references. 4. If the difference is in the header or footer, open the nav or footer page URL directly (for example `/en/references/nav`).

Resolution Steps 1. Publish the page again from Universal Editor or the Sites console, including changed references. 2. Publish the changed fragment, nav, footer, sheet or asset separately. 3. If live is still stale after a confirmed publish, purge the cache for the path (Admin API cache operation) and, for a customer CDN, purge it there. 4. For inherited header or footer changes, ask the user to open a new browser session.

Validation Steps - The live URL in a private window shows the change. - `curl -I` on the live URL shows a fresh `last-modified` header.

Preventive Measures - Use the Publish dialog's reference list; publish sheets and fragments before the pages using them. - Configure push invalidation for the production CDN before go-live.

13.3 Block Not Rendering ("failed to load module")

Symptoms - A block shows raw rows and cells (unstyled text, images and links in a grid of `div`s) or nothing at all. - The console shows `failed to load module for <block>` or `failed to load block <block>`.

Root Cause - `scripts/aem.js` `loadBlock()` imports `/blocks/{name}/{name}.js` and its CSS. The error is logged when the file is missing (404), has a syntax error, or the block's `decorate()` throws (for example authored content in an unexpected structure). - The block name in content does not match a block folder (for example a block created on a feature branch not yet merged to `main`). - A dependency import failed (a renamed decorator or util).

Diagnosis Steps 1. Open the console and read the error after `failed to load module for …`; it contains the stack. 2. In the Network tab check the block JS and CSS requests (status 200, correct path). 3. `curl` the page's `.plain.html` and inspect the block's rows; compare them with the block model (`blocks/{name}/_{name}.json`). 4. Run the same page on `aem up` with the local `main` to see if the problem is code or content.

Resolution Steps 1. Code error: fix in a branch, PR, merge (or revert the faulty merge, see 12.6.1). 2. Content error: correct the block content in Universal Editor (fill required fields such as the Video's Video field or the List Item's Page or link) and republish. 3. Missing block on `main`: merge the branch that adds it, then reload Universal Editor.

Validation Steps - The console has no load errors and the block element has `data-block-status="loaded"`.

Preventive Measures - `npm run lint` and `npm test` before every PR; check the feature preview with real content. - Blocks are written to tolerate missing fields; keep it so when changing them.

13.4 Pages Published at the Wrong Path or a Locale Missing

Symptoms - A published page is reachable at `/content/tcs-corp/...` or at an unexpected path, or not at all. - A new locale's pages return 404 after publishing. - `/de-de/` (a locale Home page) returns 404 while child pages work.

Root Cause - The site configuration's `public.json paths` differ from `paths.json`, or `paths.json` was not regenerated after `utils/site.js LOCALES` changed. - The page is under an excluded path (`block-library`, `site-global-templates`). - The locale Home page mapping (`/content/tcs-corp/de/de-de:/de-de/`) is missing; the Home page is published as `/de-de/index` and served at `/de-de/`.

Diagnosis Steps 1. Compare `paths.json` (repository) with the configured paths (configuration Admin API). 2. Run `npm run build:paths` and check whether `paths.json` changes (`git diff`). 3. Check the page's AEM path against `includes` and `excludes`.

Resolution Steps 1. Regenerate `paths.json`, commit it, and apply the same content to `public.json` paths. 2. Republish the affected pages after the mapping is fixed.

Validation Steps - The page loads at `/{code}/...`; the locale Home page loads at `/{code}/`; `/query-index.json` lists the published path.

Preventive Measures - Treat `LOCALES`, `paths.json` and `public.json` as one change (12.5); never edit `paths.json` by hand.

13.5 Component Dialog Fields Missing or Outdated in Universal Editor

Symptoms - A new field, tab or option does not appear in the properties panel. - A new block is not offered in the "Add" menu of a section. - The dialog shows fields that were removed.

Root Cause - The partial model (`blocks/{name}/_{name}.json` or `models/_*.json`) was changed but `component-models.json`, `component-definition.json` or `component-filters.json` was not rebuilt (`npm run build:json`), for example when committing with `--no-verify` or without the Husky hook installed. - The change is on a feature branch: Universal Editor on AEM author uses the code of `main`. - The block is not in the section filter (`models/_section.json` filters, aggregated in `component-filters.json`) or in the container's filter (for example `cards` allows `card`). - The editor was not reloaded after Code Sync.

Diagnosis Steps 1. Search `component-models.json` on `main` for the field name. 2. Run `npm run build:json` on `main` locally; a diff means the aggregated files were not rebuilt. 3. Check `component-filters.json` for the block id in `section` (or the container filter).

Resolution Steps 1. Run `npm run build:json`, commit the three aggregated files, PR, merge. 2. Add the block id to the right filter, rebuild, merge. 3. Reload Universal Editor (full browser reload).

Validation Steps - The field appears in the properties panel; the block appears in the Add menu.

Preventive Measures - Always run `npm install` once per clone so the pre-commit hook is active; check the PR diff includes the aggregated files when partials change.

13.6 Wrong Header (Navigation) or Footer

Symptoms - A page shows another locale's navigation or footer, the default navigation instead of a special one (for example a careers navigation), or an empty header.

Root Cause - `blocks/header/header.js` and `blocks/footer/footer.js` resolve the page with `getInheritedMetadata('nav' | 'footer')`: the page's own Navigation / Footer page property (or the metadata sheet's `nav` / `footer` column), else the nearest parent page's value up to the locale Home page, else the locale's

`references/nav` / `references/footer`. - A parent page sets a Navigation value that children inherit unintentionally. - The configured value points to another locale (an absolute path such as `/en/references/nav` on a German page). - The nav or footer page of the locale is not published: the header is left empty ("no nav page published (e.g. a new locale)"). - Values are cached per browser session (13.2).

Diagnosis Steps 1. View the page source and look for `<meta name="nav">` / `<meta name="footer">`. 2. Walk the parent pages (the page's path segments up to `/{locale}/`) and check their page properties (or open each parent and look for the meta tags). 3. Open `/{locale}/references/nav.plain.html` and `/{locale}/references/footer.plain.html`.

Resolution Steps 1. Correct or clear the Navigation / Footer property on the page or the parent that sets it; use values relative to the locale (`references/nav-careers`) or the same locale's paths. 2. Publish the locale's `references/nav` and `references/footer` pages. 3. Republish the page (and the parent whose property changed).

Validation Steps - In a new private window the page shows the expected header and footer.

Preventive Measures - Set special Navigation / Footer values on the highest page of the section that needs them, not on the Home page; document special navigation pages in the locale's `references/` folder.

13.7 404 Page in the Wrong Locale or Built-in English 404

Symptoms - A missing German URL shows the English 404 page, or the built-in "404 Page Not Found" with "Go home" instead of the authored error page. - The "Go back" button is English on every locale.

Root Cause - `scripts/scripts.js` `loadErrorPage()` tries, in order: the 404 Page (`error-page`) inherited from the missing page's nearest existing parent, `/{locale}/error-pages/404`, `/en/error-pages/404`; the first that loads is shown (content from another locale is marked with its `lang`); if none loads, the built-in content of `404.html` stays. - The locale's `error-pages/404` page is not published. - The "Go back" button and the built-in texts are hard-coded in `404.html` ("Go back", "Page Not Found", "Go home") and are not translated through placeholders.

Diagnosis Steps 1. Open `/{locale}/error-pages/404.plain.html`; a 404 response means it isn't published. 2. Check the `error-page` value of the nearest parent page. 3. Inspect `<main>`: class `error-authored` means an authored page loaded; a `lang` attribute different from the page's means a fallback locale was used.

Resolution Steps 1. Create and publish `/content/tcs-corp/{country}/{language}/error-pages/404` for every locale. 2. Fix or clear wrong 404 Page values on parent pages. 3. To translate the "Go back" label, a code change to `404.html` is needed (Not implemented as of this handover).

Validation Steps - A non-existent URL in each locale (for example `/de-de/does-not-exist`) shows that locale's authored 404 page with HTTP status 404.

Preventive Measures - Include the 404 page in the locale launch checklist (14.8.1); run the monthly 404 check (14.7).

13.8 Images Not Optimised or Too Large

Symptoms - Slow pages, large image downloads, PageSpeed warnings "Properly size images" or "Serve images in next-gen formats".

Root Cause - Images authored from AEM Assets are delivered as `/media_…` resources and optimised (responsive `<picture>` with WebP and width variants) by `createOptimizedPicture` or the platform. - Images from other origins or full URLs typed into text fields (for example image paths in metadata columns or a section background typed as an external URL) are used as they are (`decorators/responsive-image-decorator.js` `buildImage` only optimises same-origin `/media_` URLs). - Images committed to the repository (`icons/`, `fonts/`) are served as committed. - Large Lottie (`.json`) animations are not optimised.

Diagnosis Steps 1. In the Network tab filter by Img; check the URL (`/media_…?width=…&format=webply` vs an external URL) and transfer size. 2. Check where the image comes from (block field, metadata column, external URL).

Resolution Steps 1. Use AEM Assets references in image fields instead of typed URLs. 2. Compress committed SVG / image files before committing. 3. Reduce the size of Lottie files with the design team.

Validation Steps - PageSpeed Insights no longer reports the image; images load as `<picture>` with `width` variants.

Preventive Measures - Author guidance (Part 6 “Image Dimensions”); check new icons in PR review.

13.9 Page Missing from the Query Index

Symptoms - A published page does not appear in a List (child pages or tagged pages), the Filter List or the Conversation Canvas menu; it is missing in `/query-index.json` and `/sitemap.xml`.

Root Cause - The page is not published to live (only preview, or unpublished). - The page is under an excluded path in `helix-query.yaml`: `/*/references/**`, `/*/error-pages/**`, `/*/data/**`, `/*/nav`, `/*/footer`, block libraries, `site-global-templates`, `/**/fragments/**`, `/**/modals/**`. - The index was not updated after a `helix-query.yaml` change. - The row exists but the consumer filters it: List hides `hide-in-nav = yes`; the Filter List filters by Root path, Page types and Content types; rows are filtered to the current locale root.

Diagnosis Steps 1. Search the path in `https://main--tcs-corp--sradha-tcs.aem.live/query-index.json?limit=500` (use `offset` for further pages). 2. Check the page’s publication status and path against the excludes. 3. Check the row’s `pageType`, `contentType`, `tags`, `hideInNav` values.

Resolution Steps 1. Publish the page; or re-index it (Admin API index operation). 2. Move the page out of an excluded folder if it should be listed. 3. Correct its metadata (page properties or the metadata sheet) and republish.

Validation Steps - The row appears in `/query-index.json` with the expected columns; the List shows it.

Preventive Measures - Keep content pages outside `references/`, `fragments/` and `modals/` folders.

13.10 List or Filter List Empty in Universal Editor

Symptoms - In the editor the List shows “The pages show on the published page (from the query index).” or “Set up the list in its properties: …”; the Filter List shows “The results show on the published page, from the pages’ metadata.”

Root Cause - Expected behaviour: the query index is not available on the AEM author host, so lists of pages (child pages, tagged pages) and the Filter List cannot load pages in the editor (`blocks/list/README.md` Known limitations). The setup message lists missing settings (no parent / root and “Use the current page when empty” off; tags source without tags; fixed list without items).

Diagnosis Steps 1. Read the message: setup problem or “show on the published page”. 2. Open the page on preview or live.

Resolution Steps 1. Complete the settings named in the message. 2. Review the result on preview after publishing the page (and the listed pages).

Validation Steps - The published page shows the cards.

Preventive Measures - Train authors (14.8.5) that page lists render on preview / live, not in the editor; use the Cards block (typed content) when the editor preview matters.

13.11 List or Filter List Empty or Incomplete on the Published Site

Symptoms - The published List shows "There are no items to show." (or is hidden), or the Filter List shows "No resources found."; some expected pages are missing.

Root Cause - The listed pages are not published or not indexed (13.9). - Child pages: the Parent page / Child depth do not include them; Tagged pages: the Search root or the tag match (any / all) excludes them; tags on the pages differ from the List's tags. - Filter List: Page types / Content types do not match the pages' `page-type` / `content-type` values; the Root path excludes them; the Datasource is Manual with no pages. - `hide-in-nav = yes` on the pages (List). - Maximum items limits the result.

Diagnosis Steps 1. Add `?debug=list` to the URL; the console shows the configuration, warnings and cards. 2. Compare the index rows of the expected pages with the List / Filter List settings.

Resolution Steps 1. Publish missing pages; correct the metadata or the block settings; republish the page with the block.

Validation Steps - The expected cards appear; the result count (Filter List with Show result count) matches.

Preventive Measures - Maintain Page Type and Content type consistently (15.6); use the controlled values of the Filter List options.

13.12 Redirects Not Working

Symptoms - An old URL listed in `/{locale}/redirects.json` returns 404 instead of redirecting. - The site root `/` returns 404 instead of redirecting to `/en/`.

Root Cause - Edge Delivery applies only a root `/redirects.json`. The per-locale redirect sheets (`/{locale}/redirects.json`) are published but not applied by the platform (by the project's choice). In code they are used only by the List, which links cards to the redirect target (`utils/redirects.js`, up to 5 hops, loops stopped). - No root redirect sheet is mapped in `paths.json`, and `/content/tcs-corp` itself is not published, so `/` has no content.

Diagnosis Steps 1. `curl -I` the old URL on live: a 404 confirms the redirect isn't applied. 2. Confirm the row exists in the locale sheet JSON.

Resolution Steps 1. Implement redirects at the CDN (rules reading the sheets) or introduce a root redirects sheet ([To be confirmed: redirect mechanism for production — owner: site administrator]). 2. Add the `/` → `/en/` redirect in that mechanism.

Validation Steps - `curl -I` returns 301 / 302 with the expected `location`.

Preventive Measures - Decide the redirect mechanism before go-live (14.9 Known limitations).

13.13 Placeholders Showing English Text

Symptoms - Buttons, labels and messages (for example "Load more", "Next slide", "Read more") are English on a non-English locale.

Root Cause - `utils/placeholders.js` loads `/{locale}/placeholders.json` once per page and falls back to the English text in code for every missing key. Causes: - The locale's placeholders sheet is not published, or the key is missing or empty. - The columns are not named `Key` and `Text` (lower-case `key` / `text` also work). - The key differs from the code key (comparison ignores case and separators: "Load More", "load-more" and "loadMore" all match `loadMore`). - In Universal Editor, the sheet is requested at its published path (`/de-de/placeholders.json`) on the author host, where it may not be available; check preview before treating it as a defect. - Some texts are not placeholders: the built-in `404.html` texts.

Diagnosis Steps 1. Open `/{locale}/placeholders.json` on live; check status and the `Key` / `Text` values. 2. Find the key in code (`t('loadMore', 'Load more')`); the full list of about 90 keys is in 14.8.3.

Resolution Steps 1. Add or correct the row in the locale's `placeholders` sheet; publish the sheet.

Validation Steps - Reload the page; the translated label appears (new page view; no session cache).

Preventive Measures - When a release adds keys, add them to all 13 sheets (12.4.1 #8).

13.14 Tags Showing IDs or Technical Labels

Symptoms - Tags show as "cloud" or "Topics / Cloud" derived from the ID instead of the translated title, or as `tcs:topics/cloud`.

Root Cause - `utils/taxonomy.js` maps tag IDs to titles from `/{locale}/taxonomy.json`. When the taxonomy is not published, the tag is missing, or the language sheet is missing, the label is derived from the tag ID (`utils/tags.js`). - With translated titles the taxonomy is multi-sheet; the sheet named after the locale's ISO 639-1 language (for example `de`, `fr`, `ja`) is used, else the default sheet. - The `paths.json` mapping `…/taxonomy:/{code}/taxonomy.json` is missing for the locale.

Diagnosis Steps 1. Open `/{locale}/taxonomy.json`; check `:type`, `:names` and that the tag ID appears in the `tag` column with a `title`.

Resolution Steps 1. Publish the locale's taxonomy page after adding or translating tags. 2. Add the missing mapping by regenerating `paths.json` (13.4).

Validation Steps - The tag label matches the taxonomy title on reload.

Preventive Measures - Follow the tag rotation procedure (14.8.4).

13.15 Lint or PageSpeed (aem-psi-check) Failures in Pull Requests

Symptoms - The Build check fails on a PR; the PageSpeed check fails or reports a low score.

Root Cause - Lint: ESLint (Airbnb base, `.js` extensions in imports, Unix line endings) or the xwalk model rules (`xwalk/max-cells` with the per-model limits in `.eslintrc.js`, `invalid-field-name`, `no-duplicate-fields`, `no-orphan-collapsible-fields`, `no-custom-resource-types`) or Stylelint (`stylelint-config-standard` for `blocks/**/*.css` and `styles/*.css`). - PageSpeed: the PR description lacks a valid preview URL, or the page regressed (render-blocking resources, large LCP image, layout shift, heavy third-party script).

Diagnosis Steps 1. Open the failing check's log; run `npm run lint` locally. 2. Run PageSpeed Insights on the feature preview URL and read the diagnostics.

Resolution Steps 1. `npm run lint:fix` for auto-fixable issues; fix the rest manually; for `max-cells` use field collapsing / element grouping (15.4) instead of raising the limit. 2. Fix the performance regression (13.16) or correct the URL in the PR description.

Validation Steps - Both checks are green on the latest commit.

Preventive Measures - Lint locally before pushing; include a representative page URL in every PR.

13.16 Layout Shift (CLS) or LCP Regression

Symptoms - Content jumps while loading; PageSpeed or RUM shows worse LCP or CLS.

Root Cause - Eager phase: `loadEager()` decorates the page, loads the template, and loads only the first section, waiting for its first image. Heavy first sections (carousels, page lists from the query index, large images) delay LCP.

- Lazy images in the first section (List / Cards "Image loading" = lazy) can delay LCP; eager images below the fold waste bandwidth. - Fonts: `fonts.css` loads eagerly only on desktop (>= 900 px) or when the session flag is set; otherwise in the lazy phase (fallback font `inter-fallback` limits shift). - Header and footer load in the lazy phase; the header height is reserved (`--nav-height: 4rem`). - Blocks that build their content from fetched data (List, Filter List, Page Header with Authors sheet) change height after loading. - Third-party scripts added outside `delayed.js`.

Diagnosis Steps 1. PageSpeed Insights: LCP element and layout shift culprits. 2. Performance panel recording; check which block decorates last in the first section.

Resolution Steps 1. Keep the LCP image authored in the first block (Hero or Page Header) and set "Image loading" to eager only for lists in the first section. 2. Move non-essential scripts to `scripts/delayed.js`. 3. Reserve space in block CSS (aspect ratios, min-heights) where content arrives later.

Validation Steps - PageSpeed Insights mobile and desktop back to target (100 per AGENTS.md); RUM CLS < 0.1 and LCP < 2.5 s over the following days.

Preventive Measures - PSI on every PR; avoid page lists as the first block of a page.

13.17 Bulk Metadata Sheet Not Applied

Symptoms - Columns of `/{locale}/metadata.json` (for example `page-type`, `og-image`, `content-type`) do not appear as `<meta>` tags; templates and Page Header do not pick them up.

Root Cause - The platform applies bulk metadata only from sheets listed in the site configuration's `metadata.source`. Until all 13 locale sheets are listed, no bulk metadata applies. - The row's `url` pattern does not start with the locale (`/de-de/**`) or does not match. - The sheet is not published. - A page property set on the page overrides the column for that page (for example Page Type).

Diagnosis Steps 1. Read the configuration (`/config/sradha-tcs/sites/tcs-corp/metadata.json` via the Admin API). 2. Open `/{locale}/metadata.json` and check the `url` column patterns. 3. View the page source for `<meta name="{column}">`.

Resolution Steps 1. POST the full source list (12.5); fix patterns; publish the sheet; republish affected pages if their index rows must change.

Validation Steps - The page head shows the column; `/query-index.json` shows the indexed value after re-publish.

Preventive Measures - Keep the metadata source list in the configuration change log; add new locales to it at launch.

13.18 Hide Block Confusion

Symptoms - An author cannot see a block in Preview mode or on the site; a block looks faded with a dashed outline in the editor; a hidden block still appears in search engine results.

Root Cause - The Additional tab **Hide block** (`classes_hideblock`) adds the class `hideblock`; `styles/styles.css` hides the block (and its wrapper) when the page is not in Universal Editor edit mode (`html:not(.adobe-ue-edit)`), and in edit mode shows it at 45 % opacity with a dashed outline so it stays selectable. - The block is hidden with CSS only: its content is still in the published HTML.

Diagnosis Steps 1. Select the block in edit mode; check Hide block in the Additional tab.

Resolution Steps 1. Clear Hide block and republish, or delete the block if it should not be on the page.

Validation Steps - The block shows in Preview mode and on the published page.

Preventive Measures - Use Hide block only temporarily; never for confidential content (it is still in the source).

13.19 Anchor ID Not Applied or Anchor Links Not Jumping

Symptoms - A link `#our-services` does not scroll to the block; the dialog rejects the value; the id ends with `-2`.

Root Cause - The Anchor ID field (`classes_anchor`) is validated by the model: `^anchor-[a-z0-9][a-z0-9-]*$` ("Start with anchor-, then lower-case letters, digits and hyphens."). The class becomes the block's id **including** the `anchor-` prefix (for example `anchor-our-services`), so links must use `#anchor-our-services`. - `decorators/block-options-decorator.js` keeps ids unique on the page: a duplicate gets `-2`, `-3`, …; a block that already has an id (for example its own Component ID) keeps it. - Sections use a different field: the Section's **Anchor name** (any text, normalised to lower-case hyphenated, `#` optional), which is what the Anchor List links to.

Diagnosis Steps 1. Inspect the block element's `id` attribute. 2. Check for duplicates on the page.

Resolution Steps 1. Use a valid unique Anchor ID; link to `#anchor-…`; for Anchor List chapters set the section's Anchor name.

Validation Steps - Opening `page#anchor-our-services` scrolls to the block (on load, `loadLazy()` scrolls to the hash element).

Preventive Measures - Keep a list of anchors on long pages; avoid duplicates.

13.20 Data Layer Events Missing

Symptoms - Analytics shows no data; `window.adobeDataLayer` lacks components or `cmp:*` events.

Root Cause - No tag manager is loaded (`scripts/delayed.js` is empty): the data layer is filled but nothing sends it to Adobe Analytics / Customer Journey Analytics. This is expected as of this handover. - Blocks are tracked when decorated (main content in the lazy phase; header, footer and late blocks after 3 s in `loadDelayed()`). - A block with `data-analytics="off"` (List, Cards or Separator with Analytics off; Separator defaults to off) has no state and no events; blocks inside `[data-analytics]` are skipped by the generic tracking (List, Cards, Separator push their own state when on). - Clicks on `<summary>` are not pushed as clicks (accordion items push `cmp:show` / `cmp:hide`). - Muted, looping background videos without controls push no `cmp:media`. - Form submit events contain field names only (no values) by design.

Diagnosis Steps 1. In the console run `window.adobeDataLayer.filter((e) => e.event)` after interacting. 2. Check the block element for `data-cmp-id`, `data-analytics`.

Resolution Steps 1. Turn Analytics on in the block's Advanced tab where available. 2. Implement Adobe Tags with the Adobe Experience Platform Web SDK (embed code in `delayed.js`) ([To be confirmed: Adobe Tags property and reporting suite — owner: analytics owner]).

Validation Steps - The expected entries and events appear with `eventInfo.path` `component.{id}`.

Preventive Measures - Include the data layer check in post-release validation (12.4.3 #7).

13.21 Form Submit Does Nothing (No Data Received)

Symptoms - Visitors submit a Form Variations form and see "Thank you, your details were sent.", but no one receives the data.

Root Cause - There is no submit endpoint. On a valid submit `blocks/form-variations/form-variations.js` dispatches a bubbling `form-variations:submit` DOM event with `{ detail: { data } }`, resets the form and shows the success message (`formSuccess` placeholder). The data is not sent anywhere.

Diagnosis Steps 1. Network tab on submit: no request is made.

Resolution Steps 1. Implement a back end connection (Salesforce, Marketo, an API) listening to the event ([To be confirmed: form back end — owner: product owner]). 2. Until then, do not publish forms that collect leads, or change the success message in the placeholders sheet to avoid promising delivery.

Validation Steps - After the integration, a test submission reaches the target system.

Preventive Measures - Track as a go-live blocker for any form page (14.9).

13.22 Code Changes Not Visible After Merge (Code Sync)

Symptoms - A merged fix doesn't show on `main--tcs-corp--sradha-tcs.aem.page`; the feature preview shows old code.

Root Cause - AEM Code Sync didn't process the push (app not installed or permissions changed, temporary failure). - Browser cache of JS / CSS. - The change is in a file ignored by `.hlxignore` (for example `_*` partials, `docs/*`, `test/*`).

Diagnosis Steps 1. Check the Code Sync check on the merge commit in GitHub. 2. Request the changed file directly on `main--tcs-corp--sradha-tcs.aem.page/blocks/...` and compare. 3. Admin API status / log for the code operation.

Resolution Steps 1. Re-trigger the sync (Admin API code operation, or push an empty commit). 2. Ask the user to hard-reload.

Validation Steps - The file on `aem.page` matches the commit.

Preventive Measures - Monitor the Code Sync check on every merge (14.7).

13.23 Content Security Policy Blocking Scripts

Symptoms - Console: "Refused to load the script … because it violates the Content Security Policy" or "Refused to execute inline script"; a third-party tag does not run.

Root Cause - `head.html` (and `404.html`) set `script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:`. With `strict-dynamic`, only scripts carrying `nonce="aem"` and scripts created by them run; host allow-lists and `'unsafe-inline'` are ignored by modern browsers. Inline scripts or `<script src>` tags added without the nonce are blocked.

Diagnosis Steps 1. Read the CSP violation in the console (blocked URI, directive).

Resolution Steps 1. Load third-party scripts from code (`loadScript()` from `scripts/aem.js` in `scripts/delayed.js`), which `strict-dynamic` allows; or add `nonce="aem"` to a script tag in `head.html`. 2. Change the policy only together in `head.html` and `404.html`, with security review.

Validation Steps - No CSP violations in the console; the script runs.

Preventive Measures - Security review for every new third-party script.

13.24 Embedded Content Not Loading (YouTube, Vimeo, X)

Symptoms - An Embed block shows only the placeholder image or an empty frame.

Root Cause - The Embed block recognises YouTube, Vimeo and X / Twitter URLs (X loads `https://platform.twitter.com/widgets.js`); other URLs are framed as they are and may refuse to be framed (`X-Frame-Options`).

- With a placeholder image, the embed loads after the Play click; otherwise when it comes near the viewport. - Privacy / consent tools or ad blockers block the provider.

Diagnosis Steps 1. Check the URI field value and the console / network errors for the provider.

Resolution Steps 1. Use the provider's public video URL; for unsupported providers, check they allow framing.

Validation Steps - The player loads on preview.

Preventive Measures - Prefer the Video block for YouTube / Vimeo / AEM Assets videos (privacy-enhanced mode by default).

13.25 Sitemap Uses the Wrong Host

Symptoms - `/sitemap.xml` lists `https://main--tcs-corp--sradha-tcs.aem.live/...` URLs on the production domain.

Root Cause - `helix-sitemap.yaml` sets `origin: https://main--tcs-corp--sradha-tcs.aem.live` until a production domain (`cdn.prod.host`) is configured. - There is one sitemap for all locales without hreflang alternates.

Diagnosis Steps 1. Open `/sitemap.xml`; check the `<loc>` host.

Resolution Steps 1. When the production domain is configured, change `origin` to it (or remove it so the configured production host is used), merge, and regenerate the sitemap.

Validation Steps - `<loc>` entries use the production host; Search Console accepts the sitemap.

Preventive Measures - Part of the go-live checklist ([To be confirmed: production domain and CDN]).

13.26 Template (Page Type) Not Applied

Symptoms - An article page has no Page Header or "About the author"; a press release has no Media contacts; the console shows `failed to load template <name>`.

Root Cause - `scripts/scripts.js` loads `templates/{name}/{name}.js` and `.css` only for Page Types in `TEMPLATES` (`home`, `article`, `blog`, `press-release`). The Page Type comes from the page property, else the metadata sheet's `page-type` (else the `template` metadata). - The page property is Default (empty) and the metadata sheet isn't applied (13.17). - "About the author" needs `author-id` matching the Authors sheet; Media contacts need the Media Contacts sheet (filtered by the page's `region`, else all).

Diagnosis Steps 1. Check `<meta name="page-type">` and the body class. 2. Check the Authors / Media Contacts sheet JSON.

Resolution Steps 1. Set the Page Type page property or the metadata sheet; publish the data sheets.

Validation Steps - The body has the page-type class and the template content appears.

Preventive Measures - Create pages by copying the Site Global Templates (they carry the right Page Type).

13.27 Page Header or Article Author Details Missing

Symptoms - The Page Header shows no author name, image or LinkedIn link; no date or reading time.

Root Cause - The Page Header takes everything from metadata: `author-id` resolved through `/{locale}/data/authors.json` (name, designation, region, image, LinkedIn), else the `author` metadata; `publish-date`; `reading-time`; `content-type`; `featured-image` (Landing layout); `listen-link`. Missing columns or a non-matching author ID (case-insensitive) leave the part out.

Diagnosis Steps 1. Check the page head metas; open the Authors sheet JSON and find the `author-id` row.

Resolution Steps 1. Add the author to the locale's Authors sheet and publish it; set the page metadata.

Validation Steps - The lockup shows on reload.

Preventive Measures - Maintain the Authors sheet in every locale where articles are published.

13.28 Site Root Returns 404

Symptoms - `https://main--tcs-corp--sradha-tcs.aem.live/` returns the 404 page.

Root Cause - Pages outside the locales (including `/content/tcs-corp` itself) are not published; there is no redirect from `/` to `/en/` (13.12).

Diagnosis Steps 1. `curl -I /` returns 404.

Resolution Steps 1. Configure the `/` → `/en/` redirect in the chosen redirect mechanism (CDN rule).

Validation Steps - `/` redirects to `/en/`.

Preventive Measures - Go-live checklist item.

13.29 Universal Editor Reloads the Page on Every Edit or Shows Stale Rendering

Symptoms - After each change in the properties panel the whole page reloads; or a block shows the old rendering until reload.

Root Cause - `scripts/editor-support.js` re-renders only the changed block, section or default content from the editor's update (sanitised with DOMPurify). When it cannot match the update to an element, it reloads the page (`window.location.reload()`). Container sections (Gen Carousel, Tabs Container) and blocks that depend on others re-render as a whole. - Blocks that read the query index or sheets cannot show published data in the editor (13.10).

Diagnosis Steps 1. Check the console for errors during the update.

Resolution Steps 1. Reload the editor; if persistent with one block, report it to development with the block and page path.

Validation Steps - Edits apply without reload for simple blocks.

Preventive Measures - Test new blocks for in-place editing during development.

13.30 Video or Animation Not Playing Automatically

Symptoms - An autoplay or background video waits for a click; a Lottie animation shows a still frame.

Root Cause - By design, visitors with the operating system setting "reduce motion" get click-to-play videos (`blocks/video/video.js`) and paused animations (`decorators/animation-decorator.js`, WCAG 2.2.2 pause button). - Motion Types (Hero, Table List) don't run in Universal Editor or with reduced motion. - The video asset is not published (assets under `/content/dam/tcs-corp/` are published with the pages that use them; check the Publish dialog references). - The Lottie player (`/scripts/lib/lottie/lottie_light.min.js`) failed: console `failed to play the Lottie animation`.

Diagnosis Steps 1. Check the OS reduced-motion setting; the console; the asset URL status.

Resolution Steps 1. Publish the asset; check the Lottie `.json` URL; accept reduced-motion behaviour.

Validation Steps - With reduced motion off, the video autoplays (muted) and the animation plays.

Preventive Measures - Never rely on motion to convey information.

13.31 Fragment, Modal or Tab Content Not Showing

Symptoms - A Fragment block, a Modal, or a Tab with a Fragment is empty.

Root Cause - `blocks/fragment/fragment.js loadFragment()` fetches `{path}.plain.html`; it shows nothing when the path doesn't start with `/` or the fragment page isn't published (non-200 response). - Links to a path containing `/modals/` open as a modal (`autolinkModals` in `scripts.js`); the modal page must be published.

Diagnosis Steps 1. Network tab: the `.plain.html` request status.

Resolution Steps 1. Publish the fragment / modal page; select it with the content picker (not a typed URL).

Validation Steps - The content appears on reload.

Preventive Measures - Keep fragments in the locale's `references/` (or `fragments/` / `modals/`) folders and publish them before the pages that use them.

13.32 Issue Index

Area	Issues
Publishing and caching	13.2, 13.4, 13.22, 13.28
Rendering and code	13.3, 13.16, 13.23, 13.29, 13.30
Authoring and models	13.5, 13.10, 13.18, 13.19
Locale and site data	13.6, 13.7, 13.13, 13.14, 13.17, 13.26, 13.27, 13.31
Index, lists, search	13.9, 13.11, 13.25
Integrations	13.12, 13.20, 13.21, 13.24
Development process	13.15, 13.22
Images and media	13.8, 13.24, 13.30

Part 14 — Operational Runbook & AMS Handover

This runbook is for the Application Management Services (AMS) team taking over the operation of the TCS Corporate website. It defines ownership, monitoring, logs, errors, escalation, recovery, regular health checks and the standard operational procedures, lists the known limitations of the implementation, and closes with the handover checklist.

14.1 Support Model and Ownership

14.1.1 Support Levels

Level	Team	Scope	Typical actions	Hand-off criterion
L1	Author support / content operations [To be confirmed: team name — owner: AMS lead]	Authoring questions, publishing, sheets, page properties, tags, images	Guide the author with Part 6 and 15.5; check publication status; republish; fix content	Issue reproduces on correctly authored content, or concerns several pages / locales
L2	AMS / site administration [To be confirmed: team name — owner: AMS lead]	Site configuration, path mapping, metadata sources, cache, index, sitemap, Code Sync status, AEM author access	Admin API status / cache / index operations; configuration checks; log review; triage of console errors	Defect in code (JS / CSS / models) or a needed code change
L3	Development team [To be confirmed: vendor / team — owner: technical lead]	Blocks, decorators, utils, templates, models, `helix-query.yaml`, `helix-sitemap.yaml`, `head.html`	Fix, PR, merge, revert; release	Platform defect (Edge Delivery, Universal Editor, AEM as a Cloud Service)
Adobe	Adobe Support (Admin Console support case)	Edge Delivery Services platform, AEM as a Cloud Service, Universal Editor, Cloud Manager	Platform investigation	—

14.1.2 RACI

R = Responsible, A = Accountable, C = Consulted, I = Informed.

Activity	Content team	L1	L2 / AMS	L3 Development	Product owner	Adobe
Authoring and publishing pages	R	C	I	—	A	—
Maintaining sheets (placeholders, metadata, taxonomy, data, redirects)	R	C	C	I	A	—
Author onboarding and access requests	I	R	C	—	A	—
Site configuration (paths, metadata sources, CDN, robots)	I	I	R	C	A	C
Cache purge, re-index, sitemap regeneration	I	C	R	C	I	—
Code changes and releases	I	I	C	R	A	—
Rollback of code	I	I	C	R	A	—
Rollback of content	R	C	C	—	A	—
Monitoring (RUM, PageSpeed, uptime)	I	I	R	C	A	—
Platform incidents	I	I	R	C	I	R (resolution)

Activity	Content team	L1	L2 / AMS	L3 Development	Product owner	Adobe
Accessibility and SEO compliance	C	I	C	R	A	—
Analytics (data layer, future Adobe Tags)	I	I	C	R	A	—

14.1.3 Contacts and Service Levels

Item	Value
Product owner	[To be confirmed: name and contact — owner: TCS digital marketing]
Technical lead	[To be confirmed: name and contact — owner: development vendor]
AMS lead	[To be confirmed: name and contact — owner: AMS provider]
Site administrator	[To be confirmed: name and contact — owner: AMS provider]
Analytics owner	[To be confirmed: name and contact — owner: TCS marketing analytics]
Adobe account / support contact	[To be confirmed: Adobe customer success manager and support entitlement — owner: TCS IT]
Support hours	[To be confirmed: support hours and time zones — owner: AMS lead]
Severity definitions and response / resolution SLAs	[To be confirmed: SLA per severity (P1–P4) — owner: AMS contract]
Ticketing tool and queue	[To be confirmed: tool and queue names — owner: AMS lead]

Proposed severity scale (to be confirmed in the AMS contract):

Severity	Definition	Examples
P1 Critical	The live site or a locale is unavailable or seriously wrong for all visitors	All pages return errors; wrong content published site-wide; CSP blocking all scripts
P2 High	A key function or a key page is broken	Header or footer missing in a locale; a home page block not rendering; forms published with no back end
P3 Medium	A non-critical function is degraded, with a workaround	One List empty; tags showing ID labels; placeholders English in one locale
P4 Low	Cosmetic issues, questions, improvements	Spacing, wording, documentation requests

14.2 Monitoring Requirements

Area	What to monitor	Tool / source (implemented?)	Frequency	Threshold / alert
Real User Monitoring (RUM)	Core Web Vitals (LCP, CLS, INP), page views, 404s, JavaScript errors	Implemented: `scripts/aem.js` `sampleRUM()` runs on every page (sampling weight 100 = about 1 in 100 page views; `?rum=on` forces sampling; collector `https://ot.aem.live`; the RUM enhancer is loaded for selected views); `404.html` sends a `404` checkpoint with the referrer; errors and CSP violations are sent as `error` checkpoints. View the data in the Edge Delivery operational telemetry / RUM explorer ([To be confirmed: RUM do-	Daily review, weekly trend	LCP <= 2.5 s, CLS <= 0.1, INP <= 200 ms at the 75th percentile; spikes in 404 or error checkpoints

Area	What to monitor	Tool / source (implemented?)	Frequency	Threshold / alert
		main key and dashboard access — owner: site administrator])		
PageSpeed / Lighthouse	Lab performance, accessibility, best practices, SEO	PageSpeed Insights (manual); aem-psi-check on PRs	Each PR; monthly on key templates	Target 100 (AGENTS.md); investigate any score below 90
Uptime	Availability of the live host and key URLs per locale	Not implemented in the repository: [To be confirmed: uptime tool (for example synthetic checks) — owner: AMS lead]	Every 1–5 minutes	Non-200 on `/en/` or any locale Home; response time above the agreed limit
Publishing pipeline	Preview / live status of recently published paths	Admin API status and log	On incident; weekly sample	Pages published in AEM but missing on live
Code Sync	Sync of each push	GitHub check of the AEM Code Sync app on commits	Each merge	Missing or failed check on a `main` commit
Query index and sitemap	Row count, new pages present, sitemap host	`/query-index.json`, `/sitemap.xml`	Daily / weekly (14.7)	Unexpected drop in rows; pages missing
Site data	Sheets JSON reachable per locale	`/{locale}/placeholders.json`, `/{locale}/taxonomy.json`, `/{locale}/data/*.json`	Weekly	Non-200 responses
AEM author health	Availability and performance of the author environment, replication / publication queues	Cloud Manager (environment status, logs), AEM author (Sites console, workflow and publication status), Adobe status page	Daily	Author unavailable; publications stuck
Analytics	Data layer populated	Browser console checks (`window.adobeDataLayer`); reporting once Adobe Tags is implemented	Weekly	Missing page entry or `cmp:click` events
Security	CSP violations, dependency updates	RUM error checkpoints; Renovate PRs	Weekly	New CSP violations; unreviewed dependency PRs

14.3 Logs

14.3.1 Browser Console Messages Emitted by the Code

Message	Level	Source	Meaning	Action
`failed to load module for <block>`	error	`scripts/aem.js loadBlock()`	The block JS failed to load or its `decorate()` threw	13.3
`failed to load block <block>`	error	`scripts/aem.js loadBlock()`	Loading the block (CSS or JS) failed	13.3
`failed to load template <name>`	error	`scripts/scripts.js loadTemplate()`	The Page Type template JS / CSS failed	13.26
`Auto Blocking failed`	error	`scripts/scripts.js buildAutoBlocks()`	Auto blocking (currently empty) threw	L3
`failed to play the Lottie animation`	error	`decorators/animation-decorator.js`	The Lottie player or animation JSON failed	13.30

Message	Level	Source	Meaning	Action
`Editable parent not found or richtext property …`	warn	`scripts/editor-support-rte.js` (Universal Editor only)	Rich text instrumentation mismatch in the editor	L3 if editing is impaired
`Found orphan elements of a richtext, that were not consecutive siblings of …`	warn	`scripts/editor-support-rte.js` (Universal Editor only)	Rich text markup split unexpectedly	L3 if editing is impaired
`[list] config …`, `[list] cards …`, `[list] component id "…" is used twice`	debug	`blocks/list/list.js`, only with `?debug=list`	List configuration, warnings, resulting cards	Diagnosis aid (13.11)
`[separator] …`, `[separator] component id "…" is used twice`	debug	`blocks/separator/separator.js`, only with `?debug=separator`	Separator configuration and warnings	Diagnosis aid
`ping:<checkpoint>`	debug	`scripts/aem.js` RUM, sampled views	RUM data sent	None
CSP violation messages	error (browser)	Browser	A script was blocked by the policy	13.23
404 responses for `*.json` or `*.plain.html`	network	Browser network	A sheet, fragment, nav, footer or error page isn't published	13.6, 13.7, 13.13, 13.31

Tip: Enable the "Verbose" level in the console to see `console.debug` output.

14.3.2 Other Logs

Log	Where	Access	Use
Edge Delivery Admin API log	`GET https://admin.hlx.page/log/sradha-tcs/tcs-corp/main`	Site administrators (authenticated)	Who previewed, published, unpublished or synced what, and when
Admin API status	`GET https://admin.hlx.page/status/sradha-tcs/tcs-corp/main/{path}`	Authenticated	State of one path
AEM author logs	Cloud Manager → environment `e2271469` (program `p220207`) → Download logs (error, request, access logs) or log forwarding	Cloud Manager roles [To be confirmed: log forwarding to a SIEM — owner: TCS IT]	Publication errors, Universal Editor back-end errors
AEM workflow / publication status	AEM author: Tools → Workflow → Instances; Sites console publication status	AEM administrators	Stuck or failed publications
Code Sync and CI	GitHub → commit checks (AEM Code Sync, Build) and Actions logs (`.github/workflows/main.yaml`)	Repository members	Failed lint, failed sync
PageSpeed (aem-psi-check)	PR checks	Repository members	Performance regressions
RUM	Operational telemetry / RUM explorer	Domain key holders	Real-user performance and errors

14.4 Common Errors

#	Error / symptom	Likely cause	First check	Fix	Level
1	Old content on live	Not published, reference not published, cache	Admin API status; private window	Republish; purge cache	L1 / L2
2	`failed to load module for …`	Code error or unexpected content	Console stack; `.plain.html`	Fix content or code; revert	L2 / L3
3	New field missing in the dialog	`component-*.json` not rebuilt or not on `main`	`component-models.json` on `main`	`npm run build:json`, merge, reload editor	L3
4	Wrong header / footer	Inherited Navigation / Footer value	`<meta name="nav">`, parents	Fix property; publish nav / footer	L1
5	English 404 on another locale	Locale 404 page not published	`/{locale}/error-pages/404.plain.html`	Publish the 404 page	L1
6	Page missing from List / Filter List	Not published, excluded or filtered	`/query-index.json`	Publish; fix metadata or settings	L1 / L2
7	List empty in the editor	Index not available on author (expected)	Message text	Check on preview	L1
8	English UI labels	Placeholders key missing / sheet not published	`/{locale}/placeholders.json`	Add key; publish sheet	L1
9	Tag shows ID-like label	Taxonomy missing / not published	`/{locale}/taxonomy.json`	Add tag title; publish	L1
10	Bulk metadata ignored	`metadata.source` not configured / pattern wrong	Config; sheet patterns	Configure; fix patterns	L2
11	Old URL 404	Locale redirect sheets not applied	`curl -I`	CDN redirect mechanism	L2 / L3
12	Form “sent” but no data	No form back end	Network tab	Integration project	L3
13	No analytics data	No Adobe Tags loaded	`window.adobeDataLayer`; `delayed.js`	Implement Adobe Tags	L3
14	Script blocked	CSP `strict-dynamic`	Console CSP message	Load via `delayed.js`	L3
15	Build check red	Lint error	Actions log	`npm run lint:fix`; fix	L3
16	Hidden block confusion	Hide block on	Additional tab	Clear Hide block	L1
17	`#anchor` doesn't jump	Anchor ID without `anchor-` prefix in the link, or duplicate	Element id	Use `#anchor-…`	L1
18	Locale pages 404 after launch	Paths config not updated	`public.json` vs `paths.json`	Apply generated paths	L2

14.5 Escalation Matrix

From	To	When	Information to include
Author	L1	Any authoring or publishing problem the author guide does not solve	Page path (AEM and published), locale, steps, screenshot, time

From	To	When	Information to include
L1	L2	Correct content still renders wrongly; several pages or locales affected; configuration suspected	Above + publication status, Admin API status output, console messages, `.plain.html` excerpt
L2	L3	Code defect (console errors from block code, model problem, CSS defect), a code change is needed, or a configuration file in the repository must change	Above + reproduction on `aem.page`, browser / device, breakpoint, related merge or release
L2 / L3	Adobe Support	Platform behaviour (preview / publish not working for correct content, Admin API errors, Universal Editor failures, AEM author outage, Code Sync outage)	Program `p220207`, environment `e2271469`, org / site `sradha-tcs/tcs-corp`, paths, timestamps (UTC), request ids if shown
Any	Product owner	P1 / P2 incidents; decisions on rollback; content takedown	Impact, workaround, plan
Any	Security contact [To be confirmed: security incident contact — owner: TCS IT]	Suspected security incident (defacement, injected script, data exposure)	Evidence, time, affected URLs

Severity	L1 → L2	L2 → L3	L3 → Adobe	Management notification
P1	Immediately	Immediately	Immediately if platform-related	Immediately
P2	Within [To be confirmed]	Within [To be confirmed]	When platform-related	Same business day
P3	Next business day	As scheduled	As needed	Weekly report
P4	Backlog	Backlog	—	Monthly report

14.6 Recovery Procedures

Scenario	Procedure	Reference
Faulty code release	Revert the merge PR in GitHub; confirm Code Sync; verify key pages per locale	12.6.1
Wrong content published	Restore the page version in AEM; republish; purge cache if needed	12.6.2
Content must be taken down urgently	Unpublish in AEM (Manage Publication); if AEM is unavailable, Admin API unpublish (`DELETE /live/…`) for the path	12.1.9
Locale paths broken	Re-apply the previous `paths.json` to the configuration; republish affected pages	12.6.3
Bulk metadata broken	Re-POST the previous metadata source list; fix and republish the sheet	13.17
Query index wrong or incomplete	Fix `helix-query.yaml` if changed; re-index affected paths (Admin API index)	13.9
Sitemap wrong	Fix `helix-sitemap.yaml`; regenerate the sitemap	13.25
Header / footer missing site-wide	Check and republish `/{locale}/references/nav` and `/footer`; check the Navigation / Footer values on the Home pages	13.6
Placeholders / taxonomy sheet corrupted	Restore the sheet version in AEM and publish it	12.6.2
AEM author unavailable	The live site keeps serving the last published content; raise an Adobe case; freeze publishing	14.5

Scenario	Procedure	Reference
Code Sync unavailable	No code deployment possible; live site unaffected; retry later or raise an Adobe case	13.22
GitHub repository access lost	Restore access through the organisation owner; the code bus keeps serving the last sync	[To be confirmed: GitHub organisation owners — owner: TCS IT]

14.7 Health Checks

Base URL for the checks: `https://main--tcs-corp--sradha-tcs.aem.live` (replace with the production domain once configured).

14.7.1 Daily

#	Check	How	Expected
1	Home page of every locale	Open `/en/`, `/de-de/`, `/fr-fr/`, `/fr-ca/`, `/es-es/`, `/pt-pt/`, `/nl-nl/`, `/sv-se/`, `/da-dk/`, `/no-no/`, `/zh-cn/`, `/ja-jp/`, `/ko-kr/`	HTTP 200, header and footer present, no console errors
2	Uptime alerts	Uptime tool	No open alerts
3	RUM errors / 404 checkpoints	RUM explorer	No unusual spikes
4	Publication queue	AEM author publication / workflow status	No stuck items
5	Code Sync on yesterday's merges	GitHub checks	Green

14.7.2 Weekly

#	Check	How	Expected
1	Query index	`/query-index.json?limit=500` (and further `offset` pages): total row count and a sample of recently published pages	Count consistent with the previous week plus new pages; new pages present with `pageType`, `contentType`, `tags`
2	Sitemap	`/sitemap.xml`	Recently published pages present; `<loc>` host as expected
3	Sheets JSON per locale	`/{locale}/placeholders.json`, `/{locale}/taxonomy.json`, `/{locale}/metadata.json`, `/{locale}/redirects.json`, `/{locale}/data/authors.json`, `locations.json`, `partners.json`, `social-links.json`, `media-contacts.json`	HTTP 200 and valid JSON (`data` array, or `:type` multi-sheet for a translated taxonomy)
4	Data layer	On a sample page run `window.adobeDataLayer` in the console; click a link	A `page` entry, `component` entries for the blocks, a `cmp:click` event with `xdm:linkURL` and `breakpoint`
5	Dependency PRs	GitHub (Renovate)	Reviewed
6	Admin API log review	`/log/...`	Publications by expected users only

14.7.3 Monthly

#	Check	How	Expected
1	404 page per locale	Open `/{locale}/health-check-404` (any non-existent path) for all 13 locales	HTTP 404, the locale's authored 404 page (`main.error-authored`, no foreign `lang` on `main`)
2	PageSpeed	PageSpeed Insights on Home, an article, a press release, a listing page (Filter List) per main locale; mobile and desktop	Target 100; investigate below 90
3	Accessibility spot check	Keyboard-only navigation of header (desktop dropdowns, mobile menu, Escape), carousel, tabs, accordion, modal, forms; a screen reader pass on one page; automated scan (for example axe)	No new WCAG 2.1 AA failures
4	Responsive check	Key pages at 1440, 1024 and 402 px	Layout as designed
5	Placeholder completeness	Compare the keys of `/en/placeholders.json` with each locale	Same key set
6	Taxonomy completeness	Tags used on pages vs `/{locale}/taxonomy.json`	All tags have titles
7	Redirect sheets	Review entries pending a redirect mechanism	Tracked
8	Access review	AEM author users and groups (Admin Console), GitHub collaborators, Admin API roles	Leavers removed
9	Hidden blocks review	Pages with Hide block on (search the editor)	Still intended

14.8 Operational Procedures

14.8.1 Adding a Locale

1. Agree the AEM path (`/content/tcs-corp/{country}/{language}`), the published code (for example `it-it`) and the language tag (`it-IT`).
2. Development: add the entry to `LOCALES` in `utils/site.js` (`{ country: 'it', code: 'it-it', lang: 'it-IT' }`), run `npm run build:paths`, commit `utils/site.js` and `paths.json`, PR, merge.
3. Site administrator: apply the new `paths.json` to the site configuration's `public.json` `paths`; add `/{code}/metadata.json` to the `metadata.source` list.
4. Content team: create the locale page and the standard structure: `references/nav`, `references/footer`, `error-pages/404` (and 403 / 500 if used by the CDN), `block-library/`, sheets `placeholders` (all keys, 14.8.3), `metadata` (URL patterns starting with `/{code}/`), `taxonomy` (with the language sheet if translated), `redirects`, and `data/authors`, `data/locations`, `data/partners`, `data/social-links`, `data/media-contacts`.
5. Publish the sheets, the nav and footer pages and the 404 page, then the Home page and content.
6. Verify: Home at `/{code}/`, header, footer, 404 in the locale, `<html lang>` = the language tag, placeholders translated, `/query-index.json` rows under `/{code}/`.
7. Add the locale to the monitoring and health checks.

> **Note:** If the redirect mechanism (13.12) and the sitemap strategy (one sitemap, no hreflang) are extended later, include the new locale there too.

14.8.2 Adding a Block

1. Development: create `blocks/{name}/{name}.js`, `{name}.css` and `_{name}.json` (definition, model with the Additional tab copied from an existing block, filter for items), following AGENTS.md (shared decorators and utils, rem units, placeholders for UI text, data layer through `data-cmp-*` attributes).
2. Add the block id to the `section` filter in `models/_section.json` (or a container filter).
3. Run `npm run build:json`, `npm run lint`, `npm test`.
4. Add a test page in the `block-library` of `us/en` (and the other locales as needed).
5. PR with the feature preview link; merge; authors reload Universal Editor.
6. Update the documentation (Part 6 chapter, component inventory 15.8).

14.8.3 Updating Placeholders

1. Open `/content/tcs-corp/{country}/{language}/placeholders` in AEM.
2. Edit the `Text` of the key (columns: Key, Text, Description, Category; code reads Key and Text; keys match without case or separators). Keep `{n}`, `{total}`, `{title}`, `{source}`, `{platform}`, `{label}`, `{text}`, `{problems}`, `{value}` variables.
3. Publish the sheet; check `/{locale}/placeholders.json` and the page.

Keys used by the code (English fallbacks in code):

Key	English fallback	Key	English fallback
aboutTheAuthor	About the author	noPagesFound	No pages found
advancedFilters	Advanced filters	noResources	No resources found.
all	All	noResourcesHint	Try selecting another filter.
back	Back	onThisPage	On this page
cards	Cards	open	Open
cardsEditorHint	Add a Card to the block and give it a title, a description or an image.	openNavigation	Open navigation
carousel	Carousel	opensInNewTab	(opens in a new tab)
carouselRole	carousel	pauseAnimation	Pause animation
clearAll	Clear all	pauseSlides	Pause slides
close	Close	pauseVideo	Pause video
closeNavigation	Close navigation	play	Play
contentFrom	Content from {source}	playAnimation	Play animation
dialog	Dialog	playSlides	Play slides
download	Download	playVideo	Play video
downloadThisPage	Download this page	playVideoTitle	Play video: {title}
explore	Explore	previousSlide	Previous slide
exploreItem	Explore {title}	published	Published
fieldEmail	Enter a valid email address.	readMore	Read more
fieldFormat	Enter a valid value.	readTheTranscript	Read the transcript
fieldRequired	This field is required.	requiredNote	* Required fields

Key	English fallback	Key	English fallback
fieldTooLong	This text is too long.	resetFilters	Reset filters
filterListEditor	The results show on the published page, from the pages' metadata.	resultCount	{n} results
filters	Filters	resultCountOne	1 result
followUsOn	Follow us on {platform}	searchByVoice	Search by voice
footerLinks	Footer links	searchPlaceholder	Ask us a question
formSuccess	Thank you, your details were sent.	selectAnOption	Select an option
itemLinkLabel	{text}: {title}	selectedCount	{n} selected
linkCopied	Link copied to clipboard	send	Send
list	List	separatorHidden	Separator hidden on all devices
listEditorHint	Set up the list in its properties: {problems}.	separatorInvisible	Semantic separator without a line: screen readers announce a break nobody sees
listEditorPages	The pages show on the published page (from the query index).	shareThisPage	Share this page
listEmpty	There are no items to show.	showSlide	Show slide {n} of {total}
listNext	Next items	showSubmenu	Show {label} menu
listPrevious	Previous items	showingCount	Showing {n} of {total}
listenToThisPage	Listen to this page	shownWhenValue	Shown when a dropdown value is "{value}"
loadMore	Load more	siteNavigation	Site
mainNavigation	Main navigation	slideOf	{n} of {total}
mediaContacts	Media contacts	slideRole	slide
menu	Menu	stopVideo	Stop video
minRead	{n} min read	submit	Submit
minutesShort	{n}m	suggestions	Suggestions
nextSlide	Next slide	tabNumber	Tab {n}
table	Table	tabs	Tabs
tabsBlockEmpty	Add a Tab, then choose the Fragment it shows in the Tab properties.	tabsEmpty	Add a Tab, then the components of the tab after it.
video	Video	viewAll	View all
viewLinkedinProfile	View LinkedIn profile		

Note: The list was extracted from the `t('key', 'fallback')` calls in `blocks/`, `decorators/`, `templates/` and `scripts/` (91 keys). Editor-only messages (for example `listEditorHint`, `tabsEmpty`, `separatorHidden`) are shown to authors in Universal Editor.

14.8.4 Rotating or Changing the Tag Taxonomy

1. Create or change the tags in the AEM tag namespace (Tools → Tagging). Tag IDs have the form `namespace:path` (for example `tcs:topics/cloud`).
2. Update the taxonomy page of every locale (`/content/tcs-corp/{country}/{language}/taxonomy`) so each tag ID has a title (and a translated title on the language sheet named after the ISO 639-1 language, for example `de`).
3. Publish the taxonomy pages; verify `/{locale}/taxonomy.json`.
4. Re-tag pages (page properties) and blocks with tag fields (Teaser, Cards, List Items, List, Filter List primary tags and advanced categories) that used retired tags; republish the pages so the index (`tags` column) updates.
5. Check the Filter List pills and advanced filter categories (they read tags below taxonomy roots).
6. Retire old tags only after no published page uses them (search the query index `tags` column).

> **Warning:** Renaming a tag ID changes the IDs stored on pages and in the query index; Lists with tag sources and Filter List filters stop matching until pages are re-tagged and republished.

14.8.5 Onboarding Authors

1. Request access: Adobe Admin Console user with the AEM author product profile for program `p220207` / environment `e2271469` ([To be confirmed: AEM groups for authors, publishers and locale owners — owner: AEM administrator]).
2. Grant locale permissions (which `/content/tcs-corp/{country}/{language}` trees the author may edit and publish) [To be confirmed: permission model].
3. Install AEM Sidekick in the browser (optional, for viewing preview / live and opening the editor).
4. Training sessions: Universal Editor basics; page properties (Title, Description, Keywords, Page Type, Navigation, Footer, 404 Page); Additional tab (Anchor ID, Hide block, spacing); sheets; tags; publishing and verification; the block library of the locale; this guide's Part 6 and cheat sheets (15.5–15.7).
5. Give the author the support route (L1) and the definition of done for a publication (12.4.3).

14.9 Known Limitations

#	Limitation	Impact	Source	Workaround / next step
1	Per-locale redirect sheets (`/{locale}/redirects.json`) are not applied by the platform; only a root `/redirects.json` is	Old URLs return 404; `/` doesn't redirect to `/en/`	AGENTS.md, `utils/redirects.js`	CDN rules or a root redirects sheet [To be confirmed]
2	No tag manager loaded (`scripts/delayed.js` is empty)	Data layer populated but no analytics collected; Adobe Analytics / CJA, Target, Journey Optimizer not implemented	`scripts/delayed.js`	Add the Adobe Tags embed (Web SDK) in `delayed.js`
3	No form back end	Form Variations shows success but the data goes nowhere (`form-variations:submit` event only)	`blocks/form-variations/form-variations.js`	Integration [To be confirmed: form back end]
4	No search service	Conversation Canvas search dispatches `conversation-canvas:search` only; no results page	`blocks/conversation-canvas/conversation-canvas.js`	Search integration [To be confirmed]

#	Limitation	Impact	Source	Workaround / next step
5	Content Fragment lists not supported (by decision)	Lists are built from pages only	`blocks/list/README.md`	Use pages with metadata, or Cards
6	Page lists come from the query index	Unpublished / excluded pages not listed; manual order = path order; no full-text search; empty in Universal Editor	`blocks/list/README.md`	Fixed list order or Cards
7	Legacy Cards blocks are not key-value blocks	Cards created before the new version use default settings; to use block settings, add a new Cards block and move the cards	`blocks/cards/README.md`	Recreate the block
8	Cards: no author lockup on the feature card; internal links not resolved through redirect sheets; text fields edited in the properties panel (description in place)	Minor presentation differences from List	`blocks/cards/README.md`	Use List for pages
9	List: statistic cards' smaller units ("bn") and Figma's two-tone title (Variation 6) are not styled separately	Design deviation	`blocks/list/README.md`	Single-colour titles
10	Separator: site tokens until the Figma designs are applied; no brand-icon decoration (dot, square, diamond only); no free-form classes or colours; Accessible label not read by every screen reader	Design deviation	`blocks/separator/README.md`	Design follow-up
11	Interactive Dashboard is a placeholder	Renders nothing on the site (only a labelled placeholder in the editor); only the Additional tab exists	`blocks/interactive-dashboard/interactive-dashboard.js`	Future development
12	Built-in 404 texts ("Page Not Found", "Go home", "Go back") are English, not placeholders	Untranslated fallback 404 and the Go back button	`404.html`	Author 404 pages per locale; code change for "Go back"
13	One sitemap for all locales, no hreflang alternates; origin is the `.aem.live` host	SEO for multilingual pages	`helix-sitemap.yaml`	Configure per-locale sitemaps and production origin
14	Inherited Navigation / Footer / 404 Page values cached per browser session	Changes to a parent's property show in a new session	`utils/inherited-metadata.js`	Communicate; new session
15	Bulk metadata only applies once all locale sheets are listed in `metadata.source`	Metadata sheet columns ignored until configured	AGENTS.md	Configure (12.5)
16	Locations and Partners data sheets have loaders (`getLocations`, `getPartners`) but no block uses them	Sheet content has no effect yet	`utils/data.js`	Future components
17	`persona` and `campaign-id` metadata are indexed and readable but not used by any block	No personalisation by persona	`helix-query.yaml`, `utils/metadata.js`	Future personalisation; Teaser's Personalization flag only adds a marker class / data attribute
18	Sheets and the query index are requested at published paths; in Universal Editor they may not resolve	Editor shows English fallbacks, ID-derived tags, empty page lists	`utils/site.js` `localeSheetPath`, `utils/query-index.js`	Validate on preview

#	Limitation	Impact	Source	Workaround / next step
19	Unit tests are not run in CI	Regressions in List / Cards / Separator logic caught only if run locally	`.github/workflows/main.yaml`	Add `npm test` to the workflow
20	Production domain, CDN, robots.txt not configured	Site only on `*.aem.live`	Project facts	[To be confirmed: production domain and CDN]
21	Boilerplate leftovers: Sidekick project name "AEM XWalk Boilerplate", PR template placeholders `{repo}` / `{owner}`, `README.md` placeholders, `package.json` name / version	Cosmetic, can confuse	Repository	Clean up in a code change
22	403 and 500 error pages are authored for the CDN but not used by code	Need CDN rules to be served	AGENTS.md, `scripts/scripts.js`	CDN configuration

14.10 AMS Handover Checklist

14.10.1 Access

#	Item	Owner	Status
1	Adobe Admin Console: AEM as a Cloud Service author access (program `p220207`, environment `e2271469`) for AMS users	TCS IT	[ ]
2	Cloud Manager access (logs, environment status)	TCS IT	[ ]
3	GitHub `sradha-tcs/tcs-corp`: read / write for L3, read for L2	Repository administrator	[ ]
4	Edge Delivery site configuration / Admin API role (admin or publish) for L2	Site administrator	[ ]
5	RUM / operational telemetry dashboard access (domain key)	Site administrator	[ ]
6	Uptime monitoring tool	AMS lead	[ ]
7	Production CDN console [To be confirmed]	TCS IT	[ ]
8	Ticketing queue and escalation contacts	AMS lead	[ ]

14.10.2 Knowledge Transfer Sessions

#	Session	Audience	Content	Duration	Status
1	Architecture overview	L2, L3	Edge Delivery Services with Universal Editor, three-phase loading, code bus, content bus, configuration service	[To be confirmed]	[ ]
2	Locale model and site data	L1, L2	13 locales, path mapping, references, sheets, inheritance of Navigation / Footer / 404	[To be confirmed]	[ ]
3	Authoring and publishing	L1	Universal Editor, page properties, Additional tab, publishing, verification	[To be confirmed]	[ ]
4	Components walkthrough	L1, L3	Part 6 components and their variations, block library	[To be confirmed]	[ ]
5	Lists, index and metadata	L1, L2	Query index, List, Filter List, metadata sheet, taxonomy	[To be confirmed]	[ ]

#	Session	Audience	Content	Duration	Status
6	Development workflow	L3	Branching, PRs, checks, models and `build:json`, `build:paths`, tests	[To be confirmed]	[ ]
7	Operations	L2	Admin API operations, cache, re-index, sitemap, logs, health checks, recovery	[To be confirmed]	[ ]
8	Analytics and integrations	L3, analytics owner	Data layer schema and events, Adobe Tags readiness, form and search events	[To be confirmed]	[ ]
9	Shadowing period	L1, L2	AMS handles tickets with the development team on call	[To be confirmed]	[ ]

14.10.3 Documentation

#	Document	Location	Status
1	This handover document (version 1.0)	[To be confirmed: document repository]	[ ]
2	Project instructions	`AGENTS.md`	Available
3	Shared code references	`decorators/README.md`, `utils/README.md`, `templates/README.md`	Available
4	Block references	`blocks/list/README.md`, `blocks/cards/README.md`, `blocks/separator/README.md`	Available
5	Earlier guides	`docs/handover/AUTHOR-GUIDE.pdf`, `DEVELOPER-GUIDE.pdf`, `ADMIN-GUIDE.pdf` (11 October 2026; not served, `docs/` is in `.hlxignore`)	Available (superseded by this document where they differ)
6	Screenshots for this document	`docs/handover/screenshots/` (see 15.9)	[ ]
7	Site configuration export (paths, metadata sources, CDN)	[To be confirmed]	[ ]
8	Runbook contacts and SLAs	14.1.3	[ ]

14.10.4 Open Items at Handover

#	Open item	Owner	Target date
1	Production domain, CDN and push invalidation	[To be confirmed]	[To be confirmed]
2	Redirect mechanism for the per-locale sheets and `/` → `/en/`	[To be confirmed]	[To be confirmed]
3	Bulk metadata sources configured for all 13 locales	Site administrator	[To be confirmed]
4	Adobe Tags / Web SDK implementation and analytics reporting	Analytics owner	[To be confirmed]
5	Form back end	Product owner	[To be confirmed]
6	Search service for the Conversation Canvas	Product owner	[To be confirmed]
7	Interactive Dashboard implementation	Product owner	[To be confirmed]
8	Sitemap origin and multilingual sitemap strategy	Site administrator	[To be confirmed]
9	Unit tests in CI; boilerplate clean-up (Sidekick label, PR template, README)	Technical lead	[To be confirmed]
10	Support model, SLAs and contacts (14.1.3)	AMS lead	[To be confirmed]
11	Release tagging convention	Technical lead	[To be confirmed]

#	Open item	Owner	Target date
12	Uptime monitoring set-up	AMS lead	[To be confirmed]

Part 15 — Appendix

15.1 Glossary

Term	Definition (as used on this project)
Additional tab	The last tab of every block dialog except Header and Footer: Anchor ID, Hide block, Padding left / right / top / bottom, Margin left / right / top / bottom (`classes_*` fields applied by `decorators/block-options-decorator.js` and `styles/styles.css`)
Adobe Client Data Layer	The `window.adobeDataLayer` array the site fills with page and component state and `cmp:*` events (AEM Core Components schema) for Adobe Tags and the Web SDK
Admin API	The Edge Delivery Services HTTP API at `admin.hlx.page` for preview, publish, cache, index, sitemap, code, status, log and configuration operations
AEM Code Sync	GitHub app that copies every pushed commit of the repository to the Edge Delivery code bus
AEM Sidekick	Browser extension for Edge Delivery URLs: environment switch, status, open in editor
Anchor ID	Additional tab field; a value `anchor-…` becomes the block's HTML id for # links
Anchor name	Section property; the id of a section, used by the Anchor List
Block	A reusable component of a page (`blocks/{name}/`), decorated by its JavaScript from authored rows and cells
Block library	Author-only pages per locale (`block-library/`) showing every block variation; excluded from publishing
Bulk metadata	Metadata set for many pages at once by URL pattern in the locale's `metadata` sheet
Code bus	Edge Delivery storage of the code of each branch
Collapsed field	A model field whose name ends with Alt, Title, Type, Text or MimeType and is rendered into the same element as its base field (for example `imageAlt` becomes the image's `alt`)
Component definition	Entry in `component-definition.json` that makes a component insertable in Universal Editor
Component filter	Entry in `component-filters.json` listing which components a container (section, block) accepts
Component model	Entry in `component-models.json` defining the dialog fields of a component
Container block	A block with child items (Accordion, Cards, Columns, Conversation Canvas, Form Variations, List, Table, Table List, Tabs block)
Container section	A section-based component that wraps the section content in a synthetic block (Gen Carousel, Tabs Container)
Content bus	Edge Delivery storage of the previewed and published content
Decorator	A shared module in `decorators/` that creates or decorates page elements
Default content	Text, Title, Image and Button authored directly in a section, not in a block
Delayed phase	The third loading phase (3 s after load): data layer tracking of late blocks and `scripts/delayed.js` (empty; the place for Adobe Tags)
Eager phase	The first loading phase: decorate the page, load the template and the first section to reach LCP
Element grouping	Model fields sharing a prefix before _ (for example `content_title`, `content_description`) rendered into one cell
Fragment	A page whose content is included in other pages (Fragment block, nav, footer, modals, tab fragments)

Term	Definition (as used on this project)
Inherited metadata	Navigation, Footer and 404 Page values taken from the nearest parent page up to Home (`utils/inherited-metadata.js`)
Key-value block	A block whose model is rendered as rows of property name and value (template `"key-value": true`)
Lazy phase	The second loading phase: the remaining sections, the header and footer, lazy styles and fonts
Locale	One of the 13 country / language sites (`/content/tcs-corp/{country}/{language}`, published at `/{code}/`)
Lotto / Lottie	The "Lotto" image type plays a Lottie animation (`.json`) with the vendored player `scripts/lib/lottie/lottie_light.min.js`
Page Type	Page property (or `page-type` column) selecting the template: Default, Home, Article, Blog, Press Release
Placeholder	A UI text from the locale's placeholders sheet (`Key`, `Text`), with an English fallback in code
Preview tier	Edge Delivery content state visible on `*.aem.page`
Live tier	Edge Delivery content state visible on `*.aem.live` and the production domain
Query index	`/query-index.json`, the published pages and their metadata columns defined in `helix-query.yaml`
References	The locale folder `references/` holding nav, footer and other fragments
Section	A horizontal part of a page holding default content and blocks; has its own properties (Style, Width Type, Rounded edge, Background Image, Anchor name)
Sheet	An AEM spreadsheet page published as JSON (placeholders, metadata, redirects, data sheets)
Site configuration	The Edge Delivery configuration service entry for `sradha-tcs/tcs-corp` (paths, metadata sources, CDN, access)
Site Global Templates	Author-only template pages (`/content/tcs-corp/site-global-templates/`) that authors copy
Taxonomy	The locale's `taxonomy` page mapping tag IDs to titles (`/{code}/taxonomy.json`)
Template	Code in `templates/{name}/` loaded for a Page Type (home, article, blog, press-release)
Universal Editor	The AEM visual editor used to author the site in context
Util	A shared module in `utils/` that builds no UI
xwalk	The AEM authoring (Universal Editor) flavour of Edge Delivery Services ("crosswalk"), based on `aem-boilerplate-xwalk`

15.2 Acronyms

Acronym	Meaning
ACDL	Adobe Client Data Layer
AEM	Adobe Experience Manager
AMS	Application Management Services
API	Application Programming Interface
CDN	Content Delivery Network
CI	Continuous Integration

Acronym	Meaning
CJA	Customer Journey Analytics
CLS	Cumulative Layout Shift
CMS	Content Management System
CSP	Content Security Policy
CSS	Cascading Style Sheets
CTA	Call to Action
DAM	Digital Asset Management (AEM Assets)
DOM	Document Object Model
EDS	Edge Delivery Services
IMS	Adobe Identity Management System
INP	Interaction to Next Paint
ISO	International Organization for Standardization (language codes ISO 639-1)
JCR	Java Content Repository
JSON	JavaScript Object Notation
L1 / L2 / L3	Support levels 1, 2 and 3
LCP	Largest Contentful Paint
OG	Open Graph (social sharing metadata)
PR	Pull Request
PSI	PageSpeed Insights
RACI	Responsible, Accountable, Consulted, Informed
RTE	Rich Text Editor
RUM	Real Use Monitoring (Edge Delivery operational telemetry)
SEO	Search Engine Optimisation
SLA	Service Level Agreement
UE	Universal Editor
URL	Uniform Resource Locator
WCAG	Web Content Accessibility Guidelines
WebVTT	Web Video Text Tracks (caption file format)
XDM	Experience Data Model (Adobe Experience Platform)

15.3 Reference Documents

15.3.1 Adobe Documentation

Topic	URL
Edge Delivery Services documentation	`https://www.aem.live/docs/`
Developer tutorial	`https://www.aem.live/developer/tutorial`
Universal Editor tutorial (AEM authoring)	`https://www.aem.live/developer/ue-tutorial`
AEM authoring with Edge Delivery Services	`https://www.aem.live/docs/aem-authoring`
Markup, sections and blocks	`https://www.aem.live/developer/markup-sections-blocks`
Markup reference	`https://www.aem.live/developer/markup-reference`
Component model definitions	`https://www.aem.live/developer/component-model-definitions`
Authoring path mapping	`https://www.aem.live/developer/authoring-path-mapping`
Keeping it 100 (performance)	`https://www.aem.live/developer/keeping-it-100`
Anatomy of a project	`https://www.aem.live/developer/anatomy-of-a-project`
David's Model (content modelling)	`https://www.aem.live/docs/davidsmodel`
Taxonomy	`https://www.aem.live/docs/authoring-taxonomy`
Block Collection	`https://www.aem.live/developer/block-collection`
Content modeling for AEM authoring (field collapse, element grouping)	`https://experienceleague.adobe.com/en/docs/experience-manager-cloud-service/content/edge-delivery/aem-authoring/content-modeling`
Universal Editor documentation	`https://experienceleague.adobe.com/en/docs/experience-manager-cloud-service/content/implementing/developing/universal-editor/introduction`
xwalk lint rules	`https://github.com/adobe-rnd/eslint-plugin-xwalk`
Boilerplate	`https://github.com/adobe-rnd/aem-boilerplate-xwalk`
Documentation search	`curl -s https://www.aem.live/docpages-index.json` (filter by keyword)

15.3.2 Repository Documents

Document	Content
`AGENTS.md`	Project instructions: structure, code style (rem units, breakpoints), Additional tab, data layer, accessibility, shared code, site structure and data, publishing process
`decorators/README.md`	Decorator modules, exports, users; data layer schema
`utils/README.md`	Util modules, exports, users; site data
`templates/README.md`	Site Global Templates and Page Types; adding a template
`blocks/list/README.md`	List: sources, field matrix, data flow, metadata and index contracts, Figma variations, accessibility, analytics, limitations
`blocks/cards/README.md`	Cards: manual cards, relation to List, item fields, backward compatibility, limitations
`blocks/separator/README.md`	Separator: variants, settings, accessibility, limitations

Document	Content
`docs/handover/*.pdf`	Earlier author, developer and admin guides (11 October 2026)
`helix-query.yaml`, `helix-sitemap.yaml`, `paths.json`, `fstab.yaml`, `head.html`, `.hlxignore`, `.eslintrc.js`	Index, sitemap, path mapping, content source, head and CSP, served-file exclusions, lint rules

15.4 Content Modeling Standards

15.4.1 Principles

1. The authored structure (rows and cells) is the contract between author and code. Change it only with backward compatibility (several blocks read rows made before their current model: Cards, List, Video, Teaser).
2. Prefer default content (Text, Title, Image, Button) where a block isn't needed.
3. Keep blocks semantic and few-celled: combine fields with **field collapsing** and **element grouping** instead of adding cells.
4. Use **key-value** models for settings-heavy blocks (14 blocks: anchor-list, cards, conversation-canvas, filter-list, list, page-header, quote, separator, tab, table-list, tabs, teaser, testimonial, video).
5. Use **`classes`** fields (select / multiselect, or `classes_*` names) for presentation options: they arrive as block classes and never shift content rows.
6. Never hard-code user-facing strings: use placeholders with an English fallback.
7. Page-level data belongs in page properties or the metadata sheet, not in blocks; page-specific values that aren't metadata (CTAs, overrides) are block fields.

15.4.2 xwalk Rendering Rules Used in the Models

Rule	Naming	Effect	Examples in this project
Block options	Field named `classes` (select or multiselect), or `classes_<name>`	Values become CSS classes on the block	Teaser Layout `classes` (no-image, background, card); Hero `classes_alignment`, `classes_background`, `classes_motion`; every Additional tab field (`classes_anchor`, `classes_hideblock`, `classes_paddingleft`, …)
Field collapse, `Alt`	`<field>Alt`	Image `alt` of `<field>`	`imageAlt`, `authorImageAlt`, `content_iconAlt`, `placeholder_imageAlt`
Field collapse, `Text`	`<field>Text`	Link text of `<field>`	`linkText` (Button, Modal, List Item, Card)
Field collapse, `Title`	`<field>Title`	Link `title` of `<field>`	`linkTitle` (Button)
Field collapse, `Type`	`<field>Type`	Heading level of a title field, or link type (primary / secondary)	`titleType`, `content_titleType`, `listHeadingType`, `linkType`
Field collapse, `MimeType`	`<field>MimeType`	Mime type hint for a reference	Not used in this project
Element grouping	`<group>_<field>`	All fields of the group rendered in one cell	Hero `content_eyebrow`, `content_title`, `content_titleType`, `content_description`, `content_icon`; Embed `embed_placeholder`, `embed_uri`; Video `placeholder_image`

Rule	Naming	Effect	Examples in this project
Composite multifield	`container` with `multi: true` (enabled in `xwalk.json`)	Repeated groups of sub-fields	Hero and Page Header CTA List, Teaser Links, Testimonial / Teaser Stats, Footer Link List, Anchor List Links, Filter List Primary filter tags and Advanced filter categories
Conditional fields	`condition` (JsonLogic)	Field shown only when the condition holds	List source-specific fields; Cards / List layout fields per mode; Teaser single vs multiple links
Validation	`validation` (`regExp`, `numberMin`, `numberMax`, `maxLength`, `customErrorMsg`), `required`	Enforced in the dialog	Anchor ID regExp; Component ID regExp; Table List Maximum items 1–10; Card Analytics label max 120; required Video, List source, Tab title, form field names and labels
Shared field groups	`"...": "../../models/_card-list.json#/fields/<group>"`	One definition reused by several models (resolved by `build:json`)	`models/_card-list.json` groups `header`, `cards`, `layout`, `scroll`, `advanced`, used by List and Cards

15.4.3 Lint Rules for Models (`npm run lint`)

`.eslintrc.js` extends `plugin:xwalk/recommended`:

Rule	Checks
`xwalk/max-cells`	Number of cells a non-key-value model renders. Limits configured: default `*` = 4; `section` 6; `tabs` 8; `table-list` 10; `table-list-card` 6; `list-item` 13; `card` 16; `table-col-5` 5; `field-text` 6; `field-select` 6; `field-textarea` 6; `field-checkbox` 5; `contact-card` 5
`xwalk/invalid-field-name`	Names start with a letter; letters, digits, `_`, `-`, `:` only; no spaces; at most one `_` and one `:`
`xwalk/no-duplicate-fields`	No duplicate field names in a model
`xwalk/no-orphan-collapsible-fields`	A suffixed field (…`Alt`, …`Text`, …`Title`, …`Type`, …`MimeType`) needs its base field
`xwalk/no-custom-resource-types`	Only the standard `core/franklin/components/...` resource types

The `.eslintrc.js` comment explains the higher limits: section properties render as section-metadata rows; the Table List settings and the form field, table and card items need one cell per value.

15.4.4 Build and Commit Rules

Rule	Detail
Edit partials only	`blocks/{name}/_{name}.json` and `models/_*.json`; never edit `component-models.json`, `component-definition.json`, `component-filters.json` by hand
Rebuild	`npm run build:json` (`merge-json-cli` from `models/_component-models.json`, `models/_component-definition.json`, `models/_component-filters.json`, which include `../blocks/*/_*.json`)
Pre-commit	Husky runs `build:json` automatically when a staged file matches `_*.json` and stages the three aggregated files
Filters	Add new blocks to the `section` filter (`models/_section.json`); item components to their container's filter
Additional tab	Copy the Additional tab from an existing block model into every new block model (not Header / Footer)

Rule	Detail
Served files	Partials start with _ and are not served (`.hlxignore _*`)

15.4.5 Component Groups in the Editor

Group (`component-definition.json`)	Components
Default Content	Text, Title, Image, Button
Sections	Section
Blocks	All block definitions from `blocks/*/_*.json` (including the container sections Gen Carousel and Tabs Container)

Top-level filter `main`: Section, Gen Carousel, Tabs Container. Section filter: text, image, button, title, page-header, hero, teaser, quote, testimonial, anchor-list, accordion, tabs-block, table, table-list, list, separator, embed, video, modal, form-variations, filter-list, interactive-dashboard, conversation-canvas, cards, columns, fragment, footer.

15.5 Authoring Cheat Sheets

15.5.1 Create a Page

Step	Action
1	In the Sites console go to `/content/tcs-corp/site-global-templates/` and copy the template page you need (home, article, blog, press release)
2	Paste it into the target locale folder (for example `/content/tcs-corp/de/de-de/insights/`)
3	Rename the page (the name becomes the URL segment: lower case, hyphens)
4	Open page properties: set Title, Description, Keywords; check Page Type; leave Navigation / Footer / 404 Page empty unless the page needs its own
5	Edit the content in Universal Editor; publish (15.5.3)

15.5.2 Set Page Properties

Property	Use	Inherited by child pages
Title	Page title (`og:title`, card title fallback)	No
Description	Meta description (card description fallback)	No
Keywords	Meta keywords (multi)	No
Page Type	Template: Default, Home, Article, Blog, Press Release (overrides the metadata sheet)	No
Navigation	Header page; empty = inherit, else `references/nav`	Yes
Footer	Footer page; empty = inherit, else `references/footer`	Yes
404 Page	Error page for missing pages below this page; empty = inherit, else `error-pages/404`	Yes

15.5.3 Publish

Task	Action	Verify
One page	Universal Editor → Publish (include changed references)	Live URL in a private window
Several pages / a branch	Sites console → Manage Publication (include children, schedule)	Sample URLs, `/query-index.json`
A sheet	Sites console → select the sheet → Quick Publish	`/{locale}/{sheet}.json`
Take down	Manage Publication → Unpublish	URL shows the 404 page
Undo a change	Timeline → Revert to version → Publish	Live URL

15.5.4 Additional Tab (every block except Header and Footer)

Field	Values	Effect
Anchor ID	`anchor-` + lower-case letters, digits, hyphens (enforced)	Block id for `#anchor-...` links (unique: `-2`, `-3` added)
Hide block	On / off	Hidden on the site and in Preview; faded with a dashed outline in edit mode; still in the HTML
Padding left / right / top / bottom	Empty, 0, 100, 200, 300, 400, 500, 600, 800, 1000, 1200, 1600, 2000, 2400, 3000	`--space-*` token: 0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 7.5 rem
Margin left / right / top / bottom	Same values	Same tokens as margin

15.5.5 Link to Part of a Page

Target	Set on the target	Link to
A section	Section properties → Anchor name (for example "Our services"; normalised to `our-services`)	`#our-services`; or add it to an Anchor List
A block	Additional tab → Anchor ID `anchor-our-services`	`#anchor-our-services`
A tab	Tab → ID	`#<id>` opens that tab (deep link)

15.5.6 Sheets and Data

Task	Sheet (per locale)	Columns
Translate a UI label	`placeholders`	Key, Text (Description, Category for authors)
Set metadata for many pages	`metadata`	`url` pattern (starting with `/{code}/`) and metadata columns (15.6)
Add or translate a tag title	`taxonomy`	`tag`, `title`; a sheet per language for translations
Add an author	`data/authors`	author-id, name, designation, bio, image, linkedin, region
Add a social link	`data/social-links`	platform, url, icon, sort-order
Add a media contact	`data/media-contacts`	id, name, designation, email, phone, region
Add a location / partner	`data/locations`, `data/partners`	location-id, location-name, country, region, latitude, longitude / partner-id, partner-name, logo, website, tier, description (not used by any block yet)
Record a redirect	`redirects`	Source, Destination (not applied by the platform; used for List card links)

15.5.7 Lists of Content

Need	Use	Key settings
Cards of child pages	List	Source: Child pages; Parent page (or current page); Direct children only / Child depth
Cards of tagged pages	List	Source: Tagged pages; Search root; Tags; Match any / all
Hand-picked pages	List	Source: Fixed list; add List Items (Page or link, overrides)
Curated cards that aren't pages (statistics, services)	Cards	Card items with typed content; Card style
Filterable resource library	Filter List	Datasource; Root path; Page types; Content types; Primary filter tags; Advanced filter categories
Numbered table of industries / solutions / products	Table List	Variation; Table List Items

15.5.8 Navigation, Footer and Error Pages

Task	Action
Change the main navigation of a locale	Edit and publish `/content/tcs-corp/{country}/{language}/references/nav` (sections: brand, sections, tools)
Special navigation for a section of the site	Create `references/nav-<name>`, set it as Navigation on the section's top page, publish both
Change the footer	Edit and publish `references/footer` (Footer block: Conversation Canvas Page, Copyright Text with `{year}`, Link List); social icons come from `data/social-links`
Locale 404 page	Create and publish `error-pages/404` (a Button in it gets the "Go back" button next to it)

15.6 Metadata Cheat Sheet

15.6.1 Page Properties (model `page-metadata`, `models/_page.json`)

Property	Name	Type	Notes
Title	`jcr:title`	text	Published as the page title / `og:title`
Description	`jcr:description`	text	Published as `description`
Keywords	`keywords`	text, multi	Published as `keywords`
Page Type	`page-type`	select: Default (empty), Home, Article, Blog, Press Release	Overrides the sheet's `page-type`
Navigation	`nav`	content reference	Inherited
Footer	`footer`	content reference	Inherited
404 Page	`error-page`	content reference	Inherited

Page tags (`<meta property="article:tag">`) are read by `utils/metadata.js getPageTags`, with a comma-separated `tags` metadata value as a fallback.

15.6.2 Bulk Metadata Columns (`/{locale}/metadata.json`)

Column	Format / example	Used by
`url`	URL pattern, `/de-de/insights/**`	Row matching (must start with the locale)
`page-type`	`article`, `blog`, `press-release`, `home` (templates); `landing`, `listing` (Page Header default Landing layout); `report`, `case-study`, news, `event` (Filter List Page types)	Template, body class, Page Header default layout, data layer `xdm:template`, Filter List, index `pageType`
`meta-title`	text	`<title>` (SEO decorator), index
`meta-description`	text	`description` meta, index
`meta-keywords`	text	`keywords` meta
`author-id`	ID matching the Authors sheet (case-insensitive)	Page Header lockup, article / blog About the author, Filter List and List feature card
`author`	name	Author fallback (Page Header, Filter List Author source)
`publish-date`	`YYYY-MM-DD`	Page Header, Filter List date and Latest sort, List date fallback
`modified-date`	`YYYY-MM-DD`	List date and Last modified sort
`reading-time`	number of minutes (6 → "6 min read" / "6m") or text	Page Header, List, Filter List
`og-title`, `og-description`, `og-image`, `og-type`	text / image URL	Open Graph tags (fallback to meta-*)
`twitter-card`, `twitter-title`, `twitter-description`, `twitter-image`	text / image URL	Twitter tags (fallback to og-, *meta-*)
`canonical-url`	absolute URL	`<link rel="canonical">`, `og:url`
`robots`	`noindex, nofollow`	Standard robots meta (server side); indexed
`region`	text	Press release Media contacts filter; indexed
`country`	text	Page metadata (default: the locale's country); indexed
`language`	language code, `de-DE`	`<html lang>` and data layer `xdm:language` when a valid code (else the locale's tag)
`persona`	text	Indexed; not used by blocks
`campaign-id`	text	Indexed; not used by blocks
`thumbnail-image`	image URL	Filter List fallback image, List card image fallback
`featured-image`	image URL	Page Header (Landing), Filter List image, List fallback
`content-type`	`Article`, `Report`, `Whitepaper`, … (Filter List Content types: article, blog, report, case-study, news, event, whitepaper, podcast, video)	Page Header eyebrow / label, Filter List eyebrow and filter, List content type
`featured`	`yes` / `true`	Filter List hero and featured cards
`priority`	number (featured cards: <= 3 when no page is featured)	Filter List featured and priority sort

Column	Format / example	Used by
`feature-card`, `featured-priority`	as featured / priority	Earlier names, read as fallbacks
`listen-link`	URL of an audio version	Page Header listen action
`card-title`, `card-description`, `card-image`, `card-image-alt`	text / image URL	List card overrides
`image-alt`	text	List card image alt fallback
`eyebrow`, `category`	text	List card eyebrow (category as fallback)
`hide-in-nav`	`yes`	Page left out of Lists
`nav`, `footer`, `error-page`	path (AEM, published or locale-relative)	Inherited Navigation, Footer and 404 Page (same as the page properties)

Note: Every column is published as `<meta name="{column}">`. The sheet applies only after the site configuration lists all locale sheets in `metadata.source` (13.17).

15.7 Taxonomy Cheat Sheet

Topic	Rule
Tag IDs	AEM tags `namespace:path` (for example `tcs:topics/cloud`); stored on pages and in the index `tags` column
Labels	`/{locale}/taxonomy.json`: column `tag` (the ID) and `title`; translated titles on a sheet named after the ISO 639-1 language (`de`, `fr`, `es`, `pt`, `nl`, `sv`, `da`, `no`, `zh`, `ja`, `ko`, `en`), else the default sheet
Fallback	Missing tags show a label derived from the ID (`utils/tags.js`); authored titles show as they are
Hierarchy	Child tags match their parent in List tag matching and the Filter List (`tag` or `tag/...`)
Where tags are authored	Page tags (page properties); Teaser Tags; Card Tags; List Item Tags; List Tags (Tagged pages); Filter List Primary filter category, Primary filter tags, Advanced filter categories (Taxonomy root)
Where labels show	Teaser, Page Header, Cards, List, Filter List pills and advanced filter, Conversation Canvas
Data layer	Page tags in the page entry `xdm:tags`
Changing tags	14.8.4; republish pages to update the index
Publishing	Publish the taxonomy page of each locale after every tag change

15.8 Component Inventory

No.	Component	Chapter	Type	Items/models	Main presentation modes	Documented variations (Part 6)	Data sources	Status
1	Header	6.1	Block (page header area, loaded by `aem.js`; no dialog)	None (the nav page: brand, sections, tools)	1: standard header (mobile menu below 900 px, desktop dropdowns from 900 px)	8	Navigation page (inherited `nav`, else `references/nav`); placeholders	Implemented
2	Footer	6.2	Block (page footer, and the authored Footer block on the footer page)	`footer` (Conversation Canvas Page, Copyright Text, Link List multifield)	1: standard footer (Figma desktop / tablet / mobile frames)	7	Footer page (inherited `footer`, else `references/footer`); Social Links sheet; Conversation Canvas page; placeholders	Implemented
3	Fragment	6.3	Block	`fragment` (Reference)	1: included page content	5	Referenced page (`.plain.html`)	Implemented
4	Page Header	6.4	Block	`page-header` (Layout, CTA List multifield)	5: Default (landing for landing and listing page types, else article); Landing; Secondary; Article; Display	13	Page metadata; Authors sheet; taxonomy; placeholders	Implemented
5	Section	6.5	Section	`section` (Section Name, Style, Width Type, Rounded edge, Background Image, Image alt text, Anchor name)	5: Default (full width); Highlight style; Fixed width; Rounded edge; Background image	6	None	Implemented
6	Text	6.6	Default content	Rich text	1: rich text	7	None	Implemented
7	Title	6.7	Default content	`title` (Title, Title Type)	1: heading (levels H1–H6)	6	None	Implemented
8	Image	6.8	Default content	`image` (Image, Alt Text)	1: image	2	AEM Assets	Implemented
9	Button	6.9	Default content	`button` (Link, Text, Title, Views, Type, Icon, Alignment)	5: Primary button; Secondary button; Tertiary button; Icon button (arrow-right, chevron-right, chevron-down, search); Link	7	None	Implemented
10	Hero	6.10	Block	`hero` (Image, Content, Links with CTA List, Style tabs)	6: Default (full-bleed image); Media contained; Lotto (Lottie) image type; Background colour (light, dark, dark subtle, black); Motion (fade, slide, zoom);	29	AEM Assets; Lottie JSON	Implemented

No.	Component	Chapter	Type	Items/models	Main presentation modes	Documented variations (Part 6)	Data sources	Status
					Alignment (vertical, horizontal, text)			
11	Teaser	6.11	Block (key-value)	`teaser` (Content, Image, Style, Links, Stats tabs)	4 layouts: Image & Text (default, image left / right); Text split (no-image); Background image; Card — with BG colour (grey, dark, dark green), Lotto image type, stats, multiple links	66	Taxonomy; placeholders; AEM Assets	Implemented
12	Quote	6.12	Block (key-value)	`quote`	3: Default; Grey background; Image position (desktop right / left, mobile top / bottom)	5	AEM Assets	Implemented
13	Testimonial	6.13	Block (key-value)	`testimonial` (Content, Author, Style, Show / Hide, Stats)	3: Default; Grey background; With stats (author details can be hidden)	9	AEM Assets	Implemented
14	Columns	6.14	Container block	`columns` (Columns, Rows); `column` items (text, image, button, title)	1: n columns (`columns-{n}-cols`; image-only columns styled)	4	None	Implemented
15	Cards	6.15	Container block (key-value)	`cards` (shared `models/_card-list.json` groups); `card` items	6 card presentations: Image; Panel; Report; Statistic; Mixed; rows of links (Display as teaser off) — Figma List variations 1–14; Style options compact statistic title, dividers, header centred	46	Taxonomy; placeholders	Implemented
16	List	6.16	Container block (key-value)	`list` (shared `models/_card-list.json` groups); `list-item` items	3 sources (Child pages, Fixed list, Tagged pages) × the 6 card presentations of Cards — Figma List variations 1–14	69	Query index; redirects sheet; Authors sheet; taxonomy; placeholders	Implemented
17	Filter List	6.17	Block (key-value)	`filter-list` (Content, Source, Filters, Advanced, Cards, Mapping)	7: Layout (masonry, grid, list); Card style (default, compact, featured, mixed); plus hero card and featured cards options	25	Query index (or manual pages); taxonomy; Authors sheet; placeholders	Implemented
18	Table List	6.18	Container block (key-value)	`table-list`; `table-list-card` items (`table-list-header` legacy model)	3: Industries; Solutions; Products	17	Placeholders	Implemented
19	Anchor List	6.19	Block (key-value)	`anchor-list` (Links multifield, Sticky)	2: Default; Sticky	2	Section Anchor names on the page; placeholders	Implemented
20	Accordion	6.20	Container block	`accordion`; `accordion-item` (Summary, Text)	1: collapsible items (`<details>`)	1	None	Implemented
21	Tabs	6.21	Container section (Tabs Container) and	`tabs` (section: ID, Width Type, Label, Style, Alignment, Orientation); `tabs-block`; `tabs-item` (Ti-	3 styles (underline, boxed, pills) × 3 alignments (left, center, stretch) × 2 orientations (horizontal, vertical); 2 sources (Tabs	12	Fragments; placeholders	Implemented

No.	Component	Chapter	Type	Items/models	Main presentation modes	Documented variations (Part 6)	Data sources	Status
			container block (Tabs)	tle, ID, Active tab, Fragment)	Container section, Tabs block with fragments)			
22	Tab	6.22	Item block (in a Tabs Container section)	`tab` (Title, ID, Active tab)	1: tab marker (shows nothing on the page)	3	None	Implemented
23	Modal	6.23	Block	`modal` (Modal Content, Button Text)	2: Modal block (button opener); automatic modal for links to `/modals/` pages	3	Fragment page; placeholders	Implemented
24	Gen Carousel	6.24	Container section	`gen-carousel` (Style, Navigation, Autoplay interval); slides: Hero, Teaser	3: Manual; Auto; Manual and auto	6	Placeholders	Implemented
25	Table	6.25	Container block	`table` (Options, Columns); `table-row`, `table-col-2` … `table-col-5`	5 column counts (1–5); options Striped, Bordered, No Header, First Column Header	12	Placeholders	Implemented
26	Separator	6.26	Block (key-value)	`separator` (General, Appearance, Responsive, Spacing, Advanced)	10 variants: default, subtle, strong, inset, full-width, dotted, dashed, gradient, branded, custom-design; 4 themes (auto, light, dark, brand)	43	Placeholders	Implemented
27	Embed	6.27	Block	`embed` (Placeholder Image, alt, URl)	4: YouTube; Vimeo; X / Twitter; other URL (iframe) — each with or without a placeholder image	5	External providers	Implemented
28	Video	6.28	Block (key-value)	`video` (Video, Playback, Captions, Display, Privacy)	3 playback modes: Click to play; Autoplay; Background — with pop-up option, 5 aspect ratios, 4 sizes, 3 alignments	33	AEM Assets, YouTube, Vimeo; WebVTT captions; placeholders	Implemented
29	Form Variations	6.29	Container block	`form-variations`; items `field-text`, `field-select`, `field-textarea`, `field-checkbox`, `contact-card`, `field-submit`	2 themes: Default; Dark — with conditional fields (Show only if dropdown value equals)	13	Placeholders (no submit endpoint)	Implemented (no back end)
30	Conversation Canvas	6.30	Container block (key-value)	`conversation-canvas` (Menu, Search bar groups); `conversation-canvas-button` items	3: Default; Primary (sticky dock); Secondary — Title size bigger / medium	18	Query index (menu); page metadata; taxonomy; placeholders (no search service)	Implemented (no search service)
31	Interactive Dashboard	6.31	Block	`interactive-dashboard` (Additional tab only)	0: placeholder	2	None	Placeholder (renders nothing on

No.	Component	Chapter	Type	Items/models	Main presentation modes	Documented variations (Part 6)	Data sources	Status
								the site)

> **Note:** "Main presentation modes" summarises the author-selectable modes; "Documented variations" counts the variation sections of each Part 6 chapter, which document every option that changes the rendering separately and are authoritative.

15.9 Screenshot Register

15.9.1 Conventions

Item	Convention
Folder	`docs/handover/screenshots/` (referenced in the document as `screenshots/<file>`; `docs/` is not served, `.hlxignore` `docs/*`)
File names	Lower case, hyphen-separated, `.png`: `<component-slug>-<purpose>.png`
Component slugs	`header`, `footer`, `fragment`, `page-header`, `section`, `text`, `title`, `image`, `button`, `hero`, `teaser`, `quote`, `testimonial`, `columns`, `cards`, `list`, `filter-list`, `table-list`, `anchor-list`, `accordion`, `tabs`, `tab`, `modal`, `gen-carousel`, `table`, `separator`, `embed`, `video`, `form-variations`, `conversation-canvas`, `interactive-dashboard`
Content	Real locale content from the `us/en` block library or a representative published page; no personal data, no credentials, no unpublished confidential content
Format	PNG, full width of the viewport, cropped to the component plus some context; annotation letters (A, B, C, …) added for authoring views

15.9.2 Standard Screenshots per Component

File name pattern	Purpose	Capture in	Viewport width
`<slug>-desktop-view.png`	Desktop rendering	Preview `https://main--tcs-corp--sradha-tcs.aem.page/en/...`	1440 px
`<slug>-tablet-view.png`	Tablet rendering	Preview	1024 px
`<slug>-mobile-view.png`	Mobile rendering	Preview (device emulation)	402 px
`<slug>-<variation-slug>.png`	One per variation (6.X.3)	Preview	1440 px (plus 402 px where the variation differs on mobile)
`<slug>-authoring-view.png`	Universal Editor canvas with the block selected; annotation letters per authorable element	Universal Editor (edit mode)	1440 px browser window

File name pattern	Purpose	Capture in	Viewport width
`<slug>-properties-panel.png`	Properties panel with the block's tabs	Universal Editor	1440 px
`<slug>-step-NN-<slug>.png`	Authoring guide steps (two-digit step number)	Universal Editor / Sites console	1440 px

15.9.3 Capture Procedure

1. Use a clean browser profile (no extensions except AEM Sidekick), zoom 100 %, light system theme.
2. Preview captures: open the page on `main--tcs-corp--sradha-tcs.aem.page`, open developer tools device toolbar, set the width (1440, 1024 or 402 px) and a height that shows the whole component, wait until the page has fully loaded (header, footer, lazy images), then capture the node or area screenshot.
3. Universal Editor captures: open the page in Universal Editor in a browser window 1440 px wide, select the block, open the properties panel on the right tab, capture.
4. Disable motion where it hides content (operating system "reduce motion" on) for static captures of carousels and animations; capture the motion state separately if needed.
5. Save under `docs/handover/screenshots/` with the exact file name used in the document.
6. Record the capture in the register below (date, page path, who).

15.9.4 Register

File name	Chapter	Page path used	Width	Captured by	Date	Status
`<slug>-desktop-view.png` (31 components)	6.1–6.31	[To be confirmed]	1440	[To be confirmed]	[To be confirmed]	Pending
`<slug>-tablet-view.png` (31 components)	6.1–6.31	[To be confirmed]	1024	[To be confirmed]	[To be confirmed]	Pending
`<slug>-mobile-view.png` (31 components)	6.1–6.31	[To be confirmed]	402	[To be confirmed]	[To be confirmed]	Pending
`<slug>-<variation-slug>.png` (per variation)	6.X.3	[To be confirmed]	1440 / 402	[To be confirmed]	[To be confirmed]	Pending
`<slug>-authoring-view.png` (31 components)	6.X.4	[To be confirmed]	1440	[To be confirmed]	[To be confirmed]	Pending
`<slug>-properties-panel.png` (31 components)	6.X.4	[To be confirmed]	1440	[To be confirmed]	[To be confirmed]	Pending
`<slug>-step-NN-<slug>.png` (per step)	6.X.5	[To be confirmed]	1440	[To be confirmed]	[To be confirmed]	Pending
`publishing-ue-publish-dialog.png`	12.1.3	Any `us/en` page	1440	[To be confirmed]	[To be confirmed]	Pending
`publishing-manage-publication.png`	12.1.4	Sites console, `us/en`	1440	[To be confirmed]	[To be confirmed]	Pending

Tip: Before capturing, generate the exact list of placeholder file names from the assembled document (search for `**File name:**`) and copy it into this register, one row per file.

Part 16 — Handover Findings and Open Items Register

16.1 How to Use This Register

While this document was written, every component, model and configuration file was read and checked against the published preview. This part collects what that review found:

- **16.2 Findings** — behaviour in the code or content that differs from expectations, documented as it is today in the relevant chapters, with a recommendation.
- **16.3 Open Items** — facts that weren't available at handover (`[To be confirmed]` in the text).
- **16.4 Not Implemented** — capabilities that are out of the current implementation.

Priorities: **P1** blocks go-live of the affected feature; **P2** should be fixed before or soon after go-live; **P3** improvement. Owners are roles until names are confirmed.

16.2 Findings

ID	Area	Finding	Impact	Recommendation	Priority	Owner	Chapter
F-01	Form Variations	A valid submit shows "Thank you, your details were sent." although there is no submit endpoint (only a `form-variations:submit` DOM event)	Visitors believe their request was sent; leads are lost	Connect a back end (Salesforce, Marketo or an API) before publishing any form, or hide forms	P1	Product Owner, Development	6.29
F-02	Analytics	`cmp:submit` is pushed before the form validates (failed attempts count) and lists all field names, including hidden conditional ones	Over-counted submissions	Push the event after validation; exclude disabled fields	P2	Development	11.4, 6.29
F-03	Analytics	`cmp:search` is pushed for empty search terms	Noise in search reports	Skip empty queries	P3	Development	11.4, 6.30
F-04	Analytics	Card clicks in List and Cards carry no `position` (the attribute is on the list item, the listener reads the link); feature card clicks report under the block id	Position reports empty for cards	Read `data-cmp-position` from the closest list item	P2	Development	6.15, 6.16, 11.6
F-05	Analytics	`mailto:` and `tel:` links are pushed in full as exit links	Contact details in analytics	Mask or classify mailto / tel links	P2	Development, Privacy	11.9
F-06	Analytics	No tag manager is loaded (`scripts/delayed.js` is empty); the Adobe Client Data Layer library isn't loaded, so `adobeDataLayer` is a plain array	No analytics collected yet	Add the Adobe Tags embed (with the Web SDK and the ACDL extension) in `delayed.js`; implement consent	P1 for analytics go-live	Analytics Lead	11.7

ID	Area	Finding	Impact	Recommendation	Priority	Owner	Chapter
F-07	Metadata	AEM may publish page tags as `cq-tags` while the code and the query index read `article:tag`	Page tags may not reach Page Header, List, Filter List and the data layer	Verify on a tagged page; align the code or the publishing	P1 for tag features	Development	7.9, 8.2
F-08	Taxonomy	The published `/en/taxonomy.json` uses the `tcs-corp:` namespace; code examples and field help use `tcs:`	Authors typing `tcs:…` match nothing	Align help texts and examples with `tcs-corp:`	P2	Content Lead, Development	7.1
F-09	Taxonomy	Only the English taxonomy is published; other locales show labels derived from tag IDs; translated titles may need `?sheet=` requests	Untranslated tag labels	Publish the taxonomy for every locale; verify language sheets	P2	Content Lead	7.1, 4.9
F-10	Query index	The live query index doesn't match `helix-query.yaml` (old columns; card, eyebrow, category, hide-in-nav, author and feature columns missing)	Card overrides, hide-in-nav and featured rules have no effect in List / Filter List	Re-index the site after the configuration update	P1 for List / Filter List	Site Administrator	6.16, 6.17, 12.5
F-11	Metadata	`/en/metadata.json` has 29 of the 39 documented columns and no rows; other locales' metadata sheets aren't published; `metadata.source` isn't confirmed	Bulk metadata doesn't apply	Add the missing columns, publish all 13 sheets, configure `metadata.source`	P2	Content Lead, Site Administrator	8.1, 9.2
F-12	Page types	Page Type offers home, article, blog, press-release; Filter List page-type filter offers article, blog, report, case-study, news, event; Page Header Landing layout needs landing / listing (sheet only)	Filters and layouts can't select some pages	Agree one list of page types and content types	P2	Product Owner	4.2, 6.4, 6.17
F-13	Header	A section label that is a direct link (`<li><a>…</a><ul>`) is replaced by a toggle and loses its link; the extra toggle for wrapped links has no visible size (mouse can't open it on desktop); on mobile, sub-lists stay visible while toggles change state	Navigation usability	Fix label detection and toggle sizing; define the mobile sub-menu behaviour	P2	Development	6.1
F-14	Page Header	A featured image that isn't same-origin site media is lazy-loaded (the eager / high priority setting is skipped)	Slower LCP on landing pages	Apply loading attributes to plain `img` too	P3	Development	6.4
F-15	Fragment	No protection against a fragment that includes itself; full URLs in Reference are ignored	A misconfigured fragment never loads	Add a recursion guard; accept same-site URLs	P3	Development	6.3
F-16	Footer	An icon name without a file gives a broken icon; the bar renders when empty; new-tab footer links aren't announced	Visual and accessibility issues	Validate icons; skip empty bars; use `announceNewTab`	P3	Development	6.2
F-17	Hero, Teaser	New-window links don't announce "(opens in a new tab)"; the Hero eyebrow stays brand blue on dark backgrounds	Accessibility (3.2.5 advisory, contrast)	Use `announceNewTab`; adjust the eyebrow colour on dark	P3	Development, Design	6.10, 6.11

ID	Area	Finding	Impact	Recommendation	Priority	Owner	Chapter
F-18	Teaser	Some options have no effect in some layouts (text alignment Center in Text split, top spacing in Background image and Card, image ratio Original for animations)	Author confusion	Hide the options with conditions where they don't apply	P3	Development	6.11
F-19	Dates	Month abbreviations are English in every locale (`utils/dates.js`)	Untranslated dates	Use `Intl.DateTimeFormat` with the page language	P2	Development	4.9
F-20	404 page	`404.html` has its title, "Page Not Found", "Go home" and "Go back" hard-coded in English	Untranslated fallback 404	Read the texts from placeholders	P3	Development	10.2
F-21	Sections	The Section dialog offers only "Highlight"; the dark, brand and light styles exist in CSS but can't be selected; Separator Theme Auto therefore always uses the light palette	Designs needing dark / brand sections can't be authored	Add the styles to the section model	P2	Development, Design	6.5, 6.26
F-22	Additional tab	Anchor ID, Hide block and spacing apply to blocks only; on section-based components (Tabs container, Gen Carousel) their effect isn't verified	Options may do nothing there	Verify in Universal Editor; remove the tab from section models or support it	P3	Development	5.6, 6.21, 6.24
F-23	Columns	The Columns / Rows dialog fields aren't read by the code (it counts the first row's cells); an empty Columns block logs `failed to load module for columns`; column widths are always equal	Author expectations; console error	Guard empty blocks; document or implement widths	P3	Development	6.14
F-24	Quote	The caption sits inside a `div` within the `figure`	HTML validation warning	Make the caption a direct child of the figure	P3	Development	6.12
F-25	Testimonial	Description shows only when Show Description is exactly `true` (empty hides it); in stats the last paragraph is always the value	Content silently hidden	Default Show Description to on; document stats entry	P3	Development	6.13
F-26	Embed	With a placeholder, YouTube gets `muted=1` (YouTube expects `mute`); no privacy-enhanced domain; Vimeo only accepts `vimeo.com/{id}` URLs	Autoplay may play with sound or be blocked; privacy	Use `mute=1` and `youtube-nocookie.com`; parse Vimeo URLs fully	P2	Development	6.27
F-27	Video	The caption colour (#999 on white, about 2.8:1) is below WCAG AA; End At isn't applied to Vimeo	Accessibility; behaviour	Darken the caption colour; implement Vimeo end	P2	Development, Design	6.28
F-28	Cards, List, Filter List	Secondary text #999 on white (about 2.8:1) and white on brand blue #4673db (about 4.2:1) are below 4.5:1 for small text	Accessibility	Design sign-off or darker tokens	P2	Design	6.15–6.17
F-29	List, Cards	The "(opens in a new tab)" hint is lost on call-to-action buttons (their generated label replaces it); Maximum items has no upper limit in code; an empty Child depth counts as 1 though the dialog shows 2	Accessibility; minor inconsistencies	Append the hint to generated labels; align defaults	P3	Development	6.15, 6.16

ID	Area	Finding	Impact	Recommendation	Priority	Owner	Chapter
F-30	Filter List	Doesn't exclude hide-in-nav pages and doesn't apply redirects (unlike List); Authors lookup is case-sensitive; downloads the whole index; the report card's Download opens the page	Inconsistent results	Align with the List's data rules	P3	Development	6.17
F-31	Table List	An item row with exactly two cells may be read as the legacy header item	An item could disappear	Verify with published content; rely on the item model	P3	Development	6.18
F-32	Form Variations	Desktop: the "* Required fields" note takes a grid column; a checkbox error appears beside the consent text; field width depends on the field name ("name" = half width)	Layout	Span the note across columns; fix the error position; use an explicit width field	P3	Development	6.29
F-33	Conversation Canvas	The footer canvas sample has the search button label "sradha" (tooltip and accessible name); Button Alignment and BG Style have no effect; the menu ignores hide-in-nav	Content and authoring confusion	Correct the label in the footer reference; remove unused options	P2	Content Lead	6.30
F-34	Cards	Cards blocks created before the key-value model can't reliably use the new block settings	Older blocks keep the defaults	Recreate those blocks (move the cards into a new Cards block)	P3	Content Lead	6.15
F-35	Inheritance	Inherited Navigation, Footer and 404 Page values are cached per browser session (sessionStorage)	Changes show in a new session only	Document for authors; consider a version key	P3	Development	3.5
F-36	Root URL	`/` returns 404 (no root mapping or redirect to `/en/`)	Visitors to the root see an error	Configure a CDN or root redirect	P1 for go-live	Site Administrator	3.8
F-37	Release	CI runs lint only (`npm test` isn't run); no release tags; `package.json` keeps the boilerplate version; the PR template and Sidekick project name keep boilerplate values	Weaker release control	Run tests in CI; tag releases; update the template and Sidekick config	P3	Development	12.2, 12.3
F-38	Fonts	The Inter font covers Latin scripts only; Chinese, Japanese and Korean use system fonts	Visual consistency in CJK locales	Add CJK font strategy	P3	Design	4.9
F-39	Localisation	No hreflang alternates, no language switcher, no translation automation in the code	SEO across locales; navigation between locales	Add hreflang in the head and a locale switcher	P2	Product Owner, Development	4.9
F-40	Data	Locations and Partners sheets have loaders but no component reads them; `persona`, `country` and `campaign-id` are indexed but unused	Unused content effort	Confirm the roadmap or retire the sheets / columns	P3	Product Owner	9.6, 9.7, 8.2

16.3 Open Items (To Be Confirmed)

ID	Item	Owner	Where used
O-01	Stakeholder names, contact matrix, RACI	Product Owner	1.4, 1.11, 14.1
O-02	Business objectives, KPIs and success metrics	Product Owner	1.2, 1.10
O-03	Production domain, CDN vendor and configuration (redirects, 403 / 500 / maintenance pages, `/` → `/en/`)	Site Administrator	2.6, 3.8, 10
O-04	Site configuration: `public.json` paths from `paths.json`, `metadata.source` for the 13 sheets	Site Administrator	3.2, 8.1, 12.5
O-05	Access restrictions of the preview and live hosts; GitHub branch protection	Site Administrator	2.7, 12.2
O-06	AEM user groups, approval workflow, Universal Editor publish destinations	AEM Administrator	5.1, 5.7
O-07	Adobe Tags property, Web SDK datastream, consent platform, XDM schema	Analytics Lead	11.7
O-08	Form back end (Salesforce, Marketo or API) and privacy notices	Product Owner	6.29
O-09	Search service for the Conversation Canvas	Product Owner	6.30
O-10	Business taxonomy (namespaces, owners, translation process)	Content Lead	7.3–7.6
O-11	Master list of page types and content types	Product Owner	4.2
O-12	Translation process (AEM translation, language copies) and per-locale legal requirements	Content Lead	4.9
O-13	SLAs, support hours, escalation contacts, monitoring tools	AMS Lead	14.1–14.5
O-14	Design sign-off for colour contrast (secondary text, white on brand blue, video caption)	Design Lead	16.2 F-27, F-28
O-15	Whether 403 and 500 pages exist in AEM for every locale (only `/en` samples exist locally)	Content Lead	10.3, 10.4
O-16	Maximum video asset size, approved embed domains, consent for third-party players	Product Owner	6.27, 6.28
O-17	Formal browser support matrix	QA Lead	1.7
O-18	Whether AEM omits never-set fields in item rows of positional blocks (Form Variations reads fields by position)	Development	6.29
O-19	Screenshots for every placeholder in this document	Content Lead, QA	15.9

16.4 Not Implemented (as of this handover)

Capability	State	Where the extension point is
Adobe Tags, Web SDK, Adobe Analytics, Customer Journey Analytics	Data layer ready; no tag loaded	`scripts/delayed.js` (Part 11.7)
Adobe Target, Adobe Journey Optimizer, personalisation	Not implemented; Teaser has a personalisation marker class only	Data layer, Tags (Part 11)
Form submission back end (Salesforce, Marketo, API)	Event only (`form-variations:submit`)	Form Variations block (6.29)
Site search service	Event only (`conversation-canvas:search`)	Conversation Canvas (6.30)

Capability	State	Where the extension point is
Locale redirects applied automatically	Sheets exist per locale; the platform applies only a root `/redirects.json`	CDN rules (3.8)
hreflang, language switcher	Not implemented	`head.html`, header (4.9)
Maintenance page	Not implemented	CDN (10.5)
Content Fragment lists in List	Not supported by decision	List (6.16)
Consent management	Not implemented	Tags / CMP (11.7)